PASSBOOK · CA INTERMEDIATE · GROUP II

Paper 4

Cost and Management Accounting

ICAI Question Bank

Every ICAI past-exam, RTP and MTP question for this paper, Nov 2019 to Sep 2026, sorted by chapter and sub-topic, each with ICAI's own answer, word for word.

1,482 questions (452 MCQs, 1,030 descriptive) from 64 ICAI documents · A4 edition · made 2026-10-21

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How this book works

- **Sorted by the syllabus.** Chapters and sub-topics follow the January 2027 syllabus (PassBook's lessons). Each question sits under the sub-topic it mainly tests. That placement is PassBook's judgement, most of all for old-scheme papers, which used different chapters.
- **Order inside a sub-topic:** exam papers first, then RTPs, then MTPs, newest attempt first.
- **Every card** shows where it comes from (EXAM RTP MTP), the attempt, ICAI's question number and marks. RTPs print no marks. old scheme marks papers from before May 2024.
- **Questions ICAI repeated** word for word, with the same answer, are printed once. The other documents are listed under *Also asked in* (52 repeats). Where ICAI answered the same question differently in two documents, both are printed, and each card points to the other: *Same question, different ICAI answer*.
- **A case scenario** stays together: the case, all its questions, then ICAI's answers in ICAI's order.
- **ICAI's words, unchanged.** Questions, options and answers are copied from ICAI's PDFs. Each card ends with the PDF pages. Lines marked *PassBook's note* are ours: why a question is out of syllabus, or where the Study Material now teaches a newer law (CARO 2020 for CARO 2016, say). ICAI's own slips are kept as printed.
- **Missing answers.** ICAI has not yet published Suggested Answers for the September 2026 exam (marked * below), and one old MTP's answers are no longer online. Those questions appear without an answer.
- **Out of syllabus** questions are in an appendix at the end (25), so they never get in the way.

Most asked sub-topics

How many separate ICAI documents asked about each sub-topic (repeats counted in every document that printed them). Revise the top rows first.

#	Sub-topic	Documents	Exams
1	Product costs under ABC versus traditional absorption Ch 6	60	16
2	Cost sheet for a manufacturer Ch 2	59	16
3	Reconciling cost and financial profit Ch 7	43	11
4	Break-even point, margin of safety and angle of incidence Ch 14	43	9
5	Apportioning joint costs: physical units, sales value and net realisable value Ch 11	39	12
6	Batch costing and the economic batch quantity Ch 8	38	7
7	Material variances Ch 13	38	7
8	Time rate, piece rate and the Halsey and Rowan premium plans Ch 4	37	13
9	Functional budgets and the master budget Ch 15	37	9
10	Transport costing: passenger-km and tonne-km Ch 12	37	7
11	Equivalent units: FIFO and weighted average Ch 10	37	6
12	Economic Order Quantity and quantity discounts Ch 3	35	12
13	Process accounts, normal loss, abnormal loss and abnormal gain Ch 10	33	14
14	Hotels, hospitals, power, IT and other services Ch 12	32	11
15	Absorption rates and the machine hour rate Ch 5	32	9

Documents in this book

32 new-scheme and 32 old-scheme ICAI documents. A dash means ICAI published none for that attempt, or it is not in the bank.

Attempt	Exam	RTP	MTP
September 2026	✓*	✓	I, II
May 2026	✓	✓	I, II
January 2026	✓	✓	I, II
September 2025	✓	✓	I, II
May 2025	✓	✓	I, II
January 2025	✓	✓	I, II
September 2024	✓	✓	I, II
May 2024	✓	✓	I, II
November 2023 Old P3	✓	✓	I, II
May 2023 Old P3	✓	✓	I, II
November 2022 Old P3	✓	✓	I, II
May 2022 Old P3	✓	✓	I, II
December 2021 Old P3	✓	✓	I, II
July 2021 Old P3	✓	✓	I, II
January 2021 Old P3	✓	–	–
November 2020 Old P3	✓	✓	I
May 2020 Old P3	–	✓	I
November 2019 Old P3	✓	–	I

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CHAPTER 1

Introduction to Cost and Management Accounting

1.1 Objectives, scope and users of cost and management accounting

3 questions · asked in 3 ICAI documents · 1 exam paper

1 EXAM **May 2025** Q6(c) 4 marks

Contemplate the list of functions given below and identify each one of them with the most relevant scope of Cost Accounting:

(i) It involves a detailed examination of each cost.

(ii) Helps in planning and control, performance appraisal and managerial decision making.

(iii) Cost involved in alternative courses of action.

(iv) To find out factors responsible for variance in actual costs from the budgeted costs.

ICAI'S ANSWER

Function	Scope of Cost Accounting
(i)	Cost Control
(ii)	Cost Report
(iii)	Cost Comparison
(iv)	Cost Analysis

ICAI · P4 · May 2025 exam · question p. 31-32 · answer p. 33

2 RTP **Old P3 Nov 2023** Q14(b) old scheme

Narrate the objectives of cost accounting.

ICAI'S ANSWER

The main objectives of introduction of a Cost Accounting System in a manufacturing organization are as follows:

(i) Ascertainment of cost: The main objective of a Cost Accounting system is to ascertain cost for cost objects. Costing may be post completion or continuous but the aim is to arrive at a complete and accurate cost figure to assist the users to compare, control and make various decisions.

(ii) Determination of selling price: Cost Accounting System in a manufacturing organisation enables to determine desired selling price after adding expected profit margin with the cost of the goods manufactured.

(iii) Cost control and Cost reduction: Cost Accounting System equips the cost controller to adhere and control the cost estimate or cost budget and assist them to identify the areas of cost reduction.

(iv) Ascertainment of profit of each activity: Cost Accounting System helps to classify cost on the basis of activity to ascertain activity wise profitability.

(v) Assisting in managerial decision making: Cost Accounting System provides relevant cost information and assists managers to make various decisions.

ICAI · Old P3 · RTP Nov 2023 · question p. 9 · answer p. 24

3 MTP Jan 2025 S-I Q6(a) 5 marks

Cost and Management Accounting information is used by different stakeholders. The users of the information can be broadly categorised into internal and external to the entity.

GIVE two examples of internal users and three examples of external users and EXPLAIN how they are concerned with the Cost and Management Accounting information.

ICAI'S ANSWER

Internal Users

Internal users, who use the Cost and Management Accounting information may include the followings:

(a) Policy Makers- The policy makers are those who formulate strategies

(i) to achieve the goals (short & long term both) to fulfil the objectives of the organisation.

(ii) to position the organisation into the competitive market environment.

(iii) to design the organisational structure to get the policy and strategies implemented. etc.

(b) Managers- The managers use the information

(i) to know the cost of a cost object and cost centre

(ii) to know the price for the product or service

(iii) to measure and evaluate performance of responsibility centres

(iv) to the know the profitability-product-wise, department-wise, customer-wise etc.

(v) to evaluate the strategic options and to make decisions

(c) Operational level staff- The operational level staff like supervisors, foreman, team leaders require information

(i) to know the objectives and performance goals for them

(ii) to know product and service specifications like volume, quality and process etc.

(iii) to know the performance parameters against which their performance is measured and evaluated.

(iv) to know divisional (responsibility centre) profitability etc.

(d) Employees- Employees are concerned with the information related with time and attendance, incentives for work, performance standards etc.

External Users

External users, who use the Cost and Management Accounting information may include the followings:

(a) **Regulatory Authorities-** Regulatory Authorities are concerned with cost accounting data and information for different purpose which includes tariff determination, providing subsidies, rate fixation etc. To do this the regulatory bodies require information on the basis of some standards and format in this regard.

(b) **Auditors-** The auditors while conducting audit of financial accounts or for some other special purpose audit like cost audit etc. require information related with costing and reports reviewed by management etc.

(c) **Shareholders-** Shareholders are concerned with information that effect their investment in the entity. Management communicates to the shareholders through periodic communique, annual reports etc. regarding new orders received, product expansion, market share for products etc.

(d) **Creditors and Lenders-** Creditors and lenders are concerned with data and information which affects an entity's ability to serve lenders or creditors. For example, any financial institutions which provides loan to an entity against book debts and inventories are more concerned with regular reporting on net debt position and stock balances.

ICAI · P4 · MTP Jan 2025 · Series I · question p. 16 · answer p. 18-20

1.2 Cost accounting department and installing a costing system

9 questions · asked in 10 ICAI documents · 1 exam paper

4 EXAM **Old P3 May 2022** Q6(a) 5 marks old scheme

Briefly explain the essential features of a good Cost Accounting System.

ICAI'S ANSWER

The essential features, which a good cost accounting system should possess, are as follows:

(a) Informative and simple: Cost accounting system should be tailor-made, practical, simple and capable of meeting the requirements of a business concern. The system of costing should not sacrifice the utility by introducing inaccurate and unnecessary details.

(b) Accurate and authentic: The data to be used by the cost accounting system should be accurate and authenticated; otherwise it may distort the output of the system and a wrong decision may be taken.

(c) Uniformity and consistency: There should be uniformity and consistency in classification, treatment and reporting of cost data and related information. This is required for benchmarking and comparability of the results of the system for both horizontal and vertical analysis.

(d) Integrated and inclusive: The cost accounting system should be integrated with other systems like financial accounting, taxation, statistics and operational research etc. to have a complete overview and clarity in results.

(e) Flexible and adaptive: The cost accounting system should be flexible enough to make necessary amendment and modifications in the system to incorporate changes in technological, reporting, regulatory and other requirements.

(f) Trust on the system: Management should have trust on the system and its output. For this, an active role of management is required for the development of such a system that reflects a strong conviction in using information for decision making.

ICAI · Old P3 · May 2022 exam · question p. 26 · answer p. 27-28

5 RTP **Sep 2026** Q16(a)

You have been asked to install a costing system in a manufacturing company. What practical difficulties will you expect and how will you propose to overcome the same?

ICAI'S ANSWER

The practical difficulties with which one usually confronted with while installing a costing system in

a manufacturing company are as follows:

(i) Lack of top management support: Installation of a costing system does not receive the adequate support of top management. They consider it as interference in their work. They believe that such, a system will involve additional paperwork. They also have a misconception in their minds that the system is meant for keeping a check on their activities.

(ii) Resistance from cost accounting departmental staff: The staff resists because of fear of loosing their jobs and importance after the implementation of the new system.

(iii) Non-co-operation from user departments: The foremen, supervisor and other staff members may not co-operate in providing requisite data, as this would not only add to their responsibilities but will also increase paper work of the entire team as well.

(iv) Shortage of trained staff: Since cost accounting system's installation involves specialised work, there may be a shortage of trained staff.

To overcome these practical difficulties, necessary steps required are:

- To sell the idea to top management – To convince them of the utility of the system.
- Resistance and non-co-operation can be overcome by behavioral approach. To deal with the staff concerned effectively.
- Proper training should be given to the staff at each level

Regular meetings should be held with the cost accounting staff, user departments, staff and top management to clarify their doubts/misgivings.

ICAI · P4 · RTP Sep 2026 · question p. 19 · answer p. 47-48

6 RTP **Old P3 Dec 2021** Q15(b) old scheme

'Like other branches of accounting, cost accounting also has certain limitations'. EXPLAIN the limitations.

ICAI'S ANSWER

"Like other branches of accounting, cost accounting also has certain limitations". The limitations of cost accounting are as follows:

(i) **Expensive:** It is expensive because analysis, allocation and absorption of overheads requires considerable amount of additional work, and hence additional money.

(ii) **Requirement of reconciliation:** The results shown by cost accounts differ from those shown by financial accounts. Thus, preparation of reconciliation statements is necessary to verify their accuracy.

(iii) **Duplication of work**: It involves duplication of work as organization has to maintain two sets of accounts i.e. Financial Accounts and Cost Accounts.

ICAI · Old P3 · RTP Dec 2021 · question p. 13 · answer p. 34

7 MTP **Sep 2026 S-I** Q6(b) 5 marks

Also asked in MTP May 2024 S-II Q6(b)

Same question, different ICAI answer: #12 Old P3 MTP Nov 2019 S-I

DISCUSS the essential features of a good cost accounting system?

ICAI'S ANSWER

The essential features which a good Cost Accounting System should possess, are as follows:

(a) Informative and Simple: Cost Accounting System should be tailor-made, practical, simple and capable of meeting the requirements of a business concern. The system of costing should not sacrifice the utility by introducing meticulous and unnecessary details.

(b) Accuracy: The data to be used by the Cost Accounting System should be accurate; otherwise it may distort the output of the system, and a wrong decision may be taken.

(c) Support from Management and subordinates: Necessary cooperation and participation of executives from various departments of the concern is essential for developing a good system of Cost Accounting.

(d) Cost-Benefit: The Cost of installing and operating the system should justify the results.

(e) Procedure: A carefully phased programme should be prepared by using network analysis for the introduction of the system.

(f) Trust: Management should have faith in the Costing System and should also provide a helping hand for its development and success.

ICAI · P4 · MTP Sep 2026 · Series I · question p. 15 · answer p. 21

8 MTP **May 2025 S-II** Q6(a) 5 marks

ENUMERATE the factors which are to be considered before installing a system of cost accounting in a manufacturing organization.

ICAI'S ANSWER

Before installation of a system of cost accounting in a manufacturing organisation the under mentioned factors should be studied:

(a) **Objective:** The objective of costing system, for example whether it is being introduced for fixing prices or for insisting a system of cost control.

(b) **Nature of Business or Industry:** The Industry in which business is operating. Every business industry has its own peculiar feature and costing objectives. According to its cost information requirement cost accounting methods are followed. For example Indian Oil Corporation Ltd. has to maintain process wise cost accounts to find out cost incurred on a particular process say in crude refinement process etc.

(c) **Organisational Hierarchy:** Costing system should fulfill the requirement of different level of management. Top management is concerned with the corporate strategy, strategic level management is concerned with marketing strategy, product diversification, product pricing etc. Operational level management needs the information on standard quantity to be consumed, report on idle time etc.

(d) **Knowing the product:** Nature of product determines the type of costing system to be implemented. The product which has by-products requires costing system which account for by-products as well. In case of perishable or short self- life, marginal costing method is required to know the contribution and minimum price at which it can be sold.

(e) **Knowing the production process:** A good costing system can never be established without the complete knowledge of the production process. Cost apportionment can be done on the most appropriate and scientific basis if a cost accountant can identify degree of effort or resources

consumed in a particular process. This also includes some basic technical know-how and process peculiarity.

(f) **Information synchronisation:** Establishment of a department or a system requires substantial amount of organisational resources. While drafting a costing system, information needs of various other departments should be taken into account. For example in a typical business organisation accounts department needs to submit monthly stock statement to its lender bank, quantity wise stock details at the time filing returns to tax authorities etc.

(g) **Method of maintenance of cost records:** The manner in which Cost and Financial accounts could be inter-locked into a single integral accounting system and in which results of separate sets of accounts, cost and financial, could be reconciled by means of control accounts.

(h) **Statutory compliances and audit:** Records are to be maintained to comply with statutory requirements, standards to be followed (Cost Accounting Standards and Accounting Standards).

(i) **Information Attributes**: Information generated from the Costing system should be possess all the attributes of an information i.e. complete, accurate, timeliness, confidentiality etc. This also meets the requirements of management information system.

ICAI · P4 · MTP May 2025 · Series II · question p. 14 · answer p. 17-19

9 MTP **Jan 2025 S-II** Q6(a) 5 marks

As a consultant hired by a manufacturing company, HOW would you go about assessing the critical factors for designing and implementing a cost accounting system?

ICAI'S ANSWER

Before installation of a system of cost accounting in a manufacturing organisation the under mentioned factors should be studied:

(a) **Objective:** The objective of costing system, for example whether it is being introduced for fixing prices or for insisting a system of cost control.

(b) **Nature of Business or Industry:** The Industry in which business is operating. Every business industry has its own peculiar feature and costing objectives. According to its cost information requirement cost accounting methods are followed. For example Indian Oil Corporation Ltd. has to maintain process wise cost accounts to find out cost incurred on a particular process say in crude refinement process etc.

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(g) **Method of maintenance of cost records:** The manner in which Cost and Financial accounts could be inter-locked into a single integral accounting system and in which results of separate sets of accounts, cost and financial, could be reconciled by means of control accounts.

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(i) **Information Attributes:** Information generated from the Costing system should be possess all the attributes of an information i.e. complete, accurate, timeliness, confidentiality etc. This also meets the requirements of management information system.

ICAI · P4 · MTP Jan 2025 · Series II · question p. 13 · answer p. 18-19

10 MTP **Sep 2024 S-II** Q6(a) 5 marks

Management of Tillu manufacturing co. is thinking of installing a costing system its company. What practical DIFFICULTIES management will expect and how management will OVERCOME the same?

ICAI'S ANSWER

The Practical difficulties with which a Cost Accountant is usually confronted with while installing a costing system in a manufacturing company are as follows:

(i) Lack of top management support: Installation of a costing system does not receive the support of top management. They consider it as interference in their work. They believe that such, a system will involve additional paperwork. They also have a misconception in their minds that the system is meant for keeping a check on their activities.

(ii) Resistance from cost accounting departmental staff: The staff resist because of fear of loosing their jobs and importance after the implementation of the new system.

(iii) Non co-operation from user departments: The foremen, supervisor and other staff members may not cooperate in providing requisite data, as this would not only add to their responsibilities but will also increase paper work of the entire team as well.

(iv) Shortage of trained staff: Since cost accounting system's installation involves specialised work, there may be a shortage of trained staff.

To overcome these practical difficulties, necessary steps required are:

- Sell the idea to top management and convince them of the utility of the system.
- Resistance and non co-operation can be overcome by behavioural approach. To deal with the staff concerned effectively.
- Proper training should be given to the staff at each level
- Regular meetings should be held with the cost accounting staff, user departments, staff and top management to clarify their doubts/ misgivings.

ICAI · P4 · MTP Sep 2024 · Series II · question p. 12 · answer p. 19-20

11 MTP **Old P3 Nov 2022 S-II** Q6(a) 5 marks old scheme

DISCUSS the essentials of good Cost Accounting System.

ICAI'S ANSWER

The essential features, which a good cost and management accounting system should possess, are as follows:

(a) Informative and simple: Cost and management accounting system should be tailor-made, practical, simple and capable of meeting the requirements of a business concern. The system of costing should not sacrifice the utility by introducing meticulous and unnecessary details.

(b) Accurate and authentic: The data to be used by the cost and management accounting system should be accurate and authenticated; otherwise it may distort the output of the system and a wrong decision may be taken.

(c) Uniformity and consistency: There should be uniformity and consistency in classification, treatment and reporting of cost data and related information. This is required for benchmarking and comparability of the results of the system for both horizontal and vertical analysis.

(d) Integrated and inclusive: The cost and management accounting system should be integrated with other systems like financial accounting, taxation, statistics and operational research etc. to have a complete overview and clarity in results.

(e) Flexible and adaptive: The cost and management accounting system should be flexible enough to make necessary amendments and modification in the system to incorporate changes in technological, reporting, regulatory and other requirements.

(f) Trust on the system: Management should have trust on the system and its output. For this, an active role of management is required for the development of such a system that reflect a strong conviction in using information for decision making.

ICAI · Old P3 · MTP Nov 2022 · Series II · question p. 7 · answer p. 14

12 MTP **Old P3 Nov 2019 S-I** Q6(a) 5 marks old scheme

Same question, different ICAI answer: #7 MTP Sep 2026 S-I

DISCUSS the essential features of a good cost accounting system.

ICAI'S ANSWER

The essential features, which a good cost and management accounting system should possess, are as follows:

(i) **Informative and simple:** Cost and management accounting system should be tailor-made, practical, simple and capable of meeting the requirements of a business concern. The system of costing should not sacrifice the utility by introducing meticulous and unnecessary details.

(ii) **Accurate and authentic:** The data to be used by the cost and management accounting system should be accurate and authenticated; otherwise it may distort the output of the system and a wrong decision may be taken.

(iii) **Uniformity and consistency:** There should be uniformity and consistency in classification, treatment and reporting of cost data and related information. This is required for benchmarking and comparability of the results of the system for both horizontal and vertical analysis.

(iv) **Integrated and inclusive:** The cost and management accounting system should be integrated with other systems like financial accounting, taxation, statistics and operational research etc. to have

a complete overview and clarity in results.

(v) **Flexible and adaptive:** The cost and management accounting system should be flexible enough to make necessary amendments and modification in the system to incorporate changes in technological, reporting, regulatory and other requirements.

(vi) **Trust on the system:** Management should have trust on the system and its output. For this, an active role of management is required for the development of such a system that reflects a strong conviction in using information for decision making.

ICAI · Old P3 · MTP Nov 2019 · Series I · question p. 8 · answer p. 11

1.3 Cost, financial and management accounting, and financial management

4 questions · asked in 6 ICAI documents · 1 exam paper

13 EXAM **Old P3 Nov 2020** Q6(a) 5 marks old scheme

Also asked in Old P3 RTP May 2020 Q13(a) · Old P3 MTP May 2023 S-I Q6(b)

Differentiate between "Cost Accounting and Management Accounting".

ICAI'S ANSWER

Difference between Cost Accounting and Management Accounting

	Basis	Cost Accounting	Management Accounting
(i)	**Nature**	It records the quantitative aspect only.	It records both qualitative and quantitative aspect.
(ii)	**Objective**	It records the cost of producing a product and providing a service.	It Provides information to management for planning and co-ordination.
(iii)	**Area**	It only deals with cost Ascertainment.	It is wider in scope as it includes financial accounting, budgeting, taxation, planning etc.
(iv)	**Recording of data**	It uses both past and present figures.	It is focused with the projection of figures for future.
(v)	**Development**	Its development is related to industrial revolution.	It develops in accordance to the need of modern business world.
(vi)	**Rules and Regulation**	It follows certain principles and procedures for recording costs of different products.	It does not follow any specific rules and regulations.

ICAI · Old P3 · Nov 2020 exam · question p. 22 · answer p. 22-23

14 RTP **Sep 2025** Q18(a)

Though Cost Accounting and Management Accounting are used synonymously, but there are a few differences. ENUMERATE any five differences.

ICAI'S ANSWER

Difference between Cost Accounting and Management Accounting

	Basis	Cost Accounting	Management Accounting
(i)	**Nature**	It records the quantitative aspect only.	It records both qualitative and quantitative aspect.
(ii)	**Objective**	It records the cost of producing a product and providing a service.	It provides information to management for planning and co-ordination.
(iii)	**Area**	It only deals with cost Ascertainment.	It is wider in scope as it includes financial accounting, budgeting, taxation, planning etc.
(iv)	**Recording of data**	It uses both past and present figures.	It is focused with the projection of figures for future.
(v)	**Development**	Its development is related to industrial revolution.	Its development is related to the need of modern business world.
(vi)	**Rules and Regulation**	It follows certain principles and procedures for recording costs of different products.	It does not follow any specific rules and regulations.

ICAI · P4 · RTP Sep 2025 · question p. 22 · answer p. 54

15 RTP **Old P3 Nov 2022** Q15(c) old scheme

Though Cost Accounting and Management Accounting is used synonymously but there are a few differences. Elaborate those differences.

ICAI'S ANSWER

	Basis	Cost Accounting	Management Accounting
(i)	**Nature**	It records the quantitative aspect only.	It records both qualitative and quantitative aspect.
(ii)	**Objective**	It records the cost of producing a product and providing a service.	It provides information to management for planning and co-ordination.
(iii)	**Area**	It only deals with cost Ascertainment.	It is wider in scope as it includes financial accounting, budgeting, taxation, planning etc.
(iv)	**Recording of data**	It uses both past and present figures.	It is focused with the projection of figures for future.
(v)	**Development**	Its development is related to industrial revolution.	Its development is related to the need of modern business world.
(vi)	**Rules and Regulation**	It follows certain principles and procedures for recording costs of different products.	It does not follow any specific rules and regulations.

ICAI · Old P3 · RTP Nov 2022 · question p. 13 · answer p. 35

16 MTP **Old P3 Dec 2021 S-I** Q6(b) 5 marks old scheme

EXPLAIN the difference between Cost Accounting and Management Accounting

ICAI'S ANSWER

Difference between Cost Accounting and Management Accounting

	Basis	Cost Accounting	Management Accounting
(i)	Nature	It records the quantitative aspect only.	It records both qualitative and quantitative aspect.
(ii)	Objective	It records the cost of producing a product and providing a service.	It Provides information to management for planning and co-ordination.
(iii)	Area	It only deals with cost Ascertainment.	It is wider in scope as it includes financial accounting, budgeting, taxation, planning etc.
(iv)	Recording of data	It uses both past and present figures.	It is focused with the projection of figures for future.
(v)	Development	Its development is related to industrial revolution.	It develops in accordance to the need of modern business world.
(vi)	Rules and Regulation	It follows certain principles and procedures for recording costs of different products.	It does not follow any specific rules and regulations.

ICAI · Old P3 · MTP Dec 2021 · Series I · question p. 6 · answer p. 11-12

1.4 Cost units, cost centres, cost objects and responsibility centres

19 questions · asked in 20 ICAI documents · 6 exam papers

17 EXAM **Sep 2025** Q6(a) 5 marks

Define Responsibility Centre and discuss the types of Responsibility Centres.

ICAI'S ANSWER

As the organization grows, its functions, organisational structure and other related functions also grow in terms of volume and complexity. To have a better control over the organisation, management delegates its responsibility and authority to various departments or persons. These departments or persons are known as responsibility centres and are held responsible for performance in terms of expenditure, revenue, profitability and return on investment.

Types of Responsibility Centres are as follows:

(i) **Cost Centres:** The responsibility centre which is held accountable for incurrence of costs which are under its control.

(ii) **Revenue Centres:** The responsibility centres which are accountable for generation of revenue for the entity.

(iii) **Profit Centres:** These are the responsibility centres which have both responsibility of generation of revenue and incurrence of expenditures. Profitability is the basis for measurement of performance of these responsibility centres.

(iv) **Investment Centres:** These are the responsibility centres which are not only responsible for profitability but also have the authority to make capital investment decisions. The performance of these responsibility centres are measured on the basis of Return on Investment (ROI) besides profit.

ICAI · P4 · Sep 2025 exam · question p. 26 · answer p. 27

18 EXAM **Jan 2025** Q6(b) 5 marks

Also asked in MTP May 2026 S-II Q6(a)

State cost unit of the following Industry Sector:

Industry Sector	Cost Unit
Oil	
Professional services	
Education	
Brick-making	
Education	
Brick-making	
Engineering	
Electricity	
Hotel/Catering	
Coal mining	
Brewing	
Hospitals	

ICAI'S ANSWER

Service industry	Cost Unit
Oil	Cost per barrel, Cost per gallon/litre
Professional Services	Per service/project, per hour
Education	Per course, per student, per batch, per lecture etc.
Brick-making	Per brick, per thousand bricks
Engineering	Per project, per hour, per job, per contract
Electricity	Per kilowatt-hour (kWh)
Hotel/Catering	Guest Days or Room Days, Per item, per meal etc.
Coal Mining	Per ton, per quintal

Service industry	Cost Unit
Brewing	per gallon/litre, per barrel
Hospital	Patient per day, room per day or per bed, per operation etc.

ICAI · P4 · Jan 2025 exam · question p. 27 · answer p. 28-29

19 EXAM **Old P3 May 2023** Q6(a) 5 marks old scheme

Define cost objects and give examples of any four cost objects,

ICAI'S ANSWER

Definition of cost objects

Cost object is anything for which a separate measurement of cost is required. Cost object may be a product, a service, a project, a customer, a brand category, an activity, a department or a programme etc.

Examples of cost objects

Product	Smart phone, Tablet computer, SUV Car, Book etc.
Service	An airline flight from Delhi to Mumbai, Concurrent audit assignment, Utility bill payment facility etc.
Project	Metro Rail project, Road projects etc.
Activity	Quality inspection of materials, Placing of orders etc.
Process	Refinement of crudes in oil refineries, melting of billets or ingots in rolling mills etc.
Department	Production department, Finance & Accounts, Safety etc.

ICAI · Old P3 · May 2023 exam · question p. 28 · answer p. 29

20 EXAM **Old P3 Nov 2022** Q6(e) 5 marks old scheme

Mention the cost units (physical measurements) for the following Industry/product:

(i) Automobile

(ii) Gas

(iii) Brick works

(iv) Power

(v) Steel

(vi) Transport (by road)

(vii) Chemical

(viii) Oil

(ix) Brewing

(x) Cement

ICAI'S ANSWER

Industry or Product	Cost Units
Automobile	Number
Gas	Cubic feet
Brick works	1,000 bricks
Power	Kilo-watt hour (kWh)
Steel	Tonne
Transport (by road)	Passenger- kilometer or Tonne-kilometer
Chemical	Litre, gallon, kilogram, tonne etc.
Oil	Barrel, tonne, litre
Brewing	Barrel
Cement	Ton/ per bag etc.

ICAI · Old P3 · Nov 2022 exam · question p. 23 · answer p. 26

21 EXAM **Old P3 Jul 2021** Q6(a) 5 marks old scheme

Specify the types of Responsibility centres under the following situations:

(i) Purchase of bonds, stocks, or real estate property.

(ii) Ticket counter in a Railway station.

(iii) Decentralized branches of an organization.

(iv) Maharana, Navratna and Miniratna public sector undertaking (PSU) of Central Government.

(v) Sales Department of an organization.

ICAI'S ANSWER

Particulars	Types of Responsibility Centre
(i) Purchase of bonds, stocks, or real estate property.	Investment Centre
(ii) Ticket counter in a Railway station.	Revenue Centre
(iii) Decentralized branches of an organization.	Profit Centre
(iv) Maharatna, Navratna and Miniratna public sector undertaking (PSU) of Central Government.	Investment Centre
(v) Sales Department of an organization.	Revenue Centre

ICAI · Old P3 · Jul 2021 exam · question p. 78 · answer p. 79

22 EXAM Old P3 Nov 2019 Q6(d) 5 marks old scheme

Mention the Cost Unit of the following Industries:

(i) Electricity

(ii) Automobile

(iii) Cement

(iv) Steel

(v) Gas

(vi) Brick Making

(vii) Coal Mining

(viii) Engineering

(ix) Professional Services

(x) Hospital

ICAI'S ANSWER

Cost Unit of Industries:

S. No.	Industry	Cost Unit Basis
(i)	Electricity	Kilowatt-hour (kWh)
(ii)	Automobile	Number
(iii)	Cement	Ton/ per bag etc.
(iv)	Steel	Ton
(v)	Gas	Cubic feet
(vi)	Brick-making	1,000 bricks
(vii)	Coal mining	Tonne/ton
(viii)	Engineering	Contract, job
(ix)	Professional services	Chargeable hour, job, contract
(x)	Hospitals	Patient day

ICAI · Old P3 · Nov 2019 exam · question p. 23-24 · answer p. 26

23 RTP **Sep 2025** Q18(d)

DESCRIBE cost unit and give EXAMPLE(S) of cost unit for the following Industry or Product:

Industry or Product
Construction
Transport
Technology
Oil
Gas
Electricity

ICAI'S ANSWER

Cost unit is a unit of product, service or time (or combination of these) in relation to which costs may be ascertained or expressed. These are usually the units of physical measurement like number, weight, area, volume, length, time and value.

Industry or Product	Cost unit(s)
Construction	Square meter, cubic meter
Transport	Passenger- kilometer, tonne-kilometer
Technology	Software license, hardware component
Oil	Barrel, tonne, litre
Gas	Cubic feet
Electricity	Kilowatt-hour (kWh)

ICAI · P4 · RTP Sep 2025 · question p. 23 · answer p. 56

24 RTP **Sep 2024** Q19(d)

Same question, different ICAI answer: #35 Old P3 MTP Jul 2021 S-I

DISCUSS the Standard and Discretionary Cost Centres.

ICAI'S ANSWER

(i) **Standards Cost Centres:** Cost Centre where output is measurable and input required for the output can be specified. Based on a well-established study, an estimate of standard units of input to produce a unit of output is set. The actual cost for inputs is compared with the standard cost. Any deviation (variance) in cost is measured and analysed into controllable and uncontrollable cost. The manager of the cost centre is supposed to comply with the standard and held responsible for adverse cost variances. The input-output ratio for a standard cost centre is clearly identifiable.

(ii) **Discretionary Cost Centre:** The cost centre whose output cannot be measured in financial terms, thus input-output ratio cannot be defined. The cost of input is compared with allocated budget for the activity. Example of discretionary cost centres are Research & Development department,

Advertisement department where output of these department cannot be measured with certainty and co-related with cost incurred on inputs.

ICAI · P4 · RTP Sep 2024 · question p. 16 · answer p. 37-38

25 RTP **Old P3 May 2023** Q15(a) old scheme

SUGGEST the unit of cost for following industries:

(a) Transport

(b) Power

(c) Hotel

(d) Hospital

(e) Steel

(f) Coal mining

(g) Professional Services

(h) Gas

(i) Engineering

(j) Oil

ICAI'S ANSWER

Cost units are as follows:

Industry or Product	Cost Unit Basis
Transport	Passenger- kilometer
Power	Kilo-watt hour (kWh)
Hotel	Room
Hospitals	Patient day
Steel	Ton
Coal mining	Tonne/ton
Professional services	Chargeable hour, job, contract
Gas	Cubic feet
Engineering	Contract, job
Oil	Barrel, tonne, litre

ICAI · Old P3 · RTP May 2023 · question p. 11 · answer p. 26

26 RTP **Old P3 Nov 2022** Q15(d) old scheme

What are cost units? Write the cost unit basis against each of the following Industry/Product-Automobile, Steel, Cement, Chemicals, Power and Transport.

ICAI'S ANSWER

Cost units are usually the units of physical measurement like number, weight, area, volume, length, time and value.

Industry or Product	Cost Unit Basis
Automobile	Number
Steel	Ton
Cement	Ton/ per bag etc.
Chemicals	Litre, gallon, kilogram, ton etc.
Power	Kilo-watt hour (kWh)
Transport	Passenger- kilometer

ICAI · Old P3 · RTP Nov 2022 · question p. 13 · answer p. 36

27 RTP **Old P3 Nov 2020** Q15(a) old scheme

DISCUSS short notes on (i) Discretionary Cost Centre and (ii) Investment Centre

ICAI'S ANSWER

(i) **Discretionary Cost Centre:** The cost centre whose output cannot be measured in financial terms, thus input-output ratio cannot be defined. The cost of input is compared with allocated budget for the activity. Example of discretionary cost centres are Research & Development department, Advertisement department where output of these department cannot be measured with certainty and co-related with cost incurred on inputs.

(ii) **Investment Centres:** These are the responsibility centres which are not only responsible for profitability but also has the authority to make capital investment decisions. The performance of these responsibility centres are measured on the basis of Return on Investment (ROI) besides profit. Examples of investment centres are Maharatna, Navratna and Miniratna companies of Public Sector Undertakings of Central Government.

ICAI · Old P3 · RTP Nov 2020 · question p. 12 · answer p. 30

28 MTP **May 2026 S-I** Q6(a) 5 marks

SPECIFY the responsibility centre type for each of the following scenarios:

Sr. No.	Particulars	Type of Responsibility Centre
(i)	Decentralised branches of an organisation	
(ii)	Public sector undertaking of Central Government	
(iii)	Customer support department of an e-commerce company	
(iv)	Restaurant unit in a hotel	
(v)	Reservation department of airlines	

ICAI'S ANSWER

Sr. No.	Particulars	Type of Responsibility Centre
(i)	Decentralised branches of an organisation	Profit Centre
(ii)	Public sector undertaking of Central Government	Investment Centre
(iii)	Customer support department of an e-commerce company	Cost Centre
(iv)	Restaurant unit in a hotel	Profit Centre
(v)	Reservation department of airlines	Revenue Centre

ICAI · P4 · MTP May 2026 · Series I · question p. 14 · answer p. 19

29 MTP **Jan 2026 S-I** Q6(c) 4 marks

WRITE DOWN the cost units for the following industry-

Power, Steel, Transport, Brewing, Electricity, Oil, Hotel, Hospital

ICAI'S ANSWER

Industry	Cost Unit
Power	Kilo-watt hour (kWh)
Steel	Ton
Transport	Passenger- kilometer
Brewing	Barrel
Electricity	Kilowatt-hour (kWh)
Oil	Barrel, tonne, litre
Hotel	Room
Hospital	Patient day

ICAI · P4 · MTP Jan 2026 · Series I · question p. 16 · answer p. 21

30 MTP **Jan 2026 S-II** Q3(b) 5 marks

Phoenix Manufacturing Ltd. has grown significantly and now operates through several specialised departments. The management wants to assign each department as an appropriate responsibility centre:

1. The Assembly Department works with fixed input–output norms. Standard labour hours and material quantities are established for each unit, and any deviation is analysed as variance.
2. The Corporate R&D Lab focuses on product innovation. Its output cannot be measured financially, and its performance is evaluated based only on the annual budget allocated.
3. The West Zone Sales Department is responsible for achieving sales targets and generating revenue. It does not control production costs, but it incurs selling expenses such as sales commission.

4. The Global Operations Division has authority over revenue, cost, and investment decisions, and its performance is evaluated based on ROI.

Required:

IDENTIFY the type of responsibility centre for each of the four departments above and SHOW the reason for your classification.

ICAI'S ANSWER

1. Assembly Department – Standard Cost Centre

The Assembly Department operates with fixed input–output norms, and standard quantities of materials and labour are established for each unit. Any deviation (variance) in cost is measured and analysed into controllable and uncontrollable cost. Since its output is measurable and the input-output relationship is clearly defined, it qualifies as a Standard Cost Centre, where the manager is responsible for controlling costs and complying with established standards.

2. Corporate R&D Lab – Discretionary Cost Centre

The R&D Lab focuses on product innovation, and its output cannot be measured in financial terms. The performance of this department is evaluated only against the budget allocated. Because the input-output ratio cannot be defined and the cost cannot be directly linked to output, it is classified as a Discretionary Cost Centre.

3. West Zone Sales Department – Revenue Centre

This department is responsible for generating revenue and achieving sales targets but has no control over production or administrative costs, except for selling expenses like commissions. As its primary responsibility is revenue generation, it is classified as a Revenue Centre, where performance is measured based on revenue achieved.

4. Global Operations Division – Investment Centre

The Global Operations Division has authority over revenue, cost, and capital investment decisions. Its performance is evaluated on profitability as well as Return on Investment (ROI). Therefore, it is classified as an Investment Centre, where managers are not only responsible for profitability but also have the authority to make capital investment decisions.

ICAI · P4 · MTP Jan 2026 · Series II · question p. 13 · answer p. 15-16

31 MTP **Sep 2025 S-I** Q6(a) 5 marks

EXPLAIN Responsibility Centre and its types in brief.

ICAI'S ANSWER

Responsibility Centres: To have a better control over the organisation, management delegates its responsibility and authority to various departments or persons. These departments or persons are known as responsibility centres and are held responsible for performance in terms of expenditure, revenue, profitability and return on investment.

Four types of responsibility centres are explained below:

- **Cost Centres:** The responsibility centre which is held accountable for incurrence of costs which are under its control. The performance of this responsibility centre is measured against pre-determined standards or budgets.

- **Revenue Centres**: The responsibility centres which are accountable for generation of revenue for the entity.
- **Profit Centres**: These are the responsibility centres which have both responsibility of generating revenue and incurrence of expenditures. Since, managers of profit centres are accountable for both costs as well as revenue, profitability is the basis for measurement of performance of these responsibility centres.
- **Investment Centres:** These are the responsibility centres which are not only responsible for profitability but also has the authority to make capital investment decisions. The performance of these responsibility centres are measured on the basis of Return on Investment (ROI) besides profit.

ICAI · P4 · MTP Sep 2025 · Series I · question p. 15 · answer p. 18-19

32 MTP **Sep 2024 S-I** Q6(a) 5 marks

EXPLAIN the types of responsibility centres.

ICAI'S ANSWER

The various types of responsibility centres are as follows:

(i) Cost Centres: The responsibility centre which is held accountable for *incurrence of costs* which are under its control. The performance of this responsibility centre is measured against pre-determined standards or budgets. The cost centres are of two types:

(a) Standard Cost Centre and (b) Discretionary Cost Centre

(a) Standard Cost Centre: Cost Centre where *output is measurable and input required for the output can be specified*. Based on a well-established study, an estimate of standard units of input to produce a unit of output is set. The actual cost for inputs is compared with the standard cost. Any deviation (variance) in cost is measured and analysed into controllable and uncontrollable cost. The manager of the cost centre is expected to comply with the standard and held responsible for adverse cost variances. The input-output ratio for a standard cost centre is clearly identifiable.

(b) Discretionary Cost Centre: The cost centre *whose output cannot be measured in financial terms, thus input-output ratio cannot be defined*. The cost of input is compared with allocated budget for the activity. Examples of discretionary cost centres are Research & Development department, Advertisement department where output of these department cannot be measured with certainty and co-related with cost incurred on inputs.

(ii) Revenue Centres: The responsibility centres which are accountable for *generation of revenue for the entity*. Sales Department for example, is responsible for achievement of sales target and revenue generation. Though, revenue centres do not have control on expenditures it incurs but sometimes expenditures related with selling activities like commission to sales person etc. are incurred by revenue centres.

(iii) Profit Centres: These are the responsibility centres which have *both responsibility of generation of revenue and incurrence of expenditures*. Since, managers of profit centres are accountable for both costs as well as revenue, profitability is the basis for measurement of performance of these responsibility centres. Examples of profit centres are decentralised branches of an organisation.

(iv) Investment Centres: These are the responsibility centres which are *not only responsible for profitability but also have the authority to make capital investment decisions*. The performance of these responsibility centres are measured on the basis of Return on Investment (ROI) besides profit.

Examples of investment centres are Maharatna, Navratna and Miniratna companies of Public Sector Undertakings of Central Government.

ICAI · P4 · MTP Sep 2024 · Series I · question p. 12 · answer p. 16-17

33 MTP **Old P3 Nov 2023 S-II** Q6(b) 5 marks old scheme

Explain Standard cost centre and Discretionary cost centre.

ICAI'S ANSWER

(i) Standard Cost Centre: Cost Centre where output is measurable and input required for the output can be specified. Based on a well-established study, an estimate of standard units of input to produce a unit of output is set. The actual cost for inputs is compared with the standard cost. Any deviation (variance) in cost is measured and analysed into controllable and uncontrollable cost. The manager of the cost centre is expected to comply with the standard and held responsible for adverse cost variances. The input-output ratio for a standard cost centre is clearly identifiable.

(ii) Discretionary Cost Centre: The cost centre whose output cannot be measured in financial terms; thus input-output ratio cannot be defined. The cost of input is compared with allocated budget for the activity. Examples of discretionary cost centres are Research & Development department, Advertisement department where output of these department cannot be measured with certainty and co-related with cost incurred on inputs.

ICAI · Old P3 · MTP Nov 2023 · Series II · question p. 7 · answer p. 13-14

34 MTP **Old P3 May 2023 S-I** Q6(d) 5 marks old scheme

DEFINE cost units? WRITE the cost unit basis against each of the following Industry/Product- Automobile, Steel, Cement, Chemicals, Power and Transport.

ICAI'S ANSWER

Cost units are usually the units of physical measurement like number, weight, area, volume, length, time and value.

Industry or Product	Cost Unit Basis
Automobile	Number
Steel	Ton
Cement	Ton/ per bag etc.
Chemicals	Litre, gallon, kilogram, ton etc.
Power	Kilo-watt hour (kWh)
Transport	Passenger- kilometer

ICAI · Old P3 · MTP May 2023 · Series I · question p. 6 · answer p. 11

35 MTP **Old P3 Jul 2021 S-I** Q6(d) 5 marks old scheme

Same question, different ICAI answer: #24 RTP Sep 2024

DISCUSS the Standard and Discretionary Cost Centres.

ICAI'S ANSWER

(i) **Standards Cost Centre:** Cost Centre where output is measurable and input required for the output can be specified. Based on a well-established study, an estimate of standard units of input to produce a unit of output is set. The actual cost for inputs is compared with the standard cost. Any deviation (variance) in cost is measured and analysed into controllable and uncontrollable cost. The manager of the cost centre is supposed to comply with the standard and held responsible for adverse cost variances. The input-output ratio for a standard cost centre is clearly identifiable.

(ii) **Discretionary Cost Centre:** The cost centre whose output cannot be measured in financial terms, thus input-output ratio cannot be defined. The cost of input is compared with allocated budget for the activity. Example of discretionary cost centres are Research & Development department, Advertisement department where output of these department cannot be measured with certainty and co-related with cost incurred on inputs.

ICAI · Old P3 · MTP Jul 2021 · Series I · question p. 8 · answer p. 14

1.5 Elements and classification of cost

24 questions · asked in 18 ICAI documents · 4 exam papers

36 **EXAM** **Sep 2026** Q6(a) **5 marks**

NS Ltd. has listed some of the costs incurred during the current period.

I	II	III Classification
(i)	Salary to office attendants	
(ii)	Product blueprint cost	
(iii)	Hire charges paid for factory machinery for a certain product	
(iv)	Salary paid to marketing manager	
(v)	One time licence fee paid for software used to make computerised graphics for a particular job	
(vi)	Wages paid to factory workers	
(vii)	Overtime wages paid to finish a certain job due to urgency created by customer	
(viii)	Specialised packing at customer's request	
(ix)	Advertisement Expenses	
(x)	Royalty paid for production	

Classify the above items under appropriate functions of cost.

(Present only columns I and III in your answer books.)

ICAI'S ANSWER

ICAI has not published its answers to this paper yet.

ICAI · P4 · Sep 2026 exam · question p. 22

37 EXAM **Jan 2026** Q5(c) 3 marks

A company can make any one of the 2 products M or S during the year. It can exercise its option only at the beginning of each year.

Relevant information about the products for the next year is given below:

Particulars	M	S
Selling Price (₹ /unit)	75	45
Variable Cost (₹/unit)	45	30
Market Demand (in units)	3,000	8,000
Production Capacity (in units)	4,000	7,000
Fixed Cost (₹)		1,20,000

You are required to compute the opportunity cost for each of the product.

ICAI'S ANSWER

Computation of opportunity cost for each product

Particulars	M	S
I. Contribution per unit (₹)	30 (75-45)	15 (45-30)
II. Units (Lower of Production/Market Demand)	3,000	7,000
III. Possible Contribution (₹) {I x II}	90,000	1,05,000
IV. Opportunity Cost (₹)	1,05,000	90,000

(*) Opportunity cost is the maximum possible contribution forgone by not producing alternative products i.e. if product M is produced then opportunity cost will be ₹ 1,05,000(Contribution of S).

ICAI · P4 · Jan 2026 exam · question p. 25 · answer p. 28

38 EXAM **Jan 2026** Q6(a) 5 marks

Classify the following items of expenses by functions and variability:

S. No.	Item	Function	Variability/ Behaviour
(i)	Consumable Store		
(ii)	General Manager's salary		
(iii)	Delivery van expenses		
(iv)	Compensation (fixed salary plus commission on sales)		
(v)	Rent of finished goods warehouse		

ICAI'S ANSWER

Classification of Expenses by Function and Variability

S. No.	Item	Function	Variability / Behaviour
(i)	Consumable stores	Factory/Production/ manufacturing overhead	Variable
(ii)	General Manager's salary	Administration	Fixed
(iii)	Delivery van expenses	Selling & Distribution/ Distribution	Semi-variable/ Variable
(iv)	Compensation (fixed salary plus commission on sales)	Selling / Selling & Distribution	Semi-variable
(v)	Rent of finished goods warehouse	Selling & Distribution/ Distribution	Fixed

ICAI · P4 · Jan 2026 exam · question p. 28 · answer p. 29

39 EXAM **May 2025** Q6(a) 5 marks

In the following independent situations, identify the type of cost and state whether it is relevant/non-relevant in managerial decision making:

(i) A Limited owns a commercial space of 1500 square feet and uses the same for its own office accommodation purposes. Similar office is available nearby on rent of ₹ 30,000 per month.

(ii) MNC Limited has paid ₹ 1 Lakh as rent for a factory shed which is temporarily closed for the last two months.

(iii) Beta Company has paid ₹ 3 Lakhs to a market research agency to find out the market demand of the innovative product developed by the company.

(iv) Zen LLP has paid incentive of ₹ 5 Lakhs @ 1% on sales to the salesmen for achieving sales beyond the expected sales of ₹ 25 Lakhs per month per salesman.

(v) A start-up company has invested ₹ 50 Lakhs in Project P. The company could have earned interest of ₹ 3 Lakhs by investing the amount in a bank fixed deposit @ 6% per annum.

ICAI'S ANSWER

Situation	Type of Cost	Relevant / Non-Relevant
(i)	Opportunity Cost/ Imputed Cost/ Notional Cost	Relevant
(ii)	Committed Cost/Shut Down Cost	Non-Relevant
(iii)	Sunk Cost	Non-Relevant
(iv)	Out of Pocket Cost	Relevant
(v)	Opportunity Cost	Relevant

ICAI · P4 · May 2025 exam · question p. 30-31 · answer p. 32

40 EXAM Old P3 Nov 2023 Question 6 · Q6(a) to Q6(e) 25 marks old scheme

QUESTION 6

Answer any four of the following:

Q6(a) 5 marks

Explain very briefly the following terms used in Cost and Management Accounting:

(i) Pre-determined Cost

(ii) Estimated Cost

(iii) Imputed Cost

(iv) Discretionary Cost

Q6(b) 5 marks

State with reasons whether the following independent statements are true or false:

(i) Under LIFO method, in the period of falling prices, lower income is reported and income-tax liability is reduced.

(ii) Under VED analysis, inventories are classified on the basis of cost of individual items.

(iii) Material requisition note is prepared by the store keeper.

(iv) Simple average pricing method is suitable when quantity purchased under each lot is different and prices fluctuate considerably.

(v) Bin card and stores ledger are maintained by the purchasing department.

Q6(c) 5 marks

What do you mean by employee productivity? Point out the factors which must be taken into consideration for increasing employee productivity.

Q6(d) 5 marks

Explain very briefly the following terms:

(i) Retention Money

(ii) Escalation Clause

(iii) Co-Products

(iv) Job Costing

(v) Process Costing

Q6(e) 5 marks

What is meant by cost driver? Give its different categories. Suggest suitable cost drivers (at least two) in the following business functions:

(i) Distribution

(ii) Research and Development

(iii) Customer services

ICAI'S ANSWERS

Q6(a)

(i) Pre- Determined Cost

A cost which is computed in advance before production or operations start, on the basis of specification of all the factors affecting cost, is known as a pre -determined cost.

(ii) Estimated Cost

Estimated cost is "the expected cost of manufacture, or acquisition, often in terms of a unit of product computed on the basis of information available in advance of actual production or purchase". Estimated costs are prospective costs since they refer to prediction of costs.

(iii) Imputed Cost

Imputed costs do not involve any immediate cash payment. Implicit costs are not recorded in the books of account but yet, they are important for certain types of managerial decisions such as equipment replacement and relative profitability of two alternative courses of action. They are also known as economic costs. These cost are similar to opportunity cost.

(iv) Discretionary Cost

Discretionary costs are not tied to a clear cause and effect relationship between inputs and outputs. They arise from periodic decisions regarding the maximum outlay to be incurred. Examples are - advertising, public relations, training etc.

Q6(b)

Statement No.	True/False	Reason
(i)	**False**	Under LIFO method, in case of falling prices profit tends to rise due to lower material cost, thus income tax liability is increased.
(ii)	**False**	Under VED Analysis, inventories are classified on the basis of its criticality for the production function and final product.
(iii)	**False**	Material Requisition Note is prepared by the production or other consuming department. It is a voucher used to get material issued from store.
(iv)	**False**	Simple average pricing method is suitable when the materials are received in uniform lots of similar quantity, and prices do not fluctuate considerably.
(v)	**False**	Bin card is maintained by the storekeeper in the store. While Stores ledger is maintained in cost accounting department.

Q6(c)

Meaning of employee productivity

Productivity is generally determined by the input/output ratio.

In case of employees, it is calculated as: $\frac{\text{Standard time for doing actual work}}{\text{Actual time taken}}$

Employee productivity is used for measuring the efficiency of individual workers. It is an index of efficiency in the utilisation of human resources, materials, capital, power and all kinds of services and facilities.

It is measured by the output in relation to input. Productivity can be improved by reducing the input for a certain quantity or value of output or by increasing the output from the same given quantity or value of input.

Factors for increasing Employee productivity: The important factors which must be taken into consideration for increasing employee productivity are as follows:

1. Employing only those workers who possess the right type of skill.

2. Placing a right type of person to a right job.
3. Training young and old workers by providing them the right types of opportunities.
4. Taking appropriate measures to avoid the situation of excess or shortage of employees.
5. Carrying out work study for fixation of wages and for the simplification and standardisation of work.

Q6(d)

(i) **Retention Money:** Retention money is a part of the value of work certified which though certified but is not paid by the contractee. Retention amount is kept by the contractee as security amount against any damage.

(ii) **Escalation Clause:** Escalation clause is a clause written in the agreement (contract) between the contractor and contractee which states that in case of increase in the prices of materials, wages or other supplies beyond a certain level the contract price will be increased by an agreed amount.

(iii) **Co-Products:** Co-products may be defined as Two or more products which are contemporary but do not emerge necessarily from the same material in the same process.

(iv) **Job Costing:** Job costing is the method of costing required to be done for unique products manufactured done against specific orders. In this method of costing, cost of each job is ascertained separately.

(v) **Process Costing:** Process costing is a method of costing used in industries where the material has to pass through two or more process for being converted into a final product. Here the cost of completing each stage of work is ascertained, like cost of making pulp and cost of making paper from pulp.

PassBook's note: Mixed scope: (iii) co-products, (iv) job costing and (v) process costing are in the current Study Material (chapters 11, 9 and 10).

Q6(e)

Meaning of Cost Driver: A Cost driver is a factor or variable which effect level of cost. Generally, it is an activity which is responsible for cost incurrence. Level of activity or volume of production is the example of a cost driver. An activity may be an event, task, or unit of work etc.

There are two categories of cost driver.

- **Resource Cost Driver -** It is a measure of the quantity of resources consumed by an activity. It is used to assign the cost of a resource to an activity or cost pool.
- **Activity Cost Driver -** It is a measure of the frequency and intensity of demand, placed on activities by cost objects. It is used to assign activity costs to cost objects.

Business Function	Cost drivers
Distribution	Number of units distributed, Number of customers
Research and Development	Number of research projects, personnel hours on a project, technical complexities of the projects.
Customer service	Number of service calls, number of products serviced, hours spent in servicing of products.

ICAI · Old P3 · Nov 2023 exam · question p. 28 · answer p. 29-30, 30, 30-31, 31, 31-32

41 RTP **Sep 2026** Q16(b)

State the types of cost in the following cases:

(i) Interest paid on own capital not involving any cash outflow.

(ii) Withdrawing money from bank deposit for the purpose of purchasing new machine for expansion purpose.

(iii) Rent paid for the factory building which is temporarily closed

(iv) Cost associated with the acquisition and conversion of material into finished product.

ICAI'S ANSWER

	Type of costs
Interest paid on own capital not involving any cash outflow.	Imputed Cost
Withdrawing money from bank deposit for the purpose of purchasing new machine for expansion purpose.	Opportunity Cost
Rent paid for the factory building which is temporarily closed	Shut Down Cost
Cost associated with the acquisition and conversion of material into finished product.	Product Cost

ICAI · P4 · RTP Sep 2026 · question p. 19 · answer p. 48

42 RTP **May 2026** Q17(a)

A manufacturing firm is planning to introduce a new product while simultaneously considering changes in its production methods, advertising budget, and investment plans. As the management accountant, you are required to advise the management on how different types of costs should be treated for effective decision-making. Explain how pre-determined cost, Differential Cost, product cost, out-of-pocket cost, shutdown cost, sunk cost, and implicit cost would influence managerial decisions in this scenario.

ICAI'S ANSWER

Following cost would influence managerial decisions:

1. **Pre-determined Cost** - A cost which is computed in advance before production or operations start, on the basis of specification of all the factors affecting cost, is known as a pre-determined cost.
2. **Differential Cost** - (Incremental and decremental costs). It represents the change (increase or decrease) in total cost (variable as well as fixed) due to change in activity level, technology, process or method of production, etc. For example, if any change is proposed in the existing level or in the existing method of production, the increase or decrease in total cost or in specific elements of cost as a result of this decision will be known as incremental cost or decremental cost.
3. **Product Costs -** These are the costs which are associated with the purchase and sale of goods (in the case of merchandise inventory). In the production scenario, such costs are associated with the acquisition and conversion of materials and all other manufacturing inputs into finished product for sale. Hence, under marginal costing, variable manufacturing costs and under absorption costing, total manufacturing costs (variable and fixed) constitute inventoriable or product costs.
4. **Out-of-pocket Cost** - It is that portion of total cost, which involves cash outflow. This cost concept is a short-run concept and is used in decisions relating to fixation of selling price in recession,

make or buy, etc. Out–of–pocket costs can be avoided or saved if a particular proposal under consideration is not accepted.

5. **Shut down Costs** - Those costs, which continue to be, incurred even when a plant is temporarily shut-down e.g. rent, rates, depreciation, etc. These costs cannot be eliminated with the closure of the plant. In other words, all fixed costs, which cannot be avoided during the temporary closure of a plant, will be known as shut down costs.
6. **Sunk Costs** - Historical costs incurred in the past are known as sunk costs. They play no role in decision making in the current period. For example, in the case of a decision relating to the replacement of a machine, the written down value of the existing machine is a sunk cost and therefore, not considered.
7. **Implicit Costs -** These costs do not involve any immediate cash payment. They are not recorded in the books of account. They are also known as economic costs.

ICAI · P4 · RTP May 2026 · question p. 21 · answer p. 49-50

43 RTP **Jan 2026** Q19(a)

Also asked in MTP May 2025 S-I Q6(a)

Same question, different ICAI answer: #48 MTP Sep 2026 S-II

Standard costs, marginal costs are some of the types of cost which helps the management in decision making. DISCUSS any five types of costs categorised based on its use in Managerial Decision Making.

ICAI'S ANSWER

Type of costs categorised based on its use in Managerial Decision Making

(i) Pre-determined Cost - A cost which is computed in advance before production or operations start, on the basis of specification of all the factors affecting cost, is known as a pre-determined cost.

(ii) Standard Cost - A pre-determined cost, which is calculated from managements 'expected standard of efficient operation' and the relevant necessary expenditure. It may be used as a basis for price fixation and for cost control through variance analysis.

(iii) Marginal Cost - The amount at any given volume of output by which aggregate costs increases if the volume of output is increased or decreased by one unit.

(iv) Estimated Cost - Kohler defines estimated cost as "the expected cost of manufacture, or acquisition, often in terms of a unit of product computed on the basis of information available in advance of actual production or purchase". Estimated costs are prospective costs since they refer to prediction of costs.

(v) Differential Cost - (Incremental and decremental costs). It represents the change (increase or decrease) in total cost (variable as well as fixed) due to change in activity level, technology, process or method of production, etc. For example, if any change is proposed in the existing level or in the existing method of production, the increase or decrease in total cost or in specific elements of cost as a result of this decision will be known as incremental cost or decremental cost.

(vi) Imputed Costs - These costs are notional costs which do not involve any cash outlay. Interest on capital, the payment for which is not actually made, is an example of imputed cost. These costs are similar to opportunity costs.

(vii) Capitalized Costs - These are costs which are initially recorded as assets and subsequently treated as expenses. Example, installation expenses on the erection of a machine are added to the

cost of a machine.

(viii) Product Costs - These are the costs which are associated with the purchase and sale of goods (in the case of merchandise inventory). In the production scenario, such costs are associated with the acquisition and conversion of materials and all other manufacturing inputs into finished product for sale. Hence, under marginal costing, variable manufacturing costs and under absorption costing, total manufacturing costs (variable and fixed) constitute inventoriable or product costs.

(ix) Opportunity Cost - This cost refers to the value of sacrifice made or benefit of opportunity foregone in accepting an alternative course of action. For example, a firm financing its expansion plan by withdrawing money from its bank deposits. In such a case the loss of interest on the bank deposit is the opportunity cost for carrying out the expansion plan.

(x) Out-of-pocket Cost - It is that portion of total cost, which involves cash outflow. This cost concept is a short-run concept and is used in decisions relating to fixation of selling price in recession, make or buy, etc. Out–of–pocket costs can be avoided or saved if a particular proposal under consideration is not accepted.

(xi) Shut down Costs - Those costs, which continue to be, incurred even when a plant is temporarily shut-down e.g. rent, rates, depreciation, etc. These costs cannot be eliminated with the closure of the plant. In other words, all fixed costs, which cannot be avoided during the temporary closure of a plant, will be known as shut down costs.

(xii) Sunk Costs - Historical costs incurred in the past are known as sunk costs. They play no role in decision making in the current period. For example, in the case of a decision relating to the replacement of a machine, the written down value of the existing machine is a sunk cost and therefore, not considered.

(xiii) Absolute Cost - These costs refer to the cost of any product, process or unit in its totality. When costs are presented in a statement form, various cost components may be shown in absolute amount or as a percentage of total cost or as per unit cost or all together. Here the costs depicted in absolute amount may be called absolute costs and are base costs on which further analysis and decisions are made.

(xiv) Discretionary Costs - Such costs are not tied to a clear cause and effect relationship between inputs and outputs. They usually arise from periodic decisions regarding the maximum outlay to be incurred. Examples include advertising, public relations, executive training etc.

(xv) Period Costs - These are the costs, which are not assigned to the products but are charged as expenses against the revenue of the period in which they are incurred. All non-manufacturing costs such as general & administrative expenses, selling and distribution expenses are recognised as period costs.

(xvi) Engineered Costs - These are costs that result specifically from a clear cause and effect relationship between inputs and outputs. The relationship is usually personally observable. Examples of inputs are direct material costs, direct labour costs etc. Examples of output are cars, computers etc.

(xvii) Explicit Costs - These costs are also known as out-of-pocket costs and refer to costs involving immediate payment of cash. salaries, wages, postage and telegram, printing and stationery, interest on loan etc. are some examples of explicit costs involving immediate cash payment.

(xviii)Implicit Costs - These costs do not involve any immediate cash payment. They are not recorded in the books of account. They are also known as economic costs.

44 RTP **May 2025** Case scenario · Q1(i) to Q1(v)

CASE SCENARIO

Mr. Linde is a German national, who came to India again on 1st April, 2024. He represents his company and wants to start business in India as well. His company expertise in the manufacturing of Industrial machines. Recently launched "Make in India" movement has motivated Mr. Linde thinks that this might be the perfect opportunity for his company to establish his company in India.

Last, Mr. Linde came to India on 1st April, 2012. He purchased a land for ₹ 50,00,000 and constructed a building by spending ₹ 16,00,000. After that he opened a Private limited company in that building. He spent another ₹ 2,80,000 for this. He also employed 3 people for survey and to understand the need of Indian customers and spent ₹ 1,50,000 in salaries.

He was disappointed in the response of market, who were importing everything from China back then. He closed the office & went back to Germany. All these years the office was closed and only an amount of ₹ 12,500 per month was paid to a guard and property tax was also paid. Property tax was paid on an average of ₹ 18,000 per year.

Now when Mr. Linde is back, he opens the office and starts to plan on how this time he will capture the Indian market.

Expenses started to incur as soon as the office opened:

- Salaries of staff ₹ 2,50,000 per month.
- Electricity, water, & maintenance of office at ₹ 50,000 per month.
- Security staff at ₹ 15,000 per month.

Linde plans to purchase a land in Manesar which will be used for the factory. After a search he found an appropriate land and purchased a land for ₹ 1.50 crores. He handed over the land to a SPV company of a REIT to build a state of the art facility for their factory. Factory will be built in 2 years. They will spend ₹ 85 lacs each year for this construction.

Linde, back in the Noida office, made 3 departments:

(1) Office and administration

(2) Sales and marketing

(3) Account and Finance

Expenses for these departments (except for salaries) are expected to be:-

- Office and administration = ₹ 95,000 per year
- Sales and marketing = ₹ 1,12,000 per year
- Accounts and Finance = ₹ 88,000 per year

Office overheads are to be bifurcated in these departments on the basis of their individual spending ratio.

Technology is developed in Germany but at present its execution is not required. Therefore, they do not require any expert as of now and also because the factory is not ready.

Mr. Linde, being the only person representing his company and lone German in the Indian office feels difficult to manage everything as he finds Indian corporate environment very challenging. He asked

his company to deploy another German manager to India. This will cost the company additional two million Indian national rupee per year to relocate this additional manager in India. The German management is divided on this decision. The ones who disagree say “Mr. Linde is competent enough to run a small extension of our company in India. We will allocate more resources to Indian subsidiary when actual operations will start, till then everything can be managed by Mr. Linde alone. Right Indian Company is itself a cost centre and we are already paying him 3.5 million INR annually, therefore we are not ready to invest until it starts generating revenue”.

Linde has another opportunity to relocate the head office, also, to Manesar, where the factory building is being constructed. The distance between head office and factory will reduce greatly, which will be highly beneficial when the factory will become operative. He will have to sale the old office in Noida, which will be sold at ₹ 2.50 crores and purchase a ready-made building in Manesar for ₹ 3.75 crores. This new building will have larger space that can accommodate the future needs for space, when company will grow. It seems to be like perfect investment opportunity to Linde.

Expenses in this new building are expected to be:-

- Salaries of staff ₹ 3,00,000 per month
- Electricity, water, & maintenance of office at ₹ 80,000 per month.
- Security at ₹ 30,000 per month.

Indexed cost of building in Noida is ₹ 2.25 crores and tax on long-term capital gain is 12.5%.

On the basis of above information, answer the following 5 MCQs:

Q1(i)

Find out an avoidable cost till the factory becomes operative. What is its value?

A 20,00,000

B 49,20,000

C 98,40,000

D 40,00,000

Q1(ii)

Find out the total of Sunk and shut down cost in the given case study. Select the correct option from below.

A 4,30,000

B 70,30,000

C 24,46,000

D 90,46,000

Q1(iii)

What is total out-of-pocket cost for the company in Noida branch, after factory land in Manesar is purchased, till the factory operation begins?

A 3,21,50,000

B 1,51,50,000

C 81,50,000

D 40,75,000

Q1(iv)

What will be out of pocket expenses incurred in relocation of Head office to Manesar?

- A 3,75,00,000
- B 1,28,12,500
- C 1,25,00,000
- D 4,24,20,000

Q1(v)

How much is the unexpired cost of the Noida office as on 1st October, 2024, if salaries to all the employees are paid till 31st March, 2025?

- A 33,40,000
- B 30,00,000
- C 15,90,000
- D 15,00,000

ICAI'S ANSWERS

Q1(i)

(D)

Only avoidable cost is a new managers salary for 2 years

= ₹ 20,00,000 x 2 = ₹ 40,00,000

Q1(ii)

(D)

Shut down cost is the cost spent when the company was shut down for 12 years in India

= 12,500 x 12 x 12 + 18,000 x 12 = ₹ 20,16,000

Sunk cost are all the costs that was spent in 2012

= 50,00,000+16,00,000+2,80,000+1,50,000

= ₹ 70,30,000

Total = ₹ 90,46,000

Q1(iii)

(A)

Calculation

Particulars		Amount (₹)
Salary	2,50,000 × 12 × 2	60,00,000
Electricity, etc	50,000 × 12 × 2	12,00,000
Security	15,000 × 12months x 2 years	3,60,000
O&A	95,000 × 2 years	1,90,000
Sales	1,12,000 × 2 years	2,24,000

Particulars		Amount (₹)
Accounts	88,000 × 2 years	1,76,000
Salary of linde	35,00,000 × 2	70,00,000
Construction	85,00,000 × 2	1,70,00,000
Total		**3,21,50,000**

Q1(iv)

(B)

Cost of new office = ₹ 3,75,00,000

Money received from sale of Noida office

= 2,50,00,000 – (2,50,00,000 – 2,25,00,000) x 12.5%

= ₹ 2,46,87,500

Out of pocket expenses for relocation of head office

= 3,75,00,000 – 2,46,87,500 = ₹ 1,28,12,500

Q1(v)

(A)

Unexpired cost = advance salary paid till march of next year

= (2,50,000+15,000) x 6months + 35,00,000/2 = ₹ 33,40,000

ICAI · P4 · RTP May 2025 · question p. 1-3 · answer p. 25, 25-26, 26

45 RTP **May 2025** Q20(a)

DEFINE Product costs. Describe three different purposes for computing product costs.

ICAI'S ANSWER

Definition of product costs: Product costs are inventoriable costs. These are the costs, which are assigned to the product. Under marginal costing variable manufacturing costs and under absorption costing, total manufacturing costs constitute product costs.

Purposes for computing product costs:

The three different purposes for computing product costs are as follows:

(i) *Preparation of financial statements:* Here focus is on inventoriable costs.

(ii) *Product pricing:* It is an important purpose for which product costs are used. For this purpose, the cost of the areas along with the value chain should be included to make the product available to the customer.

(iii) *Contracting with government agencies:* For this purpose government agencies may not allow the contractors to recover research and development and marketing costs under cost plus contracts.

ICAI · P4 · RTP May 2025 · question p. 25 · answer p. 55-56

46 RTP **May 2024** Q13(c)

Also asked in Old P3 RTP May 2022 Q14(a) · Old P3 MTP Nov 2023 S-I Q6(a) · Old P3 MTP May 2023 S-II Q6(a)

Explain the difference between controllable & uncontrollable costs?

ICAI'S ANSWER

Controllable costs and Uncontrollable costs: Cost that can be controlled, typically by a cost, profit or investment centre manager is called controllable cost. Controllable costs incurred in a particular responsibility centre can be influenced by the action of the executive heading that responsibility centre.

Costs which cannot be influenced by the action of a specified member of an undertaking are known as uncontrollable costs.

ICAI · P4 · RTP May 2024 · question p. 17 · answer p. 43

47 RTP **Old P3 Nov 2022** Q15(a) old scheme

Health Wealth Hospital is interested in estimating the cost for each patient stay. The hospital offers general health care facility i.e. only basic services.

You are required to:

(i) CLASSIFY each of the following costs as either direct or indirect with respect to each patient.

(ii) CLASSIFY each of the following costs as either fixed or variable with respect to hospital costs per day.

	Direct	Indirect	Fixed	Variable
Electronic monitoring	____	____	____	____
Meals for patients	____	____	____	____
Nurses' salaries	____	____	____	____
Parking maintenance	____	____	____	____
Security	____	____	____	____

ICAI'S ANSWER

Item	Direct	Indirect	Fixed	Variable
Electronic monitoring	YES			YES
Meals for patients	YES			YES
Nurses' salaries		YES	YES	
Parking maintenance		YES	YES	
Security		YES	YES	

ICAI · Old P3 · RTP Nov 2022 · question p. 13 · answer p. 34

48 MTP **Sep 2026 S-II** Q6(b) 5 marks

Same question, different ICAI answer: #43 RTP Jan 2026

Standard costs, marginal costs are some of the types of cost which helps the management in decision making. DISCUSS any five types of costs categorised based on its use in Managerial Decision Making.

ICAI'S ANSWER

Type of costs categorised based on its use in Managerial Decision Making:

(i) **Pre-determined Cost -** A cost which is computed in advance before production or operations start, on the basis of specification of all the factors affecting cost, is known as a pre-determined cost.

(ii) **Standard Cost -** A pre-determined cost, which is calculated from managements 'expected standard of efficient operation' and the relevant necessary expenditure. It may be used as a basis for price fixation and for cost control through variance analysis.

(iii) **Marginal Cost -** The amount at any given volume of output by which aggregate costs increases if the volume of output is increased or decreased by one unit.

(iv) **Estimated Cost -** Kohler defines estimated cost as "the expected cost of manufacture, or acquisition, often in terms of a unit of product computed on the basis of information available in advance of actual production or purchase". Estimated costs are prospective costs since they refer to prediction of costs.

(v) **Differential Cost -** (Incremental and decremental costs). It represents the change (increase or decrease) in total cost (variable as well as fixed) due to change in activity level, technology, process or method of production, etc. For example, if any change is proposed in the existing level or in the existing method of production, the increase or decrease in total cost or in specific elements of cost as a result of this decision will be known as incremental cost or decremental cost.

(vi) **Imputed Costs -** These costs are notional costs which do not involve any cash outlay. Interest on capital, the payment for which is not actually made, is an example of imputed cost. These costs are similar to opportunity costs.

(vii) **Capitalized Costs -** These are costs which are initially recorded as assets and subsequently treated as expenses. Example, installation expenses on the erection of a machine are added to the cost of a machine.

(viii) **Product Costs -** These are the costs which are associated with the purchase and sale of goods (in the case of merchandise inventory). In the production scenario, such costs are associated with the acquisition and conversion of materials and all other manufacturing inputs into finished product for sale. Hence, under marginal costing, variable manufacturing costs and under absorption costing, total manufacturing costs (variable and fixed) constitute inventoriable or product costs.

(ix) **Opportunity Cost -** This cost refers to the value of sacrifice made or benefit of opportunity foregone in accepting an alternative course of action. For example, a firm financing its expansion plan by withdrawing money from its bank deposits. In such a case the loss of interest on the bank deposit is the opportunity cost for carrying out the expansion plan.

(x) **Out-of-pocket Cost -** It is that portion of total cost, which involves cash outflow. This cost concept is a short-run concept and is used in decisions relating to fixation of selling price in recession, make or buy, etc. Out–of–pocket costs can be avoided or saved if a particular proposal under consideration is not accepted.

(xi) **Shut down Costs -** Those costs, which continue to be, incurred even when a plant is temporarily shut-down e.g. rent, rates, depreciation, etc. These costs cannot be eliminated with the closure of the plant. In other words, all fixed costs, which cannot be avoided during the temporary closure of a plant, will be known as shut down costs.

(xii) **Sunk Costs -** Historical costs incurred in the past are known as sunk costs. They play no role in decision making in the current period. For example, in the case of a decision relating to the

replacement of a machine, the written down value of the existing machine is a sunk cost and therefore, not considered.

(xiii) **Absolute Cost -** These costs refer to the cost of any product, process or unit in its totality. When costs are presented in a statement form, various cost components may be shown in absolute amount or as a percentage of total cost or as per unit cost or all together. Here the costs depicted in absolute amount may be called absolute costs and are base costs on which further analysis and decisions are made.

(xiv) **Discretionary Costs –** Such costs are not tied to a clear cause and effect relationship between inputs and outputs. They usually arise from periodic decisions regarding the maximum outlay to be incurred. Examples include advertising, public relations, executive training etc.

(xv) **Period Costs -** These are the costs, which are not assigned to the products but are charged as expenses against the revenue of the period in which they are incurred. All non-manufacturing costs such as general & administrative expenses, selling and distribution expenses are recognised as period costs.

(xvi) **Engineered Costs -** These are costs that result specifically from a clear cause and effect relationship between inputs and outputs. The relationship is usually personally observable. Examples of inputs are direct material costs, direct labour costs etc. Examples of output are cars, computers etc.

(xvii) **Explicit Costs -** These costs are also known as out-of-pocket costs and refer to costs involving immediate payment of cash. salaries, wages, postage and telegram, printing and stationery, interest on loan etc. are some examples of explicit costs involving immediate cash payment.

(xviii) **Implicit Costs -** These costs do not involve any immediate cash payment. They are not recorded in the books of account. They are also known as economic costs.

*** Any 5 Points**

ICAI · P4 · MTP Sep 2026 · Series II · question p. 14 · answer p. 21-23

49 MTP **May 2024 S-I** Q6(c) 4 marks

As per the controllability, cost can be classified as controllable & uncontrollable costs. How will you DIFFERENTIATE them?

ICAI'S ANSWER

Controllable costs and Uncontrollable costs: Cost that can be controlled, typically by a cost, profit or investment centre manager is called controllable cost. Controllable costs incurred in a particular responsibility centre can be influenced by the action of the executive heading that responsibility centre.

Costs which cannot be influenced by the action of a specified member of an undertaking are known as uncontrollable costs.

ICAI · P4 · MTP May 2024 · Series I · question p. 11 · answer p. 16

50 MTP **Old P3 Nov 2022 S-II** Q6(d) 5 marks old scheme

EXPLAIN the difference between product cost and period cost.

ICAI'S ANSWER

Product costs are those costs that are identified with the goods purchased or produced for resale. In a manufacturing organisation they are attached to the product and that are included in the inventory valuation for finished goods, or for incomplete goods. Product cost is also known as inventoriable cost. Under absorption costing method it includes direct material, direct labour, direct expenses, directly attributable costs (variable and non-variable) and other production (manufacturing) overheads. Under marginal costing method Product Costs includes all variable production costs and the all fixed costs are deducted from the contribution.

Periods costs are the costs, which are not assigned to the products but are charged as expense against revenue of the period in which they are incurred. General Administration, marketing, sales and distributor overheads are recognized as period costs.

ICAI · Old P3 · MTP Nov 2022 · Series II · question p. 7 · answer p. 15

51 MTP **Old P3 May 2022 S-I** Q6(b) 5 marks old scheme

Some of the items of PR Company, a manufacturer of corporate office furniture, are provided below. As the company is in the process of developing a formal cost accounting system, you are required to CLASSIFY the items into three categories namely: (i) Cost tracing (ii) Cost allocation (iii) Non-manufacturing item.

Carpenter wages, Depreciation - office building, Glue for assembly, Lathe department supervisor, Metal brackets for drawers, Factory washroom supplies, Lumber, Samples for trade shows, Lathe depreciation, Lathe operator wages.

ICAI'S ANSWER

Item	Cost Tracing	Cost Allocation	Non-manufacturing
Carpenter wages	√		
Depreciation - office building			√
Glue for assembly		√	
Lathe department supervisor		√	
Metal brackets for drawers	√		
Factory washroom supplies		√	
Lumber	√		
Samples for trade shows			√
Lathe depreciation		√	
Lathe operator wages		√	

ICAI · Old P3 · MTP May 2022 · Series I · question p. 7 · answer p. 10-11

1.6 Cost behaviour: fixed, variable, semi-variable and step costs

6 questions · asked in 3 ICAI documents

52 RTP **Old P3 Jul 2021** Q13(c) old scheme

Also asked in Old P3 MTP Dec 2021 S-II Q6(b)

DISCUSS cost classification based on variability and controllability.

ICAI'S ANSWER

Cost classification based on variability

(i) Fixed Costs – These are the costs which are incurred for a period, and which, within certain output and turnover limits, tend to be unaffected by fluctuations in the levels of activity (output or turnover). They do not tend to increase or decrease with the changes in output. For example, rent, insurance of factory building etc., remain the same for different levels of production.

(ii) Variable Costs – These costs tend to vary with the volume of activity. Any increase in the activity results in an increase in the variable cost and vice-versa. For example, cost of direct labour, etc.

(iii) Semi-variable Costs – These costs contain both fixed and variable components and are thus partly affected by fluctuations in the level of activity. Examples of semi variable costs are telephone bills, gas and electricity etc.

Cost classification based on controllability

(i) Controllable Costs - Cost that can be controlled, typically by a cost, profit or investment centre manager is called controllable cost. Controllable costs incurred in a particular responsibility centre can be influenced by the action of the executive heading that responsibility centre. For example, direct costs comprising direct labour, direct material, direct expenses and some of the overheads are generally controllable by the shop level management.

(ii) Uncontrollable Costs - Costs which cannot be influenced by the action of a specified member of an undertaking are known as uncontrollable costs. For example, expenditure incurred by, say, the tool room is controllable by the foreman in-charge of that section but the share of the tool-room expenditure which is apportioned to a machine shop is not to be controlled by the machine shop foreman.

ICAI · Old P3 · RTP Jul 2021 · question p. 11 · answer p. 32

53 MTP **May 2025 S-I** Case Scenario I · MCQ 1 to MCQ 5 10 marks

CASE SCENARIO I

Rinku Ltd is a manufacturing company which is producing bags of different varieties. The company is planning to establish a new plant in the neighboring country to produce the bags. New plant has a production capacity of 2,00,000 units per year. As per the studies, normal capacity utilization is 90% of the production capacity. The company will be able to sell the whole production after making price adjustments.

The following are the annual cost data on the basis of cost studies for the new plant in the neighboring country:

Material Cost	= ₹ 42,00,000 (100% variable)
Labour	= ₹ 40,00,000 (70% variable)
Factory Overheads	= ₹ 35,00,000 (60% variable)
Administrative Overheads	= ₹ 10,00,000 (30% variable)

Bags are being produced and sold on steady basis. It is estimated that it costs ₹ 1 as inventory holding cost per bag per month and that the set up cost per run of bag manufacture is ₹ 1,000. The production of the new plant will be sold only the sales agent in the neighboring country who will receive ₹ 5 per bag. There is no other selling expenses other than commission. Fixed cost are calculated on the basis of normal capacity utilization of the plant.

Assume 365 days in a year.

Being a cost manager of the company, you are required answer the following questions being asked by the management of the company:

MCQ 1 2 marks

What is the total variable cost of the bags being produced?

A ₹ 80,00,000

B ₹ 93,00,000

C ₹ 1,03,00,000

D ₹ 33,00,000

MCQ 2 2 marks

What is the total fixed cost of the bags being produced?

A ₹ 40,00,000

B ₹ 35,00,000

C ₹ 53,00,000

D ₹ 33,00,000

MCQ 3 2 marks

Calculate the break-even point if the sales price is ₹100 per bag?

A 65,159 bags

B 77,139 bags

C 93,000 bags

D 86,503 bags

MCQ 4 2 marks

Calculate the optimum run size and number of runs for bag manufacturing?

A 7,746 bags and 24 runs

B 8,000 bags and 23 runs

C 6,503 bags and 28 runs

D 5,478 bags and 33 runs

MCQ 5 2 marks

What is the interval between two consecutive runs?

A 11 days

B 10 days

C 15 days

D 19 days

ICAI'S ANSWERS

MCQ 1

(C) ₹ 1,03,00,000

MCQ 2

(D) ₹ 33,00,000

Working note

Particulars	₹
Variable Cost:	
Material	42,00,000
Labour (40,00,000 × 70%)	28,00,000
Factory Overheads (35,00,000 × 60%)	21,00,000
Administrative Overheads (10,00,000 × 30%)	3,00,000
Commission (1,80,000 × 5)	9,00,000
Total Variable cost	1,03,00,000
Fixed Cost:	
Labour (40,00,000 × 30%)	12,00,000
Factory Overheads (35,00,000 × 40%)	14,00,000
Administrative Overheads (10,00,000 × 70%)	7,00,000
Total Fixed Cost	33,00,000

MCQ 3

(B) 77,139 bags

Variable cost per bag = ₹ 1,03,00,000/1,80,000

= ₹ 57.22

Contribution per bag = ₹ 100 – ₹ 57.22 = ₹ 42.78

Break-even point (in number of bags)

= Fixed cost/Contribution per bag

= ₹33,00,000/₹42.78

= 77,139 bags

MCQ 4

(D) 5,478 bags and 33 runs

$$\text{EBQ} = \sqrt{\frac{2DS}{C}}$$

Where,

D = no. of bags to be produced annually

S = Set up cost per production run

C = Carrying cost per unit per annum

$$\text{EBQ} = \sqrt{\frac{2 \text{ x } 1,80,000 \text{ x } 1000}{1 \text{ x } 12}}$$

= 5,478 bags

No. of optimum runs = 1,80,000/5,478

= 32.86 or 33 runs

MCQ 5

(A) 11 days

Interval between 2 runs (in days) = 365 days/33 = 11 days

ICAI · P4 · MTP May 2025 · Series I · question p. 1-2 · answer p. 1, 1-2, 2

1.7 Cost control and cost reduction

8 questions · asked in 10 ICAI documents · 2 exam papers

54 **EXAM** **May 2026** Q6(b) **5 marks**

Also asked in Old P3 MTP Jul 2021 S-II Q6(a)

Same question, different ICAI answer: #55 Exam May 2024

Distinguish between cost control and cost reduction.

ICAI'S ANSWER

Difference between Cost Control and Cost Reduction

Cost Control	Cost Reduction
1. Cost control aims at *maintaining* the costs in accordance with the established standards.	1. Cost reduction is concerned with *reducing* costs. It challenges all standards and endeavours to improvise them continuously
2. Cost control seeks to *attain lowest possible cost* under existing conditions.	2. Cost reduction recognises no condition as permanent, since a *change will result in lower cost.*
3. In case of cost control, emphasis is on *past and present*	3. In case of cost reduction, it is on *present and future.*
4. Cost control is a *preventive* function	4. Cost reduction is a *corrective* function. It operates even when an efficient cost control system exists.
5. Cost control ends when targets are achieved.	5. Cost reduction has no visible end and is a continuous process.

ICAI · P4 · May 2026 exam · question p. 28 · answer p. 29

55 EXAM **May 2024** Q6(a) 5 marks

Same question, different ICAI answer: #54 Exam May 2026

Distinguish between cost control and cost reduction.

ICAI'S ANSWER

Cost Control	Cost Reduction
1. Cost control aims at maintaining the costs in accordance with the established standards.	1. Cost reduction is concerned with reducing costs. It challenges all standards and endeavours to improvise them continuously
2. Cost control seeks to attain lowest possible cost under existing conditions.	2. Cost reduction recognises no condition as permanent, since a change will result in lower cost.
3. In case of cost control, emphasis is on past and present	3. In case of cost reduction, it is on present and future.
4. Cost control is a preventive function	4. Cost reduction is a corrective function. It operates even when an efficient cost control system exists.
5. Cost control ends when targets are achieved.	5. Cost reduction has no visible end and is a continuous process.

ICAI · P4 · May 2024 exam · question p. 24 · answer p. 25

56 RTP **Jan 2026** Q19(d)

Also asked in MTP May 2025 S-I Q6(d)

Cost control emphasis is on past and present, while cost reduction emphasis is on present and future. DISCUSS some more differences between Cost control and Cost reduction.

ICAI'S ANSWER

Cost Control	Cost Reduction
1. Cost control aims at maintaining the costs in accordance with the established standards.	1. Cost reduction is concerned with reducing costs. It challenges all standards and endeavours to improvise them continuously
2. Cost control seeks to attain lowest possible cost under existing conditions.	2. Cost reduction recognises no condition as permanent, since a change will result in lower cost.
3. Cost control is a preventive function.	3. Cost reduction is a corrective function. It operates even when an efficient cost control system exists.
4. Cost control ends when targets are achieved.	4. Cost reduction has no visible end and is a continuous process.

ICAI · P4 · RTP Jan 2026 · question p. 23 · answer p. 59

57 RTP **Old P3 May 2023** Q15(d) old scheme

DISCUSS the difference between cost control and cost reduction.

ICAI'S ANSWER

S. No.	Cost Control	Cost Reduction
1	Cost control aims at maintaining the costs in accordance with the established standards.	Cost reduction is concerned with reducing costs. It challenges all standards and endeavours to improvise them continuously
2	Cost control seeks to attain lowest possible cost under existing conditions.	Cost reduction recognises no condition as permanent, since a change will result in lower cost.
3	In case of cost control, emphasis is on past and present	In case of cost reduction, it is on present and future.
4	Cost control is a preventive function	Cost reduction is a corrective function. It operates even when an efficient cost control system exists.
5	Cost control ends when targets are achieved.	Cost reduction has no visible end and is a continuous process.

ICAI · Old P3 · RTP May 2023 · question p. 11 · answer p. 27

58 RTP **Old P3 Nov 2022** Q15(b) old scheme

Same question, different ICAI answer: #59 Old P3 RTP Dec 2021

Differentiate between Cost Control and Cost Reduction.

ICAI'S ANSWER

Cost Control		Cost Reduction	
1.	Cost control aims at **maintaining** the costs in accordance with the established standards.	1.	Cost reduction is concerned with **reducing** costs. It challenges all standards and endeavours to improvise them continuously
2.	Cost control seeks to **attain lowest possible cost** under existing conditions.	2.	Cost reduction recognises no condition as permanent, since a **change will result in lower cost.**
3.	In case of cost control, emphasis is on **past and present**	3.	In case of cost reduction, it is on **present and future.**
4.	Cost control is a **preventive** function	4.	Cost reduction is a **corrective** function. It operates even when an efficient cost control system exists.
5.	Cost control ends when targets are achieved.	5.	Cost reduction has no visible end and is a continuous process.

ICAI · Old P3 · RTP Nov 2022 · question p. 13 · answer p. 35

59 RTP **Old P3 Dec 2021** Q15(a) old scheme

Same question, different ICAI answer: #58 Old P3 RTP Nov 2022

DIFFERENTIATE between Cost Control and Cost Reduction.

ICAI'S ANSWER

S. No.	Cost Control	Cost Reduction
1	Cost control aims at maintaining the costs in accordance with the established standards.	Cost reduction is concerned with reducing costs. It challenges all standards and endeavours to improvise them continuously.
2	Cost control seeks to attain lowest possible cost under existing conditions.	Cost reduction recognises no condition as permanent, since a change will result in lower cost.
3	In case of cost control, emphasis is on past and present.	In case of cost reduction, it is on present and future.
4	Cost control is a preventive function.	Cost reduction is a corrective function. It operates even when an efficient cost control system exists.
5	Cost control ends when targets are achieved.	Cost reduction has no visible end and is a continuous process.

ICAI · Old P3 · RTP Dec 2021 · question p. 13 · answer p. 33-34

60 MTP **Old P3 Nov 2023 S-II** Q6(a) 5 marks old scheme

State the difference between cost control and cost reduction.

ICAI'S ANSWER

Difference between Cost Control and Cost Reduction

Cost Control	Cost Reduction
1. Cost control aims at maintaining the costs in accordance with the established standards.	1. Cost reduction is concerned with reducing costs. It challenges all standards and endeavours to better them continuously.
2. Cost control seeks to attain lowest possible cost under existing conditions.	2. Cost reduction recognises no condition as permanent, since a change will result in lower cost.
3. In case of Cost Control, emphasis is on past and present.	3. In case of cost reduction, it is on present and future.
4. Cost Control is a preventive function.	4. Cost reduction is a corrective function. It operates even when an efficient cost control system exists.
5. Cost control ends when targets are achieved.	5. Cost reduction has no visible end.

ICAI · Old P3 · MTP Nov 2023 · Series II · question p. 7 · answer p. 13

61 MTP **Old P3 Nov 2020 S-I** Q6(a) 5 marks old scheme

DISCUSS the steps to be followed to exercise control over cost.

ICAI'S ANSWER

To exercise control over cost, following steps are followed:

(i) *Determination of pre-determined standard or results*: Standard cost or performance targets for a cost object or a cost centre is set before initiation of production or service activity. These are desired cost or result that need to be achieved.

(ii) *Measurement of actual performance*: Actual cost or result of the cost object or cost centre is measured. Performance should be measured in the same manner in which the targets are set i.e. if the targets are set up operation-wise, and then the actual costs should also be collected and measured operation-wise to have a common basis for comparison.

(iii) *Comparison of actual performance with set standard or target*: The actual performance so measured is compared against the set standard and desired target. Any deviation (variance) between the two is noted and reported to the appropriate person or authority.

(iv) *Analysis of variance and action*: The variance in results so noted are further analysed to know the reasons for variance and appropriate action is taken to ensure compliance in future. If necessary, the standards are further amended to take developments into account.

ICAI · Old P3 · MTP Nov 2020 · Series I · question p. 8 · answer p. 14

1.8 Methods and techniques of costing, and digital costing

19 questions · asked in 15 ICAI documents · 5 exam papers

62 EXAM **Sep 2024** Q6(a) 5 marks

Describe any five benefits of the Digital Costing System.

ICAI'S ANSWER

Benefits of Digital Costing System are as follows:

(i) Ascertainment of cost with certainty on a cost object. This helps to analyse the activities for cost allocation and apportionment.

(ii) Analysis of data on time spent on each activity to study and formulate incentive plans.

(iii) Helps in material requirement planning and scheduling the material procurement. Data on resource consumption can be analysed for resource optimisation and finding the possibilities for zero wastage and Just-in Time (JIT).

(iv) Helps to identify and eliminate the non-value-added activities.

(v) Data on resource consumption is helpful in setting the standards and measurement of variances on real time basis.

(vi) Data on current market prices of material and consumables helps to estimate cost and setting standards on Marked to Market (M2M) basis.

(vii) Extrapolation of data on customer behaviour towards the products to predict the market demand. It is helpful is preparation of budgets and planning of production.

(viii) A better analysis of cost behaviour improves the cost benefit analysis and equipping the management in informed decision making.

ICAI · P4 · Sep 2024 exam · question p. 36 · answer p. 37

63 EXAM Old P3 May 2022 Q6(e) 5 marks old scheme

Identify the methods of costing from the following statements:

(i) Costs are directly charged to a group of products.

(ii) Nature of the product is complex and method cannot be ascertained.

(iii) Costs ascertained for a single product.

(iv) All costs are directly charged to a specific job.

(v) Costs are charged to operations and averaged over units produced.

ICAI'S ANSWER

Method of costing followed:

	Situation	Method of costing
(i)	Costs are directly charged to a group of products.	Batch costing
(ii)	Nature of the product is complex and method cannot be ascertained.	Multiple costing
(iii)	Cost is ascertained for a single product.	Unit/ Single/Output costing
(iv)	All costs are directly charged to a specific job.	Job costing
(v)	Costs are charged to operations and averaged over units produced.	Process costing

ICAI · Old P3 · May 2022 exam · question p. 27 · answer p. 30

64 EXAM Old P3 Dec 2021 Question 6 · Q6(a) to Q6(e) 25 marks old scheme

QUESTION 6

Answer any four of the following:

Q6(a) 5 marks

Briefly explain the 'techniques of costing'.

Q6(b) 5 marks

Narrate the terms 'Joint Products' and 'By-Products' with an example of each term.

Q6(c) 5 marks

Discuss the steps involved in setting labour time standards.

Q6(d) 5 marks

What is 'Budgetary Control System' and discuss the components of the same.

Q6(e) 5 marks

Describe the difference between 'Cost Control' and 'Cost Reduction'.

ICAI'S ANSWERS

Q6(a)

Techniques	Description
Uniform Costing	When a number of firms in an industry agree among themselves to follow the same system of costing in detail, adopting common terminology for various items and processes they are said to follow a system of uniform costing. Advantages of such a system are: i. A comparison of the performance of each of the firms can be made with that of another, or with the average performance in the industry. ii. Under such a system, it is also possible to determine the cost of production of goods which is true for the industry as a whole. It is found useful when tax-relief or protection is sought from the Government.
Marginal Costing	It is defined as the ascertainment of marginal cost by differentiating between fixed and variable costs. It is used to ascertain effect of changes in volume or type of output on profit.
Standard Costing and Variance Analysis	It is the name given to the technique whereby *standard costs are pre-determined and subsequently compared with the recorded actual costs.* It is thus a technique of cost ascertainment and cost control. This technique may be used in conjunction with any method of costing. However, it is especially suitable where the manufacturing method involves production of standardised goods of repetitive nature.
Historical Costing	It is the ascertainment of costs after they have been incurred. This type of costing has limited utility. • *Post Costing*: It means ascertainment of cost after production is completed. • *Continuous costing*: Cost is ascertained as soon as the job is completed or even when the job is in progress.
Absorption Costing	It is the practice of charging all costs, both variable and fixed to operations, processes or products. This differs from marginal costing where fixed costs are excluded.
Direct costing	Direct costing is a specialized form of cost analysis that only uses variable costs to make decisions. It does not consider fixed costs, which are assumed to be associated with the time periods in which they are incurred.

Q6(b)

(i) Joint Products - Joint products represent "two or more products separated in the course of the same processing operation usually requiring further processing, each product being in such proportion that no single product can be designated as a major product".

In other words, two or more products of equal importance, produced, simultaneously from the same process, with each having a significant relative sale value are known as joint products.

For example, in the oil industry, gasoline, fuel oil, lubricants, paraffin, coal tar, asphalt and kerosene are all produced from crude petroleum. These are known as joint products.

(ii) By-Products - These are defined as "products recovered from material discarded in a main process, or from the production of some major products, where the material value is to be considered at the time of severance from the main product." Thus, by-products emerge as a result of processing operation of another product or they are produced from the scrap or waste of materials of a process. In short, a by-product is a secondary or subsidiary product which emanates as a result of manufacture of the main product.

The point at which they are separated from the main product or products is known as split-off point. The expenses of processing are joint till the split –off point.

Examples of by-products are molasses in the manufacture of sugar, tar, ammonia and benzole obtained on carbonisation of coal and glycerine obtained in the manufacture of soap.

Q6(c)

Procedure of Setting Labour Time Standards

The following are the steps involved in setting labour standards:

(a) **Standardisation:** Products to be produced are decided based on production plan and customer's order.

(b) **Labour specification:** Types of labour and labour time is specified. Labour time specification is based on past records and it takes into account normal wastage of time.

(c) **Standardisation of methods:** Selection of proper machines to use proper sequence and method of operations.

(d) **Manufacturing layout:** A plan of operation for each product listing the operations to be performed is prepared.

(e) **Time and motion study:** It is conducted for selecting the best way of completing the job or motions to be performed by workers and the standard time which an average worker will take for each job. This also takes into account the learning efficiency and learning effect.

(f) **Training and trial:** Workers are trained to do the work and time spent at the time of trial run is noted down.

Q6(d)

Budgetary Control System: It is the system of management control and accounting in which all the operations are forecasted and planned in advance to the extent possible and the actual results compared with the forecasted and planned results.

Components of Budgetary Control System: The policy of a business for a defined period is represented by the master budget, the detailed components of which are given in a number of

individual budgets called functional budgets. These functional budgets are broadly grouped under the following heads:

1. **Physical budgets:** Those budgets which contain information in quantitative terms such as the physical units of sales, production etc. This may include quantity of sales, quantity of production, inventories, and manpower budgets are physical budgets.
2. **Cost budgets:** Budgets which provides cost information in respect of manufacturing, administration, selling and distribution, etc. for example, manufacturing costs, selling costs, administration cost, and research and development cost budgets are cost budgets.
3. **Profit budgets:** A budget which enables the ascertainment of profit. For example, sales budget, profit and loss budget, etc.
4. **Financial budgets:** A budget which facilitates in ascertaining the financial position of a concern, for example, cash budgets, capital expenditure budget, budgeted balance sheet etc.

Q6(e)

Cost Control	Cost Reduction
1. Cost control aims at *maintaining* the costs in accordance with the established standards.	1. Cost reduction is concerned with *reducing* costs. It challenges all standards and endeavours to improvise them continuously
2. Cost control seeks to *attain lowest possible cost* under existing conditions.	2. Cost reduction recognises no condition as permanent, since a *change will result in lower cost.*
3. In case of cost control, emphasis is on *past and present*	3. In case of cost reduction, it is on *present and future.*
4. Cost control is a *preventive* function	4. Cost reduction is a *corrective* function. It operates even when an efficient cost control system exists.
5. Cost control ends when targets are achieved.	5. Cost reduction has no visible end and is a continuous process.

ICAI · Old P3 · Dec 2021 exam · question p. 23 · answer p. 23-24, 24-25, 25, 25-26, 26

65 EXAM **Old P3 Jan 2021** Q6(d) 5 marks old scheme

State the method of costing that would be most suitable for:

(i) Oil Refinery

(ii) Interior Decoration

(iii) Airlines Company

(iv) Advertising

(v) Car Assembly

ICAI'S ANSWER

Method of Costing

S.No.	Industry	Method of Costing
(i)	Oil Refinery	Process Costing
(ii)	Interior Decoration	Job Costing
(iii)	Airlines Company	Operation/ Service Costing
(iv)	Advertising	Job Costing
(v)	Car Assembly	Multiple Costing

ICAI · Old P3 · Jan 2021 exam · question p. 25 · answer p. 27

66 EXAM Old P3 Jan 2021 Q6(e) 5 marks old scheme

Give any five examples of the impact of use of Information Technology in Cost Accounting.

ICAI'S ANSWER

Example of Impact of Information Technology in cost accounting may include the following:

(i) After the introduction of ERPs, different functional activities get integrated and as a consequence a *single entry into the accounting system* provides custom made reports for every purpose and saves an organisation from preparing different sets of documents. Reconciliation process of results of both cost and financial accounting systems become simpler and less sophisticated.

(ii) A move towards *paperless environment* can be seen where documents like Bill of Material, Material Requisition Note, Goods Received Note, labour utilisation report etc. are no longer required to be prepared in multiple copies, the related department can get *e-copy* from the system.

(iii) Information Technology with the help of internet (including intranet and extranet) helping in resource procurement and mobilisation. For example, production department can get materials from the stores without issuing material requisition note physically. Similarly, purchase orders can be initiated to the suppliers with the help of extranet. This enables an entity to shift towards Just-in-Time (JIT) approach of inventory management and production.

(iv) Cost information for a cost centre or cost object is ascertained with accuracy in timely manner. Each cost centre and cost object is codified and all related costs are assigned to the cost objects or cost centres using assigned codes. This automates the cost accumulation and ascertainment process. *The cost information can be customised as per the requirement*. For example, when an entity manufacture or provide services, are able to know information job-wise, batch-wise, process-wise, cost centre wise etc.

(v) Uniformity in preparation of report, budgets and standards can be achieved with the help of IT. ERP software plays an important role in bringing uniformity irrespective of location, currency, language and regulations.

(vi) Cost and revenue variance reports are generated in *real time basis* which enables the management to take control measures immediately.

(vii) IT enables an entity to monitor and analyse each process of manufacturing or service activity closely to eliminate non value added activities.

ICAI · Old P3 · Jan 2021 exam · question p. 25 · answer p. 28

67 EXAM Old P3 Nov 2020 Q6(d) 5 marks old scheme

State the Method of Costing to be used in the following industries:

(i) Real Estate

(ii) Motor repairing workshop

(iii) Chemical Industry

(iv) Transport service

(v) Assembly of bicycles

(vi) Biscuits manufacturing Industry

(vii) Power supply Companies

(viii) Car manufacturing Industry

(ix) Cement Industry

(x) Printing Press

ICAI'S ANSWER

Method of costing used in different industries:

S. No.	Industries	Method of Costing
(i)	Real Estate	Contract Costing
(ii)	Motor Repairing Workshop	Job Costing
(iii)	Chemical Industry	Process Costing
(iv)	Transport Service	Service/Operating Costing
(v)	Assembly of Bicycles	Unit/ Single/Output/Multiple Costing
(vi)	Biscuits Manufacturing Industry	Batch Costing
(vii)	Power Supply Companies	Service/Operating Costing
(viii)	Car Manufacturing Industry	Multiple Costing
(ix)	Cement Industry	Unit/Single/Output Costing
(x)	Printing Press	Job Costing

ICAI · Old P3 · Nov 2020 exam · question p. 22 · answer p. 24

68 RTP May 2024 Q13(b)

Same question, different ICAI answer: #69 Old P3 RTP May 2022 · #75 Old P3 MTP Nov 2023 S-I

Discuss the impact of Information Technology in Cost Accounting.

ICAI'S ANSWER

The impact of IT in cost accounting may include the followings:

(i) After the introduction of ERPs, different functional activities get integrated and as a consequence a single entry into the accounting system provides custom made reports for every purpose and saves

an organisation from preparing different sets of documents. Reconciliation process of results of both cost and financial accounting systems become simpler and less sophisticated.

(ii) A move towards paperless environment can be seen where documents like Bill of Material, Material Requisition Note, Goods Received Note, labour utilisation report etc. are no longer required to be prepared in multiple copies, the related department can get e-copy from the system.

(iii) Information Technology with the help of internet (including intranet and extranet) helps in resource procurement and mobilisation. For example, production department can get materials from the stores without issuing material requisition note physically. Similarly, purchase orders can be initiated to the suppliers with the help of extranet. This enables an entity to shift towards Just-in-Time (JIT) approach of inventory management and production.

(iv) Cost information for a cost centre or cost object is ascertained with accuracy in timely manner. Each cost centre and cost object is codified and all related costs are assigned to the cost object or cost centre. This process automates the cost accumulation and ascertainment process. The cost information can be customised as per the requirement. For example, when an entity manufactures or provide services, it can know information job-wise, batch-wise, process-wise, cost centre wise etc.

(v) Uniformity in preparation of report, budgets and standards can be achieved with the help of IT. ERP software plays an important role in bringing uniformity irrespective of location, currency, language and regulations.

(vi) Cost and revenue variance reports are generated in real time basis which enables the management to take control measures immediately.

(vii) IT enables an entity to monitor and analyse each process of manufacturing or service activity closely to eliminate non-value-added activities.

The above are examples of few areas where Cost Accounting is done with the help of IT.

ICAI · P4 · RTP May 2024 · question p. 17 · answer p. 42-43

69 RTP **Old P3 May 2022** Q14(d) old scheme

Same question, different ICAI answer: #68 RTP May 2024 · #75 Old P3 MTP Nov 2023 S-I

DISCUSS the impact of Information Technology in Cost Accounting.

ICAI'S ANSWER

The impact of IT in cost accounting may include the following:

(i) After the introduction of ERPs, different functional activities get integrated and as a consequence a single entry into the accounting system provides custom made reports for every purpose and saves an organisation from preparing different sets of documents. Reconciliation process of results of both cost and financial accounting systems become simpler and less sophisticated.

(ii) A move towards paperless environment can be seen where documents like Bill of Material, Material Requisition Note, Goods Received Note, labour utilisation report etc. are no longer required to be prepared in multiple copies, the related department can get e-copy from the system.

(iii) Information Technology with the help of internet (including intranet and extranet) helps in resource procurement and mobilisation. For example, production department can get materials from the stores without issuing material requisition note physically. Similarly, purchase orders can be initiated to the suppliers with the help of extranet. This enables an entity to shift towards Just-in-Time (JIT) approach of inventory management and production.

(iv) Cost information for a cost centre or cost object is ascertained with accuracy in timely manner. Each cost centre and cost object is codified and all related costs are assigned to the cost object or cost centre. This process automates the cost accumulation and ascertainment process. The cost information can be customised as per the requirement. For example, when an entity manufactures or provide services, it can know information job-wise, batch-wise, process-wise, cost centre wise etc.

(v) Uniformity in preparation of report, budgets and standards can be achieved with the help of IT. ERP software plays an important role in bringing uniformity irrespective of location, currency, language and regulations.

(vi) Cost and revenue variance reports are generated in real time basis which enables the management to take control measures immediately.

(vii) IT enables an entity to monitor and analyse each process of manufacturing or service activity closely to eliminate non-value-added activities.

The above are examples of few areas where Cost Accounting is done with the help of IT.

ICAI · Old P3 · RTP May 2022 · question p. 10 · answer p. 32

70 RTP **Old P3 May 2020** Q13(b) old scheme

DISCUSS the impact of Information Technology (IT) on cost accounting system.

ICAI'S ANSWER

The impact of IT in cost accounting system may include the following:

(i) After the introduction of ERPs, different functional activities get integrated and as a consequence a single entry into the accounting system provides custom made reports for every purpose and saves an organisation from preparing different sets of documents. Reconciliation process of results of both cost and financial accounting systems become simpler and less sophisticated.

(ii) A move towards paperless environment can be seen where documents like Bill of Material, Material Requisition Note, Goods Received Note, labour utilisation report etc. are no longer required to be prepared in multiple copies, the related department can get e-copy from the system.

(iii) Information Technology with the help of internet (including intranet and extranet) helps in resource procurement and mobilisation. For example, production department can get materials from the stores without issuing material requisition note physically. Similarly, purchase orders can be initiated to the suppliers with the help of extranet. This enables an entity to shift towards Just-in-Time (JIT) approach of inventory management and production.

(iv) Cost information for a cost centre or cost object is ascertained with accuracy in timely manner. Each cost centre and cost object is codified and all related costs are assigned to the cost object or cost centre. This process automates the cost accumulation and ascertainment process. The cost information can be customised as per the requirement. For example, when an entity manufacture or provide services, it can know information job-wise, batch-wise, process-wise, cost centre wise etc.

(v) Uniformity in preparation of report, budgets and standards can be achieved with the help of IT. ERP software plays an important role in bringing uniformity irrespective of location, currency, language and regulations.

(vi) Cost and revenue variance reports are generated in real time basis which enables the management to take control measures immediately.

(vii) IT enables an entity to monitor and analyse each process of manufacturing or service activity closely to eliminate non value added activities.

The above are examples of few areas where Cost Accounting is done with the help of IT.

ICAI · Old P3 · RTP May 2020 · question p. 10 · answer p. 28

71 MTP **Jan 2026 S-I** Q6(a) 5 marks

You are a cost accountant working with a consultancy that advises different types of businesses. A few clients have approached you seeking advice on the best costing methods for their operations. Analyze each of the following businesses and recommend the most suitable method of costing for each, explaining your reasoning:"

(i) A company that manufactures cement in large quantities.

(ii) A custom furniture workshop that designs unique pieces per customer.

(iii) A pharmaceutical company that produces medicine in batches.

(iv) A food processing unit producing packaged snacks on a continuous basis.

(v) A transport company that operates a fleet of buses across cities.

ICAI'S ANSWER

Business Scenario	Costing Method	Key Reason
Cement Manufacturing	Unit/Output Costing	Uniform products, mass production
Custom Furniture Workshop	Job Costing	Unique, custom jobs per client
Pharmaceutical Batch Production	Batch Costing	Production in identifiable batches
Food Processing (Snacks)	Process Costing	Continuous and standardized production
Bus Transport Company	Service/Operating Costing	Costing per service unit (e.g., km, passenger-km)

ICAI · P4 · MTP Jan 2026 · Series I · question p. 15-16 · answer p. 18

72 MTP **Sep 2025 S-II** Q6(a) 5 marks

STATE the method of costing and the suggested unit of cost for the following nature of output along with the one example of industry.

Nature of Output	Method	Cost	Examples of Industries
A Series of Processes			
Similar units of a Single Product, produced by Single Process			
Customer Specifications: single Unit			
Consisting of multiple varieties of activities and processes			

ICAI'S ANSWER

The following table summarises the various methods of costing applied in different industries:

Nature of Output	Method	Cost	Examples of Industries
A Series of Processes	Process costing or Operation Costing	For each process	Sugar
Similar units of a Single Product, produced by Single Process	Unit or output or Single Costing	For the entire activity, but averaged for the output	Cold Drinks
Customer Specifications: single Unit	Job Costing	For each order/ assignment/job	Advertising
Consisting of multiple varieties of activities and processes	Multiple Costing	Combination of any method	Car Assembly

ICAI · P4 · MTP Sep 2025 · Series II · question p. 16 · answer p. 25

73 MTP **Jan 2025 S-I** Q6(d) 4 marks

IDENTIFY the method of costing in the following cases and give one example of industry where this method is followed:

(i) Cost of each job is ascertained separately. It is suitable in all cases where work is undertaken on receiving a customer's order.

(ii) Cost of completing each stage of work is ascertained.

(iii) Each group is treated as a unit of cost and thus separately costed. Here cost per unit is determined by dividing the cost of the group by the number of units produced.

(iv) A combination of two or more methods of costing.

ICAI'S ANSWER

S. No.	Method of costing	Example of industry where this method is followed
(i)	Job Costing	Printing press
(ii)	Process Costing	Paper and Pulp
(iii)	Batch Costing	Bakery
(iv)	Multiple Costing	Bicycles

ICAI · P4 · MTP Jan 2025 · Series I · question p. 17 · answer p. 22

74 MTP **May 2024 S-I** Q6(b) 5 marks

"Technology has played a significant role in cost accounting enabling business to automate their process."

EXPLAIN the impact of Information Technology in Cost Accounting in the light of above statement.

ICAI'S ANSWER

The impact of IT in cost accounting may include the followings:

(i) After the introduction of ERPs, different functional activities get integrated and as a consequence a single entry into the accounting system provides custom made reports for every purpose and saves an organisation from preparing different sets of documents. Reconciliation process of results of both cost and financial accounting systems become simpler and less sophisticated.

(ii) A move towards paperless environment can be seen where documents like Bill of Material, Material Requisition Note, Goods Received Note, labour utilisation report etc. are no longer required to be prepared in multiple copies, the related department can get e-copy from the system.

(iii) Information Technology with the help of internet (including intranet and extranet) helps in resource procurement and mobilisation. For example, production department can get materials from the stores without issuing material requisition note physically. Similarly, purchase orders can be initiated to the suppliers with the help of extranet. This enables an entity to shift towards Just-in-Time (JIT) approach of inventory management and production.

(iv) Cost information for a cost centre or cost object is ascertained with accuracy in timely manner. Each cost centre and cost object is codified and all related costs are assigned to the cost object or cost centre. This process automates the cost accumulation and ascertainment process. The cost information can be customised as per the requirement. For example, when an entity manufactures or provide services, it can know information job-wise, batch-wise, process-wise, cost centre wise etc.

(v) Uniformity in preparation of report, budgets and standards can be achieved with the help of IT. ERP software plays an important role in bringing uniformity irrespective of location, currency, language and regulations.

(vi) Cost and revenue variance reports are generated in real time basis which enables the management to take control measures immediately.

(vii) IT enables an entity to monitor and analyse each process of manufacturing or service activity closely to eliminate non-value-added activities.

The above are examples of few areas where Cost Accounting is done with the help of IT.

ICAI · P4 · MTP May 2024 · Series I · question p. 11 · answer p. 15-16

75 MTP **Old P3 Nov 2023 S-I** Q6(d) 5 marks old scheme

Also asked in Old P3 MTP May 2023 S-II Q6(d)

Same question, different ICAI answer: #68 RTP May 2024 · #69 Old P3 RTP May 2022

DISCUSS the impact of Information Technology in Cost Accounting.

ICAI'S ANSWER

The impact of IT in cost accounting may include the followings:

(i) After the introduction of ERPs, different functional activities get integrated and as a consequence a single entry into the accounting system provides custom made reports for every purpose and saves an organisation from preparing different sets of documents. Reconciliation process of results of both cost and financial accounting systems become simpler and less sophisticated.

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(iii) Information Technology with the help of internet (including intranet and extranet) helps in resource procurement and mobilisation. For example, production department can get materials from the stores without issuing material requisition note physically. Similarly, purchase orders can be initiated to the suppliers with the help of extranet. This enables an entity to shift towards Just-in-Time (JIT) approach of inventory management and production.

(iv) Cost information for a cost centre or cost object is ascertained with accuracy in timely manner. Each cost centre and cost object is codified and all related costs are assigned to the cost object or cost centre. This process automates the cost accumulation and ascertainment process. The cost information can be customised as per the requirement. For example, when an entity manufacture or provide services, it can know information job-wise, batch-wise, process-wise, cost centre wise etc.

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(vi) Cost and revenue variance reports are generated in real time basis which enables the management to take control measures immediately.

(vii) IT enables an entity to monitor and analyse each process of manufacturing or service activity closely to eliminate non value added activities.

The above are examples of few areas where Cost Accounting is done with the help of IT.

ICAI · Old P3 · MTP Nov 2023 · Series I · question p. 7 · answer p. 13

76 MTP **Old P3 May 2022 S-II** Q6(b) 5 marks old scheme

STATE the method of costing for the following industries:

(i) Sugar manufacturing

(ii) Bridge Construction

(iii) Advertising

(iv) Car Assembly

ICAI'S ANSWER

S. No.	Industry	Method of costing
(i)	Sugar manufacturing	Process costing
(ii)	Bridge Construction	Contract Costing
(iii)	Advertising	Job costing
(iv)	Car Assembly	Multiple Costing (Combination of any method)

ICAI · Old P3 · MTP May 2022 · Series II · question p. 7 · answer p. 13

CHAPTER 2

Cost Sheet

2.1 Cost sheet for a manufacturer

90 questions · asked in 59 ICAI documents · 16 exam papers

77 EXAM **Sep 2026** Q5(a) 7 marks

The following data is given in the books of Vikram Industries :

Particulars	Amount (₹)
Opening Inventories :	
- Raw materials	15,00,000
- Work-in-process	20,00,000
- Finished goods (12,000 units)	10,80,000
Closing Inventories :	
- Raw materials	16,50,000
- Work-in-process	18,00,000
Raw materials purchase	1,68,00,000
Wages paid to production workers	40,00,000
Expenses paid for utilities	2,50,000
Office and administrative expenses paid (General)	30,00,000
Travelling allowance paid to office staff	1,50,000
Selling expenses	7,00,000
Machine hours worked	24,000 hours
Machine hour rate	₹ 10 per hour
Units sold (on FIFO basis)	1,80,000 units
Units produced	2,10,000 units
Desired profit	20% on sales

Compute the values of the following figures in rupees as they will appear in the Cost Sheet :

(i) Prime Cost

(ii) Cost of Production

(iii) Cost of Goods Sold

(iv) Cost of Sales

(v) Sales Revenue

(You are not required to present the entire cost sheet. Present relevant figures only.)

ICAI'S ANSWER

ICAI has not published its answers to this paper yet.

ICAI · P4 · Sep 2026 exam · question p. 19-20

78 EXAM **Sep 2026** Q6(c) 4 marks

State how you would treat the following items in the Cost Sheet :

(i) Wages paid to idle workers during a sudden government-imposed lockdown.

(ii) A subsidy received from the State Government for using eco-friendly raw materials.

(iii) Interest paid on a term loan taken for purchase of plant and machinery.

(iv) GST paid on purchase of raw materials, for which input tax credit is eligible.

ICAI'S ANSWER

ICAI has not published its answers to this paper yet.

ICAI · P4 · Sep 2026 exam · question p. 23

79 EXAM **May 2026** Q2(a) 7 marks

The cost particulars of cotton yarn manufactured by a textile mill in April 2026 are as follows:

Direct Material:	- Cotton Bales: 2,000 kgs @ ₹ 120 per kg - Other materials : 600 kgs @ ₹ 85 per kg
Direct Wages:	- 75 skilled men @ ₹ 60 per day for 25 days - 40 unskilled men @ ₹ 30 per day for 25 days
Direct Expenses:	Hire charge for Special equipment ₹ 9,000 Power and fuel ₹ 3,000
Works Overhead	160% of direct wages
Administration overhead related with production	₹ 45,000
Selling and distribution expenses	₹ 37,500
Selling Price per kg of cotton yarn	₹ 600

Additional Information:

- 1,500 kgs. of Cotton yarn was produced.
- Waste cotton and chemical scrap realized during the manufacturing process amounted to ₹ 23,500.
- There were no opening or closing stocks of raw materials and finished stock.

You are required to prepare a cost statement for the month of April 2026 showing:

(i) Prime Cost

(ii) Net Works Cost

(iii) Cost of Production

(iv) Cost of Sales

(v) Total Profit & Sales

ICAI'S ANSWER

Cost Sheet of Cotton Yarn Manufacturing for the month of April 2026

Particulars	(₹)	(₹)
Direct Material:		
Cotton Bales (2,000 kgs × ₹ 120)	2,40,000	
Other materials (600 kgs × ₹ 85)	51,000	2,91,000
Add: Direct Wages		
75 skilled men @ ₹ 60 per day for 25 days	1,12,500	
40 unskilled men @ ₹ 30 per day for 25 days	30,000	1,42,500
Add: Direct Expenses		
Hire charges for special equipment	9,000	
Power and fuel	3,000	12,000
Prime Cost		**4,45,500**
Add: Works / Factory Overheads (160% of direct wages)		2,28,000
Works Cost / Factory Cost		6,73,500
Less: Sale of waste cotton and cotton scrap		(23,500)
Net Works Cost		**6,50,000**
Add: Administration overhead related to production		45,000
Cost of Production		**6,95,000**
Add: Selling and distribution expenses		37,500
Cost of Sales		**7,32,500**
Sales (1,500 kgs × ₹ 600)		**9,00,000**
Profit		**1,67,500**

ICAI · P4 · May 2026 exam · question p. 12-13 · answer p. 14

80 EXAM **Jan 2026** Q2(a) 9 marks

Dhol Makers Ltd. sold 6,000 units of a musical instrument "Dholak",

The cost data for the year ended 31st March, 2025 is as follows:

Particulars	(₹)
Expenses paid for pollution control and maintenance	93,400
Expenses paid for quality control check activities	70,000
Lease rent of production assets	1,20,000
Hire charges paid for hiring specific equipment	315 per hour
Primary packing cost	51,600
Direct Labour Wages	6,00,000
Fee paid for technical assistance	1,26,000
Opening stock of finished goods as on 1st April, 2024 (2,000 units)	10,10,000
Closing stock of finished goods as on 31st March, 2025 (1,000 units)	?
Cost of production	27,50,000

It is further ascertained that:

(1) Labour is paid @ ₹ 1,000 per labour hour.

(2) Specific equipment is hired for the number of labour hours worked.

(3) The company follows the 'First-In, First-Out' (FIFO) method for closing stock valuation.

(4) Selling and distribution overhead is ₹ 25 per dholak.

(5) Profit is 20% on sales.

You are required to prepare a cost sheet showing:

(i) Raw material consumed

(ii) Prime cost

(iii) Works/Factory cost

(iv) Cost of goods sold

(v) Cost of sales

(vi) Net profit & Sales

ICAI'S ANSWER

Cost Sheet of Dhol Makers Ltd. for the year ended 31st March, 2025

Sl. No.	Particulars	(₹)	(₹)
(i)	Material Consumed (see working note 3)		15,00,000
(ii)	Direct Labour Wages		6,00,000
(iii)	Direct expenses:		

Sl. No.	Particulars	(₹)	(₹)
	Hire Charges paid for hiring specific equipment (₹ 315 per hour x 600 labour hours worked) (see working note 1)	1,89,000	
	Fees paid for technical assistance	1,26,000	3,15,000
	Prime Cost		**24,15,000**
(iv)	Works/Factory overheads:		
	Lease rent of production assets	1,20,000	
	Expenses paid for pollution control and maintenance	93,400	2,13,400
	Works/Factory Cost		**26,28,400**
(v)	Add: Expenses paid for quality control check activities		70,000
(vi)	Add: Primary packing cost		51,600
	Cost of Production (Given)		**27,50,000**
	Add: Opening stock of finished goods (2,000 units)		10,10,000
	Less: Closing stock of finished goods (₹ 27,50,000/5,000 units X 1,000 units)		(5,50,000)
	Cost of Goods Sold		**32,10,000**
(vii)	Selling and Distribution Overheads (₹ 25 per unit x 6,000 units sold)		1,50,000
	Cost of Sales		**33,60,000**
	Profit (20% on sales or 25% on cost)		**8,40,000**
	Sales		**42,00,000**

Working Notes:

1. Calculation of No. of Direct Labour Hours worked

= Direct Labour Wages/Labour Rate per hour

= ₹ 6,00,000/₹ 1,000 per hour

= 600 Direct Labour Hours

2. Calculation of No. of units produced:

Particulars	Units
No. of units sold	6,000
Add: No. of units of closing stock of finished goods	1,000
Less: No. of units opening stock of finished goods	(2,000)
No. of units produced	5,000

3. Value of Raw Material Consumed:

Particulars	(₹)	(₹)
Cost of Production		27,50,000
Less: Expenses paid for quality control check activities		(70,000)
Less: Primary packing cost		(51,600)
Works/Factory Cost		26,28,400
Less: Works/ Factory overheads:		
Lease rent of production assets	1,20,000	
Expenses paid for pollution control and maintenance	93,400	(2,13,400)
Prime Cost		24,15,000
Less: Direct expenses:		
Hire Charges paid for hiring specific equipment (₹ 315 per hour x 600 labour hours worked) (see working note 1)	1,89,000	
Fees paid for technical assistance	1,26,000	(3,15,000)
Less: Direct Labour Wages		(6,00,000)
Value of Material Consumed		**15,00,000**

ICAI · P4 · Jan 2026 exam · question p. 12-13 · answer p. 14-16

81 EXAM **Sep 2025** Case Scenario - I · MCQ 1 to MCQ 5 10 marks

CASE SCENARIO - I

The following information pertains to ABC Limited for the period 1st April, 2024 to 31st March, 2025:

Sl. No.	Particulars	Amount (₹)	Amount (₹)
1	Royalty paid for production		7,76,400
2	Amount paid for power & fuel		2,15,200

Sl. No.	Particulars	Amount (₹)	Amount (₹)
3	Packing cost paid for re-distribution of Finished Goods		70,500
4	Repairs & Maintenance paid for: - Plant & Machinery - Sales office building	 65,000 80,000	 1,45,000
5	Insurance premium paid for Plant & Machinery		1,17,520
6	Research & Development cost paid for improvement in production process		17,800
7	Depreciation on office building		75,500
8	Salary paid to General Manager		15,50,000
9	Salary & bonus paid to Sales & Marketing staff		11,40,500
10	Receipt from sale of scrap and waste generated during production		20,000
11	Value of Stock as on 1st April,2024: - Raw materials - Work-in- process - Finished goods	 	 5,00,000 8,40,000 ?
12	Value of stock as on 31st March,2025: - Raw materials - Work-in- process - Finished goods	 	 1,10,000 6,50,000 ?

Other information are as follows:

(i) Raw materials purchased were 15,000 kgs @₹ 300 per kg.

(ii) Freight inwards paid at 4% of the cost of raw materials purchased.

(iii) Wages paid to factory workers was 30% of raw material consumed.

(iv) Closing stock of finished goods was ₹ 9,90,000 more than the opening stock. The average stock of finished goods was ₹ 11,55,000.

(v) Sales during the period 1st April, 2024 to 31st March, 2025 was ₹ 1,10,00,000.

On the basis of above Case Scenario, you are required to answer the following MCQs 1 to 5:

MCQ 1 2 marks

What is the Direct employee (labour) cost?

A ₹ 15,21,000

B ₹ 14,67,000

C ₹ 14,62,500

D ₹ 15,29,500

MCQ 2 2 marks

What is the Prime cost?

A ₹ 78,88,000

B ₹ 65,91,600

C ₹ 75,82,600

D ₹ 72,92,000

MCQ 3 2 marks

What is the Factory cost?

A ₹ 78,85,520

B ₹ 79,55,120

C ₹ 75,75,650

D ₹ 76,45,840

MCQ 4 2 marks

What is the cost of goods sold?

A ₹ 76,55,520

B ₹ 78,35,850

C ₹ 69,62,920

D ₹ 70,82,500

MCQ 5 2 marks

What is the amount of Profit?

A ₹ 12,25,350

B ₹ 11,33,000

C ₹ 10,95,500

D ₹ 11,20,580

ICAI'S ANSWERS

MCQ 1

Answer key: **(A)**

MCQ 2

Answer key: **(C)**

MCQ 3

Answer key: **(B)**

MCQ 4

Answer key: **(C)**

MCQ 5

Answer key: **(D)**

ICAI · P4 · Sep 2025 exam · question p. 1-2 · answer p. 7

82 EXAM **May 2025** Q2(a) 7 marks

Kidz Company manufactures and sells two models of baby toys namely, Max and Pro. During the Financial Year 2024-25, 1,500 units of Max and 3,600 units of Pro were manufactured. However, only 60% of Max and 80% of Pro were sold during the year. Labour cost per unit of Max is two times that of Pro. There was no opening stock of finished goods or work-in progress.

The cost particulars of the two models of Baby Toys are given below:

Particulars	Max (₹)	Pro (₹)	Total (₹)
Material Cost	42,000	63,000	1,05,000
Labour Cost	-	-	1,21,000

Further, the cost controller of the factory informed that:

- Works overhead is 50% of labour cost
- Office overhead is recovered at 20% of works cost.
- Selling and distribution overhead is ₹ 20 and ₹ 30 per unit sold for Model Pro and Model Max respectively.

Required:

(i) Prepare a cost sheet for the financial year 2024-25, showing the various elements of cost for each model of Baby Toys (Prime cost, work cost, cost of production, cost of goods sold and cost of sales).

(ii) Calculate the per unit selling price of each model of Baby Toys if profit is charged at 20 percent on sales.

ICAI'S ANSWER

Preparation of Cost Sheet for FY 2024-25

Particulars	Max (₹)	Pro (₹)	Total (₹)
Direct materials	42,000	63,000	1,05,000
Direct wages (Working note- (i))	55,000	66,000	1,21,000
Prime cost	**97,000**	**1,29,000**	**2,26,000**
Work overhead (50% of labour cost)	27,500	33,000	60,500
Factory Cost/Work Cost	**1,24,500**	**1,62,000**	**2,86,500**
Administration/Office Overhead (20 % of work cost)	24,900	32,400	57,300
Cost of production	**1,49,400**	**1,94,400**	**3,43,800**
Less: Closing stock (Working note- (ii))	(59,760)	(38,880)	(98,640)
Cost of goods sold	**89,640**	**1,55,520**	**2,45,160**
Selling and distribution overhead	27,000 (₹ 30 x 900)	57,600 (₹ 20 x 2,880)	84,600
Cost of sales/ Total cost	**1,16,640**	**2,13,120**	**3,29,760**
Profit (20% on sales or 25% of cost)	29,160	53,280	82,440
Sales value	**1,45,800**	**2,66,400**	**4,12,200**

Particulars	Max (₹)	Pro (₹)	Total (₹)
Selling price per unit	**162**	**92.5**	

Working Notes:

(i) Direct labour cost per unit of Pro = M

Direct labour cost per unit of Max	= 2M
Total Direct labour cost	= M x 3,600 units + 2M × 1,500 units
₹ 1,21,000	= 3,600 M + 3,000 M
M	= ₹ 18.333

Therefore, Direct labour Cost per unit of Max

= 2 x ₹ 18.33 = ₹ 36.667

Total labour cost of Pro	= ₹ 18.333 × 3,600 units
	= ₹ 66,000
Total labour cost of Max	= ₹ 36.667 × 1,500 units
	= ₹ 55,000

(ii) Value of closing stock

Particulars	Max (units)	Pro (units)
No. of units produced	1,500	3,600
No. of units sold	60% of 1,500 = 900	80% of 3,600 = 2,880
Closing Stock	600	720

Max = (1,49,400/1,500) x 600 = ₹ 59,760

Pro = (1,94,400/3,600) x 720 = ₹ 38,880

Note: If a student assumes Office overheads as general overheads, then it will be part of cost of sales.

ICAI · P4 · May 2025 exam · question p. 13 · answer p. 14-16

83 EXAM **Jan 2025** Q2(a) 9 marks

The following information relates to a manufacturing concern A Ltd. for the year ended 31st March, 2024.

Particulars	As on 1st April,2023	As on 31st March,2024
Raw Material (in ₹)	3,40,000	1,80,000
Work in Progress (in ₹)	5,50,000	3,50,000

Particulars	Amount
Raw Material Purchased [Inclusive of GST@18% (Ineligible for ITC)]	8,00,000
Packaging Cost (primary)	3,00,000
Fee Paid to Independent Directors	5,00,000
Production bonus paid to factory workers	10% of Wages paid to factory workers
Job charges paid to job workers	41,000
Salary paid to Supervisor	6,17,900
Wages paid to factory workers	6,30,000
Salary paid to Production Control Manager	7,20,000
Sale of Scrap generated during Manufacturing	50,000
Selling Overheads per unit	2
Salary paid to General Manager	12,40,000
Freight Inwards	2% on Raw Material Purchased
Expenses Paid for Quality Control check activities	4,30,000

Particulars	Cost Price (₹)	WDV as on 1st April, 2023 (₹)	Depreciation Rate	Insurance Cost per annum
Factory Building	25,00,000	21,87,000	10%	2% of Cost Price
Plant and Machinery	15,00,000	11,56,000	15%	2% of Cost Price
Office Building	40,00,000	36,00,000	10%	Nil

Additional information:

(i) Depreciation is charged on the written down value method.

(ii) Stock of finished goods as on 1st April, 2023 was 80,000 units having a total cost of 8,00,000. The entire stock of opening finished goods is sold during the year, closing stock is 70,000 units. During the period, 4,50,000 units were sold.

(iii) A Ltd. wants a profit of 20% on Total Sales.

Required:

Prepare a Cost statement showing the various elements of cost and profit earned for the year ended 31st March, 2024.

ICAI'S ANSWER

Cost Sheet

Particulars	(₹)	(₹)
Material Consumed:		
Raw materials purchased	8,00,000	
Freight inwards (2% of Raw materials purchased)	16,000	
Add: Opening stock of raw materials	3,40,000	
Less: Closing stock of raw materials	(1,80,000)	9,76,000
Direct employee (labour) cost:		
Wages paid to factory workers	6,30,000	
Production bonus paid to factory workers (10%)	63,000	6,93,000
Direct expenses:		
Job charges paid to job workers	41,000	41,000
Prime Cost		**17,10,000**
Works/ Factory overheads:		
Depreciation on factory building (21,87,000 × 10%)	2,18,700	
Depreciation on plant & machinery (11,56,000 × 15%)	1,73,400	
Insurance premium paid for plant & machinery (15,00,000 × 2%)	30,000	
Insurance premium paid for factory building (25,00,000 × 2%)	50,000	
Salary paid to supervisors	6,17,900	10,90,000
Gross factory cost		**28,00,000**
Add: Opening value of W-I-P		5,50,000
Less: Closing value of W-I-P		(3,50,000)
Factory Cost		**30,00,000**
Quality control cost:		
Expenses paid for quality control check activities	4,30,000	4,30,000
Administration cost related with production:		
Salary paid to Production control manager	7,20,000	7,20,000
Less: Realisable value on sale of scrap and waste		(50,000)
Add: Primary packing cost		3,00,000

Particulars	(₹)	(₹)
Cost of Production		**44,00,000**
Add: Opening stock of finished goods (80,000 units)		8,00,000
Less: Closing stock of finished goods (70,000 units)		(7,00,000)
Cost of Goods Sold		**45,00,000**
Administrative overheads:		
Depreciation on office building (36,00,000 × 10%)	3,60,000	
Salary paid to General Manager	12,40,000	
Fee paid to independent directors	5,00,000	
Selling Overheads (4,50,000 units x ₹2):	9,00,000	30,00,000
Cost of Sales		**75,00,000**
Profit (25% on cost)		18,75,000
Sales		**93,75,000**

Working Note: Calculation of value of closing finished goods.

Number of units produced = units sold + closing stock – opening stock

= 4,50,000 + 70,000 – 80,000

= 4,40,000 units

$$\text{Per unit cost} = \frac{\text{Cost of production}}{\text{Number of units produced}}$$

$$= \frac{₹44,00,000}{4,40,000 \text{ units}} = ₹ 10 \text{ per unit}$$

Value of Closing stock = ₹ 10 x 70,000 units = ₹ 7,00,000

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84 EXAM **Sep 2024** Q2(a) 8 marks

MNP Limited have the capacity to produce 84,000 units of a product very month. Its prime cost per unit at various levels of production is as follows:

Level	Prime Cost per unit (₹)
10%	50
20%	48
30%	46
40%	44
50%	42

Level	Prime Cost per unit (₹)
60%	40
70%	38
80%	36
90%	34
100%	32

Its prime cost consists of raw material consumed, direct wages and direct expenses in the ratio of 3 : 2 : 1. In the month of January 2024, the company worked at 40% capacity and raw material purchased amounting to ₹ 8,40,000. In the month of February 2024, the company worked at 100% capacity and raw material purchased for ₹ 16,46,400.

It is the policy of the company to maintain opening stock of raw material equal to 1/3 of closing stock of raw material. Factory overheads are recovered at 60% of direct wages cost. Fixed administration expenses (as part of production cost) and fixed selling and distribution expenses are ₹ 2,01,600 and ₹ 1,68,000 per month respectively. During the month of January 2024 company sold 33,600 units @ ₹ 68.8 per unit. The variable distribution cost amounts to ₹ 1.5 per unit sold.

The management of the company chalks out a pl for the month of February 2024 to sell its whole output @ ₹ 61 per unit by incurring following further expenditure :

(i) Company sponsors a television programme on every Sunday at a cost of ₹ 26,250 per week. There are 4 Sundays in February 2024.

(ii) Hi-tea programme every month for its potential customers at a cost of ₹ 1,05,000.

(iii) Special gift item costing ₹ 105 on sale of a dozen units.

(iv) Lucky draws scheme is introduced every month by giving the first prize of ₹ 1,00,000; second prize of ₹ 80,000; third prize of ₹ 40,000 and four consolation prizes of ₹ 8,000 each.

Note: (In the month of February 2024, there is a significant saving in material cost per unit due to entry of new suppliers in the market and saving in per unit cost of Direct wages and Direct expenses due to introduction of new· policy by the management.)

Prepare a cost sheet for the month of January 2024 and February 2024 showing prime cost (with different elements of prime cost), factory cost, cost of production, total cost and profit earned.

ICAI'S ANSWER

Cost Sheet

Particulars	January 2024	February 2024
	33,600 Units	**84,000 Units**
Opening Stock of Raw Material	50,400	1,51,200
Add: Purchases	8,40,000	16,46,400
Less: Closing stock of Raw Material	(1,51,200)	(4,53,600)

Particulars	January 2024	February 2024
Direct materials consumed:	**7,39,200**	**13,44,000**
Direct Wages	4,92,800	8,96,000
Direct expenses	2,46,400	4,48,000
Prime Cost	**14,78,400**	**26,88,000**
Factory overheads (60% of direct wages)	2,95,680	5,37,600
Factory / Works Cost	**17,74,080**	**32,25,600**
Add: Administration overhead (Production)	2,01,600	2,01,600
Cost of Production / Cost of goods sold	**19,75,680**	**34,27,200**
Add: Fixed selling and distribution Overhead	1,68,000	1,68,000
Variable distribution overheads (₹ 1.5 per unit)	50,400	1,26,000
- Sponsorship cost	-	1,05,000
- Hi tea programme	-	1,05,000
- Special gifts (84,000 × 1/12 × 105)	-	7,35,000
- Lucky draw prize *	-	2,52,000
Cost of sales / Total Cost	**21,94,080**	**49,18,200**
Profit (Balancing figure)	1,17,600	2,05,800
Sales revenue	**23,11,680**	**51,24,000**

***Lucky draw prize:**

	Amount (₹)
1st Prize	1,00,000
2nd Prize	80,000
3rd Prize	40,000
Consolation Prizes (4 × ₹ 8,000)	32,000
Total	2,52,000

Working note :

Calculation of opening and costing stock of Raw Material

January

Units Manufactured = 84,000 x 40% = 33,600 units

Prime Cost = 33,600 x 44 = ₹ 14,78,400

Raw Material consumed = ₹ 14,78,400 X 3/6 = ₹ 7,39,200

Raw Material purchase (given) = ₹ 8,40,000

Let closing stock of Raw Material be x

Opening stock of Raw Material be 1/3x

Opening Stock + Purchase – closing stock = Raw Material consumed

1/3x + ₹ 8,40,000 – x = ₹ 7,39,200

1/3x – x = ₹ 7,39,200 – ₹ 8,40,000

2/3x = ₹ 1,00,800

x = ₹ 1,51,200 (closing stock)

Opening stock = ₹ 1,51,200 x 1/3 = ₹ 50,400

February

Prime Cost = 84,000 x 32 = ₹ 26,88,000

Raw Material consumed = ₹ 26,88,000 x 3/6 = ₹ 13,44,000

Raw Material purchased (given) = ₹ 16,46,400

Opening Stock + Purchase – closing stock = Raw Material consumed

₹ 1,51,200 + ₹ 16,46,400 – closing stock = ₹ 13,44,000

Closing stock = ₹ 4,53,600

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85 EXAM **Old P3 Nov 2023** Q2(a) 10 marks old scheme

The following data relates to the manufacture of product BXE for the year ended 31st March,2023:

	Amount (₹)
Value of stock as on 1st April,2022	
Raw materials	27,00,000
Work in progress	10,60,000
Finished Goods	25,00,000
Material purchased	2,48,00,000
Freight inward	7,50,000
Direct wages	42,00,000
Power & Fuel	18,75,000
Cost of special drawings	3,60,000
Trade Discount	4,50,000

	Amount (₹)
Insurance on material procured	15,000
Rent of Factory Building (1/5th used for office purpose)	7,00,000
Depreciation on machinery	6,25,000
Depreciation on Delivery Vans	1,20,000
Consumable stores and indirect wages	15,20,000
Quality Control cost	9,00,000
Primary packing cost	12,90,000
General Administrative overheads (excluding rent of building)	17,50,000
Salary paid to Marketing Staff	9,60,000
Packing cost for transportation	1,84,000
Value of stock as on 31st March, 2023	
Raw materials	32,60,000
Work in progress	11,80,000
Finished Goods	28,38,000

Additional Information:

- Further, some of the finished product was found defective and the defective products were rectified by incurring expenditure of additional factory overheads to the extent of ₹ 33,600. The cost of rectification is not included in details mentioned above.
- An amount of ₹ 1,20,600 was realised by selling scrap and waste generated during the year.

Prepare Cost sheet for the year ended 31stMarch, 2023 showing:

(i) Prime cost,

(ii) Factory cost,

(iii) Cost of production.

(iv) Cost of goods sold, and

(v) Cost of sales.

ICAI'S ANSWER

Cost Sheet for the product BXE

Sl. No.	Particulars	(₹)	(₹)
(i)	Material Consumed:		
	Raw materials purchased	2,48,00,000	

Sl. No.	Particulars	(₹)	(₹)
	Freight inwards	7,50,000	
	Insurance on material procured	15,000	
	Less: Trade discount	(4,50,000)	
	Add: Opening stock of raw materials	27,00,000	
	Less: Closing stock of raw materials	(32,60,000)	2,45,55,000
(ii)	Direct wages		42,00,000
(iii)	Direct expenses:		
	Power & fuel	18,75,000	
	Cost of special drawings	3,60,000	22,35,000
	Prime Cost		**3,09,90,000**
(iv)	Works/ Factory overheads:		
	Rent of factory building (4/5th of 7,00,000)	5,60,000	
	Depreciation on machinery	6,25,000	
	Defective rectification cost	33,600	
	Consumable stores & indirect wages	15,20,000	27,38,600
	Gross works cost		3,37,28,600
	Add: Opening work in process		10,60,000
	Less: Closing work in process		(11,80,000)
	Factory cost		**3,36,08,600**
(v)	Quality control cost		9,00,000
(vi)	Primary packing cost		12,90,000
(vii)	Less: Amount realised from scrap sale		(1,20,600)
	Cost of production		**3,56,78,000**
	Add: Opening stock of finished goods		25,00,000
	Less: Closing stock of finished goods		(28,38,000)
	Cost of Goods Sold		**3,53,40,000**
	Administrative overheads:		

Sl. No.	Particulars	(₹)	(₹)
(viii)	Rent of factory building (1/5th of 7,00,000)		1,40,000
	General administrative overheads		17,50,000
	Selling and Distribution overheads:		
(x)	Salary paid to marketing staff		9,60,000
(xi)	Packing cost for transportation		1,84,000
(xii)	Depreciation on delivery vans		1,20,000
	Cost of Sales		**3,84,94,000**

Alternatively, Power and fuel expenses of ₹ 18,75,000 can be taken as a part of factory overhead. Accordingly, prime cost will be 2,91,15,000. However, there will be no change in factory cost, cost of production, cost of goods sold and cost of sales.

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86 EXAM **Old P3 Nov 2023** Q5(b) 6 marks old scheme

The following data relate to the manufacture of a product 'VD-100* during the month of October 2023:

Good units produced	12,600
Units Sold	11,800
Direct wages	₹ 8,82,000
Administrative Overheads	₹ 4,72,000
Selling price per unit	₹ 416

Each unit produced requires 2 kg. of material 'Z'. Cost of material 'Z' is ₹ 72 per kg. 10% of the production has been scrapped as bad and fetches ₹ 45 per unit. Factory overheads are 80% of wages. Selling and distribution overheads are ₹ 54 per unit sold. There is no opening or closing stock of material and work in progress.

You are required to find out total cost of sales and profit for the month of October 2023.

ICAI'S ANSWER

Since 10% units are scrapped.

Units produced (total) is 14,000 (12,600/90%)

Calculation of cost of sales and profit

Particulars	₹
Raw Material (28,000 × ₹ 72)	20,16,000
Wages	8,82,000

Particulars	₹
Prime Cost	**28,98,000**
Factory overheads	7,05,600
Factory Cost	**36,03,600**
Sale of Scrap (1,400 × ₹ 45)	(63,000)
Cost of Production	**35,40,600**
Less: Closing Stock of finished goods (₹35,40,600/12,600 ×800)	2,24,800
Cost of goods sold	33,15,800
Add: Administration overheads	4,72,000
Add: Selling & Distribution overheads (₹ 54 × 11,800)	6,37,200
Cost of Sales	**44,25,000**
Sales (11,800 × ₹ 416)	49,08,800
Profit	**4,83,800**

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87 EXAM **Old P3 May 2023** Q3(b) 10 marks old scheme

The following information is available from SN Manufacturing Limited's for the month of April 2023.

	April 1	April 30
Opening and closing inventories data:		
Stock of finished goods	2,500 units	?
Stock of raw materials	₹ 42,500	₹ 38,600
Work-in progress	₹ 42,500	₹ 42,800
Other data are:		
Raw materials Purchased		₹ 6,95,000
Carriage inward		₹ 36,200
Direct wages paid		₹ 3,22,800
Royalty paid for production		₹ 35,800

	April 1	April 30
Purchases of special designs, moulds and patterns (estimated life 12 Production cycles)		₹ 1,53,600
Power, fuel and haulage (factory)		₹ 70,600
Research and development costs for improving the production process (amortized)		₹ 31,680
Primary packing cost (necessary to maintain quality)		₹ 6920
Administrative Overhead		₹ 46,765
Salary and wages for supervisor and foremen		₹ 28,000

Other information:

- Opening stock of finished goods is to be valued at ₹ 8.05 per unit.
- During the month of April, 1,52,000 units were produced and 1,52,600 units were sold. The closing stock of finished goods is to be valued at the relevant month's cost of production. The company follows the FIFO method.
- Selling and distribution expenses are to be charged at 20 paisa per unit.
- Assume that one production cycle is completed in one month.

Required:

(i) Prepare a cost sheet for the month ended on April 30, 2023, showing the various elements of cost (raw material consumed, prime cost, factory cost, cost of production, cost of goods sold, and cost of sales).

(ii) Calculate the selling price per unit if profit is charged at 20 percent on sales.

ICAI'S ANSWER

Cost Sheet for the month of April 2023

Particulars	Amount (₹)	Amount (₹)
Raw materials consumed:		
Raw materials purchased	6,95,000	
Add: Carriage inward	36,200	
Add: Value of opening stock of raw materials	42,500	
Less: Value of closing stock of raw materials	(38,600)	**7,35,100**
Direct wages paid		3,22,800
Royalty paid for production		35,800
Amortised cost of special designs, moulds and patterns (₹153,600 ÷ 12)		12,800
Power, fuel and haulage (factory)*		70,600

Particulars	Amount (₹)	Amount (₹)
Prime Cost*		**11,77,100**
Salary and wages of supervisor and foremen		28,000
Gross Works Cost		12,05,100
Add: Opening stock of WIP		42,500
Less: Closing stock of WIP		(42,800)
Factory/ Works Cost		**12,04,800**
Research and development cost	31,680	
Primary packing cost	6,920	38,600
Cost of Production		**12,43,400**
Add: Opening stock of finished goods (₹ 8.05 × 2,500 units)		20,125
Less: Value of closing stock [(2,500+152,000 -1,52,600) × (12,43,400÷152000)		(15,542)
Cost of Goods Sold		**12,47,983**
Add: Administrative overheads		46,765
Add: Selling and distribution expenses (₹ 0.20 × 1,52,600)		30,520
Cost of Sales		**13,25,268**
Add: Profit (20% on Sales or 25% on cost of sales)		3,31,317
Sales value		16,56,585
Selling price per unit (₹ 16,56,585 ÷ 1,52,600 units)		**10.86**

*May be taken as part of Factory / Works cost, however Total Factory Cost will remain the same. If taken as part of factory cost then prime cost will be ₹ 11,06,500.

Alternative Solution (Based on work-in-progress figure of ₹ 45,500 as on 1st April 2023 as per Hindi part of Question paper)

Particulars	Amount (₹)	Amount (₹)
Raw materials consumed:		
Raw materials purchased	6,95,000	
Add: Carriage inward	36,200	
Add: Value of opening stock of raw materials	42,500	
Less: Value of closing stock of raw materials	(38,600)	7,35,100

Particulars	Amount (₹)	Amount (₹)
Direct wages paid		3,22,800
Royalty paid for production		35,800
Amortised cost of special designs, moulds and patterns (₹ 153,600 ÷ 12)		12,800
Power, fuel and haulage (factory)*		70,600
Prime Cost		**11,77,100**
Salary and wages of supervisor and foremen		28,000
Gross Works Cost		12,05,100
Add: Opening stock of WIP		45,500
Less: Closing stock of WIP		(42,800)
Factory/ Works Cost		**12,07,800**
Research and development cost	31,680	
Primary packing cost	6,920	38,600
Cost of Production		**12,46,400**
Add: Opening stock of finished goods (₹ 8.05 × 2,500 units)		20,125
Less: Value of closing stock [(2,500+1,52,000 -1,52,600) × (12,46,400÷1,52,000)		(15,580)
Cost of Goods Sold		**12,50,945**
Add: Administrative overheads		46,765
Add: Selling and distribution expenses (₹ 0.20 × 1,52,600)		30,520
Cost of Sales		**13,28,230**
Add: Profit (20% on Sales or 25% on cost of sales)		3,32,058
Sales value		16,60,288
Selling price per unit (₹ 16,60,288 ÷ 1,52,600 units)		**10.88**

*May be taken as part of Factory / Works cost, however Total Factory Cost will remain the same. *If taken as part of factory cost then prime cost will be ₹ 11,06,500.*

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88 EXAM **Old P3 Nov 2022** Q4(b) 10 marks old scheme

PNME Ltd. manufactures two types of masks- 'Disposable Masks' and ‘Cloth Masks'. The cost data for the

year ended 31stMarch, 2022 is as follows:

	₹
Direct Materials	12,50,000
Direct Wages	7,00,000
Production Overhead	4,00,000
Total	23,50,000

It is further ascertained that:

- Direct material cost per unit of Cloth Mask was twice as much of Direct material cost per unit of Disposable Mask.
- Direct wages per unit for Disposable Mask were 60% of those for Cloth Mask.
- Production overhead per unit was at same rate for both the types of the masks.
- Administration overhead was 50% of Production overhead for each type of mask.
- Selling cost was ₹ 2 per Cloth Mask.
- Selling Price was ₹ 35 per unit of Cloth Mask.
- No. of units of Cloth Masks sold- 45,000
- No. of units of Production of

Cloth Masks:	50,000
Disposable Masks:	1,50,000

You are required to prepare a cost sheet for Cloth Masks showing:

(i) Cost per unit and Total Cost.

(ii) Profit per unit and Total Profit.

ICAI'S ANSWER

Preparation of Cost Sheet for Cloth Masks

No. of units produced = 50,000 units

No. of units sold = 45,000 units

Particulars	Per unit (₹)	Total (₹)
Direct materials (Working note- (i))	10.00	5,00,000
Direct wages (Working note- (ii))	5.00	2,50,000
Prime cost	15.00	7,50,000
Production overhead (Working note- (iii))	2.00	1,00,000
Factory Cost	17.00	8,50,000
Administration Overhead* (50% of Production Overhead)	1.00	50,000

Particulars	Per unit (₹)	Total (₹)
Cost of production	18.00	9,00,000
Less: Closing stock (50,000 units – 45,000 units)	-	(90,000)
Cost of goods sold i.e. 45,000 units	18.00	8,10,000
Selling cost	2.00	90,000
Cost of sales/ Total cost	20.00	9,00,000
Profit	15.00	6,75,000
Sales value (₹ 35 × 45,000 units)	35.00	15,75,000

Working Notes:

(i) Direct material cost per unit of Disposable Mask = M

Direct material cost per unit of Cloth Mask = 2M

Total Direct Material cost = 2M × 50,000 units + M × 1,50,000 units

Or, ₹ 12,50,000 = 1,00,000 M + 1,50,000 M

Or, $M = \frac{₹\ 12,50,000}{2,50,000} = ₹\ 5$

Therefore, Direct material Cost per unit of Cloth Mask = 2 × ₹ 5 = ₹ 10

(ii) Direct wages per unit for Cloth Mask = W

Direct wages per unit for Disposable Mask= 0.6W

So, (W x 50,000) + (0.6W x 1,50,000) = ₹ 7,00,000

W = ₹5 per unit

Therefore, Direct material Cost per unit of Cloth Mask = ₹ 5

(iii) Production overhead per unit = $\frac{₹\ 4,00,000}{(50,000+1,50,000)} = ₹\ 2$

Production overhead for Cloth Mask = ₹ 2 × 50,000 units = ₹ 1,00,000

* Administration overhead is related to production overhead in the question and hence to be considered in cost of production only.

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89 EXAM **Old P3 May 2022** Q3(b) 10 marks old scheme

The following data are available from the books and records of A Ltd. for the month of April 2022:

Particulars	Amount (₹)
Stock of raw materials on 1st April 2022	10,000
Raw materials purchased	2,80,000
Manufacturing wages	70,000

Particulars	Amount (₹)
Depreciation on plant	15,000
Expenses paid for quality control check activities	4,000
Lease Rent of Production Assets	10,000
Administrative Overheads (Production)	15,000
Expenses paid for pollution control and engineering & maintenance	1,000
Stock of raw materials on 30th April 2022	40,000
Primary packing cost	8,000
Research & development cost (Process related)	5,000
Packing cost for redistribution of finished goods	1,500
Advertisement expenses	1,300

Stock of finished goods as on 1st April 2022 was 200 units having a total cost of ₹ 28,000. The entire opening stock of finished goods has been sold during the month. Production during the month of April, 2022 was 3,000 units. Closing stock of finished goods as on 30th April, 2022 was 400 units.

You are required to:

I. Prepare a Cost Sheet for the above period showing the:

(i) Cost of Raw Material consumed

(ii) Prime Cost

(iii) Factory Cost

(iv) Cost of Production

(v) Cost of goods sold

(vi) Cost of Sales

II. Calculate selling price per unit, if sale is made at a profit of 20% on sales.

ICAI'S ANSWER

I. Statement of Cost (for the month of April, 2022)

S. No.	Particulars	Amount (₹)	Amount (₹)
	Opening stock of Raw material	10,000	
	Add: Purchase of Raw material	2,80,000	
	Less: Closing stock of raw materials	(40,000)	
(i)	**Raw material consumed**		**2,50,000**
	Manufacturing wages		70,000

S. No.	Particulars	Amount (₹)	Amount (₹)
(ii)	**Prime Cost**		**3,20,000**
	Factory/work overheads:		
	Depreciation on plant	15,000	
	Lease rent of production Asset	10,000	
	Expenses paid for pollution control and engineering & Maintenance	1,000	26,000
(iii)	**Factory/Work Cost**		**3,46,000**
	Expenses paid for quality control check activity		4,000
	Research and Development Cost		5,000
	Administration Overheads (Production)		15,000
	Primary Packing Cost		8,000
(iv)	**Cost of Production**		**3,78,000**
	Add: Opening stock of finished goods		28,000
	Less: Closing stock of finished goods		**(50,400)**
(v)	**Cost of Goods Sold**		**3,55,600**
	Advertisement expenses		1,300
	Packing cost for re-distribution of finished goods sold		1,500
(vi)	**Cost of Sales**		**3,58,400**

Note: Valuation of Closing stock of finished goods

$= \frac{3,78,000}{3000 \text{ units}} \times 400 \text{ units}$

$= ₹50,400$

II. Cost per unit sold $= \frac{3,58,400}{200+3,000-400} = ₹ 128$ per unit

$\therefore$ Selling Price $= \frac{128}{80\%} =$ **₹160 per unit**

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90 EXAM **Old P3 Dec 2021** Q2(a) 10 marks old scheme

G Ltd. manufactures leather bags for office and school purposes.

The following information is related with the production of leather bags for the month of September, 2021.

(1) Leather sheets and cotton clothes are the main inputs and the estimated requirement per bag is two metres of leather sheets and one metre of cotton cloth. 2,000 metre of leather sheets and 1,000 metre of

cotton cloths are purchased at ₹ 3,20,000 and ₹ 15,000 respectively. Freight paid on purchases is ₹ 8,500.

(2) Stitching and finishing need 2,000 man hours at ₹ 80 per hour.

(3) Other direct costs of ₹ 10 per labour hour is incurred.

(4) G Ltd. have 4 machines at a total cost of ₹ 22,00,000. Machines have a life of 10 years with a scrap value of 10% of the original cost. Depreciation is charged on a straight-line method.

(5) The monthly cost of administration and sales office staffs are ₹ 45,000 and ₹ 72,000 respectively. G Ltd. pays ₹ 1,20,000 per month as rent for a 2,400 sq. feet factory premises. The administrative and sales office occupies 240 sq. feet and 200 sq. feet respectively of factory space.

(6) Freight paid on delivery of finished bags is ₹ 18,000.

(7) During the month, 35 kgs of scrap (cuttings of leather and cotton) are sold at ₹ 150 per kg.

(8) There are no opening and closing stocks of input materials. There is a finished stock of 100 bags in stock at the end of the month.

You are required to prepare a cost sheet in respect of above for the month of September 2021 showing:

(i) Cost of Raw Material Consumed

(ii) Prime Cost

(iii) Works/Factory Cost

(iv) Cost of Production

(v) Cost of Goods Sold

(vi) Cost of Sales

ICAI'S ANSWER

No. of bags manufactured = 1,000 units

Cost sheet for the month of September 2021

	Particulars	Total Cost (₹)	Cost per unit (₹)
1.	Direct materials consumed:		
	- Leather sheets	3,20,000	320.00
	- Cotton cloths	15,000	15.00
	Add: Freight paid on purchase	8,500	8.50
	(i) Cost of material consumed	3,43,500	343.50
2.	Direct wages (₹80 × 2,000 hours)	1,60,000	160.00
3.	Direct expenses (₹10 × 2,000 hours)	20,000	20.00
4.	**(ii) Prime Cost**	5,23,500	523.50
5.	Factory Overheads: Depreciation on machines {(₹ 22,00,000 × 90%) ÷ 120 months}	16,500	16.50
	Apportioned cost of factory rent	98,000	98.00

	Particulars	Total Cost (₹)	Cost per unit (₹)
6.	**(iii) Works/ Factory Cost**	6,38,000	638.00
7.	*Less:* Realisable value of cuttings (₹150×35 kg.)	(5,250)	(5.25)
8.	**(iv) Cost of Production**	6,32,750	632.75
9.	*Add:* Opening stock of bags	0	
10.	*Less:* Closing stock of bags (100 bags × ₹632.75)	(63,275)	
11.	**(v) Cost of Goods Sold**	5,69,475	632.75
12.	*Add:* Administrative Overheads:		
	- Staff salary	45,000	50.00
	- Apportioned rent for administrative office	12,000	13.33
13.	*Add:* Selling and Distribution Overheads		
	- Staff salary	72,000	80.00
	- Apportioned rent for sales office	10,000	11.11
	- Freight paid on delivery of bags	18,000	20.00
14.	**(vi) Cost of Sales**	7,26,475	807.19

Apportionment of Factory rent:

To factory building {(₹ 1,20,000 ÷ 2400 sq. feet) × 1,960 sq. feet} = ₹ 98,000

To administrative office {(₹ 1,20,000 ÷ 2400 sq. feet) × 240 sq. feet} = ₹ 12,000

To sale office {(₹ 1,20,000 ÷ 2400 sq. feet) × 200 sq. feet} = ₹ 10,000

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91 EXAM **Old P3 Jul 2021** Q2(a) 10 marks old scheme

The following data relates to manufacturing of a standard product during the month of March, 2021:

Particulars	Amount (in ₹)
Stock of Raw material as on 01-03-2021	80,000
Work in Progress as on 01-03-2021	50,000
Purchase of Raw material	2,00,000
Carriage Inwards	20,000
Direct Wages	1,20,000
Cost of special drawing	30,000

Particulars	Amount (in ₹)
Hire charges paid for Plant	24,000
Return of Raw Material	40,000
Carriage on return	6,000
Expenses for participation in Industrial exhibition	8,000
Legal charges	2,500
Salary to office staff	25,000
Maintenance of office building	2,000
Depreciation on Delivery van	6,000
Warehousing charges	1,500
Stock of Raw material as on 31-03-2021	30,000
Stock of Work in Progress as on 31-03-2021	24,000

- Store overheads on materials are 10% of material consumed.
- Factory overheads are 20% of the Prime cost.
- 10% of the output was rejected and a sum of ₹ 5,000 was realized on sale of scrap.
- 10% of the finished product was found to be defective and the defective products were rectified at an additional expenditure which is equivalent to 20% of proportionate direct wages.
- The total output was 8000 units during the month.

You are required to prepare a Cost Sheet for the above period showing the :

(i) Cost of Raw Material consumed.

(ii) Prime Cost

(iii) Work Cost

(iv) Cost of Production

(v) Cost of Sales

ICAI'S ANSWER

Statement of Cost for the month of March, 2021

	Particulars	Amount (₹)	Amount (₹)
(i)	**Cost of Material Consumed:**		
	Raw materials purchased (₹ 2,00,000 – ₹ 40,000)	1,60,000	
	Carriage inwards	20,000	
	Add: Opening stock of raw materials	80,000	

	Particulars	Amount (₹)	Amount (₹)
	Less: Closing stock of raw materials	(30,000)	**2,30,000**
	Direct Wages		1,20,000
	Direct expenses:		
	Cost of special drawing	30,000	
	Hire charges paid for Plant	24,000	54,000
(ii)	**Prime Cost**		**4,04,000**
	Carriage on return	6,000	
	Store overheads (10% of material consumed)	23,000	
	Factory overheads (20% of Prime cost)	80,800	
	Additional expenditure for rectification of defective products (refer working note)	2,160	1,11,960
	Gross factory cost		5,15,960
	Add: Opening value of W-I-P		50,000
	Less: Closing value of W-I-P		(24,000)
(iii)	**Works/ Factory Cost**		**5,41,960**
	Less: Realisable value on sale of scrap		(5,000)
(iv)	**Cost of Production**		**5,36,960**
	Add: Opening stock of finished goods		-
	Less: Closing stock of finished goods		-
	Cost of Goods Sold		**5,36,960**
	Administrative overheads:		
	Maintenance of office building	2,000	
	Salary paid to Office staff	25,000	
	Legal Charges	2,500	29,500
	Selling overheads:		
	Expenses for participation in Industrial exhibition	8,000	8,000
	Distribution overheads:		
	Depreciation on delivery van	6,000	

	Particulars	Amount (₹)	Amount (₹)
	Warehousing charges	1,500	7,500
(v)	**Cost of Sales**		**5,81,960**

Alternative Solution

(considering Hire charges paid for Plant as indirect expenses)

Statement of Cost for the month of March, 2021

Particulars	Amount (₹)	Amount (₹)
Cost of Material Consumed:		
Raw materials purchased (₹ 2,00,000 – ₹ 40,000)	1,60,000	
Carriage inwards	20,000	
Add: Opening stock of raw materials	80,000	
Less: Closing stock of raw materials	(30,000)	**2,30,000**
Direct Wages		1,20,000
Direct expenses:		
Cost of special drawing	30,000	30,000
Prime Cost		**3,80,000**
Hire charges paid for Plant	24,000	
Carriage on return	6,000	
Store overheads (10% of material consumed)	23,000	
Factory overheads (20% of Prime cost)	76,000	
Additional expenditure for rectification of defective products (refer working note)	2,160	1,31,160
Gross factory cost		**5,11,160**
Add: Opening value of W-I-P		50,000
Less: Closing value of W-I-P		(24,000)
Works/ Factory Cost		**5,37,160**
Less: Realisable value on sale of scrap		(5,000)
Cost of Production		**5,32,160**
Add: Opening stock of finished goods		-

Particulars	Amount (₹)	Amount (₹)
Less: Closing stock of finished goods		-
Cost of Goods Sold		**5,32,160**
Administrative overheads:		
Maintenance of office building	2,000	
Salary paid to Office staff	25,000	
Legal Charges	2,500	29,500
Selling overheads:		
Expenses for participation in Industrial exhibition	8,000	8,000
Distribution overheads:		
Depreciation on delivery van	6,000	
Warehousing charges	1,500	7,500
Cost of Sales		**5,77,160**

Working Notes:

1. Number of Rectified units

Total Output	8,000 units
Less: Rejected 10%	800 units
Finished product	7,200 units
Rectified units (10% of finished product)	**720 units**

2. Proportionate additional expenditure on 720 units

= 20% of proportionate direct wages

= 0.20 x (₹ 1,20,000/8,000) x 720

= ₹ 2,160

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92 EXAM **Old P3 Jan 2021** Q2(b) 10 marks old scheme

The following data are available from the books and records of Q Ltd. for the month of April 2020:

Direct Labour Cost = ₹ 1,20,000 (120% of Factory Overheads)

Cost of Sales = ₹ 4,00,000

Sales = ₹ 5,00,000

Accounts show the following figures:

	1st April, 2020 (₹)	30th April, 2020 (₹)
Inventory:		
Raw material	20,000	25,000
Work-in-progress	20,000	30,000
Finished goods	50,000	60,000
Other details:		
Selling expenses		22,000
General & Admin. expenses		18,000

You are required to prepare a cost sheet for the month of April 2020 showing:

(i) Prime Cost

(ii) Works Cost

(iii) Cost of Production

(iv) Cost of Goods sold

(v) Cost of Sales and Profit earned.

ICAI'S ANSWER

Cost Sheet for the Month of April 2020

Particulars	(₹)
Opening stock of Raw Material	20,000
Add: Purchases [Refer Working Note-2]	1,65,000
Less: Closing stock of Raw Material	(25,000)
Raw material consumed	1,60,000
Add: Direct labour cost	1,20,000
Prime cost	**2,80,000**
Add: Factory overheads	1,00,000
Gross Works cost	**3,80,000**
Add: Opening work-in-progress	20,000
Less: Closing work-in-progress	(30,000)
Works Cost	**3,70,000**
Cost of Production	**3,70,000**

Particulars	(₹)
Add: Opening stock of finished goods	50,000
Less: Closing stock of finished goods	(60,000)
Cost of goods sold	**3,60,000**
Add: General and administration expenses*	18,000
Add: Selling expenses	22,000
Cost of sales	**4,00,000**
Profit {Balancing figure (₹ 5,00,000 – ₹ 4,00,000)}	1,00,000
Sales	5,00,000

*General and administration expenses have been assumed as not relating to the production activity.

Working Note:

1. Computation of the raw material consumed

Particulars	(₹)
Cost of Sales	4,00,000
Less: General and administration expenses	(18,000)
Less: Selling expenses	(22,000)
Cost of goods sold	3,60,000
Add: Closing stock of finished goods	60,000
Less: Opening stock of finished goods	(50,000)
Cost of production/Gross works cost	3,70,000
Add: Closing stock of work-in-progress	30,000
Less: Opening stock of work-in-progress	(20,000)
Works cost	3,80,000
Less: Factory overheads (₹ 1,20,000/120 × 100)	(1,00,000)
Prime cost	2,80,000
Less: Direct labour	(1,20,000)
Raw material consumed	1,60,000

2. Computation of the raw material purchased

Particulars	(₹)
Closing stock of Raw Material	25,000
Add: Raw Material consumed	1,60,000
Less: Opening stock of Raw Material	(20,000)
Raw Material purchased	1,65,000

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93 EXAM **Old P3 Nov 2020** Q2(a) 10 marks old scheme

X Ltd. manufactures two types of pens 'Super Pen' and 'Normal Pen'.

The cost data for the year ended 30th September, 2019 is as follows:

	(₹)
Direct Materials	8,00,000
Direct Wages	4,48,000
Production Overhead	1,92,000
Total	14,40,000

It is further ascertained that :

(1) Direct materials cost in Super Pen was twice as much of direct material in Normal Pen.

(2) Direct wages for Normal Pen were 60% of those for Super Pen.

(3) Production overhead per unit was at same rate for both the types.

(4) Administration overhead was 200% of direct labour for each.

(5) Selling cost was ₹ 1 per Super pen.

(6) Production and sales during the year were as follow :

Production		Sales	
	No. of units		**No. of units**
Super Pen	40,000	Super Pen	36,000
Normal Pen	1,20,000		

(7) Selling price was ₹ 30 per unit for Super Pen.

Prepare a Cost Sheet for 'Super Pen' showing:

(i) Cost per unit and Total Cost

(ii) Profit per unit and Total Profit

ICAI'S ANSWER

Preparation of Cost Sheet for Super Pen

No. of units produced = 40,000 units

No. of units sold = 36,000 units

Particulars	Per unit (₹)	Total (₹)
Direct materials (Working note- (i))	8.00	**3,20,000**
Direct wages (Working note- (ii))	4.00	**1,60,000**
Prime cost	12.00	4,80,000
Production overhead (Working note- (iii))	1.20	**48,000**
Factory Cost	13.20	5,28,000
Administration Overhead* (200% of direct wages)	8.00	**3,20,000**
Cost of production	21.20	8,48,000
Less: Closing stock (40,000 units – 36,000 units)	-	**(84,800)**
Cost of goods sold i.e. 36,000 units	21.20	7,63,200
Selling cost	1.00	**36,000**
Cost of sales/ Total cost	**22.20**	**7,99,200**
Profit	**7.80**	**2,80,800**
Sales value (₹ 30 × 36,000 units)	30.00	**10,80,000**

Working Notes:

(i) Direct material cost per unit of Normal pen = M

Direct material cost per unit of Super pen = 2M

Total Direct Material cost = 2M × 40,000 units + M × 1,20,000 units

Or, ₹ 8,00,000 = 80,000 M + 1,20,000 M

Or, $M = \frac{₹\ 8,00,000}{2,00,000} = ₹\ 4$

Therefore, Direct material Cost per unit of Super pen = 2 × ₹ 4 = ₹ 8

(ii) Direct wages per unit for Super pen = W

Direct wages per unit for Normal Pen = 0.6W

So, (W x 40,000) + (0.6W x 1,20,000) = ₹ 4,48,000

W = ₹ 4 per unit

(iii) Production overhead per unit = $\frac{₹\ 1,92,000}{(40,000 + 1,20,000)} = ₹\ 1.20$

Production overhead for Super pen = ₹ 1.20 × 40,000 units = ₹ 48,000

* Administration overhead is specific to the product as it is directly related to direct labour as mentioned in the question and hence to be considered in cost of production only.

> **Assumption:** It is assumed that in point (1) and (2) of the Question, direct materials cost and direct wages respectively is related to per unit only.
>
> Note: Direct Material and Direct wages can be calculated in alternative ways.

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94 EXAM **Old P3 Nov 2019** Q3(b) 10 marks old scheme

XYZ a manufacturing firm, has revealed following information for September ,2019:

	1st September (₹)	30th September (₹)
Raw Materials	2,42,000	2,92,000
Works-in-progress	2,00,000	5,00,000

The firm incurred following expenses for a targeted production of 1,00,000 units during the month :

	(₹)
Consumable Stores and spares of factory	3,50,000
Research and development cost for process improvements	2,50,000
Quality control cost	2,00,000
Packing cost (secondary) per unit of goods sold	2
Lease rent of production asset	2,00,000
Administrative Expenses (General)	2,24,000
Selling and distribution Expenses	4,13,000
Finished goods (opening)	Nil
Finished goods (closing)	5000 units

Defective output which is 4% of targeted production, realizes ₹ 61 per unit.

Closing stock is valued at cost of production (excluding administrative expenses)

Cost of goods sold, excluding administrative expenses amounts to ₹ 78,26,000.

Direct employees cost is 1/2 of the cost of material consumed.

Selling price of the output is ₹ 110 per unit.

You are required to :

(i) Calculate the Value of material purchased

(ii) Prepare cost sheet showing the profit earned by the firm.

ICAI'S ANSWER

Workings:

1. Calculation of Sales Quantity:

Particular	Units
Production units	1,00,000
Less: Defectives (4%×1,00,000 units)	4,000
Less: Closing stock of finished goods	5,000
No. of units sold	91,000

2. Calculation of Cost of Production

Particular	Amount (₹)
Cost of Goods sold (given)	78,26,000
Add: Value of Closing finished goods (₹ 78,26,000/91,000 units × 5,000 units)	4,30,000
Cost of Production	82,56,000

3. Calculation of Factory Cost

Particular	Amount (₹)
Cost of Production	82,56,000
Less: Quality Control Cost	(2,00,000)
Less: Research and Development Cost	(2,50,000)
Add: Credit for Recoveries/Scrap/By-Products/ misc. income (1,00,000 units × 4% × ₹ 61)	2,44,000
Factory Cost	80,50,000

4. Calculation of Gross Factory Cost

Particular	Amount (₹)
Cost of Factory Cost	80,50,000
Less: Opening Work in Process	(2,00,000)
Add: Closing Work in Process	5,00,000
Cost of Gross Factory Cost	83,50,000

5. Calculation of Prime Cost

Particular	Amount (₹)
Cost of Gross Factory Cost	83,50,000
Less: Consumable stores & spares	(3,50,000)
Less: Lease rental of production assets	(2,00,000)
Prime Cost	78,00,000

6. Calculation of Cost of Materials Consumed & Labour cost

Let Cost of Material Consumed = M and Labour cost = 0.5M

Prime Cost = Cost of Material Consumed + Labour Cost

78,00,000 = M + 0.5M

M = 52,00,000

Therefore, Cost of Material Consumed = ₹ 52,00,000 and

Labour Cost = ₹ 26,00,000

(i) Calculation of Value of Materials Purchased

Particular	Amount (₹)
Cost of Material Consumed	52,00,000
Add: Value of Closing stock	2,92,000
Less: Value of Opening stock	(2,42,000)
Value of Materials Purchased	52,50,000

Cost Sheet

Sl.	Particulars	Total Cost (₹)
1.	Direct materials consumed:	
	Opening Stock of Raw Material	2,42,000
	Add: Additions/ Purchases [balancing figure as per requirement (i)]	52,50,000
	Less: Closing stock of Raw Material	(2,92,000)
	Material Consumed	52,00,000
2.	Direct employee (labour) cost	26,00,000
3.	Prime Cost (1+2)	78,00,000
4.	Add: Works/ Factory Overheads	
	Consumable stores and spares	3,50,000

Sl.	Particulars	Total Cost (₹)
	Lease rent of production asset	2,00,000
5.	Gross Works Cost (3+4)	83,50,000
6.	Add: Opening Work in Process	2,00,000
7.	Less: Closing Work in Process	(5,00,000)
8.	Works/ Factory Cost (5+6-7)	80,50,000
9.	Add: Quality Control Cost	2,00,000
10.	Add: Research and Development Cost	2,50,000
11.	Less: Credit for Recoveries/Scrap/By-Products/misc. income	(2,44,000)
12.	Cost of Production (8+9+10-11)	82,56,000
13.	Add: Opening stock of finished goods	-
14.	Less: Closing stock of finished goods (5000 Units)	(4,30,000)
15.	Cost of Goods Sold (12+13-14)	78,26,000
16.	Add: Administrative Overheads (General)	2,24,000
17.	Add: Secondary packing	1,82,000
18.	Add: Selling Overheads& Distribution Overheads	4,13,000
19.	Cost of Sales (15+16+17+18)	86,45,000
20.	Profit	13,65,000
21.	Sales 91,000 units@ ₹ 110 per unit	1,00,10,000

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95 RTP **Sep 2026** Q8

The following details are available from the books of Nova Manufacturing Ltd. for the year ending 31st March 2026:

Nova Manufacturing Ltd. produces high-precision metal components for industrial machinery. During the year, the company invested in new automation facilities, expanded factory supervision, and adopted new inventory control systems. The following data has been extracted from the cost accounting records:

Particulars	Amount (₹)
Purchase of raw materials	1,02,50,000
Consumable materials	6,20,000
Direct wages	72,40,000

Particulars	Amount (₹)
Carriage inward	2,10,000
Wages to foreman and storekeeper	10,20,000
Other indirect wages to factory staff	1,95,000
Expenditure on research & development for new automation systems	12,80,000
Salary to accountants	8,40,000
Employer's contribution to EPF & ESI	9,00,000
Cost of power & fuel	33,20,000
Production planning office expenses	15,50,000
Salary to delivery staff	18,20,000
Income tax for the assessment year 2025–26	3,60,000
Fees to statutory auditor	2,20,000
Fees to cost auditor	1,10,000
Fees to independent directors	11,80,000
Factory maintenance & repairs	7,30,000
Depreciation on plant & machinery	9,20,000
Quality control department expenses	5,40,000
Office rent	11,50,000
Insurance on finished goods warehouse	3,40,000
Packing materials	4,80,000
Donation to State Disaster Relief Fund	1,50,000
Value of sales	3,48,90,000

Opening Inventories (as on 01-04-2025):

- Raw Materials: ₹ 8,10,000
- Work-in-Progress: ₹ 9,25,000
- Finished Goods: ₹ 16,90,000

Closing Inventories (as on 31-03-2026):

- Raw Materials: ₹ 5,90,000
- Work-in-Progress: ₹ 7,80,000
- Finished Goods: ₹ 12,40,000

During the year, 2,000 defective units were produced due to machine malfunction.

The abnormal loss (not includible in cost) amounted to ₹ 4,60,000 and the scrap value realized was ₹ 40,000.

From the above information, PREPARE a COST SHEET for the year ended 31st March 2026.

ICAI'S ANSWER

Statement of Cost of Nova Manufacturing Ltd. for the year ended 31st March, 2026:

Sl. No.	Particulars	Amount (₹)	Amount (₹)
(i)	**Material Consumed:**		
	Purchase of Raw Materials	1,02,50,000	
	Carriage Inward	2,10,000	
	Add: Opening Stock of Raw Materials	8,10,000	
	Less: Closing Stock of Raw Materials	(5,90,000)	
	Raw Material Consumed		**1,06,80,000**
(ii)	**Direct Employee (Labour) Cost:**		
	Direct Wages	72,40,000	
	Employer's Contribution to EPF & ESI	9,00,000	
	Total Direct Employee Cost		**81,40,000**
(iii)	**Direct Expenses:**		
	Consumable Materials	6,20,000	
	Cost of Power & Fuel	33,20,000	
	Total Direct Expenses		**39,40,000**
	Prime Cost		**2,27,60,000**
(iv)	**Works / Factory Overheads:**		
	Wages to Foreman & Storekeeper	10,20,000	
	Other Indirect Wages to Factory Staff	1,95,000	
	Factory Maintenance & Repairs	7,30,000	
	Depreciation on Plant & Machinery	9,20,000	
	(Abnormal Loss – Not included in cost)	—	
	Total Factory Overheads		**28,65,000**
	Gross Factory Cost		**2,56,25,000**

Sl. No.	Particulars	Amount (₹)	Amount (₹)
	Add: Opening WIP	9,25,000	
	Less: Closing WIP	(7,80,000)	
	Factory Cost		**2,57,70,000**
(v)	Quality Control Department Expenses		5,40,000
(vi)	**Research & Development Cost (Automation Systems)**		**12,80,000**
(vii)	**Production Planning Office Expenses**		**15,50,000**
	Cost Of Production		**2,91,40,000**
	Add: Opening Stock of Finished Goods	16,90,000	
	Less: Closing Stock of Finished Goods	(12,40,000)	
	Cost Of Goods Sold		**2,95,90,000**
(viii)	**Administrative Overheads:**		
	Salary to Accountants	8,40,000	
	Fees to Statutory Auditor	2,20,000	
	Fees to Cost Auditor	1,10,000	
	Fees to Independent Directors	11,80,000	
	Office Rent	11,50,000	
	Total Administrative Overheads		**35,00,000**
(ix)	**Selling & Distribution Overheads:**		
	Salary to Delivery Staff	18,20,000	
	Insurance on Finished Goods Warehouse	3,40,000	
	Packing Materials	4,80,000	
	Total Selling & Distribution Overheads		**26,40,000**
	Cost Of Sales		**3,57,30,000**
	Profit / (Loss) (Balancing Figure)		**(8,40,000)**
	Sales		**3,48,90,000**

Notes:

- Abnormal loss of ₹ 4,60,000 (Scrap ₹ 40,000) is excluded from the cost sheet.
- Income tax and donation are not included in cost sheets.

96 RTP **May 2026** Q10

Also asked in MTP Jan 2025 S-I Q3(a)

Following information is available from the books of YSPP Ltd. for the current year ending 31st March:

S. No.	Particulars	(₹)	(₹)
(i)	Raw materials purchased		35,00,00,000
(ii)	Freight inwards		39,22,100
(iii)	Wages paid to factory workers		1,02,20,000
(iv)	Contribution made towards employees' PF & ESIS		12,60,000
(v)	Hire charges paid for hiring specific equipment		8,40,000
(vi)	Amount paid for power & fuel		16,17,000
(vii)	Amount paid for purchase of moulds and patterns (life is equivalent to four years production)		31,36,000
(viii)	Job charges paid to job workers		28,42,000
(ix)	Lease rent paid for production assets		3,92,000
(x)	Depreciation on:		
	Factory building	2,94,000	
	Office building	1,96,000	
	Plant & Machinery	4,41,000	
	Delivery vehicles	3,01,000	12,32,000
(xi)	Salary paid to supervisors		4,41,000
(xii)	Repairs & Maintenance paid for: Plant & Machinery	1,68,000	
	Sales office building	63,000	2,31,000
(xiii)	Insurance premium paid for:		
	Plant & Machinery	1,09,200	
	Factory building	63,350	
	Stock of raw materials & WIP	1,26,000	2,98,550
(xiv)	Expenses paid for quality control check activities		68,600
(xv)	Salary paid to quality control staffs		3,36,700

S. No.	Particulars	(₹)	(₹)
(xvi)	Research & development cost paid for improvement in production process		63,700
(xvii)	Expenses paid for administration of factory work		4,15,100
(xviii)	Salary paid to functional mangers:		
	Production control	33,60,000	
	Finance & Accounts	32,13,000	
	Sales & Marketing	35,42,000	1,01,15,000
(xix)	Salary paid to General Manager		43,96,000
(xx)	Packing cost paid for:		
	Primary packing necessary to maintain quality	3,36,000	
	For re-distribution of finished goods	3,92,000	7,28,000
(xxi)	Fee paid to auditors		6,30,000
(xxii)	Fee paid to independent directors		7,70,000
(xxiii)	Value of stock as on 1st April (beginning):		
	Raw materials	63,00,000	
	Work-in-process	32,20,000	
	Finished goods	38,50,000	1,33,70,000
(xxiv)	Value of stock as on 31st March (ending):		
	Raw materials	33,60,000	
	Work-in-process	30,45,000	
	Finished goods	63,00,000	1,27,05,000

Due to delay in picking up cargo from the port, YSPP Ltd. had to pay ₹ 15,000 as demurrage in the month of March.

From the above data you are required to PREPARE Statement of cost for YSPP Ltd. for the year ended 31st March, showing (i) Prime cost, (ii) Factory cost, (iii) Cost of Production, (iv) Cost of sales.

ICAI'S ANSWER

Statement of Cost of YSPP Ltd.

for the year ended 31st March:

S. No.	Particulars	(₹)	(₹)
(I)	Material consumed:		

S. No.	Particulars	(₹)	(₹)
	Raw materials purchased	35,00,00,000	
	Freight inwards	39,22,100	
	Add: opening stock of raw materials	63,00,000	
	Less: closing stock of raw materials	(33,60,000)	35,68,62,100
(II)	Direct employee (labour) cost:		
	Wages paid to factory workers	1,02,20,000	
	Contribution made towards employees' PF & ESIS	12,60,000	1,14,80,000
(III)	Direct expenses:		
	Hire charges paid for hiring specific equipment	8,40,000	
	Amount paid for power & fuel	16,17,000	
	Amortised cost of moulds and patterns	7,84,000	
	Job charges paid to job workers	28,42,000	60,83,000
	Prime cost		37,44,25,100
(IV)	Works/ factory overheads:		
	Lease rent paid for production assets	3,92,000	
	Depreciation on factory building	2,94,000	
	Depreciation on plant & machinery	4,41,000	
	Repairs & maintenance paid for plant & machinery	1,68,000	
	Insurance premium paid for plant & machinery	1,09,200	
	Insurance premium paid for factory building	63,350	
	Insurance premium paid for stock of raw materials & WIP	1,26,000	
	Salary paid to supervisors	4,41,000	20,34,550
	Gross factory cost		37,64,59,650
	Add: opening value of w-i-p		32,20,000
	Less: closing value of w-i-p		(30,45,000)
	Factory cost		37,66,34,650
(V)	Quality control cost:		

S. No.	Particulars	(₹)	(₹)
	Expenses paid for quality control check activities	68,600	
	Salary paid to quality control staffs	3,36,700	4,05,300
(VI)	Research & development cost paid for improvement in production process		63,700
(VII)	Administration cost related with production:		
	- Expenses paid for administration of factory work	4,15,100	
	- Salary paid to production control manager	33,60,000	37,75,100
(VIII)	Add: primary packing cost		3,36,000
	Cost of production		38,12,14,750
	Add: opening stock of finished goods		38,50,000
	Less: closing stock of finished goods		(63,00,000)
	Cost of goods sold		37,87,64,750
(IX)	Administrative overheads:		
	Depreciation on office building	1,96,000	
	Salary paid to manager- finance & accounts	32,13,000	
	Salary paid to general manager	43,96,000	
	Fee paid to auditors	6,30,000	
	Fee paid to independent directors	7,70,000	92,05,000
(X)	Selling overheads:		
	Repairs & maintenance paid for sales office building	63,000	
	Salary paid to manager- sales & marketing	35,42,000	36,05,000
(XI)	Distribution overheads:		
	Depreciation on delivery vehicles	3,01,000	
(XII)	Packing cost paid for re-distribution of finished goods	3,92,000	6,93,000
	Cost of sales		39,22,67,750

Note: Demurrage is a type of penalty, thus will not form part of cost.

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97 RTP Jan 2026 Case Scenario 2 · Q2(i) to Q2(v)

CASE SCENARIO 2

In order to provide employment opportunities in rural areas of Madhya Pradesh, State government decided to give interest free loan up to ₹ 30,00,00,000 to Khadi cloth industrialists. In return these industrialists are expected to set up the factory and produce khadi cloth in such rural areas. The loan is refundable after 20 years in lump sum. Madhya Pradesh government kept a stipulation that 40% of cloth production shall be supplied to government at a nominal rate of ₹ 10 per metre. This supply shall be used by the government for making uniforms of class IV employees of state government.

Haryana cloth mill of Hissar took loan of ₹ 30 crores and commenced business in Dindori, a rural district of Madhya Pradesh. Production commenced at Dindori and following are the details:

Production during the year 2023-24	40,00,000 metres of Khadi cloth
Raw cotton used	₹ 1,20,00,000
Direct wages	₹ 1,80,00,000
Freight inwards	₹ 8,00,000
Factory overheads estimated at 30% of prime cost	
Office and administration overheads estimated at 20% of factory cost	
Donations	₹ 2,00,000
Fines & Penalties	₹ 5,00,000
Selling and distribution overhead estimated at	₹ 1,69,52,000

The selling price to be charged from general public is to be fixed in such a way as to earn a profit of 30% on total cost of sales and price should be in multiple of Re.1.

Actual overheads incurred are as under:-

(1) Factory overheads = ₹ 89,50,000

(2) Office and administration overheads = ₹ 90,50,000

(3) Selling and distribution overheads = ₹ 1,70,00,000

The company has been requested to submit a cost audit report to the government detailing the production activities in its factory. As part of this process, the government has provided a draft format containing several key questions that must be addressed within the report. These questions are designed to ensure the accuracy and transparency of the company's production costs, and they play a critical role in shaping the audit report. Before proceeding with the preparation of the final cost audit report, it is essential that the company carefully reviews and works out the answers to these questions.

Q2(i)

Determine cost of sales per metre of cloth produced?

A ₹ 13.25

B ₹ 16.25

C ₹ 18.25

D ₹ 15.25

Q2(ii)

Determine the loss element per metre in government supply?

A ₹ 7.25

B ₹ 5.75

C ₹ 4.50

D ₹ 6.25

Q2(iii)

What will be the selling price per metre charged from general public?

A ₹ 29

B ₹ 27

C ₹ 31

D ₹ 33

Q2(iv)

What will be the rate of return on overall sales proceeds as per financial accounts?

A 22.313%

B 18.781%

C 24.387%

D 20.564%

Q2(v)

What will be the net profit in financial accounts from the factory, if government fixed the price of the cloth to be sold to general public at ₹ 28?

A ₹ 1,67,20,000

B ₹ 1,66,00,000

C ₹ 1,66,20,000

D ₹ 1,67,00,000

ICAI'S ANSWERS

Q2(i)

(B) ₹ 16.25

Particulars	Amount (₹)	Amount (₹)
Raw material	1,20,00,000	
Add: Freight inwards	8,00,000	
Direct material:		1,28,00,000
Direct labour (wages)		1,80,00,000
Prime cost		**3,08,00,000**

Particulars	Amount (₹)	Amount (₹)
Add: Factory overhead 30% of 3,08,00,000		92,40,000
Factory cost		**4,00,40,000**
Add: Office and administration overheads 20% of 4,00,40,000		80,08,000
Cost of production		**4,80,48,000**
Add: Selling and distribution overheads		1,69,52,000
Cost of sales		**6,50,00,000**
Add: Profit 30% of 6,50,00,000		1,95,00,000
Desirable sales		**8,45,00,000**

$$\text{Cost of sales per metre} = \frac{₹6,50,00,000}{40,00,000 \text{ metres}} = ₹ 16.25 \text{ per metre}$$

Q2(ii)

(D) ₹ 6.25

Loss element per metre on government supply

Cost of sales of khadi cloth per metre	₹ 16.25
Less: Supply rate to government per metre	(₹ 10)
Loss element per metre	₹ 6.25

Q2(iii)

(A) ₹ 29

Particulars	
Desirable sale as per cost sheet	₹ 8,45,00,000
Less: Amount at which supply is made to MP govt. (40,00,000 m x 40% x ₹ 10)	(₹ 1,60,00,000)
Amount to be recouped from general public	₹ 6,85,00,000
Units to be sold to general public	24,00,000 m
Selling price to general public (₹ 6,85,00,000/24,00,000 m)	₹ 28.5416

This will be rounded-off to Re. 1, so selling price will be ₹ 29 per metre.

Q2(iv)

(A) 22.313%

Profit & loss account

Particulars	Amount (₹)	Particulars	Amount (₹)
To direct material	1,20,00,000	By sales (₹ 29 × 24,00,000 + 10 × 16,00,000)	8,56,00,000
To direct labour	1,80,00,000		
To freight inwards	8,00,000		
To factory overheads	89,50,000		
To office and administration	90,50,000		
To selling and distribution overhead	1,70,00,000		
To Donations	2,00,000		
To fines and penalties	5,00,000		
To profit	1,91,00,000		
Total	**8,56,00,000**	**Total**	**8,56,00,000**

$$\text{Rate of return} = \frac{\text{Profit}}{\text{Sales}} = \frac{1,91,00,000}{8,56,00,000} = 22.313\%$$

Q2(v)

(D) ₹ 1,67,00,000

Particulars	Amount (₹)
New sales (₹ 28 × 24,00,000 + ₹ 10 × 16,00,000)	8,32,00,000
Less: Total costs as per P&L made above	(6,65,00,000)
New profit	**1,67,00,000**

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98 RTP **Sep 2025** Q10

The accounts of Ranu Ltd for the year ended 31st March,2025 , shows the following:

Particulars	(₹)
Carriage Inward	7,150
Administrative Office Salaries	12,600
Bad debts written off	6,500
Carriage Outward	4,300
GST (ITC Allowed)	10,000

Particulars	(₹)
Salaries related to factory office	7,000
Production Wages	1,20,000
Repairs - Plant and Machinery	4,450
Rent, Rates, Taxes, Insurance etc.	
Factory	8,500
Office	2,000
Sales	4,50,000
Stock of Raw materials:	
1st April, 2024	50,000
31st March, 2025	64,500
Materials Purchased (excluding GST)	2,00,000
Travelling Expenses	2,100
Manager's Salary (1/4 Office and 3/4 Factory)	10,000
Depreciation on Plant and Machinery	6,500
Depreciation on Office Furniture	500
Director's Fees	6,000
Gas and Water (Factory)	1,500
Gas and Water (Office)	500
General Expenses	3,400

You are required to PREPARE a cost statement for the year ended 31st March, 2025.

ICAI'S ANSWER

Ranu Ltd.

Cost Statement for the year ended 31st March, 2025

Particular	(₹)	(₹)
Raw Materials Consumed:		
Stock of Raw Materials as on 1st Apr, 2024	50,000	
Add: Materials Purchased	2,00,000	
Add: Carriage Inward	7,150	

Particular	(₹)	(₹)
Less: Stock of Raw Materials as on 31st March, 2025	(64,500)	
Raw Materials Consumed		1,92,650
Production Wages		1,20,000
Prime Cost		3,12,650
Add: Works/Factory Overheads:		
Salaries related to factory office	7,000	
Repairs to Plant and Machinery	4,450	
Rent, Rates, Taxes, Insurance etc. – Factory	8,500	
Depreciation on Plant and Machinery	6,500	
Gas and Water (Factory)	1,500	
Manager's Salary (3/4)	7,500	
Works or Factory Overheads		35,450
Works/Factory Cost		3,48,100
Add: Office and Administration Overheads:		
Administrative Office Salaries	12,600	
Rent, Rates, Taxes, Insurance etc. – Office	2,000	
Depreciation on Office Furniture	500	
Director's Fees	6,000	
Gas and Water (Office)	500	
Manager's Salary (1/4)	2,500	
General Expenses	3,400	
Office and Administration Overheads		27,500
Cost of Production/Cost of Goods Sold		3,75,600
Add: Selling and Distribution Overheads		
Carriage Outward	4,300	
Travelling Expenses	2,100	
Selling and Distribution Overheads		6,400

Particular	(₹)	(₹)
Total Cost of Sales		3,82,000
Add: Profit (Balancing Figure)		68,000
Sales		4,50,000

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99 RTP **May 2025** Q11(a)

The figures listed below are derived from the Trial Balance of EVS & Co. as of 31st March:

	Dr. (₹)	Cr. (₹)
Opening Inventories:		
Finished Stock	10,96,000	
Raw Materials	19,18,000	
Work-in-Process	27,40,000	
Office Appliances	2,38,380	
Plant & Machinery	63,08,850	
Building	27,40,000	
Sales		1,35,21,600
Sales Return and Rebates	1,91,800	
Cash discount allowed on sales	1,17,820	
Materials Purchased	43,84,000	
Freight incurred on Materials	2,19,200	
Purchase Returns		65,760
Direct employee cost	21,92,000	
Indirect employee cost	2,46,600	
Drawing and Designing cost	1,37,000	
Repairs and maintenance of factory	1,91,800	
Heat, Light and Power expenses	8,90,500	
Pollution Control Expenses	2,56,190	
Sales Commission	4,60,320	

	Dr. (₹)	Cr. (₹)
Sales Promotion	3,08,250	
Distribution Deptt. - Salaries and Expenses	2,46,600	
Office - Salaries and Expenses	1,17,820	
Packing Cost to make the product marketable	3,15,100	
Printing and Stationery expenses	89,050	
Bank Charges paid	8,220	

Further details are available as follows:

(i)	**Closing Inventories:**	
	Finished Goods	₹ 15,75,500
	Raw Materials	₹ 24,66,000
	Work-in-Process	₹ 26,30,400
(ii)	Outstanding direct employee cost	₹ 1,09,600
(iii)	Depreciation to be provided on:	
	Office Appliances	15%
	Plant and Machinery	40%
	Buildings	10%
(iv)	70% of the Heat, Light and Power expenses is related to the Factory and the remaining 30% is equally shared between the Office and Selling Department.	
	Depreciation on Buildings is to be distributed at a similar percentage between the Factory, Office and the Selling Department as that of the Heat, Light and Power.	

With the help of the above information, you are required to PREPARE a condensed Profit and Loss Statement of EVS & Co. for the year ended 31st March along with the following schedules:

(i) Cost of Sales (showing Prime Cost, Gross Works Cost, Cost of production, Cost of Goods Sold and Cost of Sales)

(ii) Selling and Distribution Expenses.

(iii) Administration Expenses

ICAI'S ANSWER

Profit and Loss Statement of EVS & Co. for the year ended 31st March

Particulars	(₹)	(₹)
Gross Sales	1,35,21,600	
Less: Returns and rebates	(1,91,800)	1,33,29,800
Less: Cost of Sales [See Schedule (i)]		**(1,20,21,887)**
Net Operating Profit		13,07,913
Less: Cash discount allowed on sales		(1,17,820)
Net Profit		**11,90,093**

(i) Schedule of Cost of Sales

Particulars	(₹)	(₹)
Raw Material (Inventory opening balance)		19,18,000
Add: Material Purchased	43,84,000	
Add: Freight on Material	2,19,200	
Less: Purchase Returns	(65,760)	45,37,440
		64,55,440
Less: Closing Raw Material Inventory		(24,66,000)
Materials consumed in Production		39,89,440
Direct employee cost (₹ 21,92,000 + ₹ 1,09,600)		23,01,600
Prime Cost		**62,91,040**
Factory Overheads:		
Indirect employee cost	2,46,600	
Drawing and Designing cost	1,37,000	
Repairs and maintenance of factory	1,91,800	
Heat, Light and Power (₹ 8,90,500 × 70%)	6,23,350	
Pollution Control Expenses	2,56,190	
Depreciation of Plant (40% of ₹ 63,08,850)	25,23,540	
Depreciation of Building (10% of ₹ 27,40,000 × 70%)	1,91,800	41,70,280
Gross Works Cost		**1,04,61,320**
Add: Opening Work-in-Process inventory		27,40,000

Particulars	(₹)	(₹)
Less: Closing Work-in-Process inventory		(26,30,400)
Cost of production		**1,05,70,920**
Add: Opening Finished Goods inventory		10,96,000
Less: Closing Finished Goods inventory		(15,75,500)
Cost of Goods Sold		**1,00,91,420**
Add: Administration Expenses [See Schedule (iii)]		4,25,522
Add: Selling and Distribution Expenses [See Schedule (ii)]		15,04,945
Cost of Sales		**1,20,21,887**

Note: Cash discount allowed on sales will not form part of Cost Sheet.

(ii) Schedule of Selling and Distribution Expenses

Particulars	(₹)
Sales Commission	4,60,320
Sales Promotion	3,08,250
Distribution Deptt. - Salaries and Expenses	2,46,600
Heat, Light and Power (₹ 8,90,500 × 15%)	1,33,575
Depreciation of Building (10% of ₹ 27,40,000 × 15%)	41,100
Packing Cost to make the product marketable	3,15,100
	15,04,945

(iii) Schedule of Administration Expenses

Particulars	(₹)
Office Salaries and Expenses	1,17,820
Depreciation of Office Appliances (₹ 2,38,380 × 15%)	35,757
Depreciation of Building (10% of ₹ 27,40,000 × 15%)	41,100
Heat, Light and Power (₹ 8,90,500 × 15%)	1,33,575
Printing and Stationery expenses	89,050
Bank Charges paid	8,220
	4,25,522

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100 RTP Jan 2025 Q5 MCQ

Following information is available for the month of March relating to manufacturing of a product:

Particulars	Amount (₹)
Cost of Sales	37,51,540
Stock of Raw material as on 01st March	6,50,000
Direct Wages	11,44,000
Hire charges paid for Plant (indirect expenses)	3,24,740
Salary to office staff	1,78,750
Maintenance of office building	13,000
Depreciation on Delivery van	39,000
Warehousing charges	61,750
Stock of Raw material as on 31st March	1,95,000
Realisable value on sale of scrap	32,500

Factory overheads are 20% of the Prime cost.

FIND OUT the value of Raw Material purchased with the help of Statement of Cost.

A ₹ 10,40,000

B ₹ 14,95,000

C ₹ 26,39,000

D ₹ 34,91,540

ICAI'S ANSWER

(A) Statement of Cost for the month of March

Particulars	Amount (₹)	Amount (₹)
Cost of Material Consumed:		
Raw materials purchased	10,40,000**	
Add: Opening stock of raw materials	6,50,000	
Less: Closing stock of raw materials	(1,95,000)	14,95,000
Direct Wages		11,44,000
Prime Cost		26,39,000*
Hire charges paid for Plant (indirect expenses)	3,24,740	
Factory overheads (20% of Prime cost)	5,27,800	8,52,540

Particulars	Amount (₹)	Amount (₹)
Works/ Factory Cost		34,91,540
Less: Realisable value on sale of scrap		(32,500)
Cost of Production/ Cost of Goods Sold		34,59,040
Administrative overheads:		
Maintenance of office building	13,000	
Salary paid to Office staff	1,78,750	1,91,750
Distribution overheads:		
Depreciation on delivery van	39,000	
Warehousing charges	61,750	1,00,750
Cost of Sales		37,51,540

(Reverse calculation to be done to find out the value of Raw materials purchased)

* Prime Cost + 3,24,740 + 20% of Prime Cost = 34,91,540

1.2 Prime Cost = 34,91,540 - 3,24,740 = 31,66,800

Prime Cost = 26,39,000

** Raw materials purchased = 14,95,000 - 6,50,000 + 1,95,000

= 10,40,000

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101 RTP **Jan 2025** Q11

IC Ltd. manufactures two types of phone covers, one is 'plastic' phone cover and another is 'silicon' phone cover.

The cost data relating to the manufacturing of both the phone covers for the year ended 31st March is provided below:

Particulars	Amount (₹)
Direct Materials	1,00,00,000
Direct Wages	56,00,000
Production Overhead	32,00,000
Total	1,88,00,000

Other information relating to the production of the phone covers is as follows:

- Direct material cost per unit of 'silicon' phone cover was twice than that of 'plastic' phone cover.
- Direct wages per unit for 'plastic' phone cover were 60% of those for 'silicon' phone cover.

- Production overhead per unit was at same rate for both the type of phone covers.
- Administration overhead being part of cost of production was 50% of Production overhead.
- Selling cost and Selling Price of 'silicon' phone cover were ₹ 8 and ₹ 140 per unit respectively.
- No. of units of 'silicon' phone covers sold- 90,000
- No. of units of Production of -
  'silicon' phone cover: 1,00,000

  'plastic' phone cover: 3,00,000

You are required to PREPARE a cost sheet for 'silicon' phone cover showing Cost and Profit (per unit and Total).

ICAI'S ANSWER

Preparation of Cost Sheet for 'silicon' phone covers

No. of units produced = 1,00,000 units

No. of units sold = 90,000 units

Particulars	Per unit (₹)	Total (₹)
Direct Materials (Working note- (i))	40.00	40,00,000
Direct Wages (Working note- (ii))	20.00	20,00,000
Prime Cost	60.00	60,00,000
Production Overhead (Working note- (iii))	8.00	8,00,000
Factory Cost	68.00	68,00,000
Administration Overhead (50% of Production Overhead)	4.00	4,00,000
Cost of Production	72.00	72,00,000
Less: Closing stock (1,00,000 units – 90,000 units)	-	(7,20,000)
Cost of Goods Sold i.e. 90,000 units	72.00	64,80,000
Selling cost	8.00	7,20,000
Cost of Sales/ Total Cost	**80.00**	**72,00,000**
Profit	**60.00**	**54,00,000**
Sales Value (₹ 140 × 90,000 units)	140.00	1,26,00,000

Working Notes:

(i) Direct material cost per unit of 'plastic' phone cover = M

Direct material cost per unit of 'silicon' phone cover = 2M

Total Direct Material Cost = 2M × 1,00,000 units + M × 3,00,000 units

Or, ₹ 1,00,00,000 = 2,00,000 M + 3,00,000 M

Or, $M = \frac{₹1,00,00,000}{5,00,000} = ₹ 20$

Therefore, Direct material Cost per unit of 'silicon' phone cover = 2 × ₹ 20 = ₹ 40

(ii) Direct wages per unit for 'silicon' phone cover = W

Direct wages per unit for 'plastic' phone cover = 0.6W

So, (W x 100,000) + (0.6W x 3,00,000) = ₹ 56,00,000

Or, 1,00,000 W + 1,80,000 W = ₹ 56,00,000

Or, $W = \frac{₹56,00,000}{2,80,000} = ₹ 20$ per unit

Therefore, Direct wages per unit of 'silicon' phone cover = ₹ 20

(iii) Production overhead per unit = $\frac{₹32,00,000}{(1,00,000 + 3,00,000)} = ₹8$

Production overhead for 'silicon' phone cover = ₹ 8 × 1,00,000 units = ₹ 8,00,000

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102 RTP **Sep 2024** Q11

From the following data of Appu Ltd., CALCULATE (i) Material Consumed; (ii) Prime Cost and (iii) Cost of production.

		Amount (₹)
(i)	Repair & maintenance paid for plant & machinery	9,80,500
(ii)	Insurance premium paid for inventories	26,000
(iii)	Insurance premium paid for plant & machinery	96,000
(iv)	Raw materials purchased	64,00,000
(v)	Opening stock of raw materials	2,88,000
(vi)	Closing stock of raw materials	4,46,000
(vii)	Wages paid	23,20,000
(viii)	Value of opening Work-in-process	4,06,000
(ix)	Value of closing Work-in-process	6,02,100
(x)	Quality control cost for the products in manufacturing process	86,000
(xi)	Research & development cost for improvement in production process	92,600
(xii)	Administrative cost for:	
	- Factory & production	9,00,000
	- Others	11,60,000
(xiii)	Amount realised by selling scrap generated during the manufacturing process	9,200

		Amount (₹)
(xiv)	Packing cost necessary to preserve the goods for further processing	10,200
(xv)	Salary paid to Director (Technical)	8,90,000

ICAI'S ANSWER

Calculation of Cost of Production of Appu Ltd.

Particulars	Amount (₹)
Raw materials purchased	64,00,000
Add: Opening stock	2,88,000
Less: Closing stock	(4,46,000)
Material consumed	62,42,000
Wages paid	23,20,000
Prime cost	85,62,000
Repair and maintenance cost of plant & machinery	9,80,500
Insurance premium paid for inventories	26,000
Insurance premium paid for plant & machinery	96,000
Quality control cost	86,000
Research & development cost	92,600
Administrative overheads related with factory and production	9,00,000
	1,07,43,100
Add: Opening value of W-I-P	4,06,000
Less: Closing value of W-I-P	(6,02,100)
	1,05,47,000
Less: Amount realised by selling scrap	(9,200)
Add: Primary packing cost	10,200
Cost of Production	1,05,48,000

Notes:

(i) Other administrative overhead does not form part of cost of production.

(ii) Salary paid to Director (Technical) is an administrative cost.

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103 RTP **May 2024** Q5

P Ltd. has gathered cost information from ledgers and other sources for the year ended 31st December 2023. The information are tabulated below:

Sl. No.		Amount (₹)	Amount (₹)
(i)	Raw materials purchased		5,00,00,000
(ii)	Freight inward		9,20,600
(iii)	Wages paid to factory workers		25,20,000
(iv)	Royalty paid for production		1,80,000
(v)	Amount paid for power & fuel		3,50,000
(vi)	Job charges paid to job workers		3,10,000
(vii)	Stores and spares consumed		1,10,000
(viii)	Depreciation on office building		50,000
(ix)	Repairs & Maintenance paid for:		
	- Plant & Machinery	40,000	
	- Sales office building	20,000	60,000
(x)	Insurance premium paid for:		
	- Plant & Machinery	28,200	
	- Factory building	18,800	47,000
(xi)	Expenses paid for quality control check activities		18,000
(xii)	Research & development cost paid for improvement in production process		20,000
(xiii)	Expenses paid for pollution control and engineering & maintenance		36,000
(xiv)	Salary paid to Sales & Marketing managers		5,60,000
(xv)	Salary paid to General Manager		6,40,000
(xvi)	Packing cost paid for:		
	- Primary packing necessary to maintain quality	46,000	
	- For re-distribution of finished goods	80,000	1,26,000
(xvii)	Fee paid to independent directors		1,20,000
(xviii)	Performance bonus paid to sales staffs		1,20,000
(xix)	Value of stock as on 1st January, 2023:		

Sl. No.		Amount (₹)	Amount (₹)
	- Raw materials	10,00,000	
	- Work-in-process	8,60,000	
	- Finished goods	12,00,000	30,60,000
(xx)	Value of stock as on 31st December, 2023:		
	- Raw materials	8,40,000	
	- Work-in-process	6,60,000	
	- Finished goods	10,50,000	25,50,000

Amount realized by selling of scrap and waste generated during manufacturing process – ₹ 48,000/-

The board meeting is scheduled to be held in next week and you being an associate to the chief cost controller of the company, has been asked to PREPARE a cost sheet.

ICAI'S ANSWER

Statement of Cost of P Ltd. for the year ended 31st December, 2023:

Sl. No.	Particulars	Amount (₹)	Amount (₹)
(i)	Material Consumed:		
	- Raw materials purchased	5,00,00,000	
	- Freight inward	9,20,600	
	Add: Opening stock of raw materials	10,00,000	
	Less: Closing stock of raw materials	(8,40,000)	5,10,80,600
(ii)	Direct employee (labour) cost:		
	- Wages paid to factory workers		25,20,000
(iii)	Direct expenses:		
	- Royalty paid for production	1,80,000	
	- Amount paid for power & fuel	3,50,000	
	- Job charges paid to job workers	3,10,000	8,40,000
	Prime Cost		5,44,40,600
(iv)	Works/ Factory overheads:		
	- Stores and spares consumed	1,10,000	
	- Repairs & Maintenance paid for plant & machinery	40,000	

Sl. No.	Particulars	Amount (₹)	Amount (₹)
	- Insurance premium paid for plant & machinery	28,200	
	- Insurance premium paid for factory building	18,800	
	- Expenses paid for pollution control and engineering & maintenance	36,000	2,33,000
	Gross factory cost		5,46,73,600
	Add: Opening value of W-I-P		8,60,000
	Less: Closing value of W-I-P		(6,60,000)
	Factory Cost		5,48,73,600
(v)	Quality control cost:		
	- Expenses paid for quality control check activities		18,000
(vi)	Research & development cost paid for improvement in production process		20,000
(vii)	Less: Realisable value on sale of scrap and waste		(48,000)
(viii)	Add: Primary packing cost		46,000
	Cost of Production		5,49,09,600
	Add: Opening stock of finished goods		12,00,000
	Less: Closing stock of finished goods		(10,50,000)
	Cost of Goods Sold		5,50,59,600
(ix)	Administrative overheads:		
	- Depreciation on office building	50,000	
	- Salary paid to General Manager	6,40,000	
	- Fee paid to independent directors	1,20,000	8,10,000
(x)	Selling overheads:		
	- Repairs & Maintenance paid for sales office building	20,000	
	- Salary paid to Manager- Sales & Marketing	5,60,000	
	- Performance bonus paid to sales staffs	1,20,000	7,00,000
(xi)	Distribution overheads:		
	- Packing cost paid for re-distribution of finished goods		80,000

Sl. No.	Particulars	Amount (₹)	Amount (₹)
	Cost of Sales		5,66,49,600

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104 RTP **Old P3 Nov 2023** Q5 old scheme

A Ltd. produces a single product X. During the month of July 2023, the company has produced 14,560 tonnes of X. The details for the month of July 2023 are as follows:

(i) Materials consumed ₹ 15,00,000

(ii) Power consumed in operating production machinery 13,000 Kwh @ ₹ 7 per Kwh

(iii) Diesels consumed in operating production machinery 1,000 litres @ ₹ 93 per litre

(iv) Wages & salary paid – ₹ 64,00,000

(v) Gratuity & leave encashment paid – ₹ 44,20,000

(vi) Hiring charges paid for Heavy Earth Moving machines (HEMM) engaged in production - ₹ 13,00,000. Hiring charges is paid on the basis of production.

(vii) Hiring charges paid for cars used for official purpose – ₹ 80,000

(viii) Reimbursement of diesel cost for the cars – ₹ 20,000

(ix) The hiring of cars attracts GST under RCM @5% without credit.

(x) Maintenance cost paid for weighing bridge (used for weighing of final goods at the time of despatch) – ₹ 7,000

(xi) AMC cost of CCTV installed at weighing bridge (used for weighing of final goods at the time of despatch) and factory premises is ₹ 6,000 and ₹ 18,000 per month respectively.

(xii) TA/ DA and hotel bill paid for sales manager- ₹ 16,000

(xiii) The company has 180 employees works for 26 days in a month.

Required:

PREPARE a Cost sheet for the month of July 2023.

ICAI'S ANSWER

(a) Cost Sheet of A Ltd. for the month of July 2023

Particulars	Amount (₹)	Amount (₹)
Materials consumed		15,00,000
Wages & Salary	64,00,000	
Gratuity & leave encashment	44,20,000	1,08,20,000
Power cost (13,000 kwh × ₹ 7)	91,000	
Diesel cost (1,000 ltr × ₹ 93)	93,000	1,84,000
HEMM hiring charges		13,00,000

Particulars	Amount (₹)	Amount (₹)
Prime Cost		**1,38,04,000**
AMC cost of CCTV installed at factory premises		18,000
Cost of Production/ Cost of Goods Sold		**1,38,22,000**
Hiring charges of cars	80,000	
Reimbursement of diesel cost	20,000	
	1,00,000	
Add: GST @5% on RCM basis	5,000	1,05,000
Maintenance cost for weighing bridge	7,000	
AMC cost of CCTV installed at weigh bridge	6,000	13,000
TA/ DA & hotel bill of sales manager		16,000
Cost of Sales		**1,39,56,000**

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105 RTP **Old P3 May 2023** Q5 old scheme

From the following data of Motilal Ltd., CALCULATE Cost of production:

		(₹)
(i)	Repair & maintenance paid for plant & machinery	9,80,500
(ii)	Insurance premium paid for inventories	26,000
(iii)	Insurance premium paid for plant & machinery	96,000
(iv)	Raw materials purchased	64,00,000
(v)	Opening stock of raw materials	2,88,000
(vi)	Closing stock of raw materials	4,46,000
(vii)	Wages paid	23,20,000
(viii)	Value of opening Work-in-process	4,06,000
(ix)	Value of closing Work-in-process	6,02,100
(x)	Quality control cost for the products in manufacturing process	86,000
(xi)	Research & development cost for improvement in production process	92,600
(xii)	Administrative cost for:	

		(₹)
	- Factory & production	9,00,000
	- Others	11,60,000
(xiii)	Amount realised by selling scrap generated during the manufacturing process	9,200
(xiv)	Packing cost necessary to preserve the goods for further processing	10,200
(xv)	Salary paid to Director (Technical)	8,90,000

ICAI'S ANSWER

Calculation of Cost of Production of Motilal Ltd for the period.....

Particulars	(₹)
Raw materials purchased	64,00,000
Add: Opening stock	2,88,000
Less: Closing stock	(4,46,000)
Material consumed	62,42,000
Wages paid	23,20,000
Prime cost	85,62,000
Repair and maintenance cost of plant & machinery	9,80,500
Insurance premium paid for inventories	26,000
Insurance premium paid for plant & machinery	96,000
Quality control cost	86,000
Research & development cost	92,600
Administrative overheads related with factory and production	9,00,000
	1,07,43,100
Add: Opening value of W-I-P	4,06,000
Less: Closing value of W-I-P	(6,02,100)
	1,05,47,000
Less: Amount realised by selling scrap	(9,200)
Add: Primary packing cost	10,200
Cost of Production	1,05,48,000

Notes:

(i) Other administrative overhead does not form part of cost of production.

(ii) Salary paid to Director (Technical) is an administrative cost.

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106 RTP **Old P3 Nov 2022** Q5 old scheme

CT Limited is engaged in producing medical equipment. It has furnished following details related to its products produced during a month:

	Units	Amount (₹)
Raw materials		
Opening stock	1,000	90,00,000
Purchases	49,000	44,10,00,000
Closing stock	1,750	1,57,50,000
Works-in-progress		
Opening	2,000	1,75,50,000
Closing	1,000	94,50,000
Direct employees' wages, allowances etc.		6,88,50,000
Primary packaging cost (per unit)		1,440
R&D expenses & Quality control expenses		2,10,60,000
Consumable stores, depreciation on plant		3,42,00,000
Administrative overheads related to production		3,15,00,000
Selling expenses		4,84,30,800
Royalty paid for production		3,64,50,000
Cost of web-site (for online sale) maintenance		60,75,000
Secondary packaging cost (per unit)		225

There was a normal scrap of 250 units of direct material which realized ₹ 5,400 per unit.

The entire finished product was sold at a profit margin of 20% on sales.

You are required to PREPARE a cost sheet showing:

(i) Prime cost

(ii) Gross works cost

(iii) Factory costs

(iv) Cost of production

(v) Profit

(vi) Sales

ICAI'S ANSWER

Cost Sheet

Particulars	Units	Amount (₹)
Material		
Opening stock	1,000	90,00,000
Add: Purchases	49,000	44,10,00,000
Less: Closing stock	(1,750)	(1,57,50,000)
	48,250	**43,42,50,000**
Less: Normal wastage of materials realized @ ₹ 5,400 per unit	(250)	**(13,50,000)**
Material consumed		43,29,00,000
Direct employee's wages and allowances		6,88,50,000
Direct expenses- Royalty paid for production		3,64,50,000
Prime cost	48,000	**53,82,00,000**
Factory overheads - Consumable stores, depreciation etc.		3,42,00,000
Gross Works Cost	48,000	**57,24,00,000**
Add: Opening WIP	2,000	1,75,50,000
Less: Closing WIP	(1,000)	(94,50,000)
Factory/Works Cost	49,000	**58,05,00,000**
Administration Overheads related to production		3,15,00,000
R&D expenses and Quality control cost		2,10,60,000
Add: Primary packaging cost @ ₹ 1,440 per unit		**7,05,60,000**
Cost of production	49,000	**70,36,20,000**
Selling expenses		4,84,30,800
Cost of maintaining website for online sale		60,75,000
Secondary packaging cost @ ₹ 225 per unit	49,000	**1,10,25,000**
Cost of sales		76,91,50,800
Add: Profit @ 20% on sales or 25% of cost		**19,22,87,700**

Particulars	Units	Amount (₹)
Sales value		**96,14,38,500**

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107 RTP **Old P3 May 2022** Q5 old scheme

A Ltd. produces a single product X. During the month of December 2021, the company has produced 14,560 tonnes of X. The details for the month of December 2021 are as follows:

(i) Materials consumed ₹ 15,00,000

(ii) Power consumed 13,000 Kwh @ ₹ 7 per Kwh

(iii) Diesels consumed 1,000 litres @ ₹ 93 per litre

(iv) Wages & salary paid – ₹ 64,00,000

(v) Gratuity & leave encashment paid – ₹ 44,20,000

(vi) Hiring charges paid for HEMM- ₹ 13,00,000

(vii) Hiring charges paid for cars used for official purpose – ₹ 80,000

(viii) Reimbursement of diesel cost for the cars – ₹ 20,000

(ix) The hiring of cars attracts GST under RCM @5% without credit.

(x) Maintenance cost paid for weighing bridge (used for weighing of final goods at the time of despatch) – ₹ 7,000

(xi) AMC cost of CCTV installed at weighing bridge (used for weighing of final goods at the time of despatch) and factory premises is ₹ 6,000 and ₹ 18,000 per month respectively.

(xii) TA/ DA and hotel bill paid for sales manager- ₹ 16,000

(xiii) The company has 180 employees works for 26 days in a month.

Required:

(a) PREPARE a Cost sheet for the month of December 2021.

(b) COMPUTE Earnings per manshift (EMS) and Output per manshift (OMS) for the month of December 2021.

ICAI'S ANSWER

(a) Cost Sheet of A Ltd. for the month of December 2021

Particulars	Amount (₹)	Amount (₹)
Materials consumed		15,00,000
Wages & Salary	64,00,000	
Gratuity & leave encashment	44,20,000	1,08,20,000
Power cost (13,000 kwh × ₹ 7)	91,000	
Diesel cost (1,000 ltr × ₹ 93)	93,000	1,84,000

Particulars	Amount (₹)	Amount (₹)
HEMM hiring charges		13,00,000
Prime Cost		**1,38,04,000**
AMC cost of CCTV installed at factory premises		18,000
Cost of Production/ Cost of Goods Sold		**1,38,22,000**
Hiring charges of cars	80,000	
Reimbursement of diesel cost	20,000	
	1,00,000	
Add: GST @5% on RCM basis	5,000	1,05,000
Maintenance cost for weighing bridge	7,000	
AMC cost of CCTV installed at weigh bridge	6,000	13,000
TA/ DA & hotel bill of sales manager		16,000
Cost of Sales		**1,39,56,000**

(b) Manshift = 180 employees × 26 days = 4,680 manshifts

Computation of earnings per manshift (EMS):

$$\text{EMS} = \frac{\text{Total employee benefits paid}}{\text{Manshift}}$$

$$= \frac{₹\ 1,08,20,000}{4,680} = ₹\ 2,312$$

Computation of Output per manshift (OMS):

$$\text{OMS} = \frac{\text{Total Output/ Production}}{\text{Manshift}}$$

$$= \frac{14,560\ \text{Tonne}}{4,680} = 3.11\ \text{tonne}$$

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108 RTP **Old P3 Dec 2021** Q5 old scheme

Impact Ltd. provides you the following details of its expenditures for the year ended 31st March, 2021:

S. No.	Particulars	Amount (₹)	Amount (₹)
(i)	Raw materials purchased		5,00,00,000
(ii)	GST paid under Composition scheme		10,00,000
(iii)	Freight inwards		5,20,600
(iv)	Trade discounts received		10,00,000
(v)	Wages paid to factory workers		15,20,000

S. No.	Particulars	Amount (₹)	Amount (₹)
(vi)	Contribution made towards employees' PF & ESIS		1,90,000
(vii)	Production bonus paid to factory workers		1,50,000
(viii)	Fee for technical assistance		1,12,000
(ix)	Amount paid for power & fuel		2,62,000
(x)	Job charges paid to job workers		4,50,000
(xi)	Stores and spares consumed		1,10,000
(xii)	Depreciation on:		
	Factory building	64,000	
	Office building	46,000	
	Plant & Machinery	86,000	1,96,000
(xiii)	Salary paid to supervisors		1,20,000
(xiv)	Repairs & Maintenance paid for:		
	Plant & Machinery	58,000	
	Sales office building	50,000	
	Vehicles used by directors	20,600	1,28,600
(xv)	Insurance premium paid for:		
	Plant & Machinery	31,200	
	Factory building	28,100	59,300
(xvi)	Expenses paid for quality control check activities		25,000
(xvii)	Research & development cost paid for improvement in production process		48,200
(xviii)	Expenses paid for administration of factory work		1,38,000
(xix)	Salary paid to functional mangers:		
	Production control	4,80,000	
	Finance & Accounts	9,60,000	
	Sales & Marketing	12,00,000	26,40,000
(xx)	Salary paid to General Manager		13,20,000
(xxi)	Packing cost paid for:		

S. No.	Particulars	Amount (₹)	Amount (₹)
	Primary packing necessary to maintain quality	1,06,000	
	For re-distribution of finished goods	1,12,000	2,18,000
(xxii)	Interest and finance charges paid (for usage of non- equity fund)		3,50,000
(xxiii)	Fee paid to auditors		1,80,000
(xxiv)	Fee paid to legal advisors		1,20,000
(xxv)	Fee paid to independent directors		2,40,000
(xxvi)	Payment for maintenance of website for online sales		1,80,000
(xxvii)	Performance bonus paid to sales staffs		2,40,000
(xxviii)	Value of stock as on 1st April, 2020:		
	Raw materials	9,00,000	
	Work-in-process	4,00,000	
	Finished goods	7,00,000	20,00,000
(xxix)	Value of stock as on 31st March, 2021:		
	Raw materials	5,60,000	
	Work-in-process	2,50,000	
	Finished goods	11,90,000	20,00,000

Amount realized by selling of waste generated during manufacturing process – ₹ 66,000/-

From the above data, you are required to PREPARE Statement of cost of Impact Ltd. for the year ended 31st March, 2021, showing (i) Prime cost, (ii) Factory cost, (iii) Cost of Production, (iv) Cost of goods sold and (v) Cost of sales.

ICAI'S ANSWER

Statement of Cost of Impact Ltd. for the year ended 31st March, 2021:

Sl. No.	Particulars	Amount (₹)	Amount (₹)
(i)	Material Consumed:		
	Raw materials purchased	5,00,00,000	
	GST paid under Composition scheme*	10,00,000	
	Freight inwards	5,20,600	
	Less: Trade discounts received	(10,00,000)	
	Add: Opening stock of raw materials	9,00,000	

Sl. No.	Particulars	Amount (₹)	Amount (₹)
	Less: Closing stock of raw materials	(5,60,000)	5,08,60,600
(ii)	Direct employee (labour) cost:		
	Wages paid to factory workers	15,20,000	
	Contribution made towards employees' PF & ESIS	1,90,000	
	Production bonus paid to factory workers	1,50,000	18,60,000
(iii)	Direct expenses:		
	Fee for technical assistance	1,12,000	
	Amount paid for power & fuel	2,62,000	
	Job charges paid to job workers	4,50,000	8,24,000
	Prime Cost		**5,35,44,600**
(iv)	Works/ Factory overheads:		
	Stores and spares consumed	1,10,000	
	Depreciation on factory building	64,000	
	Depreciation on plant & machinery	86,000	
	Repairs & Maintenance paid for plant & machinery	58,000	
	Insurance premium paid for plant & machinery	31,200	
	Insurance premium paid for factory building	28,100	
	Salary paid to supervisors	1,20,000	4,97,300
	Gross factory cost		**5,40,41,900**
	Add: Opening value of W-I-P		4,00,000
	Less: Closing value of W-I-P		(2,50,000)
	Factory Cost		**5,41,91,900**
(v)	Quality control cost:		
	Expenses paid for quality control check activities		25,000
(vi)	Research & development cost paid for improvement in production process		48,200
(vii)	Administration cost related with production:		
	-Expenses paid for administration of factory work	1,38,000	

Sl. No.	Particulars	Amount (₹)	Amount (₹)
	-Salary paid to Production control manager	4,80,000	6,18,000
(viii)	Less: Realisable value on sale of scrap and waste		(66,000)
(ix)	Add: Primary packing cost		1,06,000
	Cost of Production		**5,49,23,100**
	Add: Opening stock of finished goods		7,00,000
	Less: Closing stock of finished goods		(11,90,000)
	Cost of Goods Sold		**5,44,33,100**
(x)	Administrative overheads:		
	Depreciation on office building	46,000	
	Repairs & Maintenance paid for vehicles used by directors	20,600	
	Salary paid to Manager- Finance & Accounts	9,60,000	
	Salary paid to General Manager	13,20,000	
	Fee paid to auditors	1,80,000	
	Fee paid to legal advisors	1,20,000	
	Fee paid to independent directors	2,40,000	28,86,600
(xi)	Selling overheads:		
	Repairs & Maintenance paid for sales office building	50,000	
	Salary paid to Manager- Sales & Marketing	12,00,000	
	Payment for maintenance of website for online sales	1,80,000	
	Performance bonus paid to sales staffs	2,40,000	16,70,000
(xii)	Packing cost paid for re-distribution of finished goods		1,12,000
(xiii)	Interest and finance charges paid		3,50,000
	Cost of Sales		**5,94,51,700**

* GST paid under Composition scheme would be included under cost of material as it is not eligible for input tax credit.

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109 RTP Old P3 Jul 2021 Q5 old scheme

RTA Ltd. has the following expenditures for the year ended 31st December, 2020:

Sl. No.		Amount (₹)	Amount (₹)
(i)	Raw materials purchased		5,00,00,000
(ii)	Freight inward		9,20,600
(iii)	Wages paid to factory workers		25,20,000
(iv)	Royalty paid for production		1,80,000
(v)	Amount paid for power & fuel		3,50,000
(vi)	Job charges paid to job workers		3,10,000
(vii)	Stores and spares consumed		1,10,000
(viii)	Depreciation on office building		50,000
(ix)	Repairs & Maintenance paid for:		
	- Plant & Machinery	40,000	
	- Sales office building	20,000	60,000
(x)	Insurance premium paid for:		
	- Plant & Machinery	28,200	
	- Factory building	18,800	47,000
(xi)	Expenses paid for quality control check activities		18,000
(xii)	Research & development cost paid for improvement in production process		20,000
(xiii)	Expenses paid for pollution control and engineering & maintenance		36,000
(xiv)	Salary paid to Sales & Marketing mangers		5,60,000
(xv)	Salary paid to General Manager		6,40,000
(xvi)	Packing cost paid for:		
	- Primary packing necessary to maintain quality	46,000	
	- For re-distribution of finished goods	80,000	1,26,000
(xvii)	Fee paid to independent directors		1,20,000
(xviii)	Performance bonus paid to sales staffs		1,20,000
(xix)	Value of stock as on 1st January, 2020:		
	- Raw materials	10,00,000	

Sl. No.		Amount (₹)	Amount (₹)
	- Work-in-process	8,60,000	
	- Finished goods	12,00,000	30,60,000
(xx)	Value of stock as on 31st December, 2020:		
	- Raw materials	8,40,000	
	- Work-in-process	6,60,000	
	- Finished goods	10,50,000	25,50,000

Amount realized by selling of scrap and waste generated during manufacturing process – ₹ 48,000/-

From the above data you are requested to PREPARE Statement of Cost for RTA Ltd. for the year ended 31st December, 2020, showing (i) Prime cost, (ii) Factory cost, (iii) Cost of Production, (iv) Cost of goods sold and (v) Cost of sales.

ICAI'S ANSWER

Statement of Cost of RTA Ltd. for the year ended 31st December, 2020:

Sl. No.	Particulars	Amount (₹)	Amount (₹)
(i)	Material Consumed:		
	- Raw materials purchased	5,00,00,000	
	- Freight inward	9,20,600	
	Add: Opening stock of raw materials	10,00,000	
	Less: Closing stock of raw materials	(8,40,000)	5,10,80,600
(ii)	Direct employee (labour) cost:		
	- Wages paid to factory workers		25,20,000
(iii)	Direct expenses:		
	- Royalty paid for production	1,80,000	
	- Amount paid for power & fuel	3,50,000	
	- Job charges paid to job workers	3,10,000	8,40,000
	Prime Cost		5,44,40,600
(iv)	Works/ Factory overheads:		
	- Stores and spares consumed	1,10,000	
	- Repairs & Maintenance paid for plant & machinery	40,000	
	- Insurance premium paid for plant & machinery	28,200	

Sl. No.	Particulars	Amount (₹)	Amount (₹)
	- Insurance premium paid for factory building	18,800	
	- Expenses paid for pollution control and engineering & maintenance	36,000	2,33,000
	Gross factory cost		5,46,73,600
	Add: Opening value of W-I-P		8,60,000
	Less: Closing value of W-I-P		(6,60,000)
	Factory Cost		5,48,73,600
(v)	Quality control cost:		
	- Expenses paid for quality control check activities		18,000
(vi)	Research & development cost paid for improvement in production process		20,000
(vii)	Less: Realisable value on sale of scrap and waste		(48,000)
(viii)	Add: Primary packing cost		46,000
	Cost of Production		5,49,09,600
	Add: Opening stock of finished goods		12,00,000
	Less: Closing stock of finished goods		(10,50,000)
	Cost of Goods Sold		5,50,59,600
(ix)	Administrative overheads:		
	- Depreciation on office building	50,000	
	- Salary paid to General Manager	6,40,000	
	- Fee paid to independent directors	1,20,000	8,10,000
(x)	Selling overheads:		
	- Repairs & Maintenance paid for sales office building	20,000	
	- Salary paid to Manager- Sales & Marketing	5,60,000	
	- Performance bonus paid to sales staffs	1,20,000	7,00,000
(xi)	Distribution overheads:		
	- Packing cost paid for re-distribution of finished goods		80,000
	Cost of Sales		5,66,49,600

110 RTP **Old P3 Nov 2020** Q5 old scheme

The following details are available from the books of R Ltd. for the year ending 31st March 2020:

Particulars	Amount (₹)
Purchase of raw materials	84,00,000
Consumable materials	4,80,000
Direct wages	60,00,000
Carriage inward	1,72,600
Wages to foreman and store keeper	8,40,000
Other indirect wages to factory staffs	1,35,000
Expenditure on research and development on new production technology	9,60,000
Salary to accountants	7,20,000
Employer's contribution to EPF & ESI	7,20,000
Cost of power & fuel	28,00,000
Production planning office expenses	12,60,000
Salary to delivery staffs	14,30,000
Income tax for the assessment year 2019-20	2,80,000
Fees to statutory auditor	1,80,000
Fees to cost auditor	80,000
Fees to independent directors	9,40,000
Donation to PM-national relief fund	1,10,000
Value of sales	2,82,60,000
Position of inventories as on 01-04-2019:	
- Raw Material	6,20,000
- W-I-P	7,84,000
- Finished goods	14,40,000
Position of inventories as on 31-03-2020:	
- Raw Material	4,60,000
- W-I-P	6,64,000

Particulars	Amount (₹)
- Finished goods	9,80,000

From the above information PREPARE a cost sheet for the year ended 31st March 2020.

ICAI'S ANSWER

Statement of Cost of R Ltd. for the year ended 31st March, 2020:

Sl. No.	Particulars	Amount (₹)	Amount (₹)
(i)	Material Consumed:		
	- Raw materials purchased	84,00,000	
	- Carriage inward	1,72,600	
	Add: Opening stock of raw materials	6,20,000	
	Less: Closing stock of raw materials	(4,60,000)	87,32,600
(ii)	Direct employee (labour) cost:		
	- Direct wages	60,00,000	
	- Employer's Contribution towards PF & ESIS	7,20,000	67,20,000
(iii)	Direct expenses:		
	- Consumable materials	4,80,000	
	- Cost of power & fuel	28,00,000	32,80,000
	Prime Cost		1,87,32,600
(iv)	Works/ Factory overheads:		
	- Wages to foreman and store keeper	8,40,000	
	- Other indirect wages to factory staffs	1,35,000	9,75,000
	Gross factory cost		1,97,07,600
	Add: Opening value of W-I-P		7,84,000
	Less: Closing value of W-I-P		(6,64,000)
	Factory Cost		1,98,27,600
(v)	Research & development cost paid for improvement in production process		9,60,000
(vi)	Production planning office expenses		12,60,000
	Cost of Production		2,20,47,600

Sl. No.	Particulars	Amount (₹)	Amount (₹)
	Add: Opening stock of finished goods		14,40,000
	Less: Closing stock of finished goods		(9,80,000)
	Cost of Goods Sold		2,25,07,600
(vii)	Administrative overheads:		
	- Salary to accountants	7,20,000	
	- Fees to statutory auditor	1,80,000	
	- Fees to cost auditor	80,000	
	- Fee paid to independent directors	9,40,000	19,20,000
(viii)	Selling overheads& Distribution overheads:		
	- Salary to delivery staffs		14,30,000
	Cost of Sales		2,58,57,600
	Profit (balancing figure)		24,02,400
	Sales		2,82,60,000

Note: Income tax and Donation to PM National Relief Fund is avoided in the cost sheet.

ICAI · Old P3 · RTP Nov 2020 · question p. 4-5 · answer p. 17-18

111 RTP **Old P3 May 2020** Q5 old scheme

From the following data of Arnav Metallic Ltd., CALCULATE Cost of production:

		Amount (₹)
(i)	Repair & maintenance paid for plant & machinery	9,80,500
(ii)	Insurance premium paid for plant & machinery	96,000
(iii)	Raw materials purchased	64,00,000
(iv)	Opening stock of raw materials	2,88,000
(v)	Closing stock of raw materials	4,46,000
(vi)	Wages paid	23,20,000
(vii)	Value of opening Work-in-process	4,06,000
(viii)	Value of closing Work-in-process	6,02,100
(ix)	Quality control cost for the products in manufacturing process	86,000
(x)	Research & development cost for improvement in production process	92,600

		Amount (₹)
(xi)	Administrative cost for:	
	- Factory & production	9,00,000
	- Others	11,60,000
(xii)	Amount realised by selling scrap generated during the manufacturing process	9,200
(xiii)	Packing cost necessary to preserve the goods for further processing	10,200
(xiv)	Salary paid to Director (Technical)	8,90,000

ICAI'S ANSWER

Calculation of Cost of Production of Arnav Metallic Ltd. for the period.....

Particulars	Amount (₹)
Raw materials purchased	64,00,000
Add: Opening stock	2,88,000
Less: Closing stock	(4,46,000)
Material consumed	62,42,000
Wages paid	23,20,000
Prime cost	85,62,000
Repair and maintenance cost of plant & machinery	9,80,500
Insurance premium paid for plant & machinery	96,000
Quality control cost	86,000
Research & development cost	92,600
Administrative overheads related with factory and production	9,00,000
	1,07,17,100
Add: Opening value of W-I-P	4,06,000
Less: Closing value of W-I-P	(6,02,100)
	1,05,21,000
Less: Amount realised by selling scrap	(9,200)
Add: Primary packing cost	10,200
Cost of Production	**1,05,22,000**

Notes:

(i) Other administrative overhead does not form part of cost of production.

(ii) Salary paid to Director (Technical) is an administrative cost.

ICAI · Old P3 · RTP May 2020 · question p. 4 · answer p. 15-16

112 MTP **Sep 2026 S-I** MCQ 13 2 marks MCQ

Same question, different ICAI answer: #127 MTP May 2024 S-II

What would be Prime cost from below information?

Direct materials Purchased	:	₹ 75,000
Direct labour	:	₹ 45,000
Direct expenses	:	₹ 15,000
Manufacturing overheads	:	₹ 22,500
Direct materials consumed	:	₹ 67,500

A ₹ 1,35,000

B ₹ 1,27,500

C ₹ 1,57,500

D ₹ 1,50,000

ICAI'S ANSWER

(B) ₹ 1,27,500

Direct materials consumed	:	₹ 67,500
Direct labour	:	₹ 45,000
Direct expenses	:	₹ 15,000
Prime Cost		₹ 1,27,500

ICAI · P4 · MTP Sep 2026 · Series I · question p. 6 · answer p. 6

113 MTP **Sep 2026 S-I** Q1(b) 5 marks

Krishna Auto Components Pvt. Ltd., an auto-ancillary unit based in Pune manufacturing precision metal parts for two-wheeler assembly lines, prepares a quarterly cost sheet for internal management review and for pricing discussions with its OEM clients. The cost accountant has compiled the following data for the quarter ending 31st March 2026 and has been asked to present a complete cost sheet — from raw material consumption all the way down to profit — before the upcoming board review.

The following details have been gathered:

Particulars		Amount
(i)	Direct expenses	₹ 1,65,000
(ii)	Direct wages, being 180% of factory overheads	₹ 2,70,000
(iii)	Cost of goods sold	₹ 17,40,000
(iv)	Selling & distribution overheads	₹ 52,000
(v)	Sales	₹ 20,60,000
(vi)	Administration overheads related to production, being 8% of factory overheads	

Stock details as per Stock Register:

Particulars	31.12.2025 (₹)	31.03.2026 (₹)
Raw material	2,20,000	1,95,000
Work-in-progress	1,55,000	1,72,500
Finished goods	2,90,000	2,60,000

You are required to PREPARE a Cost Sheet showing:

(i) Raw material consumed

(ii) Prime cost

(iii) Factory cost

(iv) Cost of goods sold

(v) Cost of sales and profit

ICAI'S ANSWER

Cost Sheet

(for the quarter ending 31st March 2026)

		Amount (₹)
(i)	**Raw materials consumed**	
	Opening stock of raw materials	2,20,000
	Add: Purchase of materials*	11,05,500
	Less: Closing stock of raw materials	(1,95,000)
	Raw materials consumed	**11,30,500**
	Add: Direct wages (1,50,000 × 180%)	2,70,000
	Direct Expenses	1,65,000

		Amount (₹)
(ii)	**Prime cost**	**15,65,500**
	Add: Factory overheads (2,70,000/180%)	1,50,000
	Gross Factory cost	17,15,500
	Add: Opening work-in-process	1,55,000
	Less: Closing work-in-process	(1,72,500)
(iii)	**Factory cost**	**16,98,000**
	Add: Administration overheads (8% of factory overheads)	12,000
	Add: Opening stock of finished goods	2,90,000
	Less: Closing stock of finished goods	(2,60,000)
(iv)	**Cost of goods sold**	**17,40,000**
	Add: Selling & distribution overheads	52,000
(v)	**Cost of sales**	**17,92,000**
Net Profit		**2,68,000**
Sales		20,60,000

***Purchase of raw materials** = (17,40,000 + 2,60,000 – 2,90,000 – (1,50,000 × 8%) + 1,72,500 – 1,55,000 – (2,70,000 × 100/180) – 1,65,000 – 2,70,000 + 1,95,000 – 2,20,000) = **₹ 11,05,500**

Working Notes

1. Purchase of raw materials = Raw material consumed + Closing stock – Opening stock of raw material
2. Raw material consumed = Prime cost – Direct wages – Direct expenses
3. Factory Overheads = 2,70,000 × 100/180 = ₹ 1,50,000
4. Prime cost = Factory cost + Closing WIP – Opening WIP – Factory overheads
5. Factory Cost = Cost of goods sold + Closing stock of Finished goods – Opening stock of finished goods – Administrative overheads
6. Net Profit = Sales – Cost of sales

ICAI · P4 · MTP Sep 2026 · Series I · question p. 8-9 · answer p. 7-8

114 MTP **Sep 2026 S-I** Q5(b) 4 marks

EXPLAIN the treatment of following items in cost sheet.

(i) Credit for Recoveries

(ii) Packing Cost (primary)

(iii) Joint Products and By-Products Cost

(iv) Quality Control Cost

ICAI'S ANSWER

Treatment is as follows:

(i) Credit for Recoveries: The realised or realisable value of scrap or waste is deducted as it reduces the cost of production.

(ii) Packing Cost (primary): Packing material which is essential to hold and preserve the product for its use by the customer is added in the factory cost.

(iii) Joint Products and By-Products: Joint costs are allocated between/among the products on a rational and consistent basis. In case of by-products, the net realisable value of by-products is deducted from the cost of production.

(iv) Quality Control Cost: It is added in the factory cost as this is the cost of resources consumed towards quality control procedures.

ICAI · P4 · MTP Sep 2026 · Series I · question p. 15 · answer p. 19-20

115 MTP **May 2026 S-I** Case Scenario 2 · MCQ 6 to MCQ 10 10 marks

CASE SCENARIO 2

A manufacturing company is in the process of preparing its cost sheet for the financial year. The management wants to analyze the efficiency of production operations and requires detailed computation of key cost components such as material consumed, prime cost, cost of production, etc.

During the year, the company incurred various expenditures related to raw materials, labour, factory operations, and administrative activities. It also maintained records of inventory levels and work-in-process to ensure accurate cost determination.

The following information has been gathered from the company's records:

		Amount (₹)
(i)	Repair & maintenance paid for plant & machinery	9,80,500
(ii)	Insurance premium paid for inventories	26,000
(iii)	Insurance premium paid for plant & machinery	96,000
(iv)	Raw materials purchased	64,00,000
(v)	Opening stock of raw materials	2,88,000
(vi)	Closing stock of raw materials	4,46,000
(vii)	Wages paid	23,20,000
(viii)	Value of opening Work-in-process	4,06,000
(ix)	Value of closing Work-in-process	6,02,100
(x)	Quality control cost for the products in manufacturing process	86,000
(xi)	Research & development cost for improvement in production process	92,600

		Amount (₹)
(xii)	Administrative cost for:	
	- Factory & production	9,00,000
	- Others	11,60,000
(xiii)	Amount realised by selling scrap generated during the manufacturing process	9,200
(xiv)	Packing cost necessary to preserve the goods for further processing	10,200
(xv)	Salary paid to Director (Technical)	8,90,000
(xvi)	Television programme sponsorship cost	5,00,000

As a production manager you are required to answer the following questions (MCQs 6 to 10):

MCQ 6 2 marks

What is the material consumed by the company?

A ₹ 60,42,000

B ₹ 62,42,000

C ₹ 64,00,000

D ₹ 59,88,000

MCQ 7 2 marks

What is the prime cost of the company?

A ₹ 82,62,000

B ₹ 85,62,000

C ₹ 88,82,000

D ₹ 90,00,000

MCQ 8 2 marks

Calculate the work cost after adjusting Work-in-Progress.

A ₹ 96,64,500

B ₹ 98,70,500

C ₹ 92,58,400

D ₹ 94,68,400

MCQ 9 2 marks

What is the cost of production?

A ₹ 1,02,48,000

B ₹ 1,03,86,000

C ₹ 1,05,48,000

D ₹ 1,07,20,000

MCQ 10 2 marks

What is the cost of sales?

A ₹ 1,30,98,000

B ₹ 1,25,98,000

C ₹ 1,20,48,000

D ₹ 1,28,90,000

ICAI'S ANSWERS

MCQ 6

(B) ₹ 62,42,000

MCQ 7

(B) ₹ 85,62,000

MCQ 8

(D) ₹ 94,68,400

MCQ 9

(C) ₹ 1,05,48,000

MCQ 10

(A) ₹ 1,30,98,000

Working Note:

Calculation of Cost of Production of A Ltd. for the period.....

Particulars	Amount (₹)
Raw materials purchased	64,00,000
Add: Opening stock	2,88,000
Less: Closing stock	(4,46,000)
Material consumed	**62,42,000.00**
Wages paid	23,20,000.00
Prime cost	**85,62,000.00**
Repair and maintenance cost of plant & machinery	9,80,500.00
Insurance premium paid for inventories	26,000.00
Insurance premium paid for plant & machinery	96,000.00
Gross Work Cost	**96,64,500.00**
Add: Opening value of W-I-P	4,06,000.00
Less: Closing value of W-I-P	(6,02,100)
Work Cost	**94,68,400.00**

Particulars	Amount (₹)
Quality control cost	86,000.00
Research & development cost	92,600.00
Administrative overheads related with factory and production	9,00,000.00
Less: Amount realised by selling scrap	(9,200)
Add: Primary packing cost	10,200.00
Cost of Production	**1,05,48,000.00**
Administrative overheads (Other)	11,60,000.00
Salary paid to Director (Technical)	8,90,000
Television programme sponsorship cost	5,00,000
Cost of Sales	**1,30,98,000.00**

ICAI · P4 · MTP May 2026 · Series I · question p. 3-4 · answer p. 3, 3-4

116 MTP **Jan 2026 S-I** Q5(a) 7 marks

Priyank Ltd. produces two types of bags — 'Premium' and 'Standard'. The summarized cost data for the year ended 31st March, 2025 is given below:

Particulars	Amount (₹)
Direct Materials	12,00,000
Direct Wages	6,72,000
Production Overheads	2,88,000
Total Cost	21,60,000

There was no work-in-progress at the beginning or at the end of the year. The following additional information is available:

1. The direct material cost per unit of 'Premium' bags was twice that of 'Standard' bags.
2. A 2% cash discount was received for payments made within 30 days to suppliers of direct materials.
3. The direct labour cost per unit of 'Standard' bags was 60% of that of 'Premium' bags.
4. The rate of production overhead per unit was identical for both types of bags.
5. Administration overheads were charged at 200% of direct labour cost for each product type.
6. The selling cost was ₹ 1 per 'Premium' bag.
7. The details of production and sales during the year were as follows:

Type	Units Produced	Units Sold
Premium	60,000	54,000
Standard	1,80,000	—

8. The selling price of each 'Premium' bag was ₹ 30.
9. During the year, the company had to pay a penalty of ₹ 50,000 due to a copyright infringement case related to the manufacturing process of 'Premium' bags.

PREPARE a Cost Sheet of Priyank Ltd. for 'Premium' bags showing:

(i) Cost per unit and Total Cost

(ii) Profit per unit and Total Profit

ICAI'S ANSWER

Cost Sheet of 'Premium'

Particulars	Per unit (₹)	Total (₹)
Direct materials (Working note- (i))	8.00	4,80,000
Direct wages (Working note- (ii))	4.00	2,40,000
Prime cost	12.00	7,20,000
Production overhead (Working note- (iii))	1.20	72,000
Factory Cost	13.20	7,92,000
Administration Overhead (200% of direct wages)	8.00	4,80,000
Cost of production	21.20	12,72,000
Less: Closing stock (60,000 units – 54,000 units)	-	1,27,200
Cost of goods sold i.e. 54,000 units	21.20	11,44,800
Selling cost	1.00	54,000
Cost of sales/ Total cost	22.20	11,98,800
Profit	7.80	4,21,200
Sales value (₹ 30 × 54,000 units)	30.00	16,20,000

Working Notes:

(i) Direct material cost per unit of 'Standard' = M

Direct material cost per unit of 'Premium' = 2M

Total Direct Material cost = 2M × 60,000 units + M × 1,80,000 units

Or, ₹ 12,00,000 = 1,20,000 M + 1,80,000 M

Or, $M = \frac{₹\ 12,00,000}{3,00,000} = ₹\ 4$

Therefore, Direct material Cost per unit of 'Premium' = 2 × ₹ 4 = ₹ 8

(ii) Direct wages per unit for 'Premium' = W

Direct wages per unit for 'Standard' = 0.6W

So, (W x 60,000) + (0.6W x 1,80,000) = ₹ 6,72,000

W = ₹ 4 per unit

(iii) Production overhead per unit = $\frac{₹\ 2,88,000}{(60,000 + 1,80,000)}$ = ₹ 1.20

Production overhead for 'Premium' = ₹ 1.20 × 60,000 units = ₹ 72,000

Notes:

1. Administration overhead is specific to the product as it is directly related to direct labour as mentioned in the question and hence to be considered in cost of production only.
2. Cash discount is treated as interest and finance charges; hence, it is ignored.
3. Penalty paid against the copyright infringement case is an abnormal cost; hence, not included.

ICAI · P4 · MTP Jan 2026 · Series I · question p. 14-15 · answer p. 15-16

117 MTP **Jan 2026 S-II** Q4(a) 9 marks

XYZ Ltd. manufactures four types of products namely P, Q, R and S. The details for the month of April 2025 are as follows:

Particulars	P	Q	R	S
Actual Production & Sales (in units)	7,000	17,500	45,000	1,60,000
Direct Material Cost (in ₹)	3,50,000	4,75,000	5,80,000	9,60,000
Direct Wages (in ₹)	1,05,000	1,40,000	1,70,000	2,65,000
Selling price per unit (in ₹)	120.00	75.00	38.00	15.00

Additional Information:

If the company limits manufacturing to just one type of product, the monthly production will be as follows:

Product	:	Quantity
P	:	60,000 units
Q	:	75,000 units
R	:	1,00,000 units
S	:	3,00,000 units

Additional Information

- Factory overhead expenditure for the month is ₹ 10,00,000. Total available hours are 3,00,000 hours.
- Factory overhead expenses need to be allocated to each product based on the number of units, which could have been produced in a month when single product production was in operation.

- Administration cost (relating to production) has been allocated to each product on appropriate basis.
- Selling and distribution cost of ₹ 5,60,000 needs to be allocated among the products P, Q, R and S in the ratio of 1 : 2 : 3 : 4 respectively.
- The company requires a profit margin @ 20% on sales.
- There is no opening & closing inventory.

You are required to SHOW for each product :

(i) Prime Cost

(ii) Works Cost

(iii) Cost of Goods Sold

(iv) Cost of Sales

(v) Profit earned and Sales for the month

ICAI'S ANSWER

If the company limits the manufacture to just one product, the monthly production will be as follows:

Products	Units	Weights
P	60,000	5
Q	75,000	4
R	1,00,000	3
S	3,00,000	1

Statement showing calculation of weighted Production

Products	Weights	Actual Production (in units)	Weighted Production
P	5	7,000	35,000
Q	4	17,500	70,000
R	3	45,000	1,35,000
S	1	1,60,000	1,60,000
		Total	4,00,000

Factory overhead recovery rate = Factory Overhead / Weighted Production

= ₹ 10,00,000 / ₹ 4,00,000 = ₹ 2.50 per unit

Overhead recovery rate and Factory overhead allocated to each brand will be:

Product	Overhead Recovery rate per unit	Factory overhead allocation in ₹
P	2.50 × 5 = ₹ 12.50	₹ 12.50 × 7,000 = ₹ 87,500
Q	2.50 × 4 = ₹ 10.00	₹ 10 × 17,500 = ₹ 1,75,000
R	2.50 × 3 = ₹ 7.50	₹ 7.50 × 45,000 = ₹ 3,37,500
S	2.50 × 1 = ₹ 2.50	₹ 2.5 x ₹ 1,60,000 = ₹ 4,00,000

Cost Sheet of XYZ Ltd for April 2025

Particulars	P (in ₹)	Q (in ₹)	R (in ₹)	S (in ₹)
Direct Material Cost	3,50,000	4,75,000	5,80,000	9,60,000
Add: Direct Wages	1,05,000	1,40,000	1,70,000	2,65,000
Prime Cost	4,55,000	6,15,000	7,50,000	12,25,000
Add: Factory Overheads	87,500	1,75,000	3,37,500	4,00,000
Works Cost	5,42,500	7,90,000	10,87,500	16,25,000
Office Overheads (Balancing figure)	73,500	1,48,000	1,12,500	71,000
Cost of Goods Sold	6,16,000	9,38,000	12,00,000	16,96,000
Add: Selling Overheads in ratio of 1:2:3:4	56,000	1,12,000	1,68,000	2,24,000
Cost of Sales	6,72,000	10,50,000	13,68,000	19,20,000
Profit @ 25% on cost (1/5 the of sales)	1,68,000	2,26,500	3,42,000	4,80,000
Sales (Sales units X Selling price p.u.)	8,40,000	13,12,500	17,10,000	24,00,000

ICAI · P4 · MTP Jan 2026 · Series II · question p. 14-15 · answer p. 16-17

118 MTP **Sep 2025 S-I** Case Scenario 1 · MCQ 1 to MCQ 5 10 marks

CASE SCENARIO 1

SpeedX Automotive Pvt. Ltd. is a well-known manufacturer of automotive components focusing on producing high-quality fuel-efficient engine parts and transmission systems for both commercial and passenger vehicles. With a strong presence in the Indian market and an expanding footprint in Southeast Asia, SpeedX Automotive has established itself as a reliable supplier in the automotive industry. Headquartered in Pune, Maharashtra, the company is dedicated to meticulous cost management and strategic financial oversight, ensuring operational efficiency and profitability.

SpeedX Automotive Pvt. Ltd. allocates significant investments to direct costs, the lifeblood of their manufacturing process. The company spends ₹ 130,000 on purchasing direct materials, ensuring that the quality of raw materials meets their high standards. Direct labor, a critical component of their production, accounts for ₹ 90,000, reflecting their commitment to a skilled workforce. Additionally,

direct expenses are managed at ₹ 30,000, covering various essential costs that directly impact their manufacturing operations.

Overhead costs are another crucial aspect of SpeedX Automotive's financial strategy. Factory overheads amount to ₹ 60,000, covering all indirect manufacturing expenses. Administration overheads are managed efficiently at ₹ 20,000. Selling and distribution overheads are maintained at ₹ 10,000, ensuring that products reach the market effectively. Quality control, vital for maintaining the company's reputation, is allocated ₹ 5,000, while ₹ 10,000 is dedicated to research and development, highlighting their focus on innovation.

Inventory management is key to SpeedX Automotive's financial health. The opening stock of raw materials stands at ₹ 50,000. The work-in-process inventory starts at ₹ 15,000 and is increased to ₹ 35,000, indicating a smooth production flow. For finished goods, the company maintains an opening stock worth ₹ 30,000, adjusted to a closing stock value of ₹ 25,000, showcasing their agile response to market demands.

Throughout the period, SpeedX Automotive's manufacturing expertise is evident as they successfully produce 4,000 units. Of these, 3,800 units are sold at a price of ₹ 100 per unit.

Additional Information:

During the period, the price of raw materials fluctuated. At the beginning of the period, the price was ₹ 10 per unit. Midway through the period, the price increased to ₹ 12 per unit. The company could only purchase 10,000 units at the lower price of ₹ 10 per unit. Raw material consumed was 14,000 units. Company follows FIFO method for inventory valuation.

Based on the above information, answer the following MCQs (1 to 5):

MCQ 1 2 marks

What is the value of closing stock of raw materials?

A ₹ 35,000

B ₹ 42,000

C ₹ 40,000

D ₹ 40,845

MCQ 2 2 marks

Calculate the cost of production.

A ₹ 325,000

B ₹ 315,000

C ₹ 350,000

D ₹ 370,000

MCQ 3 2 marks

What is the cost of goods sold?

A ₹ 320,000

B ₹ 335,000

C ₹ 325,000

D ₹ 345,000

MCQ 4 2 marks

Determine the cost of sales.

A ₹ 350,000

B ₹ 360,000

C ₹ 370,000

D ₹ 380,000

MCQ 5 2 marks

What is the cost per unit of goods sold if 3,800 units were sold?

A ₹ 91.11

B ₹ 94.74

C ₹ 84.21

D ₹ 97.45

ICAI'S ANSWERS

MCQ 1

Answer key: **(C)**

(c) ₹ 40,000

Calculate units at each price:

Units at ₹ 10 per unit: 10,000 units

Cost for these units: ₹ 1,00,000

Remaining cost for units at ₹ 12 per unit: ₹ 130,000 - ₹ 100,000 = ₹ 30,000

Units at ₹ 12 per unit: $\frac{₹\ 30,000}{₹\ 12}$ = 2,500 units

Total units purchased: 10,000 + 2,500 = 12,500 units

Opening Stock: $\frac{₹\ 50,000}{₹\ 10}$ = 5,000 units

Total units available: 5,000 + 12,500 = 17,500 units

Units consumed in production: 14,000 units

Closing Stock Units: 17,500 - 14,000 = 3,500 units

Closing Stock Value (FIFO) = (2500 units at ₹ 12 + 1,000 units at ₹ 10)

= (2500 x ₹ 12) + (1,000 x ₹ 10) = ₹ 30,000 + ₹ 10,000 = ₹ 40,000

MCQ 2

Answer key: **(B)**

(b) ₹ 315,000

Cost of Production = Prime Cost + Factory Overheads + Quality Control Costs + Research and Development Costs + Opening WIP – Closing WIP

Prime Cost = Direct Materials Consumed + Direct Labor + Direct Expenses

= (₹ 140,000 + ₹ 90,000 + ₹ 30,000) + ₹ 60,000 + ₹ 5,000 + ₹ 10,000 - ₹ 20,000 (WIP adjustment) = ₹ 315,000

MCQ 3

Answer key: **(A)**

(a) ₹ 320,000

Cost of Goods Sold = Cost of Production + Opening Stock of Finished Goods - Closing Stock of Finished Goods

= ₹ 315,000 + ₹ 30,000 - ₹ 25,000 = ₹ 320,000

MCQ 4

Answer key: **(A)**

(a) ₹ 350,000

Cost of Sales = Cost of Goods Sold + Administration Overheads + Selling and Distribution Overheads

= ₹ 320,000 + ₹ 20,000 + ₹ 10,000 = ₹ 350,000

MCQ 5

Answer key: **(C)**

(c) ₹ 84.21

Cost per Unit of Goods Sold = Cost of Goods Sold / Units Sold

= ₹ 320,000 / 3,800 units = ₹ 84.21

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119 MTP **Sep 2025 S-II** Case Scenario 1 · MCQ 1 to MCQ 5 10 marks

CASE SCENARIO 1

ZEN Manufacturing Pvt. Ltd., headquartered in Pune, Maharashtra, is a mid-sized but rapidly growing enterprise specializing in the production of precision-engineered automobile components tailored specifically for the electric vehicle (EV) industry. With India's EV segment witnessing exponential growth driven by sustainability goals and government incentives, ZEN has emerged as a trusted Tier-2 supplier to some of the country's leading electric vehicle manufacturers.

The company operates with a mission to deliver high-quality, durable, and technologically advanced components that meet stringent safety and efficiency standards. Its operations encompass R&D, material procurement, in-house manufacturing, and logistics support — all backed by a strong team of engineers, production planners, and administrative personnel.

With a focus on continuous process improvement and cost competitiveness, ZEN undertakes annual internal cost audits to evaluate operational efficiency, identify cost leakages, and support strategic decision-making. As part of this year's financial review for the period ending 31st March, 2025, the company has appointed you, a newly hired Cost Consultant, to analyze its cost performance, prepare a comprehensive cost sheet, and evaluate the profitability of its operations based on the financial data provided.

Particulars	Amount (₹)
Purchase of raw materials	96,50,000
Consumable materials used	5,25,000
Direct labour wages	72,80,000

Particulars	Amount (₹)
Freight and carriage inward	2,15,000
Wages to floor supervisors and store assistants	9,60,000
Indirect wages to factory staff	1,75,000
R&D expenses related to enhancing production methods	11,20,000
Salaries to accounts and admin staff	8,50,000
Penalty paid for the late payment of interest	80,000
Employer's contribution to EPF & ESI	8,00,000
Electricity and fuel used in production	30,50,000
Production scheduling & planning office expenses	13,50,000
Delivery personnel salaries	15,20,000
Income tax for AY 2024-25	3,10,000
Statutory audit fees	2,20,000
Cost audit fees	95,000
Honorarium to independent board members	10,50,000
Contribution to disaster relief fund	1,40,000
Total Value of Sales	3,12,80,000

Inventory details:

Particulars	As on 01-04-2024 (₹)	As on 31-03-2025 (₹)
Raw Materials	7,10,000	5,40,000
Work-in-Progress	8,50,000	7,25,000
Finished Goods	16,20,000	11,80,000

Based on the above information, as a Cost Consultant you are required to answer the following 5 MCQs:

MCQ 1 2 marks

What is the Prime Cost of ZEN Manufacturing Pvt. Ltd. for the year ended 31st March 2025?

A ₹ 2,16,90,000

B ₹ 2,18,90,000

C ₹ 2,13,10,000

D ₹ 2,09,60,000

MCQ 2 2 marks

What is the Factory Cost of ZEN Manufacturing Pvt. Ltd.?

- A ₹ 2,29,50,000
- B ₹ 2,28,25,000
- C ₹ 2,54,20,000
- D ₹ 2,16,90,000

MCQ 3 2 marks

What is the Cost of Production of ZEN Manufacturing Pvt. Ltd?

- A ₹ 2,28,25,000
- B ₹ 2,29,50,000
- C ₹ 2,54,20,000
- D ₹ 2,13,10,000

MCQ 4 2 marks

What is the Cost of Goods Sold (COGS) for ZEN Manufacturing Pvt. Ltd. in FY 2024-25?

- A ₹ 2,54,20,000
- B ₹ 2,58,60,000
- C ₹ 2,95,95,000
- D ₹ 3,12,80,000

MCQ 5 2 marks

What is the Net Profit for ZEN Manufacturing Pvt. Ltd.?

- A ₹ 22,15,000
- B ₹ 16,85,000
- C ₹ 19,20,000
- D ₹ 2,95,95,000

ICAI'S ANSWERS

MCQ 1

(A) ₹ 2,16,90,000

MCQ 2

(A) ₹ 2,29,50,000

MCQ 3

(C) ₹ 2,54,20,000

MCQ 4

(B) ₹ 2,58,60,000

MCQ 5

(B) ₹ 16,85,000

Working Note:

Sl. No.	Particulars	Amount (₹)	Amount (₹)
(i)	Material Consumed:		
	- Raw materials purchased	96,50,000	
	- Carriage inward	2,15,000	
	Add: Opening stock of raw materials	7,10,000	
	Less: Closing stock of raw materials	(5,40,000)	1,00,35,000
(ii)	Direct employee (labour) cost:		
	- Direct wages	72,80,000	
	- Employer's Contribution towards PF & ESIS	8,00,000	80,80,000
(iii)	Direct expenses:		
	- Consumable materials	5,25,000	
	- Electricity & fuel	30,50,000	35,75,000
	Prime Cost		2,16,90,000
(iv)	Works/ Factory overheads:		
	- Wages to floor supervisors and store assistants	9,60,000	
	- Other indirect wages to factory staffs	1,75,000	11,35,000
	Gross factory cost		2,28,25,000
	Add: Opening value of W-I-P		8,50,000
	Less: Closing value of W-I-P		(7,25,000)
	Factory Cost		2,29,50,000
(v)	Research & development cost paid for improvement in production process		11,20,000
(vi)	Production planning office expenses		13,50,000
	Cost of Production		2,54,20,000
	Add: Opening stock of finished goods		16,20,000
	Less: Closing stock of finished goods		(11,80,000)
	Cost of Goods Sold		2,58,60,000
(vii)	Administrative overheads:		

Sl. No.	Particulars	Amount (₹)	Amount (₹)
	- Salary to accountants	8,50,000	
	- Fees to statutory auditor	2,20,000	
	- Fees to cost auditor	95,000	
	- Fee paid to independent directors	10,50,000	22,15,000
(viii)	Selling overheads & Distribution overheads:		
	- Salary to delivery staffs		15,20,000
	Cost of Sales		2,95,95,000
	Profit (balancing figure)		16,85,000
	Sales		3,12,80,000

Note: Income tax for AY 2024-25, Penalty paid for the late payment of interest and Contribution to disaster relief fund are avoided in cost sheet.

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120 MTP **May 2025 S-I** Q2(a) 8 marks

Electraunika Ltd. manufactures two types of cables- 'USB Type C Cable' and 'USB Lightning Cable'. As the market is currently dominated by 'USB Type C Cable', it is produced 3 times the 'USB Lightning Cable'.

The combined cost data of both the cables for the month of March is given below:

Particulars	(₹)
Direct Materials	2,06,25,000
Direct Wages	1,15,50,000
Production Overhead	66,00,000
Administration overhead (related to production)	33,00,000
Selling Cost	59,40,000

Cost bifurcation between both the cables is ascertained as follows:

(i) Direct material cost per unit of 'USB Lightning Cable' was twice than that of 'USB Type C Cable'.

(ii) Direct wages per unit for 'USB Lightning Cable' were 5/3 times the rate of 'USB Type C Cable'.

(iii) Production overhead, Administration overhead and selling cost per unit were at same rate for both the types of cable.

Other information:

(i) 75,000 units of 'USB Lightning Cable' were produced.

(ii) 90% of both the cables produced were sold.

(iii) Selling Price was ₹ 385 per unit for 'USB Lightning Cable'.

You are required to PREPARE a cost sheet for 'USB Lightning Cable' showing per unit and Total Sales Value.

ICAI'S ANSWER

Preparation of Cost Sheet for 'USB Lightning Cable'

No. of units produced = 75,000 units

No. of units sold = 90% of 75,000 units

= 67,500 units

Particulars	Per unit (₹)	Total (₹)
Direct materials (Working note- (ii))	110.00	82,50,000
Direct wages (Working note- (iii))	55.00	41,25,000
Prime cost	**165.00**	**1,23,75,000**
Production overhead (Working note- (iv))	22.00	16,50,000
Factory Cost	**187.00**	**1,40,25,000**
Administration Overhead (Working note- (v))	11.00	8,25,000
Cost of production	**198.00**	**1,48,50,000**
Less: Closing stock (10% of 75,000 units)	-	(14,85,000)
Cost of goods sold i.e. 67,500 units	**198.00**	**1,33,65,000**
Selling cost (Working note- (vi))	22.00	14,85,000
Cost of sales/ Total cost	**220.00**	**1,48,50,000**
Profit	165.00	1,11,37,500
Sales value (₹ 385 × 67,500 units)	**385.00**	**2,59,87,500**

Working Notes:

(i) No. of units produced for 'USB Type C Cable' is 3 times the 'USB Lightning Cable'

= 3 x 75,000 units

= 2,25,000 units

(ii) Direct material cost per unit of 'USB Type C Cable' = M

Direct material cost per unit of 'USB Lightning Cable' = 2M

Total Direct Material cost = 2M × 75,000 units + M × 2,25,000 units

₹ 2,06,25,000 = 1,50,000 M + 2,25,000 M

$$M = \frac{₹2,06,25,000}{3,75,000}$$

= ₹ 55

Therefore, Direct material Cost per unit of 'USB Lightning Cable'

= 2 × ₹ 55 = ₹ 110

(iii) Direct wages per unit for 'USB Lightning Cable' were 5/3 times the rate of 'USB Type C Cable'.

Direct wages per unit for 'USB Type C Cable' = W

Direct wages per unit for 'USB Lightning Cable' = $\frac{5}{3}$ W

So, ($\frac{5}{3}$ W x 75,000) + (W x 2,25,000) = ₹ 1,15,50,000

Or, W (3,75,000 + 6,75,000) = ₹ 3,46,50,000

Or, W = ₹ 33 per unit

Therefore, Direct material Cost per unit of 'USB Lightning Cable' -

= $\frac{5}{3}$ x ₹ 33 = ₹ 55 per unit

(iv) Production overhead per unit = $\frac{₹66,00,000}{(75,000 + 2,25,000)}$ = ₹ 22

(v) Administration overhead per unit = $\frac{₹33,00,000}{(75,000 + 2,25,000)}$ = ₹ 11

(vi) Selling cost per unit = $\frac{₹59,40,000}{90\% \text{ of } (75,000 + 2,25,000)}$ = ₹ 22

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121 MTP **May 2025 S-II** Q2(a) 6 marks

The following information is made available:

Opening stock of Work in progress – ₹ 60,000

Opening Raw Materials – ₹ 1,20,000

Opening Finished Goods – ₹ 1,13,250

Purchase of Materials – ₹ 7,50,000

Indirect manufacturing costs – 40% of conversion costs

Sales Revenue – ₹ 22,50,000

Direct Labour – ₹ 6,66,750

Prime Costs – ₹ 11,93,250

Gross margin – 30% of revenue

Cost of goods available for sale – ₹ 16,67,325

From the above DETERMINE the following:

a. Closing Raw Materials

b. Closing Work in progress

c. Closing Finished Goods.

ICAI'S ANSWER

Prime cost = 11,93,250

Direct Labour = 6,66,750

Direct Material consumed (Prime cost – Direct Labour) = 5,26,500

Opening Raw Material (RM) = 1,20,000

Purchase of RM = 7,50,000

Closing RM (Op RM + Purchases – Materials Consumed) = 3,43,500

Sales 22,50,000

Less: Gross profit @ 30% 6,75,000

Cost of Sales 15,75,000

Cost of goods available for sales 16,67,325

Hence Closing Finished goods (FG) 92,325 (16,67,325 – 15,75,000)

Opening Finished goods 1,13,250

Cost of Production (Cost of Sales + Closing FG – Opening FG) =15,54,075

Conversion costs = Labour + Production Overhead

Overhead = 40% and hence labour = 60%

Hence total conversion cost = 6,66,750/60% = 11,11,250

Hence POH (Total conversion – Labour cost) = 4,44,500

Works Cost = 16,37,750

Opening Work in progress (WIP) = 60,000

Closing Work in progress (Work Cost + Opening WIP – Closing WIP) = 1,43,675

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122 MTP **May 2025 S-II** Q3(b) 4 marks

EXPLAIN the treatment of given items of Cost in Cost Sheet/Statement

(i) Abnormal costs

(ii) Subsidy/Grant/Incentives

(iii) Penalty, fine, damages, and demurrage

(iv) Interest and other finance costs

ICAI'S ANSWER

Treatment of various items of Cost in Cost Sheet/Statement

(i) **Abnormal costs:** Any abnormal cost, where it is material and quantifiable, shall not form part of cost of production or acquisition or supply of goods or provision of service. Examples of abnormal costs are:

(a) Cost pertaining to or arising out of a pandemic e.g. COVID-19

(b) Cost associated with employees due to sudden lockdown.

(ii) **Subsidy/Grant/Incentives:** Any such type of payment received/ receivable are reduced from the cost objects to which such amount pertains.

(iii) **Penalty, fine, damages, and demurrage:** These types of expenses are not form part of cost.

(iv) **Interest and other finance costs:** Interest, including any payment in the nature of interest for use of non-equity funds and incidental cost that an entity incurs in arranging those funds. Interest and finance charges are not included in cost of production. Interest and Financing Charges shall be presented in the cost statement as a separate item of cost of sales.

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123 MTP **Sep 2024 S-I** Q1(a) 5 marks

From the following data of Meta Ltd., CALCULATE Cost of production:

		Amount (₹)
(i)	Repair & maintenance paid for plant & machinery	9,80,500
(ii)	Insurance premium paid for inventories	26,000
(iii)	Insurance premium paid for plant & machinery	96,000
(iv)	Raw materials purchased	64,00,000
(v)	Opening stock of raw materials	2,88,000
(vi)	Closing stock of raw materials	4,46,000
(vii)	Wages paid	23,20,000
(viii)	Value of opening Work-in-process	4,06,000
(ix)	Value of closing Work-in-process	6,02,100
(x)	Quality control cost for the products in manufacturing process	86,000
(xi)	Research & development cost for improvement in production process	92,600
(xii)	Administrative cost for:	
	- Factory & production	9,00,000
	- Others	11,60,000
(xiii)	Amount realised by selling scrap generated during the manufacturing process	9,200
(xiv)	Packing cost necessary to preserve the goods for further processing	10,200
(xv)	Salary paid to Director (Technical)	8,90,000
(xvi)	Expenses paid for pollution control and engineering & maintenance	22,000

ICAI'S ANSWER

Calculation of Cost of Production of Meta Ltd for the period.....

Particulars	Amount (₹)
Raw materials purchased	64,00,000

Particulars	Amount (₹)
Add: Opening stock	2,88,000
Less: Closing stock	(4,46,000)
Material consumed	62,42,000
Wages paid	23,20,000
Prime cost	85,62,000
Repair and maintenance cost of plant & machinery	9,80,500
Insurance premium paid for inventories	26,000
Insurance premium paid for plant & machinery	96,000
Quality control cost	86,000
Research & development cost	92,600
Administrative overheads related with factory and production	9,00,000
	1,07,43,100
Add: Opening value of W-I-P	4,06,000
Less: Closing value of W-I-P	(6,02,100)
	1,05,47,000
Less: Amount realised by selling scrap	(9,200)
Add: Primary packing cost	10,200
Add: Expenses paid for pollution control and engineering & maintenance	22,000
Cost of Production	1,05,70,000

Notes:

(i) Other administrative overhead does not form part of cost of production.

(ii) Salary paid to Director (Technical) is an administrative cost.

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124 MTP **Sep 2024 S-I** Q2(b) 4 marks

EXPLAIN the treatment of following items in cost sheet.

(i) Credit for Recoveries

(ii) Packing Cost (primary)

(iii) Joint Products and By-Products

(iv) Quality Control Cost

ICAI'S ANSWER

Treatment is as follows:

(i) Credit for Recoveries: The realised or realisable value of scrap or waste is deducted as it reduces the cost of production.

(ii) Packing Cost (primary): Packing material which is essential to hold and preserve the product for its use by the customer is added in the factory cost.

(iii) Joint Products and By-Products: Joint costs are allocated between/among the products on a rational and consistent basis. In case of by-products, the net realisable value of by-products is deducted from the cost of production.

(iv) Quality Control Cost: It is added in the factory cost as this is the cost of resources consumed towards quality control procedures.

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125 MTP **Sep 2024 S-II** Q3(b) 7 marks

ABC Ltd is engaged in producing electronic equipments. It has furnished following details related to its products produced during a month:

	Units	Amount (₹)
Opening stock	10,000	5,00,00,000
Purchases	4,90,000	25,20,00,000
Closing stock	17,500	85,00,000
Works-in-progress		
Opening	20,000	1,20,00,000
Closing	10,000	60,50,000
Direct employees' wages, allowances etc.		5,50,50,000
Primary packaging cost (per unit)		140
R&D expenses & Quality control expenses		1,90,00,000
Guards' salaries		20,00,000
Directors' salaries		60,00,000
Consumable stores, depreciation on plant related to factory overhead		3,42,00,000
Product inspection (before primary packaging)		22,00,000
Rearrangement design of factory machine		75,00,000
Administrative overheads related to production		3,45,00,000
Selling expenses		3,94,50,000

	Units	Amount (₹)
Royalty paid for production		3,10,50,000
Cost of web-site (for online sale) maintenance		60,75,000
Gifts & Snacks		30,50,000
GST (credit allowed)		5,50,00,000
AMC cost of CCTV		10,00,000
Hiring of cars for the transportation of employees and guests		25,00,000
Audit and Legal Fees		29,00,000
Secondary packaging cost (per unit)		20

Distribution of the following costs:
Guard's salaries to Factory, Office and Distribution in the ratio 7: 2:1.
Hiring of cars is only for selling and distribution
AMC of CCTV to Factory, Office and Selling in the ratio 6 : 2 : 2.

The company paid EPF of 12% over above basic pay. However, Guards will not receive any incentive or EPF.

It has lucky draws every month giving the first prize of ₹ 1,00,000; 2nd prize of ₹ 50,000, 3rd prize of ₹ 20,000 and three consolation prizes of ₹ 10,000 each to customers buying the product.

It also sponsors a television programme every week at a cost of ₹ 20,00,000 per month.

The hiring of cars attracts GST under RCM @5% without credit.

There was a normal scrap of 2,000 units of direct material which realized ₹ 350 per unit. The entire finished product was sold at a profit margin of 25% on sales.

You are required to PREPARE a cost sheet

ICAI'S ANSWER

Cost Sheet

Particulars	Units	Amount (₹)
Material		
Opening stock	10,000	5,00,00,000
Add: Purchases	4,90,000	25,20,00,000
Less: Closing stock	(17,500)	(85,00,000)
	4,82,500	**29,35,00,000**
Less: Normal wastage of materials realized @ ₹ 350 per unit	(2,000)	(7,00,000)

Particulars	Units	Amount (₹)
Material consumed		29,28,00,000
Direct employee's wages and allowances		5,50,50,000
Direct expenses- Royalty paid for production		3,10,50,000
Prime cost	4,80,500	**37,89,00,000**
Factory overheads - Consumable stores, depreciation etc.		3,42,00,000
Rearrangement design of factory machine		75,00,000
Gross Works Cost	4,80,500	**38,64,00,000**
Add: Opening WIP	20,000	1,20,00,000
Less: Closing WIP	(10,000)	(60,50,000)
Factory/Works Cost	4,90,500	39,23,50,000
Administration Overheads related to production		3,45,00,000
R&D expenses and Quality control cost		1,90,00,000
AMC cost of CCTV installed at factory premises		6,00,000
Guard Salaries for factory premises		14,00,000
Product Inspection		22,00,000
Add: Primary packaging cost @ ₹ 140 per unit		6,86,70,000
Cost of production	4,90,500	**51,87,20,000**
Administration Overheads		
Guard salaries for office		4,00,000
Audit and legal fees		29,00,000
Director's Salaries		60,00,000
EPF Director's Salaries @12%		7,20,000
AMC cost for CCTV installed at office.		2,00,000
Selling and Distribution Overheads		
Cost of maintaining website for online sale		60,75,000
Secondary packaging cost @ ₹ 20 per unit	4,90,500	98,10,000
Gift and snacks		30,50,000

Particulars	Units	Amount (₹)
Guard salaries for selling department		2,00,000
AMC cost for CCTV installed at selling department		2,00,000
Hiring charges of cars		25,00,000
Add: GST @5% on RCM basis		1,25,000
Television programme sponsorship cost		20,00,000
Customers' prize cost*		2,00,000
Selling expenses		3,94,50,000
Cost of sales		**58,64,75,000**
Add: Profit @ 25% on sales or 33.333% of cost		19,54,89,712
Sales value		**78,19,64,712**

***Customers' prize cost:**

	Amount (₹)
1st Prize	1,00,000
2nd Prize	50,000
3rd Prize	20,000
Consolation Prizes (3 × ₹10,000)	30,000
Total	**2,00,000**

***Customers' prize cost:**

	Amount (₹)
1st Prize	1,00,000
2nd Prize	50,000
3rd Prize	20,000
Consolation Prizes (3 × ₹10,000)	30,000
Total	**2,00,000**

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126 MTP May 2024 S-I Case scenario · MCQ 2(i) to MCQ 2(v) 10 marks

CASE SCENARIO

M Ltd. is producing a single product and may expand into product diversification in next one to two years. M Ltd. is amongst a labour-intensive company where majority of processes are done manually. Employee cost is a major cost element in the total cost of the company. The company conventionally uses performance parameters Earnings per manshift (EMS) to measure cost paid to an employee for a shift of 8 hours, and Output per manshift (OMS) to measure an employee's output in a shift of 8 hours.

The Chief Manager (Finance) of the company has emailed you few information related to the last month. The email contains the following data related to the last month:

During the last month, the company has produced 2,34,000 tonnes of output.

Expenditures for the last months are:

(i) Raw materials consumed ₹ 50,00,000

(ii) Power consumed 13,000 Kwh @ ₹ 8 per Kwh to run the machines for production.

(iii) Diesels consumed 2,000 litres @ ₹ 93 per litre to run power generator used as alternative or backup for power cuts.

(iv) Wages & salary paid – ₹ 6,40,00,000

(v) Gratuity & leave encashment paid – ₹ 64,20,000

(vi) Hiring charges paid for HEMM- ₹ 30,00,000. HEMM are directly used in production.

(vii) Hiring charges paid for cars used for official purpose – ₹ 66,000

(viii) Reimbursement of diesel cost for the cars – ₹ 22,000

(ix) The hiring of cars attracts GST under RCM @5% without credit.

(x) Maintenance cost paid for weighing bridge (used for weighing of final goods at the time of dispatch) – ₹ 12,000

(xi) AMC cost of CCTV installed at weighing bridge (used for weighing of final goods at the time of dispatch) and factory premises is ₹ 8,000 and ₹ 18,000 per month respectively.

(xii) TA/ DA and hotel bill paid for sales manager- ₹ 36,000

(xiii) The company has 1,800 employees works for 26 days in a month.

You are asked to calculate the followings:

MCQ 2(i) 2 marks

What is the amount of prime cost incurred during the last month:

A ₹ 7,54,20,000

B ₹ 7,57,10,000

C ₹ 7,56,06,000

D ₹ 7,87,10,000

MCQ 2(ii) 2 marks

What is the total and per shift cost of production for last month:

A ₹ 7,87,10,000 and ₹ 336.37 respectively

B ₹ 7,87,10,000 and ₹ 1,681.84 respectively

C ₹ 7,87,28,000 and ₹ 1,682.22 respectively

D ₹ 7,87,28,000 and ₹ 336.44 respectively

MCQ 2(iii) 2 marks

What is the value of administrative cost incurred during the last month:

A ₹ 92,400

B ₹ 88,000

C ₹ 1,48,400

D ₹ 1,44,000

MCQ 2(iv) 2 marks

What is the value of selling and distribution cost and total cost of sales:

A ₹ 36,000 & ₹ 7,88,76,400 respectively

B ₹ 56,000 & ₹ 7,88,76,400 respectively

C ₹ 36,000 & ₹ 7,88,72,000 respectively

D ₹ 56,000 & ₹ 7,88,72,000 respectively

MCQ 2(v) 2 marks

What is the value EMS and OMS for the last month:

A ₹ 1,504.70 & 5 tonnes respectively

B ₹ 1,367.52 & 5 tonnes respectively

C ₹ 1,504.70 & 4.37 tonnes respectively

D ₹ 1,367.52 & 4.37 tonnes respectively

ICAI'S ANSWERS

MCQ 2(i)

Answer key: **(D)**

D

MCQ 2(ii)

Answer key: **(C)**

C Please refer cost sheet below for cost of production

Cost of production per manshift =

Cost of production ÷ Total manshift

₹ 7,87,28,000 ÷ 46,800 = ₹1,682.22

MCQ 2(iii)

Answer key: **(A)**

A Car hire charges including GST @5%, please refer the cost sheet

MCQ 2(iv)

Answer key: **(B)**

B Selling and distribution cost includes the following:

Maintenance cost for weighing bridge	12,000
AMC cost of CCTV installed at weigh bridge	8,000
TA/ DA & hotel bill of sales manager	36,000
	56,000

For Cost of Sale please refer the cost sheet

MCQ 2(v)

Answer key: **(A)**

A Manshift = 1,800 employees × 26 days = 46,800 manshifts

Computation of earnings per manshift (EMS):

$$\text{EMS} = \frac{\text{Total employee benefits paid}}{\text{Manshift}}$$

$$= \frac{₹7,04,20,000}{46,800} = ₹\ 1504.70$$

Computation of Output per manshift (OMS):

$$\text{OMS} = \frac{\text{Total Output/ Production}}{\text{Manshift}}$$

$$= \frac{2,34,000\ \text{Tonne}}{46,800} = 5\ \text{tonnes}$$

Workings

Cost Sheet of M Ltd. for the last month

Particulars	Amount (₹)	Amount (₹)
Materials consumed		50,00,000
Wages & Salary	6,40,00,000	
Gratuity & leave encashment	64,20,000	7,04,20,000
Power cost (13,000 kwh × ₹8)	1,04,000	
Diesel cost (2,000 ltr × ₹93)	1,86,000	2,90,000
HEMM hiring charges		30,00,000
Prime Cost		**7,87,10,000**
AMC cost of CCTV installed at factory premises		18,000
Cost of Production/ Cost of Goods Sold		**7,87,28,000**
Hiring charges of cars	66,000	
Reimbursement of diesel cost	22,000	
	88,000	

Particulars	Amount (₹)	Amount (₹)
Add: GST @5% on RCM basis	4,400	92,400
Maintenance cost for weighing bridge	12,000	
AMC cost of CCTV installed at weigh bridge	8,000	20,000
TA/ DA & hotel bill of sales manager		36,000
Cost of Sales		**7,88,76,400**

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Same question, different ICAI answer: #112 MTP Sep 2026 S-I

What would be Prime cost from below information?

Direct materials Purchased : ₹ 75,000

Direct labour : ₹ 45,000

Direct expenses : ₹ 15,000

Manufacturing overheads : ₹ 22,500

Direct materials consumed : ₹ 67,500

A ₹ 1,35,000

B ₹ 1,27,500

C ₹ 1,57,500

D ₹ 1,50,000

ICAI'S ANSWER

Answer key: **(B)**

B

Direct labour : ₹ 45,000

Direct expenses : ₹ 15,000

Direct materials consumed : ₹ 67,500

Prime Cost ₹ 1,27,500

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Following information relate to a manufacturing concern for the year ended 31st March, 2023:

	(₹)
Raw Material (opening)	2,28,000
Raw Material (closing)	3,05,000

	(₹)
Purchases of Raw Material	43,50,000
Freight Inwards	1,20,000
Direct wages paid	12,56,000
Direct wages-outstanding at the end of the year	1,50,000
Factory Overheads	20% of prime cost
Work-in-progress (opening)	1,92,500
Work-in-progress (closing)	1,40,700
Administrative Overheads (related to production)	1,73,000
Distribution Expenses	₹ 16 per unit
Finished Stock (opening)- 1,320 Units	6,08,500
Sale of scrap of material	7,000

The firm produced 14,350 units of output during the year. The stock of finished goods at the end of the year is valued at cost of production. The firm sold 14,903 units at a price of ₹579 per unit during the year.

PREPARE cost sheet of the firm.

ICAI'S ANSWER

Cost sheet for the year ended 31st March, 2023.

Units produced - 14,000 units

Units sold - 14,153 units

Particulars	Amount (₹)
Raw materials purchased	43,50,000
Add: Freight Inward	1,20,000
Add: Opening value of raw materials	2,28,000
Less: Closing value of raw materials	(3,05,000)
	43,93,000
Less: Sale of scrap of material	(7,000)
Materials consumed	43,86,000
Direct Wages (12,56,000 + 1,50,000)	14,06,000
Prime Cost	57,92,000
Factory overheads (20% of Prime Cost)	11,58,400

Particulars	Amount (₹)
Add: Opening value of W-I-P	1,92,500
Less: Closing value of W-I-P	(1,40,700)
Factory Cost	70,02,200
Add: Administrative overheads	1,73,000
Cost of Production	71,75,200
Add: Value of opening finished stock	6,08,500
Less: Value of closing finished stock [₹ 500(71,75,200/14,350) × 767] (1,320 + 14,350 – 14,903 = 767 units)	(3,83,500)
Cost of Goods Sold	74,00,200
Distribution expenses (₹16 × 14,903 units)	2,38,448
Cost of Sales	76,38,648
Profit (Balancing figure)	9,90,189
Sales (₹ 579 × 14,903 units)	86,28,837

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129 MTP **Old P3 Nov 2023 S-I** Q4(b) 10 marks old scheme

G Ltd. has the following expenditures for the year ended 31st March, 2023:

Sl. No.		Amount (₹)	Amount (₹)
(i)	Raw materials purchased		20,00,00,000
(ii)	Freight inward		22,41,200
(iii)	Wages paid to factory workers		58,40,000
(iv)	Royalty paid for production		3,45,200
(v)	Amount paid for power & fuel		9,24,000
(vi)	Job charges paid to job workers		16,24,000
(vii)	Stores and spares consumed		2,24,000
(viii)	Depreciation on office building		1,12,000
(ix)	Repairs & Maintenance paid for:	96,000	
	- Plant & Machinery		
	- Sales office building	36,000	1,32,000

Sl. No.		Amount (₹)	Amount (₹)
(x)	Insurance premium paid for:		
	- Plant & Machinery	62,400	
	- Factory building	36,200	98,600
(xi)	Expenses paid for quality control check activities		39,200
(xii)	Research & development cost paid improvement in production process		36,400
(xiii)	Expenses paid for pollution control and engineering & maintenance		53,200
(xiv)	Salary paid to Sales & Marketing mangers:		20,24,000
(xv)	Salary paid to General Manager		25,12,000
(xvi)	Packing cost paid for:		
	- Primary packing necessary to maintain quality	1,92,000	
	- For re-distribution of finished goods	2,24,000	4,16,000
(xvii)	Performance bonus paid to sales staffs		7,20,000
(xviii)	Value of stock as on 1st April, 2022:		
	- Raw materials	36,00,000	
	- Work-in-process	18,40,000	
	- Finished goods	22,00,000	76,40,000
(xix)	Value of stock as on 31st March, 2023:		
	- Raw materials	19,20,000	
	- Work-in-process	17,40,000	
	- Finished goods	36,40,000	73,00,000

Amount realized by selling of scrap and waste generated during manufacturing process – ₹1,72,000/-

From the above data you are requested to PREPARE Statement of cost for G Ltd. for the year ended 31st March, 2023, showing (i) Prime cost, (ii) Factory cost, (iii) Cost of Production, (iv) Cost of goods sold and (v) Cost of sales.

ICAI'S ANSWER

Statement of Cost of G Ltd. for the year ended 31st March, 2023:

Sl. No.	Particulars	Amount (₹)	Amount (₹)
(i)	Material Consumed:		

Sl. No.	Particulars	Amount (₹)	Amount (₹)
	- Raw materials purchased	20,00,00,000	
	- Freight inward	22,41,200	
	Add: Opening stock of raw materials	36,00,000	
	Less: Closing stock of raw materials	(19,20,000)	20,39,21,200
(ii)	Direct employee (labour) cost:		
	- Wages paid to factory workers		58,40,000
(iii)	Direct expenses:		
	- Royalty paid for production	3,45,200	
	- Amount paid for power & fuel	9,24,000	
	- Job charges paid to job workers	16,24,000	28,93,200
	Prime Cost		21,26,54,400
(iv)	Works/ Factory overheads:		
	- Stores and spares consumed	2,24,000	
	- Repairs & Maintenance paid for plant & machinery	96,000	
	- Insurance premium paid for plant & machinery	62,400	
	- Insurance premium paid for factory building	36,200	
	- Expenses paid for pollution control and engineering & maintenance	53,200	4,71,800
	Gross factory cost		21,31,26,200
	Add: Opening value of W-I-P		18,40,000
	Less: Closing value of W-I-P		(17,40,000)
	Factory Cost		21,32,26,200
(v)	Quality control cost:		
	- Expenses paid for quality control check activities		39,200
(vi)	Research & development cost paid improvement in production process		36,400
(vii)	Less: Realisable value on sale of scrap and waste		(1,72,000)
(viii)	Add: Primary packing cost		1,92,000

Sl. No.	Particulars	Amount (₹)	Amount (₹)
	Cost of Production		21,33,21,800
	Add: Opening stock of finished goods		22,00,000
	Less: Closing stock of finished goods		(36,40,000)
	Cost of Goods Sold		21,18,81,800
(ix)	Administrative overheads:		
	- Depreciation on office building	1,12,000	
	- Salary paid to General Manager	25,12,000	26,24,000
(x)	Selling overheads:		
	- Repairs & Maintenance paid for sales office building	36,000	
	- Salary paid to Manager- Sales & Marketing	20,24,000	
	- Performance bonus paid to sales staffs	3,60,000	24,20,000
(xi)	Distribution overheads:		
	- Packing cost paid for re-distribution of finished goods		2,24,000
	Cost of Sales		21,71,49,800

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Following figures has been extracted from the books of M/s A&R Brothers:

	Amount (₹)
Stock on 1st March, 2023	
- Raw materials	6,06,000
- Finished goods	3,59,000
Stock on 31st March, 2023	
- Raw materials	7,50,000
- Finished goods	3,09,000
Work-in-process:	
- On 1st March, 2023	12,56,000
- On 31st March, 2023	14,22,000

	Amount (₹)
Purchase of raw materials	28,57,000
Sale of finished goods	1,34,00,000
Direct wages	37,50,000
Factory expenses	21,25,000
Office and administration expenses	10,34,000
Selling and distribution expenses	7,50,000
Sale of scrap	26,000

You are required to compute:

(i) Value of material consumed

(ii) Prime cost

(iii) Cost of production

(iv) Cost of goods sold

(v) Cost of sales

(vi) Profit/ loss

ICAI'S ANSWER

Cost Sheet of M/s A&R Brothers for the month ended March 2023:

	Particulars	Amount (₹)	Amount (₹)
(i)	**Materials consumed:**		
	- Opening stock	6,06,000	
	- Add: Purchases	28,57,000	
		34,63,000	
	- Less: Closing stock	(7,50,000)	**27,13,000**
	Direct wages		37,50,000
(ii)	**Prime cost**		**64,63,000**
	Factory expenses		21,25,000
			85,88,000
	Add: Opening W-I-P		12,56,000
	Less: Closing W-I-P		(14,22,000)
	Factory cost		84,22,000

	Particulars	Amount (₹)	Amount (₹)
	Less: Sale of scrap		(26,000)
(iii)	**Cost of Production**		**83,96,000**
	Add: Opening stock of finished goods		3,59,000
	Less: Closing stock of finished goods		(3,09,000)
(iv)	**Cost of Goods Sold**		**84,46,000**
	Office and administration expenses		10,34,000
	Selling and distribution expenses		7,50,000
(v)	**Cost of Sales**		**1,02,30,000**
(vi)	**Profit** (balancing figure)		**31,70,000**
	Sales		1,34,00,000

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Following information obtained from the records of a Manufacturing Company for the month of March:

Direct labour cost ₹ 25,000 being 150% of works overheads.

Cost of goods sold excluding administrative expenses ₹ 75,000.

Inventory accounts showed the following opening and closing balances:

	March 1 (₹)	March 31 (₹)
Raw materials	11,600	15,370
Work-in-progress	15,225	21,025
Finished goods	25,520	27,550

Other information is as follows:	
	(₹)
Selling expenses	6,125
General and administration expenses	4,375
Sales for the month	1,05,250

Required to:

(i) FIND out the value of materials purchased.

(ii) PREPARE a cost statement showing the various elements of cost and also the profit earned.

ICAI'S ANSWER

(i) Computation of the value of materials purchased

To find out the value of materials purchased, reverse calculations from the given data can be presented as below:

Particulars	(₹)
Cost of goods sold	**75,000**
Add: Closing stock of finished goods	27,550
Less: Opening stock of finished goods	(25,520)
Cost of production	**77,030**
Add: Closing stock of work-in-progress	21,025
Less: Opening stock of work-in-progress	(15,225)
Works cost	**82,830**
Less: Factory overheads: [₹25,000×100/150]	(16,667)
Prime cost	**66,163**
Less: Direct labour	(25,000)
Raw material consumed	**41,163**
Add: Closing stock of raw materials	15,370
Raw materials available	56,533
Less: Opening stock of raw materials	(11,600)
Value of materials purchased	**44,933**

(ii) Cost statement

	(₹)
Raw material consumed [Refer to statement (i) above]	41,163
Add: Direct labour cost	25,000
Prime cost	**66,163**
Add: Factory overheads	16,667
Works cost	**82,830**
Add: Opening work-in-progress	15,225
Less: Closing work-in-progress	(21,025)

	(₹)
Cost of production	**77,030**
Add: Opening stock of finished goods	25,520
Less: Closing stock of finished goods	(27,550)
Cost of goods sold	**75,000**
Add: General and administration expenses	4,375
Add: Selling expenses	6,125
Cost of sales	**85,500**
Profit (sales i.e ₹1,05,250 – Cost of sales i.e ₹ 85,500)	19,750
Sales	**1,05,250**

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132 MTP **Old P3 May 2023 S-II** Q2(b) 10 marks old scheme

K Ltd. has the following expenditures for the year ended 31st March, 2023:

Sl. No.		Amount (₹)	Amount (₹)
(i)	Raw materials purchased		10,00,00,000
(ii)	Freight inward		11,20,600
(iii)	Wages paid to factory workers		29,20,000
(iv)	Royalty paid for production		1,72,600
(v)	Amount paid for power & fuel (Directly related to production)		4,62,000
(vi)	Job charges paid to job workers		8,12,000
(vii)	Stores and spares consumed		1,12,000
(viii)	Depreciation on office building		56,000
(ix)	Repairs & Maintenance paid for:		
	- Plant & Machinery	48,000	
	- Sales office building	18,000	66,000
(x)	Insurance premium paid for:		
	- Plant & Machinery	31,200	
	- Factory building	18,100	49,300

Sl. No.		Amount (₹)	Amount (₹)
(xi)	Expenses paid for quality control check activities		19,600
(xii)	Research & development cost paid improvement in production process		18,200
(xiii)	Expenses paid for pollution control and engineering & maintenance		26,600
(xiv)	Salary paid to Sales & Marketing mangers:		10,12,000
(xv)	Salary paid to General Manager		12,56,000
(xvi)	Packing cost paid for:		
	- Primary packing necessary to maintain quality	96,000	
	- For re-distribution of finished goods	1,12,000	2,08,000
(xvii)	Fee paid to independent directors		2,20,000
(xviii)	Performance bonus paid to sales staffs		1,80,000
(xix)	Value of stock as on 1st April, 2022:		
	- Raw materials	18,00,000	
	- Work-in-process	9,20,000	
	- Finished goods	11,00,000	38,20,000
(xx)	Value of stock as on 31st March, 2023:		
	- Raw materials	9,60,000	
	- Work-in-process	8,70,000	
	- Finished goods	18,20,000	36,50,000

Amount realized by selling of scrap and waste generated during manufacturing process – ₹86,000/-

From the above data you are requested to PREPARE Statement of cost for K Ltd. for the year ended 31st March, 2023, showing (i) Prime cost, (ii) Factory cost, (iii) Cost of Production, (iv) Cost of goods sold and (v) Cost of sales.

ICAI'S ANSWER

Statement of Cost of K Ltd. for the year ended 31st March, 2023:

Sl. No.	Particulars	Amount (₹)	Amount (₹)
(i)	Material Consumed:		
	- Raw materials purchased	10,00,00,000	
	- Freight inward	11,20,600	

Sl. No.	Particulars	Amount (₹)	Amount (₹)
	Add: Opening stock of raw materials	18,00,000	
	Less: Closing stock of raw materials	(9,60,000)	10,19,60,600
(ii)	Direct employee (labour) cost:		
	- Wages paid to factory workers		29,20,000
(iii)	Direct expenses:		
	- Royalty paid for production	1,72,600	
	- Amount paid for power & fuel	4,62,000	
	- Job charges paid to job workers	8,12,000	14,46,600
	Prime Cost		10,63,27,200
(iv)	Works/ Factory overheads:		
	- Stores and spares consumed	1,12,000	
	- Repairs & Maintenance paid for plant & machinery	48,000	
	- Insurance premium paid for plant & machinery	31,200	
	- Insurance premium paid for factory building	18,100	
	- Expenses paid for pollution control and engineering & maintenance	26,600	2,35,900
	Gross factory cost		10,65,63,100
	Add: Opening value of W-I-P		9,20,000
	Less: Closing value of W-I-P		(8,70,000)
	Factory Cost		10,66,13,100
(v)	Quality control cost:		
	- Expenses paid for quality control check activities		19,600
(vi)	Research & development cost paid improvement in production process		18,200
(vii)	Less: Realisable value on sale of scrap and waste		(86,000)
(viii)	Add: Primary packing cost		96,000
	Cost of Production		10,66,60,900
	Add: Opening stock of finished goods		11,00,000

Sl. No.	Particulars	Amount (₹)	Amount (₹)
	Less: Closing stock of finished goods		(18,20,000)
	Cost of Goods Sold		10,59,40,900
(ix)	Administrative overheads:		
	- Depreciation on office building	56,000	
	- Salary paid to General Manager	12,56,000	
	- Fee paid to independent directors	2,20,000	15,32,000
(x)	Selling overheads:		
	- Repairs & Maintenance paid for sales office building	18,000	
	- Salary paid to Manager- Sales & Marketing	10,12,000	
	- Performance bonus paid to sales staffs	1,80,000	12,10,000
(xi)	Distribution overheads:		
	- Packing cost paid for re-distribution of finished goods		1,12,000
	Cost of Sales		10,87,94,900

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The following information pertains to A Limited for the year 1st April 2021 to 31st March 2022:

Particulars	Amount (₹)
Sales	50,00,000
Direct labour	10,50,000
Administrative overheads (relating to production activity)	1,50,000
Selling expenses	2,50,000

Inventory details are as follows:

	As on 1st April 2021 (Amount in ₹)	As on 31st March 2022 (Amount in ₹)
Raw materials	5,00,000	6,30,000
Finished goods	9,80,000	10,50,000
Work in Progress	6,00,000	8,00,000

Additional Information:

- Direct labour would be 175% of works overheads.

- Cost of goods sold would be ₹ 6,900 per unit
- Selling expenses would be ₹ 500 per unit.

You are required to PREPARE a cost sheet for the year ended 31st March, 2022 showing:

(i) Value of material purchased

(ii) Prime cost

(iii) Works cost

(iv) Cost of production

(v) Cost of goods sold

(vi) Cost of Sales

(vii) Profit earned

(viii) Profit as a percentage of sales

ICAI'S ANSWER

Cost Sheet of A Limited for the year ended 31st March 2022

Particulars	Amount (₹)	Amount (₹)
Opening Stock of Raw materials	5,00,000	
Add: **Purchases** (balancing figure)	**20,50,000**	
Less: Closing stock of raw materials	6,30,000	
Direct material consumed (balancing figure)		19,20,000
Direct labour		10,50,000
Prime Cost		**29,70,000**
Add: Factory Overheads (10,50,000 / 175%)		**6,00,000**
Add: Opening Stock of Work in Progress		6,00,000
		41,70,000
Less: Closing Stock of Work in Progress		8,00,000
Works Cost		**33,70,000**
Add: Administrative Overheads (relating to production activity)		1,50,000
COST OF PRODUCTION		**35,20,000**
Add: Opening stock of finished goods		9,80,000
Cost of Goods available for sale		45,00,000
Less: Closing Stock of finished goods		10,50,000
COST OF GOODS SOLD (Working Note: (iv))		**34,50,000**

Particulars	Amount (₹)	Amount (₹)
Add: Selling and Distribution Overhead		2,50,000
COST OF SALES		**37,00,000**
Add: **Profit** (Balancing figure) [Sales - Cost of Sales]		13,00,000
SALES		**50,00,000**

Profit as a % of sales = $\frac{13 \text{ Lakhs}}{50 \text{ Lakhs}}$ x 100 = 26%

Working Notes:

(i) The cost sheet is completed by Reverse Working. Purchases amount is the balancing figure.

(ii) Direct labour = 175% of factory overhead (given). Hence, if direct labour = 10,50,000, then Factory Overhead = 10,50,000 / 175% = **₹ 6,00,000**

(iii) Selling Overhead ₹ 2,50,000 (total), selling per unit ₹ 500.

Number of units sold = ₹ 2,50,000/ ₹ 500 = 500 units

(iv) Cost of goods sold = 500 units x ₹ 6,900 = ₹ 34,50,000

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The following data relates to manufacturing of a standard product during the month of February, 2022:

Particulars	Amount (in ₹)
Stock of Raw material as on 01-02-2022	1,20,000
Work in Progress as on 01-02-2022	75,000
Purchase of Raw material	3,00,000
Carriage Inwards	30,000
Direct Wages	1,80,000
Cost of special drawing	45,000
Hire charges paid for Plant (Direct)	36,000
Return of Raw Material	60,000
Carriage on return	9,000
Expenses for participation in Industrial exhibition	12,000
Maintenance of office building	3,000
Salary to office staff	37,500
Legal charges	3,750

Particulars	Amount (in ₹)
Depreciation on Delivery van	9,000
Warehousing charges	2,250
Stock of Raw material as on 28-02-2022	45,000
Stock of Work in Progress as on 28-02-2022	36,000

- Store overheads on materials are 10% of material consumed.
- Factory overheads are 20% of the Prime cost.
- 10% of the output was rejected and a sum of ₹ 7,500 was realized on sale of scrap.
- 10% of the finished product was found to be defective and the defective products were rectified at an additional expenditure which is equivalent to 20% of proportionate direct wages.
- The total output was 8,000 units during the month.

You are required to PREPARE a Cost Sheet for the above period showing the:

(i) Cost of Raw Material consumed.

(ii) Prime Cost

(iii) Work Cost

(iv) Cost of Production

(v) Cost of Sales

ICAI'S ANSWER

Statement of Cost for the month of February, 2022

Particulars	Amount (₹)	Amount (₹)
(i) Cost of Material Consumed:		
Raw materials purchased (₹ 3,00,000 – ₹ 60,000)	2,40,000	
Carriage inwards	30,000	
Add: Opening stock of raw materials	1,20,000	
Less: Closing stock of raw materials	(45,000)	3,45,000
Direct Wages		1,80,000
Direct expenses:		
Cost of special drawing	45,000	
Hire charges paid for Plant (Direct)	36,000	81,000
(ii) Prime Cost		**6,06,000**
Carriage on return	9,000	

Particulars	Amount (₹)	Amount (₹)
Store overheads (10% of material consumed)	34,500	
Factory overheads (20% of Prime cost)	1,21,200	
Additional expenditure for rectification of defective products (refer working note)	3,240	1,67,940
Gross factory cost		7,73,940
Add: Opening value of W-I-P		75,000
Less: Closing value of W-I-P		(36,000)
(iii) Works/ Factory Cost		**8,12,940**
Less: Realisable value on sale of scrap		(7,500)
(iv) Cost of Production		**8,05,440**
Add: Opening stock of finished goods		-
Less: Closing stock of finished goods		-
Cost of Goods Sold		**8,05,440**
Administrative overheads:		
Maintenance of office building	3,000	
Salary paid to Office staff	37,500	
Legal Charges	3,750	44,250
Selling overheads:		
Expenses for participation in Industrial exhibition	12,000	12,000
Distribution overheads:		
Depreciation on delivery van	9,000	
Warehousing charges	2,250	11,250
(v) Cost of Sales		**8,72,940**

Working Notes:

1. Number of Rectified units

Total Output	8,000 units
Less: Rejected 10%	800 units
Finished product	7,200 units
Rectified units (10% of finished product)	**720 units**

2. Proportionate additional expenditure on 720 units

= 20% of proportionate direct wages

= 0.20 x (₹ 1,80,000/8,000) x 720

= ₹ 3,240

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Comput Ltd. has capacity to produce 1,00,000 units of a product every month. Its fixed general administration expenses amount to ₹ 7,50,000 and fixed marketing expenses amount to ₹ 12,50,000 per month respectively. The variable distribution cost amounts to ₹ 150 per unit.

Its works cost at varying levels of production is as under:

Level	Works cost per unit (₹)
10%	2,000
20%	1,950
30%	1,900
40%	1,850
50%	1,800
60%	1,750
70%	1,700
80%	1,650
90%	1,600
100%	1,550

It can sell 100% of its output at ₹ 2,500 per unit provided it incurs the following additional expenditure:

(i) it spends ₹ 5,00,000 on refreshments served every month to its customers;

(ii) it gives gift items costing ₹ 150 per unit of sale;

(iii) it sponsors a television programme every week at a cost of ₹ 1,00,00,000 per month.

(iv) it has lucky draws every month giving the first prize of ₹ 2,50,000; 2nd prize of ₹ 1,25,000, 3rd prize of ₹ 50,000 and three consolation prizes of ₹ 25,000 each to customers buying the product.

However, it can market 30% of its output at ₹ 2,750 per unit without incurring any of the expenses referred to in (i) to (iv) above.

PREPARE a cost sheet for the month showing total cost and profit at 30% and 100% capacity level.

ICAI'S ANSWER

Cost Sheet (For the month)

Level of Capacity	30%		100%	
	30,000 units		**1,00,000 units**	
	Per unit (₹)	**Total (₹)**	**Per unit (₹)**	**Total (₹)**
Works Cost	**1,900.00**	**5,70,00,000**	**1,550.00**	**15,50,00,000**
Add: Fixed general administration expenses	25.00	7,50,000	7.50	7,50,000
Add: Fixed marketing expenses	41.67	12,50,000	12.50	12,50,000
Add: Variable distribution cost	150.00	45,00,000	150.00	1,50,00,000
Add: Special Costs:				
- Refreshments	-	-	5.00	5,00,000
- Gift items costs	-	-	150.00	1,50,00,000
- Television programme sponsorship cost	-	-	100.00	1,00,00,000
- Customers' prizes*	-	-	5.00	5,00,000
Cost of sales	**2,116.67**	**6,35,00,000**	**1,980.00**	**19,80,00,000**
Profit (Balancing figure)	633.33	1,90,00,000	520.00	5,20,00,000
Sales revenue	**2,750.00**	**8,25,00,000**	**2,500.00**	**25,00,00,000**

***Customers' prize cost:**

	Amount (₹)
1st Prize	2,50,000
2nd Prize	1,25,000
3rd Prize	50,000
Consolation Prizes (3 × ₹ 25,000)	75,000
Total	**5,00,000**

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Xim Ltd. manufactures two types of boxes 'Super' and 'Normal'. The cost data for the year ended 31st March, 2021 is as follows:

	(₹)
Direct Materials	12,00,000
Direct Wages	6,72,000
Production Overhead	2,88,000
Total	21,60,000

There was no work-in-progress at the beginning or at the end of year. It is further ascertained that:

1. Direct materials cost per unit in 'Super' was twice as much of direct material in 'Normal'.
2. 2% cash discount was received for payment made within 30 days to the creditors of Direct materials.
3. Direct wages per unit for 'Normal' were 60% of those of 'Super'.
4. Production overhead per unit was at same rate for both the types of boxes.
5. Administration overhead was 200% of direct labour for each type.
6. Selling cost was ₹ 1 per 'Super' type.
7. Production and sales during the year were as follows:

Production		Sales	
Type	**No. of units**	**Type**	**No. of units**
Super	60,000	Super	54,000
Normal	1,80,000		

8. Selling price was ₹ 30 per unit for 'Super'.
9. Company was also involved in a copyright infringement case related to the manufacturing process of 'Super' production. As per the verdict, it had to pay penalty of ₹ 50,000.

PREPARE Cost Sheet of Xim Ltd. for 'Super' showing:

(i) Cost per unit and Total Cost

(ii) Profit per unit and Total Profit

ICAI'S ANSWER

Cost Sheet of 'Super'

Particulars	Per unit (₹)	Total (₹)
Direct materials (Working note- (i))	8.00	4,80,000
Direct wages (Working note- (ii))	4.00	2,40,000
Prime cost	12.00	7,20,000
Production overhead (Working note- (iii))	1.20	72,000

Particulars	Per unit (₹)	Total (₹)
Factory Cost	13.20	7,92,000
Administration Overhead (200% of direct wages)	8.00	4,80,000
Cost of production	21.20	12,72,000
Less: Closing stock (60,000 units – 54,000 units)	-	1,27,200
Cost of goods sold i.e. 54,000 units	21.20	11,44,800
Selling cost	1.00	54,000
Cost of sales/ Total cost	22.20	11,98,800
Profit	7.80	4,21,200
Sales value (₹ 30 × 54,000 units)	30.00	16,20,000

Working Notes:

(i) Direct material cost per unit of 'Normal' = M

Direct material cost per unit of 'Super' = 2M

Total Direct Material cost = 2M × 60,000 units + M × 1,80,000 units

Or, ₹ 12,00,000 = 1,20,000 M + 1,80,000 M

Or, $M = \frac{₹\ 12,00,000}{3,00,000} = ₹\ 4$

Therefore, Direct material Cost per unit of 'Super' = 2 × ₹ 4 = ₹ 8

(ii) Direct wages per unit for 'Super' = W

Direct wages per unit for 'Normal' = 0.6W

So, (W x 60,000) + (0.6W x 1,80,000) = ₹ 6,72,000

W = ₹ 4 per unit

(iii) Production overhead per unit = $\frac{₹\ 2,88,000}{(60,000 + 1,80,000)} = ₹\ 1.20$

Production overhead for 'Super' = ₹ 1.20 × 60,000 units = ₹ 72,000

Notes:

1. Administration overhead is specific to the product as it is directly related to direct labour as mentioned in the question and hence to be considered in cost of production only.
2. Cash discount is treated as interest and finance charges; hence, it is ignored.
3. Penalty paid against the copyright infringement case is an abnormal cost; hence, not included.

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137 MTP Old P3 Dec 2021 S-II Q4(a) 10 marks old scheme

G Ltd. has the following expenditures for the year ended 31st March, 2021:

Sl. No.		Amount (₹)	Amount (₹)
(i)	Raw materials purchased		20,00,00,000
(ii)	Freight inward		22,41,200
(iii)	Wages paid to factory workers		58,40,000
(iv)	Royalty paid for production		3,45,200
(v)	Amount paid for power & fuel		9,24,000
(vi)	Job charges paid to job workers		16,24,000
(vii)	Stores and spares consumed		2,24,000
(viii)	Depreciation on office building		1,12,000
(ix)	Repairs & Maintenance paid for:		
	- Plant & Machinery	96,000	
	- Sales office building	36,000	1,32,000
(x)	Insurance premium paid for:		
	- Plant & Machinery	62,400	
	- Factory building	36,200	98,600
(xi)	Expenses paid for quality control check activities		39,200
(xii)	Research & development cost paid improvement in production process		36,400
(xiii)	Expenses paid for pollution control and engineering & maintenance		53,200
(xiv)	Salary paid to Sales & Marketing Managers:		20,24,000
(xv)	Salary paid to General Manager		25,12,000
(xvi)	Packing cost paid for:		
	- Primary packing necessary to maintain quality	1,92,000	
	- For re-distribution of finished goods	2,24,000	4,16,000
(xvii)	Performance bonus paid to sales staffs		7,20,000
(xviii)	Value of stock as on 1st April, 2020:		
	- Raw materials	36,00,000	
	- Work-in-process	18,40,000	

Sl. No.		Amount (₹)	Amount (₹)
	- Finished goods	22,00,000	76,40,000
(xix)	Value of stock as on 31st March, 2021:		
	- Raw materials	19,20,000	
	- Work-in-process	17,40,000	
	- Finished goods	36,40,000	73,00,000

Amount realized by selling of scrap and waste generated during manufacturing process – ₹1,72,000/-

From the above data you are requested to PREPARE Statement of cost for G Ltd. for the year ended 31st March, 2021, showing (i) Prime cost, (ii) Factory cost, (iii) Cost of Production, (iv) Cost of goods sold and (v) Cost of sales.

ICAI'S ANSWER

Statement of Cost of G Ltd. for the year ended 31st March, 2021:

Sl. No.	Particulars	Amount (₹)	Amount (₹)
(i)	Material Consumed:		
	- Raw materials purchased	20,00,00,000	
	- Freight inward	22,41,200	
	Add: Opening stock of raw materials	36,00,000	
	Less: Closing stock of raw materials	(19,20,000)	20,39,21,200
(ii)	Direct employee (labour) cost:		
	- Wages paid to factory workers		58,40,000
(iii)	Direct expenses:		
	- Royalty paid for production	3,45,200	
	- Amount paid for power & fuel	9,24,000	
	- Job charges paid to job workers	16,24,000	28,93,200
	Prime Cost		21,26,54,400
(iv)	Works/ Factory overheads:		
	- Stores and spares consumed	2,24,000	
	- Repairs & Maintenance paid for plant & machinery	96,000	
	- Insurance premium paid for plant & machinery	62,400	
	- Insurance premium paid for factory building	36,200	

Sl. No.	Particulars	Amount (₹)	Amount (₹)
	- Expenses paid for pollution control and engineering & maintenance	53,200	4,71,800
	Gross factory cost		21,31,26,200
	Add: Opening value of W-I-P		18,40,000
	Less: Closing value of W-I-P		(17,40,000)
	Factory Cost		21,32,26,200
(v)	Quality control cost:		
	- Expenses paid for quality control check activities		39,200
(vi)	Research & development cost paid improvement in production process		36,400
(vii)	Less: Realisable value on sale of scrap and waste		(1,72,000)
(viii)	Add: Primary packing cost		1,92,000
	Cost of Production		21,33,21,800
	Add: Opening stock of finished goods		22,00,000
	Less: Closing stock of finished goods		(36,40,000)
	Cost of Goods Sold		21,18,81,800
(ix)	Administrative overheads:		
	- Depreciation on office building	1,12,000	
	- Salary paid to General Manager	25,12,000	26,24,000
(x)	Selling overheads:		
	- Repairs & Maintenance paid for sales office building	36,000	
	- Salary paid to Manager- Sales & Marketing	20,24,000	
	- Performance bonus paid to sales staffs	7,20,000	27,80,000
(xi)	Distribution overheads:		
	- Packing cost paid for re-distribution of finished goods		2,24,000
	Cost of Sales		**21,75,09,800**

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A Ltd. has the following expenditures for the year ended 31st March 2021:

Sl. No.		Amount (Rs.)	Amount (Rs.)
(i)	Raw materials purchased		10,00,00,000
(ii)	Freight inward		11,20,600
(iii)	Wages paid to factory workers		29,20,000
(iv)	Royalty paid for production		1,72,600
(v)	Amount paid for power & fuel		4,62,000
(vi)	Job charges paid to job workers		8,12,000
(vii)	Stores and spares consumed		1,12,000
(viii)	Depreciation on office building		56,000
(ix)	Repairs & Maintenance paid for:		
	- Plant & Machinery	48,000	
	- Sales office building	18,000	66,000
(x)	Insurance premium paid for:		
	- Plant & Machinery	31,200	
	- Factory building	18,100	49,300
(xi)	Expenses paid for quality control check activities		19,600
(xii)	Research & development cost paid for improvement in production process		18,200
(xiii)	Expenses paid for pollution control and engineering & maintenance		26,600
(xiv)	Salary paid to Sales & Marketing mangers:		10,12,000
(xv)	Salary paid to General Manager		12,56,000
(xvi)	Packing cost paid for:		
	- Primary packing necessary to maintain quality	96,000	
	- For re-distribution of finished goods	1,12,000	2,08,000
(xvii)	Fee paid to independent directors		2,20,000
(xviii)	Performance bonus paid to sales staffs		1,80,000
(xix)	Value of stock as on 1st April, 2020:		
	- Raw materials	18,00,000	

Sl. No.		Amount (Rs.)	Amount (Rs.)
	- Work-in-process	9,20,000	
	- Finished goods	11,00,000	38,20,000
(xx)	Value of stock as on 31st March, 2021:		
	- Raw materials	9,60,000	
	- Work-in-process	8,70,000	
	- Finished goods	18,20,000	36,50,000

Amount realized by selling of scrap and waste generated during manufacturing process – Rs. 86,000/-

From the above data you are requested to PREPARE Statement of cost for A Ltd. for the year ended 31st March, 2021, showing (i) Prime cost, (ii) Factory cost, (iii) Cost of Production, (iv) Cost of goods sold and (v) Cost of sales.

ICAI'S ANSWER

Statement of Cost of A Ltd. for the year ended 31st March, 2021:

Sl. No.	Particulars	Amount (Rs.)	Amount (Rs.)
(i)	Material Consumed:		
	- Raw materials purchased	10,00,00,000	
	- Freight inward	11,20,600	
	Add: Opening stock of raw materials	18,00,000	
	Less: Closing stock of raw materials	(9,60,000)	10,19,60,600
(ii)	Direct employee (labour) cost:		
	- Wages paid to factory workers		29,20,000
(iii)	Direct expenses:		
	- Royalty paid for production	1,72,600	
	- Amount paid for power & fuel	4,62,000	
	- Job charges paid to job workers	8,12,000	14,46,600
	Prime Cost		10,63,27,200
(iv)	Works/ Factory overheads:		
	- Stores and spares consumed	1,12,000	
	- Repairs & Maintenance paid for plant & machinery	48,000	
	- Insurance premium paid for plant & machinery	31,200	

Sl. No.	Particulars	Amount (Rs.)	Amount (Rs.)
	- Insurance premium paid for factory building	18,100	
	- Expenses paid for pollution control and engineering & maintenance	26,600	2,35,900
	Gross factory cost		10,65,63,100
	Add: Opening value of W-I-P		9,20,000
	Less: Closing value of W-I-P		(8,70,000)
	Factory Cost		10,66,13,100
(v)	Quality control cost:		
	- Expenses paid for quality control check activities		19,600
(vi)	Research & development cost paid for improvement in production process		18,200
(vii)	Less: Realisable value on sale of scrap and waste		(86,000)
(viii)	Add: Primary packing cost		96,000
	Cost of Production		10,66,60,900
	Add: Opening stock of finished goods		11,00,000
	Less: Closing stock of finished goods		(18,20,000)
	Cost of Goods Sold		10,59,40,900
(ix)	Administrative overheads:		
	- Depreciation on office building	56,000	
	- Salary paid to General Manager	12,56,000	
	- Fee paid to independent directors	2,20,000	15,32,000
(x)	Selling overheads:		
	- Repairs & Maintenance paid for sales office building	18,000	
	- Salary paid to Manager- Sales & Marketing	10,12,000	
	- Performance bonus paid to sales staffs	1,80,000	12,10,000
(xi)	Distribution overheads:		
	- Packing cost paid for re-distribution of finished goods		1,12,000
	Cost of Sales		10,87,94,900

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Mix Soap Pvt. Ltd., manufactures three brands of soap – Luxury, Herbal and Beauty. The following information has been obtained for the period from June 1 to June 30, 2021 relating to three brands:

	Luxury	Herbal	Beauty
Actual Production (units)	6,750	14,000	77,500
Wages paid (Rs.)	7,500	18,750	1,15,000
Raw materials consumed (Rs.)	20,000	47,000	2,40,000
Selling price per unit (Rs.)	25	15	8

Other data are:

Factory overheads	Rs. 80,000
General & administration overheads (equal for all)	Rs. 48,000
Selling overheads	20% of Works cost

If the company limits the manufacture to just one brand of soap adopting a single brand production, then monthly production will be:

	Units
Luxury	5,000
Herbal	15,000
Beauty	30,000

Further, factory overheads are to be allocated to each brand on the basis of the units which could have been produced when single brand production was in operation.

You are required to:

(i) FIND out the Factory overhead rate for all the brands.

(ii) PREPARE a cost statement for the month of June showing the various elements of cost and also the profit earned.

ICAI'S ANSWER

(i) Calculation of Factory overhead rate.

If the single brand production was in operation, then

1 unit of Luxury = 3 units of Herbal = 6 units of Beauty. Therefore, the factory overhead ratio in the reverse order would be 5,000:15,000:30,000 or 1:3:6.

The overhead rate will be lowest in case of brand which will be produced in high number. Therefore, in case of Beauty soap brand, the overhead rate will be:

$$= \frac{80,000}{6 \text{ x } 6,750 + 3 \text{ x } 14,000 + 1 \text{ x } 77,500}$$

$$= \frac{80,000}{40,500 + 42,000 + 77,500}$$

$$= \frac{80,000}{1,60,000} = 0.5$$

So, the overhead rate will be:

Luxury = 0.5 x 6 = Rs. 3

Herbal = 0.5 x 3 = Rs. 1.5

Beauty = 0.5 x 1 = Rs. 0.5

(ii) Statement of Cost of Mix Soap Pvt. Ltd. for the month of June 2021:

	Luxury (Rs.)	Herbal (Rs.)	Beauty (Rs.)	Total (Rs.)
Raw material consumed	20,000	47,000	2,40,000	3,07,000
Add: Wages paid	7,500	18,750	1,15,000	1,41,250
Prime cost	27,500	65,750	3,55,000	4,48,250
Add: Factory overheads	20,250 (Rs.3 x 6,750)	21,000 (Rs.1.5 x 14,000)	38,750 (Rs.0.5 x 77,500)	80,000
Works cost	47,750	86,750	3,93,750	5,28,250
Add: General & administration oveheads (1:1:1)	16,000	16,000	16,000	48,000
Add: Selling expenses	9,550 (Rs.47,750 x 0.20)	17,350 (Rs.86,750 x 0.20)	78,750 (Rs. 3,93,750 × 0.20)	1,05,650
Cost of sales	73,300	1,20,100	4,88,500	6,81,900
Profit (Balancing figure)	95,450	89,900	1,31,500	3,16,850
Sales	1,68,750 (Rs.25 x 6,750)	2,10,000 (Rs.15 x 14,000)	6,20,000 (Rs.8 x 77,500)	9,98,750

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Aloe Ltd. has the capacity to produce 2,00,000 units of a product every month. Its works cost at varying levels of production is as under:

Level	Works cost per unit (₹)
10%	400
20%	390
30%	380

Level	Works cost per unit (₹)
40%	370
50%	360
60%	350
70%	340
80%	330
90%	320
100%	310

Its fixed administration expenses amount to ₹ 3,60,000 and fixed marketing expenses amount to ₹ 4,80,000 per month respectively. The variable distribution cost amounts to ₹ 30 per unit.

It can sell 100% of its output at ₹ 500 per unit provided it incurs the following further expenditure:

(i) It gives gift items costing ₹ 30 per unit of sale;

(ii) It has lucky draws every month giving the first prize of ₹ 60,000; 2nd prize of ₹ 50,000, 3rd prize of ₹ 40,000 and ten consolation prizes of ₹ 5,000 each to customers buying the product.

(iii) It spends ₹ 2,00,000 on refreshments served every month to its customers;

(iv) It sponsors a television programme every week at a cost of ₹ 20,00,000 per month.

It can market 50% of its output at ₹ 560 by incurring expenses referred from (ii) to (iv) above and 30% of its output at ₹ 600 per unit without incurring any of the expenses referred from (i) to (iv) above.

PREPARE a cost sheet for the month showing total cost and profit at 30%, 50% and 100% capacity level & COMPARE its profit.

ICAI'S ANSWER

Cost Sheet (For the month)

Level of Capacity	30%		50%		100%	
	60,000 units		**1,00,000 units**		**2,00,000 units**	
	Per unit (₹)	**Total (₹)**	**Per unit (₹)**	**Total (₹)**	**Per unit (₹)**	**Total (₹)**
Works Cost	**380.00**	**2,28,00,000**	**360.00**	**3,60,00,000**	**310.00**	**6,20,00,000**
Add: Fixed administration expenses	6.00	3,60,000	3.60	3,60,000	1.80	3,60,000
Add: Fixed marketing expenses	8.00	4,80,000	4.80	4,80,000	2.40	4,80,000

Level of Capacity	30%		50%		100%	
Add: Variable distribution cost	30.00	18,00,000	30.00	30,00,000	30.00	60,00,000
Add: Special Costs:						
- Gift items costs	-	-	-	-	30.00	60,00,000
- Customers' prizes*	-	-	2.00	2,00,000	1.00	2,00,000
- Refreshments	-	-	2.00	2,00,000	1.00	2,00,000
- Television programme sponsorship cost	-	-	20.00	20,00,000	10.00	20,00,000
Cost of sales	**424.00**	**2,54,40,000**	**422.40**	**4,22,40,000**	**386.20**	**7,72,40,000**
Profit (Bal. fig.)	176.00	1,05,60,000	137.60	1,37,60,000	113.80	2,27,60,000
Sales revenue	**600.00**	**3,60,00,000**	**560.00**	**5,60,00,000**	**500.00**	**10,00,00,000**

* Customers' prize cost:

Particulars	Amount (₹)
1st Prize	60,000
2nd Prize	50,000
3rd Prize	40,000
Consolation Prizes (10 × ₹ 5,000)	50,000
Total	**2,00,000**

Comparison of Profit

30% capacity	50% capacity	100% capacity
$\frac{\text{Rs.}176}{\text{Rs.}600} \times 100$	$\frac{\text{Rs.}137.6}{\text{Rs.}560} \times 100$	$\frac{\text{Rs.}113.8}{\text{Rs.}500} \times 100$
29.33 %	**24.57%**	**22.76%**

Profit (in value as well as in percentage) is higher at 30% level of capacity than that at 50% and 100% level of capacity.

ICAI · Old P3 · MTP Nov 2020 · Series I · question p. 5-6 · answer p. 8

141 MTP **Old P3 May 2020 S-I** Q2(b) 10 marks old scheme

Following figures has been extracted from the books of M/s A&R Brothers:

	Amount (Rs.)
Stock on 1st March, 2020	
- Raw materials	6,06,000
- Finished goods	3,59,000
Stock on 31st March, 2020	
- Raw materials	7,50,000
- Finished goods	3,09,000
Work-in-process:	
- On 1st March, 2020	12,56,000
- On 31st March, 2020	14,22,000
Purchase of raw materials	28,57,000
Sale of finished goods	1,34,00,000
Direct wages	37,50,000
Factory expenses	21,25,000
Office and administration expenses	10,34,000
Selling and distribution expenses	7,50,000
Sale of scrap	26,000

You are required to COMPUTE:

(i) Value of material consumed

(ii) Prime cost

(iii) Cost of production

(iv) Cost of goods sold

(v) Cost of sales

(vi) Profit/ loss

ICAI'S ANSWER

Cost Sheet of M/s A&R Brothers for the month ended March 2020:

	Particulars	Amount (Rs.)	Amount (Rs.)
(i)	**Materials consumed:**		
	- Opening stock	6,06,000	

	Particulars	Amount (Rs.)	Amount (Rs.)
	- Add: Purchases	28,57,000	
		34,63,000	
	- Less: Closing stock	(7,50,000)	**27,13,000**
	Direct wages		37,50,000
(ii)	**Prime cost**		**64,63,000**
	Factory expenses		21,25,000
			85,88,000
	Add: Opening W-I-P		12,56,000
	Less: Closing W-I-P		(14,22,000)
	Factory cost		84,22,000
	Less: Sale of scrap		(26,000)
(iii)	**Cost of Production**		**83,96,000**
	Add: Opening stock of finished goods		3,59,000
	Less: Closing stock of finished goods		(3,09,000)
(iv)	**Cost of Goods Sold**		**84,46,000**
	Office and administration expenses		10,34,000
	Selling and distribution expenses		7,50,000
(v)	**Cost of Sales**		**1,02,30,000**
(vi)	**Profit** (balancing figure)		**31,70,000**
	Sales		1,34,00,000

ICAI · Old P3 · MTP May 2020 · Series I · question p. 3-4 · answer p. 4-5

142 MTP **Old P3 Nov 2019 S-I** Q4(b) 10 marks old scheme

From the following data of A Ltd., CALCULATE (i) Material Consumed; (ii) Prime Cost and (iii) Cost of production.

		Amount (Rs.)
(i)	Repair & maintenance paid for plant & machinery	9,80,500
(ii)	Insurance premium paid for inventories	26,000
(iii)	Insurance premium paid for plant & machinery	96,000

		Amount (Rs.)
(iv)	Raw materials purchased	64,00,000
(v)	Opening stock of raw materials	2,88,000
(vi)	Closing stock of raw materials	4,46,000
(vii)	Wages paid	23,20,000
(viii)	Value of opening Work-in-process	4,06,000
(ix)	Value of closing Work-in-process	6,02,100
(x)	Quality control cost for the products in manufacturing process	86,000
(xi)	Research & development cost for improvement in production process	92,600
(xii)	Administrative cost for:	
	- Factory & production	9,00,000
	- Others	11,60,000
(xiii)	Amount realised by selling scrap generated during the manufacturing process	9,200
(xiv)	Packing cost necessary to preserve the goods for further processing	10,200
(xv)	Salary paid to Director (Technical)	8,90,000

ICAI'S ANSWER

Calculation of Cost of Production of A Ltd. for the period.....

Particulars	Amount (Rs.)
Raw materials purchased	64,00,000
Add: Opening stock	2,88,000
Less: Closing stock	(4,46,000)
Material consumed	62,42,000
Wages paid	23,20,000
Prime cost	85,62,000
Repair and maintenance cost of plant & machinery	9,80,500
Insurance premium paid for inventories	26,000
Insurance premium paid for plant & machinery	96,000
Quality control cost	86,000
Research & development cost	92,600

Particulars	Amount (Rs.)
Administrative overheads related with factory and production	9,00,000
	1,07,43,100
Add: Opening value of W-I-P	4,06,000
Less: Closing value of W-I-P	(6,02,100)
	1,05,47,000
Less: Amount realised by selling scrap	(9,200)
Add: Primary packing cost	10,200
Cost of Production	1,05,48,000

Notes:

(i) Other administrative overhead does not form part of cost of production.

(ii) Salary paid to Director (Technical) is an administrative cost.

ICAI · Old P3 · MTP Nov 2019 · Series I · question p. 7 · answer p. 9

2.2 Estimated cost sheets, tenders and quotations

10 questions · asked in 9 ICAI documents · 3 exam papers

143 EXAM **Sep 2026** Q2(b) 7 marks

Arrow Precision Components Ltd. is a manufacturer of high-quality aerospace and defence parts. The following are the details of expenses incurred for the year ended 31st March, 2026 :

Particulars	Amount (₹)
(i) Opening Stock of Raw Materials	28,50,000
(ii) Closing Stock of Raw Materials	32,75,000
(iii) Purchase of Raw Materials	3,45,80,000
(iv) Direct Labour	1,68,40,000
(v) Factory Overhead	67,36,000
(vi) Administrative Overhead	42,10,000

During the financial year 2026-27, the company has received a special order from a leading aircraft manufacturer for the supply of one type of component manufactured by the company. The estimated cost of materials and direct labour for this order are 25% of last year's actual consumption and ₹ 72,50,000 respectively. Assume no work-in-process for both the years.

The company follows the policy of charging Factory Overhead as a percentage of Direct Labour and Administrative Overhead as a percentage of Factory Cost based on the previous year's actual costs. The cost of delivery and installation at the customer's site is estimated at ₹ 8,40,000. The company wants to

safely consider an additional 10% on the calculated estimates for material and labour. Overheads are also to be taken after considering this factor. The company will decide on the profit margin based on total costs considered as per its policy and market conditions.

You are required to :

Prepare a Job Cost Sheet for the order received and determine the total cost to be considered as per the company's policy.

(Present figures in rupees.)

(Round off percentages to two decimal places and figures to the next rupee.)

ICAI'S ANSWER

ICAI has not published its answers to this paper yet.

ICAI · P4 · Sep 2026 exam · question p. 14-15

144 EXAM **Sep 2025** Q1(a) 5 marks

RST Ltd. manufactures a standard line of office desks. The company operates with a monthly manufacturing capacity of 5,000 units. The following data relates to the output and cost of two consecutive months of production:

Month	Units Manufactured	Direct Material (₹)	Direct Wages (₹)	Factory Overheads (₹)
April	3,000	15,00,000	6,00,000	3,50,000
May	3,800	19,00,000	7,60,000	4,30,000

In the month of June, the number of units manufactured will be 4,000 units. However, the prices of direct material will increase by 10% and direct wages will increase by 15%. The fixed factory overheads will reduce by 20%.

The company desires to earn a profit of 11% on selling price.

Calculate the selling price per desk in the month of June when the monthly output is 4,000 units.

ICAI'S ANSWER

Statement of total cost and Sales

Particulars	Amount (₹)
Direct Material (15,00,000/3,000 × 4,000) x 110%	22,00,000
Direct Labour (6,00,000/3,000 × 4,000) x 115%	9,20,000
Factory Overheads:	
Variable (4,000 × 100)	4,00,000
Fixed (50,000 × 80%)	40,000
Total Cost	35,60,000
Profit (Bal. Fig)	4,40,000
Sales (35,60,000/89%)	40,00,000

Particulars	Amount (₹)
No of units	4,000
Selling Price per unit	**1,000**

Calculation of fixed and variable factory overhead

Variable Factory overhead (per unit) = $\frac{4,30,000 - 3,50,000}{3,800 - 3,000}$ = ₹ 100

Fixed factory overheads {3,50,000 - (3,000 x 100)} = ₹ 50,000

ICAI · P4 · Sep 2025 exam · question p. 8 · answer p. 9-10

145 **EXAM** **Old P3 Jan 2021** Q3(b) 10 marks old scheme

XYZ Ltd. is engaged in the manufacturing of toys. It can produce 4,20,000 toys at its 70% capacity on per annum basis. Company is in the process of determining sales price for the financial year 2020-21. It has provided the following information:

Direct Material	₹ 60 per unit
Direct Labour	₹ 30 per unit
Indirect Overheads:	
Fixed	₹ 65,50,000 per annum
Variable	₹ 15 per unit
Semi-variable	₹ 5,00,000 per annum up to 60% capacity and ₹ 50,000 for every 5% increase in capacity or part thereof up to 80% capacity and thereafter ₹ 75,000 for every 10% increase in capacity or part thereof.

Company desires to earn a profit of ₹ 25,00,000 for the year. Company has planned that the factory will operate at 50% of capacity for first six months of the year and at 75% of capacity for further three months and for the balance three months, factory will operate at full capacity.

You are required to :

(1) Determine the average selling price at which each of the toy should be sold to earn the desired profit.

(2) Given the above scenario, advise whether company should accept an offer to sell each Toy at:

(a) ₹ 130 per Toy

(b) ₹ 129 per Toy

ICAI'S ANSWER

(1) Statement of Cost

	For first 6 months	For further 3 months	For remaining 3 months	Total
	6,00,000 × 6/12 x 50% = 1,50,000 units	**6,00,000 × 3/12 x 75% = 1,12,500 units**	**6,00,000 × 3/12 = 1,50,000 units**	**4,12,500 units**

	For first 6 months	For further 3 months	For remaining 3 months	Total
Direct Material	90,00,000	67,50,000	90,00,000	2,47,50,000
Direct labour	45,00,000	33,75,000	45,00,000	1,23,75,000
Indirect – Variable Expenses	22,50,000	16,87,500	22,50,000	61,87,500
Indirect – Fixed Expenses	32,75,000	16,37,500	16,37,500	65,50,000
Indirect Semi-variable expenses				
- For first six months @ 5,00,000 per annum	2,50,000			
- For further three months @ 6,50,000* per annum		1,62,500		
- For further three months @ 8,50,000** per annum			2,12,500	6,25,000
Total Cost	1,92,75,000	1,36,12,500	1,76,00,000	5,04,87,500
Desired Profit				25,00,000
Sales value				5,29,87,500
Average Sales price per Toy				**128.45**

* ₹ 5,00,000+ [3 times (from 60% to 75%) x 50,000] = ₹ 6,50,000

** ₹ 6,50,000+ [1 time (from 75% to 80%) x 50,000] + [2 times (from 80% to 100%) × 75,000] = ₹ 8,50,000

(2) (a) Company Should accept the offer as it is above its targeted sales price of ₹ 128.45 per toy.

(b) Company Should accept the offer as it is above its targeted sales price of ₹ 128.45 per toy.

ICAI · Old P3 · Jan 2021 exam · question p. 12 · answer p. 14-15

146 RTP **Jan 2026** Q4 MCQ

A manufacturing firm has produced 10000 units of a standard product by incurring the following

expenses:

Material costs	:	₹ 80,000
Labour costs	:	₹ 40,000
Production overhead	:	₹ 30,000
Office overhead	:	20% of factory cost

Estimated selling and distribution overhead is ₹ 4 per unit. If the firm has no inventory at the beginning of the period and expects to maintain 20% of output as closing inventory, at what price per unit the product should be sold to earn a profit of 20% on sales?

A ₹ 27.50

B ₹ 23.00

C ₹ 26.40

D ₹ 28.75

ICAI'S ANSWER

(A) ₹ 27.50

Particulars	Amount (₹)
Material Costs	80,000
Labour Costs	40,000
Production Overheads	30,000
Factory Cost	**1,50,000**
Add: Office Overheads (20% of Factory Cost)	30,000
Cost of Production (10,000 units)	**1,80,000**
Less: Closing Stock (2,000 units @ ₹ 18)	(36,000)
Cost of Goods Sold (8,000 units)	**1,44,000**
Add: Selling & Distribution (8,000 × ₹ 4)	32,000
Total Cost of Sales	**1,76,000**
Add: Profit (20% of Sales = 44,000)	44,000
Sales Revenue	**2,20,000**

Selling Price per Unit = 2,20,000/8,000 = ₹ 27.50 per unit

ICAI · P4 · RTP Jan 2026 · question p. 7 · answer p. 28

147 MTP Sep 2026 S-II Q4(b) 6 marks

A factory incurred the following expenditure during the year 2025:

		₹
Direct material consumed		12,00,000
Manufacturing Wages		7,00,000
Manufacturing overhead:		
Fixed	3,60,000	
Variable	2,50,000	6,10,000
		25,10,000

In the year 2026, following changes are expected in production and cost of production.

i. Production will increase due to recruitment of 60% more workers in the factory.

ii. Overall efficiency will decline by 10% on account of recruitment of new workers.

iii. There will be an increase of 20% in Fixed overhead and 60% in Variable overhead rate.

iv. The cost of direct material will be decreased by 6%.

v. The company desire to earn a profit of 10% on selling price.

Ascertain the cost of production and selling price.

ICAI'S ANSWER

Budgeted Cost Sheet for the year 2026

Particulars			Amount ₹
Direct material consumed		12,00,000	
Add: 44% due to increased output		5,28,000	
		17,28,000	
Less: 6% for decline in price		1,03,680	16,24,320
Direct wages (manufacturing)		7,00,000	
Add: 60% increase		4,20,000	11,20,000
Prime cost			27,44,320
Manufactured Overhead:			
Fixed	3,60,000		
Add: 20% increase	72,000		
		4,32,000	
Variable	2,50,000		

Particulars			Amount ₹
Add: 60% increase	3,26,000		
		5,76,000	10,08,000
Cost of production			37,52,320
Add: 1/9 of Cost or 10% on selling price			4,16,924
Selling price			41,69,244

Working Note:

1. Production will increase by 60% but efficiency will decline by 10%.

160 – 10% of 160 = 144%

So, increase by 44%.

2. Variable overhead once will change because of increase in production by 60% (from 2,50,000 to 4,00,000) then with efficiency declining by 10% it shall be 3,60,000 and then again as mentioned in point No. (iii) of this question it will increase by 60% then variable overhead shall be ₹ 3,60,000 × 160% = ₹ 5,76,000.

ICAI · P4 · MTP Sep 2026 · Series II · question p. 12-13 · answer p. 17-18

148 MTP **May 2026 S-II** Q1(a) 5 marks

Alpha Tech Industries Pvt. Ltd. is a mid-sized company that manufactures electronic components for solar inverters. In the financial year 2024, the company produced 10,000 units of its flagship controller unit. Due to rising demand for renewable energy products, the company is planning to scale operations in 2025.

Below are the actual production costs for the year 2024:

Particulars	Amount (₹)
Direct Material Consumed	15,00,000
Direct Labour (Wages)	9,00,000
Factory Overheads:	
- Fixed Overheads (Depreciation, Salaries, etc.)	4,50,000
- Variable Overheads (Utilities, Consumables)	3,00,000
Total Production Cost (2024)	31,50,000

Due to a long-term supply contract and growing market demand, the company is planning the following changes in 2025:

1. Target production will rise due to the addition of a second shift and hiring of 50% more factory workers.
2. Due to the higher number of new hires and training time, the overall efficiency is expected to drop by 15%.

3. Cost Adjustments:
   - The company signed a new bulk procurement deal, reducing direct material costs by 8%.
   - Due to inflation and increased scale, variable overheads are expected to increase by 70%.
   - Fixed overheads (including salaries, rent, etc.) will rise by 10% due to new administrative hires and expansion.
4. Alpha Tech wants to maintain a 12.5% profit margin on selling price in 2025 to remain competitive and cover expansion-related risks.

ASCERTAIN the cost of production and selling price.

ICAI'S ANSWER

Budgeted Cost Sheet for the Year 2025 – Alpha Tech Industries Pvt. Ltd.

Particulars	Amount (₹)	Amount (₹)
Direct material consumed	15,00,000	
Add: 27.5% due to increased output	4,12,500	
	19,12,500	
Less: 8% for decline in material price	1,53,000	
Total Material Cost		**17,59,500**
Direct wages (manufacturing)	9,00,000	
Add: 50% increase	4,50,000	
Total Wages		**13,50,000**
Prime cost		**31,09,500**
Manufacturing Overhead:		
Fixed Overhead	4,50,000	
Add: 10% increase	45,000	
Total Fixed Overhead		**4,95,000**
Variable Overhead	3,00,000	
Add: 27.5% due to increased output	82,500	
	3,82,500	
Add: 70% increase as per inflation & scale	2,67,750	
Total Variable Overhead		**6,50,250**
Total Overhead		**11,45,250**
Cost of Production		**42,54,750**

Particulars	Amount (₹)	Amount (₹)
Add: 1/7 of Cost or 12.5% on selling price		6,07,821
Selling Price		**48,62,571**

Working Note:

1. Production will increase by 50%, but efficiency will decline by 15%.
   So, 150 – 15% of 150 = 127.5% of previous output

   Increase = 27.5%
2. Material cost is increased by 27.5% for volume, then reduced by 8% for lower prices.
3. Labour cost increases by 50% directly due to hiring.
4. Variable overhead first increases by 27.5% due to higher production, then further
5. increases by 70% due to inflation and scale, so:
6. ₹ 3,00,000 x 127.5% = ₹ 3,82,500, then ₹ 3,82,500 x 170% = ₹ 6,50,250
7. Profit is 12.5% of selling price → i.e. 1/7 of cost

ICAI · P4 · MTP May 2026 · Series II · question p. 7 · answer p. 6-7

149 MTP **May 2026 S-II** Q5(a) 8 marks

M/s Nova Private Limited has a normal production capacity of 48,000 units of toys per annum. The estimated costs of production are as under:

(i) Direct Material	₹ 55 per unit
(ii) Direct Labour	₹ 32 per unit (subject to a minimum of ₹ 84,000 p.m.)
(iii) Factory Overheads:	
(a) Fixed	₹ 4,80,000 per annum
(b) Variable	₹ 14 per unit
(c) Semi-variable	₹ 1,50,000 per annum up to 60% capacity and additional ₹ 60,000 for every 20% increase in capacity or proportionately.

(iv) Administrative Overheads related to factory ₹ 6,72,000 per annum (fixed)

(v) Selling overheads are incurred at ₹ 10 per unit.

(vi) Each unit of raw material yields scrap which is sold at the rate of ₹ 7 per unit.

(vii) In year 2025, the factory worked at 60% capacity for the first four months but it was expected that it would work at 90% capacity for the remaining eight months.

(viii) During the first four months, the selling price per unit was ₹ 180.

You are required to:

(i) Prepare a cost sheet showing Prime Cost, Works Cost, Cost of Production and Cost of Sales.

(ii) Calculate the selling price per unit for remaining eight months to achieve the total annual profit of ₹ 15,60,000.

ICAI'S ANSWER

(i) Cost Sheet of M/s Nova Pvt. Ltd. for the year 2025.

Normal Capacity: 48,000 units p.a.

Particulars	4 Months 9,600 Units		8 Months 28,800 units	
	Amount (₹)	**Cost per unit (₹)**	**Amount (₹)**	**Cost per unit (₹)**
Direct material	5,28,000		15,84,000	
Less: Scrap	(67,200)		(2,01,600)	
Materials consumed	4,60,800	48	13,82,400	48
Direct Wages	3,36,000	35	9,21,600	32
Prime Cost	**7,96,800**	83	**23,04,000**	80
Factory overheads:				
- Fixed	1,60,000	16.67	3,20,000	11.11
- Variable	1,34,400	14.00	4,03,200	14.00
- Semi variable	50,000	5.21	1,90,000	6.60
Works Cost	**11,41,200**	118.875	**32,17,200**	111.71
Add: Administrative overheads	2,24,000	23.33	4,48,000	15.56
Cost of Production	**13,65,200**	142.21	**36,65,200**	127.26
Selling Overheads	96,000	10.00	2,88,000	10.00
Cost of Sales	**14,61,200**	152.21	**39,53,200**	137.26

Working Notes:

1. Calculation of Costs

Particulars	9,600 units	28,800 units
	Amount (₹)	**Amount (₹)**
Material	5,28,000 (₹ 55 × 9,600 units)	15,84,000 (₹ 55 × 28,800 units)
Wages	3,36,000 (Max. of ₹ 32 × 9,600 units = ₹ 3,07,200 and ₹ 84,000 × 4 months = ₹ 3,36,000)	9,21,600 (28,800 Units × 32)

Particulars	9,600 units	28,800 units
Variable Cost	1,34,400 (₹ 14 × 9,600 units)	4,03,200 (₹ 14 × 28,800 units)
Semi-variable Cost	50,000 $\left(\frac{₹\ 1,50,000}{12\ \text{Months}} \times 4\ \text{Months}\right)$	1,00,000 $\left(\frac{₹\ 1,50,000}{12\ \text{Months}} \times 8\ \text{Months}\right)$ + 60,0000(for 20 % increase) + 30,000 (for 10% increase)
Selling Overhead	96,000 (₹ 10 × 9,600 units)	2,88,000 (₹ 10 × 28,800 units)

Notes: Alternatively scrap of raw material can also be reduced from Work cost.

(ii) Calculation of Selling price for eight months period

Particulars	Amount (₹)
Total Cost of sales ₹ (14,61,200+39,53,200)	54,14,400
Add: Desired profit	15,60,000
Total sales value	69,74,400
Less: Sales value realised in first four months ₹ 180 × 9,600 units)	(17,28,000)
Sales Value to be realised in next eight months	**52,46,400**
No. of units to be sold in next eight months	28,800
Selling price per unit (₹ 52,46,400 ÷ 28,800 units)	**182.17**

ICAI · P4 · MTP May 2026 · Series II · question p. 14-15 · answer p. 18-19

150 MTP **Old P3 Nov 2022 S-II** Q2(b) 10 marks old scheme

A factory can produce 1,80,000 units per annum at its 60% capacity. The estimated costs of production are as under:

Direct material	₹300 per unit
Direct employee cost	₹160 per unit
Indirect expenses:	
- Fixed	₹32,50,000 per annum
- Variable	₹50 per unit
- Semi-variable	₹20,000 per month up to 50% capacity and ₹2,500 for every 20% increase in the capacity or part thereof.

If production program of the factory is as indicated below and the management desires to ensure a profit of ₹1,00,00,000 for the year, DETERMINE the average selling price at which each unit should be

quoted:

First three months of the year- 50% of capacity;

Remaining nine months of the year- 75% of capacity.

ICAI'S ANSWER

Statement of Cost

	First three months (Rs.)	Remaining nine months (Rs.)	Total (Rs.)
	37,500 units	**1,68,750 units**	**2,06,250 units**
Direct material	1,12,50,000	5,06,25,000	6,18,75,000
Direct employee cost	60,00,000	2,70,00,000	3,30,00,000
Indirect- variable expenses	18,75,000	84,37,500	1,03,12,500
Indirect – fixed expenses	8,12,500	24,37,500	32,50,000
Indirect- semi-variable expenses			
- For first three months @ Rs.20,000 p.m.	60,000		
- For remaining nine months @ Rs.25,000 p.m.		2,25,000	2,85,000
Total cost	1,99,97,500	8,87,25,000	10,87,22,500
Desired profit	-	-	1,00,00,000
Sales value	-	-	11,87,22,500
Average selling price per unit			575.62

ICAI · Old P3 · MTP Nov 2022 · Series II · question p. 3 · answer p. 5

151 MTP **Old P3 May 2022 S-I** Q2(a) 10 marks old scheme

The following data relates to the manufacturing project received for the budgeted output of 19,600 units. You are required to CALCULATE the selling price per unit covering a profit of 25% on the selling price.

Direct materials: 40 sq. m. per unit @ ₹ 10.60 per sq. m.

Direct wages: Bonding department 48 hours per unit @ ₹ 25 per hour

Finishing department 30 hours per unit @ ₹ 19 per hour

Budgeted costs and hours per annum-

Variable overhead:

	(₹)	Total hours
Bonding department	15,00,000	10,00,000
Finishing department	6,00,000	6,00,000

Fixed overhead-

	(₹)
Production	15,68,000
Selling and distribution	7,84,000
Administration (General)	3,92,000

ICAI'S ANSWER

Decision making Cost Sheet (per unit)

Particulars	(Amount in ₹)	(Amount in ₹)
Direct materials 40 m^2 at ₹ 10.60 per m^2		424
Direct wages:		
Bonding department- 48 hours at ₹ 25 per hour	1,200	
Finishing department- 30 hours at ₹ 19 per hour	570	1,770
Prime Cost		**2,194**
Variable overhead:*		
Bonding department- 48 hours at ₹ 1.50 per hour	72	
Finishing department- 30 hours at ₹ 1.00 per hour	30	102
Variable production cost		**2,296**
Fixed production overhead#		80
Total production cost		**2,376**
Selling and distribution cost$	40	
Administration cost$	20	60
Total Cost		**2,436**

Selling price per unit = ₹ $2{,}436 \times \frac{100}{75}$ = **₹ 3,248**

Working Notes:

* Variable overhead rates-

Bonding: $\frac{15,00,000}{10,00,000 \text{ hours}} = ₹ 1.50$

Finishing: $\frac{6,00,000}{6,00,000 \text{ hours}} = ₹ 1.00$

Fixed production overhead rate per unit of output = $\frac{15,68,000}{19,600 \text{ units}} = ₹ 80$

$ Selling and production cost per unit of output = $\frac{7,84,000}{19,600 \text{ units}} = ₹ 40$

Administration cost per unit of output = $\frac{3,92,000}{19,600 \text{ units}} = ₹ 20$

ICAI · Old P3 · MTP May 2022 · Series I · question p. 3 · answer p. 2-3

152 MTP **Old P3 Dec 2021 S-II** Q1(c) 5 marks old scheme

A factory can produce 1,80,000 units per annum at its 60% capacity. The estimated costs of production are as under:

Direct material	₹ 50 per unit
Direct employee cost	₹ 16 per unit
Indirect expenses:	
- Fixed	₹ 32,50,000 per annum
- Variable	₹ 10 per unit
- Semi-variable	₹ 40,000 per month up to 50% capacity and ₹ 15,000 for every 20% increase in the capacity or part thereof.

If production program of the factory is as indicated below and the management desires to ensure a profit of ₹10,00,000 for the year, DETERMINE the average selling price at which each unit should be quoted:

First three months of the year- 50% of capacity;

Remaining nine months of the year- 75% of capacity.

ICAI'S ANSWER

Statement of Cost

	First three months (₹)	Remaining nine months (₹)	Total (₹)
	37,500 units	**1,68,750 units**	**2,06,250 units**
Direct material	18,75,000	84,37,500	1,03,12,500
Direct employee cost	6,00,000	27,00,000	33,00,000
Indirect- variable expenses	3,75,000	16,87,500	20,62,500
Indirect – fixed expenses	8,12,500	24,37,500	32,50,000

	First three months (₹)	Remaining nine months (₹)	Total (₹)
Indirect- semi-variable expenses			
- For first three months @ ₹ 40,000 p.m.	1,20,000		1,20,000
- For remaining nine months @ ₹ 70,000* p.m.		6,30,000	6,30,000
Total cost	37,82,500	1,58,92,500	1,96,75,000
Desired profit	-	-	10,00,000
Sales value	-	-	2,06,75,000
Average selling price per unit			100.24

* ₹ 40,000 for 50% capacity + ₹ 15,000 for 20% increase in capacity + ₹ 15,000 for 5% increase in capacity (because cost is increased for every 20% increase in capacity or part thereof)

ICAI · Old P3 · MTP Dec 2021 · Series II · question p. 1-2 · answer p. 3

2.3 Cost sheet for the service sector

1 question · asked in 1 ICAI document · 1 exam paper

153 EXAM Sep 2024 Q1(a) 5 marks

Language Achievers, a renowned institute specializing in TOEFL preparation, has secured a spacious hall for 20,000 on weekly basis with a seating capacity of 250 students. The instructor, highly qualified and experienced, is compensated generously with an honorarium of 1,500 per lecture. Additionally, he receives reimbursement for travel expenses of ₹ 200 per day along with refreshments costing 1,500 per week to ensure his comfort and focus during teaching sessions. Administrative and miscellaneous expenses, covering essential utilities and materials are, 500 per week. Language Achievers has meticulously planned its curriculum, scheduling batches of 2 lectures per day, 5 days a week for 30 weeks, ensuring comprehensive coverage of the TOEFL syllabus.

Required:

(i) Calculate the total cost per batch.

(ii) Determine the minimum fee per student in a batch to cover costs, if the batch is fully occupied.

(iii) Calculate the fee to be charged from. each student if batch is 80% filled and institute aims to achieve a profit margin of 25% on the fee.

ICAI'S ANSWER

(i) Calculation of Total cost per batch

Particulars	Amount (₹)
Hall Charges (₹ 20,000 × 30)	6,00,000
Honorarium of instructor (₹ 1,500 × 2 × 5 × 30)	4,50,000
Reimbursement of travel expenses (₹ 200 × 5 × 30)	30,000
Refreshment (₹ 1,500 × 30)	45,000
Administrative and miscellaneous expenses (₹ 500 × 30)	15,000
Total Cost	11,40,000
No. of Batches	1
Total cost per batch	11,40,000

(ii) Minimum fee per student in a batch to cover costs

$$= \frac{\text{Total cost per batch}}{\text{Students per batch}}$$

$$= \frac{11,40,000}{250} = ₹\ 4,560$$

(iii) Number of Students if batch is 80% filled

= 250 students x 80% = 200 students

Total Fee to be recovered to achieve 25% profit margin on the fee

= ₹ 11,40,000 + (₹ 11,40,000 x 1/4th of sales or 1/3rd of the cost)

= ₹ 15,20,000

Fee per student

$$= \frac{\text{Total Fee per batch}}{\text{Students per batch}}$$

$$= \frac{15,20,000}{200} = ₹\ 7,600$$

ICAI · P4 · Sep 2024 exam · question p. 8 · answer p. 10-11

CHAPTER 3

Material Cost

3.1 Procurement procedure, GeM, ERP and stock verification

7 questions · asked in 8 ICAI documents · 3 exam papers

154 EXAM **Sep 2026** Q6(c) 4 marks

What are the main distinctions between a Bill of Materials and a Material Requisition Note in terms of their contents and utility ?

ICAI'S ANSWER

ICAI has not published its answers to this paper yet.

ICAI · P4 · Sep 2026 exam · question p. 23

155 EXAM **Sep 2025** Q6(c) 4 marks

Discuss the uses of Bill of Material in the following departments:

(i) Marketing (Purchase) Department

(ii) Production Department

(iii) Stores Department

(iv) Cost/Accounting Department

ICAI'S ANSWER

Uses of Bill of Material

Marketing (Purchase) Dept.	Production Dept.	Stores Dept.	Cost/ Accounting Dept.
Materials are procured (purchased) on the basis of specifications mentioned in it.	Production is planned according to the nature, volume of the materials required to be used. Accordingly, material requisition lists are prepared.	It is used as a reference document while issuing materials to the requisitioning department.	It is used to estimate cost and profit. Any purchase, issue and usage are compared/verified against this document.

ICAI · P4 · Sep 2025 exam · question p. 27 · answer p. 28

156 EXAM **Old P3 Dec 2021** Q4(c) 5 marks old scheme

What is Bill of Material? Describe the uses of Bill of Material in following departments:

(i) Purchases Department

(ii) Production Department

(iii) Stores Department

(iv) Cost/Accounting Department

ICAI'S ANSWER

Bill of Material: It is a detailed list specifying the standard quantities and qualities of materials and components required for producing a product or carrying out of any job.

Uses of Bill of Material in different department:

Purchase Department	Production Department	Stores Department	Cost/ Accounting Department
Materials are procured (purchased) on the basis of specifications mentioned in it.	Production is planned according to the nature, volume of the materials required to be used. Accordingly, material requisition lists are prepared.	It is used as a reference document while issuing materials to the requisitioning department.	It is used to estimate cost and profit. Any purchase, issue and usage are compared/ verified against this document.

ICAI · Old P3 · Dec 2021 exam · question p. 16 · answer p. 18

157 MTP **May 2025 S-II** Q6(b) 5 marks

EXPLAIN the difference between Bill of Materials and Material Requisition Note.

ICAI'S ANSWER

Difference between Bill of Materials and Material Requisition Note

Bill of Materials	Material Requisition Note
1. It is the document prepared by the engineering or planning dept.	1. It is prepared by the production or other consuming department.
2. It is a complete schedule of component parts and raw materials required for a particular job or work order.	2. It is a document asking Store-keeper to issue materials to the consuming department.
3. It often serves the purpose of a material requisition as it shows the complete schedule of materials required for a particular job i.e. it can replace material requisition.	3. It cannot replace a bill of materials.
4. It can be used for the purpose of quotations.	4. It is useful in arriving historical cost only.
5. It helps in keeping a quantitative control on materials drawn through material requisition.	5. It shows the material actually drawn from stores.

ICAI · P4 · MTP May 2025 · Series II · question p. 14 · answer p. 19

158 MTP **Old P3 May 2023 S-I** Q6(e) 5 marks old scheme

Also asked in Old P3 MTP May 2022 S-I Q6(a)

DISTINGUISH clearly between Bin cards and Stores Ledger.

ICAI'S ANSWER

Bin Card	Stores Ledger
It is maintained by the storekeeper in the store.	It is maintained in cost accounting department.
It contains only quantitative details of material received, issued and returned to stores.	It contains information both in quantity and value.
Entries are made when transaction takes place.	It is always posted after the transaction.
Each transaction is individually posted.	Transactions may be summarized and then posted.
Inter-department transfers do not appear in Bin Card.	Material transfers from one job to another job are recorded for costing purposes.

ICAI · Old P3 · MTP May 2023 · Series I · question p. 6 · answer p. 11

159 MTP **Old P3 Nov 2020 S-I** Q6(b) 5 marks old scheme

Same question, different ICAI answer: #160 Old P3 MTP Nov 2019 S-I

DISTINGUISH between Bill of Materials and Material Requisition Note.

ICAI'S ANSWER

Bill of Materials	Material Requisition Note
1. It is the document prepared by the engineering or planning department.	1. It is prepared by the production or other consuming department.
2. It is a complete schedule of component parts and raw materials required for a particular job or work order.	2. It is a document authorizing Store-keeper to issue materials to the consuming department.
3. It often serves the purpose of a material requisition as it shows the complete schedule of materials required for a particular job i.e. it can replace material requisition.	3. It cannot replace a bill of materials.
4. It can be used for the purpose of quotations.	4. It is useful in arriving historical cost only.
5. It helps in keeping a quantitative control on materials drawn through material requisition.	5. It shows the material actually drawn from stores.

ICAI · Old P3 · MTP Nov 2020 · Series I · question p. 8 · answer p. 14

160 MTP **Old P3 Nov 2019 S-I** Q6(b) 5 marks old scheme

Same question, different ICAI answer: #159 Old P3 MTP Nov 2020 S-I

DISTINGUISH between Bill of Materials and Material Requisition Note.

ICAI'S ANSWER

Bills of Material	Material Requisition Note
1.It is document or list of materials prepared by the engineering/ drawing department.	1.It is prepared by the foreman of the consuming department.
2.It is a complete schedule of component parts and raw materials required for a particular job or work order.	2.It is a document authorizing Store-Keeper to issue material to the consuming department.
3.It often serves the purpose of a Store Requisition as it shows the complete schedule of materials required for a particular job i.e. it can replace stores requisition.	3.It cannot replace a bill of material.
4.It can be used for the purpose of quotation.	4.It is useful in arriving historical cost only.
5.It helps in keeping a quantitative control on materials drawn through Stores Requisition.	5.It shows the material actually drawn from stores.

ICAI · Old P3 · MTP Nov 2019 · Series I · question p. 8 · answer p. 12

3.2 Valuing material receipts

16 questions · asked in 9 ICAI documents · 2 exam papers

161 EXAM **Old P3 May 2022** Q6(b) 5 marks old scheme

Write down the treatment of following items associated with purchase of materials.

(i) Cash discount

(ii) IGST

(iii) Demurrage

(iv) Shortage

(v) Basic Custom Duty

ICAI'S ANSWER

Treatment of items associated with purchase of materials is tabulated as below

S. No.	Items	Treatment
(i)	Cash Discount	Cash discount is not deducted from the purchase price. It is treated as interest and finance charges. It is ignored.
(ii)	Integrated Goods and Service Tax (IGST)	Integrated Goods and Service Tax (IGST) is paid on inter-state supply of goods and provision of services and collected from the buyers. It is excluded from the cost of purchase if credit for the same is available. Unless mentioned specifically it should not form part of cost of purchase.
(iii)	Demurrage	Demurrage is a penalty imposed by the transporter for delay in uploading or offloading of materials. It is an abnormal cost and not included with cost of purchase

S. No.	Items	Treatment
(iv)	Shortage	Shortage in materials are treated as follows: **Shortage due to normal reasons:** Good units absorb the cost of shortage due to normal reasons. Losses due to breaking of bulk, evaporation, or due to any unavoidable conditions etc. are the reasons of normal loss. **Shortage due to abnormal reasons:** Shortage arises due to abnormal reasons such as material mishandling, pilferage, or due to any avoidable reasons are not absorbed by the good units. Losses due to abnormal reasons are debited to costing profit and loss account.
(v)	Basic Custom Duty	Basic Custom duty is paid on import of goods from outside India. It is added with the purchase cost.

ICAI · Old P3 · May 2022 exam · question p. 26 · answer p. 28

162 EXAM **Old P3 Jan 2021** Q6(a) 5 marks old scheme

State how the following items are treated in arriving at the value of cost of material purchased:

(i) Detention Charges/Fines

(ii) Demurrage

(iii) Cost of Returnable containers

(iv) Central Goods and Service Tax (CGST)

(v) Shortage due to abnormal reasons.

ICAI'S ANSWER

Treatment of items in arriving at the value of cost of material Purchased

S. No.	Items	Treatment
(i)	**Detention charges/ Fine**	Detention charges/ fines imposed for non-compliance of rule or law by any statutory authority. It is an abnormal cost and **not included** with cost of purchase.
(ii)	**Demurrage**	Demurrage is a penalty imposed by the transporter for delay in uploading or offloading of materials. It is an abnormal cost and **not included** with cost of purchase.
(iii)	**Cost of returnable containers**	Treatment of cost of returnable containers are as follows: Returnable Containers: If the containers are returned and their costs are refunded, then cost of containers should not be considered in the cost of purchase. If the amount of refund on returning the container is less than the amount paid, then, only the short fall is added with the cost of purchase.
(iv)	**Central Goods and Service Tax (CGST)**	Central Goods and Service Tax (CGST) is paid on manufacture and supply of goods and collected from the buyer. It **is excluded** from the cost of purchase if the input credit is available for the same. Unless mentioned specifically CGST is not added with the cost of purchase.

S. No.	Items	Treatment
(v)	**Shortage due to abnormal reasons**	Shortage arises due to abnormal reasons such as material mishandling, pilferage, or due to any avoidable reasons are not absorbed by the good units. Losses due to abnormal reasons are debited to costing profit and loss account.

ICAI · Old P3 · Jan 2021 exam · question p. 24-25 · answer p. 25-26

163 RTP **Jan 2026** Q8

XYZ Enterprises manufactures different types of beverages. The monthly demand pattern for these beverages is as follows:

- Orange Juice: 60,000 litres
- Apple Juice: 20,000 litres
- Grape Juice: 5,000 litres

To produce one litre of Apple Juice, Orange Juice, and Grape Juice, 6 kg of apples, 3 kg of oranges, and 5 kg of grapes are required, respectively. There is no opening or closing stock of raw materials.

XYZ Enterprises can source the raw materials either from local farmers (import in case or grapes) or from wholesale suppliers. The company has the flexibility to purchase any quantity of materials from the wholesale suppliers, but when buying from the farmers or in case of import, it must adhere to a minimum quantity that is specified for each type of material. Below is a summary of the purchasing conditions:

Material	Wholesale Supplier	Farmers
Oranges	Minimum purchase: Any quantity	7,20,000 kg
	Price per kg: ₹ 18.00	Price per kg: ₹ 15.00
	Transportation: ₹ 5,000	Transportation: ₹ 20,000
	-	Sorting and piling cost: ₹ 1,500
	Loading cost: ₹ 10/50 kg	Loading cost: ₹5/50 kg
	Unloading cost: ₹ 2/50 kg	Unloading cost: ₹2/50 kg
Apples	Minimum purchase: Any quantity	3,60,000 kg
	Price per kg: ₹ 12.00	Price per kg: ₹ 10.00
	Transportation: ₹ 11,000	Transportation: ₹ 15,000
	-	Sorting and piling cost: ₹ 1,200
	Loading cost: ₹10/50 kg	Loading cost: ₹ 3/50 kg
	Unloading cost: ₹2/50 kg	Unloading cost: ₹ 2/50 kg
	Wholesale Supplier	**Import**

Material	Wholesale Supplier	Farmers
Grapes	Minimum purchase: Any quantity	1,50,000 kg
	Price per kg: ₹ 32.00	Price per kg: ₹ 25.00
	Transportation: ₹ 3,000	Transportation: ₹ 15,000
	Loading cost: ₹ 10/50 kg	Loading cost: ₹ 25/50 kg
	Unloading cost: ₹2/50 kg	Unloading cost: ₹2/50 kg

Transportation and Sorting and piling cost is given for per purchase order.

8% import duty is applicable is purchase of grapes for which ITC is not available.

XYZ Enterprises is also paying an annual interest rate of 15% on cash credit facility and ₹1,000 per 1,000 kg of rent to its warehouse.

You are required to:

1. Calculate the total purchase cost for each material (Apple, Orange, and Grape) under the following two options:

(a) Purchase from Wholesale Supplier

(b) Purchase from Farmers/Import

2. Calculate the Economic Order Quantity (EOQ) for each material under both options.
3. Provide a Recommendation on the best purchase option for the "Grapes" material if minimum purchase quantity for import is 1,50,000 kg.

ICAI'S ANSWER

1. (i) Calculation of Purchase Cost per Kg. of Materials

	Wholesale Market (₹)	Farmers (₹)
Orange:		
Purchase price	18.00	15.00
Add: Loading Cost	0.20 (₹ 10 ÷ 50 Kg.)	0.10 (₹ 5 ÷ 50 Kg.)
Add: Unloading Cost	0.04 (₹ 2 ÷ 50 Kg.)	0.04 (₹ 2 ÷ 50 Kg.)
	18.24	15.14
Apple:		
Purchase price	12.00	10.00
Add: Loading Cost	0.20 (₹ 10 ÷ 50 Kg.)	0.06 (₹ 3 ÷ 50 Kg.)
Add: Unloading Cost	0.04 (₹ 2 ÷ 50 Kg.)	0.04 (₹ 2 ÷ 50 Kg.)
	12.24	10.10

	Wholesale Market (₹)	Farmers (₹)
Grape:		
Purchase price	32.00	25.00
Add: Import duty @ 8%	---	2.00
Add: Loading Cost	0.20 (₹ 10 ÷ 50 Kg.)	0.50 (₹ 25 ÷ 50 Kg.)
Add: Unloading Cost	0.04 (₹ 2 ÷ 50 Kg.)	0.04 (₹ 2 ÷ 50 Kg.)
	32.24	27.54

(ii) Economic Order Quantity (E.O.Q)

$$= \sqrt{\frac{2 \text{ x Annual requirement x Ordering cost}}{\text{Carrying cost per kg. per annum}}}$$

Annual Requirement (A)

Commodity		Quantity (Kg.)
Orange	(60,000 Ltr. x 3 Kg. x 12 months)	21,60,000
Apple	(20,000 Ltr. x 6 Kg. x 12 months)	14,40,000
Grape	(5,000 Ltr. x 5 Kg. x 12 months)	3,00,000

Cost per Order (O)

	Wholesale Market (₹)	Farmers (₹)
Orange:		
- Transportation cost	5,000	20,000
- Sorting and piling cost	---	1,500
	5,000	21,500
Apple:		
- Transportation cost	11,000	15,000
- Sorting and piling cost	---	1,200
	11,000	16,200
Grape:		
- Transportation cost	3,000	15,000
- Sorting and piling cost	---	---
	3,000	15,000

Carrying Cost per Kg. per annum (C x i)

	Wholesale Market (₹)	Farmers (₹)
Orange:		
- Interest on cash credit	2.736 (₹ 18.24 × 15%)	2.271 (₹ 15.14 × 15%)
- Warehouse rent*	1.0000	1.0000
	3.736	3.271
Apple:		
- Interest on cash credit	1.836 (₹ 12.24 × 15%)	1.515 (₹ 10.10 × 15%)
- Warehouse rent	1.0000	1.0000
	2.836	2.515
Grape:		
- Interest on cash credit	4.836 (₹ 32.24 × 15%)	4.131 (₹ 27.54 × 15%)
- Warehouse rent	1.0000	1.0000
	5.836	5.131

* Warehouse rent per Kg. $= \frac{₹1{,}000}{1{,}000\text{Kg.}} = ₹1$

Calculation of E.O.Q for each material under the both options

	Wholesale Market (Kg.)	Farmers/Import (Kg.)
Orange	√(2 × 21,60,000Kg. x ₹ 5,000/₹ 3.736) = 76,036.72	√(2 × 21,60,000Kg. x ₹21,500/₹ 3.271) = 1,68,508.11
Apple	√(2 × 14,40,000Kg. x ₹11,000/₹ 2.836) = 1,05,691.36	√(2 × 14,40,000Kg. x ₹16,200/₹ 2.515) = 1,36,202.40
Grape	√(2×3,00,000Kg. x ₹3,000/₹ 5.836) = 17,562.19	√(2 × 3,00,000Kg. x ₹15,000/₹ 5.131) = 41,881.31

(iii) Selection of best purchase option for the purchase of Grapes

	Wholesale Market	Import
Annual Requirement (A) (Kg.)	3,00,000	3,00,000
Order Quantity (Q)	17,562.19	1,50,000
No. of orders (A/Q)	17	2
Average Inventory (Q/2) (Kg.)	8,781	75,000

	Wholesale Market	Import
Ordering Cost (₹) (I)	51,000 (17 Orders x ₹ 3,000)	30,000 (2 Order x ₹ 15,000)
Carrying Cost (₹) (II) (Average Inventory x Carrying cost per kg.)	51,245.92 (8,781 Kg. x ₹ 5.836)	3,84,825 (75,000 Kg. x ₹ 5.131)
Purchase Cost (₹) (III)	96,72,000 (3,00,000 Kg. x ₹ 32.24)	82,62,000 (3,00,000 Kg. x ₹ 27.54)
Total Cost (I) + (II) + (III)	97,74,245.92	86,76,825

Importing grapes is the best purchase option for the XYZ enterprises.

ICAI · P4 · RTP Jan 2026 · question p. 8-10 · answer p. 29-33

164 RTP **Sep 2024** Case scenario · Q1(i) to Q1(v)

CASE SCENARIO

'Axe Trade', an unregistered supplier under GST, purchased material from Vye Ltd. which is registered supplier under GST. During the month of June 2024, the Axe Traders has purchased a lot of 5,000 units on credit from Vye Ltd. The information related to the purchase are as follows:

Listed price of one lot of 5,000 units	- ₹ 2,50,000
Trade discount	- @ 10% on listed price
CGST and SGST (Credit available)	- 18% (9% CGST + 9% SGST)
Cash discount	- @ 10%

(Will be given only if payment is made within 30 days.)

Toll Tax paid	₹ 5,000
Freight and Insurance	₹ 17,220
Demurrage paid to transporter	₹ 5,000
Commission and brokerage on purchases	₹ 10,000
Amount deposited for returnable containers	₹ 30,000
Amount of refund on returning the container	₹ 20,000
Other Expenses	@ 2% of total cost

A 20% shortage in material on receipt is expected considering the nature of the raw material.

The payment to the supplier was made within 21 days of the purchases.

Q1(i)

If Axe Traders pays the supplier within 30 days of purchase, then, what is the total amount of cash discount received from the supplier and how it is treated to calculate material cost?

A ₹ 25,000 & it will not be deducted from the material cost

B ₹ 26,550 & it will be deducted from the material cost

C ₹ 26,550 & it will not be deducted from the material cost

D ₹ 22,500 & it will not be deducted from the material cost

Q1(ii)

What will be the amount of other expenses and how it is treated in material cost?

A ₹ 6,154.40 & it will be added with the material cost

B ₹ 6,280.00 & it will be added with the material cost

C ₹ 5,344.40 & it will be added with the material cost

D ₹ 5,453.47 & it will not be added with the material cost

Q1(iii)

What is the amount of GST and how will it be treated in cost sheet of Axe Traders?

A ₹ 40,500 & it will not be added with material cost

B ₹ 40,500 & it will be added with material cost

C ₹ 45,000 & it will not be added with material cost

D ₹ 45,000 & it will be added with material cost

Q1(iv)

What is the total material cost chargeable in the cost sheet of Axe Traders?

A ₹ 3,14,000

B ₹ 2,73,500

C ₹ 2,72,673

D ₹ 3,13,874

Q1(v)

The number of good units and cost per unit of the materials received are:

A 5,000 units & ₹ 62.80

B 5,000 units & ₹ 54.70

C 4,000 units & ₹ 78.50

D 4,000 units & ₹ 68.38

ICAI'S ANSWERS

Q1(i)

(D) Cash discount is received when credit amount is paid within the stipulated period of 30 days. The

amount of cash discount to be received from the supplier is:

	Particulars	Amount (₹)
A.	Listed price	2,50,000
B.	Less: Trade Discount @10%	(25,000)
C.	Taxable value (A-B)	2.25,000
D.	Add: GST@18% (18% of C)	40,500
E.	Total amount payable to the supplier	2,65,500
F.	Cash discount @10% (10% of C)	**(22,500)**
G.	Net amount to be paid to the supplier (E-F)	2,43,000

Q1(ii)

(B)

Particulars	Units	(₹)
Listed Price of Materials	5,000	2,50,000
Less: Trade discount @ 10% on invoice price		(25,000)
		2,25,000
Add: GST @ 18% of ₹ 2,25,000		40,500
		2,65,500
Add: Toll Tax		5,000
Freight and Insurance		17,220
Commission and Brokerage Paid		10,000
Add: Cost of returnable containers: Amount deposited ₹ 30,000 Less: Amount refunded ₹ 20,000		10,000
		3,07,720
Add: Other Expenses @ 2% of Total Cost (₹ 3,07,720/98 ×2)		6,280
Total cost of material		3,14,000
Less: Shortage material due to normal reasons @ 20%	1,000	-
Total cost of material of good units	4,000	3,14,000
Cost per unit (₹ 3,14,000/4,000 units)		78.5

Q1(iii)

(B) Axe Traders is an unregistered supplier in the GST; thus, GST credit is not applicable for it. GST paid on the purchase of the material will be the part of the material cost.

Q1(iv)

(A) Please refer the solution above

Q1(v)

(C) Please refer the solution above

ICAI · P4 · RTP Sep 2024 · question p. 1-2 · answer p. 16, 17

165 **RTP** **Old P3 May 2022** Q1 old scheme

Also asked in Old P3 MTP Nov 2023 S-I Q1(c)

Sky & Co., an unregistered supplier under GST, purchased material from Vye Ltd. which is registered under GST. The following information is available for one lot of 5,000 units of material purchased:

Listed price of one lot	₹ 2,50,000
Trade discount	@ 10% on listed price
CGST and SGST (Credit Not available)	12% (6% CGST + 6% SGST)
Cash discount	@ 10%
(Will be given only if payment is made within 30 days.)	
Toll Tax paid	₹ 5,000
Freight and Insurance	₹ 17,000
Demurrage paid to transporter	₹ 5,000
Commission and brokerage on purchases	₹ 10,000
Amount deposited for returnable containers	₹ 30,000
Amount of refund on returning the container	₹ 20,000
Other Expenses	@ 2% of total cost

20% of material shortage is due to normal reasons.

The payment to the supplier was made within 21 days of the purchases.

You are required to CALCULATE cost per unit of material purchased by Sky & Co.

ICAI'S ANSWER

Calculation of cost per unit:

Particulars	Units	(₹)
Listed Price of Materials	5,000	2,50,000
Less: Trade discount @ 10% on invoice price		(25,000)

Particulars	Units	(₹)
		2,25,000
Add: CGST @ 6% of ₹ 2,25,000		13,500
Add: SGST @ 6% of ₹ 2,25,000		13,500
		2,52,000
Add: Toll Tax		5,000
Freight and Insurance		17,000
Commission and Brokerage Paid		10,000
Add: Cost of returnable containers:		
Amount deposited ₹ 30,000		
Less: Amount refunded ₹ 20,000		10,000
		2,94,000
Add: Other Expenses @ 2% of Total Cost (₹ 2,94,000/98 ×2)		6,000
Total cost of material		3,00,000
Less: Shortage material due to normal reasons @ 20%	1,000	-
Total cost of material of good units	4,000	3,00,000
Cost per unit (₹ 3,00,000/4,000 units)		75

Note:

1. GST is payable on net price i.e., listed price less discount.
2. Cash discount is treated as interest and finance charges; hence it is ignored.
3. Demurrage is penalty imposed by the transporter for delay in uploading or off-loading of materials. It is an abnormal cost and not included.
4. Shortage due to normal reasons should not be deducted from cost to ascertain total cost of good units.

ICAI · Old P3 · RTP May 2022 · question p. 1 · answer p. 11

166 MTP **Sep 2024 S-II** Case scenario · MCQ 1 to MCQ 5 10 marks

CASE SCENARIO

Mr. Vikas, a toy importer has understood the importance of manufacturing in India. He is backed up by the new govt. policies that motivate him to manufacture in India. As per the custom department any import made for the manufacturing under “Made in India”, custom duty will be refunded upto 80%. Vikas decided not to import toy from China anymore, instead import raw material from Srilanka, for the manufacturing of toys in India. Under an agreement of Govt. Of India with Srilankan Govt., any impo8rt from Srilanka will receive tax benefits.

Vikas ordered material Xendga & material Zenga from Srilanka. Details are given below:-

	Srilankan Rupees (SLR)
Material Xendga (12,000 units * 125 SLR)	15,00,000
Material Zenga (8,000 units * 225 SLR)	18,00,000
Factory cost	33,00,000
Add: Containers cost	2,00,000
Add: Freight upto loading shipment on ship (paid by exporter)	50,000
F.O.B.	35,50,000

- Ocean Freight is $ 2,000
- Insurance is $ 1,500

When shipment reached India, it was unloaded at Chennai port. Vikas requested to put the goods in custom port's warehouse. Vikas due to cash crunch was not in a position to pay custom duty and therefore did not file the bill of exchange (B.O.E.). Custom authorities charged a penalty of INR 15,000.

Finally, after a month Vikas filled B.O.E. and paid custom duty of 20% on CIF value of the shipment. IGST was also applicable @ 18% on the combined value of CIF & custom duty paid.

He spent further a sum of INR 12,500 to bring the imported goods to his factory. An inspection was done on the goods and it was found that 5% of the goods were broken. This came to management as a surprise because generally such rate of defects on imports is 8%.

Additional Information:

- Exchange rates:
  1. 1 SLR = 0.25 INR
  2. 1 USD = 75 INR
- IGST credits are available.
- Containers were refunded at INR 38,000.
- Indian and Srilankan brokers were paid commission by Vikas on factory cost. Indian broker charged 6% whereas Srilankan broker charged 12%.
- CIF (cost, insurance and Freight) includes F.O.B (Free on Board)., Insurance & Ocean freight.

You are required to answer the following 5 questions:

MCQ 1 2 marks

What is the total cost of shipment to be recorded by Vikas?

A INR 13,17,000

B INR 13,04,500

C INR 13,54,500

D INR 13,32,500

MCQ 2 2 marks

What is the absorption rate of total cost per unit of Zenga?

A INR 90.28

B INR 84.44

C INR 93.62

D INR 85.77

MCQ 3 2 marks

What is the absorption rate of total cost per unit of Xendga?

A INR 52.01

B INR 54.24

C INR 58.13

D INR 68.65

MCQ 4 2 marks

Amount of refundable taxes?

A INR 4,13,600

B INR 4,57,600

C INR 2,20,000

D INR 2,37,600

MCQ 5 2 marks

If loss of goods was 9% instead of 5%, what will be the amount that will be charged to statement of profit & loss?

A INR 13,045

B INR 19,898.4

C INR 14,178.4

D INR 24,045

ICAI'S ANSWERS

MCQ 1

(A)

Working notes:

Factory cost (33,00,000 × 0.25)	INR 8,25,000
Add: Freight (50,000 × 0.25)	INR 12,500
F.O.B. (Free On Board)	INR 8,37,500
Containers (2,00,000 × 0.25)	INR 50,000
Insurance (1,500 × 75)	INR 1,12,500
Ocean freight (2,000 × 75)	INR 1,50,000

CIF (Cost, Insurance and Freight) = 8,37,500 + 1,12,500 + 1,50,000

= INR 11,00,000

Custom duty = 20% x 11,00,000 = INR 2,20,000

IGST = 18% x (11,00,000 + 2,20,000)

= INR 2,37,600

Penalty = INR 15,000

Commission

Indian = 6% x 8,25,000 = INR 49,500

Srilankan = 12% x 8,25,000 = INR 99,000

Particulars	Amount (INR)
Factory cost	8,25,000
Containers (50,000-38,000)	12,000
Insurance	1,12,500
Ocean freight	1,50,000
Freight inwards	12,500
Commission (49,500+99,000)	1,48,500
Custom duty non-refundable 20%* 2,20,000	44,000
TOTAL	**13,04,500**

PassBook's note: ICAI's key is printed '1. (a)' (INR 13,04,500 is option (b)'s figure, the total of ICAI's own working).

MCQ 2

(A)

Good units = 8,000* (1-5%) = 7,600 UNITS

Normal loss to be absorbed in good units. No abnormal loss.

Particulars	Product Zenga (INR)
Factory cost	4,50,000
Other cost except commission, insurance and custom duty to be absorbed on the basis of quantity i.e. 12:8 or 3:2 (12,000+1,50,000+12,500)*2/5	69,800
Commission, insurance and custom duty to be absorbed on value basis 15:18 or 5:6 (1,48,500+1,12,500+44,000)*6/11	1,66,363.63
Total Cost	6,86,163.63
Number of good units	7,600 units
Per unit Cost	**90.28**

MCQ 3

(B)

Good units = 12000 * (1-5%) = 11400 units

Particulars	Product Xendga (INR)
Factory cost	3,75,000
Other cost (12,000+1,50,000+12,500)*3/5	1,04,700
Commission, insurance and custom duty (1,48,500+1,12,500+44,000)*5/11	1,38,636.36
Total Cost	618,336.36
Number of good units	11,400 units
Per unit Cost	**54.24**

MCQ 4

(A)

Custom duty 80% x 2,20,000	= 1,76,000
Add: IGST	= 2,37,600
	4,13,600

MCQ 5

(C)

Normal loss upto 8%

Abnormal loss 1%

Total cost of xendga INR 6,18,336.36

Total cost of zenga INR 6,86,163.63

Particulars	XENGDA (INR)	ZENGA (INR)	(INR)
Normal loss of 8%	960 units	640 units	
Good units after normal loss	11,040 units	7,360 units	
Per unit cost to be absorbed in good units (total costs/no of good units after normal loss)	56 (6,18,336.36/11,040)	93.23 (6,86,163.63/7,360)	
Abnormal loss in units 1%	120 units	80 units	
Loss in Profit & Loss	56 × 120 = 6,720	93.23 × 80= 7,458.4	14,178.4

ICAI · P4 · MTP Sep 2024 · Series II · question p. 1-2 · answer p. 1, 2, 2-3

167 MTP **May 2024 S-I** Q2(b) 5 marks

S & Sons, an unregistered supplier under GST, purchases material from V Ltd. which is a GST registered

supplier. The following information is available for one lot of 5,000 units of material purchased:

Listed price of one lot	₹ 5,00,000
Trade discount	@ 10% on listed price
CGST and SGST (Credit Not available)	18% (9% CGST + 9% SGST)
Cash discount	@ 10%
(Will be given only if payment is made within 30 days.)	
Toll Tax paid	₹ 1,800
Freight and Insurance	₹ 36,000
Demurrage paid to transporter	₹ 5,000
Commission and brokerage on purchases	₹ 10,000
Amount deposited for returnable containers	₹ 30,000
Amount of refund on returning the container	₹ 26,000
Other Expenses	@ 2% of total cost

5% of material shortage is due to normal reasons.

The payment to the supplier was made within 21 days of the purchases.

You are required to calculate cost per unit of material purchased by S & Sons.

ICAI'S ANSWER

Calculation of cost per unit:

Particulars	Units	(₹)
Listed Price of Materials	5,000	5,00,000
Less: Trade discount @ 10% on invoice price		(50,000)
		4,50,000
Add: GST @18% of ₹ 4,50,000		81,000
		5,31,000
Add: Toll Tax		1,800
Freight and Insurance		36,000
Commission and Brokerage Paid		10,000
Add: Cost of returnable containers:		
Amount deposited ₹ 30,000		

Particulars	Units	(₹)
Less: Amount refunded ₹ 26,000		4,000
		5,82,800
Add: Other Expenses @ 2% of Total Cost (₹ 5,82,800/98 ×2)		11,894
Total cost of material		5,94,694
Less: Shortage material due to normal reasons @ 5%	250	-
Total cost of material of good units	4,750	5,94,694
Cost per unit (₹ 5,94,694/4,750 units)		**125.20**

Note:

1. GST is payable on net price i.e., listed price less discount.
2. GST paid on purchase is added with cost as ITC on GST cannot be claimed
3. Cash discount is treated as interest and finance item; hence it is ignored.
4. Demurrage is penalty imposed by the transporter for delay in uploading or off-loading of materials. It is an abnormal cost and not included.
5. Shortage due to normal reasons should not be deducted from cost to ascertain total cost of good units.

ICAI · P4 · MTP May 2024 · Series I · question p. 7-8 · answer p. 6-7

168 MTP **Old P3 May 2022 S-I** Q1(a) 5 marks old scheme

SKY Company Ltd., not registered under GST, purchased material 'RPP' from a company, registered under GST. The following information is available for one lot of 5,000 units of material purchased:

Listed price of one lot	₹ 7,50,000
Trade discount	@ 10% on Listed price.
CGST and SGST (Credit Not available)	12% (6% CGST + 6% SGST)
Road Tax paid	₹ 15,000
Freight and Insurance	₹ 51,000
Detention Charges	₹ 15,000
Commission and brokerage on purchases	₹ 30,000
Amount deposited for returnable containers	₹ 90,000
Amount of refund on returning the container	₹ 60,000
Other Expenses	@ 2% of total cost

20% of material shortage is due to normal reasons.

You are required to CALCULATE cost per unit of material purchased to SKY Company Ltd.

ICAI'S ANSWER

Computation of Total cost of material purchased of SKY Manufacturing Company

Particulars	Units	(Amount in ₹)
Listed Price of Materials	5,000	7,50,000
Less: Trade discount @ 10% on invoice price		(75,000)
		6,75,000
Add: CGST @ 6% of ₹ 6,75,000		40,500
SGST @ 6% of ₹ 6,75,000		40,500
		7,56,000
Add: Road Tax paid		15,000
Freight and Insurance		51,000
Commission and Brokerage Paid		30,000
Add: Cost of returnable containers:		
Amount deposited ₹ 90,000		
Less: Amount refunded ₹ 60,000		30,000
		8,82,000
Add: Other Expenses @ 2% of Total Cost (8,82,000/98 ×2)		18,000
Total cost of material		9,00,000
Less: Shortage due to Normal Loss @ 20%	1,000	-
Total cost of material of good units	**4,000**	**9,00,000**
Cost per unit (₹ 9,00,000/4,000 units)		**225**

Notes:

1. GST is payable on net price i.e., listed price less discount.
2. Detention charges/ fines imposed for non-compliance of rule or law by any statutory authority. It is an abnormal cost and not included with cost of purchase.
3. Shortage due to normal reasons should not be deducted from cost to ascertain total cost of good units.

ICAI · Old P3 · MTP May 2022 · Series I · question p. 1 · answer p. 1

3.3 Pricing material issues: FIFO, LIFO and average methods

6 questions · asked in 6 ICAI documents · 3 exam papers

169 EXAM **Sep 2026** MCQ 13 2 marks MCQ

The following information is available in respect of an item of inventory in a factory that uses FIFO (First In First Out) method of valuation of issues. The opening stock was 150 units @ ₹ 40 per unit. The following transactions have taken place :

Day	Received Units (in Numbers)	Rate (₹)	Issued Units (in Numbers)
1	25	50	–
2	300	52	–
3	–	–	340
4	200	50	–
5	–	–	300
6	100	40	–

The value of material issued on Day 5 is ₹ :

A 15,200

B 15,300

C 15,270

D 15,230

ICAI'S ANSWER

Answer key: **ICAI has not published the key yet.**

ICAI · P4 · Sep 2026 exam · question p. 8

170 EXAM **Jan 2026** MCQ 14 2 marks MCQ

In a factory, during the month of October 2025, the following transactions have occurred in respect of purchase and issue of "Material A."

Material A

Stock on 01.10.2025, 100 units at ₹ 50 per unit

Purchases:

05-10-2025, 2,500 units at ₹ 55 per unit

08-10-2025, 600 units at ₹ 56 per unit

Issues:

15-10-2025, 1,500 units

What is the value of material A consumed during the period, using LIFO method of pricing issues?

A ₹ 83,100

B ₹ 82,000

C ₹ 82,500

D ₹ 84,000

ICAI'S ANSWER

Answer key: **(A)**

ICAI · P4 · Jan 2026 exam · question p. 6 · answer p. 7

171 EXAM **Old P3 May 2023** Q2(a) 10 marks old scheme

A Limited has furnished the following information for the months from 1st January to 30th April, 2023:

	January	February	March	April
Number of Working days	25	24	26	25
Production (in units) per working day	50	55	60	52
Raw Material Purchases (% by weights to total of 4 months)	21%	26%	30%	23%
Purchase price of raw material (per kg)	₹ 10	₹ 12	₹ 13	₹ 11

Quantity of raw material per unit of product: 4 kg.

Opening stock of raw material on 1st January: 6,020 kg. (Cost ₹ 63, 210)

Closing stock of raw material on 30th April: 5,100 kg.

All the purchases of material are made at the start of each month.

Required:

(i) Calculate the consumption of raw materials (in kgs) month-by- month and in total.

(ii) Calculate the month-wise quantity and value of raw materials purchased.

(iii) Prepare the priced stores ledger for each month using the FIFO method.

ICAI'S ANSWER

(i) Calculation of consumption of Raw Material (in kgs) month by month and total

Particulars	Jan	Feb	March	April	Total
No. of working days	25	24	26	25	-
Production (Per day)	50	55	60	52	-
Production	1,250	1,320	1,560	1,300	5,430
Raw Material Consumed (in kgs)	**5,000**	**5,280**	**6,240**	**5,200**	21,720

Calculation of Raw Material Purchased

Purchased	(Kg)
Closing stock on 30th April	5,100
Add: Raw Material consumed	21,720
Less: Opening stock on 1st January	(6,020)
Raw Material purchased	20,800

(ii) Calculation of month wise quantity and value of raw material purchased

	%	Purchased (Kg)	Price (₹)	Value (₹)
January	21	4,368	10	43,680
February	26	5,408	12	64,896
March	30	6,240	13	81,120
April	23	4,784	11	52,624
Total		20,800		2,42,320

(iii) Store Price Ledger by using FIFO method.

Months	Particulars	Receipts			Issue			Balance		
		Qty	Rate	Amount (₹)	Qty	Rate	Amount (₹)	Qty	Rate	Amount (₹)
Jan	Opening							6,020	10.5	63,210
	Purchases	4,368	10	43,680				6,020	10.5	63,210
								4,368	10	43,680
	Consumption				**5,000**	**10.5**	**52,500**	**1,020**	**10.5**	**10,710**
								4,368	**10**	**43,680**
Feb	Purchases	5,408	12	64,896				1,020	10.5	10,710
								4,368	10	43,680
								5,408	12	64,896
	Consumption				**1,020**	**10.5**	**10,710**	**108**	**10**	**1,080**
					4,260	**10**	**42,600**	**5,408**	**12**	**64,896**
March	Purchase	6,240	13	81,120				108	10	1,080
								5,408	12	64,896
								6,240	13	81,120
	Consumption				**108**	**10**	**1,080**			
					5,408	**12**	**64,896**			

Months	Particulars	Receipts			Issue			Balance		
					724	**13**	**9,412**	**5,516**	**13**	**71,708**
April	Purchases	4,784	11	52,624				5,516	13	71,708
								4,784	11	52,624
	Consumption				**5,200**	**13**	**67,600**	**316**	**13**	**4,108**
								4,784	**11**	**52,624**
										56,732

ICAI · Old P3 · May 2023 exam · question p. 7-8 · answer p. 9-10

172 MTP **Old P3 Nov 2022 S-II** Q5(b) 10 marks old scheme

The following are the details of receipts and issues of a material of stores in a manufacturing company for the period of three months ending 30th June, 2022:

Receipts:

Date	Quantity (kg.)	Rate per kg. (₹)
April 10	1,600	50.00
April 20	2,400	49.00
May 5	1,000	51.00
May 17	1,100	52.00
May 25	800	52.50
June 11	900	54.00
June 24	1,400	55.00

There was 1,500 kg. in stock at April 1, 2022 which was valued at ₹48.00 per kg.

Issues:

Date	Quantity (kg.)
April 4	1,100
April 24	1,600
May 10	1,500
May 26	1,700
June 15	1,500
June 21	1,200

Issues are to be priced on the basis of weighted average method.

The stock verifier of the company reported a shortage of 80 kgs. on 31st May, 2022 and 60 kgs. on 30th June, 2022.

You are required to PREPARE a Stores Ledger Account.

ICAI'S ANSWER

Stores Ledger Account

for the three months ending 30th June, 2022

(Weighted Average Method)

Date	Receipts				Issues				Balance		Rate for further Issue (Rs.)
	GRN No.	Qty. (Kg.)	Rates (Rs.)	Amounts	MR No.	Qty. (Kg.)	Rates (Rs.)	Amount (Rs.)	Qty. (Kg.)	Amount (Rs.)	
April 1									1,500	72,000	48.00
April 4						1,100	48.00	52,800	400	19,200	48.00
April 10		1,600	50.00	80,000					2,000	99,200	$\frac{99,200}{2,000}$ = 49.60
April 20		2,400	49.00	1,17,600					4,400	216,800	$\frac{2,16,800}{4,400}$ = 49.30
April 24						1,600	49.30	78,880	2,800	137,920	$\frac{1,37,920}{2,800}$ = 49.30
May 5		1,000	51.00	51,000					3,800	188,920	$\frac{1,88,920}{3,800}$ = 49.70
May 10						1,500	49.70	74,550	2,300	114,370	$\frac{1,14,370}{2,300}$ = 49.70
May 17		1,100	52.00	57,200					3,400	171,570	$\frac{1,71,570}{3,400}$ = 50.50
May 25		800	52.50	42,000					4,200	213,570	$\frac{2,13,570}{4,200}$ = 50.90
May 26						1,700	50.90	86,530	2,500	127,040	$\frac{1,27,040}{2,500}$ = 50.90
May 31					Shortage	80			2,420	127,040	$\frac{1,27,040}{2,420}$ = 52.50
June 11		900	54.00	48,600					3,320	175,640	$\frac{1,75,640}{3,320}$ = 52.90
June 15						1,500	52.90	79,350	1,820	96,290	$\frac{96,290}{1,820}$ = 52.90
June 21						1,200	52.90	63,480	620	32,810	$\frac{32,810}{620}$ = 52.90
June 24		1,400	55.00	77,000					2,020	109,810	$\frac{1,09,810}{2,020}$ = 54.40
June 30					Shortage	60			1,960	109,810	$\frac{1,09,810}{1,960}$ = 56.00

ICAI · Old P3 · MTP Nov 2022 · Series II · question p. 6-7 · answer p. 13

173 MTP **Old P3 May 2022 S-II** Q3(a) 10 marks old scheme

M/s SE Traders is a distributor of an electronic items. A periodic inventory of electronic items on hand is taken when books are closed at the end of each quarter. The following information is available for the

quarter ended on 30th September, 2021:

Sales	₹ 2,19,30,000
Opening Stock	12,500 units @ ₹ 600 per unit
Administrative Expenses	₹ 5,62,500
Purchases (including freight inward):	
- July 1, 2021	25,000 units @ ₹ 573 per unit
- September 30, 2021	12,500 units @ ₹ 630 per unit
Closing stock- September 30, 2021	16,000 units

You are required to COMPUTE the following by WAM (Weighted Average Method), FIFO method and LIFO method assuming issue/ consumption pattern was even throughout the quarter:

(i) Value of Inventory on 30th September, 2021.

(ii) Profit or loss for the quarter ended 30th September, 2021.

ICAI'S ANSWER

(i) Computation of Value of Inventory as on 30th September 2021:

Date	Particulars	Units	WAM (₹)	FIFO (₹)	LIFO (₹)
01-07-21	Opening Stock	12,500	75,00,000 (₹600×12,500)	75,00,000 (₹600×12,500)	75,00,000 (₹600×12,500)
01-07-21	Purchases	25,000	1,43,25,000 (₹573×25,000)	1,43,25,000 (₹573×25,000)	1,43,25,000 (₹573×25,000)
30-09-21	Purchases	12,500	78,75,000 (₹630×12,500)	78,75,000 (₹630×12,500)	78,75,000 (₹630×12,500)
01-07-21 to 30-09-21	Issues/ Consumption (Balancing figure)	34,000	**2,01,96,000***	**1,98,19,500****	**2,01,94,500*****
30-09-21	Closing Stock	16,000	**95,04,000**	**98,80,500**	**95,05,500**

$$\text{Weighted average rate} = \frac{₹\ 75{,}00{,}000 + ₹\ 1{,}43{,}25{,}000 + ₹\ 78{,}75{,}000}{(12{,}500 + 25{,}000 + 12{,}500)\ \text{units}} = ₹\ 594$$

*	₹ 594 × 34,000 = ₹ 2,01,96,000
**	₹ 600 × 12,500 + ₹ 573 × 21,500 = ₹ 1,98,19,500
***	₹ 630 × 12,500 + ₹ 573 × 21,500 = ₹ 2,01,94,500

(ii) Computation of Profit or Loss for the Quarter ended 30th September 2021

Particulars	WAM (₹)	FIFO (₹)	LIFO (₹)
Sales	2,19,30,000	2,19,30,000	2,19,30,000
Less: Consumption	2,01,96,000	1,98,19,500	2,01,94,500
Less: Administrative Exp.	5,62,500	5,62,500	5,62,500
Profit or Loss	**11,71,500**	**15,48,000**	**11,73,000**

ICAI · Old P3 · MTP May 2022 · Series II · question p. 3-4 · answer p. 5-6

174 MTP **Old P3 Jul 2021 S-II** Q6(b) 5 marks old scheme

EXPLAIN the advantages that would accrue in using the LIFO method of pricing for the valuation of raw material stock.

ICAI'S ANSWER

The advantages that would accrue in using the LIFO method of pricing for the valuation of raw material stock are as follows:

- The cost of materials issued will be either nearer to and or will reflect the current market price. Thus, the cost of goods produced will be related to the trend of the market price of materials. Such a trend in price of materials enables the matching of cost of production with current sales revenues.
- The use of the method during the period of rising prices does not reflect undue high profit in the income statement as it was under the first-in-first-out or average method. In fact, the profit shown here is relatively lower because the cost of production takes into account the rising trend of material prices.
- In the case of falling prices profit tends to rise due to lower material cost, yet the finished products appear to be more competitive and are at market price.
- Over a period, the use of LIFO helps to iron out the fluctuations in profits.
- In the period of inflation LIFO will tend to show the correct profit and thus avoid paying undue taxes to some extent.

ICAI · Old P3 · MTP Jul 2021 · Series II · question p. 7 · answer p. 11

3.4 Stock levels: re-order, minimum, maximum, average and danger levels

28 questions · asked in 17 ICAI documents · 3 exam papers

175 EXAM **May 2026** MCQ 11 2 marks MCQ

The management of Gamma Limited has provided the following information for Product X:

Maximum Stock Level	1125 units
Re-Ordering Level (ROL)	750 units
Minimum rate of consumption per day	25 units
Minimum lead time	5 units

What is the Re-Order Quantity (ROQ)?

- A 375 units
- B 450 units
- C 500 units
- D 600 units

ICAI'S ANSWER

Answer key: **(C)**

ICAI · P4 · May 2026 exam · question p. 5 · answer p. 7

176 EXAM **Jan 2025** Case scenario · Q1(c)(1) to Q1(c)(2) 4 marks

CASE SCENARIO

The Cost Accountant of a Manufacturing concern has given the following details in respect of a raw material X:

Difference between Minimum lead time and Maximum lead time is 4 days.

Average Lead time to procure the Raw Material X is 7 days.

Reorder Level	1,80,000 units
Reorder Quantity	90,000 units
Minimum Stock Level	1,00,000 units
Maximum Stock Level	1,90,000 units

Required to Calculate:

Q1(c)(1) 2 marks

Maximum Consumption per day

Q1(c)(2) 2 marks

Minimum Consumption per day

ICAI'S ANSWERS

ICAI's working for this case

Let the Minimum lead time be A and Maximum lead time be B

B - A = 4 ...1

Average lead time of X = 7

$$\frac{A+B}{2} = 7$$

A + B = 14 ...2

From equation 1 and 2, we get

A (Minimum lead time) = 5

B (Maximum lead time) = 9

Q1(c)(1)

Re-order level = Maximum re-order period/lead time × Maximum consumption

1,80,000 units = 9 × Maximum consumption

Maximum consumption = 20,000 units

Q1(c)(2)

Maximum stock of A = Re-order level + Re-order quantity – (Minimum consumption × Minimum re-order period/lead time)

1,90,000 unit = 1,80,000 units+90,000 units – (Min. Consumption x 5)

Min. Consumption = $\frac{80{,}000 \text{ units}}{5}$ = 16,000 units

Part (ii) can also be done in following way:

(ii) Minimum Stock Level = Re-order level – (Average Consumption X Average Lead time

1,00,000 = 1,80,000 – (Average Consumption X 7)

Average Consumption per day = 11,428.57

Average Consumption per day = (Maximum consumption + Minimum consumption)/ 2

11,428.57 = (20,000 + X) /2

Minimum Consumption per day = **2,858 units**

ICAI · P4 · Jan 2025 exam · question p. 8 · answer p. 10, 11

177 EXAM **Sep 2024** Case Scenario-II · MCQ 6 to MCQ 10

CASE SCENARIO-II

FW Limited manufactures various types of footwear and covers a considerable market share. The footwear made by company are stylish and durable. The management calls for an urgent meeting because it has come to their notice that two of their old permanent customers have moved on to its competitors.

Marketing Manager has stated that there are circumstances when company cannot fulfill the demand of their customers due to shortage of supply and this is the main reason for move on.

Production Manager has stated that production team is working efficiently but workers have to wait long enough for raw material which leads to idle time and low production.

The cost accounts department of FW Limited has furnished the following data for the component B :

Purchase Price	₹ 4,800 per unit
Trade Discount	2% of purchase price
Total duties (No Credit availed)	8% of purchase price
Insurance Charges	₹ 62,000 per year
Units purchased during the year	60,000 units
Opening Stock	5,000 units @ ₹ 5,150 per unit
Closing Stock	4,500 units

Usages per week		Delivery period	
Minimum	1,050 units	Minimum	5 weeks
Maximum	1,200 units	Maximum	9 weeks
Average	1,125 units	Average	7 weeks

Lead time for emergency purchases is 2 weeks.

Additional Information :

- Normal wastage during the storage is 80 units (no realizable value) and abnormal wastage is 40 units.
- Factory works for 365 days in a year.

You are required to calculate the followings (MCQs 6 to 10):

MCQ 6

Calculate per unit cost of material by using Average Price Method.

A ₹ 5,100

B ₹ 5,119

C ₹ 5,094

D ₹ 5,133

MCQ 7

Calculate minimum stock level.

A 10,800 units

B 7,825 units

C 5,250 units

D 2,925 units

MCQ 8

What will be danger level of stock ?

A 2,400 units

B 7,875 units

C 2,250 units

D 2,240 units

MCQ 9

Calculate average number of days (round off) for which average inventory level to be held.

A 27 days

B 29 days

C 26 days

D 30 days

MCQ 10

Calculate amount of Abnormal Loss during storage to be transferred to Costing Profit & Loss Account (based on average price)

- A ₹ 2,04,000
- B ₹ 2,04,760
- C ₹ 2,03,760
- D ₹ 2,05,320

ICAI'S ANSWERS

MCQ 6

Answer key: **(A)**

MCQ 7

Answer key: **(D)**

MCQ 8

Answer key: **(C)**

MCQ 9

Answer key: **(B)**

MCQ 10

Answer key: **(A)**

ICAI · P4 · Sep 2024 exam · question p. 3-4 · answer p. 7

178 RTP **May 2024** Case scenario · Q1(i) to Q1(v)

CASE SCENARIO

The purchase committee of A Ltd. has been entrusted to review the material procurement policy of the company. The chief marketing manager has appraised the committee that the company at present produces a single product X by using two raw materials A and B in the ratio of 3:2. Material A is perishable in nature and has to be used within 10 days from Goods received note (GRN) date otherwise material becomes obsolete. Material B is durable in nature and can be used even after one year. Material A is purchased from the local market within 1 to 2 days of placing order. Material B, on the other hand, is purchased from neighbouring state and it takes 2 to 4 days to receive the material in the store.

The purchase price of per kilogram of raw material A and B is ₹30 and ₹44 respectively exclusive of taxes. To place an order, the company has to incur an administrative cost of ₹1,200. Carrying cost for Material A and B is 15% and 5% respectively. At present material A is purchased in a lot of 15,000 kg. to avail 10% discount on market price. GST applicable for both the materials is 18% and the input tax credit is availed.

The sales department has provided an estimate that the company could sell 30,000 kg. in January 2024 and also projected the same trend for the entire year.

The ratio of input and output is 5:3. Company works for 25 days in a month and production is carried out evenly.

The following queries/ calculations to be kept ready for purchase committees' reference:

Q1(i)

For the month of January 2024, what would be the quantity of the materials to be requisitioned for both material A and B:

A 9,000 kg & 6,000 kg respectively

B 18,000 kg & 12,000 kg respectively

C 27,000 kg & 18,000 kg respectively

D 30,000 kg & 20,000 kg respectively.

Q1(ii)

The economic order quantity (EOQ) for both the material A & B:

A 13,856 kg & 16,181 kg respectively

B 16,197 kg & 17,327 kg respectively

C 16,181 kg & 17,165 kg respectively

D 13,197 kg & 17,165 kg respectively

Q1(iii)

What would the maximum stock level for material A:

A 18,200 kg.

B 12,000 kg.

C 16,000 kg.

D 16,200 kg.

Q1(iv)

Calculate saving/ loss in purchase of Material A if the purchase order quantity is equal to EOQ.

A Profit of Rs. 3,21,201.

B Loss of Rs. 3,21,201.

C Profit of Rs. 2,52,500.

D Loss of Rs. 2,52,500.

Q1(v)

What would the minimum stock level for material A:

A 1,800 kg.

B 1,200 kg.

C 600 kg.

D 2,400 kg.

ICAI'S ANSWERS

Q1(i)

Answer key: **(D)**

(d) Monthly Production of X = 30,000 kgs.

Raw Material Required = $\frac{30,000}{3} \times 5 = 50,000$ kgs.

Material A = $\frac{50,000}{5} \times 3 =$ **30,000 kg.**

Material B = $\frac{50,000}{5} \times 2 =$ **20,000 kg.**

Q1(ii)

Answer key: **(A)**

(a) Calculation of Economic Order Quantity (EOQ):

$$\text{Material A} = \sqrt{\frac{2 \times \text{Annual consumption} \times \text{Order cost}}{\text{Carrying cost per unit p.a.}}}$$

$$= \sqrt{\frac{2 \times (30{,}000 \times 12) \times 1{,}200}{15\% \text{ of } 30}} = \textbf{13,856 kg.}$$

$$\text{Material B} = \sqrt{\frac{2 \times (20{,}000 \times 12) \times 1{,}200}{5\% \text{ of } 44}} = \textbf{16,181 kg.}$$

Q1(iii)

Answer key: **(B)**

(b) Calculation of Maximum Stock level: Since, the Material A is perishable in nature and it required to be used within 10 days, hence, the Maximum Stock Level shall be lower of two:

(a) Stock equal to 10 days consumption

$= \frac{30000}{25} \times 10$ days = 12,000 kg.

(b) Maximum Stock Level for Material A:

Re-order Quantity + Re-order level – (Min consumption* × Min. lead time)

Where, Re-order Quantity = 15,000 kg.

Re-order level = Max. Consumption* × Max. Lead time

= 30,000/25 × 2 days = 2,400 kg.

Maximum stock Level = 15,000 kg. + 2,400 kg. - (30,000/25 × 1 day)

= 17,400 – 1,200 = 16,200 kg.

Stock required for 10 days consumption is lower than the maximum stock level calculated through the formula. Therefore, Maximum Stock Level will be **12,000 kg.**

(*Since, production is processed evenly throughout the month hence material consumption will also be even.)

Q1(iv)

Answer key: **(B)**

(b) Calculation of Savings/ loss in Material A if purchase quantity equals to EOQ.

	Purchase Quantity = 15,000 kg.	Purchase Quantity = EOQ i.e. 13,856 kg.
Annual consumption	3,60,000 kg. (30,000 × 12 months)	3,60,000 kg. (30,000 × 12 months)
No. of orders [Note- (i)]	30 (3,60,000 ÷ 12,000)	30 (3,60,000 ÷ 12,000)
Ordering Cost (a)	₹36,000 (₹1200 × 30)	₹36,000 (₹1200 × 30)
Carrying Cost (b) [Note- (ii)]	₹30,375 (15% of ₹27 × 7,500)	₹31,176 (15% of ₹30 × 6,928)
Purchase Cost (c) (for good portion)	₹97,20,000 (₹27 × 3,60,000)	₹1,08,00,000 (₹30 × 3,60,000)

	Purchase Quantity = 15,000 kg.	Purchase Quantity = EOQ i.e. 13,856 kg.
Loss due to obsolescence (d) [Note- (iii)]	₹24,30,000 [₹27 × (30 × 3,000)]	₹16,70,400 [₹30 × (30 × 1,856)]
Total Cost [(a) + (b) + (c) + (d)]	₹ 1,22,16,375	₹ 1,25,37,576

Purchasing of material -A at present policy of 15,000 kg. saves ₹ **3,21,201.**

Notes: (i) Since, material gets obsolete after 10 days, the quantity in excess of 10 days consumption i.e. 12,000 kg. are wasted. Hence, after 12,000 kg. a fresh order needs to be given.

(ii) Carrying cost is incurred on average stock of Materials purchased.

(iii) the excess quantity of material becomes obsolete and loss has to be incurred.

Q1(v)

Answer key: **(C)**

(c) Minimum Stock Level for Material A

= Re-order level – (Average Consumption Rate x Average Re-order Period)

= 2400 – (1200 x 1.5) = **600 kgs**

Re-order level = Max. Consumption* × Max. Lead time

= 30,000/25 × 2 days = 2,400 kg.

Average Consumption Rate = (30,000/25 + 30,000/25)/2

= 1,200 Kg

Average Re-order Period = (1 + 2)/2 =1.5 Days

Stock required for 10 days consumption is lower than the maximum stock level calculated through the formula. Therefore, Maximum Stock Level will be 12,000 kg.

(*Since, production is processed evenly throughout the month hence material consumption will also be even.)

ICAI · P4 · RTP May 2024 · question p. 1-2 · answer p. 17, 17-18, 18-19, 19

179 RTP **Old P3 Nov 2023** Q1 old scheme

Following details are related to a manufacturing concern:

Re-order Level	1,60,000 units
Economic Order Quality	90,000
Minimum Stock Level	1,00,000 units
Maximum Stock Level	1,90,000 units
Average Lead Time	6 days
Difference between minimum lead time and Maximum lead time	4 days

Calculate:

(i) Maximum consumption per day

(ii) Minimum consumption per day

ICAI'S ANSWER

Difference between Minimum lead time Maximum lead time = 4 days

Max. lead time – Min. lead time = 4 days

Or, Max. lead time = Min. lead time + 4 days.....(i)

Average lead time is given as 6 days i.e.

$\frac{\text{Max. lead time} + \text{Min. lead time}}{2}$ = 6 days..... (ii)

Putting the value of (i) in (ii),

$\frac{\text{Min. lead time} + 4 \text{ days} + \text{Min. lead time}}{2}$ = 6 days

Or, Min. lead time + 4 days + Min. lead time = 12 days

Or, 2 Min. lead time = 8 days

Or, Minimum lead time = $\frac{8 \text{ days}}{2}$ = 4 days

Putting this Minimum lead time value in (i), we get

Maximum lead time = 4 days + 4 days = 8 days

(i) **Maximum consumption per day:**

Re-order level = Max. Re-order period × Maximum Consumption per day

1,60,000 units = 8 days × Maximum Consumption per day

Or, Maximum Consumption per day = $\frac{1,60,000 \text{ units}}{8 \text{ days}}$ = 20,000 units

(ii) **Minimum Consumption per day:**

Maximum Stock Level =

Re-order level + Re-order Quantity – (Min. lead time × Min. Consumption per day)

Or, 1,90,000 units = 1,60,000 units + 90,000 units – (4 days × Min. Consumption per day)

Or, 4 days × Min. Consumption per day = 2,50,000 units – 1,90,000 units

Or, Minimum Consumption per day = $\frac{60,000 \text{ units}}{4 \text{ days}}$ = 15,000 units

ICAI · Old P3 · RTP Nov 2023 · question p. 1 · answer p. 9-10

180 RTP **Old P3 Jul 2021** Q1 old scheme

A Ltd. produces a product 'X' using a raw material 'D'. To produce one unit of X, 4 kg of D is required. As per the sales forecast conducted by the company, it will be able to sale 20,000 units of X in the coming year.

The following are the information related to the raw material D:

(i) The Re-order quantity is 400 kg. less than the Economic Order Quantity (EOQ).

(ii) Maximum consumption per day is 40 kg. more than the average consumption per day.

(iii) There is an opening stock of 2,000 kg.

(iv) Time required to get the raw materials from the suppliers is 4 to 8 days.

(v) The purchase price is ₹ 250 per kg.

There is an opening stock of 1,800 units of the finished product X.

The carrying cost of inventory is 14% p.a.

To place an order company has to incur ₹ 1,340 on paper and documentation work.

From the above information FIND OUT the followings in relation to raw material D:

(a) Re-order Quantity

(b) Maximum Stock level

(c) Minimum Stock level

(d) Calculate the impact on the profitability of the company by not ordering the EOQ.

[Take 300 days for a year]

ICAI'S ANSWER

Working Notes:

(i) Computation of Annual consumption & Annual Demand for raw material 'D':

Sales forecast of the product 'X'	20,000 units
Less: Opening stock of 'X'	1,800 units
Fresh units of 'X' to be produced	18,200 units
Raw material required to produce 18,200 units of 'X' (18,200 units × 4 kg.)	72,800 kg.
Less: Opening Stock of 'D'	2,000 kg.
Annual demand for raw material 'D'	70,800 kg.

(ii) Computation of Economic Order Quantity (EOQ):

$$\text{EOQ} = \sqrt{\frac{2 \times \text{Annual demand of 'D'} \times \text{Ordering cost}}{\text{Carrying cost per unit per annum}}}$$

$$= \sqrt{\frac{2 \times 70{,}800 \text{ kg.} \times ₹\ 1{,}340}{₹\ 250 \times 14\%}} = \sqrt{\frac{2 \times 70{,}800 \text{ kg.} \times ₹\ 1{,}340}{₹\ 35}} = 2{,}328 \text{ kg.}$$

(iii) Re- Order level:

= (Maximum consumption per day × Maximum lead time)

$$= \left\{\left(\frac{\text{Annual Consumption of 'D'}}{300 \text{ days}} + 40 \text{ kg.}\right) \times 8 \text{ days}\right\}$$

$$= \left\{\left(\frac{70{,}800 \text{ kg.}}{300 \text{ days}} + 40 \text{ kg.}\right) \times 8 \text{ days}\right\} = 2{,}208 \text{ kg.}$$

(iv) Minimum consumption per day of raw material 'D':

Average Consumption per day = 236 Kg.

Hence, Maximum Consumption per day = 236 kg. + 40 kg. = 276 kg.

So Minimum consumption per day will be

$$\text{Average Consumption} = \frac{\text{Min. consumption} + \text{Max. consumption}}{2}$$

Or, 236 kg. = $\frac{\text{Min. consumption} + 276 \text{ kg.}}{2}$

Or, Min. consumption = 472 kg – 276 kg. = 196 kg.

(a) Re-order Quantity :

EOQ – 400 kg. = 2,328 kg. – 400 kg.= 1,928 kg.

(b) Maximum Stock level:

= Re-order level + Re-order Quantity – (Min. consumption per day × Min. lead time)

= 2,208 kg. + 1,928 kg. – (196 kg. × 4 days) = 4,136 kg. – 784 kg. = 3,352 kg.

(c) Minimum Stock level:

= Re-order level – (Average consumption per day × Average lead time)

= 2,208 kg. – (236 kg. × 6 days) = 792 kg.

(d) Impact on the profitability of the company by not ordering the EOQ.

		When purchasing the ROQ	When purchasing the EOQ
I	Order quantity	1,928 kg.	2,328 kg.
II	No. of orders a year	70,800 kg./1,928 kg. = 36.72 or 37 orders	70,800 kg./2,328 kg. = 30.41 or 31 orders
III	Ordering Cost	37 orders × ₹ 1,340 = ₹ 49,580	31 orders × ₹ 1,340 = ₹ 41,540
IV	Average Inventory	1,928 kg./2 = 964 kg.	2,328 kg./2 = 1,164 kg.
V	Carrying Cost	964 kg. × ₹ 35 = ₹ 33,740	1,164 kg. × ₹ 35 = ₹ 40,740
VI	Total Cost	₹ 83,320	₹ 82,280

Extra Cost incurred due to not ordering EOQ = ₹83,320 - ₹82,280 = ₹1,040

ICAI · Old P3 · RTP Jul 2021 · question p. 1 · answer p. 11-13

181 RTP **Old P3 Nov 2020** Q1 old scheme

A company uses four raw materials A, B, C and D for a particular product for which the following data apply :–

Raw Material	Usage per unit of product (Kg.)	Re-order Quantity (Kg.)	Price per Kg. (₹)	Delivery period (in weeks)			Re-order level (Kg.)	Minimum level (Kg.)
				Minimum	Average	Maximum		
A	12	12,000	12	2	3	4	60,000	?
B	8	8,000	22	5	6	7	70,000	?
C	6	10,000	18	3	5	7	?	25,500
D	5	9,000	20	1	2	3	?	?

Weekly production varies from 550 to 1,250 units, averaging 900 units of the said product. What would be the following quantities:–

(i) Minimum Stock of A?

(ii) Maximum Stock of B?

(iii) Re-order level of C?

(iv) Average stock level of A?

(v) Re-order level of D?

(vi) Minimum Stock level of D?

ICAI'S ANSWER

(i) Minimum stock of A

Re-order level – (Average consumption × Average time required to obtain delivery)

= 60,000 kg. – (900units × 12 kg. × 3 weeks) = 27,600 kg.

(ii) Maximum stock of B

Re-order level + Re-order quantity– (Min. Consumption × Min. Re-order period)

= 70,000 kg.+ 8,000 kg– (550units ×8 kg.× 5 weeks).

=78,000–22,000 = 56,000 kg.

(iii) Re-order level of C

Maximum re-order period × Maximum Usage

= 7 weeks × (1,250units × 6 kg.) = 52,500 kg.

OR

= Minimum stock of C+(Average consumption × Average delivery time)

= 25,500 kg.+ [(900 units ×6 kg.)×5 weeks] =52,500 kg.

(iv) Average stock level of A

$= \frac{\text{Minimum stock} + \text{Maximum stock}}{2}$ (Refer to Working Note)

$= \frac{27,600 + 58,800}{2} = 43,200$ kg.

Working note

Maximum stock of A = ROL + ROQ – (Minimum consumption × Minimum re-order period)

= 60,000 kg. + 12,000 kg. – [(550units × 12 kg.) × 2 weeks] = 58,800 kg.

(v) Re-order level of D

Maximum re-order period × Maximum Usage

= 3 weeks × (1,250 units × 5 kg.) = 18,750 kg

(vi) Minimum stock of D

Re-order level – (Average consumption × Average time required to obtain delivery)

= 18,750 kg. – (900units × 5 kg. × 2 weeks) = 9,750 kg.

ICAI · Old P3 · RTP Nov 2020 · question p. 1 · answer p. 12-13

182 RTP **Old P3 May 2020** Q1 old scheme

Arnav Electronics manufactures electronic home appliances. It follows weighted average Cost method

for inventory valuation. Following are the data of component X:

Date	Particulars	Units	Rate per unit (₹)
15-12-19	Purchase Order- 008	10,000	9,930
30-12-19	Purchase Order- 009	10,000	9,780
01-01-20	Opening stock	3,500	9,810
05-01-20	GRN*-008 (against the Purchase Order- 008)	10,000	-
05-01-20	MRN**-003 (against the Purchase Order- 008)	500	-
06-01-20	Material Requisition-011	3,000	-
07-01-20	Purchase Order- 010	10,000	9,750
10-01-20	Material Requisition-012	4,500	-
12-01-20	GRN-009 (against the Purchase Order- 009)	10,000	-
12-01-20	MRN-004 (against the Purchase Order- 009)	400	-
15-01-20	Material Requisition-013	2,200	-
24-01-20	Material Requisition-014	1,500	-
25-01-20	GRN-010 (against the Purchase Order- 010)	10,000	-
28-01-20	Material Requisition-015	4,000	-
31-01-20	Material Requisition-016	3,200	-

*GRN- Goods Received Note; **MRN- Material Returned Note

Based on the above data, you are required to CALCULATE:

(i) Re-order level

(ii) Maximum stock level

(iii) Minimum stock level

(iv) PREPARE Store Ledger for the period January 2020 and DETERMINE the value of stock as on 31-01-2020.

(v) Value of components used during the month of January, 2020.

(vi) Inventory turnover ratio.

ICAI'S ANSWER

Workings:

Consumption is calculated on the basis of material requisitions:

Maximum component usage = 4,500 units (Material requisition on 10-01-20)

Minimum component usage = 1,500 units (Material requisition on 24-01-20)

Lead time is calculated from purchase order date to material received date

Maximum lead time = 21 days (15-12-2019 to 05-01-2020)

Minimum lead time = 14 days (30-12-2019 to 12-01-2020)

Calculations:

(i) Re-order level

= Maximum usage × Maximum lead time

= 4,500 units × 21 days = 94,500 units

(ii) Maximum stock level

= Re-order level + Re-order Quantity – (Min. Usage × Min. lead time)

= 94,500 units + 10,000 units – (1,500 units × 14 days)

= 1,04,500 units – 21,000 units = 83,500 units

(iii) Minimum stock level

= Re-order level – (Avg. consumption × Avg. lead time)

= 94,500 units – (3,000 units × 17.5 days)

= 94,500 units – 52,500 units

= 42,000 units

(iv) Store Ledger for the month of January 2020:

Date	Receipts				Issue				Balance		
	GRN/ MRN	Units	Rate ₹	Amt. (₹ '000)	MRN/ MR	Units	Rate ₹	Amt. (₹ '000)	Units	Rate ₹	Amt. (₹ '000)
01-01-20	-	-	-	-	-	-	-	-	3,500	9,810	34,335
05-01-20	008	10,000	9,930	99,300	003	500	9,930	4,965	13,000	9,898	1,28,670
06-01-20	-	-	-	-	011	3,000	9,898	29,694	10,000	9,898	98,980
10-01-20	-	-	-	-	012	4,500	9,898	44,541	5,500	9,898	54,439
12-01-20	009	10,000	9,780	97,800	004	400	9,780	3,912	15,100	9,823	1,48,327
15-01-20	-	-	-	-	013	2,200	9,823	21,611	12,900	9,823	1,26,716
24-01-20	-	-	-	-	014	1,500	9,823	14,734	11,400	9,823	1,11,982

Date	Receipts				Issue				Balance		
25-01-20	010	10,000	9,750	97,500	-	-	-	-	21,400	9,789	2,09,482
28-01-20	-	-	-	-	015	4,000	9,789	39,156	17,400	9,789	1,70,326
31-01-20	-	-	-	-	016	3,200	9,789	31,325	14,200	9,789	1,39,001

[Note: Decimal figures may be rounded-off to the nearest rupee value wherever required)

Value of stock as on 31-01-2020 ('000) = ₹1,39,001

(v) Value of components used during the month of January 2020:

Sum of material requisitions 011 to 016 ('000)

= ₹ 29,694 + ₹ 44,541 + ₹ 21,611 + ₹ 14,734 + ₹ 39,156 + ₹ 31,325 = ₹ 1,81,061

(vi) Inventory Turnover Ratio

$$= \frac{\text{Value of materials used}}{\text{Average stock value}}$$

$$= \frac{₹\ 1,81,061}{₹\ (1,39,001 + 34,335)/2} = \frac{₹\ 1,81,061}{₹\ 86,668} = 2.09$$

ICAI · Old P3 · RTP May 2020 · question p. 1 · answer p. 10-12

183 MTP Jan 2026 S-I MCQ 11 2 marks MCQ

A company observes that the consumption of raw materials varies between 300 units and 900 units per week. The lead time ranges from 3 weeks to 7 weeks, and the reorder quantity is 2000 units. What is the maximum stock level?

A 8,100 units

B 7,700 units

C 6,800 units

D 7,400 units

ICAI'S ANSWER

(D) 7,400 units

Maximum Stock Level

= (Maximum Consumption x Maximum Lead Time) + Reorder Quantity – Minimum Consumption x Minimum Lead Time)

= 900 x 7 + 2,000 - 300 x 3

= 7,400 units

ICAI · P4 · MTP Jan 2026 · Series I · question p. 5 · answer p. 4

184 MTP Jan 2026 S-I Q6(b) 5 marks

"You are appointed as an inventory manager in a manufacturing firm. The company is facing problems of both overstocking and frequent stockouts, which is affecting production. Your task is to apply inventory control methods by setting appropriate quantitative stock levels. Management asked you to Explain any 5 methods of inventory control by setting quantitative levels.

ICAI'S ANSWER

Different method of Inventory Control by Setting Quantitative Levels are explained below:

(i) **Re-order Stock Level (ROL):** This level lies between minimum and the maximum levels in such a way that before the material ordered is received into the stores, there is sufficient quantity in hand to cover both normal and abnormal consumption situations. In other words, it is the level at which fresh order should be placed for replenishment of stock.

It is calculated as:

ROL = Maximum Consumption x Maximum Re-order Period

Or

ROL = Minimum Stock Level + (Average Rate of Consumption x Average Re-order period)

(ii) **Re-Order Quantity:** Re-order quantity is the quantity of materials for which purchase requisition is made by the store department. While setting the quantity to be re-ordered, consideration is given to the maintenance of minimum level of stock, re-order level, minimum delivery time and the most important the cost. Hence, the quantity should be where, the total of carrying cost and ordering cost is at minimum. For this purpose, an economic order quantity should be calculated.

Economic Order Quantity (EOQ): The size of an order for which total of ordering and carrying cost are minimum.

$$\text{EOQ} = \sqrt{\frac{2 \times \text{Annual Requirement (A)} \times \text{Cost per order (O)}}{\text{Carrying Cost per unit per annum (C)}}}$$

(iii) **Minimum Stock Level:** It is lowest level of material stock, which must be maintained in hand at all times, so that there is no stoppage of production due to non-availability of inventory.

It is calculated as below:

Minimum Stock Level = Re-order Stock Level - (Average Consumption Rate x Average Re-order Period)

(iv) **Maximum Stock Level:** It is the highest level of quantity for any material which can be held in stock at any time. Any quantity beyond this level cause extra amount of expenditure due to engagement of fund, cost of storage, obsolescence etc.

It can be calculated as below:

Maximum Stock Level = Re-order Level + Re-order Quantity - (Minimum Consumption Rate x Minimum Re-order Period)

Here, Re-order Quantity may be EOQ

(v) **Average Inventory Level:** This is the quantity of material that is normally held in stock over a period. It is also known as normal stock level.

It can be calculated as below:

Average Stock Level = Minimum Stock Level + 1/2 Re-order Quantity

Alternatively, it can be calculated as below:

Average Stock Level = $\frac{\text{Maximum Stock Level} + \text{Minimum StockLevel}}{2}$

(vi) **Danger level:** It is the level at which normal issues of the raw material inventory are stopped and emergency issues are only made.

It can be calculated as below:

Danger Level = Average Consumption* x Lead time for emergency purchase

*Some time minimum consumption is also used.

(vii) **Buffer Stock:** Some quantity of stock may be kept for contingency to be used in case of sudden order, such stock is known as buffer stock.

ICAI · P4 · MTP Jan 2026 · Series I · question p. 16 · answer p. 19-20

185 MTP Jan 2026 S-II MCQ 12 2 marks MCQ

The cost of placing each purchase order is ₹ 20, and the company plans to purchase 5,000 units during the year at a purchase price of ₹ 50 per unit, inclusive of transportation charges. The annual storage cost per unit is ₹ 5. The lead time varies, with an average of 10 days, a maximum of 15 days, and a minimum of 5 days. The rate of consumption also fluctuates, with an average usage of 15 units per day and a maximum usage of 20 units per day. The maximum stock level is:

A 450 units

B 300 units

C 375 units

D 500 units

ICAI'S ANSWER

(A) 450 units

Maximum level = ROL + ROQ – [Min. rate of consumption × Min. lead time] *(Refer to working notes1, 2 and 3)*

= 300 units + 200 units – [10 units per day × 5 days]

= **450 units**

Working Notes:

1. Minimum rate of consumption per day

Av. rate of consumption = $\frac{\text{Minimum rate of consumption} + \text{Maximum rate of consumption}}{2}$

15 units per day = $\frac{X \text{ units / day} + 20 \text{ units per day}}{2}$ or X = 10 units per day.

2. Re-order Quantity (ROQ) or Economic Order Quantity (EOQ)

= $\sqrt{\frac{2 \times 5{,}000 \text{ units} \times ₹20}{5}}$ = 200 units

3. Re-ordering level (ROL) = Maximum usage per period × Maximum lead time

= 20 units per day × 15 days

= 300 units

ICAI · P4 · MTP Jan 2026 · Series II · question p. 5-6 · answer p. 6

186 MTP Sep 2025 S-II Q3(a) 8 marks

A Ltd. produces a product 'X' using a raw material D. To produce one unit of X, 2 kg of D is required. As per the sales forecast conducted by the company, it will able to sale 10,000 units of X in the coming year. The following is the information regarding the raw material D:

(i) The Re-order quantity is 200 kg. less than the Economic Order Quantity (EOQ).

(ii) Maximum consumption per day is 20 kg. more than the average consumption per day.

(iii) There is an opening stock of 1,000 kg.

(iv) Time required to get the raw materials from the suppliers is 4 to 8 days.

(v) The purchase price is ₹ 125 per kg.

There is an opening stock of 900 units of the finished product X.

The rate of interest charged by bank on Cash Credit facility is 13.76%.

To place an order company has to incur ₹ 720 on paper and documentation work.

From the above information, FIND out the followings in relation to raw material D:

(i) Re-order Quantity

(ii) Maximum Stock level

(iii) Minimum Stock level

(iv) Calculate the impact on the profitability of the company by not ordering the EOQ.

[Take 364 days for a year]

ICAI'S ANSWER

Working Notes:

(1) Computation of Annual consumption & Annual Demand for raw material 'D'

Sales forecast of the product 'X'	10,000 units
Less: Opening stock of 'X'	900 units
Fresh units of 'X' to be produced	9,100 units
Raw material required to produce 9,100 units of 'X' (9,100 units × 2 kg.)	18,200 kg.
Less: Opening Stock of 'D'	1,000 kg.
Annual demand for raw material 'D'	17,200 kg.

(2) Computation of Economic Order Quantity (EOQ)

$$\text{EOQ} = \sqrt{\frac{2 \times \text{Annual demand of 'Dee'} \times \text{Ordering cost}}{\text{Carrying cost per unit per annum}}}$$

$$= \sqrt{\frac{2 \times 17{,}200 \text{ kg.} \times ₹\ 720}{₹\ 125 \times 13.76\%}} = \sqrt{\frac{2 \times 17{,}200 \text{ kg.} \times ₹\ 720}{₹\ 17.2}} = 1{,}200 \text{ kg.}$$

(3) Re- Order level

= (Maximum consumption per day × Maximum lead time)

$$= \left\{\left(\frac{\text{Annual Consumption of 'Dee'}}{364 \text{ days}} + 20\text{kg.}\right) \times 8 \text{ days}\right\}$$

$= \left\{ \left(\frac{18{,}200\text{kg.}}{364 \text{ days}} + 20\text{kg.} \right) \times 8 \text{ days} \right\} = 560 \text{ kg.}$

(4) Minimum consumption per day of raw material 'D'

Average Consumption per day = 50 Kg.

Hence, Maximum Consumption per day = 50 kg. + 20 kg. = 70 kg.

So, Minimum consumption per day will be

Average Consumption = $\frac{\text{Min.consumption} + \text{Max.consumption}}{2}$

Or, 50 kg. = $\frac{\text{Min.consumption} + 70\text{kg.}}{2}$

Or, Min. consumption = 100 kg – 70 kg. = 30 kg.

(i) Re-order Quantity

EOQ – 200 kg. = 1,200 kg. – 200 kg. = 1,000 kg.

(ii) Maximum Stock level

= Re-order level + Re-order Quantity – (Min. consumption per day × Min. lead time)

= 560 kg. + 1,000 kg. – (30 kg. × 4 days) = 1,560 kg. – 120 kg. = 1,440 kg.

(iii) Minimum Stock level

= Re-order level – (Average consumption per day × Average lead time)

= 560 kg. – (50 kg. × 6 days) = 260 kg.

(iv) Impact on the profitability of the company by not ordering the EOQ

		When purchasing the ROQ	When purchasing the EOQ
I	Order quantity	1,000 kg.	1,200 kg.
II	No. of orders a year	$\frac{17{,}200\text{kg.}}{1{,}000\text{kg.}}$ =17.2 or 18 orders	$\frac{17{,}200\text{kg.}}{1{,}200\text{kg.}}$ =14.33 or 15 orders
III	Ordering Cost	18 orders × ₹ 720 = ₹ 12,960	15 orders × ₹ 720 = ₹ 10,800
IV	Average Inventory	$\frac{1{,}000\text{kg.}}{2}$ = 500 kg.	$\frac{1{,}200\text{kg.}}{2}$ = 600 kg.
V	Carrying Cost	500 kg. × ₹ 17.2 = ₹ 8,600	600 kg. × ₹ 17.2 = ₹ 10,320
VI	Total Cost	₹ 21,560	₹ 21,120

Extra Cost incurred due to not ordering EOQ = ₹ 21,560 - ₹ 21,120 = ₹ 440

ICAI · P4 · MTP Sep 2025 · Series II · question p. 11-12 · answer p. 14-16

187 MTP **May 2025 S-I** MCQ 13 2 marks MCQ

A company manufactures 5,000 units of a product per month. The cost of placing an order is ₹ 100. The purchase price of the raw material is ₹ 10 per kg. The re-order period is 4 to 8 weeks. The consumption of raw materials varies from 100 kg to 450 kg per week, the average consumption being 275 kg. The carrying cost of inventory is 20% per annum. What is Maximum level of stock?

A 4,396 kg.

B 5,500 kg

C 6,210 kg

D 3,956 kg

ICAI'S ANSWER

(A) 4,396 kg.

Annual consumption of raw material (A) = (275 kg. × 52 weeks) = 14,300 kg.

Cost of placing an order (O) = ₹ 100

Carrying cost per kg. Per annum (c × i) = ₹ 10 × 20% = ₹ 2

Economic order quantity (EOQ)/ Reorder Quantity (ROQ) = $\sqrt{\frac{2AO}{C \times i}}$

= $\sqrt{\frac{2 \times 14{,}300 \text{ kgs.} \times ₹100}{₹2}}$

= 1,196 Kg. (Approx)

Reorder level (ROL) = Maximum usage × Maximum re-order period

= 450 kg. × 8 weeks = 3,600 kg.

Maximum level = ROL + ROQ – (Min. usage × Min. re-order period)

= 3,600 kg. + 1,196 kg. – (100 kg. × 4 weeks)

= 4,396 kg.

ICAI · P4 · MTP May 2025 · Series I · question p. 6 · answer p. 6

188 MTP **May 2025 S-I Q1(a)** 5 marks

A company uses three raw materials A, B and C for a particular product for which the following data apply:

Raw Material	Usage per unit of product (Kg.)	Re-order Quantity (Kg.)	Price per Kg. (₹)	Delivery period (in weeks)			Re-order level (Kg.)	Minimum level (Kg.)
				Minimum	Average	Maximum		
A	10	10,000	0.10	1	2	3	8,000	?
B	4	5,000	0.30	3	4	5	4,750	?
C	6	10,000	0.15	2	3	4	?	2,000

Weekly production varies from 175 to 225 units, averaging 200 units of the said product.

WHAT would be the following quantities:

(i) Minimum Stock of A?

(ii) Maximum Stock of B?

(iii) Re-order level of C?

(iv) Average stock level of A?

ICAI'S ANSWER

(i) Minimum stock of A

Re-order level – (Average consumption × Average time required to obtain delivery)

= 8,000 kg. – (200 units × 10 kg. × 2 weeks) = 4,000 kg.

(ii) Maximum stock of B

Re-order level – (Min. Consumption × Min. Re-order period) + Re-order quantity

= 4,750 kg. – (175 units × 4 kg. × 3 weeks) + 5,000 kg.

= 9,750 - 2,100 = 7,650 kg.

(iii) Re-order level of C

Maximum re-order period × Maximum Usage

= 4 weeks × (225 units × 6 kg.) = 5,400 kg.

OR

= Minimum stock of C + (Average consumption × Average delivery time)

= 2,000 kg. + [(200 units × 6 kg.) × 3 weeks] = 5,600 kg.

(iv) Average stock level of A

= Minimum stock level of A + $\frac{1}{2}$ Re-order quantity

= 4,000 kg. + $\frac{1}{2}$ 10,000 kg. = 4,000 + 5,000 = 9,000 kg.

OR

$$= \frac{\text{Minimum stock} + \text{Maximum stock}}{2}$$ (Refer to Working Note)

$$= \frac{4,000 + 16,250}{2} = 10,125 \text{ kg.}$$

Working note

Maximum stock of A = ROL + ROQ – (Minimum consumption × Minimum re-order period)

= 8,000 kg. + 10,000 kg. – [(175 units × 10 kg.) × 1 week] = 16,250 kg.

ICAI · P4 · MTP May 2025 · Series I · question p. 7 · answer p. 7-8

189 MTP Jan 2025 S-I Q3(b) 6 marks

Following information is extracted from the purchase department of A Ltd.:

(i) Number of units to be purchased during the year is 10,000

(ii) Cost of placing a purchase order is ₹ 40

(iii) Purchase price per unit is ₹ 80

(iv) Insurance charges to be paid for protecting goods during transit is ₹ 20 per unit

(v) Cash discount to be received is 2%

(vi) Annual cost of storage per unit is ₹ 5

(vii) Details of lead time:

Average-	20 days
Maximum-	30 days
Minimum-	10 days
For emergency purchases-	8 days.

(viii) Rate of consumption:

Average-	30 units per day
Maximum-	40 units per day.

From the information given above, you are required to CALCULATE:

(i) Re-ordering level

(ii) Maximum level

(iii) Minimum level

(iv) Danger level.

ICAI'S ANSWER

Basic Data:

A	(Number of units to be purchased annually)	= 10,000 units
O	(Ordering cost per order)	= ₹ 40
C	(Annual cost of storage per unit)	= ₹ 5

Purchase price per unit = ₹ 80 + ₹ 20 (Insurance charges)

= ₹ 100

(Note: Cash discount is treated as an interest and finance item and thus, it is ignored.)

Computations:

(i)	**Re-ordering level**	= Maximum usage per period × Maximum lead time
	(ROL)	= 40 units per day × 30 days
		= **1,200 units**
(ii)	**Maximum level**	= ROL + ROQ – [Min. rate of consumption × Min. lead time] (Refer to working notes 1 and 2)
		= 1200 units + 400 units – [20 units per day × 10 days]
		= **1,400 units**
(iii)	**Minimum level**	= ROL – [Average rate of consumption × Average re-order-period]

		= 1,200 units – (30 units per day × 20 days)
		= **600 units**
(iv)	**Danger level**	= Average consumption × Lead time for emergency purchases
		= 30 units per day × 8 days
		= **240 units**

Working Notes:

1. Minimum rate of consumption per day (X)

$$\text{Av. rate of consumption} = \frac{\text{Minimum rate of consumption} + \text{Maximum rate of consumption}}{2}$$

$$30 \text{ units per day} = \frac{\text{X units/day} + 40 \text{ units per day}}{2}$$

Or, X = 20 units per day.

2. Re-order Quantity (ROQ) or Economic Order Quantity (EOQ)

$$= \sqrt{\frac{2 \text{x} 10{,}000 \text{units} \text{x} \text{Rs}.40}{\text{Rs}.5}}$$

= 400 units

ICAI · P4 · MTP Jan 2025 · Series I · question p. 13 · answer p. 12-13

190 MTP **Jan 2025 S-II** Q4(a) 6 marks

Following information is available from the purchase books of a company:

Cost of placing a purchase order	₹ 10,000
Number of units to be purchased during the year	12,50,000
Purchase price per unit	₹ 125
Annual cost of storage per unit	₹ 62.50

Details of lead time:

Maximum	20 days
Minimum	10 days
Average	15 days
Emergency	3 days

Rate of consumption:

Average	1,500 units per day
Maximum	2,000 units per day

From the details given above, you are required to CALCULATE:

(i) Re-ordering level

(ii) Maximum level

(iii) Minimum level

(iv) Danger level

ICAI'S ANSWER

(i) Re-ordering level = Maximum usage per period × Maximum lead time

(ROL) = 2,000 units per day × 20 days

= 40,000 units

(ii) Maximum level= ROL+ROQ–[Min. rate of consumption× Min. lead time] (Refer to working notes 1 and 2)

= 40,000 units + 20,000 units – [1,000 units per day x 10 days]

= 50,000 units

(iii) Minimum level = ROL – Average rate of consumption × Average re-order-period

= 40,000 units – (1,500 units per day × 15 days)

= 17,500 units

(iv) Danger level = Average consumption × Lead time for emergency purchases

= 1,500 units per day × 3 days

= 4,500 units

Working Notes:

1. Minimum rate of consumption per day

$$\text{Average rate of consumption} = \left(\frac{\text{Minimum rate of consumption} + \text{Maximum rate of consumption}}{2}\right)$$

$$\text{1,500 units per day} = \left(\frac{\text{X units per day} + \text{2,000 units per day}}{2}\right)$$

Or, X = 1,000 units per day

$$\text{2. Re-order Quantity (ROQ)} = \sqrt{\frac{\text{2x12,50,000units x ₹10,000}}{62.50}}$$

= 20,000 units

ICAI · P4 · MTP Jan 2025 · Series II · question p. 10-11 · answer p. 13-14

191 MTP **Old P3 May 2023 S-II** Q1(d) 5 marks old scheme

A company manufactures 10,000 units of a product per month. The cost of placing an order is ₹200. The purchase price of the raw material is ₹20 per kg. The re-order period is 4 to 8 weeks. The consumption of raw materials varies from 200 kg to 900 kg per week, the average consumption being 550 kg. The carrying cost of inventory is 20% per annum.

You are required to CALCULATE:

(i) Re-order quantity

(ii) Re-order level

(iii) Maximum level

(iv) Minimum level

(v) Average stock level

ICAI'S ANSWER

(i) Reorder Quantity (ROQ) = 1,691 kg. *(Refer to working note)*

(ii) Reorder level (ROL) = Maximum usage × Maximum re-order period

= 900 kg. × 8 weeks = 7,200 kg.

(iii) Maximum level = ROL + ROQ – (Min. usage × Min. re-order period)

= 7,200 kg. + 1,691 kg. – (200 kg.× 4 weeks)

= 8,091 kg.

(iv) Minimum level = ROL – (Normal usage × Normal re-order period)

= 7,200 kg. – (550 kg. × 6 weeks)

= 3,900 kg.

(v) Average stock level = $\frac{1}{2}$ (Maximum level + Minimum level)

= $\frac{1}{2}$ (8,091 kg. + 3,900 kg.) = 5,995.5 kg.

OR

= Minimum level + $\frac{1}{2}$ ROQ

= 3,900 kg. + $\frac{1}{2}$ × 1,691 kg. = 4,745.5 kg.

Working Note

Annual consumption of raw material (A) = (550 kg. × 52 weeks) = 28,600 kg.

Cost of placing an order (O) = ₹ 200

Carrying cost per kg. Per annum (c × i) = ₹ 20 × 20% = ₹4

Economic order quantity (EOQ) = $\sqrt{\frac{2AO}{C \times i}}$

= $\sqrt{\frac{2 \times 28{,}600 \text{ kgs.} \times ₹200}{₹4}}$ = 1,691 Kg. (Approx)

ICAI · Old P3 · MTP May 2023 · Series II · question p. 2 · answer p. 2-3

192 MTP **Old P3 Jul 2021 S-I** Q1(c) 5 marks old scheme

A company manufactures 10,000 units of a product per month. The cost of placing an order is Rs. 200. The purchase price of the raw material is Rs. 20 per kg. The re-order period is 4 to 8 weeks. The consumption of raw materials varies from 200 kg to 900 kg per week, the average consumption being 550 kg. The carrying cost of inventory is 20% per annum.

You are required to CALCULATE:

(i) Re-order quantity

(ii) Re-order level

(iii) Maximum level

(iv) Minimum level

(v) Average stock level

ICAI'S ANSWER

(i) Reorder Quantity(ROQ) = 1,691 kg. *(Refer to working note)*

(ii) Reorder level (ROL) = Maximum usage × Maximum re-order period

= 900 kg. × 8 weeks = 7,200 kg.

(iii) Maximum level = ROL + ROQ – (Min. usage × Min. re-order period)

= 7,200 kg. + 1,691 kg. – (200 kg.× 4 weeks)

= 8,091 kg.

(iv) Minimum level = ROL – (Normal usage × Normal re-order period)

= 7,200 kg. – (550 kg. × 6 weeks)

= 3,900 kg.

(v) Average stock level = $\frac{1}{2}$ (Maximum level + Minimum level)

= $\frac{1}{2}$(8,091 kg. + 3,900 kg.) = 5,995.5 kg.

OR

= Minimum level + $\frac{1}{2}$ ROQ

= 3,900 kg. + $\frac{1}{2}$ × 1,691 kg. = 4,745.5 kg.

Working Note:

Annual consumption of raw material (A) = (550 kg. × 52 weeks) = 28,600 kg.

Cost of placing an order (O) = Rs. 200

Carrying cost per kg. per annum (C) = Rs. 20 × 20% = Rs. 4

Economic order quantity (EOQ) = $\sqrt{\frac{2AO}{C}}$

= $\sqrt{\frac{2 \times 28{,}600 \text{ kgs.} \times \text{Rs.}200}{\text{Rs.}4}}$ = 1,691 Kg. (Approx)

ICAI · Old P3 · MTP Jul 2021 · Series I · question p. 1-2 · answer p. 2

193 MTP **Old P3 Jul 2021 S-II** Q1(b) 5 marks old scheme

A company uses three raw materials Pi, Qu and Ar for a particular product for which the following data applies:

Raw Material	Usage per unit of product (Kg.)	Re-order Quantity (Kg.)	Price per Kg. (Rs.)	Delivery period (in weeks)			Re-order level (Kg.)	Minimum level (Kg.)
				Minimum	Average	Maximum		
Pi	5	10,000	0.10	1	2	3	8,000	?
Qu	2	5,000	0.30	3	4	5	4,750	?
Ar	3	10,000	0.15	2	3	4	?	2,000

Weekly production varies from 350 to 450 units, averaging 400 units of the said product.

WHAT would be the following quantities:

(i) Minimum Stock of Pi?

(ii) Maximum Stock of Qu?

(iii) Re-order level of Ar?

(iv) Average stock level of Pi?

ICAI'S ANSWER

(i) Minimum stock of Pi

Re-order level – (Average consumption × Average time required to obtain delivery)

= 8,000 kg. – (400 units × 5 kg. × 2 weeks) = 4,000 kg.

(ii) Maximum stock of Qu

Re-order level – (Min. Consumption × Min. delivery period) + Re-order quantity

= 4,750 kg. – (350 units × 2 kg. × 3 weeks) + 5,000 kg.

= 9,750 - 2,100 = 7,650 kg.

(iii) Re-order level of Ar

Maximum delivery period × Maximum Usage

= 4 weeks × (450 units × 3 kg.) = 5,400 kg.

OR

= Minimum stock of Ar + (Average consumption × Average delivery time)

= 2,000 kg. + [(400 units × 3 kg.) × 3 weeks] = 5,600 kg.

(iv) Average stock level of Pi

$= \text{Minimum stock level of Pi} + \frac{1}{2} \text{ Re-order quantity}$

$= 4{,}000 \text{ kg.} + \frac{1}{2} 10{,}000 \text{ kg.} = 4{,}000 + 5{,}000 = 9{,}000 \text{ kg.}$

OR

$= \frac{\text{Minimum stock} + \text{Maximum stock}}{2}$ (Refer to Working Note)

$= \frac{4{,}000 + 16{,}250}{2} = 10{,}125 \text{ kg.}$

Working note

Maximum stock of Pi = ROL + ROQ – (Minimum consumption × Minimum delivery period)

= 8,000 kg. + 10,000 kg. – [(350 units × 5 kg.) × 1 week] = 16,250 kg.

ICAI · Old P3 · MTP Jul 2021 · Series II · question p. 1 · answer p. 1-2

3.5 Economic Order Quantity and quantity discounts

52 questions · asked in 35 ICAI documents · 12 exam papers

194 EXAM **Sep 2026** Case scenario · Q1(b)(i) to Q1(b)(ii) 4 marks

CASE SCENARIO

Ultramagic Ltd. manufactures a product TM for tourist recreational purposes. As per the estimates provided by the tourism industry, there will be a market demand of 12,500 units of TM in the coming year. Ultramagic Ltd. is expected to have a market share of 16% of the total market demand of TM.

The company produces one unit of TM from one unit of raw material RW which is purchased at ₹ 2,500 per unit.

Other Information :

Order cost per order	₹ 4,000
Storage rate	9% per annum
Interest rate	5% per annum
Obsolescence rate	2% per annum

Required :

Q1(b)(i) 2 marks

Compute the Economic Order Quantity (EOQ).

Q1(b)(ii) 2 marks

Compute the corresponding ordering cost per annum.

ICAI'S ANSWERS

Q1(b)(i)

ICAI has not published its answers to this paper yet.

Q1(b)(ii)

ICAI has not published its answers to this paper yet.

ICAI · P4 · Sep 2026 exam · question p. 11-12

195 EXAM **Jan 2026** Case Scenario - II · MCQ 6 to MCQ 10 10 marks

CASE SCENARIO - II

Spice Guard Ltd. manufactures 'Pepper Spray' that is used by women for their self-defence. As per the estimates, there will be an average quarterly total market demand of 3,125 units of 'Pepper Spray'. Spice Guard Ltd. is expected to have 8% market share of the total market demand of the 'Pepper Spray'.

The company produces the 'Pepper Spray' using two raw materials - 'OC' and 'OE'. Each unit of pepper spray requires 4 kgs. of 'OC' and 1.6 kgs. of 'OE'.

	'OC'	'OE'
Ordering Cost per order	₹ 3,125	₹ 500
Storage rate	5% per annum	₹ 2.5 per kg. per month
Interest rate	13% per annum	1.25% per quarter
Obsolescence rate	2% per annum	-
Raw material price	₹ 2,000 per kg.	₹200 per kg.

On the basis of above case scenario, you are required to answer the following (MCQs 6 to 10):

MCQ 6 2 marks

What is the annual demand for raw material 'OC' and 'OE'?

A 1,000 kgs & 400 kgs respectively.

B 4,000 kgs & 1,600 kgs respectively

C 12,500 kgs & 5,000 kgs respectively.

D 50,000 kgs & 20,000 kgs respectively

MCQ 7 2 marks

What is the Economic Order Quantity (EOQ) in kgs. for raw material 'OE' required by Spice Guard Ltd.?

A 200 kgs.

B 500 kgs.

C 400 kgs.

D 231 kgs.

MCQ 8 2 marks

What is the Economic Order Quantity (EOQ) in kgs. for raw material 'OC' required by Spice Guard Ltd. and number of orders to be placed in a year?

A 210 kgs. and 19 orders

B 250 kgs. and 16 orders

C 200 kgs. and 20 orders

D 400 kgs and 10 orders

MCQ 9 2 marks

What is the total annual cost of raw material 'OC' at Economic Order Quantity (EOQ) level?

A ₹ 81,20,000

B ₹ 80,00,000

C ₹ 81,00,000

D ₹ 80,40,000

MCQ 10 2 marks

What is the total annual cost of 'OC,' if the company proposes to keep order size to 200 kgs. of 'OC' per order?

A ₹ 81,00,000

B ₹ 80,80,000

C ₹ 81,25,000

D ₹ 81,02,500

ICAI'S ANSWERS

MCQ 6

Answer key: **(B)**

MCQ 7

Answer key: **(A)**

MCQ 8

Answer key: **(B)**

MCQ 9

Answer key: **(C)**

MCQ 10

Answer key: **(D)**

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196 **EXAM** **Sep 2025** Q5(b) **3 marks**

A company manufactures electronic gadgets and uses a specialized component. The company incurs an ordering cost of ₹ 1,250 per order. The carrying cost for storing the specialized components is ₹ 25 per unit per annum. The company's annual production is 90,000 gadgets, and each gadget requires one component for its assembly.

You are required to calculate:

(i) Economic Order Quantity

(ii) Number of orders to be placed in a year

ICAI'S ANSWER

$$\text{EOQ} = \sqrt{\frac{2 \times A \times O}{C}}$$

A = Units consumed during year = 90,000 components

O = Ordering cost per order = ₹ 1,250

C = Inventory carrying cost per unit per annum = ₹ 25

$$\text{EOQ} = \sqrt{\frac{2 \times 90{,}000 \times 1{,}250}{25}} = 3{,}000 \text{ components}$$

No. of orders to be placed in a year

$$= \frac{\text{Total consumption of materials per annum}}{\text{EOQ}}$$

$$= \frac{90{,}000}{3{,}000} = 30 \text{ orders per year}$$

ICAI · P4 · Sep 2025 exam · question p. 24 · answer p. 26

197 EXAM May 2025 Case Scenario - I · MCQ 1 to MCQ 5

CASE SCENARIO - I

Skylark Electronics Company assembles and sells laptops in India. An important component of laptop is it's rechargeable battery. The company buys its monthly requirement of 4,500 batteries and it would buy its annual requirement in 10 equal instalments. The purchase cost of one battery is ₹ 800.

The batteries are used evenly throughout the year in the assembling process on 360 days per year. The ordering cost is ₹ 9000 per order and the inventory carrying cost is 37.50% per annum. The high carrying cost results from the need to keep the batteries in carefully controlled temperature under humid conditions along with high cost of insurance.

Delivery of the batteries from the vendor generally takes 6 days but it may go up to as much as 10 days. The days of delivery time and percentage of their occurrence are shown in the table below:

Delivery Time (Days)	6	7	8	9	10
Percentage of Occurrence (%)	70	15	5	5	5

On the basis of above case scenario, you are required to answer the following MCQs 1 to 5.

MCQ 1

At what quantity of purchase of batteries, the ordering costs will be equal to the inventory carrying costs?

A 1,600

B 1,700

C 1,800

D 1,900

MCQ 2

What will be the total annual cost of purchases as per the quantity calculated in Q-1 above?

A ₹ 3,84,80,000

B ₹ 4,37,40,000

C ₹ 4,29,30,000

D ₹ 5,80,84,000

MCQ 3

Assuming that the company is willing to take a 15% risk of being out of stock, what would be the safety stock and the Re-order point?

A Safety stock 1050 batteries and Re-order point 2250 batteries

B Safety stock 2250 batteries and Re-order point 1050 batteries

C Safety stock 1450 batteries and Re-order point 2850 batteries

D Safety stock 1250 batteries and Re-order point 2650 batteries

MCQ 4

Assuming that the company is willing to take a 5% risk of being out of stock what would be the safety stock and Re-order point?

A Safety stock 1100 batteries and Re-order point 2800 batteries

B Safety stock 1350 batteries and Re-order point 2550 batteries

C Safety stock 1280 batteries and Re-order point 2900 batteries

D Safety stock 1550 batteries and Re-order point 3280 batteries

MCQ 5

Assuming 5% risk of out of stock what would be the total cost of ordering and carrying inventory for one year?

A ₹ 5,40,000

B ₹ 8,15,000

C ₹ 9,45,000

D ₹ 10,80,000

ICAI'S ANSWERS

MCQ 1

Answer key: **(C)**

MCQ 2

Answer key: **(B)**

MCQ 3

Answer key: **ICAI's key gives none of the four options.**

PassBook's note: ICAI's Answer Key prints "None" as the correct answer for MCQ 3 (no option is correct). marks: none because ICAI prints no marks for the Part I MCQs: the Suggested Answers give none, and the published question-paper scan is the Part II booklet only (its instructions print "PART - II 70 Marks" and nothing per MCQ).

MCQ 4

Answer key: **ICAI's key gives none of the four options.**

PassBook's note: ICAI's Answer Key prints "None" as the correct answer for MCQ 4 (no option is correct). marks: none because ICAI prints no marks for the Part I MCQs: the Suggested Answers give none, and the published question-paper scan is the Part II booklet only (its instructions print "PART - II 70 Marks" and nothing per MCQ).

MCQ 5

Answer key: **ICAI's key gives none of the four options.**

PassBook's note: ICAI's Answer Key prints "None" as the correct answer for MCQ 5 (no option is correct). marks: none because ICAI prints no marks for the Part I MCQs: the Suggested Answers give none, and the published question-paper scan is the Part II booklet only (its instructions print "PART - II 70 Marks" and nothing per MCQ).

ICAI · P4 · May 2025 exam · question p. 1 · answer p. 6

198 EXAM **Jan 2025** MCQ 15 MCQ

A company is analysing its inventory management practices and has determined that the Economic Order Quantity (EOQ) is 400 units. The cost incurred for placing a single order is ₹ 25, while the total demand for the year amounts to 8,000 units.

Calculate the Carrying Cost per unit.

A ₹ 2.80 per unit

B ₹ 1.85 per unit

C ₹ 1.58 per unit

D ₹ 2.50 per unit

ICAI'S ANSWER

Answer key: **(D)**

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199 EXAM **Sep 2024** MCQ 12 MCQ

In the automotive machine manufacturing sector, a component is manufactured. The Economic Order Quantity (EOQ) for the component is 1,500 units. The cost of placing an order is ₹ 100, and the carrying cost per annum is 10%. The cost per unit of component is ₹ 20.

Calculate the annual demand for this specific automative component.

A 45,500 units

B 75,000 units

C 36,000 units

D 22,500 units

ICAI'S ANSWER

Answer key: **(D)**

ICAI · P4 · Sep 2024 exam · question p. 5 · answer p. 7

200 EXAM **May 2024** Q1(a) 5 marks

Tesco cycles Ltd. used about 3,60,000 cycle locks per annum and the usage is fairly constant at 30,000 per month. The cycle lock costs ₹ 240 each at wholesale rate and carrying cost is estimated to be 10% of the annual average inventory value. The cost to place an order is ₹ 1200. It takes 45 days to receive delivery from the date of order. In order to avoid any kind of disruption in assembly line, safety stock of 6,500 cycle locks is always maintained by Tesco Cycles Ltd.

(Assume 360 days in a year).

Compute:

(i) E.O.Q.

(ii) The re-order level.

(iii) The company has been offered a quantity discount of 2% on the purchase of cycle locks provided the order size is 30,000 units at a time. Advise whether quantity discount offer can be accepted?

ICAI'S ANSWER

(i) Calculation of Economic Order Quantity

$$\text{EOQ} = \sqrt{\frac{2AO}{C}} = \sqrt{\frac{2 \times 3{,}60{,}000 \text{ units} \times ₹1200}{₹24}} = \textbf{6,000 units}$$

Where,

A = Annual Demand = 3,60,000 units

O = Ordering cost per order = ₹1200

C = Inventory carrying cost per unit per annum = 10% of ₹240 = ₹ 24

(ii) Re-order Level = Safety Stock + Lead Time Consumption

= 6,500 + (1,000 x 45) units = **51,500 units**

Or,

Minimum level of cycle locks + [Average rate of consumption × Average time required to obtain fresh delivery]

= 6,500 + (1,000 x 45) units = **51,500 units**

(iii) Evaluation of Profitability of Different Options of Order Quantity

(a) When EOQ is ordered (order size of 6,000 units)

		(₹)
Purchase Cost	(3,60,000 units × ₹ 240)	8,64,00,000
Ordering Cost	[(3,60,000 units/6,000 units) × ₹ 1,200]	72,000
Carrying Cost	(6,000 units × ₹240 ×½ × 10/100)	72,000
Total Cost		**8,65,44,000**

(b) When Quantity Discount is accepted (order size of 30,000 units)

		(₹)
Purchase Cost	[3,60,000 units × ₹ 235.2 (240-4.8)]	8,46,72,000
Ordering Cost	[(3,60,000 units/30,000 units) × ₹1,200]	14,400
Carrying Cost	(30,000 units × ₹ 235.2 ×½ × 10/100)	3,52,800
Total Cost		**8,50,39,200**

Advise – The total cost of inventory is lower if discount is accepted. Hence, the company is advised to accept the quantity discount.

ICAI · P4 · May 2024 exam · question p. 1 · answer p. 2-3

201 EXAM **Old P3 Nov 2023** Q1(a) 5 marks old scheme

ABC Limited manufactures a product 'AM25' using material 'CEE'.

The following information is available regarding material 'CEE':

Purchase price per unit	₹ 300
Cost of placing an order	₹ 150
Carrying cost per unit per annum	6% of purchase price
Consumption of material 'CEE' per annum	1,94,400 units
Lead time	Average 6 days, Maximum 8 days, Minimum 4 days

Maximum consumption of material 'CEE' per day is 200 kg more than the average consumption per day.

Required:

Calculate the following in relation to material 'CEE':

(i) Economic Order Quantity.

(ii) Reorder Level

(iii) Maximum Stock Level. (Assume 360 days in a year)

ICAI'S ANSWER

(i) Economic Order Quantity (EOQ) = $\sqrt{\frac{2AO}{C}}$

Where, A= Annual demand for the material CEE = 1,94,400 Kgs

O = Ordering cost = ₹ 150

C = Carrying cost per unit per annum = 6% of ₹ 300 = 18

EOQ = $\sqrt{\frac{2 \times 1,94,400 \times 150}{18}}$ = **1,800 Units (Kgs.)**

(ii) Re-order level (ROL) = Maximum consumption# × Maximum lead time

ROL = 740 × 8 = **5,920 Kg.**

Maximum Consumption = Average consumption +200 kg

= $\frac{1,94,400}{360}$ + 200 = 540 + 200 Kg = 740 Kg.

Maximum lead time = 8 days

(iii) Maximum Stock level = Re-order quantity + Re-order level – (Min. consumption* × Min. lead time)

= 1,800 + 5,920 – (340×4)

= 7,720 – 1,360 = **6,360 Kg**

*Minimum consumption = 2 × Average consumption – Maximum Consumption

= 2 × 540 – 740

= 1080 – 740 = 340 kg.

ICAI · Old P3 · Nov 2023 exam · question p. 1 · answer p. 3

202 EXAM **Old P3 Nov 2022** Q1(c) 5 marks old scheme

MM Ltd. uses 7500 valves per month which is purchased at a price of ₹ 1.50 per unit. The carrying cost is estimated to be 20% of average inventory investment on an annual basis. The cost to place an order and getting the delivery is ₹ 15. It takes a period of 1.5 months to receive a delivery from the date of placing an order and a safety stock of 3200 valves is desired.

You are required to determine:

(i) The Economic Order Quantity (EOQ) and the frequency of orders.

(ii) The re-order point.

(iii) The Economic Order Quantity (EOQ) if the valve cost ₹ 4.50 each instead of 1.50 each.

(Assume a year consists of 360 days)

ICAI'S ANSWER

(i) Calculation of Economic Order Quantity

Annual requirement (A)	= 7500×12= 90,000 Valves
Cost per order (O)	= ₹ 15
Inventory carrying cost (i)	= 20%
Cost per unit of spare (c)	= ₹ 1.5
Carrying cost per unit (i × c)	= ₹ 1.5 × 20% = ₹ 0.30

Economic Order Quantity (EOQ) = $\sqrt{\frac{2 \times A \times O}{i \times c}}$

= $\sqrt{\frac{2 \times 90{,}000 \times 15}{0.3}}$ = 3,000 Valves

Frequency of order or Number of Orders = 90,000/3,000 = 30 orders.

So Order can be placed in every 12 (360days/30) days

(ii) Re-order Quantity = {Maximum Consumption X Maximum lead time} + safety Stock

= {7500X1.5} + 3200 = 14,450 Valves

(iii) Calculation of Economic Order Quantity if valve costs ₹ 4.50

Carrying cost is 20% of ₹ 4.50 = ₹ 0.90

Economic Order Quantity (EOQ) = $\sqrt{\frac{2 \times A \times O}{i \times c}}$

= $\sqrt{\frac{2 \times 90{,}000 \times 15}{0.9}}$

= 1732.0508 units or 1733 Valves

ICAI · Old P3 · Nov 2022 exam · question p. 2 · answer p. 4

203 EXAM **Old P3 May 2022** Q1(a) 5 marks old scheme

A Limited a toy company purchases its requirement of raw material from S Limited at ₹ 120 per kg. The company incurs a handling cost of ₹ 400 plus freight of ₹ 350 per order. The incremental carrying cost of inventory of raw material is ₹ 0.25 per kg per month. In addition the cost of working capital finance on the investment in inventory of raw material is ₹ 15 per kg per annum. The annual production of the toys is 60,000 units and 5 units of toys are obtained from one kg. of raw material.

Required:

(i) Calculate the Economic Order Quantity (EOQ) of raw materials.

(ii) Advise, how frequently company should order to minimize its procurement cost. Assume 360 days in a year.

(iii) Calculate the total ordering cost and total inventory carrying cost per annum as per EOQ.

ICAI'S ANSWER

Annual requirement of raw material in kg. (A) = $\frac{60{,}000 \text{ units}}{5 \text{ units per kg.}}$ = **12,000 kg.**

Ordering Cost (Handling & freight cost) (O) = ₹ 400 + ₹ 350 = ₹ 750

Carrying cost per unit per annum i.e. inventory carrying cost + working capital cost (c × i)

= (₹ 0.25 × 12 months) + ₹15

= ₹ 18 per kg.

(i) E.O.Q.= $\sqrt{\frac{2 \times 12{,}000\text{kgs.} \times ₹\ 750}{₹\ 18}}$ **= 1,000 kg.**

(ii) Frequency of orders for procurement:

Annual consumption (A) = 12,000 kg.

Quantity per order (EOQ) = 1,000 kg.

No. of orders per annum $(\frac{A}{EOQ}) = \frac{12{,}000\text{kg.}}{1{,}000\text{kg.}} = 12$

Frequency of placing orders (in months) = $\frac{12\text{months}}{12\text{orders}}$ **= 1 months**

Or, (in days) = $\frac{360\text{days}}{12\text{orders}}$ = **30 days**

(iii) Calculation of total ordering cost and total inventory carrying cost as per EOQ:

	Amount/Quantity
Size of the order	1,000 kg.
No. of orders	12
Cost of placing orders	**₹ 9,000** (12 orders × ₹ 750)
Inventory carrying cost	**₹ 9,000** (1,000 kg. × ½ × ₹ 18)
Total Cost	₹18,000

ICAI · Old P3 · May 2022 exam · question p. 1 · answer p. 2-3

204 EXAM **Old P3 Nov 2020** Q3(b) 10 marks old scheme

An automobile company purchases 27,000 spare parts for its annual requirements. The cost per order is ₹ 240 and the annual carrying cost of average inventory is 12.5%. Each spare part costs ₹ 50.

At present, the order size is 3,000 spare parts.

(Assume that number of days in a year = 360 days)

Find out:

(i) How much the company's cost would be saved by opting EOQ model?

(ii) The Re-order point under EOQ model if lead time is 12 days.

(iii) How frequently should orders for procurement be placed under EOQ model?

ICAI'S ANSWER

Working Notes:

Annual requirement (A)	= 27,000 units
Cost per order (O)	= ₹ 240
Inventory carrying cost (i)	= 12.5%
Cost per unit of spare (c)	= ₹ 50
Carrying cost per unit (i × c)	= ₹ 50 × 12.5% = **₹ 6.25**

Economic Order Quantity (EOQ) = $\sqrt{\frac{2 \times A \times O}{i \times c}}$

= $\sqrt{\frac{2 \times 27{,}000 \times 240}{6.25}}$ = **1440 units**

(i) Calculation of saving by opting EOQ:

	Existing Order policy	EOQ Model
No. of orders	9 $\left(\frac{27{,}000}{3{,}000}\right)$	**18.75 or 19** $\left(\frac{27{,}000}{1{,}440}\right)$
A. Ordering Cost (₹)	**2,160** (₹ 240 × 9)	**4,500** $\left\{₹\ 240 \times \left(\frac{27{,}000}{1{,}440}\right)\right\}$
B. Carrying cost (₹)	**9,375** $\left(\frac{3{,}000 \times ₹\ 6.25}{2}\right)$	**4,500** $\left(\frac{1{,}440 \times ₹\ 6.25}{2}\right)$
Total cost (A+B) (₹)	**11,535**	**9,000**

Savings of Cost by opting EOQ Model = ₹ 11,535 – ₹ 9,000 = ₹ 2,535

(ii) Re-order point under EOQ:

Re-order point/ Re-order level = Maximum consumption × Maximum lead time

Consumption per day = $\frac{27{,}000 \text{ units}}{360 \text{ days}}$ = **75 units**

Re-order point/ Re-order level = 75 units × 12 days **= 900 units**

(iii) Frequency of Orders (in days):

$\frac{360 \text{ days}}{\text{No.of orders a year}} = \frac{360 \text{ days}}{19}$ = 18.95 days or 19 days

ICAI · Old P3 · Nov 2020 exam · question p. 11 · answer p. 12-13

205 EXAM **Old P3 Nov 2019** Q1(a) 5 marks old scheme

Surekha Limited produces 4,000 Litres of paints on a quarterly basis. Each Litre requires 2 kg of raw material. The cost of placing one order for raw material is ₹ 40 and the purchasing price of raw material is ₹ 50 per kg. The storage cost and interest cost is 2% and 6% per annum respectively. The lead time for procurement of raw material is 15 days.

Calculate Economic Order Quantity and Total Annual Inventory Cost in respect of the above raw material.

ICAI'S ANSWER

Working:

Calculation of Annual demand of raw material

= 4,000 Litres (per quarter) x 4 (No. of Quarter in a year) x 2 kg. (raw material required for each Litre of paint)

= **32,000 kg.**

Calculation of Carrying cost

Storage rate = 2%

Interest Rate = 6%

Total = 8% per annum

Carrying cost per unit per annum = 8% of ₹ 50 = ₹ 4 per unit per annum

(i) EOQ = $\sqrt{\frac{2 \times \text{Annual demand (A)} \times \text{Ordering Cost per order(O)}}{\text{Carrying cost per unit per annum (C)}}}$

= $\sqrt{\frac{2 \times 32{,}000\text{kg} \times ₹\ 40}{₹\ 4}}$ = **800** Kg

(ii) Total Annual Inventory Cost

Purchasing cost of 32,000 kg @ ₹ 50 per kg = ₹ 16,00,000

Ordering Cost $\left(\frac{32{,}000\text{kg}}{800\text{kg}} \times ₹\ 40\right)$ = ₹ 1,600

Carrying Cost of Inventory $\left(\frac{15 \text{ days}}{30 \text{ days}} \times 800 \text{ kg} \times ₹\ 4\right)$ = ₹ 1,600

₹ 16,03,200

ICAI · Old P3 · Nov 2019 exam · question p. 1 · answer p. 3

206 **RTP** **Sep 2026** **Q5**

M/s Esha Materials Private Limited produces a product which names "ESS". The consumption of raw material for the production of "ESS" is 210 Kgs to 350 Kgs per week. Other information is as follows:

Procurement Time:	5 to 9 Days
Purchase price of Raw Materials:	₹ 100 per kg
Ordering Cost per Order:	₹ 200
Storage Cost:	1% per month plus ₹ 2 per unit per annum

Consider 365 days a year.

You are required to CALCULATE:

(a) Economic Order Quantity

(b) Re-Order Level (ROL)

(c) Maximum Stock Level

(d) Minimum Stock Level

(e) Average Stock Level

(f) Number of Orders to be placed per year

(g) Total Inventory Cost

(h) If the supplier is willing to offer 1% discount on purchase of total annual quantity in two orders, whether offer is acceptable?

(i) If the answer is no, what should be the counteroffer w.r.t. percentage of discount?

ICAI'S ANSWER

As procurement time is given in days, consumption should also be calculated in days:

Maximum Consumption per Day: $\frac{350}{7}$ = 50 Kgs

Minimum Consumption per Day: $\frac{210}{7}$ = 30 Kgs.

Average Consumption per Day: $\frac{(50+30)}{2}$ = 40 Kgs

(a) Calculation of Economic Order Quantity (EOQ)

Annual consumption of Raw Materials (A): 40 Kgs x 365 days = 14,600 Kgs

Storage or Carrying Cost per unit per annum (C):(₹ 100 x 1% x 12 months) + ₹ 2 = ₹ 14

Ordering Cost (O): ₹ 200 per Order

EOQ = $\sqrt{\frac{2 \times A \times O}{C}}$

= $\sqrt{\frac{2 \times 14{,}600 \times 200}{14}}$ = 646 Kgs.

(b) Re-Order Level (ROL) = (Maximum consumption Rate × Maximum Procurement Time)

= 50 kgs per day × 9 days

= 450 kgs

(c) Maximum Stock Level = Recorder Level + Recorder Quantity – (Minimum Consumption Rate × Minimum Procurement Time)

= 450 kgs + 646 kgs - (30 kgs x 5 days)

= 946 kgs

(d) Minimum Stock Level = Recorder Level – (Average consumption Rate × Average Procurement Time)

= 450 kgs – (40 kgs X 7 days)

= 170 kgs

(e) Average Stock Level = $\frac{\text{Maximum Stock Level} + \text{Minimum Stock Level}}{2}$

= $\frac{946 \text{ kgs} + 170 \text{ kgs}}{2}$

= 558 kgs

(f) Number of Orders to be placed per year

= $\frac{\text{Annual Consumption of Raw Materials}}{\text{EOQ}}$

$= \frac{14600 \text{ kgs}}{646 \text{ kgs}}$

= 22.60 Orders or 23 Orders

(g) Total Inventory Cost

Cost of Materials (A x Purchase Price) (14600 kgs x ₹ 100)	=₹ 14,60,000
Total Ordering Cost (No. of Orders x O) (23 Orders x 200)	= ₹ 4,600
Total Carrying Cost (EOQ / 2 x C) (646 kgs / 2 x ₹ 14)	= ₹ 4,522
Total Inventory Cost	₹ 14,69,122

(h) If the supplier is willing to offer 1% discount on purchase of total annual quantity in two orders:

Offer Price	= ₹ 100 × 99%	= ₹ 99
Revised Carrying Cost	= (₹ 99 × 1% x 12 months) + ₹ 2	= ₹ 13.88
Revised Order Quantity	= 14600 kgs / 2 Orders	= 7300 kgs

Total Inventory Cost at Offer Price

Cost of Materials (A x Purchase Price) (14600 kgs x ₹ 99)	= ₹ 14,45,400
Total Ordering Cost (No. of Orders x O) (2 Orders x 200)	= ₹ 400
Total Carrying Cost (EOQ / 2 x C) (7300 kgs / 2 x ₹ 13.88)	= ₹ 50,662
Total Inventory Cost	₹ 14,96,462

Advice: As total inventory cost at offer price is ₹ 27,340 (14,96,462 – 14,69,122) higher, offer should not be accepted.

(i) Counter-offer:

Let Discount Rate	= z%
Counter-Offer Price	= ₹ 100 – z% = ₹ 100 – z
Revised Carrying Cost	= [(₹ 100 – z) x 1% x 12 months] + ₹ 2
	= ₹ 12 -0.12z + ₹ 2
	= ₹ 14 – 0.12z

Total Inventory Cost at Counter-Offer Price

Cost of Materials (A x Purchase Price) [14600 kgs x (₹ 100 – z)] = ₹ 14,60,000 – 14,600z

Total Ordering Cost (No. of Orders x O) (2 Orders x 200) = ₹ 400

Total Carrying Cost (EOQ / 2 x C) [7300 kgs / 2 x (₹ 14 – 0.12z)]

= ₹ 51,100 – 438z

Total Inventory Cost	= ₹ 15,11,500 – 15038z
₹ 14,69,122	= ₹ 15,11,500 – 15038z
Or 15038z	= 42,378
Or z	= 2.82

Therefore, discount should be at least 2.82% in offer price.

ICAI · P4 · RTP Sep 2026 · question p. 7-8 · answer p. 26-28

207 RTP **May 2026** Q6

Hari Udyog is a small-scale manufacturing unit producing a single product X using two raw materials PK and BK in the input ratio of 3 : 2. The input–output ratio in production is 5 : 3.

Raw material PK is perishable and becomes obsolete if not used within 5 days of purchase, whereas raw material BK is durable and can be stored for more than one year. Material PK is procured locally with a lead time of 1 to 2 days, while material BK is purchased from a neighbouring state and takes 2 to 4 days to reach the factory.

The estimated sales volume for April 2025 is 30,000 kg, and the same level of sales is expected to continue throughout the year. The company operates for 25 days a month, and production is carried out evenly.

The purchase price per kilogram (excluding taxes) is ₹ 15 for material PK and ₹ 22 for material BK. To place an order, the company incurs an administrative cost of ₹ 120 per order. The annual carrying cost is estimated at 15% of the value for material PK and 5% for material BK.

Currently, material PK is purchased in lots of 8,000 kg to avail a 10% trade discount on the market price. GST at 5% is applicable on material PK, for which input tax credit is available. Material BK is subject to 2% CST, for which no input tax credit is available.

You are required to calculate:

(i) Economic Order Quantity (EOQ) for each material;

(ii) Maximum stock level for Material PK;

(iii) Calculate saving/ loss in Material PK if purchase quantity equals to EOQ.

ICAI'S ANSWER

Workings:

Monthly Production of X = 30,000 kgs.

Raw Material Required = $\frac{30,000}{3} \times 5 = 50,000$ kgs.

Material PK = $\frac{50,000}{5} \times 3 = 30,000$ kg.

Material BK = $\frac{50,000}{5} \times 2 = 20,000$ kg.

(i) Calculation of Economic Order Quantity (EOQ):

Material PK = $\sqrt{\frac{2 \times \text{Annual consumption} \times \text{Order cost}}{\text{Carrying cost per unit p.a.}}}$

$$= \sqrt{\frac{2 \times (30{,}000 \text{ kg.} \times 12 \text{ months}) \times ₹120}{(15\% \text{ of } ₹15)}}$$

$$= \sqrt{\frac{8{,}64{,}00{,}000}{2.25}} = 6{,}196.77 \text{ kg. or } 6{,}197 \text{ kg.}$$

$$\text{Material BK} = \sqrt{\frac{2 \times (20{,}000 \text{ kg.} \times 12 \text{ months}) \times ₹120}{(5\% \text{ of } ₹22.44*)}}$$

$$= \sqrt{\frac{5{,}76{,}00{,}000}{1.122}} = 7{,}164.97 \text{ or } 7{,}165 \text{ kg.}$$

*Purchase price + 2% CST = ₹ 22 + 2% of ₹ 22 = ₹ 22.44

(ii) Calculation of Maximum Stock level: Since, the Material PK is perishable in nature and it required to be used within 5 days, hence, the Maximum Stock Level shall be lower of two:

(a) Stock equal to 5 days consumption

$$= \frac{30{,}000 \text{ kg.}}{25 \text{ days}} \times 5 \text{ days} = 6{,}000 \text{ kg.}$$

(a) Maximum Stock Level for Material PK:

Re-order Quantity + Re-order level – (Min consumption* × Min. lead time)

Where, Re-order Quantity = 8,000 kg.

Re-order level = Max. Consumption* × Max. Lead time

= 30,000/25 × 2 days = 2,400 kg.

Maximum stock Level = 8,000 kg. + 2,400 kg. - (30,000/25 × 1 day)

= 10,400 – 1,200 = 9,200 kg.

Stock required for 5 days consumption is lower than the maximum stock level calculated through the formula. Therefore, Maximum Stock Level will be 6,000 kg.

(*Since, production is processed evenly throughout the month hence material consumption will also be even.)

(iii) Calculation of Savings/ loss in Material PK if purchase quantity equals to EOQ.

	Purchase Quantity = 8,000 kg.	Purchase Quantity = EOQ i.e. 6,197 kg.
Annual consumption	3,60,000 kg. (30,000 × 12 months)	3,60,000 kg. (30,000 × 12 months)
No. of orders [Note- (i)]	60 (3,60,000 ÷ 6,000)	60 (3,60,000 ÷ 6,000)
Ordering Cost (a)	₹ 7,200 (₹ 120 × 60)	₹ 7,200 (₹ 120 × 60)
Carrying Cost (b) [Note- (ii)]	₹ 8,100 (15% of ₹ 13.50 × 4,000)	₹ 6,972 (15% of ₹ 15 × 3,098.5)
Purchase Cost (c) (for good portion)	₹ 48,60,000 (₹ 13.50 × 3,60,000)	₹ 54,00,000 (₹ 15 × 3,60,000)
Loss due to obsolescence (d) [Note- (iii)]	₹ 16,20,000 [₹ 13.5 × (60 × 2,000)]	₹ 1,77,300 [₹ 15 × (60 × 197)]

	Purchase Quantity = 8,000 kg.	Purchase Quantity = EOQ i.e. 6,197 kg.
Total Cost [(a) + (b) + (c) + (d)]	₹ 64,95,300	₹ 55,91,472

If purchase quantity equals to EOQ, there will be a saving of ₹ 9,03,828 i.e. ₹ 64,95,300 - ₹ 55,91,472.

Notes: (i) As after 5 days of purchase the Material PK gets obsolete, the quantity in excess of 5 days consumption i.e. 6,000 kg. are wasted. Hence, after 6,000 kg. a fresh order needs to be given.

(ii) Carrying cost is incurred on average stock of Materials purchased.

(iii) the excess quantity of material gets obsolete and loss has to be incurred.

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208 RTP **Sep 2025** Q7

XYZ Ltd is engaged in the business of manufacturing product "A". The material required for the manufacture of Product A is Material "B". Each unit of Product A requires 2 Kgs of Material B, however the yield of each unit of Material B is 80% of the input.

The Details governing the production requirements for the year is as follows:

a. Projected sales of product A for the period - 10,000 units.

b. Opening stock of product A - 948 units

c. Opening stock of Raw Material B - 2,630 Kgs

d. Ordering cost per order - ₹500

e. Rate of Overdraft - 12.5%

f. Obsolescence Rate - 12.5%

g. Purchase price of Material B - ₹ 20 per Kg

h. Current purchase policy - Quarterly

i. Max usage is more than minimum usage by - 100 Kgs

j. Waiting time from date of order - 4 to 8 days

Based on the above information determine the following:

i. Economic Order Quantity

ii. Re-Order Quantity

iii. Re-Order Level

iv. Maximum Stock

v. Minimum Stock

vi. Profitability analysis on EOQ v. ROQ

vii. If a volume discount of 5% is provided for a quarterly purchase by the supplier, will your answer to (vi) above change?

ICAI'S ANSWER

Projected sales of product A	-	10,000

Unit Requirement of Material B	-	2 Kgs of B per unit of A
Yield of Material B	-	80%
Hence the revised requirement of Material B	-	2.5 Kgs (2/80%)
Opening stock of Product A	-	948
Product A to be produced	-	9,052
Material B Required @ 2.5 Kgs per unit of A	-	22,630 Kgs
Less: Opening stock of Material B	-	2,630 Kgs
Purchase Requirement (A) of Material B	-	20,000 Kgs
Order cost per order	-	₹ 500
Purchase price per unit	-	₹ 20
Carrying Cost (Interest + Obsolescene)	-	25% of 20 = ₹ 5

(i) EOQ = $\sqrt{\frac{2AO}{C}} = \sqrt{\frac{2 \times 20{,}000 \text{ units} \times ₹500}{₹5}}$ - **2,000 Kgs**

(ii) ROQ (Quarterly Purchase policy) - 5,000 Kgs

Material Requirement	-	22,630
Average Consumption	-	62 Kgs

Max usage – Min Usage = 100Kgs

Max usage + Min Usage = 124Kgs (Formulating average usage with max & min)

Hence Minimum Usage = 12 Kgs and Maximum Usage = 112 Kgs

Lead time – 4 to 8 days

(iii) Re-Order level = Max Usage x Max Lead Time

= 112 Kgs x 8 Days **= 896 Kgs**

(iv) Maximum Stock = Re-order Level + Economic order Quantity – (Min Use x Min Lead time)

= 896 + 2000 – (12x4) **= 2,848 Kgs**

(If ROQ is assumed as 5,000 the answer will be different)

Maximum Stock = Re-order Level + Re-order Quantity – (Min Use x Min Lead time)

= 896 + 5,000 – (12x4) **= 5,848 Kgs**

(v) Minimum Stock = Re-Order level – (Average Use x Average Lead)

= 896 – (62x6) **= 524 Kgs**

(vi) Profitability analysis on EOQ v. ROQ

Particulars	EOQ	ROQ
Annual Requirement (A)	20,000	20,000
Quantity Ordered(B)	2,000	5,000
No. of Orders (C) = (A)/(B)	10	4
Order cost @ ₹500	5,000	2,000
Average Inventory	1,000	2,500
Carrying Cost @ ₹5	5,000	12,500
Associated Costs	10,000	14,500
Savings in EOQ		4,500

(vii) After 5% Discount

Particulars	EOQ	ROQ
Annual Requirement (A)	20,000	20,000
Quantity Ordered(B)	2,000	5,000
No of Orders (C) = (A)/(B)	10	4
Order cost @ ₹500	5,000	2,000
Average Inventory	1,000	2,500
Carrying Cost @ ₹5 & 4.75	5,000	11,875
Discount	0	-20000
Net Costs	10,000	-6,125
Benefit of ROQ over EOQ		16,125

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209 RTP **May 2025** Q8

Catalist Ltd. is a distributor of industrial chemicals, providing the chemical in drum packaging.

Each drum of the chemical costs ₹ 200 from a supplier and the company sells it for ₹ 240.

Annual demand is estimated to be for 2,50,000 drums.

The cost of delivery is estimated at ₹ 100 per order and the annual variable holding cost per drum at ₹ 180 plus 10% of purchase cost.

Based on above data, the managing director calculates the economic order quantity and suggests that it should serve as the foundation for purchasing decisions in upcoming periods.

However, the purchasing manager states that the managing director has ignored his share of bonus of 10% of the amount by which total annual inventory holding and order costs (before such remuneration) are below ₹ 2,00,000. He further points out that the suppliers also offer quantity discounts on purchase orders, i.e. if the order size is 1,000 drums or above, the price per drum is only ₹ 199.60, compared to ₹ 200 when an order is between 500 and 999 drums.

You are required to:

(i) CALCULATE the economic order quantity as calculated by the company's managing director.

(ii) COMMENT whether Catalist Ltd. can look forward to the quantity discount offered for purchasing 1,000 drums, after CALCULATING total cost considering purchasing manager's bonus along with supplier quantity discounts.

ICAI'S ANSWER

(i) Economic order quantity (EOQ) as calculated by the company's managing director

EOQ = $\sqrt{\frac{2AO}{C}}$

where A = annual inventory requirement,

O = ordering cost per order and

C = carrying cost per unit per annum

= $\sqrt{\frac{2 \text{x} 2{,}50{,}000 \text{ drums x } ₹ 100}{[₹ 180 + (10\% \text{ of } ₹ 200)]}}$

= 500 units

(ii) Comparison of total cost considering purchasing manager's bonus and supplier quantity discounts

Particulars		At EOQ of 500 units (₹)	If considered quantity discount at 1000 units (₹)
Ordering Cost	[(2,50,000 units/500 units) × ₹ 100]	50,000	
	[(2,50,000 units/1000 units) × ₹ 100]		25,000
Carrying Cost	{500 units/ 2 x [(₹ 180 + (10% of ₹ 200)]}	50,000	
	{1,000 units/ 2 x [(₹ 180 + (10% of ₹ 199.60)]}		99,980
		1,00,000	1,24,980
Purchasing manager's bonus	10% of (₹ 2,00,000 - ₹ 1,00,000)	10,000	

Particulars		At EOQ of 500 units (₹)	If considered quantity discount at 1000 units (₹)
	10% of (₹ 2,00,000 - ₹ 1,24,980)		7,502
Annual inventory cost	2,50,000 units x ₹ 200	5,00,00,000	
	2,50,000 units x ₹ 199.60		4,99,00,000
Total Cost		**5,01,10,000**	**5,00,32,482**

In above comparison, the potential savings from purchasing in bulk outweigh the higher carrying costs associated with holding more inventory. Thus, Catalist Ltd. may look forward to the quantity discount offered at 1,000 units.

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210 RTP **Jan 2025** Q8

Ani Ltd. uses 6 kg. of Material 'EXE' to produce 1 finished unit of Product 'EME'. The current demand of Product 'EME' is 16,000 units quarterly. 1 kg. of Material 'EXE' costs ₹ 40. The cost relating to quotations, documentation works, employee cost directly attributable to the procurement of material, every-time the order is made, is ₹ 2,000. The cost of fund invested in inventories, cost of storage, insurance cost, etc. is estimated to be 15% per annum of average inventory.

You are required the following:

(i) CALCULATE the Economic Order Quantity for Material 'EXE'.

(ii) COMMENT, should Ani Ltd. accept an offer of 2.5% discount by the supplier of Material 'EXE', if supply of the annual requirement of the Material is made in 4 equal installments?

ICAI'S ANSWER

Annual demand of material 'EXE'

= 16,000 units (per quarter) x 4 (No. of Quarter in a year) x 6 kg. (for every finished product)

= 3,84,000 kg.

(i) Calculation of Economic Order Quantity (EOQ) for material 'EXE'

$$EOQ = \sqrt{\frac{2 \text{ x Annual demand x ordering cost}}{\text{Carrying cost per unit per annum}}}$$

$$= \sqrt{\frac{2x3,84,000kg.x\ ₹2,000}{₹40x15\%}} = 16,000 \text{ kg.}$$

(ii) Evaluation of Cost under different options of 'order quantity'.

Particulars	When EOQ is ordered	When discount of 2.5% is accepted and supply is in 4 equal installments
Order size	16,000 kg.	96,000 kg. $= \frac{3,84,000 \text{ kg.}}{4}$

Particulars	When EOQ is ordered	When discount of 2.5% is accepted and supply is in 4 equal installments
No. of orders	24 $= \frac{3,84,000 \text{ kg.}}{16,000 \text{ kg.}}$	4
Purchase Cost per kg.	₹ 40	₹ 39 {₹ 40 - (₹ 40 × 2.5%)}
Total Purchase Cost (A)	₹ 1,53,60,000 (3,84,000 kg. x ₹ 40)	₹ 1,49,76,000 (3,84,000 kg. x ₹ 39)
Ordering Cost (B)	₹ 48,000 (24 orders x ₹ 2,000)	₹ 8,000 (4 orders x ₹ 2,000)
Carrying Cost (C)	₹ 48,000 $= \frac{16,000 \text{ kg.}}{2}$ x 15% x ₹ 40	₹ 2,80,800 $= \frac{96,000 \text{ kg.}}{2}$ x 15% x ₹ 39
Total Cost (A + B + C)	**₹ 1,54,56,000**	**₹ 1,52,64,800**

COMMENT – The total cost is lower if Ani Ltd. accept an offer of 2.5% discount by the supplier, when supply of the annual requirement of material 'EXE' is made in 4 equal installments.

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211 RTP **Old P3 May 2023** Q1 old scheme

Reliable India Pvt Ltd is a startup company engaged in manufacturing of Agro Tech product from a raw material, which is purchased at ₹190 per kg. The company incurs a handling cost of ₹1,470 plus, freight of ₹770 per order. The incremental carrying cost of inventory of raw material is ₹3 per kg per month. In addition, the cost of working capital finance on the investment in inventory of raw material is ₹20 per kg per annum. The annual production of the product is 1,50,000 units and 3 units are obtained from one kg. of raw material. Assume 360 days in a year.

Required:

(i) Calculate the economic order quantity of raw materials.

(ii) Determine, how frequently company should order for procurement be placed.

(iii) If the company proposes to rationalize placement of orders on quarterly basis, determine the percentage of discount in the price of raw materials should be negotiated?

ICAI'S ANSWER

(i) Calculation of Economic Order Quantity (E.O.Q)

Annual requirement (usage) of raw material in kg. (A) = $\frac{1,50,000 \text{units}}{3 \text{unitsperkg.}}$ = 50,000kg.

Ordering Cost (Handling & freight cost) (O) = ₹1,470 + ₹770 = ₹2,240

Carrying cost per unit per annum (C) i.e. inventory carrying cost + working capital cost= (₹3 × 12 months) + ₹20 = ₹56 per kg.

E.O.Q = $\sqrt{\frac{2AO}{C}} = \sqrt{\frac{2 \times 50{,}000 \text{ kg.} \times ₹\ 2{,}240}{₹\ 56}} = 2{,}000$ kg.

(ii) Frequency of placing orders for procurement :

Annual consumption (A) = 50,000 kg.

Quantity per order (E.O.Q) = 2,000 kg.

No. of orders per annum = $\frac{A}{E.O.Q} = \frac{50{,}000\text{kg.}}{2{,}000\text{kg.}} = 25$ orders

Frequency of placing orders (in days) = $\frac{360\text{days}}{25\text{orders}} = 14.4$ Days

(iii) Percentage of discount in the price of raw materials to be negotiated:

Particulars	On Quarterly Basis	On E.O.Q Basis
1. Annual Usage (in Kg.)	50,000 kg.	50,000 kg.
2. Size of the order	12,500 kg.	2,000 kg.
3. No. of orders (1 ÷ 2)	4	25
4. Cost of placing orders or Ordering cost	₹ 8,960	₹ 56,000
(No. of orders × Cost per order)	(4 order × ₹ 2,240)	(25 orders × ₹ 2,240)
5. Inventory carrying cost	₹3,50,000	₹56,000
(Average inventory × Carrying cost per unit)	(12,500 kg. × ½ × ₹ 56)	(2,000 kg. × ½ × ₹ 56)
6. Total Cost (4 + 5)	₹ 3,58,960	₹ 1,12,000

When order is placed on quarterly basis the ordering cost and carrying cost increased by ₹2,46,960 (₹3,58,960 - ₹1,12,000). So, discount required = ₹ 2,46,960

Total annual purchase = 50,000 kg. × ₹190 = ₹95,00,000 So, Percentage of discount to be negotiated = = $\frac{₹\ 2{,}46{,}960}{₹\ 95{,}00{,}000} \times 100 = 2.60\%$

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212 RTP **Old P3 Nov 2022** Q1 old scheme

M/s Tanishka Materials Private Limited produces a product which names "ESS". The consumption of raw material for the production of "ESS" is 210 Kgs to 350 Kgs per week. Other information is as follows:

Procurement Time:	5 to 9 Days
Purchase price of Raw Materials:	₹ 100 per kg
Ordering Cost per Order:	₹ 200
Storage Cost:	1% per month plus ₹ 2 per unit per annum

Consider 365 days a year.

You are required to CALCULATE:

(a) Economic Order Quantity

(b) Re-Order Level (ROL)

(c) Maximum Stock Level

(d) Minimum Stock Level

(e) Average Stock Level

(f) Number of Orders to be placed per year

(g) Total Inventory Cost

(h) If the supplier is willing to offer 1% discount on purchase of total annual quantity in two orders, whether offer is acceptable?

(i) If the answer is no, what should be the counteroffer w.r.t. percentage of discount?

ICAI'S ANSWER

As procurement time is given in days, consumption should also be calculated in days:

Maximum Consumption per Day: $\frac{350}{7}$ = 50 Kgs

Minimum Consumption per Day: $\frac{210}{7}$ = 30 Kgs.

Average Consumption per Day: $\frac{(50+30)}{2}$ = 40 Kgs

(a) Calculation of Economic Order Quantity (EOQ)

Annual consumption of Raw Materials (A): 40 Kgs x 365 days = 14,600 Kgs

Storage or Carrying Cost per unit per annum (C):(₹ 100 x 1% x 12 months) + ₹ 2 = ₹ 14

Ordering Cost (O): ₹ 200 per Order

EOQ = $\sqrt{\frac{2 \times A \times O}{C}}$

= $\sqrt{\frac{2 \times 14{,}600 \times 200}{14}}$ = 646 Kgs.

(b) Re-Order Level (ROL) = (Maximum consumption Rate × Maximum Procurement Time)

= 50 kgs per day × 9 days

= 450 kgs

(c) Maximum Stock Level = Recorder Level + Recorder Quantity – (Minimum Consumption Rate × Minimum Procurement Time)

= 450 kgs + 646 kgs - (30 kgs X 5 days)

= 946 kgs

(d) Minimum Stock Level = Recorder Level – (Average consumption Rate × Average Procurement Time)

= 450 kgs – (40 kgs X 7 days)

= 170 kgs

(e) Average Stock Level = $\frac{\text{Maximum Stock Level} + \text{Minimum Stock Level}}{2}$

= $\frac{946 \text{ kgs} + 170 \text{ kgs}}{2}$

= 558 kgs

(f) Number of Orders to be placed per year

$$= \frac{\text{Annual Consumption of Raw Materials}}{\text{EOQ}}$$

$$= \frac{14600 \text{ kgs}}{646 \text{ kgs}}$$

= 22.60 Orders or 23 Orders

(g) Total Inventory Cost

Cost of Materials (A x Purchase Price) (14600 kgs x ₹ 100)= ₹ 14,60,000

Total Ordering Cost (No. of Orders x O) (23 Orders x 200) = ₹ 4,600

Total Carrying Cost (EOQ / 2 x C) (646 kgs / 2 x ₹ 14) = ₹ 4,522

Total Inventory Cost ₹ 14,69,122

(h) If the supplier is willing to offer 1% discount on purchase of total annual quantity in two orders:

Offer Price = ₹ 100 x 99% = ₹ 99

Revised Carrying Cost = (₹ 99 x 1% x 12 months) + ₹2 = ₹ 13.88

Revised Order Quantity = 14600 kgs / 2 Orders = 7300 kgs

Total Inventory Cost at Offer Price

Cost of Materials (A x Purchase Price) (14600 kgs x ₹ 99) = ₹ 14,45,400

Total Ordering Cost (No. of Orders x O) (2 Orders x 200) = ₹ 400

Total Carrying Cost (EOQ / 2 x C) (7300 kgs / 2 x ₹13.88) = ₹ 50,662

Total Inventory Cost ₹ 14,96,462

Advice: As total inventory cost at offer price is ₹ 27,340 (14,96,462 – 14,69,122) higher, offer should not be accepted.

(i) Counter-offer:

Let Discount Rate = z%

Counter-Offer Price = ₹ 100 – z% = ₹ 100 – z

Revised Carrying Cost = [(₹ 100 – z) x 1% x 12 months] + ₹ 2 = ₹ 12 -0.12z + ₹ 2

= ₹ 14 – 0.12z

Total Inventory Cost at Counter-Offer Price

Cost of Materials (A x Purchase Price) [14600 kgs x (₹ 100 – z)] = ₹ 14,60,000 – 14,600z

Total Ordering Cost (No. of Orders x O) (2 Orders x 200) = ₹ 400

Total Carrying Cost (EOQ / 2 x C) [7300 kgs / 2 x (₹ 14 – 0.12z)] = ₹ 51,100 – 438z

Total Inventory Cost ₹ 15,11,500 – 15038z

₹ 14,69,122 = ₹ 15,11,500 – 15038z

Or 15038z = 42,378

Or z = 2.82

Therefore, discount should be at least 2.82% in offer price.

213 MTP **Sep 2026 S-I** Case Scenario 2 · MCQ 6 to MCQ 10 10 marks

CASE SCENARIO 2

Himalayan Bakers Pvt. Ltd., a mid-sized bakery chain headquartered in Dehradun, produces its signature product "Multigrain Loaf" for retail and hotel supply. Each loaf requires 2.5 kg of Multigrain Flour Mix (MFM) as the key raw material.

Based on a sales forecast prepared by the company's regional sales heads (factoring in new hotel contracts and an upcoming bakery outlet expansion), Himalayan Bakers expects to sell 12,000 units of Multigrain Loaf in the coming financial year.

The purchase manager, Ms. Tanya Bhatt, has compiled the following details regarding MFM from the stores and finance departments:

Particulars		Details
(i)	Re-order Quantity	280 kg less than the EOQ
(ii)	Maximum consumption per day	25 kg more than the average consumption per day
(iii)	Opening stock of MFM	1,200 kg
(iv)	Lead time from suppliers	Minimum 3 days, Maximum 7 days
(v)	Purchase price of MFM	₹ 95 per kg
(vi)	Opening stock of finished Multigrain Loaves	1,100 units
(vii)	Bank's Cash Credit interest rate (treated as carrying cost %)	14.4% per annum
(viii)	Documentation/paperwork cost per order	₹ 540

(Assume 364 days in a year for all calculations and also assume that henceforth, the closing stocks are not required.)

You are required to answer the following questions (MCQs 6 to 10)

MCQ 6 2 marks

What is the Re-order Quantity (ROQ) of MFM?

A 1,434 kg

B 1,154 kg

C 1,716 kg

D 999 kg

MCQ 7 2 marks

What is the Maximum Stock Level of MFM?

A 1,690 kg

B 1,720 kg

C 1,217 kg

D 699 kg

MCQ 8 2 marks

What is the Minimum Stock Level of MFM?

A 699 kg

B 374 kg

C 323 kg

D 318 kg

MCQ 9 2 marks

What is the total cost of purchase of raw material if the company orders at ROQ?

A ₹ 25,88,750

B ₹ 24,95,063

C ₹ 26,09,166

D ₹ 27,00,000

MCQ 10 2 marks

What is the extra annual cost the company bears by ordering at ROQ instead of the true EOQ? (Number of orders should be rounded off to next whole number.)

A ₹ 244

B ₹ 351

C ₹ 540

D ₹ 695

ICAI'S ANSWERS

ICAI's working for this case

Working Notes:

(i) Computation of Annual Consumption & Annual Demand for raw material 'MFM':

Particulars	Amount
Sales forecast of the product 'Multigrain Loaf'	12,000 units
Less: Opening stock of 'Multigrain Loaf'	1,100 units
Fresh units of 'Multigrain Loaf' to be produced	10,900 units
Raw material required to produce 10,900 units (10,900 units × 2.5 kg.)	27,250 kg.
Less: Opening Stock of 'MFM'	1,200 kg.
Annual demand for raw material 'MFM'	**26,050 kg.**

(ii) Computation of Economic Order Quantity (EOQ)

$$\text{EOQ} = \sqrt{\frac{2 \times A \times O}{C}}$$

Where:

- A = Annual demand = 26,050 kg.
- O = Ordering cost = ₹540 per order
- C = Carrying cost per kg p.a. = ₹95 × 14.4% = ₹13.68

$$EOQ = \sqrt{\frac{2 \times 26{,}050 \times 540}{13.68}}$$

$$= \sqrt{\frac{2{,}81{,}34{,}000}{13.68}}$$

EOQ = 1,434 kg. (approx.)

(iii) Re-order Level

= Maximum consumption per day × Maximum lead time

Average Consumption per day = 26,050 kg. ÷ 364 days = 71.57 kg.

Maximum Consumption per day = 71.57 kg. + 25 kg. = 96.57 kg.

= 96.57 kg. × 7 days

Re-order Level = 676kg. (approx.)

(iv) Minimum Consumption per Day of Raw Material 'MFM'

Average Consumption per day = 71.57 kg.

Since:

Average Consumption = (Min. consumption + Max. consumption) / 2

71.57 kg. = (Min. consumption + 96.57 kg.) / 2

Minimum Consumption = 143.14 kg. – 96.57 kg. = 46.57 kg.

MCQ 6

(B) 1,154 kg

Re-order Quantity

= EOQ – 280 kg.

= 1,434 kg. – 280 kg.

Re-order Quantity = 1,154 kg.

MCQ 7

(A) 1,690 kg

Maximum Stock Level

= Re-order level + Re-order Quantity – (Minimum consumption per day × Minimum lead time)

= 676 kg. + 1,154 kg. – (46.57 kg. × 3 days)

= 1,830 kg. – 139.71 kg.

Maximum Stock Level = 1,690 kg. (approx.)

MCQ 8

(D) 318 kg

Minimum Stock Level

= Re-order level – (Average consumption per day × Average lead time)

Average lead time = (3+7)/2 = 5 days

= 676 kg. – (71.57 kg. × 5 days)

= 676 kg. – 357.85 kg.

Minimum Stock Level = 318 kg. (approx.)

MCQ 9

(B) ₹ 24,95,063

Total Cost of Purchase of Raw Material if the company orders at ROQ

- Purchase Cost = 26,050 kg. × ₹ 95 = ₹ **24,74,750**
- No. of orders = 26,050 kg. ÷ 1,154 kg. = 22.57 → **23 orders**
- Ordering Cost = 23 orders × ₹ 540 = ₹ **12,420**
- Average Inventory = 1,154 kg. ÷ 2 = 577 kg.
- Carrying Cost = 577 kg. × ₹ 13.68 = ₹ **7,893**

Total Cost of Purchase = ₹ 24,74,750 + ₹ 12,420 + ₹ 7,893 = ₹ 24,95,063

MCQ 10

(A) ₹ 244

Extra Annual Cost of ordering at ROQ instead of the true EOQ

	Particulars	When purchasing the ROQ	When purchasing the EOQ
I	Order quantity	1,154 kg.	1,434 kg.
II	No. of orders a year	26,050/1,154 = 22.57 → 23 orders	26,050/1,434 = 18.17 → 19 orders
III	Ordering Cost	23 orders × ₹ 540 = ₹ 12,420	19 orders × ₹ 540 = ₹ 10,260
IV	Average Inventory	1,154/2 = 577 kg.	1,434/2 = 717 kg.
V	Carrying Cost	577 kg. × ₹ 13.68 = ₹ 7,893	717 kg. × ₹ 13.68 = ₹ 9,809
VI	Total Cost	₹ 20,313	₹ 20,069

Extra Cost incurred due to not ordering EOQ = ₹ 20,313 – ₹ 20,069 = ₹ 244

ICAI · P4 · MTP Sep 2026 · Series I · question p. 4 · answer p. 4, 4-5, 5

214 MTP **Sep 2026 S-II** Q1(b) 5 marks

Vaibhav Industries Ltd. uses chemical-A in one of its finished products. The chemical-A is purchased from a vendor outside India. The company purchases 36,000 lit. of chemical-A per year at the rate of ₹ 900 per lit. plus import duty@10%on such purchases.

The chemical-A is used evenly throughout the year in the production process on a 360- day-per-year basis. The company incurs ₹ 1, 75,000 on one year agreement for material supply with the vendor and it estimates that ₹ 35,000 will be incurred to place a single purchase order. The chemical-A is needed to be kept in a very carefully controlled temperature and humidity conditions. Vaibhav Industries Ltd. incurs 1.5% and 0.2676% of the value of inventory as storage cost and as insurance cost respectively.

Delivery from the vendor generally takes 12 days, but it can take as much as 16 days. The days of delivery time and percentage of their occurrence are shown in the following tabulation:

Delivery time (days):	12	13	14	15	16
Percentage of occurrence:	70	10	10	5	5

Required:

(i) Compute the economic order quantity (EOQ).

(ii) Assume the company is willing to assume a 10% risk of being out of stock. What would be the safety stock? The re-order point?

(iii) Assume 5% stock-out risk. What would be the total cost of ordering and carrying inventory for one year?

ICAI'S ANSWER

Working Note 1:

Risk of being out of stock

Delivery Time	Percentage of currency (%)	Cumulative percentage of occurrence (%)	Risk Of non Occurrence (%)
12 days	70	70	30
13 days	10	80	20
14 days	10	90	10
15 days	5	95	5
16 days	5	100	0

Working Note 2:

(a) Ordering cost per order (0)- ₹ 35,000

(b) Cost per liter of Chemical - A

Rate per liter	₹ 900
Add: import duty@10%	₹ 90
	₹ 990

Carrying cost per liter per annum of Chemical-A

1.7676% (1.5% + 0.2676%) of 990 = ₹17.50

(i) Economic Order Quantity (E.O.Q)

$$\sqrt{\frac{2AC}{C}} = \sqrt{\frac{2 \times 36{,}000 \text{ ltrs.} \times ₹\ 35{,}000}{₹\ 17.50}} = 12{,}000 \text{ litres}$$

(ii) Safety Stock at 10% risk of being out of stock

Safety stock required for two days i.e. for 13th and 14th day

Per day consumption $\frac{36{,}000 \text{ Ltr.}}{360 \text{ days.}} = 100$ litres

Safety stock = 100 × 2 days = 200 litres

Re-Order Point = Safety Stock + Average lead time x Average consumption

= 200 +12 x 100

= 1400 liters.

(iii) At 5% risk of being out stock, safety stock will be safety stock for three days

100 liters x 3 days = 300 liters

Total Ordering cost = $\frac{36,000 \text{ Ltr.}}{12,000 \text{ Ltr.}} \times ₹ 35,000 = ₹ 1,05,000$

Total Carrying cost of inventory = (Safety stock. + Average inventory) × Carrying cost per liter per annum

= $(300 + \frac{1}{2} \times 12,000 \text{ ltr.}) ₹ 17.50 = ₹ 1,10,250$

:. Total cost of ordering & carrying inventory = 1,05,000 + 1,10,250 = ₹2,15,250

[**Note:** Amount of ₹ 1,75,000 incurred on making agreement for material supply will be apportioned over the entire quantity of 36,000 ltr and included with cost of Chemical-A. However, for the purpose of calculating carrying cost i.e. storage cost and insurance cost, only invoice cost of material is taken. Invoice cost consists of cost per liter of chemical-A plus import duty.]

ICAI · P4 · MTP Sep 2026 · Series II · question p. 8-9 · answer p. 7-9

215 MTP **Jan 2026 S-I** Q4(b) 6 marks

The yearly production of a company's product which has a steady market is 40,000 units. Each unit of a product requires 1 kg. of raw material. The cost of placing one order for raw material is ₹ 1,000 and the inventory carrying cost is ₹ 20 per annum. The lead time for procurement of raw material is 36 days and a safety stock of 1,000 kg. of raw materials is maintained by the company.

The company has been able to negotiate the following discount structure with the raw material supplier:

Order quantity (kg.)	Discount (₹)
Upto 6,000	NIL
6,001 – 8,000	4,000
8,001 – 16,000	20,000
16,001 – 30,000	32,000
30,001 – 45,000	40,000

You are required to:

(i) CALCULATE the re-order point considering 30 days in a month.

(ii) PREPARE a statement showing the total cost of procurement and storage of raw material after considering the discount of the company elects to place one, two, four or five orders in the year.

(iii) STATE the number of orders which the company should place to minimize the costs after taking EOQ also into consideration.

ICAI'S ANSWER

Working notes

1. Annual production = 40,000 units

2. Raw material required for 40,000 units (40,000 units × 1 kg.) = 40,000 kg.

3. EOQ = $\sqrt{\frac{2 \times 40{,}000 \text{ kgs.} \times ₹\ 1{,}000}{₹\ 20}}$ = 2,000 kgs.

4. Total cost of procurement and storage when the order size is equal to EOQ or 2,000 kg.

No. of orders (40,000 kg. ÷ 2,000 kg.)	= 20 times
Ordering cost (20 orders × ₹1,000)	= ₹ 20,000
Carrying cost (₹) (½ × 2,000 kg. × ₹ 20)	= ₹ 20,000
Total cost	₹ 40,000

(i) **Re-order point** = Safety stock + Lead time consumption

= 1,000 kg. + $\frac{40{,}000\text{kg.}}{360\text{days}} \times 36\text{days}$

= 1,000 kg. + 4,000 kg. = 5,000 kg.

(ii) **Statement showing the total cost of procurement and storage of raw materials** *(after considering the discount)*

Order size	No. of orders	Total cost of procurement	Average stock	Total cost of storage of raw materials	Discount	Total cost
Kg.		**(₹)**	**Kg.**	**(₹)**	**(₹)**	**(₹)**
(1)	**(2)**	**(3)=(2)×₹1,000**	**(4)=½×(1)**	**(5)=(4)×₹ 20**	**(6)**	**(7)=[(3)+ (5)– (6)**
40,000	1	1,000	20,000	4,00,000	40,000	3,61,000
20,000	2	2,000	10,000	2,00,000	32,000	1,70,000
10,000	4	4,000	5,000	1,00,000	20,000	84,000
8,000	5	5,000	4,000	80,000	4,000	81,000

(iii) Number of orders which the company should place to minimize the costs after taking EOQ also into consideration is 20 orders each of size 2,000 kg. The total cost of procurement and storage in this case comes to ₹ 40,000, which is minimum. *(Refer to working notes 3 and 4)*

ICAI · P4 · MTP Jan 2026 · Series I · question p. 13-14 · answer p. 14-15

216 MTP **Jan 2026 S-II Q1(c)** 4 marks

A Company manufactures a special product which requires a component 'Alpha'. The following

particulars are collected for the year 2022-23:

(i)	Annual demand of Alpha	8,000 units
(ii)	Cost of placing an order	₹ 200 per order
(iii)	Cost per unit of Alpha	₹ 400
(iv)	Carrying cost p.a.	20%

The company has been offered a quantity discount of 4% on the purchase of 'Alpha' provided the order size is 4,000 components at a time.

Required:

(i) COMPUTE the economic order quantity

(ii) STATE whether the quantity discount offer can be accepted.

ICAI'S ANSWER

(i) Calculation of Economic Order Quantity

$$\text{EOQ} = \sqrt{\frac{2AO}{C}} = \sqrt{\frac{2 \times 8{,}000 \text{ units} \times ₹200}{₹400 \times 20/100}} = \textbf{200 units}$$

(ii) Evaluation of Profitability of Different Options of Order Quantity

(a) When EOQ is ordered

		(₹)
Purchase Cost	(8,000 units × ₹ 400)	32,00,000
Ordering Cost	[(8,000 units/200 units) × ₹ 200]	8,000
Carrying Cost	(200 units × ₹ 400 × ½ ×20/100)	8,000
Total Cost		**32,16,000**

(b) When Quantity Discount is accepted

		(₹)
Purchase Cost	(8,000 units × ₹ 384*)	30,72,000
Ordering Cost	[(8,000 units/4000 units) × ₹ 200]	400
Carrying Cost	(4000 units × ₹ 384 ×½ ×20/100)	1,53,600
Total Cost		**32,26,000**

*Unit Cost = ₹ 400

Less: Quantity Discount @ 4% = 16

Purchase Cost = 400 – 16 = ₹ 384

Advise – The total cost of inventory is lower if EOQ is adopted. Hence, the company is advised not to accept the quantity discount.

ICAI · P4 · MTP Jan 2026 · Series II · question p. 9 · answer p. 9-10

217 MTP Sep 2025 S-I Q1(a) 5 marks

A Ltd. manufactures a product X which requires two raw materials A and B in a ratio of 1:4. The sales department has estimated a demand of 5,00,000 units for the product for the year. To produce one unit of finished product, 4 units of material A is required.

Stock position at the beginning of the year is as below:

Product- X	12,000 units
Material A	24,000 units
Material B	52,000 units

To place an order the company has to spend ₹ 15,000. The company is financing its working capital using a bank cash credit @13% p.a.

Product X is sold at ₹ 1,040 per unit. Material A and B are purchased at ₹ 150 and ₹ 200 respectively.

Required:

COMPUTE economic order quantity (EOQ):

(i) If purchase order for the both materials is placed separately.

(ii) If purchase order for the both materials is not placed separately.

ICAI'S ANSWER

(i) Workings:

Annual production of Product X = Annual demand – Opening stock

= 5,00,000 – 12,000 = 4,88,000 units

Annual requirement for raw materials = Annual production × Material per unit – Opening stock of material

Material A = 4,88,000 × 4 units – 24,000 units = 19,28,000 units

Material B = 4,88,000 × 16 units – 52,000 units = 77,56,000 units

(i) Computation of EOQ when purchase order for the both materials is placed separately

$$\text{EOQ} = \sqrt{\frac{2 \times \text{Annual Requirement for material} \times \text{Ordering cost}}{\text{Carrying cost per unit per annum}}}$$

$$\text{Material A} = \sqrt{\frac{2 \times 19{,}28{,}000 \text{ units} \times ₹\ 15{,}000}{13\% \text{ of } ₹\ 150}}$$

$$= \sqrt{\frac{38{,}56{,}000 \times ₹15{,}000}{₹19.5}} = 54{,}462 \text{ units}$$

$$\text{Material B} = \sqrt{\frac{2 \times 77{,}56{,}000 \text{ units} \times ₹\ 15{,}000}{13\% \text{ of } ₹\ 200}}$$

$$= \sqrt{\frac{1{,}55{,}12{,}000 \times ₹15{,}000}{₹26}}$$

= 94,600 units

(ii) Computation of EOQ when purchase order for the both materials is not placed separately

$$\text{Material A \& B} = \sqrt{\frac{2 \times (19{,}28{,}000 + 77{,}56{,}000) \text{ units} \times ₹15{,}000}{13\% \text{ of } ₹190}}$$

$$= \sqrt{\frac{1,93,68,000 \times ₹15,000}{₹24.7}} = 1,08,452 \text{ units}$$

$$\text{Material A} = \frac{1,08,452 \times 19,28,000}{96,84,000} = 21,592 \text{ units}$$

$$\text{Material A} = \frac{1,08,452 \times 77,56,000}{96,84,000} = 86,860 \text{ units}$$

$$* \frac{(₹150 \times 19,28,000) + (₹200 \times 77,56,000)}{(19,28,000 + 77,56,000)} = ₹ 190$$

ICAI · P4 · MTP Sep 2025 · Series I · question p. 7-8 · answer p. 5-6

218 MTP **May 2025 S-II** Q1(a) 5 marks

Aditya Ltd. has a monthly requirement for an item of raw material is 1,000 units. The purchase price per unit of material is ₹60. The cost of processing an order is ₹ 540 and the carrying cost is 20%. There is a single supplier for the material which offers quantity discounts as under:

Order Quantity (in units)	Price per unit (₹)
Less than 2,000 units	60.00
2,000 units and less than 4,000 units	59.80
4,000 units and less than 6,000 units	59.50
6,000 units and less than 8,000 units	58.90
8,000 units and above	58.40

The company uses the cash credit facility provided by the company's banker to finance its raw material purchase. The bank due to its own infrastructural constraint, can accommodate a maximum of five fund transfer (NEFT/ RTGS) requests for any single beneficiary per annum. The company in short term is unable to arrange any other source of finance.

Required:

(i) CALCULATE the optimum purchase order size for the company;

(ii) CALCULATE the order level where the company could have minimised its total cost;

(iii) The amount of loss that the company has to bear due to bank's inability to process fund transfer requests.

ICAI'S ANSWER

(i) **Calculation of optimum purchase order size or Economic Order Quantity (EOQ):**

$$\text{EOQ} = \sqrt{\frac{2 \times A \times O}{C \times i}}$$

Where, A = Annual requirement for inventory = 1,000 units × 12 months = 12,000 units

O = Ordering cost = ₹ 540

C = Cost per unit = ₹ 60

C × i = Carrying cost per unit per annum = 20% × ₹ 60 = ₹ 12

$$\text{EOQ} = \sqrt{\frac{2 \times 12,000\text{units} \times ₹540}{₹12}} = \sqrt{\frac{1,29,60,000}{12}} = 1,039.23 \text{ or } 1,039 \text{ units.}$$

(ii)

Order Size (in units)	1,500	2,000	4,000	6,000	8,000
No. of order	8	6	3	2	1.5*
Cost per order (₹)	540	540	540	540	540
Average inventory	750	1,000	2,000	3,000	4,000
Cost per unit (₹)	60.00	59.80	59.50	58.90	58.40
Carrying cost per unit @ 20% (₹)	12.00	11.96	11.90	11.78	11.68
(a) Ordering Cost (₹)	4,320	3,240	1,620	1,080	810
(b) Carrying cost (₹)	9,000	11,960	23,800	35,340	46,720
(c) Material cost (₹)	7,20,000	7,17,600	7,14,000	7,06,800	7,00,800
Total Cost {(a) + (b) + (c)} (₹)	7,33,320	7,32,800	7,39,420	7,43,220	7,48,330

*(This may also be taken as 2 orders)

At order level of 2,000 units, the total cost to the company is least.

(iii) **Calculation of amount of loss due to bank's inability to process more than five fund transfer requests:**

No. of orders	5
Purchase quantity per order (12,000 units ÷ 5)	2,400 units
Cost per unit	₹ 59.80
(a) Ordering Cost (₹ 540 × 5 orders)	₹ 2,700
(b) Carrying Cost (20% of ₹ 59.80 ×1,200 units)	₹ 14,352
(c) Material Cost (₹ 59.80 × 12,000 units)	₹ 7,17,600
Total Cost {(a) + (b) + (c)}	₹ 7,34,652
Minimum cost at 2,000 units order level	₹ 7,32,800
Loss	₹ 1,852

ICAI · P4 · MTP May 2025 · Series II · question p. 7-8 · answer p. 6-7

219 MTP **Sep 2024 S-I** Case scenario · MCQ 1 to MCQ 5 10 marks

CASE SCENARIO

Tropic Pvt Ltd was engaged in the business of manufacturing Product P. The product P required 2 units of Material R. The company intends to sell 24,000 units of Product P and does not wish to retain any closing stock. However the opening stock of Product P is 4,000 units. Raw Material R has to be

procured after considering the opening stock of R amounting to 10,000 units. The technical team further confirms that the yield in the course of manufacture of Product P is 80% of the input.

The company presently procures its annual requirement of materials on a quarterly basis from its regular supplier enjoying a discount of 2.5% on the invoice price of the material of ₹ 20 per unit. Every time the company places orders for Material R, it incurs ₹ 125 for each of the order placed. The company also has taken a rented warehouse for storing material R and the annual cost of storage is ₹ 10 per unit.

The company appointed Mr. T a Chartered Accountant to review the cost of inventory and provide measures of improvement of cost. After reviewing the material purchase and consumption pattern, Mr. T suggested that the implementation of Wilson's EOQ would be beneficial to the company. He emphasized that the change in the quantity ordered would result in reduction of inventory carrying costs.

Mr. T further reviewed the labour costing and identified that the employees were paid overtime wages to ensure timely completion of projects. Overtime wages comprised of daily wage and 100% of daily wages as overtime premium. Based on the cost record it was understood that every month had 180 hours of regular working hours which was remunerated at ₹ 200 per hour and Overtime of 20 hours which was remunerated at ₹ 400 per hour. Mr. T suggested that the above time taken may be considered as standard and a scheme of Incentive be introduced to reduce overtime cost. He further indicated that Rowan scheme of incentive be used to measure performance and the improved productivity per hour would be 125 units per hour.

In this regard, address the following queries in line with the suggestions provided by Mr. T to Tropic Pvt Ltd.

MCQ 1 2 marks

The annual requirement of Material R to meet the target sales of 24,000 units of Product P is:

A 48,000 units

B 60,000 units

C 40,000 units

D 50,000 units

MCQ 2 2 marks

The ordering quantity as per the current inventory policy and the proposed Wilson's Economic order quantity of Material R are:

A Order Quatity as per the current inventory policy – 10,000 units & Economic Order Quantity – 1,000 units

B Order Quantity as per the current inventory policy – 15,000 units & Economic Order Quantity – 1,225 units

C Order Quantity as per the current inventory policy – 12,000 units & Economic Order Quantity – 1,095 units

D Order Quantity as per the current inventory policy – 12,500 units & Economic Order Quantity – 1,118 units

MCQ 3 2 marks

The net savings to inventory cost on migration from the current inventory policy to the Wilson's Economic Order Quantity policy would be:

A Savings from EOQ as compared to current discount policy – ₹ 26,820

B Savings from EOQ as compared to current discount policy – ₹ 20,500

C Savings from EOQ as compared to current discount policy – ₹ 33,253

D Savings from EOQ as compared to current discount policy – ₹ 25,546

MCQ 4 2 marks

Incentive payable under the Rowan Incentive scheme amounts to:

A ₹ 7,500

B ₹ 6,400

C ₹ 6,000

D ₹ 8,000

MCQ 5 2 marks

The savings in labour cost achieved by implementation of incentive scheme over the overtime payments amounts to:

A ₹ 9,600

B ₹ 5,600

C ₹ 8,000

D ₹ 3,200

ICAI'S ANSWERS

MCQ 1

(C) 40,000 units.

Projected Sales of Product P – 24,000 units

Less: Opening stock of Product P- (4,000 units)

Product P to be produced- 20,000 units

Raw Material required- 50,000 units (20,000 x 2/80% yield)

Opening stock of Material R available- 10,000 units

Material to be procured - 40,000 units.

MCQ 2

(A) Order Quantity as per the current inventory policy – 10,000 units and EOQ – 1,000 units

Annual requirement - Procurement- 40,000 units

Order Quantity as per the current inventory policy (Quarterly) - 10,000 units

Ordering Cost- ₹125 per order

Carrying Cost- ₹ 10 per unit p.a.

EOQ - 1,000 units.

MCQ 3

(B) Savings from EOQ as Compared to current discount policy – ₹ 20,500

Associated Costs under EOQ:

Ordering Costs = No. of orders x Ordering cost per order

No of orders = Annual Requirement/ EOQ (or) current order quantity

Hence No of orders = 40

Therefore Ordering Cost = 40 x 125 = ₹ 5,000.

Carrying cost = Average Inventory x Carrying cost per unit per annum

Average Inventory = (EOQ/ current order quantity)/2

= 1,000/2 = 500

Carrying cost = 500 x 10 = ₹ 5,000

Associated Costs under EOQ = Ordering cost + Carrying Cost

= ₹ 10,000 ---------- A

Associated Costs under current inventory policy:

No of orders = 4 (Quarterly)

Ordering cost = 4 x 125 = ₹ 500

Average inventory = 10,000/2 = 5,000

Carrying cost = 5,000x10 = 50,000

Associated Costs = 50,000+500 = 50,500

Less: Discount = 20,000

Net cost = 30,500. ---------- B

Incremental Cost = B – A = 20,500

MCQ 4

(B) ₹ 6,400

Time taken under the Overtime regime 180 Hours + 20 Hours overtime = 200 Hours

Time to be taken under the Incentive regime

Units to be produced = 20,000 units

Units produced per hour under incentive scheme = 125 units

Time taken = 160 Hours

Time saved = 200 – 160 = 40 hours.

Incentive under Rowan scheme = (Time saved/Time allowed) x time taken x Rate

= (40/200) x 160x200 = ₹ 6,400.

MCQ 5

(B) ₹ 5,600

Cost under the Overtime scheme:

Base wage = 200 x 200 = 40,000

OT Premium = 20 x 200 = 4,000

Total Wages under Overtime scheme = 44,000

Cost under Incentive scheme:

Base Wage = 160 hours x 200 = 32,000

Incentive = 6,400

Total wages paid = 38,400

Savings in Incentive scheme over Overtime scheme = ₹ 5,600.

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220 MTP **May 2024 S-II** Q1(b) 5 marks

DSM Ltd manufactures speed boats which require propeller TP-M4. The following particulars are collected for the year 2023-24:

(i) Annual demand of TP-M4 12,000 units

(ii) Cost of placing an order ₹1,200 per order

(iii) Cost per unit of TP-M4 is ₹1,740/-

(iv) Carrying cost p.a. 12%

The company has been offered a quantity discount of 5 % on the purchase of TP-M4, provided the order size is 6,000 units at a time.

Required to:

(i) COMPUTE the economic order quantity (EOQ)

(ii) ADVISE whether the quantity discount offer can be accepted.

ICAI'S ANSWER

(i) Calculation of Economic Order Quantity

$$EOQ = \sqrt{\frac{2 \times \text{Annual Demand} \times \text{Ordering Cost}}{\text{Carrying cost per unit per annum}}}$$

$$= \sqrt{\frac{2 \times 12{,}000 \text{ units} \times ₹1{,}200}{₹1{,}740 \times 0.12}} = 371 \text{ units (Approx)}$$

(ii) Evaluation of Profitability of Different Options of Order Quantity

(a) When EOQ is ordered

		(₹)
Purchase Cost	(12,000 units × ₹ 1,740)	2,08,80,000.00
Ordering Cost*	[(12,000 units ÷ 371 units) i.e. 33 × ₹ 1,200]	39,600.00
Carrying Cost**	(371 units × ₹ 1,740 × ½ × 12/100)	38,732.40
Total Cost		2,09,58,332.40

(b) When Quantity Discount of 5% is offered.

		(₹)
Purchase Cost	(12,000 units × ₹ 1,740 × 0.95)	1,98,36,000.00
Ordering Cost*	[(12,000 units ÷ 6,000 units) × ₹1,200]	2,400.00
Carrying Cost**	(6,000 units × ₹1,653 × ½ × 12/100)	5,95,080.00
Total Cost		2,04,33,480.00

Advise – The total cost of inventory is lower if quantity discount offer is accepted. Hence, the company is advised to accept the quantity discount.

* Ordering Cost = $\frac{\text{Annual Demand}}{\text{Order Quantity}} \times$ Cost of placing an order

** Carrying Cost = $\frac{\text{Cost per unit} \times \text{Quantity ordered} \times \text{Carrying Cost}}{2}$

ICAI · P4 · MTP May 2024 · Series II · question p. 6 · answer p. 5

221 MTP **Old P3 Nov 2023 S-II** Q1(c) 5 marks old scheme

P Limited produces product 'P'. It uses annually 60,000 units of a material 'Rex' costing ₹ 10 per unit. Other relevant information are:

Cost of placing an order	:	₹ 800 per order
Carrying cost	:	15% per annum of average inventory
Re-order period	:	10 days
Safety stock	:	600 units

The company operates 300 days in a year.

You are required to calculate:

(i) Economic Order Quantity for material 'Rex'.

(ii) Re-order Level.

(iii) Maximum Stock Level.

(iv) Average Stock Level.

ICAI'S ANSWER

(i) Economic Order Quantity (E.O.Q)

$$= \sqrt{\frac{2 \times \text{Annual requirement of 'Rex'} \times \text{Ordering cost per order}}{\text{Annual carrying cost per unit per annum}}}$$

$$= \sqrt{\frac{2 \times 60{,}000 \text{ units} \times ₹\ 800}{₹10 \times 15\%}} = \sqrt{\frac{9{,}60{,}00{,}000}{₹1.5}} = \textbf{8,000 units}$$

(ii) Re-order Level = Safety Stock + (Normal daily Usage × Re-order period)

$$= 600 + \left(\frac{60{,}000 \text{ units}}{300 \text{ days}} \times 10 \text{ days}\right)$$

= 600 + 2,000

= 2,600 units

(iii) Maximum Stock Level = E.O.Q (Re-order Quantity) + Safety Stock

= 8,000 units + 600 units

= 8,600 units

(iv) Average Stock Level = Minimum Stock level + $\frac{1}{2}$ Re-order Quantity

= 600* + $\frac{1}{2}$ 8,000 units

= 4,600 units

OR

$$\text{Average Stock Level} = \frac{\text{Maximum Stock level} + \text{Minimum Stock level}}{2}$$

$$= \frac{8{,}600 \text{ units} + 600 \text{ units}}{2}$$

= 4,600 units

* Minimum Stock Level = Re-order level – (Normal daily usage × Re-order period)

$$= 2{,}600 - \left(\frac{60{,}000 \text{ units}}{300 \text{ days}} \times 10 \text{ days}\right)$$

= 2,600 – 2,000

= 600 units

OR

Minimum Stock Level = Safety Stock level = 600 units

ICAI · Old P3 · MTP Nov 2023 · Series II · question p. 1-2 · answer p. 2-3

222 MTP **Old P3 May 2023 S-I** Q1(a) 5 marks old scheme

Joy Toy Limited deals in trading of 'superhero' toy figure. The annual demand for the toy car is 14,400 units. The company incurs fixed order placement and transportation cost of ₹212 each time an order is placed. Each toy costs ₹ 450 and the trader has a carrying cost of 25 percent p.a. The company has been offered a quantity discount of 8% on the purchase of 'superhero' toy figure provided the order size is 5,000 units at a time.

Required:

(i) COMPUTE the economic order quantity

(ii) STATE whether the quantity discount offer can be accepted.

ICAI'S ANSWER

(i) Calculation of Economic Order Quantity (EOQ) = $\sqrt{\frac{2AO}{C}}$

$$= \sqrt{\frac{2 \text{ x} 14{,}400 \text{ units x } ₹212}{₹450 \text{ x } 25\%}} = 233 \text{ units}$$

(ii) Evaluation of Profitability of Different Options of Order Quantity

(A)	When EOQ is ordered	(₹)
Purchase Cost	(14,400 units x Rs. 450)	64,80,000
Ordering Cost	[(14,400 units/233 units) x Rs. 212]	13,102
Carrying Cost	(233 units x 1/2 × 450 × 25%)	13,106
Total Cost		**65,06,208**

(B) When Quantity Discount of 8% is accepted

		(₹)
Purchase Cost	(14,400 units x Rs. 414)	59,61,600
Ordering Cost	[(14,400 units/5,000 units) x Rs212]	611
Carrying Cost	(5,000 units x 1/2 x Rs.414 × 25%)	2,58,750
Total Cost		**62,20,961**

Advise – The total cost of inventory is lower if quantity discount is accepted.

The company would save Rs. 2,85,247 (Rs. 65,06,208 - Rs. 62,20,961).

Note: Figures may change slightly because of approximation and decimals)

ICAI · Old P3 · MTP May 2023 · Series I · question p. 1 · answer p. 1

223 MTP **Old P3 Nov 2022 S-I** Q1(d) 5 marks old scheme

A company produces a product 'AB' by using two raw materials - 'Material Ae' and 'Material Be' in the ratio of 5:3.

A sales volume of 50,000 kgs is estimated for the month of December by the managers expecting the trend will continue for entire year. The ratio of input and output is 8:5.

Other Information about Raw Material Ae is as follows:

Purchase Price	₹ 150 per kg
Re-order period	2 to 3 days
Carrying Cost	12%

Note: Material Ae is perishable in nature and if not used within 3.5 days of purchase it becomes obsolete.

To place an order for material 'Ae', the company has to incur an administrative cost of ₹ 375 per order. At present, material 'Ae' is purchased in a lot of 7,500 kgs. to avail the discount on purchase. Company works for 25 days in a month and production is carried out evenly.

You are required to CALCULATE:

(i) Economic Order Quantity (EOQ) for Material Ae;

(ii) Maximum stock level for Material Ae.

ICAI'S ANSWER

(i) Monthly production of AB = 50,000 kgs

Raw material required = 50,000/5 x 8 = 80,000 kgs

Material Ae and Material Be ratio = 5:3

Therefore, material Ae = 80,000/8 x 5 =50,000 kgs

Calculation of EOQ = $\sqrt{\frac{2 \text{ x (Annual demand x cost per order)}}{\text{Annual holding cost per unit}}}$

EOQ = $\sqrt{\frac{2 \text{ x } 50{,}000 \text{ kgs x } 12 \text{ x } 375}{12\% \text{ of } ₹150}}$ = **5,000 kgs**

(ii) Calculation of maximum stock level of Material Ae which is perishable in nature and is required to be used within 3.5 days.

(a) Stock equals to 3.5 days consumption = 50,000 kgs/ 25 days x 3.5 days = **7,000 kgs**

(b) Maximum stock level for Material Ae

Maximum stock = Reorder quantity + reorder level – (minimum consumption x minimum lead time)

Where, reorder quantity = 7,500 kgs

Reorder level = maximum consumption* x maximum lead time

= 50,000/ 25 x 3 days = 6,000 kgs

Now, Maximum stock level = 7,500 kgs + 6,000 kgs – (50,000 /25 days x 2 days) = **9,500 kgs**

Stock required for 3.5 days consumption is lower than the maximum stock level calculated above. Therefore, **maximum stock level will be 7,000 kgs.**

(*since production is processed evenly throughout the month hence material consumption will also be even.)

ICAI · Old P3 · MTP Nov 2022 · Series I · question p. 2-3 · answer p. 3

224 MTP **Old P3 Dec 2021 S-I** Q2(a) 10 marks old scheme

The yearly production of a company's product which has a steady market is 40,000 units. Each unit of a product requires 1 kg. of raw material. The cost of placing one order for raw material is ₹ 1,000 and the inventory carrying cost is ₹ 20 per annum. The lead time for procurement of raw material is 36 days and a safety stock of 1,000 kg. of raw materials is maintained by the company. The company has been able to negotiate the following discount structure with the raw material supplier:

Order quantity (kg.)	Discount (₹)
Upto 6,000	NIL
6,001 – 8,000	4,000
8,001 – 16,000	20,000
16,001 – 30,000	32,000
30,001 – 45,000	4,0000

You are REQUIRED to:

(i) Calculate the re-order point considering 30 days in a month.

(ii) Prepare a statement showing the total cost of procurement and storage of raw material after considering the discount of the company elects to place one, two, four or five orders in the year.

(iii) State the number of orders which the company should place to minimize the costs after taking EOQ also into consideration.

ICAI'S ANSWER

Working notes

1. Annual production = 40,000 units
2. Raw material required for 40,000 units (40,000 units × 1 kg.) = 40,000 kg.

3. EOQ = $\sqrt{\frac{2 \times 40{,}000 \text{ kgs.} \times ₹\ 1{,}000}{₹\ 20}}$ = 2,000 kgs.

4. Total cost of procurement and storage when the order size is equal to EOQ or 2,000 kg.

No. of orders (40,000 kg. ÷ 2,000 kg.)	= 20 times
Ordering cost (20 orders × ₹1,000)	= ₹ 20,000
Carrying cost (₹) (½ × 2,000 kg. × ₹ 20)	= ₹ 20,000
Total cost	₹ 40,000

(i) **Re-order point** = Safety stock + Lead time consumption

= 1,000 kg. + $\frac{40{,}000\text{kg.}}{360\text{days}}$ ×36days

= 1,000 kg. + 4,000 kg. = 5,000 kg.

(ii) **Statement showing the total cost of procurement and storage of raw materials**

(after considering the discount)

Order size	No. of orders	Total cost of procurement	Average stock	Total cost of storage of raw materials	Discount	Total cost
Kg.		**(₹)**	**Kg.**	**(₹)**	**(₹)**	**(₹)**
(1)	**(2)**	**(3)=(2)×₹1,000**	**(4)=½×(1)**	**(5)=(4)×₹20**	**(6)**	**(7)=[(3)+ (5)– (6)**
40,000	1	1,000	20,000	4,00,000	40,000	3,61,000
20,000	2	2,000	10,000	2,00,000	32,000	1,70,000
10,000	4	4,000	5,000	1,00,000	20,000	84,000
8,000	5	5,000	4,000	80,000	4,000	81,000

(iii) Number of orders which the company should place to minimize the costs after taking EOQ also into consideration is 20 orders each of size 2,000 kg. The total cost of procurement and storage in this case comes to ₹ 40,000, which is minimum. *(Refer to working notes 3 and 4)*

ICAI · Old P3 · MTP Dec 2021 · Series I · question p. 2 · answer p. 3-4

225 MTP **Old P3 Dec 2021 S-II** Q1(b) 5 marks old scheme

The annual demand for an item of raw material is 48,000 units and the purchase price is ₹ 80 per unit. The cost of processing an order is ₹ 1,350 and the annual cost of storage is ₹ 15 per unit.

(i) DETERMINE is the optimal order quantity and total relevant cost for the order?

(ii) If the cost of processing an order is ₹ 800 and all other data remain same, then DETERMINE the differential cost?

(iii) If the supplier offers bulk purchase of 48,000 units at a price of ₹ 72 and cost of placing the is Nil, SHOULD the order be accepted?

ICAI'S ANSWER

(i) Optimal order quantity i.e. **E.O.Q.**

$= \sqrt{\frac{2 \times 48{,}000 \times 1{,}350}{15}} = \sqrt{86,40,000} = 2{,}939$ units

Relevant Cost of this order quantity	₹
Ordering cost = $\frac{48{,}000}{2{,}939}$ =16.33, say 17 orders at ₹1,350	22,950.00
Carrying Cost = $\frac{1}{2}$ × 2,939 × 15	22,042.50
Relevant cost	**44,992.50**

(ii) Revised EOQ = $\sqrt{\frac{2 \times 48{,}000 \times 800}{15}}$ = 2,263 units

Relevant Cost of this order quantity	₹
Ordering cost = $\frac{48{,}000}{2{,}263}$ = 21.21, say 22 orders at ₹ 800	17,600.00
Carrying cost = $\frac{1}{2}$ × 2,263 × 15	16,972.50
Relevant cost	**34,572.50**

Differential cost = 44,992.50 – 34,572.50 = **₹ 10,420**

(iii) In case of discount in purchase price, the total cost of Purchase cost, ordering cost and carrying cost should be compared.

Original offer at ₹ 80 per unit		Supplier offered at ₹ 72 per unit	
	₹		₹
Purchase Cost (48,000 × 80)	38,40,000.00	Purchase cost (48,000 × 72)	34,56,000.00
Ordering cost	22,950.00	Ordering cost	0.00
Carrying cost	22,042.50	Carrying cost $\frac{1}{2}$ × 48,000 × 15	3,60,000.00
Total cost	38,84,992.50		38,16,000.00

This special offer at ₹ 72 per unit should be accepted as it saves ₹ 68,992.50 as compared to original offer.

ICAI · Old P3 · MTP Dec 2021 · Series II · question p. 1 · answer p. 2-3

226 MTP **Old P3 Nov 2020 S-I** Q1(b) 5 marks old scheme

A company deals in trading of a toy car *'Terminato'*. The annual demand for the toy car is 9,680 units. The company incurs fixed order placement and transportation cost of ₹ 200 each time an order is placed. Each toy costs ₹ 400 and the trader has a carrying cost of 20 percent p.a.

The company has been offered a quantity discount of 5% on the purchase of *'Terminato'* provided the order size is 4,840 units at a time.

Required:

(i) COMPUTE the economic order quantity

(ii) STATE whether the quantity discount offer can be accepted.

ICAI'S ANSWER

(i) Calculation of Economic Order Quantity

$$\text{EOQ} = \sqrt{\frac{2AO}{C}} = \sqrt{\frac{2 \times 9{,}680\text{units} \times \text{Rs.}200}{\text{Rs.}400 \times 20\%}} = \textbf{220 units}$$

(ii) Evaluation of Profitability of Different Options of Order Quantity

(A) When EOQ is ordered

		(₹)
Purchase Cost	(9,680 units × ₹ 400)	38,72,000
Ordering Cost	[(9,680 units/220 units) × ₹ 200]	8,800
Carrying Cost	(220 units ×½ × ₹ 400 × 20%)	8,800
Total Cost		**38,89,600**

(B) When Quantity Discount is accepted

		(₹)
Purchase Cost	(9,680 units × ₹ 380)	36,78,400
Ordering Cost	[(9,680 units/4,840 units) × ₹ 200]	400
Carrying Cost	(4,840 units ×½ × ₹ 380 ×20%)	1,83,920
Total Cost		**38,62,720**

Advise – The total cost of inventory is lower if quantity discount is accepted. The company would save ₹ 26,880 (₹ 38,89,600 - ₹ 38,62,720).

ICAI · Old P3 · MTP Nov 2020 · Series I · question p. 1-2 · answer p. 1-2

227 MTP **Old P3 May 2020 S-I** Q3(a) 10 marks old scheme

A company manufactures a product from a raw material, which is purchased at Rs.180 per kg. The company incurs a handling cost of Rs.1,460 plus freight of Rs.940 per order. The incremental carrying cost of inventory of raw material is Rs.2.5 per kg per month. In addition, the cost of working capital finance on the investment in inventory of raw material is Rs.18per kg per annum. The annual production of the product is 1,00,000 units and 2.5 units are obtained from one kg. of raw material.

Required:

(i) CALCULATE the economic order quantity of raw materials.

(ii) DETERMINE, how frequently company should order for procurement be placed.

(iii) If the company proposes to rationalize placement of orders on quarterly basis, DETERMINE the percentage of discount in the price of raw materials should be negotiated?

Assume 360 days in a year.

ICAI'S ANSWER

(i) Calculation of Economic Order Quantity (E.O.Q)

Annual requirement (usage) of raw material in kg. (A) = $\frac{1,00,000\text{units}}{2.5\text{units per kg.}}$ = 40,000kg.

Ordering Cost (Handling & freight cost) (O) = Rs.1,460 + Rs.940 = Rs.2,400

Carrying cost per unit per annum (C) i.e. inventory carrying cost + working capital cost

= (Rs.2.5 × 12 months) + Rs.18 = Rs.48 per kg.

E.O.Q. = $\sqrt{\frac{2AO}{C}} = \sqrt{\frac{2 \times 40,000\text{kg.} \times \text{Rs.}2,400}{\text{Rs.}48}}$ = 2,000 kg.

(ii) Frequency of placing orders for procurement:

Annual consumption (A) = 40,000 kg.

Quantity per order (E.O.Q) = 2,000 kg.

No. of orders per annum $(\frac{A}{E.O.Q})$ = $\frac{40,000\text{kg.}}{2,000\text{kg.}}$ = 20 orders

Frequency of placing orders (in days) = $\frac{360\text{days}}{20\text{orders}}$ = 18 days

(iii) Percentage of discount in the price of raw materials to be negotiated:

Particulars	On Quarterly Basis	On E.O.Q Basis
1. Annual Usage (in Kg.)	40,000 kg.	40,000 kg.
2. Size of the order	10,000 kg.	2,000 kg.
3. No. of orders (1 ÷ 2)	4	20
4. Cost of placing orders or Ordering cost (No. of orders × Cost per order)	Rs.9,600 (4 order × Rs2,400)	Rs.48,000 (20 orders × Rs2,400)
5. Inventory carrying cost (Average inventory × Carrying cost per unit)	Rs.2,40,000 (10,000 kg. × ½ × Rs.48)	Rs.48,000 (2,000 kg. × ½ × Rs.48)
6. Total Cost (4 + 5)	Rs.2,49,600	Rs.96,000

When order is placed on quarterly basis the ordering cost and carrying cost increased by Rs.1,53,600 (Rs.2,49,600 - Rs.96,000).

So, discount required = Rs.1,53,600

Total annual purchase = 40,000 kg. × Rs.180 = Rs.72,00,000

So, Percentage of discount to be negotiated = $\frac{\text{Rs.}1,53,600}{\text{Rs.}72,00,000} \times 100$ = 2.13%

ICAI · Old P3 · MTP May 2020 · Series I · question p. 4 · answer p. 5-6

228 MTP **Old P3 Nov 2019 S-I** Q1(c) 5 marks old scheme

A Ltd. manufactures a product X which requires two raw materials A and B in a ratio of 1:4. The sales department has estimated a demand of 5,00,000 units for the product for the year. To produce one unit of finished product, 4 units of material A is required.

Stock position at the beginning of the year is as below:

Product- X	12,000 units
Material A	24,000 units
Material B	52,000 units

To place an order the company has to spend Rs.15,000. The company is financing its working capital using a bank cash credit @13% p.a.

Product X is sold at Rs.1,040 per unit. Material A and B are purchased at Rs.150 and Rs.200 respectively.

Required:

COMPUTE economic order quantity (EOQ):

(i) If purchase order for the both materials is placed separately.

(ii) If purchase order for the both materials is not placed separately.

ICAI'S ANSWER

Workings:

Annual production of Product X = Annual demand – Opening stock

= 5,00,000 – 12,000 = 4,88,000 units

Annual requirement for raw materials = Annual production × Material per unit – Opening stock of material

Material A = 4,88,000 × 4 units – 24,000 units = 19,28,000 units

Material B = 4,88,000 × 16 units – 52,000 units = 77,56,000 units

(i) Computation of EOQ when purchase order for the both materials is placed separately

$$\text{EOQ} = \sqrt{\frac{2 \times \text{Annual Requirement for material} \times \text{Ordering cost}}{\text{Carrying cost per unit per annum}}}$$

$$\text{Material A} = \sqrt{\frac{2 \times 19{,}28{,}000\text{units} \times \text{Rs.}15{,}000}{13\%\text{of Rs.}150}} = \sqrt{\frac{38{,}56{,}000 \times \text{Rs.}15{,}000}{\text{Rs.}19.5}}$$

= 54,462 units

$$\text{Material B} = \sqrt{\frac{2 \times 77{,}56{,}000\text{units} \times \text{Rs.}15{,}000}{13\%\text{of Rs.}200}} = \sqrt{\frac{1{,}55{,}12{,}000 \times \text{Rs.}15{,}000}{\text{Rs.}26}}$$

= 94,600 units

(ii) Computation of EOQ when purchase order for the both materials is not placed separately

$$\text{Material A \& B} = \sqrt{\frac{2 \times (19{,}28{,}000 + 77{,}56{,}000)\text{units} \times \text{Rs.}15{,}000}{13\% \text{ of Rs.}190^*}}$$

$$= \sqrt{\frac{1{,}93{,}68{,}000 \times \text{Rs.}15{,}000}{\text{Rs.}24.7}} = 1{,}08{,}452 \text{ units}$$

$$\text{Material A} = \frac{1{,}08{,}452 \times 19{,}28{,}000}{96{,}84{,}000} = 21{,}592 \text{ units}$$

$$\text{Material A} = \frac{1{,}08{,}452 \times 77{,}56{,}000}{96{,}84{,}000} = 86{,}860 \text{ units}$$

$$^* \frac{(\text{Rs.}150 \times 19{,}28{,}000) + (\text{Rs.}200 \times 77{,}56{,}000)}{(19{,}28{,}000 + 77{,}56{,}000)} = \text{Rs.}190$$

ICAI · Old P3 · MTP Nov 2019 · Series I · question p. 1-2 · answer p. 2-3

3.6 ABC, FSN, HML, VED, JIT and perpetual inventory

7 questions · asked in 6 ICAI documents · 5 exam papers

229 EXAM **May 2026** Q6(a) 5 marks

Same question, different ICAI answer: #231 Old P3 Exam Nov 2022

Which system of inventory management is known as 'Demand Pull' or 'Pull through' system of production? Explain. Also, specify the two principles on which this system is based.

ICAI'S ANSWER

Just In Time (JIT) Inventory Management is also known as 'Demand pull' or 'Pull through' system of production. JIT is a system of inventory management with an approach to have zero inventories in stores. According to this approach material should only be purchased when it is actually required for production.

In this system, production process actually starts after the order for the products is received. Based on the demand, production process starts and the requirement for raw materials is sent to the purchase department for purchase.

JIT is based on two principles

(i) Produce goods only when it is required and

(ii) the products should be delivered to customers at the time only when they want.

ICAI · P4 · May 2026 exam · question p. 28 · answer p. 28-29

230 EXAM **Sep 2025** Q5(a) 8 marks

TS Limited is suffering from material deterioration and finds that their valuable stocks are not properly stored. The company furnishes following information:

Serial No.	Material Name	Units	Total Cost (₹)
1	MA	54,105	14,855
2	MB	32,300	12,823
3	MC	28,600	13,972
4	MD	10,250	47,685
5	ME	23,410	39,015
6	MF	2,580	1,08,260
7	MG	8,900	89,410
8	MH	4,855	98,980

Store-keeper of the company argues that he has taken proper care in storing three types of material named MA, MB and MC as they are in bulk quantity. He further argues that only a few units of material MG and MH has been deteriorated due to bad weather.

The management of TS Limited wants to get him aware about value of different items.

Required:

Rank the materials and draw a plan of ABC selective control by using the following basis of selective control.

₹ 50,000 & above	'A' category items
₹ 15,000 to ₹ 50,000	'B' category items
Below ₹ 15.000	'C' category items

ICAI'S ANSWER

Statement of Total Cost and Ranking

Item	Material Name	Units	% of Total units	Total cost (₹)	% of Total cost	Ranking
1	MA	54,105	32.80	14,855	3.49	6
2	MB	32,300	19.58	12,823	3.02	8
3	MC	28,600	17.33	13,972	3.29	7
4	MD	10,250	6.21	47,685	11.22	4
5	ME	23,410	14.19	39,015	9.18	5
6	MF	2,580	1.56	1,08,260	25.47	1
7	MG	8,900	5.39	89,410	21.04	3
8	MH	4,855	2.94	98,980	23.29	2
		1,65,000	100	4,25,000	100	

Basis for selective control

₹ 50,000 & above — 'A' items

₹ 15,000 to 50000 — 'B' items

Below ₹ 15,000 — 'C' items

On this basis, a plan of A B C selective control is given below:

Ranking	Item Nos.	% of Total units	Cost (₹)	% of Total Cost	Category
1	6	1.56	1,08,260	25.47	
2	8	2.94	98,980	23.29	
3	7	5.39	89,410	21.04	
Total	**3**	**9.89**	**2,96,650**	**69.80**	**A**
4	4	6.21	47,685	11.22	
5	5	14.19	39,015	9.18	

Ranking	Item Nos.	% of Total units	Cost (₹)	% of Total Cost	Category
Total	**2**	**20.40**	**86,700**	**20.40**	**B**
6	1	32.80	14,855	3.49	
7	3	17.33	13,972	3.29	
8	2	19.58	12,823	3.02	
Total	**3**	**69.71**	**41,650**	**9.80**	**C**
Grand Total	**8**	**100**	**4,25,000**	**100**	

ICAI · P4 · Sep 2025 exam · question p. 23-24 · answer p. 24-25

231 EXAM **Old P3 Nov 2022** Q6(a) 5 marks old scheme

Same question, different ICAI answer: #229 Exam May 2026

Which system of inventory management is known as 'Demand pull' or 'Pull through' system of production? Explain. Also, specify the two principles on which this system is based.

ICAI'S ANSWER

Just in Time (JIT) Inventory Management is also known as 'Demand pull' or 'Pull through' system of production. In this system, production process actually starts after the order for the products is received. Based on the demand, production process starts and the requirement for raw materials is sent to the purchase department for purchase.

It is a system of inventory management with an approach to have a zero inventories in stores. According to this approach material should only be purchased when it is actually required for production.

JIT is based on two principles

(i) Produce goods only when it is required and

(ii) the products should be delivered to customers at the time only when they want.

ICAI · Old P3 · Nov 2022 exam · question p. 22 · answer p. 23

232 EXAM **Old P3 Jul 2021** Q1(a) 5 marks old scheme

MM Ltd. has provided the following information about the items in its inventory.

Item Code Number	Units	Unit Cost (₹)
101	25	50
102	300	01
103	50	80
104	75	08
105	225	02
106	75	12

MM Ltd. has adopted the policy of classifying the items constituting 15% or above of Total Inventory Cost as 'A' category, items constituting 6% or less of Total Inventory Cost as 'C' category and the remaining items as 'B' category.

You are required to:

(i) Rank the items on the basis of % of Total Inventory Cost.

(ii) Classify the items into A, B and C categories as per ABC Analysis of Inventory Control adopted by MM Ltd.

ICAI'S ANSWER

(i) Statement of Total Inventory Cost and Ranking of items

Item code no.	Units	% of Total units	Unit cost (₹)	Total Inventory cost (₹)	% of Total Inventory cost	Ranking
101	25	3.33	50	1,250	16.67	**2**
102	300	40.00	1	300	4.00	**6**
103	50	6.67	80	4,000	53.33	**1**
104	75	10.00	8	600	8.00	**4**
105	225	30.00	2	450	6.00	**5**
106	75	10.00	12	900	12.00	**3**
	750	100	153	7,500	100	

(ii) Classifying items as per ABC Analysis of Inventory Control

Basis for ABC Classification as % of Total Inventory Cost

15% & above	--	'A' items
7% to 14%	--	'B' items
6% & Less	--	'C' items

Ranking	Item code No.	% of Total units	Total Inventory cost (₹)	% of Total Inventory Cost	Category
1	103	6.67	4,000	53.33	

Ranking	Item code No.	% of Total units	Total Inventory cost (₹)	% of Total Inventory Cost	Category
2	101	3.33	1,250	16.67	
Total	2	10.00	5,250	70.00	A
3	106	10.00	900	12.00	
4	104	10.00	600	8.00	
Total	2	20.00	1,500	20.00	B
5	105	30.00	450	6.00	
6	102	40.00	300	4.00	
Total	2	70.00	750	10.00	C
Grand Total	6	100	7,500	100	

ICAI · Old P3 · Jul 2021 exam · question p. 53 · answer p. 55

233 EXAM **Old P3 Jul 2021** Q6(e) 5 marks old scheme

Write a short note on VED analysis of Inventory Control.

ICAI'S ANSWER

Vital, Essential and Desirable (VED): Under this system of inventory analysis, **inventories are classified on the basis of its criticality for the production function and final product**. Generally, this classification is done for spare parts which are used for production.

(i) Vital- Items are classified as vital when its **unavailability can interrupt the production process and cause a production loss.** Items under **this category are strictly controlled by setting re-order level.**

(ii) Essential- Items under this category are essential but not vital. **The unavailability may cause sub standardisation and loss of efficiency in production process.** Items under this category are reviewed periodically and get the second priority.

(iii) Desirable- Items under this category are optional in nature; **unavailability does not cause any production or efficiency loss.**

ICAI · Old P3 · Jul 2021 exam · question p. 78 · answer p. 80

234 EXAM **Old P3 Nov 2019** Q6(c) 5 marks old scheme

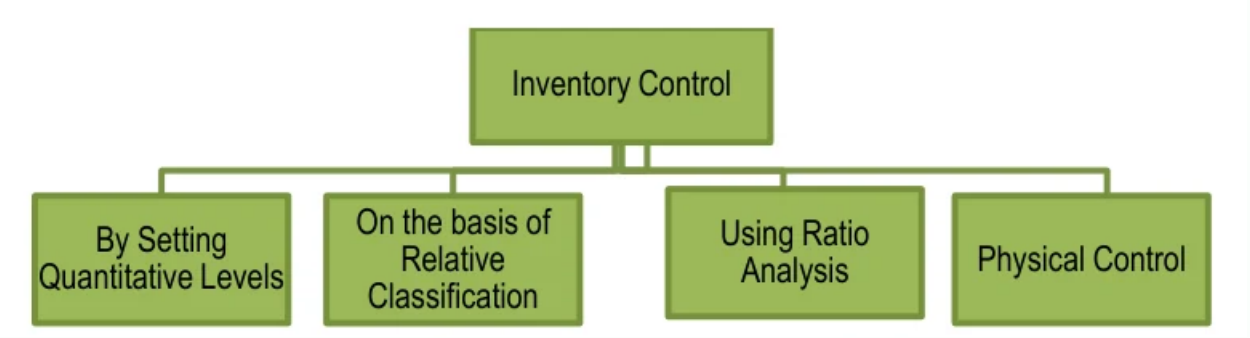
Define Inventory Control and give its objectives.

List down the basis to be adopted for Inventory Control.

ICAI'S ANSWER

Inventory Control: The Chartered Institute of Management Accountants (CIMA) defines Inventory Control as "The function of ensuring that sufficient goods are retained in stock to meet all requirements without carrying unnecessarily large stocks."

The objective of inventory control is to make a balance between sufficient stock and over-stock. The stock maintained should be sufficient to meet the production requirements so that uninterrupted production flow can be maintained. Insufficient stock not only pause the production but also cause a loss of revenue and goodwill. On the other hand, Inventory requires some funds for purchase, storage, maintenance of materials with a risk of obsolescence, pilferage etc. A trade-off between Stock-out and Over-stocking is required. The management may employ various methods of Inventory control to have a balance. Management may adopt the following basis for Inventory control:



ICAI · Old P3 · Nov 2019 exam · question p. 23 · answer p. 25-26

235 MTP **Jan 2025 S-II** Q6(c) 4 marks

You are managing the inventory for a manufacturing company and notice that certain items in the store are not being utilized frequently, leading to increased holding costs. HOW would you identify slow-moving and non-moving items, and WHAT strategies would you implement to minimize such stocks effectively?

ICAI'S ANSWER

Detection of slow moving and non-moving item of stores:

The existence of slow moving and non-moving item of stores can be detected in the following ways.

(i) By preparing and *perusing periodic reports* showing the status of different items or stores.

(ii) By calculating the *inventory turnover period* of various items in terms of number of days/ months of consumption.

(iii) By computing *inventory turnover ratio* periodically, relating to the issues as a percentage of average stock held.

(iv) By implementing the use of a well-designed information system.

Necessary steps to reduce stock of slow moving and non-moving item of stores:

(i) Proper procedure and guidelines should be laid down for the disposal of non-moving items, before they further deteriorate in value.

(ii) Diversify production to use up such materials.

(iii) Use these materials as substitute, in place of other materials.

ICAI · P4 · MTP Jan 2025 · Series II · question p. 13 · answer p. 19

3.7 Inventory turnover ratios and stock discrepancies

5 questions · asked in 5 ICAI documents · 2 exam papers

236 EXAM **Sep 2025** MCQ 13 2 marks MCQ

The management of Y Limited has provided the following information:

Opening stock of raw material - ₹ 25,000

Closing stock of raw material - ₹ 75,000

Raw material consumed during the year - ₹ 7,50,000

What will be the number of days for which average inventory of raw material is to be held by Y Limited? Assume 360 days in a year.

A 14 days

B 15 days

C 24 days

D 26 days

ICAI'S ANSWER

Answer key: **(C)**

ICAI · P4 · Sep 2025 exam · question p. 6 · answer p. 7

237 EXAM **Old P3 Dec 2021** Q1(a) 5 marks old scheme

XYZ Ltd. uses two types of raw materials – 'Material A' and 'Material B' in the production process and has provided the following data for the year ended on 31st March, 2021:

Particulars	Material A (₹)	Material B (₹)
Opening stock as on 01.04.2020	30,000	32,000
Purchase during the year	90,000	51,000
Closing stock as on 31.03.2021	20,000	14,000

(i) You are required to calculate:

(a) The inventory turnover ratio of 'Material A' and 'Material B'.

(b) The number of days for which the average inventory is held for both materials 'A' and 'B'.

(ii) Based on above calculations, give your comments.

(Assume 360 days in a year.)

ICAI'S ANSWER

(i) Calculation of Inventory Turnover ratios and number of days:

	Material A (₹)	Material B (₹)
Opening stock	30,000	32,000
Add: Purchases	90,000	51,000
	1,20,000	83,000
Less: Closing stock	20,000	14,000
Materials consumed	1,00,000	69,000
Average inventory: (Opening Stock + Closing Stock) ÷ 2	25,000	23,000
(a) Inventory Turnover ratio: (Consumption ÷ Average inventory)	4 times	3 times
(b) Number of days for which the average inventory held (Number of Days in a year/IT ratio)	90 days	120 days

(ii) Comments: Material A is moving faster than Material B. Or Material A has a less holding period.

ICAI · Old P3 · Dec 2021 exam · question p. 1 · answer p. 2-3

238 RTP **Old P3 Dec 2021 Q1** old scheme

The following data are available in respect of material X for the year ended 31st March, 2021:

	(₹)
Opening stock	9,00,000
Purchases during the year	1,70,00,000
Closing stock	11,00,000

(i) CALCULATE:

(a) Inventory turnover ratio, and

(b) The number of days for which the average inventory is held.

(ii) INTERPRET the ratio calculated as above if the industry inventory turnover rate is 10.

ICAI'S ANSWER

(i) (a) Inventory turnover ratio (Refer to working note)

$$= \frac{\text{Cost of stock of raw material consumed}}{\text{Average stock of raw material}}$$

$$= \frac{₹\ 1,68,00,000}{₹\ 10,00,000} = 16.8$$

(b) Average number of days for which the average inventory is held

$$= \frac{365}{\text{Inventory turnover ratio}} = \frac{365\text{days}}{16.8} = 21.73 \text{ days}$$

Working Note:

Particulars	(₹)
Opening stock of raw material	9,00,000
Add: Material purchases during the year	1,70,00,000
Less: Closing stock of raw material	11,00,000
	1,68,00,000

(ii) The Inventory turnover ratio for material X is 16.8 which mean an inventory item takes only 21.73 or 22 days to issue from stores for production process. The rate is better than the industry rate which is 10 time or 36.5 days. This inventory turnover ratio indicates better inventory management system and good demand for the final product in market.

ICAI · Old P3 · RTP Dec 2021 · question p. 1 · answer p. 13-14

239 MTP **May 2026 S-I** Q3(a) 5 marks

The following information has been extracted from the books of AB Ltd. for the year ended 31st March, 2025:

Particulars	Material X	Material Y
Average stock (kg.)	59,400 kgs	1,00,800 kgs
Position of stock	Closing stock is more than opening stock by 10,800 kg.	Opening stock is less than closing stock by 28,800 kg.
Inventory turnover ratio	6 times	8 times

You are required to :

(i) CALCULATE the quantity of each material purchased during the year and value of opening and closing stock.

(ii) CALCULATE the number of days for which the average inventory is held in respect of each material (Assume 360 days in a year).

(iii) COMMENT on the movement of inventory of each material.

ICAI'S ANSWER

(i) Inventory Turnover Ratio (X) = Material Consumed / Average Stock

6 = Material Consumed / 59,400

Material consumed (X) = 3,56,400 kg. (59,400 x 6)

Material Y = Material Consumed / Average Stock

8 = Material Consumed / 1,00,800

Material Consumed = 8,06,400 kg. (1,00,800 x 8)

$$\text{Average Stock} = \frac{\text{Opening stock} + \text{Closing Stock}}{2}$$

Let opening stock is assumed to A Kg.

Closing stock will be = A+10,800

Material X = $\frac{A + A + 10{,}800}{2}$ = 59,400 Kgs

2A = 1,08,000

A = 54,000 Kg.

Opening Stock X = 54,000 Kg.

Closing Stock X = 54,000 + 10,800 = 64,800 Kgs.

Material Y = $\frac{A + A + 28{,}800}{2}$ = 1,00,800 Kgs.

2A =1,72,800

A = 86,400 = Opening Stock Y

Closing Stock Y = 86,400 + 28,800 = 1,15,200 Kg.

Purchases = Material Consumed + Closing Stock - Opening Stock

Material X = 3,56,400 + 64,800-54,000

= 3,67,200 Kgs.

Material Y = 8,06,400 + 1,15,200 -86,400

= 8,35,200 Kgs.

(ii) Number of days for which the average inventory is held:

Number of Days in a year/ Inventory turnover ratio

Material X = 360/6 = 60 days

Material Y = 360/8 = 45 days

(iii) Comments:

Material Y is fast moving than that of Material X because it was held for 45 days against 60 days of X material.

ICAI · P4 · MTP May 2026 · Series I · question p. 10 · answer p. 10-11

240 MTP **May 2024 S-II** Q4(b) 4 marks

HOW is slow moving and non-moving item of stores detected and WHAT steps are necessary to reduce such stocks?

ICAI'S ANSWER

Detection of slow moving and non-moving item of stores:

The existence of slow moving and non-moving item of stores can be detected in the following ways.

(i) By preparing and *perusing periodic reports* showing the status of different items or stores.

(ii) By calculating the *inventory turnover period* of various items in terms of number of days/ months of consumption.

(iii) By computing *inventory turnover ratio* periodically, relating to the issues as a percentage of average stock held.

(iv) By implementing the use of a well designed information system.

Necessary steps to reduce stock of slow moving and non-moving item of stores:

(i) Proper procedure and guidelines should be laid down for the disposal of non-moving items, before they further deteriorates in value.

(ii) Diversify production to use up such materials.

(iii) Use these materials as substitute, in place of other materials.

ICAI · P4 · MTP May 2024 · Series II · question p. 9 · answer p. 11-12

3.8 Waste, scrap, spoilage, defectives and obsolescence

6 questions · asked in 6 ICAI documents · 3 exam papers

241 EXAM **May 2026** Q6(c) 4 marks

Explain the meaning of spoiled work and defective work. Discuss the treatment of defective work under the following circumstances:

(i) Where a percentage of defective work is allowed in a particular batch, as it cannot be avoided.

(ii) Where the defect is due to bad workmanship.

(iii) Where the defect is due to the Inspection Department wrongly accepting incoming material of poor quality.

ICAI'S ANSWER

Spoiled work is the quantity of production that has been totally rejected and cannot be rectified.

Defective work refers to production that is not as perfect as the saleable product but is capable of being rectified and brought to the required degree of perfection provided some additional expenditure is incurred.

Treatment of Defects arise in the following circumstances:

Circumstances	Treatment
Where a percentage of defective work is allowed in a particular batch as it cannot be avoided.	If the actual number of defectives is within the normal limit or is near thereto the cost of rectification will be charged to the whole job and spread over the entire output of the batch. If, on the other hand, the number of defective units substantially exceeds the normal, the cost of rectification of the number which exceeds the normal will be written off as a loss in the Costing Profit and Loss Account.
Where the defect is due to bad workmanship.	In this case cost of rectification will be abnormal cost and shall be written off as a loss. However, if the management did provide for a certain proportion of defectives on account of bad workmanship as an unavoidable feature of production, the cost of rectifying to the extent provided for by the management will be treated as a normal cost and charged to the batch.
Where defect is due to the Inspection Department wrongly accepting incoming material of poor quality.	In this case the cost of rectification will be charged to the department and will not be considered as cost of manufacture of the batch. Being an abnormal cost, it will be written off to the Costing Profit and Loss Account.

ICAI · P4 · May 2026 exam · question p. 28 · answer p. 30-31

242 EXAM **Jan 2025** Q5(c) 4 marks

Define spoiled work and defective work and discuss the treatment of defective work in the following circumstances:

Circumstances	Treatment
Where a percentage of defective work is allowed in a particular batch as it cannot be avoided.	
Where the defect is due to bad workmanship	
Where defect is due to the Inspection Department wrongly accepting incoming material of poor quality.	

ICAI'S ANSWER

Spoiled work is the quantity of production that has been totally rejected and cannot be rectified.

Defective work refers to production that is not as perfect as the saleable product but is capable of being rectified and brought to the required degree of perfection provided some additional expenditure is incurred

Circumstances	Treatment
Where a percentage of defective work is allowed in a particular batch as it cannot be avoided.	If the actual number of defectives is within the normal limit or is near thereto **the cost of rectification will be charged to the whole job and spread over the entire output of the batch**. If, on the other hand, the number of defective units substantially exceeds the normal, the cost of rectification of the number which exceeds the normal will be written off as a loss in the Costing Profit and Loss Account.
Where the defect is due to bad workmanship.	In this case cost of rectification will be abnormal cost and **shall be written off as a loss.** However, if the management did provide for a certain proportion of defectives on account of bad workmanship as an unavoidable feature of production. If that be the case, the cost of rectifying to the extent provided for by the management will be treated as a normal cost and charged to the batch.
Where defect is due to the Inspection Department wrongly accepting incoming material of poor quality.	In this case the **cost of rectification will be charged to the department and will not be considered as cost of manufacture of the batch.** Being an abnormal cost, it will be written off to the Costing Profit and Loss Account.

ICAI · P4 · Jan 2025 exam · question p. 24 · answer p. 26-27

243 EXAM **May 2024** Q6(b) 5 marks

Distinguish between Waste and Scrap. Discuss the treatment of normal and abnormal scrap in Cost Accounts.

ICAI'S ANSWER

Difference between Waste and Scrap

Waste	Scrap
1. The portion of raw material which is lost during storage/ production and discarded.	1. The output which is discarded and disposed off without further treatment.
2. It is connected with raw material or inputs to the production process.	2. It is the loss connected with the output
3. Waste of materials may be visible or invisible.	3. Scraps are generally identifiable and has physical substance.
4. Generally, waste has no recoverable value.	4. Scraps are termed as by-products and has small recoverable value.

Treatment of Scrap

Normal- The cost of scrap is borne by good units and income arises on account of realisable value is deducted from the cost.

Abnormal- The scrap account should be charged with full cost. The credit is given to the job or process concerned. The profit or loss in the scrap account, on realisation, will be transferred to the Costing Profit and Loss Account.

ICAI · P4 · May 2024 exam · question p. 24 · answer p. 26

244 RTP **May 2026** Q17(b)

A furniture manufacturing company produces wooden tables. During production in June, the supervisor reports two issues:

- Some wooden pieces get completely damaged during cutting and can only be sold as low-value waste.
- A few finished tables have minor defects like rough edges or uneven legs, which can be corrected with extra labour and polishing.

As the cost accountant, explain:

1. How scrap differs from defective products in this scenario.
2. accounting treatment of scrap and defectives in the cost accounts.

ICAI'S ANSWER

Difference between Scrap and Defectives

Scrap	Defectives
1. It is the loss connected with the output	1. This type of loss is connected with the output as well as the input.
2. Scraps are not intended but cannot be eliminated due to the nature of material or process itself.	2. Defectives also are not intended but can be eliminated through a proper control system.
3. Generally, scraps are not used or rectified.	3. Defectives can be used after rectification.
4. Scraps have insignificant recoverable value.	4. Defectives are sold at a lower value from that of the good one.

Treatment of Scrap

Normal- The cost of scrap is borne by good units and income arises on account of realisable value is deducted from the cost.

Abnormal- The scrap account should be charged with full cost. The credit is given to the job or process concerned. The profit or loss in the scrap account, on realisation, will be transferred to the Costing Profit and Loss Account.

Treatment of Defectives:

Normal- An amount equal to the cost less realisable value on sale of defectives are charged to material cost of good production.

Abnormal- Material cost of abnormal defectives are not included in material cost but treated as loss after giving credit to the realisable value of such defectives. The material cost of abnormal loss is transferred to costing profit and loss account.

ICAI · P4 · RTP May 2026 · question p. 21 · answer p. 50-51

245 MTP **Sep 2025 S-I** Q6(b) 5 marks

EXPLAIN Defectives unit and its treatment. Also EXPLAIN the process of Reclamation of loss from defective units.

ICAI'S ANSWER

Defectives: It signifies those units or portions of production which do not meet the quality standards. Defectives arise due to sub-standard materials, bad-supervision, bad-planning, poor workmanship, inadequate-equipment and careless inspection.

The defectives which can be re-made as per the quality standard by using additional materials are known as reworks. Reworks include repairs, reconditioning and refurbishing.

Defectives which cannot be brought up to the quality standards are known as rejects. The rejects may either be disposed-off or re-cycled for production process.

Treatment of Defectives:

Normal - An amount equal to the cost less realisable value on sale of defectives are charged to material cost of good production.

Abnormal - Material cost of abnormal defectives are not included in material cost but treated as loss after giving credit to the realisable value of such defectives. The material cost of abnormal loss is transferred to costing profit and loss account.

Reclamation of loss from defective units

In the case of articles that have been spoiled, it is necessary to take steps to reclaim as much of the loss as possible. For this purpose:

(i) All defective units should be sent to a place fixed for the purpose;

(ii) These should be dismantled;

(iii) Goods and serviceable parts should be separated and taken back into the stock;

(iv) Parts which can be made serviceable by further work should be separated and sent to the workshop for the purpose and taken into stock after the defects have been removed; and

(v) Parts which cannot be made serviceable should be collected in one place for being melted or sold off.

Printed forms should be used to record quantities for all purposes aforementioned.

ICAI · P4 · MTP Sep 2025 · Series I · question p. 15 · answer p. 19-20

246 MTP **Sep 2025 S-II** Q6(c) 4 marks

EXPLAIN spoiled and defective work. Also, EXPLAIN the treatment of defect under the following circumstances:

Circumstances	Treatment
(1) Where a percentage of defective work is allowed in a particular batch as it cannot be avoided.	
(2) Where defect is due to bad workmanship.	
(3) Where defect is due to the Inspection Department wrongly accepting incoming material of poor quality.	

ICAI'S ANSWER

Spoiled work is the quantity of production that has been totally rejected and cannot be rectified.

Defective work refers to production that is not as perfect as the saleable product but is capable of being rectified and brought to the required degree of perfection provided some additional expenditure is incurred.

Circumstances	Treatment
(1) Where a percentage of defective work is allowed in a particular batch as it cannot be avoided.	If the actual number of defectives is within the normal limit or is near thereto the cost of rectification will be charged to the whole job and spread over the entire output of the batch. If, on the other hand, the number of defective units substantially exceeds the normal, the cost of rectification of the number which exceeds the normal will be written off as a loss in the Costing Profit and Loss Account.
(2) Where defect is due to bad workmanship.	In this case cost of rectification will be abnormal cost, *i.e.,* not a legitimate element of the cost. Therefore, the cost of rectification shall be written off as a loss, unless by an arrangement, it is to be recovered as a penalty from the workman concerned.
(3) Where defect is due to the Inspection Department wrongly accepting incoming material of poor quality.	In this case the cost of rectification will be charged to the department and will not be considered as cost of manufacture of the batch. Being an abnormal cost, it will be written off to the Costing Profit and Loss Account.

ICAI · P4 · MTP Sep 2025 · Series II · question p. 17 · answer p. 26-27

CHAPTER 4

Employee Cost and Direct Expenses

4.1 Attendance, payroll and the elements of wages

16 questions · asked in 14 ICAI documents · 3 exam papers

247 EXAM **May 2026** MCQ 12 2 marks MCQ

The following information pertains to a worker G:

Basic Pay	₹ 25,000 per month
Dearness Allowance	₹ 5,000 per month
Other Allowances	₹ 2,000 per month

The number of working days in a year is 325 of 8 hours each. G has availed 25 days as holidays on full pay in a year.

What is the wage rate per hour of worker G?

A ₹ 160

B ₹ 175

C ₹ 140

D ₹ 185

ICAI'S ANSWER

Answer key: **(A)**

ICAI · P4 · May 2026 exam · question p. 5-6 · answer p. 7

248 EXAM **May 2026** Q6(c) 4 marks

A factory employing 3000 workers faces issues of proxy attendance and payroll errors under the manual system. The Plant Head has recommended using an automated method, the Bio-Metric Attendance system. Discuss briefly the features of this Bio-Metric Attendance System.

ICAI'S ANSWER

Bio-Metric Attendance system-

1. Under bio-metric attendance system attendance is marked by recognizing an employee on the basis of physical and behavioural traits.
2. An employee's unique identity like finger print, face and retina image etc. are kept in a database which is matched at the time of marking of attendance before the attendance device for this purpose.
3. Bio-metric attendance system includes fingerprint recognition system, face recognition system, Time and attendance tracking technology etc.
4. This system reduces the risk of time manipulation and proxy attendance. However, it may not be suitable for small organisations due to cost associated with set-up and maintenance.

ICAI · P4 · May 2026 exam · question p. 28 · answer p. 30

249 EXAM **Sep 2025** Q6(c) 4 marks

List the important factors which need consideration for controlling employee costs.

ICAI'S ANSWER

The main points which need consideration for controlling employee costs are the following:

(i) Assessment of manpower requirements.

(ii) Control over time-keeping and time-booking.

(iii) Time & Motion Study.

(iv) Control over idle time and overtime.

(v) Control over employee turnover.

(vi) Wage and Incentive systems.

(vii) Job Evaluation and Merit Rating.

(viii) Employee productivity.

ICAI · P4 · Sep 2025 exam · question p. 26 · answer p. 28

250 EXAM **Old P3 Nov 2020** Q4(b) 6 marks old scheme

Following are the particulars of two workers 'R' and 'S' for a month:

Particulars	R	S
(i) Basic Wages (₹)	15,000	30,000
(ii) Dearness Allowance	50%	50%
(iii) Contribution to EPF (on basic wages)	7%	7.5%
(iv) Contribution to ESI (on basic wages)	2%	2%
(v) Overtime (hours)	20	-

The normal working hours for the month are 200 hrs. Overtime is paid at double the total of normal wages and dearness allowance. Employer's contribution to State Insurance and Provident Fund are at equal rates with employees' contributions.

Both workers were employed on jobs A, B and C in the following proportions :

Jobs	A	B	C
R	75%	10%	15%
S	40%	20%	40%

Overtime was done on job 'A'.

You are required to :

(i) Calculate ordinary wage rate per hour of 'R' and 'S'.

(ii) Allocate the worker's cost to each job 'A', 'B' and 'C'.

ICAI'S ANSWER

(i) Calculation of Net Wages paid to Worker 'R' and 'S'

Particulars	R (₹)	S (₹)
Basic Wages	15,000.00	30,000.00
Dearness Allowance (DA) (50% of Basic Wages)	7,500.00	15,000.00
Overtime Wages (Refer to Working Note 1)	4,500.00	----
Gross Wages earned	27,000.00	45,000.00
Less: Provident Fund (7% × ₹ 15,000); (7.5% × ₹ 30,000)	(1,050.00)	(2,250.00)
Less: ESI (2% × ₹ 15,000); (2% × ₹ 30,000)	(300.00)	(600.00)
Net Wages paid	**25,650.00**	**42,150.00**

Calculation of ordinary wage rate per hour of Worker 'R' and 'S'

	R (₹)	S (₹)
Gross Wages (Basic Wages + DA) (excluding overtime)	22,500.00	45,000.00
Employer's contribution to P.F. and E.S.I.	1,350.00	2,850.00
	23,850.00	47,850.00
Ordinary wages Labour Rate per hour (₹ 23,850 ÷ 200 hours); (₹ 47,850 ÷ 200 hours)	**119.25**	**239.25**

(ii) Statement Showing Allocation of workers cost to each Job

	Total Wages	Jobs		
		A	**B**	**C**
Worker R				
Ordinary Wages (15:2:3)	23,850.00	17,887.50	2,385.00	3577.50
Overtime	4500.00	**4500.00**	-	--
Worker S				
Ordinary Wages (2:1:2)	47,850.00	19,140.00	9,570.00	19,140.00
	76,200.00	**41,527.50**	**11,955.00**	**22,717.50**

Working Note:

Normal Wages are considered as basic wages.

$$\text{Over time} = \frac{2 \text{ x } (\text{Basic wage} + \text{D.A.}) \text{ x } 20 \text{ hours}}{200 \text{ hours}}$$

$$= 2\text{x } \frac{₹22{,}500}{200} \text{ x } 20\text{hours}$$

= ₹ 4,500

ICAI · Old P3 · Nov 2020 exam · question p. 14-15 · answer p. 17-18

251 EXAM **Old P3 Nov 2020** Q4(c) 4 marks old scheme

Discuss any four objectives of 'Time keeping' in relation to attendance and payroll procedures.

ICAI'S ANSWER

The objectives of time-keeping in relation to attendance and payroll procedures are as follows:

(i) For the preparation of payrolls.

(ii) For calculating overtime.

(iii) For ascertaining and controlling employee cost.

(iv) For ascertaining idle time.

(v) For disciplinary purposes.

(vi) For overhead distribution

ICAI · Old P3 · Nov 2020 exam · question p. 15 · answer p. 18

252 RTP **Jan 2026** Q9

Tommy HL Ltd. pays the following to a skilled worker engaged in production works. The following are the employee benefits paid to the employee:

(a) Basic salary per day	₹1,000
(b) Dearness allowance (DA)	20% of basic salary
(c) House rent allowance	16% of basic salary
(d) Transport allowance	₹50 per day of actual work
(e) Overtime	Twice the hourly rate (considers basic and DA), only if works more than 9 hours a day otherwise no overtime allowance. If works for more than 9 hours a day then overtime is considered after 8th hours.
(f) Work of holiday and Sunday	Double of per day basic rate provided works atleast 4 hours. The holiday and Sunday basic is eligible for all allowances and statutory deductions.
(h) Earned leave & Casual leave	These are paid leave.
(h) Employer's contribution to Provident fund	12% of basic and DA
(i) Employer's contribution to Pension fund	7% of basic and DA

The company normally works 8-hour a day and 26-day in a month. The company provides 30 minutes lunch break in between.

During the month of August 2025, Mr. Shyam works for 23 days including 15th August and a Sunday and applied for 3 days of casual leave. On 15th August and Sunday he worked for 5 and 6 hours respectively without lunch break.

On 5th and 13th August he worked for 10 and 9 hours respectively.

During the month Mr. Shyam worked for 100 hours on Job no. AB100.

You are required to CALCULATE:

(i) Earnings per day

(ii) Effective wages rate per hour of Mr. Shyam.

(iii) Wages to be charged to Job no. AB100.

ICAI'S ANSWER

Workings

1. Normal working hours in a month = (Daily working hours – lunch break) × no. of days

= (8 hours – 0.5 hours) × 26 days = 195 hours

2. Hours worked by Mr. Shyam = No. of normal days worked + Overtime + holiday / Sunday worked

= (21 days × 7.5 hours) + (9.5 hours + 8.5 hours) + (5 hours + 6 hours)

= 157.5 hours + 18 hours + 11 hours = 186.50 hours.

(i) Calculation of earnings per day

Particulars	Amount (₹)
Basic salary (₹ 1,000 × 26 days)	26,000
Dearness allowance (20% of basic salary)	5,200
	31,200
House rent allowance (16% of basic salary)	4,160
Employer's contribution to Provident fund (12% × ₹ 31,200)	3,744
Employer's contribution to Pension fund (7% × ₹ 31,200)	2,184
	41,288
No. of working days in a month (days)	26
Rate per day	1,588
Transport allowance per day	50
Earnings per day	**1,638**

(ii) Calculation of effective wage rate per hour of Mr. Shyam:

Particulars	Amount (₹)
Basic salary (₹ 1,000 × 26 days)	26,000
Additional basic salary for Sunday & holiday (₹ 1,000 × 2 days)	2,000
Dearness allowance (20% of basic salary)	5,600
	33,600
House rent allowance (16% of basic salary)	4,480
Transport allowance (₹ 50 × 23 days)	1,150
Overtime allowance (₹ 160 × 2 × 2 hours)*	640
Employer's contribution to Provident fund (12% × ₹33,600)	4,032
Employer's contribution to Pension fund (7% × ₹ 33,600)	2,352
Total monthly wages	46,254
Hours worked by Mr. Shyam (hours)	186.5
Effective wage rate per hour	**248**

*(Daily Basic + DA) ÷ 7.5 hours

= (1,000+200) ÷ 7.5 = ₹160 per hour

(iii) Calculation of wages to be charged to Job no. AB100

= ₹ 248 × 100 hours = ₹ 24,800

ICAI · P4 · RTP Jan 2026 · question p. 10-11 · answer p. 33-34

253 RTP **May 2025** Q9

Pi and Qu are part of a manufacturing team that handles multiple jobs within the production process. Detailed description of their respective monthly wages is provided below:

The standard working hours for the month are 210. Overtime is compensated at twice the sum of basic wages and dearness allowance. The employer's contributions to State Insurance and the Provident Fund

are equal to the employees' contributions. Employees Pi and Qu were assigned to jobs A, S, and D in the following proportions:

Particulars	Pi	Qu
(i) Basic Wages (₹)	12,000	15,000
(ii) Dearness Allowance	50%	50%
(iii) Contribution to provident Fund (on basic wages)	10%	10%
(iv) Contribution to Employee's State Insurance (on basic wages)	1.75%	1.75%
(v) Overtime (Hours)	8	--

Jobs	A	S	D
Worker Pi	30%	30%	40%
Worker Qu	30%	50%	20%

Overtime was done on job S.

You are required to CALCULATE the earnings of Pi and Qu and allocate the employee cost to each job A, S, and D.

ICAI'S ANSWER

Statement showing Earnings of Workers Pi and Qu

	Pi (₹)	Qu (₹)
Basic wages	12,000	15,000
Dearness Allowance (50% of Basic Wages)	6,000	7,500
Overtime wages (Refer to Working Note)	1,371	--
Gross wages earned	19,371	22,500
Less: Contribution to Provident fund	(1,200)	(1,500)
Less: Contribution to ESI	(210)	(263)
Net wages earned	**17,961**	**20,737**

Statement of Employee Cost:

	Pi (₹)	Qu (₹)
Gross Wages (excluding overtime)	18,000	22,500
Add: Employer's contribution to PF	1,200	1,500
Add: Employer's contribution to ESI	210	263
Gross wages earned	19,410	24,263

Statement Showing Allocation of Wages to Jobs

	Total Wages (₹)	Jobs		
		A (₹)	S (₹)	D (₹)
Worker Pi:				
- Ordinary Wages (3: 3 : 4)	19,410	5,823	5,823	7,764
- Overtime	1,371	--	1,371	--
Worker Qu:				
- Ordinary Wages (3 : 5 : 2)	24,263	7,279	12,131	4,853
	45,044	13,102	19,325	12,617

Working Note:

$$\text{Over time} = 2 \text{ x } (\frac{\text{Basic wage} + \text{DA}}{210 \text{ hours}}) \text{ X } 8 \text{ hours}$$

$$= 2 \text{ x } (\frac{₹\ 18{,}000}{210 \text{ hours}}) \text{ X } 8 \text{ hours}$$

$$= ₹\ 1{,}371$$

ICAI · P4 · RTP May 2025 · question p. 12-13 · answer p. 33-34

254 RTP **Old P3 Nov 2020** Q2 old scheme

GZ Ld. pays the following to a skilled worker engaged in production works. The following are the employee benefits paid to the employee:

(a)	Basic salary per day	₹1,000
(b)	Dearness allowance (DA)	20% of basic salary
(c)	House rent allowance	16% of basic salary
(d)	Transport allowance	₹50 per day of actual work
(e)	Overtime	Twice the hourly rate (considers basic and DA), only if works more than 9 hours a day otherwise no overtime allowance. If works for more than 9 hours a day then overtime is considered after 8th hours.
(f)	Work of holiday and Sunday	Double of per day basic rate provided works atleast 4 hours. The holiday and Sunday basic is eligible for all allowances and statutory deductions.
(h)	Earned leave & Casual leave	These are paid leave.
(h)	Employer's contribution to Provident fund	12% of basic and DA

(i)	Employer's contribution to Pension fund	7% of basic and DA

The company normally works 8-hour a day and 26-day in a month. The company provides 30 minutes lunch break in between.

During the month of August 2020, Mr.Z works for 23 days including 15th August and a Sunday and applied for 3 days of casual leave. On 15th August and Sunday he worked for 5 and 6 hours respectively without lunch break.

On 5th and 13th August he worked for 10 and 9 hours respectively.

During the month Mr. Z worked for 100 hours on Job no.HT200.

You are required to CALCULATE:

(i) Earnings per day

(ii) Effective wages rate per hour of Mr. Z.

(iii) Wages to be charged to Job no.HT200.

ICAI'S ANSWER

Workings:

1. Normal working hours in a month = (Daily working hours – lunch break) × no. of days

= (8 hours – 0.5 hours) × 26 days = 195 hours

2. Hours worked by Mr.Z = No. of normal days worked + Overtime + holiday/ Sunday worked

= (21 days × 7.5 hours) + (9.5 hours + 8.5 hours) + (5 hours + 6 hours)

= 157.5 hours + 18 hours + 11 hours = 186.50 hours.

(i) Calculation of earnings per day

Particulars	Amount (₹)
Basic salary (₹1,000 × 26 days)	26,000
Dearness allowance (20% of basic salary)	5,200
	31,200
House rent allowance (16% of basic salary)	4,160
Employer's contribution to Provident fund (12% × ₹31,200)	3,744
Employer's contribution to Pension fund (7% × ₹31,200)	2,184
	41,288
No. of working days in a month (days)	26
Rate per day	1,588
Transport allowance per day	50

Particulars	Amount (₹)
Earnings per day	**1,638**

(ii) Calculation of effective wage rate per hour of Mr. Z:

Particulars	Amount (₹)
Basic salary (₹1,000 × 26 days)	26,000
Additional basic salary for Sunday & holiday (₹1,000 × 2 days)	2,000
Dearness allowance (20% of basic salary)	5,600
	33,600
House rent allowance (16% of basic salary)	4,480
Transport allowance (₹50 × 23 days)	1,150
Overtime allowance (₹160 × 2 × 2 hours)*	640
Employer's contribution to Provident fund (12% × ₹33,600)	4,032
Employer's contribution to Pension fund (7% × ₹33,600)	2,352
Total monthly wages	46,254
Hours worked by Mr. Z (hours)	186.5
Effective wage rate per hour	**248**

*(Daily Basic + DA) ÷ 7.5 hours

= (1,000+200) ÷ 7.5 = ₹160 per hour

(iii) Calculation of wages to be charged to Job no. HT200

= ₹ 248 × 100 hours = ₹ 24,800

ICAI · Old P3 · RTP Nov 2020 · question p. 1-2 · answer p. 13-14

255 MTP **Sep 2026 S-I** Q1(a) 5 marks

Greenfield Apparel Manufacturing Ltd., a garment export unit located in Tirupur, Tamil Nadu, employs a team of skilled tailoring and finishing-line workers. As part of its annual cost audit, the HR and Costing department has been asked to determine the true cost of employing each worker — including hidden costs like idle time, welfare schemes, and statutory contributions — so that pricing for export orders can be set accurately.

The following data has been extracted from the company's payroll and cost records for the year:

Particulars		Details
(i)	Salary (each employee, per month)	₹ 28,000
(ii)	Bonus	20% of salary

Particulars		Details
(iii)	Employer's contribution to PF, ESI, etc.	12% of salary
(iv)	Total cost of employees' welfare activities	₹ 5,28,000 per annum
(v)	Total leave permitted during the year	24 days
(vi)	No. of employees	220
(vii)	Normal idle time	60 hours per annum
(viii)	Abnormal idle time (due to machine breakdown)	40 hours
(ix)	Working days without leave per annum	312 days of 8 hours
(x)	Overtime worked	100 hours per annum, paid at double the normal hourly rate

You are required to CALCULATE:

(i) The Annual Cost of each employee

(ii) The Employee Cost per Hour

(iii) The Cost of Abnormal Idle Time per employee, arising from the machine breakdown — information the production manager needs to justify a compensation claim against the equipment supplier

(iv) The Overtime Premium per employee, and whether it should be treated as part of normal employee cost or charged separately.

ICAI'S ANSWER

(i) Annual Cost of Each Employee

	Particulars	₹
1.	Salary (₹ 28,000 × 12)	3,36,000
2.	Bonus (20% of Salary)	67,200
3.	Employer's Contribution to PF, ESI, etc. (12% of Salary)	40,320
4.	Employees' Welfare cost (₹ 5,28,000 / 220)	2,400
	Total Annual Cost	**4,45,920**

(ii) Employee Cost per Hour

Particulars	Hours
Effective Working hours (312 days × 8 hours)	2,496
Less: Leave days (24 days × 8 hours)	192
Available Working hours	**2,304**
Less: Normal Idle time	60
Net Working Hours	**2,244**

Employee cost = ₹ 4,45,920

Employee cost per hour = ₹ 4,45,920 / 2,244 hours = ₹ 198.72

Employee Cost per hour = ₹ 198.72

(iii) Cost of Abnormal Idle Time, per Employee

= 40 hours × ₹ 198.72 = ₹ 7,948.80

(iv) Overtime Premium per Employee

Normal hourly rate = ₹ 198.72 (employee cost per hour, as computed above)

Overtime is paid at double the normal hourly rate.

Particulars	₹
Overtime wages (100 hours × ₹ 198.72 × 2)	39,744.00
Less: Normal wages for those hours (100 hours × ₹ 198.72)	19,872.00
Overtime Premium	**19,872.00**

Treatment: The overtime premium of ₹ 19,872 should generally be charged to Costing Profit & Loss Account as an abnormal/separate cost (i.e., not included as part of normal employee cost) if the overtime arose due to a customer's specific rush order or abnormal circumstances. However, if overtime is a regular feature of production policy (e.g., to meet normal capacity shortfall), the premium is treated as part of normal employee cost and absorbed through the overhead/production cost of the related cost centre.

Since the question does not specify a customer-driven rush order, the overtime premium here is best treated as part of normal employee cost, to be recovered as a factory overhead rather than charged off entirely to Costing P&L.

ICAI · P4 · MTP Sep 2026 · Series I · question p. 7-8 · answer p. 6-7

256 MTP **Sep 2026 S-II** Q3(b) 4 marks

Distinguish between:

(i) Casual worker and Outworker

(ii) Job and Batch Costing

ICAI'S ANSWER

i. **Casual worker and outworker**

A worker who is appointed for a short duration to carry on normal business activities in place of a regular but temporarily absent worker. Such a worker is also know as daily wager or 'badlies'. A casual worker do not enjoy the facilities available to a regular worker.

A worker who do not work in the factory premises but either he works in his home or at a site outside the factory is know as an outworker. An outworker who works in a home is usually compensated on the basis of his output. He is supplied with raw materials and tools necessary for carrying out the job. An outworker (outside to factory) is usually engaged on specialized jobs/contract work.

ii. **Job and Batch Costing**

Job Costing: The objective under this method of costing is to ascertain the cost of each job order. A job card is prepared for each job to accumulate costs. The cost of the job is determined by adding all costs against the job it is incurred. This method of costing is used in printing press, foundries and general engineering workshops, advertising etc.

Batch Costing: This system of costing is used where small components/parts of the same kind are required to be manufactured in large quantities. Here batch of similar products is treated as a job and cost of such a job is ascertained as discussed under 1, above. For Example, if in a cycle manufacturing unit, rims are produced in batches of 2,500 units each, then the cost will be determined in relation to a batch of 2,500 units.

ICAI · P4 · MTP Sep 2026 · Series II · question p. 11 · answer p. 15

257 MTP **Jan 2026 S-I** Q1(a) 5 marks

Assuming a man-day of 8 hours, you are required to CALCULATE the labour cost per man-day. The following data has been provided.

(i)	Basic Salary	₹ 80 per day
(ii)	Dearness Allowance	80 paise per every point over 100 cost of living index for working class. Current cost of living index is 785 points.
(iii)	Leave Salary	10% of (i) and (ii)
(iv)	Employer's contribution to Provident Fund	10% of (i), (ii) and (iii)
(v)	Employer's contribution to State Insurance	2.5% of (i), (ii) and (iii)
(vi)	Expenditure on amenities to labour	₹ 30 per head per month
(vii)	Number of working days in a month	25 days of 8 hours each

ICAI'S ANSWER

Statement of Labour Cost (per man-day of 8 hours)

Particulars	(₹)
(a) Basic Salary	80.00

Particulars	(₹)
(b) Dearness Allowance @ 80 paise per every point over 100 cost of living index for a month of 25 days (785 – 100)x0.80/25	21.92
	101.92
(c) Leave Salary – 10% of (a) and (b) 101.92×10/100	10.19
	112.11
(d) Employer's contribution to Provident Fund 10% of (a), (b) and (c) 112.11×10/100	11.21
(e) Employer's contribution to State Insurance 2.5% of (a), (b) and (c) 112.11×2.5/100	2.80
(f) Amenities to labour @ ₹ 30 per head per month of 25 working days ₹ 30/25 days	1.20
Total	127.32

ICAI · P4 · MTP Jan 2026 · Series I · question p. 6-7 · answer p. 5

258 MTP **Jan 2026 S-II** MCQ 13 2 marks MCQ

A worker receives the following monthly earnings:

Basic Pay: ₹ 12,000 per month

Dearness Allowance (D.A.): ₹ 4,000 per month

Fringe Benefits: ₹ 1,500 per month

There are 305 working days in a year, of which 25 days are paid holidays. Assume 8 working hours per day.

What is the employee hour rate?

A ₹ 93.75 per hour

B ₹ 85.50 per hour

C ₹ 97.25 per hour

D ₹ 102.00 per hour

ICAI'S ANSWER

(A) ₹ 93.75 per hour

(i)	Effective working days in a year	305
	Less: Leave days on full pay	25
	Effective working days	280 days
	Total effective working hours (280 days × 8 hours)	2,240 hours
(ii)	Total wages paid in a year (₹)	
	Basic Pay	1,44,000
	D.A.	48,000
	Fringe Benefits	18,000
	Total Wages	2,10,000

(iii) Hourly Rate = ₹ 2,10,000 ÷ 2,240 hours = ₹ 93.75 per hour

ICAI · P4 · MTP Jan 2026 · Series II · question p. 6 · answer p. 6-7

259 MTP **Sep 2025 S-I** Q3(c) 4 marks

The following are the emoluments and benefits received by Mr. A, an employee of XYZ Co.:

Particulars	
Basic pay	₹ 9,000 p.m.
Dearness allowance	₹ 1,800 p.m.
Bonus	20% of salary and D.A.
Other allowances	₹ 2,250 p.m.
Employer's contribution to P.F.	10% of salary and D.A.

Out of the total 2,160 annual working hours of Mr. A, 360 hours are classified as non-productive and are treated as normal idle time.

You are required to COMPUTE the effective hourly cost of employee 'A'.

ICAI'S ANSWER

Statement showing computation of effective hourly cost of employee 'A'

Particulars		Per month (₹)	Per annum (₹)
(A)	Earning of Employee 'A':		
	Basic pay	9,000	1,08,000
	Dearness Allowance	1,800	21,600
	Bonus	2,160	25,920

Particulars		Per month (₹)	Per annum (₹)
	Employer's contribution to P.F.	1,080	12,960
	Other allowances	2,250	27,000
		16,290	**1,95,480**
(B)	Effective working hours (refer working)		1,800 hours
(C)	**Effective hourly cost {(A) ÷ (B)}**		**₹108.60**

Working:

Calculation of effective working hours = Annual working hours - Normal idle time

= 2,160 hours – 360 hours

= 1,800 hours

ICAI · P4 · MTP Sep 2025 · Series I · question p. 12 · answer p. 12

260 MTP **Sep 2024 S-II** Q6(c) 4 marks

DISCUSS the objectives of time keeping & time booking.

ICAI'S ANSWER

Objectives of time keeping and time booking: Time keeping has the following two objectives:

(i) Preparation of Payroll: Wage bills are prepared by the payroll department on the basis of information provided by the time keeping department.

(ii) Computation of Cost: Labour cost of different jobs, departments or cost centers are computed by costing department on the basis of information provided by the time keeping department.

The objectives of time booking are as follows:

(i) To ascertain the labour time spent on a job and the idle labour hours.

(ii) To ascertain labour cost of various jobs and products.

(iii) To calculate the amount of wages and bonus payable under the wage incentive scheme.

(iv) To compute and determine overhead rates and absorption of overheads under the labour and machine hour method.

(v) To evaluate the performance of labour by comparing actual time booked with standard or budgeted time.

ICAI · P4 · MTP Sep 2024 · Series II · question p. 12 · answer p. 21

261 MTP **Old P3 May 2023 S-II** Q5(a) 10 marks old scheme

MT Ltd. pays the followings to skilled workers engaged in production works. The following are the employee benefits paid to the employees:

(a)	Basic salary per day	₹1,000
(b)	Dearness allowance (DA)	20% of basic salary

(c)	House rent allowance	16% of basic salary
(d)	Transport allowance	₹50 per day of actual work
(e)	Overtime	Twice the hourly rate (considers basic and DA), only if works more than 9 hours a day otherwise no overtime allowance. If works for more than 9 hours a day then overtime is considered after 8th hours.
(f)	Work of holiday and Sunday	Double of per day basic rate provided works atleast 4 hours. The holiday and Sunday basic is eligible for all allowances and statutory deductions.
(h)	Earned leave & Casual leave	These are paid leave.
(h)	Employer's contribution to Provident fund	12% of basic and DA
(i)	Employer's contribution to Pension fund	7% of basic and DA

The company normally works 8-hour a day and 26-day in a month. The company provides 30 minutes lunch break in between.

During the month of August 2020, Mr. Z works for 23 days including 15th August and a Sunday and applied for 3 days of casual leave. On 15th August and Sunday he worked for 5 and 6 hours respectively without lunch break.

On 5th and 13th August he worked for 10 and 9 hours respectively.

During the month Mr. Z worked for 100 hours on Job no.HT200.

You are required to CALCULATE:

(i) Earnings per day

(ii) Effective wages rate per hour of Mr. Z.

(iii) Wages to be charged to Job no.HT200

ICAI'S ANSWER

Workings:

1. Normal working hours in a month = (Daily working hours – lunch break) × no. of days

= (8 hours – 0.5 hours) × 26 days = 195 hours

2. Hours worked by Mr.Z = No. of normal days worked + Overtime + holiday/ Sunday worked

= (21 days × 7.5 hours) + (9.5 hours + 8.5 hours) + (5 hours + 6 hours)

= 157.5 hours + 18 hours + 11 hours = 186.50 hours.

(i) Calculation of earnings per day

Particulars	Amount (₹)
Basic salary (₹1,000 × 26 days)	26,000

Particulars	Amount (₹)
Dearness allowance (20% of basic salary)	5,200
	31,200
House rent allowance (16% of basic salary)	4,160
Employer's contribution to Provident fund (12% × ₹31,200)	3,744
Employer's contribution to Pension fund (7% × ₹31,200)	2,184
	41,288
No. of working days in a month (days)	26
Rate per day	1,588
Transport allowance per day	50
Earnings per day	**1,638**

(ii) Calculation of effective wage rate per hour of Mr. Z:

Particulars	Amount (₹)
Basic salary (₹1,000 × 26 days)	26,000
Additional basic salary for Sunday & holiday (₹1,000 × 2 days)	2,000
Dearness allowance (20% of basic salary)	5,600
	33,600
House rent allowance (16% of basic salary)	4,480
Transport allowance (₹50 × 23 days)	1,150
Overtime allowance (₹160 × 2 × 2 hours)*	640
Employer's contribution to Provident fund (12% × ₹33,600)	4,032
Employer's contribution to Pension fund (7% × ₹33,600)	2,352
Total monthly wages	46,254
Hours worked by Mr.Z (hours)	186.5
Effective wage rate per hour	**248**

*(Daily Basic+DA) ÷ 7.5 hours

= (1,000+200) ÷ 7.5 = ₹₹160 per hour

(iii) Calculation of wages to be charged to Job no. HT200

= ₹248 × 100 hours = ₹24,800

262 MTP **Old P3 Nov 2022 S-I** Q4(a) 10 marks old scheme

Archika Tyre Manufacturing Private Limited has four workers Ram, Shyam, Mohan & Kundan who are paid wages on the basis of ₹ 100 per day, ₹ 120 per day, ₹ 130 per day & ₹ 2500 per month respectively.

Standard working days in a week are six of 8 hours per day. For the month of October 2022, there was only one holiday other than Sunday for which no payment was made to employees except Kundan who was paid for full month. Sundays are considered paid holidays i.e. employees are paid for Sunday also even there is no working on that day. Provident fund contribution is 8% of monthly wages by employer and employee each. ESI contribution is 5% of monthly wages by employer and 4% of monthly wages by employee.

On the basis of above information, you are required to CALCULATE (regarding the month of October 2022):

(i) Amount of net wages receivable by each employee from the employer.

(ii) What is the total amount of Provident Fund required to be deposited by employer?

(iii) What is the total amount of ESI required to be deposited by employer?

(iv) What is the total labour cost to employer?

(v) If total material cost is ₹ 20,000 for October 2022 and overheads are charged equal to labour cost, calculate total cost for the month.

ICAI'S ANSWER

(i) Calculation of net wages receivable by each employee from the employer (October 2022):

	Ram (₹)	Shyam (₹)	Mohan (₹)	Kundan (₹)	Total (₹)
Wages for October 2022	3,000 (₹ 100 x 30 days)	3,600 (₹ 120 x 30 days)	3,900 (₹ 130 x 30 days)	2,500	13,000
Less: Employee Contribution to PF @ 8%	240	288	312	200	1,040
Less: Employee Contribution to ESI @ 4%	120	144	156	100	520
Net Wages Receivable	2,640	3,168	3,432	2,200	11,440

(ii) Calculation of total amount of Provident Fund required to be deposited by employer (October 2022):

	(₹)
Total Wages for the month	13,000
Employer's Contribution to Provident Fund @8% of ₹ 13,000	1,040
Add: Employee's Contribution to Provident Fund @8% of ₹ 13,000	1,040
Total amount of Provident Fund required to be deposited by employer	2,080

(iii) Calculation of total amount of ESI required to be deposited by employer (October 2022):

	(₹)
Total Wages for the month	13,000
Employer's Contribution to ESI @5% of ₹ 13,000	650
Add: Employee's Contribution to ESI @4% of ₹ 13,000	520
Total amount of ESI required to be deposited by employer	1,170

(iv) Total labour cost to employer (October 2022):

	(₹)
Total Wages for the month	13,000
Add: Employer's Contribution to Provident Fund @8% of ₹ 13,000	1,040
Add: Employer's Contribution to ESI @5% of ₹ 13,000	650
Total labour cost to employer	14,690

(v) Calculation of Total Cost for October 2022

	(₹)
Total Material Cost	20,000
Total Labour Cost	14,690
Total Overheads (Equal to Labour Cost)	14,690
Total Cost	49,380

ICAI · Old P3 · MTP Nov 2022 · Series I · question p. 5 · answer p. 7-8

4.2 Idle time and overtime

16 questions · asked in 17 ICAI documents · 5 exam papers

263 EXAM Sep 2026 MCQ 14 2 marks MCQ

A company sells handcrafted labour-intensive gift items. A certain urgent order at a fixed selling price entailed 200 overtime hours. The overtime premium per hour was ₹ 300. Of these hours, 150 were due to extra work above capacity, whereas 50 hours were due to rework that arose due to faulty material purchased and not inspected by the "Raw Materials" Section, a cost centre. What will be the appropriate treatment of overtime premium ?

A The entire premium of ₹ 60,000 is charged to the urgent order.

B Premium of ₹ 45,000 is charged to the order and ₹ 15,000 is charged to the Materials Section.

C ₹ 60,000 is charged to the Costing P and L A/c.

D ₹ 60,000 is treated as material cost and added to cost of defective material.

ICAI'S ANSWER

Answer key: **ICAI has not published the key yet.**

ICAI · P4 · Sep 2026 exam · question p. 9

264 EXAM **Jan 2025** MCQ 13 MCQ

The data pertaining to the worker C in a factory depicts that he is paid at a rate of ₹ 100 per hour and a week comprises 48 hours for a 6 days' work. The allowed absence time is 15 minutes per day for maintenance etc. The job card of C indicates, his chargeable time is scattered for 2 different jobs J1-21 hours and J2-24 hours. Any unaccounted time is attributable for power failure.

Calculate cost of normal idle time and abnormal idle time.

A ₹ 100 and ₹ 150

B ₹ 150 and ₹ 150

C ₹ 150 and ₹ 100

D ₹ 100 and ₹ 100

ICAI'S ANSWER

Answer key: **(B)**

ICAI · P4 · Jan 2025 exam · question p. 5 · answer p. 6

265 EXAM **Sep 2024** Q2(b) 6 marks

In a factory there are 50 workers, working 8 hours per day including 30 minutes for lunch break, worked for 160 days during a period of six months ended on 31st December, 2023. During this period total employee's cost was recorded ₹ 3,90,000. The management of the factory decided the overtime premium rates for the month of January 2024 as under :

Sundays and holidays	180% of basic wages rate
Before and after normal working hours	160% of basic wages rate

During the last six months (ended on 31st December, 2023), the following hours were worked:

Normal time	56,250
Sundays and holidays	750
Before and after normal working hours	3,000
Total hours·	60,000

During the month of January 2024, the factory worked on a job BX in the following manner.

Normal working	2,400	men hours
Overtime on Sundays and holidays	200	men hours
Overtime before and after normal working	400	men hours
Total hours	3,000	

You are required to calculate the labour cost chargeable to job BX and overheads in each of the following situations:

(i) Where overtime is worked regularly in whole year as a policy on account of shortage of workers.

(ii) Where overtime is worked irregularly to meet the requirement of production.

(iii) Where overtime is worked at the request of the customer to complete the job in time.

(iv) Where overtime is worked on account of flood in the area.

ICAI'S ANSWER

Workings

Calculation of Basic Wage Rate

Effective Working Hours for six months ended 31st December, 2023

= 50 workers x (8-0.5) hours per day x 160 days = 60,000 hours

Basic Wage Rate = $\frac{₹3,90,000}{60,000}$

= ₹ 6.50 per hour

Basic wage rate : ₹ 6.50 per hour

Overtime wage rate for Sundays and holidays : ₹ 6.50 × 180%

= ₹ 11.70 per hour

Overtime wage rate before and after working hours : ₹ 6.50 × 160%

= ₹ 10.40 per hour

Computation of average inflated wage rate (including overtime premium):

Particulars	(₹)
Annual wages for the normal time (56,250 hrs × ₹ 6.50)	3,65,625
Wages for overtime on Sundays and holidays (750 hrs × ₹ 11.70)	8,775
Wages for overtime before and after working hours (3,000 hrs × ₹ 10.40)	31,200
Total wages for 60,000 hrs	**4,05,600**

Average inflated wage rate = $\frac{₹4,05,600}{60,000 \text{ hours}}$ = ₹ 6.76 per hour

(a) Where overtime is worked regularly as a policy due to shortage of workers:

The overtime premium is treated as a part of employee cost and job is charged at an inflated wage rate. Hence, employee cost chargeable to job BX

= Total hours × Inflated wage rate = 3,000 hrs. × ₹ 6.76 = ₹ 20,280

(b) Where overtime is worked irregularly to meet the requirements of production:

Basic wage rate is charged to the job and overtime premium is charged to factory overheads as under:

Employee cost chargeable to Job BX: 3,000 hours @ ₹ 6.50 per hour = **₹ 19,500**

Factory overhead: {200 hrs. × ₹ (11.70 – 6.50)} + {400hrs × ₹ (10.40 – 6.50)} = {₹ 1,040 + ₹ 1,560} = **₹ 2,600**

(c) Where overtime is worked at the request of the customer to complete the job in time.

Basic wage rate is charged to the job and overtime premium is charged to job as under:

Employee cost chargeable to Job BX: 3,000 hours @ ₹ 6.50 per hour = **₹ 19,500**

Overtime premium: {200 hrs × ₹ (11.70 – 6.50)} + {400hrs × ₹ (10.40 – 6.50)} = {₹ 1,040 + ₹ 1,560} = **₹ 2,600**

Total = ₹ 19,500 + ₹ 2,600 = ₹ 22,100

(d) Where overtime is worked on account of flood in the area

Basic wage rate is charged to the job and overtime premium is charged to costing P & L Account as under:

Employee cost chargeable to Job BX: 3,000 hours @ ₹ 6.50 per hour = **₹ 19,500**

Costing P&L A/c : {200 hrs × ₹ (11.70 – 6.50)} + {400hrs × ₹ (10.40 – 6.50)} = {₹ 1,040 + ₹ 1,560} = **₹ 2,600**

Total = ₹ 19,500 + ₹ 2,600 = ₹ 22,100

ICAI · P4 · Sep 2024 exam · question p. 14-15 · answer p. 18-19

266 EXAM **May 2024** Q6(c) 4 marks

Also asked in MTP May 2026 S-II Q6(c)

Describe briefly idle time and explain the treatment of idle time in cost accounts in following situations:

(i) The setting up time for the machine in case of Direct Worker Mr. A.

(ii) Normal break time for lunch in case of Indirect Worker Mr. B.

(iii) Time lost due to breakdown of machine in case of Worker Mr. C.

ICAI'S ANSWER

The time during which no production is carried-out because the worker remains idle but are paid. In other words, it is the difference between the time paid and the time booked. Idle time can be normal or abnormal.

Situation	Idle Time	Treatment
The setting up time for the machine in case of Direct Worker Mr. A	Normal idle time	It is treated as a part of cost of production. It is to be considered while setting of standard hours or standard rate.
Normal rest time, break time for lunch in case of Indirect Worker Mr. B	Normal idle time	It is to be considered for the computation of overhead rate.
Time lost due to break-down of machines in case of Worker Mr. C	Abnormal idle time	It is to be shown as a separate item in the Costing Profit and Loss Account.

ICAI · P4 · May 2024 exam · question p. 25 · answer p. 27

267 EXAM **Old P3 May 2022** Q6(c) 5 marks old scheme

Explain the treatment of Overtime Premium in following situations:

(i) SV & Co. wants to grab some special orders, and overtime is required to meet the same.

(ii) Dept. X has to work overtime to make up a shortfall in production due to some fault of management in dept. Y.

(iii) S Ltd. has to work overtime regularly throughout the year as a policy due to the workers' shortage.

(iv) Due to flood in Odisha, RS Ltd. has to work overtime to complete the job.

(v) A customer requested the company MN Ltd. to expedite the job because of his urgency of work.

ICAI'S ANSWER

Treatment of Overtime premium in different situations

	Situation	Treatment
(i)	SV & Co. wants to grab some special orders, and overtime is required to meet the same.	If overtime is required to cope with general production programmes or for meeting urgent orders, the overtime premium should be treated as overhead cost of the particular department or cost centre which works overtime.
(ii)	Dept. X has to work overtime to make up a shortfall in production due to some fault of management in dept. Y.	If overtime is worked in a department due to the fault of another department, the overtime premium should be charged to the latter department (Y).
(iii)	S Ltd. has to work overtime regularly throughout the year as a policy due to the workers' shortage.	The overtime premium is treated as a part of employee cost and job is charged at an effective average wage rate.
(iv)	Due to flood in Odisha, RS Ltd. has to work overtime to complete the job.	Overtime worked on account of abnormal conditions such as flood, earthquake etc., should not be charged to cost, but to Costing Profit and Loss Account.
(v)	A customer requested the company MN Ltd. to expedite the job because of his urgency of work.	Where overtime is worked at the request of the customer, overtime premium is also charged to the job/ customer directly.

ICAI · Old P3 · May 2022 exam · question p. 27 · answer p. 29

268 RTP **Sep 2025** Q3 MCQ

Suppose the units of a product are produced at the rate of 4 units per useful direct labour hour. Direct labour idle time is 10% of hours paid for. Sales of 1,000 units are budgeted and finished goods stock is expected to rise by 100 units. The budgeted direct labour hours for the production would be:

A 306

B 250

C 275

D 400

ICAI'S ANSWER

Answer key: **(A)**

(a) 306

Total production required = Budgeted sales + Increase in finished goods stock

= 1,000+100=1,100 units

Production rate = 4 units per useful direct labour hour

Useful hours = $\frac{1{,}100 \text{ units}}{4 \text{ units/hour}}$ = 275 hours

Useful hours = 90% of paid hours:

Paid hours = 275/90% = 305.55 = 306 hours

ICAI · P4 · RTP Sep 2025 · question p. 7-8 · answer p. 29

269 RTP **Old P3 May 2022** Q2 old scheme

A total of 108 labour hours have been put in a particular job card for repair work engaging a semi-skilled and skilled labour (Mr. Deep and Mr. Sam respectively).

The hours devoted by both the workers individually on daily basis for this particular job are given below:

Monday	Tuesday	Wednesday	Thursday	Friday
10.5	8.0	10.5	9.5	10.5

The skilled labour also worked on Saturday for 10 hours.

Sunday is a weekly holiday and each worker has to work for 8 hours on all week days and 5 hours on Saturdays; the workers are however paid full wages for Saturday (8 hours for 5 hours worked).

Semi-skilled and skilled worker is paid ordinary wage @ ₹ 400 and ₹ 600 respectively per day of 8 hours labour. Further, the workers are also paid dearness allowance @ 20%.

Extra hours worked over and above 8 hours are also paid at ordinary wage rate however, overtime premium of 100% of ordinary wage rate is paid if a worker works for more than 9 hours in a day AND 48 hours in a week.

You are required to COMPUTE the wages payable to Mr. Deep (Semi-skilled) and Mr. Sam (Skilled).

ICAI'S ANSWER

Calculation of total normal hours to be paid for Mr. Deep (Semi-skilled):

Day	Normal hours	Extra hours	Overtime hours	Equivalent normal hours for overtime worked	Total normal hours
	A	**B**	**C**	**D = C×2**	**E = A+B+D**
Monday	8	1	1½	3	12
Tuesday	8	--	--	--	8
Wednesday	8	1	1½	3	12
Thursday	8	1	½	1	10
Friday	8	1	1½	3	12
Saturday	--	--	--	--	--
Total	40	4	5	10	54

Calculation of total normal hours to be paid for Mr. Sam (Skilled):

Day	Normal hours	Extra hours	Overtime hours	Equivalent normal hours for overtime worked	Total normal hours
	A	**B**	**C**	**D = C×2**	**E = A+B+D**
Monday	8	1	1½	3	12
Tuesday	8	---	---	---	8
Wednesday	8	1	1½	3	12
Thursday	8	1	½	1	10
Friday	8	1	1½	3	12
Saturday	5	3* + 1	1**	2	11
Total	**45**	**8**	**6**	**12**	**65**

*Mr. Sam will be paid for equivalent 8 normal working hours at ordinary wage rate, though 5 hours of working is required on Saturday. Further, extra 9th hour worked will also be paid at ordinary wage rate.

** Overtime of 1 hour worked over and above 9 hours will be paid at overtime rate.

Wages payable:

	Mr. Deep	Mr. Sam
Basic Wages per hour (₹ 400/8, ₹ 600/8) (₹)	50	75
Dearness allowance per hour (@ 20%) (₹)	10	15
Hourly rate (₹)	60	90
Total equivalent normal hours	54	65
Total Wages payable (₹)	**3,240**	**5,850**

ICAI · Old P3 · RTP May 2022 · question p. 1-2 · answer p. 12-13

270 RTP Old P3 Dec 2021 Q2 old scheme

Textile Ltd. pays following overtime premium for its labour beside normal wages of ₹ 100 per hour:

Before and after normal working hours	80% of basic wage rate
Sundays and holidays	150% of basic wage rate

During the previous year 2019-20, the following hours were worked:

Normal time	3,00,000 hours
Overtime before and after normal working hours	60,000 hours
Overtime on Sundays and holidays	15,000 hours
Total	3,75,000 hours

During the current year 2020-21, the following hours have been worked on job 'Spinning':

Normal	4,000 hours
Overtime before and after normal working hours	400 hours
Overtime on Sundays and holidays	100 hours
Total	4,500 hours

You are required to CALCULATE the labour cost chargeable to job 'Spinning' and overhead in each of the following instances:

(a) Where overtime is worked regularly throughout the year as a policy due to the workers' shortage.

(b) Where overtime is worked irregularly to meet the requirements of production.

(c) Where overtime is worked at the request of the customer to expedite the job.

ICAI'S ANSWER

Workings:

Basic wage rate	= ₹ 100 per hour
Overtime wage rate before and after working hours	= ₹ 100 + (₹ 100 × 80%)
	= ₹ 180 per hour
Overtime wage rate for Sundays and holidays	= ₹ 100 + (₹ 100 × 150%)
	= ₹ 250 per hour

Computation of average inflated wage rate (including overtime premium):

Particulars	Amount (₹)
Annual wages for the previous year for normal time (3,00,000 hrs. × ₹ 100)	3,00,00,000
Wages for overtime before and after normal working hours (60,000 hrs. × ₹ 180)	108,00,000
Wages for overtime on Sundays and holidays (15,000 hrs. × ₹ 250)	37,50,000
Total wages for 3,75,000 hrs.	**4,45,50,000**

Average inflated wage rate = $\frac{₹\ 4,45,50,000}{3,75,000\ \text{hours}}$ = ₹ 118.80

(a) **Where overtime is worked regularly as a policy due to workers' shortage**

The overtime premium is treated as a part of employee cost and job is charged at an inflated wage rate. Hence, employee cost chargeable to job 'Spinning'

= Total hours × Inflated wage rate = 4,500hrs. × ₹ 118.80 = **₹ 5,34,600**

(b) **Where overtime is worked irregularly to meet the requirements of production**

Basic wage rate is charged to the job and overtime premium is charged to factory overheads as under:

Employee cost chargeable to Job 'Spinning' = 4,500hours @ ₹ 100 per hour

= **₹ 4,50,000**

Factory overhead = {400 hrs. × (₹ 100 × 80%)} + {100 hrs. × (₹ 100 × 150%)}

= {₹ 32,000 + ₹ 15,000} = **₹ 47,000**

(c) **Where overtime is worked at the request of the customer, overtime premium is also charged to the job as under:**

		(₹)
Job 'Spinning' Employee cost: 4,500hrs. @ ₹ 100	=	4,50,000
Overtime premium: 400 hrs. @ (₹ 100 × 80%)	=	32,000
100 hrs. @ (₹ 100 × 150%)	=	15,000
Total		**4,97,000**

ICAI · Old P3 · RTP Dec 2021 · question p. 1-2 · answer p. 14-15

271 MTP **Sep 2025 S-I** Q4(b) 4 marks

Also asked in Old P3 MTP May 2020 S-I Q6(a)

DISCUSS the accounting treatment of Idle time and overtime wages.

ICAI'S ANSWER

Accounting treatment of idle time wages & overtime wages in cost accounts: Normal idle time is treated as a part of the cost of production. Thus, in the case of direct workers, an allowance for

normal idle time is built into the labour cost rates. In the case of indirect workers, normal idle time is spread over all the products or jobs through the process of absorption of factory overheads.

Under Cost Accounting, the overtime premium is treated as follows:

- If overtime is resorted to at the desire of the customer, then the overtime premium may be charged to the job directly.
- If overtime is required to cope with general production program or for meeting urgent orders, the overtime premium should be treated as overhead cost of particular department or cost center which works overtime.
- Overtime worked on account of abnormal conditions should be charged to costing Profit & Loss Account.
- If overtime is worked in a department due to the fault of another department the overtime premium should be charged to the latter department.

ICAI · P4 · MTP Sep 2025 · Series I · question p. 13 · answer p. 13-14

272 MTP **Sep 2025 S-II** MCQ 14 2 marks MCQ

In a factory, a worker is paid ₹ 200 per day as basic pay, along with a Dearness Allowance (D.A.) of 120% of the basic pay, for working 6 days a week, 8 hours each day. He is allowed a 25 minutes lunch break and a 15 minutes tea break during his shift. Throughout the week, his time card recorded a total of 13 hours charged to Job A, 12 hours to Job B, and 15 hours to Job C. These hours reflect the time spent on each specific task during the workweek.

The time that was not assigned to specific jobs was considered abnormal idle time, as it was spent waiting for the next job to become available. This waiting time, which wasn't booked to any particular job, represents lost time that could have been used more efficiently if there had been a continuous flow of work.

CALCULATE the amount of wages to be allocated to different jobs and STATE the treatment of abnormal idle time.

A Job A - ₹ 780, Job B - ₹ 720, Job C- 900 and wages paid for Abnormal idle time - ₹ 240 to be charged to Job C as maximum time is devoted to this Job.

B Job A - ₹ 900, Job B - ₹ 720, Job C- 780 and wages paid for Abnormal idle time - ₹ 240 to be charged to Costing Profit & Loss A/c.

C Job A - ₹ 780, Job B - ₹ 720, Job C- 900 and wages paid for Abnormal idle time - ₹ 240 to be charged to Costing Profit & Loss A/c.

D Job A - ₹ 900, Job B - ₹ 720, Job C- 780 and wages paid for Abnormal idle time - ₹ 240 to be charged to Job C as maximum time is devoted to this Job.

ICAI'S ANSWER

(C) Job A - ₹ 780, Job B - ₹ 720, Job C- 900 and wages paid for Abnormal idle time - ₹ 240 to be charged to Costing Profit & Loss A/c.

Working notes:

(i) Total effective hours in a week:

[(8 hrs. – (25 minutes + 15 minutes)] × 6 days = 44 hours

(ii) Total wages for a week:

(₹ 200 + 120% of ₹ 200) × 6 days = ₹ 2,640

(iii) Wage rate per hour = $\left(\frac{₹\ 2{,}640}{44\ \text{Hours}}\right)$ = ₹ 60

(iv) Time wasted waiting for job (Abnormal idle time):

= 44 hrs. – (13 hrs. + 12 hrs. + 15 hrs.) = 4 hrs.

Allocation of wages		(₹)
Job A	13 hours × ₹ 60	780
Job B	12 hours × ₹ 60	720
Job C	15 hours × ₹ 60	900
Charged to Costing Profit & Loss A/c	4 hours × ₹ 60	240
Total		2,640

ICAI · P4 · MTP Sep 2025 · Series II · question p. 7-8 · answer p. 7

273 MTP **May 2025 S-I** Q1(c) 4 marks

PR Ltd. totally understands the importance of employee motivation and retention, thus, it has implemented a structured wage policy. The basic wage rate is set at ₹ 100 per hour along with Dearness Allowance (DA) of 50%, which is competitive within the industry. To encourage productivity and compensate for the demands of overtime, the company offers enhanced rates for hours worked beyond the standard schedule. The normal working days are from Monday to Saturday, and the worker would be paid overtime premium of-

75% of basic wage rate (inclusive of DA) for working before and after normal working hours, and

125% of basic wage rate (inclusive of DA) for working on Sundays.

Throughout the previous year, the following hours were worked-

Normal time	4,00,000 hours
Overtime before and after normal working hours	80,000 hours
Overtime on Sundays	20,000 hours
Total	5,00,000 hours

However, the hours recorded for job 'PR123' are as follows:

Normal time	4,000 hours
Overtime before and after normal working hours	400 hours
Overtime on Sundays	100 hours
Total	4,500 hours

You are required to CALCULATE the employee cost chargeable to job 'PR123' and the treatment of overtime premium:

(i) where overtime is worked regularly throughout the year as a policy due to the workers' shortage.

(ii) where overtime is worked irregularly to meet the requirements of production.

ICAI'S ANSWER

Workings

Basic wage rate = ₹ (100 + ₹ 50 for DA) per hour

= ₹ 150 per hour

Overtime wage rate before and after working hours

= ₹ 150 + 75% of ₹ 150

= ₹ 262.50 per hour

Overtime wage rate for Sundays = ₹ 150 + 125% of ₹ 150

= ₹ 337.50 per hour

Computation of average inflated wage rate (including overtime premium)

Particulars	(₹)
Annual wages for the previous year for normal time (4,00,000 hrs. × ₹ 150)	6,00,00,000
Wages for overtime before and after working hours (80,000 hrs. × ₹ 262.50)	2,10,00,000
Wages for overtime on Sundays (20,000 hrs. × ₹ 337.50)	67,50,000
Total wages for 5,00,000 hrs.	**8,77,50,000**

Average inflated wage rate = $\frac{₹8,77,50,000}{5,00,000 \text{ hrs.}}$ = ₹ 175.50

(i) Where overtime is worked regularly as a policy due to workers' shortage:

The overtime premium is treated as a part of employee cost and job is charged at an inflated wage rate.

Hence, employee cost chargeable to job 'PR123'

= Total hours × Inflated wage rate

= 4,500 hrs. × ₹ 175.50 = **₹ 7,89,750**

(ii) Where overtime is worked irregularly to meet the requirements of production:

Basic wage rate is charged to the job and overtime premium is charged to factory overheads as under:

Employee cost chargeable to Job 'PR123'

= 4,500 hours @ ₹ 150 per hour

= ₹ 6,75,000

Factory overhead = {400 hrs. × ₹ (75% of ₹ 150)} + {100 hrs. × ₹ (125% of ₹ 150)}

= {₹ 45,000 + ₹ 18,750}

= ₹ 63,750

ICAI · P4 · MTP May 2025 · Series I · question p. 7-8 · answer p. 9-10

274 MTP **May 2025 S-I** Q2(b) 4 marks

DISCUSS the effect of overtime payment on productivity.

ICAI'S ANSWER

Effect of overtime payment on productivity: Overtime work should be resorted to only when it is extremely essential because it involves extra cost. The overtime payment increases the cost of production in the following ways:

1. The overtime premium paid is an extra payment in addition to the normal rate.
2. The efficiency of operators during overtime work may fall and thus output may be less than normal output.
3. In order to earn more the workers may not concentrate on work during normal time and thus the output during normal hours may also fall.
4. Reduced output and increased premium of overtime will bring about an increase in cost of production.

ICAI · P4 · MTP May 2025 · Series I · question p. 9 · answer p. 12

275 MTP **May 2025 S-II** Q6(c) 4 marks

EXPLAIN the meaning of Idle time and SHOW the treatment for the following causes of Idle time.

Causes	Treatment
1. The time lost between factory gate and the place of work,	
2. The interval between one job and another,	
3. Idle time may also arise due to abnormal factors like lack of coordination	
4. Power failure, Breakdown of machines	

ICAI'S ANSWER

Idle Time - The time during which no production is carried-out because the worker remains idle but are paid. In other words, it is the difference between the time paid and the time booked. *Idle time can be normal or abnormal.* The time for which employees are paid includes holidays, paid leaves, allowable rest or off time etc.

Causes	Treatment
1. The time lost between factory gate and the place of work, 2. The interval between one job and another,	It is treated as a part of cost of production. Thus, in the case of direct workers an allowance for normal idle time is considered setting of standard hours or standard rate. In case of indirect workers, normal idle time is considered for the computation of overhead rate
3. Idle time may also arise due to abnormal factors like lack of coordination 4. Power failure, Break-down of machines	*Abnormal idle time* cost is not included as a part of production cost and is shown as a separate item in the Costing Profit and Loss Account.

ICAI · P4 · MTP May 2025 · Series II · question p. 14-15 · answer p. 19-20

276 MTP **Jan 2025 S-II** Q4(b) 4 marks

Idle time is the time during which no production is carried-out because the worker remains idle but are paid. It can be normal or abnormal. LIST OUT some of the causes/examples of normal and abnormal idle time.

ICAI'S ANSWER

Causes/examples of normal idle time:

1. The time lost between factory gate and the place of work.
2. The interval between one job and another.
3. The setting up time for the machine.
4. Normal rest time, **break for lunch etc.**

Causes/examples of abnormal idle time:

1. Lack of coordination.
2. Power failure, Breakdown of machines.
3. Non-availability of raw materials, strikes, lockouts, poor supervision, fire, flood etc.

ICAI · P4 · MTP Jan 2025 · Series II · question p. 11 · answer p. 14

277 MTP **Old P3 Nov 2022 S-II** Q6(b) 5 marks old scheme

STATE various causes of and treatment of Overtime Premium in Cost Accounting.

ICAI'S ANSWER

Causes and Treatment of Overtime premium in cost accounting

Causes	*Treatment*
(1) The customer may agree to bear the entire charge of overtime because urgency of work.	(1) If overtime is resorted to at the desire of the customer, then overtime premium may be charged to the job directly.

Causes	*Treatment*
(2) Overtime may be called for to make up any shortfall in production due to some unexpected development.	(2) If overtime is required to cope with general production programmes or for meeting urgent orders, the overtime premium should be treated as overhead cost of the particular department or cost centre which works overtime.
(3) Overtime work may be necessary to make up a shortfall in production due to some fault of management.	(3) If overtime is worked in a department due to the fault of another department, the overtime premium should be charged to the latter department.
(4) Overtime work may be resorted to, to secure an out-turn in excess of the normal output to take advantage of an expanding market or of rising demand	(4) Overtime worked on account of abnormal conditions such as flood, earthquake etc., should not be charged to cost, but to Costing Profit and Loss Account.

ICAI · Old P3 · MTP Nov 2022 · Series II · question p. 7 · answer p. 14-15

278 MTP **Old P3 Nov 2020 S-I** Q3(b) 10 marks old scheme

In a factory, the basic wage rate is ₹ 300 per hour and overtime rates are as follows:

Before and after normal working hours	180% of basic wage rate
Sundays and holidays	230% of basic wage rate

During the previous year, the following hours were worked	
- Normal time	1,00,000 hours
- Overtime before and after working hours	20,000 hours
Overtime on Sundays and holidays	5,000 hours
Total	1,25,000 hours

The following hours have been worked on job 'A'

Normal	1,000 hours
Overtime before and after working hrs.	100 hours.
Sundays and holidays	25 hours.
Total	1,125 hours

You are required to CALCULATE the labour cost chargeable to job 'A' and overhead in each of the following instances:

(i) Where overtime is worked regularly throughout the year as a policy due to the workers' shortage.

(ii) Where overtime is worked irregularly to meet the requirements of production.

(iii) Where overtime is worked at the request of the customer to expedite the job.

ICAI'S ANSWER

Workings

Basic wage rate	: ₹ 300 per hour
Overtime wage rate before and after working hours	: ₹ 300 × 180% = ₹ 540 per hour
Overtime wage rate for Sundays and holidays	: ₹ 300 × 230% = ₹ 690 per hour

Computation of average inflated wage rate (including overtime premium):

Particulars	Amount (₹)
Annual wages for the previous year for normal time (1,00,000 hrs. × ₹ 300)	3,00,00,000
Wages for overtime before and after working hours (20,000 hrs. × ₹ 540)	1,08,00,000
Wages for overtime on Sundays and holidays (5,000 hrs. × ₹ 690)	34,50,000
Total wages for 1,25,000 hrs.	**4,42,50,000**

Average inflated wage rate = $\frac{\text{Rs.}4,42,50,000}{1,25,000\text{hours}}$ = ₹354

(i) **Where overtime is worked regularly as a policy due to workers' shortage:**

The overtime premium is treated as a part of employee cost and job is charged at an inflated wage rate. Hence, employee cost chargeable to job 'A'

= Total hours × Inflated wage rate = 1,125 hrs. × ₹ 354 = ₹ 3,98,250

(ii) **Where overtime is worked irregularly to meet the requirements of production:**

Basic wage rate is charged to the job and overtime premium is charged to factory overheads as under:

Employee cost chargeable to Job 'A': 1,125 hours @ ₹300 per hour = ₹3,37,500

Factory overhead: {100 hrs. × ₹ (540 – 300)} + {25 hrs. × ₹ (690 – 300)}

= {₹ 24,000 + ₹ 9,750} = ₹ 33,750

(iii) **Where overtime is worked at the request of the customer, overtime premium is also charged to the job as under:**

			(₹)
Job 'A' Employee cost	1,125 hrs. @ ₹ 300	=	3,37,500
Overtime premium	100 hrs. @ ₹ (540 – 300)	=	24,000
	25 hrs. @ ₹ (690 – 300)	=	9,750
Total			**3,71,250**

ICAI · Old P3 · MTP Nov 2020 · Series I · question p. 4-5 · answer p. 7

4.3 Employee turnover: measurement, causes, effects and cost

28 questions · asked in 24 ICAI documents · 7 exam papers

279 EXAM **May 2026** Q4(b) 5 marks

The labour turnover rates for the quarter ended 31st March, 2026 are computed as 12%, 6% and 4% under Flux method, Replacement method and Separation method respectively. The number of workers replaced during that quarter is 54.

You are required to ascertain -

(i) Average number of workers on roll for the quarter

(ii) Number of workers recruited and joined

(iii) Number of workers left and discharged

ICAI'S ANSWER

Working Note:

(i) Average number of workers on roll (for the quarter):

Employee Turnover rate using Replacement method

$$= \frac{\text{No. of replacements}}{\text{Average number of workers on roll}} \times 100$$

Or, $\frac{6}{100} = \frac{54}{\text{Average number of workers on roll}}$

Or, Average number of workers on roll = $\frac{54 \times 100}{6} = 900$

(ii) Number of workers recruited and joined:

Employee turnover rate (Flux method)

$$= \frac{\text{No. of Separations * (S)} + \text{No. of Accessions (A)}}{\text{Average number of workers on roll}}$$

Or, $\frac{12}{100} = \frac{36^* + A}{900}$ Or, A = $\left[\frac{10{,}800}{100} - 36\right] = 72$

No. of workers recruited and joined 72.

(iii) Number of workers left and discharged:

Employee turnover rate (Separation method)

$$\frac{4}{100} = \frac{\text{No. of Separations (S)}}{\text{Average number of workers on roll}} \times 100$$

$\frac{4}{100} = \frac{S}{900}$ Or, S* = 36

Hence, number of workers left and discharged comes to 36.

ICAI · P4 · May 2026 exam · question p. 21 · answer p. 22-23

280 EXAM **May 2025** Case scenario · Q1(c)(i) to Q1(c)(ii) 4 marks

CASE SCENARIO

Following information is given of a newly setup organization for the year ended on 31st March, 2025:

Number of workers replaced during the period	78
Number of workers left and discharged during the period	28
Employee turnover rates using separation method	3.5%

Required:

Compute the employee turnover rates using:

Q1(c)(i) 2 marks

Replacement method and

Q1(c)(ii) 2 marks

Flux method

ICAI'S ANSWERS

ICAI's working for this case

Employee turnover rate (Separation method)

$$= \frac{\text{No. of Separations(S)}}{\text{Average number of workers on roll}} \times 100$$

$$\frac{3.5}{100} = \frac{28}{\text{Average number of workers on roll}}$$

$$\text{Average number of workers on roll} = \frac{28 \times 100}{3.5} = 800$$

Q1(c)(i)

Employee Turnover rate using Replacement method

$$= \frac{\text{No. of replacements}}{\text{Average number of workers on roll}} \times 100$$

$$= \frac{78 \times 100}{800} = 9.75\%$$

Q1(c)(ii)

Employee turnover rate (Flux method)

$$= \frac{(\text{Number of employees separated} + \text{Number of employees replaced during the period})}{\text{Average number of employees during the period on roll}} \times 100$$

$$= \frac{28 + 78}{800} \text{ x } 100 = 13.25\%$$

ICAI · P4 · May 2025 exam · question p. 8 · answer p. 12

281 EXAM **Sep 2024** MCQ 14 MCQ

On 01-04-2023 number of workers employed in a factory was 150. During the year 30 workers resigned and 5 workers were discharged. Due to resignation and discharge, 15 workers were replaced. For the year 2023-24, labour turnover rate by separation method will be :

A 21.43%

B 18%

C 25%

D 30%

ICAI'S ANSWER

Answer key: **(C)**

ICAI · P4 · Sep 2024 exam · question p. 6 · answer p. 7

282 **EXAM** **May 2024** Q5(a) **6 marks**

Super Ltd, a manufacturing company is facing the problem of high labour turnover in the factory. Before analysing the causes and taking remedial steps, the management of the company wants to ascertain the profit lost for the year 2022-23 on account of labour turnover. For this purpose, it has given you the following information:

(i) Sales for the last year 2022-23 was ₹ 2,16,80,000 and P/V ratio was 15%.

(ii) The total number of actual hours worked by the direct labour force was 5,00,000 hours. The actual direct labour hours included 60,000 hours attributable to training new recruits, out of which 40% of the hours were unproductive.

(iii) Due to delays by the Personnel Department in filling vacancies on account of labour turnover, 95,000 potential productive hours (excluding unproductive training hours) were lost.

(iv) 1,500 units of the output produced during training period were defective. Cost of rectification of defective units was ₹ 40 per unit.

(v) Settlement cost of the workers leaving the organization was ₹ 2,37,880.

(vi) Recruitment and Selection cost was ₹ 1,40,000.

(vii) Cost of Training and Induction was ₹ 1,61,950.

Assuming that the potential production lost as a consequence of labour turnover could have been sold at prevailing prices, find the profit lost for the year 2022-23 on account of labour turnover.

ICAI'S ANSWER

Workings:

(i) Computation of productive hours

Actual hours worked (given)	5,00,000
Less: Unproductive training hours	24,000
Actual productive hours	4,76,000

(ii) Productive hours lost:

Loss of potential productive hours+ Unproductive training hours

= 95,000 + 24,000 = 1,19,000 hours

(iii) Loss of contribution due to unproductive hours :

$$= \frac{\text{Sales value}}{\text{Actual productive hours}} \times \text{Total unproductive hours}$$

$$= \frac{₹2,16,80,000}{4,76,000 \text{ hrs}} \times 1,19,000 \text{ hours} = ₹ 54,20,000$$

Contribution lost for 1,19,000 hours= ₹ 54,20,000 x 15% = ₹8,13,000

Computation of profit forgone on account of employee turnover

	(₹)
Contribution foregone (as calculated above)	8,13,000
Settlement cost due to leaving	2,37,880
Recruitment and Selection cost	1,40,000
Training and Induction costs	1,61,950
Cost of Rectification (1500 units x ₹40)	60,000
Profit foregone	**14,12,830**

The above question can also be solved in alternative way after taking proper assumptions

Workings:

(i) Computation of productive hours

Actual hours worked (given)	5,00,000
Less: Unproductive training hours	24,000
Actual productive hours	4,76,000

(ii) Productive hours lost:

Loss of potential productive hours

= 95,000 hours

(iii) Loss of contribution due to unproductive hours :

$$= \frac{\text{Sales value}}{\text{Actual productive hours}} \times \text{Total unproductive hours}$$

$$= \frac{₹2,16,80,000}{4,76,000 \text{ hrs}} \times 95,000 \text{ hours} = ₹ 43,26,891$$

Contribution lost for 95,000 hours= ₹ 43,26,891 x 15%

= ₹ 6,49,034(approx.)

Computation of profit forgone on account of employee turnover

	(₹)
Contribution foregone (as calculated above)	6,49,034
Settlement cost due to leaving	2,37,880
Recruitment and Selection cost	1,40,000
Training and Induction costs	1,61,950
Cost of Rectification (1500 units x ₹40)	60,000
Profit foregone	**12,48,864**

283 EXAM **Old P3 May 2023** Q5(c) 5 marks old scheme

How does the high employee turnover increase the cost of production? Explain.

ICAI'S ANSWER

High Employee Turnover increases the cost of production

Replacement costs are the costs which arise due to employee turnover. If employees leave soon after they acquire the **necessary training and experience** of good work, additional costs will have to be incurred on new workers, i.e., **cost of recruitment, training and induction**, **abnormal breakage and scrap** and **extra wages and overheads due to the inefficiency** of new workers.

It is obvious that a company will incur very high replacement costs if the rate of employee turnover is high. Similarly, only adequate preventive costs can keep Employee turnover at a low level. Each company must, therefore, work out the optimum level of Employee turnover keeping in view its personnel policies and the behaviour of replacement cost and preventive costs at various levels of Employee turnover rates.

ICAI · Old P3 · May 2023 exam · question p. 24 · answer p. 28

284 EXAM **Old P3 May 2022** Q1(b) 5 marks old scheme

PQR Limited has replaced 72 workers during the quarter ended 31st March 2022. The labour rates for the quarter are as follows:

Flux method	16%
Replacement method	8%
Separation method	5%

You are required to ascertain:

(i) Average number of workers on roll (for the quarter),

(ii) Number of workers left and discharged during the quarter,

(iii) Number of workers recruited and joined during the quarter,

(iv) Equivalent employee turnover rates for the year.

ICAI'S ANSWER

Working Note:

(i) Average number of workers on roll (for the quarter):

Employee Turnover rate using Replacement method

$$= \frac{\text{No. of replacements}}{\text{Average number of workers on roll}} \times 100$$

$$\text{Or, } \frac{8}{100} = \frac{72}{\text{Average number of workers on roll}}$$

$$\text{Or, Average number of workers on roll} = \frac{72 \times 100}{8} = \mathbf{900}$$

(ii) Number of workers left and discharged:

Employee turnover rate (Separation method)

$= \frac{\text{No. of Separations(S)}}{\text{Average number of workers on roll}} \times 100 = \frac{5}{100} = \frac{S}{900}$ Or, S = 45

Hence, number of workers left and discharged comes **to 45**

(iii) Number of workers recruited and joined:

Employee turnover rate (Flux method)

$= \frac{\text{No. of Separations*(S)+No. of Accessions(A)}}{\text{Average number of workers on roll}}$

Or, $\frac{16}{100} = \frac{45 + A}{900}$ Or, A = $\left[\frac{14400}{100} - 45\right]$ = 99

No. of workers recruited and joined 99

(iv) Calculation of Equivalent employee turnover rates:

$= \frac{\text{EmployeeTurnove rate for the quarter(s)}}{\text{Number of quarter(s)}} \times 4$ quarters

Using Flux method = $\frac{16\%}{1}$ ×4 = 64%

Using Replacement method = $\frac{8\%}{1}$ ×4 = 32%

Using Separation method = $\frac{5\%}{1}$ ×4 = 20%

ICAI · Old P3 · May 2022 exam · question p. 1 · answer p. 3-4

285 EXAM **Old P3 Jul 2021** Q2(c) 5 marks old scheme

Following information is given of a newly setup organization for the year ended on 31st March, 2021.

Number of workers replaced during the period	50
Number of workers left and discharged during the period	25
Average number of workers on the roll during the period	500

You are required to:

(i) Compute the Employee Turnover Rates using Separation Method and Flux Method.

(ii) Equivalent Employee Turnover Rates for (i) above, given that the organization was setup on 31st January, 2021.

ICAI'S ANSWER

(i) Employee Turnover rate

Using Separation method:

$= \frac{\text{Number of employees Separated during the period}}{\text{Average number of employees during the period on roll}} \times 100$

$= \frac{25}{500} \times 100$ = **5%**

Using Flux method:

$= \frac{\text{Number of employees Separated} + \text{Number of employees Replaced during the period}}{\text{Average number of employees during the period on roll}} \times 100$

$= \frac{50 + 25}{500} \times 100$ = **15%**

(ii) Equivalent Employee Turnover rate:

$$= \frac{\text{Employee Turnover rate for the period}}{\text{Number of days in the period}} \times 365$$

Using Separation method	$= \frac{5}{60} \times 365$	= 30.42%
Or,	$= \frac{5}{60} \times 360$	= 30%
Or,	$= \frac{5}{2} \times 12$	= 30%
Using Flux method	$= \frac{15}{60} \times 365$	= 91.25%
Or,	$= \frac{15}{60} \times 360$	= 90%
Or,	$= \frac{15}{2} \times 12$	= 90%

ICAI · Old P3 · Jul 2021 exam · question p. 59 · answer p. 63-64

286 **RTP** **Sep 2024** Q8

The labour turnover rates for the quarter ended 30th June, 2024 are computed as 14%, 8% and 6% under Flux method, Replacement method and Separation method respectively. If the number of workers replaced during 1st quarter of the financial year 2024-25 is 36, COMPUTE the following:

(i) The number of workers recruited and joined; and

(ii) The number of workers left and discharged.

ICAI'S ANSWER

Labour Turnover Rate (Replacement method) $= \frac{\text{No. of workers replaced}}{\text{Average No. of workers}}$

Or, $\frac{8}{100} = \frac{36}{\text{Average No. of workers}}$

Or, Average No. of workers = 450

Labour Turnover Rate (Separation method) $= \frac{\text{No. of workers separated}}{\text{Average No. of workers}}$

Or, $\frac{6}{100} = \frac{\text{No. of workers separated}}{450}$

Or, No. of workers separated = 27

Labour Turnover Rate (Flux Method) $= \frac{\text{No. of Separations} + \text{No. of accession (Joinings)}}{\text{Average No. of workers}}$

Or, $\frac{14}{100} = \frac{27 + \text{No. of accessions (Joinings)}}{450}$

Or, 100 (27 + No. of Accessions) = 6,300

Or, No. of Accessions = 36

(i) The No. of workers recruited and Joined = 36

(ii) The No. of workers left and discharged = 27

ICAI · P4 · RTP Sep 2024 · question p. 7 · answer p. 21-22

287 RTP May 2024 Case scenario · Q2(i) to Q2(v)

CASE SCENARIO

The board of the J Ltd. has been appraised by the General Manager (HR) that the employee attrition rate in the company has increased. The following facts has been presented by the GM(HR):

(1) Training period of the new recruits is 50,000 hours. During this period their productivity is 60% of the experienced workers. Time required by an experienced worker is 10 hours per unit.

(2) 20% of the output during training period was defective. Cost of rectification of a defective unit was ₹ 25.

(3) Potential productive hours lost due to delay in recruitment were 1,00,000 hours.

(4) Selling price per unit is ₹ 180 and P/V ratio is 20%.

(5) Settlement cost of the workers leaving the organization was ₹ 1,83,480.

(6) Recruitment cost was ₹ 1,56,340

(7) Training cost was ₹ 1,13,180

You being an associate finance to GM(HR), has been asked the following questions:

Q2(i)

How much quantity of output is lost due to labour turnover?

A 10,000 units

B 8,000 units

C 12,000 units

D 12,600 units

Q2(ii)

How much loss in the form of contribution, the company incurred due to labour turnover?

A ₹ 4,32,000

B ₹ 4,20,000

C ₹ 4,36,000

D ₹ 4,28,000

Q2(iii)

What is the cost repairing of defective units?

A ₹ 75,000

B ₹ 15,000

C ₹ 50,000

D ₹ 25,000

Q2(iv)

Calculate the profit lost by the company due to increased labour turnover.

A ₹ 7,50,000

B ₹ 15,00,000

C ₹ 5,00,000

D ₹ 9,00,000

Q2(v)

How much quantity of output is lost due to inexperience of the new worker?

A 1,000 units

B 2,600 units

C 2,000 units

D 12,600 units

ICAI'S ANSWERS

Q2(i)

Answer key: **(C)**

(c) Output by experienced workers in 50,000 hours = $\frac{50,000}{10}$

= 5,000 units

∴ Output by new recruits = 60% of 5,000 = 3,000 units

Loss of output = 5,000 – 3,000 = 2,000 units

Total loss of output = Due to delay recruitment + Due to inexperience

= 10,000 +2,000=**12,000 units**

Q2(ii)

Answer key: **(A)**

(a) Contribution per unit = 20% of ₹ 180 = ₹ 36

Total contribution lost = ₹ 36 × 12,000 units = **₹ 4,32,000**

Q2(iii)

Answer key: **(B)**

(b) Cost of repairing defective units = 3,000 units × 0.2 × ₹ 25

= **₹ 15,000**

Q2(iv)

Answer key: **(D)**

(d) Calculation of loss of profit due to labour turnover

	(₹)
Loss of Contribution	4,32,000
Cost of repairing defective units	15,000
Recruitment cost	1,56,340
Training cost	1,13,180
Settlement cost of workers leaving	1,83,480
Profit forgone in 2022-23	9,00,000

Q2(v)

Answer key: **(C)**

(c) Output by experienced workers in 50,000 hours = $\frac{50,000}{10}$

= 5,000 units

∴ Output by new recruits = 60% of 5,000 = 3,000 units

Loss of output = 5,000 – 3,000 = **2,000 units**

ICAI · P4 · RTP May 2024 · question p. 3 · answer p. 20

288 RTP **Old P3 May 2023** Q2 old scheme

Following information are available from the cost records of BMR Limited, CALCULATE Labour turnover rate and Labour flux rate:

No. of Employees as on 01.04.2021 = 9,400

No. of Employees as on 31.03.2022 = 10,600

During the year, 160 Employees left while 640 Employees were discharged and 1,500 Employees were recruited during the year; of these, 400 Employees were recruited because of exits and the rest were recruited in accordance with expansion plans.

ICAI'S ANSWER

Employee turnover rate:

It comprises of computation of Employee turnover by using following methods:

(i) Separate Method: $= \frac{\text{Number of employees seperated during the period}}{\text{Average number of employees during the period on roll}} \text{ x } 100$

OR, $= \frac{\text{Number of employees left} + \text{Number of employees discharged}}{\text{Average number of employees during the period on roll}} \text{ x } 100$

$= \frac{(160 + 640)}{(9{,}400 + 10{,}600) \div} \text{ x}100$

$= \frac{800}{10{,}000} \text{ x } 100 = 8\%$

(ii) Replacement Method $= \frac{\text{Number of employees replaced during the period}}{\text{Average number of employees during the period on roll}} \text{ x } 100$

$= \frac{400}{10{,}000} \text{ x } 100 = 4\%$

(iii) New Recruitment $= \frac{\text{Number of employees joining in a period (excluding replacement)}}{\text{Average number of employees during the period on roll}} \text{ x } 100$

$= \frac{\text{Number of Recruitments - Number of Replacements}}{\text{Average number of employees during the period on roll}} \text{ x } 100$

$= \frac{1500 - 400}{10{,}000} \text{ x } 100$

$= \frac{1{,}100}{10{,}000} \text{ x } 100 = 11\%$

Flux Method $= \frac{\text{Number of separation} + \text{Number of replacement} + \text{Number of new joining}}{\text{Average number of employees during the period on roll}} \text{ x } 100$

$= \frac{(800 + 400 + 1{,}100)}{(9{,}400 + 10{,}600) \div 2} \text{ x } 100$

$= \frac{2{,}300}{10{,}000} \text{ x } 100 = 23\%$

ICAI · Old P3 · RTP May 2023 · question p. 1 · answer p. 12-13

289 RTP Old P3 Nov 2022 Q2 old scheme

HR Ltd. is progressing in its legal industry. One of its trainee executives, Mr. H, in the Personnel department has calculated labour turnover rate 24.92% for the last year using Flux method.

Following is the data provided by the Personnel department for the last year:

Employees	At the beginning	Joined	Left	At the end
Records clerk	810	1,620	90	2,340
Human Resource Manager	?	30	90	60
Legal Secretary	?	90	---	?
Staff Attorney	?	30	30	?
Associate Attorney	?	30	---	45
Senior Staff Attorney	6	---	---	18
Senior Records clerk	12	---	---	51
Litigation attorney	?	---	---	?
Employees transferred from the Subsidiary Company				
Senior Staff Attorney	---	12	---	---
Senior Records clerk	---	39	---	---
Employees transferred to the Subsidiary Company				
Litigation attorney	---	---	90	---
Associate Attorney	---	---	15	---

At the beginning of the year there were total 1,158 employees on the payroll of the company. The opening strength of the Legal Secretary, Staff Attorney and Associate Attorney were in the ratio of 3 : 3 : 2.

The company has decided to abandon the post of Litigation attorney and consequently all the Litigation attorneys were transferred to the subsidiary company.

The company and its subsidiary are maintaining separate set of books of account and separate Personnel Department.

You are required to:

(a) CALCULATE Labour Turnover rate using Replacement method and Separation method.

(b) VERIFY the Labour turnover rate calculated under Flux method by Mr. H

ICAI'S ANSWER

Working Notes:

(i) Calculation of no. of employees at the beginning and end of the year

	At the Beginning of the year	At the end of the year
Records clerk	810	2,340
Human Resource Manager [Left- 90 + Closing- 60 – Joined- 30]	120	60
Legal Secretary*	45	135
Staff Attorney*	45	45
Associate Attorney*	30	45
Senior Staff Attorney	6	18
Senior Records clerk	12	51
Litigation attorney	90	0
Total	**1,158**	**2,694**

(*) At the beginning of the year:

Strength of Legal Secretary, Staff Attorney and Associate Attorney =

[1158 – {810 + 120 + 6 + 12 + 90} employees] or [1158 – 1038 = 120 employees]

[{Legal Secretary - $120 \times \frac{3}{8} = 45$, Staff Attorney - $120 \times \frac{3}{8} = 45$ & Associate Attorney - $120 \times \frac{2}{8} = 30$} employees]

At the end of the year:

[Legal Secretary -(Opening 45 + 90 Joining) = 135; Staff Attorney - (Opening 45 + 30 Joined – 30 Left) = 45]

(ii) No. of Employees Separated, Replaced and newly recruited during the year

Particulars	Separations	New Recruitment	Replacement	Total Joining
Records clerk	90	1,530	90	1,620
Human Resource Manager	90	--	30	30
Legal Secretary	--	90	--	90
Staff Attorney	30	--	30	30
Associate Attorney	15	15	15	30
Senior Staff Attorney	--	12	--	12
Senior Records clerk	--	39	--	39
Litigation attorney	90	--	--	--
Total	**315**	**1,686**	**165**	**1,851**

(Since, HR Ltd. and its subsidiary are maintaining separate Personnel Department, so transfer-in and transfer-out are treated as recruitment and separation respectively.)

(a) Calculation of Labour Turnover rate:

$$\text{Replacement Method} = \frac{\text{No. of employees replaced during the year}}{\text{Average no. of employees on roll}} \times 100$$

$$= \frac{165}{(1{,}158 + 2{,}694)/2} \times 100 = \frac{165}{1{,}926} \times 100 = 8.57\%$$

$$\text{Separation Method} = \frac{\text{No. of employees separated during the year}}{\text{Average no. of employees on roll}} \times 100$$

$$= \frac{315}{1{,}926} \times 100 = 16.36\%$$

(b) Labour Turnover rate under Flux Method:

$$= \frac{\text{No. of employees (Joined + Separated) during the year}}{\text{Average no. of employees on roll}} \times 10$$

$$= \frac{\text{No. of employees (Replaced + New recruited + Separated) during the year}}{\text{Average no. of employees on roll}} \times 100$$

$$= \frac{1{,}851 + 315}{1{,}926} \times 100 = 112.46\%$$

Labour Turnover rate calculated by Mr. H is incorrect as it seems he has not taken the No. of new recruitment while calculating the labour turnover rate under Flux method.

ICAI · Old P3 · RTP Nov 2022 · question p. 1-2 · answer p. 16-17

290 RTP **Old P3 May 2020 Q2** old scheme

From the following information, CALCULATE employee turnover rate using – (i) Separation Method, (ii) Replacement Method, (iii) New Recruitment Method, and (iv) Flux Method:

No. of workers as on 01.01.2019 = 3,600

No. of workers as on 31.12.2019 = 3,790

During the year, 40 workers left while 120 workers were discharged. 350 workers were recruited during the year, of these 150 workers were recruited because of exits and the rest were recruited in accordance with expansion plans.

ICAI'S ANSWER

Employee turnover rate using:

(i) Separation Method:

$$= \frac{\text{No. of workers left} + \text{No. of workers discharged}}{\text{Average number of workers}} \times 100$$

$$= \frac{(40 + 120)}{(3{,}600 + 3{,}790)/2} \times 100 = \frac{160}{3{,}695} \times 100 = 4.33\%$$

(ii) Replacement Method:

$$= \frac{\text{No. of workers replaced}}{\text{Average number of workers}} \times 100 = \frac{150}{3{,}695} \times 100 = 4.06\%$$

(iii) New Recruitment Method:

$$= \frac{\text{No. of workers newly recruited}}{\text{Average number of workers}} \times 100$$

$$= \frac{\text{No. Recruitments - No. of Replacements}}{\text{Average number of workers}} \times 100$$

$$= \frac{350 - 150}{3{,}695} \times 100 = \frac{200}{3{,}695} \times 100 = 5.41\%$$

(iv) Flux Method:

$$= \frac{\text{No. of separations} + \text{No. of accessions}}{\text{Average number of workers}} \times 100$$

$$= \frac{(160 + 350)}{(3{,}600 + 3{,}790)/2} \times 100 = \frac{510}{3{,}695} \times 100 = 13.80\%$$

ICAI · Old P3 · RTP May 2020 · question p. 2 · answer p. 12

291 MTP **May 2026 S-I** Q6(b) 5 marks

XYZ Manufacturing Ltd. is facing high employee turnover and a rise in idle time in its Assembly Department. On investigation, it was found that employees often left due to dissatisfaction with working conditions and lack of incentives. The Management decided to revise the wage payment system and introduced regular training sessions to enhance skills and morale.

As the Cost Accountant, answer the following:

(i) IDENTIFY and EXPLAIN two types of employee cost that occur due to employee turnover

(ii) SUGGEST any two factors that can be taken by the company to improve employee productivity apart from training and wage payment revision.

ICAI'S ANSWER

(i) Two types of costs which are associated with employee turnover are:

(a) Preventive Costs: The cost incurred to prevent employee turnover or keep it as lowest as possible. Cost incurred for prevention of employee turnover includes the following:

(i) Cost of medical benefit provided to the employees;

(ii) Cost incurred on employees' welfare like pension etc.

(iii) Cost on other benefits with an objective to retain employees.

(b) Replacement Costs: These are the costs which arise due to employee turnover. If employees leave soon after they acquire the necessary training and experience of good work, additional costs will have to be incurred on new workers, i.e., cost of recruitment, training and induction, abnormal breakage and scrap and extra wages and overheads due to the inefficiency of new workers.

(ii) The important factors which must be taken into consideration for increasing employee productivity are as follows:

1. Employing only those workers who possess the right type of skill.
2. Placing a right type of person to a right job.
3. Taking appropriate measures to avoid the situation of excess or shortage of employees.
4. Carrying out work study for fixation of wages and for the simplification and standardisation of work.

ICAI · P4 · MTP May 2026 · Series I · question p. 14-15 · answer p. 19-20

292 MTP **Jan 2026 S-II** Q6(b) 5 marks

Also asked in Old P3 MTP Nov 2019 S-I Q6(c)

DISCUSS the remedial steps to be taken to minimize the labour turnover.

ICAI'S ANSWER

The following steps are useful for minimizing labour turnover:

(a) *Exit interview*: An interview to be arranged with each outgoing employee to ascertain the reasons of his leaving the organization.

(b) *Job analysis and evaluation*: to ascertain the requirement of each job.

(c) Organization should make use of a scientific system of recruitment, placement and promotion for employees.

(d) Organization should create healthy atmosphere, providing education, medical and housing facilities for workers.

(e) Committee for settling workers grievances.

ICAI · P4 · MTP Jan 2026 · Series II · question p. 18 · answer p. 21-22

293 MTP **Sep 2025 S-II** Q6(c) 4 marks

XYZ Ltd. is experiencing a high rate of employee turnover in its production department, resulting in increased training costs, recruitment expenses, and loss of productive hours. As the cost accountant, you are asked to analyze the situation and recommend two practical measures the company can implement to reduce these costs and improve employee retention. IDENTIFY and EXPLAIN two major cost impacts of high employee turnover on the organization.

ICAI'S ANSWER

Two types of costs which are associated with employee turnover are:

(a) **Preventive Costs:** The cost incurred to prevent employee turnover or keep it as lowest as possible. Cost incurred for prevention of employee turnover includes the following:

(i) Cost of medical benefit provided to the employees;

(ii) Cost incurred on employees' welfare like pension etc.

(iii) Cost on other benefits with an objective to retain employees.

(b) **Replacement Costs:** These are the costs which arise due to employee turnover. If employees leave soon after they acquire the necessary training and experience of good work, additional costs will have to be incurred on new workers, *i.e.,* cost of recruitment, training and induction, abnormal breakage and scrap and extra wages and overheads due to the inefficiency of new workers.

ICAI · P4 · MTP Sep 2025 · Series II · question p. 17 · answer p. 27

294 MTP **Jan 2025 S-I** MCQ 11 2 marks MCQ

The rate of change in the composition of employee force over the average number of employees for the year is computed as 9% under 'separation method'. However, the same rate is computed as 15% and 30% under 'replacement method' and 'flux method' respectively.

Considering the average number of employees on roll during the year as 200, FIND OUT the number of employees -

(i) replaced, (ii) left and discharged and (iii) recruited and joined

A Replaced- 18 employees, left and discharged- 30 employees and recruited & joined- 42 employees

B Replaced- 30 employees, left and discharged- 42 employees and recruited & joined- 18 employees

C Replaced- 30 employees, left and discharged- 18 employees and recruited & joined- 42 employees

D Replaced- 42 employees, left and discharged- 18 employees and recruited & joined- 30 employees

ICAI'S ANSWER

(C) Replaced- 30 employees, left and discharged- 18 employees and recruited & joined- 42 employees

(i) Number of employees replaced:

Employee Turnover rate (Replacement method)

$$= \left(\frac{\text{No. of Replacements}}{\text{Average number of employees on roll}} \text{ x } 100 \right)$$

$$\text{Or, } \left(\frac{15}{100} \right) = \left(\frac{\text{No. of replacements}}{200} \right)$$

$$\text{Or, Number of Replacements} = \left(\frac{200 \text{ x } 15}{100} \right) = 30$$

(ii) Number of employees left and discharged:

Employee turnover rate (Separation method)

$$= \left(\frac{\text{No. of Separations (S)}}{\text{Average number of employees on roll}} \text{ x } 100 \right)$$

$$\text{Or, } \left(\frac{9}{100} \right) = \left(\frac{S}{200} \right)$$

Or, S = 18

Hence, number of employees left and discharged = 18

(iii) Number of employees recruited and joined:

Employee turnover rate (Flux method)

$$= \left(\frac{\text{No. of Separations (S)} + \text{No. of Accessions (A)}}{\text{Average number of employees on roll}} \text{ x } 100 \right)$$

$$\text{Or, } \left(\frac{30}{100} \right) = \left(\frac{18 + A}{200} \right)$$

$$\text{Or, A} = \left(\frac{6000}{100} - 18 \right) = 42$$

Hence, number of employees recruited and joined = 42

ICAI · P4 · MTP Jan 2025 · Series I · question p. 5 · answer p. 3-4

295 MTP **May 2024 S-II** Q6(c) 4 marks

ENUMERATE the remedial steps to be taken to minimize the labour turnover

ICAI'S ANSWER

The following steps are useful for minimizing labour turnover:

(a) *Exit interview*: An interview to be arranged with each outgoing employee to ascertain the reasons of his leaving the organization.

(b) *Job analysis and evaluation*: to ascertain the requirement of each job.

(c) Organization should make use of a scientific system of recruitment, placement and promotion for employees.

(d) Organization should create healthy atmosphere, providing education, medical and housing facilities for workers.

(e) Committee for settling workers grievances.

ICAI · P4 · MTP May 2024 · Series II · question p. 10 · answer p. 15

296 MTP **Old P3 Nov 2023 S-I** Q1(d) 5 marks old scheme

J Ltd. wants to ascertain the profit lost during the year 2022-23 due to increased labour turnover. For this purpose, they have given you the following information:

(1) Training period of the new recruits is 50,000 hours. During this period their productivity is 60% of the experienced workers. Time required by an experienced worker is 10 hours per unit.

(2) 20% of the output during training period was defective. Cost of rectification of a defective unit was ₹ 25.

(3) Potential productive hours lost due to delay in recruitment were 1,00,000 hours.

(4) Selling price per unit is ₹ 180 and P/V ratio is 20%.

(5) Settlement cost of the workers leaving the organization was ₹ 1,83,480.

(6) Recruitment cost was ₹ 1,56,340

(7) Training cost was ₹ 1,13,180

You are required to calculate the profit lost by the company due to increased labour turnover during the year 2022-23.

ICAI'S ANSWER

Output by experienced workers in 50,000 hours = $\frac{50,000}{10}$ = 5,000 units

∴ Output by new recruits = 60% of 5,000 = 3,000 units

Loss of output = 5,000 – 3,000 = 2,000 units

Total loss of output = Due to delay recruitment + Due to inexperience

= 10,000 + 2,000 = 12,000 units

Contribution per unit = 20% of ₹180 = ₹ 36

Total contribution lost = ₹36 × 12,000 units = ₹ 4,32,000

Cost of repairing defective units = 3,000 units × 0.2 × ₹ 25 = ₹ 15,000

Profit forgone due to labour turnover

	(₹)
Loss of Contribution	4,32,000
Cost of repairing defective units	15,000
Recruitment cost	1,56,340
Training cost	1,13,180
Settlement cost of workers leaving	1,83,480
Profit forgone in 2022-23	9,00,000

ICAI · Old P3 · MTP Nov 2023 · Series I · question p. 2 · answer p. 3

297 MTP Old P3 Nov 2023 S-II Q1(d) 5 marks old scheme

The rate of change of labour force in a company during the year ending 31st March, 2023 was calculated as 13%,8% and 5% respectively under 'Flux Method', 'Replacement method' and 'Separation method'. The number of workers separated during the year is 40.

You are required to calculate:

(i) Average number of workers on roll.

(ii) Number of workers replaced during the year.

(iii) Number of new accessions i.e. new recruitment.

(iv) Number of workers at the beginning of the year.

ICAI'S ANSWER

(i) Labour Turnover Rate (Separation method)

$$= \frac{\text{No. of workers separated}}{\text{Average no. of workers on roll}}$$

$$\text{Or, } \frac{5}{100} = \frac{40}{\text{Average no. of workers on roll}}$$

Or, Average no. of workers on roll = 800

(ii) Labour Turnover Rate (Replacement method)

$$= \frac{\text{No. of workers replaced}}{\text{Average no. of workers on roll}}$$

$$\text{Or, } \frac{8}{100} = \frac{\text{No. of workers replaced}}{800}$$

Or, No. of workers replaced = 64

(iii) Labour Turnover Rate (Flux Method)

$$= \frac{\text{No. of Separations} + \text{No. of accessions (new recruitments)}}{\text{Average No. of workers on roll}}$$

$$\text{Or, } \frac{13}{100} = \frac{40 + \text{No. of accessions (New recruitments)}}{800}$$

Or, 100 (40 + No. of Accessions) = 10,400

Or, No. of new accessions = 64

(iv) No. of workers at the beginning of the year

Let workers at the beginning of the year were 'X'

$$\text{Average no. of workers on roll} = \frac{\text{Workers at the begining} + \text{Workers at the end}}{2}$$

$$800 = \frac{X + (X + \text{New accessions} - \text{Separations})}{2}$$

$$800 = \frac{X + (X + 64 - 40)}{2}$$

$$800 = \frac{X + (X + 24)}{2}$$

2 X = 1,600 – 24 or, X = 788 workers

ICAI · Old P3 · MTP Nov 2023 · Series II · question p. 2 · answer p. 3-4

298 MTP Old P3 Nov 2022 S-II Q1(c) 5 marks old scheme

R Ltd. has computed labour turnover rates for the quarter ended 31st March, 2022 as 20%, 10% and 5% under flux method, replacement method and separation method respectively. If the number of workers replaced during that quarter is 50,

FIND OUT (i) Workers recruited and joined

(ii) Workers left and discharged and

(iii) Average number of workers on roll.

ICAI'S ANSWER

$$\text{Labour Turnover Rate (Replacement method)} = \frac{\text{No.of workers replaced}}{\text{Average no.of workers}} \times 100$$

$$\text{Or, } \frac{10}{100} = \frac{50}{\text{Average no.of workers}}$$

Thus, Average No. of workers = 500

$$\text{Labour Turnover Rate (Separation method)} = \frac{\text{No.of workers separated}}{\text{Average No.of workers}} \times 100$$

$$\text{Or, } \frac{5}{100} = \frac{\text{Number of workers separated}}{500}$$

Thus, No. of workers separated = 25

Labour Turnover Rate (Flux Method)

$$= \frac{\text{No.of Separations} + \text{No.of Accession (Joinings)}}{\text{Average no.of workers}} \times 100$$

$$\text{Or, } \frac{20}{100} = \frac{25 + \text{No.of accessions (Joinings)}}{500}$$

Or, 100 (25 + No. of Accessions) = 10,000

Or, 25 + No. of Accessions =100

Thus, No. of Accessions = 100 - 25 =75

Accordingly,

(i) Workers recruited and joined = 75

(ii) Workers left and discharged = 25

(iii) Average number of workers on roll = 500

ICAI · Old P3 · MTP Nov 2022 · Series II · question p. 2 · answer p. 2-3

299 MTP Old P3 May 2022 S-II Q1(a) 5 marks old scheme

R Ltd. is facing increasing employee turnover in the factory and before analyzing the causes and taking remedial steps; the management wants to have an idea of the profit foregone as a result of employee turnover in the last year.

Last year sales amounted to ₹ 99,63,960 and P/V ratio was 20%.

The total number of actual hours worked by the direct employee force was 5.34 lakhs. The actual direct employee hours included 36,000 hours attributable to training new recruits, out of which half of the hours were unproductive. As a result of the delays by the Personnel Department in filling vacancies due to employee turnover, 1,20,000 potentially productive hours (excluding unproductive training hours) were lost.

The costs incurred consequent on employee turnover revealed, on analysis, the following:

Settlement cost due to leaving	₹ 52,584
Recruitment costs	₹ 32,088
Selection costs	₹ 15,300
Training costs	₹ 36,588

Assuming that the potential production lost as a consequence of employee turnover could have been sold at prevailing prices, FIND the profit foregone last year on account of employee turnover.

ICAI'S ANSWER

Workings:

(i) Computation of productive hours

Actual hours worked	5,34,000
Less: Unproductive training hours	18,000
Actual productive hours	5,16,000

(ii) Productive hours lost:

Loss of potential productive hours + Unproductive training hours

= 1,20,000 + 18,000 = 1,38,000 hours

(iii) Loss of contribution due to unproductive hours:

$$= \frac{\text{Sales value}}{\text{Actual productive hours}} \times \text{Total unproductive hours}$$

$$= \frac{₹\ 99,63,960}{5,16,000\ \text{hrs}} \times 1,38,000\ \text{hours} = ₹\ 26,64,780$$

$$\text{Contribution lost for } 1,38,000 \text{ hours} = \frac{₹\ 26,64,780}{100} \times 20 = ₹\ 5,32,956$$

Computation of profit forgone on account of employee turnover

	(₹)
Contribution foregone (as calculated above)	5,32,956
Settlement cost due to leaving	52,584
Recruitment cost	32,088
Selection cost	15,300
Training costs	36,588
Profit foregone	**6,69,516**

ICAI · Old P3 · MTP May 2022 · Series II · question p. 1 · answer p. 1

300 MTP Old P3 Jul 2021 S-I Q1(a) 5 marks old scheme

The labour turnover rates for the quarter ended 30th September, 2020 are computed as 14%, 8% and 6% under Flux method, Replacement method and Separation method respectively. If the number of workers replaced during 2nd quarter of the financial year 2020-21 is 36, COMPUTE the following:

(i) The number of workers recruited and joined; and

(ii) The number of workers left and discharged.

ICAI'S ANSWER

$$\text{Labour Turnover Rate (Replacement method)} = \frac{\text{No. of workers replaced}}{\text{Average No. of workers}}$$

$$\text{Or, } \frac{8}{100} = \frac{36}{\text{Average No. of workers}}$$

Or, Average No. of workers = 450

$$\text{Labour Turnover Rate (Separation method)} = \frac{\text{No. of workers separated}}{\text{Average No. of workers}}$$

$$\text{Or, } \frac{6}{100} = \frac{\text{No. of workers separated}}{450}$$

Or, No. of workers separated = 27

$$\text{Labour Turnover Rate (Flux Method)} = \frac{\text{No. of Separations} + \text{No. of accession (Joinings)}}{\text{Average No. of workers}}$$

$$\text{Or, } \frac{14}{100} = \frac{27 + \text{No. of accessions (Joinings)}}{450}$$

Or, 100 (27 + No. of Accessions) = 6,300

Or, No. of Accessions = 36

(i) The No. of workers recruited and Joined = 36

(ii) The No. of workers left and discharged = 27

ICAI · Old P3 · MTP Jul 2021 · Series I · question p. 1 · answer p. 1

301 MTP Old P3 Jul 2021 S-II Q1(a) 5 marks old scheme

From the following information, CALCULATE employee turnover rate using – (i) Separation Method, (ii) Replacement Method, (iii) New Recruitment Method, and (iv) Flux Method:

No. of workers as on 01.04.2020 = 3,800

No. of workers as on 31.03.2021 = 4,200

During the year, 40 workers left while 160 workers were discharged and 600 workers were recruited during the year; of these, 150 workers were recruited because of exits and the rest were recruited in accordance with expansion plans.

ICAI'S ANSWER

Employee turnover rate using:

(i) Separation Method:

$$= \frac{\text{No. of workers left} + \text{No. of workers discharged}}{\text{Average number of workers}} \times 100$$

$$= \frac{(40 + 160)}{(3{,}800 + 4{,}200) \div 2} \times 100 = \frac{200}{4{,}000} \times 100 = 5\%$$

(ii) Replacement Method:

$$= \frac{\text{No. of workers replaced}}{\text{Average number of workers}} \times 100 = \frac{150}{4{,}000} \times 100 = 3.75\%$$

(iii) New Recruitment Method:

$$= \frac{\text{No. of workers newly recruited}}{\text{Average number of workers}} \times 100$$

$$= \frac{\text{No. of Recruitments-No. of Replacements}}{\text{Average number of workers}} \times 100$$

$$= \frac{600 - 150}{4{,}000} \times 100 = \frac{450}{4{,}000} \times 100 = 11.25\%$$

(iv) Flux Method:

$$= \frac{\text{No. of separations} + \text{No. of accessions}}{\text{Average number of workers}} \times 100$$

$$= \frac{(200 + 600)}{(3{,}800 + 4{,}200) \div 2} \times 100 = \frac{800}{4{,}000} \times 100 = 20\%$$

ICAI · Old P3 · MTP Jul 2021 · Series II · question p. 1 · answer p. 1

4.4 Time rate, piece rate and the Halsey and Rowan premium plans

47 questions · asked in 37 ICAI documents · 13 exam papers

302 EXAM **Sep 2026** MCQ 12 2 marks MCQ

In a factory, standard time for a job is 72 hours. The hourly-rate of wages is ₹ 60. Halsey Premium Plan is in operation at the factory. A worker completes the job at less than standard time and his effective hourly-rate wages was ₹ 75. How many hours did he actually take to complete the work ?

A 60

B 72

C 48

D 64

ICAI'S ANSWER

Answer key: **ICAI has not published the key yet.**

ICAI · P4 · Sep 2026 exam · question p. 8

303 EXAM **Jan 2026** Q1(c) 4 marks

Worker 'X' of M/s ABC Manufacturing Ltd, was assigned a Job no 101. He began the job on 1st June at 9:00 A.M. and completed on 4th June at 12:00 P.M. There are 9 working hours in a day from 9:00 A.M. onwards. Work done and approved was 1,800 units.

Other information:

Standard quantity per hour	: 45 units per hour
Wage rate	: ₹ 150 per hour
Bonus (Halsey plan)	: 50% of time saved

You are required to calculate the remuneration of worker X on the basis of Halsey plan and Rowan Plan.

ICAI'S ANSWER

Time allowed (Standard Time) = $\frac{1{,}800 \text{ units}}{45 \text{ units per hour}} = 40 \text{ hours}$

Time taken = 30 hours (9+9+9+3)

Time saved = 10 hour

Rate = ₹ 150 per hour

Total Remuneration of Worker X (under 50% Halsey Scheme)

= Hours worked × Rate per hour + ½ × time saved × Rate per hour

= 30 hours × ₹ 150 + ½ × 10 hour × ₹150

= ₹ 5,250

Total Remuneration of Worker X (under Rowan Scheme)

= Hours worked × Rate per hour + $\frac{\text{Time taken}}{\text{Time allowed}}$ × Time saved × Rate

= 30 hours × ₹ 150 + $\frac{30 \text{ hours}}{40 \text{ hours}}$ × 10 hour × ₹150

= ₹ 5,625

ICAI · P4 · Jan 2026 exam · question p. 9 · answer p. 12

304 EXAM **Jan 2026** Q6(b) 5 marks

Suggest a suitable incentive scheme to a factory for its workers, keeping in view the following:

(i) The entire gains of improved production should not go to the workers.

(ii) In the name of speed, quality should not suffer.

(iii) The rate setting department being newly established are liable to commit mistakes.

ICAI'S ANSWER

Rowan Scheme of premium bonus (variable sharing plan) is a suitable incentive scheme for the workers of the factory. If this scheme is adopted, the entire gains due to time saved by a worker will not pass to him.

Another feature of this scheme is that a worker cannot increase his earnings or bonus by merely increasing its work speed. The reason for this is that the bonus under Rowan Scheme is maximum when the time taken by a worker on a job is half of the time allowed. As this fact is known to the workers, therefore, they work at such a speed which helps them to maintain the quality of output too.

Lastly, Rowan System provides a safeguard in the case of any loose fixation of the standards by the rate-setting department. Workers cannot take undue advantage of such a situation.

ICAI · P4 · Jan 2026 exam · question p. 28 · answer p. 29

305 EXAM **Sep 2025** Q1(b) 5 marks

The following information relates to two workers - Ajoy and Bijoy who are engaged in producing the

same product by using the same material:

Time allowed to make the product:	40 hours
Actual time taken to complete the product	32 hours by Ajoy 30 hours by Bijoy
Normal Wage Rate	Same for both
Bonus payment plan	Halsey 50% plan for Ajoy Rowan plan for Bijoy
Factory overhead recovered	@₹ 360 per hour for actual time taken by each worker.
Factory cost for the product for each worker	₹ 1,24,800 Ajoy ₹ 1,24,800 Bijoy

Required:

(i) Compute the normal hourly wage rate.

(ii) Compute the cost of material used.

ICAI'S ANSWER

Step 1: Let X be the cost of material and Y be the normal rate of wages per hour.

Step 2: Factory Cost of Worker 'Ajoy'

	(₹)
A. Material Cost	X
B. Wages (Halsey Plan)	32 Y
C. Bonus = 50% of (SH - AH) × R	4 Y
= 50% of (40 - 32) × R	
D. Overheads (32 hours × ₹ 360 per hour)	11,520
E. Factory Cost	1,24,800
Or X + 36 Y = ₹1,24,800 (Given) – ₹11,520 = ₹ 1,13,280.....equation (i)	

Step 3 : Factory Cost of Worker 'Bijoy'

	(₹)
A. Material Cost	X
B. Wages (Rowan Plan)	30 Y
C. Bonus = 30/40 × (40 - 30) × Y	7.5 Y
D. Overheads (30 hours × ₹ 360 per hour))	10,800
E. Factory Cost	1,24,800
Or X + 37.5 Y = ₹ 1,24,800 (Given) – ₹ 10,800 = ₹ 1,14,000.....equation (ii)	

Step 4: Subtracting equation (i) from equation (ii)

1.5Y = ₹ 720

Y = ₹ 480 per hour

(a) The normal rate of wages: ₹ 480 per hour

(b) The cost of material: X + 36 × ₹ 480 = ₹ 1,13,280

Or X = ₹ 1,13,280 – ₹ 17,280 = ₹ 96,000

Statement of the Factory Cost of the product made by the two workers

	'Ajoy' (₹)	'Bijoy' (₹)
Material cost	96,000	96,000
Direct Wages	15,360 (32 x ₹ 480)	14,400 (30 x ₹ 480)
Bonus	1,920 (4 x ₹ 480)	3,600 (7.5 x ₹ 480)
Factory Overhead	11,520	10,800
Factory Cost	1,24,800	1,24,800

ICAI · P4 · Sep 2025 exam · question p. 8-9 · answer p. 10-11

306 EXAM **May 2025** Q2(b) 7 marks

A plastic manufacturing company is operating with an employment of 128 skilled workers. The product is in great demand. The company desires to increase production to meet market demand but is short of skilled workers. The company finds extremely difficult to find new skilled workers to fulfil its demands. The company is considering the introduction of an incentive scheme – either Halsey scheme (with 50% bonus) or Rawan scheme of wage payment for increasing the labour productivity to cope up the increasing demand.

The company believes that if the proposed incentive scheme could bring about an average 15% increase over the present earnings of the workers, it could act as sufficient incentive for them to produce more with increased efficiency.

The following data is worth consideration, in measuring the increase in productivity for the month of April 2025.

Hourly rate of wages (guaranteed)	₹ 30
Maximum time allowed to produce one unit by one worker	2.5 hours
Number of working days in the month	25
Number of workers hours per day of each worker	8
Actual production during the month (units)	12,500

Required:

(i) Calculate the effective rate of earnings under the Halsey scheme and the Rowan scheme.

(ii) Calculate the increased labour efficiency on introduction of the incentive schemes.

(iii) Calculate the savings to the plastic company in terms of direct labour cost per unit under both the schemes.

(iv) Advise the company about the selection of the scheme to fulfil their assurance.

ICAI'S ANSWER

Working Notes:

1. Actual time taken to produce 12,500 pieces

= No. of working days in the month × No. of working hours per day of each worker × No. of workers

= 25 days × 8 hrs. × 128 workers = 25,600 hours

2. Total time wages of 128 workers per month:

= No. of working days in the month × No. of working hours per day of each worker × Hourly rate of wages × No. of workers

= 25 days × 8 hrs. × ₹ 30 × 128 workers = ₹ 7,68,000

3. Time saved per month:

Time allowed per piece to a worker	2.5 hours
No. of units produced during the month by 128 workers	12,500 pieces
Total time allowed to produce 12,500 pieces (12,500 × 2.5 hours)	31,250 hours
Actual time taken to produce 12,500 pieces	25,600 hours
Time saved (31,250 hours – 25,600 hours)	5,650 hours

4. Bonus under Halsey scheme to be paid to 128 workers:

Bonus = (50% of time saved) × hourly rate of wages

= 50/100 × 5,650 hours × ₹ 30 = ₹ 84,750

Total wages to be paid to 128 workers are (₹ 7,68,000 + ₹ 84,750) ₹ 8,52,750, if company considers the introduction of Halsey Incentive Scheme to increase the employee productivity.

5. Bonus under Rowan Scheme to be paid to 128 workers:

$$\text{Bonus} = \frac{\text{Time taken}}{\text{Time allowed}} \times \text{Time saved} \times \text{hourly rate}$$

$$= \frac{25{,}600 \text{ hours}}{31{,}250 \text{ hours}} \times 5{,}650 \text{ hours} \times ₹\ 30 = ₹1{,}38{,}854.4$$

Total wages to be paid to 128 workers are (₹ 7,68,000 + ₹ 1,38,854.4) ₹ 9,06,854.4, if company considers the introduction of Rowan Incentive Scheme to increase the Employee productivity.

(i) (a) Effective hourly rate of earnings under Halsey scheme:

(Refer to Working Notes 1, 2, 3 and 4)

$$= \frac{\text{Total time wages of 128 workers} + \text{Total bonus under Halsey scheme}}{\text{Total hours worked}}$$

$$= \frac{₹8{,}52{,}750}{25{,}600 \text{ hours}} = ₹\ 33.310$$

(b) Effective hourly rate of earnings under Rowan scheme:

(Refer to Working Notes 1, 2, 3 and 5)

$$= \frac{\text{Total time wages of 128 workers} + \text{Total bonus under Rowan scheme}}{\text{Total hours worked}}$$

$$= \frac{₹9{,}06{,}854.4}{25{,}600 \text{ hours}} = ₹35.424$$

(ii) Efficiency in % on introduction of the incentive schemes

$$= \frac{\text{Time allowed as per standard}}{\text{Time Taken}} \times 100$$

$$= \frac{31{,}250 \text{ hours}}{25{,}600 \text{ hours}} = 122.07\%$$

Labour efficiency has increased by 22.07%

(iii) (a) Saving in terms of direct Employee cost per piece under Halsey scheme:

(Refer to Working Note 4)

Employee cost per piece (under time wage scheme)

= 2.5hours × ₹ 30 = ₹ 75.

Employee cost per piece (under Halsey scheme)

$$= \frac{\text{Total wages paid under the scheme}}{\text{Total number of units produced}} = \frac{₹8{,}52{,}750}{12{,}500} = ₹\ 68.22$$

Saving per piece: (₹75 – ₹68.22) = ₹6.78

(b) Saving in terms of direct Employee cost per piece under Rowan Scheme:

(Refer to Working Note 5)

Employee cost per piece under Rowan scheme

$$= \frac{₹9{,}06{,}854.4}{12{,}500} = ₹72.548$$

Saving per piece = ₹ 75 – ₹ 72.548 = ₹ 2.452

(iv) Since the company has assured 15% increase over present earnings i.e. ₹ 7,68,000 which comes to ₹ 8,83,200 and total wages under halsey scheme and rowan scheme are ₹8,52,750 and ₹ 9,06,854.4 respectively, so the company is advised to select the rowan scheme of wage payment to fulfill its requirement of 15% increase.

ICAI · P4 · May 2025 exam · question p. 13-14 · answer p. 16-18

307 EXAM **Jan 2025** Q2(b) 5 marks

A skilled worker has assigned a work. The relevant data is given as follows:

Time rate per hour	₹ 25
Time allowed	9 hours
Time taken	6 hours

The worker has given an option to choose either Halsey (50% plan) or Rowan plan.

You are required to calculate earnings under both plans and which plan is more beneficial for a worker.

ICAI'S ANSWER

Earning under Halsey Plan

Total Wages = (Time taken × Time rate) + (50% of time saved × time rate)

= (6 hrs x ₹ 25) + ((9 hrs - 6 hrs) x 50% x ₹ 25)

= ₹150 + ₹37.5 = ₹187.5

Earning under Rowan Plan

Total Wages

$$= (\text{Time taken} \times \text{Time Rate}) + \left(\frac{\text{Time saved}}{\text{Time allowed}} \times \text{Time taken} \times \text{Time Rate}\right)$$

$$= (6 \text{ hrs} \times ₹ 25) + \left(\frac{(9 \text{ hrs} - 6 \text{ hrs})}{9 \text{ hrs}} \times 6 \text{ hrs} \times ₹ 25\right)$$

= ₹150 + ₹50 = ₹200

Note: Rowan Plan is more beneficial for a worker since he is more earning under this plan i.e. 200-187.5 = ₹ 12.50

ICAI · P4 · Jan 2025 exam · question p. 13 · answer p. 15

308 EXAM **Sep 2024** MCQ 13 MCQ

In a mutual project, both Raj and Bhuvan are contributing their efforts, using identical materials. Raj receives a bonus based on the Rowan plan, while the Halsey plan determines Bhuvan's bonus. The standard time allocated for the project is 150 hours. Raj completes the project in 90 hours, while Bhuvan finishes it in 120 hours. The normal hourly wage rate for Raj is ₹ 30. The total earnings for both workers are equal.

Calculate the normal hourly wage rate to be paid to Bhuvan.

A ₹ 26.50

B ₹ 24.00

C ₹ 22.50

D ₹ 28.00

ICAI'S ANSWER

Answer key: **(D)**

ICAI · P4 · Sep 2024 exam · question p. 6 · answer p. 7

309 EXAM Old P3 Nov 2023 Q1(b) 5 marks old scheme

A worker took 60 hours to complete a job in a factory. The normal rate of wages is ₹ 80 per hour. The worker is entitled to receive bonus according to the Halsey Premium Plan. Factory overhead is recovered on the job at ₹ 60 per man hour actually worked. The factory cost of the job is ₹ 37,280 and material cost of the job is ₹ 28,400.

Required:

(i) Calculate the standard time for completing the job and effective hourly rate under the Halsey Premium plan.

(ii) Calculate the effective rate of earnings per hour if wages would have been paid under the Rowan Plan.

ICAI'S ANSWER

(i) Calculation of standard time and effective hourly rate:

Standard time = Actual hours worked + time saved = 60 +12 = 72 hours

Effective hourly rate under Halsey premium plan = $\frac{\text{Total labour cost}}{\text{Actual hour worked}} = \frac{5,280}{60}$

= ₹ 88

(ii) Calculation of effective rate earnings under Rowan plan:

(Rate × Actual hours worked) + Rate × $\frac{\text{Time Saved}}{\text{Std. Time}}$ × Time taken

₹ 80 × 60 hours + ₹ 80 × $\frac{12}{72}$ × 60

₹ 4,800 + 800 = ₹ 5,600

Effective rate per hour = 5,600 ÷ 60 hour = ₹ 93.33

Working Note:

(1) Calculation of labour cost = Factory cost – Material cost – Factory Overhead

= 37,280 – 28,400 – (₹ 60 × 60 hours)

= 37,280 – 28,400 – 3,600 = ₹ 5,280

(2) Calculation of bonus and time saved

Total labour cost = Normal Rate × Actual hours worked + ½ time saved × normal rate

₹ 5,280 = (₹ 80 × 60 hours) + ½ (time saved × ₹ 80)

40 × time saved = ₹ 5,280 – ₹ 4,800

Time saved = (5,280 - 4,800) ÷ 40

Time saved = 12 hours

The solution can also be presented in following way:

(b)

Particulars	(₹)
Factory Cost	37,280
Less: Factory Overheads 60 x ₹ 60	3,600
Prime Cost	33,680
Direct material	28,400
Direct wages (Balancing Figure)	5,280

(i) Wages under Halsey Plan (Rate × Actual hours worked) + Rate × $\frac{\text{Time Saved}}{\text{Std. Time}}$ ×Time taken

₹ 5,280 = 60 x ₹ 80 + (S* – 60)/2 x ₹ 80

₹ 5,280 = ₹ 4,800 + 40S – 2,400

S = ₹ 2,880/40 = 72 hours

*Standard time

Effective rate of earnings per hour = 5,280/60 = ₹ 88

(ii) Wages under Rowan Plan: (Rate × Actual hours worked) + Rate × $\frac{\text{Time Saved}}{\text{Std. Time}}$ ×Time taken

= 60 x 80 + $\frac{72 - 60}{72}$ x 60 x 80 = ₹ 5,600

Effective rate of earnings per hour = 5,600/60 = ₹ 93.33

ICAI · Old P3 · Nov 2023 exam · question p. 1 · answer p. 3-5

310 EXAM **Old P3 May 2023** Q1(b) 5 marks old scheme

SMC Company Limited is producing a particular design of toys under the following existing incentive system:

Normal working hours in the week	48 hours
Late shift hours in the week	12 hours
Rate of payment	Normal working: ₹ 150 per hour Late shift: ₹ 300 per hour

Average output per operator for 60 hours per week (including late shift hours): 80 toys.

The company's management has now decided to implement a system of labour cost payment with either the Rowan Premium Plan or the Halsey Premium Plan in order to increase output, eliminate late shift overtime, and reduce the labour cost.

The following information is obtained:

The standard time allotted for ten toys is seven and half hours.

Time rate: ₹ 150 per hour (as usual).

Assuming that the operator works for 48-hours in a week and produces 100 toys, you are required to calculate the weekly earnings for one operator under-

(i) The existing Time Rate,

(ii) Rowan Premium Plan and,

(iii) Halsey Premium Plan (50%).

ICAI'S ANSWER

Working Notes:

(1) Effective rate per hour:

Incentive for 60 hours = (₹ 150 × 48 hours + ₹ 300 × 12 hours)

= 7,200 + 3,600 = ₹ 10,800

= ₹ 10,800 ÷ 60 hours = ₹ 180 per hour

(2) Time taken/ Allowed to produce 100 toys:

= (60 hours ÷ 80 toys) × 100 toys = 75 hours

(3) Time saved = Time Allowed – Time Taken

= 75 hours – 48 hours = 27 hours

(i) Calculation of weekly earnings for one operator under the existing time rate:

= (48 hours x ₹ 150) + (12 hours x ₹ 300) **= ₹ 10,800**

Alternative solution

= Effective rate per hour (WN-1) × Time required for 100 toys (WN-2)

= ₹ 180 × 75 hours = **₹ 13,500**

(ii) Calculation of weekly earnings for one operator under Rowan Premium plan:

(Time taken × Rate per hour) + (Time Saved/ Time Allowed × Time taken × Rate per hour)

= (48 hours × ₹ 150) + [(27 ÷ 75) × 48 × ₹ 150]

= 7,200 + 2,592 = **₹ 9,792**

(iii) Calculation of weekly earnings for one operator under Halsey Premium plan:

(Time taken × Rate per hour) + (50% of Time Saved × Rate per hour)

= (48 hours × ₹ 150) + (50% of 27 hours × ₹ 150)

= ₹ 7,200 + ₹ 2,025 = **₹ 9,225**

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311 EXAM **Old P3 Nov 2022** Q2(b) 6 marks old scheme

A skilled worker, in PK Ltd., is paid a guaranteed wage rate of ₹ 15.00 per hour in a 48-hour week. The standard time to produce a unit is 18 minutes. During a week, a skilled worker -Mr. 'A' has produced 200 units of the product. The Company has taken a drive for cost reduction and wants to reduce its labour cost.

You are required to:

(i) Calculate wages of Mr. 'A' under each of the following methods:

(A) Time rate,

(B) Piece -rete with a guaranteed weekly wage,

(C) Halsey Premium Plan

(D) Rowan Premium Plan

(ii) Suggest which bonus plan i.e. Halsey Premium Plan or Rowan Premium Plan, the company should follow.

ICAI'S ANSWER

(i) Calculation of wages of Mr. 'A' under different wage schemes:

A. Time rate

Wages = Time Worked × Rate for the time

= 48 hours x ₹ 15

= ₹ 720

B. Piece rate with a guaranteed weekly wage

Wages = Number of units produced × Rate per unit

= 200 units x ₹ 4.50*

= ₹ 900

*(₹ 15 / 60 minutes) x 18 minutes = ₹ 4.50

C. Halsey Premium Plan

Wages = Time taken × Time rate + 50% of time saved × Time rate

Wages = Time taken × Time rate + 50% (Standard time – Actual time) × Time rate

= (48 hours x ₹ 15) + 50% of (60 hours# – 48 hours) x ₹ 15

= ₹ 720 + ₹ 90

= ₹ 810

#(200 units x 18 minutes) / 60 minutes = 60 hours

D. Rowan Premium Plan

$$\text{Wages} = \text{Time taken} \times \text{Rate per hour} + \frac{\text{Time Saved}}{\text{Time Allowed}} \times \text{Time taken} \times \text{Rate per hour}$$

$$= (48 \text{ hours x ₹ } 15) + \left(\frac{60 - 48 \text{ hours}}{60 \text{ hours}} \text{ x } 48 \text{ hours x ₹ } 15\right)$$

= ₹ 720 + ₹ 144

= ₹ 864

(ii) The company may follow Halsey Premium Plan over Rowan Premium Bonus Plan as the total wages paid is lower than that of Rowan Premium Bonus Plan.

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312 EXAM **Old P3 Dec 2021** Q1(c) 5 marks old scheme

A skilled worker is paid a guaranteed wage rate of ₹ 150 per hour. The standard time allowed for a job is 10 hours. He took 8 hours to complete the job. He has been paid the wages under Rowan Incentive Plan.

You are required to:

(i) Calculate an effective hourly rate of earnings under Rowan Incentive Plan.

(ii) Calculate the time in which he should complete the job, if the worker is placed under Halsey Incentive Scheme (50%) and he wants to maintain the same effective hourly rate of earnings.

ICAI'S ANSWER

(i) Calculation of Effective hourly rate of earnings under Rowan Incentive Plan:

Standard time allowed = 10 hours

Time taken = 8 hours; Time saved = 2 hours

	Particulars	Amount (₹)
A	Basic guaranteed wages (₹150×8 hours)	1,200
B	Add: Bonus for time saved (2/10 × 8 × ₹ 150)	240
C	Total earnings (A+B)	1,440
D	Hours worked	8 hours
E	Effective hourly rate (C÷D)	180

(ii) Let the time taken to complete the job is "T" and the time saved is 10-T

Effective hourly rate under the Halsey Incentive scheme

$$= \frac{(\text{Rate} \times \text{Hours Worked}) + (\text{Rate} \times 50\% \text{ of Time Saved})}{\text{Hours Worked}} = ₹\ 180$$

$$\frac{(₹150 \times T) + ₹150 \times 50\%\ (10 - T)}{T} = ₹\ 180$$

150T + 750 -75T = 180T

180T-75T = 750

$$T = \frac{750}{105} = 7.14 \text{ hours}$$

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313 EXAM Old P3 Jul 2021 Q6(c) 5 marks old scheme

Rowan Premium Bonus system does not motivate a highly efficient worker as a less efficient worker and a highly efficient worker can obtain same bonus under this system. Discuss with an example.

ICAI'S ANSWER

Rowan Premium Plan: According to this system a standard time allowance is fixed for the performance of a job and bonus is paid if time is saved.

Under Rowan System, the bonus is that proportion of the time wages as time saved bears to the standard time.

$$\text{Bonus} = \frac{\text{Time Saved}}{\text{Time Allowed}} \times \text{Time taken} \times \text{Rate per hour}$$

Example explaining highly efficient worker and less efficient worker obtaining same bonus:

Time rate (per Hour)	₹ 60
Time allowed	8 hours.
Time taken by 'X'	6 hours.
Time taken by 'Y'	2 hours.

Bonus = $\frac{\text{Time Saved}}{\text{Time Allowed}}$ × Time taken × Rate per hour

For 'X' = $\frac{2 \text{ hours}}{8 \text{ hours}}$ × 6 hours × ₹ 60 = **₹ 90**

For 'Y' = $\frac{6 \text{ hours}}{8 \text{ hours}}$ × 2 hours × ₹ 60 = **₹ 90**

From the above example, it can be concluded that a highly efficient worker may obtain same bonus as less efficient worker under this system.

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314 EXAM **Old P3 Jan 2021** Q2(a) 10 marks old scheme

Z Ltd is working by employing 50 skilled workers. It is considering the introduction of an incentive scheme - either Halsey Scheme (with 50% Bonus) or Rowan Scheme - of wage payment for increasing the labour productivity to adjust with the increasing demand for its products by 40%. The company feels that if the proposed incentive scheme could bring about an average 20% increase over the present earnings of the workers, it could act as sufficient incentive for them to produce more and the company has accordingly given assurance to the workers.

Because of this assurance, an increase in productivity has been observed as revealed by the figures for the month of April, 2020:

Hourly rate of wages (guaranteed)	₹ 50
Average time for producing one unit by one worker at the previous performance (this may be taken as time allowed)	1.975 hours
Number of working days in a month	24
Number of working hours per day of each worker	8
Actual production during the month	6,120 units

Required:

(i) Calculate the effective increase in earnings of workers in percentage terms under Halsey and Rowan scheme.

(ii) Calculate the savings to Z Ltd in terms of direct labour cost per unit under both the schemes.

(iii) Advise Z Ltd about the selection of the scheme that would fulfil its assurance of incentivising workers and also to adjust with the increase in demand.

ICAI'S ANSWER

Working Notes:

1. Total time wages of 50 workers per month:

= No. of working days in the month × No. of working hours per day of each worker × Hourly rate of wages × No. of workers

= 24 days × 8 hrs. × ₹ 50 × 50 workers = ₹ 4,80,000

2. Time saved per month:

Time allowed per unit to a worker	1.975 hours
No. of units produced during the month by 50 workers	6,120 units
Total time allowed to produce 6,120 units (6,120 × 1.975 hrs)	12,087 hours
Actual time taken to produce 6,120 units (24 days × 8 hrs. × 50 workers)	9,600 hours
Time saved (12,087 hours – 9,600 hours)	2,487 hours

3. Bonus under Halsey scheme to be paid to 50 workers:

Bonus = (50% of time saved) × hourly rate of wages

= 50/100 × 2,487 hours × ₹ 50 = ₹ 62,175

Total wages to be paid to 50 workers are (₹ 4,80,000 + ₹ 62,175) ₹ 5,42,175, if Z Ltd. considers the introduction of Halsey Incentive Scheme to increase the worker productivity.

4. Bonus under Rowan Scheme to be paid to 50 workers:

$$\text{Bonus} = \frac{\text{Time taken}}{\text{Time allowed}} \times \text{Time saved} \times \text{hourly rate}$$

$$= \frac{9{,}600 \text{ hours}}{12{,}087 \text{ hours}} \times 2{,}487 \text{ hours} \times ₹\ 50 = ₹\ 98{,}764$$

Total wages to be paid to 50 workers are (₹ 4,80,000 + ₹ 98,764) ₹ 5,78,764, if Z Ltd. considers the introduction of Rowan Incentive Scheme to increase the worker productivity.

(i) (a) Effective hourly rate of earnings under Halsey scheme:

(Refer to Working Notes 1, 2 and 3)

$$= \frac{\text{Total time wages of 50 workers} + \text{Total bonus under Halsey scheme}}{\text{Total hours worked}}$$

$$= \frac{₹\ 4{,}80{,}000 + ₹\ 62{,}175}{9{,}600 \text{ hours}} = \textbf{₹ 56.48}$$

Effective increase in earnings of worker (in %) = $\frac{₹\ 56.48 - ₹\ 50}{₹\ 50}$ x 100 = **2.96%**

(b) Effective hourly rate of earnings under Rowan scheme:

(Refer to Working Notes 1, 2 and 4)

$$= \frac{\text{Total time wages of 50 workers} + \text{Total bonus under Rowan scheme}}{\text{Total hours worked}}$$

$$= \frac{₹\ 4{,}80{,}000 + ₹\ 96{,}875}{9{,}600 \text{ hours}} = \textbf{₹ 60.29}$$

Effective increase in earnings of worker (in %)= $\frac{₹\ 60.29 - ₹\ 50}{₹\ 50}$ x 100 =**20.58%**

(ii) (a) Saving in terms of direct labour cost per unit under Halsey scheme:

(Refer to Working Note 3)

Labour cost per unit (under time wage scheme)

= 1.975 hours × ₹ 50 = ₹ 98.75

Labour cost per unit (under Halsey scheme)

$$= \frac{\text{Total wages paid under the scheme}}{\text{Total number of units produced}} = \frac{₹\ 5{,}42{,}175}{6{,}120} = \text{s}₹\ 88.60$$

Saving per unit = ₹ 98.75 – ₹ 88.60 = **₹ 10.15**

(b) Saving in terms of direct worker cost per unit under Rowan Scheme:

(Refer to Working Note 4)

Labour cost per unit under Rowan scheme = ₹ 5,78,764/6,120 units= ₹ 94.57

Saving per unit = ₹ 98.75 – ₹ 94.57 = **₹ 4.18**

(iii) Calculation of Productivity:

Normal Production Hours worked/Unit per Hour (9,600/1.975)	4,861
Actual Production Units	6,120
Increase in labour productivity	1,259
% Productivity i.e. increase in production/Normal production	**25.9%**

Advice: Rowan plan fulfils the company's assurance of 20% increase over the present earnings of workers. This would increase productivity by 25.9% only. It will not adjust with the increase in demand by 40%.

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315 EXAM **Old P3 Nov 2019** Q4(a) 10 marks old scheme

Zico Ltd. has its factory at two locations viz Nasik and Satara. Rowan plan is used at Nasik factory and Halsey plan at Satara factory.

Standard time and basic rate of wages are same for a job which is similar and is carried out on similar machinery. Normal working hours is 8 hours per day in a 5 day week.

Job at Nasik factory is completed in 32 hours while at Satara factory it has taken 30 hours.

Conversion costs at Nasik and Satara are ₹ 5,408 and ₹ 4,950 respectively. Overheads account for ₹ 25 per hour.

Required:

(i) To find out the normal wage; and

(ii) To compare the respective conversion costs.

ICAI'S ANSWER

Particulars	Nasik	Satara
Hours worked	32 hr.	30 hr.
Conversion Costs	₹5,408	₹4,950
Less: Overheads	₹800 (₹25×32 hr.)	₹750 (₹25×30 hr.)
Labour Cost	₹4,608	₹4,200

(i) Finding of Normal wage rate:

Let Wage rate be ₹R per hour, this is same for both the Nasik and Satara factory.

Normal wage rate can be found out taking total cost of either factory.

Nasik: Rowan Plan

Total Labour Cost = Wages for hours worked + Bonus as per Rowan plan

₹ 4,608 = Hours worked × Rate per hour + $\left(\frac{\text{Time saved}}{\text{Time allowed}} \times \text{Hours worked} \times \text{Rate per hour}\right)$

Or, ₹ 4,608 = 32 hr. × R + $\left(\frac{40-32}{40} \times 32 \times R\right)$

Or, ₹ 4,608 = 32R + 6.4R

R = ₹ 120

Normal wage = 32 hrs × ₹ 120 = ₹ 3,840

OR

Satara: Halsey Plan

Total Labour Cost = Wages for hours worked + Bonus as per Halsey plan

₹ 4,200 = Hours worked × Rate per hour + (50% ×Hours saved×Rate per hour)

₹ 4,200 = 30 hr. × R + 50% × (40 hr. – 30 hr.) × R

₹ 4,200 = 35 R

Or R = ₹ 120

Normal Wage = 30 hrs × ₹ 120 = ₹ 3,600

(ii) Comparison of conversion costs:

Particulars	Nasik (₹)	Satara (₹)
Normal Wages (32 × 120)	3,840	
(30×120)		3,600
Bonus (6.4 × 120)	768	
(5 × 120)		600
Overhead	800	750
	5,408	4,950

ICAI · Old P3 · Nov 2019 exam · question p. 16 · answer p. 17-18

316 RTP Sep 2026 Q6

The existing incentive system of Alpha Limited is as under:

Normal working week	5 days of 8 hours each plus 3 late shifts of 3 hours each
Rate of Payment	Day work: ₹ 160 per hour Late shift: ₹ 225 per hour
Average output per operator for 49-hours week i.e. including 3 late shifts	240 articles

In order to increase output and eliminate overtime, it is decided to switch on to a system of payment by results. The following information is obtained:

Time-rate (as usual)	:	₹ 160 per hour
Basic time allowed for 15 articles	:	2.5 hours
Piece-work rate	:	*Add:* 20% to basic piece-rate
Premium Bonus	:	*Add:* 50% to time.

If during the last week 270 articles are produced in a 40-hour week.

Required:

(i) CALCULATE weekly earnings, number of articles produced and labour cost per article for one operator under the following systems:

(a) Existing time-rate

(b) Straight piece-work

(c) Rowan system

(d) Halsey premium system

(ii) PREPARE a Statement showing hours worked, weekly earnings, number of articles produced and labour cost per article for one operator under the above systems.

ICAI'S ANSWER

(i) (a) Existing time rate

Weekly wages:

Normal shift	(40 hours × ₹ 160):	₹ 6,400
Late shift	(9 hours × ₹ 225)	₹ 2,025
		₹ 8,425

(b) Piece Rate System

15 articles are produced in 2.5 hours

Therefore, to produce 270 articles, hours required is $\frac{2.5 \text{ hours}}{15 \text{ articles}} \times 270 \text{ articles} = 45 \text{ hours}$.

Cost of producing 270 articles:	
At basic time rate (45 hours × ₹160) =	₹7,200
Add: Bonus @ 20% on basic Piece rate	₹1,440
Earning for the week	₹8,640

(c) Rowan Premium System

(i) Time allowed for producing 270 articles $\left(\frac{2.5\text{hours}}{15\text{articles}} \times 270\text{articles} \times 150\%\right)$ = 67.5 hours

(ii) Time taken to produce 270 articles = 40.0 hours

(iii) Time Saved = 67.5 – 40 = 27.5 hours

Earnings under Rowan Premium system:

$= (\text{Time taken} \times \text{Rate per hour}) + \left(\frac{\text{Time saved}}{\text{Time allowed}} \times \text{Time taken} \times \text{Rate per hour}\right)$

$= (40 \text{ hours} \times ₹160) + \left(\frac{27.5 \text{ hours}}{67.5 \text{ hours}} \times 40 \text{ hours} \times ₹160\right)$ = ₹ 9,007.41

(d) Halsey Premium System

$= (\text{Time taken} \times \text{Rate per hour}) + (\frac{1}{2} \times \text{Time saved} \times \text{Rate per hour})$

$= (40 \text{ hours} \times ₹160) + \left(\frac{1}{2} \times 27.5 \text{ hours} \times ₹160\right)$ = ₹6,400 + ₹2,200

= ₹8,600

(ii) Statement showing hours worked, weekly earnings, number of articles produced and cost per article

Method of Payment	Hours worked	Weekly earnings (₹)	Number of articles produced	Labour cost per article (₹)
Existing time rate	49	8,425.00	240	35.10
Straight piece rate system	40	8,640.00	270	32.00
Rowan Premium System	40	9,007.41	270	33.36
Halsey Premium System	40	8,600.00	270	31.85

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317 RTP **May 2026** Q7

Rivaan Industries manufactures and sells wooden study tables. The demand for these tables has been steadily increasing due to the company's strong distribution network and reputation for durable products. Despite the rising market potential, the company continues to rely on its long-established wage practices. At present, Rivaan Industries pays wages under a piece–rate system, with a standard piece rate of ₹ 18 per table produced.

The company policy for payment under the piece-rate system is as follows:

- For efficiency less than 100%: Workers are paid 80% of the standard piece rate
- For efficiency equal to or more than 100%: Workers are paid 125% of the standard piece rate

The standard output per hour is 3 tables.

Each worker works 9 hours per day from Monday to Friday and 4 hours on Saturday.

A total of 132 workers are currently eligible for this wage plan.

However, the cutting and finishing machines used in production were purchased many years ago and now limit workers' efficiency. As a result, workers have demanded a shift to a time-rate wage system at ₹65 per hour, along with incentives under the Halsey Premium Plan (50% sharing of time saved).

To evaluate the financial impact, the management has prepared the following estimated production levels for January 2026 under three different scenarios:

Production Estimates (January 2026)

Scenario	Worst Case	Optimal Case	Best Case
Production (units)	36,900	73,800	1,10,700

In addition, the company is assessing a potential Technology Upgrade Case involving partial automation, which may influence both output levels and worker productivity.

Required:

(a) Compute the Total wages, and Average wages per worker per month, for all four scenarios (Worst, Optimal, Best, and Technology Upgrade Case) under each of the following wage systems:

(i) Current piece–rate wage system (based on differential piece rates)

(ii) Workers' proposed system: Time-rate wages @ ₹ 65 per hour + Halsey Premium (50%)

(b) Mr. R, one of the workers, produced 900 tables in December 2025. If he produces the same number of tables in January 2026, compute:

- His earnings under the current piece-rate system
- His earnings under the proposed time-rate + Halsey incentive system
- The difference in his earnings

Based on your computation, should Mr. R support the workers' demand?

(Assume 4 working weeks in a month.)

ICAI'S ANSWER

(a) Calculation of Total wages and average wages per worker per month.

(i) When Current system of wages and incentive payment system is followed:

Standard Production per Month

Standard production per worker = 196 hours × 3 units = 588 units

Standard production for 132 workers = 588 × 132 = 77,616 units

Particulars	Worst Case	Optimal Case	Best Case
I. Standard Production (units) (588 units × 132 workers)	77,616	77,616	77,616
II. Actual Production (units)	36,900	73,800	1,10,700
III. Efficiency (%) (II ÷ I × 100)	47.54%	95.08%	142.63%
IV. Applicable Piece Rate	₹ 14.40 (₹ 18 x 0.8)	₹ 14.40 (₹ 18 x 0.8)	₹ 22.50 (₹ 18 x 1.25)
V. Total Wages (II × IV)	₹ 5,31,360	₹ 10,62,720	₹ 24,90,750
VI. Average Wage per Worker (V ÷ 132)	₹ 4,025.45	₹ 8,050.91	₹ 18,869.32

(ii) When workers' demand for time rate wages and Halsey premium plan is accepted:

Particulars	Worst Case	Optimal Case	Best Case
I. Actual Production (units)	36,900	73,800	1,10,700
II. Standard Units per hour	3	3	3
III. Standard Hours Required (I ÷ II)	12,300	24,600	36,900
IV. Expected working hours (132 workers × 196 hrs.)	25,872	25,872	25,872
V. Time Saved (IV − III)	—	—	11,028
VI. Time Wages (IV × ₹ 65)	₹ 16,81,680	₹ 16,81,680	₹ 16,81,680
VII. Halsey Bonus (50% × Time Saved × 65)	—	—	₹ 3,58,410
VIII. Total Wages (VI + VII)	₹16,81,680	₹16,81,680	₹20,40,090
IX. Average Wage per Worker (VIII ÷ 132)	₹12,740	₹12,740	₹15,455.23

(b) Calculation of gain or loss in the current monthly income of Mr. R:

1. Under Current Piece-Rate System

Standard production = 588 units

Production in December = 900 units

Efficiency = 900 ÷ 588 × 100 = 153.06%

Rate = ₹ 22.50

Total wages = 900 × 22.50 = ₹ 20,250

2. Under Proposed Time Rate + Halsey Plan

Standard hours = 900 units ÷ 3 units = 300 hours

Actual hours available = 196 hours

Time saved = 300 – 196 = 104 hrs ,

Time wages = 196 × ₹ 65 = ₹ 12,740

Halsey Bonus (50% × Time Saved × 65)

= 104 hrs x 50% x 65 = ₹ 3,380

Total expected wages = ₹ 16,120

3. Loss To Mr. R

Loss = 20,250 – 16,120 = ₹ 4,130

Supporting the demand of colleague workers will cost ₹ 4,130 in the next month to Mr. R.

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318 **RTP** **Sep 2025** Q8

The existing incentive system of Sakshi Limited is as under:

Normal working week	5 days of 8 hours each plus 3 late shifts of 3 hours each
Rate of Payment	Day work: ₹ 160 per hour Late shift: ₹ 225 per hour
Average output per operator for 49-hours week i.e. including 3 late shifts	240 articles

In order to increase output and eliminate overtime, it is decided to switch on to a system of payment by results. The following information is obtained:

Time-rate (as usual)	:	₹ 160 per hour
Basic time allowed for 15 articles	:	2.5 hours
Piece-work rate	:	Add 20% to basic piece-rate
Premium Bonus	:	Add 50% to time.

If during the last week 270 articles are produced in a 40-hour week.

Required:

(i) CALCULATE weekly earnings for one operator under the following systems:

(a) Existing time-rate

(b) Straight piece-work

(c) Rowan system

(d) Halsey premium system

(ii) PREPARE a Statement showing hours worked, weekly earnings, number of articles produced and labour cost per article for one operator under the above systems.

ICAI'S ANSWER

(i) (a) Existing time rate

Weekly wages:

Normal shift	(40 hours × ₹160):	₹ 6,400
Late shift	(9 hours × ₹225)	₹ 2,025
		₹ 8,425

(b) Piece Rate System

15 articles are produced in 2.5 hours

Therefore, to produce 270 articles, hours required is $\frac{2.5 \text{ hours}}{15 \text{ articles}} \times 270 \text{ articles} = 45 \text{ hours}$.

Cost of producing 270 articles:

At basic time rate (45 hours × ₹160)	=	₹ 7,200
Add: Bonus @ 20% on basic Piece rate		₹ 1,440
Earning for the week		₹ 8,640

(c) Rowan Premium System

(i) Time allowed for producing 270 articles $\left(\frac{2.5 \text{ hours}}{15 \text{ articles}} \times 270 \text{ articles} \times 150\%\right) = 67.5 \text{ hours}$

(ii) Time taken to produce 270 articles = 40.0 hours

(iii) Time Saved = 67.5 – 40 = 27.5 hours

Earnings under Rowan Premium system:

$= (\text{Time taken} \times \text{Rate per hour}) + \left(\frac{\text{Time saved}}{\text{Time allowed}} \times \text{Time taken} \times \text{Rate per hour}\right)$

$= (40 \text{ hours} \times ₹ 160) + \left(\frac{27.5 \text{ hours}}{67.5 \text{ hours}} \times 40 \text{ hours} \times ₹ 160\right)$

=₹ 6,400 + ₹ 2,607.41 ₹ 9,007.41

(d) Halsey Premium System

$= (\text{Time taken} \times \text{Rate per hour}) + (\frac{1}{2} \times \text{Time saved} \times \text{Rate per hour})$

$= (40\text{hours} \times ₹ 160) + \left(\frac{1}{2} \times 27.5\text{hours} \times ₹ 160\right)$

= ₹ 6,400 + ₹ 2,200 = ₹ 8,600

(ii) Statement showing hours worked, weekly earnings, number of articles produced and cost per article

Method of Payment	Hours worked	Weekly earnings (₹)	Number of articles produced	Labour cost per article (₹)
Existing time rate	49	8,425.00	240	35.10
Straight piece rate system	40	8,640.00	270	32.00
Rowan Premium System	40	9,007.41	270	33.36
Halsey Premium System	40	8,600.00	270	31.85

ICAI · P4 · RTP Sep 2025 · question p. 11 · answer p. 33-34

319 RTP **May 2025** Q3 MCQ

Same question, different ICAI answer: #324 MTP Sep 2026 S-II

Mr. A works in a manufacturing company where he is paid bonus according to the Halsey 50% plan, besides the normal wages. The relevant data is as below:

Time Rate (per hour)	₹ 100
Time allowed	10 hours
Time taken	5 hours
Time saved	5 hours

Mr. A believes that his bonus under Halsey system is getting reduced by 50%, thus intending to shift towards Rowan Premium Plan.

You are required to CALCULATE the total earnings of Mr. A as per Halsey plan and Rowan Premium plan & ENUMERATE the reason for difference in both the earnings.

A. Total earnings as per Halsey plan- ₹ 500 and as per Rowan Premium plan- ₹ 750. Earnings under Halsey Plan is lower than that of Rowan Premium plan as the bonus is getting reduced by 50%.

B. Total earnings as per Halsey plan- ₹ 750 and as per Rowan Premium plan- ₹ 500. Earnings under Rowan Premium plan is lower than that of Halsey plan as the actual time taken is 50% of the time allowed.

C. Total earnings as per Halsey plan- ₹ 750 and as per Rowan Premium plan- ₹ 750. When the actual time taken is 50% of the time allowed, the earnings under Halsey and Rowan Plans are equal.

D. Total earnings as per Halsey plan- ₹ 250 and as per Rowan Premium plan- ₹ 250. Total earnings under both the Plans are equal as the time taken under both the plans are same.

ICAI'S ANSWER

(C)

Calculation of total earnings:

As per Halsey 50% plan

= Time taken × Time rate + (50% of Time saved × Time rate)

= 5 hrs. × ₹ 100 + [1/2 × (5 hrs. × ₹ 100)]

= ₹ 500 + ₹ 250 = **₹ 750**

As per Rowan Premium plan

$$= \text{Time taken} \times \text{Rate per hour} + \left(\frac{\text{Time Saved}}{\text{Time Allowed}} \times \text{Time taken} \times \text{Rate per hour}\right)$$

$$= 5 \text{ hours} \times ₹\ 100 + \left[\left(\frac{5 \text{ Hours}}{10 \text{ Hours}}\right) \times 5 \text{ hours} \times ₹\ 100\right]$$

= ₹ 500 + ₹ 250 = **₹ 750**

When the actual time taken is 50% of the time allowed, the earnings under Halsey and Rowan Plans are equal.

ICAI · P4 · RTP May 2025 · question p. 7-8 · answer p. 29-30

320 RTP **Jan 2025** Q3 MCQ

Phalsa Ltd. pays its workers on time-basis because their services cannot be tangibly measured. The company's normal working week includes 5 days of 8 hours each. Sometimes, the workers needs to work late at night which was 3 nights of 3 hours each for the current week. The average output produced per worker for the week is 120 units.

Information regarding incentive rate is as follows:

Rate of Payment	Day shift: ₹ 320 per hour Night shift: ₹ 450 per hour

However, this time-basis payment made workers lazy, making their expected output lower. As workers started doing more of the night shifts for higher earnings with minimal impact on the outputs, the company decided to shift on to a system of payments on output basis.

Information regarding amended incentive rate is as follows:

Time-rate (as usual)	:	₹ 320 per hour
Basic time allowed for 15 units	:	5 hours
Piece-work rate	:	Add 15% to basic piece-rate

In the amended incentive system, the normal weekly working hours remained the same while production increased to 135 units.

CALCULATE the labour cost per unit as per the existing incentive system, along with the amended incentive system.

A ₹ 140.42 and ₹ 122.67 respectively

B ₹ 124.81 and ₹ 138.00 respectively

C ₹ 124.81 and ₹ 122.67 respectively

D ₹ 140.42 and ₹ 138.00 respectively

ICAI'S ANSWER

(A) Calculation of existing labour cost per unit (time basis)

Normal weekly hours = 5 days x 8 hours = 40 hours

Night shift hours = 3 nights x 3 hours = 9 hours

Average production per week = 120 units

Weekly wages:		
Normal shift	(40 hours × ₹ 320)	₹ 12,800
Night shift	(9 hours × ₹ 450)	₹ 4,050
Total wages		₹ 16,850

Labour cost per unit = $\left(\frac{₹16,850}{120 \text{ units}}\right)$

= ₹ 140.42

Calculation of amended labour cost per unit (piece basis)

15 units are produced in 5 hours

Therefore, to produce 135 units, hours required is $\left(\frac{5 \text{ hours}}{15 \text{ units}}\right)$ X 135 units = 45 hours.

Labour cost of producing 135 units:

At basic time rate (45 hours × ₹ 320) = ₹ 14,400

Add: Bonus @ 15% on basic Piece rate

$\left[\left(\frac{₹14,400}{135 \text{ units}}\right) \text{ x } 15\%\right]$ x 135 units = ₹ 2,160

Earning for the week ₹ 16,560

Labour cost per unit = $\left(\frac{₹16,560}{135 \text{ units}}\right)$

= ₹ 122.67

ICAI · P4 · RTP Jan 2025 · question p. 5-6 · answer p. 25-26

321 RTP **Jan 2025** Q9

AeBee Publishers works for various educational institutes for editing, binding, printing of various books and magazines on job work basis. Currently, the company has employed 30 workers and pays them on hour rate basis for each job assigned. To complete one of the process of binding, the average time allowed to an employee is 8 hours for a 10 pages magazine.

In the month of March, two employees 'Cee' and 'Dee' were given 21 and 30 units of magazines respectively for binding work. The following are the details of the work assigned:

Particulars	'Cee'	'Dee'
Work assigned	21 units	30 units
Time taken	78 hours	114 hours

The existing rate of wages is ₹ 60 per hour along with bonus as per Halsey System.

However, a new wage agreement has been signed between the employees and the company where, employees will be paid ₹ 65 per hour with effect from the April month. But, inadvertently, for the month of March, the accountant of the company paid the wages to these employees considering rate of wages as ₹ 65 per hour.

You are required to CALCULATE the following:

(i) Amount of loss that the company has incurred due to incorrect rate selection in the month of March.

(ii) Loss incurred by the company due to incorrect rate selection if it had followed Rowan scheme of bonus payment.

(iii) Amount that could have been saved if Rowan Scheme of bonus payment were followed.

ICAI'S ANSWER

Particulars	'Cee'	'Dee'
No. of binding work assigned (units)	21	30
Hour allowed per magazine (Hours)	8	8
Total hours allowed (Hours)	168	240
Hours Taken (Hours)	78	114
Hours Saved (Hours)	90	126

(i) Calculation of loss incurred due to incorrect rate selection

(While calculating loss only excess rate per hour has been taken)

Particulars	'Cee' (₹)	'Dee' (₹)	Total (₹)
Basic Wages	390 (78 Hrs. ×₹ 5)	570 (114 Hrs. × ₹ 5)	960
Bonus (as per Halsey Scheme) (50% of Time Saved × Excess Rate)	225 (50% of 90 Hrs. × ₹ 5)	315 (50% of 126 Hrs. × ₹ 5)	540
Excess Wages Paid	**615**	**885**	**1,500**

(ii) Amount of loss if Rowan scheme of bonus payment were followed

Particulars	'Cee' (₹)	'Dee' (₹)	Total (₹)
Basic Wages	390.00 (78 Hrs. × ₹ 5)	570.00 (114 Hrs. × ₹ 5)	960.00
Bonus (as per Rowan Scheme) $(\frac{\text{Time Taken}}{\text{Time Allowed}}$ × Time Saved x Excess Rate)	208.93 $=(\frac{78}{168}$ × 90 × ₹ 5)	299.25 $=(\frac{114}{240}$ × 126 × ₹ 5)	508.18
Excess Wages Paid	**598.93**	**869.25**	**1,468.18**

(iii) Calculation of amount that could have been saved if Rowan Scheme were followed

Particulars	'Cee' (₹)	'Dee' (₹)	Total (₹)
Wages paid under Halsey Scheme	615.00	885.00	1,500.00
Wages paid under Rowan Scheme	598.93	869.25	1,468.18
Difference (Savings)	**16.07**	**15.75**	**31.82**

322 RTP **Old P3 Nov 2023** Q2 old scheme

A skilled worker is paid a guaranteed wage rate of ₹ 120 per hour. The standard time allowed for a job is 6 hour. He took 5 hours to complete the job. He is paid wages under Rowan Incentive Plan.

(i) Calculate his effective hourly rate of earnings under Rowan Incentive Plan.

(ii) If the worker is placed under Halsey Incentive Scheme (50%) and he wants to maintain the same effective hourly rate of earnings, calculate the time in which he should complete the job.

ICAI'S ANSWER

(i) Effective hourly rate of earnings under Rowan Incentive Plan

Earnings under Rowan Incentive plan =

$$\text{(Actual time taken} \times \text{wage rate)} + \frac{\text{Time Saved}}{\text{Time Allowed}} \times \text{Time taken} \times \text{Wage rate}$$

$$= (5 \text{ hours} \times ₹\ 120) + \left(\frac{1 \text{ hour}}{6 \text{ hours}} \times 5 \text{ hours} \times ₹\ 120\right)$$

= ₹ 600 + ₹ 100 = ₹ 700

Effective hourly rate = ₹ 700/5 hours = ₹ 140 /hour

(ii) Let time taken = X

$$\therefore \text{ Effective hourly rate} = \frac{\text{Earnings under Halsay Scheme}}{\text{Time Taken}}$$

Or, Effective hourly rate under Halsey Incentive plan =

$$\frac{(\text{Time taken} \times \text{Rate}) + 50\% \text{ of Rate} \times (\text{Time allowed} - \text{Time taken})}{\text{TimeTaken}}$$

$$\text{Or, } ₹140 = \frac{(X \times ₹120) + 50\% \text{ of } ₹\ 120 \times (6 - X)}{X}$$

Or, 140X = 120X + 360 – 60X

Or, 80X = 360

$$\text{Or, } X = \frac{360}{80} = 4.5 \text{ hours}$$

Therefore, to earn effective hourly rate of ₹140 under Halsey Incentive Scheme worker has to complete the work in 4.5 hours.

ICAI · Old P3 · RTP Nov 2023 · question p. 1 · answer p. 10-11

323 RTP **Old P3 Jul 2021** Q2 old scheme

JBL Sisters operates a boutique which works for various fashion houses and retail stores. It has employed 26 workers and pays them on time rate basis. On an average an employee is allowed 8 hours for boutique work on a piece of garment. In the month of December 2020, two workers M and J were given 15 pieces and 21 pieces of garments respectively for boutique work. The following are the details of their work:

	M	J
Work assigned	15 pcs.	21 pcs.
Time taken	100 hours	140 hours

Workers are paid bonus as per Halsey System. The existing rate of wages is ₹ 60 per hour. As per the new wages agreement the workers will be paid ₹ 72 per hour w.e.f. 1st January 2021. At the end of the month December 2020, the accountant of the company has wrongly calculated wages to these two workers taking ₹ 72 per hour.

Required:

(i) CALCULATE the loss incurred due to incorrect rate selection.

(ii) CALCULATE the loss incurred due to incorrect rate selection, had Rowan scheme of bonus payment followed.

(iii) CALCULATE the loss/ savings if Rowan scheme of bonus payment had followed.

(iv) DISCUSS the suitability of Rowan scheme of bonus payment for JBL Sisters?

ICAI'S ANSWER

Workings Notes:

Calculation of Total hours saved:

	M	J
No. of garments assigned (Pieces.)	15	21
Hour allowed per piece (Hours)	8	8
Total hours allowed (Hours)	120	168
Hours Taken (Hours)	100	140
Hours Saved (Hours)	20	28

(i) Calculation of loss incurred due to incorrect rate selection:

(While calculating loss only excess rate per hour has been taken)

	M (₹)	J (₹)	Total (₹)
Basic Wages	1,200 (100 Hrs. × ₹12)	1,680 (140 Hrs. × ₹12)	2,880
Bonus (as per Halsey Scheme) (50% of Time Saved × Excess Rate)	120 (50% of 20 Hrs. × ₹12)	168 (50% of 28 Hrs. × ₹12)	288
Excess Wages Paid	1,320	1,848	3,168

(ii) Calculation of loss incurred due to incorrect rate selection had Rowan scheme of bonus payment followed:

	M (₹)	J (₹)	Total (₹)
Basic Wages	1,200 (100 Hrs. × ₹12)	1,680 (140 Hrs. × ₹12)	2,880
Bonus (as per Rowan Scheme) (Time Taken/Time Allowed × Time Saved × Excess Rate)	200 (100/120 × 20 × ₹12)	280 (140/168 × 28 × ₹12)	480
Excess Wages Paid	1,400	1,960	3,360

(iii) Calculation of amount that could have been saved if Rowan Scheme were followed

	M (₹)	J (₹)	Total (₹)
Wages paid under Halsey Scheme	1,320	1,848	3,168
Wages paid under Rowan Scheme	1,400	1,960	3,360
Difference (loss)	(80)	(112)	(192)

(iv) Rowan Scheme of incentive payment has the following benefits, which is suitable with the nature of business in which JBL Sisters operates:

(a) Under Rowan Scheme of bonus payment, workers cannot increase their earnings or bonus by merely increasing its work speed. Bonus under Rowan Scheme is maximum when the time taken by a worker on a job is half of the time allowed. As this fact is known to the workers, therefore, they work at such a speed which helps them to maintain the quality of output too.

(b) If the rate setting department commits any mistake in setting standards for time to be taken to complete the works, the loss incurred will be relatively low.

ICAI · Old P3 · RTP Jul 2021 · question p. 1-2 · answer p. 13-14

324 MTP **Sep 2026 S-II** MCQ 11 2 marks MCQ

Same question, different ICAI answer: #319 RTP May 2025

Mr. A works in a manufacturing company where he is paid bonus according to the Halsey 50% plan, besides the normal wages. The relevant data is as below:

Time Rate (per hour)	₹ 100
Time allowed	10 hours
Time taken	5 hours
Time saved	5 hours

Mr. A believes that his bonus under Halsey system is getting reduced by 50%, thus intending to shift towards Rowan Premium Plan.

You are required to CALCULATE the total earnings of Mr. A as per Halsey plan and Rowan Premium plan & ENUMERATE the reason for difference in both the earnings.

A Total earnings as per Halsey plan- ₹ 500 and as per Rowan Premium plan- ₹ 750. Earnings under Halsey Plan is lower than that of Rowan Premium plan as the bonus is getting reduced by 50%.

B Total earnings as per Halsey plan- ₹ 750 and as per Rowan Premium plan- ₹ 500. Earnings under Rowan Premium plan is lower than that of Halsey plan as the actual time taken is 50% of the time allowed.

C Total earnings as per Halsey plan- ₹ 750 and as per Rowan Premium plan- ₹ 750. When the actual time taken is 50% of the time allowed, the earnings under Halsey and Rowan Plans are equal.

D Total earnings as per Halsey plan- ₹ 250 and as per Rowan Premium plan- ₹ 250. Total earnings under both the Plans are equal as the time taken under both the plans are same.

ICAI'S ANSWER

(C) Calculation of total earnings:

As per Halsey 50% plan

= Time taken × Time rate + (50% of Time saved × Time rate)

= 5 hrs. × ₹ 100 + [1/2 × (5 hrs. × ₹ 100)]

= ₹ 500 + ₹ 250 = **₹ 750**

As per Rowan Premium plan

$$= \text{Time taken} \times \text{Rate per hour} + \left(\frac{\text{Time Saved}}{\text{Time Allowed}} \times \text{Time taken} \times \text{Rate per hour}\right)$$

$$= 5 \text{ hours} \times ₹\ 100 + \left[\left(\frac{5 \text{ Hours}}{10 \text{ Hours}}\right) \times 5 \text{ hours} \times ₹\ 100\right]$$

= ₹ 500 + ₹ 250 = **₹ 750**

When the actual time taken is 50% of the time allowed, the earnings under Halsey and Rowan Plans are equal.

ICAI · P4 · MTP Sep 2026 · Series II · question p. 6 · answer p. 5

325 MTP **May 2026 S-II** Case Scenario 1 · MCQ 1 to MCQ 5 10 marks

CASE SCENARIO 1

In a factory, both direct and indirect employees are entitled to production bonus in accordance with a group incentive scheme, the outline of which is as follows:

(a) For any production in excess of the standard production which is fixed at 22,400 tonnes per month (of 28 days), a general incentive of ₹ 2,000 per tonne is to be paid in aggregate. The amount payable to each separate group of employees is determined based on an assumed percentage of such excess production being contributed by them which is as follows:

Employee Group	% Contribution in excess production over standard
Direct Employee	60%
Inspection Staff	20%
Maintenance Staff	15%
Supervisory Staff	5%

(b) Moreover, if the excess production (without considering rejections) is more than 15% above the standard, the direct employees also get a special bonus @ ₹ 400 per tonne for all production in excess

of 115% of standard production.

(c) Inspection staff are penalised @ ₹ 2,500 per tonne for rejection by customers in excess of 3% of production.

(d) Maintenance staffs are also penalised @ ₹ 2,500 per hour for machine breakdown.

(e) Actual working days are 25 days in a month.

(f) Actual production achieved during the month is 27,000 tonnes, of which the customers have rejected 1,000 tonnes.

(g) Machine breakdown (in 25 days) : 75 hours

On the basis of above Case Scenario, you are required to answer the following MCQs 1 to 5:

MCQ 1 2 marks

What is the percentage of excess production to standard production?

A 20%

B 15%

C 35%

D 45%

MCQ 2 2 marks

What will be the general incentive to Direct Employees, Inspection Staff, Maintenance Staff & Supervisory Staff?

A ₹ 55,20,000, ₹ 18,40,000, ₹ 13,80,000 & ₹ 4,60,000 respectively.

B ₹ 84,00,000, ₹ 28,00,000, ₹ 21,00,000 & ₹ 7,00,000 respectively.

C ₹ 55,20,000, ₹ 18,40,000, ₹ 21,00,000 & ₹ 7,00,000 respectively.

D ₹ 84,00,000, ₹ 28,00,000, ₹ 13,80,000 & ₹ 4,60,000 respectively.

MCQ 3 2 marks

What will be the special bonus to Direct Employees?

A ₹ 28,00,000

B ₹ 12,00,000

C ₹ 16,00,000

D ₹ 15,00,000

MCQ 4 2 marks

What will be the amount of penalty that is to be imposed on the Inspection Staff and the Maintenance Staff?

A ₹ 4,75,000 & ₹ 1,87,500 respectively

B ₹ 1,87,500 & ₹ 4,75,000 respectively

C ₹ 7,45,000 & ₹ 1,87,500 respectively

D ₹ 4,75,000 & ₹ 1,78,500 respectively

MCQ 5 2 marks

What will be the aggregate production bonus (net of penalty) earned by Direct Employees, Inspection Staff, Maintenance Staff & Supervisory Staff?

A ₹ 1,40,00,000

B ₹ 1,33,37,500

C ₹ 1,24,00,000

D ₹ 1,49,37,500

ICAI'S ANSWERS

MCQ 1

(C) 35%

Working Notes

Standard production for 28 days = 22,400 tonnes

Standard production Per day = 22,400 ÷ 28 = 800 tonnes

For 25 days:

Standard = 800 × 25 = 20,000 tonnes

Actual production = 27,000 tonnes

Excess production = 27,000 − 20,000 = 7,000 tonnes

% Excess Production = $\frac{7,000}{20,000} \times 100 = 35\%$

MCQ 2

(B) ₹ 84,00,000, ₹ 28,00,000, ₹ 21,00,000 & ₹ 7,00,000 respectively.

General Incentive

Total incentive = 7,000 × 2,000 = ₹ 1,40,00,000

Group	%	Amount (₹)
Direct	60%	84,00,000
Inspection	20%	28,00,000
Maintenance	15%	21,00,000
Supervisory	5%	7,00,000

MCQ 3

(C) ₹ 16,00,000

Special Bonus (Direct Employees)

115% of standard = 20,000 × 115% = 23,000 tonnes

Eligible production = 27,000 − 23,000 = 4,000 tonnes

Bonus = 4,000 × 400 = ₹ 16,00,000

MCQ 4

(A) ₹ 4,75,000 & ₹ 1,87,500 respectively

Penalties

Inspection Staff:

Allowed rejection = 3% × 27,000 = 810 tonnes

Excess rejection = 1,000 − 810 = 190 tonnes

Penalty = 190 × 2,500 = ₹4,75,000

Maintenance Staff:

Penalty = 75 × 2,500 = ₹ 1,87,500

MCQ 5

(D) ₹ 1,49,37,500

Net Aggregate Bonus

Total Incentives = General (1,40,00,000) + Special (16,00,000)

= ₹ 1,56,00,000

Total Penalties = 4,75,000 + 1,87,500 = ₹ 6,62,500

Net Bonus = 1,56,00,000 − 6,62,500 = ₹ 1,49,37,500

ICAI · P4 · MTP May 2026 · Series II · question p. 1-2 · answer p. 1, 1-2, 2

326 MTP **May 2026 S-II** Q3(a) 6 marks

ZED Limited is working by employing 50 skilled workers, it is considering the introduction of incentive scheme-either Halsey scheme (with 50% bonus) or Rowan scheme of wage payment for increasing the labour productivity to cope up the increasing demand for the product by 40%. It is believed that proposed incentive scheme could bring about an average 20% increase over the present earnings of the workers; it could act as sufficient incentive for them to produce more.

Because of assurance, the increase in productivity has been observed as revealed by the figures for the month of April, 2025.

Hourly rate of wages (guaranteed)	₹ 30
Average time for producing one unit by one worker at the previous performance (This may be taken as time allowed)	1.975 hours
Number of working days in the month	24
Number of working hours per day of each worker	8
Actual production during the month	6,120 units

Required:

(i) CALCULATE the effective rate of earnings under the Halsey scheme and the Rowan scheme.

(ii) CALCULATE the savings to the ZED Limited in terms of direct labour cost per piece.

(iii) ADVISE ZED Limited about the selection of the scheme to fulfill their assurance.

ICAI'S ANSWER

Working notes:

1. Computation of time saved (in hours) per month:

(Standard production time for 6,120 units) – (Actual time taken by the workers)

= (6,120 units × 1.975 hours) – (24 days × 8 hours per day × 50 skilled workers)

= (12,087 hours – 9,600 hours)

= 2,487 hours

2. Computation of bonus for time saved under Halsey and Rowan schemes:

Time saved = 2,487 hours

(Refer to working note 1)

Wage rate per hour = ₹ 30

Bonus under Halsey Scheme = ½ × 2,487 hours × ₹ 30

(With 50% bonus) = ₹ 37,305

Bonus under Rowan Scheme = $\frac{\text{Time saved}}{\text{Time allowed}}$ × Time taken × Rate per hour

= $\frac{2,487 \text{ hours}}{12,087\text{hours}}$ × 9,600 hours × ₹ 30

= ₹ 59,258.38

(i) Computation of effective rate of earnings under the Halsey and Rowan scheme:

Total earnings (under Halsey scheme) (Refer to working note 2)

= Time wages + Bonus

= (24 days × 8 hours + 50 skilled workers × ₹ 30) + ₹ 37,305

= ₹ 2,88,000 + ₹ 37,305 = ₹ 3,25,305

Total earnings (under Rowan scheme) (Refer to working note 2)

= Time wages + Bonus

= ₹ 2,88,000 + ₹ 59,258.38

= ₹ 3,47,258.38

Effective rate of earnings per hour (under Halsey Plan)

= $\frac{₹3,25,305}{9,600\text{hours}}$ = ₹ 33.89

Effective rate of earnings per hour (under Rowan Plan)

= $\frac{₹3,47,258.38}{9,600\text{hours}}$ = ₹ 36.17

(ii) Savings to the ZED Ltd., in terms of direct labour cost per piece:

	(₹)
Direct labour cost (per unit) under time wages system (1.975 hours per unit × ₹ 30)	59.25
Direct labour cost (per unit) under Halsey Plan $\left(\frac{₹3,25,305}{6,120\text{units}}\right)$	53.15
Direct labour cost (per unit) under Rowan Plan $\left(\frac{₹3,47,258.38}{6,120\text{units}}\right)$	56.74

Saving of direct labour cost under:

Halsey Plan (₹ 59.25 – ₹ 53.15) ₹ 6.10

Rowan Plan (₹ 59.25 – ₹ 56.74) ₹ 2.51

(iii) Advise to ZED Ltd.: (about the selection of the scheme to fulfill assurance)

Halsey scheme brings more savings to the management of ZED Ltd., over the present earnings of ₹ 2,88,000 but the other scheme i.e. Rowan scheme fulfils the promise of 20% increase over the present earnings of ₹ 2,88,000 by paying 20.58% in the form of bonus. Hence Rowan Plan may be adopted.

ICAI · P4 · MTP May 2026 · Series II · question p. 11-12 · answer p. 11-12

327 MTP Jan 2026 S-I MCQ 15 2 marks MCQ

The relevant data is as below:

Time Rate (per hour)	₹ 60
Time allowed	8 hours
Time taken	6 hours
Time saved	2 hours

CALCULATE the earnings of a worker under Halsey System

A ₹ 360

B ₹ 420

C ₹ 500

D ₹ 600

ICAI'S ANSWER

(B) ₹ 420

Calculation of total earnings:

= Time taken × Time rate + 50% (Time Allowed – Time Taken) × Time rate

= 6 hrs. × ₹ 60 + 1/2 × (2 hrs. × ₹ 60) or ₹ 360 + ₹ 60 = ₹ 420

ICAI · P4 · MTP Jan 2026 · Series I · question p. 6 · answer p. 4

328 MTP Jan 2026 S-II Q2(b) 6 marks

Rivaan Industries manufactures and sells wooden study tables. The demand for these tables has been steadily increasing due to the company's strong distribution network and reputation for durable products. Despite the rising market potential, the company continues to rely on its long-established wage practices. At present, Rivaan Industries pays wages under a piece–rate system, with a standard piece rate of ₹ 18 per table produced.

The company policy for payment under the piece-rate system is as follows:

- For efficiency less than 100%: Workers are paid 80% of the standard piece rate
- For efficiency equal to or more than 100%: Workers are paid 125% of the standard piece rate

The standard output per hour is 3 tables.

Each worker works 9 hours per day from Monday to Friday and 4 hours on Saturday.

A total of 132 workers are currently eligible for this wage plan.

However, the cutting and finishing machines used in production were purchased many years ago and now limit workers' efficiency. As a result, workers have demanded a shift to a time-rate wage system at ₹ 65 per hour, along with incentives under the Halsey Premium Plan (50% sharing of time saved).

To evaluate the financial impact, the management has prepared the following estimated production levels for January 2026 under three different scenarios:

Production Estimates (January 2026)

Scenario	Worst Case	Optimal Case	Best Case
Production (units)	36,900	73,800	1,10,700

In addition, the company is assessing a potential Technology Upgrade Case involving partial automation, which may influence both output levels and worker productivity.

Required:

(a) COMPUTE the Total wages, and Average wages per worker per month, for all three scenarios (Worst, Optimal, Best Case) under each of the following wage systems:

(i) Current piece–rate wage system (based on differential piece rates)

(ii) Workers' proposed system: Time-rate wages @ ₹ 65 per hour + Halsey Premium (50%)

(b) Mr. R, one of the workers, produced 900 tables in December 2025. If he produces the same number of tables in January 2026, COMPUTE:

- His earnings under the current piece-rate system
- His earnings under the proposed time-rate + Halsey incentive system
- The difference in his earnings

Based on your computation, SHOULD Mr. R support the workers' demand?

(Assume 4 working weeks in a month.)

ICAI'S ANSWER

(a) Calculation of Total wages and average wages per worker per month.

(i) When Current system of wages and incentive payment system is followed:

Standard hours per Month = (9 hrs x 4 days + 4 hrs) x 4 = 196 hours

Standard production per worker = 196 hours × 3 units = 588 units

Standard production for 132 workers = 588 × 132 = 77,616 units

Particulars	Worst Case	Optimal Case	Best Case
I. Standard Production (units) (588 units × 132 workers)	77,616	77,616	77,616
II. Actual Production (units)	36,900	73,800	1,10,700
III. Efficiency (%) (II ÷ I × 100)	47.54%	95.08%	142.63%
IV. Applicable Piece Rate	₹ 14.40 (₹ 18 x 0.8)	₹ 14.40 (₹ 18 x 0.8)	₹ 22.50 (₹ 18 x 1.25)
V. Total Wages (II × IV)	₹ 5,31,360	₹ 10,62,720	₹ 24,90,750
VI. Average Wage per Worker (V ÷ 132)	₹ 4,025.45	₹ 8,050.91	₹ 18,869.32

(ii) When workers' demand for time rate wages and Halsey premium plan is accepted:

Particulars	Worst Case	Optimal Case	Best Case
I. Actual Production (units)	36,900	73,800	1,10,700
II. Standard Units per hour	3	3	3
III. Standard Hours Required (I ÷ II)	12,300	24,600	36,900
IV. Actual working hours (132 workers × 196 hrs)	25,872	25,872	25,872
V. Time Saved (III − IV)	—	—	11,028
VI. Time Wages (IV × ₹ 65)	₹ 16,81,680	₹ 16,81,680	₹ 16,81,680
VII. Halsey Bonus (50% × Time Saved × 65)	—	—	₹ 3,58,410
VIII. Total Wages (VI + VII)	₹ 16,81,680	₹ 16,81,680	₹ 20,40,090
IX. Average Wage per Worker (VIII ÷ 132)	₹ 12,740	₹ 12,740	₹ 15,455.23

(b) Calculation of gain or loss in the current monthly income of Mr. R:

1. Under Current Piece-Rate System

Standard production = 588 units

Production in December = 900 units

Efficiency = 900 ÷ 588 × 100 = 153.06%

Rate = ₹ 22.50

Total wages = 900 × 22.50 = ₹ 20,250

2. Under Proposed Time Rate + Halsey Plan

Standard hours = 900 units ÷ 3 units = 300 hours

Actual hours available = 196 hours

Time saved = 300 – 196 = 104 hrs , so bonus = 0

Time wages = 196 × ₹ 65 = ₹ 12,740

Halsey Bonus (50% × Time Saved × 65) = 104 hrs x 50% x 65 = ₹3,380

Total expected wages = ₹ 16,120

3. Loss To Mr. R

Loss = 20,250 – 16,120 = ₹ 4,130

Supporting the demand of colleague workers will cost ₹ 4,130 in the next month to Mr. R.

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329 MTP **Sep 2025 S-II** Q1(b) 5 marks

The assembly department of a manufacturing company employs 50 direct workers. Each worker earns a wage of ₹500 per week for a standard 40-hour workweek. When required, the workers may perform overtime up to 10 hours per week per worker, with overtime paid at time rate plus 50% premium.

Currently, the workers produce 5 units per hour on average, which is considered the standard output rate. The company is planning to introduce an incentive scheme expecting this rate to increase to 7 units per hour. Under the new scheme, workers will still be permitted to work overtime if needed, but no incentive will be paid on overtime hours.

The management is evaluating the introduction of either the Halsey Plan or the Rowan Plan to improve productivity. The company has a target weekly output of 17,500 units. The selling price per unit is ₹15, while the direct material cost is ₹10 per unit. The variable overheads are ₹0.60 per direct labour hour, and fixed overheads are ₹12,000 per week.

PREPARE a comparative statement showing the effect on the company's weekly profit under the following alternatives:

(a) Continuing with the existing wage system (no incentive)

(b) Introducing the Halsey Plan

(c) Introducing the Rowan Plan

ICAI'S ANSWER

Working Notes:

1. Total available hours per week (normal)
   = 50 workers x 40 hours = 2,000 hours
2. Total standard hours required to produce 17,500 units at 5 units/hr
   = 17,500/5 = 3,500 hours
3. Total hours needed after bonus scheme at 7 units/hr
   = 17,500 ÷ 7 = 2,500 hours
4. Time saved = 3,500 – 2,500 = 1,000 hours
5. Wage rate per hour = ₹ 500 ÷ 40 = ₹ 12.50/hour
6. Overtime limit = 10 hours/worker/week → max overtime = 50 x 10 = 500 hours
7. **Bonus**:
   (i) Halsey Scheme = $\frac{1}{2}$ x Time saved x Wage rate per hour
   = $\frac{1}{2}$ x 1,000 x ₹ 12.50 = ₹6,250

(ii) Rowan Scheme = $\frac{\text{Time saved}}{\text{Time allowed}}$ x Time taken x Wage rate per hour

= $\frac{1{,}000 \text{ hours}}{3{,}500 \text{ hours}}$ x 2,500 hours x ₹ 12.50 = ₹ 8,929

8. **Overtime premium (only in present system)**:
   - Overtime hours = 3,500 – 2,000 = 1,500 hrs
   - Limited to 500 hrs (as only 10 hrs per worker allowed), rest must be absorbed or not worked

→ Overtime premium = 500 hrs x ₹ 6.25 (half rate) = ₹ 3,125

Statement Showing the Weekly Profit Under Present, Halsey, and Rowan Schemes

Particulars	Present (₹)	Halsey (₹)	Rowan (₹)
Sales Revenue (17,500 x ₹15)	2,62,500	2,62,500	2,62,500
***Less:* Costs**			
- Direct Material (17,500 x ₹ 10)	1,75,000	1,75,000	1,75,000
- Direct Wages	43,750 (3,500 x ₹ 12.50)	31,250 (2,500 x ₹ 12.50)	31,250 (2,500 x ₹ 12.50)
- Overtime Premium (500 x ₹ 6.25)	3,125	–	–
- Bonus	–	6,250	8,929
- Variable Overheads	2,100 (3,500 x ₹ 0.60)	1,500 (2,500 x ₹ 0.60)	1,500 (2,500 x ₹ 0.60)
- Fixed Overheads	12,000	12,000	12,000
Total Cost (B)	**2,35,975**	**2,26,000**	**2,28,679**
Profit (A – B)	**26,525**	**36,500**	**33,821**

ICAI · P4 · MTP Sep 2025 · Series II · question p. 9 · answer p. 9-10

330 MTP **May 2025 S-II** Q1(b) 5 marks

A company has 120 direct labourers. Each labourer is paid ₹ 800 per 40 hour week. Owing to customer demands and timelines, Overtime is resorted to a maximum of 30 hours per week compensated at the weekly time rate with a premium of 50% over the regular wage. The current output works out to 6 units per man hour which is the standard output. The introduction of an incentive scheme could enhance productivity to 8 units per man hour. However, the incentive scheme if introduced would not have an overtime policy in place for extra hours worked. Budgeted weekly production is 38,400 units. The selling price is ₹ 22 per unit and direct costs other than labour is ₹ 16 per unit. The variable overheads is ₹ 1 per labour hour and Fixed overhead is ₹ 18,000 per week.

PREPARE a statement to demonstrate the effect of shifting from an overtime policy to:

(i) Halsey Incentive Scheme

(ii) Rowan Incentive Scheme

ICAI'S ANSWER

Projected weekly output - 38,400 units

No of units per man hour - 6 units

No of Manhours needed - 6,400 hours

No of hours available - 4,800 hours

Overtime needed - 1,600 hours

Therefore, total wages during overtime:

Normal Wages (6,400 x 20) - 1,28,000

Overtime premium (1,600 x 10) - 16,000

Total Wages - 1,44,000

Upon introduction of incentive scheme

No of units per man hour - 8 units

No of manhours needed - 4,800 hours

Time saved (OT saved) - 1,600 hours

Halsey Scheme Incentive

= 50% of time saved × Time rate

= 1,600 hours x 20 per hour x 50% = 16,000

Total Labour Cost = 96,000 + 16,000 = 1,12,000

Rowan Scheme Incentive

$$= \frac{\text{Time taken}}{\text{Time allowed}} \times \text{Time saved} \times \text{hourly rate}$$

= (1,600/6,400) x 4,800 x 20 = 24,000

Total Labour Cost = 96,000 + 24,000 = 1,20,000

Statement of profitability

Particulars	Overtime	Halsey	Rowan
Sales @ ₹ 22	8,44,800	8,44,800	8,44,800
Less: Direct costs	6,14,400	6,14,400	6,14,400
Less: Labour	1,44,000	1,12,000	1,20,000
Less: Variable Overhead	6,400	4,800	4,800
Less: Fixed Overhead	18,000	18,000	18,000
Profit	62,000	95,600	87,600

ICAI · P4 · MTP May 2025 · Series II · question p. 8 · answer p. 7-8

331 MTP Jan 2025 S-I Q1(c) 4 marks

The management of a company wants to formulate an incentive plan for the workers with a view to increase productivity. The following particulars have been extracted from the books of company:

Piece Wage rate	₹ 10
Weekly working hours	40
Hourly wages rate	₹ 40 (guaranteed)
Standard/normal time per unit	15 minutes.

Actual output for a week:

Worker A:	176 pieces
Worker B:	140 pieces

Under Halsey scheme, worker gets a bonus equal to 50% of Wages of time saved.

CALCULATE earning of workers under Halsey's and Rowan's premium scheme.

ICAI'S ANSWER

Calculation of earnings for workers under different incentive plans:

Halsey's Premium Plan:

	Worker – A	Worker – B
Actual time taken	40 hours	40 hours
Standard time for actual Production	44 hours (176 Pcs×15 Min./60 Min.)	35 hours (140 Pcs×15 Min./60 Min.)
Minimum Wages	₹ 1,600 (40 hours x ₹ 40)	₹ 1,600 (40 hours x ₹ 40)
Bonus	₹ 80 {50% (44-40) x ₹40}	No bonus
Earning	₹ 1,680	₹ 1,600
Rowan's Premium Plan:		
Minimum Wages (as above)	₹ 1,600	₹ 1,600
Bonus	= ₹ 145.45 (4 hours/44 hours×40hours×₹ 40)	No bonus
Earning	₹ 1,745.45	₹ 1,600

ICAI · P4 · MTP Jan 2025 · Series I · question p. 9 · answer p. 8

332 MTP Jan 2025 S-II MCQ 11 2 marks MCQ

Mr. Ben is paid higher wages than Mr. Akon. Though their normal wage rate is same, Mr. Ben gets higher payment as under Halsey system than that to Mr. Akon as under Rowan System.

The total time allowed to make the same product is 75 hours, however, Mr. Ben takes 60 hours while Mr. Akon takes 45 hours.

The production of the product also involve other costs that are not traced directly to the product like salary to quality assurance manager, factory rent, supplies, salary to production supervisor, electricity consumed, etc. which comes to ₹ 2,26,800 leading to factory overhead rate being ₹ 120 per man-hour actually worked.

The total factory cost for the product produced by Mr. Akon comes to ₹ 1,25,640 and by Mr. Ben comes to ₹ 1,29,600.

From the information given above, COMPUTE the normal wage rate along with the cost of material.

A Normal wage rate- ₹ 63 per hour and cost of material- ₹ 1,20,240

B Normal wage rate- ₹ 67.5 per hour and cost of material- ₹ 1,22,400

C Normal wage rate- ₹ 480 per hour and cost of material- ₹ 90,000

D Normal wage rate- ₹ 450 per hour and cost of material- ₹ 87,840

ICAI'S ANSWER

(C) Let X be the cost of material and Y be the normal rate of wages per hour.

Factory Cost of Mr. Akon (Rowan System) = $X + 45Y + \frac{45}{75}$ x (75 - 45) Y + (45 x ₹ 120)

₹ 1,25,640 = X + 63Y + ₹ 5,400

X + 63Y = ₹ 1,20,240 ... (i)

Factory Cost of Mr. Ben (Halsey System) = X + 60Y + 50% (75 - 60) Y + (60 x ₹ 120)

₹ 1,29,600 = X + 67.5Y + ₹ 7,200

X + 67.5Y = ₹ 1,22,400 ... (ii)

From subtracting (i) from (ii), we get,

4.5Y = ₹ 2,160

Y = ₹ 480 per hour

Or, **normal wage rate = ₹ 480 per hour**

Therefore, X = ₹ 1,20,240 - 63Y

X = ₹ 1,20,240 - (63 x ₹ 480)

X = ₹ 90,000

Or, **cost of material = ₹ 90,000**

ICAI · P4 · MTP Jan 2025 · Series II · question p. 5 · answer p. 5

333 MTP Jan 2025 S-II Q1(c) 4 marks

Shivi is working by employing 10 skilled workers. It is considering the introduction of some incentive scheme – either Halsey scheme (with 50% bonus) or Rowan scheme of wage payment for increasing the labour productivity to cope with the increased demand for the product by 25%. She feels that, if the

proposed incentive scheme could bring about an average 20% increase over the present earnings of the workers, it would act as sufficient incentive for them to produce more and she has accordingly given this assurance to the workers.

As a result of this assurance, the increase in productivity has been observed as revealed by the following figures for the current month:

Hourly rate of wages (guaranteed)	₹ 2.00
Average time for producing 1 piece by one worker at the previous performance (this may be taken as time allowed)	2 hours
Number of working days in the month	25
Number of working hours per day for each worker	8
Actual production during the month	1,250 units

Required:

(1) CALCULATE effective rate of earnings per hour under Halsey scheme and Rowan scheme.

(2) CALCULATE the savings of Navya in terms of direct labour cost per piece under the above schemes.

(3) ADVISE Navya about the selection of the scheme to fulfill her assurance

ICAI'S ANSWER

Production during the month	1,250 units
Time allowed for 1,250 units @ 2 hours per unit (1,250 × 2 hours)	2,500 hours
Actual time taken 25 days x 8 hours x 10 workers	2,000 hours
Time saved	500 hours

Labour cost per piece under time rate scheme: 2 hours x ₹ 2 = ₹ 4

Calculation of effective hourly rate under:

Halsey Scheme:

	(₹)
Basic wages of 10 workers: 2,000 hours @ ₹ 2 per hour	4,000
Bonus 50% x (500 hours x ₹ 2)	500
Total wages for 2,000 hours	4,500

Effective hourly rate of earning = $\frac{₹4,500}{2,000 \text{ hours}}$ = ₹ 2.25

Labour cost per piece = $\frac{₹4,500}{1,250 \text{ units}}$ = ₹ 3.60

Saving in terms of direct labour cost per piece (₹ 4.00 – ₹ 3.60) = ₹ 0.40

Rowan Scheme:

	(₹)
Basic wages (as calculated under Halsey scheme)	4,000
Bonus: 500 hours × 2,000 hours/2,500 hours × ₹ 2	800
Total wages for 2,000 hours	4,800

Effective hourly rate of earnings $\frac{₹4,800}{2,000 \text{ hours}}$ = ₹ 2.40

Labour cost per piece $\frac{₹4,800}{1,250 \text{ units}}$ = ₹ 3.84

Saving in terms of direct labour cost per piece (₹ 4.00 – ₹ 3.84) = ₹ 0.16

Advise: Shivi should introduce Halsey incentive scheme, as it gives more saving than the Rowan incentive scheme.

ICAI · P4 · MTP Jan 2025 · Series II · question p. 8 · answer p. 8-9

334 MTP **Sep 2024 S-I** Q1(c) 4 marks

A job can be executed either through workman A or B. A takes 32 hours to complete the job while B finishes it in 30 hours. The standard time to finish the job is 40 hours.

The hourly wage rate is same for both the workers. In addition workman A is entitled to receive bonus according to Halsey plan (50%) sharing while B is paid bonus as per Rowan plan. The works overheads are absorbed on the job at ₹ 7.50 per labour hour worked. The factory cost of the job comes to ₹ 2,200 irrespective of the workman engaged.

FIND out the hourly wage rate and cost of raw materials input. Also SHOW cost against each element of cost included in factory cost.

ICAI'S ANSWER

Calculation of:

(i) Time saved and wages:

Workmen	A	B
Standard time (hrs.)	40	40
Actual time taken (hrs.)	32	30
Time saved (hrs.)	08	10
Wages paid @ ₹ x per hr. (₹)	32x	30x

(ii) Bonus Plan:

	Halsey	Rowan
Time saved (hrs.)	8	10
Bonus (₹)	4x	7.5x
	[8 hrs × ₹ x/2]	[10 hrs/40 hrs × 30hrs × ₹ x]

(iii) Total wages:

Workman A: 32x + 4x = ₹ 36x

Workman B: 30x + 7.5x = ₹ 37.5x

Statement of factory cost of the job

Workmen	A	B
	₹	₹
Material cost (assumed)	y	y
Wages (shown above)	36x	37.5x
Works overhead	240	225
Factory cost (given)	2,200	2,200

The above relations can be written as follows:

36x + y + 240 = 2,200 (i)

37.5x+ y+ 225 = 2,200 (ii)

Subtracting (i) from (ii) we get

1.5x – 15 = 0

or 1.5 x = 15

or x = ₹ 10 per hour

On substituting the value of x in (i) we get y = ₹ 1,600

Hence the wage rate per hour is ₹ 10 and the cost of raw material is ₹ 1,600 on the job.

ICAI · P4 · MTP Sep 2024 · Series I · question p. 7 · answer p. 7

335 MTP **Sep 2024 S-I** Q6(b) 5 marks

EXPLAIN the efficiency rating procedures of the employees.

ICAI'S ANSWER

Efficiency is usually related with performance and may be computed by comparing the time taken with the standard time allotted to perform the given job/task.

If the time taken by a worker on a job equals or less than the standard time, then he is rated efficient.

In case he takes more time than the standard time he is rated as inefficient.

$$\text{Efficiency in \%} = \frac{\text{Time allowed as per standard}}{\text{Time Taken}} \times 100$$

For efficiency rating of employees the following procedures may be followed:

1. **Determining standard time/performance standards:** The first step is to determine the standard time taken by a worker for performing a particular job/task. *The standard time can be determined by using Time & Motion study or Work study techniques*. While determining the standard time for a job/task a heterogeneous group of workers is taken and contingency allowances are added for determining standard time.
2. **Measuring Actual Performance of workers:** For computing efficiency rating it is necessary to develop a procedure for recording the actual performance of workers. The system developed should record the output of each worker along with the time taken by him.
3. **Computation of efficiency rating:** The efficiency rating of each worker can be computed by using the above mentioned Formula.

ICAI · P4 · MTP Sep 2024 · Series I · question p. 12 · answer p. 17

336 MTP **Sep 2024 S-II** Q1(a) 5 marks

A skilled worker is paid a guaranteed wage rate of ₹ 150.00 per hour. The standard time allowed for a job is 50 hours. He gets an effective hourly rate of wages of ₹ 180.00 under Rowan Incentive Plan due to saving in time. For the same saving in time, CALCULATE the hourly rate of wages he will get, if he is placed under Halsey Premium Scheme (50%).

ICAI'S ANSWER

Increase in hourly rate of wages under Rowan Plan is ₹ 30 i.e. (₹180 – ₹ 150)

$\frac{\text{Time Saved}}{\text{Time Allowed}} \times$ ₹ 150 = ₹30 (Please refer Working Note)

Or, $\frac{\text{Time Saved}}{50\text{hours}} \times$ ₹ 150 = ₹ 30

Or, Time saved = $\frac{1{,}500}{150}$ = 10 hours

Therefore, Time Taken is 40 hours i.e. (50 hours – 10 hours)

Effective Hourly Rate under Halsey System:

Time saved = 10 hours

Bonus @ 50% = 10 hours × 50% × ₹ 150 = Rs 750

Total Wages = (₹150 × 40 hours + ₹ 750) = Rs 6,750

Effective Hourly Rate = ₹ 6,750 ÷ 40 hours = ₹ 168.75

Working Note:

Effective hourly rate

$$= \frac{(\text{Time Taken} \times \text{Rate per hour}) + \frac{\text{Time Taken}}{\text{Time Allowed}} \times \text{Time Saved} \times \text{Rate per hour}}{\text{Time Taken}}$$

$$\text{Or, ₹ 180} = \frac{\text{Time Taken} \times \text{Rate per hour}}{\text{Time Taken}} + \frac{\frac{\text{Time Taken}}{\text{Time Allowed}} \times \text{Time Saved} \times \text{Rate per hour}}{\text{Time Taken}}$$

$$\text{Or, ₹ 180 - } \frac{\text{Time Taken} \times \text{Rate per hour}}{\text{Time Taken}} = \frac{\text{Time Taken}}{\text{Time Allowed}} \times \text{Time Saved} \times \text{Rate per hour} \times \frac{1}{\text{Time Taken}}$$

$$\text{Or, ₹ 180 – ₹ 150} = \frac{\text{Time Saved}}{\text{Time Allowed}} \times \text{₹ 150}$$

ICAI · P4 · MTP Sep 2024 · Series II · question p. 5-6 · answer p. 5-6

337 MTP **May 2024 S-I** Q3(b) 10 marks

The following particulars have been compiled in respect of three workers, which are under consideration of the management.

	I	II	III
Actual hours worked	380	100	540
Hourly rate of wages (in ₹)	40	50	60
Productions in units:			
- Product X	210	-	600
- Product Y	360	-	1350
- Product Z	460	250	-
Standard time allowed per unit of each product is:			
	X	**Y**	**Z**
Minutes	15	20	30

For the purpose of piece rate, each minute is valued at ₹ 1/-

You are required to calculate the wages of each worker under:

(i) Guaranteed hourly rate basis

(ii) Piece work earning basis, but guaranteed at 75% of basic pay (Guaranteed hourly rate if his earnings are less than 50% of basic pay.)

(iii) Premium bonus basis where the worker received bonus based on Rowan scheme.

ICAI'S ANSWER

(i) Computation of wages of each worker under guaranteed hourly rate basis

Worker	Actual hours worked (Hours)	Hourly wage rate (₹)	Wages (₹)
I	380	40	15,200
II	100	50	5,000
III	540	60	32,400

(ii) Computation of Wages of each worker under piece work earning basis

Product	Piece rate per unit	Worker-I		Worker-II		Worker-III	
	(₹)	Units	Wages (₹)	Units	Wages (₹)	Units	Wages (₹)
X	15	210	3,150	-	-	600	9,000
Y	20	360	7,200	-	-	1,350	27,000
Z	30	460	13,800	250	7,500	-	-
Total			24,150		7,500		36,000

Since each worker's earnings are more than 50% of basic pay. Therefore, worker-I, II and III will be paid the wages as computed i.e. ₹24,150, ₹7,500 and ₹36,000 respectively.

Working Notes:

1. Piece rate per unit

Product	Standard time per unit in minute	Piece rate each minute (₹)	Piece rate per unit (₹)
X	15	1	15
Y	20	1	20
Z	30	1	30

2. Time allowed to each worker

Worker	Product-X	Product-Y	Product-Z	Total Time (Hours)
I	210 units × 15 = 3,150	360 units × 20 = 7,200	460 units × 30 = 13,800	24,150/60 = 402.50
II	-	-	250 units × 30 = 7,500	7,500/60 = 125
III	600 units × 15 = 9,000	1, 350 units × 20 = 27,000	-	36,000/60 = 600

(iii) Computation of wages of each worker under Premium bonus basis (where each worker receives bonus based on Rowan Scheme)

Worker	Time Allowed (Hr.)	Time Taken (Hr.)	Time saved (Hr.)	Wage Rate per hour (₹)	Earnings (₹)	Bonus (₹)*	Total Earning (₹)
I	402.5	380	22.5	40	15,200	850	16,050
II	125	100	25	50	5,000	1,000	6,000
III	600	540	60	60	32,400	3,240	35,640

$$* \frac{\text{Time Taken}}{\text{Time Allowed}} \times \text{Time Saved} \times \text{Wage Rate}$$

$$\text{Worker-I} = \frac{380}{402.5} \times 22.5 \times 40 = 850 \; ; \text{Worker-II} = \frac{100}{125} \times 25 \times 50 = 1{,}000$$

$$\text{Worker-III} = \frac{540}{600} \times 60 \times 60 = 3{,}240$$

ICAI · P4 · MTP May 2024 · Series I · question p. 8 · answer p. 8-9

338 MTP May 2024 S-II MCQ 3 2 marks MCQ

If the amount of wages under Halsey plan is ₹ 420, total time allowed is 8 hours and the guaranteed time rate is ₹ 60 per hour. What is the total time saved by the worker?

A 2 hours

B 3 hours

C 6 hours

D 3.5 hours

ICAI'S ANSWER

Answer key: **(A)**

A

(TT x 60) + [0.50 x (8-TT) x 60] = 420 TT* = 6 hours

Time saved = 8-6 = 2

* TT=Total Time Taken

ICAI · P4 · MTP May 2024 · Series II · question p. 4 · answer p. 3

339 MTP May 2024 S-II Q1(c) 4 marks

A skilled worker in Shanu Ltd. is paid a guaranteed wage rate of ₹ 30 per hour. The standard time per unit for a particular product is 4 hours. Sam, a machine-man, has been paid wages under the Rowan Incentive Plan and he had earned an effective hourly rate of ₹ 37.50 on the manufacture of that particular product.

WHAT could have been his total earnings and effective hourly rate, had he been put on Halsey Incentive Scheme (50%)?

ICAI'S ANSWER

Let T hours be the total time worked in hours by the skilled worker (machine-man Sam); ₹ 30/- is the rate per hour; standard time is 4 hours per unit and effective hourly earning rate is ₹ 37.50 then

Earning = Hours worked × Rate per hour

$$+ \frac{\text{Time saved}}{\text{Time allowed}} \times \text{Time taken} \times \text{Rate per hour}$$

(Under Rowan incentive plan)

$$₹\ 37.5\ T = (T \times ₹\ 30) + \frac{(4 - T)}{4} \times T \times ₹\ 30$$

₹ 37.5 = ₹ 30 + (4 – T) × ₹ 7.5

Or ₹ 7.5 T = ₹ 22.5

Or T= 3 hours

Total earnings and effective hourly rate of skilled worker (machine man Sam) under Halsey Incentive Scheme (50%)

Total earnings = (Hours worked × Rate per hour) + (½ Time saved × Rate per hour)

(under 50% Halsey Incentive Scheme)

= (3 hours × ₹ 30) + (½ × 1 hour × ₹ 30)

$$\text{Effective hourly rate} = \frac{\text{Total earnings}}{\text{Hours taken}} = \frac{₹105}{3\ \text{hours}} = ₹\ 35$$

340 MTP **Old P3 May 2022 S-I** Q3(a) 10 marks old scheme

The standard time allowed for a certain piece of work is 240 hours. Normal wage rate is ₹ 75 per hour.

The bonus system applicable to the work is as follows:

Percentage of time saved to time allowed (slab rate)	Bonus
(i) Up to the first 20% of time allowed	25% of the corresponding saving in time.
(ii) For and within the next 30% of time allowed	40% of the corresponding saving in time.
(iii) For and within the next 30% of time allowed	30% of the corresponding saving in time.
(iv) For and within the next 20% of time allowed	10% of the corresponding saving in time.

CALCULATE the total earnings of a worker over the piece of work and his earnings per hour when he takes-

(a) 256 hours,

(b) 120 hours, and

(c) 24 hours respectively.

ICAI'S ANSWER

Calculation of total earnings and earnings per hour:

	Particulars	(a) Time taken is 256 hours	(b) Time taken is 120 hours	(c) Time taken is 24 hours
A.	Time Allowed	240 hours	240 hours	240 hours
B.	Time taken	256 hours	120 hours	24 hours
C.	Time Saved (A-B)	Nil	120 hours	216 hours
D.	Bonus hours (Refer workings)	Nil	40.80 hours	64.80 hours
E.	Hours to be paid (B+D)	256 hours	160.80 hours	88.80 hours
F.	Wages rate per hour	₹ 75	₹ 75	₹ 75
G.	**Total earnings (E×F)**	**₹ 19,200**	**₹ 12,060**	**₹ 6,660**
H.	**Earnings per hour (G÷B)**	**₹ 75**	**₹ 100.50**	**₹ 277.50**

Working Notes:

Calculation of bonus hours:

	Time saved 120 hours	Time saved 216 hours
For first 20% of time allowed i.e. 48 hours	12 (25% of 48 hours)	12 (25% of 48 hours)
For next 30% of time allowed i..e. 72 hours	28.80 (40% of 72 hours)	28.80 (40% of 72 hours)
For next 30% of time allowed i..e. 72 hours	-	21.60 (30% of 72 hours)
For next 20% of time allowed i..e. 48 hours	-	2.40 (10% of 24 hours)
Bonus hours	**40.80**	**64.80**

ICAI · Old P3 · MTP May 2022 · Series I · question p. 4 · answer p. 4

341 MTP **Old P3 May 2022 S-II** Q6(a) 5 marks old scheme

BRIEF OUT advantages and disadvantages of Halsey Premium Plan.

ICAI'S ANSWER

Advantages	Disadvantages
1. Time rate is guaranteed while there is opportunity for increasing earnings by increasing production. 2. The system is equitable in as much as the employer gets a direct return for his efforts in improving production methods and providing better equipment.	1. Incentive is not so strong as with piece rate system. In fact the harder the worker works, the lesser he gets per piece. 2. The sharing principle may not be liked by employees.

ICAI · Old P3 · MTP May 2022 · Series II · question p. 7 · answer p. 12

342 MTP **Old P3 Dec 2021 S-I** Q4(b) 10 marks old scheme

ABC Ltd. has its factory at two locations viz Noida and Patparganj. Rowan plan is used at Noida factory and Halsey plan at Patparganj factory.

Standard time and basic rate of wages are same for a job which is similar and is carried out on similar machinery. Normal working hours is 9 hours per day in a 5 day week.

Job at Noida factory is completed in 36 hours while at Patparganj factory it has taken 33 hours 45 minutes. Conversion costs at Noida and Patparganj are ₹ 6,084 and ₹ 5,569 respectively. Overheads account for ₹ 25 per hour.

REQUIRED:

(i) To find out the normal wage; and

(ii) To compare the respective conversion costs.

ICAI'S ANSWER

Particulars	Noida	Patparganj
Hours worked	36 hr.	33.75 hr.
Conversion Costs	₹ 6,084	₹ 5,569
Less: Overheads	₹ 900 (₹25 × 36 hr.)	₹ 844 (₹ 25 × 33.75 hr.)
Labour Cost	₹ 5,184	₹ 4,725

(i) **Finding of Normal wage rate:**

Let Wage rate be ₹ R per hour, this is same for both the Noida and Patparganj factory.

Normal wage rate can be found out taking total cost of either factory.

Noida: Rowan Plan

Total Labour Cost = Wages for hours worked + Bonus as per Rowan plan

₹ 5,184 = Hours worked × Rate per hour + $\left(\frac{\text{Time saved}}{\text{Time allowed}} \times \text{Hours worked} \times \text{Rate per hour}\right)$

Or, ₹ 5,184 = 36 hr. × R + $\left(\frac{45-36}{45} \times 36 \times \text{R}\right)$

Or, ₹ 5,184 = 36R + 7.2R

R = ₹ 120

Normal wage = 36 hrs × ₹ 120 = ₹ 4,320

OR

Patparganj: Halsey Plan

Total Labour Cost = Wages for hours worked + Bonus as per Halsey plan

₹ 4,725 = Hours worked × Rate per hour + (50% ×Hours saved×Rate per hour)

₹ 4,725 = 33.75 hr. × R + 50% × (45 hr. – 33.75 hr.) × R

₹ 4,725 = 39.375 R

R = ₹ 120

Normal Wage = 33.75 hrs × ₹ 120 = ₹ 4,050

(ii) **Comparison of conversion costs:**

Particulars	Noida (₹)	Patparganj (₹)
Normal Wages (36 × 120)	4,320	
(33.75 × 120)		4,050
Bonus (7.2 × 120)	864	
(5.625 × 120)		675
Overhead	900	844
	6,084	5,569

ICAI · Old P3 · MTP Dec 2021 · Series I · question p. 5 · answer p. 8-9

343 MTP **Old P3 Dec 2021 S-II** Q1(a) 5 marks old scheme

The following particulars have been compiled in respect of three workers:

	M	N	O
Actual hours worked	380	100	540
Hourly rate of wages (in ₹)	90	100	110
Productions in units:			
- Product A	210	-	600
- Product B	360	-	1350
- Product C	460	250	-
Standard time allowed per unit of each product is:			
	A	**B**	**C**
Minutes	15	20	30

For the purpose of piece rate, each minute is valued at ₹ 1.50.

You are required to CALCULATE the wages of each worker under:

(i) Guaranteed hourly rate basis.

(ii) Piece work earning basis but guaranteed at 75% of basic pay (Guaranteed hourly rate if his earnings are less than 50% of basic pay.)

ICAI'S ANSWER

(i) Computation of wages of each worker under guaranteed hourly rate basis

Worker	Actual hours worked (Hours)	Hourly wage rate (₹)	Wages (₹)
M	380	90	34,200
N	100	100	10,000
O	540	110	59,400

(ii) Computation of Wages of each worker under piece work earning basis

Product	Piece rate per unit	Worker-M		Worker-N		Worker-O	
	(₹)	Units	Wages (₹)	Units	Wages (₹)	Units	Wages (₹)
A	22.50	210	4,725	-	-	600	13,500
B	30.00	360	10,800	-	-	1,350	40,500
C	45.00	460	20,700	250	11,250	-	-
Total			36,225		11,250		54,000

Since each worker's earnings are more than 50% of basic pay. Therefore, worker-M, N and O will be paid the wages as computed i.e. ₹ 36,225, ₹ 11,250 and ₹ 54,000 respectively.

Working Notes:

1. Piece rate per unit

Product	Standard time per unit (in minutes)	Piece rate each minute (₹)	Piece rate per unit (₹)
A	15	1.5	22.50
B	20	1.5	30.00
C	30	1.5	45.00

ICAI · Old P3 · MTP Dec 2021 · Series II · question p. 1 · answer p. 1

344 MTP **Old P3 May 2020 S-I** Q1(c) 5 marks old scheme

A company is undecided as to what kind of wage scheme should be introduced. The following particulars have been compiled in respect of three workers, which are under consideration of the management.

	I	II	III
Actual hours worked	380	100	540
Hourly rate of wages (in Rs.)	40	50	60
Productions in units:			
- Product A	210	-	600

	I	II	III
- Product B	360	-	1350
- Product C	460	250	-
Standard time allowed per unit of each product is:			
	A	**B**	**C**
Minutes	15	20	30

For the purpose of piece rate, each minute is valued at Rs. 1/-

You are required to COMPUTE the wages of each worker under:

(i) Guaranteed hourly rate basis.

(ii) Piece work earning basis, but guaranteed at 75% of basic pay (Guaranteed hourly rate if his earnings are less than 50% of basic pay.)

(iii) Premium bonus basis where the worker received bonus based on Rowan scheme.

ICAI'S ANSWER

(i) Computation of wages of each worker under guaranteed hourly rate basis

Worker	Actual hours worked (Hours)	Hourly wage rate (Rs.)	Wages (Rs.)
I	380	40	15,200
II	100	50	5,000
III	540	60	32,400

(ii) Computation of Wages of each worker under piece work earning basis

Product	Piece rate per unit (Rs.)	Worker-I		Worker-II		Worker-III	
		Units	**Wages (Rs.)**	**Units**	**Wages (Rs.)**	**Units**	**Wages (Rs.)**
A	15	210	3,150	-	-	600	9,000
B	20	360	7,200	-	-	1,350	27,000
C	30	460	13,800	250	7,500	-	-
Total			24,150		7,500		36,000

Since each worker's earnings are more than 50% of basic pay. Therefore, worker-I, II and III will be paid the wages as computed i.e. Rs. 24,150, Rs. 7,500 and Rs. 36,000 respectively.

Working Note:

1. Piece rate per unit

Product	Standard time per unit in minute	Piece rate each minute (Rs.)	Piece rate per unit (Rs.)
A	15	1	15
B	20	1	20
C	30	1	30

(iii) Computation of wages of each worker under Premium bonus basis (where each worker receives bonus based on Rowan Scheme)

Worker	Time Allowed (Hr.)	Time Taken (Hr.)	Time saved (Hr.)	Wage Rate per hour (Rs.)	Earnings (Rs.)	Bonus (Rs.)*	Total Earning (Rs.)
I	402.5	380	22.5	40	15,200	850	16,050
II	125	100	25	50	5,000	1,000	6,000
III	600	540	60	60	32,400	3,240	35,640

Working Note:

1. Time allowed to each worker

Worker	Product-A	Product-B	Product-C	Total Time (Hours)
I	210 units × 15 = 3,150	360 units × 20 = 7,200	460 units × 30 = 13,800	24,150/60 = 402.50
II	-	-	250 units × 30 = 7,500	7,500/60 = 125
III	600 units × 15 = 9,000	1, 350 units × 20 = 27,000	-	36,000/60 = 600

$$* \frac{\text{Time Taken}}{\text{Time Allowed}} \times \text{Time Saved} \times \text{Wage Rate}$$

$$\text{Worker-I} = \frac{380}{402.5} \times 22.5 \times 40 = 850$$

$$\text{Worker-II} = \frac{100}{125} \times 25 \times 50 = 1{,}000$$

$$\text{Worker-III} = \frac{540}{600} \times 60 \times 60 = 3{,}240$$

ICAI · Old P3 · MTP May 2020 · Series I · question p. 1-2 · answer p. 2-3

4.5 Direct expenses

4 questions · asked in 4 ICAI documents

345 RTP Sep 2026 Q7

The following expenditures were incurred in Aditya Ltd. For the month of March 20X8:

		Amount (₹)
(i)	Paid for power & fuel	4,80,200
(ii)	Wages paid to factory workers	8,44,000
(iii)	Bill paid to job workers	9,66,000
(iv)	Royalty paid for production	8,400
(v)	Fee paid to technician hired for the job	96,000
(vi)	Administrative overheads	76,000
(vii)	Commission paid to sales staffs	1,26,000

You are required to CALCULATE direct expenses for the month.

ICAI'S ANSWER

Calculation of Direct Expenses

		Amount (₹)
(i)	Paid for power & fuel	4,80,200
(ii)	Bill paid to job workers	9,66,000
(iii)	Royalty paid for production	8,400
(iv)	Fee paid to the technician	96,000
	Total Direct expenses	15,50,600

Notes:

(i) Wages paid to factory workers is direct employee cost.

(ii) Administrative overhead is indirect expense.

(iii) Commission paid to sales staffs comes under selling expenses.

ICAI · P4 · RTP Sep 2026 · question p. 9 · answer p. 30

346 MTP May 2026 S-I MCQ 11 2 marks MCQ

The following expenditures were incurred in P.L. Ltd. for the month of March, 2025:

	Particulars	Amount (₹)
(i)	Product blueprint cost	1,90,000
(ii)	Paid for power & fuel	3,00,000
(iii)	Wages paid to factory workers	80,000
(iv)	Hire charges paid for machinery used	50,000
(v)	Commission paid to marketing manager	35,000
(vi)	Salary to office attendants	45,000
(vii)	Fee paid to technician hired in factory	96,000
(viii)	Administrative overheads	89,000

Calculate the direct expenses for the month.

A ₹ 7,16,000

B ₹ 6,36,000

C ₹ 7,40,000

D ₹ 6,71,000

ICAI'S ANSWER

(B) ₹ 6,36,000

Calculation of Direct Expenses:

	Particulars	Amount (₹)
(i)	Product blueprint cost	1,90,000
(ii)	Paid for power & fuel	3,00,000
(iii)	Wages paid to factory workers	NIL
(iv)	Hire charges paid for machinery used	50,000
(v)	Commission paid to marketing manager	NIL
(vi)	Salary to office attendants	NIL
(vii)	Fee paid to technician hired in factory	96,000
(viii)	Administrative overheads	NIL
		6,36,000

Notes:

a. Wages paid to factory workers is direct employee cost.

b. Commission paid to marketing manager come under selling expenses.

c. Salary to office attendants is classified under office and administration cost.

d. Administrative overhead is indirect expense.

ICAI · P4 · MTP May 2026 · Series I · question p. 5 · answer p. 4-5

347 MTP **Old P3 Nov 2022 S-II** Q6(c) 5 marks old scheme

STATE Direct Expenses with examples.

ICAI'S ANSWER

Expenses other than direct material cost and direct employee cost, which are incurred to manufacture a product or for provision of service and can be directly traced in an economically feasible manner to a cost object. The following costs are examples for direct expenses:

(a) Royalty paid/ payable for production or provision of service;

(b) Hire charges paid for hiring specific equipment;

(c) Cost for product/ service specific design or drawing;

(d) Cost of product/ service specific software;

(e) Other expenses which are directly related with the production of goods or provision of service.

ICAI · Old P3 · MTP Nov 2022 · Series II · question p. 7 · answer p. 15

348 MTP **Old P3 Jul 2021 S-I** Q5(b)(i) 5 marks old scheme

A Ltd. is an engineering manufacturing company producing job orders on the basis of specifications provided by the customers. During the last month it has completed three jobs namely A, B and C. The following are the items of expenditures which are incurred in addition to direct materials and direct employee cost:

(i) Office and administration cost - Rs. 6,00,000

(ii) Product blueprint cost for job A - Rs. 2,80,000

(iii) Hire charges paid for machinery used in job work B - Rs. 80,000

(iv) Salary to office attendants - Rs. 1,00,000

(v) One time license fee paid for software used to make computerised graphics for job C - Rs. 1,00,000.

(vi) Salary paid to marketing manager - Rs. 2,40,000.

Required:

CALCULATE direct expenses attributable to each job.

ICAI'S ANSWER

Calculation of Direct expenses

Particulars	Job A (Rs.)	Job B (Rs.)	Job C (Rs.)
Product blueprint cost	2,80,000	--	--
Hire charges paid for machinery	--	80,000	--
License fee paid for software	--	--	1,00,000
Total Direct expenses	2,80,000	80,000	1,00,000

ICAI · Old P3 · MTP Jul 2021 · Series I · question p. 7 · answer p. 12

CHAPTER 5

Overheads: Absorption Costing

5.1 Classifying overheads, allocation and apportionment

1 question · asked in 1 ICAI document · 1 exam paper

349 EXAM **Jan 2026** Q1(a) 5 marks

ABC Limited manufactures four products A, B, C and D in the same factory. The following information is given for a certain period:

Particulars	A	B	C	D
Input Materials (in units)	600	400	400	200
Average Yield	80%	90%	96%	72%

The finished goods' packaging cost is ₹ 4,600.

Other information:

(1) All the products are packed in boxes where each box is having capacity of 24 units.

(2) Product A and product B are packed in larger boxes whereas product C and product D are packed in smaller boxes, which cost half the price of bigger box.

(3) Each box contains only one type of product. There is no product mix up in packaging.

You are required to:

(i) Find out how many large and small boxes are used.

(ii) Find out the cost of per large box and per small box.

(iii) Allocate the packaging cost among the four finished products.

ICAI'S ANSWER

(i) Calculation of Output Units and Boxes required

Product	A	B	C	D
Input Units	600	400	400	200
Yield %	80%	90%	96%	72%
Output Units	480	360	384	144
Boxes Required*	20	15	16	6

* Capacity per box = 24 units

Large boxes used = 20 + 15 = 35

Small boxes used = 16 + 6 = 22

(ii) Let Cost of one small box = ₹ x and Cost of one large box = ₹ 2x

35(2x) + 22x = ₹4,600

70x + 22x = ₹4,600

Cost of one small box ‘x’ = ₹ 50

Cost of one large box ‘2x’ = ₹ 100

(iii) Allocation of Packaging Cost

Product	Boxes	Cost per Box	Packaging Cost (₹)
A	20 (Large)	100	2,000
B	15 (Large)	100	1,500
C	16 (Small)	50	800
D	6 (Small)	50	300
Total			**4,600**

ICAI · P4 · Jan 2026 exam · question p. 8 · answer p. 10

5.2 Primary distribution of factory overheads

7 questions · asked in 5 ICAI documents

350 RTP **Jan 2026** Q10

BrightTech Appliances Ltd. is a leading manufacturer of smart home devices such as the SmartCool AC, PowerLite Inverter, and EcoDry Dryer. Amid increasing operational complexity, the company’s finance controller has initiated a review of overhead allocations to gain better control over departmental efficiency and product profitability.

The company operates through four major departments:

1. **Production** – handles manufacturing across three plants using advanced machinery including stores management.
2. **Administration** – supports HR, finance, legal, and compliance.
3. **Sales & Distribution** – manages retail tie-ups, e-commerce, and logistics.
4. **General Management** – oversees strategy, innovation, and executive-level operations.

The extract from the budget for the next financial year is as follows:

Particulars	Amount (₹)	Notes
Raw Material Cost	5,75,00,000	Consumed in a ratio of 3:4:3 for the three products
Indirect Material Cost	12,50,000	Manufacturing ₹7,20,000, Stores ₹25,000, General admin ₹3,80,000, Personnel ₹95,000, Sales ₹30,000
Salary & Wages	4,20,00,000	Includes direct wages of ₹1,60,00,000 (temporary contract workers)
Rent & Property Tax	₹1,20,000	₹90,000 Warehouse Rent, ₹30,000 Property Tax
Depreciation on Non-Current Assets	₹25,00,000	Factory/office building ₹10,50,000, ACs ₹2,00,000, Machinery ₹12,50,000

Particulars	Amount (₹)	Notes
Power & Fuel	₹4,90,000	₹4,70,000 for manufacturing, ₹20,000 for delivery vans
Machinery Insurance Premium	₹5,00,000	At 4% of WDV of machinery
Group Employee Insurance	₹3,10,000	Based on gross salary (excluding direct workers and top management)
Printing & Stationery	₹8,20,000	Manufacturing ₹21,000, Finance ₹5,50,000, Legal ₹28,000, Sales ₹2,21,000
Audit Fees	₹1,40,000	Statutory and internal audit
Electricity Expense	₹4,00,000	Metered use: Production 5,600, Admin 10,000, Sales & Distribution 4,000, General Management 1,600 units
Telephone & Mobile	₹4,95,000	Production ₹1,25,000, Personnel ₹50,000, General Mgmt ₹30,000, Sales ₹75,000, Customer Support ₹2,15,000
Travelling Expenses	₹24,00,000	Production ₹5,50,000, General Mgmt ₹12,00,000, Sales ₹6,50,000
Meal Coupon Subsidy	₹2,25,000	₹3,000 per employee on roll
Software License Renewals	₹16,50,000	Stores Management ₹8,50,000, Finance ₹1,50,000, Stores ₹30,000, Customer Support ₹6,20,000
Misc. Expenditures	₹9,50,000	Allocated based on direct expenses

Additional Information:

- **Employee Count:** Production 22, Admin 20, Sales & Distribution 28
- **Average Gross Salary:** Production ₹6,30,000, Admin ₹4,20,000, Sales & Distribution ₹3,50,000, General Mgmt ₹2,80,000
- **Floor Area:** Production 9,900 m^2, Admin 3,600 m^2, Sales & Distribution 2,700 m^2, General Mgmt 1,800 m^2
- **AC Tonnage:** Production 6,000 RT, Admin 3,000 RT, S&D 3,000 RT, General Mgmt 1,500 RT
- **Depreciation Rates:** Building 5%, AC 15%, Machinery 10%
- **Department Roles:**
  - General Mgmt: 70% Sales strategy, 20% Production/ Marketing, 10% Admin
  - Admin: 50% Production, 50% Sales.

Required:

(i) Prepare a schedule of Cost Allocation for Production Dept., Administration Dept., Sales & Distribution Dept. and General Management.

(ii) Prepare a schedule of Cost Apportionment (Primary Distribution) for Production Dept., Administration Dept., Sales & Distribution Dept. and General Management.

(iii) Prepare a schedule of Secondary distribution of Administration Dept. and General Management costs.

ICAI'S ANSWER

(i) Schedule of Cost Allocation among the Departments

	Total	Production Dept.	Administration Dept.	Selling & Distribution Dept.	General Management
Raw Material Cost	5,75,00,000	5,75,00,000	-	-	-
Indirect Material Cost:	12,50,000	7,45,000	4,75,000	30,000	-
Salary & Wages	1,60,00,000	1,60,00,000	-	-	-
Rent & Property tax	90,000	-	-	90,000	-
Depreciation on Machinery	12,50,000	12,50,000	-	-	-
Power & fuel	4,90,000	4,70,000	-	20,000	-
Insurance premium on machinery	5,00,000	5,00,000	-	-	-
Printing & Stationery:	8,20,000	21,000	5,78,000	2,21,000	-
Audit fees	1,40,000	-	-	-	1,40,000
Telephone & Mobile expenses:	4,95,000	1,25,000	50,000	2,90,000	30,000
Travelling expenses	24,00,000	5,50,000	-	6,50,000	12,00,000
Software license renewal fees:	16,50,000	8,80,000	1,50,000	6,20,000	-
Total Allocated Direct Expenses	8,25,85,000	7,80,41,000	12,53,000	19,21,000	13,70,000

(ii) Schedule of Cost Apportionment (Primary Distribution)

	Basis	Total	Production Dept.	Administration Dept.	Selling & Distribution Dept.	General Management
Allocated Cost	Direct	8,25,85,000	7,80,41,000	12,53,000	19,21,000	13,70,000
Salary & Wages	Gross Salary (9:6:5:4)	2,60,00,000	97,50,000	65,00,000	54,16,667	43,33,333
Rent & Property tax	Floor Area (11:4:3:2)	30,000	16,500	6,000	4,500	3,000
Depreciation						
– Building	Floor Area (11:4:3:2)	10,50,000	5,77,500	2,10,000	1,57,500	1,05,000

	Basis	Total	Production Dept.	Administration Dept.	Selling & Distribution Dept.	General Management
– AC	RT (4:2:2:1)	2,00,000	88,889	44,444	44,444	22,222
Group Employee Insurance	Gross Salary (9:6:5:4)	3,10,000	1,16,250	77,500	64,583	51,667
Electricity Expenses	Units (14:25:10:4)	4,00,000	1,05,660	1,88,679	75,472	30,189
Meal Coupon Subsidy	No. of Employees (22,20,28)	2,25,000	66,000	60,000	84,000	15,000 (b.f.)
Miscellaneous Expenditure	Direct Expenses (Allocated Expenses)	9,50,000	8,97,729	14,414	22,098	15,760
Total		11,17,50,000	8,96,59,528	83,54,037	77,90,264	59,46,170

(iii) Schedule of Secondary Distribution:

	Basis	Total	Production Dept.	Administration Dept.	Selling & Distribution Dept.	General Management
Total Allocated and Apportioned costs		11,17,50,000	8,96,59,528	83,54,037	77,90,264	59,46,170
General Management	(2:1:7)	59,46,170	11,89,234	5,94,617	41,62,319	(59,46,170)
			9,08,48,762	89,48,654	1,19,52,583	
Administration dept.	(1:1)	89,48,654	44,74,327	(89,48,654)	44,74,327	
Total			9,53,23,089	-	1,64,26,910	

ICAI · P4 · RTP Jan 2026 · question p. 11-14 · answer p. 35-36

351 RTP **Sep 2025** Q9

SA Ltd. has three production (M1, M2 and A1) and three service departments (Stores, Engineering services and General service). Engineering department serves the M1 and M2 only.

The relevant information related with Product X and Y are as follows:

	Product X	Product Y
M1	10 Machine hours	6 Machine hours
M2	4 Machine hours	14 Machine hours
A1	14 Direct labour hours	18 Direct labour hours

The annual budgeted overhead cost for the year is

	Indirect Wages (₹)	Consumable Supplies (₹)
M1	9,30,400	2,52,000
M2	8,26,800	3,64,000
A1	3,24,400	84,000
Stores	1,64,000	56,000
Engineering Service	1,06,800	84,000
General Service	1,50,400	64,000

	(₹)	
– Depreciation on Machinery	7,92,000	
– Insurance of Machinery	1,44,000	
– Insurance of Building	64,800	(Total building insurance cost for M1 is one third of annual premium)
– Power	1,29,600	
– Light	1,08,000	
– Rent	2,53,500	(The general service deptt. is located in a building owned by the company. It is valued at ₹ 1,20,000 and is charged into cost at notional value of 8% per annum. This cost is additional to the rent shown above)
The value of issues of materials to the production departments are in the same proportion as shown above for the Consumable supplies.		

The following data are also available:

Department	Book value Machinery (₹)	Area (Sq. ft.)	Effective H.P. hours %	Production Direct Labour hour	Capacity Machine hour
M1	24,00,000	5,000	50	2,00,000	40,000
M2	18,00,000	6,000	35	1,50,000	50,000
A1	6,00,000	8,000	05	3,00,000	–
Stores	2,40,000	2,000	–	–	–
Engg. Service	7,20,000	2,500	10	–	–
General Service	2,40,000	1,500	–	–	–

Required:

(i) PREPARE an overhead analysis sheet, showing the bases of apportionment of overhead to departments.

(ii) PREPARE a statement allocating service department overheads to production department ignoring the apportionment of service department costs among service departments.

(iii) CALCULATE suitable overhead absorption rate for the production departments.

(iv) CALCULATE the overheads to be absorbed by two products, X and Y.

ICAI'S ANSWER

(i) Summary of Apportionment of Overheads (₹)

Items	Basis of Apportionment	Total Amount	Production Deptt.	Service Deptt.				
			M1	M2	A1	Store Service	Engineering Service	General Service
Indirect wages	Allocation given	25,02,800	9,30,400	8,26,800	3,24,400	1,64,000	1,06,800	1,50,400
Consumable supplies	Allocation given	9,04,000	2,52,000	3,64,000	84,000	56,000	84,000	64,000
Depreciation	Capital value of machine (20:15:5:2:6:2)	7,92,000	3,16,800	2,37,600	79,200	31,680	95,040	31,680
Insurance of Machine	Capital value of machine (20:15:5:2:6:2)	1,44,000	57,600	43,200	14,400	5,760	17,280	5,760
Insurance on Building	1/3rd to M1 Balance area basis (-:12:16:4:5:3)	64,800	21,600	12,960	17,280	4,320	5,400	3,240
Power	HP Hr% (10:7:1:-:2:-)	1,29,600	64,800	45,360	6,480	–	12,960	–
Light	Area (10:12:16:4:5:3)	1,08,000	21,600	25,920	34,560	8,640	10,800	6,480
Rent*	Area (10:12:16:4:5:-)	2,53,500	53,940	64,720	86,300	21,580	26,960	--
Total		48,98,700	17,18,740	16,20,560	6,46,620	2,91,980	3,59,240	2,61,560

*Rent to be apportioned among the departments which actually use the rented building. The notional rent is imputed cost and is not included in the calculation.

(ii) Allocation of service departments overheads

Service Deptt.	Basis of Apportionment	Production Deptt.			Service Deptt.		
		M1	M2	A1	Store Service	Engineering Service	General Service
Store	Ratio of consumable value (126 : 182 : 42)	1,05,120	1,51,820	35,040	(2,91,980)	–	–
Engineering service	In Machine hours Ratio of M1 and M2 (4 : 5)	1,59,660	1,99,580	–	–	(3,59,240)	–
General service	Labour hour Basis (20 : 15 : 30)	80,480	60,360	1,20,720	–	–	(2,61,560)
Production Department allocated in (i)		17,18,740	16,20,560	6,46,620			
Total		20,64,000	20,32,320	8,02,380			

(iii) Overhead Absorption rate

	M1	M2	A1
Total overhead allocated	20,64,000	20,32,320	8,02,380
Machine hours	40,000	50,000	–
Labour hours	–	–	3,00,000
Rate per machine hour	51.60	40.65	–
Rate per Direct labour	–	–	2.67

(iv) Statement showing overhead absorption for Product X and Y

Machine Deptt.	Absorption Rate	Product X		Product Y	
		Hours	(₹)	Hours	(₹)
M1	51.60	10	516.00	6	309.60
M2	40.65	4	162.60	14	569.10
A1	2.67	14	37.38	18	48.06
			715.98		926.76

ICAI · P4 · RTP Sep 2025 · question p. 12-13 · answer p. 35-37

352 MTP **Sep 2026 S-II** Q1(a) 5 marks

In a factory, the average annual charges for direct wages amounts to ₹ 4,80,000. The following are some of the expenses (average figure for a year) incurred inside the factory:

(a) Works Manager's salary and other emoluments ₹ 50,000.

(b) Factory rent and rates ₹ 36,000. The total area of the factory is 45,000 sq. ft. of which the shops occupy 40,000 sq. ft.

(c) Wages of caretakers, sweepers, etc., ₹ 7,000.

(d) Other factory overhead charges ₹ 53,000.

A work order is executed in a shop occupying 3,000 sq. ft. and costs ₹ 10,000 in wages. If the total wages for all the work orders executed in this shop amount to ₹ 160,000, calculate the total amount of factory overhead charges for this work order.

ICAI'S ANSWER

Working Note 1:

Primary distribution of Rent amongst Production Department and Service Department:

Production Department = (36,000/45000) × 40,000 = 32000

Service Department = (36,000/45000) × 5000 = 4,000

Working Note 2:

Total of Overheads:

Works Manager's salary	₹ 50,000
Wages of caretakers, sweepers	₹ 7,000
Other factory overhead charges	₹ 53,000
Add: Share of Rent from Primary Distribution	₹ 4000
Total	₹ 1,14,000

The total amount of factory overhead charges for this work order:

Total Rent for this shop floor = (32,000/40,000) × 3,000 = ₹2,400

Share of rent (In proportion to wages in this shop) = (2,400/1,60,000) × 10,000 = ₹ 150

Share of other overheads (In proportion to total wages) = (1,14,000/4,80,000) × 10,000 = ₹ 2,375

Total Overheads for this Work order = ₹ 2,525 (150 +2,375)

ICAI · P4 · MTP Sep 2026 · Series II · question p. 8 · answer p. 7

353 MTP **Jan 2026 S-I** Q6(c) 4 marks

Show the basis of apportionment of following overhead cost:

Overhead Cost
1. Lighting and heating (conditioning), Fire precaution service
2. Time keeping, labour welfare expense
3. Holiday pay and ESI and PF contribution
4. General overhead
5. (i) Depreciation of plant and machinery (ii) Insurance of stock
6. (i) Power/steam consumption (ii) Internal transport (iii) Managerial salaries
7. Lighting expenses (light)
8. Electric power (machine operation)

ICAI'S ANSWER

The basis apportionment of expenses are stated below:

Overhead Cost	Bases of Apportionment
1. Lighting and heating (conditioning), Fire precaution service	Floor area, or volume of department
2. Time keeping, labour welfare expense	Number of workers
3. Holiday pay and ESI and PF contribution	Direct wages
4. General overhead	Direct labour hour, or Direct wages, or Machine hours.
5. (i) Depreciation of plant and machinery (ii) Insurance of stock	Capital values
6. (i) Power/steam consumption (ii) Internal transport (iii) Managerial salaries	Technical estimates
7. Lighting expenses (light)	No. of light points, or Area or Metered units
8. Electric power (machine operation)	Horse power of machines, or Number of machine hour, or value of machines or units consumed.

ICAI · P4 · MTP Jan 2026 · Series I · question p. 16 · answer p. 20-21

354 MTP **Jan 2025 S-I** Case scenario · Q2(a)(i) to Q2(a)(iii) 12 marks

CASE SCENARIO

Baba Ltd. belongs to an automotive industry, manufacturing hybrid bicycles. The production of bicycles passes through three departments, viz. X1, Y2, Z3. The bicycles being equipped with gears needs quality check from time to time. Thus, the company also operates two service departments, namely quality control (QC) and maintenance (M), for its bicycle.

Following information is extracted from the accounting books regarding expenses as incurred/ charged:

Particulars	(₹)
Rent and Rates	40,00,000
General Lighting	4,80,000
Indirect Wages	15,51,200
Power	12,00,000
Depreciation on Machines	80,00,000
Sundries	77,56,000

Additional information:

	Production Departments			Service Departments	
	X1	**Y2**	**Z3**	**QC**	**M**
Direct wages (₹)	24,00,000	16,00,000	24,00,000	12,00,000	1,56,000
Working hours	6,140	8,950	4,838	-	-
Value of machines (₹)	4,80,00,000	6,40,00,000	8,00,00,000	40,00,000	40,00,000
H.P. of machines	120	60	100	20	-
Light points	20	30	40	20	10
Floor space (sq. ft.)	4,000	5,000	6,000	4,000	1,000

A technical assessment unveiled the following basis for the apportionment of expenses of service departments:

	X1	Y2	Z3	QC	M
QC	20%	30%	40%	-	10%
M	40%	20%	30%	10%	-

You are required to DETERMINE the following:

Q2(a)(i) 5 marks

Overheads distributed to all the departments, viz. X1, Y2, Z3, QC and M.

Q2(a)(ii) 5 marks

Overheads total and rate per hour under all the Production Departments after redistribution of Service Department's Overhead.

Q2(a)(iii) 2 marks

Total cost of a bicycle, considering the Direct Material and labour Cost of ₹ 20,000 and ₹ 12,000 respectively, which is being processed for manufacturing in Departments X1, Y2 and Z3 for 4, 5 and 3 hours respectively.

ICAI'S ANSWERS

Q2(a)(i)

Statement Showing Distribution of Overheads of Baba Ltd.

Particulars	Basis	Total	Production Departments			Service Departments	
			X1	Y2	Z3	QC	M
		(₹)	(₹)	(₹)	(₹)	(₹)	(₹)
Direct wages	Actual	13,56,000	-	-	-	12,00,000	1,56,000
Rent & rates	Area	40,00,000	8,00,000	10,00,000	12,00,000	8,00,000	2,00,000
General lighting	Light points	4,80,000	80,000	1,20,000	1,60,000	80,000	40,000
Indirect wages	Direct wages	15,51,200	4,80,000	3,20,000	4,80,000	2,40,000	31,200
Power	H.P.	12,00,000	4,80,000	2,40,000	4,00,000	80,000	–
Depreciation of machines	Value of machines	80,00,000	19,20,000	25,60,000	32,00,000	1,60,000	1,60,000
Sundries	Direct wages	77,56,000	24,00,000	16,00,000	24,00,000	12,00,000	1,56,000
		2,43,43,200	**61,60,000**	**58,40,000**	**78,40,000**	**37,60,000**	**7,43,200**

Q2(a)(ii)

Redistribution of Service Department's Expenses over Production Departments

	X1	Y2	Z3	QC	M
	(₹)	(₹)	(₹)	(₹)	(₹)
Total overhead distributed as above	61,60,000	58,40,000	78,40,000	37,60,000	7,43,200
Dept. QC Overheads apportioned (20:30:40:—:10)	7,52,000	11,28,000	15,04,000	-37,60,000	3,76,000
Dept. M overheads apportioned (40:20:30:10:—)	4,47,680	2,23,840	3,35,760	1,11,920	-11,19,200
Dept. QC Overheads apportioned (20:30:40:—:10)	22,384	33,576	44,768	-1,11,920	11,192
Dept. M overheads apportioned (40:20:30:10:—)	4,477	2,238	3,358	1,119	-11,192

	X1	Y2	Z3	QC	M
Dept. QC Overheads apportioned (20:30:40:—:10)	224	336	448	-1,119	112
Dept. M overheads apportioned (40:20:30:10:—)	45	22	34	11	-112
Dept. QC Overheads apportioned (20:30:40:—:10)	2	3	5	-11	-
Total	**73,86,812**	**72,28,015**	**97,28,373**		
Working hours	6,140	8,950	4,838		
Rate per hour	**1,203**	**808**	**2,011**		

Q2(a)(iii)

Determination of total cost of a bicycle:

Particulars	(₹)
Direct material cost	20,000
Direct labour cost	12,000
Overhead cost (See working note)	14,885
	46,885

Working Note:

Overhead cost

= (₹ 1,203 × 4 hrs.) + (₹ 808 × 5 hrs.) + (₹ 2,011 × 3 hrs.)

= ₹ 4,812 + ₹ 4,040 + ₹ 6,033 = ₹ 14,885

ICAI · P4 · MTP Jan 2025 · Series I · question p. 9-10 · answer p. 8-9, 9, 9-10

5.3 Secondary distribution: direct, step, repeated and simultaneous methods

28 questions · asked in 22 ICAI documents · 7 exam papers

355 EXAM Sep 2026 Q1(c) 3 marks

Textiles Manufacturing Co. has two production departments : Fibers (FIB) and Fabrics (FAB) and two service departments – FC and DD which also render services to each other.

The following information is given :

Service Departments' expenses before allocation :

	Amount (₹)
FC	3,00,000
DD	1,20,000

Usage of Services :

	Fibers (FIB)	Fabrics (FAB)	FC	DD
FC	40%	40%	-	20%
DD	40%	30%	30%	-

Use simultaneous equation method and find out the total expenses of FC.

ICAI'S ANSWER

ICAI has not published its answers to this paper yet.

ICAI · P4 · Sep 2026 exam · question p. 12

356 **EXAM** **Sep 2025** Q2(b) **4 marks**

A factory has three production departments, P1, P2, P3 and two service departments S1, and S2 Both the service departments are independent and provide services to each other. Following is the detail of expenses of each service department:

Department	Amount (₹)
S1	1,60,000
S2	2,40,000

Further the expenses of department S1 and S2 are apportioned on the following basis:

	P1	P2	P3	S1	S2
S1	25%	35%	20%	-	20%
S2	35%	30%	25%	10%	-

You are required to apportion the expenses of departments S1 and S2 to production departments P1, P2 and P3 using Simultaneous Equation Method.

ICAI'S ANSWER

The total expenses of the two service departments will be determined as follows:

S1 = 1,60,000 + 0.10 S2

S2 = 2,40,000 + 0.20 S1

Substituting the value of S1,

S2 = 2,40,000 + 0.20 (1,60,000 + 0.1 S2)

= 2,40,000 + 32,000 + 0.02 S2

= 2,72,000 + 0.02 S2

S2 - 0.02 S2 = 2,72,000

$$S2 = \left(\frac{2,72,000}{0.98}\right) = ₹\ 2,77,551$$

The total of expenses of the S2 is ₹ 2,77,551 and that of the S1 is ₹ 1,87,755 *i.e.*, ₹ 1,60,000 + 0.1 x ₹ 2,77,551.

The expenses will be allocated to the production departments as under:

	Production Department		
	Dept.-P1	**Dept.-P2**	**Dept.-P3**
S1 (25%, 35% and 20% of ₹1,87,755)	46,939	65,714	37,551
S2(35%, 30% and 25% of ₹ 2,77,551)	97,143	83,265	69,388
Total	1,44,082	1,48,979	1,06,939

The total of expenses apportioned to P1, P2 and P3 is ₹ 4,00,000.

ICAI · P4 · Sep 2025 exam · question p. 13-14 · answer p. 15-16

357 **EXAM** **Jan 2025** Q6(c) **4 marks**

Suggest any one basis of re-apportionment of service department overheads over production departments in the following contexts:

Cost of the Service Departments	Basis
Planning and progress	
Transport Department	
Personnel Department	
Fire Protection	
Power House (electric lighting cost)	
Computer Section	
Canteen and Welfare	
Hospital and Dispensary	

ICAI'S ANSWER

Cost of the Service Departments	Basis
Planning and progress	Direct labour hours, Machine hours, Direct labour wages, Asset value x Hours worked
Transport Department	Crane hours, Truck hours, Truck mileage, Truck tonnage, Truck ton-hours, Tonnage handled. No. of packages of Standard size
Personnel Department	No of direct workers No. of employees etc.
Fire Protection	Capital values
Power House (electric lighting cost)	Floor area, Cubic content, No. of electric Points, Wattage.

Cost of the Service Departments	Basis
Computer Section	Computer hours, Specific allocation to departments
Canteen and Welfare	No of direct workers No. of employees etc.
Hospital and Dispensary	No of direct workers No. of employees etc.

ICAI · P4 · Jan 2025 exam · question p. 28 · answer p. 29-30

358 EXAM Old P3 Nov 2023 Q3(a) 10 marks old scheme

HCP Ltd. is a manufacturing company having two production departments, P and Q and two service departments, R and S. The budgeted cost information for the month of October 2023 is furnished below:

		Production Departments		Service Departments	
	(₹)	P (₹)	Q (₹)	R (₹)	S (₹)
Indirect material	1,77,500	94,750	49,750	18,270	14,730
Indirect Labour	1,55,000	35,000	75,000		
Factory Rent	75,000				
Depreciation on machinery	37,500				
Power	96,000				
Security Expenses for Factory Premises	24,000				
Insurance- machinery	12,000				
Supervisor Expenses	48,000				
Additional information					
Floor Area (Sq. meters)		1250	750	200	300
Net book value of machinery (₹)		21,00,000	5,00,000	1,00,000	3,00,000
H.P. of machines		800	200	80	120
Machine hours		4,000	1,000	600	800
Number of employees		10	30	6	4
Labour hours		2,000	6,000	1,200	600

The overhead costs of the two service department are distributed using step method in the same order viz. R and S respectively on the following basis:

Department R	Number of employees
Department S	Machine hours

Required:

(i) Prepare a statement showing distribution of overheads to various departments, clearly showing the basis of distribution.

(ii) Calculate the total budgeted overheads for both production departments after the service departments have been re-apportioned to them.

(iii) Calculate the most appropriate overhead absorption rate for each of the production department.

ICAI'S ANSWER

(i) Overhead Distribution Statement

Particular	Basis	Total Amount (₹)	Production Departments		Service Departments	
			P (₹)	Q (₹)	R (₹)	S (₹)
Indirect material	Direct	1,77,500	94,750	49,750	18,270	14,730
Indirect labour	Direct	1,55,000	35,000	75,000	15,000	30,000
Factory rent (125:75:20:30)	Floor Area	75,000	37,500	22,500	6,000	9,000
Depreciation of machinery (21:5:1:3)	Book value of machinery	37,500	26,250	6,250	1,250	3,750
Power (80:20:8:12)	H.P. of machines	96,000	64,000	16,000	6,400	9,600
Security expenses for factory premises (125:75:20:30)	Floor Area	24,000	12,000	7,200	1,920	2,880
Insurance-machinery (21:5:1:3)	Book value of machinery	12,000	8,400	2,000	400	1,200
Supervisor expenses (10:30:6:4)	Number of employees	48,000	9,600	28,800	5,760	3,840
Total		**6,25,000**	**2,87,500**	**2,07,500**	**55,000**	**75,000**

(ii) Redistribution of Service Department's Expenses

Particular	Production Departments		Service Departments	
	P (₹)	Q (₹)	R (₹)	S (₹)

Particular	Production Departments		Service Departments	
Overhead as per primary distribution	2,87,500	2,07,500	55,000	75,000
Expenses of service department R is apportioned among other departments P, Q & S in the ratio of number of employees (10:30:4)	12,500	37,500	(55,000)	5,000
Expenses of service department S is apportioned among other departments P & Q in the ratio of Machine hours (40:10)	64,000	16,000	-	(80,000)
Total Budgeted overheads	**3,64,000**	**2,61,000**	**-**	**-**

(iii) Calculation of overhead rates for each of the production department

Particular	Production Departments	
	P (₹)	**Q (₹)**
Total Budgeted overheads	3,64,000	2,61,000
Actual machine hours	4000 hours	-
Actual labour hours	-	6000 hours
Actual machine/labour hour rate	**91**	**43.5**

Note: Department P is assumed to be machine oriented and Department Q is assumed to be labour oriented as per information available in the question

The solution 3(a) can also be presented in following way for Distribution of Power expenses:

Overhead Distribution Statement

Particular	Basis	Total Amount (₹)	Production Departments		Service Departments	
			P (₹)	**Q (₹)**	**R (₹)**	**S (₹)**
Indirect material	Direct	1,77,500	94,750	49,750	18,270	14,730
Indirect labour	Direct	1,55,000	35,000	75,000	15,000	30,000
Factory rent (125:75:20:30)	Floor Area	75,000	37,500	22,500	6,000	9,000
Depreciation of machinery (21:5:1:3)	Book value of machinery	37,500	26,250	6,250	1,250	3,750
Power (3200:200:48:96)	H.P. x machine hours	96,000	86,682	5,418	1,300	2,600

Particular	Basis	Total Amount (₹)	Production Departments		Service Departments	
Security expenses for factory premises (125:75:20:30)	Floor Area	24,000	12,000	7,200	1,920	2,880
Insurance-machinery (21:5:1:3)	Book value of machinery	12,000	8,400	2,000	400	1,200
Supervisor expenses (10:30:6:4)	Number of employees	48,000	9,600	28,800	5,760	3,840
Total		**6,25,000**	**3,10,182**	**1,96,918**	**49,900**	**68,000**

Power can be distributed on the basis of HP of machines x machine hours

800 x 4000 = 32,00,000, 200 x 1000 = 2,00,000, 80 x 600 = 48,000, 120 x 800 = 96,000

Ratio is 3200:200:48:96

(ii) Redistribution of Service Department's Expenses

Particular	Production Departments		Service Departments	
	P (₹)	**Q (₹)**	**R (₹)**	**S (₹)**
Overhead as per primary distribution	**3,10,182**	**1,96,918**	**49,900**	**68,000**
Expenses of service department R is apportioned among other departments P, Q & S in the ratio of number of employees (10:30:4)	11,340.90	34,022.73	(49,900)	4,536.37
Expenses of service department S is apportioned among other departments P & Q in the ratio of Machine hours (40:10)	58,029.10	14,507.27	-	(72,536.37)
Total Budgeted overheads	**3,79,552**	**2,45,448**	**-**	**-**

(iii) Calculation of overhead rates for each of the production department

Particular	Production Departments	
	P (₹)	**Q (₹)**
Total Budgeted overheads	**3,79,552**	**2,45,448**
Actual machine hours	4000 hours	-
Actual labour hours	-	6000 hours
Actual machine/labour hour rate	**94.89**	**40.91**

Note: Department P is assumed to be machine oriented and Department Q is assumed to be labour oriented as per information available in the question

ICAI · Old P3 · Nov 2023 exam · question p. 13-14 · answer p. 15-18

359 EXAM Old P3 May 2023 Q6(d) 5 marks old scheme

Suggest any one basis of re-apportionment of service department overheads over production departments in the following instances:

Cost of Service Department	Basis
(i) Maintenance and Repair Shop	
(ii) Hospital and Dispensary	
(iii) Fire Protection	
(iv) Stores Department	
(v) Transport Department	
(vi) Computer Section	
(vii) Power House (Electric Power Cost)	
(viii) Inspection	
(ix) Tool Room	
(x) Time- keeping	

ICAI'S ANSWER

Basis of re-apportionment of service department overheads over production departments

Cost of the Service Departments:	Basis
(i) Maintenance and Repair shop	Direct labour hours, Machine hours, Direct labour wages, Asset value x Hours worked
(ii) Hospital and Dispensary	No. of employees, No. of direct workers etc.
(iii) Fire Protection	Capital values
(iv) Stores Department	No. of requisitions, Weight or value of Materials issued.
(v) Transport Department	Crane hours, Truck hours, Truck mileage, Truck tonnage, Truck ton-hours, Tonnage handled. No. of packages of Standard size
(vi) Computer Section	Computer hours, Specific allocation to departments
(vii) Power House (Electric Power Cost)	Horse power, Kwh, Horse power × Machine hours, Kwh × Machine hours
(viii) Inspection	Inspection hours, number of inspections.

Cost of the Service Departments:	Basis
(ix) Tool room	Direct labour hours, Machine hours, Direct labour wages, Asset value x Hours worked
(x) Time-keeping	No. of card punched, No. of employees

ICAI · Old P3 · May 2023 exam · question p. 28 · answer p. 30-31

360 EXAM Old P3 Jul 2021 Q1(b) 5 marks old scheme

SNS Trading Company has three Main Departments and two Service Departments. The data for each department is given below:

Departments	Expenses (in ₹)	Area in (Sq. Mtr)	Number of Employees
Main Department:			
Purchase Department	5,00,000	12	800
Packing Department	8,00,000	15	1700
Distribution Department	3,50,000	7	700
Service Departments:			
Maintenance Department	6,40,000	4	200
Personnel Department	3,20,000	6	250

The cost of Maintenance Department and Personnel Department is distributed on the basis of 'Area in Square Metres' and 'Number of Employees' respectively.

You are required to:

(i) Prepare a Statement showing the distribution of expenses of Service Departments to the Main Departments using the "Step Ladder method" of Overhead Distribution.

(ii) Compute the Rate per hour of each Main Department, given that, the Purchase Department, Packing Department and Distribution Department works for 12 hours a day, 24 hours a day and 8 hours a day respectively. Assume that there are 365 days in a year and there are no holidays.

ICAI'S ANSWER

(i) Schedule Showing the Distribution of Expenses of Service Departments using Step ladder method.

	Main Department			Service Department	
	Purchase (₹)	**Packing (₹)**	**Distribution (₹)**	**Maintenance (₹)**	**Personnel (₹)**
Expenses	5,00,000	8,00,000	3,50,000	6,40,000	3,20,000
Distribution of Maintenance Department (12:15:7:-:6)	1,92,000	2,40,000	1,12,000	(6,40,000)	96,000
Distribution of Personnel Department (800:1700:700:-:-)	1,04,000	2,21,000	91,000	-	(4,16,000)
Total	7,96,000	12,61,000	5,53,000	-	-

(ii) Calculation of Expenses rate per hour of Main Department

	Purchase	Packing	Distribution
Total apportioned expenses (₹)	7,96,000	12,61,000	5,53,000
Total Hours worked	4,380 (12 × 365)	8,760 (24 × 365)	2,920 (8 × 365)
Expenses rate per hour (₹)	**181.74**	**143.95**	**189.38**

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361 EXAM **Old P3 Nov 2020** Q2(b) 10 marks old scheme

TEE Ltd. is a manufacturing company having three production departments 'P', 'Q' and 'R' and two service departments 'X' and 'Y' details pertaining to which are as under :

	P	Q	R	X	Y
Direct wages (₹)	5,000	1,500	4,500	2,000	800
Working hours	13,191	7,598	14,995	-	-
Value of machine (₹)	1,00,000	80,000	1,00,000	20,000	50,000
H.P. of machines	100	80	100	20	50
Light points (Nos.)	20	10	15	5	10
Floor space (sq. ft.)	2,000	2,500	3,500	1,000	1,000

The expenses are as follows:

	(₹)
Rent and Rates	10,000
General Lighting	600
Indirect Wages	3,450
Power	3,500
Depreciation on Machines	70,000
Sundries (apportionment on the basis of direct wages)	13,800

The expenses of Service Departments are allocated as under :

	P	Q	R	X	Y
X	45%	15%	30%	-	10%
Y	35%	25%	30%	10%	-

Product 'A' is processed for manufacture in Departments P, Q and R for 6, 5 and 2 hours respectively.

Direct Costs of Product A are :

Direct material cost is ₹ 65 per unit and Direct labour cost is ₹ 40 per unit.

You are Required to:

(i) Prepare a statement showing distribution of overheads among the production and service departments.

(ii) Calculate recovery rate per hour of each production department after redistributing the service departments costs.

(iii) Find out the Total Cost of a 'Product A'.

ICAI'S ANSWER

(i) Statement showing distribution of Overheads

Primary Distribution Summary

Item of cost	Basis of apportionment	Total (₹)	P (₹)	Q (₹)	R (₹)	X (₹)	Y (₹)
Direct wages	Actual	2,800	--	--	--	2,000	800
Rent and Rates	Floor area (4:5:7:2:2)	10,000	2,000	2,500	3,500	1,000	1,000
General lighting	Light points (4:2:3:1:2)	600	200	100	150	50	100
Indirect wages	Direct wages (50:15:45:20:8)	3,450	1,250	375	1,125	500	200

Item of cost	Basis of apportionment	Total (₹)	P (₹)	Q (₹)	R (₹)	X (₹)	Y (₹)
Power	Horse Power of machines used (10:8:10:2:5)	3,500	1,000	800	1,000	200	500
Depreciation of machinery	Value of machinery (10:8:10:2:5)	70,000	20,000	16,000	20,000	4,000	10,000
Sundries	Direct wages (50:15:45:20:8)	13,800	5,000	1,500	4,500	2,000	800
Total		**1,04,150**	**29,450**	**21,275**	**30,275**	**9,750**	**13,400**

Secondary Distribution using simultaneous equation method:

Overheads of service cost centres

Let, X be the overhead of service cost centre X

Y be the overhead of service cost centre Y

$X = 9,750 + 0.10\ Y$

$Y = 13,400 + 0.10\ X$

Substituting the value of Y in X we get

$X = 9,750 + 0.10\ (13,400 + 0.10\ X)$

$X = 9,750 + 1,340 + 0.01\ X$

$0.99\ X = 11,090$

$\therefore \mathbf{X} = ₹\ 11,202$

$\therefore \mathbf{Y} = 13,400 + 0.10 \times 11,202$

$= ₹\ 14,520.20$

Secondary Distribution Summary

Particulars	Total (₹)	P (₹)	Q (₹)	R (₹)
Allocated and Apportioned over-heads as per primary distribution		29,450.00	21,275.00	30,275.00
X	11,202.00	5,040.90	1,680.30	3,360.60
Y	14,520.20	5,082.07	3,630.05	4,356.06
Total		39,572.97	26,585.35	37,991.66

(ii) Calculation of Overhead recovery rate per hour

	P (₹)	Q (₹)	R (₹)
Total overheads cost	**39,572.97**	**26,585.35**	**37,991.66**
Working hours	13,191	7,598	14,995
Rate per hour (₹)	**3**	**3.50**	**2.53**

(iii) Cost of Product A

	(₹)
Direct material	65.00
Direct labour	40.00
Prime cost	105.00
Production on overheads	
P 6 hours × ₹ 3 = ₹ 18	
Q 5 hours × ₹ 3.50 = ₹ 17.50	
R 2 hours × ₹ 2.53 = ₹ 5.06	40.56
Total cost	**145.56**

Note: Secondary Distribution can also be done using repeated distribution Method

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362 RTP **Sep 2026** Case scenario · Q1(i) to Q1(v)

CASE SCENARIO

Apex Mobility Systems Ltd. manufactures smart airport luggage carts used in major international airports. The company operates a centralized factory with two production departments—Fabrication and Assembly, and two service departments—Stores and Maintenance.

Due to rising operational costs and increased competition, management has decided to analyze overhead allocation and costing accuracy for better pricing and profitability decisions.

For the financial year ending 31st March 2026, the following budgeted data is available:

Allocated Overhead Costs (₹)

Department	Amount (₹)
Fabrication	18,20,000
Assembly	9,60,000
Stores	3,00,000
Maintenance	2,40,000

Other Overheads (₹)

Particulars	Amount (₹)
Factory Rent	18,00,000
Building Insurance	2,00,000
Plant Insurance	2,40,000
Depreciation on Machinery	3,20,000
Canteen Subsidy	5,00,000

Direct Costs (₹)

Department	Material (₹)	Labour (₹)	Total (₹)
Fabrication	75,00,000	10,00,000	85,00,000
Assembly	2,00,000	15,00,000	17,00,000

Additional Information

Basis	Fabrication	Assembly	Stores	Maintenance
Floor area (sq. m.)	30,000	12,000	3,000	5,000
Plant value (₹)	20,00,000	10,00,000	1,00,000	2,00,000
Store requisitions	4,000	2,000	—	—
Maintenance hours	3,000	2,500	500	—
No. of employees	150	100	40	20
Machine hours	35,00,000	80,000	—	—
Labour hours	80,000	30,00,000	—	—

The company also received a **special job order (Job No. AMS-101)** with the following details:

- Direct Material = ₹ 1,50,000
- Direct Labour:
  - Fabrication: 300 hours @ ₹ 20/hr
  - Assembly: 200 hours @ ₹ 20/hr
- Machine hours:
  - Fabrication: 250 hours
  - Assembly: 180 hours

You are required to answer the following questions:

Q1(i)

Total overhead allocated from Stores Department to production Department is:

A ₹ 2,53,099

B ₹ 5,43,669

C ₹ 5,06,197

D ₹ 5,01,486

Q1(ii)

Total overhead allocated from Maintenance Department to other Department is:

A ₹ 5,43,669

B ₹ 5,01,486

C ₹ 5,06,197

D ₹ 42,186

Q1(iii)

Most appropriate overhead recovery rate for Fabrication Department is:

A ₹ 1.21 per machine hour

B ₹ 12 per machine hour

C ₹ 20 per machine hour

D ₹ 4.56 per machine hour

Q1(iv)

Most appropriate overhead recovery rate for Assembly Department is:

A ₹ 18 per labour hour

B ₹ 2.50 per labour hour

C ₹ 20 per labour hour

D ₹ 0.71 per labour hour

Q1(v)

Full production cost of Job No. AMS-101 is:

A ₹ 2,20,000

B ₹ 2,50,099

C ₹ 1,60,445

D ₹ 3,62,525

ICAI'S ANSWERS

ICAI's working for this case

Working Note:

Table of Primary Distribution of Overheads and Direct Costs

Particulars	Basis of Apportionment	Total Amount	Production Department		Service Departments	
			Fabrication	Assembly	Stores	Maintenance
Overheads Allocated	Allocation	33,20,000	18,20,000	9,60,000	3,00,000	2,40,000
Direct Costs	Actual	1,02,00,000	85,00,000	17,00,000	---	---
Total			1,03,20,000	26,60,000	3,00,000	2,40,000

Other Overheads

Particulars	Basis	Total (₹)	Fabrication	Assembly	Stores	Maintenance
Factory Rent	Floor Area (30:12:3:5)	18,00,000	10,80,000	4,32,000	1,08,000	1,80,000
Building Insurance	Floor Area	2,00,000	1,20,000	48,000	12,000	20,000
Plant Insurance	Plant Value (20:10:1:2)	2,40,000	1,45,455	72,727	7,273	14,545
Depreciation	Plant Value	3,20,000	1,93,939	96,970	9,697	19,394
Canteen Subsidy	Employees (150:100:40:20)	5,00,000	2,41,935	1,61,290	64,516	32,258
Total		30,60,000	17,81,329	8,10,987	2,01,486	2,66,197

Total after Primary Distribution

Particulars	Total (₹)	Fabrication	Assembly	Stores	Maintenance
Total Overheads	63,80,000	36,01330	17,70,987	5,01,486	5,06,197

Re-distribution of Service Departments' Expenses (Step Method)

Particulars	Basis of Apportionment	Production Department	Service Departments		
		Fabrication	**Assembly**	**Stores**	**Maintenance**
Overheads as per Primary distribution	As per Primary distribution	36,01,330	17,70,987	5,01,486	5,06,197
Maintenance Department Cost	Maintenance Hours (30:25:5:-)	2,53,099	2,10,915	42,183	(5,06,197)
		38,54,429	19,81,902	5,43,669	---
Stores Department	No. of Stores Requisition (5:2:-:-)	3,88,335	1,55,334	(5,43,669)	
		42,42,764	21,37,236	---	---

Q1(i)

(B) ₹ 5,43,669

Total overhead allocated from Stores Department

= ₹ 5,43,669

Q1(ii)

(C) ₹ 5,06,197

Total overhead allocated from Maintenance Department

= ₹ 5,06,197

Q1(iii)

(A) ₹ 1.21 per machine hour

Overhead Recovery Rate - Fabrication Department

Basis: Machine Hours = 35,00,000

Rate = 42,42,764/35,00,000 = ₹ 1.21 per machine hour

Q1(iv)

(D) ₹ 0.71 per labour hour

Overhead Recovery Rate - Assembly Department

Basis: Labour Hours = 30,00,000

Rate = 21,37,236/30,00,000 = ₹ 0.71 per labour hour

Q1(v)

(C) ₹ 1,60,445

Calculation of Full Production Cost (Job AMS-101)

Particulars	Amount (₹)
Direct Materials	1,50,000
Direct Labour:	
- Fabrication Deptt. (300 hours × ₹ 20)	6,000
- Assembly Deptt. (200 hours × ₹ 20)	4,000
Production Overheads:	
- Fabrication Deptt. (250 hours × ₹ 1.21)	303
- Assembly Deptt. (200 hours × ₹ 0.71)	142
Total Production Cost	1,60,445

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363 RTP **Old P3 May 2023** Q3 old scheme

SANDY Ltd. is a manufacturing company having three production departments, 'A', 'B' and 'C' and two service departments 'X' and 'Y'. The following is the budget for December 2022:

	Total (₹)	A (₹)	B (₹)	C (₹)	X (₹)	Y (₹)
Direct material		1,60,000	3,20,000	6,40,000	3,20,000	1,60,000
Direct wages		8,00,000	3,20,000	12,80,000	1,60,000	3,20,000
Factory rent	6,40,000					
Power	4,00,000					

	Total (₹)	A (₹)	B (₹)	C (₹)	X (₹)	Y (₹)
Depreciation	1,60,000					
Other overheads	14,40,000					
Additional information:						
Area (Sq. ft.)		800	400	800	400	800
Capital value of assets (₹) lakhs)		32	64	32	16	16
Machine hours		1,600	3,200	6,400	1,600	1,600
Horsepower of machines		80	64	32	24	40

Apportionment of expenses of service departments is as under:

	A	B	C	X	Y
Service Dept. 'X'	72	24	48	–	16
Service Dept. 'Y'	96	56	–	8	–

Required:

(i) PREPARE a statement showing distribution of overheads to various departments.

(ii) PREPARE a statement showing re-distribution of service departments expenses to production departments using Repeated Distribution method. Also CALCULATE machine hour rate of the production departments 'A', 'B', 'C'.

ICAI'S ANSWER

(i) Overhead Distribution Summary

	Basis	Total (₹)	A (₹)	B (₹)	C (₹)	X (₹)	Y (₹)
Direct materials	Direct	–	–	–	–	3,20,000	1,60,000
Direct wages	Direct	–	–	–	–	1,60,000	3,20,000
Factory rent (2:1:2:1:2)	Area	6,40,000	1,60,000	80,000	1,60,000	80,000	1,60,000
Power (10:16:16:3:5)*	H.P. × Machine Hrs.	4,00,000	80,000	1,28,000	1,28,000	24,000	40,000
Depreciation (2:4:2:1:1)	Capital value of assets	1,60,000	32,000	64,000	32,000	16,000	16,000

	Basis	Total (₹)	A (₹)	B (₹)	C (₹)	X (₹)	Y (₹)
Other overheads (1:2:4:1:1)	Machine hrs.	14,40,000	1,60,000	3,20,000	6,40,000	1,60,000	1,60,000
Total		26,40,000	4,32,000	5,92,000	9,60,000	7,60,000	8,56,000

*{(1600×80) : (3200×64) : (6400×32) : (1600×24) : (1600×40)}

(1,28,000 : 2,04,800 : 2,04,800 : 38,400 : 64,000)

(10:16:16:3:5)

(ii) Redistribution of service department's expense using repeated distribution Method:

	A (₹)	B (₹)	C (₹)	X (₹)	Y (₹)
Total overheads	4,32,000	5,92,000	9,60,000	7,60,000	8,56,000
Dept. X overhead apportioned in the ratio (72:24:48: —:16)	3,42,000	1,14,000	2,28,000	-7,60,000	76,000
Dept. Y overhead apportioned in the ratio (96:56: —:8: —)	5,59,200	3,26,200	-	46,600	-9,32,000
Dept. X overhead apportioned in the ratio (72:24:48: —:16)	20,970	6,990	13,980	-46,600	4,660
Dept. Y overhead apportioned in the ratio (96:56: —:8: —)	2,796	1,631	-	233	-4,660
Dept. X overhead apportioned in the ratio (72:24:48: —:16)	105	35	70	-233	23
Dept. Y overhead apportioned in the ratio (96:56: —:8: —)	15	8	-	-	-23
	13,57,086	10,40,864	12,02,050	-	-

Calculation of machine hour rate

		A	B	C
A	Total overheads (₹)	13,57,086	10,40,864	12,02,050
B	Machine hours	1,600	3,200	6,400
C	**Machine hour rate (₹) [A ÷ B]**	**848.18**	**325.27**	**187.82**

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364 RTP **Old P3 May 2022** Q3 old scheme

Pretz Ltd. is a manufacturing company having two production departments, 'A' & 'B' and two service

departments 'X' & 'Y'. The following is the budget for March, 2022:

	Total (₹)	A (₹)	B (₹)	X (₹)	Y (₹)
Direct material		2,00,000	4,00,000	4,00,000	2,00,000
Direct wages		10,00,000	4,00,000	2,00,000	4,00,000
Factory rent	9,00,000				
Power (Machine)	5,10,000				
Depreciation	2,00,000				
General Lighting	3,00,000				
Perquisites	4,00,000				
Additional information:					
Area (Sq. ft.)		500	250	250	500
Capital value of assets (₹ lakhs)		40	80	20	20
Light Points		10	20	10	10
Machine hours		1,000	2,000	1,000	1,000
Horse power of machines		50	40	15	25

A technical assessment of the apportionment of expenses of service departments is as under:

	A	B	X	Y
Service Dept. 'X' (%)	55	25	–	20
Service Dept. 'Y' (%)	60	35	5	–

You are required to:

(a) PREPARE a statement showing distribution of overheads to various departments.

(b) PREPARE a statement showing re-distribution of service departments expenses to production departments using-

(i) Simultaneous equation method

(ii) Trial and error method

(iii) Repeated Distribution Method.

ICAI'S ANSWER

(a) Primary Distribution of Overheads

	Basis	Total (₹)	A (₹)	B (₹)	X (₹)	Y (₹)
Direct materials	Direct	6,00,000	–	–	4,00,000	2,00,000

	Basis	Total (₹)	A (₹)	B (₹)	X (₹)	Y (₹)
Direct wages	Direct	6,00,000	–	–	2,00,000	4,00,000
Factory rent (2:1:1:2)	Area	9,00,000	3,00,000	1,50,000	1,50,000	3,00,000
Power (Machine) (10:16:3:5)*	H.P. × Machine Hrs.	5,10,000	1,50,000	2,40,000	45,000	75,000
Depreciation (2:4:1:1)	Capital value	2,00,000	50,000	1,00,000	25,000	25,000
General Lighting (1:2:1:1)	Light Points	3,00,000	60,000	1,20,000	60,000	60,000
Perquisites (5:2:1:2)	Direct Wages	4,00,000	2,00,000	80,000	40,000	80,000
		35,10,000	**7,60,000**	**6,90,000**	**9,20,000**	**11,40,000**

*{(1000×50) : (2000×40) : (1000×15) : (1000×25)}

(50000 : 80000 : 15000 : 25000)

(10 : 16 : 3 : 5)

(b) (i) Redistribution of Service Department's expenses using 'Simultaneous equation method'

X = 9,20,000 + 0.05 Y

Y = 11,40,000 + 0.20 X

Substituting the value of X,

Y = 11,40,000 + 0.20 (9,20,000 + 0.05 Y)

= 13,24,000 + 0.01 Y

Y - 0.01Y = 13,24,000

$$Y = \frac{13,24,000}{0.99}$$

Y = ₹ 13,37,374

The total expense of **Y is ₹ 13,37,374** and that of **X is ₹ 9,86,869** *i.e.,* ₹ 9,20,000 + (0.05 × ₹ 13,37,374).

Distribution of Service departments' overheads to Production departments

	Production Departments	
	A (₹)	**B (₹)**
Overhead as per primary distribution	7,60,000	6,90,000
Dept- X (55% and 25% of ₹ 9,86,869)	5,42,778	2,46,717
Dept- Y (60% and 35% of ₹ 13,37,374)	8,02,424	4,68,081
	21,05,202	**14,04,798**

(ii) Redistribution of Service Department's expenses using 'Trial and Error Method':

	Service Departments	
	X (₹)	**Y (₹)**
Overheads as per primary distribution	9,20,000	11,40,000
(i) Apportionment of Dept-X expenses to Dept-Y (20% of ₹ 9,20,000)	---	1,84,000
	---	13,24,000
(ii) Apportionment of Dept-Y expenses to Dept-X (5% of ₹ 13,24,000)	66,200	---
(i) Apportionment of Dept-X expenses to Dept-Y (20% of ₹ 66,200)	---	13,240
(ii) Apportionment of Dept-Y expenses to Dept-X (5% of ₹ 13,240)	662	---
(i) Apportionment of Dept-X expenses to Dept-Y (20% of ₹ 662)		132
(ii) Apportionment of Dept-Y expenses to Dept-X (5% of ₹ 132)	7	
Total	**9,86,869**	**13,37,372**

Distribution of Service departments' overheads to Production departments

	Production Departments	
	A (₹)	**B (₹)**
Overhead as per primary distribution	7,60,000	6,90,000
Dept- X (55% and 25% of ₹ 9,86,869)	5,42,778	2,46,717
Dept- Y (60% and 35% of ₹ 13,37,372)	8,02,423	4,68,080
	21,05,201	**14,04,797**

(iii) Redistribution of Service Department's expenses using 'repeated distribution method':

	A (₹)	B (₹)	X (₹)	Y (₹)
Overhead as per primary distribution	7,60,000	6,90,000	9,20,000	11,40,000
Dept. X overhead apportioned in the ratio (55:25:—:20)	5,06,000	2,30,000	(9,20,000)	1,84,000
Dept. Y overhead apportioned in the ratio (60:35:5: —)	7,94,400	4,63,400	66,200	(13,24,000)
Dept. X overhead apportioned in the ratio (55:25:—:20)	36,410	16,550	(66,200)	13,240
Dept. Y overhead apportioned in the ratio (60:35:5: —)	7,944	4,634	662	(13,240)

	A (₹)	B (₹)	X (₹)	Y (₹)
Dept. X overhead apportioned in the ratio (55:25:—:20)	364	166	(662)	132
Dept. Y overhead apportioned in the ratio (60:35:5: —)	79	46	7	(132)
Dept. X overhead apportioned in the ratio (55:25:—:20)	4	3	(7)	-
	21,05,201	**14,04,799**	–	–

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365 RTP **Old P3 Dec 2021** Q3 old scheme

PL Ltd. has three production departments P1, P2 and P3 and two service departments S1 and S2. The following data are extracted from the records of the company for the month of October, 2020:

	(₹)
Rent and rates	12,50,000
General lighting	1,50,000
Indirect Wages	3,75,000
Power	5,00,000
Depreciation on machinery	10,00,000
Insurance of machinery	4,00,000

Other Information:

	P1	P2	P3	S1	S2
Direct wages (₹)	7,50,000	5,00,000	7,50,000	3,75,000	1,25,000
Horse Power of Machines used	60	30	50	10	–
Cost of machinery (₹)	60,00,000	80,00,000	1,00,00,000	5,00,000	5,00,000
Floor space (Sq. ft)	2,000	2,500	3,000	2,000	500
Number of light points	10	15	20	10	5
Production hours worked	6,225	4,050	4,100	–	–

Expenses of the service departments S1 and S2 are reapportioned as below:

	P1	P2	P3	S1	S2
S1	20%	30%	40%	–	10%
S2	40%	20%	30%	10%	–

Required:

(i) COMPUTE overhead absorption rate per production hour of each production department.

(ii) DETERMINE the total cost of product X which is processed for manufacture in department P1, P2 and P3 for 5 hours, 3 hours and 4 hours respectively, given that its direct material cost is ₹ 12,500 and direct labour cost is ₹ 7,500.

ICAI'S ANSWER

Primary Distribution Summary

Item of cost	Basis of apportionment	Total (₹)	P1 (₹)	P2 (₹)	P3 (₹)	S1 (₹)	S2 (₹)
Direct wages	Actual	5,00,000	--	--	--	3,75,000	1,25,000
Rent and Rates	Floor area (4 : 5 : 6 : 4 : 1)	12,50,000	2,50,000	3,12,500	3,75,000	2,50,000	62,500
General lighting	Light points (2 : 3 : 4 : 2 : 1)	1,50,000	25,000	37,500	50,000	25,000	12,500
Indirect wages	Direct wages (6 : 4 : 6 : 3 : 1)	3,75,000	1,12,500	75,000	1,12,500	56,250	18,750
Power	Horse Power of machines used (6 : 3 : 5 : 1)	5,00,000	2,00,000	1,00,000	1,66,667	33,333	–
Depreciation of machinery	Value of machinery (12 : 16 : 20 : 1 : 1)	10,00,000	2,40,000	3,20,000	4,00,000	20,000	20,000
Insurance of machinery	Value of machinery (12 : 16 : 20 : 1 : 1)	4,00,000	96,000	1,28,000	1,60,000	8,000	8,000
		41,75,000	**9,23,500**	**9,73,000**	**12,64,167**	**7,67,583**	**2,46,750**

Overheads of service cost centres

Let S1 be the overhead of service cost centre S1 and S2 be the overhead of service cost centre S2.

S1 = 7,67,583 + 0.10 S2

S2 = 2,46,750 + 0.10 S1

Substituting the value of S2 in S1 we get

S1 = 7,67,583 + 0.10 (2,46,750 + 0.10 S1)

S1 = 7,67,583 + 24,675 + 0.01 S1

0.99 S1 = 7,92,258

∴ S1 = ₹ 8,00,260

$\therefore$ S2 = 2,46,750 + 0.10 × 8,00,260

= ₹ 3,26,776

Secondary Distribution Summary

Particulars	Total (₹)	P1 (₹)	P2 (₹)	P3 (₹)
Allocated and Apportioned over-heads as per primary distribution	31,60,667	9,23,500	9,73,000	12,64,167
S1	8,00,260	1,60,052	2,40,078	3,20,104
S2	3,26,776	1,30,710	65,355	98,033
		12,14,262	**12,78,433**	**16,82,304**

(i) **Overhead rate per hour**

	P1	P2	P3
Total overheads cost (₹)	12,14,262	12,78,433	16,82,304
Production hours worked	6,225	4,050	4,100
Rate per hour (₹)	**195.06**	**315.67**	**410.32**

(ii) **Cost of Product X**

		(₹)
Direct material		12,500.00
Direct labour		7,500.00
Prime cost		**20,000.00**
Production on overheads		
P1 5 hours × ₹ 195.06 =	975.30	
P2 3 hours × ₹ 315.67 =	947.01	
P3 4 hours × ₹ 410.32 =	1,641.28	3,563.59
Factory cost		**23,563.59**

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366 RTP **Old P3 May 2020** Q3 old scheme

ABC Ltd. has three production departments P1, P2 and P3 and two service departments S1 and S2. The

following data are extracted from the records of the company for the month of January, 2020:

	(₹)
Rent and rates	6,25,000
General lighting	7,50,000
Indirect wages	1,87,500
Power	25,00,000
Depreciation on machinery	5,00,000
Insurance of machinery	2,00,000

Other Information:

	P1	P2	P3	S1	S2
Direct wages (₹)	3,75,000	2,50,000	3,75,000	1,87,500	62,500
Horse Power of Machines used	60	30	50	10	–
Cost of machinery (₹)	30,00,000	40,00,000	50,00,000	2,50,000	2,50,000
Floor space (Sq. ft)	2,000	2,500	3,000	2,000	500
Number of light points	10	15	20	10	5
Production hours worked	6,225	4,050	4,100	–	–

Expenses of the service departments S1 and S2 are reapportioned as below:

	P1	P2	P3	S1	S2
S1	20%	30%	40%	–	10%
S2	40%	20%	30%	10%	–

Required:

(i) COMPUTE overhead absorption rate per production hour for each production department.

(ii) DETERMINE the total cost of product X which is processed for manufacture in department P1, P2 and P3 for 5 hours, 3 hours and 4 hours respectively, given that its direct material cost is ₹6,250 and direct labour cost is ₹3,750.

ICAI'S ANSWER

Primary Distribution Summary

Item of cost	Basis of apportionment	Total (₹)	P1 (₹)	P2 (₹)	P3 (₹)	S1 (₹)	S2 (₹)
Direct wages	Actual	2,50,000	--	--	--	1,87,500	62,500

Item of cost	Basis of apportionment	Total (₹)	P1 (₹)	P2 (₹)	P3 (₹)	S1 (₹)	S2 (₹)
Rent and rates	Floor area (4 : 5 : 6 : 4 : 1)	6,25,000	1,25,000	1,56,250	1,87,500	1,25,000	31,250
General lighting	Light points (2 : 3 : 4 : 2 : 1)	7,50,000	1,25,000	1,87,500	2,50,000	1,25,000	62,500
Indirect wages	Direct wages (6 : 4 : 6 : 3 : 1)	1,87,500	56,250	37,500	56,250	28,125	9,375
Power	Horse Power of machines used (6 : 3 : 5 : 1)	25,00,000	10,00,000	5,00,000	8,33,333	1,66,667	–
Depreciation on machinery	Value of machinery (12:16:20:1:1)	5,00,000	1,20,000	1,60,000	2,00,000	10,000	10,000
Insurance of machinery	Value of machinery (12:16:20:1:1)	2,00,000	48,000	64,000	80,000	4,000	4,000
		50,12,500	14,74,250	11,05,250	16,07,083	6,46,292	1,79,625

Overheads of service cost centres:

Let S1 be the overhead of service cost centre S1 and S2 be the overhead of service cost centre S2.

S1 = 6,46,292 + 0.10 S2

S2 = 1,79,625 + 0.10 S1

Substituting the value of S2 in S1 we get

S1 = 6,46,292 + 0.10 (1,79,625 + 0.10 S1)

S1 = 6,46,292 + 17,962.5 + 0.01 S1

0.99 S1 = 6,64,254.5

∴ S1 = ₹6,70,964

∴ S2 = 1,79,625 + 0.10 × 6,70,964

= ₹2,46,721.4

Secondary Distribution Summary

Particulars	Total (₹)	P1 (₹)	P2 (₹)	P3 (₹)
Allocated and Apportioned overheads as per primary distribution	41,86,583	14,74,250	11,05,250	16,07,083
S1	6,70,964	1,34,192.8	2,01,289.2	2,68,385.6
S2	2,46,721.4	98,688.6	49,344.3	74,016.5
		17,07,131.4	13,55,883.5	19,49,485.1

(i) Overhead rate per hour

	P1	P2	P3
Total overheads cost (₹)	17,07,131.4	13,55,883.5	19,49,485.1
Production hours worked	6,225	4,050	4,100
Rate per hour (₹)	274.24	334.79	475.48

(ii) Cost of Product X

	(₹)
Direct material	6,250.00
Direct labour	3,750.00
Prime cost	10,000.00
Production on overheads	
P1 5 hours × ₹ 274.24 = 1,371.20	
P2 3 hours × ₹ 334.79 = 1,004.37	
P3 4 hours × ₹ 475.48 = 1,901.92	4,277.49
Factory cost	14,277.49

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367 MTP **May 2026 S-I** Q3(b) 9 marks

The Union Ltd. has the following account balances and distribution of direct charges on 31st March, 2025.

	Total	Production Depts.		Service Depts.	
		Machine Shop	Packing	General Plant	Stores
Allocated Overheads:	(₹)	(₹)	(₹)	(₹)	(₹)
Indirect labour	29,000	8,000	6,000	4,000	11,000
Maintenance Material	9,900	3,400	1,600	2,100	2,800
Misc. supplies	5,900	1,500	2,900	900	600
Supervisor's salary	16,000	--	--	16,000	--
Cost & payroll salary	80,000	--	--	80,000	--
Overheads to be apportioned:					
Power	78,000				
Rent	72,000				

	Total	Production Depts.	Service Depts.
Fuel and Heat	60,000		
Insurance	12,000		
Taxes	8,400		
Depreciation	1,20,000		

The following data were compiled by means of the factory survey made in the previous year:

	Floor Space	Radiator Section	No. of employees	Investment	H.P. hours
Machine Shop	2,000 Sq. ft.	45	20	8,00,000	3,500
Packing	800 Sq. ft.	90	12	2,40,000	500
General Plant	400 Sq. ft.	30	4	80,000	-
Stores & maintenance	1,600 Sq. ft.	60	8	1,60,000	1,000

Expenses charged to the stores departments are to be distributed to the other departments by the following percentages:

Machine shop 50%; Packing 20%; General Plant 30%;

General Plant overheads is distributed on the basis of number of employees.

(a) PREPARE an overhead distribution statement with supporting schedules to show computations and basis of distribution.

(b) DETERMINE the service department distribution by simultaneous equation method.

ICAI'S ANSWER

Overhead Distribution Statement

	Production Departments		Service Departments	
	Machine Shops	**Packing**	**General Plant**	**Stores**
Allocated Overheads:	(₹)	(₹)	(₹)	(₹)
Indirect labour	8,000	6,000	4,000	11,000
Maintenance Material	3,400	1,600	2,100	2,800
Misc. supplies	1,500	2,900	900	600
Supervisor's salary	--	--	16,000	--
Cost & payroll salary	--	--	80,000	--
Total allocated overheads	12,900	10,500	1,03,000	14,400
Add: Apportioned Overheads (As per Schedule below)	1,84,350	70,125	22,775	73,150

	Production Departments		Service Departments	
	1,97,250	80,625	1,25,775	87,550

Schedule of Apportionment of Overheads

Item of Cost	Basis	Production Departments		Service Departments	
		Machine Shops (₹)	Packing (₹)	General Plant (₹)	Stores (₹)
Power	HP hours (7 : 1 : - : 2)	54,600	7,800	--	15,600
Rent	Floor space (5 : 2 : 1 : 4)	30,000	12,000	6,000	24,000
Fuel & Heat	Radiator sec. (3 : 6 : 2 : 4)	12,000	24,000	8,000	16,000
Insurance	Investment (10 : 3 : 1 : 2)	7,500	2,250	750	1,500
Taxes	Investment (10 : 3 : 1 : 2)	5,250	1,575	525	1,050
Depreciation	Investment (10 : 3 : 1 : 2)	75,000	22,500	7,500	15,000
		1,84,350	70,125	22,775	73,150

(b) Re-distribution of Overheads of Service Departments to Production Departments:

Let, the total overheads of General Plant = 'a' and the total overheads of Stores = 'b'

a = 1,25,775 + 0.3b .....(i)

b = 87,550 + 0.2a .....(ii)

Putting the value of 'b' in equation no. (i)

a = 1,25,775 + 0.3 (87,550 + 0.2a)

Or a = 1,25,775 + 26,265 + 0.06a

Or 0.94a = 1,52,040 Or a = 1,61,745 (appx.)

Putting the value of a = 1,61,745 in equation no. (ii) to get the value of 'b'

b = 87,550 + 0.2 × 1,61,745 = 1,19,899

Secondary Distribution Summary

Particulars	Total (₹)	Machine Shops (₹)	Packing (₹)
Allocated and Apportioned overheads as per Primary distribution	2,77,875	1,97,250.00	80,625.00
- General Plant	1,61,745	80,872.50 (1,61,745 × 5/10)	48,523.50 (1,61,745 × 3/10)
- Stores	1,19,899	59,949.50 (1,19,899 × 50%)	23,979.80 (1,19,899 × 20%)
		3,38,072.00	1,53,128.30

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368 MTP **May 2026 S-II** Q4(a) 9 marks

ABC Steel Ltd has its own Captive Power Generation Unit, which has two users, Rolling Mill Department and Finishing Department. When the plans were prepared for the power plant, top management decided that its practical capacity should be 2,00,000 machine hours. Annual budgeted practical capacity fixed costs are ₹ 15,00,000 and budgeted variable costs are ₹ 5 per machine-hour. The following data are available:

	Rolling Mill Department	Finishing Department	Total
Actual Usage in 2025-26 (Machine hours)	65,000	45,000	1,10,000
Practical capacity for each department (Machine hours)	1,10,000	90,000	2,00,000

Required

(i) ALLOCATE the Power Generation Unit cost to the rolling mill and the finishing department using a single rate method in which the budgeted rate is calculated using practical capacity and costs are allocated based on actual usage.

(ii) ALLOCATE the Power Generation Unit cost to the rolling mill and the finishing department, using the dual -rate method in which fixed costs are allocated based on practical capacity and variable costs are allocated based on actual usage.

(iii) ALLOCATE the Power Generation Unit cost to the rolling mill and the finishing department using the dual-rate method in which the fixed-cost rate is calculated using practical capacity, but fixed costs are allocated to the cutting and welding department based on actual usage. Variable costs are allocated based on actual usage.

(iv) COMMENT on your results in requirements (i), (ii) and (iii).

ICAI'S ANSWER

Working Notes:

1. *Fixed practical capacity cost per machine hour:*

Practical capacity (machine hours)	2,00,000
Practical capacity fixed costs ₹)	15,00,000
Fixed practical capacity cost per machine hour	₹ 7.50
₹ 15,00,000 ÷ 2,00,000 hours)	

2. Budgeted rate per machine hour (using practical capacity):

= Fixed practical capacity cost per machine hour + Budgeted variable cost per machine hour

= ₹ 7.50 + ₹ 5 = ₹ 12.50

(i) Statement showing Power Plant's cost allocation to the Cutting & Welding departments by using single rate method on actual usage of machine hours.

	Rolling Mill Department (₹)	Finishing Department (₹)	Total (₹)
Power plants cost allocation by using actual usage (machine hours) (Refer to Working Note 2)	8,12,500 (65,000 hours × ₹ 12.50)	5,62,500 (45,000 hours × ₹ 12.50)	13,75,000

(ii) Statement showing Power Plant's cost allocation to the Cutting & Welding departments by using dual rate method.

	Rolling Mill Department (₹)	Finishing Department (₹)	Total (₹)
Fixed Cost	8,25,000	6,75,000	15,00,000
(Allocated on practical capacity for each department i.e.): (1,10,000 hours : 90,000 hours)	$\left(\frac{₹\ 15,00,000 \times 11}{20}\right)$	$\left(\frac{₹\ 15,00,000 \times 9}{20}\right)$	
Variable cost	3,25,000	2,25,000	5,50,000
(Based on actual usage of machine hours)	(65,000 hours × ₹ 5)	(45,000 hours × ₹ 5)	
Total cost	11,50,000	9,00,000	20,50,000

(iii) Statement showing Power Plant's cost allocation to the Cutting & Welding Departments using dual rate method

	Rolling Mill Department (₹)	*Finishing Department* (₹)	Total (₹)
Fixed Cost	4,87,500	3,37,500	8,25,000
Allocation of fixed cost on actual usage basis (Refer to Working Note 1)	(65,000 hours × ₹ 7.50)	(45,000 hours × ₹ 7.50)	
Variable cost	3,25,000	2,25,000	5,50,000
(Based on actual usage)	(65,000 hours × ₹ 5)	(45,000 hours × ₹ 5)	
Total cost	8,12,500	5,62,500	13,75,000

(iv) Comments: Under dual rate method, under (iii) and single rate method under (i), the allocation of fixed cost of practical capacity of plant over each department are based on single rate. The major advantage of this approach is that the user departments are allocated fixed capacity costs only for the capacity used. The unused capacity cost ₹ 6,75,000 ₹ 15,00,000 – ₹ 8,25,000) will not be allocated to the user departments. This highlights the cost of unused capacity.

Under (ii) fixed cost of capacity are allocated to operating departments on the basis of practical capacity, so all fixed costs are allocated and there is no unused capacity identified with the power plant.

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369 MTP **Sep 2025 S-II** Case scenario · Q2(a)(i) to Q2(a)(iii) 12 marks

CASE SCENARIO

IOT Solutions Tech Pvt. Ltd., a leading provider of innovative IoT solutions and cutting-edge technology products, outlines the following budgeted and actual hours used for its two central corporate support departments, Finance and IT. These departments collaborate with one another and provide essential support to the company's two manufacturing divisions: Mobile Devices (MD) and Desktop Solutions (DS).

Particulars	Support Department		Operating Department	
	Finance	**IT**	**MD**	**DS**
Budgeted Usage				
Finance (hours)		750	4,500	2,250
(%)		10%	60%	30%
IT (hours)	7,500		67,500	75,000
(%)	5%		45%	50%
Actual Usage				
Finance (hours)		1,200	1,200	3,600
(%)		20%	20%	60%
IT (hours)	6,000		79,800	34,200
(%)	5%		66.5%	28.5%

The overhead costs under these two support departments before any inter-department cost allocations are also outlined as below:

Particulars	Support Department	
	Finance (₹)	IT (₹)
Budgeted fixed overhead costs	22,96,800	30,30,500
Actual variable overhead costs	12,76,000	38,28,000

The company uses dual- rate method of apportioning support departments cost to its operating departments where fixed overhead costs are apportioned using budgeted hours used by other departments, and variable overhead costs are apportioned using actual hours used by other departments.

You are required to PREPARE a statement apportioning the costs of support departments over the operating departments using:

Q2(a)(i) 4 marks

Direct method,

Q2(a)(ii) 4 marks

Step-down method (apportioning the Finance department costs first), and

Q2(a)(iii) 4 marks

Reciprocal method (using simultaneous equation).

ICAI'S ANSWERS

Q2(a)(i)

Direct method of apportioning the costs of support departments over the operating departments.

Items of cost	Basis of apportionment	Total (₹)	Operating Departments	
			MD (₹)	DS (₹)
Fixed overhead costs:				
Finance	4,500 : 2,250	22,96,800	15,31,200	7,65,600
IT	67,500 : 75,000	30,30,500	14,35,500	15,95,000
Total fixed overhead costs apportioned (A)		53,27,300	29,66,700	23,60,600
Variable overhead costs:				
Finance	1,200 : 3,600	12,76,000	3,19,000	9,57,000
IT	79,800 : 34,200	38,28,000	26,79,600	11,48,400
Total variable overhead costs apportioned (B)		51,04,000	29,98,600	21,05,400
Total overhead costs (A + B)		1,04,31,300	59,65,300	44,66,000

Q2(a)(ii)

Step-down method of apportioning the costs of support departments over the operating departments (apportioning the finance department costs first)

Items of cost	Basis of apportionment	Total (₹)	Support Departments		Operating Departments	
			Finance (₹)	IT (₹)	MD (₹)	DS (₹)
Fixed overhead costs		53,27,300	22,96,800	30,30,500		
Distribution of Finance Department	750:4,500:2,250	--	(22,96,800)	2,29,680	13,78,080	6,89,040
		53,27,300	--	32,60,180	13,78,080	6,89,040
Distribution of IT Department	67,500: 75,000	--	--	(32,60,180)	15,44,296	17,15,884
Total fixed overhead costs apportioned (A)		53,27,300	--	--	29,22,376	24,04,924
Variable overhead costs		51,04,000	12,76,000	38,28,000		
Distribution of Finance Department	1,200:1,200:3,600	--	(12,76,000)	2,55,200	2,55,200	7,65,600
		51,04,000	--	40,83,200	2,55,200	7,65,600
Distribution of IT Department	79,800: 34,200	--	--	(40,83,200)	28,58,240	12,24,960
Total variable overhead costs apportioned (B)		51,04,000	--	--	31,13,440	19,90,560
Total overhead costs (A + B)		1,04,31,300	--	--	60,35,816	43,95,484

Q2(a)(iii)

Reciprocal method (using simultaneous equation) of apportioning the costs of support departments over the operating departments

Let F stand for Finance department expenses and T for IT department expenses.

Then, **for fixed overhead costs**-

F = ₹ 22,96,800 + 0.05T

T = ₹ 30,30,500 + 0.10F

Substituting the value of F,

T = ₹ 30,30,500 + 0.10 (22,96,800 + 0.05T)

= ₹ 30,30,500 + 2,29,680 + 0.005T

= ₹ 32,60,180 + 0.005T

T - 0.005T = ₹ 32,60,180

$$T = \frac{₹\ 32,60,180}{0.995}$$

= ₹ 32,76,563

Therefore,

F = ₹ 22,96,800 + 0.05T

= ₹ 22,96,800 + (0.05 x 32,76,563)

= ₹ 24,60,628

Now, **for Variable overhead costs**-

F = ₹ 12,76,000 + 0.05T

T = ₹ 38,28,000 + 0.20F

Substituting the value of F,

T = ₹ 38,28,000 + 0.20 (12,76,000 + 0.05T)

= ₹ 38,28,000 + 2,55,200 + 0.01T

= ₹ 40,83,200 + 0.01T

T - 0.01T = ₹ 40,83,200

$$T = \frac{₹\ 40,83,200}{0.99}$$

= ₹ 41,24,444

Therefore,

F = ₹ 12,76,000 + 0.05T

= ₹ 12,76,000 + (0.05 x 41,24,444)

= ₹ 14,82,222

Items of cost	Basis of apportionment	Total (₹)	Operating Departments	
			MD (₹)	**DS (₹)**
Fixed overhead costs:				
Distribution of Finance Department (₹ 24,60,628)	60%, 30%	22,14,565	14,76,377	7,38,188
Distribution of IT Department (₹ 32,76,563)	45%, 50%	31,12,735	14,74,453	16,38,282
Total fixed overhead costs apportioned (A)		**53,27,300**	**29,50,830**	**23,76,470**
Variable overhead costs:				

Items of cost	Basis of apportionment	Total (₹)	Operating Departments	
Distribution of Finance Department (₹ 14,82,222)	20%, 60%	11,85,778	2,96,445	8,89,333
Distribution of IT Department (₹ 41,24,444)	66.5%, 28.5%	39,18,222	27,42,755	11,75,467
Total variable overhead costs apportioned (B)		**51,04,000**	**30,39,200**	**20,64,800**
Total overhead costs (A + B)		**1,04,31,300**	**59,90,030**	**44,41,270**

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370 MTP **May 2025 S-I** Q5(b) 8 marks

Arnav Ltd. has three production departments M, N and O and two service departments P and Q. The following particulars are available for the month of September, 2024:

	(₹)
Lease rental	35,000
Power & Fuel	4,20,000
Wages to factory supervisor	6,400
Electricity	5,600
Depreciation on machinery	16,100
Depreciation on building	18,000
Payroll expenses	21,000
Canteen expenses	28,000
ESI and Provident Fund Contribution	58,000

Followings are the further details available:

Particulars	M	N	O	P	Q
Floor space (square meter)	1,200	1,000	1,600	400	800
Light points (nos.)	42	52	32	18	16
Cost of machines (₹)	12,00,000	10,00,000	14,00,000	4,00,000	6,00,000
No. of employees (nos.)	48	52	45	15	25
Direct Wages (₹)	1,72,800	1,66,400	1,53,000	36,000	53,000
HP of Machines	150	180	120	-	-
Working hours (hours)	1,240	1,600	1,200	1,440	1,440

The expenses of service department are to be allocated in the following manner:

	M	N	O	P	Q
P	30%	35%	25%	-	10%
Q	40%	25%	20%	15%	-

You are required to CALCULATE the overhead absorption rate per hour in respect of the three production departments.

ICAI'S ANSWER

Primary Distribution Summary

Item of cost	Basis of apportionment	Total	Production Dept.			Service Dept.	
		(₹)	M (₹)	N (₹)	O (₹)	P (₹)	Q (₹)
Lease rental	Floor space (6 : 5 : 8 : 2 : 4)	35,000	8,400	7,000	11,200	2,800	5,600
Power & Fuel	HP of Machines × Working hours (93: 144 : 72)	4,20,000	1,26,408	1,95,728	97,864	-	-
Supervisor's wages*	Working hours (31 : 40 : 30)	6,400	1,964	2,535	1,901	-	-
Electricity	Light points (21: 26: 16 : 9 : 8)	5,600	1,470	1,820	1,120	630	560
Depreciation on machinery	Value of machinery (6 : 5 : 7 : 2 : 3)	16,100	4,200	3,500	4,900	1,400	2,100
Depreciation on building	Floor space (6 : 5 : 8 : 2 : 4)	18,000	4,320	3,600	5,760	1,440	2,880
Payroll expenses	No. of employees (48: 52: 45: 15: 25)	21,000	5,448	5,903	5,108	1,703	2,838

Item of cost	Basis of apportionment	Total	Production Dept.		Service Dept.		
Canteen expenses	No. of employees (48: 52: 45: 15: 25)	28,000	7,265	7,870	6,811	2,270	3,784
ESI and PF contribution	Direct wages (864: 832: 765: 180: 265)	58,000	17,244	16,606	15,268	3,593	5,289
		6,08,100	**1,76,719**	**2,44,562**	**1,49,932**	**13,836**	**23,051**

* Wages to supervisor is to be distributed to production departments only.

Let 'P' be the overhead of service department P and 'Q' be the overhead of service department Q.

P = 13,836 + 0.15 Q

Q = 23,051 + 0.10 P

Substituting the value of Q in P we get

P = 13,836 + 0.15 (23,051 + 0.10 P)

P = 13,836 + 3,457.65 + 0.015 P

0.985 P = 17,293.65

∴P = ₹ 17,557

∴Q = 23,051 + 0.10 × 17,557

= ₹ 24,806.70 or ₹ 24,807

Secondary Distribution Summary

Particulars	Total (₹)	M (₹)	N (₹)	O (₹)
Allocated and Apportioned over-heads as per primary distribution	5,71,213	1,76,719	2,44,562	1,49,932
P (90% of ₹17,557)	15,801	5,267	6,145	4,389
Q (85% of ₹24,807)	21,086	9,923	6,202	4,961
		1,91,909	2,56,909	1,59,282

Overhead rate per hour

	M	N	O
Total overheads cost (₹)	1,91,909	2,56,909	1,59,282
Working hours	1,240	1,600	1,200
Rate per hour (₹)	154.77	160.57	132.74

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371 MTP **Jan 2025 S-II** Q6(d) 4 marks

DISCUSS in brief three main methods of allocating support departments costs to operating departments.

ICAI'S ANSWER

The three main methods of allocating support departments costs to operating departments are:

(i) ***Direct re-distribution method:*** Under this method, support department costs are directly apportioned to various production departments only. This method does not consider the service provided by one support department to another support department.

(ii) ***Step method:*** Under this method the cost of the support departments that serves the maximum numbers of departments is first apportioned to other support departments and production departments. After this the cost of support department serving the next largest number of departments is apportioned. In this manner we finally arrive on the cost of production departments only.

(iii) ***Reciprocal service method:*** This method recognises the fact that where there are two or more support departments they may render services to each other and, therefore, these inter-departmental services are to be given due weight while re-distributing the expenses of the support departments. The methods available for dealing with reciprocal services are:

(a) Simultaneous equation method

(b) Repeated distribution method

(c) Trial and error method.

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372 MTP **Old P3 May 2023 S-II** Q4(a) 10 marks old scheme

KV Ltd. manufactures luggage trolleys for airports. The factory, in which the company undertakes all of its production, has two production departments- 'Fabrication' and 'Assembly', and two service departments- 'Stores' and 'Maintenance'.

The following information have been extracted from the company's budget for the financial year ended 31st March, 2023:

Particulars	₹
Allocated Overhead Costs	
Fabrication Department	15,52,000
Assembly Department	7,44,000
Stores Department	2,36,000
Maintenance Department	1,96,000
Other Overheads	
Factory rent	15,28,000
Factory building insurance	1,72,000
Plant & machinery insurance	1,96,000
Plant & Machinery Depreciation	2,65,000

Particulars	₹
Subsidy for staffs' canteen	4,48,000

Direct Costs	₹	₹
Fabrication Department:		
Material	63,26,000	
Labour	8,62,000	71,88,000
Assembly Department:		
Material	1,42,000	
Labour	13,06,000	14,48,000

The following additional information is also provided:

	Fabrication Department	Assembly Department	Stores Department	Maintenance Department
Floor area (square meters)	24,000	10,000	2,500	3,500
Value of plant & machinery (₹)	16,50,000	7,50,000	75,000	1,75,000
No. of stores requisitions	3,600	1,400	---	---
Maintenance hours required	2,800	2,300	400	---
No. of employees	120	80	38	12
Machine hours	30,00,000	60,000		
Labour hours	70,000	26,00,000		

Required:

(i) PREPARE a table showing the distribution of overhead costs of the two service departments to the two production departments using step method; and

(ii) CALCULATE the most appropriate overhead recovery rate for each department.

ICAI'S ANSWER

(i) Table of Primary Distribution of Overheads

Particulars	Basis of Apportionment	Total Amount	Production Department		Service Departments	
			Fabrication	Assembly	Stores	Maintenance

Particulars	Basis of Apportionment	Total Amount	Production Department	Service Departments		
Overheads Allocated		27,28,000	15,52,000	7,44,000	2,36,000	1,96,000
Direct Costs	Actual	86,36,000	71,88,000	14,48,000	---	---
Other Overheads:						
Factory rent	Floor Area (48:20:5:7)	15,28,000	9,16,800	3,82,000	95,500	1,33,700
Factory building insurance	Floor Area (48:20:5:7)	1,72,000	1,03,200	43,000	10,750	15,050
Plant & Machinery insurance	Value of Plant & Machinery (66:30:3:7)	1,96,000	1,22,038	55,472	5,547	12,943
Plant & Machinery Depreciation	Value of Plant & Machinery (66:30:3:7)	2,65,000	1,65,000	75,000	7,500	17,500
Canteen Subsidy	No. of employees (60:40:19:6)	4,48,000	2,15,040	1,43,360	68,096	21,504
		1,39,73,000	1,02,62,078	28,90,832	4,23,393	3,96,697

Re-distribution of Service Departments' Expenses:

Particulars	Basis of Apportionment	Production Department	Service Departments		
		Fabrication	**Assembly**	**Stores**	**Maintenance**
Overheads as per Primary distribution	As per Primary distribution	1,02,62,078	28,90,832	4,23,393	3,96,697
Maintenance Department Cost	Maintenance Hours (28:23:4:-)	2,01,955	1,65,891	28,851	(3,96,697)
		1,04,64,033	30,56,723	4,52,244	---
Stores Department	No. of Stores Requisition (18:7:-:-)	3,25,616	1,26,628	(4,52,244)	
		1,07,89,649	31,83,351	---	---

(ii) Overhead Recovery Rate

Department	Apportioned Overhead (₹)	Basis of Overhead Recovery Rate	Overhead Recovery Rate (₹)
	(I)	**(II)**	**[(I) ÷ (II)]**
Fabrication	1,07,89,649	30,00,000 Machine Hours	3.60 per Machine Hour
Assembly	31,83,351	26,00,000 Labour Hours	1.22 per Labour Hour

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373 MTP Old P3 May 2022 S-I Q5(b) 10 marks old scheme

PM Ltd. has three Production Departments P1, P2, P3 and two Service Departments S1 and S2 details pertaining to which are as under:

	P1	P2	P3	S1	S2
Direct wages (₹)	60,000	40,000	60,000	30,000	3,900
Working hours	3,070	4,475	2,419	-	-
Value of machines (₹)	12,00,000	16,00,000	20,00,000	1,00,000	1,00,000
H.P. of machines	60	30	50	10	-
Light points	10	15	20	10	5
Floor space (sq. ft.)	2,000	2,500	3,000	2,000	500

The following figures extracted from the accounting records are relevant:

	(₹)
Rent and Rates	1,00,000
General Lighting	12,000
Indirect Wages	38,780
Power	30,000
Depreciation on Machines	2,00,000
Sundries	1,93,900

The expenses of the service departments are allocated as under:

	P1	P2	P3	S1	S2
S1	20%	30%	40%	-	10%
S2	40%	20%	30%	10%	-

DETERMINE the total cost of product X which is processed for manufacture in Departments P1, P2 and P3 for 4, 5 and 3 hours respectively, given that its Direct Material Cost is ₹ 1,000 and Direct Labour Cost is ₹ 600.

ICAI'S ANSWER

Statement Showing Distribution of Overheads of PM Ltd.

Particulars	Basis	Total	Production Departments			Service Departments	
			P1	P2	P3	S1	S2
		(₹)	(₹)	(₹)	(₹)	(₹)	(₹)
Direct wages	Actual	33,900	-	-	-	30,000	3,900

Particulars	Basis	Total	Production Departments			Service Departments	
Rent & rates	Area	1,00,000	20,000	25,000	30,000	20,000	5,000
General lighting	Light points	12,000	2,000	3,000	4,000	2,000	1,000
Indirect wages	Direct wages	38,780	12,000	8,000	12,000	6,000	780
Power	H.P.	30,000	12,000	6,000	10,000	2,000	-
Depreciation of machines	Value of machines	2,00,000	48,000	64,000	80,000	4,000	4,000
Sundries	Direct wages	1,93,900	60,000	40,000	60,000	30,000	3,900
		6,08,580	**1,54,000**	**1,46,000**	**1,96,000**	**94,000**	**18,580**

Redistribution of Service Department's Expenses over Production Departments

	P1	P2	P3	S1	S2
	(₹)	**(₹)**	**(₹)**	**(₹)**	**(₹)**
Total overhead distributed as above	1,54,000	1,46,000	1,96,000	94,000	18,580
Dept. S1 Overheads apportioned (20:30:40:—:10)	18,800	28,200	37,600	(94,000)	9,400
Dept. S2 overheads apportioned (40:20:30:10:—)	11,192	5,596	8,394	2,798	(27,980)
Dept. S1 Overheads apportioned (20:30:40:—:10)	560	839	1,119	(2,798)	280
Dept. S2 overheads apportioned (40:20:30:10:—)	124	63	93	-	(280)
	1,84,676	**1,80,698**	**2,43,206**	**-**	**-**
Working hours	3,070	4,475	2,419		
Rate per hour	**60.16**	**40.38**	**100.54**		

Determination of total cost of Product 'X'

	(₹)
Direct material cost	1,000.00
Direct labour cost	600.00
Overhead cost (See working note)	744.14
	2,344.14

Working Note:

Overhead cost

= (₹ 60.16 × 4 hrs.) + (₹ 40.38 × 5 hrs.) + (₹ 100.54 × 3 hrs.)

= ₹ 240.62 + ₹ 201.90 + ₹ 301.62 = ₹ 744.14

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374 MTP **Old P3 Dec 2021 S-II** Q2(b) 10 marks old scheme

The following account balances and distribution of indirect charges are taken from the accounts of a manufacturing concern for the year ending on 31st March, 2021:

Item	Total Amount	Production Departments		Service Departments		
	(₹)	X (₹)	Y (₹)	Z (₹)	A (₹)	B (₹)
Indirect Material	5,00,000	80,000	1,20,000	1,80,000	1,00,000	20,000
Indirect Labour	10,40,000	1,80,000	2,00,000	2,80,000	2,40,000	1,40,000
Supervisor's Salary	3,84,000	-	-	3,84,000	-	-
Fuel & Heat	60,000					
Power	7,20,000					
Rent & Rates	6,00,000					
Insurance of Assets	72,000					
Canteen Charges	2,40,000					
Depreciation	10,80,000					

The following departmental data are also available:

	Production Departments		Service Departments		
	X	Y	Z	A	B
Area (Sq. ft.)	4,400	4,000	3,000	2,400	1,200
Capital Value of Assets (₹)	40,00,000	60,00,000	50,00,000	10,00,000	20,00,000
Kilowatt Hours	3,500	4,000	3,000	1,500	-
Radiator Sections	20	40	60	50	30
No. of Employees	60	70	120	30	20

Expenses charged to the service departments are to be distributed to other departments by the following percentages:

	X	Y	Z	A	B
Department A (%)	30	30	20	-	20
Department B (%)	25	40	25	10	-

PREPARE an overhead distribution statement to show the total overheads of production departments after re-apportioning service departments' overhead by using simultaneous equation method. Show all the calculations to the nearest rupee.

ICAI'S ANSWER

Primary Distribution of Overheads

Item	Basis	Total Amount (₹)	Production Departments			Service Departments	
			X (₹)	Y (₹)	Z (₹)	A (₹)	B (₹)
Indirect Material	Actual	5,00,000	80,000	1,20,000	1,80,000	1,00,000	20,000
Indirect Labour	Actual	10,40,000	1,80,000	2,00,000	2,80,000	2,40,000	1,40,000
Supervisor's Salary	Actual	3,84,000	-	-	3,84,000	-	-
Fuel & Heat	Radiator Sections {2:4:6:5:3}	60,000	6,000	12,000	18,000	15,000	9,000
Power	Kilowatt Hours {7:8:6:3:-}	7,20,000	2,10,000	2,40,000	1,80,000	90,000	-
Rent & Rates	Area (Sq. ft.) {22:20:15:12:6}	6,00,000	1,76,000	1,60,000	1,20,000	96,000	48,000
Insurance	Capital Value of Assets {4:6:5:1:2}	72,000	16,000	24,000	20,000	4,000	8,000
Canteen Charges	No. of Employees {6:7:12:3:2}	2,40,000	48,000	56,000	96,000	24,000	16,000
Depreciation	Capital Value of Assets {4:6:5:1:2}	10,80,000	2,40,000	3,60,000	3,00,000	60,000	1,20,000
Total overheads		**46,96,000**	**9,56,000**	**11,72,000**	**15,78,000**	**6,29,000**	**3,61,000**

Re-distribution of Overheads of Service Department A and B

Total overheads of Service Departments may be distributed using simultaneous equation method

Let, the total overheads of A = 'a' and the total overheads of B = 'b'

a = 6,29,000 + 0.10 b	(i)
or, 10a - b = 62,90,000	[(i) x10]
b = 3,61,000 + 0.20 a	(ii)
or, -0.20a + b = 3,61,000	

Solving equation (i) & (ii)

10a - b = 62,90,000

-0.20a + b = 3,61,000

9.8a = 66,51,000

a = 6,78,673

Putting the value of 'a' in equation (ii), we get

b = 3,61,000 + 0.20 × 6,78,673

b = 4,96,735

Secondary Distribution of Overheads

	Production Departments		
	X (₹)	**Y (₹)**	**Z (₹)**
Total overhead as per primary distribution	9,56,000	11,72,000	15,78,000
Service Department A (80% of 6,78,673) (3:3:2)	2,03,602	2,03,602	1,35,734
Service Department B (90% of 4,96,735) (5:8:5)	1,24,184	1,98,694	1,24,184
Total	**12,83,786**	**15,74,296**	**18,37,918**

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375 MTP **Old P3 Jul 2021 S-I** Q2(b) 10 marks old scheme

The following account balances and distribution of indirect charges are taken from the accounts of a manufacturing concern for the year ending on 31st March 2021:

Item	Total Amount	Production Departments			Service Departments	
	(Rs.)	**X (Rs.)**	**Y (Rs.)**	**Z (Rs.)**	**A (Rs.)**	**B (Rs.)**
Indirect Material	2,50,000	40,000	60,000	90,000	50,000	10,000
Indirect Labour	5,20,000	90,000	1,00,000	1,40,000	1,20,000	70,000
Supervisor's Salary	1,92,000	-	-	1,92,000	-	-
Fuel & Heat	30,000					

Item	Total Amount	Production Departments			Service Departments	
Power	3,60,000					
Rent & Rates	3,00,000					
Insurance	36,000					
Canteen Charges	1,20,000					
Depreciation	5,40,000					

The following departmental data are also available:

	Production Departments			Service Departments	
	X	**Y**	**Z**	**A**	**B**
Area (Sq. ft.)	4,400	4,000	3,000	2,400	1,200
Capital Value of Assets (Rs.)	40,00,000	60,00,000	50,00,000	10,00,000	20,00,000
Kilowatt Hours	3,500	4,000	3,000	1,500	-
Radiator Sections	20	40	60	50	30
No. of Employees	60	70	120	30	20

Expenses charged to the service departments are to be distributed to other departments by the following percentages:

	X	Y	Z	A	B
Department A (%)	30	30	20	-	20
Department B (%)	25	40	25	10	-

PREPARE an overhead distribution statement to show the total overheads of production departments after re-apportioning service departments' overhead by using simultaneous equation method. Show all the calculations to the nearest rupee.

ICAI'S ANSWER

Primary Distribution of Overheads

Item	Basis	Total Amount (Rs.)	Production Departments			Service Departments	
			X (Rs.)	**Y (Rs.)**	**Z (Rs.)**	**A (Rs.)**	**B (Rs.)**
Indirect Material	Actual	2,50,000	40,000	60,000	90,000	50,000	10,000
Indirect Labour	Actual	5,20,000	90,000	1,00,000	1,40,000	1,20,000	70,000
Supervisor's Salary	Actual	1,92,000	-	-	1,92,000	-	-

Item	Basis	Total Amount (Rs.)	Production Departments			Service Departments	
Fuel & Heat	Radiator Sections {2:4:6:5:3}	30,000	3,000	6,000	9,000	7,500	4,500
Power	Kilowatt Hours {7:8:6:3:-}	3,60,000	1,05,000	1,20,000	90,000	45,000	-
Rent & Rates	Area (Sq. ft.) {22:20:15:12:6}	3,00,000	88,000	80,000	60,000	48,000	24,000
Insurance	Capital Value of Assets {4:6:5:1:2}	36,000	8,000	12,000	10,000	2,000	4,000
Canteen Charges	No. of Employees {6:7:12:3:2}	1,20,000	24,000	28,000	48,000	12,000	8,000
Depreciation	Capital Value of Assets {4:6:5:1:2}	5,40,000	1,20,000	1,80,000	1,50,000	30,000	60,000
Total overheads		23,48,000	4,78,000	5,86,000	7,89,000	3,14,500	1,80,500

Re-distribution of Overheads of Service Department A and B

Total overheads of Service Departments may be distributed by simultaneous equation.

Let, the total overheads of A = a and the total overheads of B = b

a = 3,14,500 + 0.10 b	(i)
or, 10a - b = 31,45,000	[(i) x10]
b = 1,80,500 + 0.20 a	(ii)
or, -0.20a + b = 1,80,500	

Solving equation (i) & (ii)

10a - b = 31,45,000

-0.20a + b = 1,80,500

9.8a = 33,25,500

a = Rs. 3,39,337

Putting the value of 'a' in equation (ii), we get

b = 1,80,500 + 0.20 × 3,39,337

b = Rs. 2,48,367

Secondary Distribution of Overheads

	Production Departments		
	X (Rs.)	Y (Rs.)	Z (Rs.)
Total overhead as per primary distribution	4,78,000	5,86,000	7,89,000
Service Department A (80% of Rs.3,39,337)	1,01,801	1,01,801	67,867
Service Department B (90% of Rs.2,48,367)	62,092	99,347	62,092
Total	6,41,893	7,87,148	9,18,959

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376 MTP Old P3 Nov 2019 S-I Q3(b) 10 marks old scheme

V Ltd. manufactures luggage trolleys for airports. The factory, in which the company undertakes all of its production, has two production departments- 'Fabrication' and 'Assembly', and two service departments- 'Stores' and 'Maintenance'.

The following information have been extracted from the company's budget for the financial year ended 31st March, 2019:

Particulars	Rs.
Allocated Overhead Costs	
Fabrication Department	15,52,000
Assembly Department	7,44,000
Stores Department	2,36,000
Maintenance Department	1,96,000
Other Overheads	
Factory rent	15,28,000
Factory building insurance	1,72,000
Plant & machinery insurance	1,96,000
Plant & Machinery Depreciation	2,65,000
Subsidy for staffs' canteen	4,48,000

Direct Costs	Rs.	Rs.
Fabrication Department:		
Material	63,26,000	
Labour	8,62,000	71,88,000
Assembly Department:		
Material	1,42,000	
Labour	13,06,000	14,48,000

The following additional information is also provided:

	Fabrication Department	Assembly Department	Stores Department	Maintenance Department
Floor area (square meters)	24,000	10,000	2,500	3,500
Value of plant & machinery (Rs.)	16,50,000	7,50,000	75,000	1,75,000
No. of stores requisitions	3,600	1,400	---	---
Maintenance hours required	2,800	2,300	400	---
No. of employees	120	80	38	12
Machine hours	30,00,000	60,000		
Labour hours	70,000	26,00,000		

Required:

(i) PREPARE a table showing the distribution of overhead costs of the two service departments to the two production departments using step method; and

(ii) CALCULATE the most appropriate overhead recovery rate for each department.

(iii) Using the rates calculated in part (ii) above, CALCULATE the full production costs of the following job order:

Job number IGI2019

Direct Materials	Rs. 2,30,400
Direct Labour:	
Fabrication Department	240 hours @ Rs. 50 per hour
Assembly Department	180 hours @ Rs. 50 per hour
Machine hours required:	
Fabrication Department	210 hours
Assembly Department	180 hours

ICAI'S ANSWER

(i) Table of Primary Distribution of Overheads

Particulars	Basis of Apportionment	Total Amount	Production Department		Service Departments	
			Fabrication	**Assembly**	**Stores**	**Maintenance**
Overheads Allocated		27,28,000	15,52,000	7,44,000	2,36,000	1,96,000
Direct Costs	Actual	86,36,000	71,88,000	14,48,000	---	---
Other Overheads:						
Factory rent	Floor Area (48:20:5:7)	15,28,000	9,16,800	3,82,000	95,500	1,33,700
Factory building insurance	Floor Area (48:20:5:7)	1,72,000	1,03,200	43,000	10,750	15,050
Plant & Machinery insurance	Value of Plant & Machinery (66:30:3:7)	1,96,000	1,22,038	55,472	5,547	12,943
Plant & Machinery Depreciation	Value of Plant & Machinery (66:30:3:7)	2,65,000	1,65,000	75,000	7,500	17,500
Canteen Subsidy	No. of employees (60:40:19:6)	4,48,000	2,15,040	1,43,360	68,096	21,504
		1,39,73,000	1,02,62,078	28,90,832	4,23,393	3,96,697

Re-distribution of Service Departments' Expenses:

Particulars	Basis of Apportionment	Production Department		Service Departments	
		Fabrication	**Assembly**	**Stores**	**Maintenance**
Overheads as per Primary distribution	As per Primary distribution	1,02,62,078	28,90,832	4,23,393	3,96,697
Maintenance Department Cost	Maintenance Hours (28:23:4:-)	2,01,955	1,65,891	28,851	(3,96,697)
		1,04,64,033	30,56,723	4,52,244	---
Stores Department	No. of Stores Requisition (18:7:-:-)	3,25,616	1,26,628	(4,52,244)	
		1,07,89,649	31,83,351	---	---

(ii) Overhead Recovery Rate

Department	Apportioned Overhead (Rs.)	Basis of Overhead Recovery Rate	Overhead Recovery Rate (Rs.)
	(I)	**(II)**	**[(I) ÷ (II)]**
Fabrication	1,07,89,649	30,00,000 Machine Hours	3.60 per Machine Hour
Assembly	31,83,351	26,00,000 Labour Hours	1.22 per Labour Hour

(iii) Calculation of full production costs of Job no. IGI2019.

Particulars	Amount (Rs.)
Direct Materials	2,30,400
Direct Labour:	
Fabrication Deptt. (240 hours × Rs.50)	12,000
Assembly Deptt. (180 hours × Rs.50)	9,000
Production Overheads:	
Fabrication Deptt. (210 hours × Rs. 3.60)	756
Assembly Deptt. (180 hours × Rs. 1.22)	220
Total Production Cost	2,52,376

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5.4 Absorption rates and the machine hour rate

51 questions · asked in 32 ICAI documents · 9 exam papers

377 EXAM **Sep 2026** MCQ 15 2 marks MCQ

A factory has installed a new machine which has a brochure claiming that it can produce 60,000 units of product X per annum. However, the company estimates that due to Sundays and holidays, stock taking and minor repairs, production can be achieved only to the extent of 50,000 units. Further, time lost on batch change will reduce this to 40,000 units. Which of the following is not correct ?

A Normal Capacity is 40,000 units.

B Installed Capacity is 60,000 units.

C Practical Capacity is not 60,000 units.

D Normal Idle Capacity is 10,000 units.

ICAI'S ANSWER

Answer key: **ICAI has not published the key yet.**

ICAI · P4 · Sep 2026 exam · question p. 9

378 EXAM **Sep 2026** Case scenario · Q5(b)(i) to Q5(b)(iii) 7 marks

CASE SCENARIO

A manufacturing company operates three production departments and it has established a policy to allocate factory overheads using a single blanket rate calculated as total budgeted overheads divided by budgeted direct labour hours. Below are the relevant data for one month :

Department	Budgeted			Actual			Job 1023	
	Factory Overheads (₹)	Direct Labour Hours	Machine Hours	Factory Overheads (₹)	Direct Labour Hours	Machine Hours	Direct Labour Hours	Machine Hours
Cutting	4,00,000	20,000	60,000	4,20,000	22,000	62,000	40	120
Assembly	3,00,000	2,000	25,000	2,80,000	2,000	24,000	4	40
Finishing	2,00,000	14,000	1,000	2,10,000	14,000	1,000	30	2

The company has a margin of 30% on sales price to cover other administrative and selling overheads and profit.

Required :

Q5(b)(i) 2 marks

Compute the value of factory overhead that will be used in the pricing of Job No. 1023 as per the company's existing policy.

Q5(b)(ii) 3 marks

Recommend the most appropriate method of absorbing factory overheads and calculate the value of overhead that will be used in the pricing of Job No. 1023.

Q5(b)(iii) 2 marks

Calculate the total under/over recovery of overheads in the Finishing Department based on your recommendation.

(Round off rates up to two decimal points and the final value to the next higher rupee.)

ICAI'S ANSWERS

Q5(b)(i)

ICAI has not published its answers to this paper yet.

Q5(b)(ii)

ICAI has not published its answers to this paper yet.

Q5(b)(iii)

ICAI has not published its answers to this paper yet.

ICAI · P4 · Sep 2026 exam · question p. 21

379 EXAM **May 2026** Case Scenario - II · MCQ 6 to MCQ 10 10 marks

CASE SCENARIO - II

Safety First Tyre Company is the manufacturer of 'Puncture Resistant' motorcycle tyres. In the manufacturing process, it undertakes three different jobs namely, Belting (B), Extrusion (E) and Vulcanizing (V). All the three jobs require the use of a special machine and also the use of Artificial Intelligence (AI) system when necessary. The Al system is hired from outside and the hire charges paid for every quarter is ₹ 4,08,000. An estimate of overhead expenses relating to the special machine is given below:

- The cost of the special machine is ₹ 66,00,000 and depreciation is charged @ 10% per annum on straight line basis.
- Rent paid for every six months is ₹ 1,38,000.
- Other indirect expenses per annum ₹ 4,92,000.

During the first month of operation, the following details are available from the job book of the company:

Jobs

	Belting (B)	Extrusion (E)	Vulcanizing (v)
Number of hours the special machine was used			
Without the use of Artificial Intelligence (AI) system	600	900	-
With the use of Artificial Intelligence (AI) system	450	550	1000

Any fraction in calculations may be rounded off up to two decimal places.

On the basis of the above Case Scenario, you are required to answer the following MCQs 6 to 10:

MCQ 6 2 marks

What is the amount of overheads per month for using the special machine without the aid of Artificial Intelligence (AI) system?

A ₹ 36,000

B ₹ 44,000

C ₹ 51,000

D ₹ 58,000

MCQ 7 2 marks

What is the amount of overheads per month for using the special machine with the aid of Artificial Intelligence (Al) system?

- A ₹ 1,82,000
- B ₹ 2,04,000
- C ₹ 1,95,714
- D ₹ 2,25,000

MCQ 8 2 marks

What is the machine hour rate for the company as a whole for a month when the Artificial Intelligence (AI) system was used?

- A ₹ 102 per hour
- B ₹ 122.55 per hour
- C ₹ 130 per hour
- D ₹ 133.75 per hour

MCQ 9 2 marks

What is the machine hour rate for the company as a whole for a month when the Artificial Intelligence (AI) was not used?

- A ₹ 41 per hour
- B ₹ 29.45 per hour
- C ₹ 38.25 per hour
- D ₹ 34 per hour

MCQ 10 2 marks

What is the machine hour rate for the month in respect of the individual jobs -Belting (B), Extrusion (E) and Vulcanizing (V) respectively?

- A ₹ 45.37 (B), 68.19 (E) and ₹ 98.95 (V) per hour
- B ₹ 63.14 (B), 59.79 (E) and ₹ 102 (V) per hour
- C ₹ 55.25 (B), 48.33 (E) and ₹ 105 (V) per hour
- D ₹ 49.41 (B), 78.54 (E) and ₹ 94.35 (V) per hour

ICAI'S ANSWERS

MCQ 6

Answer key: **(C)**

MCQ 7

Answer key: **(B)**

MCQ 8

Answer key: **(A)**

MCQ 9

Answer key: **(D)**

MCQ 10

Answer key: **(B)**

ICAI · P4 · May 2026 exam · question p. 3-4 · answer p. 7

380 EXAM Jan 2026 MCQ 12 2 marks MCQ

T Ltd., a textile manufacturing company, recovers factory overheads on a fixed percentage basis on direct wages and administrative overheads at 25% on factory cost.

The company has furnished the following data for Job 201:

Direct materials are ₹ 56,350, Direct wages are ₹ 45,000, Sales are ₹ 1,90,000 and profit is 25% on total cost.

What is the percentage recovery rate for factory overheads for Job 201?

A 35%

B 40%

C 45%

D 50%

ICAI'S ANSWER

Answer key: (C)

ICAI · P4 · Jan 2026 exam · question p. 5-6 · answer p. 7

381 EXAM Jan 2026 Q5(a) 6 marks

A machine shop has 6 identical drilling machines manned by 4 operators.

The machine cannot be worked without an operator wholly engaged on it.

The original cost of all these machines works out to ₹ 3,00,000.

These particulars are furnished pertaining to a period of 4 months:

Normal available hours per month	156 hrs.
Absenteeism (without pay) hours	9 hrs.
Leave (with pay) hours	10 hrs.
Normal idle time unavoidable hours	5 hrs.
Average rate of wages per worker for 6 hours a day	₹ 450
Production bonus estimated	20% on wages
Value of power consumed	₹ 20,500
Supervision and indirect labour	₹ 8,500
Lighting and electricity	₹ 4,500

These particulars are furnished for a year:

Repairs and maintenance including consumables - 2% per annum of value of machines.

Depreciation - 15% per annum of original cost.

Other sundry works expenses - ₹ 12,000 p.a.

General management expenses allocated - ₹ 24,000 p.a.

The machine shop has taken an annual insurance for all 6 machines also.

Given that the comprehensive machine hour rate for 4 months period is ₹ 125 per machine hour.

You are required to find out the total amount of annual insurance premium.

ICAI'S ANSWER

Computation of Amount of Annual Insurance Premium

Particulars	Amount (₹)
Operator's wages (Refer Working Note 2)	1,84,500
Production bonus (20% on wages)	36,900
Power consumed	20,500
Supervision and indirect labour	8,500
Lighting and electricity	4,500
Repairs & maintenance (2% × ₹ 3,00,000 × 1/3)	2,000
Depreciation (15% × ₹ 3,00,000 × 1/3)	15,000
Other sundry works expenses (₹ 12,000 × 1/3)	4,000
General management expenses (₹ 24,000 × 1/3)	8,000
Total Overhead of Machine shop without Insurance Premium	2,83,900
Total Insurance Premium for 4 months (B.F.)	**₹ 16,100**
Total Overhead of Machine shop for 4 months	3,00,000

$$\text{Machine hour rate} = \frac{\text{Total overheads of machine shop}}{\text{Hours of machines operation}}$$

Total Overhead of Machine shop = ₹125 x 2,400 hrs = 3,00,000

Annual Insurance Premium = ₹ 16,100 x 3 = ₹ 48,300

Working notes

1. Computation of hours, for which 4 operators are available for 4 months.

	For 4 months and 4 operators
Normal available hours (156 × 4 months x 4 operators)	2,496
Less: Absenteeism hours (9 × 4 operators)	(36)
Paid hours	**2,460**
Less: Leave hours (10 × 4 operators)	(40)
Less: Idle time hours (5 × 4 operators)	(20)
Effective working hours	**2,400**

As machines cannot be worked without an operator wholly engaged on them therefore, hours for which 4 operators are available for 4 months are the hours for which machines can be used. Hence 2,400 hours represent effective working hours.

2. Computation of operator's wages

Average rate of wages: $\frac{₹450}{6 \text{hours}}$ = ₹ 75 per hour

Total wages paid to 4 operators for 4 months = 2,460 hours × ₹ 75 = ₹ 1,84,500

ICAI · P4 · Jan 2026 exam · question p. 23-24 · answer p. 25-26

382 EXAM **Jan 2025** Case scenario · Q4(b)(i) to Q4(b)(ii) 6 marks

CASE SCENARIO

LMN Foods is a manufacturer of organic snacks. For the year ending 2023, the company compiled the following financial data:

Item	Amount (in ₹)
Opening inventory of raw materials	2,00,000
Closing inventory of raw materials	2,50,000
Raw material purchases	12,00,000
Labour costs	5,00,000
Production overheads	2,50,000
Marketing and distribution expenses	1,52,000

In 2024, LMN Foods accepted a request for a bulk supply of their best-selling snacks. The estimated costs for fulfilling this order are as follows:

- Estimated raw material cost: ₹ 3,00,000
- Estimated labour cost: ₹ 1,50,000
- Packaging and transportation costs: ₹ 49,400

LMN Foods allocates production overhead based on direct labour costs and marketing and distribution expenses as a percentage of the total production cost based on the previous year's data.

Required:

Q4(b)(i) 2 marks

Calculate the overhead recovery rates for 2023 based on actual costs.

Q4(b)(ii) 4 marks

Prepare a comprehensive cost statement for the bulk order and determine the Sales required for achieving a profit margin of 20% on the final sales amount.

ICAI'S ANSWERS

Q4(b)(i)

Calculation of Overhead Recovery Rate:

Production Overhead recovery rate based on Direct Labour Costs

$$= \frac{\text{Production Overhead}}{\text{Direct Labour Cost}} \times 100$$

$$= \frac{2,50,000}{5,00,000} \times 100 = 50\% \text{ of Direct Labour}$$

Marketing & Distribution Overhead recovery rate based on Total Production Costs

$$= \frac{\text{Marketing \& Distribution Overhead}}{\text{Total Production Cost}} \times 100$$

$$= \frac{1,52,000}{19,00,000} \times 100 = 8\% \text{ of Total Production Costs}$$

Working Note:

Statement showing Total Cost for 2023

Particulars	Amount (₹)
Opening stock of raw material	2,00,000
Add: Purchases	12,00,000
Less: Closing stock of raw material	(2,50,000)
Direct Material Consumed	11,50,000
Direct Labour Cost	5,00,000
Prime Cost	16,50,000
Add: Production Overhead	2,50,000
Production cost	19,00,000
Add: Marketing & Distribution Cost	1,52,000
Total Cost	20,52,000

Q4(b)(ii)

Cost Statement for the Bulk Order and Determination of Sales in 2024:

Particulars	Amount (₹)
Direct Material Cost	3,00,000
Direct Labour Cost	1,50,000
Prime Cost	4,50,000
Add: Production Overhead (50% of Direct Labour Cost)	75,000
Production cost	5,25,000
Add: Marketing & Distribution Cost (8% of Production Cost)	42,000
Packaging and Transportation Costs	49,400
Total Cost	6,16,400

Particulars	Amount (₹)
Add: Profit @ 25% on cost	1,54,100
Sales value (Price to be quoted for the order)	7,70,500

Hence the price to be quoted is ₹7,70,500

ICAI · P4 · Jan 2025 exam · question p. 20 · answer p. 22-23, 23

383 EXAM Sep 2024 Q5(a) 7 marks

This data pertains to the three machines operating in the manufacturing division of PQR Corp for the financial year 2023-2024:

Particulars	Estimated Expenses			
	TOTAL (₹)	Machines		
		X (₹)	Y (₹)	Z (₹)
Direct Labour Expenses (per quarter)	2,50,000			
Oil Expenses (per quarter)	1,03,125	37,500	37,500	28,125
Machine Insurance Expenses (per quarter)	60,000			
Depreciation (per annum)	6,00,000	1,00,000	2,00,000	3,00,000
Building Maintenance Expenses (per quarter)	1,00,000			
Wages of Operator (per quarter)	2,25,000			
Electricity Expenses (per quarter)	3,00,000			
Rent and Rates (per month)	80,000			
Salary of Technician (per month)	62,500			

(The Technician works only on machines X and Y and the Operator controls all three machines and both spend equal time on each of the machines worked upon by them.)

There are 14 holidays besides Sundays in the year, of which six are on Saturdays. There was a Strike of workers for 5 working days (including one Saturday). The manufacturing department operates for 8 hours per day on regular week days, while on Saturdays, the operating hours are reduced by 2 hours per day. All machines operate at 80% capacity throughout the year. Assume 366 days in a year.

The following additional information is also available:

(i) A 20% hike in the price of oil.

(ii) A 10% rise in Oil consumption for machines 'X' and 'Y' only.

(iii)

Particulars	Machines		
	X	Y	Z
No. of Workers	5	3	2
Ratio of K.W. Rating	3	3	4
Ratio of Floor space utilized	1	2	1

Required:

Prepare a Statement detailing the allocation of expenses to each machine on an annual basis and thereafter, compute the comprehensive machine hour rate for each of the specified machine.

ICAI'S ANSWER

Computation of Comprehensive Machine Hour Rate

Particulars	Basis of Apportionment	Total (₹)	Machines		
			X (₹)	Y (₹)	Z (₹)
Standing Charges (Per Annum)					
Direct Labour Expenses	No. of Workers (5:3:2)	10,00,000	5,00,000	3,00,000	2,00,000
Wages of Operator	Equal	9,00,000	3,00,000	3,00,000	3,00,000
Machine Insurance Expenses	Depreciation (1:2:3)	2,40,000	40,000	80,000	1,20,000
Building Maintenance Expenses	Floor Space (1:2:1)	4,00,000	1,00,000	2,00,000	1,00,000
Rent and Rates	Floor Space (1:2:1)	9,60,000	2,40,000	4,80,000	2,40,000
Salary of Technician	(1:1:0)	7,50,000	3,75,000	3,75,000	-
Total Standing Charges		42,50,000	15,55,000	17,35,000	9,60,000
Hourly Rate (A)			856.28	955.40	528.63
Machine Expenses (Per Annum)					
Oil Expenses	Direct (W.N.1)	5,31,000	1,98,000	1,98,000	1,35,000
Depreciation	Direct	6,00,000	1,00,000	2,00,000	3,00,000
Electricity Expenses	K. W. Rating (3:3:4)	12,00,000	3,60,000	3,60,000	4,80,000
Total Machine Expenses		23,31,000	6,58,000	7,58,000	9,15,000
Hourly Rate (B)			362.33	417.40	503.85

Particulars	Basis of Apportionment	Total (₹)	Machines		
Total Expenses		65,81,000	22,13,000	24,93,000	18,75,000
(A) + (B)			**1,218.61**	**1,372.80**	**1,032.49**

Calculation of Effective Working Hours.

	Per Machine Per Annum
Full Day Working Hours (250 days x 8 hours)	2,000
6 hours per day Working Hours (45 days x 6 hours)	270
Total Hours	**2,270**
Capacity Utilization	80%
Effective working hours	**1,816**

No. of Shutdown Days

Sundays	52
Holidays	14
Strike Period	5
	71

No. of 6 Hours Working Days

Saturdays	52
Less: Holidays	(6)
Less: Strike Period	(1)
	45

No. of Full Working Days

Total No of Days in a year	366
Less: No of Shutdown Days	(71)
Less: No. of 6 Hours Working Days	(45)
	250

Working notes

1. Calculation of Oil Expenses.

Particulars	Machines		
	X (₹)	Y (₹)	Z (₹)
Oil Expenses	1,50,000	1,50,000	1,12,500
Add: Increase in Price @20%	30,000	30,000	22,500
	1,80,000	1,80,000	1,35,000
Add: Increase in Consumption @10%	18,000	18,000	-
Total	**1,98,000**	**1,98,000**	**1,35,000**

ICAI · P4 · Sep 2024 exam · question p. 28-29 · answer p. 30-32

384 EXAM **Old P3 May 2023** Q6(b) 5 marks old scheme

Explain what is meant by Practical capacity and Normal capacity. How is normal capacity determined?

ICAI'S ANSWER

Meaning of Practical capacity and Normal capacity

Practical capacity is defined as actually utilised capacity of a plant. It is also known as operating capacity. This capacity takes into account loss of time due to repairs, maintenance, minor breakdown, idle time, set up time, normal delays, Sundays and holidays, stock taking etc. Generally, practical capacity is taken between 80 to 90% of the rated capacity. It is also used as a base for determining overhead rates. Practical capacity is also called net capacity or available capacity.

Normal capacity is the volume of production or services achieved or achievable on an average over a period under normal circumstances taking into account the reduction in capacity resulting from planned maintenance.

Normal capacity is determined as under:

Installed capacity		xxx
Adjustments for:		
(i) Time lost due to scheduled preventive or planned maintenance	xxx	
(ii) Number of shifts or machine hours or man hours		
(iii) Holidays, normal shut down days, normal idle time	xxx	
(iv) Normal time lost in batch change over	xxx	xxx
Normal Capacity		**xxx**

ICAI · Old P3 · May 2023 exam · question p. 28 · answer p. 29

385 EXAM **Old P3 Nov 2022** Q2(a) 10 marks old scheme

USP Ltd. is the manufacturer of 'double grip motorcycle tyres'. In the manufacturing process, it undertakes three different jobs namely, Vulcanising, Brushing and Striping. All of these jobs require the

use of a special machine and also the aid of a robot when necessary. The robot is hired from outside and the hire charges paid for every six months is₹ 2,70,000. An estimate of overhead expenses relating to the special machine is given below:

- Rent for a quarter is ₹ 18,000.
- The cost of the special machine is ₹ 19,20,000 and depreciation is charged @10% per annum on straight line basis.
- Other indirect expenses are recovered at 20% of direct wages.

The factory manager has informed that in the coming year, the total direct wages will be ₹ 12,00,000 which will be incurred evenly throughout the year.

During the first month of operation, the following details are available from the job book:

Number of hours the special machine was used

Jobs	Without the aid of the robot	With the of the robot
Vulcanising	500	400
Brushing	1000	400
Striping	-	1200

You are required to :

(i) Compute the Machine Hour Rate for the company as a whole for a month (A) when the robot is used and (B) when the robot is not used.

(ii) Compute the Machine Hour Rate for the individual jobs i.e. Vulcanising, Brushing and Striping.

ICAI'S ANSWER

Working notes:

(I)	Total machine hours use (500 + 1,000 + 400 + 400 + 1,200)	3,500
(II)	Total machine hours without the use of robot (500 + 1,000)	1,500
(III)	Total machine hours with the use of robot (400 + 400 + 1,200)	2,000
(IV)	Total overheads of the machine per month	
	Rent (₹ 18,000 ÷ 3 months)	6,000
	Depreciation [(₹ 19,20,000 × 10%) ÷ 12 months]	16,000
	Indirect expenses [(₹ 12,00,000 × 20%) ÷ 12 months]	20,000
	Total	42,000
(V)	Robot hire charges for a month (₹ 2,70,000 ÷ 6 months)	₹ 45,000
(VI)	Overheads for using machines without robot = ₹ 42,000/3,500 hrs. × 1,500 hrs. =	₹ 18,000

(VII)	Overheads for using machines with robot = ₹ 42,000/3,500 hrs. ×2,000 hrs. + ₹ 45,000 =	₹ 69,000

(i) Computation of Machine hour rate for the firm as a whole for a month.

(A) When the robot was used: $\frac{₹\ 69{,}000}{2{,}000\ \text{hours}} = ₹\ 34.50\ \text{per hour}$

(B) When the robot was not used: $\frac{₹\ 18{,}000}{1{,}500\ \text{hrs.}} = ₹\ 12\ \text{per hour}$

(ii) Computation of Machine hour rate for the individual job

	Rate per hour	Job					
		Vulcanising		Brushing		Striping	
	(₹)	Hrs.	(₹)	Hrs.	(₹)	Hrs.	(₹)
Overheads							
Without robot	12.00	500	6,000	1,000	12,000	-	-
With robot	34.50	400	13,800	400	13,800	1,200	41,400
Total		900	19,800	1,400	25,800	1,200	41,400
Machine hour rate			22		18.43		34.50

ICAI · Old P3 · Nov 2022 exam · question p. 5-6 · answer p. 7-8

386 EXAM **Old P3 May 2022** Q2(a) 10 marks old scheme

In a manufacturing company, the overhead is recovered as follows:

Factory Overheads: a fixed percentage basis on direct wages and

Administrative overheads: a fixed percentage basis on factory cost.

The company has furnished the following data relating to two jobs undertaken by it in a period.

	Job 1 (₹)	Job 2 (₹)
Direct materials	1,08,000	75,000
Direct wages	84,000	60,000
Selling price	3,33,312	2,52,000
Profit percentage on total cost	12%	20%

You are required to:

(i) Compute the percentage recovery rates of factory overheads and administrative overheads.

(ii) Calculate the amount of factory overheads, administrative overheads and profit for each of the two jobs.

(iii) Using the above recovery rates, determine the selling price to be quoted for job 3. Additional data pertaining to Job 3 is as follows:

Direct materials	₹ 68,750
Direct wages	₹ 22,500
Profit percentage on selling price	15%

ICAI'S ANSWER

(i) Computation of percentage recovery rates of factory overheads and administrative overheads.

Let the factory overhead recovery rate as percentage of direct wages be F and administrative overheads recovery rate as percentage of factory cost be A.

Factory Cost of Jobs:

Direct materials + Direct wages + Factory overhead

For Job 1 = ₹ 1,08,000 +₹ 84,000 + ₹ 84,000F

For Job 2 = ₹ 75,000 +₹ 60,000 + ₹ 60,000F

Total Cost of Jobs:

Factory cost + Administrative overhead

For Job 1 = (₹ 1,92,000 + ₹ 84,000F) + (₹ 1,92,000 + ₹ 84,000F) A = ₹ 2,97,600*

For Job-2 = (₹ 1,35,000 + ₹ 60,000F) + (₹1,35,000+ ₹ 60,000F) A = ₹ 2,10,000**

The value of F & A can be found using following equations

1,92,000 + 84,000F + 1,92,000A + 84,000AF = ₹ 2,97,600 .....eqn (i)

1,35,000 + 60,000F + 1,35,000A + 60,000AF = ₹ 2,10,000 .....eqn (ii)

Multiply equation (i) by 5 and equation (ii) by 7

9,60,000 + 4,20,000F + 9,60,000A + 4,20,000AF = ₹14,88,000 ...eqn (iii)

9,45,000 + 4,20,000F + 9,45,000A + 4,20,000AF = ₹ 14,70,000 ...eqn (iv)

- - - - -

15,000 + 15,000A = ₹18,000

15,000 A = 18,000 – 15,000

A = 0.20

Now putting the value of A in equation (i) to find the value of F

1,92,000 + 84,000F + (1,92,000 × 0.20) + (84,000 F × 0.20)= ₹ 2,97,600

Or

1,92,000 + 84,000F+38,400+16,800 F = ₹2,97,600

1,00,800 F = 67,200

F = 0.667

On solving the above relations: F = 0.667 and A = 0.20

Hence, percentage recovery rates of:

Factory overheads = 66.7% or 2/3rd of wages and

Administrative overheads = 20% of factory cost.

Working note:

$$\text{Total Cost} = \frac{\text{Selling price}}{(100\% + \text{Percentage of profit})}$$

$$\text{*For Job 1} = \frac{₹\ 3{,}33{,}312}{(100\% + 12\%)} = ₹\ 2{,}97{,}600$$

$$\text{**For Job 2} = \frac{₹2{,}52{,}000}{(100\% + 20\%)} = ₹\ 2{,}10{,}000$$

(ii) Statement of jobs, showing amount of factory overheads, administrative overheads and profit:

	Job 1 (₹)	Job 2 (₹)
Direct materials	1,08,000	75,000
Direct wages	84,000	60,000
Prime cost	1,92,000	1,35,000
Factory overheads		
2/3rd of direct wages	**56,000**	**40,000**
Factory cost	2,48,000	1,75,000
Administrative overheads		
20% of factory cost	**49,600**	**35,000**
Total cost	**2,97,600**	**2,10,000**
Profit (12% & 20% respectively)	**35,712**	**42,000**
Selling price	3,33,312	2,52,000

(iii) Selling price of Job 3

	(₹)
Direct materials	68,750
Direct wages	22,500
Prime cost	91,250
Factory overheads (2/3rd of Direct Wages)	15,000
Factory cost	1,06,250
Administrative overheads (20% of factory cost)	21,250
Total cost	**1,27,500**

	(₹)
Profit margin (balancing figure)	22,500
Selling price [Total Cost/85%]	**1,50,000**

ICAI · Old P3 · May 2022 exam · question p. 6-7 · answer p. 8-10

387 EXAM **Old P3 Jan 2021** Q1(b) 5 marks old scheme

A machine shop has 8 identical machines manned by 6 operators. The machine cannot work without an operator wholly engaged on it. The original cost of all the 8 machines works out to ₹ 32,00,000. The following particulars are furnished for a six months period:

Normal available hours per month per operator	208
Absenteeism (without pay) hours per operator	18
Leave (with pay) hours per operator	20
Normal unavoidable idle time-hours per operator	10
Average rate of wages per day of 8 hours per operator	₹ 100
Production bonus estimated	10% on wages
Power consumed	₹ 40,250
Supervision and Indirect Labour	₹ 16,500
Lighting and Electricity	₹ 6,000

The following particulars are given for a year:

Insurance	₹ 3,60,000
Sundry work Expenses	₹ 50,000
Management Expenses allocated	₹ 5,00,000
Depreciation	10% on the original cost

Repairs and Maintenance (including consumables): 5% of the value of all the machines.

Prepare a statement showing the comprehensive machine hour rate for the machine shop.

ICAI'S ANSWER

Workings:

Particulars	Six months 6 operators (Hours)
Normal available hours per month (208 × 6 months x 6 operators)	7,488
Less: Absenteeism hours (18 × 6 operators)	(108)
Paid hours (A)	7,380
Less: Leave hours (20 × 6 operators)	(120)
Less: Normal idle time (10 × 6 operators)	(60)
Effective working hours	**7,200**

Computation of Comprehensive Machine Hour Rate

Particulars	Amount for six months (₹)
Operators' wages (7,380/8 ×100)	92,250
Production bonus (10% on wages)	9,225
Power consumed	40,250
Supervision and indirect labour	16,500
Lighting and Electricity	6,000
Repair and maintenance {(5% × ₹ 32,00,000)/2}	80,000
Insurance (₹ 3,60,000/2)	1,80,000
Depreciation {(₹ 32,00,000 × 10%)/2}	1,60,000
Sundry Work expenses (₹ 50,000/2)	25,000
Management expenses (₹ 5,00,000/2)	2,50,000
Total Overheads for 6 months	**8,59,225**
Comprehensive Machine Hour Rate = ₹ 8,59,225/7,200 hours	**₹ 119.33**

(Note: Machine hour rate may be calculated alternatively. Further, presentation of figures may also be done on monthly or annual basis.)

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388 EXAM **Old P3 Jan 2021** Q6(c) 5 marks old scheme

Explain Blanket Overhead Rate and Departmental Overhead Rate. How they are calculated? State the conditions required for the application of Blanket Overhead Rate.

ICAI'S ANSWER

Blanket Overhead Rate: Blanket overhead rate refers to the computation of one single overhead rate for the whole factory.

This overhead rate is computed as follows:

$$\text{Blanket Rate} = \frac{\text{Total overheads for the factory}}{\text{Total number of units of base for the factory}}$$

Departmental Overhead Rate: It refers to the computation of one single overhead rate for a particular production unit or department.

This overhead rate is determined by the following formula:

$$\text{Departmental overhead Rate} = \frac{\text{Overheads of department or cost centre}}{\text{Corresponding base}}$$

Conditions required for the Application of Blanket Overhead:

A blanket rate should be applied in the following cases:

(1) Where only one major product is being produced.

(2) Where several products are produced, but

(a) All products pass through all departments; and

(b) All products are processed for the same length of time in each department.

ICAI · Old P3 · Jan 2021 exam · question p. 25 · answer p. 27

389 RTP May 2026 Q8

Omega Engineering Pvt. Ltd., a company manufacturing precision components for the automotive sector, operates three major CNC machines in its machining department—Machines X, Y, and Z. Due to a surge in customer demand and rising input costs, the company plans to revise its predetermined machine hour rates for the next financial year. The management has provided the following annual cost data for the three machines:

Preliminary Estimates of Annual Expenses

Particulars	Total (₹)	Machines		
		X (₹)	Y (₹)	Z (₹)
Depreciation	3,00,000	1,20,000	1,00,000	80,000
Spare parts	1,50,000	60,000	50,000	40,000
Power	5,50,000	—	—	—
Consumable stores	1,20,000	50,000	40,000	30,000
Insurance of machinery	1,00,000	—	—	—
Indirect labour	2,40,000	—	—	—
Building maintenance	2,50,000	—	—	—
Annual interest on capital outlay	1,20,000	50,000	40,000	30,000

Additional monthly information:

- Rent & rates: ₹ 30,000 per month
- Salary of Production Supervisor: ₹ 60,000 per month
- Salary of Machine Operator: ₹ 18,000 per month

The supervisor and operator oversee all three machines and spend equal time on each.

The following additional information is also available:

Machine	Estimated Direct Labour Hours	K.W. Rating Ratio	Floor Space (sq. ft.)
X	1,20,000	4	50,000
Y	1,60,000	3	40,000
Z	1,40,000	2	30,000

Working Conditions

- The plant operates 8 hours per day.
- Saturdays are half-days.
- There are 15 holidays besides Sundays, of which 3 fall on Saturdays.
- Machines run at 92% capacity.
- A 3% allowance for breakdown is considered reasonable.

Management expects the following increases for the coming financial year:

1. 10% increase in spare parts prices.
2. 30% increase in spare parts consumption for Machines Y & Z only.

You are required to :

CALCULATE the predetermined machine hour rates for Machines X, Y, and Z after incorporating all the revisions.

ICAI'S ANSWER

(a) Computation of Machine Hour Rate

(A) Standing Charges

Particulars	Basis of Apportionment	Total (₹)	Machine X (₹)	Machine Y (₹)	Machine Z (₹)
Insurance	Depreciation Basis (6:5:4)	1,00,000	40,000	33,333	26,667
Indirect Labour	Direct Labour Hours (6:8:7)	2,40,000	68,571	91,429	80,000
Building Maintenance	Floor Space (5:4:3)	2,50,000	1,04,167	83,333	62,500
Rent & Rates	Floor Space (5:4:3)	3,60,000	1,50,000	1,20,000	90,000

Particulars	Basis of Apportionment	Total (₹)	Machine X (₹)	Machine Y (₹)	Machine Z (₹)
Salary of Supervisor	Equal	7,20,000	2,40,000	2,40,000	2,40,000
Salary of Operator	Equal	2,16,000	72,000	72,000	72,000
Total Standing Charges	—	—	6,74,738	6,40,095	5,71,167
Hourly Rate (Standing)	—	—	345.49	327.75	292.46

(B) Machine Expenses

Particulars	Basis of Apportionment	Total (₹)	Machine X (₹)	Machine Y (₹)	Machine Z (₹)
Depreciation	Direct	—	1,20,000	1,00,000	80,000
Spare Parts	Revised Estimates	—	66,000	71,500	57,200
Power	KW Rating (4:3:2)	5,50,000	2,44,444	1,83,333	1,22,222
Consumable Stores	Direct	1,20,000	50,000	40,000	30,000
Total Machine Expenses	—	—	4,80,444	3,94,833	2,89,422
Hourly Rate (Machine)	—	—	246.00	202.17	148.19

Total (A + B)

Machine	Machine Hour Rate (₹)
X	591.49
Y	529.92
Z	440.65

Working Notes:

(i) Calculation of effective working hours:

No. of full off-days = No. of Sunday + No. of holidays

= 52 + 15 = 67 days

No. of half working days = 52 days – 3 holidays = 49 days

No. of full working days = 365 days – 67 days – 49 days

= 249 days

Total Working Hours = {(249 days × 8 hours) + (49 days × 4 hours)}

= 1,992 + 196

= 2,188 hours

Effective Working Hours = Total working hours × 92% – 3% for breakdown

= 2,188 × 92% – 3% × (2,188 × 92%)

= 2,012.96 – 60.39

= 1,953 hours

(ii) Amount of spare parts is calculated as under:

Machine	X	Y	Z
Preliminary estimates	60,000	50,000	40,000
Add: Increase in price @ 10%	6,000	5,000	4,000
	66,000	55,000	44,000
Add: Increase in consumption @ 30%	—	16,500	13,200
Estimated cost	66,000	71,500	57,200

(iii) Interest on capital outlay is a finance cost, therefore it has been excluded from the cost accounts.

ICAI · P4 · RTP May 2026 · question p. 10-11 · answer p. 33-34

390 RTP **May 2025** Q10

X Corp. produces two products, X and Y. The production division is made up of two manufacturing departments, P1 and P2, along with two support departments, S1 and S2.

Pre-determined Overhead Rates are applied in the production departments to allocate factory overhead costs to the products. The rate for Department P1 is determined by direct machine hours, while Department P2's rate is based on direct labor hours.

A technical assessment of the apportionment of expenses of service departments is as under:

	Production Department	
	P1	**P2**
Service Dept. 'S1' (ratio)	3	3
Service Dept. 'S2' (ratio)	30	15

Following budgeted data is available:

Factory overheads for the year:

Production Departments		Service Departments	
P1	**P2**	**S1**	**S2**
₹ 2,80,50,000	₹ 2,39,25,000	₹ 66,00,000	₹ 49,50,000

Details relating to production of the Products X and Y is as follows:

	Products	
	X	**Y**
Budgeted output (units)	1,00,000	60,000
Budgeted raw-material cost per unit (All materials are used in Department P1 only.)	₹ 660	₹ 825
Budgeted time required for production per unit	P1: 1.5 machine hours P2: 2 Direct labour hours	P1: 1.0 machine hours P2: 2.5 Direct labour hours
Average wage rate budgeted in Department P2	₹ 396 per hour	₹ 412.50 per hour

Following actual data is available (for the month of December, 2024):

	Products	
	X	**Y**
Actual output (units)	8,000	6,000
Actual hours worked for production	P1: 12,200 machine hours P2: 16,400 Direct labour hours	P1: 8,300 machine hours P2: 14,800 Direct labour hours
Raw materials cost	₹ 53,79,000	₹ 50,16,000
Wages paid	₹ 65,10,900	₹ 60,72,000

Actual factory overheads incurred is as follows:

Production Departments		Service Departments	
P1	**P2**	**S1**	**S2**
₹ 25,41,000	₹ 22,44,000	₹ 6,60,000	₹ 5,28,000

You are required to:

(i) COMPUTE the pre-determined overhead rate for department P1 and P2.

(ii) PREPARE a comparative statement reflecting Budgeted cost and Actual cost for production of the Products X and Y during the month of December, 2024.

(iii) CALCULATE the amount of under/ over-absorption of production overheads

ICAI'S ANSWER

(i) Computation of predetermined overhead rate for each production departments from budgeted data

	Production Department		Service Department	
	P1	**P2**	**S1**	**S2**

	Production Department		Service Department	
Budgeted factory overheads for the year (₹)	2,80,50,000	2,39,25,000	66,00,000	49,50,000
Allocation of service department S1's costs to production departments P1 and P2 equally (₹)	33,00,000	33,00,000	(66,00,000)	—
Allocation of service department S2's costs to production departments P1 and P2 in the ratio of 2:1 (₹)	33,00,000	16,50,000	—	(49,50,000)
Total	3,46,50,000	2,88,75,000	—	—
Budgeted machine hours in department P1 (working note 1)	2,10,000	—		
Budgeted labour hours in department P2 (working note 1)	—	3,50,000		
Budgeted machine/ labour hour rate (₹)	165	82.50		

(ii) Comparative statement reflecting Budgeted cost and Actual cost for production of the Products X and Y during the month of December, 2024

(When 8,000 and 6,000 units of products X and Y respectively were actually produced)

	Budgeted (₹)	Actual (₹)
Raw materials used in Dept. P1:		
X : 8,000 units × ₹ 660	52,80,000	53,79,000
Y : 6,000 units × ₹ 825	49,50,000	50,16,000
Direct labour cost (on the basis of labour hours worked in department P2)		
X : 8,000 units × 2 hrs. × ₹ 396	63,36,000	65,10,900
Y : 6,000 units × 2.5 hrs. × ₹ 412.50	61,87,500	60,72,000
Overhead absorbed on machine hour basis in Dept. P1:		
X : 8,000 units × 1.5 hrs. × ₹ 165	19,80,000	19,18,084*
Y : 6,000 units × 1 hr. × ₹ 165	9,90,000	13,04,926*
Overhead absorbed on labour hour basis in Dept. P2:		
X : 8,000 units × 2 hrs. × ₹ 82.50	13,20,000	14,45,496**

	Budgeted (₹)	Actual (₹)
Y : 6,000 units × 2.5 hrs. × ₹ 82.50	12,37,500	13,04,472**

* (Refer to working note 4)

**(Refer to working note 5)

(iii) Amount of under/ over-absorption of production overheads

	Overhead absorbed	Overhead actually incurred	Overhead under/ over-absorbed
Overhead in Dept. P1			
Product X	19,80,000	19,18,084	**₹ 61,916 over-absorbed**
Product Y	9,90,000	13,04,926	**₹ 3,14,926 under-absorbed**
Overhead in Dept. P2			
Product X	13,20,000	14,45,496	**₹ 1,25,496 under-absorbed**
Product Y	12,37,500	13,04,472	**₹ 66,972 under-absorbed**

Working notes:

1.

	Product X	Product Y	Total
Budgeted output (in units)	1,00,000	60,000	
Budgeted machine hours in Dept. P1	1,50,000 (1,00,000×1.5 hrs.)	60,000 (60,000×1 hr.)	2,10,000
Budgeted labour hours in Dept. P2	2,00,000 (1,00,000×2 hrs.)	1,50,000 (60,000×2.5 hrs.)	3,50,000

2.

	Product X	Product Y	Total
Actual output (in units)	8,000	6,000	
Actual machine hours utilized in Dept. P1	12,200	8,300	20,500
Actual labour hours utilised in Dept. P2	16,400	14,800	31,200

3. Computation of actual overhead rates

	Production Department		Service Department	
	P1	**P2**	**S1**	**S2**
Actual factory overheads for the month of December, 2024 (₹)	25,41,000	22,44,000	6,60,000	5,28,000

	Production Department		Service Department	
Allocation of service Dept. S1's costs to production Dept. P1 and P2 equally (₹)	3,30,000	3,30,000	(6,60,000)	–
Allocation of service Dept. S2's costs to production Dept. P1 and P2 in the ratio of 2:1 (₹)	3,52,000	1,76,000	–	(5,28,000)
Total	32,23,000	27,50,000	--	--
Actual machine hours in Dept. P1 (working note 2)	20,500	--		
Actual labour hours in Dept. P2 (working note 2)	--	31,200		
Actual machine/ labour hour rate (₹)	157.22	88.14		

4. Actual overheads absorbed in Department P1 (based on machine hours)

X : 12,200 hrs × ₹ 157.22 = ₹ 19,18,084

Y : 8,300 hrs × ₹ 157.22 = ₹ 13,04,926

5. Actual overheads absorbed in Department P2 (based on labour hours)

X : 16,400 hrs × ₹ 88.14 = ₹ 14,45,496

Y : 14,800 hrs × ₹ 88.14 = ₹ 13,04,472

ICAI · P4 · RTP May 2025 · question p. 13-14 · answer p. 35-38

391 RTP **Jan 2025** Q4 MCQ

Gaarmentz Ltd. run a sewing factory for medical garments. But, the company suffers from the limiting factor i.e. labor. Each sewing machine needs 100% attention of one person at a particular point of time to operate it. The company has 8 number of alike sewing machines on which 8 operators work separately. The following particulars are furnished for a six months period:

Paid hours for all the 8 operators	9,594 hours
Effective working hours for all the 8 operators	9,360 hours
Average rate of wages per day of 8 hours per operator	₹ 110
Power consumed	₹ 60,125
Supervision and Indirect Labour	₹ 21,450

The following particulars are given for a year:

Insurance	₹ 4,68,000
Sundry Expenses	₹ 7,15,000

Depreciation charged is 10% on the original cost of all the sewing machines.

Repairs and Maintenance comes to 5% of the value of all the sewing machines.

The original cost of all the sewing machines works out to ₹ 41,60,000

CALCULATE the Comprehensive Machine Hour Rate.

A ₹ 215.86

B ₹ 217.99

C ₹ 116.43

D ₹ 119.34

ICAI'S ANSWER

(D) Computation of Comprehensive Machine Hour Rate

Particulars	Amount for six months (₹)
Operators' wages paid [(9,594 hrs./ 8 hrs.) x ₹ 110]	1,31,918
Power consumed	60,125
Supervision and indirect labour	21,450
Insurance (₹ 4,68,000/2)	2,34,000
Sundry expenses (₹ 7,15,000/2)	3,57,500
Depreciation {(₹ 41,60,000 × 10%)/2}	2,08,000
Repair and maintenance {(5% × ₹ 41,60,000)/2}	1,04,000
Total Overheads for 6 months	**11,16,993**
Comprehensive Machine Hour Rate = $\left(\frac{₹11,16,993}{9,360 \text{ hours}}\right)$	**119.34**

ICAI · P4 · RTP Jan 2025 · question p. 6-7 · answer p. 26

392 RTP **Sep 2024** Q9

From the details furnished below you are required to compute a comprehensive machine-hour rate:

Original purchase price of the machine (subject to depreciation at 10% per annum on original cost)	₹ 12,96,000
Normal working hours for the month (The machine works for only 75% of normal capacity)	200 hours
Wages to Machine-man	₹ 800 per day (of 8 hours)
Wages to Helper (machine attendant)	₹ 500 per day (of 8 hours)
Power cost for the month for the time worked	₹ 1,30,000

Supervision charges apportioned for the machine centre for the month	₹ 18,000
Electricity & Lighting (fixed in nature) for the month	₹ 9,500
Repairs & maintenance (machine) including consumable stores per month	₹ 17,500
Insurance of Plant & Building (apportioned) for the year	₹ 18,000
Other general expense per annum	₹ 18,000

The workers are paid a fixed dearness allowance of ₹ 4,500 per month. Production bonus payable to workers in terms of an award is equal to 10% of basic wages and dearness allowance. Add 10% of the basic wage and dearness allowance against leave wages and holidays with pay to arrive at a comprehensive labour-wage for debit to production.

ICAI'S ANSWER

Effective machine hours = 200 hours × 75% = 150 hours

Computation of Comprehensive Machine Hour Rate

	Per month (₹)	Per hour (₹)
Fixed cost		
Supervision charges	18,000.00	
Electricity and lighting	9,500.00	
Insurance of Plant and building (₹ 18,000 ÷12)	1,500.00	
Other General Expenses (₹ 18,000÷12)	1,500.00	
Depreciation (₹ 1,29,600÷12)	10,800.00	
	41,300.00	275.33
Direct Cost		
Repairs and maintenance	17,500.00	116.67
Power	1,30,000.00	866.67
Wages of machine man		196.00
Wages of Helper		136.00
Machine Hour rate (Comprehensive)		**1,590.67**

Wages per machine hour

	Machine man	Helper
Wages for 200 hours		
Machine-man (₹ 800 × 25)	₹ 20,000.00	---

	Machine man	Helper
Helper (₹ 500 × 25)	---	₹ 12,500.00
Dearness Allowance (DA)	₹ 4,500.00	₹ 4,500.00
	₹ 24,500.00	₹ 17,000.00
Production bonus (10% of Basic and DA)	2,450.00	1,700.00
Leave wages (10% of Basic and DA)	2,450.00	1,700.00
	29,400.00	20,400.00
Effective wage rate per machine hour	**196.00**	**136.00**

ICAI · P4 · RTP Sep 2024 · question p. 7-8 · answer p. 22-23

393 RTP **Old P3 Nov 2023** Q3 old scheme

The following particulars refer to process used in the treatment of material subsequently incorporated in a component forming part of an electrical appliance:

(i) The original cost of the machine used (Purchased in June 2018) was ₹ 10,00,000. Its estimated life is 10 years, the estimated scrap value at the end of its life is ₹ 10,000, and the estimated working time per year (50 weeks of 44 hours) is 2,200 hours . Out of which machine maintenance etc., is estimated to take up 200 hours.

No other loss of working time expected, setting up time, estimated at 100 hours, is regarded as productive time. (Holiday to be ignored).

(ii) Electricity used by the machine during production is 16 units per hour at cost of a ₹ 7 per unit. No power is consumed during maintenance or setting up.

(iii) The machine required a chemical solution which is replaced at the end of week at a cost of ₹ 2,000 each time.

(iv) The estimated cost of maintenance per year is ₹ 1,20,000.

(v) Two attendants control the operation of machine together with five other identical machines. Their combined weekly wages, insurance and the employer's contribution to holiday pay amount is ₹ 9,000.

(vi) Departmental and general works overhead allocated to this machine for the current year amount to ₹ 20,000.

You are required to calculate the machine hour rate of operating the machine.

ICAI'S ANSWER

Working Notes:

(i) Total Productive hours = Estimated Working hours – Machine Maintenance hours

= 2,200 hours – 200 hours = 2,000 hours

(ii) Depreciation per annum = $\frac{10,00,000 - 10,000}{10}$ = ₹ 99,000

(iii) Chemical solution cost per annum = ₹ 2,000 × 50 weeks = ₹ 1,00,000

(iv) Wages of attendants (per annum) = $\frac{9{,}000 \times 50 \text{ weeks}}{6 \text{ machines}}$ = ₹ 75,000

Calculation of Machine hour rate

Particulars	Amount (per annum)	Amount (per hour)
A. Standing Charge		
(i) Wages of attendants	75,000	
(ii) Departmental and general works overheads	20,000	
Total Standing Charge	95,000	
Standing Charges per hour		47.50
B. Machine Expense		
(iii) Depreciation	99,000	49.50
(iv) Electricity (7 × 1,900 × 16 units) ÷ 2,000	-	106.40
(v) Chemical solution	1,00,000	50.00
(vi) Maintenance cost	1,20,000	60.00
Machine operating cost per hour (A + B)		313.40

ICAI · Old P3 · RTP Nov 2023 · question p. 1-2 · answer p. 11

394 RTP **Old P3 Nov 2023** Q14(c) old scheme

How would you account for idle capacity cost in Cost Accounting?

ICAI'S ANSWER

Idle capacity costs are treated in the following ways in Cost Accounts:

(i) **If the idle capacity cost is due to unavoidable reasons:** A supplementary overhead rate may be used to recover the idle capacity cost. In this case, the costs are charged to the production capacity utilised.

(ii) **If the idle capacity cost is due to avoidable reasons:** Such as faulty planning, etc. the cost should be charged to Costing Profit and Loss Account.

(iii) **If the idle capacity cost is due to trade depression, etc.:** Being abnormal in nature the cost should also be charged to the Costing Profit and Loss Account.

(iv) **If the idle capacity cost is due to seasonal factors**, then the cost should be charged to cost of production by inflating overhead rate.

ICAI · Old P3 · RTP Nov 2023 · question p. 9 · answer p. 25

395 RTP **Old P3 Nov 2023** Q14(e) old scheme

Distinguish between cost allocation and cost absorption.

ICAI'S ANSWER

Distinguish between Cost allocation and Cost absorption:

Cost allocation is the allotment of whole item of cost to a cost centre or a cost unit. In other words, it is the process of identifying, assigning or allowing cost to a cost centre or a cost unit.

Cost absorption is the process of absorbing all indirect costs or overhead costs allocated or apportioned over particular cost center or production department by the units produced.

ICAI · Old P3 · RTP Nov 2023 · question p. 9 · answer p. 25

396 RTP **Old P3 Nov 2022** Q3 old scheme

SE Limited manufactures two products- A and B. The company had budgeted factory overheads amounting to ₹ 36,72,000 and budgeted direct labour hour of 1,80,000 hours. The company uses pre-determined overhead recovery rate for product costing purposes.

The department-wise break-up of the overheads and direct labour hours were as follows:

Particulars	Budgeted overheads	Budgeted direct labour hours	Rate per direct labour hour
Department Pie	₹ 25,92,000	90,000 hours	₹ 28.80
Department Qui	₹ 10,80,000	90,000 hours	₹ 12.00
Total	₹ 36,72,000	1,80,000 hours	

Additional Information:

Each unit of product A requires 4 hours in department Pie and 1 hour in department Qui. Also, each unit of product B requires 1 hour in department Pie and 4 hours in department Qui.

This was the first year of the company's operation. There was no WIP at the end of the year. However, 1,800 and 5,400 units of Products A and B were on hand at the end of the year.

The budgeted activity has been attained by the company. You are required to:

(i) DETERMINE the production and sales quantities of both products 'A' and 'B' for the above year.

(ii) ASCERTAIN the effect of using a pre-determined overhead rate instead of department-wise overhead rates on the company's income due to its effect on stock value.

(iii) CALCULATE the difference in the selling price due to the use of pre-determined overhead rate instead of using department-wise overhead rates. Assume that the direct costs (material and labour costs) per unit of products A and B were ₹ 25 and ₹ 40 respectively and the selling price is fixed by adding 40% over and above these costs to cover profit and selling and administration overhead.

ICAI'S ANSWER

(i) Computation of production and sales quantities:

The products processing times are as under –

Product	A	B	Total
Department Pie	4 hours	1 hour	90,000 hours
Department Qui	1 hour	4 hours	90,000 hours

Let X and Y be the number of units (production quantities) of the two products.

Converting these into equations, we have –

$4X + Y = 90,000$ &

$X + 4Y = 90,000$

Solving the above, we get X = 18,000; Y = 18,000

Hence, the Production and Sales Quantities are determined as under –

Product	Production Quantity	Closing Stock (Given)	Sales Quantity (Balancing Figure)
A	**18,000 units**	1,800 units	**16,200 units**
B	**18,000 units**	5,400 units	**12,600 units**

(ii) Effect of using pre-determined rate of overheads on the company's profit

Product	Closing Stock Quantity	Overhead included using pre-determined rate	Overhead included using department rate	Difference in overhead in closing stock value / Effect on closing stock value
A	1,800 units	1,800 × 5 hours x ₹ 20.40 = ₹ 1,83,600	Pie = 1,800 units x 4 hours x ₹ 28.80 = ₹ 2,07,360 Qui = 1,800 units x 1 hour x ₹ 12 = ₹ 21,600	**(-) ₹ 45,360**
B	5,400 units	5,400 × 5 hours x ₹ 20.40 = ₹ 5,50,800	Pie = 5,400 units x 1 hour x ₹ 28.80 = ₹ 1,55,520 Qui = 5,400 units x 4 hours x ₹12 = ₹ 2,59,200	**(+) ₹ 1,36,080**
Total		₹ 7,34,400	₹ 6,43,680	**(+) ₹ 90,720**

Use of pre-determined overhead rate has resulted in over valuation of stock by ₹ 90,720 due to which the company's income would be affected (increase) by ₹ 90,720. Profit would be affected only to the extent of Overhead contained in closing finished goods and closing WIP, if any.

(iii) Effect of using pre-determined on the products' selling prices

Particulars	Product A	Product B
Selling Price per unit if pre-determined overhead rate is used	₹177.80	₹ 198.80
Selling Price per unit if department wise rate is used	₹ 213.08	₹163.52
Difference	**₹ 35.28 Under-Priced**	**₹ 35.28 Over-Priced**

Workings:

(1) Pre-determined overhead recovery rate = $\frac{₹\ 36,72,000}{1,80,000\ \text{hours}} = ₹\ 20.40$ per direct labour hour

(2) If pre-determined recovery rate is used

Particulars	Product A in ₹	Product B in ₹
Materials & Labour	25.00	40.00
Add: Production Overhead A = 5 hours x ₹ 20.40 per hour B = 5 hours x ₹ 20.40 per hour	102.00	102.00
Cost of production	127.00	142.00
Add: 40% of margin	50.80	56.80
	177.80	**198.50**

(3) If department-wise recovery rate is used

Particulars	Product A in ₹	Product B in ₹
Materials & Labour	25.00	40.00
Add: Production Overhead A = Pie = 4 hours x ₹ 28.80 Qui = 1 hour x ₹ 12 B = Pie = 1 hour x ₹ 28.80 Qui = 4 hours x ₹ 12	127.20	76.80
Cost of production	152.20	116.80
Add: 40% of margin	60.88	46.72
Selling Price per unit	**213.08**	**163.52**

ICAI · Old P3 · RTP Nov 2022 · question p. 2-3 · answer p. 18-20

397 RTP **Old P3 Jul 2021** Q3 old scheme

A manufacturing unit has purchased and installed a new machine at a cost of ₹ 24,90,000 to its fleet of 5 existing machines. The new machine has an estimated life of 12 years and is expected to realise ₹ 90,000 as scrap value at the end of its working life.

Other relevant data are as follows:

(i) Budgeted working hours are 2,496 based on 8 hours per day for 312 days. Plant maintenance work is carried out on weekends when production is totally halted. The estimated maintenance hours are 416. During the production hours machine set-up and change over works are carried out. During the set-up hours no production is done. A total 312 hours are required for machine set-ups and change overs.

(ii) An estimated cost of maintenance of the machine is ₹ 2,40,000 p.a.

(iii) The machine requires a component to be replaced every week at a cost of ₹ 2,400.

(iv) There are three operators to control the operations of all the 6 machines. Each operator is paid ₹ 30,000 per month plus 20% fringe benefits.

(v) Electricity: During the production hours including set-up hours, the machine consumes 60 units per hour. During the maintenance the machine consumes only 10 units per hour. Rate of electricity per unit of consumption is ₹ 6.

(vi) Departmental and general works overhead allocated to the operation during last year was ₹ 5,00,000. During the current year it is estimated to increase by 10%.

Required:

COMPUTE the machine hour rate.

ICAI'S ANSWER

Working Note:

1. Effective machine hour:

= Budgeted working hours – Machine Set-up time

= 2,496 hours – 312 hours = 2,184 hours.

2. Operators' salary per annum:

Salary (3 operators × ₹30,000 × 12 months)	₹ 10,80,000
Add: Fringe benefits (20% of ₹10,80,000)	₹ 2,16,000
	₹ 12,96,000

3. Depreciation per annum

$$\frac{₹\ 24,90,000 - ₹\ 90,000}{12\ \text{years}} = ₹\ 2,00,000$$

Computation of Machine hour Rate

	Amount p.a. (₹)	Amount per hour (₹)
Standing charges		
Operators' Salary (₹12,96,000/6 machines × 1/2,184 hours)	12,96,000	98.90
Departmental and general overheads: (₹ 5,00,000 × 110%) (₹5,50,000/6 machines × 1/2,184 hours)	5,50,000	41.97
(A)	18,46,000	140.87
Machine Expenses		
Depreciation (₹2,00,000/2,184 hours)	2,00,000	91.58
Electricity:		
During working hours (2,496 hours × 60 units ×₹6)	8,98,560	411.43
During maintenance hours (416 hours × 10 units ×₹6)	24,960	11.43
Component replacement cost (2,400 × 52 weeks)	1,24,800	57.14
Machine maintenance cost	2,40,000	109.89
(B)	14,88,320	681.47
Machine Hour Rate (A + B)		822.34

398 RTP **Old P3 Nov 2020** Q3 old scheme

You are given the following information of the three machines of a manufacturing department of X Ltd.:

Preliminary estimates of expenses (per annum)				
	Total (₹)	Machines		
		A (₹)	B (₹)	C (₹)
Depreciation	2,00,000	75,000	75,000	50,000
Spare parts	1,00,000	40,000	40,000	20,000
Power	4,00,000			
Consumable stores	80,000	30,000	25,000	25,000
Insurance of machinery	80,000			
Indirect labour	2,00,000			
Building maintenance expenses	2,00,000			
Annual interest on capital outlay	1,00,000	40,000	40,000	20,000
Monthly charge for rent and rates	20,000			
Salary of foreman (per month)	42,000			
Salary of Attendant (per month)	12,000			

(The foreman and the attendant control all the three machines and spend equal time on them.)

The following additional information is also available:

	Machines		
	A	B	C
Estimated Direct Labour Hours	1,00,000	1,50,000	1,50,000
Ratio of K.W. Rating	3	2	3
Floor space (sq. ft.)	40,000	40,000	20,000

There are 12 holidays besides Sundays in the year, of which two were on Saturdays. The manufacturing department works 8 hours in a day but Saturdays are half days. All machines work at 90% capacity throughout the year and 2% is reasonable for breakdown.

You are required to :

CALCULATE predetermined machine hour rates for the above machines after taking into consideration the following factors:

- An increase of 15% in the price of spare parts.
- An increase of 25% in the consumption of spare parts for machine 'B' & 'C' only.
- 20% general increase in wages rates.

ICAI'S ANSWER

(a) Computation of Machine Hour Rate

Particulars	Basis of apportionment	Total (₹)	Machines		
			A (₹)	B (₹)	C (₹)
(A) Standing Charges					
Insurance	Depreciation Basis (3:3:2)	80,000	30,000	30,000	20,000
Indirect Labour	Direct Labour (2:3:3)	2,40,000	60,000	90,000	90,000
Building maintenance expenses	Floor Space (2:2:1)	2,00,000	80,000	80,000	40,000
Rent and Rates	Floor Space (2:2:1)	2,40,000	96,000	96,000	48,000
Salary of foreman	Equal	5,04,000	1,68,000	1,68,000	1,68,000
Salary of attendant	Equal	1,44,000	48,000	48,000	48,000
Total standing charges		14,08,000	4,82,000	5,12,000	4,14,000
Hourly rate for standing charges			247.43	262.83	212.53
(B) Machine Expenses:					
Depreciation	Direct	2,00,000	75,000	75,000	50,000
Spare parts	Final estimates	1,32,250	46,000	57,500	28,750
Power	K.W. rating (3:2:3)	4,00,000	1,50,000	1,00,000	1,50,000
Consumable Stores	Direct	80,000	30,000	25,000	25,000
Total Machine expenses		8,12,250	3,01,000	2,57,500	2,53,750
Hourly Rate for Machine expenses			154.52	132.19	130.26
Total (A + B)		22,20,250	7,83,000	7,69,500	6,67,750
Machine Hour rate			**401.95**	**395.02**	**342.79**

Working Notes:

(i) Calculation of effective working hours:

No. of full off-days = No. of Sunday + No. of holidays

= 52 + 12 = 64 days

No. of half working days = 52 days – 2 holidays = 50 days

No. of full working days = 365 days – 64 days – 50 days = 251 days

Total working Hours = {(251 days × 8 hours) + (50 days × 4 hours)}

= 2,008 hours + 200 = 2,208 hours.

Total effective hours = Total working hours × 90% - 2% for break-down

= 2,208 hours × 90% - 2% (2,208 hours × 90%)

= 1,987.2 hours – 39.74 hours

= 1947.46 or Rounded up to 1948 hours.

(ii) Amount of spare parts is calculated as under:

	A (₹)	B (₹)	C (₹)
Preliminary estimates	40,000	40,000	20,000
Add: Increase in price @ 15%	6,000	6,000	3,000
	46,000	46,000	23,000
Add: Increase in consumption @ 25%	–	11,500	5,750
Estimated cost	46,000	57,500	28,750

(iii) Amount of Indirect Labour is calculated as under:

	(₹)
Preliminary estimates	2,00,000
Add: Increase in wages @ 20%	40,000
	2,40,000

(iv) Interest on capital outlay is a finance cost, therefore it has been excluded from the cost accounts.

ICAI · Old P3 · RTP Nov 2020 · question p. 2-3 · answer p. 15-16

399 MTP Sep 2026 S-I Q6(c) 4 marks

EXPLAIN Single and Multiple Overhead Rates.

ICAI'S ANSWER

Single and Multiple Overhead Rates: Single overhead rate: It is one single overhead absorption rate for the whole factory.

It may be computed as follows:

$$\text{Single overhead rate} = \frac{\text{Overhead costs for the entire factory}}{\text{Total quantity of the base selected}}$$

The base can be total output, total labour hours, total machine hours, etc.

The single overhead rate may be applied in factories which produces only one major product on a continuous basis. It may also be used in factories where the work performed in each department is fairly uniform and standardized.

Multiple overhead rate: It involves computation of separate rates for each production department, service department, cost center and each product for both fixed and variable overheads. It may be computed as follows:

$$\text{Multiple overhead rate} = \frac{\text{Overhead allocated / appportioned to each department / cost centre or product}}{\text{Corresponding base}}$$

Under multiple overheads rate, jobs or products are charged with varying amount of factory overheads depending on the type and number of departments through which they pass. However, the number of overheads rate which a firm may compute would depend upon two opposing factors viz. the degree of accuracy desired and the clerical cost involved.

ICAI · P4 · MTP Sep 2026 · Series I · question p. 15 · answer p. 22

400 **MTP** **Jan 2026 S-II** Q1(b) **5 marks**

A machine shop cost centre contains three machines of equal capacities. To operate these three machines nine operators are required i.e. three operators on each machine. Operators are paid ₹ 20 per hour. The factory works for fourty eight hours in a week which includes 4 hours set up time. The work is jointly done by operators. The operators are paid fully for the forty eight hours. In additions they are paid a bonus of 10 per cent of productive time. Costs are reported for this company on the basis of thirteen four-weekly period.

The company for the purpose of computing machine hour rate includes the direct wages of the operator and also recoups the factory overheads allocated to the machines. The following details of factory overheads applicable to the cost centre are available:

- Depreciation 10% per annum on original cost of the machine. Original cost of the each machine is ₹ 52,000.
- Maintenance and repairs per week per machine is ₹ 60.
- Consumable stores per week per machine are ₹ 75.
- Power : 20 units per hour per machine at the rate of 80 paise per unit.
- Apportionment to the cost centre : Rent per annum ₹ 5,400, Heat and Light per annum ₹ 9,720, foreman's salary per annum ₹ 12,960 and other miscellaneous expenditure per annum ₹ 18,000.

Required:

(i) CALCULATE the cost of running one machine for a four week period.

(ii) CALCULATE machine hour rate.

ICAI'S ANSWER

Effective Machine hour for four-week period

= Total working hours – unproductive set-up time

= {(48 hours × 4 weeks) – {(4 hours × 4 weeks)}

= (192 – 16) hours) =176 hours.

(i) Computation of cost of running one machine for a four week period

		(₹)	(₹)
(A)	**Standing charges (per annum)**		
	Rent	5,400.00	
	Heat and light	9,720.00	
	Forman's salary	12,960.00	
	Other miscellaneous expenditure	18,000.00	
	Standing charges (per annum)	46,080.00	
	Total expenses for one machine for four week period (₹46,080 / (3 machines × 13 four – week period))		1,181.54
	Wages (48 hours × 4 weeks × ₹ 20 × 3 operators)		11,520.00
	Bonus {(176 hours × ₹ 20 × 3 operators) × 10%}		1,056.00
	Total standing charges		13,757.54
(B)	Machine Expenses		
	Depreciation = (₹52,000 × 10% × 1/13 four - week period)		400.00
	Repairs and maintenance (₹ 60 × 4 weeks)		240.00
	Consumable stores (₹ 75 × 4 weeks)		300.00
	Power (176 hours × 20 units ×₹ 0 .80)		2,816.00
	Total machine expenses		3,756.00
(C)	Total expenses (A) + (B)		17,513.54

(ii) Machine hour rate = $\frac{₹17,513.54}{176 \text{ hours}}$ = ₹99.51

ICAI · P4 · MTP Jan 2026 · Series II · question p. 8 · answer p. 8-9

401 MTP **Jan 2026 S-II** Q6(a) 5 marks

Divit Electronics manufactures three types of home appliances. The management notes that the plant has an installed capacity of 50,000 units per product per year. Because of preventive maintenance, setup time, holidays, and training sessions, the plant can only produce 45,000 units per product annually. Over the past year, the average output per product was 42,000 units, while unexpected machine breakdowns reduced actual output to 40,500 units.

DEFINE AND IDENTIFY the installed, practical, normal, and actual capacities for the plant. EXPLAIN the role of practical capacity in determining overhead absorption rates.

ICAI'S ANSWER

1. Identification of Capacities

Capacity Type	Definition	Units (per product)
Installed (Rated or Theoretical) Capacity	It refers to the maximum capacity of producing goods or providing services.	50,000 units
Practical (Operating) Capacity	It is defined as actually utilised capacity of a plant after considering planned losses such as preventive maintenance, setup times, holidays, and training	45,000 units
Normal Capacity	Normal capacity is the volume of production or services achieved or achievable on an average over a period under normal circumstances taking into account the reduction in capacity resulting from planned maintenance.	42,000 units
Actual Capacity	It is the Capacity actually achieved during a given period	40,500 units

2. Role of Practical Capacity in Overhead Absorption

- It reflects the **available productive capacity** after accounting for planned downtime, allowing management to allocate overhead costs fairly and accurately to products.
- Using practical capacity **avoids over- or under-absorbing overheads** that would occur if installed capacity (theoretical maximum) or actual capacity were used.

ICAI · P4 · MTP Jan 2026 · Series II · question p. 17 · answer p. 21

402 **MTP** **May 2025 S-II** Q1(c) **4 marks**

A company having a factory in Chennai has a 8 machineries in the process of manufacture. The company has purchased a new machinery costing ₹ 19,05,000 with a useful life of 12 years and a salvage value of ₹ 1,05,000 at the end of its useful life. The following data are as follows:

(i) The factory works for 324 days a year with 8 hours shift daily. Plant maintenance of 300 hours and set up of 92 hours are included in the above.

(ii) Estimated cost of maintenance of a machine is ₹ 37,500 per annum.

(iii) Operators wages amounts to ₹ 3,630 per week with additional benefits of 15%. 4 Operators are required to operate the machinery.

(iv) Electricity consumed during production is 16 units per hour at a cost of ₹ 4.5 per unit. No power is consumed during maintenance and set up.

(v) General overheads allocated amounts to 75,000 upto the prior year is expected to increase by 10% during the current year.

(vi) Special performance chemical at ₹ 600 for every 6 days of operation.

DETERMINE Machine hour rate if set up time is (a) productive (b) unproductive.

ICAI'S ANSWER

Statement of Overhead costs

Particulars	Computation	Amount (₹)
Depreciation	(19,05,000 – 1,05,000)/12	1,50,000
Operator's wages	(₹ 3,630 × 4 Operators x 54 Weeks x 115%) / 8 Machines	1,12,711.5
Maintenance Cost	Given – ₹ 37,500	37,500
Electricity	(2,592 hours – 392 hours) x 16 x ₹ 4.5	1,58,400
General Overhead allocated	(75,000 × 110%/8 machines)	10,312.5
Performance Chemical	(₹ 600/6 days) x 324	32,400
Total Overheads		5,01,324

Computation of Machine Hour Rate

When set up time is productive,

Effective machine hours = 2,592 hours – 300 hours maintenance = 2,292 hours

Hence Machine hour rate = 5,01,324/2,292 = **₹ 218.73**

When set up time is unproductive,

Effective machine hours = 2,592 hours – 300 hours maintenance – 92 hours set up = 2,200 hours.

Machine hour rate = 5,01,324/2,200 = **₹ 227.87 per hour**

ICAI · P4 · MTP May 2025 · Series II · question p. 8 · answer p. 8-9

403 MTP **May 2025 S-II** Q4(c) 2 marks

EXPLAIN how would you treat the idle capacity costs in Cost Accounts.

ICAI'S ANSWER

Idle capacity costs can be treated in product costing, in the following ways:

(a) If the idle capacity cost is due to unavoidable reasons such as repairs, maintenance, changeover of job etc., a supplementary overhead rate may be used to recover the idle capacity cost. In this case, the costs are charged to the production capacity utilised.

(b) If the idle capacity cost is due to avoidable reasons such as faulty planning, power failure etc.; the cost should be charged to costing profit and loss account.

(c) If the idle capacity cost is due to seasonal factors, then, the cost should be charged to the cost of production by inflating overhead rates.

ICAI · P4 · MTP May 2025 · Series II · question p. 12 · answer p. 14-15

404 MTP **Jan 2025 S-II** Q1(a) 5 marks

Shanu Ltd has calculated a predetermined overhead rate of ₹ 22 per machine hour for its Quality Check (QC) department. This rate has been calculated for the budgeted level of activity and is considered as appropriate for absorbing overheads. The following overhead expenditures at various activity levels had been estimated.

Total overheads	Number of machine hours
₹ 3,38,875	14,500
₹ 3,47,625	15,500
₹ 3,56,375	16,500

You are required to:

(i) CALCULATE the variable overhead absorption rate per machine hour.

(ii) CALCULATE the estimated total fixed overheads.

(iii) CALCULATE the budgeted level of activity in machine hours.

(iv) CALCULATE the amount of under/over absorption of overheads if the actual machine hours were 14,970 and actual overheads were ₹ 3,22,000.

(v) STATE the arguments for and against using departmental absorption rates as opposed to a single or blanket factory wide rate.

ICAI'S ANSWER

(i) Variable overhead absorption rate: = $\frac{\text{Difference in Total Overheads}}{\text{Difference in levels in terms of machine hours}}$

= $\frac{₹3,47,625 - ₹3,38,875}{15,500 \text{ hours} - 14,500 \text{ hours}}$ = ₹ 8.75 per machine hour.

(ii) Calculation of Total fixed overheads:

	(₹)
Total overheads at 14,500 hours	3,38,875
Variable overheads = ₹ 8.75 × 14,500	1,26,875
Total fixed overheads	2,12,000

(iii) Calculation of Budgeted level of activity in machine hours:

Let budgeted level of activity = X

Then, $\frac{(₹8.75X + ₹2,12,000)}{X}$ = ₹ 22

8.75X + ₹ 2,12,000 = 22X

13.25X = 2,12,000

X =16,000

Thus, budgeted level of activity = 16,000 machine hours.

(iv) Calculation of Under / Over absorption of overheads:

	(₹)
Actual overheads	3,22,000
Absorbed overheads = 14,970 hours × ₹ 22 per hour	3,29,340
Over-absorption (3,29,340 – 3,22,000)	7,340

(v) Departmental absorption rates provide costs which are more precise than those provided by the use of blanket absorption rates. Departmental absorption rates facilitate variance analysis and cost control. The application of these rates makes the task of stock and work-in-process (WIP) valuation easier and more precise. However, the setting up and monitoring of these rates can be time-consuming and expensive.

ICAI · P4 · MTP Jan 2025 · Series II · question p. 7 · answer p. 6-7

405 MTP **Sep 2024 S-I Q4(a)** 6 marks

The following particulars refer to process used in the treatment of material subsequently, incorporated in a component forming part of an electrical appliance:

(i) The original cost of the machine used (Purchased in June 2023) was ₹ 10,000. Its estimated life is 10 years, the estimated scrap value at the end of its life is ₹ 1,000, and the estimated working time per year (50 weeks of 44 hours) is 2,200 hours of which machine maintenance etc., is estimated to take up 200 hours.

No other loss of working time expected. Setting up time, estimated at 100 hours, is regarded as productive time. (Holiday to be ignored).

(ii) Electricity used by the machine during production is 16 units per hour at cost of a 9 paisa per unit. No current is taken during maintenance or setting up.

(iii) The machine required a chemical solution which is replaced at the end of week at a cost of ₹ 20 each time.

(iv) The estimated cost of maintenance per year is ₹ 1,800.

(v) Two attendants control the operation of machine together with five other identical machines. Their combined weekly wages, insurance and the employer's contribution to holiday pay amount ₹ 120.

(vi) Departmental and general works overhead allocated to this machine for the current year amount to ₹ 3,000.

You are required to CALCULATE the machine hour rate of operating the machine.

ICAI'S ANSWER

Working Notes:

(i) Total Productive hours = Estimated Working hours – Machine Maintenance hours

= 2,200 hours – 200 hours = 2,000 hours

(ii) Depreciation per annum = $\frac{₹\ 10,000 - ₹\ 1,000}{10\ \text{years}}$ = ₹ 900

(iii) Chemical solution cost per annum = ₹ 20 × 50 weeks = ₹ 1,000

(iv) Wages of attendants (per annum) = $\frac{₹\ 120 \times 50\ \text{weeks}}{6\ \text{machines}}$ = ₹ 1,000

Calculation of Machine hour rate

Particulars	Amount (per annum)	Amount (per hour)
A. Standing Charge		
(i) Wages of attendants	1,000	
(ii) Departmental and general works overheads	3,000	
Total Standing Charge	4,000	
Standing Charges per hour (4,000/2,000)		2.0
B. Machine Expense		
(iii) Depreciation	900	0.45
(iv) Electricity (₹ 0.09×16units×1,900hours/2,000hours)	-	1.37
(v) Chemical solution	1,000	0.50
(vi) Maintenance cost	1,800	0.90
Machine operating cost per hour (A + B)		5.22

ICAI · P4 · MTP Sep 2024 · Series I · question p. 9-10 · answer p. 11-12

406 MTP **Sep 2024 S-II** MCQ 13 2 marks MCQ

ABC Manufacturing allocates its factory overhead costs based on machine hours. The total estimated overhead cost for the year is ₹ 6,00,000, and the company expects to use 30,000 machine hours. During the year, job A used 300 machine hours. What amount of overhead costs should be allocated to this job?

A ₹ 4,000

B ₹ 6,000

C ₹ 10,000

D ₹ 8,000

ICAI'S ANSWER

(B)

Overhead Rate = Total Estimated Machine Hours/Total Estimated Overhead Cost

= ₹ 6,00,000/30,000 = ₹ 20

Allocated Overhead = Overhead Rate x Machine Hours Used by the Job

= ₹ 20 x 300 hrs = ₹ 6,000

ICAI · P4 · MTP Sep 2024 · Series II · question p. 5 · answer p. 5

407 MTP Sep 2024 S-II Case scenario · Q4(a) to Q4(d) 14 marks

CASE SCENARIO

Allurgy Ltd. is into metallic tools manufacturing. It has four production departments. The work performed in every department is fairly uniform, thus the manager of the company created a policy to recover the production overheads of the entire company by adopting a single blanket rate.

The relevant data for a month are given below:

Departments	Direct Materials (₹)	Direct Wages (₹)	Factory Overheads (₹)	Direct Labour Hours	Machine Hours
Budget:					
Operating	64,35,000	7,92,000	35,64,000	1,98,000	7,92,000
Assembly	11,73,000	24,15,000	9,66,000	6,90,000	69,000
Quality Control	5,10,000	10,50,000	4,20,000	3,00,000	30,000
Packing	9,90,000	6,93,000	12,37,500	4,95,000	-
Actual:	-	-	-	-	-
Operating	77,22,000	9,50,400	38,61,000	2,37,600	9,50,400
Assembly	9,38,400	18,63,000	5,79,600	6,21,000	75,900
Quality Control	4,08,000	8,10,000	2,52,000	2,70,000	33,000
Packing	11,88,000	8,91,000	13,36,500	5,94,000	-

Additional details relating to one of the jobs during the month are also provided below:

Job No. 157

Departments	Direct Materials (₹)	Direct Wages (₹)	Direct Labour Hours	Machine Hours
Operating	11,880	2,376	594	1,782
Assembly	4,140	2,484	828	207
Quality Control	1,800	1,080	360	90
Packing	2,970	594	396	-

During Quality Control phase of this particular Job, the company incurred certain additional expenditure of ₹ 495 on direct wages as there were certain production that was not as perfect as the saleable product. The defective units were normal in nature and after rectification have been brought to the required degree of perfection.

The company adds 25% on the factory cost to cover administration overheads and profit.

You are required to figure out the following:

Q4(a) 3 marks

COMPUTE the overhead absorption rate as per the blanket rate based on the percentage of total factory overheads to total factory wages and determine the selling price of the Job No. 157.

Q4(b) 4 marks

The new manager thinks that the machinery is used to a varying degree in the different departments. Thus, it is not appropriate to follow one blanket rate for the whole company. Therefore, suggest an alternative method of absorption of the factory overheads and CALCULATE the overhead rates based on the method so suggested.

Q4(c) 3 marks

DETERMINE the selling price of Job 157 based on the overhead rates calculated in (b) above.

Q4(d) 4 marks

CALCULATE the department-wise under or over recovery of overheads based on the company's current policy and the method suggested in (b) above.

ICAI'S ANSWERS

Q4(a)

Computation of overhead absorption rate

(as per the blanket rate)

Department	Budgeted factory Overheads (₹)	Budgeted direct wages (₹)
Operating	35,64,000	7,92,000
Assembly	9,66,000	24,15,000
Quality Control	4,20,000	10,50,000
Packing	12,37,500	6,93,000
Total	**61,87,500**	**49,50,000**

$$\text{Overhead absorption rate} = \frac{\text{Budgeted factory Overheads}}{\text{Budgeted direct wages}} \text{ x } 100$$

$$= \frac{61,87,500}{49,50,000} \text{ x } 100$$

= 125% of Direct wages

Selling Price of the Job No. 157

Particulars	Operating (₹)	Assembly (₹)	Quality Control (₹)	Packing (₹)	Total (₹)
Direct Materials	11,880	4,140	1,800	2,970	20,790
Direct Wages	2,376	2,484	1,080	594	6,534
Rectification cost of normal defectives			495		495
Overheads [(125% x (6,534 + 495)]					8,786.25
Total Factory Cost					**36,605.25**
Add: Mark-up (25% x ₹ 36,605.25)					9,151.31
Selling Price					**45,756.56**

Q4(b)

As the machinery is used to a varying degree in different departments, the use of departmental rates is to be preferred. The overhead recovery rates in different departments would be as follows:

(i) Operating Department: The use of machine hours is the predominant factor of production in Operating Department. Hence, machine hour rate should be used to recover overheads.

The overhead recovery rate based on machine hours would be calculated as follows:

$$\text{Machine hour rate} = \frac{\text{Budgeted factory Overheads}}{\text{Budgeted machine hours}}$$

$$= \frac{₹\ 35,64,000}{7,92,000} = ₹\ 4.50 \text{ per hour}$$

(ii) Assembly Department: Direct labour hours is the main factor of production in Assembly Department. Hence, direct labour hour rate should be used to recover overheads.

The overhead recovery rate based on direct labour hours would be calculated as follows:

$$\text{Direct labour hour rate} = \frac{\text{Budgeted factory Overheads}}{\text{Budgeted direct labour hours}}$$

$$= \frac{₹\ 9,66,000}{6,90,000} = ₹\ 1.40 \text{ per hour}$$

(iii) Quality Control Department: Direct labour hours is the main factor of production in Quality Control Department. Hence, direct labour hour rate should be used to recover overheads.

The overhead recovery rate based on direct labour hours would be calculated as follows:

$$\text{Direct labour hour rate} = \frac{\text{Budgeted factory Overheads}}{\text{Budgeted direct labour hours}}$$

$$= \frac{₹\ 4,20,000}{3,00,000} = ₹\ 1.40 \text{ per hour}$$

(iv) Packing Department: Direct labour hours is the main factor of production in Packing Department. Hence, direct labour hour rate should be used to recover overheads.

The overhead recovery rate based on direct labour hours would be calculated as follows:

$$\text{Direct labour hour rate} = \frac{\text{Budgeted factory Overheads}}{\text{Budgeted direct labour hours}}$$

$= \frac{₹\ 12{,}37{,}500}{4{,}95{,}000} = ₹\ 2.50 \text{ per hour}$

Q4(c)

Selling Price of Job No. 157

[based on the overhead rates calculated in (b) above]

Particulars	Operating (₹)	Assembly (₹)	Quality Control (₹)	Packing (₹)	Total (₹)
Direct Materials	11,880	4,140	1,800	2,970	20,790
Direct Wages	2,376	2,484	1,080	594	6,534
Rectification cost of normal defectives			495		495
Overheads (refer working note)					10,672
Total Factory Cost					**38,491**
Add: Mark-up (25% x ₹ 38,491)					9,622.75
Selling Price					**48,113.75**

Working note:

Overhead Statement

Department	Basis	Hours	Rate (₹)	Overheads (₹)
Operating	Machine hour	1,782	4.50	8,019
Assembly	Direct labour hour	828	1.40	1,159
Quality Control	Direct labour hour	360	1.40	504
Packing	Direct labour hour	396	2.50	990
			Total	10,672

Q4(d)

Department-wise statement of under or over recovery of overheads

(i) As per the current policy

Particulars	Operating (₹)	Assembly (₹)	Quality Control (₹)	Packing (₹)	Total (₹)
Direct wages (Actual)	9,50,400	18,63,000	8,10,000	8,91,000	45,14,400
Overheads recovered @ 125% of Direct wages: (A)	11,88,000	23,28,750	10,12,500	11,13,750	56,43,000
Actual overheads: (B)	38,61,000	5,79,600	2,52,000	13,36,500	60,29,100
(Under)/Over recovery of overheads: (A–B)	**(26,73,000)**	**17,49,150**	**7,60,500**	**(2,22,750)**	**(3,86,100)**

(ii) As per the method suggested

	Machine hours (Operating)	Direct labour hours (Assembly)	Direct labour hours (Quality Control)	Direct labour hours (Packing)	Total (₹)
Hours worked	9,50,400	6,21,000	2,70,000	5,94,000	
Rate/hour (₹)	4.50	1.40	1.40	2.50	
Overhead recovered (₹): (A)	42,76,800	8,69,400	3,78,000	14,85,000	70,09,200
Actual overheads (₹): (B)	38,61,000	5,79,600	2,52,000	13,36,500	60,29,100
(Under)/Over recovery: (A–B)	**4,15,800**	**2,89,800**	**1,26,000**	**1,48,500**	**9,80,100**

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The following information are available for the three machines of a manufacturing department of KBC Ltd.:

Preliminary estimates of expenses				
	Total	**(per annum)**		
		Machines		
		P	**Q**	**R**
	(₹)	**(₹)**	**(₹)**	**(₹)**
Depreciation	20,000	7,500	7,500	5,000
Spare parts	10,000	4,000	4,000	2,000
Power	40,000			
Consumable stores	10,000	4,000	3,000	3,000
Insurance of machinery	8,000			
Indirect labour	20,000			

Preliminary estimates of expenses				
Building maintenance expenses	20,000			
Annual interest on capital outlay	60,000	25,000	25,000	10,000
Monthly charge for rent and rates	10,000			
Salary of foreman (per month)	20,000			
Salary of Attendant (per month)	5,000			

(The foreman and the attendant control all the three machines and spend equal time on them.)

The following additional information is also available:

	Machines		
	P	Q	R
Estimated Direct Labour Hours	1,00,000	1,50,000	1,50,000
Ratio of K.W. Rating	3	2	3
Floor space (sq. ft.)	40,000	40,000	20,000

There are 14 holidays besides Sundays in the year, of which two were on Saturdays. The manufacturing department works 8 hours in a day but Saturdays are half days. All machines work at 85% capacity throughout the year and 2% is reasonable for breakdown.

You are required to :

CALCULATE predetermined machine hour rates for the above machines after taking into consideration the following factors:

- An increase of 15% in the price of spare parts.
- An increase of 25% in the consumption of spare parts for machine 'Q' & 'R' only.
- 20% general increase in wages rates.
- An 10% decrease in the consumption of consumable stores.

ICAI'S ANSWER

Computation of Machine Hour Rate

		Basis of apportionment	Total	Machines		
				P	Q	R
			(₹)	(₹)	(₹)	(₹)
(A)	**Standing Charges**					
	Insurance	Depreciation Basis	8,000	3,000	3,000	2,000
	Indirect Labour	Direct Labour	24,000	6,000	9,000	9,000

		Basis of apportionment	Total	Machines		
	Building Maintenance expenses	Floor Space	20,000	8,000	8,000	4,000
	Rent and Rates	Floor Space	1,20,000	48,000	48,000	24,000
	Salary of foreman	Equal	2,40,000	80,000	80,000	80,000
	Salary of attendant	Equal	60,000	20,000	20,000	20,000
	Total standing charges		4,72,000	1,65,000	1,68,000	1,39,000
	Hourly rate for standing charges			90.36	92.00	76.12
(B)	**Machine Expenses:**					
	Depreciation	Direct	20,000	7,500	7,500	5,000
	Spare parts	Final estimates	13,225	4,600	5,750	2,875
	Power	K.W. rating	40,000	15,000	10,000	15,000
	Consumable Stores	Direct	9,000	3,600	2,700	2,700
	Total Machine expenses		82,225	30,700	25,950	25,575
	Hourly Rate for Machine expenses			16.81	14.21	14.01
	Total (A + B)		5,54,225	1,95,700	1,93,950	1,64,575
	Machine Hour rate			107.17	106.22	90.13

Working Notes:

(i) Calculation of effective working hours:

No. of holidays 52 (Sundays) + 14 (other holidays) = 66

Saturday (52 – 2) = 50

No. of days (Work full time) = 365 – 66 – 50 = 249

	Hours
Full days work 249 × 8	= 1,992
Half days work 50 × 4	= 200
	2,192

	Hours
Effective capacity 85% of 2,192	1,863 (Rounded off)
Less: Normal loss of time (Breakdown) 2%	37 (Rounded off)
Effective running hour	1,826

(ii) Amount of spare parts is calculated as under:

	P	Q	R
	₹	₹	₹
Preliminary estimates	4,000	4,000	2,000
Add: Increase in price @ 15%	600	600	300
	4,600	4,600	2,300
Add: Increase in consumption @ 25%	–	1,150	575
Estimated cost	4,600	5,750	2,875

(iii) Amount of Indirect Labour is calculated as under:

	₹
Preliminary estimates	20,000
Add: Increase in wages @ 20%	4,000
	24,000

(iv) Amount of Consumables Stores is calculated as under:

	₹
Preliminary estimates	10,000
Less: Decrease in consumption @ 10%	1,000
	9,000

(v) Interest on capital outlay is a financial matter and, therefore it has been excluded from the cost accounts.

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409 MTP **May 2024 S-II** Q4(a) 8 marks

The following are the budgeted details are available from the records of a manufacturing company SP

Ltd.:

	₹	₹
Direct Materials		2,13,000
Direct Wages:		
Machine Shop (12,000 hours)	63,000	
Assembly Shop (10,000 hours)	48,000	1,11,000
Works Overhead:		
Machine Shop	88,200	
Assembly Shop	51,800	1,40,000
Administrative Overhead		92,800
Selling Overhead		81,000
Distribution Overhead		62,100

You are required to:

(a) PREPARE a Schedule of Overhead Rates from the figures available stating the basis of overhead recovery rates used under the given circumstances.

(b) WORK OUT a Cost Estimate for the following job based on overhead calculated on above basis.

Direct Material:	25 kg @ ₹ 17.20/kg
	15 kg @ ₹ 21.00/kg
Direct labour: (On the basis of hourly rate	Machine shop 30 hours
For machine shop and assembly shop)	Assembly shop 42 hours

ICAI'S ANSWER

Job Cost Sheet for the period.....

			₹
Direct materials			2,13,000
Direct wages:			
Machine shop		63,000	
Assembly shop		48,000	1,11,000
	Prime Cost		3,24,000
Works overhead:			
Machine shop		88,200	

			₹
Assembly shop		51,800	1,40,000
	Work Cost		4,64,000
Administration overhead			92,800
	Cost of Production		5,56,800
Selling overhead			81,000
Distribution overhead			62,100
	Total Cost		6,99,900

Schedule of Overhead Rate

(i) Works Overhead: Hourly rate = (Overhead amount ÷ Hours)

Machine shop = (88,200 ÷ 12,000) = ₹ 7.35 per hour

Assembly shop = (51,800 ÷ 10,000) = ₹ 5.18 per hour

(ii) Administrative Overhead as a % of works cost

$= \frac{92,800}{4,64,000} \times 100 = 20\%$

(iii) Selling and distribution overhead as % of works cost

$= \frac{81,000 + 62,100}{4,64,000} \times 100 = 30.84\%$

Labour hour rates are calculated as under:

Machine shop = ₹ 63,000 ÷ 12,000 hrs. = ₹ 5.25

Assembly shop = ₹ 48,000÷10,000 hrs. = ₹ 4.80

(b) **Cost Estimate for Job**

Direct Materials	₹	₹
(i) 25 kg @ ₹ 17.20 per kg	430	
(ii) 15 kg @ ₹ 21 per kg	315	745.00
Direct Labour		
Machine shop (30 hrs. @ ₹ 5.25)	157.50	
Assembly shop (42 hrs. @ ₹ 4.80)	201.60	359.10
Prime Cost		1104.10
Works Overhead		
Machine shop (30 hours @ ₹ 7.35)	220.50	
Assembly shop (42 hours @ ₹ 5.18)	217.56	438.06

Direct Materials	₹	₹
Works Cost		1542.16
Administration overhead (20% of works cost)		308.43
Cost of Production		1850.59
Selling and distribution cost (30.84% of works cost)		475.60
Total Estimated Cost		2326.19

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410 MTP **Old P3 Nov 2023 S-I** Q4(a) 10 marks old scheme

From the details furnished below you are required to COMPUTE a comprehensive machine-hour rate:

Original purchase price of the machine (subject to depreciation at 10% per annum on original cost)	₹ 6,48,000
Normal working hours for the month (The machine works for only 75% of normal capacity)	200 hours
Wages to Machine-man	₹ 400 per day (of 8 hours)
Wages to Helper (machine attendant)	₹ 275 per day (of 8 hours)
Power cost for the month for the time worked	₹ 65,000
Supervision charges apportioned for the machine centre for the month	₹ 18,000
Electricity & Lighting for the month	₹ 9,500
Repairs & maintenance (machine) including Consumable stores per month	₹ 17,500
Insurance of Plant & Building (apportioned) for the year	₹ 18,250
Other general expense per annum	₹ 17,500

The workers are paid a fixed Dearness allowance of ₹ 4,575 per month. Production bonus payable to workers in terms of an award is equal to 33.33% of basic wages and dearness allowance. Add 10% of the basic wage and dearness allowance against leave wages and holidays with pay to arrive at a comprehensive labour-wage for debit to production.

ICAI'S ANSWER

Effective machine hours = 200 hours × 75% = 150 hours

Computation of Comprehensive Machine Hour Rate

	Per month (₹)	Per hour (₹)
Fixed cost		

	Per month (₹)	Per hour (₹)
Supervision charges	18,000.00	
Electricity and lighting	9,500.00	
Insurance of Plant and building (₹18,250 ÷12)	1,520.83	
Other General Expenses (₹17,500÷12)	1,458.33	
Depreciation (₹64,800÷12)	5,400.00	
	35,879.16	239.19
Direct Cost		
Repairs and maintenance	17,500.00	116.67
Power	65,000.00	433.33
Wages of machine man		139.27
Wages of Helper		109.41
Machine Hour rate (Comprehensive)		1,037.87

Wages per machine hour

	Machine man	Helper
Wages for 200 hours		
Machine-man (₹400 × 25)	₹10,000.00	---
Helper (₹275 × 25)	---	₹6,875.00
Dearness Allowance (DA)	₹4,575.00	₹4,575.00
	₹14,575.00	₹11,450.00
Production bonus (1/3 of Basic and DA)	4,858.33	3,816.67
Leave wages (10% of Basic and DA)	1,457.50	1,145.00
	20,890.83	16,411.67
Effective wage rate per machine hour	₹139.27	₹109.41

ICAI · Old P3 · MTP Nov 2023 · Series I · question p. 5 · answer p. 8-9

411 MTP **Old P3 Nov 2022 S-I** Q5(a) 10 marks old scheme

A work-shop has 8 identical machines operated by 6 operators. The machine cannot work without an operator wholly engaged on it. The original cost of all the 8 machines works out to ₹ 64,00,000. The

following particulars are furnished for a six months' period:

Normal available hours per operator	1,248
Absenteeism (without pay) hours per operator	18
Leave (with pay) hours per operator	20
Normal unavoidable idle time-hours per operator	10
Production bonus estimated	10% on wages
Power consumed	₹ 80,500
Supervision and Indirect Labour	₹ 33,000
Lighting and Electricity	₹ 12,000
Average rate of wages per day of 8 hours per operator	₹ 200

The following particulars are given for a year:

Insurance	₹ 7,20,000
Sundry work Expenses	₹ 1,00,000
Management Expenses allocated	₹ 10,00,000
Depreciation	10% on the original cost

Repairs and Maintenance (including consumables): 5% of the value of all the machines.

Prepare a statement showing the comprehensive machine hour rate for the machine shop.

ICAI'S ANSWER

Workings:

Particulars	Six months 6 operators (Hours)
Normal available hours half yearly (1,248 × 6 operators)	7,488
Less: Absenteeism hours (18 × 6 operators)	(108)
Paid hours (A)	7,380
Less: Leave hours (20 × 6 operators)	(120)
Less: Normal idle time (10 × 6 operators)	(60)
Effective working hours	**7,200**

Computation of Comprehensive Machine Hour Rate

Particulars	Amount for six months (₹)
Operators' wages (7,380/8 ×200)	1,84,500

Particulars	Amount for six months (₹)
Production bonus (10% on wages)	18,450
Power consumed	80,500
Supervision and indirect labour	33,000
Lighting and Electricity	12,000
Repair and maintenance {(5% × ₹ 64,00,000)/2}	1,60,000
Insurance (₹ 7,20,000/2)	3,60,000
Depreciation {(₹ 64,00,000 × 10%)/2}	3,20,000
Sundry Work expenses (₹ 1,00,000/2)	50,000
Management expenses (₹ 10,00,000/2)	5,00,000
Total Overheads for 6 months	**17,18,450**
Comprehensive Machine Hour Rate = ₹ 17,18,450/7,200 hours	**₹ 238.67**

ICAI · Old P3 · MTP Nov 2022 · Series I · question p. 6 · answer p. 9-10

412 MTP **Old P3 Nov 2022 S-II** Q1(b) 5 marks old scheme

Madhu Ltd has calculated a predetermined overhead rate of ₹22 per machine hour for its Quality Check (QC) department. This rate has been calculated for the budgeted level of activity and is considered as appropriate for absorbing overheads. The following overhead expenditures at various activity levels had been estimated.

Total overheads	Number of machine hours
₹3,38,875	14,500
₹3,47,625	15,500
₹3,56,375	16,500

You are required to:

(i) COMPUTE the variable overhead absorption rate per machine hour.

(ii) COMPUTE the estimated total fixed overheads.

(iii) CALCULATE the budgeted level of activity in machine hours.

(iv) CALCULATE the amount of under/over absorption of overheads if the actual machine hours were 14,970 and actual overheads were ₹3,22,000.

ICAI'S ANSWER

(i) Computation of variable overhead absorption rate:

$$\text{Variable overhead absorption rate} = \frac{\text{Difference in Total Overheads}}{\text{Difference in levels in terms of machine hours}}$$

$= \frac{\text{Rs.}3{,}47{,}625 - \text{Rs.}3{,}38{,}875}{15{,}500\text{hours} - 14{,}500\text{hours}} = \text{Rs.}8.75$ per machine hour.

(ii) Computation of Total fixed overheads:

	(Rs.)
Total overheads at 14,500 hours	3,38,875
Less: Variable overheads (Rs. 8.75 × 14,500)	(1,26,875)
Total fixed overheads	2,12,000

(iii) Calculation of Budgeted level of activity in machine hours:

Let budgeted level of activity = X

Then, $\frac{(\text{Rs.}8.75\ X + \text{Rs.}2{,}12{,}000)}{X} = \text{Rs. } 22$

8.75X + Rs.2,12,000 = 22X

13.25X = 2,12,000

X =16,000

Thus, budgeted level of activity = 16,000 machine hours.

(iv) Calculation of Under / Over absorption of overheads:

	(Rs.)
Actual overheads	3,22,000
Absorbed overheads (14,970 hours × Rs. 22 per hour)	3,29,340
Over-absorption (3,29,340 – 3,22,000)	7,340

ICAI · Old P3 · MTP Nov 2022 · Series II · question p. 1 · answer p. 1-2

413 MTP **Old P3 May 2022 S-II** Q5(b) 10 marks old scheme

M/s Avyukt Automobile Parts has four identical machines in its factory. Cost of each machine is ₹ 5,00,000 with expected scrap value of 10% at the end of its effective life (9 years). The expected annual running hours of machine is expected to run for 2,200 hours. The other details in respect of the machine shop are:

(I)	Factory Rent	₹ 5,000 per month
(II)	Lighting of Factory	₹ 3,000 per month
(III)	Operator Wages (Two operators and each operator is in charge of two machines)	₹10,000 per month (per Operator)
(IV)	Fixed repairs and maintenance charges per machine	₹ 2,000 per quarter
(V)	Insurance premium for the machine (Annual)	3% of cost
(VI)	Forman's salary (Devoted 1/6th of his time to this factory)	₹ 2,500 per month

(VII)	Other factory overhead (Annual)	₹40,000
(VIII)	Power Consumption per machine per hour	80 units
(IX)	Rate of Power	₹ 150 for 100 units
(X)	Unproductive Hours lost during repairs	50 per annum
(XI)	Unproductive Hours Lost while Job Setting	650 per annum

You are required to COMPUTE a comprehensive machine hour rate assuming power is used during operating time only.

ICAI'S ANSWER

Computation of Comprehensive Machine Hour Rate per Machine

Particulars	Per Annum (₹)	Per Hour (₹)
Standing Charges:		
Depreciation (Working Note 2)	50,000	
Factory Rent (₹ 5,000 × 12 months / 4)	15,000	
Lighting of Factory (₹ 3,000 × 12 months / 4)	9,000	
Operator Wages (₹ 10,000 × 12 months / 2)	60,000	
Repairs and maintenance (₹ 2,000 × 4)	8,000	
Insurance premium (₹ 5,00,000 × 3%)	15,000	
Forman's salary (₹ 2,500 × 12 x ⅙ / 4)	1,250	
Other factory overhead (₹ 40,000 / 4)	10,000	
	1,68,250	
Standing Charges per hour (₹ 1,68,250 / 1,500 hours)		112.17
Running Charges:		
Power (80 units x ₹ 150 / 100)		120.00
Comprehensive Machine Hour Rate		232.17

Working Notes:

1. Computation of Total Operative Hours

Total Running Hours:	2,200
Less: Unproductive hours lost during repairs	50
Less: Unproductive hours Lost while Job Setting	650
Total Operative Hours	**1,500 per annum**

2. Calculation of Annual Depreciation

$$\text{Annual Depreciation} = \frac{\text{Purchase Cost} - \text{Estimated Scrap Value}}{\text{Effective Life in Years}}$$

$$= \frac{₹\ 5{,}00{,}000 - ₹\ 50{,}000}{9\ \text{Years}}$$

$$= ₹\ 50{,}000$$

ICAI · Old P3 · MTP May 2022 · Series II · question p. 6-7 · answer p. 12

414 MTP **Old P3 Dec 2021 S-I** Q1(c) 5 marks old scheme

A machine costing ₹ 10 lakhs, was purchased on 01-04-2021. The expected life of the machine is 10 years. At the end of this period its scrap value is likely to be ₹ 10,000. The total cost of all the machines including new one was ₹ 90 lakhs.

The other information is given as follows:

(i) Working hours of the machine for the year was 4,200 including 200 non-productive hours.

(ii) Repairs and maintenance for the new machine during the year was ₹ 6,000.

(iii) Insurance Premium was paid for all the machine ₹ 9,000.

(iv) New machine consumes 8 units of electricity per hour, the rate per unit being ₹ 3.75

(v) The new machine occupies 1/10th area of the department. Rent of the department is ₹ 2,400 per month.

(vi) Depreciation is charged on straight line basis.

COMPUTE machine hour rate for the new machine.

ICAI'S ANSWER

Computation of machine hour rate of new Machine

	Total (₹)	Per hour (₹)
A. Standing Charges		
I. Insurance Premium ₹ 9,000 × $\frac{1}{9}$	1,000	
II. Rent $\frac{1}{10}$ × ₹2,400 ×12 months	2,880	
	3,880	0.97*
B. Machine expenses		
I. Repairs and Maintenance (₹ 6,000 ÷ 4,000 hours)		1.50

	Total (₹)	Per hour (₹)
II. Depreciation $\left[\frac{₹10,00,000 - ₹10,000}{10 \text{ years} \times 4,000 \text{ hours}}\right]$		24.75
III. Electricity (8 units x ₹ 3.75)		30.00
Machine hour rate		57.22

Working Note

Calculation of productive Machine hour rate

Total hours	4,200
Less: Non-Productive hours	200
Effective machine hours	4,000

* ₹ 3,880 ÷ 4,000 hours = ₹ 0.97

ICAI · Old P3 · MTP Dec 2021 · Series I · question p. 1-2 · answer p. 2-3

415 MTP **Old P3 Jul 2021 S-II** Q1(c) 5 marks old scheme

The following particulars refer to process used in the treatment of material subsequently, incorporated in a component forming part of an electrical appliance:

(i) The original cost of the machine used (Purchased in June 2013) was Rs. 1,00,000. Its estimated life is 10 years, the estimated scrap value at the end of its life is Rs.10,000, and the estimated working time per year (50 weeks of 44 hours) is 2,200 hours of which machine maintenance etc., is estimated to take up 200 hours.

No other loss of working time expected, setting up time, estimated at 100 hours, is regarded as productive time. (Holiday to be ignored).

(ii) Electricity used by the machine during production is 16 units per hour at cost of a 90 paisa per unit. No current is taken during maintenance or setting up.

(iii) The machine required a chemical solution which is replaced at the end of week at a cost of Rs. 200 each time.

(iv) The estimated cost of maintenance per year is Rs.12,000.

(v) Two attendants control the operation of machine together with five other identical machines. Their combined weekly wages, insurance and the employer's contribution to holiday pay amount Rs. 1,200.

(vi) Departmental and general works overhead allocated to this machine for the current year amount to Rs. 20,000.

You are required to CALCULATE the machine hour rate of operating the machine.

ICAI'S ANSWER

Working Notes:

(i) Total Productive hours = Estimated Working hours – Machine Maintenance hours

= 2,200 hours – 200 hours = 2,000 hours

(ii) Depreciation per annum = $\frac{\text{Rs. } 1{,}00{,}000\text{- Rs. } 10{,}000}{10\text{years}}$ = Rs. 9000

(iii) Chemical solution cost per annum = Rs. 200 × 50 weeks = Rs.10,000

(iv) Wages of attendants (per annum) = $\frac{\text{Rs. } 1{,}200 \times 50 \text{ weeks}}{6 \text{ machines}}$ = Rs.10,000

Calculation of Machine hour rate

Particulars	Amount (Rs.) (per annum)	Amount (Rs.) (per hour)
A. Standing Charge		
(i) Wages of attendants	10,000	
(ii) Departmental and general works overheads	20,000	
Total Standing Charge	30,000	
Standing Charges per hour (30,000/2,000)		15.00
B. Machine Expense		
(iii) Depreciation	9,000	4.50
(iv) Electricity (Rs. 0.9×16units×1,900hours/2,000hours)	-	13.68
(v) Chemical solution	10,000	5.00
(vi) Maintenance cost	12,000	6.00
Machine operating cost per hour (A + B)		**44.18**

ICAI · Old P3 · MTP Jul 2021 · Series II · question p. 1-2 · answer p. 2

416 MTP **Old P3 Nov 2020 S-I** Q5(a) 10 marks old scheme

A Ltd. manufactures two products- A and B. The manufacturing division consists of two production departments P1 and P2 and two service departments S1 and S2.

Budgeted overhead rates are used in the production departments to absorb factory overheads to the products. The rate of Department P1 is based on direct machine hours, while the rate of Department P2 is based on direct labour hours. In applying overheads, the pre-determined rates are multiplied by actual hours.

For allocating the service department costs to production departments, the basis adopted is as follows:

(i) Cost of Department S1 to Department P1 and P2 equally, and

(ii) Cost of Department S2 to Department P1 and P2 in the ratio of 2 : 1 respectively.

The following budgeted and actual data are available:

Annual profit plan data:

Factory overheads budgeted for the year:

Departments	P1	27,51,000	S1	8,00,000
	P2	24,50,000	S2	6,00,000

Budgeted output in units:

Product A 50,000; B 30,000.

Budgeted raw-material cost per unit:

Product A ₹ 120; Product B ₹ 150.

Budgeted time required for production per unit:

Department P1 :	Product A : 1.5 machine hours
	Product B : 1.0 machine hour
Department P2 :	Product A : 2 Direct labour hours
	Product B : 2.5 Direct labour hours

Average wage rates budgeted in Department P2 are:

Product A - ₹ 72 per hour and Product B – ₹ 75 per hour.

All materials are used in Department P1 only.

Actual data (for the month of Jan, 2020):

Units actually produced:	Product A : 4,000 units
	Product B : 3,000 units

Actual direct machine hours worked in Department P1:

On Product A 6,100 hours, Product B 4,150 hours.

Actual direct labour hours worked in Department P2:

On Product A 8,200 hours, Product B 7,400 hours.

Costs actually incurred:		Product A		Product B
		₹		₹
Raw materials		4,89,000		4,56,000
Wages		5,91,900		5,52,000
Overheads: Department	P1	2,50,000	S1	80,000
	P2	2,25,000	S2	60,000

You are required to:

(i) COMPUTE the pre-determined overhead rate for each production department.

(ii) PREPARE a performance report for Jan, 2020 that will reflect the budgeted costs and actual costs.

ICAI'S ANSWER

(i) **Computation of pre-determined overhead rate for each production department from budgeted data**

	Production Department		Service Department	
	P1	**P2**	**S1**	**S2**
Budgeted factory overheads for the year (₹)	27,51,000	24,50,000	8,00,000	6,00,000
Allocation of service department S1's costs to production departments P1 and P2 equally (₹)	4,00,000	4,00,000	(8,00,000)	--
Allocation of service department S2's costs to production departments P1 and P2 in the ratio of 2:1 (₹)	4,00,000	2,00,000	–	(6,00,000)
Total	35,51,000	30,50,000	--	--
Budgeted machine hours in department P1 (working note-1)	1,05,000	--		
Budgeted labour hours in department P2 (working note-1)	--	1,75,000		
Budgeted machine/ labour hour rate (₹)	33.82	17.43		

(ii) **Performance report for Jan, 2020**

(When 4,000 and 3,000 units of Products A and B respectively were actually produced)

	Budgeted (₹)	Actual (₹)
Raw materials used in Dept. P1:		
A : 4,000 units × ₹ 120	4,80,000	4,89,000
B : 3,000 units × ₹ 150	4,50,000	4,56,000
Direct labour cost (on the basis of labour hours worked in department P2)		
A : 4,000 units × 2 hrs. × ₹ 72	5,76,000	5,91,900
B : 3,000 units × 2.5 hrs. × ₹ 75	5,62,500	5,52,000
Overhead absorbed on machine hour basis in Dept. P1:		

	Budgeted (₹)	Actual (₹)
A : 4,000 units × 1.5 hrs. × ₹ 33.82	2,02,920	1,96,420*
B : 3,000 units × 1 hr. × ₹ 33.82	1,01,460	1,33,630*
Overhead absorbed on labour hour basis in Dept. P2:		
A : 4,000 units × 2 hrs. × ₹ 17.43	1,39,440	1,49,814**
B : 3,000 units × 2.5 hrs. × ₹ 17.43	1,30,725	1,35,198**
	26,43,045	27,03,962

* *(Refer to working note 4)*

** *(Refer to working note 5)*

Working notes:

1.

	Product A	Product B	Total
Budgeted output (units)	50,000	30,000	
Budgeted machine hours in Dept. P1	75,000 (50,000×1.5 hrs.)	30,000 (30,000×1 hr.)	1,05,000
Budgeted labour hours in Dept. P2	1,00,000 (50,000×2 hrs.)	75,000 (30,000×2.5 hrs.)	1,75,000

2.

	Product A	Product B	Total
Actual output (units)	4,000	3,000	
Actual machine hours utilized in Dept. P1	6,100	4,150	10,250
Actual labour hours utilised in Dept. P2	8,200	7,400	15,600

3. Computation of actual overhead rates for each production department from actual data

	Production Department		Service Department	
	P1	**P2**	**S1**	**S2**
Actual factory overheads for the month of Jan, 2020 (₹)	2,50,000	2,25,000	80,000	60,000
Allocation of service Dept. S1's costs to production Dept. P1 and P2 equally (₹)	40,000	40,000	(80,000)	–

	Production Department		Service Department	
Allocation of service Dept. S2's costs to production Dept. P1 and P2 in the ratio of 2:1 (₹)	40,000	20,000	–	(60,000)
Total	3,30,000	2,85,000	--	--
Actual machine hours in Dept. P1 (working note 2)	10,250	--		
Actual labour hours in Dept. P2 (working note 2)	--	15,600		
Actual machine/ labour hour rate (₹)	32.20	18.27		

4. Actual overheads absorbed (based on machine hours)

A : 6,100 hrs × ₹ 32.20 = ₹ 1,96,420

B : 4,150 hrs × ₹ 32.20 = ₹ 1,33,630

5. Actual overheads absorbed (based on labour hours)

A : 8,200 hrs × ₹ 18.27 = ₹ 1,49,814

B : 7,400 hrs × ₹ 18.27 = ₹ 1,35,198

ICAI · Old P3 · MTP Nov 2020 · Series I · question p. 6-7 · answer p. 10-12

417 MTP **Old P3 May 2020 S-I** Q1(d) 5 marks old scheme

A Ltd has calculated a predetermined overhead rate of Rs.22 per machine hour for its Quality Check (QC) department. This rate has been calculated for the budgeted level of activity and is considered as appropriate for absorbing overheads. The following overhead expenditures at various activity levels had been estimated.

Total overheads	Number of machine hours
Rs.3,38,875	14,500
Rs.3,47,625	15,500
Rs.3,56,375	16,500

You are required to:

(i) CALCULATE the variable overhead absorption rate per machine hour.

(ii) CALCULATE the estimated total fixed overheads.

(iii) CALCULATE the budgeted level of activity in machine hours.

(iv) CALCULATE the amount of under/over absorption of overheads if the actual machine hours were 14,970 and actual overheads were Rs.3,22,000.

(v) ANALYSE the arguments for and against using departmental absorption rates as opposed to a single or blanket factory wide rate.

ICAI'S ANSWER

(i) Variable overhead absorption rate = $\frac{\text{Difference in Total Overheads}}{\text{Difference in levels in terms of machine hours}}$

= $\frac{\text{Rs.3,47,625 - Rs.3,38,875}}{\text{15,500 hours - 14,500 hours}}$ = Rs.8.75 per machine hour.

(ii) Calculation of Total fixed overheads:

	(Rs.)
Total overheads at 14,500 hours	3,38,875
Less: Variable overheads (Rs. 8.75 × 14,500)	(1,26,875)
Total fixed overheads	2,12,000

(iii) Calculation of Budgeted level of activity in machine hours:

Let budgeted level of activity = X

Then, $\frac{(\text{Rs. } 8.75\ X + \text{Rs.2,12,000})}{X}$ = Rs.22

8.75X + Rs.2,12,000 = 22X

13.25X = 2,12,000

X =16,000

Thus, budgeted level of activity = 16,000 machine hours.

(iv) Calculation of Under / Over absorption of overheads:

	(Rs.)
Actual overheads	3,22,000
Absorbed overheads (14,970 hours × Rs. 22 per hour)	3,29,340
Over-absorption (3,29,340 – 3,22,000)	7,340

(v) Departmental absorption rates provide costs which are more precise than those provided by the use of blanket absorption rates. Departmental absorption rates facilitate variance analysis and cost control. The application of these rates makes the task of stock and work-in-process (WIP) valuation easier and more precise. However, the setting up and monitoring of these rates can be time consuming and expensive.

ICAI · Old P3 · MTP May 2020 · Series I · question p. 2 · answer p. 3-4

5.5 Under- and over-absorption of overheads

19 questions · asked in 15 ICAI documents · 4 exam papers

418 EXAM **Sep 2025** MCQ 15 2 marks MCQ

PMP Limited manufactures a single product and absorbs the production overheads at a pre-determined rate of ₹ 20 per machine hour. At the end of financial year 2024-25, it has been found that actual production overheads incurred were ₹ 12,00,000. It included ₹ 80,000 on account of 'written off obsolete stores and ₹ 30,000 being the wages paid for the strike period under an award. The actual machine hours worked during the period were 50,000 hours. The production and sale was 20,000 units and 18,000 units

respectively. What will be the amount of under-absorbed production overhead to be charged to Cost of Sales?

A ₹ 91,000

B ₹ 90,000

C ₹ 81,000

D ₹ 80,000

ICAI'S ANSWER

Answer key: **(C)**

ICAI · P4 · Sep 2025 exam · question p. 6-7 · answer p. 7

419 EXAM **Sep 2025** Q5(c) 3 marks

Complete the table regarding accounting entries pertaining to Over/Under 3 absorption of overheads:

Absorption of overhead	Accounts	Dr/Cr	Calculation of Amount Formula
Under-absorption	Stock of Finished goods Account		
Over-absorption	Stock of Semi-finished goods (WIP) Account		
Under-absorption	Cost of Sales Account		

ICAI'S ANSWER

Accounting entries pertaining to over/under absorption of overheads

Absorption of Overhead	Accounts	Dr/Cr	Calculation of Amount Formula
Under-absorption	Stock of Finished Goods Account	Dr	Units of Finished Stock x Supplementary rate per unit
Over-absorption	Stock of Semi-finished Goods (WIP) Account	Cr	Equivalent Completed Units x Supplementary rate per unit
Under-absorption	Cost of Sales Account	Dr	Units sold x Supplementary rate per unit

ICAI · P4 · Sep 2025 exam · question p. 24 · answer p. 26

420 EXAM **May 2024** Q4(b) 6 marks

The cost variance report was being discussed at a review meeting where in Cost Accountant of the company reported under-absorption of production overheads.

The following information was available from the cost records of the company at the end of financial year 2023-24:

- Actual production overheads incurred were ₹ 4,50,000 which included ₹ 42,000 on account of 'written off obsolete stores.
- 18,000 units were produced during the year out of which 10,000 units were sold and 8,000 units of finished goods were in stock.
- There were also 5,000 units in progress which may be reckoned as 40% complete.
- The actual machine hours worked during the period were 43,000.

ABC Ltd. absorbs the production overheads at a predetermined rate of ₹ 8 per machine hour.

On investigation, it has been found that 20% of the under-absorption of production overheads was due to defective planning and the rest was attributable to normal increase in costs of indirect materials and indirect labour.

You are required to:

(i) Calculate the amount of under-absorption of production overheads during the year 2023-24; and

(ii) Show the treatment of under-absorption of production overheads in cost accounts.

ICAI'S ANSWER

(i) Amount of under-absorption of production overheads during the current year

	(₹)
Total production overheads actually incurred during the current year	4,50,000
Less : 'Written off' obsolete stores	42,000
Net production overheads actually incurred : **(A)**	4,08,000
Production overheads absorbed by 43,000 machine hours@ ₹ 8 per hour : **(B)**	3,44,000
Amount of under – absorption of production overheads : **[(A) – (B)]**	**64,000**

(ii) Accounting treatment of under absorption of production overheads

It is given in the statement of the question that 18,000 units were produced, and 5,000 units were 40% complete, 20% of the under-absorbed overheads were due to defective planning and the rest were attributable to normal increase in costs of indirect materials and indirect labour.

1.	(20 % of ₹ 64,000) i.e., ₹ 12,800 of under-absorbed overheads were due to defective planning. This being abnormal, should be debited to the Costing Profit and Loss A/c.	₹ 12,800
2.	Balance (80% of ₹ 64,000) i.e., ₹ 51,200 of under-absorbed overheads should be distributed over work-in-progress, finished goods and cost of sales by using supplementary rate.	₹ 51,200
	Total under-absorbed overheads	₹ 64,000

Apportionment of unabsorbed overheads of ₹ 51,200 over, work-in progress, finished goods and cost of sales

	Equivalent Completed Units	(₹)
Work-in-Progress		
(5,000 units × 40%×₹ 2.56)	2,000	5,120
(Refer to working note)		
Finished goods		
(8,000 units × ₹ 2.56)	8,000	20,480
Cost of sales		
(10,000 units × ₹ 2,56)	10,000	25,600
	20,000	51,200

Working Note

Supplementary rate per unit = $\frac{51,200}{20,000}$ = ₹ 2.56

ICAI · P4 · May 2024 exam · question p. 16 · answer p. 18-20

421 EXAM Old P3 Dec 2021 Q5(b) 10 marks old scheme

XYZ Ltd. manufactures a single product. It recovers factory overheads at a pre-determined rate of ₹ 20 per man-day.

During the year 2020-21, the total factory overheads incurred and the man-days actually worked were ₹ 35.50 lakhs and 1.50 lakh days respectively. Out of the amount of ₹ 35.50 lakhs, ₹ 2.00 lakhs were in respect of wages for stick period and ₹ 1.00 lakh was in respect of expenses of previous year booked in this current year. During the period, 50,000 units were sold. At the end of the period, 12,000 completed units were held in stock but there was no opening stock of finished goods. Similarly, there was no stock of uncompleted units at the beginning of the period but at the end of the period there were 20,000 uncompleted units which may be treated as 65% complete in all respects.

On investigation, it was found that 40% of the unabsorbed overheads were due to factory inefficiency and the rest were attributable to increase in the cost of indirect materials and indirect labour. You are required to:

(i) Calculate the amount of unabsorbed overheads during the year 2020-21.

(ii) Show the accounting treatment of unabsorbed overheads in cost accounts and pass journal entry.

ICAI'S ANSWER

(i) Amount of under-absorption of overheads during the year 2020-21

		(₹)
Total production overheads actually incurred during the year 2020-21		35,50,000
Less: Wages paid during strike period	₹2,00,000	
Wages of previous year booked in current year	₹ 1,00,000	3,00,000
Net production overheads actually incurred: (A)		32,50,000
Production overheads absorbed by 1.50 lakh man-days @ ₹ 20 per man-day: (B)		30,00,000
Amount of under-absorption of production overheads: [(A)–(B)]		2,50,000

(ii) Accounting treatment of under absorption of production overheads: It is given in the statement of the question that 62,000 units (50,000 sold + 12,000 closing stock – 0 opening stock) were completely finished and 20,000 units were 65% complete, 40% of the under-absorbed overheads were due to factory inefficiency and the rest were attributable to increase in cost of indirect materials and indirect labour.

	(₹)
1. (40% of ₹2,50,000) i.e. ₹ 1,00,000 of under – absorbed overheads were due to factory inefficiency. This being abnormal, should be debited to the Costing Profit and Loss A/c	1,00,000
2. Balance (60% of ₹ 2,50,000) i.e. ₹ 1,50,000 of under – absorbed overheads should be distributed over work-in-progress, finished goods and cost of sales by using supplementary rate	1,50,000
Total under-absorbed overheads	2,50,000

Apportionment of unabsorbed overheads of ₹1,50,000 over work-in-progress, finished goods and cost of sales.

	Equivalent Completed units	(₹)
Work-in-progress (13,000 units × ₹ 2) (Refer to Working Note)	20000 * 65% = 13,000	26,000
Finished goods (12,000 units × ₹ 2)	12,000	24,000
Cost of sales (50,000 units × ₹ 2)	50,000	1,00,000
	75,000	1,50,000

Journal entry:

Work-in-progress control A/c	Dr.	₹ 26,000	
Finished goods control A/c	Dr.	₹ 24,000	
Cost of Sales A/c	Dr.	₹ 1,00,000	
Costing Profit & Loss A/c	Dr.	₹ 1,00,000	
To Overhead control A/c			₹ 2,50,000

Working Note:

$$\text{Supplementary overhead absorption rate} = \frac{₹1,50,000}{75,000 \text{ units}} = ₹ 2 \text{ per unit}$$

ICAI · Old P3 · Dec 2021 exam · question p. 19-20 · answer p. 22-23

422 EXAM **Old P3 Nov 2019** Q2(b) 10 marks old scheme

ABS Enterprises produces a product and adopts the policy to recover factory overheads applying blanket rate based on machine hours. The cost records of the concern reveal the following information:

Budgeted production overheads	₹ 10,35,000
Budgeted machine hours	₹ 90,000
Actual machine hours worked	₹ 45,000
Actual production overheads	₹ 8,80,000
Production overheads (actual) include-	
Paid to worker as per court's award	₹ 50,000
Wages paid for strike period	₹ 38,000
Stores written off	₹ 22,000
Expenses of previous year booked in current year	₹ 18,500
Production -	
Finished goods	30,000 units
Sale of finished goods	27,000 units

The analysis of cost information reveals that 1/3 of the under absorption of overheads was due to defective production planning and the balance was attributable to increase in costs.

You are required:

(i) To find out the amount of under absorbed production overheads.

(ii) To give the ways of treating it in Cost Accounts.

(iii) To apportion the under absorbed overheads over the items.

ICAI'S ANSWER

(i) Amount of under absorption of production overheads:

Particular	Amount (₹)	Amount (₹)
Total production overheads actually incurred		8,80,000
Less: Amount paid to worker as per court order	50,000	
Wages paid for the strike period under an award	38,000	
Stores written off	22,000	
Expenses of previous year booked in the current year	18,500	1,28,500
		7,51,500
Less: Production overheads absorbed as per machine hour rate (45,000 hours × ₹**11.50***)		**5,17,500**
Amount of under- absorbed production overheads		**2,34,000**

*Budgeted Machine hour rate (Blanket rate) = $\frac{₹\ 10,35,000}{90,000}$ = ₹ 11.50 per hour

(ii) Accounting treatment of under absorbed production overheads:

(a) As 1/3rd of the under absorbed overheads were due to defective production planning, this being abnormal, hence should be debited to Costing Profit and Loss Account.

Amount to be debited to Costing Profit and Loss Account

= ₹ 2,34,000 × 1/3 = ₹ **78,000**.

(b) Balance of under absorbed production overheads should be distributed over Finished goods and Cost of sales by applying supplementary rate*.

Amount to be distributed = ₹ 2,34,000 × 2/3 = **₹1,56,000**

*Supplementary rate = $\frac{₹\ 1,56,000}{30,000\ \text{units}}$ = ₹ 5.20 per unit

(iii) Apportionment of under absorbed production overheads over Finished goods and Cost of sales:

Particular	Units	Amount (₹)
Finished goods (3,000 units × ₹5.20)	3,000	15,600
Cost of sales (27,000 units × ₹5.20)	27,000	1,40,400
Total	30,000	1,56,000

ICAI · Old P3 · Nov 2019 exam · question p. 7 · answer p. 10-11

423 RTP Sep 2024 Q3 MCQ

The accountant for Brilliant Tools Ltd applies overhead based on machine hours. The budgeted overhead and machine hours for the year are ₹ 1,30,000 and 8,000 hours, respectively. The actual overhead and

machine hours incurred were ₹ 1,37,500 and 10,000 hours. The cost of goods sold and inventory data compiled for the year is as follows:

Direct Material ₹ 25,000

Cost of Goods Sold ₹ 2,25,000

Units: WIP 50,000 and Finished Goods 75,000

What is the amount of over/under absorbed overhead for the year?

A Over absorbed by ₹ 25,000

B Under absorbed by ₹ 25,000

C Over a absorbed by ₹ 32,500

D Under absorbed by ₹ 32,500

ICAI'S ANSWER

(A) Overabsorbed by ₹ 25,000

Predetermined Overhead Rate = Budgeted Overhead / Budgeted hours i.e. 130,000 / 8,000 = ₹ 16.25 per hour.

Hence, absorbed overhead = 10,000 X 16.25 = ₹ 1,62,500.

Since actual overhead incurred were ₹ 1,37,500

Hence the overhead were over absorbed by 1,62,500 – 1,37,500 = ₹ 25,000.

ICAI · P4 · RTP Sep 2024 · question p. 5 · answer p. 19

424 RTP May 2024 Case scenario · Q3(i) to Q3(v)

CASE SCENARIO

During half year ending inter departmental review meeting of P Ltd., cost variance report was discussed and the performance of the departments were assessed. The following figures were presented.

For a period of first six months of the financial year, following information were extracted from the books:

Actual production overheads	₹ 34,08,000
The above amount is inclusive of the following payments made:	
Paid as per court's order	₹ 4,50,000
Expenses of previous year booked in current year	₹ 1,00,000
Paid to workers for strike period under an award	₹ 4,20,000
Obsolete stores written off	₹ 36,000

Production and sales data for the six months are as under:

Production:	
Finished goods	1,10,000 units
Works-in-progress (50% complete in every respect)	80,000 units
Sale:	
Finished goods	90,000 units

Machine worked during the period was 3,000 hours.

At the of preparation of revenue budget, it was estimated that a total of ₹ 50,40,000 would be required for budgeted machine hours of 6,000 as production overheads for the entire year.

During the meeting, a data analytic report revealed that 40% of the over/under-absorption was due to defective production policies and the balance was attributable to increase in costs.

You were also present at the meeting; the chairperson of the meeting has asked you to be ready with the followings for the performance appraisal of the departmental heads:

Q3(i)

How much was the budgeted machine hour rate used to recover overhead?

A ₹ 760

B ₹ 820

C ₹ 780

D ₹ 840

Q3(ii)

How much amount of production overhead has been recovered (absorbed) upto the end of half year end?

A ₹ 25,20,000

B ₹ 34,08,000

C ₹ 24,00,000

D ₹ 24,60,000

Q3(iii)

What is the amount of overhead under/ over absorbed?

A 1,18,000 over-absorbed

B 1,18,000 under- absorbed

C 18,000 over-absorbed

D 18,000 under-absorbed

Q3(iv)

What is the supplementary rate for apportionment of over/under absorbed overheads over WIP, Finished goods and Cost of sales?

A ₹ 0.315 per unit

B ₹ 0.472 per unit

C ₹ 0.787 per unit

D ₹ 1 per unit

Q3(v)

What is the amount of over/under absorbed overhead apportioned to Work in Progress?

A ₹ 9,440

B ₹ 42,480

C ₹ 18,880

D ₹ 70,800

ICAI'S ANSWERS

Q3(i)

Answer key: **(D)**

(d) Budgeted Machine hour rate (Blanket rate)

$= \frac{₹\ 50{,}40{,}000}{6{,}000\ \text{hours}} =$ **₹ 840 per hour**

Q3(ii)

Answer key: **(A)**

(a) ₹ 25,20,000

Q3(iii)

Answer key: **(A)**

(a)

	Amount (₹)	Amount (₹)
Total production overheads actually incurred during the period		34,08,000
Less: Amount paid to worker as per court order	4,50,000	
Expenses of previous year booked in the current year	1,00,000	
Wages paid for the strike period under an award	4,20,000	
Obsolete stores written off	36,000	10,06,000
		24,02,000
Less: Production overheads absorbed as per machine hour rate (3,000 hours × ₹ 840*)		25,20,000
Amount of over absorbed production overheads		**1,18,000**

* Budgeted Machine hour rate (Blanket rate) calculated in part (i)

Q3(iv)

Answer key: **(B)**

(b) Accounting treatment of over absorbed production overheads: As, 40% of the over absorbed overheads were due to defective production policies, this being abnormal, hence should be credited

to Costing Profit and Loss Account.

Amount to be credited to Costing Profit and Loss Account

= ₹ 1,18,000× 40% = ₹ 47,200.

Balance of over absorbed production overheads should be distributed over Works in progress, Finished goods and Cost of sales by applying supplementary rate*.

Amount to be distributed = ₹ 1,18,000× 60% = ₹ 70,800

Supplementary rate = $\frac{₹\ 70{,}800}{1{,}50{,}000\ \text{units}}$ = **₹ 0.472 per unit**

Q3(v)

Answer key: **(C)**

(c) Apportionment of over absorbed production overheads over WIP, Finished goods and Cost of sales:

	Equivalent completed units	Amount (₹)
Work-in-Progress (80,000 units × 50% ×0.472)	40,000	**18,880**
Finished goods (20,000 units × 0.472)	20,000	9,440
Cost of sales (90,000 units × 0.472)	90,000	42,480
Total	1,50,000	70,800

ICAI · P4 · RTP May 2024 · question p. 4-5 · answer p. 20, 21, 21-22, 22

425 RTP **Old P3 Nov 2023** Q14(d) old scheme

Also asked in MTP May 2024 S-I Q6(a)

Explain the treatment of over and under absorption of overheads in cost accounts.

ICAI'S ANSWER

Treatment of over and under absorption of overheads are:

(i) Writing off to costing P&L A/c: Small difference between the actual and absorbed amount should simply be transferred to costing P&L A/c, if difference is large then investigate the causes and after that abnormal loss/ gain shall be transferred to costing P&L A/c.

(ii) Use of supplementary Rate: Under this method the balance of under and over absorbed overheads may be charged to cost of W.I.P., finished stock and cost of sales proportionately with the help of supplementary rate of overhead.

(iii) Carry Forward to Subsequent Year: Difference should be carried forward in the expectation that next year the position will be automatically corrected.

ICAI · Old P3 · RTP Nov 2023 · question p. 9 · answer p. 25

426 MTP **Sep 2026 S-I** MCQ 14 2 marks MCQ

Same question, different ICAI answer: #432 MTP May 2024 S-I

Based on the data below, what is the amount of the overhead under-/over-absorbed?

Budgeted overhead – ₹ 5,25,000

Budgeted machine hours - 17,500

Actual machine hours - 17,040

Actual overheads - ₹ 5,20,000

A ₹ 5,000 under-absorbed

B ₹ 8,800 under-absorbed

C ₹ 8,800 over-absorbed

D ₹ 5,000 over-absorbed

ICAI'S ANSWER

(B) ₹ 8,800 under-absorbed

Actual Overhead – (Actual machine hours × machine hour rate)

5,20,000 – (17040 × 30) = ₹ 8,800 under absorbed

ICAI · P4 · MTP Sep 2026 · Series I · question p. 6-7 · answer p. 6

427 MTP **Sep 2026 S-II** MCQ 15 2 marks MCQ

Also asked in MTP May 2025 S-II MCQ 12

Pre-determined factory overhead rate was ₹ 15 per labour hour. Actual labour hour worked 60,000. Actual factory overhead was ₹ 11,00,000 however it includes ₹ 26,000 being the wages paid for strike period and overtime wages amounting to ₹ 9,000. It was observed that 2/3 of the under absorbed were due to inflation and rest were due to faulty planning. The amount of over/under absorbed factory overhead transferred to costing P&L will be:

A 58,000 under-absorbed

B 55,000 over-absorbed

C 58,000 over-absorbed

D 55,000 under-absorbed

ICAI'S ANSWER

(A) 58,000 under-absorbed

Particulars	Amount (₹)
Absorption rate	15
Actual hours	60,000
Absorbed Overheads	9,00,000
Actual overheads (11,00,000 - 26,000)	10,74,000
Under absorption	1,74,000
1/3 of 1,74,000	58,000

ICAI · P4 · MTP Sep 2026 · Series II · question p. 7 · answer p. 6

428 MTP **May 2026 S-II** MCQ 15 2 marks MCQ

XYZ Ltd. manufactures a single product 'Z'. The company by using traditional costing method absorbs all production overheads based on machine hours. The machine hour rate of overheads is ₹ 50 per hour. Actual production overhead incurred were ₹ 12,00,000 (including ₹ 1,30,000 on account of written off obsolete stores and ₹ 75,000 wages being paid for the strike period). Actual machine hours worked during the year were 18,500 hours.

Calculate the amount of under-absorption of production overheads during the year.

A ₹ 70,000

B ₹ 2,75,000

C ₹ 1,45,000

D ₹ 2,00,000

ICAI'S ANSWER

(A) ₹ 70,000

Amount of under – absorption of production overheads during the year:

Particulars	Amount (in ₹)
Total production overheads actually incurred during the year	12,00,000
Less: Written off obsolete stores	1,30,000
Less: Wages being paid for the strike period	75,000
Net production overheads actually incurred (A)	9,95,000
Production overhead absorbed by 18,500 machine hours @ ₹ 50 per hour:(B)	9,25,000
Amount of under absorption of production overheads (A-B)	70,000

ICAI · P4 · MTP May 2026 · Series II · question p. 6 · answer p. 5

429 MTP **Jan 2026 S-I** Q1(b) 5 marks

Your company uses a historical cost system and applies overheads on the basis of "pre-determined" rates. The following are the figure from the Trial Balance as at 30th September, 2025:

Manufacturing overheads	₹ 21,32,720 Dr.
Manufacturing overheads applied	₹ 18,29,520 Cr.
Work-in-progress	₹ 7,07,400 Dr.
Finished goods stocks	₹ 11,53,660 Dr.
Cost of goods sold	₹ 42,02,940 Dr.

Required:

(i) COMPUTE the over/ under absorption of the manufacturing overhead.

(ii) STATE the treatment of the over/ under absorbed overhead computed in (i) above.

ICAI'S ANSWER

(i) Computation of over/ under absorption of manufacturing overheads:

	(₹)
Actual overheads	21,32,720
Overhead recovered (applied)	18,29,520
Under absorption (recovery) of overhead	3,03,200

(ii) Treatment of under-absorbed overhead

The under-absorption of overheads is disposed-off using two methods:

(1) Write off the under – absorbed overhead to Costing Profit & Loss Account.

(2) Use supplementary rate, to recover the under-absorbed overhead.

According to first method, the total unabsorbed overhead amount of ₹ 3,03,200 will be written off to Costing Profit & Loss Account. The use of this method will reduce the profits of the concern by ₹ 3,03,200 for the period.

According to second method, a supplementary rate may be used to adjust the overhead cost of each cost unit. The under-absorbed amount in total may, at the end of the accounting period, be apportioned on ratio basis to the three control accounts, viz, Work-in-progress, Finished goods stock and Cost of goods sold account. Apportioning of under-absorbed overhead can be carried out by using direct labour hours/ machine hours/ the value of the balances in each of these accounts, as the basis. Prorated figures of under-absorbed overhead over Work-in-progress, Finished goods stock and Cost of goods sold in this question on the basis of values, of the balances in each of these accounts are as follows:

	Additional Overhead (Under-absorbed)		Total
	(₹)	(₹)	(₹)
Work-in-progress	7,07,400	35,370	7,42,770
Finished Goods Stock	11,53,660	57,683	12,11,343
Cost of Goods Sold	42,02,940	2,10,147	44,13,087
	60,64,000	3,03,200	63,67,200

By using this method, the profit for the period will be reduced by ₹2,10,147 and the value of stock will increase by ₹93,053. The latter will affect the profit of the subsequent period.

Working Notes

The apportionment of under-absorbed overhead over Work-in-progress, Finished goods stock and Cost of goods sold on the basis of their value in the respective account is as follows:

Overhead to be absorbed by work-in-progress	= ₹ 3,03,200/₹60,64,000 × 7,07,400 = ₹ 35,370
Overhead to be absorbed by finished goods	= ₹ 3,03,200/₹60,64,000 × 11,53,660 = ₹ 57,683
Overhead to be absorbed by cost of goods sold	= ₹ 3,03,200/₹60,64,000 × 42,02,940 = ₹ 2,10,147

ICAI · P4 · MTP Jan 2026 · Series I · question p. 7 · answer p. 5-7

430 MTP **Sep 2025 S-I** Q4(c) 4 marks

A Ltd has calculated a predetermined overhead rate of ₹ 22 per machine hour for its Quality Check (QC) department. This rate has been calculated for the budgeted level of activity and is considered as appropriate for absorbing overheads. The following overhead expenditures at various activity levels had been estimated.

Total overheads	Number of machine hours
₹ 3,38,875	14,500
₹ 3,47,625	15,500
₹ 3,56,375	16,500

You are required to:

(i) CALCULATE the variable overhead absorption rate per machine hour.

(ii) CALCULATE the estimated total fixed overheads.

(iii) CALCULATE the budgeted level of activity in machine hours.

(iv) CALCULATE the amount of under/over absorption of overheads if the actual machine hours were 14,970 and actual overheads were ₹ 3,22,000.

ICAI'S ANSWER

(i) Variable overhead absorption rate

$$= \frac{\text{Difference in Total Overheads}}{\text{Difference in levels in terms of machine hours}}$$

$$= \frac{₹\ 3,47,625 - ₹\ 3,38,875}{15,500 \text{ hours} - 14,500 \text{ hours}} = ₹\ 8.75 \text{ per machine hour.}$$

(ii) Calculation of Total fixed overheads:

	(₹)
Total overheads at 14,500 hours	3,38,875
Less: Variable overheads (₹ 8.75 × 14,500)	(1,26,875)
Total fixed overheads	2,12,000

(iii) Calculation of Budgeted level of activity in machine hours:

Let budgeted level of activity = X

Then, $\frac{(₹\ 8.75X + ₹\ 2,12,000)}{X} = ₹\ 22$

8.75X + ₹ 2,12,000 = 22X

13.25X = 2,12,000

X =16,000

Thus, budgeted level of activity = 16,000 machine hours.

(iv) Calculation of Under / Over absorption of overheads:

	(₹)
Actual overheads	3,22,000
Absorbed overheads (14,970 hours × ₹ 22 per hour)	3,29,340
Over-absorption (3,29,340 – 3,22,000)	7,340

ICAI · P4 · MTP Sep 2025 · Series I · question p. 13 · answer p. 14-15

431 MTP **May 2025 S-I** MCQ 15 2 marks MCQ

The accountant for Brilliant Tools Ltd applies overhead based on machine hours. The budgeted overhead and machine hours for the year are ₹ 1,30,000 and 8,000 hours, respectively. The actual overhead and machine hours incurred were ₹ 1,37,500 and 10,000 hours.

What is the amount of over/under absorbed overhead for the year?

A Over absorbed by ₹ 25,000

B Under absorbed by ₹ 25,000

C Over a absorbed by ₹ 32,500

D Under absorbed by ₹ 32,500

ICAI'S ANSWER

(A) Overabsorbed by ₹ 25,000

Predetermined Overhead Rate = Budgeted Overhead / Budgeted hours i.e. 1,30,000 / 8,000 = ₹ 16.25 per hour.

Hence, absorbed overhead = 10,000 X 16.25 = ₹ 1,62,500.

Since actual overhead incurred were ₹ 1,37,500

Hence the overhead were over absorbed by 1,62,500 – 1,37,500 = ₹ 25,000.

ICAI · P4 · MTP May 2025 · Series I · question p. 6 · answer p. 7

432 MTP **May 2024 S-I** MCQ 6 2 marks MCQ

Same question, different ICAI answer: #426 MTP Sep 2026 S-I

Based on the data below, what is the amount of the overhead under-/over-absorbed?

Budgeted overhead – ₹ 5,25,000

Budgeted machine hours- 17,500

Actual machine hours- 17,040

Actual overheads- ₹ 5,20,000

A 5,000 under-absorbed

B 8,800 under-absorbed

C 8,800 over-absorbed

D 5,000 over-absorbed

ICAI'S ANSWER

Answer key: **(B)**

B Actual Overhead – (Actual machine hours × machine hour rate)

5,20,000 – (17040 × 30) = 8,800 under absorbed

ICAI · P4 · MTP May 2024 · Series I · question p. 5 · answer p. 4

5.6 Administration, selling, distribution and R&D overheads

6 questions · asked in 6 ICAI documents · 1 exam paper

433 EXAM **Jan 2025** Q6(c) 4 marks

Explain the methods that can be used for controlling Selling and Distribution Overheads.

ICAI'S ANSWER

The problem of controlling selling & distribution overheads can be tackled by adopting the following steps:

(a) Comparison with past performance - Comparing the figures of selling & distribution overhead with the figures of previous period.

(b) Budgetary Control-Selling & distribution overhead budgets may be used to control such overhead expenses by making a comparison of budgetary figures with actual figures of overhead expenses, ascertaining variances and finally taking suitable actions,

(c) Standard Costing - Standards of selling & distribution expenses may be set up for salesmen, territories, products etc. The laid down standards on comparison with actual overhead expenses will reveal variances, which can be controlled by suitable action.

ICAI · P4 · Jan 2025 exam · question p. 27 · answer p. 29

434 RTP **May 2025** Q20(d)

EXPLAIN:

(i) Pre-production Costs

(ii) Research and Development Costs

(iii) Training Costs

ICAI'S ANSWER

(i) **Pre-production Costs:** These costs forms the part of development cost, incurred in making a trial production run, preliminary to formal production. These costs are incurred when a new factory is in the process of establishment or a new project is undertaken or a new product line or product is taken up, but there is no established or formal production to which such costs may be charged.

(ii) **Research and Development Costs:** Research costs are the costs incurred for the original and planned investigation undertaken with a prospect of gaining new scientific or technical knowledge

and understanding.

Development costs are the cost incurred in applying research findings or other knowledge to a plan or design for the production of new or substantially improved materials, devices, products, processes, systems or services prior to the commencement of commercial production or use.

(iii) **Training Costs:** Costs which are incurred in and in relation to providing training to the workers, apprentices, executives etc. Training cost consists of wages and salaries paid to new trainees, fees paid to trainers, cost of materials and properties used to train the trainees, costs associated with training centre, loss suffered due to lower production and extra spoilage etc. The total cost of training section is thereafter apportioned to production centers.

ICAI · P4 · RTP May 2025 · question p. 25 · answer p. 57-58

435 **RTP** **Jan 2025** **Q10**

Han Ltd. sells three products namely 'A', 'B' and 'C'. The following information is available regarding sales, costs and activity for the year ended 31st March:

Particulars	A	B	C
Sales (₹)	60,00,000	90,00,000	54,00,000
Cost of Sales (₹)	30,00,000	78,00,000	27,00,000
Area of storage (sq.ft.)	72,000	1,08,000	36,000
Number of parcels sent	2,40,000	3,00,000	2,10,000
Number of invoices sent	60,000	90,000	1,44,000

Selling and Distribution overheads and the basis of allocation are as follows :

Fixed Cost	Amount (₹)	Basis of allocation to Products
Rent and Insurance	6,00,000	Square feet
Depreciation	2,70,000	Parcel
Salesman's salaries & expenses	11,40,000	Sales Volume
Administrative wages and salaries	9,00,000	No. of Invoices
Variable Costs:		
Packing wages & materials		₹ 4.80 per parcel
Commission		2.40% of sales
Stationery		₹ 1.80 per invoice

Finance Manager of the Company has recommended to discontinue the Product 'C' since it's sales is less compared to other products.

You are required to PREPARE the profitability statement of each product, showing the percentage of profit/ (loss) on sales for each product, and also EXAMINE the recommendation of Finance Manager.

ICAI'S ANSWER

Profitability statement of each product for the year ended 31st March

Particulars	Total (₹)	Products		
		A (₹)	B (₹)	C (₹)
Sales	2,04,00,000	60,00,000	90,00,000	54,00,000
Variable Costs:				
Cost of sales	1,35,00,000	30,00,000	78,00,000	27,00,000
Commission @ 2.40% of sales	4,89,600	1,44,000	2,16,000	1,29,600
Packaging wages and materials @ ₹ 4.80 per parcel	36,00,000	11,52,000	14,40,000	10,08,000
Stationery @ ₹ 1.80 per invoice	5,29,200	1,08,000	1,62,000	2,59,200
Total Variable Costs	1,81,18,800	44,04,000	96,18,000	40,96,800
Contribution (sales - variable cost)	22,81,200	15,96,000	(6,18,000)	13,03,200
Fixed costs:				
Rent and insurance	6,00,000	2,00,000	3,00,000	1,00,000
Depreciation	2,70,000	86,400	1,08,000	75,600
Salesman's salary and expenses	11,40,000	3,35,294	5,02,941	3,01,765
Administrative wages and salaries	9,00,000	1,83,674	2,75,510	4,40,816
Total Fixed Costs	29,10,000	8,05,368	11,86,451	9,18,181
Profit or loss (Contribution – Fixed costs)	(6,28,800)	7,90,632	(18,04,451)	3,85,019
Percentage of profit or loss on sales (%)	(3.08)%	13.18%	(20.05)%	7.13%

Recommendation of finance manager is not correct. Product 'C' should not be discontinued as it is profitable.

ICAI · P4 · RTP Jan 2025 · question p. 11-12 · answer p. 32-33

436 MTP **Sep 2025 S-I** Q6(d) 4 marks

EXPLAIN Research and Development Expenses and its treatment of in costing.

ICAI'S ANSWER

Research and Development Expenses: The Terminology defines research expenses as "the expenses of searching for new or improved products, new application of materials, or new or improved methods." Similarly, development expenses are defined as "the expenses of the process which begins with the implementation of the decision to produce a new or improved product."

If research is conducted in the methods of production, the research expenses should be taken separately while computing cost of production; while the expenditure becomes a part of the administration overhead if research relates to administration. Similarly, market research expenses are charged to the selling and distribution overhead.

Development costs incurred in connection with a particular product should be charged directly to that product. Such expenses are usually treated as "deferred revenue expenses," and recovered as a cost per unit of the product when production is fully established.

General research expenses of a routine nature incurred on new or improved methods of manufacture or the improvement of the existing products should be charged to the general overhead.

Even in this case, if the amount involved is substantial it may be treated as a deferred revenue expenditure, and spread over the period during which the benefit would accrue. Expenses on fundamental research, not relating to any specific product, are treated as a part of the administration overhead. Where research proves a failure, the cost associated with it should be excluded from costs and charged to the costing Profit and Loss Account.

ICAI · P4 · MTP Sep 2025 · Series I · question p. 15 · answer p. 21

437 MTP **Jan 2025 S-II Q3(b)** 6 marks

From the following data CALCULATE (i) Administration cost, (ii) Selling cost and (iii) Distribution cost:

		Amount (₹)
(i)	Rent paid for factory building	96,000
(ii)	Salary paid to office staffs	8,20,000
(iii)	Fees paid to auditors	92,000
(iv)	Salary paid to sales manager	8,00,000
(v)	Vehicle hire charges paid for directors attending general meeting	10,200
(vi)	Wages paid to workers engaged in storing goods at sales depot	7,200
(vii)	Travelling allowance paid to sales staffs	9,600
(viii)	Cost paid for secondary packing	8,200
(ix)	Electricity bill paid for sales office	1,800
(x)	Depreciation on goods delivery vehicles	13,000
(xi)	Bonus paid to sales staffs for achieving targets	96,000
(xii)	Fees paid to independent directors	1,02,000

ICAI'S ANSWER

(i) **Calculation of Administration cost:**

Particulars	Amount (₹)
Salary paid to office staffs	8,20,000
Fees paid to auditors	92,000
Vehicle hire charges paid for directors attending general meeting	10,200
Fees paid to independent directors	1,02,000
	10,24,200

(ii) **Calculation of Selling cost:**

Particulars	Amount (₹)
Salary paid to sales manager	8,00,000
Wages paid to workers engaged in storing goods at sales depot	7,200
Travelling allowance paid to sales staffs	9,600
Electricity bill paid for sales office	1,800
Bonus paid to sales staffs for achieving targets	96,000
	9,14,600

(iii) **Calculation of Distribution cost:**

Particulars	Amount (₹)
Cost paid for secondary packing	8,200
Depreciation on goods delivery vehicles	13,000
	21,200

ICAI · P4 · MTP Jan 2025 · Series II · question p. 10 · answer p. 13

438 MTP **Old P3 Dec 2021 S-I** Q6(d) 5 marks old scheme

HOW do you deal with the following in cost accounts?

(i) Fringe benefits

(ii) Bad debts.

ICAI'S ANSWER

(i) **Fringe benefits:** These are the additional payments or facilities provided to the workers apart from their salary and direct cost-allowances like house rent, dearness and city compensatory allowances. These benefits are given in the form of overtime, extra shift duty allowance, holiday pay, pension facilities etc.

These indirect benefits stand to improve the morale, loyalty and stability of employees towards the organisation. If the amount of fringe benefit is considerably large, it may be recovered as direct charge by means of a supplementary wage or labour rate; otherwise, these may be collected as part of production overheads.

(ii) **Bad debts:** There is no unanimity among different authors of Cost Accounting about the treatment of bad debts. One view is that 'bad debts' should be excluded from cost. According to this view bad debts are financial losses and therefore, they should not be included in the cost of a particular job or product.

According to another view it should form part of selling and distribution overheads, especially when they arise in the normal course of trading. Therefore, bad debts should be treated in cost accounting in the same way as any other selling and distribution cost. However extra ordinarily large bad debts should not be included in cost accounts.

PassBook's note: The Study Material teaches both items in Chapter 4, "Treatment of certain items in costing" (bad debts with selling and distribution overheads, fringe benefits with the classification of overheads). Kept as printed: "However extra ordinarily large bad debts" (answer).

ICAI · Old P3 · MTP Dec 2021 · Series I · question p. 6 · answer p. 12

CHAPTER 6

Activity Based Costing

6.1 Activities, cost pools, cost drivers and the cost hierarchy

6 questions · asked in 9 ICAI documents · 3 exam papers

439 EXAM May 2025 Q6(b) 5 marks

Cosmos Limited uses activity based costing and accumulates overhead costs in the following cost pools:

(1) Human Resources

(2) Maintenance of buildings

(3) Parts Management

(4) Plant security

(5) Purchasing

(6) Floor manager's salary

(7) Quality control

(8) Machine set-up

(9) Designing the product

(10) Receiving Department

Classify each cost pool as per cost hierarchy i.e. unit level, batch level, product level or facility level.

ICAI'S ANSWER

S. No.	Cost Pool	Cost Hierarchy Level
(1)	Human Resources	Facility-level
(2)	Maintenance of buildings	Facility-level
(3)	Parts Management	Batch-level/Product Level
(4)	Plant security	Facility-level
(5)	Purchasing	Batch-level
(6)	Floor manager's salary	Facility-level
(7)	Quality control	Batch-level / Unit Level
(8)	Machine set-up	Batch-level
(9)	Designing the product	Product-level
(10)	Receiving Department	Batch-level

ICAI · P4 · May 2025 exam · question p. 31 · answer p. 32

440 EXAM **Old P3 Nov 2022** Q6(c) 5 marks old scheme

PP Limited is in the process of implementation of Activity Based Costing System in the organisation. For this purpose, it has identified the following Business Functions in its organisation:

(i) Research and Development

(ii) Design of Products, Services and Procedures

(iii) Customer Service

(iv) Marketing

(v) Distribution

You are required to specify two cost drivers for each Business Function Identified above.

ICAI'S ANSWER

Business functions	Cost Driver
Research and Development	• Number of research projects • Personnel hours on a project • Technical complexities of the project
Design of products, services and procedures	• Number of products in design • Number of parts per product • Number of engineering hours
Customer Service	• Number of service calls • Number of products serviced • Hours spent on servicing products
Marketing	• Number of advertisements • Number of sales personnel • Sales revenue
Distribution	• Number of units distributed • Number of customers • Weight of items distributed

(Any two cost drivers of each business function)

ICAI · Old P3 · Nov 2022 exam · question p. 22 · answer p. 24

441 EXAM **Old P3 Nov 2020** Q5(c) 4 marks old scheme

Describe the various levels of activities under 'ABC' methodology.

ICAI'S ANSWER

Various Level of Activities under ABC Methodology

Level of Activities	Meaning
1. Unit level activities	These are those activities for which the consumption of resources can be identified with the number of units produced.
2. Batch level activities	The activities such as setting up of a machine or processing a purchase order are performed each time a batch of goods is produced. The cost of batch related activities varies with number of batches made, but is common (or fixed) for all units within the batch.
3. Product level activities	These are the activities which are performed to support different products in product line.
4. Facilities level activities	These are the activities which cannot be directly attributed to individual products. These activities are necessary to sustain the manufacturing process and are common and joint to all products manufactured.

ICAI · Old P3 · Nov 2020 exam · question p. 19 · answer p. 21-22

442 RTP **Sep 2026** Q16(d)

Also asked in RTP Jan 2026 Q19(b) · MTP May 2025 S-I Q6(b)

Provide example(s) of the cost driver for the following cost pools:

Cost Pool
Quality Control
Research and Development
Machine Maintenance
Employee Training Costs
Customer Service

ICAI'S ANSWER

Cost Pool	Cost Driver
Quality Control	• Number of Inspections • Product Units Produced
Research and Development	• Number of research projects • Personnel hours on a project
Machine Maintenance	• Machine Hours
Employee Training Costs	• Number of Training Hours • Employees Trained
Customer Service	• Number of service calls • Number of products serviced • Hours spent on servicing products

443 MTP **Sep 2025 S-II** Q5(b) 5 marks

STATE with a brief reason whether you would recommend an activity based system of costing in each of the following independent situations:

(i) Company K produces one product. The overhead costs mainly consist of depreciation.

(ii) Company L produces 5 different products using different production facilities.

(iii) A consultancy firm consisting of lawyers, accountants and computer engineers provides management consultancy services to clients.

(iv) Company S produces two different labour intensive products. The contribution per unit in both products is very high. The BEP is very low. All the work is carried on efficiently to meet the target costs.

ICAI'S ANSWER

Sl. No	Description	Recommend ABC (Yes/ No)	Reasons
(i)	K produces one product. Overhead is mainly depreciation.	No	• One product situation. For allocation of overhead, ABC is not required. • ABC for cost reduction not beneficial since most of the overhead is depreciation.
(ii)	L produces 5 different products with different facilities.	Yes	• Multi product situation. ABC is required for allocation of overhead. • ABC is necessary for pricing. • Cost drivers are likely to be different. • Cost reduction may be possible. • Production facilities are different.
(iii)	Professional services - lawyers/accountants/computer engineers.	Yes	• Variety of services. Hence ABC is required for cost allocation. • Services are very different. • ABC is necessary for pricing. • Cost reduction possible.

Sl. No	Description	Recommend ABC (Yes/ No)	Reasons
(iv)	S produces 2 different labour intensive products. High unit contribution and efficient operations.	No	• Different products, but labour intensive. Hence, overhead allocation based on readily traceable direct labour cost will be accurate. Hence, ABC not required for cost allocation. • Low BEP level implies low level of fixed cost as a % of sale price or as a % of total cost. • Many fixed cost activity drivers are likely to align with the direct labour costs. Hence not required for cost allocation. • Efficient operation. Hence ABC not required even for cost reduction or ABC management.

ICAI · P4 · MTP Sep 2025 · Series II · question p. 16 · answer p. 24-25

444 MTP **Jan 2025 S-I** Q5(b) 4 marks

Also asked in MTP Sep 2024 S-II Q2(b)

EXPLAIN the Usefulness/Suitability of ABC.

ICAI'S ANSWER

ABC is particularly needed by organisations for product costing in the following situations:

1. High amount of overhead: When production overheads are high and form significant costs, ABC is more useful than traditional costing system.
2. Wide range of products: ABC is most suitable, when, there is diversity in the product range or there are multiple products.
3. Presence of non-volume related activities: When non-volume related activities e.g. material handling, inspection set-up, are present significantly and traditional system cannot be applied, ABC is a superior and better option. ABC will identify non-value-adding activities in the production process that might be a suitable focus for attention or elimination.
4. Stiff competition: When the organisation is facing stiff competition and there is an urgent requirement to compute cost accurately and to fix the selling price according to the market situation, ABC is very useful. ABC can also facilitate in reducing cost by identifying non-value-adding activities in the production process that might be a suitable focus for attention or elimination.

ICAI · P4 · MTP Jan 2025 · Series I · question p. 16 · answer p. 18

6.2 Product costs under ABC versus traditional absorption

92 questions · asked in 60 ICAI documents · 16 exam papers

445 EXAM Sep 2026 Case Scenario – I · MCQ 1 to MCQ 5 10 marks

CASE SCENARIO – I

GNPL is a food processing company specializing in agricultural products. It purchases produce directly from farmers and makes packaged food products. GNPL is interested in understanding the profitability of its products based on customer type. Three main customer categories have been identified by the Marketing Department: Distributors, Supermarkets and Online Customers. The following information relates to the year just ended :

Table – I

Activity	Customer level operating cost (₹)	Cost Driver
Product handling	10,08,000	Quantity sold (kgs)
Processing of customer orders	20,80,000	Number of sales orders
Cost of Transportation to customer	59,20,000	Kilometres travelled
Customer Visits	19,20,000	Number of customer visits

Table – II

	Distributors	Supermarkets	Online
Quantity Sold (kg)	1,25,000	65,000	62,000
No. of sales orders	42	238	370
Kilometres travelled	8,975	8,800	5,350
No. of customer visits	35	179	86
Listed Selling price (₹ per kg)	660	660	660
Discount	10%	15%	20%
Cost of goods sold (excluding customer level operating cost) (₹ per kg)	460	460	460

On the basis of the above Case Scenario, you are required to answer the following MCQs **1 to 5** :

MCQ 1 2 marks

What is the cost of customer visits incurred on customer category 'Distributors' ?

A ₹ 2,61,400

B ₹ 2,14,600

C ₹ 2,24,000

D ₹ 2,42,000

MCQ 2 2 marks

What is the cost of transport for 'Online' customer category ?

A ₹ 13,69,600

B ₹ 13,74,400

C ₹ 13,55,600

D ₹ 13,84,400

MCQ 3 2 marks

What is the cost of activities (tabulated in Table – I) as a percentage of the sales revenue net of discount for 'Supermarkets' category ?

A 12.12%

B 10.30%

C 9.19%

D 8.65%

MCQ 4 2 marks

What is the total customer-level operating cost of all activities for 'Distributors' category ?

A ₹ 32,65,000

B ₹ 31,56,000

C ₹ 30,25,000

D ₹ 29,95,000

MCQ 5 2 marks

What is the customer level operating income for 'Online' Category ?

A ₹ 8,64,000

B ₹ 9,12,000

C ₹ 12,25,000

D ₹ 12,65,000

ICAI'S ANSWERS

MCQ 1

Answer key: **ICAI has not published the key yet.**

MCQ 2

Answer key: **ICAI has not published the key yet.**

MCQ 3

Answer key: **ICAI has not published the key yet.**

MCQ 4

Answer key: **ICAI has not published the key yet.**

MCQ 5

Answer key: **ICAI has not published the key yet.**

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446 EXAM **May 2026** Q5(a) 10 marks

XYZ Private Limited manufactures two products, Mini and Maxi, utilising the same production methods. The company operates under the Absorption Costing System. The details of the products for financial

year 2025-26 are provided below:

Particulars	Products	
	Mini	**Maxi**
Number of units produced	8000 units	10000 units
Number of units sold	6400 units	8000 units
Direct Material Cost per unit	₹ 250	₹400
Direct Labour Hours per unit	4 Hours	5 Hours
Selling price per unit sold during the year	₹ 950	₹ 1180

Using the traditional method, all the Production overheads were of ₹ 54,00,000, which were directly distributed on the basis of units produced. Direct labour hour rate is ₹ 50 per hour.

Further, the Board of Directors is considering transitioning to an Activity - Based Costing (ABC) system to determine the production overheads.

The total production overheads are analyzed into the following:

Particulars	Amount (₹)
Set -up costs	9,00,000
Machine Running costs	18,00,000
Inspection costs	27,00,000
Total	**54,00,000**

The following data on cost drivers was provided by the production manager relating to the bifurcation of production overheads:

Particulars	Mini	Maxi
Number of Setups	100	200
Number of Machine Runs	750	1,050
Number of Inspections	60	210

You are required to calculate:

(i) Total Sales and Total Profit earned for each product as per Absorption Costing System.

(ii) Total Sales and Total Profit earned for each product as per Activity - Based Costing System.

ICAI'S ANSWER

(i) Total sales and total profit for each product under absorption costing system

Factory overhead rate per unit = ₹ 54,00,000/18,000 units = ₹ 300 per unit

Particulars	Mini	Maxi
No. of units produced	8000 units	10000 units
No. of units sold	6400 units	8000 units
	₹	₹
Direct Material Cost per unit	250	400
Direct Labour Cost per unit (4 hours per unit @ ₹ 50 per hour) (5 hours per unit @ ₹ 50 per hour)	200	250
Factory overheads @ ₹ 300 per unit*	300	300
Cost per unit	**750**	**950**
Total Cost of Production (Cost per unit x No. of units produced)	**60,00,000**	**95,00,000**
Less: Closing stock of finished goods (Cost per unit x no. of units of closing stock)	12,00,000	19,00,000
Cost of Goods sold	48,00,000	76,00,000
Profit (Balancing figure)	**12,80,000**	**18,40,000**
Total Sales (Selling price per unit x No. of units sold)	**60,80,000**	**94,40,000**

*Overhead rate $= \frac{\text{Budgeted overheads}}{\text{Units Produced}}$

$= \frac{₹54,00,000}{18,000 \text{ units}} = ₹ 300$

(ii) Activity cost-Driver rates

Activity	Activity Costs (₹)	Cost-driver	Cost-driver rate
Set-up cost	9,00,000	300 set-ups	₹ 3,000 per set-up
Machine running costs	18,00,000	1,800 machine runs	₹ 1000 per machine run
Inspection costs	27,00,000	270 inspections	₹ 10,000 per inspection

Total sales and total profit for each product as per Activity Based Costing Method

Particulars	Mini	Maxi
No. of units produced	8,000 units	10,000 units
No. of units sold	6,400 units	8,000 units
	₹	₹
Direct Material Cost per unit	250	400

Particulars	Mini	Maxi
Direct Labour Cost per unit (4 hours per unit @ ₹ 50 per hour) (5 hours per unit @ ₹ 50 per hour)	200	250
Prime cost per unit	450	650
Total Prime Cost	**36,00,000**	**65,00,000**
Activity Costs:		
Set-up cost (₹ 3,000 per set-up)	3,00,000 (3,000 × 100 set-ups)	6,00,000 (3,000 × 200 set-ups)
Machine running costs (₹ 1,000 per machine run)	7,50,000 (1,000 × 750 machine runs)	10,50,000 (1,000 × 1050 machine runs)
Inspection costs (₹ 10,000 per inspection)	6,00,000 (10,000 × 60 inspections)	21,00,000 (10,000 × 210 inspections)
Total Overhead Cost	**16,50,000**	**37,50,000**
Overhead per unit	206.25	375
Cost per unit	656.25	1,025
Total Cost of Production (Cost per unit x No. of units produced)	**52,50,000**	**1,02,50,000**
Less: Closing stock of finished goods (Cost per unit x no. of units of closing stock)	10,50,000	20,50,000
Cost of Goods sold	42,00,000	82,00,000
Profit (Balancing figure)	**18,80,000**	**12,40,000**
Total Sales (Selling price per unit x No. of units sold)	**60,80,000**	**94,40,000**

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447 EXAM **Sep 2025** Q3(a) 8 marks

A consultancy firm provides project management service in three sectors Technology, Healthcare and Education. The Project management service covers development and implementation of software for various MIS requirements of its clients. The fees charged per project is as follows:

Technology: ₹ 90,000

Healthcare: ₹ 1,20,000

Education: ₹ 1,10,000

The company uses Activity-Based Costing (ABC) to allocate its overhead costs. For the month of August 2025, the following informations are provided:

Service Sector	Number of Projects	Software Development Hours	Consulting Hours	Number of Client Meetings
Technology	20 projects	10,000 hours	6,400 hours	30
Healthcare	10 projects	7,000 hours	5,600 hours	20
Education	10 projects	5,000 hours	2,000 hours	40

Overhead Costs and Activities:

Activity	Total Cost (₹)	Cost Driver
Management of Projects	8,10,000	Number of projects
Consulting Service Delivery	4,20,000	Consulting hours
Client Interaction & Meetings	6,30,000	Number of client meetings
Administration and Support	15,40,000	Software development hours

You are required to:

(i) Prepare a statement showing the total cost and per project cost of project management service for each service sector Technology, Healthcare and Education using Activity Based Costing Approach.

(ii) Identify the most profitable sector based on profitability percentage on fees charged.

ICAI'S ANSWER

(i) **Statement Showing "Total cost and Cost *per project* - Activity Based Costing"**

Particulars	Technology	Healthcare	Education
Management of Project @ ₹ 20,250 per project (20,10,10)	4,05,000	2,02,500	2,02,500
Consulting Service Delivery @ ₹30 per hour (6,400, 5,600, 2,000)	1,92,000	1,68,000	60,000
Client Interaction and Meeting @ ₹ 7,000 per meeting (30,20,40)	2,10,000	1,40,000	2,80,000
Administration and Support @ ₹ 70 per hour (10,000, 7,000, 5,000)	7,00,000	4,90,000	3,50,000
Total Cost	15,07,000	10,00,500	8,92,500
Project	20	10	10
Cost per Project	75,350	1,00,050	89,250
Fee Charged	90,000	1,20,000	1,10,000
% of fee Charged	119.44	119.94	123.25
OR			

Particulars	Technology	Healthcare	Education
Profit	14,650	19,950	20,750
Profitability %	16.28%	16.63%	18.86%

(ii) Education is the most profitable sector.

Working Note:

Statement showing calculation of Cost Driver rate

Activity	Total Cost	Cost Driver Basis	Cost Driver	Cost Driver Rate
	(₹)		**(Units)**	**(₹)**
Management of Project	8,10,000	No of Projects	40	20,250 per project
Consulting Service Delivery	4,20,000	Consulting Hours	14,000	30 per hour
Client Interaction and Meeting	6,30,000	No of client meeting	90	7,000 per meeting
Administration and Support	15,40,000	Software Development Hours	22,000	70 per hour

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448 EXAM May 2025 Q3(a) 8 marks

Meri Chai Teri Chai Ltd., is engaged in manufacturing three products:

- Ginger Chai
- Masala Chai
- Saffron Chai

It calculates activity cost rates based on cost driver capacity.

Activity	Cost driver	Capacity	Cost (₹)
Machine Setup	Number of setups	64	7,68,000
Machine Processing	Machine hours	1,40,000	7,00,000
Quality inspection	Number of inspections	544	6,80,000
Packaging	Number of packings	600	7,20,000

For the year ended 31st March 2025, the following consumption of cost drivers was reported:

Product	Number of setups	Machine hours	Number of inspections	Number of packings
Ginger Chai	21	45,000	190	190
Masala Chai	22	50,000	204	250
Saffron Chai	17	40,000	150	150

Required:

(i) Compute the costs allocated to each product from each activity on the basis of Activity- Based Costing method.

(ii) Calculate the cost of unused capacity for each activity.

ICAI'S ANSWER

Working Note:

Cost Driver Rates-

Activity	Cost	Cost Driver Basis	Cost Driver	Cost Driver Rate
	(₹)		**(Units)**	**(₹)**
Setup	7,68,000	Number of setups	64	12,000 per setup
Processing	7,00,000	Machine Hours	1,40,000	5 per machine hour
Inspection	6,80,000	Number of inspections	544	1,250 per inspection
Packaging	7,20,000	Number of packings	600	1,200 per packing

(i) Computation of cost allocated to each product - Activity Based Costing

Products	Ginger Chai (₹)	Masala Chai (₹)	Saffron Chai (₹)
Setup Costs @ ₹ 12,000 per setup (21,22,17)	2,52,000	2,64,000	2,04,000
Machine Processing Costs @ ₹ 5 per hour (45,000, 50,000, 40,000)	2,25,000	2,50,000	2,00,000
Inspection Costs @ ₹ 1,250 per inspection (190,204,150)	2,37,500	2,55,000	1,87,500
Packaging Costs @ ₹ 1,200 per packing (190,250,150)	2,28,000	3,00,000	1,80,000
Total Costs	9,42,500	10,69,000	7,71,500

(ii) Computation of cost of unused capacity for each activity

Activity	Unused Capacity	Total Cost (₹)
Setup Costs @ ₹ 12,000 per setup	4 setups	48,000
Machine Processing Costs @ ₹ 5 per hour	5,000 machine hours	25,000
Inspection Costs @ ₹ 1,250 per inspection	-	-
Packaging Costs @ ₹ 1,200 per packing	10 packings	12,000
Total Costs		85,000

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449 EXAM Jan 2025 Case Scenario-I · MCQ 1 to MCQ 5

CASE SCENARIO-I

ABC Company produces three products X. Y and Z. Similar type of material is used in the production of all the three products. The company has been using traditional absorption costing method, using direct labour hours to allocate overheads to its products. The Cost Accountant has suggested considering an activity based costing system. The following information is available in the records of the company.

	X	Y	Z
Production Volume p.a. (In units)	16,000	17,000	15,000
Direct Material per unit	3 kg	4 kg	5 kg
Labour hours per unit	0.10	0.15	0.20
Machine hours per unit	0.5	0.7	0.9
No. of Production runs p.a	50	65	60
No. of purchase orders p.a	5	10	15
No. of order shipped p.a	25	35	32

Activity	Cost (₹)	Cost Driver
Machine setup costs	49,000	Production Runs
Machine running costs	64,128	Machine hours
Purchase cost	52,050	Purchase orders
Delivery cost	46,460	Orders shipped

The price of Raw Material is ₹ 2 per kg.

Direct labour cost per hour is ₹ 20.

On the basis of above Case Scenario, you are required to answer the following MCQs 1 to 5:

MCQ 1

What is overhead absorption rate per hour as per traditional absorption costing method?

A ₹ 29.60

B ₹ 29.32

C ₹ 13.78

D ₹ 15.82

MCQ 2

What is full cost per unit of product Y under traditional absorption costing method?

A ₹ 19.92

B ₹ 4.44

C ₹ 46.32

D ₹ 15.44

MCQ 3

Under an activity based costing system, what is the cost driver rate for machine set up costs?

A ₹ 280

B ₹ 1.467

C ₹ 6.85

D ₹ 230

MCQ 4

Under an activity based costing system, the amount of allocated overheads attributable to machine running hours to product X is:

A ₹ 22,848

B ₹ 15,360

C ₹ 25,920

D ₹ 14,000

MCQ 5

The total cost of product Z as per activity based costing method is :

A ₹ 2,86,073

B ₹ 2,94,905

C ₹ 84,905

D ₹ 2,60,660

ICAI'S ANSWERS

MCQ 1

Answer key: **(A)**

MCQ 2

Answer key: **(D)**

MCQ 3

Answer key: **(A)**

MCQ 4

Answer key: **(B)**

MCQ 5

Answer key: **(B)**

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450 **EXAM** **Sep 2024** Q3(a) **8 marks**

GST Limited is a multi-product company. The production and cost details of its two products P and Q are given as follows:

Particulars	Product	
	P	**Q**
Quantity produced (No.)	9,000	7,200
Direct material cost (₹)	72,000	50,000
Direct labour hours	800	600
Purchase requisition (No.)	180	144
Production runs (No.)	144	108
Quality inspections (No.)	27	18

Direct wages rate is ₹ 14.50 per hour. Presently the company uses a single overhead recovery rate based on direct labour hours. Overhead incurred by the company during the year 2023-24 are as follows:

Technical staff salary	₹ 45,000
Machine operation expenses	₹ 1,62,000
Machine maintenance expenses	₹ 27,000
Wages and salary of stores staff	₹ 36,000

During this period direct labour hours worked 72,000.

Now the Company wants to adopt Activity Based Costing. For this purpose, following activities are identified:

- Quality control
- Setup of machine for production runs
- Store receiving

It is also decided that salary of technical staff should be distributed among machine maintenance, setup and quality control in the ratio of 1 : 2 : 2. Machine maintenance expenses and machine operation expenses should be distributed in the ratio of 2 : 3 in between stores and production setup activities.

During this period cost drivers for these activities are identified as under:

- Requisition raised 5,760

- Production setup 7,200
- No. of quality test 720

You are required to compute:

(i) The cost of products P and Q based on traditional absorption costing system.

(ii) The cost of products P and Q based on ABC Costing system.

ICAI'S ANSWER

(i) Statement Showing "Total Cost - Traditional Method"

Particulars of Costs	P	Q
	(₹)	**(₹)**
Direct Materials	72,000	50,000
Direct Labour [(800,600 hours) × ₹ 14.5]	11,600	8,700
Production Overheads [(800,600 hours) × ₹ **3.75**] (WN1)	3,000	2,250
Total Cost	**86,600**	**60,950**
Cost per unit (9,000, 7,200)	9.62	8.47

WN1: Calculation of Production Overhead:

	(₹)
Technical staff salary	45,000
Machine operation expenses	1,62,000
Machine maintenance expenses	27,000
Wages and salary of stores staff	36,000
Total Production Overhead	2,70,000
Total direct labour hours worked	72,000 hours
Production Overhead rate per hour	3.75

(ii) Statement Showing "Total Cost - Activity Based Costing"

Products	P	Q
Production (units)	9,000	7,200
	(₹)	**(₹)**
Direct Materials	72,000	50,000
Direct Labour [(800,600 hours) × ₹ 14.5]	11,600	8,700
Requisition Related Costs @ ₹ 20 *per requisition raised* (180,144) (WN2)	3,600	2,880

Products	P	Q
Production Setup Costs @ ₹ 19 *per production runs* (144,108) (WN2)	2,736	2,052
Quality Inspection Costs @ ₹ 25 *per* quality test (27,18) (WN2)	675	450
Total Costs	**90,611**	**64,082**
Cost per unit (9,000, 7,200)	10.07	8.90

WN2: Statement Showing Distribution of Expenses

	Machine maintenance expenses	Total Stores	Production Setup	Quality Control
Technical staff salary of ₹ 45,000 (1:2:2)	9,000	-	18,000	18,000
Machine operation expenses of ₹ 1,62,000 (2:3)	-	64,800	97,200	-
Machine maintenance expenses of ₹ 36,000 (2:3)	(9,000)	14,400	21,600	-
Wages and salary of stores staff	-	36,000	-	-
Total	-	**1,15,200**	**1,36,800**	**18,000**

WN3: Cost for each activity cost driver:

Activity (1)	Total cost (₹) (2)	Cost allocation base (3)	Cost driver rate (4) = [(2) ÷ (3)]
Stores Receiving	1,15,200	5,760 Requisitions Raised	**₹ 20 per requisition raised**
Production Setup	1,36,800	7,200 Production Setup	**₹ 19 per production setup**
Quality Control	18,000	720 Quality Test	**₹ 25 per quality test**

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451 **EXAM** **May 2024** Q2(a) **8 marks**

Luxury Designer Pvt. Ltd. is a manufacturing company, which manufactures readymade designer shirts. It has four customers: two wholesale category customers and two retail category customers. It has

developed the following Activity- Based Costing system:

Activity	Cost Driver Rate (₹)
Order Processing	1,260 per purchase order
Customer Visits	1,500 per customer visit
Regular Delivery	30 per delivery Km. travelled
Expedited Delivery	4,490 per expedited delivery

List selling price per shirt is ₹ 1,000 and average cost per shirt is ₹ 600. CEO of Luxury Designer Pvt. Ltd. wants to evaluate the profitability of each of the four customers for the year 2023, to explore opportunities for increasing profitability of his Company in the next year 2024. The following data in context of four customers are available for 2023:

	Wholesale Customers		Retail Customers	
	WC-1	**WC-2**	**RC-1**	**RC-2**
Number of Purchase orders	50	65	224	245
Number of Customer visits	10	13	25	22
Regular Deliveries	46	52	175	198
Kilometers travelled per delivery	20	15	10	25
Expedited Deliveries	5	16	50	62
Average Number of Shirts per Shirt	215	110	18	15
Average Selling Price per Shirt	₹ 700	₹ 800	₹ 900	₹ 950

You are required to:

Calculate the customer-level operating income and operating income as a % of revenues in 2023 and rank them on the basis of relative profitability.

ICAI'S ANSWER

Working note:

Computation of revenues (at listed price), discount, cost of goods sold and customer level operating activities costs:

	Wholesale Category Customers		Retail Category Customers	
	WC-1	**WC-2**	**RC-1**	**RC-2**
Number of shirts sold (a)	10,750 (215×50)	7,150 (110×65)	4,032 (18×224)	3,675 (15×245)

	Wholesale Category Customers		Retail Category Customers	
Revenues (at listed price) (₹): (b) {(a) × ₹ 1,000)}	1,07,50,000	71,50,000	40,32,000	36,75,000
Discount (₹): (c) {(a) × Discount per shirt}	32,25,000	14,30,000	4,03,200	1,83,750
Cost of shirts (₹) : (d) {(a) × ₹ 600}	64,50,000	42,90,000	24,19,200	22,05,000
Order taking costs (₹): (No. of purchase × ₹1,260)	63,000	81,900	2,82,240	3,08,700
Customer visits costs (₹) (No. of customer visits × ₹ 1,500)	15,000	19,500	37,500	33,000
Delivery vehicles travel costs (₹) (Kms travelled by delivery vehicles × ₹ 30 per km.)	27,600	23,400	52,500	1,48,500
Cost of expediting deliveries (₹) {No. of expedited deliveries × ₹ 4,490}	22,450	71,840	2,24,500	2,78,380
Total cost of customer level operating activities (₹)	1,28,050	1,96,640	5,96,740	7,68,580

Computation of Customer level operating income

	Wholesale Category Customers		Retail Category Customers	
	WC-1	**WC-2**	**RC-1**	**RC-2**
Revenues (At list price) (Refer to working note)	1,07,50,000	71,50,000	40,32,000	36,75,000
Less: Discount (Refer to working note)	32,25,000	14,30,000	4,03,200	1,83,750

	Wholesale Category Customers		Retail Category Customers	
Revenue (At actual price)	75,25,000	57,20,000	36,28,800	34,91,250
Less: Cost of shirts (Refer to working note)	64,50,000	42,90,000	24,19,200	22,05,000
Gross margin	10,75,000	14,30,000	12,09,600	12,86,250
Less: Customer level operating activities costs (Refer to working note)	1,28,050	1,96,640	5,96,740	7,68,580
Customer level operating income	9,46,950	12,33,360	6,12,860	5,17,670
Operating income as a % of revenues	12.584%	21.562%	16.889%	14.828%
Rank	IV	I	II	III

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452 EXAM **Old P3 Nov 2023** Q4(a) 10 marks old scheme

JH Plastics Limited manufactures three products S, M and L. To date, simple traditional absorption costing system has been used to allocate overheads to products. Total production overheads are allocated on the basis of machine hours. The machine hour rate for allocating production overheads is ₹ 240 per machine hour under the traditional absorption costing system. Selling prices are calculated by adding mark up of 40% of the product cost. Information related to products for the most recent year is as under :

	Products		
	S	**M**	**L**
Units produced and sold	7,500	12,500	9,000
Direct material cost per unit (₹)	158	179	250
Direct labour cost per unit (₹)	40	45	60
Machine hours per unit	0.30	0.45	0.50
Number of Machine setups	120	120	160
Number of purchase orders	90	135	125
Number of inspections	100	160	140

The management wishes to introduce activity-based method (ABC) system of attributing production overheads to products and has identified major cost pools for production overheads and their associated cost drivers as follows:

Cost pool	Amount	Cost driver
Purchasing Department Cost	₹ 7,00,000	Number of Purchase orders
Machine setup Cost	₹ 9,00,000	Number of Machine setups
Quality Control Cost	₹ 6,56,000	Number of inspections
Machining Cost	₹ 5,64,000	Machine hours

Required:

(i) Calculate the total cost per unit and selling price per unit for each of the three products using:

(a) The traditional costing approach currently used by JH Plastics Limited;

(b) Activity based costing (ABC) approach.

(ii) Calculate the difference in selling price per unit as per (a) and (b) above and show which product is under-priced or over-priced.

ICAI'S ANSWER

(i) (a) Statement showing 'Cost per unit & Selling price per unit – Traditional Method'.

Particular	Products		
	S (₹)	**M (₹)**	**L (₹)**
Direct material cost per unit	158	179	250
Direct labour cost per unit	40	45	60
Production overhead @ ₹ 240 per machine hour	72 (₹ 240 x 0.3)	96 (₹ 240 x 0.4)	120 (₹ 240 x 0.5)
Cost per unit	**270**	**320**	**430**
Add: Profit @ 40%	108	128	172
Selling price per unit	**378**	**448**	**602**

(b) Statement showing 'Cost per unit & Selling price per unit – Activity Based Costing'.

Particular	Activity Drivers	Total Amount (₹)	Products		
			S	**M**	**L**
Production (units)	-	-	7500	12500	9000
Machine hours	-	-	2250 (7500 x 0.3)	5000 (12500 x 0.4)	4500 (9000 x 0.5)

Particular	Activity Drivers	Total Amount (₹)	Products		
			(₹)	(₹)	(₹)
Direct material cost per unit (i)			158	179	250
Direct labour cost per unit (ii)			40	45	60
Overheads					
Purchasing department cost (90:135:125)	Number of purchase orders	7,00,000	1,80,000	2,70,000	2,50,000
Machine setup cost (120:120:160)	Number of machine setups	9,00,000	2,70,000	2,70,000	3,60,000
Quality control cost (100:160:140)	Number of inspections	6,56,000	1,64,000	2,62,400	2,29,600
Machining cost (225:500:450)	Machine hours	5,64,000	1,08,000	2,40,000	2,16,000
Total Overhead			7,22,000	10,42,400	10,55,600
Overhead Cost per unit (iii)			96.27	83.39	117.29
Total Cost per unit (i+ii+iii)			**294.27**	**307.39**	**427.29**
Add: Profit @ 40%			117.71	122.96	170.92
Selling price per unit			**411.98**	**430.35**	**598.21**

Note: The question may also be solved by calculating cost driver rate & allocating various cost based on cost driver rate. However, there will be no change in any of the answer.

(ii)

Particular	Products		
	S (₹)	**M (₹)**	**L (₹)**
Selling price per unit as per Traditional Costing	378	448	602
Selling price per unit as per Activity Based Costing	411.98	430.35	598.21
Difference	**(33.98)**	**17.65**	**3.79**

Product S is underpriced while product M and L is overpriced using Traditional costing approach.

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453 EXAM Old P3 May 2023 Q4(b) 5 marks old scheme

Beta Limited produces 50,000 Units, 45,000 Units and 62,000 Units of product 'A', 'B' and 'C' respectively. At present the company follows absorption costing method and absorbs overhead on the basis of direct labour hours. Now, the company wants to adopt Activity Based Costing

The information provided by Beta Limited is follows:

	Product A	Product B	Product C
Floor Space Occupied	5,000 Sq.Ft.	4,500 Sq.Ft.	6,200 Sq.Ft.
Direct Labour Hours	7,500 Hours	7,200 Hours	7,800 Hours
Direct Machine Hours	6,000 Hours	4,500 Hours	4,650 Hours
Power consumption	32%	28%	40%

Overhead for year are as follows:

	₹
Rent & Taxes	8,63,500
Electricity Expenses	10,66,475
Indirect labour	13,16,250
Repair & Maintenance	1,28,775
	33,75,000

Required:

(i) Calculate the overhead rate per labour hour under Absorption Costing.

(ii) Prepare a cost statement showing overhead cost per unit for each product - 'A', 'B' and 'C' as per Activity based Costing.

ICAI'S ANSWER

(i) Calculation of Overhead rate per hour

$$\frac{\text{Total Overheads}}{\text{Total hours}}$$

$$\frac{33,75,000}{22,500} = \textbf{₹ 150 per hour}$$

(ii) Statement showing overhead cost per unit as per Activity Based Costing

Overheads	Cost Driver	Total	Product		
			A	B	C
		₹	₹	₹	₹
Rent & Taxes	Floor space (50:45:62)	8,63,500	2,75,000	2,47,500	3,41,000
Electricity	Power Consumption (32:28:40)	10,66,475	3,41,272	2,98,613	4,26,590

Overheads	Cost Driver	Total	Product		
Indirect labour	Labour hours (75:72:78)	13,16,250	4,38,750	4,21,200	4,56,300
Repair & Maintenance	Machine hours (600:450:465)	1,28,775	51,000	38,250	39,525
Total Cost		33,75,000	11,06,022	10,05,563	12,63,415
Units			50,000	45,000	62,000
Cost per Unit			**22.12**	**22.35**	**20.38**

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454 **EXAM** **Old P3 Nov 2022** Q2(c) **4 marks** old scheme

XYZ Ltd. is engaged in manufacturing two products- Express Coffee and Instant Coffee. It furnishes the following data for a year:

Product	Actual Output (units)	Total Machine hours	Total Number of Purchase orders	Total Number of set ups
Express Coffee	5,000	20,000	160	20
Instant Coffee	60,000	1,20,000	384	44

The annual overheads are as under:

Particulars	₹
Machine Processing costs	7,00,000
Set up related costs	7,68,000
Purchase related costs	6,80,000

You are required to:

(i) Compute the costs allocated to each product – Express Coffee and Instant Coffee from each activity on the basis of Activity- Based Costing (ABC) method.

(ii) Find out the overhead cost per unit of each product – Express coffee and Instant coffee based on (i) above.

ICAI'S ANSWER

(i) Estimation of Cost-Driver rate

Activity	Overhead cost (₹)	Cost-driver level	Cost driver rate (₹)
Machine processing	7,00,000	1,40,000 Machine hours	5
Set up Costs	7,68,000	64 Number of set up	12,000
Purchase related Costs	6,80,000	544 Number of purchase order	1250

Cost Allocation under Activity based Costing

	Express Coffee (₹)	Instant Coffee (₹)
Overhead Cost		
Machine processing (Cost Driver rate - ₹ 5) (or 20,000:1,20,000)	5 × 20,000 = 1,00,000	5 × 1,20,000 = 6,00,000
Set up Costs (Cost Driver rate - ₹ 12,000)) (or 20:44)	12,000 × 20 = 2,40,000	12,000 × 44 = 5,28,000
Purchase related Costs (Cost Driver rate - ₹ 1250) (or 160:384)	1,250 × 160 = 2,00,000	1,250 × 384 = 4,80,000
Total overhead cost	5,40,000	16,08,000

(ii) Overhead Cost per unit

Per unit Overhead cost	(₹)	(₹)
5,40,000 /5,000	108	
16,08,000/60,000		26.80

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455 EXAM **Old P3 May 2022** Q5(a) 10 marks old scheme

Star Limited manufacture three products using the same production methods. A conventional product costing system is being used currently. Details of the three products for a typical period are:

Product	Labour Hrs. per unit	Machine Hrs. per unit	Materials per Unit[1]	Volume in Units
AX	1.00	2.00	35	7,500
BX	0.90	1.50	25	12,500
CX	1.50	2.50	45	25,000

Direct Labour costs ₹ 20 per hour and production overheads are absorbed on a machine hour basis. The overhead absorption rate for the period is ₹ 30 per machine hour.

Management is considering using Activity Based Costing system to ascertain the cost of the products. Further analysis shows that the total production overheads can be divided as follows:

Particulars	%
Cost relating to set-ups	40
Cost relating to machinery	10
Cost relating to material handling	30
Costs relating to inspection	20
Total production overhead	100

The following activity volumes are associated with the product line for the period as a whole. Total activities for the period:

Product	No. of set-ups	No. of movements of Materials	No. of inspections
AX	350	200	200
BX	450	280	400
CX	740	675	900
Total	1,540	1,155	1,500

Required:

(i) Calculate the cost per unit for each product using the conventional method.

(ii) Calculate the cost per unit for each product using activity based costing method.

[1] Material cost per unit

ICAI'S ANSWER

(i) Statement showing "Cost per unit" using "conventional method"

Particulars of Costs	AX (₹)	BX (₹)	CX (₹)
Direct Materials	35	25	45
Direct Labour	20	18	30
Production Overheads	60	45	75
Cost per unit	115	88	150

(ii) Statement Showing "Cost per unit using "Activity Based Costing"

Products	AX	BX	CX
Production (units)	7,500	12,500	25,000
	(₹)	(₹)	(₹)

Products	AX	BX	CX
Direct Materials	2,62,500	3,12,500	11,25,000
Direct Labour	1,50,000	2,25,000	7,50,000
Machine Related Costs	45,000	56,250	1,87,500
Setup Costs	**2,62,500**	**3,37,500**	**5,55,000**
Material handling Cost	**1,50,000**	**2,10,000**	**5,06,250**
Inspection Costs	**77,000**	**1,54,000**	**3,46,500**
Total Costs	9,47,000	12,95,250	34,70,250
Cost *per unit* (Total Cost ÷ Units)	**126.267**	**103.62**	**138.81**

Working Notes:

Calculation of Total Machine hours

Particulars	AX	BX	CX
(A) Machine hours per unit	2	1.5	2.5
(B) Production (units)	7,500	12,500	25,000
(C) Total Machine hours (A× B)	15,000	18,750	62,500

Total Machine hours = 96,250

Total Production overheads = 96,250 × 30 = ₹ 28,87,500

Calculation of Cost Driver Rate

Cost Pool	%	Overheads (₹)	Cost Driver (Basis)	Cost Driver (Units)	Cost Driver Rate (₹)
Set up	40	11,55,000	No of set ups	1,540	750 per set up
Machine Operation	10	2,88,750	Machine hours	96,250	3 per machine hour
Material Handling	30	8,66,250	No of material movement	1,155	750 per material movement
Inspection	20	5,77,500	No of inspection	1,500	385 per inspection

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456 EXAM **Old P3 Dec 2021** Q3(b) 10 marks old scheme

A Drug Store is presently selling three types of drugs namely 'Drug A', 'Drug B' and 'Drug C'. Due to some constraints, it has decided to go for only one product line of drugs. It has provided the following data for

year 2020-21 for each product line:

	Drugs Types		
	A	**B**	**C**
Revenues (in ₹)	74,50,000	1,11,75,000	1,86,25,000
Cost of goods sold (in ₹)	41,44,500	68,16,750	1,20,63,750
Number of purchase orders placed (in nos.)	560	810	630
Number of deliveries received	950	1,000	850
Hours of shelf-stocking time	900	1,250	2,350
Units sold (in Nos.)	1,75,200	1,50,300	1,44,500

Following additional information is also provided:

Activity	Description of activity	Total Cost (₹)	Cost-allocation base
Drug Licence fee	Drug Licence fee	5,00,000	To be distributed in ratio 2:3:5 between A, B and C
Ordering	Placing of orders for purchases	8,30,000	2,000 purchase orders
Delivery	Physical delivery and receipt of foods	18,20,000	2,800 deliveries
Shelf stocking	Stocking of goods	32,40,000	4,500 hours of shelf-stocking time
Customer Support	Assistance provided to customers	28,20,000	4,70,000 units sold

You are required to:

(i) Calculate the operating income and operating income as a percentage (%) of revenue of each product line if:

(a) All the support costs (Other than cost of goods sold) are allocated in the ratio of cost of goods sold.

(b) All the support costs (Other than cost of goods sold) are allocated using activity-based costing system.

(ii) Give your opinion about choosing the product line on the basis of operating income as a percentage (%) of revenue of each product line under both the situations as above.

ICAI'S ANSWER

(i) (a) Statement of Operating income and Operating income as a percentage of revenues for each product line

(When support costs are allocated to product lines on the basis of cost of goods sold of each product)

	Drug A (₹)	Drug B (₹)	Drug C (₹)	Total (₹)
Revenues: (A)	74,50,000	1,11,75,000	1,86,25,000	3,72,50,000
Cost of Goods sold (COGS): (B)	41,44,500	68,16,750	1,20,63,750	2,30,25,000
Support cost (40% of COGS): (C) (Refer working notes)	16,57,800	27,26,700	48,25,500	92,10,000
Total cost: (D) = {(B) + (C)}	58,02,300	95,43,450	1,68,89,250	3,22,35,000
Operating income: E = {(A)-(D)}	16,47,700	16,31,550	17,35,750	50,15,000
Operating income as a % of revenues: (E/A) × 100)	22.12%	14.60%	9.32%	13.46%

Working notes:

1. Total support cost:

	(₹)
Drug Licence Fee	5,00,000
Ordering	8,30,000
Delivery	18,20,000
Shelf stocking	32,40,000
Customer support	28,20,000
Total support cost	92,10,000

2. Percentage of support cost to cost of goods sold (COGS):

$$= \frac{\text{Total support cost}}{\text{Total cost of goods sold}} \times 100$$

$$= \frac{₹92,10,000}{₹2,30,25,000} \times 100 = 40\%$$

3. Cost for each activity cost driver:

Activity	Total cost (₹)	Cost allocation base	Cost driver rate
(1)	**(2)**	**(3)**	**(4) = [(2) ÷ (3)]**
Ordering	8,30,000	2,000 purchase orders	₹ 415 per purchase order
Delivery	18,20,000	2,800 deliveries	₹ 650 per delivery
Shelf-stocking	32,40,000	4,500 hours	₹ 720 per stocking hour
Customer support	28,20,000	4,70,000 units sold	₹ 6 per unit sold

(b) Statement of Operating income and Operating income as a percentage of revenues for each product line

(When support costs are allocated to product lines using an activity-based costing system)

	Drug A (₹)	Drug B (₹)	Drug C (₹)	Total (₹)
Revenues: (A)	74,50,000	1,11,75,000	1,86,25,000	3,72,50,000
Cost & Goods sold	41,44,500	68,16,750	1,20,63,750	2,30,25,000
Drug Licence Fee	1,00,000	1,50,000	2,50,000	5,00,000
Ordering cost* (560:810:630)	2,32,400	3,36,150	2,61,450	8,30,000
Delivery cost* (950:1000:850)	6,17,500	6,50,000	5,52,500	18,20,000
Shelf stocking cost* (900:1250:2350)	6,48,000	9,00,000	16,92,000	32,40,000
Customer Support cost* (175200:150300:144500)	10,51,200	9,01,800	8,67,000	28,20,000
Total cost: (B)	67,93,600	97,54,700	1,56,86,700	3,22,35,000
Operating income C: {(A) - (B)}	6,56,400	14,20,300	29,38,300	50,15,000
Operating income as a % of revenues	8.81%	12.71%	15.78%	13.46%

* Refer to working note 3

(ii) Comparison on the basis of operating income as per the percentage (%) of revenue:

(a) When support costs are allocated to product lines on the basis of cost of goods sold of each product

	Drug A (₹)	Drug B (₹)	Drug C (₹)	Total (₹)
Operating income as a % of revenues	22.12%	14.60%	9.32%	13.46%

On comparing the operating income as a % of revenue of each product, Drug A is the most profitable product line, though its revenue is least but with highest units sold.

(b) When support costs are allocated to product lines using an activity-based costing system

	Drug A (₹)	Drug B (₹)	Drug C (₹)	Total (₹)
Operating income as a % of revenues	8.81%	12.71%	15.78%	13.46%

On comparing the operating income as a % of revenue of each product, Drug C is the most profitable product line, though its unit sold is least but with highest revenue.

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457 EXAM **Old P3 Jul 2021** Q3(b) 10 marks old scheme

PQR Ltd. is engaged in the production of three products P, Q and R. The company calculates Activity Cost

Rates on the basis of Cost Driver capacity which is provided as below:

Activity	Cost Driver	Cost Driver Capacity	Cost (₹)
Direct Labour hours	Labour hours	30,000 Labour hours	3,00,000
Production runs	No. of Production runs	600 Production runs	1,80,000
Quality Inspections	No. of Inspection	8000 Inspections	2,40,000

The consumption of activities during the period is as under:

Activity / Products	P	Q	R
Direct Labour hours	10,000	8,000	6,000
Production runs	200	180	160
Quality Inspection	3,000	2,500	1,500

You are required to:

(i) Compute the costs allocated to each Product from each Activity.

(ii) Calculate the cost of unused capacity for each Activity.

(iii) A potential customer has approached the company for supply of 12,000 units of a new product. 'S' to be delivered in lots of 1500 units per quarter. This will involve an initial design cost of ₹ 30,000 and per quarter production will involve the following:

Direct Material	₹ 18,000
Direct Labour hours	1,500 hours
No. of Production runs	15
No. of Quality Inspection	250

Prepare cost sheet segregating Direct and Indirect costs and compute the Sales value per quarter of product 'S' using ABC system considering a markup of 20% on cost.

ICAI'S ANSWER

(i) Statement of cost allocation to each product from each activity

	Product			
	P (₹)	**Q (₹)**	**R (₹)**	**Total (₹)**
Direct Labour hours (Refer to working note)	1,00,000 (10,000 Labour hours × ₹10)	80,000 (8,000 Labour hours × ₹10)	60,000 (6,000 Labour hours × ₹10)	2,40,000
Production runs (Refer to working note)	60,000 (200 Production runs × ₹ 300)	54,000 (180 Production runs × ₹ 300)	48,000 (160 Production runs × ₹ 300)	1,62,000
Quality Inspections (Refer to working note)	90,000 (3,000 Inspections × ₹30)	75,000 (2,500 Inspections × ₹ 30)	45,000 (1,500 Inspections × ₹ 30)	2,10,000

Working note:

Rate per unit of cost driver

Direct Labour hours	(₹ 3,00,000/30,000 Labour hours)	₹ 10 per Labour hour
Production runs	(₹ 1,80,000/600 Production runs)	₹ 300 per Production run
Quality Inspection	(₹ 2,40,000/8,000 Inspections)	₹ 30 per Inspection

(ii) Computation of cost of unused capacity for each activity

Particulars	(₹)
Direct Labour hours [(₹ 3,00,000 – ₹ 2,40,000) or (6,000 x ₹ 10)]	60,000
Production runs [(₹ 1,80,000 – ₹ 1,62,000) or (60 x ₹ 300)]	18,000
Quality Inspection [(₹ 2,40,000 – ₹ 2,10,000) or (1,000 x ₹ 30)]	30,000
Total cost of unused capacity	**1,08,000**

(iii) Cost sheet and Computation of Sales value per quarter of product 'S' using ABC system

Particulars	(₹)
1500 units of product 'S' to be delivered per quarter	
Initial design cost per quarter (₹ 30,000 / 8 quarters)	3,750
Direct Material Cost	18,000
Direct Labour Cost (1,500 Labour hours x ₹ 10)	15,000
Direct Costs (A)	**36,750**
Set up Cost (15 Production runs × ₹ 300)	4,500
Inspection Cost (250 Inspections × ₹ 30)	7,500

Particulars	(₹)
Indirect Costs (B)	**12,000**
Total Cost (A + B)	48,750
Add: Mark-up (20% on cost)	**9,750**
Sale Value	**58,500**
Selling Price per unit 'S' (₹ 58,500/1500 units)	**39**

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458 EXAM **Old P3 Jan 2021** Q4(b) 10 marks old scheme

ABC Ltd. manufactures three products X, Y and Z using the same plant and resources. It has given the following information for the year ended on 31st March, 2020:

	X	Y	Z
Production Quantity (units)	1200	1440	1968
Cost per unit:			
Direct Material (₹)	90	84	176
Direct Labour (₹)	18	20	30

Budgeted direct labour rate was ₹ 4 per hour and the production overheads, shown in table below, were absorbed to products using direct labour hour rate. Company followed Absorption Costing Method. However, the company is now considering adopting Activity Based Costing Method.

	Budgeted Overheads (₹)	Cost Driver	Remarks
Material Procurement	50,000	No. of orders	No. of orders was 25 units for each product.
Set-up	40,000	No. of production Runs	All the three products are produced in production runs of 48 units.
Quality Control	28,240	No. of Inspections	Done for each production run.
Maintenance	1,28,000	Maintenance hours	Total maintenance hours were 6,400 and was allocated in the ratio of 1:1:2 between X, Y & Z.

Required:

1. Calculate the total cost per unit of each product using the Absorption Costing Method.
2. Calculate the total cost per unit of each product using the Activity Based Costing Method.

ICAI'S ANSWER

1. Traditional Absorption Costing

	X	Y	Z	Total
(a) Quantity (units)	1,200	1,440	1,968	4608
(b) Direct labour per unit (₹)	18	20	30	-
(c) Direct labour hours (a × b)/₹ 4	5,400	7,200	14,760	27,360

Overhead rate per direct labour hour:

= Budgeted overheads ÷Budgeted labour hours

= (₹ 50,000 + ₹ 40,000 + ₹ 28,240 + ₹ 1,28,000) ÷ 27,360 hours

= ₹ 2,46,240 ÷ 27,360 hours

= ₹ 9 per direct labour hour

Unit Costs:

	X	Y	Z
Direct Costs:			
- Direct Labour (₹)	18.00	20.00	30.00
- Direct Material (₹)	90.00	84.00	176.00
Production Overhead: (₹)	40.50 (9 × 18/4)	45.00 (9 × 20/4)	67.50 (9 × 30/4)
Total cost per unit (₹)	**148.50**	**149.00**	**273.50**

2. Calculation of Cost-Driver level under Activity Based Costing

	X	Y	Z	Total
Quantity (units)	1,200	1,440	1,968	-
No. of orders (to be rounded off for fraction)	48 (1200 / 25)	58 (1440 / 25)	79 (1968 / 25)	185
No. of production runs	25 (1200 / 48)	30 (1440 / 48)	41 (1968 / 48)	96
No. of Inspections (done for each production run)	25	30	41	96
Maintenance hours	1,600	1,600	3,200	6400

Calculation of Cost-Driver rate

Activity	Budgeted Cost (₹) (a)	Cost-driver level (b)	Cost Driver rate (₹) (c) = (a) / (b)
Material procurement	50,000	185	270.27
Set-up	40,000	96	416.67
Quality control	28,240	96	294.17
Maintenance	1,28,000	6,400	20.00

Calculation of total cost of products using Activity Based Costing

Particulars	Product		
	X (₹)	**Y (₹)**	**Z (₹)**
Direct Labour	18.00	20.00	30.00
Direct Material	90.00	84.00	176.00
Prime Cost per unit (A)	108.00	104.00	206.00
Material procurement	10.81 [(48 × 270.27)/1200]	10.89 [(58 × 270.27)/1440]	10.85 [(79 × 270.27)/1968]
Set-up	8.68 [(25 × 416.67)/1200]	8.68 [(30 × 416.67)/ 1440]	8.68 [(41 × 416.67)/ 1968]
Quality control	6.13 [(25 × 294.17)/1200]	6.13 [(30 × 294.17)/ 1440]	6.13 [(41 × 294.17)/ 1968]
Maintenance	26.67 [(1,600 × 20)/1200]	22.22 [(1,600 × 20)/ 1440]	32.52 [(3,200 × 20)/ 1968]
Overhead Cost per unit (B)	52.29	47.92	58.18
Total Cost per unit (A + B)	**160.29**	**151.92**	**264.18**

Note: Question may also be solved assuming no. of orders for material procurement to be 25 for each product.

ICAI · Old P3 · Jan 2021 exam · question p. 15-16 · answer p. 18-20

459 EXAM **Old P3 Nov 2020** Q5(b) 6 marks old scheme

ABC Ltd. is engaged in production of three types of Fruit Juices:

Apple, Orange and Mixed Fruit.

The following cost data for the month of March 2020 are as under:

Particulars	Apple	Orange	Mixed Fruit
Units produced and sold	10,000	15,000	20,000
Material per unit (₹)	8	6	5
Direct Labour per unit (₹)	5	4	3
No. of Purchase Orders	34	32	14
No. of Deliveries	110	64	52
Shelf Stocking Hours	110	160	170

Overheads incurred by the company during the month are as under :

	(₹)
Ordering costs	64,000
Delivery costs	1,58,200
Shelf Stocking costs	87,560

Required:

(i) Calculate cost driver's rate.

(ii) Calculate total cost of each product using Activity Based Costing.

ICAI'S ANSWER

(i) Calculation Cost-Driver's rate

Activity	Overhead cost (₹)	Cost-driver level	Cost driver rate (₹)
	(A)	**(B)**	**(C) = (A)/(B)**
Ordering	64,000	34 + 32 + 14 = 80 no. of purchase orders	**800**
Delivery	1,58,200	110 + 64 + 52 = 226 no. of deliveries	**700**
Shelf stocking	87,560	110 + 160 + 170 = 440 shelf stocking hours	**199**

(ii) Calculation of total cost of products using Activity Based Costing

Particulars	Fruit Juices		
	Apple (₹)	**Orange (₹)**	**Mixed Fruit (₹)**
Material cost	80,000 (10,000 x ₹ 8)	90,000 (15,000 x ₹ 6)	1,00,000 (20,000 x ₹ 5)
Direct labour cost	50,000 (10,000 x ₹ 5)	60,000 (15,000 x ₹ 4)	60,000 (20,000 x ₹ 3)
Prime Cost (A)	1,30,000	1,50,000	1,60,000

Particulars	Fruit Juices		
Ordering cost	**27,200** (800 × 34)	**25,600** (800 × 32)	**11,200** (800 × 14)
Delivery cost	**77,000** (700 × 110)	**44,800** (700 × 64)	**36,400** (700 × 52)
Shelf stocking cost	**21,890** (199 × 110)	**31,840** (199 × 160)	**33,830** (199 × 170)
Overhead Cost (B)	1,26,090	1,02,240	81,430
Total Cost (A + B)	**2,56,090**	**2,52,240**	**2,41,430**

ICAI · Old P3 · Nov 2020 exam · question p. 19 · answer p. 21

460 EXAM **Old P3 Nov 2019** Q2(a) 10 marks old scheme

PQR Ltd has decided to analyse the profitability of its five new customers. It buys soft drink bottles in cases at ₹ 45 per case and sells them to retail customers at a list price of ₹ 54 per case. The data pertaining to five customers are given below:

Particulars	A	B	C	D	E
Number of Cases Sold	9,360	14,200	62,000	38,000	9,800
List Selling Price (₹)	54	54	54	54	54
Actual Selling Price (₹)	54	53.40	49	50.20	48.60
Number of Purchase Orders	30	50	60	50	60
Number of Customers visits	4	6	12	4	6
Number of Deliveries	20	60	120	80	40
Kilometers travelled per delivery	40	12	10	20	60
Number of expediate Deliveries	0	0	0	0	2

Its five activities and their cost drivers are:

Activity	Cost Driver
Order taking	₹ 200 per purchase order
Customer visits	₹ 300 per each visit
Deliveries	₹ 4.00 per delivery km travelled
Product Handling	₹ 2.00 per case sold
Expedited deliveries	₹ 100 per such delivery

You are required to :

(i) Compute the customer level operating income of each of five retail customers by using the Cost Driver rates.

(ii) Examine the results to give your comments on Customer 'D' in comparison with Customer 'C' and on Customer 'E' in comparison with Customer 'A'.

ICAI'S ANSWER

Working note:

Computation of revenues (at listed price), discount, cost of goods sold and customer level operating activities costs:

Particular	Customers				
	A	**B**	**C**	**D**	**E**
Cases sold: (a)	9,360	14,200	62,000	38,000	9,800
Revenues (at listed price) (₹): (b) {(a) × ₹ 54)}	5,05,440	7,66,800	33,48,000	20,52,000	5,29,200
Discount (₹): (c) {(a) × Discount per case}	-	8,520 (14,200 cases × ₹ 0.6)	3,10,000 (62,000 cases × ₹ 5)	1,44,400 (38,000 cases × ₹ 3.80)	52,920 (9,800 cases × ₹ 5.40)
Cost of goods sold (₹): (d) {(a) × ₹ 45}	4,21,200	6,39,000	27,90,000	17,10,000	4,41000
Customer level operating activities costs					
Order taking costs (₹): (No. of purchase × ₹ 200)	6,000	10,000	12,000	10,000	12,000
Customer visits costs (₹) (No. of customer visits × ₹ 300)	1,200	1,800	3,600	1,200	1,800
Delivery vehicles travel costs (₹) (Kms travelled by delivery vehicles × ₹ 4 per km.)	3,200	2,880	4,800	6,400	9,600
Product handling costs (₹) {(a) ×₹ 2}	18,720	28,400	1,24,000	76,000	19,600
Cost of expediting deliveries (₹) {No. of expedited deliveries × ₹ 100}	-	-	-	-	200
Total cost of customer level operating activities (₹)	29,120	43,080	1,44,400	93,600	43,200

(i) Computation of Customer level operating income

Particular	Customers				
	A (₹)	B (₹)	C (₹)	D (₹)	E (₹)
Revenues (At list price) (Refer to working note)	5,05,440	7,66,800	33,48,000	20,52,000	5,29,200
Less: Discount (Refer to working note)	-	8,520	3,10,000	1,44,400	52,920
Revenue (At actual price)	5,05,440	7,58,280	30,38,000	19,07,600	4,76,280
Less: Cost of goods sold (Refer to working note)	4,21,200	6,39,000	27,90,000	17,10,000	4,41000
Gross margin	84,240	1,19280	2,48,000	1,97,600	35,280
Less: Customer level operating activities costs (Refer to working note)	29,120	43,080	1,44,400	93,600	43,200
Customer level operating income	55,120	76,200	1,03,600	1,04,000	(7,920)

(ii) Comments

Customer D in comparison with Customer C: Operating income of Customer D is more than of Customer C, despite having only 61.29% (38,000 units) of the units volume sold in comparison to Customer C (62,000 units). Customer C receives a higher percent of discount i.e. 9.26% (₹ 5) while Customer D receive a discount of 7.04% (₹ 3.80). Though the gross margin of customer C (₹ 2,48,000) is more than Customer D (₹ 1,97,600) but total cost of customer level operating activities of C (₹ 1,44,400) is more in comparison to Customer D (₹ 93,600). As a result, operating income is more in case of Customer D.

Customer E in comparison with Customer A: Customer E is not profitable while Customer A is profitable. Customer E receives a discount of 10% (₹ 5.4) while Customer A doesn't receive any discount. Sales Volume of Customer A and E is almost same. However, total cost of customer level operating activities of E is far more (₹ 43,200) in comparison to Customer A (₹ 29,120). This has resulted in occurrence of loss in case of Customer E.

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461 RTP **Sep 2026** Case scenario · Q2(i) to Q2(v)

CASE SCENARIO

NutritionCrunch Pvt. Ltd. is a fast-growing company in the health food industry, specializing in the production of nutritious snack bars. The company manufactures three popular variants—Alpha, Beta, and Gamma—in a fully automated production facility.

Due to increasing competition and pressure on margins, management has become concerned that the existing conventional costing system, which allocates overheads simply based on units sold, may not be accurately reflecting the true cost and profitability of each product.

To improve cost accuracy, the company is considering the adoption of an Activity-Based Costing (ABC) system, which allocates overheads based on actual resource consumption such as inspection hours, machine usage, and special activities like packaging and advertising.

The following data has been compiled for analysis for one month:

Overheads (Per Month)

- Inspection Cost = ₹ 84,000
- Machine Repairs & Maintenance = ₹ 1,56,000
- Mold (Dye) Cost = ₹ 14,500
- Total Selling Overheads = ₹ 1,80,000

Product Data

Particulars	Alpha	Beta	Gamma
Prime Cost per unit (₹)	14	10	9
Selling Price per unit (₹)	22	16	13
Units per Production Run	2,440	3,020	3,220
Defective Units per Run	40	20	20
Inspection Hours per Run	2	5	3
Mold Cost per Run (₹)	300	400	350
Machine Hours per Run	25	15	35
Units Sold per Month	24,000	60,000	32,000

The following Additional Information is provided:

1. The company follows a just-in-time approach, meaning all non-defective units produced are sold during the same month. There is no opening or closing inventory.
2. Under the current system, all production and selling overheads are allocated purely based on the number of units sold, without considering the actual consumption of resources by each product.
3. Out of total selling overheads, ₹ 90,000 relates to advertising expenses, which are incurred only for Beta and Gamma, as these products face stiff competition in the market.
4. Alpha does not require advertising due to its established demand.
5. Product Beta requires special packaging to maintain its competitive position.
6. A cost of ₹ 60,000 per month is incurred exclusively for this purpose and is included in the total selling overheads.

As a production cum finance manager of the company, you are required to answer the following questions:

Q2(i)

What is the total combined gross margin of all the products?

A ₹ 6,12,000

B ₹ 6,48,500

C ₹ 6,30,000

D ₹ 6,68,600

Q2(ii)

What is the total prime cost excluding cost of defective for all the products?

- A ₹ 19,18,000
- B ₹ 12,16,000
- C ₹ 12,24,000
- D ₹ 10,10,000

Q2(iii)

Ranking of Products under Conventional System is:

- A Beta – I, Alpha – II, Gamma – III
- B Alpha – I, Beta – II, Gamma – III
- C Beta – I, Gamma – II, Alpha – III
- D Gamma – I, Beta – II, Alpha – III

Q2(iv)

What is the total net profit for all the products:

- A ₹ 2,34,100
- B ₹ 2,28,000
- C ₹ 2,04,000
- D ₹ 1,98,000

Q2(v)

Ranking of Products under ABC System is:

- A Alpha – I, Beta – II, Gamma – III
- B Beta – I, Alpha – II, Gamma – III
- C Alpha – I, Gamma – II, Beta – III
- D Gamma – I, Beta – II, Alpha – III

ICAI'S ANSWERS

Q2(i)

(D) ₹ 6,68,600

Q2(ii)

(C) ₹ 12,24,000

Statement Showing Total Gross Margin

Particulars	Alpha (₹)	Beta (₹)	Gamma (₹)	Total (₹)
Sales Units	24,000	60,000	32,000	1,16,000
Selling Price per unit	22	16	13	—
Sales Value (A)	5,28,000	9,60,000	4,16,000	19,04,000
Prime Cost per unit	14	10	9	—
Prime Cost (B)	3,36,000	6,00,000	2,88,000	12,24,000
Gross Margin (A – B)	**1,92,000**	**3,60,000**	**1,28,000**	**6,80,000**
Cost of Defective units	5,600	4,000	1,800	11,400
Net Gross Margin (B – C)	**1,86,400**	**3,56,000**	**1,26,200**	**6,68,600**

Q2(iii)

(A) Beta – I, Alpha – II, Gamma – III

Statement Showing Rank and Net Profit as per Conventional Accounting System

Particulars	Alpha	Beta	Gamma	Total
Sales Units/ Production (good units)	24,000	60,000	32,000	1,16,000
Gross Margin (₹) ...(A)	1,92,000	3,60,000	1,28,000	6,80,000
Cost of Defective units...(B)	5,600	4,000	1,800	11,400
Production Overheads (₹) (2,54,500/116 × 24/60/32)	52,655	1,31,638	70,207	2,54,500
Selling Overheads (₹) (1,80,000/116 × 24/60/32)	37,241	93,103	49,656	1,80,000
Sub-Total Overheads (₹) ...(C)	89,896	2,24,741	1,19,863	4,34,500
Net Profit (₹) ...(A – B - C)	**96,504**	**1,31,259**	**6,337**	**2,34,100**
Ranking	II	I	III	—

Q2(iv)

(A) ₹ 2,45,500

Q2(v)

(A) Alpha – I, Beta – II, Gamma – III

Workings

No.	Particulars	Alpha	Beta	Gamma	Total
(1)	Gross Production (Units / Run)	2,440	3,020	3,220	—
(2)	Defectives / Run	40	20	20	—
(3)	Good Units / Run	2,400	3,000	3,200	—

No.	Particulars	Alpha	Beta	Gamma	Total
(4)	Sales Units	24,000	60,000	32,000	—
(5)	No. of Runs	10	20	10	—
(6)	Gross Production 1 × 5)	24,400	60,400	32,200	—
(7)	Prime Cost / Unit	14	10	9	—
(8)	Prime Cost (₹)	3,41,600	6,04,000	2,89,800	12,35,400
(9)	Inspection Hours / Run	2	5	3	—
(10)	Total Inspection Hours .... (9 × 5)	20	100	30	150
(11)	Machine Hours / Run	25	15	35	—
(12)	Total Machine Hours ..... (11 × 5)	250	300	350	900
(13)	Mold Cost / Run	300	400	350	—
(14)	Total Mold Cost ..... (13 × 5)	3,000	8,000	3,500	14,500

Statement Showing Production & Selling Overheads

Particulars	Alpha	Beta	Gamma	Total
Inspection Cost (84,000/150 × 20/100/30)	11,200	56,000	16,800	84,000
Machine Maintenance (1,56,000/900 × 250/300/350)	43,333	52,000	60,667	1,56,000
Mold Cost	3,000	8,000	3,500	14,500
Production Overheads	57,533	1,16,000	80,967	2,54,500
Advertisement (90,000/92 × 60/32)	—	58,696	31,304	90,000
Packing	—	60,000	—	60,000
Other Overheads(30,000/116 × 24/60/32)	6,207	15,517	8,276	30,000
Selling Overheads	6,207	1,34,213	39,580	1,80,000

Statement Showing Rank and Net Profit as per Activity Based Costing System

Particulars	Alpha	Beta	Gamma	Total
Sales Units	24,000	60,000	32,000	1,16,000
Gross Margin (₹) ...(A)	1,92,000	3,60,000	1,28,000	6,80,000
Cost of Defective units ....(B)	5,600	4,000	1,800	11,400
Production Overheads (₹)	57,533	1,16,000	80,967	2,54,500
Selling Overheads (₹)	6,207	1,34,213	39,580	1,80,000
Sub-Total Overheads (₹) ....(C)	63,740	2,50,213	1,20,547	4,34,500
Net Profit (₹) ....(A – B - C)	1,22,660	1,05,787	5,653	2,34,100
Ranking	I	II	III	—

Calculate Cost of Defective Units:

Product	Defectives per Run	Runs	Total Defectives	Per unit cost	Total cost (₹)
Alpha	40	10	400	14	5,600
Beta	20	20	400	10	4,000
Gamma	20	10	200	9	1,800

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462 RTP **May 2026** Case scenario · Q9(a)-(b) to Q9(c)

CASE SCENARIO

A XYZ company supplies and maintains a range of printers for business customers. The sales department is responsible for selling the printers and related consumables such as ink and paper. The service department provides after-sales support, including routine preventive maintenance and repair services in the event of equipment breakdown. Customers are charged a fee per print, and this rate varies depending on the type and capacity of the printer.

There is one customer per machine.

The company's existing cost system uses a single overhead rate, based on total sales revenue from print charges, to charge the cost of the service department's support activities to each type of printer.

The service manager has suggested that the print charge should more accurately reflect the costs involved. The company's accountant has decided to implement an activity-based costing (ABC) system and has obtained the following information about the support activities of the service department:

Activity	Cost driver	Overheads per annum ₹ 000
Customer account handling	No. of Customers	126
Planned maintenance Scheduling	No. of planned maintenance visit	480
Unplanned maintenance scheduling	No. of unplanned maintenance visit	147
Spare part procurement	No. of purchase orders	243
Other overheads	No. of machines	600
Total overheads		1,596

The following data have also been collected for each machine size:

	Desktop Inkjet Printers	Office Laser Printers	High-Volume Production colour Printers
Charge per copy	₹ 0.03	₹ 0.04	₹ 0.05
Average number of copies per year per machine	60,000	1,20,000	1,80,000
Number of machines	300	800	500
Planned maintenance visits per machine per year	4	6	12
Unplanned maintenance visits per machine per year	1	1	2
Total number of purchase orders per year	500	1,200	1,000
Cost of parts per maintenance visit	₹ 100	₹ 300	₹ 400
Labour cost per maintenance visit	₹ 60	₹ 80	₹ 100

Q9(a)-(b)

(a) Calculate the annual profit per machine for each of the three sizes of machine, using the current basis for charging the costs of support activities to machines.

(b) Calculate the annual profit per machine for each of the three sizes of machine using activity-based costing (ABC).

Q9(c)

Explain the potential benefits to the company of using an activity-based costing system.

ICAI'S ANSWERS

Q9(a)-(b)

(a) Calculation of single overhead rate based on sales revenue

	Desktop Inkjet Printers	Office Laser Printers	High-Volume Production colour Printers	Total
	₹	₹	₹	₹
Sales revenue	₹ 1,800 × 300 = ₹ 5,40,000	₹ 4,800 × 800 = ₹ 38,40,000	₹ 9,000 × 500 = ₹ 45,00,000	₹ 88,80,000
Overheads				₹ 15,96,000
Overheads/ Sales revenue				18%
Cost per machine	₹ 1,800 × 18% = ₹ 324	4,800 × 18% = ₹ 864	9,000 × 18% = ₹ 1,620	

(b) Profit per machine

	Desktop Inkjet Printers	Office Laser Printers	High-Volume Production colour Printers
	₹	₹	₹
Copy charge per machine	(60,000 × ₹ 0.03) 1,800	(1,20,000 × 0.04) 4,800	(180,000 × ₹ 0.05) 9,000
Cost of parts per machine	(₹ 100 × 5) (500)	(₹ 300 × 7) (2,100)	(₹ 400 × 14) (5,600)
Labour cost per machine	(₹ 60 × 5) (300)	(₹ 80 × 7) (560)	(₹ 100 × 14) (1,400)
Overhead cost	(324)	(864)	(1,620)
Profit per machine	676	1,276	380

Cost driver rates

Activity	Cost Driver	Overheads ₹ 000	No. of cost drivers	Cost per driver ₹
Customer account handling	Number of customers	126	300 + 800 + 500 = 1,600	₹79 per customers*
Planned maintenance scheduling	Number of planned maintenance visits	480	(300 × 4) + (800 × 6) + (500 × 12) = 12,000	₹ 40 per planned maintenance visit

Activity	Cost Driver	Overheads ₹ 000	No. of cost drivers	Cost per driver ₹
Unplanned maintenance scheduling	Number of unplanned maintenance visits	147	(300 × 1) + (800 × 1) + (500 × 2) = 2,100	₹ 70 per unplanned maintenance visit
Spare part procurement	Number of purchase orders	243	(500 + 1,200 +1,000) = 2,700	₹ 90 per purchase order
Other overheads	Number of machines	600	(300 + 800 + 500) = 1,600	₹ 375 per machine

*** Round off**

Overhead cost per machine

	Desktop Inkjet Printers	Office Laser Printers	High-Volume Production colour Printers
Customer account handling	= ₹ 79	= ₹ 79	= ₹ 79
Planned maintenance scheduling	(₹ 40 × 4) = ₹ 160	(₹ 40 × 6) = ₹ 240	(₹ 40 × 12) = ₹480
Unplanned maintenance scheduling	(₹ 70 × 1) = ₹ 70	(₹ 70 × 1) = ₹ 70	(₹ 70 × 2) = ₹ 140
Spare part procurement	(₹ 90 × 500/300) = ₹ 150	(₹ 90 × 1,200/800) = ₹ 135	(₹ 90 × 1,000/500) = ₹ 180
Other overheads	₹ 375	₹ 375	₹ 375
Total overhead cost per machine	₹ 834	₹ 899	₹ 1,254

ABC profitability analysis

	Desktop Inkjet Printers	Office Laser Printers	High-Volume Production colour Printers
Copy charge per machine	1,800	4,800	9,000
Parts and labour per machine	(800)	(2,660)	(7,000)
Overhead cost per machine	(834)	(899)	(1,254)
Profit per machine using ABC	166	1,241	746

Q9(c)

The answer should include the following points:

- The generation of more accurate product/service costs resulting in improved decision-making;
- Improved cost management by focusing on activity cost management
- More accurate profitability analysis
- The implementation of ABC will enable the company to adopt activity-based budgeting

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463 RTP **Jan 2026** Q11

Edward Ltd. manufactures weighing machines of standard size and sells its products to two industrial customers namely MT Ltd. and KG Ltd. and to a dealer MG Bros. having shops in different cities. The maximum retail price per unit of weighing machine is ₹ 11,000 and per unit average cost of production is ₹ 5,500 (40% is general fixed overhead cost).

The Finance Officer has been asked to undertake a customer profitability analysis and calculate and compare the profit margin per customer (before deducting general fixed overhead) to know about the real customer profitability.

Following are the additional overhead information:

Delivery costs	₹ 200 per kilometer
Emergency delivery cost (in addition to delivery cost)	₹ 21,000 per delivery
Order processing cost	₹ 6,000 per order
Specific discount and sales commission	As per negotiation
Product Advertisement cost	Actual cost

The following data are available for each customer.

	MT Ltd.	KG Ltd.	MG Bros.
Sales (in units)	2,000	1,000	800
Total delivery kilometer travelled	1,000	800	900
No. of emergency delivery	2	1	0
No. of orders processed	4	2	8
Specific Discount (percentage of sales revenue)	25%	20%	15%
Sales commission (percentage of sales revenue)	15%	10%	5%
Advertisement costs (₹)	8,75,000	6,15,000	4,30,000

You are required to analyse the profitability for each customer, which customer is the most profitable.

ICAI'S ANSWER

Customer Profitability Statement

Particulars	MT Ltd.	KG Ltd.	MG Bros.
Sales (units)	2,000	1,000	800
	(₹)	(₹)	(₹)
Sales Revenue (A)	2,20,00,000	1,10,00,000	88,00,000
Less: Average Variable Cost (B) (₹ 5,500 × 60% = 3,300 p.u.)	66,00,000	33,00,000	26,40,000
Contribution [70%of Sales] ...(A)-(B)	1,54,00,000	77,00,000	61,60,000
Less: Additional Overheads			
Delivery Cost (No. of K.M. × ₹ 200)	2,00,000	1,60,000	1,80,000
Emergency Delivery Cost (Emergency Delivery × ₹ 21,000)	42,000	21,000	-
Order Processing Cost (No. of Orders × ₹ 6,000)	24,000	12,000	48,000
Specific Discount	55,00,000	22,00,000	13,20,000
Sales Commission	33,00,000	11,00,000	4,40,000
Advertisement Cost	8,75,000	6,15,000	4,30,000
Profit *per customer**	54,59,000	35,92,000	37,42,000
Profit Margin *per customer** (%)	24.81%	32.65%	42.52%
Rank	III	II	I

***Before deducting general fixed overhead cost**

The contribution margin is 70% for each customer but when the other overheads costs per customer is included in the above profitability statement the profitability of the three customers become different. MG Ltd. is the most profitable customer.

ICAI · P4 · RTP Jan 2026 · question p. 14-15 · answer p. 36-37

464 RTP **Sep 2025** Case scenario · Q2(i) to Q2(v)

CASE SCENARIO

With the rise in carbon dioxide, a greenhouse gas, release in the environment, when fossil fuels are burned, Earth's natural greenhouse effect is becoming too weak causing global warming.

To contribute towards global environment for the betterment, various cars and scooters manufacturing companies are shifting their production towards electric vehicles (EVs) manufacturing. Companies are heavily investing in research, development and production of EVs.

Olay Ltd. is also one the companies belonging to scooter manufacturing industry. Watching its rivalries shifting their production to EVs, the management of Olay Ltd. decided to take the advantage of this open opportunity.

It had been only 4 years since the company started its production of EVs, but last year, the management of the company also decided to expand its product line to 3 variants of the scooter, viz., Olay EV Max, Olay EV Ultra and Olay EV Pro.

The following information is provided for the current year from the books of Olay Ltd.:

	Olay EV Max	Olay EV Ultra	Olay EV Pro
Average revenue per unit (₹)	84,975	1,15,500	2,17,800
Average cost of goods sold per unit (₹)	82,500	1,10,000	1,98,000

Last year, when the company initially expanded its product line, the sales were not much noticeable comparative to its earlier category of scooters. The company could only sold 528, 330 and 110 units of Olay EV Max, Olay EV Ultra and Olay EV Pro respectively against the order received of 616, 396 and 165 units respectively. However, with the increasing awareness about the EVs, more people started buying EVs and, during the current year, the company's order and sales jumped to five times the last year.

For earlier years, Olay Ltd. used gross margin percentage method to evaluate the relative profitability for all of its EVs.

However, from current year, company plans to use activity based costing for analysing the profitability.

The Activity analysis of Olay Ltd. is as under:

Activity Area	Cost-allocation base
Customer purchase order processing	Purchase orders by customers
Line-item ordering	Line-items per purchase order
Store delivery	Unit sold
Cartons dispatched to stores	Cartons dispatched to a store per Unit
Shelf-stocking at customer store	Hours of shelf-stocking

All the support costs for the current year amounts to ₹ 66,23,760. These support costs are assigned to all the activity areas. The cost in each area and the quantity of the cost allocation basis used in that area are as follows:

Activity Area	Total costs (₹)	Total Units of Cost-allocation base
Line-item ordering	14,04,480	66,110 line items
Cartons dispatched to store	16,72,000	3,39,240 cartons
Shelf-stocking at customer store	2,25,280	2,904 hours

The Customer purchase order processing costs ₹ 17,60,000 along with the store delivery cost of ₹ 15,62,000.

Some of the other information is also provided below:

	Olay EV Max	Olay EV Ultra	Olay EV Pro
Average number of line items per order	10	12	14
Average number of cartons shipped per store unit	16	80	300
Average number of hours of shelf-stocking per store delivery	0.1	0.6	3

The company wants you to FIGURE OUT the following to ascertain which of the product line is more profitable:

Q2(i)

For the current year, how much is the order received and the units sold for Olay EV Max, Olay EV Ultra and Olay EV Pro respectively?

A. Order received- 616, 396 and 165 units; sold- 528, 330 and 110 units of Olay EV Max, Olay EV Ultra and Olay EV Pro respectively.

B. Order received- 3,696, 2,376 and 990 units; sold- 3,168, 1,980 and 660 units of Olay EV Max, Olay EV Ultra and Olay EV Pro respectively.

C. Order received- 528, 330 and 110 units; sold- 616, 396 and 165 units of Olay EV Max, Olay EV Ultra and Olay EV Pro respectively.

D. Order received- 3,080, 1,980 and 825 units; sold- 2,640, 1,650 and 550 units of Olay EV Max, Olay EV Ultra and Olay EV Pro respectively.

Q2(ii)

The total gross-margin percentage and the operating income percentage, for the current year would be:

A. Gross-margin - 3.72% and Operating income - 4.96%

B. Gross-margin - 4.96% and Operating income - 3.72%

C. Gross-margin - 4.96% and Operating income - 4.96%

D. Gross-margin - 3.72% and Operating income - 3.72%

Q2(iii)

The cost driver rate relating to all the five activity areas would be:

A. Customer purchase order processing- ₹ 363.64 per order, Line item ordering- ₹ 21.24 per line item order, Store delivery- ₹ 265.42 per unit sold, Cartons dispatched- ₹ 4.93 per dispatch and Shelf-stocking at customer store- ₹ 77.58 per hour.

B. Customer purchase order processing- ₹ 299.07 per order, Line item ordering- ₹ 21.24 per line item order, Store delivery- ₹ 322.73 per unit sold, Cartons dispatched- ₹ 4.93 per dispatch and Shelf-stocking at customer store- ₹ 77.58 per hour.

C. Customer purchase order processing- ₹ 299.07 per order, Line item ordering- ₹ 77.58 per line item order, Store delivery- ₹ 322.73 per unit sold, Cartons dispatched- ₹ 4.93 per dispatch and Shelf-stocking at customer store- ₹ 21.24 per hour.

D. Customer purchase order processing- ₹ 322.73 per order, Line item ordering- ₹ 21.24 per line item order, Store delivery- ₹ 299.07 per unit sold, Cartons dispatched- ₹ 4.93 per dispatch and Shelf-stocking at customer store- ₹ 77.58 per hour.

Q2(iv)

The operating cost of the individual product line, as per the method proposed for the current year w.r.t. profitability analysis, would be:

- A Olay EV Max- ₹ 16,11,013, Olay EV Ultra- ₹ 23,56,890 and Olay EV Pro- ₹ 26,56,059
- B Olay EV Max- ₹ 23,56,890, Olay EV Ultra- ₹ 26,56,059 and Olay EV Pro- ₹ 16,11,013
- C Olay EV Max- ₹ 26,56,059, Olay EV Ultra- ₹ 16,11,013 and Olay EV Pro- ₹ 23,56,890
- D Olay EV Max- ₹ 26,56,059, Olay EV Ultra- ₹ 23,56,890 and Olay EV Pro- ₹ 16,11,013

Q2(v)

Operating income as a percentage of revenues of each product line, namely Olay EV Max, Olay EV Ultra and Olay EV Pro, when all the support costs are allocated using an activity-based costing system would be:

- A Olay EV Max- 1.73%, Olay EV Ultra- 3.53% and Olay EV Pro- 7.75%
- B Olay EV Max- 1.18%, Olay EV Ultra- 1.24% and Olay EV Pro- 1.34%
- C Olay EV Max- 2.91%, Olay EV Ultra- 4.76% and Olay EV Pro- 9.09%
- D Olay EV Max- 1.78%, Olay EV Ultra- 3.70% and Olay EV Pro- 8.52%

ICAI'S ANSWERS

Q2(i)

Answer key: **(D)**

(d)

Particulars	Olay EV Max (units)	Olay EV Ultra (units)	Olay EV Pro (units)	Total (units)
Previous year order received	616	396	165	1177
Current Year order (5 times the last year)	**3,080**	**1,980**	**825**	**5,885**
Previous year sales	528	330	110	968
Current Year sales (5 times the last year)	**2,640**	**1,650**	**550**	**4,840**

Q2(ii)

(b) Statement of gross margin percentage and operating income percentage

Particulars	Olay EV Max	Olay EV Ultra	Olay EV Pro	Total
Revenues (A) (₹)	22,43,34,000 (2,640 × ₹ 84,975)	19,05,75,000 (1,650 × ₹ 1,15,500)	11,97,90,000 (550 × ₹ 2,17,800)	53,46,99,000
Less: Cost of goods sold (B) (₹)	21,78,00,000 (2,640 × ₹ 82,500)	18,15,00,000 (1,650 × ₹ 1,10,000)	10,89,00,000 (550 × ₹ 1,98,000)	50,82,00,000
Gross Margin (A - B) (₹)	65,34,000	90,75,000	1,08,90,000	2,64,99,000
Less: Operating costs (₹)				66,23,760

Particulars	Olay EV Max	Olay EV Ultra	Olay EV Pro	Total
Operating income (₹)				1,98,75,240
Gross Margin %				4.96%
Operating income %				3.72%

Q2(iii)

Answer key: **(B)**

(b) Computation of cost driver rate relating to all the activity areas

Particulars	(₹)
Customer purchase order processing (₹ 17,60,000/ 5,885 orders)	299.07 per order
Line item ordering (₹ 14,04,480/ 66,110 line items)	21.24 per line item order
Store delivery (₹ 15,62,000/ 4,840 unit sold)	322.73 per unit sold
Cartons dispatched (₹ 16,72,000/ 3,39,240 dispatches)	4.93 per dispatch
Shelf-stocking at customer store (₹ 2,25,280/ 2,904 hours)	77.58 per hour

Q2(iv)

Answer key: **(D)**

(d) Computation of operating cost

	Olay EV Max (₹)	Olay EV Ultra (₹)	Olay EV Pro (₹)	Total (₹)
Customer purchase order processing	9,21,136 (₹ 299.07 × 3,080 orders)	5,92,159 (₹ 299.07 × 1,980 orders)	2,46,733 (₹ 299.07 × 825 orders)	17,60,027
Line item ordering	6,54,192 (₹ 21.24 × 10 x 3,080 orders)	5,04,662 (₹ 21.24 × 12 × 1,980 orders)	2,45,322 (₹ 21.24 × 14 × 825 orders)	14,04,176
Store delivery	8,52,007 (₹ 322.73 × 2,640 unit sold)	5,32,505 (₹ 322.73 × 1,650 unit sold)	1,77,502 (₹ 322.73 × 550 unit sold)	15,62,013
Cartons dispatched	2,08,243 (₹ 4.93 × 16 cartons × 2,640 units)	6,50,760 (₹ 4.93 × 80 cartons × 1,650 units)	8,13,450 (₹ 4.93 × 300 cartons × 550 units)	16,72,453

	Olay EV Max (₹)	Olay EV Ultra (₹)	Olay EV Pro (₹)	Total (₹)
Shelf stocking	20,481 (₹ 77.58 × 2,640 deliveries × 0.1 Av. hrs.)	76,804 (₹ 77.58 × 1,650 deliveries × 0.6 Av. hrs)	1,28,007 (₹ 77.58 × 550 deliveries × 3 Av. hrs)	2,25,292
Operating cost	**26,56,059**	**23,56,890**	**16,11,013**	**66,23,962***

*Difference due to rounding off.

Q2(v)

(a) Operating Income Statement

(using the Activity based Costing system)

	Olay EV Max	Olay EV Ultra	Olay EV Pro
Revenues (₹) (A)	22,43,34,000 (2,640 × ₹ 84,975)	19,05,75,000 (1,650 × ₹ 1,15,500)	11,97,90,000 (550 × ₹ 2,17,800)
Less: Cost of goods sold (₹) (B)	21,78,00,000 (2,640 × ₹ 82,500)	18,15,00,000 (1,650 × ₹ 1,10,000)	10,89,00,000 (550 × ₹ 1,98,000)
Gross Margin (₹) (C) (A - B)	65,34,000	90,75,000	1,08,90,000
Operating cost (₹) (D) (Refer to (iv) part of the answer)	26,56,059	23,56,890	16,11,013
Operating income (₹) (E) (C–D)	38,77,941	67,18,110	92,78,987
Operating income (in %) (Operating income/ Revenue) × 100	**1.73%**	**3.53 %**	**7.75 %**

Summary

(i) (d) Order received- 3,080, 1,980 and 825 units; sold- 2,640, 1,650 and 550 units of Olay EV Max, Olay EV Ultra and Olay EV Pro respectively.

(ii) (b) Gross-margin - 4.96% and Operating income - 3.72%

(iii) (b) Customer purchase order processing- ₹ 299.07 per order, Line item ordering- ₹ 21.24 per line item order, Store delivery- ₹ 322.73 per unit sold, Cartons dispatched- ₹ 4.93 per dispatch and Shelf-stocking at customer store- ₹ 77.58 per hour.

(iv) (d) Olay EV Max- ₹ 26,56,059, Olay EV Ultra- ₹ 23,56,890 and Olay EV Pro- ₹ 16,11,013

(v) (a) Olay EV Max- 1.73%, Olay EV Ultra- 3.53% and Olay EV Pro- 7.75%

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465 RTP **May 2025** Case scenario · Q2(i) to Q2(v)

CASE SCENARIO

The HomeMart is the latest trending brand offering home improvement appliances with broadest selection of products with highly competitive prices. The sale is increasing year by year with huge

multiples. Current year also the sales reached triple the last year. The reason being company having good customer support where it provides after sales assistance over phone per item sold. Though it costs only Re. 1 per item sold to the company, it enhanced to ₹ 49,15,200 last year making a huge impact on the total support cost.

All the company's appliances have been majorly categorised into three product lines namely Fancy fans, Home decors, Assembled furniture. During the current year, the company's revenue as generated is ₹ 3,80,88,000, ₹ 10,08,28,800 and ₹ 5,80,75,200 respectively. However, the cost of god sold is ₹ 2,88,00,000, ₹ 7,20,00,000 and ₹ 4,32,00,000 respectively.

In business, there's a saying "The packaging sells the product the first time, but what's inside sells the product a second time". Following the saying, the company has the policy of taking back the cartons of the products sold relating to Fancy fans to reduce the packaging cost. However, for smooth returning of cartons, the company has to incur certain carrier cost on its own which is ₹ 5,76,000 for the current year and allocating the same directly to the said product.

Some other information relating to each of the product lines is provided below:

	Fancy fans	Home decors	Assembled furniture
Items sold	12,09,600	1,05,98,400	29,37,600
Number of deliveries received	600	4,380	1,320
Number of purchase orders placed	720	1,680	720
Hours of shelf-stocking time	1,080	10,800	5,400

The company also provides the following basis of cost allocation:

Activity	Description of activity	Total Cost	Cost-allocation base
Delivery	Physical delivery and receipt of products	1,20,96,000	6,300 deliveries
Ordering	Placing of orders for purchases	74,88,000	3,120 purchase orders
Shelf stocking	Stocking of products in warehouse	82,94,400	17,280 hours of shelf-stocking time

The company wants you to FIGURE OUT the following to ascertain which of the product line is more profitable:

Q2(i)

The total support cost and its percentage to the cost of goods sold would be:

A ₹ 3,33,69,600 and 23.17%

B ₹ 4,32,00,000 and 30%

C ₹ 3,33,69,600 and 30%

D ₹ 4,32,00,000 and 23.17%

Q2(ii)

Operating income as a percentage of revenues of each product line, namely Fancy fans, Home decors, Assembled furniture, when all the support costs are allocated on the basis of cost of goods sold would

be:

A 6.87%, 12.05% and 8.38% respectively

B 12.05%, 6.87% and 8.38% respectively

C 1.70%, 7.17% and 3.30% respectively

D 7.17%, 3.30% and 1.70% respectively

Q2(iii)

The cost driver rate relating to Delivery, Ordering, Shelf stocking and Customer support would be:

A Delivery- ₹ 1,920 per delivery, Ordering- ₹ 2,400 per purchase order, Shelf stocking- ₹ 480 per stocking hour and Customer support- Re. 1 per item sold

B Delivery- ₹ 2,400 per delivery, Ordering- ₹ 1,920 per purchase order, Shelf stocking- ₹ 480 per stocking hour and Customer support- Re. 1 per item sold

C Delivery- ₹ 1,920 per delivery, Ordering- ₹ 2,400 per purchase order, Shelf stocking- ₹ 480 per stocking hour and Customer support- ₹ 3 per item sold

D Delivery- ₹ 480 per delivery, Ordering- ₹ 2,400 per purchase order, Shelf stocking- ₹1,920 per stocking hour and Customer support- ₹ 3 per item sold

Q2(iv)

Operating income of each product line, namely Fancy fans, Home decors, Assembled furniture, when all the support costs are allocated using an activity-based costing system would be:

A ₹ 16,84,800, ₹ -2,05,92,000 and ₹ -7,92,000 respectively

B ₹ 3,64,03,200, ₹ 12,14,20,800 and ₹ 5,88,67,200 respectively

C ₹ 92,88,000, ₹ 2,88,28,800 and ₹ 1,48,75,200 respectively

D ₹ 41,04,000, ₹ 6,04,800 and ₹ 50,83,200 respectively

Q2(v)

Operating income as a percentage of revenues of each product line, namely Fancy fans, Home decors, Assembled furniture, when all the support costs are allocated using an activity-based costing system would be:

A 4.42%, -20.42% and -1.36% respectively

B 10.78%, 0.60% and 8.75% respectively

C 24.39%, 28.59% and 25.61% respectively

D 4.39%, 8.59% and -1.36% respectively

ICAI'S ANSWERS

Q2(i)

(B)

Total support cost	(₹)
Cartons returned	5,76,000
Delivery	1,20,96,000
Ordering	74,88,000

Total support cost	(₹)
Shelf stocking	82,94,400
Customer support (₹ 49,15,200 × 3)	1,47,45,600
Total support cost	4,32,00,000
	(₹)
Fancy fans	2,88,00,000
Home decors	7,20,00,000
Assembled furniture	4,32,00,000
Total cost of goods sold (COGS)	14,40,00,000

Percentage of support cost to the cost of goods sold (COGS):

$$= \frac{\text{Total support cost}}{\text{Total cost of goods sold (COGS)}} \text{ x } 100$$

$$= \frac{4,32,00,000}{14,40,00,000} \text{ x } 100 = 30\%$$

Q2(ii)

(C)

Particulars	Fancy Fans (₹)	Home decors (₹)	Assembled furniture (₹)
Revenue: (A)	3,80,88,000	10,08,28,800	5,80,75,200
Cost of Goods sold (COGS): (B)	2,88,00,000	7,20,00,000	4,32,00,000
Support cost (30% of COGS): (C) (as calculated in i. above)	86,40,000	2,16,00,000	1,29,60,000
Total cost: (D) = {(B) + (C)}	3,74,40,000	9,36,00,000	5,61,60,000
Operating income: E= {(A)-(D)}	6,48,000	72,28,800	19,15,200
Operating income as a percentage of revenues: (E/A) × 100)	**1.70%**	**7.17%**	**3.30%**

Q2(iii)

(A)

Activity (1)	Total cost (₹) (2)	Cost allocation base (3)	Cost driver rate (4) = [(2) ÷ (3)]
Delivery	1,20,96,000	6,300 deliveries	₹ 1,920 per delivery
Ordering	74,88,000	3,120 purchase orders	₹ 2,400 per purchase order
Shelf-stocking	82,94,400	17,280 hours of shelf-stocking time	₹ 480 per stocking hour
Customer support	1,47,45,600	1,47,45,600 items sold	Re. 1 per item sold (given)

Q2(iv)

(D)

	Fancy Fans (₹)	Home decors (₹)	Assembled furniture (₹)
Revenues: (A)	3,80,88,000	10,08,28,800	5,80,75,200
Cost & Goods sold	2,88,00,000	7,20,00,000	4,32,00,000
Carton return costs (Directly attributable to Fancy fans)	5,76,000	0	0
Delivery cost (₹ 1,920 per delivery)	11,52,000 (600 x ₹ 1,920)	84,09,600 (4,380 x ₹ 1,920)	25,34,400 (1,320 x ₹ 1,920)
Ordering cost (₹ 2,400 per purchase order)	17,28,000 (720 x ₹ 2,400)	40,32,000 (1,680 x ₹ 2,400)	17,28,000 (720 x ₹ 2,400)
Shelf stocking cost (₹ 480 per stocking hour)	5,18,400 (1,080 x ₹ 480)	51,84,000 (10,800 x ₹ 480)	25,92,000 (5,400 x ₹ 480)
Customer Support cost (₹ 1 per item sold)	12,09,600 (12,09,600 x ₹ 1)	1,05,98,400 (1,05,98,400 x ₹ 1)	29,37,600 (29,37,600 x ₹ 1)
Total Cost: (B)	3,39,84,000	10,02,24,000	5,29,92,000
Operating income: (C) = (A) - (B)	**41,04,000**	**6,04,800**	**50,83,200**

Q2(v)

(B)

	Fancy Fans (₹)	Home decors (₹)	Assembled furniture (₹)
Operating income (from iv. Above) (A)	41,04,000	6,04,800	50,83,200
Revenues (B)	3,80,88,000	10,08,28,800	5,80,75,200
Operating income as a percentage of revenues: (A/B) × 100)	**10.78%**	**0.60%**	**8.75%**

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SOFTHUG is a global brand created by Green-lush Ltd. The company manufactures three range of beauty soaps i.e. SOFTHUG- Gold, SOFTHUG- Pearl, and SOFTHUG- Diamond. The budgeted costs and production for the month of May, 2024 are as follows:

	SOFTHUG- Gold		SOFTHUG- Pearl		SOFTHUG- Diamond	
Production of soaps (Units)	4,000		3,000		2,000	
Resources per Unit:	**Qty**	**Rate**	**Qty**	**Rate**	**Qty**	**Rate**
- Essential Oils	60 ml	₹ 200/100 ml	55 ml	₹ 300/100 ml	65 ml	₹ 300/100 ml
- Cocoa Butter	20 g	₹ 200/100 g	20 g	₹ 200/100 g	20 g	₹ 200/100 g
- Filtered Water	30 ml	₹ 15/100 ml	30 ml	₹ 15/100 ml	30 ml	₹ 15/100 ml
- Chemicals	10 g	₹ 30/100 g	12 g	₹ 50/100 g	15 g	₹ 60/100 g
- Direct Labour	30 minutes	₹ 10/hour	40 minutes	₹ 10/hour	60 minutes	₹ 10 / hour

Green-lush Ltd. followed an Absorption Costing System and absorbed its production overheads, to its products using direct labour hour rate, which were budgeted at ₹ 1,98,000.

Now, Green-lush Ltd. is considering adopting an Activity Based Costing system. For this, additional information regarding budgeted overheads and their cost drivers is provided below:

Particulars	(₹)	Cost drivers
Forklifting cost	58,000	Weight of material lifted
Supervising cost	60,000	Direct labour hours
Utility cost	80,000	Number of Machine operations

The number of machine operators per unit of production are 5, 5, and 6 for SOFTHUG- Gold, SOFTHUG- Pearl, and SOFTHUG- Diamond respectively.

(Consider (i) Mass of 1 litre of Essential Oils and Filtered Water equivalent to 0.8 kg and 1 kg respectively (ii) Mass of output produced is equivalent to the mass of input materials taken together.)

You are required to:

(i) PREPARE a statement showing the unit costs and total costs of each product using the absorption costing method.

(ii) PREPARE a statement showing the product costs of each product using the ABC approach.

(iii) STATE what are the reasons for the different product costs under the two approaches?

ICAI'S ANSWER

(i) Traditional Absorption Costing

	SOFTHUG- Gold	SOFTHUG - Pearl	SOFTHUG - Diamond	Total
(a) Production of soaps (Units)	4,000	3,000	2,000	9,000
(b) Direct labour (minutes)	30	40	60	-
(c) Direct labour hours (a × b)/60 minutes	2,000	2,000	2,000	6,000

Overhead rate per direct labour hour:

= Budgeted overheads ÷ Budgeted labour hours

= ₹ 1,98,000 ÷ 6,000 hours

= ₹ 33 per direct labour hour

Unit Costs:

	SOFTHUG- Gold (₹)	SOFTHUG- Pearl (₹)	SOFTHUG- Diamond (₹)
Direct Costs:			
- Direct Labour	5.00 (10×30/60)	6.67 (10×40/60)	10.00 (10×60/60)
- Direct Material (Refer working note 1)	167.50	215.50	248.50
Production Overhead:	16.50 (33×30/60)	22.00 (33×40/60)	33.00 (33×60/60)
Total unit costs	189.00	244.17	291.50
Number of units	4,000	3,000	2,000
Total costs	**7,56,000**	**7,32,510**	**5,83,000**

Working note -1

Calculation of Direct material cost

	SOFTHUG - Gold (₹)	SOFTHUG - Pearl (₹)	SOFTHUG - Diamond (₹)
Essential oils	120.00 (200×60/100)	165.00 (300×55/100)	195.00 (300×65/100)
Cocoa Butter	40.00 (200×20/100)	40.00 (200×20/100)	40.00 (200×20/100)
Filtered water	4.50 (15×30/100)	4.50 (15×30/100)	4.50 (15×30/100)
Chemicals	3.00 (30×10/100)	6.00 (50×12/100)	9.00 (60×15/100)
Total costs	**167.50**	**215.50**	**248.50**

(ii) Activity Based Costing

	SOFTHUG- Gold	SOFTHUG- Pearl	SOFTHUG- Diamond	Total
Quantity (units)	4,000	3,000	2,000	-
Weight per unit (grams)	108 {(60 × 0.8) + 20 + 30 + 10}	106 {(55 × 0.8) + 20 + 30 + 12}	117 {(65 × 0.8) + 20 + 30 + 15}	-
Total weight (grams)	4,32,000	3,18,000	2,34,000	9,84,000
Direct labour (minutes)	30	40	60	-
Direct labour hours	2,000 (4,000×30/60)	2,000 (3,000×40/60)	2,000 (2,000×60/60)	6,000
Machine operations per unit	5	5	6	-
Total operations	20,000	15,000	12,000	47,000

Forklifting rate per gram = ₹ 58,000 ÷ 9,84,000 grams = ₹ 0.06 per gram

Supervising rate per direct labour hour = ₹ 60,000 ÷ 6,000 hours = ₹ 10 per labour hour

Utilities rate per machine operations = ₹ 80,000 ÷ 47,000 machine operations

= ₹ 1.70 per machine operations

Unit Costs under ABC:

	SOFTHUG - Gold (₹)	SOFTHUG - Pearl (₹)	SOFTHUG - Diamond (₹)
Direct Costs:			
- Direct Labour	5.00	6.67	10.00
- Direct material	167.50	215.50	248.50
Production Overheads:			
Forklifting cost	6.48 (0.06 × 108)	6.36 (0.06 × 106)	7.02 (0.06 × 117)
Supervising cost	5.00 (10×30/60)	6.67 (10×40/60)	10.00 (10×60/60)
Utilities	8.50 (1.70 × 5)	8.50 (1.70 × 5)	10.20 (1.70 × 6)
Total unit costs	192.48	243.70	285.72
Number of units	4,000	3,000	2,000
Total costs	**7,69,920**	**7,31,100**	**5,71,440**

(iii) **Comments:** The difference in the total costs under the two systems is due to the differences in the overheads borne by each of the products. The Activity Based Costs appear to be more precise.

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The sales department of A Limited is analysing the customer profitability for its Product Z. It has decided to analyse the profitability of its five new customers using activity-based costing method. It buys Product Z at ₹ 5,400 per unit and sells to retail customers at a listed price of ₹ 6,480 per unit. The data pertaining to five customers are:

	Customers				
	A	B	C	D	E
Units sold	4,500	6,000	9,500	7,500	12,750
Listed Selling Price	₹6,480	₹6,480	₹6,480	₹6,480	₹6,480
Actual Selling Price	₹6,480	₹6,372	₹5,940	₹6,264	₹5,832
Number of Purchase orders	15	25	30	25	30
Number of Customer visits	2	3	6	2	3
Number of deliveries	10	30	60	40	20
Kilometers travelled per delivery	20	6	5	10	30
Number of expedited deliveries	0	0	0	0	1

After a detailed analysis and computation, the following activities has been identified and respective cost has been calculated:

Activity	Cost Driver Rate
Order taking	₹4,500 per purchase order
Customer visits	₹ 3,600 per customer visit
Deliveries	₹ 7.50 per delivery Km travelled
Product handling	₹ 22.50 per case sold
Expedited deliveries	₹ 13,500 per expedited delivery

You are required to COMPUTE the customer-level operating income of each of five retail customers.

ICAI'S ANSWER

Working note:

1. Computation of revenues (at listed price), discount, cost of goods sold and customer level operating activities costs:

	Customers				
	A	B	C	D	E
Units sold: (a)	4,500	6,000	9,500	7,500	12,750

	Customers				
Revenues (at listed price) (₹): (b) {(a) ×₹6,480)}	2,91,60,000	3,88,80,000	6,15,60,000	4,86,00,000	8,26,20,000
Revenues (at listed price) (₹): (c) {(a) ×Actual selling price)}	2,91,60,000 (4,500 × 6,480)	3,82,32,000 (6,000 × 6,372)	5,64,30,000 (9,500 × 5,940)	4,69,80,000 (7,500 × 6,264)	7,43,58,000 (12,750 × 5,832)
Discount (₹) (d) {(b) – (c)}	0	6,48,000	51,30,000	16,20,000	82,62,000
Cost of goods sold (₹) : (e) {(a) x ₹5,400}	2,43,00,000	3,24,00,000	5,13,00,000	4,05,00,000	6,88,50,000
Customer level operating activities costs					
Order taking costs (₹): (No. of purchase orders × ₹ 4,500)	67,500	1,12,500	1,35,000	1,12,500	1,35,000
Customer visits costs (₹) (No. of customer visits x ₹ 3,600)	7,200	10,800	21,600	7,200	10,800
Delivery vehicles travel costs (₹) (Kms travelled by delivery vehicles x ₹ 7.50 per km.)	1,500	1,350	2,250	3,000	4,500
Product handling costs (₹) {(a) x ₹ 22.50}	1,01,250	1,35,000	2,13,750	1,68,750	2,86,875
Cost of expediting deliveries (₹) {No. of expedited deliveries x ₹ 13,500}	-	-	-	-	13,500
Total cost of customer level operating activities (₹)	1,77,450	2,59,650	3,72,600	2,91,450	4,50,675

Computation of Customer level operating income

	Customers				
	A	B	C	D	E
	(₹)	(₹)	(₹)	(₹)	(₹)
Revenues (At list price) (Refer to working note)	2,91,60,000	3,82,32,000	5,64,30,000	4,69,80,000	7,43,58,000
Less: Cost of goods sold (Refer to working note)	(2,43,00,000)	(3,24,00,000)	(5,13,00,000)	(4,05,00,000)	(6,88,50,000)
Gross margin	48,60,000	58,32,000	51,30,000	64,80,000	55,08,000
Less: Customer level operating activities costs (Refer to working note)	(1,77,450)	(2,59,650)	(3,72,600)	(2,91,450)	(4,50,675)
Customer level operating income	46,82,550	55,72,350	47,57,400	61,88,550	50,57,325

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468 RTP **Old P3 Nov 2023** Q4 old scheme

L Limited manufactures three products P, Q and R which are similar in nature and are usually produced in production runs of 100 units. Product P and R require both machine hours and assembly hours, whereas product Q requires only machine hours. The overheads incurred by the company during the first quarter are as under:

	₹
Machine Department expenses	18,48,000
Assembly Department expenses	6,72,000
Setup costs	90,000
Stores receiving cost	1,20,000
Order processing and dispatch	1,80,000
Inspect and Quality control cost	36,000

The data related to the three products during the period are as under:

	P	Q	R
Units produced and sold	15,000	12,000	18,000
Machine hours worked	30,000 hrs.	48,000 hrs.	54,000 hrs.
Assembly hours worked (direct labour hours)	15,000 hrs.	-	27,000 hrs.
Customers' orders executed (in numbers)	1,250	1,000	1,500
Number of requisitions raised on the stores	40	30	50

Prepare a statement showing details of overhead costs allocated to each product type using activity-based costing.

ICAI'S ANSWER

Calculation of "Activity Rate"

Cost Pool	Cost (₹) [A]	Cost Driver [B]	Cost Driver Rate (₹) [C] = [A]÷[B]
Machine Department Expenses	18,48,000	Machine Hours (1,32,000 hrs.)	14.00
Assembly Department Expenses	6,72,000	Assembly Hours (42,000 hrs.)	16.00
Setup Cost	90,000	No. of Production Runs (450*)	200.00
Stores Receiving Cost	1,20,000	No. of Requisitions Raised on the Stores (120)	1,000.00
Order Processing and Dispatch	1,80,000	No. of Customers Orders Executed (3,750)	48.00
Inspection and Quality Control Cost	36,000	No. of Production Runs (450*)	80.00
Total (₹)	29,46,000		

**Number of Production Run is 450 (150 + 120 + 180)*

Statement Showing "Overheads Allocation"

Particulars of Cost	Cost Driver	P	Q	R	Total
Machine Department Expenses	Machine Hours	4,20,000 (30,000 × ₹14)	6,72,000 (48,000 × ₹14)	7,56,000 (54,000 × ₹14)	18,48,000
Assembly Department Expenses	Assembly Hours	2,40,000 (15,000 × ₹16)	---	4,32,000 (27,000 × ₹16)	6,72,000

Particulars of Cost	Cost Driver	P	Q	R	Total
Setup Cost	No. of Production Runs	30,000 (150 × ₹200)	24,000 (120 × ₹200)	36,000 (180 × ₹200)	90,000
Stores Receiving Cost	No. of Requisitions Raised on the Stores	40,000 (40 × ₹1,000)	30,000 (30 × ₹1,000)	50,000 (50 × ₹1,000)	1,20,000
Order Processing and Dispatch	No. of Customers Orders Executed	60,000 (1,250 × ₹48)	48,000 (1,000 × ₹48)	72,000 (1,500 × ₹48)	1,80,000
Inspection and Quality Control Cost	No. of Production Runs	12,000 (150 × ₹80)	9,600 (120 × ₹80)	14,400 (180 × ₹80)	36,000
Overhead (₹)		8,02,000	7,83,600	13,60,400	29,46,000

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469 RTP **Old P3 May 2023** Q4 old scheme

Hygiene Care Ltd. is a manufacturer of a range of goods. The cost structure of its different products is as follows:

Particulars	Hand Wash	Detergent Powder	Dishwasher
Direct Materials (₹ / Pu)	150	120	120
Direct Labour @₹10/ hour (₹ / Pu)	45	60	75
Production Overheads (₹ / Pu)	40	50	40
Total Cost (₹ / Pu)	235	230	235
Quantity Produced (Units)	30,000	60,000	90,000

Hygiene Care Ltd. was absorbing overheads on the basis of direct labour hours. Management accountant has suggested that the company should introduce ABC system and has identified cost drivers and cost pools as follows:

Activity Cost Pool	Cost Driver	Associated Cost (₹)
Goods Receiving	Number of Dispatch Order	8,88,000
Inspecting and Testing costs	Number of Production Runs	26,82,000
Dispatching	Number of dispatch order	6,30,000
Storage Cost	Number of Batches of material	36,00,000

The following information is also supplied:

Details	Hand Wash	Detergent Powder	Dishwasher
Batches of material	720	780	900
Number of dispatch order	360	540	600
No. of Production Runs	1,500	2,100	2,400
Number of Dispatch Orders	600	900	1,000

Required:

CALCULATE activity-based production cost of all the three products.

ICAI'S ANSWER

1. The Total Production Overhead are 78,00,000

Items	Labour Hour	Overheads allocation on the basis of direct Labour Hour (₹)
Labour Hour Ratio	(4.5:6:7.5)	
Hand Wash	1,35,000	9,00,000
Detergent Powder	3,60,000	24,00,000
Dishwasher	6,75,000	45,00,000
Total	**11,70,000**	**78,00,000**

2. On the basis of ABC analysis this amount will be apportioned as follows:

Statement Showing "Activity Based Production Cost"

Activity Cost Pool	Cost Driver	Ratio	Total Amount (₹)	Hand Wash (₹)	Detergent Powder (₹)	Dishwasher (₹)
Goods Receiving	Dispatch order	06:09:10	8,88,000	2,13,120	3,19,680	3,55,200
Inspecting and Testing costs	Production Runs	05:07:08	26,82,000	6,70,500	9,38,700	10,72,800
Dispatching	Dispatch Order	06:09:10	6,30,000	1,51,200	2,26,800	2,52,000
Storage Cost	Batches of material	12:13:15	36,00,000	10,80,000	11,70,000	13,50,000
Total Activity Cost				**21,14,820**	**26,55,180**	**30,30,000**
Quantity Produces				30,000	60,000	90,000
Unit Cost (Overheads)				70.49	44.25	33.67

Activity Cost Pool	Cost Driver	Ratio	Total Amount (₹)	Hand Wash (₹)	Detergent Powder (₹)	Dishwasher (₹)
Add: Conversion Cost (Material + Labour)				195	180	195
Total				**265.49**	**224.25**	**228.67**

Note: This question can also be solved by using cost driver rate

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470 RTP **Old P3 Nov 2022 Q4** old scheme

The profit margin of BABY Hairclips Company were over 20% of sales producing BROWN and BLACK hairclips.

During the last year, GREEN hairclips had been introduced at 10% premium in selling price after the introduction of YELLOW hairclips earlier five years back at 10/3% premium. However, the manager of the company is disheartened with the sales figure for the current financial year as follows:

Traditional Income Statement (in ₹)

	Brown	Black	Yellow	Green	Total
Sales	1,50,00,000	1,20,00,000	27,90,000	3,30,000	3,01,20,000
Material Costs	50,00,000	40,00,000	9,36,000	1,10,000	1,00,46,000
Direct Labour	20,00,000	16,00,000	3,60,000	40,000	40,00,000
Overhead (3 times of direct labour)	60,00,000	48,00,000	10,80,000	1,20,000	1,20,00,000
Total Operating Income	20,00,000	16,00,000	4,14,000	60,000	40,74,000
Return on Sales (in %)	13.3%	13.3%	14.8%	18.2%	13.5%

It is a known fact that customers are ready to pay premium amount for YELLOW and GREEN hairclips for their attractiveness; and the percentage returns are also high on new products.

At present, all of the Plant's indirect expenses are allocated to the products at 3 times of the direct labour expenses. However, the manager is interested in allocating indirect expenses on the basis of activity cost to reveal real earner.

He provides support expenses category-wise as follows:

Support Expenses	(₹)
Indirect Labour	40,00,000
Labour Incentives	32,00,000
Computer Systems	20,00,000
Machinery depreciation	16,00,000
Machine maintenance	8,00,000
Energy for machinery	4,00,000
Total	1,20,00,000

He provides following **additional information** for accomplishment of his interest:

Incentives to be allocated @ 40% of labour expenses (both direct and indirect).

Indirect labours are involved mainly in three activities. About half of indirect labour is involved in handling production runs. Another 40% is required just for the physical changeover from one color hairclip to another because YELLOW hairclips require substantial labour for preparing the machine as compared to other colour hairclips. Remaining 10% of the time is spend for maintaining records of the products in four parts.

Another amount spent on computer system of ₹ 20,00,000 is for maintenance of documents relating to production runs and record keeping of the four products. In aggregate, approx.. 80% of the amount expend is involved in the production run activity and approx.. 20% is used to keep records of the products in four parts.

Other overhead expenses i.e. machinery depreciation, machine maintenance and energy for machinery are incurred to supply machine capacity to produce all the hairclips (practical capability of 20,000 hours).

Activity Cost Drivers:

Particulars	Brown	Black	Yellow	Green	Total
Sales Volume (units)	1,00,000	80,000	18,000	2,000	2,00,000
Selling Price (₹)	150	150	155	165	
Material cost (₹)	50	50	52	55	
Machine hours per unit (Hrs)	0.10	0.10	0.10	0.10	20,000
Production runs	100	100	76	24	300
Setup time per run (Hrs)	4	1	6	4	

You are required to –

(i) CALCULATE operating income and operating income as per percentage of sales using activity-based costing system.

(ii) STATE the reasons for different operating income under traditional income system and activity-based costing system.

ICAI'S ANSWER

(i) Calculation of operating income using Activity Based Costing

Calculation of Cost-Driver rate

Activity	Overhead cost (₹)	Allocation	Overhead cost (₹)	Cost-driver level	Cost driver rate (₹)
Indirect labour + 40% for incentives	56,00,000	50%	28,00,000	300 Production runs	9,333.33
		40%	22,40,000	1052* Setup hours	2,129.28
		10%	5,60,000	4 Number of parts	1,40,000
Computer Systems	20,00,000	80%	16,00,000	300 Production runs	5,333.33
		20%	4,00,000	4 Number of parts	1,00,000
Machinery depreciation	16,00,000	100%	16,00,000	20,000 Machine hours	80
Machine Maintenance	8,00,000	100%	8,00,000	20,000 Machine hours	40
Energy for Machinery	4,00,000	100%	4,00,000	20,000 Machine hours	20

* (100 x 4) + (100 x 1) + (76 x 6) + (24 x 4)

= (400 + 100 + 456 + 96)

= 1052 setup hours

Activity Based Costing

	Brown	Black	Red	Green	Total
Quantity (units)	1,00,000	80,000	18,000	2,000	2,00,000
	(₹)	(₹)	(₹)	(₹)	(₹)
Sales	1,50,00,000	1,20,00,000	27,90,000	3,30,000	3,01,20,000
Less: Material Costs	50,00,000	40,00,000	9,36,000	1,10,000	1,00,46,000
Less: Direct labour	20,00,000	16,00,000	3,60,000	40,000	40,00,000

	Brown	Black	Red	Green	Total
Less: 40% incentives on direct labour	8,00,000	6,40,000	1,44,000	16,000	16,00,000
(A)	**72,00,000**	**57,60,000**	**13,50,000**	**1,64,000**	**1,44,74,000**
Overheads					
Indirect labour + incentives					
- 50% based on Production runs	9,33,333 (9,333.33 x 100)	9,33,333 (9,333.33 x 100)	7,09,334 (9,333.33 x 76)	2,24,000 (9,333.33 x 24)	28,00,000
- 40% based on Setup hours	8,51,711 (2,129.28 x 400)	2,12,928 (2,129.28 x 100)	9,70,951 (2,129.28 x 456)	2,04,410 (2,129.28 x 96)	22,40,000
- 10% based on number of parts	1,40,000 (1,40,000 × 1)	1,40,000	1,40,000	1,40,000	5,60,000
Computer Systems					
- 80% based on Production runs	5,33,333 (5,333.33 x 100)	5,33,333 (5,333.33 x 100)	4,05,334 (5,333.33 x 76)	1,28,000 (5,333.33 x 24)	16,00,000
- 20% based on number of parts	1,00,000 (1,00,000 × 1)	1,00,000	1,00,000	1,00,000	4,00,000
Machinery depreciation	8,00,000 (80 x 0.1 x 1,00,000)	6,40,000 (80 x 0.1 x 80,000)	1,44,000 (80 x 0.1 x 18,000)	16,000 (80 x 0.1 × 2,000)	16,00,000
Machine Maintenance	4,00,000 (40 x 0.1 x 1,00,000)	3,20,000 (40 x 0.1 x 80,000)	72,000 (40 x 0.1 × 18,000)	8,000 (40 x 0.1 × 2,000)	8,00,000
Energy for Machinery	2,00,000 (20 x 0.1 x 1,00,000)	1,60,000 (20 x 0.1 × 80,000)	36,000 (20 x 0.1 × 18,000)	4,000 (20 x 0.1 × 2,000)	4,00,000
Total Overheads (B)	**39,58,377**	**30,39,594**	**25,77,619**	**8,24,410**	**1,04,00,000**
Operating Income (A-B)	**32,41,623**	**27,20,406**	**(12,27,619)**	**(6,60,410)**	**40,74,000**
Return on Sales (%)	**21.61**	**22.67**	**(44.00)**	**(200.12)**	**13.53**

(ii) The difference in the operating income under the two systems is due to the differences in the overheads borne by each of the products. The Activity Based Costs appear to be more accurate.

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471 RTP **Old P3 May 2022** Q4 old scheme

Also asked in Old P3 MTP Nov 2023 S-I Q2(b)

PCP Limited belongs to the apparel industry. It specializes in the distribution of fashionable garments. It buys from the industry and resells the same to the following two different supermarkets:

(i) Supermarket A dealing in Adults' garments (Age group 15 - 30)

(ii) Supermarket B dealing in Kids' garments (Age group 5 - 10)

The following data for the month of April in respect of PCP Limited has been reported:

	Supermarket A (₹)	Supermarket B (₹)
Average revenue per delivery	1,69,950	57,750
Average cost of goods sold per delivery	1,65,000	55,000
Number of deliveries	660	1,650

In the past, PCP Limited has used gross margin percentage to evaluate the relative profitability of its supermarket segments.

The company plans to use activity –based costing for analysing the profitability of its supermarket segments.

The April month's operating costs (other than cost of goods sold) of PCP Limited are ₹ 16,55,995. These operating costs are assigned to five activity areas. The cost in each area and Activity analysis including cost driver for the month of April are as follows:

Activity Area	Total costs (₹)	Cost Driver
Store delivery	3,90,500	Store deliveries
Cartons dispatched to store	4,15,250	Cartons dispatched to a store per delivery
Shelf-stocking at customer store	64,845	Hours of shelf-stocking
Line-item ordering	3,45,400	Line-items per purchase order
Customer purchase order processing	4,40,000	Purchase orders by customers

Other data for the month of April include the following:

	Supermarket A	Supermarket B
Total number of store deliveries	1,100	2,805
Average number of cartons shipped per store delivery	250	50
Average number of hours of shelf-stocking per store delivery	6	1.5
Average number of line items per order	14	12
Total number of orders	770	1,980

Required:

(i) COMPUTE gross-margin percentage for each of its supermarket segments and compute PCP Limited's operating income.

(ii) COMPUTE the operating income of each supermarket segments using the activity-based costing information.

ICAI'S ANSWER

(i)

PCP Limited's

Statement of operating income and gross margin percentage for each of its supermarket segments

Particulars	Supermarket A	Supermarket B	Total
Revenues: (₹)	11,21,67,000 (660 × ₹ 1,69,950)	9,52,87,500 (1,650 × ₹ 57,750)	20,74,54,500
Less: Cost of goods sold: (₹)	10,89,00,000 (660 × ₹ 1,65,000)	9,07,50,000 (1650 × ₹ 55,000)	19,96,50,000
Gross Margin: (₹)	**32,67,000**	**45,37,500**	**78,04,500**
Less: Other operating costs: (₹)			16,55,995
Operating income: (₹)			**61,48,505**
Gross Margin	**2.91%**	**4.76 %**	**3.76%**
Operating income %			**2.96%**

(ii) Operating Income Statement of each distribution channel in April (Using the Activity based Costing information)

	Supermarket A	Supermarket B
Gross margin (₹) : (A) (Refer to (i) part of the answer)	32,67,000	45,37,500
Operating cost (₹): (B) (Refer to working note)	6,55,600	10,00,395
Operating income (₹): (A–B)	**26,11,400**	**35,37,105**
Operating income (in %) (Operating income/Revenue) ×100	**2.33**	**3.71**

Working note:

Computation of rate per unit of the cost allocation base for each of the five activity areas for the month of April

	(₹)
Store delivery [₹ 3,90,500/ (1,100 + 2,805 store deliveries)]	100 per delivery
Cartons dispatched [₹ 4,15,250/ {(250×1,100) +(50×2,805)} carton dispatches]	1 per carton dispatch
Shelf-stocking at customer store (₹) [₹ 64,845/ {(6×1,100) + (1.5×2,805)} hours]	6 per hour
Line item ordering [₹ 3,45,400/ {(14×770) + (12×1,980)} line items]	10 per line item order
Customer purchase order processing [₹ 4,40,000/ (770 + 1,980 orders)]	160 per order

Computation of operating cost of each distribution channel:

	Supermarket A (₹)	Supermarket B (₹)
Store delivery	1,10,000 (₹ 100 × 1,100 deliveries)	2,80,500 (₹ 100 × 2,805 deliveries)
Cartons dispatched	2,75,000 (₹ 1× 250 cartons × 1,100 deliveries)	1,40,250 (₹ 1 × 50 cartons × 2,805 deliveries)
Shelf stocking	39,600 (₹ 6 × 1,100 deliveries × 6 Av. hrs.)	25,245 (₹ 6 × 2,805 deliveries × 1.5 Av. hrs)
Line item ordering	1,07,800 (₹ 10 × 14 line item x 770 orders)	2,37,600 (₹ 10 × 12 line item x 1,980 orders)
Customer purchase order processing	1,23,200 (₹ 160 × 770 orders)	3,16,800 (₹ 160 × 1,980 orders)
Operating cost	**6,55,600**	**10,00,395**

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472 RTP **Old P3 Dec 2021 Q4** old scheme

Family Store wants information about the profitability of individual product lines: Soft drinks, Fresh produce and Packaged food. Family store provides the following data for the year 2020-21 for each

product line:

	Soft drinks	Fresh produce	Packaged food
Revenues	₹ 39,67,500	₹ 1,05,03,000	₹ 60,49,500
Cost of goods sold	₹ 30,00,000	₹ 75,00,000	₹ 45,00,000
Cost of bottles returned	₹ 60,000	₹ 0	₹ 0
Number of purchase orders placed	360	840	360
Number of deliveries received	300	2,190	660
Hours of shelf-stocking time	540	5,400	2,700
Items sold	1,26,000	11,04,000	3,06,000

Family store also provides the following information for the year 2020-21:

Activity	Description of activity	Total Cost (₹)	Cost-allocation base
Bottles returns	Returning of empty bottles	60,000	Direct tracing to soft drink line
Ordering	Placing of orders for purchases	7,80,000	1,560 purchase orders
Delivery	Physical delivery and receipt of goods	12,60,000	3,150 deliveries
Shelf stocking	Stocking of goods on store shelves and on-going restocking	8,64,000	8,640 hours of shelf-stocking time
Customer Support	Assistance provided to customers including check-out	15,36,000	15,36,000 items sold

Required:

(i) Family store currently allocates support cost (all cost other than cost of goods sold) to product lines on the basis of cost of goods sold of each product line. CALCULATE the operating income and operating income as a % of revenues for each product line.

(ii) If Family Store allocates support costs (all costs other than cost of goods sold) to product lines using and activity-based costing system, CALCULATE the operating income and operating income as a % of revenues for each product line.

ICAI'S ANSWER

Working notes:

1. Total support cost:

	(₹)
Bottles returns	60,000
Ordering	7,80,000
Delivery	12,60,000
Shelf stocking	8,64,000
Customer support	15,36,000
Total support cost	45,00,000

2. Percentage of support cost to cost of goods sold (COGS):

$$= \frac{\text{Total support cost}}{\text{Total cost of goods sold}} \times 100$$

$$= \frac{₹\ 45,00,000}{₹\ 1,50,00,000} \times 100 = 30\%$$

3. Cost for each activity cost driver:

Activity (1)	Total cost (₹) (2)	Cost allocation base (3)	Cost driver rate (4) = [(2) ÷ (3)]
Ordering	7,80,000	1,560 purchase orders	₹ 500 per purchase order
Delivery	12,60,000	3,150 deliveries	₹ 400 per delivery
Shelf-stocking	8,64,000	8,640 hours	₹ 100 per stocking hour
Customer support	15,36,000	15,36,000 items sold	₹ 1 per item sold

(i) **Statement of Operating income and Operating income as a percentage of revenues for each product line**

(When support costs are allocated to product lines on the basis of cost of goods sold of each product)

	Soft Drinks (₹)	Fresh Produce (₹)	Packaged Foods (₹)	Total (₹)
Revenues: (A)	39,67,500	1,05,03,000	60,49,500	2,05,20,000
Cost of Goods sold (COGS): (B)	30,00,000	75,00,000	45,00,000	1,50,00,000
Support cost (30% of COGS): (C) (Refer working notes)	9,00,000	22,50,000	13,50,000	45,00,000
Total cost: (D) = {(B) + (C)}	39,00,000	97,50,000	58,50,000	1,95,00,000
Operating income: (E) = {(A)-(D)}	67,500	7,53,000	1,99,500	10,20,000
Operating income as a percentage of revenues: (F)= {(E)/(A) × 100}	1.70%	7.17%	3.30%	4.97%

(ii) **Statement of Operating income and Operating income as a percentage of revenues for each product line**

(When support costs are allocated to product lines using an activity-based costing system)

	Soft drinks (₹)	Fresh Produce (₹)	Packaged Food (₹)	Total (₹)
Revenues: (A)	39,67,500	1,05,03,000	60,49,500	2,05,20,000
Cost & Goods sold	30,00,000	75,00,000	45,00,000	1,50,00,000
Bottle return costs	60,000	0	0	60,000
Ordering cost* (360:840:360)	1,80,000	4,20,000	1,80,000	7,80,000
Delivery cost* (300:2,190:660)	1,20,000	8,76,000	2,64,000	12,60,000
Shelf stocking cost* (540:5,400:2,700)	54,000	5,40,000	2,70,000	8,64,000
Customer Support cost* (1,26,000:11,04,000:3,06,000)	1,26,000	11,04,000	3,06,000	15,36,000
Total cost: (B)	35,40,000	1,04,40,000	55,20,000	1,95,00,000
Operating income: (C) = {(A)- (B)}	4,27,500	63,000	5,29,500	10,20,000
Operating income as a % of revenues: (D) = {(C)/(A) × 100}	10.78%	0.60%	8.75%	4.97%

* Refer to working note 3

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473 RTP Old P3 Jul 2021 Q4 old scheme

The following budgeted information relates to N Ltd. for the year 2021:

	Products		
	X	**Y**	**Z**
Production and Sales (units)	1,00,000	80,000	60,000
	(₹)	**(₹)**	**(₹)**
Selling price per unit	90	180	140
Direct cost per unit	50	90	95
	Hours	**Hours**	**Hours**
Machine department (machine hours per unit)	3	4	5
Assembly department (direct labour hours per unit)	6	4	3

The estimated overhead expenses for the year 2021 will be as below:

Machine Department	₹ 73,60,000
Assembly Department	₹ 55,00,000

Overhead expenses are apportioned to the products on the following basis:

Machine Department	On the basis of machine hours
Assembly Department	On the basis of labour hours

After a detailed study of the activities the following cost pools and their respective cost drivers are found:

Cost Pool	Amount (₹)	Cost Driver	Quantity
Machining services	64,40,000	Machine hours	9,20,000 hours
Assembly services	44,00,000	Direct labour hours	11,00,000 hours
Set-up costs	9,00,000	Machine set-ups	9,000 set-ups
Order processing	7,20,000	Customer orders	7,200 orders
Purchasing	4,00,000	Purchase orders	800 orders

As per an estimate the activities will be used by the three products:

	Products		
	X	Y	Z
Machine set-ups	4,500	3,000	1,500
Customer orders	2,200	2,400	2,600
Purchase orders	300	350	150

You are required to PREPARE a product-wise profit statement using:

(i) Absorption costing method;

(ii) Activity-based method.

ICAI'S ANSWER

(i) Profit Statement using Absorption costing method:

	Particulars	Product			Total
		X	Y	Z	
A.	Sales Quantity	1,00,000	80,000	60,000	2,40,000
B.	Selling price per unit (₹)	90	180	140	
C.	Sales Value (₹) [A×B]	90,00,000	1,44,00,000	84,00,000	3,18,00,000
D.	Direct cost per unit (₹)	50	90	95	
E.	Direct Cost (₹) [A×D]	50,00,000	72,00,000	57,00,000	1,79,00,000
F.	Overheads:				
(i)	Machine department (₹) (Working note-1)	24,00,000	25,60,000	24,00,000	73,60,000
(ii)	Assembly department (₹) (Working note-1)	30,00,000	16,00,000	9,00,000	55,00,000
G.	Total Cost (₹) [E+F]	1,04,00,000	1,13,60,000	90,00,000	3,07,60,000
H.	Profit (C-G)	(14,00,000)	30,40,000	(6,00,000)	10,40,000

(ii) Profit Statement using Activity based costing (ABC) method:

	Particulars	Product			Total
		X	Y	Z	
A.	Sales Quantity	1,00,000	80,000	60,000	
B.	Selling price per unit (₹)	90	180	140	

	Particulars	Product			Total
C.	Sales Value (₹) [A×B]	90,00,000	1,44,00,000	84,00,000	3,18,00,000
D.	Direct cost per unit (₹)	50	90	95	
E.	Direct Cost (₹) [A×D]	50,00,000	72,00,000	57,00,000	1,79,00,000
F.	Overheads: (Refer working note-3)				
(i)	Machining services (₹)	21,00,000	22,40,000	21,00,000	64,40,000
(ii)	Assembly services (₹)	24,00,000	12,80,000	7,20,000	44,00,000
(iii)	Set-up costs (₹)	4,50,000	3,00,000	1,50,000	9,00,000
(iv)	Order processing (₹)	2,20,000	2,40,000	2,60,000	7,20,000
(v)	Purchasing (₹)	1,50,000	1,75,000	75,000	4,00,000
G.	Total Cost (₹) [E+F]	1,03,20,000	1,14,35,000	90,05,000	3,07,60,000
H.	Profit (₹) (C-G)	(13,20,000)	29,65,000	(6,05,000)	10,40,000

Working Notes:

1.

		Products			
		X	**Y**	**Z**	**Total**
A.	Production (units)	1,00,000	80,000	60,000	
B.	Machine hours per unit	3	4	5	
C.	Total Machine hours [A×B]	3,00,000	3,20,000	3,00,000	9,20,000
D.	Rate per hour (₹)	8	8	8	
E.	Machine Dept. cost [C×D]	24,00,000	25,60,000	24,00,000	73,60,000
F.	Labour hours per unit	6	4	3	
G.	Total labour hours [A×F]	6,00,000	3,20,000	1,80,000	11,00,000
H.	Rate per hour (₹)	5	5	5	
I	Assembly Dept. cost [G×H]	30,00,000	16,00,000	9,00,000	55,00,000

$$\text{Machine hour rate} = \frac{₹73,60,000}{9,20,000 \text{ hours}} = ₹8$$

$$\text{Labour hour rate} = \frac{₹55,00,000}{11,00,000 \text{ hours}} = ₹5$$

2. Calculation of cost driver rate

Cost Pool	Amount (₹)	Cost Driver	Quantity	Driver rate (₹)
Machining services	64,40,000	Machine hours	9,20,000 hours	7.00
Assembly services	44,00,000	Direct labour hours	11,00,000 hours	4.00
Set-up costs	9,00,000	Machine set-ups	9,000 set-ups	100.00
Order processing	7,20,000	Customer orders	7,200 orders	100.00
Purchasing	4,00,000	Purchase orders	800 orders	500.00

3. Calculation of activity-wise cost

		Products			
		X	**Y**	**Z**	**Total**
A.	Machining hours (Refer Working note-1)	3,00,000	3,20,000	3,00,000	9,20,000
B.	Machine hour rate (₹) (Refer Working note-2)	7	7	7	
C.	Machining services cost (₹) [A×B]	21,00,000	22,40,000	21,00,000	64,40,000
D.	Labour hours (Refer Working note-1)	6,00,000	3,20,000	1,80,000	11,00,000
E.	Labour hour rate (₹) (Refer Working note-2)	4	4	4	
F.	Assembly services cost (₹) [D×E]	24,00,000	12,80,000	7,20,000	44,00,000
G.	Machine set-ups	4,500	3,000	1,500	9,000
H.	Rate per set-up (₹) (Refer Working note-2)	100	100	100	
I.	Set-up cost (₹) [G×H]	4,50,000	3,00,000	1,50,000	9,00,000
J.	Customer orders	2,200	2,400	2,600	7,200
K.	Rate per order (₹) (Refer Working note-2)	100	100	100	
L.	Order processing cost (₹) [J×K]	2,20,000	2,40,000	2,60,000	7,20,000
M.	Purchase orders	300	350	150	800
N.	Rate per order (₹) (Refer Working note-2)	500	500	500	
O.	Purchasing cost (₹) [M×N]	1,50,000	1,75,000	75,000	4,00,000

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474 RTP Old P3 Nov 2020 Q4 old scheme

KD Ltd. is following Activity based costing. Budgeted overheads, cost drivers and volume are as follows:

Cost pool	Budgeted overheads (₹)	Cost driver	Budgeted volume
Material procurement	18,42,000	No. or orders	1,200
Material handling	8,50,000	No. of movement	1,240
Maintenance	24,56,000	Maintenance hours	17,550
Set-up	9,12,000	No. of set-ups	1,450
Quality control	4,42,000	No. of inspection	1,820

The company has produced a batch of 7,600 units, its material cost was ₹24,62,000 and wages ₹4,68,500. Usage activities of the said batch are as follows:

Material orders	56
Material movements	84
Maintenance hours	1,420 hours
Set-ups	60
No. of inspections	18

Required:

(i) CALCULATE cost driver rates.

(ii) CALCULATE the total and unit cost for the batch.

ICAI'S ANSWER

(i) Calculation of cost driver rate:

Cost pool	Budgeted overheads (₹)	Cost driver	Cost driver rate (₹)
Material procurement	18,42,000	1,200	1,535.00
Material handling	8,50,000	1,240	685.48
Maintenance	24,56,000	17,550	139.94
Set-up	9,12,000	1,450	628.97
Quality control	4,42,000	1,820	242.86

(ii) Calculation of cost for the batch:

Particulars	Amount (₹)	Amount (₹)
Material cost		24,62,000.00
Wages		4,68,500.00

Particulars	Amount (₹)	Amount (₹)
Overheads:		
- Material procurement (₹1,535×56 orders)	85,960.00	
- Material handling (₹685.48×84 movements)	57,580.32	
- Maintenance (₹139.94×1,420 hours)	1,98,714.80	
- Set-up (₹628.97×60 set-ups)	37,738.20	
- Quality control (₹242.86×18 inspections)	4,371.48	3,84,364.80
Total Cost		33,14,864.80
No. of units		7,600
Cost per units		436.17

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475 RTP **Old P3 May 2020** Q4 old scheme

Following are the data of three product lines of a departmental store for the year 2019-20:

	Soft drinks	Fresh produce	Packaged food
Revenues	₹ 39,67,500	₹ 1,05,03,000	₹ 60,49,500
Cost of goods sold	₹ 30,00,000	₹ 75,00,000	₹ 45,00,000
Cost of bottles returned	₹ 60,000	₹ 0	₹ 0
Number of purchase orders placed	360	840	360
Number of deliveries received	300	2,190	660
Hours of shelf-stocking time	540	5,400	2,700
Items sold	1,26,000	11,04,000	3,06,000

Additional information related with the store are as follows:

Activity	Description of activity	Total Cost	Cost-allocation base
Bottles returns	Returning of empty bottles	₹ 60,000	Direct tracing to soft drink line
Ordering	Placing of orders for purchases	₹ 7,80,000	1,560 purchase orders
Delivery	Physical delivery and receipt of goods	₹ 12,60,000	3,150 deliveries
Shelf stocking	Stocking of goods on store shelves and on-going restocking	₹ 8,64,000	8,640 hours of shelf-stocking time
Customer Support	Assistance provided to customers including check-out	₹ 15,36,000	15,36,000 items sold

Required:

CALCULATE the total cost and operating income using Activity Based Costing method.

ICAI'S ANSWER

Working notes:

(i) Total support cost:

	(₹)
Bottles returns	60,000
Ordering	7,80,000
Delivery	12,60,000
Shelf stocking	8,64,000
Customer support	15,36,000
Total support cost	45,00,000

(ii) Cost for each activity cost driver:

Activity (1)	Total cost (₹) (2)	Cost allocation base (3)	Cost driver rate (4) = [(2) ÷ (3)]
Ordering	7,80,000	1,560 purchase orders	₹500 per purchase order
Delivery	12,60,000	3,150 deliveries	₹400 per delivery
Shelf-stocking	8,64,000	8,640 hours	₹100 per stocking hour
Customer support	15,36,000	15,36,000 items sold	₹1 per item sold

Statement of Total cost and Operating income

	Soft drinks (₹)	Fresh Produce (₹)	Packaged Food (₹)	Total (₹)
Revenues: (A)	39,67,500	1,05,03,000	60,49,500	2,05,20,000
Cost & Goods sold	30,00,000	75,00,000	45,00,000	1,50,00,000
Bottle return costs	60,000	0	0	60,000
Ordering cost* (360:840:360)	1,80,000	4,20,000	1,80,000	7,80,000
Delivery cost* (300:2190:660)	1,20,000	8,76,000	2,64,000	12,60,000
Shelf stocking cost* (540:5400:2700)	54,000	5,40,000	2,70,000	8,64,000
Customer Support cost* (1,26,000:11,04,000:3,06,000)	1,26,000	11,04,000	3,06,000	15,36,000
Total cost: (B)	35,40,000	1,04,40,000	55,20,000	1,95,00,000
Operating income C:{(A)- (B)}	4,27,500	63,000	5,29,500	10,20,000

* Refer to working note (ii)

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476 MTP **Sep 2026 S-I** Q3(a) 9 marks

Silver Star Ltd. manufactures four products, namely X, Y, Z, and W using the same plant and process. The following information relates to a production period:

Product	X	Y	Z	W
Output in units	800	650	500	550
Cost per unit (₹)				
— Direct Material	50	55	45	60
— Direct Labour	12	10	8	9
Machine hours/unit	5 hrs.	3.5 hrs.	2.5 hrs.	1.5 hrs.

The four products are similar and are usually produced in production runs of 25 units and sold in batches of 10 units. Machine hour rate is currently used to absorb production overheads. The total overheads incurred by the company for the period are as follows:

Overhead Category	Amount (₹)
Machine operation and maintenance cost	75,000
Setup costs	25,000
Store receiving	18,000
Inspection	12,000
Material handling and dispatch	3,500

During the period the following cost drivers are to be used for the overhead costs:

Activity	Cost Driver
Setup cost	Number of production runs
Store receiving	Requisitions raised
Inspection	Number of production runs
Material handling and dispatch	Orders executed

It is also determined that:

- Machine operation and maintenance cost should be apportioned between setup cost, store receiving, and inspection activity in 5:3:2.
- Number of requisitions raised on store is 60 for each product, and the number of orders executed is 220, each order being for a batch of 10 units of a product.

Required:

(i) CALCULATE the total cost of each product, if all overhead costs are absorbed on a machine hour rate basis.

(ii) CALCULATE the total cost of each product using activity-based costing.

(iii) COMMENT briefly on the differences disclosed between overhead allocated by the present system and overhead traced under activity-based costing.

ICAI'S ANSWER

(i) Overhead Absorption on Machine Hour Basis

Statement Showing "Total Cost of Products"

Particulars	X (₹)	Y (₹)	Z (₹)	W (₹)
Direct Material	50.00	55.00	45.00	60.00
Direct Labour	12.00	10.00	8.00	9.00
Overheads	79.94	55.96	39.97	23.98
Cost of Production per unit	141.94	120.96	92.97	92.98
Output in units	800	650	500	550
Total Cost	1,13,552.10	78,622.75	46,485.03	51,140.12

[8,350 Machine Hours = (800 × 5) + (650 × 3.5) + (500 × 2.5) + (550 × 1.5)]

Total Overheads = 75,000 + 25,000 + 18,000 + 12,000 + 3,500 = 133,500

Rate per hour = 133,500 ÷ 8,350 = ₹15.99 per hour

(Overheads per unit = Rate per hour × Machine hours per unit)

(ii) Activity Based Costing System

Statement Showing "Allocation of Machine Operation and Maintenance Cost"

Particulars	Setup (₹)	Receiving (₹)	Inspection (₹)
Machine Operation & Maintenance ₹ 75,000 to be distributed in ratio 5: 3: 2	37,500	22,500	15,000

Now add these apportioned amounts to the activity pools:

Activity Pool	Original Cost (₹)	Machine-op apportioned (₹)	Total Activity Cost (₹)
Setup	25,000	37,500	62,500
Store Receiving	18,000	22,500	40,500
Inspection	12,000	15,000	27,000
Material Handling & Dispatch	—	—	3,500

Statement Showing "Activities / Drivers / Cost"

Activity	Cost (₹)	Drivers	Nos	Cost per unit of Driver (₹)
Setup	62,500	Production Runs	100	625.00
Store Receiving	40,500	Requisitions Raised	240	168.75
Inspection	27,000	Production Runs	100	270.00
Material Handling and Disp.	3,500	Orders	250	14.00

Note:

Production run size = 25 units.

Production Runs:

X (800/25) = 32

Y (650/25) = 26

Z (500/25) = 20

W (550/25) = 22

(Total = 100)

Requisitions raised = 60 for each product (Total = 240)

Orders executed (batch = 10 units): X = 80; Y = 65; Z = 50; W = 55 (Total = 250)

Statement Showing "Total Cost of Products"

Particulars of Costs	X (₹)	Y (₹)	Z (₹)	W (₹)
Direct Material (Total)	40,000	35,750	22,500	33,000
Direct Labour (Total)	9,600	6,500	4,000	4,950
Setup (Runs × 625)	20,000	16,250	12,500	13,750
Store Receiving (60 × 168.75)	10,125	10,125	10,125	10,125
Inspection (Runs × 270)	8,640	7,020	5,400	5,940
Material Handling & Dispatch (Orders × 14)	1,120	910	700	770
Total Cost	89,485.00	76,555.00	55,225.00	68,535.00
Cost per unit	111.86	117.78	110.45	124.61

(iii)

Particulars	X (₹)	Y (₹)	Z (₹)	W (₹)
Cost per unit (Traditional) ...(a)	141.94	120.96	92.97	92.98
Cost per unit (ABC) .....(b)	111.86	117.78	110.45	124.61
Difference .....(b) – (a)	(30.08)	(3.18)	17.48	31.63

Comment: The activity-based costing system reallocates overheads to products according to the true cost drivers (setups, requisitions, inspections, orders) rather than solely by machine hours. Under ABC, products X and Y show lower unit costs compared with the machine-hour method (so they were over costed by the traditional method), while products Z and W show higher unit costs (they were under costed by the traditional method). This change can materially affect pricing and product-level profitability — especially for Z and W where ABC raises unit cost significantly.

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477 MTP Sep 2026 S-II Case Scenario - 2 · MCQ 6 to MCQ 10 10 marks

CASE SCENARIO - 2

The HomeMart is the latest trending brand offering home improvement appliances with broadest selection of products with highly competitive prices. The sale is increasing year by year with huge multiples. Current year also the sales reached triple the last year. The reason being company having good customer support where it provides after sales assistance over phone per item sold. Though it costs only Re. 1 per item sold to the company, it enhanced to ₹ 49,15,200 last year making a huge impact on the total support cost.

All the company's appliances have been majorly categorised into three product lines namely Fancy fans, Home decors, Assembled furniture. During the current year, the company's revenue as generated is ₹ 3,80,88,000, ₹ 10,08,28,800 and ₹ 5,80,75,200 respectively. However, the cost of god sold is ₹ 2,88,00,000, ₹ 7,20,00,000 and ₹ 4,32,00,000 respectively.

In business, there's a saying "The packaging sells the product the first time, but what's inside sells the product a second time". Following the saying, the company has the policy of taking back the cartons of the products sold relating to Fancy fans to reduce the packaging cost. However, for smooth returning of cartons, the company has to incur certain carrier cost on its own which is ₹ 5,76,000 for the current year and allocating the same directly to the said product.

Some other information relating to each of the product lines is provided below:

	Fancy fans	Home decors	Assembled furniture
Items sold	12,09,600	1,05,98,400	29,37,600
Number of deliveries received	600	4,380	1,320
Number of purchase orders placed	720	1,680	720
Hours of shelf-stocking time	1,080	10,800	5,400

The company also provides the following basis of cost allocation:

Activity	Description of activity	Total Cost	Cost-allocation base
Delivery	Physical delivery and receipt of products	1,20,96,000	6,300 deliveries
Ordering	Placing of orders for purchases	74,88,000	3,120 purchase orders
Shelf stocking	Stocking of products in warehouse	82,94,400	17,280 hours of shelf-stocking time

The company wants you to FIGURE OUT the following to ascertain which of the product line is more profitable and answer the following question (MCQ 6 to 10):

MCQ 6 2 marks

The total support cost and its percentage to the cost of goods sold would be:

A ₹ 3,33,69,600 and 23.17%

B ₹ 4,32,00,000 and 30%

C ₹ 3,33,69,600 and 30%

D ₹ 4,32,00,000 and 23.17%

MCQ 7 2 marks

Operating income as a percentage of revenues of each product line, namely Fancy fans, Home decors, Assembled furniture, when all the support costs are allocated on the basis of cost of goods sold would be:

A 6.87%, 12.05% and 8.38% respectively

B 12.05%, 6.87% and 8.38% respectively

C 1.70%, 7.17% and 3.30% respectively

D 7.17%, 3.30% and 1.70% respectively

MCQ 8 2 marks

The cost driver rate relating to Delivery, Ordering, Shelf stocking and Customer support would be:

A Delivery- ₹ 1,920 per delivery, Ordering- ₹ 2,400 per purchase order, Shelf stocking- ₹ 480 per stocking hour and Customer support- Re. 1 per item sold

B Delivery- ₹ 2,400 per delivery, Ordering- ₹ 1,920 per purchase order, Shelf stocking- ₹ 480 per stocking hour and Customer support- Re. 1 per item sold

C Delivery- ₹ 1,920 per delivery, Ordering- ₹ 2,400 per purchase order, Shelf stocking- ₹ 480 per stocking hour and Customer support- ₹ 3 per item sold

D Delivery- ₹ 480 per delivery, Ordering- ₹ 2,400 per purchase order, Shelf stocking- ₹ 1,920 per stocking hour and Customer support- ₹ 3 per item sold

MCQ 9 2 marks

Operating income of each product line, namely Fancy fans, Home decors, Assembled furniture, when all the support costs are allocated using an activity-based costing system would be:

A ₹ 16,84,800, ₹ -2,05,92,000 and ₹ -7,92,000 respectively

B ₹ 3,64,03,200, ₹ 12,14,20,800 and ₹ 5,88,67,200 respectively

C ₹ 92,88,000, ₹ 2,88,28,800 and ₹ 1,48,75,200 respectively

D ₹ 41,04,000, ₹ 6,04,800 and ₹ 50,83,200 respectively

MCQ 10 2 marks

Operating income as a percentage of revenues of each product line, namely Fancy fans, Home decors, Assembled furniture, when all the support costs are allocated using an activity-based costing system would be:

A 4.42%, -20.42% and -1.36% respectively

B 10.78%, 0.60% and 8.75% respectively

C 24.39%, 28.59% and 25.61% respectively

D 4.39%, 8.59% and -1.36% respectively

ICAI'S ANSWERS

MCQ 6

(B)

Total support cost	(₹)
Cartons returned	5,76,000

Total support cost	(₹)
Delivery	1,20,96,000
Ordering	74,88,000
Shelf stocking	82,94,400
Customer support (₹ 49,15,200 × 3)	1,47,45,600
Total support cost	4,32,00,000
	(₹)
Fancy fans	2,88,00,000
Home decors	7,20,00,000
Assembled furniture	4,32,00,000
Total cost of goods sold (COGS)	14,40,00,000

Percentage of support cost to the cost of goods sold (COGS):

$$= \frac{\text{Total support cost}}{\text{Total cost of goods sold (COGS)}} \text{ x } 100$$

$$= \frac{4,32,00,000}{14,40,00,000} \text{ x } 100 = 30\%$$

MCQ 7

(C)

Particulars	Fancy Fans (₹)	Home Decors (₹)	Assembled Furniture (₹)
Revenue: (A)	3,80,88,000	10,08,28,800	5,80,75,200
Cost of Goods sold (COGS): (B)	2,88,00,000	7,20,00,000	4,32,00,000
Support cost (30% of COGS): (C) (as Calculated in i. above)	86,40,000	2,16,00,000	1,29,60,000
Total cost: (D) = {(B) + (C)}	3,74,40,000	9,36,00,000	5,61,60,000
Operating income: E= {(A)-(D)}	6,48,000	72,28,800	19,15,200
Operating income as a percentage of revenues: (E/A) × 100)	**1.70%**	**7.17%**	**3.30%**

MCQ 8

(A)

Activity (1)	Total cost (₹) (2)	Cost allocation base (3)	Cost driver rate (4) = [(2) ÷ (3)]
Delivery	1,20,96,000	6,300 deliveries	₹ 1,920 per delivery
Ordering	74,88,000	3,120 purchase orders	₹ 2,400 per purchase order
Shelf-stocking	82,94,400	17,280 hours of shelf-stocking time	₹ 480 per stocking hour
Customer support	1,47,45,600	1,47,45,600 items sold	₹ 1 per item sold (given)

MCQ 9

(D)

	Fancy Fans (₹)	Home decors (₹)	Assembled furniture (₹)
Revenues: (A)	3,80,88,000	10,08,28,800	5,80,75,200
Cost & Goods sold	2,88,00,000	7,20,00,000	4,32,00,000
Carton return costs (Directly attributable to Fancy fans)	5,76,000	0	0
Delivery cost (₹ 1,920 per delivery)	11,52,000 (600 x ₹ 1,920)	84,09,600 (4,380 x ₹ 1,920)	25,34,400 (1,320 x ₹ 1,920)
Ordering cost (₹ 2,400 per purchase order)	17,28,000 (720 x ₹ 2,400)	40,32,000 (1,680 x ₹ 2,400)	17,28,000 (720 x ₹ 2,400)
Shelf stocking cost (₹ 480 per stocking hour)	5,18,400 (1,080 x ₹ 480)	51,84,000 (10,800 x ₹ 480)	25,92,000 (5,400 x ₹ 480)
Customer Support cost (₹ 1 per item sold)	12,09,600 (12,09,600 x ₹ 1)	1,05,98,400 (1,05,98,400 x ₹ 1)	29,37,600 (29,37,600 x ₹ 1)
Total Cost: (B)	3,39,84,000	10,02,24,000	5,29,92,000
Operating income: (C) = (A) - (B)	41,04,000	6,04,800	50,83,200

MCQ 10

(B)

	Fancy Fans (₹)	Home decors (₹)	Assembled furniture (₹)
Operating income (from iv. Above) (A)	41,04,000	6,04,800	50,83,200
Revenues (B)	3,80,88,000	10,08,28,800	5,80,75,200
Operating income as a percentage of revenues: (A/B) × 100)	**10.78%**	**0.60%**	**8.75%**

478 MTP **May 2026 S-I** Q5(a) 9 marks

Jyoti Mfg. Co. has decided to increase the size of the store. It wants the information about the probability of the individual product lines : Lemon, Grapes and Papaya. It provides the following data for the 2025 for each product line:

Particulars	Lemon	Grapes	Papaya
Revenues (₹)	79,350	2,10,060	1,20,990
Cost of goods sold (₹)	60,000	1,50,000	90,000
Cost of bottles returned (₹)	1,200	0	0
Number of purchase orders placed	36	84	36
Number of deliveries received	30	219	66
Hours of shelf stocking time	54	540	270
Items sold	12,600	1,10,400	30,600

Jyoti Mfg. Co. also provides the following information for the year 2025:

Activity	Description of Activity	Total Costs (₹)	Cost Allocation Basis
Bottle returns	Returning of empty bottles to the store	1,200	Direct tracing to product line
Ordering	Placing of orders of purchases	15,600	156 purchase orders
Delivery	Physical delivery and the receipts of merchandise	25,200	315 deliveries
Self-stocking	Stocking of merchandise on store shelves and ongoing restocking	17,280	864 hours of time
Customer support	Assistance provided to customers including bagging and checkout	30,720	1,53,600 items sold

Required

(i) Jyoti Mfg. Co. currently allocates store support costs (all costs other than the cost of goods sold) to the product line on the basis of the cost of goods sold of each product line. CALCULATE the operating income and operating income as the percentage of revenue of each product line.

(ii) If Jyoti Mfg. Co. allocates store support costs (all costs other than the cost of goods sold) to the product lines on the basis of ABC system, CALCULATE the operating income and operating income as the percentage of revenue of each product line.

ICAI'S ANSWER

(i) Absorption Costing System

Operating Income -

Particulars	Lemon	Grapes	Papaya	Total
Revenue	79,350	2,10,060	1,20,990	4,10,400
Less: Cost of Goods Sold	60,000	1,50,000	90,000	3,00,000
Less: Store Support Cost (2:5:3)	18,000	45,000	27,000	90,000
Operating Income	1,350	15,060	3,990	20,400
Operating Income (%)	1.70	7.17	3.30	4.97

(ii) ABC System

Overhead Allocation Rate -

Activity	Total Costs (₹)	Quantity of Cost Allocation Base	Overhead Allocation Rate (₹)
Ordering	15,600	156 Purchase Orders	100.00
Delivery	25,200	315 Delivering Orders	80.00
Shelf Stocking	17,280	864 Self Stocking Hours	20.00
Customer Support	30,720	1,53,600 Items Sold	0.20

Store Support Cost -

Particulars	Cost Driver	Lemon	Grapes	Papaya	Total
Bottle Returns	Direct	1,200	0	0	1,200
Ordering	Purchase Orders	3,600	8,400	3,600	15,600
Delivery	Deliveries	2,400	17,520	5,280	25,200
Self -Stocking	Hours of time	1,080	10,800	5,400	17,280
Customer Support	Items Sold	2,520	22,080	6,120	30,720
Grand Total		10,800	58,800	20,400	90,000

Operating Income -

Particulars	Lemon	Grapes	Papaya	Total
Revenue	79,350	2,10,060	1,20,990	410,400
Less: Cost of Goods Sold	60,000	1,50,000	90,000	300,000
Less: Store Support Cost	10,800	58,800	20,400	90,000
Operating Income	8,550	1,260	10,590	20,400
Operating Income (%)	10.78	0.60	8.75	4.97

ICAI · P4 · MTP May 2026 · Series I · question p. 13-14 · answer p. 17-18

479 MTP May 2026 S-II Q5(b) 6 marks

XYZ Engg. Ltd. manufactures two types of specialized components Alpha and Beta.

Following data pertains to their operations for the month of October 2024:

Component	Units produced and sold	Direct material cost per unit (₹)	Direct labour cost per unit (₹)
Alpha	8,000	25	18
Beta	12,000	15	12

Activity	Cost Driver	Alpha	Beta	Total Overhead Cost
Material Handling	Purchase Orders	30 Orders	20 Orders	₹ 60,000
Production Set-up	Machine Set-ups	80 Set-ups	100 Set-ups	₹ 1,44,000

You are required to:

(i) CALCULATE the cost driver rates for each overhead activity.

(ii) CALCULATE the total cost and cost per unit of producing and selling each component using Activity-Based Costing (ABC).

(iii) Use ABC data calculated above and ANALYSE, whether outsourcing the production set-up for Alpha would be financially advantageous, if a specialized firm has offered to do production set-up at a contract price of ₹ 600 per machine set-up.

ICAI'S ANSWER

(i) Cost Driver Rates for Each Overhead Activity:

Activity	Overhead Cost (A)	Cost Driver Level (B)	Cost Driver Rate (in ₹) (A)/(B)
Material Handing	60,000	30 + 20 + = 50	1,200 per purchase order
Production Setup	1,44,000	80 + 100 =180	800 per machine setup

(ii) Allocation of Overheads Using ABC:

Component	Material Handing (in ₹) (A)	Production Setup (in ₹) (B)	Total overhead Cost (in ₹) (A) + (B)
Alpha	30 × 1,200 = 36,000	80 × 800 = 64,000	**1,00,000**
Beta	20 × 1,200 = 24,000	100 × 800 = 80,000	**1,04,000**

Computation of Total Cost and Cost per Unit using ABC:

Component	Direct Material Cost (₹)	Direct Labour Cost (₹)	Total Overhead Cost (₹)	Total Cost (₹)	Cost per Unit (₹)
Alpha	8,000 × 25 = 2,00,000	8,000 × 18 = 1,44,000	1,00,000	4,44,000	4,44,000 / 8,000 = **55.50**
Beta	12,000 × 15 = 1,80,000	12,000 × 12 = 1,44,000	1,04,000	4,28,000	4,28,000 / 12,000 = **35.67**

(iii) Outsourcing Analysis for Production Setup (Alpha):

Current production setup cost for Alpha = ₹ 64,000

New Outsourcing Cost = ₹ 600 × 80 setups = ₹ 48,000

Cost Savings = ₹ 64,000 - ₹ 48,000 = ₹ 16,000 or (₹ 800 – ₹ 600) x 80 setups

Outsourcing the Production Setup for Alpha would be financially advantageous for the company as it reduces costs by ₹ 16,000.

ICAI · P4 · MTP May 2026 · Series II · question p. 15 · answer p. 19-20

480 MTP **Jan 2026 S-I** Q2(a) 10 marks

VTR Tech Ltd is a precision engineering company that has experienced rapid automation in its production processes. The company now manufactures over 80 specialised components using CNC machines and automated systems. Despite this shift, VTR still allocates manufacturing overheads using a single plant-wide overhead absorption rate based on machine hours.

In the quarter ending June, VTR's manufacturing overhead costs were as follows:

Cost Item	₹ ('000)
Machine Operating Expenses	140
Maintenance Costs	30
Wages of Machine Supervisors	90
Stores Handling Staff Wages	28
Packaging & Despatch Department Wages	42

During the quarter, management initiated a review of the costing system and concluded that using machine hours to absorb all overheads is no longer meaningful. The company decided to implement an Activity-Based Costing (ABC) approach and identified the following key activities:

1. Receiving raw materials and components
2. Setting up machines for production
3. Quality assurance checks
4. Packaging and despatching finished goods

The following data was also recorded during the quarter:

- Total machine hours used: 2,500 (charged at ₹18 per hour)
- 1,100 material consignments were received from suppliers
- 950 machine setups were carried out
- 700 quality inspections were conducted
- 480 customer orders were packaged and despatched

Allocation Ratio

Machine Operation & Maintenance: 15% Receiving, 70% Setups, 15% Despatch

Supervisors: 7.5% Receiving, 70% Setups, 15% Inspections, 7.5% Despatch

VTR Tech manufactured Components A, B, and C during the quarter. The details for each are:

Particulars	Component A	Component B	Component C
Direct Material Cost (₹)	1,500	3,800	2,200
Machine Hours Used	30	520	60
Material Consignments Received	45	32	33
Machine Setups	20	22	18
Quality Inspections	12	10	20
Orders Despatched	25	90	55
Units Produced	600	13,000	2,600

Required:

(i) CALCULATE the unit cost of Components A, B, and C using VTR's existing traditional costing system (overheads absorbed on machine hours basis).

(ii) EXPLAIN how an Activity-Based Costing (ABC) system would be developed using the above data.

Then CALCULATE the unit cost of Components A, B, and C using the ABC system, showing detailed allocation of overheads and unit cost.

(iii) COMPARE the cost of component under both costing system

ICAI'S ANSWER

(i) Traditional Costing System: Overhead Absorption Based on Machine Hours

Total Manufacturing Overheads (₹ '000)

Overhead Component	Amount (₹ '000)
Machine Operating Expenses	140
Maintenance Costs	30
Wages of Machine Supervisors	90
Stores Handling Staff Wages	28
Packaging & Despatch Wages	42
Total Manufacturing Overhead	**330**

Overhead Absorption Rate = ₹ 3,30,000/2,500 hrs. = ₹ 132 per Machine Hour

Computation of Unit Cost Using Traditional System

Component	A	B	C
Machine Hrs	30	520	60
Machine Labour Cost (₹18/hr)	₹540	₹ 9,360	₹ 1,080
Direct Material	₹ 1,500	₹ 3,800	₹ 2,200
Overheads (₹132/hr)	₹ 3,960	₹ 68,640	₹ 7,920
Total Cost	₹ 6,000	₹ 81,800	₹ 11,200
Qty Produced	600	13000	2600
Unit Cost (₹)	**₹ 10.00**	**₹ 6.29**	**₹ 4.31**

(ii) ABC System Development & Cost Calculation

ABC Process:

1. Identify Activities:
   - Receiving Materials
   - Machine Setups
   - Quality Inspections
   - Packaging & Despatch
2. Allocate Overheads to Activity Pools
3. Determine Cost Drivers and Rates
4. Assigning the cost of activities to cost objects

Overhead Allocation to Activities (₹ '000):

Activity	Receiving	Setups	Inspections	Despatch	Total
Machine Operation (140)	21.00	98.00	—	21.00	140.00
Maintenance (30)	4.50	21.00	—	4.50	30.00
Supervisors' Wages (90)	6.75	63.00	13.50	6.75	90.00
Stores Staff (28)	28.00	—	—	—	28.00
Packaging & Despatch (42)	—	—	—	42.00	42.00
Total (₹ '000)	**60.25**	**182.00**	**13.50**	**74.25**	**330.00**

Allocation ratios:

- Machine Operation & Maintenance: 15% Receiving, 70% Setups, 15% Despatch
- Supervisors: 7.5% Receiving, 70% Setups, 15% Inspections, 7.5% Despatch

Determine Cost Driver Rates:

Activity	Cost (₹ '000)	Driver Volume	Rate per Unit
Receiving	60.25	1,100 consignments	₹ 54.77 per consignment
Machine Setups	182.00	950 setups	₹ 191.58 per setup
Quality Inspections	13.50	700 inspections	₹ 19.29 per inspection
Despatch	74.25	480 despatches	₹ 154.69 per order

Assign Activity Costs to Components

Component	A (₹)	B (₹)	C (₹)
Direct Labour	540	9,360	1,080
Direct Material	1,500	3,800	2,200
Receiving	45 × 54.77 = 2,464.65	32 × 54.77 = 1,752.64	33 × 54.77 = 1,807.41
Setups	20 × 191.58 = 3,831.60	22 × 191.58 = 4,214.76	18 × 191.58 = 3,448.44
Inspections	12 × 19.29 = 231.48	10 × 19.29 = 192.90	20 × 19.29 = 385.80
Despatch	25 × 154.69 = 3,867.25	90 × 154.69 = 13,922.10	55 × 154.69 = 8,507.95
Total Cost	**12,435.00**	**33,242.40**	**17,429.60**
Units	600	13,000	2,600
Unit Cost ₹	**20.73**	**2.56**	**6.70**

(iii) Comparison

Component	Traditional Unit Cost (₹)	ABC Unit Cost (₹)	Observation
A	₹ 10.00	₹ 20.73	Highly under-costed under traditional system
B	₹ 6.29	₹ 2.56	Over-costed under traditional system
C	₹ 4.31	₹ 6.70	Under-costed under traditional system

ICAI · P4 · MTP Jan 2026 · Series I · question p. 8-10 · answer p. 7-10

481 MTP Jan 2026 S-II MCQ 14 2 marks MCQ

A company manufactures Product A and Product B and uses Activity-Based Costing.

The following activity data relates to the current month:

Activity	Cost (₹)	Cost Driver	Total Driver Units	Product A	Product B
Machine Setup	60,000	Number of setups	120 setups	30 setups	90 setups
Quality Inspection	40,000	Number of inspections	200 inspections	50 inspections	150 inspections

What is the total overhead cost assigned to Product A under ABC?

A ₹ 35,000

B ₹ 25,000

C ₹ 30,000

D ₹ 40,000

ICAI'S ANSWER

(B) ₹ 25,000

Cost per setup = 60,000 / 120 = ₹ 500

Cost assigned to Product A = 30 × 500 = ₹ 15,000

Cost per inspection = 40,000 / 200 = ₹ 200

Cost assigned to Product A = 50 × 200 = ₹ 10,000

Total Overhead for Product A = 15,000 + 10,000 = ₹ 25,000

ICAI · P4 · MTP Jan 2026 · Series II · question p. 6 · answer p. 7

482 MTP **Jan 2026 S-II** Q4(b) 5 marks

Vihan Retail Co. has decided to expand the size of its store. It wants information about the profitability of its individual product lines: Pineapple, Orange, and Litchi. The company provides the following data for the year 2024 for each product line:

Particulars	Pineapple	Orange	Litchi
Revenues (₹)	79,350	2,10,060	1,20,990
Cost of goods sold (₹)	60,000	1,50,000	90,000
Cost of bottles returned (₹)	1,200	0	0
Number of purchase orders placed	36	84	36
Number of deliveries received	30	219	66
Hours of shelf stocking time	54	540	270
Items sold	12,600	1,10,400	30,600

Vihan Retail Co. also provides the following information for the year 2024:

Activity	Description of Activity	Total Costs (₹)	Cost Allocation Basis
Bottle returns	Returning of empty bottles to the store	1,200	Direct tracing to product line
Ordering	Placing of orders of purchases	15,600	156 purchase orders
Delivery	Physical delivery and the receipts of merchandise	25,200	315 deliveries
Self- stocking	Stocking of merchandise on store shelves and ongoing restocking	17,280	864 hours of time
Customer support	Assistance provided to customers including bagging and checkout	30,720	1,53,600 items sold

Required

If Vihan Retail Co. allocates store support costs (all costs other than the cost of goods sold) to the product lines on the basis of ABC system, CALCULATE the operating income and operating income as the percentage of revenue of each product line.

ICAI'S ANSWER

Overhead Allocation Rate -

Activity	Total Costs (₹)	Quantity of Cost Allocation Base	Overhead Allocation Rate (₹)
Ordering	15,600	156 Purchase Orders	100.00
Delivery	25,200	315 Delivering Orders	80.00
Shelf Stocking	17,280	864 Self Stocking Hours	20.00
Customer Support	30,720	1,53,600 Items Sold	0.20

Store Support Cost -

Particulars	Cost Driver	Pineapple	Orange	Litchi	Total
Bottle Returns	Direct	1,200	0	0	1,200
Ordering	Purchase Orders	3,600	8,400	3,600	15,600
Delivery	Deliveries	2,400	17,520	5,280	25,200
Self -Stocking	Hours of time	1,080	10,800	5,400	17,280
Customer Support	Items Sold	2,520	22,080	6,120	30,720
Grand Total		10,800	58,800	20,400	90,000

Operating Income-

Particulars	Pineapple	Orange	Litchi	Total
Revenue	79,350	2,10,060	1,20,990	410,400
Less: Cost of Goods Sold	60,000	1,50,000	90,000	300,000
Less: Store Support Cost	10,800	58,800	20,400	90,000
Operating Income	8,550	1,260	10,590	20,400
Operating Income (%)	10.78	0.60	8.75	4.97

ICAI · P4 · MTP Jan 2026 · Series II · question p. 15-16 · answer p. 17-18

483 MTP **Sep 2025 S-I** Q5(a) 8 marks

Chicago Manufacturing Co. (CMC) manufactures several product of varying levels of designs and models. It uses a single overhead recovery rate based on direct labour hours. The overheads incurred by the CMC in the half of the year are as under:

	₹
Machine operation expenses	10,12,500
Machine maintenance expenses	1,87,500
Salaries of technical staff	6,37,500
Wages and salaries of stores staff	2,62,500

During this period, CMC introduced activity based costing system and the following significant activities were identified:

— receiving materials and components

— set up of machines for production runs

— quality inspection

It is determined that:

— The machine operation and machine maintenance expenses should be apportioned between stores and production activity in 20:80 ratio.

— The technical staff salaries should be apportioned between machine maintenance, set up and quality inspection in 30:40:30 ratio.

The consumption of activities during the period under review are as under:

— Direct labour hours worked 40,000

— Direct wage rate ₹ 6 per hour

— Production set-ups 2,040

— Material and component consignments from received from suppliers 1,960

— Number of quality inspections carried out 1,280

The data relating to two product manufactured by the CMC during the period are as under:

	Product P	Product Q
Direct material costs (₹)	6,000	4,000
Direct labour hours	960	100
Direct material consignments received	48	52
Production runs	36	24
Number of quality inspections done	30	10
Quantity produced (units)	15,000	5,000

A potential customer has approached CMC for the supply of 24,000 units of a component K to be delivered in lots of 3,000 units per quarter. The job will involve an initial design cost of ₹ 60,000 and the manufacture will involve the following per quarter:

Direct material costs	₹ 12,000
Direct labour hours	300
Production runs	6
Inspections	24
Number of consignments of Direct materials to be received	20

CMC desires a mark up of 25% on cost.

Required:

(i) CALCULATE the cost of product P and Q based on the existing system of single overhead recovery rate.

(ii) DETERMINE the cost of product P and Q using activity based costing system.

(iii) COMPUTE the sales value per quarter of component K using activity based costing system.

ICAI'S ANSWER

(i) Statement Showing "Computation of Cost of Product P and Q" (Based on the Existing System of 'Single Overhead Recovery Rate')

	Product P	Product Q
Units	15,000	5,000
Direct Materials Cost (₹)	6,000	4,000
Direct Labour Cost (₹)	5,760 (960 hours x ₹ 6)	600 (100 hours x ₹ 6)
Overheads (₹) (Refer to W.N. 1)	50,400 (960 hours x ₹ 52.50)	5,250 (100 hours x ₹ 52.50)
Total Cost of Products (₹)	62,160	9,850
Cost *per unit* (₹)	4.144 (₹ 62,160 / 15,000 units)	1.97 (₹ 9,850 / 5,000 units)

(ii) Statement Showing Computation of Cost of Products P and Q (Using 'Activity Based Costing System')

	Product P	Product Q
Units	15,000	5,000
Direct Materials Cost (₹)	6,000	4,000
Direct Labour Cost (₹)	5,760	600
Receiving Cost (Refer to W.N. 4)	13,243 (48 x ₹ 275.89)	14,346 (52 x ₹ 275.89)
Setup Cost (Refer to W.N. 4)	24,141 (36 x ₹ 670.59)	16,094 (24 × 670.59)
Inspection Cost (Refer to W.N. 4)	4,482 (30 x ₹ 149.41)	1,494 (10 x ₹ 149.41)
Total Cost of Products (₹)	53,626	36,534
Cost *per unit* (₹)	3.58 (₹ 53,626 /15,000 units)	7.31 (₹ 36,534 /5,000 units)

(iii) Computation of Sales Value per Quarter 'Component K' (Using 'Activity Based Costing System')

3,000 units of 'Component K' to be delivered *per quarter*	₹
Initial Design Cost *per quarter* (₹ 60,000 / 8 quarters)	7,500
Direct Material Cost	12,000
Direct Labour Cost (300 hours x ₹ 6)	1,800
Receiving Cost (20 No. of Consignment x ₹ 275.89)	5,518
Setup Cost (6 Production Runs x ₹ 670.59)	4,024
Inspection Cost (24 Inspections x ₹ 149.41)	3,586
Total Cost	34,428
Add: Mark up (25%of cost)	8,607
Sales Value	43,035

3,000 units of 'Component K' to be delivered *per quarter*	₹
Selling Price *per unit* 'K' (₹ 43,035 / 3,000 units)	14.35

Working Notes

1. Overhead Rate per Labour Hour

$$= \frac{\text{Total Overhead Incurred by the Company in First Half Year}}{\text{Total Direct Labour Hours Worked}}$$

$$= \frac{₹21,00,000}{40,000 \text{ hours}}$$

= ₹ 52.50 *per labour hour*

2. Statement Showing Apportionment of 'Technical Staff Salaries' Over 'Machine Maintenance', 'Setup' and 'Quality Inspection' in the Ratio 30:40:30

	Total Salaries (₹)	Machine Maintenance (₹)	Setup (₹)	Quality Inspection (₹)
Technical Staff Salaries	6,37,500	1,91,250	2,55,000	1,91,250

3. Statement Showing Apportionment of 'Machine Operation' and 'Machine Maintenance' Between 'Stores' and 'Production Activity (Setup)'

	Total Expenses (₹)	Stores / Receiving (₹)	Setup (₹)
Machine Operation (20:80)	10,12,500	2,02,500	8,10,000
Machine Maintenance (20:80) [₹ 1,91,250 + ₹ 1,87,500] (Refer to W.N. 2)	3,78,750	75,750	3,03,000
Wages and Salaries of Stores Staff	2,62,500	2,62,500	----
Component of Setup Cost (Refer to W.N. 2)	2,55,000	----	2,55,000
Total	19,08,750	5,40,750	13,68,000

4. Rate per 'Activity Cost Driver'

	Stores / Receiving (₹)	Setup (₹)	Quality Inspection (₹)
Total Overheads (₹) ...(A)	5,40,750	13,68,000	1,91,250
Units of Activities Carried out ...(B)	1,960	2,040	1,280
Rate per Activity Cost Driver (₹) ...{(A) / (B)}	275.89	670.59	149.41

ICAI · P4 · MTP Sep 2025 · Series I · question p. 13-15 · answer p. 15-17

484 MTP **Sep 2025 S-II** Case Scenario 2 · MCQ 6 to MCQ 10 10 marks

CASE SCENARIO 2

SunBright Appliances Ltd., founded in 2005, is a growing Indian manufacturer of three types of home appliances: Air Conditioners (ACs), Washing Machines (WMs), and Refrigerators (RFs). The company has built a reputation for quality and durability in the mid-range market segment and has seen stable

demand across urban and semi-urban markets. In recent years, it has invested in modernizing its production facilities with semi-automated machinery and upgraded ERP systems to support cost control and performance monitoring.

SunBright is currently facing a multi-dimensional challenge:

- Increasing input costs (particularly raw materials and electricity)
- Intense competition from both multinational brands and low-cost domestic players
- Demand fluctuation due to seasonal sales cycles and changing consumer preferences
- Pressure on profit margins, especially in the washing machine segment
- Sustainability targets and pressure to reduce carbon footprint by optimizing energy usage and waste

To stay competitive, the board has asked the finance and operations teams to assess:

- Cost structures and process efficiencies
- Profitability by product line
- Feasibility of accepting foreign bulk orders
- Operational variances impacting budgets and performance

Following data is available for the quarter:

Particulars	ACs	WMs	RFs
Units Produced	5,000	6,000	4,000
Units Sold	5,000	6,000	4,000
Direct Material Cost per Unit	₹ 7,500	₹ 5,000	₹ 6,000
Direct Labour Hours per Unit	6 hrs	4 hrs	5 hrs
Direct Labour Rate	₹ 250/hr	₹ 250/hr	₹ 250/hr
Selling Price per Unit	₹ 18,000	₹ 14,000	₹ 16,000
Standard Labour Hours	5.5 hrs	3.5 hrs	5.5 hrs

Overhead Costs for the Quarter:

Activity	Total Overhead (₹)	Cost Driver	Driver Quantity (Total)
Machine Setups	₹ 9,00,000	No. of setups	180 setups
Quality Inspections	₹ 6,00,000	No. of inspections	300 inspections
Material Handling	₹ 7,50,000	No. of material moves	500 moves
Utilities & Maintenance	₹ 15,00,000	Machine hours	30,000 machine hours

Activity Driver Consumption:

Product	Setups	Inspections	Material Moves	Machine Hours
ACs	60	90	200	12,000
WMs	70	100	150	10,000
RFs	50	110	150	8,000

SunBright also received a special order from a foreign buyer for 500 WMs at ₹ 12,000 each. This order requires an Additional packaging cost of ₹ 150/unit. However, no marketing costs will be incurred. Also, it can only be accepted fully.

Budgeted overhead of the company is ₹ 37,00,000.

On the basis of above information and requirements of the board, you are required to answer the following questions:

MCQ 6 2 marks

What is the total product cost per unit using Activity-Based Costing for ACs, WMs, RFs respectively?

A ACs – ₹ 9,276.00, WMs – ₹ 6,212.50, RFs – ₹ 7,523.75

B ACs – ₹ 9,560.50, WMs – ₹ 6,500.00, RFs – ₹ 7,800.00

C ACs – ₹ 9,000.00, WMs – ₹ 6,065.55, RFs – ₹ 7,223.75

D ACs – ₹ 8,800.00, WMs – ₹ 6,100.00, RFs – ₹ 7,300.00

MCQ 7 2 marks

What is the profit per unit and total profit for ACs, WMs, RFs respectively?

A Profit/unit: ACs – ₹ 8,500.00, WMs – ₹ 7,500.00, RFs – ₹ 8,000.00; Total Profit: ACs – ₹ 4,25,00,000, WMs – ₹ 4,50,00,000, RFs – ₹ 3,20,00,000

B Profit/unit: ACs – ₹ 8,724.00, WMs – ₹ 7,787.50, RFs – ₹ 8,476.25; Total Profit: ACs – ₹ 4,36,20,000, WMs – ₹ 4,67,25,000, RFs – ₹ 3,39,05,000

C Profit/unit: ACs – ₹ 8,250.00, WMs – ₹ 7,250.00, RFs – ₹ 8,100.00; Total Profit: ACs – ₹ 4,12,50,000, WMs – ₹ 4,35,00,000, RFs – ₹ 3,24,00,000

D Profit/unit: ACs – ₹ 8,000.00, WMs – ₹ 7,000.00, RFs – ₹ 7,800.00; Total Profit: ACs – ₹ 4,00,00,000, WMs – ₹ 4,20,00,000, RFs – ₹ 3,12,00,000

MCQ 8 2 marks

Calculate Total Labour Efficiency Variance and Labour Rate Variance.

A Labour Efficiency Variance: ₹ 5,00,000 (Favorable), Labour Rate Variance: ₹ 0

B Labour Efficiency Variance: ₹ 7,50,000 (Adverse), Labour Rate Variance: ₹ 1,00,000 (Favorable)

C Labour Efficiency Variance: ₹ 8,75,000 (Adverse), Labour Rate Variance: ₹ 0

D Labour Efficiency Variance: ₹ 6,25,000 (Adverse), Labour Rate Variance: ₹ 2,50,000 (Adverse)

MCQ 9 2 marks

If it has idle capacity of only 200 WMs assuming 50% overheads is variable. Should the company accept the foreign order along with reason?

A Yes, because the offer price is above variable cost and no capacity issue exists

B Yes, because accepting the full order causes a net profit of ₹ 5,03,750

C No, because idle capacity is not sufficient for full order.

D No, because accepting the full order causes a loss of ₹ 11,87,500 due to opportunity cost of reduced regular production

MCQ 10 2 marks

What is the Total Overhead Variance?

A ₹ 25,000 (Favorable)

B ₹0

C ₹ 37,50,000 (Adverse)

D ₹ 50,000 (Adverse)

ICAI'S ANSWERS

MCQ 6

(A) ACs – ₹ 9,276.00, WMs – ₹ 6,212.50, RFs – ₹ 7,523.75

Statement Showing "Cost *per unit* - Activity Based Costing"

Product	ACs (₹)	WMs (₹)	RFs (₹)
Material	7,500	5,000	6,000
Labour Hours	6	4	5
Labour Rate/hr	250	250	250
Labour	1,500	1,000	1,250
Overhead (W.N.)	276.00	212.50	273.75
Total Cost	9,276.00	6,212.50	7,523.75

Working Notes

Cost for each activity cost driver:

Activity	Total Cost (₹)	No. of Activity Driver	Activity Rate
Machine Setups	₹ 9,00,000	180 setups	₹ 5,000/setup
Quality Inspections	₹ 6,00,000	300 inspections	₹ 2,000/inspection
Material Handling	₹ 7,50,000	500 moves	₹ 1,500/move
Utilities & Maint.	₹ 15,00,000	30,000 hrs	₹ 50/machine hour

Overhead Cost per Unit

Product	ACs (₹)	WMs (₹)	RFs (₹)
Setups	60 × 5,000 = 3,00,000	70 × 5,000 = 3,50,000	50 × 5,000 = 2,50,000
Inspections	90 × 2,000 = 1,80,000	100 × 2,000 2,00,000	110 × 2,000 = 2,20,000
Material Handling	200 × 1,500 = 3,00,000	150 × 1,500 = 2,25,000	150 × 1,500 = 2,25,000
Utilities & Maintenance	12,000 × 50 = 6,00,000	10,000 × 50 = 5,00,000	8,000 × 50 = 400,000
Total Overhead	13,80,000	12,75,000	10,95,000
Units Produced	5,000	6,000	4,000
Overhead per Unit	276.00	212.50	273.75

MCQ 7

(B) Profit/unit: ACs – ₹ 8,724.00, WMs – ₹ 7,787.50, RFs – ₹ 8,476.25; Total Profit: ACs – ₹ 4,36,20,000, WMs – ₹ 4,67,25,000, RFs – ₹ 3,39,05,000

Calculation of Profit per Unit and Total Profit

Product	ACs (₹)	WMs (₹)	RFs (₹)
Selling Price	18,000.0	14,000.0	16,000.0
Cost	9,276.0	6,212.5	7,523.75
Profit per Unit	8,724.0	7,787.5	8,476.25
Units Sold	5,000	6,000	4,000
Total Profit	4,36,20,000	4,67,25,000	3,39,05,000

MCQ 8

(C) Labour Efficiency Variance: ₹ 8,75,000 (Adverse), Labour Rate Variance: ₹ 0

Standard Labour Cost

Product	ACs	WMs	RFs
Standard Hours per unit	5.5 hrs	3.5 hrs	5.5 hrs
Rate (₹)	250	250	250
Units	5,000	6,000	4,000
Total Std Hours	27,500	21,000	22,000

Total Actual Labour Cost

Product	ACs	WMs	RFs
Actual Hours per unit	6 hrs	4 hrs	5 hrs
Units	5,000	6,000	4,000
Actual Hours	30,000	24,000	20,000

Labour Efficiency Variance

= (Standard Hours – Actual Hours) × Standard Rate

ACs = (27,500 – 30,000) × 250 = ₹ 6,25,000 (A)

WMs = (21,000 – 24,000) × 250 = ₹ 7,50,000 (A)

RFs = (22,000 – 20,000) × 250 = ₹ 5,00,000 (F)

Total Labour Efficiency Variance = ₹ 8,75,000 (A)

Labour Rate Variance

= (Standard Rate – Actual Rate) × Actual Hours

Standard and Actual Rate = ₹ 250

Labour Rate Variance = ₹ 0

MCQ 9

(B) Yes, because accepting the full order causes a net profit of ₹ 5,03,750

The incremental costs associated with the order must be considered to decide whether to accept it.

- Offered Price per WM: ₹ 12,000
- Direct Material Cost per WM: ₹ 5,000
- Direct Labour Cost per WM: ₹ 1,000
- Additional Packaging Cost per WM: ₹ 150

Because the company has idle capacity of 200 WMs, and only 50% of the overheads are variable, the remaining 50% are fixed and will not be impacted by the special order. The variable portion of the overheads for WMs is: ₹ 212.50 x 0.50 = ₹ 106.25 per unit.

- Incremental Cost per WM: 5,000 + 1,000 + 150 + 106.25 = ₹ 6,256.25
- Incremental Profit per WM: ₹ 12,000 - ₹ 6,256.25 = ₹ 5,743.75

The order requires 500 WMs, but the company only has idle capacity for 200. It must be accepted fully, implying that the company would need to reduce regular production by 300 units to fulfill the remaining 300 units of the order. SunBright would lose out on the profit of these 300 units if they accept the order.

- Profit on 200 units (idle capacity): 200 x ₹ 5,743.75 = ₹ 11,48,750
- Opportunity cost on 300 units (diverted from regular production):

= 300 x {(14,000-12,000) + 150}

= ₹ 6,45,000

Net profit on accepting the order = ₹ 5,03,750

Accepting the entire order would result in a net profit of ₹ 5,03,750. Therefore, the company should accept the foreign order.

MCQ 10

(D) ₹ 50,000 (Adverse)

Total Actual Overheads = ₹ 9,00,000 + ₹ 6,00,000 + ₹ 7,50,000 + ₹ 15,00,000

= ₹ 37,50,000

Budgeted Overheads = ₹ 37,00,000

Overhead Variance = Actual Overheads - Budgeted Overheads

= ₹ 37,50,000 - ₹ 37,00,000 = ₹ 50,000 (Adverse)

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485 MTP **May 2025 S-I** Q4(a) 6 marks

ChispyChip Ltd. manufactures 3 types of packed chips namely Crunchy Potato, Crispy Corn and Crumbly Banana and offers to sell them at wholesale price to the retailers. The following data is available for the current year:

	Crunchy Potato (₹)	Crispy corn (₹)	Crumbly Banana (₹)
Revenues	1,53,35,250	1,63,36,900	1,55,23,850
Cost of goods sold	1,12,70,000	1,27,37,400	1,18,45,000

The information regarding cost pool activities and their cost driver is also provided below:

Cost Pool Activity	Total Cost (₹)	Cost Driver
Placing of orders	17,94,000	No. of purchase orders
Physical delivery of goods	28,98,000	No. of deliveries
Stocking of goods on store shelves	19,87,200	Hours of shelf-stocking time
Supervising cost	13,80,000	Direct labour hours
Quality Inspections	9,03,900	No. of Inspections

Further, bifurcation of cost driver of the individual type of Chips is as follows:

Particulars	Crunchy Potato	Crispy corn	Crumbly Banana
Number of purchase orders placed	1,127	1,242	1,219
Number of deliveries made	2,185	2,668	2,392
Hours of shelf-stocking time	6,072	7,130	6,670
Direct Labour hours	19,159	19,435	18,906
Number of Inspections done	759	782	736

Due to banana being ripening in nature, it takes double the time during inspection than the other type of Chips.

You are required to CALCULATE the operating income and its percentage of the revenue for each product line:

(i) if the company allocates all the support cost to product lines on the basis of cost of goods sold.

(ii) if the company allocates all the support cost to product lines using an activity-based costing system.

ICAI'S ANSWER

Working notes:

1. Total support cost:

Particulars	Amount (₹)
Placing of orders	17,94,000
Physical delivery of goods	28,98,000
Stocking of goods on store shelves	19,87,200
Supervising cost	13,80,000
Quality Inspections	9,03,900
Total	**89,63,100**

2. Percentage of support cost to cost of goods sold (COGS):

$$= \frac{\text{Totalsupportcost}}{\text{Totalcostofgoodssold}} \times 100$$

$$= \left(\frac{₹89,63,100}{₹3,58,52,400} \text{ x } 100 \right)$$

$= 25\%$

3. Cost for each activity cost driver:

Activity (1)	Total cost (₹) (2)	Cost allocation base (3)	Cost driver rate (4) = [(2) ÷ (3)]
Placing of orders	17,94,000	1,127 + 1,242 + 1,219 = 3,588 purchase orders	₹ 500 per purchase order
Physical delivery of goods	28,98,000	2,185 + 2,668 + 2,392 = 7,245 deliveries	₹ 400 per delivery
Stocking of goods on store shelves	19,87,200	6,072 + 7,130 + 6,670 = 19,872 hours of shelf-stocking	₹ 100 per stocking hour
Supervising cost	13,80,000	19,159 + 19,435 + 18,906 = 57,500 direct labour hour	₹ 24 per direct labour hour
Quality Inspections	9,03,900	759 + 782 + (736 × 2*) = 3,013 inspections	₹ 300 per inspection

* As the inspection duration for Crumbly Banana Chips is double, its base would be doubled to calculate cost driver rate under Activity Based Costing.

(i) Statement of Operating income and its percentage of revenues for each product line

(When support costs are allocated to product lines on the basis of cost of goods sold of each product)

	Crunchy Potato (₹)	Crispy corn (₹)	Crumbly Banana (₹)	Total (₹)
Revenues: (A)	1,53,35,250	1,63,36,900	1,55,23,850	4,71,96,000
Cost of Goods sold (COGS): (B)	1,12,70,000	1,27,37,400	1,18,45,000	3,58,52,400
Support cost (25% of COGS): (C) (Refer working notes)	28,17,500	31,84,350	29,61,250	89,63,100
Total cost: (D) = {(B) + (C)}	1,40,87,500	1,59,21,750	1,48,06,250	4,48,15,500
Operating income: E= {(A)-(D)}	12,47,750	4,15,150	7,17,600	23,80,500
Operating income as a percentage of revenues: (E/A) × 100)	8.14%	2.54%	4.62%	5.04%

(ii) Statement of Operating income and its percentage of revenues for each product line

(When support costs are allocated to product lines using an activity-based costing system)

	Crunchy Potato (₹)	Crispy corn (₹)	Crumbly Banana (₹)	Total (₹)
Revenues: (A)	1,53,35,250	1,63,36,900	1,55,23,850	4,71,96,000
Cost & Goods sold	1,12,70,000	1,27,37,400	1,18,45,000	3,58,52,400
Placing of orders	5,63,500 (1,127 x ₹ 500)	6,21,000 (1,242 x ₹ 500)	6,09,500 (1,219 x ₹ 500)	17,94,000
Physical delivery of goods	8,74,000 (2,185 x ₹ 400)	10,67,200 (2,668 x ₹ 400)	9,56,800 (2,392 x ₹ 400)	28,98,000
Stocking of goods on store shelves	6,07,200 (6,072 x ₹ 100)	7,13,000 (7,130 x ₹ 100)	6,67,000 (6,670 x ₹ 100)	19,87,200
Supervising cost	4,59,816 (19,159 x ₹ 24)	4,66,440 (19,435 x ₹ 24)	4,53,744 (18,906 x ₹ 24)	13,80,000
Quality Inspections	2,27,700 (759 x ₹ 300)	2,34,600 (782 x ₹ 300)	4,41,600 (736 × 2 x ₹ 300)	9,03,900
Total cost: (B)	1,40,02,216	1,58,39,640	1,49,73,644	4,48,15,500
Operating income C: {(A)-(B)}	13,33,034	4,97,260	5,50,206	23,80,500
Operating income as a percentage of revenues: (C/A) × 100)	8.69%	3.04%	3.54%	5.04%

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486 MTP **May 2025 S-II** MCQ 15 2 marks MCQ

Production set up costs ₹ 3,50,000; Total production is 50,000 units of each of the products X and Y; Production in each run is 2,000 units of X or 5,000 units of Y. Company uses activity-based costing to calculate the unit cost of its products. Set-up cost per unit of Y will be

A ₹ 4.00

B ₹ 2.40

C ₹ 2.00

D ₹ 3.60

ICAI'S ANSWER

(C) ₹ 2 per unit

$$\text{Number of runs for X} = \frac{\text{Total production of X}}{\text{Batch size of X}} = 50{,}000/2{,}000 = 25 \text{ runs}$$

$$\text{Number of runs for Y} = \frac{\text{Total production of Y}}{\text{Batch size of Y}} = 50{,}000/5{,}000 = 10 \text{ runs}$$

$$\text{Set-up cost per run} = \frac{\text{Total set-up cost}}{\text{Total Runs}} = 3{,}50{,}000/35 = ₹\ 10{,}000 \text{ per run}$$

$$\text{Set-up cost per unit of Y} = \frac{\text{set-up cost per run}}{\text{Batch size of Y}} = 10{,}000/5{,}000 = ₹\ 2 \text{ per unit}$$

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487 MTP **May 2025 S-II** Q4(b) 6 marks

H-2025 Ltd. is a manufacturer of a range of goods. The cost structure of its different products is as follows:

Particulars	Product A	Product B	Product C	
Direct Materials	50	40	40	₹/u
Direct Labour @ ₹ 10/ hour	30	40	50	₹/u
Production Overheads	30	40	50	₹/u
Total Cost	110	120	140	₹/u
Quantity Produced	10,000	20,000	30,000	Units

H-2025 Ltd. was absorbing overheads on the basis of direct labour hours. A newly appointed management accountant has suggested that the company should introduce ABC system and has identified cost drivers and cost pools as follows:

Activity Cost Pool	Cost Driver	Associated Cost
Stores Receiving	Purchase Requisitions	2,96,000
Inspection	Number of Production Runs	8,94,000
Dispatch	Orders Executed	2,10,000
Machine Setup	Number of Setups	12,00,000

The following information is also supplied:

Details	Product A	Product B	Product C
No. of Setups	360	390	450
No. of Orders Executed	180	270	300
No. of Production Runs	750	1,050	1,200
No. of Purchase Requisitions	300	450	500

Required

CALCULATE activity based production cost of all the three products.

ICAI'S ANSWER

The total production overheads are ₹ 26,00,000:

Product A: 10,000 × ₹ 30 = ₹ 3,00,000

Product B: 20,000 × ₹ 40 = ₹ 8,00,000

Product C: 30,000 × ₹ 50 = ₹ 15,00,000

On the basis of ABC analysis this amount will be apportioned as follows:

Statement Showing "Activity Based Production Cost"

Activity Cost Pool	Cost Driver	Ratio	Total Amount (₹)	A (₹)	B (₹)	C (₹)
Stores Receiving	Purchase Requisition	6:9:10	2,96,000	71,040	1,06,560	1,18,400
Inspection	Production Runs	5:7:8	8,94,000	2,23,500	3,12,900	3,57,600
Dispatch	Orders Executed	6:9:10	2,10,000	50,400	75,600	84,000
Machine Setups	Setups	12:13:15	12,00,000	3,60,000	3,90,000	4,50,000
Total Activity Cost				7,04,940	8,85,060	10,10,000
Quantity Sold				10,000	20,000	30,000
Unit Cost (Overheads)				70.49	44.25	33.67
Add: Conversion Cost				80	80	90
Total				150.49	124.25	123.67

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488 MTP Jan 2025 S-I Q4(a) 6 marks

Xtyle Ltd. is a leading manufacturer in the textile industry, renowned for its commitment to quality and innovation. With decades of experience, the company specializes in producing a diverse range of textile products, including high-quality towels, designed to meet the varying needs of its customers. The company offers mainly three types of towel, viz. Hand towels, Kitchen towels and Gym towels, catering to both everyday use and specialized applications. Below are the key production data for a recent period:

Particulars	Hand towels	Kitchen towels	Gym towels
Production (units)	9,000	15,000	60,000
Machine hours per unit	10	18	14
Direct Labour hours per unit	4	12	8
Direct Material per unit (₹)	450	400	600

Currently, the company utilizes a traditional costing method, which assigns all production overhead costs based on the number of machine hours used. The overhead cost is calculated at a rate of ₹ 30 per machine hour. Additionally, the direct labor cost is charged at ₹ 100 per hour.

Now, the company plans to implement an Activity-Based Costing (ABC) system to enhance cost accuracy and provide a clearer understanding of the costs associated with each product.

The activity analysis is provided as under:

Particulars	Hand towels	Kitchen towels	Gym towels
Batch size (units)	450	1,500	3,000
Number of purchase orders per batch	3	10	8
Store delivery	45	80	125
Number of inspections per batch	5	4	3

Further, the total production overheads can be divided into several key categories. Machine setup costs account for 20% of the total, while inspection costs make up 35%. Material procurement-related costs represent 10%, and store delivery costs also constitute 10%. Finally, machine operation costs contribute 25% to the overall overheads. This breakdown provides insight into how resources are allocated across various activities within the production process.

You are required to CALCULATE the cost per unit of each product using -

(i) traditional method.

(ii) activity based costing principles.

ICAI'S ANSWER

(i) Statement Showing "Cost per unit - Traditional Method"

Particulars of Costs	Hand towels	Kitchen towels	Gym towels
	(₹)	**(₹)**	**(₹)**
Direct Materials	450	400	600
Direct Labour [(4, 12, 8 hours) × ₹ 100]	400	1,200	800
Production Overheads [(10, 18, 14 hours) × ₹ 30]	300	540	420
Cost per unit	**1,150**	**2,140**	**1,820**

(ii) Statement Showing "Cost per unit - Activity Based Costing"

Products	Hand towels	Kitchen towels	Gym towels
Production (units)	9,000	15,000	60,000
	(₹)	(₹)	(₹)
Direct Materials	40,50,000 (9,000 units x ₹ 450)	60,00,000 (15,000 units x ₹ 400)	3,60,00,000 (60,000 units x ₹ 600)
Direct Labour (refer Part (i) above)	36,00,000 (9,000 units x ₹ 400)	1,80,00,000 (15,000 units x ₹ 1,200)	4,80,00,000 (60,000 units x ₹ 800)
Setup Costs @ ₹ 1,44,000 per setup	28,80,000 (20 setups x ₹ 1,44,000)	14,40,000 (10 setups x ₹ 1,44,000)	28,80,000 (20 setups x ₹ 1,44,000)
Inspection Costs @ ₹ 63,000 per inspection	63,00,000 (100 inspections x ₹ 63,000)	25,20,000 (40 inspections x ₹ 63,000)	37,80,000 (60 inspections x ₹ 63,000)
Purchase Related Costs @ ₹ 11,250 per purchase order	6,75,000 (60 purchase orders x ₹ 11,250)	11,25,000 (100 purchase orders x ₹ 11,250)	18,00,000 (160 purchase orders x ₹ 11,250)
Store delivery costs @ ₹ 14,400 per store delivery	6,48,000 (45 store delivery x ₹ 14,400)	11,52,000 (80 store delivery x ₹ 14,400)	18,00,000 (125 store delivery x ₹ 14,400)
Machine Related Costs @ ₹ 7.5 per hour	6,75,000 (90,000 hours x ₹ 7.5)	20,25,000 (2,70,000 hours x ₹ 7.5)	63,00,000 (8,40,000 hours x ₹ 7.5)
Total Costs	**1,88,28,000**	**3,22,62,000**	**10,05,60,000**
Cost per unit (Total Cost ÷ no. of Units)	**2,092**	**2,151**	**1,676**

Working Notes:

A. Number of Batches, Purchase Orders, Inspections and Store Deliveries-

	Particulars	Hand towels	Kitchen towels	Gym towels	Total
A.	Production (units)	9,000	15,000	60,000	
B.	Batch Size (units)	450	1,500	3,000	
C.	Number of Batches (A÷B)	20	10	20	50
D.	Number of Purchase Order per batch	3	10	8	
E.	Total Purchase Orders [C × D]	60	100	160	320
F.	Number of Inspections per batch	5	4	3	
G.	Total Inspections [C × F]	100	40	60	200
H.	Total Store Deliveries	45	80	125	250

B. Total Machine Hours-

	Particulars	Hand towels	Kitchen towels	Gym towels
A.	Machine Hours per unit	10	18	14
B.	Production (units)	9,000	15,000	60,000
C.	Total Machine Hours [A × B]	90,000	2,70,000	8,40,000

Total Machine Hours = 12,00,000

Total Production Overheads-

= 12,00,000 hrs. × ₹ 30 = ₹ 3,60,00,000

C. Cost Driver Rates-

Cost Pool	%	Overheads (₹)	Cost Driver Basis	Cost Driver (Units)	Cost Driver Rate (₹)
Setup	20%	72,00,000	Number of batches	50	1,44,000 per Setup
Inspection	35%	1,26,00,000	Number of inspections	200	63,000 per Inspection
Purchases	10%	36,00,000	Number of purchase order	320	11,250 per Purchase order
Store delivery	10%	36,00,000	Number of store deliveries	250	14,400 per store delivery
Machine Operation	25%	90,00,000	Machine Hours	12,00,000	7.5 per Machine Hour

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489 MTP Jan 2025 S-II Q2(a) 10 marks

XYZ Constructions is a leading engineering and construction company providing a range of infrastructure and industrial services. Recently, they have been asked to quote for residential building construction (RBC) and industrial plant construction (IPC) projects. However, they are winning fewer RBC contracts than expected.

XYZ Constructions has a policy to price all jobs at budgeted total cost plus 50%. Overheads are currently absorbed on a labour-hour basis. The company believes that switching to activity-based costing (ABC) to absorb overheads would reduce the costs associated with RBC and make them more competitive.

You are provided with the following data:

Overhead category	Annual Overhead (₹ Lakhs)	Activity driver	Total number of activities per year
Supervisors	₹120	Site visits	600
Project Planners	₹ 80	Planning documents	300
Property related	₹400	Labour hours	50,000
Total	**₹600**		

For a typical **RBC**: Material cost: ₹ 5 lakhs, Labour hours: 1,200 hours, Site visits: 2 visits, Planning documents: 2 documents

For a typical **IPC**: Material cost: ₹ 12 lakhs, Labour hours: 2,500 hours, Site visits: 10 visits, Planning documents: 8 documents

Labour is paid at **₹ 100 per hour**.

Required:

(a) CALCULATE the cost and quoted price of an RBC and an IPC using labour hours to absorb the overheads.

(b) CALCULATE the cost and quoted price of an RBC and an IPC using ABC to absorb the overheads.

(c) Assuming that the cost of an RBC falls by nearly 7% and the price of an IPC rises by about 2% as a result of the change to ABC, SUGGEST possible pricing strategies for the two services offered by XYZ Constructions. Additionally, suggest two reasons other than high prices for the current poor sales of RBC.

ICAI'S ANSWER

(a) **Cost and Quoted Price Using Labour Hours to Absorb Overheads**

		RBC (₹ in lakhs)	IPC (₹ in lakhs)
Materials		5.00	12.00
Labour	1200 x ₹ 100; 2500 x ₹ 100	1.20	2.50
Overheads	1200 x ₹ 1200; 2500 x ₹ 1200	14.40	30.00
Total cost		20.60	44.50
Add: Profit	50% of Total Cost	10.30	22.25
Quoted Price		**30.90**	**66.75**

(b) **Cost and Quoted Price Using ABC**

Step 1: Calculate Overhead Rates for Each Activity

Overhead Category	Total Overhead (₹ Lakhs)	Activity Driver	Activity Rate
Site Engineers	₹120	Site Visits	₹ 120 / 600 = ₹ 20,000 per site visit
Project Planners	₹80	Planning Documents	₹ 80 / 300 = ₹ 26,667 per planning document
Equipment Depreciation	₹400	Labour Hours	₹ 400 / 50,000 = ₹ 800 per labour hour

Step 2: Allocate Overheads Using ABC

		RBC (in lakhs)	IPC (in lakhs)
Materials		5.00	12.00
Labour	1200 x ₹100; 2500 x ₹100	1.20	2.50
Overheads			
Site Engineers	2 x ₹ 20,000; 10 x ₹ 20,000	0.40	2.00
Project Planners	2 x ₹ 26,667; 8 x ₹ 26,667	0.53	2.13
Equipment Depreciation	1200 x ₹ 800; 2500 x ₹ 800	9.60	20.00
Total cost		16.73	38.63
Add: Profit	50% of Total Cost	8.37	19.32
Quoted Price		**25.10**	**57.95**

(c) **Possible pricing strategies for the two services offered by XYZ Constructions**

- The pricing policy is a matter for XYZ Constructions to decide. They could elect to maintain the current 50% mark-up on cost and if they did the price of the RBC would fall by around 7% in line with the costs. This should make them more competitive in the market.

- They could also reduce the prices by a little less than 7% (say 5%) in order to increase internal margins a little.

Reasons other than high prices for the current poor sales of RBC:

- If the quality of work or the reputation and reliability of the builder are questionable, lowering prices is unlikely to boost sales. While it is possible that XYZ Constructions has a strong reputation for IPC but not for RBC, it is more likely that a poor reputation would impact all their products. Poor service or inflexibility in meeting customer needs may also hurt sales and can't be fixed by lowering prices.
- Poor marketing strategies also discourage customers from selecting XYZ Constructions.
- XYZ Constructions faces competition and may need to adopt a more competitive pricing strategy, such as 'going rate pricing,' instead of simply adding a markup to costs.
- XYZ Constructions could enter the market by pricing some projects competitively to establish a foothold. Completed projects could then be leveraged to attract new customers.

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490 MTP **Sep 2024 S-I** MCQ 13 2 marks MCQ

One of Pintu Company's cost pools is parts administration. The budgeted overhead cost for that cost pool was ₹ 4,00,000 and the expected activity was 4,000 part types. The actual overhead cost for the cost pool was ₹ 4,20,000 at an actual activity of 5,000 part types. The activity rate for that cost pool was:

A ₹ 80 per part type

B ₹ 100 per part type

C ₹ 105 per part type

D ₹ 84 per part type

ICAI'S ANSWER

(B) ₹ 100 per part type

Activity rate = budgeted overhead/budgeted activity level

= 4,00,000/4,000

= ₹ 100 per part type

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491 MTP **Sep 2024 S-I** Q5(b) 7 marks

Icecold a FMCG Company manufactures and sells three flavors of ice cream:

Dark chocolate, Chocolate, and Butterscotch. The batch size for the ice cream is limited to 1,000 ice cream based on the size of the fridge and ice cream molds owned by the company. Based on budgetary projections, the information listed below is available:

	Dark chocolate	Chocolate	Butterscotch
Projected sales in units	500,000	800,000	600,000
PER UNIT data:			

	Dark chocolate	Chocolate	Butterscotch
Selling price	₹ 80	₹ 75	₹ 60
Direct materials	₹ 20	₹ 15	₹ 14
Direct labor	₹ 4	₹ 2	₹ 2
Hours per 1000-unit batch:			
Direct labor hours	20	10	10
Fridge hours	1	1	1
Packaging hours	0.5	0.5	0.5

Total overhead costs and activity levels for the year are estimated as follows:

Activity	Overhead costs	Activity levels
Direct labor		2,400 hours
Fridge	₹ 2,10,00,000	1,900 fridge hours
Packaging	₹ 1,50,00,000	950 packaging hours
	₹ 3,60,00,000	

Required:

a. With the help of ABC system, for the Chocolate ice cream:

1. Compute the activity-cost-driver rate
2. Compute the estimated overhead costs per thousand ice cream.
3. Compute the estimated operating profit per thousand ice cream.

b. With the help of traditional system (with direct labor hours as the overhead allocation base), for the Chocolate ice cream, compute the estimated operating profit per thousand ice cream.

ICAI'S ANSWER

a.

1. Estimation of cost-driver rate

Activity	Overhead cost (₹)	Cost driver	Cost driver rate (₹)
Packaging	1,50,00,000	950 Packaging hours	15,789.47
Fridge	2,10,00,000	1,900 Fridge hours	11,052.63

2. Overhead cost for chocolate ice cream

Activity	Overhead for a 1,000 ice cream batch	Amount (₹)
Packaging	1 x ₹ 11,052.63	11,052.63
Fridge	0.5 x ₹ 15,789.47	7,894.74
Total		18,947.37

3. Operating profit for chocolate ice cream

Particulars	Amount (₹)
Revenue (1,000 x ₹ 75)	75,000.00
Less: Direct Material (1,000 x ₹ 15)	15,000.00
Less: Direct Labour (10,000 x ₹ 2)	20,000.00
Less: Overhead	18,947.37
Operating Profit	21,052.63

b. Overhead per direct hour

= Total Overhead / Total Direct Labour Hours

= ₹ 3,60,00,000 / 24,000 hours

= ₹ 1,500 per direct labour hour

Since it takes 10 direct labour hour per 1,000 Chocolate ice cream, the overhead is ₹ 15,000

Particulars	Amount (₹)
Revenue (1,000 x ₹ 75)	75,000.00
Less: Direct Material (1,000 x ₹ 15)	15,000.00
Less: Direct Labour (10,000 x ₹ 2)	20,000.00
Less: Overhead	15,000
Operating Profit	25,000

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492 MTP May 2024 S-I Q5(b) 8 marks

The following budgeted information relates to Pinku Ltd. for the year 2024:

	Products		
	A	**B**	**C**
Production and Sales (units)	1,00,000	80,000	60,000
	(₹)	**(₹)**	**(₹)**
Selling price per unit	90	180	140
Direct cost per unit	50	90	95
	Hours	**Hours**	**Hours**
Machine department (machine hours per unit)	3	4	5
Assembly department (direct labour hours per unit)	6	4	3

The estimated overhead expenses for the year 2024 will be as below:

Machine Department	₹ 73,60,000
Assembly Department	₹ 55,00,000

Overhead expenses are apportioned to the products on the following basis:

Machine Department	On the basis of machine hours
Assembly Department	On the basis of labour hours

After a detailed study of the activities the following cost pools and their respective cost drivers are found:

Cost Pool	Amount (₹)	Cost Driver	Quantity
Machining services	64,40,000	Machine hours	9,20,000 hours
Assembly services	44,00,000	Direct labour hours	11,00,000 hours
Set-up costs	9,00,000	Machine set-ups	9,000 set-ups
Order processing	7,20,000	Customer orders	7,200 orders
Purchasing	4,00,000	Purchase orders	800 orders

As per an estimate the activities will be used by the three products:

	Products		
	A	B	C
Machine set-ups	4,500	3,000	1,500
Customer orders	2,200	2,400	2,600
Purchase orders	300	350	150

Prepare a product-wise profit statement using Activity-based method.

ICAI'S ANSWER

Profit Statement using Activity based costing (ABC) method:

	Particulars	Product			Total
		A	B	C	
A.	Sales Quantity	1,00,000	80,000	60,000	
B.	Selling price per unit (₹)	90	180	140	
C.	Sales Value (₹) [A×B]	90,00,000	1,44,00,000	84,00,000	3,18,00,000
D.	Direct cost per unit (₹)	50	90	95	
E.	Direct Cost (₹) [A×D]	50,00,000	72,00,000	57,00,000	1,79,00,000
F.	Overheads: (Refer working note-3)				
(i)	Machining services (₹)	21,00,000	22,40,000	21,00,000	64,40,000
(ii)	Assembly services (₹)	24,00,000	12,80,000	7,20,000	44,00,000
(iii)	Set-up costs (₹)	4,50,000	3,00,000	1,50,000	9,00,000
(iv)	Order processing (₹)	2,20,000	2,40,000	2,60,000	7,20,000
(v)	Purchasing (₹)	1,50,000	1,75,000	75,000	4,00,000
G.	Total Cost (₹) [E+F]	1,03,20,000	1,14,35,000	90,05,000	3,07,60,000
H.	Profit (₹) (C-G)	(13,20,000)	29,65,000	(6,05,000)	10,40,000

Working Notes:

1.

		Products			
		A	B	C	Total
A.	Production (units)	1,00,000	80,000	60,000	

		Products			
B.	Machine hours per unit	3	4	5	
C.	Total Machine hours [A×B]	3,00,000	3,20,000	3,00,000	9,20,000
D.	Rate per hour (₹)	8	8	8	
E.	**Machine Dept. cost [C×D]**	**24,00,000**	**25,60,000**	**24,00,000**	**73,60,000**
F.	Labour hours per unit	6	4	3	
G.	Total labour hours [A×F]	6,00,000	3,20,000	1,80,000	11,00,000
H.	Rate per hour (₹)	5	5	5	
I.	**Assembly Dept. cost [G×H]**	**30,00,000**	**16,00,000**	**9,00,000**	**55,00,000**

$$\text{Machine hour rate} = \frac{₹73,60,000}{9,20,000 \text{ hours}} = ₹8$$

$$\text{Labour hour rate} = \frac{₹55,00,000}{11,00,000 \text{ hours}} = ₹5$$

2. Calculation of cost driver rate

Cost Pool	Amount (₹)	Cost Driver	Quantity	Driver rate (₹)
Machining services	64,40,000	Machine hours	9,20,000 hours	7.00
Assembly services	44,00,000	Direct labour hours	11,00,000 hours	4.00
Set-up costs	9,00,000	Machine set-ups	9,000 set-ups	100.00
Order processing	7,20,000	Customer orders	7,200 orders	100.00
Purchasing	4,00,000	Purchase orders	800 orders	500.00

3. Calculation of activity-wise cost

		Products			
		A	**B**	**C**	**Total**
A.	Machining hours (Refer Working note-1)	3,00,000	3,20,000	3,00,000	9,20,000
B.	Machine hour rate (₹) (Refer Working note-2)	7	7	7	
C.	**Machining services cost (₹) [A×B]**	**21,00,000**	**22,40,000**	**21,00,000**	**64,40,000**
D.	Labour hours (Refer Working note-1)	6,00,000	3,20,000	1,80,000	11,00,000
E.	Labour hour rate (₹) (Refer Working note-2)	4	4	4	
F.	**Assembly services cost (₹) [D×E]**	**24,00,000**	**12,80,000**	**7,20,000**	**44,00,000**

		Products			
G.	Machine set-ups	4,500	3,000	1,500	9,000
H.	Rate per set-up (₹) (Refer Working note-2)	100	100	100	
I.	**Set-up cost (₹) [G×H]**	**4,50,000**	**3,00,000**	**1,50,000**	**9,00,000**
J.	Customer orders	2,200	2,400	2,600	7,200
K.	Rate per order (₹) (Refer Working note-2)	100	100	100	
L.	**Order processing cost (₹) [J×K]**	**2,20,000**	**2,40,000**	**2,60,000**	**7,20,000**
M.	Purchase orders	300	350	150	800
N.	Rate per order (₹) (Refer Working note-2)	500	500	500	
O.	**Purchasing cost (₹) [M×N]**	**1,50,000**	**1,75,000**	**75,000**	**4,00,000**

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493 MTP **May 2024 S-II** MCQ 4 2 marks MCQ

From the following information, calculate the Total cost of Product A and B using the ABC analysis:

	Product A	Product B
Units	5,000	5,000
Number of purchase orders placed	100	220
Number of deliveries received	70	200
Ordering Cost	₹ 4,00,000	
Delivery Cost	₹ 1,35,000	

A. A = ₹ 47,500; B = ₹ 1,27,500

B. A = ₹ 2,67,500; B = ₹ 2,67,500

C. A = ₹ 1,60,00; B = ₹ 3,75,000

D. A = ₹ 1,47,500; B = ₹ 1,47,500

ICAI'S ANSWER

Answer key: **(C)**

C

Ordering Cost = 4,00,000/320 = 1,250

Delivery Cost = 1,35,000/270 = 500

A = 1,250 x 100 + 500 x 70 = 1,60,000

B = 1,250 x 220 + 500 x 200 = 3,75,000

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494 MTP **May 2024 S-II** Q3(b) 4 marks

Anju Limited has collected the following data for its two activities. It calculates activity cost rates based on cost driver capacity.

Activity	Cost Driver	Capacity	Cost (₹)
Power	Kilowatt hours	60,000 kilowatt hours	60,00,000
Quality Inspections	Number of Inspections	10,000 Inspections	90,00,000

The company makes three products A, B and C. For the year ended March 31, 20XX, the following consumption of cost drivers was reported:

Product	Kilowatt hours	Quality Inspections
A	10,000	3,500
B	20,000	2,500
C	15,000	3,000

Required:

(i) PREPARE a statement showing cost allocation to each product from each activity.

(ii) CALCULATE the cost of unused capacity for each activity.

ICAI'S ANSWER

(i) **Statement of cost allocation to each product from each activity**

	Product			
	A (₹)	**B (₹)**	**C (₹)**	**Total (₹)**
Power (Refer to working note)	10,00,000 (10,000 kWh × ₹ 100)	20,00,000 (20,000 kWh × ₹ 100)	15,00,000 (15,000 kWh × ₹ 100)	45,00,000
Quality Inspections (Refer to working note)	31,50,000 (3,500 inspections × ₹ 900)	22,50,000 (2,500 inspections × ₹ 900)	27,00,000 (3,000 inspections × ₹ 900)	81,00,000

Working Note:

Rate per unit of cost driver:

Power : (₹ 60,00,000 ÷ 60,000 kWh) = ₹100/kWh

Quality Inspection: (₹ 90,00,000 ÷ 10,000 inspections) = ₹900 per inspection

(ii) **Calculation of cost of unused capacity for each activity:**

	(₹)
Power (₹60,00,000 – ₹45,00,000)	15,00,000
Quality Inspections (₹90,00,000 – ₹81,00,000)	9,00,000
Total cost of unused capacity	24,00,000

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495 MTP **Old P3 Nov 2023 S-II** Q2(a) 10 marks old scheme

HP bank offers three products, viz., deposits, Loans and Credit Cards. The bank has selected 4 activities for a detailed budgeting exercise, following activity-based costing methods.

The bank wants to know the product wise total cost per unit for the selected activities, so that prices may be fixed accordingly.

The following information is made available to formulate the budget:

Activity	Present Cost (₹)	Estimation for the budget period
ATM Services:		
Machine Maintenance	4,00,000	All fixed, no change.
Rents	2,00,000	Fully fixed, no change.
Currency Replenishment Cost	1,00,000	Expected to double during budget period.
	7,00,000	(This activity is driven by no. of ATM transactions)
Computer Processing	5,00,000	Half this amount is fixed and no change is expected. The variable portion is expected to increase to three times the current level. (This activity is driven by the number of computer transactions)
Issuing Statements	18,00,000	Presently, 3 lakh statements are made. In the budget period, 5 lakh statements are expected. For every increase of one lakh statement, one lakh rupees is the budgeted increase. (This activity is driven by the number of statements)
Computer Inquiries	2,00,000	Estimated to increase by 80% during the budget period. (This activity is driven by telephone minutes)

The activity drivers and their budgeted quantifies are given below:

Activity Drivers	Deposits	Loans	Credit Cards
No. of ATM Transactions	1,50,000	---	50,000
No. of Computer Processing Transactions	15,00,000	2,00,000	3,00,000
No. of Statements to be issued	3,50,000	50,000	1,00,000
Telephone Minutes	3,60,000	1,80,000	1,80,000

The bank budgets a volume of 58,600 deposit accounts, 13,000 loan accounts, and 14,000 Credit Card Accounts.

Required:

(i) Calculate the budgeted rate for each activity.

(ii) Prepare the budgeted cost statement activity wise.

(iii) Compute the budgeted product cost per account for each product using (i) and (ii) above.

ICAI'S ANSWER

Statement Showing "Budgeted Cost per unit of the Product"

Activity	Activity Cost (Budgeted) (₹)	Activity Driver	No. of Units of Activity Driver (Budget)	Activity Rate (₹)	Deposits	Loans	Credit Cards
ATM Services	8,00,000	No. of ATM Transaction	2,00,000	4.00	6,00,000	---	2,00,000
Computer Processing	10,00,000	No. of Computer processing Transaction	20,00,000	0.50	7,50,000	1,00,000	1,50,000
Issuing Statements	20,00,000	No. of Statements	5,00,000	4.00	14,00,000	2,00,000	4,00,000
Customer Inquiries	3,60,000	Telephone Minutes	7,20,000	0.50	1,80,000	90,000	90,000
Budgeted Cost	41,60,000				29,30,000	3,90,000	8,40,000
Units of Product (as estimated in the budget period)					58,600	13,000	14,000
Budgeted Cost per unit of the product					50	30	60

Working Note

Activity	Budgeted Cost (₹)	Remark
ATM Services:		
(a) Machine Maintenance	4,00,000	– All fixed, no change.
(b) Rents	2,00,000	– Fully fixed, no change.

Activity	Budgeted Cost (₹)	Remark
(c) Currency Replenishment Cost	2,00,000	– Doubled during budget period.
Total	8,00,000	
Computer Processing	2,50,000	– ₹ 2,50,000 (half of ₹5,00,000) is fixed and no change is expected.
	7,50,000	– ₹ 2,50,000 (variable portion) is expected to increase to three times the current level.
Total	10,00,000	
Issuing Statements	18,00,000	– Existing.
	2,00,000	– 2 lakh statements are expected to be increased in budgeted period. For every increase of one lakh statement, one lakh rupees is the budgeted increase.
Total	20,00,000	
Computer Inquiries	3,60,000	– Estimated to increase by 80% during the budget period.
Total	3,60,000	(₹ 2,00,000 × 180%)

ICAI · Old P3 · MTP Nov 2023 · Series II · question p. 2-3 · answer p. 4-5

496 MTP **Old P3 May 2023 S-I** Q5(a) 10 marks old scheme

Bopanna Ltd. produces three products Zm, Rm and Pm using the same plant and resources.

It has given the following information for the year ended on 31st March 2022:

	Zm	Rm	Pm
Production Quantity (units)	6000	7200	9840
Cost per unit:			
Direct Material (₹)	450	420	880
Direct Labour (₹)	80	150	200

Budgeted direct labour rate was ₹40 per hour and the production overheads, shown in table below, were absorbed to products using direct labour hour rate.

Company followed Absorption Costing Method. However, the company is now considering adopting Activity Based Costing Method.

	Budgeted Overheads (₹)	Cost Driver	Remarks
Material Procurement	2,50,000	No. of orders	No. of orders was 30 units for each product.
Set-up	1,50,000	No. of production Runs	All the three products are produced in production runs of 50 units.
Quality Control	1,00,000	No. of Inspections	Done for each production run.
Maintenance	3,00,000	Maintenance hours	Total maintenance hours were 10,000 and was allocated in the ratio of 2:1:2 between X, Y & Z.

Required:

(i) CALCULATE the total cost per unit of each product using the Absorption Costing Method.

(ii) CALCULATE the total cost per unit of each product using the Activity Based Costing Method.

ICAI'S ANSWER

(i) Traditional Absorption Costing

	Zm	*Rm*	*Pm*	Total
(a) Quantity (units)	6,000	7,200	9,840	23,040
(b) Direct labour per unit (₹)	80	150	200	-
(c) Direct labour hours (a × b)/₹ 40	12,000	27,000	49,200	88,200

Overhead rate per direct labour hour = Budgeted overheads / Budgeted labour hours

= (₹2,50,000 + ₹1,50,000 + ₹ 1,00,000 + ₹3,00,000) / 88,200 hours

= ₹8,00,000 / 88,200 hours

= ₹9 per direct labour hour(approx.)

Calculation of Cost per Unit

	Zm	*Rm*	*Pm*
Direct Costs:			
Direct Material	450	420	880
Direct Labour (₹)	80	150	200
Production Overhead: (₹)	18	33.75	45
	(80× 9/40)	(150× 9/40)	(200× 9/40)
Total cost per unit (₹)	**548**	**603.75**	**1125**

(ii) Calculation of Cost-Driver level under Activity Based Costing

	Zm	*Rm*	*Pm*	Total
Quantity (units)	6,000	7,200	9,840	-
No. of orders (to be rounded off for fraction)	200 (6,000 / 30)	240 (7,200 / 30)	328 (9,840 / 30)	768
No. of production runs	120 (6,000 / 50)	144 (7,200 / 50)	197 (9,840 / 50)	461
No. of Inspections (done for each production run)	120	144	197	461
Maintenance hours	4,000	2,000	4,000	10,000

Calculation of Cost-Driver rate

Activity	Budgeted Cost (₹) (a)	Cost-driver level (b)	Cost Driver rate (₹) (c) = (a) / (b)
Material procurement	2,50,000	768	325.5
Set-up	1,50,000	461	325.5
Quality control	1,00,000	461	217.0
Maintenance	3,00,000	10,000	30.0

Calculation of total cost of products using Activity Based Costing

Particulars	Product		
	Zm (₹)	***Rm (₹)***	***Pm (₹)***
Direct Material	450	420	880
Direct Labour	80	150	200
Prime Cost per unit (A)	530	570	1080
Material procurement	10.85 (325.5×200/6000)	10.85 (325.5×240/7200)	10.85 (325.5×328/9840)
Set-up	6.51 (325.5×120/6000)	6.51 (325.5×144/7200)	6.51 (325.5×196.8/9840)
Quality control	4.34 (217×120/6000)	4.34 (217×144/7200)	4.34 (217×196.8/9840)
Maintenance	20.0 (4000×30/6000)	8.3 (2000×30/7200)	12.2 (4000×30/9840)
Overhead Cost per unit (B)	41.7	30.0	33.9
Total Cost per unit (A + B)	**571.7**	**600.0**	**1113.9**

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497 MTP Old P3 May 2023 S-II Q3(a)(1) 5 marks old scheme

KD Ltd. is following Activity based costing. Budgeted overheads, cost drivers and volume are as follows:

Cost pool	Budgeted overheads (₹)	Cost driver	Budgeted volume
Material procurement	18,42,000	No. of orders	1,200
Material handling	8,50,000	No. of movement	1,240
Maintenance	24,56,000	Maintenance hours	17,550
Set-up	9,12,000	No. of set-ups	1,450
Quality control	4,42,000	No. of inspection	1,820

The company has produced a batch of 7,600 units, its material cost was ₹24,62,000 and wages ₹4,68,500. Usage activities of the said batch are as follows:

Material orders	56
Material movements	84
Maintenance hours	1,420 hours
Set-ups	60
No. of inspections	18

Required:

(i) CALCULATE cost driver rates.

(ii) CALCULATE the total and unit cost for the batch.

ICAI'S ANSWER

(i) Calculation of cost driver rate:

Cost pool	Budgeted overheads (₹)	Cost driver	Cost driver rate (₹)
Material procurement	18,42,000	1,200	1,535.00
Material handling	8,50,000	1,240	685.48
Maintenance	24,56,000	17,550	139.94
Set-up	9,12,000	1,450	628.97
Quality control	4,42,000	1,820	242.86

(ii) Calculation of cost for the batch:

Particulars	Amount (₹)	Amount (₹)
Material cost		24,62,000.00
Wages		4,68,500.00

Particulars	Amount (₹)	Amount (₹)
Overheads:		
- Material procurement (₹1,535×56 orders)	85,960.00	
- Material handling (₹685.48×84 movements)	57,580.32	
- Maintenance (₹139.94×1,420 hours)	1,98,714.80	
- Set-up (₹628.97×60 set-ups)	37,738.20	
- Quality control (₹242.86×18 inspections)	4,371.48	3,84,364.80
Total Cost		33,14,864.80
No. of units		7,600
Cost per units		436.17

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498 MTP **Old P3 Nov 2022 S-I** Q5(b) 10 marks old scheme

SMD Limited manufactures four products namely A, B, C and D using the same production and process facilities. The company has been following conventional method of costing and wishes to shift to activity-based costing system.

The data pertaining to four products are:

Product	Units produced	Material per unit (₹)	Labour hours per unit	Machine hours per unit
A	1,500	140	1	3
B	2,500	90	3	2
C	10,000	180	2	6
D	6,000	150	1.5	4

The following activity volumes are associated to the production process for the relevant period -

	Number of Inspections	Number of Material Movements	Number of set-ups
A	200	15	100
B	250	20	125
C	900	100	600
D	650	85	400

The cost data also states that:

- Direct Labour cost: ₹ 60 per hour
- Machine hour rate: ₹ 280 per hour

- Production overheads are absorbed on machine hour basis.
- For activity-based costing, a thorough, analysis of the production process revealed that:

Costs relating to set-ups and inspection bears the equal percentage while costs relating to machinery accounts for 20% of the production overhead.

Costs relating to material handling stands at 50% of costs relating to machinery.

You are required to:

(i) Prepare a statement showing the unit costs and total costs of each product using the absorption costing method.

(ii) Prepare a statement showing the unit costs and total costs of each product using activity-based costing system.

ICAI'S ANSWER

(i) Cost per unit - Conventional Costing: Absorption rate method

Particulars	A (₹)	B (₹)	C (₹)	D (₹)
Material	140	90	180	150
Labour @ ₹ 60 per labour hour	60	180	120	90
Overheads @ ₹ 280 per machine hour	840	560	1680	1120
Cost per unit (in ₹)	**1,040**	**830**	**1,980**	**1,360**
No of units	1,500	2,500	10,000	6,000
Total cost (₹)	**15,60,000**	**20,75,000**	**1,98,00,000**	**81,60,000**

(ii) Statement of apportionment of overheads: Amount (₹)

Type of Cost	Cost Driver	A	B	C	D
Setups	No of Setups	7,48,000 (100 x 7,480)	9,35,000 (125×7,480)	44,88,000 (600 x 7,480)	29,92,000 (400 x7,480)
Machinery	Machine hours	2,52,000 (4,500 × 56)	2,80,000 (5,000 × 56)	33,60,000 (60,000 × 56)	13,44,000 (24,000 × 56)
Material Handling	No. of Movements of material	1,78,500 (15 x 11,900)	2,38,000 (20 x 11,900)	11,90,000 (100 x 11,900)	10,11,500 (85 x 11,900)
Inspection	No. of Inspections	9,16,300 (200×4,581.50)	11,45,375 (250×4,581.50)	41,23,350 (900×4,581.50)	29,77,975 (650×4,581.50)
Total		**20,94,800**	**25,98,375**	**1,31,61,350**	**83,25,475**
Output Units		1,500	2,500	10,000	6,000
Overhead/ unit		1,396.53	1,039.35	1,316.14	1,387.58

Statement showing Cost per unit and Total cost using Activity Based Costing

Particulars	A (₹)	B (₹)	C (₹)	D (₹)
Material	140.00	90.00	180.00	150.00
Labour	60.00	180.00	120.00	90.00
Total	**200.00**	**270.00**	**300.00**	**240.00**
No. of units	1,500	2,500	10,000	6,000
Total cost (excluding overheads)	3,00,000	6,75,000	30,00,000	14,40,000
Add: Overheads (as calculated)	20,94,800	25,98,375	1,31,61,350	83,25,475
Total cost	**23,94,800**	**32,73,375**	**1,61,61,350**	**97,65,475**
Cost per unit	**1,596.53**	**1,309.35**	**1,616.14**	**1,627.58**

Working Notes:

1. Calculation of Total machine hours

Particulars	A	B	C	D
(a) Machine hours per unit	3	2	6	4
(b) Production(units)	1,500	2,500	10,000	6,000
(c) Total machine hours (a) x(b)	4,500	5,000	60,000	24,000

Total Machine hours = 93,500

Total production overheads= 93,500 x 280 = **₹ 2,61,80,000**

2. Calculation of cost driver rate

Cost pool	Amount of cost (₹)	Cost Driver (basis)	Cost Driver (units)	Cost Driver Rate (₹)
Setups	91,63,000	No. of Setups	1,225	7,480 per set up
Machinery	52,36,000	Machine Hrs.	93,500	56 per machine hour
Material Handlings	26,18,000	No. of Material Movements	220	11,900 per material movement
Inspection	91,63,000	No. of Inspections	2,000	4,581.50 per inspection
	2,61,80,000			

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499 MTP Old P3 Nov 2022 S-II Q3(b) 10 marks old scheme

ANI Limited is a trader of a Product Z. It has decided to analyse the profitability of its five new customers. It buys Z article at ₹5,400 per unit and sells to retail customers at a listed price of ₹6,480 per unit. The data pertaining to five customers are:

	Customers				
	A	B	C	D	E
Units sold	4,500	6,000	9,500	7,500	12,750
Listed Selling Price	₹6,480	₹6,480	₹6,480	₹6,480	₹6,480
Actual Selling Price	₹6,480	₹6,372	₹5,940	₹6,264	₹5,832
Number of Purchase orders	15	25	30	25	30
Number of Customer visits	2	3	6	2	3
Number of deliveries	10	30	60	40	20
Kilometers travelled per delivery	20	6	5	10	30
Number of expedited deliveries	0	0	0	0	1

Its five activities and their cost drivers are:

Activity	Cost Driver Rate
Order taking	₹4,500 per purchase order
Customer visits	₹3,600 per customer visit
Deliveries	₹7.50 per delivery Km travelled
Product handling	₹22.50 per case sold
Expedited deliveries	₹13,500 per expedited delivery

Required:

(i) COMPUTE the customer-level operating income of each of five retail customers (A, B, C, D and E).

(ii) STATE the factors ANI Limited should consider in deciding whether to drop a customer.

ICAI'S ANSWER

Working note:

1. Computation of revenues (at listed price), discount, cost of goods sold and customer level operating activities costs:

	Customers				
	A	B	C	D	E
Units sold: (a)	4,500	6,000	9,500	7,500	12,750

	Customers				
Revenues (at listed price) (Rs.): (b) {(a) ×Rs.6,480)}	2,91,60,000	3,88,80,000	6,15,60,000	4,86,00,000	8,26,20,000
Revenues (at listed price) (Rs.): (c) {(a) ×Actual selling price)}	2,91,60,000 (4,500×6,480)	3,82,32,000 (6,000×6,372)	5,64,30,000 (9,500×5,940)	4,69,80,000 (7,500×6,264)	7,43,58,000 (12,750×5,832)
Discount (Rs.) (d) {(b) – (c)}	0	6,48,000	51,30,000	16,20,000	82,62,000
Cost of goods sold (Rs.) : (d) {(a) x Rs.5,400}	2,43,00,000	3,24,00,000	5,13,00,000	4,05,00,000	6,88,50,000
Customer level operating activities costs					
Order taking costs (Rs.): (No. of purchase orders × Rs. 4,500)	67,500	1,12,500	1,35,000	1,12,500	1,35,000
Customer visits costs (Rs.) (No. of customer visits x Rs. 3,600)	7,200	10,800	21,600	7,200	10,800
Delivery vehicles travel costs (Rs.) (Kms travelled by delivery vehicles x Rs. 7.50 per km.)	1,500	1,350	2,250	3,000	4,500

	Customers				
Product handling costs (Rs.) {(a) x Rs. 22.50}	1,01,250	1,35,000	2,13,750	1,68,750	2,86,875
Cost of expediting deliveries (Rs.) {No. of expedited deliveries x Rs. 13,500}	-	-	-	-	13,500
Total cost of customer level operating activities (Rs.)	1,77,450	2,59,650	3,72,600	2,91,450	4,50,675

(i) Computation of Customer level operating income

	Customers				
	A	**B**	**C**	**D**	**E**
	(Rs.)	**(Rs.)**	**(Rs.)**	**(Rs.)**	**(Rs.)**
Revenues (At list price) (Refer to working note)	2,91,60,000	3,82,32,000	5,64,30,000	4,69,80,000	7,43,58,000
Less: Cost of goods sold (Refer to working note)	(2,43,00,000)	(3,24,00,000)	(5,13,00,000)	(4,05,00,000)	(6,88,50,000)
Gross margin	48,60,000	58,32,000	51,30,000	64,80,000	55,08,000
Less: Customer level operating activities costs (Refer to working note)	(1,77,450)	(2,59,650)	(3,72,600)	(2,91,450)	(4,50,675)
Customer level operating income	46,82,550	55,72,350	47,57,400	61,88,550	50,57,325

(ii) Factors to be considered for dropping a customer:

Dropping customers should be the last resort to be taken by an entity. Factors to be considered should include:

- What is the expected future profitability of each customer?

- Are the currently least profitable or low profitable customers are likely to be highly profitable in the future?
- What costs are avoidable if one or more customers are dropped?
- Can the relationship with the "problem" customers be restructured so that there is at "win- win" situation

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500 MTP **Old P3 May 2022 S-I** Q5(a) 10 marks old scheme

MG Ltd. manufactures three types of products namely A, B and C. The data relating to a period are as under:

Particulars	A	B	C
Machine hours per unit	10	18	14
Direct Labour hours per unit	4	12	8
Direct Material per unit (₹)	1,350	1,200	1,800
Production (units)	3,000	5,000	20,000

Currently the company uses traditional costing method and absorbs all production overheads on the basis of machine hours. The machine hour rate of overheads is ₹ 90 per hour. Direct labour hour rate is ₹ 300 per hour.

The company proposes to use activity based costing system and the activity analysis is as under:

Particulars	A	B	C
Batch size (units)	150	500	1,000
Number of purchase orders per batch	3	10	8
Number of inspections per batch	5	4	3

The total production overheads are analysed as under:

Machine set up costs	20%
Machine operation costs	30%
Inspection costs	40%
Material procurement related costs	10%

Required:

(i) CALCULATE the cost per unit of each product using traditional method of absorbing all production overheads on the basis of machine hours.

(ii) CALCULATE the cost per unit of each product using activity based costing principles.

ICAI'S ANSWER

(i) Statement Showing "Cost per unit - Traditional Method"

Particulars of Costs	A	B	C
	(₹)	**(₹)**	**(₹)**
Direct Materials	1,350	1,200	1,800
Direct Labour [(4, 12, 8 hours) × ₹ 300]	1,200	3,600	2,400
Production Overheads [(10, 18, 14 hours) × ₹ 90]	900	1,620	1,260
Cost per unit	**3,450**	**6,420**	**5,460**

(ii) Statement Showing "Cost per unit - Activity Based Costing"

Products	A	B	C
Production (units)	**3,000**	**5,000**	**20,000**
	(₹)	(₹)	(₹)
Direct Materials (1350, 1200, 1800)	40,50,000	60,00,000	3,60,00,000
Direct Labour (1200, 3600, 2400)	36,00,000	1,80,00,000	4,80,00,000
Machine Related Costs @ ₹ 27 per hour (30,000, 90,000, 2,80,000)	8,10,000	24,30,000	75,60,000
Setup Costs @ ₹ 1,44,000 per setup (20, 10, 20)	28,80,000	14,40,000	28,80,000
Inspection Costs @ ₹ 72,000 per inspection (100, 40, 60)	72,00,000	28,80,000	43,20,000
Purchase Related Costs @ ₹ 11,250 per purchase (60, 100, 160)	6,75,000	11,25,000	18,00,000
Total Costs	**1,92,15,000**	**3,18,75,000**	**10,05,60,000**
Cost per unit (Total Cost ÷ Units)	**6,405**	**6,375**	**5,028**

Working Notes:

1. Number of Batches, Purchase Orders, and Inspections-

	Particulars	A	B	C	Total
A.	Production (units)	3,000	5,000	20,000	
B.	Batch Size (units)	150	500	1,000	
C.	Number of Batches [A. ÷ B.]	20	10	20	50
D.	Number of Purchase Order per batch	3	10	8	
E.	Total Purchase Orders [C. × D.]	60	100	160	320
F.	Number of Inspections per batch	5	4	3	
G.	Total Inspections [C. × F.]	100	40	60	200

2. Total Machine Hours-

	Particulars	A	B	C
A.	Machine Hours per unit	10	18	14
B.	Production (units)	3,000	5,000	20,000
C.	Total Machine Hours [A. × B.]	30,000	90,000	2,80,000

Total Machine Hours = 4,00,000

Total Production Overheads-

= 4,00,000 hrs. × ₹ 90 = ₹ 3,60,00,000

3. Cost Driver Rates-

Cost Pool	%	Overheads (₹)	Cost Driver Basis	Cost Driver (Units)	Cost Driver Rate (₹)
Setup	20%	72,00,000	Number of batches	50	1,44,000 per Setup
Inspection	40%	1,44,00,000	Number of inspections	200	72,000 per Inspection
Purchases	10%	36,00,000	Number of purchases	320	11,250 per Purchase
Machine Operation	30%	1,08,00,000	Machine Hours	4,00,000	27 per Machine Hour

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501 MTP **Old P3 Dec 2021 S-I** Q2(b) 10 marks old scheme

Breezle Ltd has decided to analyse the profitability of its five new customers. It buys soft drink bottles in cases at ₹ 54 per case and sells them to retail customers at a list price of ₹ 64.80 per case. The data

pertaining to five customers are given below:

Particulars	Customers				
	Aey	**Bee**	**Cee**	**Dee**	**Eey**
Number of Cases Sold	9,360	14,200	62,000	38,000	9,800
List Selling Price (₹)	64.80	64.80	64.80	64.80	64.80
Actual Selling Price (₹)	64.80	64.08	58.80	60.24	58.32
Number of Purchase Orders	30	50	60	50	60
Number of Customers visits	4	6	12	4	6
Number of Deliveries	20	60	120	80	40
Kilometers travelled per delivery	40	12	10	20	60
Number of expediate Deliveries	0	0	0	0	2

Its five activities and their cost drivers are:

Activity	Cost Driver
Order taking	₹ 240 per purchase order
Customer visits	₹ 360 per each visit
Deliveries	₹ 4.80 per delivery km travelled
Product Handling	₹ 2.40 per case sold
Expedited deliveries	₹ 120 per such delivery

You are REQUIRED to :

(i) Compute the customer level operating income of each of five retail customers by using the Cost Driver rates.

(ii) Examine the results to give your comments on Customer 'Dee' in comparison with Customer 'Cee' and on Customer 'Eey' in comparison with Customer 'Aey'.

ICAI'S ANSWER

Working note:

Computation of revenues (at listed price), discount, cost of goods sold and customer level operating activities costs:

Particulars	Customers				
	Aey	**Bee**	**Cee**	**Dee**	**Eey**
Cases sold: (a)	9,360	14,200	62,000	38,000	9,800

Particulars	Customers				
Revenues (at listed price) (₹): (b) {(a) × ₹ 64.80)}	6,06,528	9,20,160	40,17,600	24,62,400	6,35,040
Discount (₹): (c) {(a) × Discount per case}	-	10,224 (14,200 cases × ₹ 0.72)	3,72,000 (62,000 cases × ₹ 6)	1,73,280 (38,000 cases × ₹ 4.56)	63,504 (9,800 cases × ₹ 6.48)
Cost of goods sold (₹): (d) {(a) × ₹ 54}	5,05,440	7,66,800	33,48,000	20,52,000	5,29,200
Customer level operating activities costs					
Order taking costs (₹): (No. of purchase × ₹ 240)	7,200	12,000	14,400	12,000	14,400
Customer visits costs (₹) (No. of customer visits × ₹ 360)	1,440	2,160	4,320	1,440	2,160
Delivery vehicles travel costs (₹) (Kms travelled by delivery vehicles × ₹ 4.80 per km.)	3,840	3,456	5,760	7,680	11,520
Product handling costs (₹) {(a) ×₹ 2.40}	22,464	34,080	1,48,800	91,200	23,520
Cost of expediting deliveries (₹) {No. of expedited deliveries × ₹ 120}	-	-	-	-	240
Total cost of customer level operating activities (₹)	34,944	51,696	1,73,280	1,12,320	51,840

(i) **Computation of Customer level operating income**

Particulars	Customers				
	Aey (₹)	**Bee (₹)**	**Cee (₹)**	**Dee (₹)**	**Eey (₹)**
Revenues (At list price) (Refer to working note)	6,06,528	9,20,160	40,17,600	24,62,400	6,35,040
Less: Discount (Refer to working note)	-	10,224	3,72,000	1,73,280	63,504
Revenue (At actual price)	6,06,528	9,09,936	36,45,600	22,89,120	5,71,536

Particulars	Customers				
Less: Cost of goods sold (Refer to working note)	5,05,440	7,66,800	33,48,000	20,52,000	5,29,200
Gross margin	1,01,088	1,43,136	2,97,600	2,37,120	42,336
Less: Customer level operating activities costs (Refer to working note)	34,944	51,696	1,73,280	1,12,320	51,840
Customer level operating income	66,144	91,440	1,24,320	1,24,800	(9,504)

(ii) **Comments**

Customer Dee in comparison with Customer Cee: Operating income of Customer Dee is more than that of Customer Cee, despite having only 61.29% (38,000 units) of the units volume sold in comparison to Customer Cee (62,000 units). Customer Cee receives a higher percent of discount i.e. 9.26% (₹ 6) while Customer Dee receive a discount of 7.04% (₹ 4.56). Though the gross margin of customer Cee (₹ 2,97,600) is more than that of Customer Dee (₹ 2,37,120) but total cost of customer level operating activities of Cee (₹ 1,73,280) is more in comparison to Customer Dee (₹ 1,12,320). As a result, operating income is more in case of Customer Dee.

Customer Eey in comparison with Customer Aey: Customer Eey is not profitable while Customer Aey is profitable. Customer Eey receives a discount of 10% (₹ 6.48) while Customer Aey doesn't receive any discount. Sales Volume of Customer Aey and Eey is almost same. However, total cost of customer level operating activities of Eey is far more (₹ 51,840) in comparison to Customer Aey (₹ 34,944). This has resulted in occurrence of loss in case of Customer Eey.

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502 MTP **Old P3 Dec 2021 S-II** Q5(a) 10 marks old scheme

The following budgeted information relates to B Ltd. for the year 2021:

	Products		
	X	**Y**	**Z**
Production and Sales (units)	1,00,000	80,000	60,000
	(₹)	(₹)	(₹)
Selling price per unit	45	90	70
Direct cost per unit	25	45	50
	Hours	**Hours**	**Hours**
Machine department (machine hours per unit)	3	4	5
Assembly department (direct labour hours per unit)	6	4	3

The estimated overhead expenses for the year 2021 will be as below:

Machine Department	₹ 36,80,000
Assembly Department	₹ 27,50,000

Overhead expenses are apportioned to the products on the following basis:

Machine Department	On the basis of machine hours
Assembly Department	On the basis of labour hours

After a detailed study of the activities the following cost pools and their respective cost drivers are found:

Cost Pool	Amount (₹)	Cost Driver	Quantity
Machining services	32,20,000	Machine hours	9,20,000 hours
Assembly services	22,00,000	Direct labour hours	11,00,000 hours
Set-up costs	4,50,000	Machine set-ups	9,000 set-ups
Order processing	3,60,000	Customer orders	7,200 orders
Purchasing	2,00,000	Purchase orders	800 orders

As per an estimate the activities will be used by the three products:

	Products		
	X	Y	Z
Machine set-ups	4,500	3,000	1,500
Customer orders	2,200	2,400	2,600
Purchase orders	300	350	150

You are required to PREPARE a product-wise profit statement using:

(i) Absorption costing method;

(ii) Activity-based method.

ICAI'S ANSWER

(i) Profit Statement using Absorption costing method:

	Particulars	Product			Total
		X	Y	Z	
A.	Sales Quantity	1,00,000	80,000	60,000	2,40,000
B.	Selling price per unit (₹)	45	90	70	
C.	Sales Value (₹) [A×B]	**45,00,000**	**72,00,000**	**42,00,000**	**1,59,00,000**

	Particulars	Product			Total
D.	Direct cost per unit (₹)	25	45	50	
E.	Direct Cost (₹) [A×D]	25,00,000	36,00,000	30,00,000	91,00,000
F.	Overheads:				
(i)	Machine department (₹) (Working note-1)	12,00,000	12,80,000	12,00,000	36,80,000
(ii)	Assembly department (₹) (Working note-1)	15,00,000	8,00,000	4,50,000	27,50,000
G.	Total Cost (₹) [E+F]	**52,00,000**	**56,80,000**	**46,50,000**	**1,55,30,000**
H.	**Profit (C-G)**	**(7,00,000)**	**15,20,000**	**(4,50,000)**	**3,70,000**

(ii) Profit Statement using Activity based costing (ABC) method:

	Particulars	Product			Total
		X	**Y**	**Z**	
A.	Sales Quantity	1,00,000	80,000	60,000	
B.	Selling price per unit (₹)	45	90	70	
C.	**Sales Value (₹) [A×B]**	**45,00,000**	**72,00,000**	**42,00,000**	**1,59,00,000**
D.	Direct cost per unit (₹)	25	45	50	
E.	Direct Cost (₹) [A×D]	25,00,000	36,00,000	30,00,000	91,00,000
F.	Overheads: (Refer working note-3)				
(i)	Machining services (₹)	10,50,000	11,20,000	10,50,000	32,20,000
(ii)	Assembly services (₹)	12,00,000	6,40,000	3,60,000	22,00,000
(iii)	Set-up costs (₹)	2,25,000	1,50,000	75,000	4,50,000
(iv)	Order processing (₹)	1,10,000	1,20,000	1,30,000	3,60,000
(v)	Purchasing (₹)	75,000	87,500	37,500	2,00,000
G.	**Total Cost (₹) [E+F]**	**51,60,000**	**57,17,500**	**46,52,500**	**1,55,30,000**
H.	**Profit (₹) (C-G)**	**(6,60,000)**	**14,82,500**	**(4,52,500)**	**3,70,000**

Working Notes:

(1)

		Products			
		X	**Y**	**Z**	**Total**
A.	Production (units)	1,00,000	80,000	60,000	
B.	Machine hours per unit	3	4	5	
C.	Total Machine hours [A×B]	3,00,000	3,20,000	3,00,000	9,20,000
D.	Rate per hour (₹)	4	4	4	
E.	**Machine Dept. cost [C×D]**	**12,00,000**	**12,80,000**	**12,00,000**	**36,80,000**
F.	Labour hours per unit	6	4	3	
G.	Total labour hours [A×F]	6,00,000	3,20,000	1,80,000	11,00,000
H.	Rate per hour (₹)	2.5	2.5	2.5	
I	**Assembly Dept. cost [G×H]**	**15,00,000**	**8,00,000**	**4,50,000**	**27,50,000**

$$\text{Machine hour rate} = \frac{₹\ 36,80,000}{9,20,000\ \text{hours}} = ₹\ 4$$

$$\text{Labour hour rate} = \frac{₹\ 27,50,000}{11,00,000\ \text{hours}} = ₹\ 2.5$$

2. Calculation of cost driver rate

Cost Pool	Amount (₹)	Cost Driver	Quantity	Driver rate (₹)
Machining services	32,20,000	Machine hours	9,20,000 hours	3.50
Assembly services	22,00,000	Direct labour hours	11,00,000 hours	2.00
Set-up costs	4,50,000	Machine set-ups	9,000 set-ups	50.00
Order processing	3,60,000	Customer orders	7,200 orders	50.00
Purchasing	2,00,000	Purchase orders	800 orders	250.00

3. Calculation of activity-wise cost

		Products			
		X	**Y**	**Z**	**Total**
A.	Machining hours (Refer Working note-1)	3,00,000	3,20,000	3,00,000	9,20,000
B.	Machine hour rate (₹) (Refer Working note-2)	3.5	3.5	3.5	
C.	**Machining services cost (₹) [A×B]**	**10,50,000**	**11,20,000**	**10,50,000**	**32,20,000**
D.	Labour hours (Refer Working note-1)	6,00,000	3,20,000	1,80,000	11,00,000

		Products			
E.	Labour hour rate (₹) (Refer Working note-2)	2	2	2	
F.	**Assembly services cost (₹) [D×E]**	**12,00,000**	**6,40,000**	**3,60,000**	**22,00,000**
G.	Machine set-ups	4,500	3,000	1,500	9,000
H.	Rate per set-up (₹) (Refer Working note-2)	50	50	50	
I.	**Set-up cost (₹) [G×H]**	**2,25,000**	**1,50,000**	**75,000**	**4,50,000**
J.	Customer orders	2,200	2,400	2,600	7,200
K.	Rate per order (₹) (Refer Working note-2)	50	50	50	
L.	**Order processing cost (₹) [J×K]**	**1,10,000**	**1,20,000**	**1,30,000**	**3,60,000**
M.	Purchase orders	300	350	150	800
N.	Rate per order (₹) (Refer Working note-2)	250	250	250	
O.	**Purchasing cost (₹) [M×N]**	**75,000**	**87,500**	**37,500**	**2,00,000**

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503 MTP **Old P3 Jul 2021 S-I** Q4(b) 10 marks old scheme

ABY Ltd. manufactures four products, namely A, B, C and D using the same plant and process. The following information relates to production period December, 2020:

Product	A	B	C	D
Output in units	1,440	1,200	960	1,008
Cost per unit:				
Direct Materials	Rs. 84	Rs. 90	Rs. 80	Rs. 96
Direct Labour	Rs. 20	Rs. 18	Rs. 14	Rs. 16
Machine hours per unit	4	3	2	1

The four products are similar and are usually produced in production runs of 48 units per batch and are sold in batches of 24 units. Currently, the production overheads are absorbed using machine hour rate. The production overheads incurred by the company for the period December, 2020 are as follows:

	(Rs.)
Machine department costs:	
Rent, deprecation and supervision	2,52,000
Set-up Costs	80,000
Store receiving costs	60,000
Inspection	40,000
Material handling and dispatch	10,368

During the period December, 2020, the following cost drivers are to be used for allocation of overheads cost:

Cost	Cost driver
Set-up Costs	Number of production runs (batches)
Stores receiving	Requisition raised
Inspection	Number of production runs (batches)
Material handling and dispatch	Orders executed

It is also determined that:

(i) Machine department costs should be apportioned among set-up, stores receiving and inspection activities in proportion of 4 : 3 : 2.

(ii) The number of requisitions raised on stores is 50 for each product. The total number of material handling and dispatch orders executed during the period are 192 and each order being for a batch size of 24 units of product.

Required:

(i) CALCULATE the total cost of each product, if all overhead costs are absorbed on machine-hour rate basis.

(ii) CALCULATE the total cost of each product using activity-based costing.

ICAI'S ANSWER

(i) Total Overhead = Rs. (2,52,000 + 80,000 + 60,000 + 40,000 + 10,368) = Rs. 4,42,368

Total machine hours = 1,440 × 4 + 1,200 × 3 + 960 × 2 + 1,008 × 1

= 5,760 + 3,600 + 1,920 + 1,008 = 12,288 M. Hrs.

$\therefore$ Overhead recovery rate / M.H. = $\frac{\text{Rs. } 4,42,368}{12,288 \text{ M.Hrs.}}$ = Rs. 36

Cost Statement when overheads are absorbed on machine hours rate basis

Product	A	B	C	D
Output in units	1,440	1,200	960	1,008
	(Rs.)	**(Rs.)**	**(Rs.)**	**(Rs.)**
Cost per unit:				
Direct material	84	90	80	96
Direct labour	20	18	14	16
Overhead (@ Rs. 36)	144 (4 × Rs.36)	108 (3 × Rs.36)	72 (2 × Rs.36)	36 (1× Rs.36)
Total cost per unit	248	216	166	148
Total cost	3,57,120	2,59,200	1,59,360	1,49,184

(ii) (1) Machine department costs of Rs. 2,52,000 to be apportioned to set-up cost, store receiving and inspection in 4 : 3 : 2 i.e. Rs. 1,12,000, Rs. 84,000 and Rs. 56,000 respectively.

(2) One production run = 48 units. Hence, the number of production runs of different products:

$A = \frac{1,440}{48} = 30$, $B = \frac{1,200}{48} = 25$, $C = \frac{960}{48} = 20$, $D = \frac{1,008}{48} = 21$ or total 96 runs.

(3) One batch order is of 24 units. So the number of batches of different products:

$A = \frac{1,440}{24} = 60$, $B = \frac{1,200}{24} = 50$, $C = \frac{960}{24} = 40$, $D = \frac{1,008}{24} = 42$ or total 192 batches.

(4) Computation of Cost driver rates

Activity	Activity Cost (Rs.)	Cost driver	Quantity	Cost driver rate
Set-up	80,000 + 1,12,000 = 1,92,000	No. of production run	96	Rs. 2,000 per production run
Store-receiving	60,000 + 84,000 = 1,44,000	Requisition raised	50 × 4 = 200	Rs. 720 per requisition
Inspection	40,000 + 56,000 = 96,000	No. of production run	96	Rs. 1,000 per production run
Material handling	10,368	Orders executed (No. of batches)	192	Rs. 54 per batch

(5) **Cost statement under Activity Based Costing:**

Product	A	B	C	D
Output in units	1,440	1,200	960	1,008
	(Rs.)	**(Rs.)**	**(Rs.)**	**(Rs.)**
Material	1,440 × 84 = 1,20,960	1,200 × 90 = 1,08,000	960 × 80 = 76,800	1,008 × 96 = 96,768

Product	A	B	C	D
Labour	1,440 × 20 = 28,800	1,200 × 18 = 21,600	960 × 14 = 13,440	1,008 × 16 = 16,128
	1,49,760	1,29,600	90,240	1,12,896
Overhead cost:				
Set up	2,000 × 30 = 60,000	2,000 × 25 = 50,000	2,000 × 20 = 40,000	2,000 × 21 = 42,000
Store receiving	720 × 50 = 36,000	720 × 50 = 36,000	720 × 50 = 36,000	720 × 50 = 36,000
Inspection	1,000 × 30 = 30,000	1,000 × 25 = 25,000	1,000 × 20 = 20,000	1,000 × 21 = 21,000
Material handling	54 × 60 = 3,240	54 × 50 = 2,700	54 × 40 = 2,160	54 × 42 = 2,268
Total overhead cost	1,29,240	1,13,700	98,160	1,01,268
Total cost	2,79,000	2,43,300	1,88,400	2,14,164
Total cost per unit (Total cost / Output)	193.75	202.75	196.25	212.46

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504 MTP **Old P3 Jul 2021 S-II** Q2(a) 10 marks old scheme

RVP Cinema provides the following data for the year 2020-21:

Particulars	Premium Hall (Rs.)	Recliner Hall (Rs.)	7D Hall (Rs.)	Cafeteria (Rs.)
Revenue	11,55,000	18,75,000	9,30,000	5,25,000
Cost of Goods sold	-	-	-	4,51,125
Digital media cost	6,19,800	9,46,875	4,02,900	-
Number of Credit Card transactions	75,000	90,000	60,000	45,000
Number of Tests	12,000	18,000	15,000	7,500
Number of Setups	225	450	150	75
Area in Square feet	3,000	4,500	2,250	750
Number of Customer contacts	2,62,500	3,00,000	1,50,000	37,500
Number of Customer online orders	2,10,000	2,47,500	1,20,000	22,500

Cost analysis has revealed the following:

Activity	Activity Cost (Rs.)	Activity Driver	Activity Capacity
Marketing Expenses	2,25,000	Number of Customer contacts	7,50,000
Website Maintenance Expenses	1,50,000	Number of Customer online orders	6,00,000
Credit Card Processing Fees	1,35,000	Number of Credit Card transactions	2,70,000
Cleaning Equipment Cost	3,15,000	Number of square feet	10,500
Inspecting and testing costs	2,62,500	Number of tests	52,500
Setting up machine's costs	4,50,000	Number of set-ups	900

Required:

(i) If RVP Cinema allocates all costs (other than Cost of Goods sold and Digital Media costs) to the departments on the basis of Activity Based Costing system, CALCULATE the operating income and percentage of operating income of each department.

(ii) RVP Cinema operated for years under the assumption that profitability can be increased by increasing net revenue from Cafeteria. However, the Supervisor of RVP Cinema wants to shut down Cafeteria. On the basis of (i) above, STATE whether the contention of the Supervisor is valid or not.

ICAI'S ANSWER

Computation showing Rates for each Activity

Activity	Activity Cost (Rs.) (A)	Activity driver	Activity Capacity (B)	Activity Rate (A/B)
Marketing Expenses	2,25,000	Number of Customer Contacts	7,50,000	0.30
Website Maintenance Expenses	1,50,000	Number of Customer Online orders	6,00,000	0.25
Credit Card Processing Fees	1,35,000	Number of Credit card transactions	2,70,000	0.50
Cleaning Equipment Cost	3,15,000	Number of Square Feet	10,500	30.00
Inspecting and Testing Cost	2,62,500	Number of Tests	52,500	5.00
Setting up machine's cost	4,50,000	Number of set-ups	900	500.00

Activity based Cost for each Department

Activity	Premium Hall (Rs.)	Recliner Hall (Rs.)	7D Hall (Rs.)	Cafeteria (Rs.)
Marketing Expenses	78,750 (2,62,500 x 0.3)	90,000 (3,00,000 x 0.3)	45,000 (1,50,000 × 0.3)	11,250 (37,500 x 0.3)
Website Maintenance Expenses	52,500 (2,10,000 x 0.25)	61,875 (2,47,500 x 0.25)	30,000 (1,20,000 x 0.25)	5,625 (22,500 x 0.25)
Credit Card Processing Fees	37,500 (75,000 x 0.5)	45,000 (90,000 x 0.5)	30,000 (60,000 x 0.5)	22,500 (45,000 x 0.5)
Cleaning Equipment Cost	90,000 (3,000 x 30)	1,35,000 (4,500 x 30)	67,500 (2,250 x 30)	22,500 (750 x 30)
Inspecting and Testing Cost	60,000 (12,000 x 5)	90,000 (18,000 x 5)	75,000 (15,000 x 5)	37,500 (7,500 x 5)
Setting up machine's cost	1,12,500 (225 x 500)	2,25,000 (450 x 500)	75,000 (150 x 500)	37,500 (75 x 500)
Total	4,31,250	6,46,875	3,22,500	1,36,875

(i) Statement of Operating Income and Operating Income percentage for each Department

Particulars	Premium Hall (Rs.)	Recliner Hall (Rs.)	7D Hall (Rs.)	Cafeteria (Rs.)
Revenues (Given) (A)	11,55,000	18,75,000	9,30,000	5,25,000
Cost of Goods Sold (given) (B1)	-	-	-	4,51,125
Digital Media Cost (given) (B2)	6,19,800	9,46,875	4,02,900	-
Activity Based Cost (as per Workings) (B3)	4,31,250	6,46,875	3,22,500	1,36,875
Operating Cost (B) (B1+ B2 + B3)	10,51,050	15,93,750	7,25,400	5,88,000
Operating Income/(Loss) (C = A – B)	1,03,950	2,81,250	2,04,600	(63,000)
Percentage of profit/(loss) on sales	**9%**	**15%**	**22%**	**(12%)**

(ii) Contention of Supervisor is valid as operating income of Cafeteria is negative i.e. (Rs. 63,000) or percentage of profit/loss is (12%).

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505 MTP **Old P3 Nov 2020 S-I** Q5(b) 10 marks old scheme

BABYSOFT is a global brand created by Bio-organic Ltd. The company manufactures three range of beauty soaps i.e. BABYSOFT- Gold, BABYSOFT- Pearl, and BABYSOFT- Diamond. The budgeted costs and

production for the month of December, 2019 are as follows:

	BABYSOFT- Gold		BABYSOFT- Pearl		BABYSOFT- Diamond	
Production of soaps (Units)	**4,000**		**3,000**		**2,000**	
Resources per Unit:	**Qty**	**Rate**	**Qty**	**Rate**	**Qty**	**Rate**
- Essential Oils	60 ml	₹ 200 / 100 ml	55 ml	₹ 300 / 100 ml	65 ml	₹ 300 / 100 ml
- Cocoa Butter	20 g	₹ 200 / 100 g	20 g	₹ 200 / 100 g	20 g	₹ 200 / 100 g
- Filtered Water	30 ml	₹ 15 / 100 ml	30 ml	₹ 15 / 100 ml	30 ml	₹ 15 / 100 ml
- Chemicals	10 g	₹ 30 / 100 g	12 g	₹ 50 / 100 g	15 g	₹ 60 / 100 g
- Direct Labour	30 minutes	₹ 10 / hour	40 minutes	₹ 10 / hour	60 minutes	₹ 10 / hour

Bio-organic Ltd. followed an Absorption Costing System and absorbed its production overheads, to its products using direct labour hour rate, which were budgeted at ₹ 1,98,000.

Now, Bio-organic Ltd. is considering adopting an Activity Based Costing system. For this, additional information regarding budgeted overheads and their cost drivers is provided below:

Particulars	(₹)	Cost drivers
Forklifting cost	58,000	Weight of material lifted
Supervising cost	60,000	Direct labour hours
Utilities	80,000	Number of Machine operations

The number of machine operators per unit of production are 5, 5, and 6 for BABYSOFT- Gold, BABYSOFT- Pearl, and BABYSOFT- Diamond respectively.

(Consider (i) Mass of 1 litre of Essential Oils and Filtered Water equivalent to 0.8 kg and 1 kg respectively (ii) Mass of output produced is equivalent to the mass of input materials taken together.)

You are requested to:

(i) PREPARE a statement showing the unit costs and total costs of each product using the absorption costing method.

(ii) PREPARE a statement showing the product costs of each product using the ABC approach.

(iii) STATE what are the reasons for the different product costs under the two approaches?

ICAI'S ANSWER

(i) **Traditional Absorption Costing**

		BABYSOFT - Gold	BABYSOFT- Pearl	BABYSOFT- Diamond	*Total*
(a)	Production of soaps (Units)	4,000	3,000	2,000	9,000
(b)	Direct labour (minutes)	30	40	60	-
(c)	Direct labour hours (a × b)/60 minutes	2,000	2,000	2,000	6,000

Overhead rate per direct labour hour:

= Budgeted overheads ÷ Budgeted labour hours

= ₹ 1,98,000 ÷ 6,000 hours

= ₹ 33 per direct labour hour

Unit Costs:

	BABYSOFT- Gold (₹)	BABYSOFT- Pearl (₹)	BABYSOFT- Diamond (₹)
Direct Costs:			
- Direct Labour	5.00 (10×30/60)	6.67 (10×40/60)	10.00 (10×60/60)
- Direct Material (Refer working note1)	167.50	215.50	248.50
Production Overhead:	16.50 (33×30/60)	22.00 (33×40/60)	33.00 (33×60/60)
Total unit costs	189.00	244.17	291.50
Number of units	4,000	3,000	2,000
Total costs	**7,56,000**	**7,32,510**	**5,83,000**

Working note-1

Calculation of Direct material cost

	BABYSOFT- Gold (₹)	BABYSOFT- Pearl (₹)	BABYSOFT- Diamond (₹)
Essential oils	120.00 (200×60/100)	165.00 (300×55/100)	195.00 (300×65/100)
Cocoa Butter	40.00 (200×20/100)	40.00 (200×20/100)	40.00 (200×20/100)
Filtered water	4.50 (15×30/100)	4.50 (15×30/100)	4.50 (15×30/100)
Chemicals	3.00 (30×10/100)	6.00 (50×12/100)	9.00 (60×15/100)
Total costs	**167.50**	**215.50**	**248.50**

(ii) **Activity Based Costing**

	BABYSOFT- Gold	BABYSOFT- Pearl	BABYSOFT- Diamond	Total
Quantity (units)	4,000	3,000	2,000	-

	BABYSOFT- Gold	BABYSOFT- Pearl	BABYSOFT- Diamond	Total
Weight per unit (grams)	108 {(60×0.8)+20+30+10}	106 {(55×0.8)+20+30+12}	117 {(65×0.8)+20+30+15}	-
Total weight (grams)	4,32,000	3,18,000	2,34,000	9,84,000
Direct labour (minutes)	30	40	60	-
Direct labour hours	2,000 (4,000×30/60)	2,000 (3,000×40/60)	2,000 (2,000×60/60)	6,000
Machine operations per unit	5	5	6	-
Total operations	20,000	15,000	12,000	47,000

Forklifting rate per gram = ₹ 58,000 ÷ 9,84,000 grams = ₹ 0.06 per gram

Supervising rate per direct labour hour = ₹ 60,000 ÷ 6,000 hours = ₹ 10 per labour hour

Utilities rate per machine operations = ₹ 80,000 ÷ 47,000 machine operations

= ₹ 1.70 per machine operations

Unit Costs under ABC:

	BABYSOFT- Gold (₹)	BABYSOFT- Pearl (₹)	BABYSOFT- Diamond (₹)
Direct Costs:			
- Direct Labour	5.00	6.67	10.00
- Direct material	167.50	215.50	248.50
Production Overheads:			
Forklifting cost	6.48 (0.06 × 108)	6.36 (0.06 × 106)	7.02 (0.06 × 117)
Supervising cost	5.00 (10×30/60)	6.67 (10×40/60)	10.00 (10×60/60)
Utilities	8.50 (1.70 × 5)	8.50 (1.70 × 5)	10.20 (1.70 × 6)
Total unit costs	192.48	243.70	285.72
Number of units	4,000	3,000	2,000
Total costs	**7,69,920**	**7,31,100**	**5,71,440**

(iii) **Comments:** The difference in the total costs under the two systems is due to the differences in the overheads borne by each of the products. The Activity Based Costs appear to be more precise.

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506 MTP Old P3 May 2020 S-I Q2(a) 10 marks old scheme

ZA Ltd. is a manufacturer of a range of goods. The cost structure of its different products is as follows:

Particulars	Product A	Product B	Product C	
Direct Materials	100	80	80	Rs./u
Direct Labour @Rs.10/ hour	30	40	50	Rs./u
Production Overheads	30	40	50	Rs./u
Total Cost	160	160	180	Rs./u
Quantity Produced	20,000	40,000	60,000	Units

ZA Ltd. was absorbing overheads on the basis of direct labour hours. A newly appointed management accountant has suggested that the company should introduce ABC system and has identified cost drivers and cost pools as follows:

Activity Cost Pool	Cost Driver	Associated Cost (Rs.)
Stores Receiving	Purchase Requisitions	5,92,000
Inspection	Number of Production Runs	17,88,000
Dispatch	Orders Executed	4,20,000
Machine Setup	Number of Setups	24,00,000

The following information is also supplied:

Details	Product A	Product B	Product C
No. of Setups	360	390	450
No. of Orders Executed	180	270	300
No. of Production Runs	750	1,050	1,200
No. of Purchase Requisitions	300	450	500

Required:

CALCULATE activity based production cost of all the three products.

ICAI'S ANSWER

The total production overheads are Rs.52,00,000:

Product A: 20,000 × Rs.30 = Rs.6,00,000

Product B: 40,000 × Rs.40 = Rs.16,00,000

Product C: 60,000 × Rs.50 = Rs.30,00,000

On the basis of ABC analysis this amount will be apportioned as follows:

Statement Showing "Activity Based Production Cost"

Activity Cost Pool	Cost Driver	Ratio	Total Amount (Rs.)	A (Rs.)	B (Rs.)	C (Rs.)
Stores Receiving	Purchase Requisition	6:9:10	5,92,000	1,42,080	2,13,120	2,36,800
Inspection	Production Runs	5:7:8	17,88,000	4,47,000	6,25,800	7,15,200
Dispatch	Orders Executed	6:9:10	4,20,000	1,00,800	1,51,200	1,68,000
Machine Setups	Setups	12:13:15	24,00,000	7,20,000	7,80,000	9,00,000
Total Activity Cost				14,09,880	17,70,120	20,20,000
Quantity Produces				20,000	40,000	60,000
Unit Cost (Overheads)				70.49	44.25	33.67
Add: Conversion Cost (Material + Labour)				130	120	130
Total				200.49	164.25	163.67

ICAI · Old P3 · MTP May 2020 · Series I · question p. 2-3 · answer p. 4

507 MTP **Old P3 Nov 2019 S-I** Q2(a) 10 marks old scheme

Asian Mfg. Co. has decided to increase the size of the store. It wants the information about the probability of the individual product lines : Lemon, Grapes and Papaya. It provides the following data for the 2018 for each product line:

Particulars	Lemon	Grapes	Papaya
Revenues (Rs.)	79,350	2,10,060	1,20,990
Cost of goods sold (Rs.)	60,000	1,50,000	90,000
Cost of bottles returned (Rs.)	1,200	0	0
Number of purchase orders placed	36	84	36
Number of deliveries received	30	219	66
Hours of shelf stocking time	54	540	270
Items sold	12,600	1,10,400	30,600

Asian Mfg. Co. also provides the following information for the year 2018:

Activity	Description of Activity	Total Costs (Rs.)	Cost Allocation Basis
Bottle returns	Returning of empty bottles to the store	1,200	Direct tracing to product line
Ordering	Placing of orders of purchases	15,600	156 purchase orders
Delivery	Physical delivery and the receipts of merchandise	25,200	315 deliveries
Self- stocking	Stocking of merchandise on store shelves and ongoing restocking	17,280	864 hours of time
Customer support	Assistance provided to customers including bagging and checkout	30,720	1,53,600 items sold

Required

(i) Asian Mfg. Co. currently allocates store support costs (all costs other than the cost of goods sold) to the product line on the basis of the cost of goods sold of each product line. CALCULATE the operating income and operating income as the percentage of revenue of each product line.

(ii) If Asian Mfg. Co. allocates store support costs (all costs other than the cost of goods sold) to the product lines on the basis of ABC system, CALCULATE the operating income and operating income as the percentage of revenue of each product line.

(iii) SHOW a comparison statement.

ICAI'S ANSWER

(i) Absorption Costing System

Operating Income-

Particulars	Lemon	Grapes	Papaya	Total
Revenue	79,350	2,10,060	1,20,990	4,10,400
Less: Cost of Goods Sold	60,000	1,50,000	90,000	3,00,000
Less: Store Support Cost	18,000	45,000	27,000	90,000
Operating Income	1,350	15,060	3,990	20,400
Operating Income (%)	1.70	7.17	3.30	4.97

(ii) ABC System

Overhead Allocation Rate-

Activity	Total Costs (Rs.)	Quantity of Cost Allocation Base	Overhead Allocation Rate (Rs.)
Ordering	15,600	156 Purchase Orders	100.00
Delivery	25,200	315 Delivering Orders	80.00
Shelf Stocking	17,280	864 Self Stocking Hours	20.00
Customer Support	30,720	1,53,600 Items Sold	0.20

Store Support Cost-

Particulars	Cost Driver	Lemon	Grapes	Papaya	Total
Bottle Returns	Direct	1,200	0	0	1,200
Ordering	Purchase Orders	3,600	8,400	3,600	15,600
Delivery	Deliveries	2,400	17,520	5,280	25,200
Self -Stocking	Hours of time	1,080	10,800	5,400	17,280
Customer Support	Items Sold	2,520	22,080	6,120	30,720
Grand Total		10,800	58,800	20,400	90,000

Operating Income-

Particulars	Lemon	Grapes	Papaya	Total
Revenue	79,350	2,10,060	1,20,990	410,400
Less: Cost of Goods Sold	60,000	1,50,000	90,000	300,000
Less: Store Support Cost	10,800	58,800	20,400	90,000
Operating Income	8,550	1,260	10,590	20,400
Operating Income (%)	10.78	0.60	8.75	4.97

(iii) Comparison

Particulars	Lemon	Grapes	Papaya	Total
Under Traditional Costing System	1.70%	7.17%	3.30%	4.97%
Under ABC System	10.78%	0.60%	8.75%	4.97%

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6.3 Activity based management and budgeting; benefits and limits

4 questions · asked in 4 ICAI documents · 2 exam papers

508 EXAM Jan 2026 Q6(c) 4 marks

Discuss the usefulness/suitability of Activity Based Costing.

ICAI'S ANSWER

ABC is particularly needed by organisations for product costing in the following situations:

1. **High amount of overhead**: When production overheads are high and form significant costs, ABC is more useful than traditional costing system.
2. **Wide range of products**: ABC is most suitable, when, there is diversity in the product range or there are multiple products.
3. **Presence of non-volume related activities**: When non-volume related activities e.g. material handling, inspection set-up, are present significantly and traditional system cannot be applied, ABC is a superior and better option. ABC will identify non-value-adding activities in the production process that might be a suitable focus for attention or elimination.
4. **Stiff competition**: When the organisation is facing stiff competition and there is an urgent requirement to compute cost accurately and to fix the selling price according to the market situation, ABC is very useful. ABC can also facilitate in reducing cost by identifying non-value-adding activities in the production process that might be a suitable focus for attention or elimination.

ICAI · P4 · Jan 2026 exam · question p. 28 · answer p. 29-30

509 EXAM Old P3 May 2023 Q6(c) 5 marks old scheme

What is meant by Activity Based Management (ABM) and discuss how Activity Based Management can be used in the business?

ICAI'S ANSWER

Meaning of Activity Based Management (ABM)

The term Activity based management (ABM) is used to describe the cost management application of ABC. The use of ABC as a costing tool to manage costs at activity level is known as Activity Based Cost Management (ABM). ABM is a discipline that focuses on the efficient and effective management of activities as the route to continuously improving the value received by customers. ABM utilizes cost information gathered through ABC.

Activity based management can be used in the following ways:

(i) Cost Reduction: ABM helps the organisation to identify costs against activities and to find opportunities to streamline or reduce the costs or eliminate the entire activity, especially if there is no value added.

(ii) Business Process Re-engineering: Business process re-engineering involves examining business processes and making substantial changes to how organisation currently operates. ABM is a powerful tool for measuring business performance, determining the cost of business output and is used as a means of identifying opportunities to improve process efficiency and effectiveness.

(iii) Benchmarking: Benchmarking is a process of comparing of ABC-derived activity costs of one segment of company with those of other segments. It requires uniformity in the definition of activities and measurement of their costs.

(iv) Performance Measurement: Many organisations are now focusing on activity performance as a means of facing competitors and managing costs by monitoring the efficiency and effectiveness of activities.

ICAI · Old P3 · May 2023 exam · question p. 28 · answer p. 30

510 RTP Sep 2025 Q18(b)

A manufacturing company based in India is planning to transition from its traditional costing system to Activity-Based Costing (ABC) in order to improve cost accuracy and gain better insights into product profitability. Given the company's diverse product portfolio and the complexity of its production processes, the implementation of ABC is expected to provide a more precise allocation of overheads.

You are required to SUGGEST the key practical stages involved in successfully implementing an Activity-Based Costing system.

ICAI'S ANSWER

A number of distinct practical stages that are required in the ABC implementation are given below:

(1) Staff Training: The co-operation of the workforce is critical to the successful implementation of ABC. Staff training should be done to create an awareness on the purpose of ABC.

(2) Process Specification: Informal, but structured interviews with key members of personnel will identify the different stages of the production process, the commitment of resources to each, processing times and bottlenecks.

(3) Activity Definition: The activities must be defined clearly in the early stage in order to manage the problems, if any, effectively. There might be overloading of information from the new data, but the same is needed in codification.

(4) Activity Driver Selection: Cost driver for each activity shall be selected.

(5) Assigning Cost: A single representative activity driver can be used to assign costs from the activity pools to the cost objects.

ICAI · P4 · RTP Sep 2025 · question p. 22-23 · answer p. 54-55

511 MTP Old P3 May 2022 S-II Q3(b) 10 marks old scheme

Equate bank offers 3 products, viz., deposits, Loans and Credit Cards. The bank has selected 4 activities for a detailed budgeting exercise, following activity-based costing methods.

The bank wants to know the product wise total cost per unit for the selected activities, so that prices may be fixed accordingly.

The following information is made available to formulate the budget:

Activity	Present Cost (₹)	Estimation for the budget period
ATM Services:		
(a) Machine Maintenance	5,20,000	All fixed, no change.
(b) Rents	2,60,000	Fully fixed, no change.
(c) Currency Replenishment Cost	1,30,000	Expected to double during budget period.
	9,10,000	

Activity	Present Cost (₹)	Estimation for the budget period
Computer Processing	6,50,000	Half this amount is fixed, and no change is expected. The variable portion is expected to increase to three times the current level.
Issuing Statements	23,40,000	Presently, 3.90 lakh statements are made. In the budget period, 6.5 lakh statements are expected. For every single increase of statement, one rupee is the budgeted increase.
Computer Inquiries	2,60,000	Estimated to increase by 80% during the budget period.

The activity drivers and their budgeted quantifies are given below:

Activity Drivers	Deposits	Loans	Credit Cards
No. of ATM Transactions	1,95,000	---	65,000
No. of Computer Processing Transactions	19,50,000	2,60,000	3,90,000
No. of Statements to be issued	4,55,000	65,000	1,30,000
Telephone Minutes	4,68,000	2,34,000	2,34,000

The bank budgets a volume of 76,180 deposit accounts, 16,900 loan accounts, and 18,200 Credit Card Accounts.

Required:

(i) CALCULATE the budgeted rate for each activity.

(ii) PREPARE the budgeted cost statement activity wise.

(iii) COMPUTE the budgeted product cost per account for each product using (i) and (ii) above.

ICAI'S ANSWER

Statement Showing "Budgeted Cost per unit of the Product"

Activity	Activity Cost (Budgeted) (₹)	Activity Driver	No. of Units of Activity Driver (Budget)	Activity Rate (₹)	Deposits	Loans	Credit Cards
ATM Services	10,40,000	No. of ATM Transaction	2,60,000	**4.00**	7,80,000	---	2,60,000
Computer Processing	13,00,000	No. of Computer processing Transaction	26,00,000	**0.50**	9,75,000	1,30,000	1,95,000
Issuing Statements	26,00,000	No. of Statements	6,50,000	**4.00**	18,20,000	2,60,000	5,20,000
Customer Inquiries	4,68,000	Telephone Minutes	9,36,000	**0.50**	2,34,000	1,17,000	1,17,000
Budgeted Cost	**54,08,000**				**38,09,000**	**5,07,000**	**10,92,000**
Units of Product (as estimated in the budget period)					76,180	16,900	18,200

Activity	Activity Cost (Budgeted) (₹)	Activity Driver	No. of Units of Activity Driver (Budget)	Activity Rate (₹)	Deposits	Loans	Credit Cards
Budgeted Cost *per unit* of the product					**50**	**30**	**60**

Working Note:

Activity	Budgeted Cost (₹)	Remark
ATM Services:		
(a) Machine Maintenance		
(b) Rents	5,20,000	– All fixed, no change.
(c) Currency Replenishment Cost	2,60,000	– Fully fixed, no change.
Total	2,60,000	– Doubled during budget period.
	10,40,000	
Computer Processing	3,25,000	– ₹ 3,25,000 (half of ₹ 6,50,000) is fixed and no change is expected.
	9,75,000	– ₹ 3,25,000 (variable portion) is expected to increase to three times the current level.
Total	13,00,000	
Issuing Statements	23,40,000	– Existing.
	2,60,000	– 2.60 lakh statements are expected to be increased in budgeted period. For every single increase of statement, one rupee is the budgeted increase.
Total	26,00,000	
Computer Inquiries	4,68,000	– Estimated to increase by 80% during the budget period. (₹ 2,60,000 × 180%)
Total	4,68,000	

ICAI · Old P3 · MTP May 2022 · Series II · question p. 4 · answer p. 6-7

CHAPTER 7

Cost Accounting Systems

7.1 Non-integrated and integrated accounting systems

27 questions · asked in 28 ICAI documents · 7 exam papers

512 EXAM **Sep 2025** Q4(b) 5 marks

Sundar Limited maintains its Cost Accounting System on the basis of Non-Integral System of Accounting. The following transactions arose during the month of August, 2025:

Particulars	Amount (₹)
Materials purchased on credit	10,25,000
Materials issued to production (Direct)	5,55,000
Direct Wages allocated to production	3,00,000
Factory Overheads over-absorbed	2,20,000
Administration Overheads under-absorbed	1,40,000

Required:

Journalize the above-mentioned transactions in Cost Books maintained on Non-Integrated System of Accounting.

ICAI'S ANSWER

Journal entries are as follows:

			Dr.	Cr.
			(₹)	(₹)
1.	Stores Ledger Control A/c	Dr.	10,25,000	
	To Cost Ledger Control A/c			10,25,000
	(Materials purchased)			
	OR			
	Material Control A/c	Dr.	10,25,000	
	To Cost Ledger Control A/c			10,25,000
	Stores Ledger Control A/c	Dr.	10,25,000	
	To Material Control A/c			10,25,000
2.	Work-in-Process Control A/c	Dr.	5,55,000	
	To Store Ledger Control A/c			5,55,000
	(Materials issued to production)			

			Dr.	Cr.
3.	Work-in-Process Control A/c	Dr.	3,00,000	
	To Wages Control A/c			3,00,000
	(Direct wages charged to production)			
4.	Factory Overheads Control A/c	Dr.	2,20,000	
	To Costing Profit & Loss A/c			2,20,000
	(Factory overheads over absorbed)			
5.	Costing Profit & Loss A/c	Dr.	1,40,000	
	To Administrative Overhead Control A/c			1,40,000
	(Admin overheads under absorbed)			

ICAI · P4 · Sep 2025 exam · question p. 21 · answer p. 22-23

513 EXAM **May 2025** MCQ 12 MCQ

A company which operates a batch costing system is fully integrated with the financial accounts.

During a particular period materials worth ₹ 30,000 and ₹ 20,000 were issued to production and Factory Maintenance respectively. The following control A/cs are being maintained.

(i) Store ledger control A/c.

(ii) Work-in-progress control A/c.

(iii) Production overhead control A/c.

(iv) Finished goods control A/c.

From the above information, identify which account/accounts will be debited to effectuate the issuance of materials:

A (i) & (ii)

B (ii) & (iii)

C (ii) & (iv)

D Only (i)

ICAI'S ANSWER

Answer key: **(B)**

ICAI · P4 · May 2025 exam · question p. 5 · answer p. 6

514 EXAM Jan 2025 Q5(b) 4 marks

Journalise the following transactions assuming that cost and financial accounts are integrated:

Particulars	Amount (in ₹)
Wages paid (20% indirect)	2,00,000
Selling and Distribution Overheads incurred	50,000
Deficiency found in stock of Raw Material (Normal)	80,000
Factory Overheads (Under Absorbed)	60,000

ICAI'S ANSWER

Journal Entries under Integrated system of accounting

	Particulars		(₹)	(₹)
(i)	Work-in Progress Ledger Control A/c	Dr.	1,60,000	
	Factory Overhead control A/c	Dr.	40,000	
	To Wages Control A/c			2,00,000
	(Being allocation of Direct and Indirect wages)			
(ii)	Wages Control A/c	Dr.	2,00,000	
	To Bank A/c			2,00,000
	(Being wages paid)			
(iii)	Cost of Sales A/c	Dr.	50,000	
	To Selling & Distribution Overhead Control A/c			50,000
	(Being selling & distribution overhead allocated)			
(iv)	Selling & Distribution Overhead Control A/c	Dr.	50,000	
	To Bank A/c			50,000
	(Being selling & distribution overhead paid)			
(v)	Factory overhead control A/c	Dr.	80,000	
	To Stores Ledger Control A/c			80,000
	(Being normal deficiency found in stock of raw material)			
(vi)	Costing Profit & Loss A/c	Dr.	60,000	
	To Factory Overhead Control A/c			60,000
	(Being transfer of under absorption of factory overhead)			

515 EXAM **Old P3 Nov 2023** Q5(c) 4 marks old scheme

Construct journal entries in the following situations assuming that cost and financial transactions are integrated:

(i)	Purchase of raw material	₹ 4,40,000
(ii)	Direct Material issued to production	₹ 3,60,000
(iii)	Wages charged to production	₹ 80,000
(iv)	Manufacturing overheads charged to production	₹ 1,32,000

ICAI'S ANSWER

Journal entries are as follows

		DR. (₹)	Cr. (₹)
Stores Ledger Control A/c	Dr.	4,40,000	
To Payables (Creditors)/ Bank A/c			4,40,000
(Materials purchased)			
Work-in-Process Control A/c	Dr.	3,60,000	
To Stores Ledger Control A/c			3,60,000
(Materials issued to production)			
Work-in-Process Control A/c	Dr.	80,000	
To Wages Control A/c			80,000
(Direct wages charged to production)			
Work-in-Process Control A/c	Dr.	1,32,000	
To Factory Overhead Control A/c			1,32,000
(Manufacturing overhead charged to production)			

ICAI · Old P3 · Nov 2023 exam · question p. 25 · answer p. 28

516 EXAM **Old P3 May 2022** Q4(c) 5 marks old scheme

Journalize the following transactions assuming the cost and financial accounts are integrated:

Particulars	Amount (₹)
Direct Materials issued to production	₹ 5,88,000
Allocation of Wages (Indirect)	₹ 7,50,000
Factory Overheads (Over absorbed)	₹ 2,25,000
Administrative Overheads (Under absorbed)	₹ 1,55,000
Deficiency found in stock of Raw material (Normal)	₹ 2,00,000

ICAI'S ANSWER

	Particulars		(₹)	(₹)
(i)	Work-in-Progress Ledger Control A/c	Dr.	5,88,000	
	To Stores Ledger Control A/c			5,88,000
	(Being issue of direct materials to production)			
(ii)	Factory Overhead control A/c	Dr.	7,50,000	
	To Wages Control A/c			7,50,000
	(Being allocation of Indirect wages)			
(iii)	Factory Overhead Control A/c	Dr.	2,25,000	
	To Costing Profit & Loss A/c			2,25,000
	(Being transfer of over absorption of Factory overhead)			
(iv)	Costing Profit & Loss A/c	Dr.	1,55,000	
	To Administration Overhead Control A/c			1,55,000
	(Being transfer of under absorption of Administration overhead)			
(v)	Factory Overhead Control A/c	Dr.	2,00,000	
	To Stores Ledger Control A/c			2,00,000
	(Being transfer of deficiency in stock of raw material)			

(Note: Costing P/&/L = P/&/L and SLC = MLC)

ICAI · Old P3 · May 2022 exam · question p. 17 · answer p. 21

517 EXAM **Old P3 Nov 2020** Q6(c) 5 marks old scheme

Explain what are the pre-requisites of integrated accounting.

ICAI'S ANSWER

The essential pre-requisites for integrated accounts include the following steps:

- The management's decision about the extent of integration of the two sets of books. Some concerns find it useful to integrate up to the stage of prime cost or factory cost while other prefer full integration of the entire accounting records.
- A suitable coding system must be made available so as to serve the accounting purposes of financial and cost accounts.
- An agreed routine, with regard to the treatment of provision for accruals, prepaid expenses, other adjustment necessary for preparation of interim accounts.
- Perfect coordination should exist between the staff responsible for the financial and cost aspects of the accounts and an efficient processing of accounting documents should be ensured.
- Under this system there is no need for a separate cost ledger. Of course, there will be a number of subsidiary ledgers; in addition to the useful Customers' Ledger and the Bought Ledger, there will be: (a) Stores Ledger; (b) Stock Ledger and (c) Job Ledger.

ICAI · Old P3 · Nov 2020 exam · question p. 22 · answer p. 23

518 EXAM **Old P3 Nov 2019** Q6(b) 5 marks old scheme

Journalise the following transactions in cost books under Non-Integrated system of Accounting.

(i)	Credit Purchase of Material	₹ 27,000
(ii)	Manufacturing overhead charged to Production	₹ 6,000
(iii)	Selling and Distribution overheads recovered from Sales	₹ 4,000
(iv)	Indirect wages incurred	₹ 8,000
(v)	Material returned from production to stores	₹ 9,000

ICAI'S ANSWER

Journal entries are as follows:

			Dr. (₹)	Cr. (₹)
(i)	Stores Ledger Control A/c.....	Dr.	27,000	
	To Cost Ledger Control A/c			27,000
(ii)	Work-in-Process Control A/c.....	Dr.	6,000	
	To Manufacturing Overhead Control A/c			6,000
(iii)	Cost of Sales A/c.....	Dr.	4,000	
	To Selling & Dist. Overhead Control A/c			4,000
(iv)	(1) Wage Control A/c.....	Dr.	8,000	
	To Cost Ledger Control A/c			8,000

			Dr. (₹)	Cr. (₹)
	(2) Manufacturing Overhead Control A/c.....	Dr.	8,000	
	To Wages Control A/c			8,000
	OR			
	Manufacturing Overhead Control A/c.....	Dr.	8,000	
	To Cost Ledger Control A/c			8,000
(v)	Stores Ledger Control A/c	Dr.	9,000	
	To Work-in-Process Control A/c			9,000

*Cost Ledger Control A/c is also known as General Ledger Control A/c

ICAI · Old P3 · Nov 2019 exam · question p. 23 · answer p. 25

519 RTP **May 2026** Q11

The following figures have been extracted from the cost records of a manufacturing unit:

	(₹)
Stores: Opening balance	32,000
Purchases of material	1,58,000
Transfer from work-in-progress	80,000
Issues to work-in-progress	1,60,000
Issues to repair and maintenance	20,000
Deficiencies found in stock taking	6,000
Work-in-progress: Opening balance	60,000
Direct wages applied	65,000
Overheads applied	2,40,000
Closing balance of W.I.P.	45,000

Finished products: Entire output is sold at a profit of 10% on actual cost from work-in-progress. Wages incurred ₹ 70,000, overhead incurred ₹ 2,50,000.

Items not included in cost records: Income from investment ₹ 10,000, Loss on sale of capital assets ₹ 20,000.

Draw up Store Control account, Work-in-progress Control account, Costing Profit and Loss account, Profit and Loss account and Reconciliation statement.

ICAI'S ANSWER

(A) Costing books

Stores Control Account

Particulars	(₹)	Particulars	(₹)
To Balance b/d	32,000	By W.I.P. Control A/c	1,60,000
To General ledger adjustment A/c	1,58,000	By Work overhead control A/c	20,000
To Work in progress control A/c	80,000	By Costing Profit and Loss A/c	6,000
		By Balance c/d	84,000
	2,70,000		2,70,000

W.I.P. Control Account

Particulars	(₹)	Particulars	(₹)
To Balance b/d	60,000	By Stores control A/c	80,000
To Stores control A/c	1,60,000	By Costing profit and loss A/c (Cost of sales)	4,00,000
To Direct wages control A/c	65,000		
To Works overhead control A/c	2,40,000	By Balance c/d	45,000
	5,25,000		5,25,000

Works Overhead Control Account

Particulars	(₹)	Particulars	(₹)
To General ledger adjustment A/c	2,50,000	By W.I.P. Control A/c	2,40,000
To Store ledger control A/c	20,000	By Costing profit & loss A/c (under recovery)	30,000
	2,70,000		2,70,000

Costing Profit & Loss Account

Particulars	(₹)	Particulars		(₹)
To W.I.P. control A/c (Cost of sales)	4,00,000	By General ledger adjustment A/c		
To Works overhead control A/c	30,000	Cost of sales	4,00,000	
To Stores control A/c (shortage)	6,000	10% profit	40,000	4,40,000
To Profit	4,000			
	4,40,000			4,40,000

(B) Financial Books

Profit & Loss Account

Particulars		(₹)	Particulars		(₹)
To Opening stock			By Sales		4,40,000
Stores	32,000		By Closing stock:		
W.I.P.	60,000	92,000	Stores	84,000	
			W.I.P.	45,000	1,29,000
To Purchases		1,58,000	By Income from investment		10,000
To Wages incurred		70,000	By Loss		11,000
To Overheads incurred		2,50,000			
To Loss on sale of capital assets		20,000			
		5,90,000			5,90,000

Reconciliation statement

	(₹)	(₹)
Profit as per Cost Accounts		4,000
Add: Income from investment recorded in Financial accounts		10,000
		14,000
Less: Under absorption of wages in Cost accounts	5,000	
Loss on sales of capital asset only included in Financial accounts	20,000	25,000
Loss as per Financial accounts		11,000

ICAI · P4 · RTP May 2026 · question p. 15-16 · answer p. 40-42

520 RTP **Sep 2025 Q4** MCQ

WHICH of the following is the correct journal entry as would appear in the cost books when overhead expenses incurred ₹ 500 (Production ₹ 250 and Administrative ₹ 250)?

A Production Overhead Control A/c Dr. ₹ 250
Administrative Overhead Control A/c Dr. ₹ 250
To Cost Ledger Control A/c ₹ 500

B Production Overhead Control A/c Dr. ₹ 250
Administrative Overhead Control A/c Dr. ₹ 250
To Overhead Expenses A/c ₹ 500

C Cost Ledger Control A/c Dr. ₹ 500
To Production Overhead Control A/c ₹ 250
To Administrative Overhead Control A/c ₹ 250

D Overhead Expenses A/c Dr. ₹ 500
To Production Overhead Control A/c ₹ 250
To Administrative Overhead Control A/c ₹ 250

ICAI'S ANSWER

Answer key: **(A)**

(a)

Production Overhead Control A/c	Dr.	₹ 250	
Administrative Overhead Control A/c	Dr.	₹ 250	
To Cost Ledger Control A/c			₹ 500

ICAI · P4 · RTP Sep 2025 · question p. 8 · answer p. 29

521 RTP **May 2025** Q4 MCQ

WHICH of the following is the correct journal entry as would appear in the cost books when Material (Direct) is issued to production?

A Store Ledger Control A/c Dr. xxx
To Work-in-Process Control A/c xxx

B Store Ledger Control A/c Dr. xxx
To Production Overhead Control A/c xxx

C Production Overhead Control A/c Dr. xxx
To Store Ledger Control A/c xxx

D Work-in-Process Control A/c Dr. xxx
To Store Ledger Control A/c xxx

ICAI'S ANSWER

(D)

Work-in-Process Control A/c Dr. xxx

To Store Ledger Control A/c xxx

ICAI · P4 · RTP May 2025 · question p. 8 · answer p. 30

522 RTP **May 2025** Q12

Amy Ltd. uses a batch costing system that is fully integrated with its financial accounts. Balances at the beginning of the period is provided below:

Particulars	(₹)
Stores Ledger Control Account	4,83,250
Work-in-Process Control Account	3,86,600
Finished Goods Control Account	6,76,550

Other information is as under:

Particulars	
Materials purchased during the period	14,49,750
Materials issued to production	5,79,900
Total wages paid (1/6th being indirect)	5,79,900
Direct wages charged to batches	3,86,600
Payment for non-productive time of direct workers	1/5th of direct wages paid
Production Overheads incurred	2,31,960
Sales	19,33,000
Cost of Finished Goods Sold	15,46,400
Cost of Goods completed and transferred into finished goods during the period	12,56,450
Physical value of Work-in-Process at the end of the period	7,73,200
Production overhead absorption rate	130% of direct wages charged to Work-in-Process

You are required to PREPARE the following accounts:

(i) Stores Ledger Control Account.

(ii) Wages Control Account.

(iii) Production Overhead Control Account.

(iv) Work-in-Process Control Account.

(v) Finished Goods Control Account.

(vi) Costing Profit and Loss Account.

ICAI'S ANSWER

(i) Stores Ledger Control Account

	(₹)		(₹)
To Balance b/d	4,83,250	By Work in Process Control A/c	5,79,900
" Creditors/ Bank A/c	14,49,750	" Balance c/d	13,53,100
	19,33,000		19,33,000

(ii) Wages Control Account

	(₹)		(₹)
To Bank A/c	5,79,900	By Work in Process Control A/c (Charged to batches)	3,86,600
		" Production Oh Control A/c (Indirect wages) (1/6th of ₹ 5,79,900)	96,650
		" Production Oh Control A/c (Non-productive wages) [1/5th of (₹ 5,79,900 -₹ 96,650)]	96,650
	5,79,900		5,79,900

(iii) Production Overhead Control Account

	(₹)		(₹)
To Bank A/c	2,31,960	By Work-in-Process Control A/c (130% of ₹ 3,86,600)	5,02,580
" Wages Control A/c (₹ 96,650 +₹ 96,650)	1,93,300		
" Costing P&L A/c (Over-absorption, balancing figure)	77,320		
	5,02,580		5,02,580

(iv) Work-in-Process Control Account

	(₹)		(₹)
To Balance b/d	3,86,600	By Finished Goods Control A/c	12,56,450
" Store Ledger Control A/c	5,79,900	" Balance c/d (Physical value)	7,73,200
" Wages Control A/c	3,86,600		
" Production OH Control A/c	5,02,580		
" Costing P&L A/c (Stock Gains)	1,73,970		
	20,29,650		20,29,650

(v) Finished Goods Control Account

	(₹)		(₹)
To Balance b/d	6,76,550	By Costing Profit & Loss A/c (Cost of Goods Sold)	15,46,400
" Work-in-Process Control A/c	12,56,450	" Balance c/d	3,86,600
	19,33,000		19,33,000

(vi) Costing Profit & Loss Account

	(₹)		(₹)
To Finished Goods Control A/c	15,46,400	By Sales A/c	19,33,000
" Balance c/d	6,37,890	" Production OH Control A/c	77,320
		" Work-in-Process Control A/c (Stock gain)	1,73,970
	21,84,290		21,84,290

ICAI · P4 · RTP May 2025 · question p. 16-17 · answer p. 40-42

523 RTP **May 2024** Q13(a)

Also asked in Old P3 RTP May 2022 Q14(c) · Old P3 MTP Nov 2023 S-I Q6(c) · Old P3 MTP May 2023 S-II Q6(c)

Same question, different ICAI answer: #532 Old P3 MTP May 2023 S-I

"Is reconciliation of cost accounts and financial accounts necessary in case of integrated accounting system?" explain.

ICAI'S ANSWER

In integrated accounting system cost and financial accounts are kept in the same set of books. Such a system will have to afford full information required for Costing as well as for Financial Accounts. In other words, information and data should be recorded in such a way so as to enable the firm to ascertain the cost (together with the necessary analysis) of each product, job, process, operation or any other identifiable activity. It also ensures the ascertainment of marginal cost, variances, abnormal losses and gains. In fact, all information that management requires from a system of Costing for doing its work properly is made available. The integrated accounts give full information in such a manner so that the profit and loss account and the balance sheet can be prepared according to the requirements of law and the management maintains full control over the liabilities and assets of its business.

Since, only one set of books are kept for both cost accounting and financial accounting purpose so there is no necessity of reconciliation of cost and financial accounts.

ICAI · P4 · RTP May 2024 · question p. 17 · answer p. 41-42

524 RTP **Old P3 May 2023** Q15(c) old scheme

EXPLAIN what are the essential pre-requisite for Integrated Accounting system?

ICAI'S ANSWER

Essential pre-requisites for Integrated Accounts: The essential pre-requisites for integrated accounts include the following steps-

1. The management's decision about the extent of integration of the two sets of books. Some concerns find it useful to integrate up to the stage of prime cost or factory cost while other prefers full integration of the entire accounting records.
2. A suitable coding system must be made available so as to serve the accounting purposes of financial and cost accounts.
3. An agreed routine, with regard to the treatment of provision for accruals, prepaid expenses, other adjustment necessary for preparation of interim accounts.

4. Perfect coordination should exist between the staff responsible for the financial and cost aspects of the accounts and an efficient processing of accounting documents should be ensured.

ICAI · Old P3 · RTP May 2023 · question p. 11 · answer p. 27

525 RTP **Old P3 May 2022** Q6 old scheme

X Ltd. maintains a non-integrated accounting system for the purpose of management information. The following are the data related with year 2021-22:

Particulars	Amount ('000)
Opening balances:	
- Stores ledger control A/c	48,000
- Work-in-process control A/c	12,000
- Finished goods control A/c	2,58,000
- Building construction A/c	6,000
- Cost ledger control A/c	3,24,000
During the year following transactions took place:	
Materials:	
- Purchased	24,000
- Issued to production	30,000
- Issued to general maintenance	3,600
- Issued to building construction	2,400
Wages:	
- Gross wages paid	90,000
- Indirect wages paid	24,000
- For building construction	6,000
Factory overheads:	
- Actual amount incurred (excluding items shown above)	96,000
- Absorbed in building construction	12,000
- Under-absorbed	4,800
Royalty paid	3,000
Selling distribution and administration overheads	15,000

Particulars	Amount ('000)
Sales	2,70,000

At the end of the year, the stock of raw material and work-in-process was ₹3,30,00,000 and ₹15,00,000 respectively. The loss arising in the raw material account is treated as factory overheads. The building under construction was completed during the year. Gross profit margin is 20% on sales.

Required:

PREPARE the relevant control accounts to record the above transactions in the cost ledger of the company.

ICAI'S ANSWER

Cost Ledger Control Account

Particulars	(₹ in '000)	Particulars	(₹ in '000)
To Costing P&L A/c	2,70,000	By Balance b/d	3,24,000
To Building Construction A/c	26,400	By Stores Ledger Control A/c	24,000
To Balance c/d	2,89,800	By Wages Control A/c	90,000
		By Factory overhead control A/c	96,000
		By Royalty A/c	3,000
		By Selling, Distribution and Administration overheads	15,000
		By Costing P&L A/c	34,200
	5,86,200		5,86,200

Stores Ledger Control Account

Particulars	(₹ in '000)	Particulars	(₹ in '000)
To Balance b/d	48,000	By WIP control A/c	30,000
To Cost Ledger control A/c	24,000	By Factory overheads control A/c	3,600
		By Building construction A/c	2,400
		By Factory overhead control A/c (loss) (Bal. fig)	3,000
		By Balance c/d	33,000
	72,000		72,000

Work-in-process Control Account

Particulars	(₹ in '000)	Particulars	(₹ in '000)
To Balance b/d	12,000	By Finished goods control A/c	1,99,800
To Stores Ledger control A/c	30,000		
To Wages Control A/c	60,000		
To Factory overhead control A/c	1,09,800		
To Royalty A/c	3,000	By Balance c/d	15,000
	2,14,800		2,14,800

Finished Goods Control Account

Particulars	(₹ in '000)	Particulars	(₹ in '000)
To Balance b/d	2,58,000	By Cost of Goods Sold A/c (Refer working note)	2,16,000
To WIP control A/c	1,99,800	By Balance c/d	2,41,800
	4,57,800		4,57,800

Cost of Sales Account

Particulars	(₹ in '000)	Particulars	(₹ in '000)
To Cost of Goods Sold A/c	2,16,000	By Costing P&L A/c	2,31,000
To Selling, Distribution and Administration A/c	15,000		
	2,31,000		2,31,000

Costing P&L Account

Particulars	(₹ in '000)	Particulars	(₹ in '000)
To Cost of Sales A/c	2,31,000	By Cost Ledger control A/c	2,70,000
To Factory overhead control A/c	4,800		
To Cost Ledger control A/c	34,200		
	2,70,000		2,70,000

Building Construction Account

Particulars	(₹ in '000)	Particulars	(₹ in '000)
To Balance b/d	6,000	By Cost Ledger control A/c	26,400
To Stores Ledger control A/c	2,400		
To Wages Control A/c	6,000		
To Factory overhead control A/c	12,000		
	26,400		26,400

Factory Overhead Control Account

Particulars	(₹ in '000)	Particulars	(₹ in '000)
To Stores Ledger control A/c	3,600	By Building Construction A/c	12,000
To Wages Control A/c	24,000	By WIP Control A/c	1,09,800
To Cost Ledger control A/c	96,000	By Costing P&L A/c	4,800
To Stores Ledger control A/c (loss)	3,000		
	1,26,600		1,26,600

Wages Control Account

Particulars	(₹ in '000)	Particulars	(₹ in '000)
To Cost Ledger control A/c	90,000	By Factory overhead control A/c	24,000
		By Building Construction A/c	6,000
		By WIP Control A/c	60,000
	90,000		90,000

Royalty Account

Particulars	(₹ in '000)	Particulars	(₹ in '000)
To Cost Ledger control A/c	3,000	By WIP Control A/c	3,000
	3,000		3,000

Cost of Goods Sold Account

Particulars	(₹ in '000)	Particulars	(₹ in '000)
To Finished Goods control A/c	2,16,000	By Cost of sales A/c	2,16,000
	2,16,000		2,16,000

Selling, Distribution and Administration Overhead Control Account

Particulars	(₹ in '000)	Particulars	(₹ in '000)
To Cost Ledger control A/c	15,000	By Cost of sales A/c	15,000
	15,000		15,000

Trial Balance

Particulars	Dr.	Cr.
	(₹ in '000)	**(₹ in '000)**
Stores Ledger Control A/c	33,000	
WIP Control A/c	15,000	
Finished Goods Control A/c	2,41,800	
Cost Ledger Control A/c		2,89,800
	2,89,800	2,89,800

Working Note:

Cost of Goods sold = 2,70,000 × 80/100 = ₹ 2,16,000

ICAI · Old P3 · RTP May 2022 · question p. 5-6 · answer p. 19-21

526 RTP **Old P3 Dec 2021** Q6 old scheme

XYZ Ltd. maintains a non-integrated accounting system for the purpose of management information. The following are the data related with year 2020-21:

Particulars	(₹ in '000)
Opening balances:	
- Stores ledger control A/c	24,000
- Work-in-process control A/c	6,000
- Finished goods control A/c	1,29,000
- Building construction A/c	3,000
- Cost ledger control A/c	1,62,000
During the year following transactions took place:	
Materials:	
- Purchased	12,000
- Issued to production	15,000
- Issued to general maintenance	1,800

Particulars	(₹ in '000)
- Issued to building construction	1,200
Wages:	
- Gross wages paid	45,000
- Indirect wages paid	12,000
- For building construction	3,000
Factory overheads:	
- Actual amount incurred (excluding items shown above)	48,000
- Absorbed in building construction	6,000
- Under-absorbed	2,400
Royalty paid	1,500
Selling, distribution and administration overheads	7,500
Sales	1,35,000

At the end of the year, the stock of raw material and work-in-process was ₹ 1,65,00,000 and ₹ 75,00,000 respectively. The loss arising in the raw material account is treated as factory overheads. The building under construction was completed during the year. Gross profit margin is 20% on sales.

Required:

PREPARE the relevant control accounts to record the above transactions in the cost ledger of the company.

ICAI'S ANSWER

Cost Ledger Control Account

Particulars	(₹ in '000)	Particulars	(₹ in '000)
To Costing P&L A/c	1,35,000	By Balance b/d	1,62,000
To Building Construction A/c	13,200	By Stores Ledger control A/c	12,000
To Balance c/d	1,44,900	By Wages Control A/c	45,000
		By Factory overhead control A/c	48,000
		By Royalty A/c	1,500
		By Selling, Distribution and Administration overheads	7,500
		By Costing P&L A/c	17,100

Particulars	(₹ in '000)	Particulars	(₹ in '000)
	2,93,100		2,93,100

Stores Ledger Control Account

Particulars	(₹ in '000)	Particulars	(₹ in '000)
To Balance b/d	24,000	By WIP control A/c	15,000
To Cost Ledger control A/c	12,000	By Factory overheads control A/c	1,800
		By Building construction A/c	1,200
		By Factory overhead control A/c (bal. fig.) (loss)	1,500
		By Balance c/d	16,500
	36,000		36,000

Wages Control Account

Particulars	(₹ in '000)	Particulars	(₹ in '000)
To Cost Ledger control A/c	45,000	By Factory overhead control A/c	12,000
		By Building Construction A/c	3,000
		By WIP Control A/c (bal. fig.)	30,000
	45,000		45,000

Factory Overhead Control Account

Particulars	(₹ in '000)	Particulars	(₹ in '000)
To Stores Ledger control A/c	1,800	By Building Construction A/c	6,000
To Wages Control A/c	12,000	By WIP Control A/c (bal. fig.)	54,900
To Cost Ledger control A/c	48,000	By Costing P&L A/c (under-absorption)	2,400
To Stores Ledger control A/c (loss)	1,500		
	63,300		63,300

Royalty Account

Particulars	(₹ in '000)	Particulars	(₹ in '000)
To Cost Ledger control A/c	1,500	By WIP Control A/c	1,500
	1,500		1,500

Work-in-process Control Account

Particulars	(₹ in '000)	Particulars	(₹ in '000)
To Balance b/d	6,000	By Finished goods control A/c (bal. fig.)	99,900
To Stores Ledger control A/c	15,000		
To Wages Control A/c	30,000		
To Factory overhead control A/c	54,900		
To Royalty A/c	1,500	By Balance c/d	7,500
	1,07,400		1,07,400

Finished Goods Control Account

Particulars	(₹ in '000)	Particulars	(₹ in '000)
To Balance b/d	1,29,000	By Cost of Goods Sold A/c (Refer working note)	1,08,000
To WIP control A/c	99,900	By Balance c/d	1,20,900
	2,28,900		2,28,900

Cost of Goods Sold Account

Particulars	(₹ in '000)	Particulars	(₹ in '000)
To Finished Goods control A/c	1,08,000	By Cost of sales A/c	1,08,000
	1,08,000		1,08,000

Selling, Distribution and Administration Overhead Control Account

Particulars	(₹ in '000)	Particulars	(₹ in '000)
To Cost Ledger control A/c	7,500	By Cost of sales A/c	7,500
	7,500		7,500

Cost of Sales Account

Particulars	(₹ in '000)	Particulars	(₹ in '000)
To Cost of Goods Sold A/c	1,08,000	By Costing P&L A/c	1,15,500
To Selling, Distribution and Administration A/c	7,500		
	1,15,500		1,15,500

Costing P&L Account

Particulars	(₹ in '000)	Particulars	(₹ in '000)
To Cost of Sales A/c	1,15,500	By Cost Ledger control A/c	1,35,000
To Factory overhead control A/c	2,400		
To Cost Ledger control A/c (bal. fig.) (Profit)	17,100		
	1,35,000		1,35,000

Building Construction Account

Particulars	(₹ in '000)	Particulars	(₹ in '000)
To Balance b/d	3,000	By Cost Ledger control A/c	13,200
To Stores Ledger control A/c	1,200		
To Wages Control A/c	3,000		
To Factory overhead control A/c	6,000		
	13,200		13,200

Trial Balance

Particulars	Dr. (₹ in '000)	Cr. (₹ in '000)
Stores Ledger Control A/c	16,500	
WIP Control A/c	7,500	
Finished Goods Control A/c	1,20,900	
Cost Ledger Control A/c		1,44,900
	1,44,900	1,44,900

Workings:

$$\text{Cost of Goods sold} = \frac{₹\ 13,50,00,000 \times 80}{100} = ₹\ 10,80,00,000$$

ICAI · Old P3 · RTP Dec 2021 · question p. 6-7 · answer p. 20-24

527 RTP Old P3 May 2020 Q6 old scheme

The following are the balances existed in the books of JPG Ltd. for the year ended, 31st March, 2019:

Particulars	Dr. (₹)	Cr. (₹)
Stores Ledger Control A/c	30,00,000	
WIP Control A/c	15,00,000	
Finished Goods Control A/c	25,00,000	
Manufacturing Overheads Control A/c		1,50,000
Cost Ledger Control A/c		68,50,000

During the year 2019-20, the following transactions took place:

Particulars	Amount (₹)
Finished product (at cost)	22,50,000
Manufacturing Overhead incurred	8,50,000
Raw material purchased	12,50,000
Factory wages	4,00,000
Indirect labour	2,00,000
Cost of sales	17,50,000
Materials issued to production	13,50,000
Sales returned (at cost)	90,000
Material returned to suppliers	1,30,000
Manufacturing overhead charged to production	8,50,000

Required:

PREPARE the following control accounts and Trial balance at the end of the year:

Cost Ledger, Stores Ledger, Work-in-process, Finished Stock, Manufacturing Overhead, Wages and Cost of Sales.

ICAI'S ANSWER

Cost Ledger Control Account

Particulars	(₹)	Particulars	(₹)
To Stores Ledger control A/c	1,30,000	By Balance b/d	68,50,000
To Costing Profit & Loss A/c	17,10,000	By Stores Ledger control A/c	12,50,000
		By Wages Control A/c	6,00,000
To Balance c/d	77,10,000	By Manufacturing overhead control A/c	8,50,000
	95,50,000		95,50,000

Store Ledger Control Account

Particulars	(₹)	Particulars	(₹)
To Balance b/d	30,00,000	By WIP Control A/c	13,50,000
To Cost Ledger control A/c	12,50,000	By Cost Ledger control A/c (return)	1,30,000
		By Balance c/d	27,70,000
	42,50,000		42,50,000

WIP Control Account

Particulars	(₹)	Particulars	(₹)
To Balance b/d	15,00,000	By Finished Stock Control A/c	22,50,000
To Wages Control A/c	4,00,000		
To Stores Ledger control A/c	13,50,000		
To Manufacturing overhead control A/c	8,50,000	By Balance c/d	18,50,000
	41,00,000		41,00,000

Finished Stock Control Account

Particulars	(₹)	Particulars	(₹)
To Balance b/d	25,00,000	By Cost of Sales A/c	17,50,000
To WIP Control A/c	22,50,000		
To Cost of Sales A/c (sales return)	90,000	By Balance c/d	30,90,000
	48,40,000		48,40,000

Manufacturing Overhead Control Account

Particulars	(₹)	Particulars	(₹)
To Cost Ledger Control A/c	8,50,000	By Balance b/d	1,50,000
To Wages Control A/c	2,00,000	By WIP Control A/c	8,50,000
		By Costing P&L A/c (under recovery)	50,000
	10,50,000		10,50,000

Wages Control Account

Particulars	(₹)	Particulars	(₹)
To Cost Ledger Control A/c	6,00,000	By WIP Control A/c	4,00,000
		By Manufacturing overhead control A/c	2,00,000
	6,00,000		6,00,000

Cost of Sales Account

Particulars	(₹)	Particulars	(₹)
To Finished Stock Control A/c	17,50,000	By Finished Stock Control A/c (sales return)	90,000
		By Costing Profit & Loss A/c	16,60,000
	17,50,000		17,50,000

Trial Balance

Particulars	Dr. (₹)	Cr. (₹)
Stores Ledger Control A/c	27,70,000	
WIP Control A/c	18,50,000	
Finished Goods Control A/c	30,90,000	
Cost Ledger Control A/c		77,10,000
	77,10,000	77,10,000

Working:

Costing P&L Account

Particulars	(₹)	Particulars	(₹)
To Cost of Sales A/c	16,60,000	By Cost Ledger control A/c	17,10,000
To Manufacturing overhead control A/c	50,000		
	17,10,000		17,10,000

ICAI · Old P3 · RTP May 2020 · question p. 4-5 · answer p. 16-18

528 MTP Jan 2025 S-I Q1(b) 5 marks

The following information have been extracted from the cost records of a manufacturing company:

		(₹)
	Stores	
*	Opening balance	9,000
*	Purchases	48,000
*	Transfer from WIP	24,000
*	Issue to work-in-progress	48,000
*	Issue for repairs	6,000
*	Deficiency found in stock	1,800
	Work-in-Progress:	
*	Opening balance	18,000
*	Direct Wages applied	18,000
*	Overhead charged	72,000
*	Closing balance	12,000
	Finished Production :	
*	Entire production is sold at a profit of 10% on cost from work-in-progress	
*	Wages paid.	21,000
*	Overhead incurred	75,000

DRAW the Stores Leger Control A/c, Work-in-Progress Control A/c, Overheads Control A/c and Costing Profit and Loss A/c.

ICAI'S ANSWER

Stores Ledger Control A/c

Particulars	(₹)	Particulars	(₹)
To Balance b/d	9,000	By Work in Process	48,000
To General Ledger Adjustment A/c	48,000	By Overhead Control A/c	6,000
To Work in Process A/c	24,000	By Overhead Control A/c (Deficiency)	1,800*
		By Balance c/d	25,200
	81,000		81,000

*Deficiency assumed as normal (alternatively can be treated as abnormal loss)

Work in Progress Control A/c

Particulars	(₹)	Particulars	(₹)
To Balance b/d	18,000	By Stores Ledger Control a/c	24,000
To Stores Ledger Control A/c	48,000	By Costing P/L A/c (Balancing figures being Cost of finished goods)	1,20,000
To Wages Control A/c	18,000	By Balance c/d	12,000
To Overheads Control a/c	72,000		
	1,56,000		1,56,000

Overheads Control A/c

Particulars	(₹)	Particulars	(₹)
To Stores Ledger Control A/c	6,000	By Work in Process A/c	72,000
To Stores Ledger Control A/c	1,800	By Balance c/d (Under absorption)	13,800
To Wages Control A/c (₹ 21,000- ₹18,000)	3,000		
To Gen. Ledger Adjust. A/c	75,000		
	85,800		85,800

Costing Profit & Loss A/c

Particulars	(₹)	Particulars	(₹)
To Work in progress	1,20,000	By Gen. ledger Adjust. A/c (Sales) (1,20,000 + 12,000)	1,32,000
To Gen. Ledger Adjust. A/c (Profit)	12,000		
	1,32,000		1,32,000

ICAI · P4 · MTP Jan 2025 · Series I · question p. 8-9 · answer p. 7-8

529 MTP **Jan 2025 S-II** MCQ 12 2 marks MCQ

WHICH of the following is the correct journal entry as would appear in the cost books when there is under recovery of overheads?

A Cost of Sales A/c..... Dr. xxx
To Administrative Overhead Control A/c xxx

B Production Overhead Control A/c.....Dr. xxx
To Work-in-Process Ledger Control A/c xxx

C Costing Profit & Loss A/c.....Dr. xxx
To Administrative Overhead Control A/c xxx

D Work-in-Process Ledger Control A/c.....Dr. xxx
To Production Overhead Control A/c xxx

ICAI'S ANSWER

(C)

ICAI · P4 · MTP Jan 2025 · Series II · question p. 5 · answer p. 5

530 **MTP** **Sep 2024 S-I** Q6(c) **4 marks**

WHAT are the essential pre-requisites for integrated accounts?

ICAI'S ANSWER

The essential pre-requisites for integrated accounts include the following steps:

1. The management's decision about the extent of integration of the two sets of books. Some concerns find it useful to integrate up to the stage of prime cost or factory cost while other prefers full integration of the entire accounting records.
2. A suitable coding system must be made available so as to serve the accounting purposes of financial and cost accounts.
3. An agreed routine, with regard to the treatment of provision for accruals, prepaid expenses, other adjustment necessary for preparation of interim accounts.
4. Perfect coordination should exist between the staff responsible for the financial and cost aspects of the accounts and an efficient processing of accounting documents should be ensured.
Under this system there is no need for a separate cost ledger. Of course, there will be a number of subsidiary ledgers; in addition to the useful Customers' Ledger and the Purchase Ledger, there will be: (a) Stores Ledger; (b) Stock Ledger and (c) Job Ledger.

ICAI · P4 · MTP Sep 2024 · Series I · question p. 12 · answer p. 17-18

531 **MTP** **May 2024 S-I** Q3(a) **4 marks**

What are the important ledgers to be maintained under non-integrated accounting system in the Cost Accounting?

ICAI'S ANSWER

The important ledgers to be maintained under non-integrated accounting system in the Cost Accounting are the followings:

(a) **Cost Ledger** - This is the principle ledger of the cost department in which impersonal accounts are recorded. This ledger is made self-balancing by maintaining therein a Control Account for each subsidiary ledger.

(b) **Stores Ledger** - It contains an account for each item of stores. The entries in each account maintained in this ledger are made from the invoice, goods received note, material requisitions, material received note etc. Accounts in respect of each item of stores show receipt, issue and balance in physical as well as in monetary terms.

(c) **Work-in-Process Ledger** - This ledger is also known as job ledger, it contains accounts of unfinished jobs and processes. All material costs, wages and overheads for each job in process are posted to the respective job accounts in this ledger. The balance in a job account represents total balance of job/work-in-process, as shown by the job account.

(d) **Finished Goods Ledger** - It contains an account for each item of finished product manufactured or the completed job. If the finished product is transferred to stock, a credit entry is made in the work-in-process ledger and a corresponding debit entry is made in this ledger.

ICAI · P4 · MTP May 2024 · Series I · question p. 8 · answer p. 7-8

532 MTP **Old P3 May 2023 S-I** Q6(c) 5 marks old scheme

Same question, different ICAI answer: #523 RTP May 2024

"Is reconciliation of cost accounts and financial accounts necessary in case of integrated accounting system?" EXPLAIN.

ICAI'S ANSWER

In integrated accounting system cost and financial accounts are kept in the same set of books. Such a system will have to afford full information required for Costing as well as for Financial Accounts. In other words, information and data should be recorded in such a way so as to enable the firm to ascertain the cost (together with the necessary analysis) of each product, job, process, operation or any other identifiable activity. It also ensures the ascertainment of marginal cost, variances, abnormal losses and gains. In fact all information that management requires from a system of Costing for doing its work properly is made available. The integrated accounts give full information in such a manner so that the profit and loss account and the balance sheet c an be prepared according to the requirements of law and the management maintains full control over the liabilities and assets of its business.

Since, only one set of books are kept for both cost accounting and financial accounting purpose so there is no necessity of reconciliation of cost and financial accounts.

ICAI · Old P3 · MTP May 2023 · Series I · question p. 6 · answer p. 11

533 MTP **Old P3 Nov 2022 S-II** Q4(b) 10 marks old scheme

The following are the balances existed in the books of JPG Ltd. for the year ended, 31st March, 2022:

Particulars	Dr.	Cr.
	(₹)	(₹)
Stores Ledger Control A/c	15,00,000	
WIP Control A/c	7,50,000	
Finished Goods Control A/c	12,50,000	
Manufacturing Overheads Control A/c		75,000
Cost Ledger Control A/c		34,25,000

During the year 2022-23, the following transactions took place:

Particulars	Amount (₹)
Finished product (at cost)	11,25,000
Manufacturing Overhead incurred	4,25,000

Particulars	Amount (₹)
Raw material purchased	6,25,000
Factory wages	2,00,000
Indirect labour	1,00,000
Cost of sales	8,75,000
Materials issued to production	6,75,000
Sales returned (at cost)	45,000
Material returned to suppliers	65,000
Manufacturing overhead charged to production	4,25,000

Required:

PREPARE the following control accounts and Trial balance at the end of the year:

Cost Ledger, Stores Ledger, Work-in-process, Finished Stock, Manufacturing Overhead, Wages and Cost of Sales.

ICAI'S ANSWER

Cost Ledger Control Account

Particulars	(Rs.)	Particulars	(Rs.)
To Stores Ledger control A/c	65,000	By Balance b/d	34,25,000
To Costing Profit & Loss A/c	8,55,000	By Stores Ledger control A/c	6,25,000
		By Wages Control A/c	3,00,000
To Balance c/d	38,55,000	By Manufacturing overhead control A/c	4,25,000
	47,75,000		47,75,000

Store Ledger Control Account

Particulars	(Rs.)	Particulars	(Rs.)
To Balance b/d	15,00,000	By WIP Control A/c	6,75,000
To Cost Ledger control A/c	6,25,000	By Cost Ledger control A/c (return)	65,000
		By Balance c/d	13,85,000
	21,25,000		21,25,000

WIP Control Account

Particulars	(Rs.)	Particulars	(Rs.)
To Balance b/d	7,50,000	By Finished Stock Control A/c	11,25,000
To Wages Control A/c	2,00,000		
To Stores Ledger control A/c	6,75,000		
To Manufacturing overhead control A/c	4,25,000	By Balance c/d	9,25,000
	20,50,000		20,50,000

Finished Stock Control Account

Particulars	(Rs.)	Particulars	(Rs.)
To Balance b/d	12,50,000	By Cost of Sales A/c	8,75,000
To WIP Control A/c	11,25,000		
To Cost of Sales A/c (sales return)	45,000	By Balance c/d	15,45,000
	24,20,000		24,20,000

Manufacturing Overhead Control Account

Particulars	(Rs.)	Particulars	(Rs.)
To Cost Ledger Control A/c	4,25,000	By Balance b/d	75,000
To Wages Control A/c	1,00,000	By WIP Control A/c	4,25,000
		By Costing P&L A/c (under recovery)	25,000
	5,25,000		5,25,000

Wages Control Account

Particulars	(Rs.)	Particulars	(Rs.)
To Cost Ledger Control A/c	3,00,000	By WIP Control A/c	2,00,000
		By Manufacturing overhead control A/c	1,00,000
	3,00,000		3,00,000

Cost of Sales Account

Particulars	(Rs.)	Particulars	(Rs.)
To Finished Stock Control A/c	8,75,000	By Finished Stock Control A/c (sales return)	45,000
		By Costing Profit & Loss A/c	8,30,000
	8,75,000		8,75,000

Trial Balance

Particulars	Dr.	Cr.
	(Rs.)	**(Rs.)**
Stores Ledger Control A/c	13,85,000	
WIP Control A/c	9,25,000	
Finished Goods Control A/c	15,45,000	
Cost Ledger Control A/c		38,55,000
	38,55,000	38,55,000

Working:

Costing P&L Account

Particulars	(Rs.)	Particulars	(Rs.)
To Cost of Sales A/c	8,30,000	By Cost Ledger control A/c	8,55,000
To Manufacturing overhead control A/c	25,000		
	8,55,000		8,55,000

ICAI · Old P3 · MTP Nov 2022 · Series II · question p. 5 · answer p. 10-12

534 MTP **Old P3 May 2022 S-I** Q6(d) 5 marks old scheme

EXPLAIN what are the essential pre-requisites of Integrated accounting system?

ICAI'S ANSWER

Essential pre-requisites for Integrated Accounts: The essential pre-requisites for integrated accounts include the following steps-

1. The management's decision about the extent of integration of the two sets of books. Some concerns find it useful to integrate up to the stage of prime cost or factory cost while other prefers full integration of the entire accounting records.
2. A suitable coding system must be made available so as to serve the accounting purposes of financial and cost accounts.
3. An agreed routine, with regard to the treatment of provision for accruals, prepaid expenses, other adjustment necessary for preparation of interim accounts.
4. Perfect coordination should exist between the staff responsible for the financial and cost aspects of the accounts and an efficient processing of accounting documents should be ensured.

ICAI · Old P3 · MTP May 2022 · Series I · question p. 7 · answer p. 11

535 MTP **Old P3 May 2022 S-II** Q6(d) 5 marks old scheme

BRIEF OUT advantages of Integrated Accounts.

ICAI'S ANSWER

Advantages of Integrated Accounts are as follows:

(i) No need for Reconciliation- The question of reconciling costing profit and financial profit does not arise, as there is only one figure of profit.

(ii) Less efforts- Due to use of one set of books, there is a significant saving in efforts made.

(iii) Less time consuming- No delay is caused in obtaining information as it is provided from books of original entry.

(iv) Economical process- It is economical also as it is based on the concept of "Centralisation of Accounting function".

ICAI · Old P3 · MTP May 2022 · Series II · question p. 7 · answer p. 13

536 MTP **Old P3 Dec 2021 S-I** Q6(a) 5 marks old scheme

JOURNALISE the following transactions in cost books under Non-Integrated system of Accounting.

(i)	Credit Purchase of Material	₹ 27,000
(ii)	Manufacturing overhead charged to Production	₹ 6,000
(iii)	Selling and Distribution overheads recovered from Sales	₹ 4,000
(iv)	Indirect wages incurred for Manufacturing department	₹ 8,000
(v)	Material returned from production to stores	₹ 9,000

ICAI'S ANSWER

Journal entries are as follows:

			Dr. (₹)	Cr. (₹)
(i)	Stores Ledger Control A/c.....	Dr.	27,000	
	To Cost Ledger Control A/c			27,000
(ii)	Work-in-Process Control A/c.....	Dr.	6,000	
	To Manufacturing Overhead Control A/c			6,000
(iii)	Cost of Sales A/c.....	Dr.	4,000	
	To Selling & Dist. Overhead Control A/c			4,000
(iv)	(1) Wage Control A/c.....	Dr.	8,000	
	To Cost Ledger Control A/c			8,000
	(2) Manufacturing Overhead Control A/c.....	Dr.	8,000	
	To Wages Control A/c			8,000
	OR			

			Dr. (₹)	Cr. (₹)
	Manufacturing Overhead Control A/c.....	Dr.	8,000	
	To Cost Ledger Control A/c			8,000
(v)	Stores Ledger Control A/c	Dr.	9,000	
	To Work-in-Process Control A/c			9,000

*Cost Ledger Control A/c is also known as General Ledger Control A/c

ICAI · Old P3 · MTP Dec 2021 · Series I · question p. 6 · answer p. 11

537 MTP **Old P3 Dec 2021 S-II** Q3(b) 10 marks old scheme

XYZ Ltd. maintains a non-integrated accounting system for the purpose of management information. The following are the data related with year 2020-21:

Particulars	Amount ('000)
Opening balances:	
- Stores ledger control A/c	48,000
- Work-in-process control A/c	12,000
- Finished goods control A/c	2,58,000
- Building construction A/c	6,000
- Cost ledger control A/c	3,24,000
During the year following transactions took place:	
Materials:	
- Purchased	24,000
- Issued to production	30,000
- Issued to general maintenance	3,600
- Issued to building construction	2,400
Wages:	
- Gross wages paid	90,000
- Indirect wages paid	24,000
- For building construction	6,000
Factory overheads:	
- Actual amount incurred (excluding items shown above)	96,000

Particulars	Amount ('000)
- Absorbed in building construction	12,000
- Under-absorbed	4,800
Royalty paid	3,000
Selling distribution and administration overheads	15,000
Sales	2,70,000

At the end of the year, the stock of raw material and work-in-process was ₹ 33,00,000, and ₹15,00,000 respectively. The loss arising in the raw material account is treated as factory overheads. The building under construction was completed during the year. Gross profit margin is 20% on sales.

Required:

PREPARE the relevant control accounts to record the above transactions in the cost ledger of the company.

ICAI'S ANSWER

Cost Ledger Control Account

Particulars	(₹ in '000)	Particulars	(₹ in '000)
To Costing P&L A/c	2,70,000	By Balance b/d	3,24,000
To Building Construction A/c	26,400	By Stores Ledger control A/c	24,000
To Balance c/d	2,89,800	By Wages Control A/c	90,000
		By Factory overhead control A/c	96,000
		By Royalty A/c	3,000
		By Selling. Distribution and Administration overheads	15,000
		By Costing P&L A/c	34,200
	5,86,200		5,86,200

Stores Ledger Control Account

Particulars	(₹ in '000)	Particulars	(₹ in '000)
To Balance b/d	48,000	By WIP control A/c	30,000
To Cost Ledger control A/c	24,000	By Factory overheads control A/c	3,600
		By Building construction A/c	2,400
		By Factory overhead control A/c (loss) (bal. fig.)	3,000
		By Balance c/d	33,000
	72,000		72,000

Wages Control Account

Particulars	(₹ in '000)	Particulars	(₹ in '000)
To Cost Ledger control A/c	90,000	By Factory overhead control A/c	24,000
		By Building Construction A/c	6,000
		By WIP Control A/c (bal. fig.)	60,000
	90,000		90,000

Factory Overhead Control Account

Particulars	(₹ in '000)	Particulars	(₹ in '000)
To Stores Ledger control A/c	3,600	By Building Construction A/c	12,000
To Wages Control A/c	24,000	By Costing P&L A/c	4,800
To Cost Ledger control A/c	96,000	By WIP Control A/c (bal. fig)	1,09,800
To Stores Ledger control A/c (loss)	3,000		
	1,26,600		1,26,600

Royalty Account

Particulars	(₹ in '000)	Particulars	(₹ in '000)
To Cost Ledger control A/c	3,000	By WIP Control A/c	3,000
	3,000		3,000

Work-in-process Control Account

Particulars	(₹ in '000)	Particulars	(₹ in '000)
To Balance b/d	12,000	By Finished goods control A/c (bal. fig)	1,99,800
To Stores Ledger control A/c	30,000		
To Wages Control A/c	60,000		
To Factory overhead control A/c	1,09,800		
To Royalty A/c	3,000	By Balance c/d	15,000
	2,14,800		2,14,800

Finished Goods Control Account

Particulars	(₹ in '000)	Particulars	(₹ in '000)
To Balance b/d	2,58,000	By Cost of Goods Sold A/c (Refer working note)	2,16,000
To WIP control A/c	1,99,800	By Balance c/d	2,41,800
	4,57,800		4,57,800

Cost of Goods Sold Account

Particulars	(₹ in '000)	Particulars	(₹ in '000)
To Finished Goods control A/c	2,16,000	By Cost of sales A/c	2,16,000
	2,16,000		2,16,000

Selling, Distribution and Administration Overhead Control Account

Particulars	(₹ in '000)	Particulars	(₹ in '000)
To Cost Ledger control A/c	15,000	By Cost of sales A/c	15,000
	15,000		15,000

Cost of Sales Account

Particulars	(₹ in '000)	Particulars	(₹ in '000)
To Cost of Goods Sold A/c	2,16,000	By Costing P&L A/c	2,31,000
To Selling, Distribution and Administration A/c	15,000		
	2,31,000		2,31,000

Costing P&L Account

Particulars	(₹ in '000)	Particulars	(₹ in '000)
To Cost of Sales A/c	2,31,000	By Cost Ledger control A/c	2,70,000
To Factory overhead control A/c	4,800		
To Cost Ledger control A/c	34,200		
	2,70,000		2,70,000

Building Construction Account

Particulars	(₹ in '000)	Particulars	(₹ in '000)
To Balance b/d	6,000	By Cost Ledger control A/c	26,400
To Stores Ledger control A/c	2,400		
To Wages Control A/c	6,000		
To Factory overhead control A/c	12,000		
	26,400		26,400

Trial Balance

Particulars	Dr.	Cr.
	(₹ in '000)	**(₹ in '000)**
Stores Ledger Control A/c	33,000	
WIP Control A/c	15,000	
Finished Goods Control A/c	2,41,800	
Cost Ledger Control A/c		2,89,800
	2,89,800	2,89,800

Working Note:

$$\text{Cost of Goods sold} = \frac{₹\ 2,70,000 \times 80}{100} = ₹\ 2,16,000$$

ICAI · Old P3 · MTP Dec 2021 · Series II · question p. 5 · answer p. 8-11

538 MTP **Old P3 May 2020 S-I** Q5(b)(i) 5 marks old scheme

SHOW Journal entries for the following transactions assuming cost and financial accounts are

integrated:

(1)	Materials issued:	
	Direct	Rs. 6,50,000
	Indirect (to factory)	Rs. 2,30,000
(2)	Allocation of wages (25% indirect)	Rs. 9,00,000
(3)	Under/Over absorbed overheads:	
	Factory (Over)	Rs. 60,000
	Administration (Under)	Rs. 50,000
(4)	Payment to Creditors (Trade payables)	Rs. 9,00,000
(5)	Collection from Debtors (Trade receivables)	Rs. 8,00,000

ICAI'S ANSWER

Journal Entries under Integrated system of accounting

	Particulars		Rs.	Rs.
(i)	Work-in-Progress Control A/c	Dr.	6,50,000	
	Factory Overhead Control A/c	Dr.	2,30,000	
	To Stores Ledger Control A/c			8,80,000
	(Being issue of Direct and Indirect materials)			
(ii)	Work-in Progress Ledger Control A/c	Dr.	6,75,000	
	Factory Overhead control A/c	Dr.	2,25,000	
	To Wages Control A/c			9,00,000
	(Being allocation of Direct and Indirect wages)			
(iii)	Factory Overhead Control A/c	Dr.	60,000	
	To Costing Profit & Loss A/c			60,000
	(Being transfer of over absorption of Factory overhead)			
	Costing Profit & Loss A/c	Dr.	50,000	
	To Administration Overhead Control A/c			50,000
	(Being transfer of under absorption of Administration overhead)			
(iv)	Trade Payables A/c	Dr.	9,00,000	
	To Cash/ Bank A/c			9,00,000

	Particulars		Rs.	Rs.
	(Being payment made to creditors)			
(v)	Cash/ Bank A/c	Dr.	8,00,000	
	To Trade receivables A/c			8,00,000
	(Being payment received from debtors)			

ICAI · Old P3 · MTP May 2020 · Series I · question p. 6 · answer p. 11-12

7.2 Reconciling cost and financial profit

46 questions · asked in 43 ICAI documents · 11 exam papers

539 **EXAM** **Sep 2026** Q3(a) **7 marks**

During the finalisation of accounts for the period ended 31st August, 2026, the Management of M/s. ZPT Ltd. observed that the profit disclosed by the Cost Accounts differed from that shown by the Financial Accounts.

The following information was extracted from the books :

Profit as per Cost Accounts amounted to ₹ 9,80,000.

On investigation, the following items were identified :

(i) Factory overheads were under-absorbed in the Cost Accounts by ₹ 16,000.

(ii) Administration overheads were over-absorbed in the Cost Accounts by ₹ 9,000.

(iii) Depreciation charged in the Financial Accounts exceeded the depreciation recovered in the Cost Accounts by ₹ 18,000.

(iv) Interest on bank deposits amounted to ₹ 28,000.

(v) Provision for Income Tax was ₹ 1,60,000.

(vi) Profit on sale of investments amounted to ₹ 15,000.

(vii) Charges in lieu of rent for owned premises were ₹ 6,000.

(viii) Donation amounted to ₹ 10,000.

Required :

Prepare a Reconciliation Statement starting from the profit as per Cost Accounts and arrive at the profit as per Financial Accounts while maintaining the serial order given above for adjustments.

(Candidates may use shorter forms of the items instead of writing the whole lines as described in the question.)

ICAI'S ANSWER

ICAI has not published its answers to this paper yet.

ICAI · P4 · Sep 2026 exam · question p. 15-16

540 EXAM May 2026 Q1(c) 4 marks

A manufacturing concern reported a net loss of ₹ 4,20,000 as per the Cost Accounts for the year ended 31st March, 2026. Upon scrutiny of both sets of accounts, the following differences were identified:

Sr. No.	Particulars	Amount (₹)
(i)	Profit on sale of investments (Financial Accounts only)	55,000
(ii)	Factory overheads under-absorbed in Cost Accounts	70,000
(iii)	Discount allowed to customers (Financial Accounts only)	30,000
(iv)	Depreciation overcharged in Cost Accounts	90,000
(v)	Selling overheads over-absorbed in Cost Accounts	65,000
(vi)	Bad debts written-off (Financial Accounts only)	1,10,000
(vii)	Over-valuation of closing stock in Cost Accounts	40,000

You are required to:

(i) Prepare a Reconciliation Statement.

(ii) Ascertain the net profit / (loss) as per the Financial Accounts.

ICAI'S ANSWER

Reconciliation Statement

Particulars	Amount (₹)	Amount (₹)
Net Loss as per Cost Accounts		**4,20,000**
Add:		
Profit on sale of investments	55,000	
Depreciation overcharged in Cost Accounts	90,000	
Selling overheads over-absorbed in Cost Accounts	65,000	**2,10,000**
		2,10,000
Less:		
Factory overheads under-absorbed in Cost Accounts	70,000	
Discount allowed to customers	30,000	
Bad debts written-off	1,10,000	
Over-valuation of closing stock in Cost Accounts	40,000	**2,50,000**
Net Loss as per Financial Accounts		**4,60,000**

ICAI · P4 · May 2026 exam · question p. 9 · answer p. 11-12

541 EXAM Jan 2026 Q4(b) 7 marks

The following financial information is available for a company with a normal production capacity of 40,000 units for the year ended 31st March, 2025:

- Sales amounted to ₹ 8,00,000, Units sold: 35,000 units.
- There was no opening or closing stock of finished units.
- Direct material and direct wages costs were ₹ 3,50,000 and ₹ 2,00,000 respectively.
- Actual factory expenses were ₹ 1,00,000, of which 55% were fixed.
- Administrative expenses related to production activities incurred during the year but not paid were ₹ 50,000, which were entirely fixed.
- Actual selling and distribution expenses amounted to ₹ 20,000, of which 35% were fixed.
- The company earned interest and dividend income totaling ₹ 10,000.

You are required to prepare:

(i) The cost sheet and ascertain the profit as per cost accounts for the year ended 31st March, 2025, assuming that the indirect expenses are absorbed on the basis of normal production capacity; and

(ii) A statement reconciling the profits shown by financial and cost accounts.

ICAI'S ANSWER

Cost Sheet for the year ended 31st March 31st March 2025

Particulars	(₹)	(₹)
Direct Material		3,50,000
Direct Wages		2,00,000
Prime cost		5,50,000
Factory expenses:		
Variable (45% of ₹ 1,00,000)	45,000	
Fixed* (₹ 55,000 × 35,000/40,000)	48,125	93,125
Works cost		6,43,125
Administrative expenses related to production activities (₹ 50,000 × 35,000/40,000)		43,750
Cost of production		6,86,875
Selling & Distribution expenses:		
Variable (65% of ₹ 20,000)	13,000	
Fixed (₹ 7,000 × 35,000/40,000)	6,125	19,125
Cost of Sales		7,06,000
Profit (Balancing figure)		94,000

Particulars	(₹)	(₹)
Sales revenue		8,00,000

Statement Reconciling profits shown by Financial and Cost Accounts

Particulars	(₹)	(₹)
Profit as per Cost Accounts		94,000
Add: Income from interest and dividends		10,000
		1,04,000
Less: Factory expenses under-charged in Cost Accounts (₹ 1,00,000 – ₹ 93,125)	6,875	
Administrative expenses under-charged in Cost Accounts (₹ 50,000 – ₹ 43,750)	6,250	
Selling & distribution expenses under—charged in Cost Accounts (₹ 20,000 – ₹ 19,125)	875	(14,000)
Profit as per Financial Accounts		90,000

ICAI · P4 · Jan 2026 exam · question p. 21 · answer p. 22-23

542 EXAM **May 2025** Q4(a) 8 marks

NT Ltd. showed a net loss of ₹ 9,000 as per their cost accounts for the year ended 31-03-2025. However, the financial books disclosed a net profit of ₹ 7,000 for the same period. The following information was revealed as a result of scrutiny of the figures of both the sets of books:

	(₹)
Factory overheads absorbed	52,000
Selling & distribution overheads over absorbed	6,000
Administrative overheads under absorbed	7,500
Interest on Loan	8,800
Dividend received	9,000
Factory overheads charged	45,000
Depreciation charged in financial accounts	42,000
Depreciation recovered in cost accounts	45,000

There is the difference in the value of closing stock of finished goods due to varying basis of valuation. 500 units of closing stock is valued at ₹ 41,000 in cost accounts whereas market price per unit of closing stock is ₹ 80 per unit.

It is found that there is still a difference in the reconciliation statement after considering all of the above which is due to the excess of raw material consumption in cost accounts.

Prepare:

(i) A reconciliation statement taking net loss as per cost accounts as base, and

(ii) Find out the excess of material consumption during the period in cost accounts effectuated the difference in reconciliation.

ICAI'S ANSWER

(i) Statement of Reconciliation of profit as obtained under Cost and Financial Accounts

		(₹)	(₹)
	Net Loss as per Cost Records		**(9,000)**
Add:	Excess of Material Consumption (see below)	8,300	
	Factory Overhead over-absorbed (₹ 52,000 - ₹ 45,000)	7,000	
	Selling & Distribution Overhead over-absorbed	6,000	
	Dividend Received	9,000	
	Depreciation over-charged (₹ 45,000 - ₹ 42,000)	3,000	33,300
			24,300
Less:	Administration Overhead under-absorbed	7,500	
	Interest on Loan	8,800	
	Over-valuation of Closing Stock of Finished Goods (₹ 41,000 - ₹ 40,000) (₹ 80 p.u. x 500 units = ₹ 40,000 in financial accounts)	1,000	(17,300)
	Profit as per Financial Records		**7,000**

(ii) Statement showing calculation of excess of material consumption in cost accounts:

		(₹)	(₹)
	Net Loss as per Cost Records		**(9,000)**
Add:	Excess of Material Consumption	X	
	Factory Overhead over-absorbed	7,000	
	Selling & Distribution Overhead over-absorbed	6,000	
	Dividend Received	9,000	
	Depreciation over-charged	3,000	X + 25,000
			X + 16,000
Less:	Administration Overhead under-absorbed	7,500	
	Interest on Loan	8,800	

		(₹)	(₹)
	Over-valuation of Closing Stock of Finished Goods	1,000	(17,300)
	Profit as per Financial Records		**X -1,300**

Therefore, x – 1,300	=	7,000
Or x	=	7,000 + 1,300
	=	8,300

Excess of material consumption during the period in cost accounts is ₹ 8,300.

ICAI · P4 · May 2025 exam · question p. 22 · answer p. 23-24

543 EXAM **Jan 2025** Q6(a) 5 marks

Explain the steps involved in procedure for reconciliation of Cost & Financial accounts. Also explain the circumstances where reconciliation statement can be avoided.

ICAI'S ANSWER

There are 3 steps involved in the procedure for reconciliation.

i. Ascertainment of profit as per Financial Accounts

ii. Ascertainment of profit as per Cost Accounts

iii. Reconciliation of both the profits

Circumstances where reconciliation statement can be avoided: When the Cost and Financial Accounts are integrated - there is no need to have a separate reconciliation statement between the two sets of accounts. Integration means that the same set of accounts fulfil the requirement of both i.e., Cost and Financial Accounts.

ICAI · P4 · Jan 2025 exam · question p. 27 · answer p. 28

544 EXAM **Sep 2024** Q1(b) 5 marks

XYZ Ltd. declared a net profit of ₹ 2,25,000 based on their financial accounts for the year ending 31st March, 2024. The profit disclosed in cost books are not matched with financial accounts. The following information were revealed during the scrutiny of the figures of both the sets of books:

Sr. No.	Particulars	₹
1.	Preliminary expenses written off in financial accounts	35,000
2.	Factory Overheads Over charged in cost accounts	20,000
3.	Expenses on issue of shares in financial accounts	30,000
4.	Undervaluation of closing stock in cost accounts	65,000
5.	Interest on Bank Deposits in financial accounts	60,000
6.	Under recovery of administration overheads in cost accounts	25,000

Sr. No.	Particulars	₹
7.	Notional Rent of own premises charged in cost accounts	30,000
8.	Under recovery of selling overheads in cost accounts	35,000
9.	Bad debts recovered in financial accounts	50,000

Required:

Prepare Reconciliation Statement to arrive at net profit/loss as per Cost Accounts.

ICAI'S ANSWER

Reconciliation Statement

	(₹)	(₹)
Profit (loss) as per Financial Accounts		2,25,000
Add:		
Preliminary expenses written off	35,000	
Expenses on issue of shares in financial accounts	30,000	
Under recovery of administration overheads in cost accounts	25,000	
Under recovery of selling overheads in cost accounts	35,000	
		1,25,000
Less:		
Factory Overheads Over charged in cost accounts	20,000	
Undervaluation of closing stock in cost accounts	65,000	
Interest on Bank Deposits	60,000	
Notional Rent of own premises charged in cost accounts	30,000	
Bad debts recovered in financial accounts	50,000	
		(2,25,000)
Net Profit as per Cost Accounts		**1,25,000**

ICAI · P4 · Sep 2024 exam · question p. 9 · answer p. 11

545 EXAM **May 2024** Q3(b) 7 marks

S.K. Manufacturing Co. Ltd. showed a net profit of ₹ 5,40,400 as per their cost accounts for the year ended 31.03.2004. However, the financial books disclosed a net profit of ₹ 2,60,500 for the same period.

The following information was revealed as a result of scrutiny of the figures of both the sets of books:

	₹
Factory overheads under absorbed	84,800
Administrative overheads over absorbed	24,000
Interest paid on bank borrowings	50,000
Interest & Divided received	65,200
Notional rent of own premises charged in cost accounts	60,000
Losses on the sales of fixed assets and investments	48,000
Donations and subscriptions	18,800
Overvaluation of closing stock of finished goods in Cost accounts	1,25,000
Store adjustments (credited in financial books)	7,500
Depreciation over charged in cost accounts	40,000
Income tax provided	1,50,000

You are required to:

(i) Prepare a reconciliation statement taking net profit as per cost accounts as base.

(ii) State when is the reconciliation statement of Cost and Financial accounts not required?

ICAI'S ANSWER

(i) Statement of Reconciliation of profit as obtained under Cost and Financial Accounts

		(₹)	(₹)
	Profit as per Cost Records		**5,40,400**
Add:	Administrative Overhead over absorbed	24,000	
	Interest & Dividend Received	65,200	
	Notional rent of own premises	60,000	
	Stores adjustments (credited in financial books)	7,500	
	Depreciation over charged in cost accounts	40,000	1,96,700
			7,37,100
Less:	Factory overheads under absorbed	84,800	
	Interest paid on bank borrowings	50,000	
	Losses on sale of fixed assets and investments	48,000	

		(₹)	(₹)
	Donations and subscriptions	18,800	
	Over-valuation of closing stock of finished goods in cost accounts	1,25,000	
	Income tax	1,50,000	(4,76,600)
	Profit as per Financial Records		**2,60,500**

(ii) Circumstances where reconciliation statement can be avoided: When the Cost and Financial Accounts are integrated - there is no need to have a separate reconciliation statement between the two sets of accounts. Integration means that the same set of accounts fulfil the requirement of both i.e., Cost and Financial Accounts.

ICAI · P4 · May 2024 exam · question p. 12 · answer p. 14-15

546 EXAM **Old P3 May 2023** Q5(b) 5 marks old scheme

The following information has been obtained from financial accounting and cost accounting records.

	Financial Accounting	Cost Accounting
	₹	₹
(i) Factory Overhead	94,750	90,000
(ii) Administrative Overhead	60,000	57,000
(iii) Selling Overhead	55,000	61,000
(iv) Opening Stock	17,500	22,500
(v) Closing Stock	12,500	15,000

Required:

Indicate under-recovery and over-recovery and their effects on cost accounting profit.

[Note: You are not required to prepare reconciliation statement.]

ICAI'S ANSWER

	Financial Accounting	Cost Accounting	Difference	Under/Over-recovery	Effect on Cost Accounting Profit	Net Effect* on Cost Accounting Profit
	₹	₹	₹			
(i) Factory Overhead	94,750	90,000	4,750	Under-recovery	Increased	To be reduced/ deducted
(ii) Administrative Overhead	60,000	57,000	3,000	Under-recovery	Increased	To be reduced/ deducted
(iii) Selling Overhead	55,000	61,500	-6,500	Over-recovery	Decreased	To be added
(iv) Opening Stock	17,500	22,500	-5,000	Over valuation	Decreased	To be added
(v) Closing Stock	12,500	15,000	-2,500	Over valuation	Increased	To be reduced/ deducted

*Taking Cost Accounting Profit as base

(Under recovery and over recovery with effect are answered by the candidate, or if under recovery and over recovery with treatment (net effect) are answered, due credit shall be given in both cases)

ICAI · Old P3 · May 2023 exam · question p. 24 · answer p. 27-28

547 EXAM **Old P3 Nov 2022** Q5(b) 5 marks old scheme

X Ltd. follows Non-Integrated Accounting System. Financial Accounts of the company show a Net Profit of ₹ 5,50,000 for the year ended 31st March, 2022. The chief accountant of the company has provided following information from the Financial Accounts and Cost Accounts:

Sr. No	Particulars	(₹)
(i)	Legal Chargers Provided in Financial accounts	15,250
(ii)	Interim Dividend received credited in financial accounts	4,50,000
(iii)	Preliminary Expenses written off in financial accounts	25,750
(iv)	Over recovery of selling overheads in cost accounts	11,380
(v)	Profit on sale of capital asset credited in financial accounts	30,000
(vi)	Under valuation of closing stock in cost accounts	25,000
(vii)	Over recovery of production overheads in cost accounts	10,200
(viii)	Interest paid on Debentures shown in financial accounts	50,000

Required:

Find out the Profit (Loss) as per Cost Accounts by preparing a Reconciliation Statement.

ICAI'S ANSWER

Reconciliation Statement

(Reconciliation the profit as per financial records with the profit as per costing records)

	Particulars	(₹)	Total (₹)
	Profit as per Financial Accounts		5,50,000
Add:	Legal Charges	15,250	
	Preliminary expenses written off	25,750	
	Interest paid	50,000	91,000
			6,41,000
Less:	Under-valuation of closing stock in cost book	25,000	
	Interim Dividend Received	4,50,000	
	Over recovery of selling overheads in cost accounts	11,380	
	Over recovery of production overhead in cost accounts	10,200	5,26,580

	Particulars	(₹)	Total (₹)
	Profit on sale of Assets	30,000	
	Profit as per Cost Accounts		1,14,420

ICAI · Old P3 · Nov 2022 exam · question p. 19 · answer p. 21

548 EXAM Old P3 Nov 2022 Q6(b) 5 marks old scheme

Indicate, for following items, whether to be shown in the Cost Accounts or Financial Accounts:

(i) Preliminary expenses written off during the year

(ii) Interest received on bank deposits

(iii) Dividend, interest received on investments

(iv) Salary for the proprietor at notional figure though not incurred

(v) Charges in lieu of rent where premises are owned

(vi) Rent receivables

(vii) Loss on sale of Fixed Assets

(viii) Interest on capital at notional figure though not incurred

(ix) Goodwill written off

(x) Notional Depreciation on the assets fully depreciated for which book value is Nil.

ICAI'S ANSWER

S. No.	Items	Accounts
(i)	Preliminary expenses written off during the year	Financial Accounts
(ii)	Interest received on bank deposits	Financial Accounts
(iii)	Dividend, interest received on investments	Financial Accounts
(iv)	Salary for the proprietor at notional figure though not incurred	Cost Accounts
(v)	Charges in lieu of rent where premises are owned	Cost Accounts
(vi)	Rent receivables	Financial Accounts
(vii)	Loss on the sales of Fixed Assets	Financial Accounts
(viii)	Interest on capital at notional figure though not incurred	Cost Accounts
(ix)	Goodwill written off	Financial Accounts

S. No.	Items	Accounts
(x)	Notional Depreciation on the assets fully depreciated for which book value is nil	Cost Accounts

ICAI · Old P3 · Nov 2022 exam · question p. 22 · answer p. 23-24

549 EXAM Old P3 Dec 2021 Q4(b) 5 marks old scheme

R Ltd. showed a Net Profit of ₹ 3,60,740 as per their cost accounts for the year ended 31st March, 2021.

The following information was revealed as a result of scrutiny of the figures from the both sets of accounts:

Sr. No.	Particulars	(₹)
i.	Over recovery of selling overheads in cost accounts	10,250
ii.	Over valuation of closing stock in cost accounts	7,300
iii.	Rent received credited in financial accounts	5,450
iv.	Bad debts provided in financial accounts	3,250
v.	Income tax provided in financial accounts	15,900
vi.	Loss on sale of capital asset debited in financial accounts	5,800
vii.	Under recovery of administration overheads in cost accounts	3,600

Required:

Prepare a reconciliation statement showing the profit as per financial records.

ICAI'S ANSWER

Statement of Reconciliation

(Reconciling the profit as per costing records with the profit as per financial records)

	(₹)	(₹)
Net Profit as per Cost Accounts		3,60,740
Add:		
Over recovery of selling overheads in cost accounts	10,250	
Rent received credited in financial accounts	5,450	15,700
		376,440
Less:		
Over valuation of closing stock in cost accounts	7,300	
Bad debts provided in financial accounts	3,250	

	(₹)	(₹)
Income tax provided in financial accounts	15,900	
Loss on sale of capital asset debited in financial accounts	5,800	
Under recovery of administration overheads in cost accounts	3,600	35,850
Profit as per Financial Accounts		3,40,590

ICAI · Old P3 · Dec 2021 exam · question p. 16 · answer p. 18

550 EXAM Old P3 Jul 2021 Q3(a) 10 marks old scheme

The Profit and Loss account of ABC Ltd. for the year ended 31st March, 2021 is given below:

Profit and Loss account

(for the year ended 31st March, 2021)

Particulars	₹	Particulars	₹
To Direct Material	6,50,000	By Sales (15000 units)	15,00,000
To Direct Wages	3,50,000	By Dividend received	9,000
To Factory overheads	2,60,000		
To Administrative overheads	1,05,000		
To Selling overheads	85,000		
To Loss on sale of investments	2,000		
To Net Profit	57,000		
	15,09,000		15,09,000

- Factory overheads are 50% fixed and 50% variable.
- Administrative overheads are 100% fixed.
- Selling overheads are completely variable.
- Normal production capacity of ABC Ltd. is 20,000 units.
- Indirect Expenses are absorbed in the cost accounts on the basis of normal production capacity.
- Notional rent of own premises charged in Cost Accounts is amounting to ₹ 12,000.

You are required to:

(i) Prepare a Cost Sheet and ascertain the Profit as per Cost Records for the year ended 31st March, 2021.

(ii) Reconcile the Profit as per Financial Records with Profit as per Cost Records.

ICAI'S ANSWER

(i) Cost Sheet

(for the year ended 31st March, 2021)

	(₹)	(₹)
Direct material		6,50,000
Direct wages		3,50,000
Prime cost		**10,00,000**
Factory Overheads:		
Variable (50% of ₹ 2,60,000)	1,30,000	
Fixed (₹ 1,30,000 × 15,000/20,000)	97,500	2,27,500
Works cost		**12,27,500**
Administrative Overheads (₹ 1,05,000 × 15,000/20,000)		78,750
Notional Rent		12,000
Cost of production		**13,18,250**
Selling Overheads		85,000
Cost of Sales		**14,03,250**
Profit (Balancing figure)		96,750
Sales revenue		**15,00,000**

(ii) Statement of Reconciliation

(Reconciling profit shown by Financial and Cost Accounts)

	(₹)	(₹)
Profit as per Cost Account		96,750
Add: Dividend received	9,000	
Add: Notional Rent	12,000	21,000
Less: Factory Overheads under-charged in Cost Accounts (₹ 2,60,000 – ₹ 2,27,500)	32,500	
Less: Administrative expenses under-charged in Cost Accounts (₹ 1,05,000 – ₹ 78,750)	26,250	
Less: Loss on sale of Investments	2,000	(60,750)
Profit as per Financial Accounts		**57,000**

(Note: Solution can be done considering base profit as per Financial Accounts)

ICAI · Old P3 · Jul 2021 exam · question p. 64-65 · answer p. 66-67

551 RTP Sep 2026 Q9

The financial books of a company reveal the following data for the year ended 31st March, 2026:

	(₹)
Opening Stock:	
Finished goods 875 units	76,525
Work-in-process	33,000
01.04.2025 to 31.3.2026	
Raw materials consumed	7,84,000
Direct Labour	4,65,000
Factory overheads	2,65,000
Goodwill written off	95,000
Administration overheads	3,15,000
Interest paid	72,000
Bad Debts	21,000
Selling and Distribution Overheads	65,000
Interest received	18,500
Rent received	72,000
Sales 14,500 units	20,80,000
Closing Stock: Finished goods 375 units	43,250
Work-in-process	48,200

The cost records provide as under:

Factory overheads are absorbed at 60% of direct wages.

Administration overheads are recovered at 20% of factory cost.

Selling and distribution overheads are charged at ₹ 5 per unit sold.

Opening Stock of finished goods is valued at ₹ 105 per unit.

The company values work-in-process at factory cost for both Financial and Cost Profit Reporting.

Required:

(i) Prepare statements for the year ended 31st March, 2026, show

- the profit as per financial records
- the profit as per costing records.

(ii) Present a statement reconciling the profit as per costing records with the profit as per Financial Records.

ICAI'S ANSWER

(i) Statement of Profit as per financial records (for the year ended March 31, 2026)

	(₹)		(₹)
To Opening stock:		By Sales	20,80,000
Finished Goods	76,525	By Closing stock:	
Work-in-process	33,000	Finished Goods	43,250
To Raw materials consumed	7,84,000	Work-in-Process	48,200
To Direct labour	4,65,000	By Rent received	72,000
To Factory overheads	2,65,000	By Interest received	18,500
To Goodwill written off	95,000		
To Administration overheads	3,15,000		
To Selling & distribution overheads	65,000		
To Interest paid	72,000		
To Bad debts	21,000		
To Profit	70,425		
	22,61,950		22,61,950

Statement of Profit as per costing records (for the year ended March 31,2026)

	(₹)	(₹)
Sales revenue (14,500 units) (A)		20,80,000
Cost of Sales:		
Opening stock (875 units x ₹ 105)	91,875	
Add: Cost of production of 14,000 units (Refer to Working Note 1& 2)	18,15,360	
Less: Closing stock $\left(\frac{₹18,15,360 \times 375 \text{ units}}{14,000 \text{ units}}\right)$	(48,626)	
Production cost of goods sold (14,500 units)	18,58,609	
Selling & distribution overheads (14,500 units x ₹ 5)	72,500	
Cost of sales: (B)	19,31,109	19,31,109
Profit: {(A) – (B)}		1,48,891

(ii) Statement of Reconciliation

(Reconciling the profit as per costing records with the profit as per financial records)

	(₹)	(₹)
Profit as per Cost Accounts		1,48,891
Add: Factory overheads over absorbed (₹ 2,79,000 – ₹ 2,65,000)	14,000	
S&D overheads over absorbed (₹ 72,500 – ₹ 65,000)	7,500	
Opening stock overvalued (₹ 91,875 – ₹ 76,525)	15,350	
Interest received	18,500	
Rent received	72,000	1,27,350
		2,76,241
Less: Administration overheads under recovery (₹ 3,15000 – ₹ 3,02,560)	12,440	
Closing stock overvalued (₹ 48,626 – ₹ 43,250)	5,376	
Goodwill written off	95,000	
Interest paid	72,000	
Bad debts	21,000	2,05,816
Profit as per financial accounts		70,425

Working Notes:

		Units
1.	**Number of units produced**	
	Sales	14,500
	Add: Closing stock	375
	Total	14,875
	Less: Opening stock	875
	Number of units produced	14,000

2. Cost Sheet

	(₹)	(₹)
Raw materials consumed		7,84,000
Direct labour		4,65,000
Prime cost		12,49,000

	(₹)	(₹)
Factory overheads (60% of direct wages)		2,79,000
Factory cost		15,28,000
Add: Opening work-in-process		33,000
Less: Closing work-in-process		(48,200)
Factory cost of goods produced		15,12,800
Administration overheads (20% of factory cost)		3,02,560
Cost of production of 14,000 units		18,15,360

Cost of production per unit:

$$= \frac{\text{Total Cost of Production}}{\text{No. of units produced}} = \frac{₹18,15,360}{14,000 \text{ units}} = ₹129.67$$

ICAI · P4 · RTP Sep 2026 · question p. 11-12 · answer p. 32-35

552 RTP **Jan 2026** Q12

XYZ Limited manufactures a standard product in a competitive market. During the year, the company experienced operational changes, including partial automation and fluctuations in input costs. Although the company continued using traditional costing methods, discrepancies were observed between financial and cost profits. Management has requested a detailed analysis to understand the variance and assess the accuracy of overhead absorption.

The following is the Trading and Profit & Loss Account of XYZ Limited:

Dr.	₹	Cr.	₹	₹
To Raw Materials	32,50,000	By Sales (40,000 units)	92,00,000	
To Direct Wages	21,75,000	By Closing Finished Goods (2,000 units)	3,90,000	
To Production Overheads	11,20,000	By Work-in-Progress:		
To Administration Overheads	6,25,000	Materials	72,000	
To Selling & Distribution Overheads	4,60,000	Wages	34,000	
To Preliminary Expenses Written Off	30,000	Production Overheads	26,500	1,32,500
To Goodwill Written Off	55,000	By Dividend Received	5,25,000	
To Fines	10,000	By Interest from Bank Deposits	1,20,000	
To Interest on Term Loan	18,000			

Dr.	₹	Cr.	₹	₹
To Loss on Sale of Equipment	20,000			
To Tax	2,25,000			
To Net Profit for the Year	23,79,500			
Total	1,03,67,500		Total	1,03,67,500

XYZ Limited manufactures a standard product and uses traditional costing systems for internal reporting.

The Cost Accounting records of XYZ Ltd. show the following:

1. Production Overheads have been absorbed into work-in-progress at 25% on Prime Cost.
2. Administration Overheads have been charged at ₹14.75 per unit of finished goods produced.
3. Selling and Distribution Overheads have been charged at ₹11.50 per unit sold.
4. The under- or over-absorption of overheads has not been adjusted in the Costing Profit & Loss Account.

You are required to:

1. Prepare a Proforma Costing Profit and Loss Account, showing the net profit as per cost accounts.
2. Prepare control accounts for:
   - Production Overheads
   - Administration Overheads
   - Selling & Distribution Overheads
3. Reconcile the profit as per cost accounts with the profit shown in the financial accounts.

ICAI'S ANSWER

(i) Costing Profit & Loss Statement

Particulars	Amount (₹)
Raw Materials	32,50,000
Direct Wages	21,75,000
Prime Cost	**54,25,000**
Add: Production Overheads (25% of Prime Cost)	13,56,250
Factory Cost	**67,81,250**
Less: Work-in-Progress	(1,32,500)
Manufacturing Cost Incurred During the Period	**66,48,750**
Add: Administration Overheads (₹14.75 × 42,000 units)	6,19,500
Cost of Production	**72,68,250**

Particulars	Amount (₹)
Less: Closing Finished Goods (2,000 units x ₹ 173.05)	(3,46,100)
Cost of Goods Sold	**69,22,150**
Add: Selling & Distribution Overheads (₹11.50 × 40,000 units)	4,60,000
Cost of Sales	**73,82,150**
Profit (Balancing Figure)	18,17,850
Sales (40,000 units)	**92,00,000**

(ii) Control Accounts

Production Overheads A/c

Particulars	Amount (₹)	Particulars	Amount (₹)
To General Ledger Adjustment A/c	11,20,000	By WIP A/c (absorbed)	13,56,250
To Overhead Adjustment A/c (Over-absorption)	2,36,250		
Total	13,56,250	Total	13,56,250

Administration Overheads A/c

Particulars	Amount (₹)	Particulars	Amount (₹)
To General Ledger Adjustment A/c	6,25,000	By Finished Goods A/c	6,19,500
		By Overhead Adjustment A/c (Under-absorption)	5,500
Total	**6,25,000**	**Total**	**6,25,000**

Selling & Distribution Overheads A/c

Particulars	Amount (₹)	Particulars	Amount (₹)
To General Ledger Adjustment A/c	4,60,000	By Cost of Sales A/c	4,60,000
Total	**4,60,000**	**Total**	**4,60,000**

(iii) Reconciliation Statement

Particulars	Amount (₹)	Amount (₹)
Profits as per Cost Accounts		18,17,850
Add:		
– Production Overheads – over-absorbed	2,36,250	

Particulars	Amount (₹)	Amount (₹)
– Dividend received	5,25,000	
– Interest from bank deposits	1,20,000	
– Write-down of Finished Stock (3,90,000 – 3,46,100)	43,900	
Total Additions		**9,25,150**
Less:		
– Administration Overheads – under-absorbed	5,500	
– Preliminary expenses written off	30,000	
– Goodwill written off	55,000	
– Fines	10,000	
– Interest on term loan	18,000	
– Loss on sale of equipment	20,000	
– Taxation	2,25,000	
Total Deductions		**3,63,500**
Profit as per Financial Accounts		**23,79,500**

ICAI · P4 · RTP Jan 2026 · question p. 15-17 · answer p. 37-39

553 RTP **Sep 2025** Q11

The following figures have been taken from the financial accounts of a manufacturing firm for the year ended 31st March, 2025:

	(₹)
Direct material consumption	20,00,000
Direct wages	12,00,000
Factory overheads	6,40,000
Repair and maintenance (Factory-related)	48,000
Advertisement expenses (for brand building – capital in nature)	40,000
Administrative overheads	2,80,000
Selling and distribution overheads	3,84,000
Bad debts	32,000
Preliminary expenses written off	16,000

	(₹)
Legal charges	4,000
Depreciation on Plant and Machinery	1,20,000
Interest on Loan	60,000
Dividends received	40,000
Interest on fixed deposit	8,000
Sales- 48,000 units	48,00,000
Closing stock:	
- Finished stock- 4,000 units	3,20,000
- Work-in-process	96,000

The cost accounts for the same period reveal that the direct material consumption was ₹ 22,40,000; Factory overhead (including factory related repairs and maintenance and depreciation) is recovered at 20% on prime cost; Administration overhead is recovered @ ₹ 4.8 per unit of cost of goods sold and selling and distribution overheads are recovered at ₹6.40 per unit sold.

Required:

PREPARE Costing and Financial Profit & Loss Accounts and RECONCILE the difference in the profit as arrived at in the two sets of accounts.

ICAI'S ANSWER

Statement of Cost and Profit

(As per Cost Records)

		(₹)
	Direct Material	22,40,000
	Direct Wages	12,00,000
	Prime Cost	**34,40,000**
	Factory Overhead (20% of ₹ 34,40,000)	6,88,000
		41,28,000
Less:	Closing Stock (WIP)	-96,000
	Works Cost or Cost of production (52,000 units)	**40,32,000**
Less:	Finished Goods (4,000 units @ ₹77.5385)	-3,10,154
	Cost of Goods Sold (48,000 units)	**37,21,846**
	Administrative Overhead (48,000 units @ ₹ 4.8 p.u.)	2,30,400

		(₹)
	Selling and Distribution Overhead (48,000 @ ₹ 6.40 p.u.)	3,07,200
	Cost of Sales	**42,59,446**
	Net profit (Balancing figure)	5,40,554
	Sales Revenue	**48,00,000**

Profit and Loss Account

(As per financial records)

Particulars	Amount (₹)	Particulars	Amount (₹)
To Direct Material consumed	20,00,000	By Sales	48,00,000
To Direct Wages	12,00,000	By Closing Work-in-process	96,000
To Factory overheads	6,40,000	By Closing Finished stock	3,20,000
To Depreciation on Plant and Machinery	1,20,000		
To Repairs & Maintenance (Factory)	48,000		
To Gross Profit c/d	12,08,000		
	52,16,000		52,16,000
To Administrative overheads	2,80,000	BY Gross Profit b/d	12,08,000
To Selling & distribution overheads	3,84,000	By Dividend received	40,000
To Bad debts	32,000	By Interest on deposit	8,000
To Preliminary expenses	16,000		
To Interest on Loan	60,000		
To Legal charges	4,000		
To Net profit (balancing figure)	4,80,000		
	12,56,000		12,56,000

Reconciliation Statement

	Particulars	Amount (₹)	Amount (₹)
	Net profit as per Financial Profit & Loss A/c		**4,80,000**
Add:	Administrative overheads (2,80,000 - 2,30,400)	49,600	
	Selling & Distribution overheads (3,84,000 - 3,07,200)	76,800	

	Particulars	Amount (₹)	Amount (₹)
	Bad debts	32,000	
	Preliminary expenses	16,000	
	Interest on loan	60,000	
	Legal charges	4,000	
	Factory overheads (6,40,000 + 1,20,000 + 48,000 - 6,88,000)	1,20,000	3,58,400
			8,38,400
Less:	Difference in value of materials consumed	2,40,000	
	Dividend received	40,000	
	Interest on deposit	8,000	
	Closing stock (3,20,000 - 3,10,154)	9,846	-2,97,846
	Profit as per Costing Profit & Loss A/c		**5,40,554**

Note: Since advertisement expense is capital in nature, it is not considered in profit and loss account.

ICAI · P4 · RTP Sep 2025 · question p. 15-16 · answer p. 38-41

554 RTP **Jan 2025** Q12

Following information is extracted as a result of scrutiny of the figures from both the financial accounts and cost accounts of CK Ltd. for the year ending 31st March:

Particulars	Amount (₹)
Net Profit (as per cost accounts)	57,71,840
Under recovery of selling overheads in cost accounts	1,16,800
Under valuation of closing stock in cost accounts	1,64,000
Rent received credited in financial accounts	87,200
Bad debts provided in financial accounts	52,000
Income tax provided in financial accounts	2,54,400
Under recovery of administration overheads in cost accounts	1,50,400

You are required to PREPARE a Statement of Reconciliation showing the profit as per financial records.

ICAI'S ANSWER

Statement of Reconciliation

(Reconciling the profit as per costing records with the profit as per financial records)

Particulars	(₹)	(₹)
Net Profit as per Cost Accounts		57,71,840
Add: Under valuation of closing stock in cost accounts	1,64,000	
Rent received credited in financial accounts	87,200	2,51,200
		60,23,040
Less: Under recovery of selling overheads in cost accounts	1,16,800	
Bad debts provided in financial accounts	52,000	
Income tax provided in financial accounts	2,54,400	
Under recovery of administration overheads in cost accounts	1,50,400	5,73,600
Profit as per Financial Accounts		**54,49,440**

ICAI · P4 · RTP Jan 2025 · question p. 13 · answer p. 34-35

555 RTP **Jan 2025** Q19(b)

LIST DOWN certain financial expenses and income included in Financial Accounts only.

ICAI'S ANSWER

Items included in Financial Accounts only-

(A) Purely Financial Expenses:

(i) Interest on loans or bank mortgages

(ii) Expenses and discounts on issue of shares, debentures etc.

(iii) Other capital losses i.e., loss by fire not covered by insurance etc.

(iv) Losses on the sales of fixed assets and investments

(v) Income tax, donations, subscriptions

(vi) Expenses of the company's share transfer office, if any.

(B) Purely Financial Income

(i) Interest received on bank deposits, loans and investments

(ii) Dividends received

(iii) Profits on the sale of fixed assets and investments

(iv) Transfer fee received

(v) Rent receivables.

ICAI · P4 · RTP Jan 2025 · question p. 20 · answer p. 45-46

556 RTP **Sep 2024** Q12

A manufacturing company disclosed a net loss of ₹ 3,47,000 as per their cost accounts for the year ended March 31,2024. The financial accounts however disclosed a net loss of ₹ 5,10,000 for the same period.

The following information was revealed as a result of scrutiny of the figures of both the sets of accounts.

		(₹)
(i)	Factory Overheads under-absorbed	40,000
(ii)	Administration Overheads over-absorbed	60,000
(iii)	Depreciation charged in Financial Accounts	3,25,000
(iv)	Depreciation charged in Cost Accounts	2,75,000
(v)	Interest on investments not included in Cost Accounts	96,000
(vi)	Income-tax provided	54,000
(vii)	Interest on loan funds in Financial Accounts	2,45,000
(viii)	Transfer fees (credit in financial books)	24,000
(ix)	Stores adjustment (credit in financial books)	14,000
(x)	Dividend received	32,000

PREPARE a memorandum Reconciliation Account

ICAI'S ANSWER

Memorandum Reconciliation Accounts

Dr.	(₹)	Cr.	(₹)
To Net Loss as per Costing books	3,47,000	By Administration overheads over recovered in cost accounts	60,000
To Factory overheads under absorbed in Cost Accounts	40,000	By Interest on investment not included in Cost Accounts	96,000
To Depreciation under charged in Cost Accounts	50,000	By Transfer fees in financial books	24,000
To Income-Tax not provided in Cost Accounts	54,000	By Stores adjustment (Credit in financial books)	14,000
To Interest on Loan Funds in Financial Accounts	2,45,000	By Dividend received in financial books	32,000
		By Net loss as per financial books	5,10,000
	7,36,000		7,36,000

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557 RTP May 2024 Q6

The financial books of a company reveal the following data for the year ended 31st March, 2023:

	(₹)
Opening Stock:	
Finished goods 875 units	76,525
Work-in-process	33,000
01.04.2022 to 31.03.2023	
Raw materials consumed	7,84,000
Direct labour	4,65,000
Factory overheads	2,65,000
Goodwill written off	95,000
Administration overheads	3,15,000
Income tax paid	72,000
Bad debts	21,000
Selling and distribution overheads	65,000
Interest received	18,500
Rent received	72,000
Sales 14,500 units	20,80,000
Closing Stock: Finished goods 375 units	43,250
Work-in-process	48,200

The management of the company, for preparing cost sheet and variance analysis uses the following cost recovery basis which has been elaborated by the cost controller of the company:

Factory overheads are absorbed at 60% of direct wages.

Administration overheads (production related) are recovered at 20% of factory cost.

Selling and distribution overheads are charged at ₹ 5 per unit sold.

Opening Stock of finished goods is valued at ₹105 per unit.

The company values work-in-process at factory cost for both financial and cost accounting purpose.

You being an associate to the cost controller of the company has been asked to:

(i) PREPARE a statement of profit as per costing records and financial records.

(ii) CALCULATE cost of production per unit.

(iii) PREPARE a statement reconciling the profit as per costing records with the profit as per financial records.

ICAI'S ANSWER

Statement of Profit as per financial records

(for the year ended March 31, 2023)

	(₹)		(₹)
To Opening stock:		By Sales	20,80,000
Finished goods	76,525	By Closing stock:	
Work-in-process	33,000	Finished Goods	43,250
To Raw materials consumed	7,84,000	Work-in-Process	48,200
To Direct labour	4,65,000	By Rent received	72,000
To Factory overheads	2,65,000	By Interest received	18,500
To Goodwill written off	95,000		
To Administration overheads	3,15,000		
To Selling & distribution overheads	65,000		
To Income tax paid	72,000		
To Bad debts	21,000		
To Profit	70,425		
	22,61,950		22,61,950

Statement of Profit as per costing records

(for the year ended March 31,2023)

	(₹)	(₹)
Sales revenue (14,500 units) (A)		20,80,000
Cost of Sales:		
Opening stock (875 units x ₹ 105)	91,875	
Add: Cost of production of 14,000 units (Refer to Working Note 1& 2)	18,15,360	
Less: Closing stock (₹18,15,360 × 375 units/14,000 units)	(48,626)	
Production cost of goods sold (14,500 units)	18,58,609	
Selling & distribution overheads (14,500 units x ₹ 5)	72,500	
Cost of sales: (B)	19,31,109	19,31,109
Profit: {(A) – (B)}		1,48,891

Workings:

1. Number of units produced	Units
Sales	14,500
Add: Closing stock	375
Total	14,875
Less: Opening stock	875
Number of units produced	14,000

Cost Sheet

	(₹)	(₹)
Raw materials consumed		7,84,000
Direct labour		4,65,000
Prime cost		12,49,000
Factory overheads (60% of direct wages)		2,79,000
Factory cost		15,28,000
Add: Opening work-in-process		33,000
Less: Closing work-in-process		(48,200)
Factory cost of goods produced		15,12,800
Administration overheads (20% of factory cost)		3,02,560
Cost of production of 14,000 units		18,15,360

Cost of production per unit:

$$= \frac{\text{Total Cost of Production}}{\text{No. of units produced}} = \frac{₹\ 18{,}15{,}360}{14{,}000\ \text{units}} = ₹129.67$$

Statement of Reconciliation

(Reconciling the profit as per costing records with the profit as per financial records)

	(₹)	(₹)
Profit as per Cost Accounts		1,48,891
Add: Factory overheads over absorbed (₹ 2,79,000 – ₹ 2,65,000)	14,000	
S & D overheads over absorbed (₹ 72,500 - ₹ 65,000)	7,500	
Opening stock overvalued (₹ 91,875 – ₹ 76,525)	15,350	
Interest received	18,500	

	(₹)	(₹)
Rent received	72,000	1,27,350
		2,76,241
Less: Administration overheads under recovery (₹ 3,15000 – ₹ 3,02,560)	12,440	
Closing stock overvalued (₹ 48,626 – ₹ 43,250)	5,376	
Goodwill written off	95,000	
Income tax paid	72,000	
Bad debts	21,000	2,05,816
Profit as per financial accounts		70,425

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558 RTP **Old P3 Nov 2023** Q6 old scheme

The financial books of a company reveal the following data for the year ended 31st March, 2023:

	(₹)
Opening Stock:	
Finished goods 625 units	1,06,250
Work-in-process	92,000
01.04.2022 to 31.03.2023	
Raw materials consumed	16,80,000
Direct Labour	12,20,000
Factory overheads	8,44,000
Administration overheads (production related)	3,96,000
Dividend paid	2,44,000
Bad Debts	36,000
Selling and Distribution Overheads	1,44,000
Interest received	76,000
Rent received	92,000
Sales 12,615 units	45,60,000
Closing Stock: Finished goods 415 units	91,300

	(₹)
Work-in-process	82,400

The cost records provide as under:

- Factory overheads are absorbed at 70% of direct wages.
- Administration overheads are recovered at 15% of factory cost.
- Selling and distribution overheads are charged at ₹ 6 per unit sold.
- Opening Stock of finished goods is valued at ₹ 240 per unit.
- The company values work-in-process at factory cost for both Financial and Cost Profit Reporting.

Required:

(i) Prepare statements for the year ended 31st March, 2023 showing:

- the profit as per financial records
- the profit as per costing records.

(ii) Prepare a statement reconciling the profit as per costing records with the profit as per financial records.

ICAI'S ANSWER

(i) Statement of Profit as per financial records

(for the year ended March 31, 2023)

	(₹)		(₹)
To Opening stock of Finished Goods	1,06,250	By Sales	45,60,000
To Work-in-process	92,000	By Closing stock of finished Goods	91,300
To Raw materials consumed	16,80,000	By Work-in-Process	82,400
To Direct labour	12,20,000	By Rent received	92,000
To Factory overheads	8,44,000	By Interest received	76,000
To Administration overheads	3,96,000		
To Selling & distribution overheads	1,44,000		
To Dividend paid	2,44,000		
To Bad debts	36,000		
To Profit	1,39,450		
	49,01,700		49,01,700

Statement of Profit as per costing records

(for the year ended March 31,2023)

	(₹)
Sales revenue (A) (12,615 units)	45,60,000
Cost of sales:	
Opening stock (625 units × ₹ 240)	1,50,000
Add: Cost of production of 12,405 units (Refer to working note 2)	43,28,140
Less: Closing stock $\left(\frac{₹\ 43{,}28{,}140 \times 415\ \text{units}}{12{,}405\ \text{units}}\right)$	(1,44,795)
Production cost of goods sold (12,615 units)	43,33,345
Selling & distribution overheads (12,615 units × ₹6)	75,690
Cost of sales: (B)	44,09,035
Profit: {(A) – (B)}	1,50,965

(ii) Statement of Reconciliation

(Reconciling the profit as per costing records with the profit as per financial records)

	(₹)	(₹)
Profit as per Cost Accounts		1,50,965
Add: Administration overheads over absorbed (₹5,64,540 – ₹3,96,000)	1,68,540	
Opening stock overvalued (₹1,50,000 – ₹ 1,06,250)	43,750	
Interest received	76,000	
Rent received	92,000	
Factory overheads over recovered (₹ 8,54,000 – ₹ 8,44,000)	10,000	3,90,290
		5,41,255
Less: Selling & distribution overheads under recovery (₹ 1,44,000 – ₹ 75,690)	68,310	
Closing stock overvalued (₹1,44,795 – ₹ 91,300)	53,495	
Dividend	2,44,000	
Bad debts	36,000	(4,01,805)
Profit as per financial accounts		1,39,450

Working notes:

1. Number of units produced

	Units
Sales	12,615
Add: Closing stock	415
Total	13,030
Less: Opening stock	(625)
Number of units produced	12,405

2. Cost Sheet

	(₹)
Raw materials consumed	16,80,000
Direct labour	12,20,000
Prime cost	29,00,000
Factory overheads (70% of direct wages)	8,54,000
Factory cost	37,54,000
Add: Opening work-in-process	92,000
Less: Closing work-in-process	(82,400)
Factory cost of goods produced	37,63,600
Administration overheads (15% of factory cost)	5,64,540
Cost of production of 12,405 units (Refer to working note 1) Cost of production per unit: $= \frac{\text{Total Cost of Production}}{\text{No. of units produced}} = \frac{₹43,28,140}{12,405 \text{ units}} = ₹ 348.90$	43,28,140

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559 RTP **Old P3 May 2023** Q6 old scheme

The financial records of Riva Private Limited showed a net profit of ₹1,69,500 for the year ended 31st March, 2022. The cost accounts, however, disclosed a net loss of ₹ 88,500 for the same period. The following information were revealed as a result of scrutiny of the figures of cost accounts and financial accounts:

		(₹)
(i)	(Administrative overhead under recovered	63,750.0
(ii)	Factory overhead over recovered	3,37,500.0
(iii)	Depreciation under charged in Cost Accounts	65,000.0
(iv)	Dividend received	50,000.0

		(₹)
(v)	Loss due to obsolescence charged in Financial Accounts	42,000.0
(vi)	Income tax provided	1,09,000.0
(vii)	Bank interest credited in Financial Accounts	34,000.0
(viii)	Value of opening stock:	
	In Cost Accounts	4,12,500.0
	In Financial Accounts	3,62,500.0
(ix)	Value of closing stock:	
	In Cost Accounts	3,13,750.0
	In Financial Accounts	3,30,000.0
(x)	Goodwill written-off in Financial Accounts	62,500.0
(xi)	Notional rent of own premises charged in Cost Accounts	1,50,000.0
(xii)	Provision for doubtful debts in Financial Accounts	37,500.0

Prepare a reconciliation statement by taking costing net loss as base.

ICAI'S ANSWER

Statement of Reconciliation

Sl. No.	Particulars	(₹)	(₹)
	Net loss as per Cost Accounts		(88,500)
	Additions		
1	Factory O/H over recovered	3,37,500	
2	Dividend Received	50,000	
3	Bank Interest received	34,000	
4	Difference in Value of Opening Stock (4,12,500 – 3,62,500)	50,000	
5	Difference in Value of Closing Stock (3,30,000 – 3,13,7500)	16,250	
6	Notional Rent of own Premises	1,50,000	6,37,750
	Deductions		
1	Administration O/H under recovered	63,750	
2	Depreciation under charged	65,000	
3	Loss due to obsolescence	42,000	

Sl. No.	Particulars	(₹)	(₹)
4	Income tax Provided	1,09,000	
5	Goodwill written-off	62,500	
6	Provision for doubtful debts	37,500	(3,79,750)
	Net Profit as per Financial A/c.		1,69,500

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560 RTP **Old P3 Nov 2022** Q6 old scheme

The financial books of a company reveal the following data for the financial year ending on 31st March, 2022:

	(₹)
Opening Stock:	
Finished goods 875 units	1,48,750
Work-in-process	64,000
01.04.2021 to 31.3.2022	
Raw materials consumed	15,60,000
Direct Labour	9,00,000
Factory overheads	6,00,000
Goodwill written off	2,00,000
Administration overheads	5,90,000
Dividend paid	1,70,000
Bad Debts	24,000
Selling and Distribution Overheads	1,22,000
Interest received	90,000
Rent received	36,000
Sales 14,500 units	41,60,000
Closing Stock: Finished goods 375 units	82,500
Work-in-process	77,334

The cost records provide as under:

- Factory overheads are absorbed at 60% of direct wages.

- Administration overheads are recovered at 20% of factory cost.
- Selling and distribution overheads are charged at ₹ 8 per unit sold.
- Opening Stock of finished goods is valued at ₹ 208 per unit.
- The company values work-in-process at factory cost for both Financial and Cost Profit Reporting.

Required:

(i) PREPARE statements for the year ended 31st March, 2022 showing-

- the profit as per financial records
- the profit as per costing records.

(ii) PRESENT a statement reconciling the profit as per costing records with the profit as per Financial Records.

ICAI'S ANSWER

(i) Statement of Profit as per financial records

(for the year ended March 31, 2022)

	(₹)		(₹)
To Opening stock:		By Sales	41,60,000
Finished Goods	1,48,750	By Closing stock:	
Work-in-process	64,000	Finished Goods	82,500
To Raw materials consumed	15,60,000	Work-in-Process	77,334
To Direct labour	9,00,000	By Rent received	36,000
To Factory overheads	6,00,000	By Interest received	90,000
To Goodwill written off	2,00,000		
To Administration overheads	5,90,000		
To Selling & distribution overheads	1,22,000		
To Dividend paid	1,70,000		
To Bad debts	24,000		
To Profit	**67,084**		
	44,45,834		44,45,834

Statement of Profit as per costing records

(for the year ended March 31,2022)

	(₹)	(₹)
Sales revenue (14,500 units) (A)		41,60,000

	(₹)	(₹)
Cost of Sales:		
Opening stock (875 units x ₹ 208)	1,82,000	
Add: Cost of production of 14,000 units (Refer to Working Note 1& 2)	35,84,000	
Less: Closing stock $\left(\frac{₹\ 35{,}84{,}000 \times 375\ \text{units}}{14{,}000\ \text{units}}\right)$	(96,000)	
Production cost of goods sold (14,500 units)	36,70,000	
Selling & distribution overheads (14,500 units x ₹ 8)	1,16,000	
Cost of sales: (B)		37,86,000
Profit: {(A) – (B)}		**3,74,000**

(ii) Statement of Reconciliation

(Reconciling the profit as per costing records with the profit as per financial records)

	(₹)	(₹)
Profit as per Cost Accounts		**3,74,000**
Add: Admin. overheads over absorbed (₹ 5,97,333 – ₹ 5,90,000)	7,333	
Opening stock overvalued (₹ 1,82,000 – ₹ 1,48,750)	33,250	
Interest received	90,000	
Rent received	36,000	**1,66,583**
		5,40,583
Less: Factory overheads under recovery (₹ 6,00,000 – ₹ 5,40,000)	60,000	
Selling & distribution overheads under recovery (₹ 1,22,000 – ₹ 1,16,000)	6,000	
Closing stock overvalued (₹ 96,000 – ₹ 82,500)	13,500	
Goodwill written off	2,00,000	
Dividend	1,70,000	
Bad debts	24,000	**4,73,500**
Profit as per financial accounts		**67,083**

Working Notes:

1. Number of units produced

	Units
Sales	14,500
Add: Closing stock	375
Total	14,875
Less: Opening stock	875
Number of units produced	14,000

2. Cost Sheet

	(₹)	(₹)
Raw materials consumed		15,60,000
Direct labour		9,00,000
Prime cost		24,60,000
Factory overheads (60% of direct wages)		5,40,000
Factory cost		30,00,000
Add: Opening work-in-process		64,000
Less: Closing work-in-process		77,334
Factory cost of goods produced		29,86,666
Administration overheads (20% of factory cost)		5,97,333
Cost of production of 14,000 units		**35,83,999**

Cost of production per unit: $= \frac{\text{Total Cost of Production}}{\text{No. of units produced}} = \frac{₹\ 35,83,999}{14,000 \text{units}} = ₹\ 256$

ICAI · Old P3 · RTP Nov 2022 · question p. 6-7 · answer p. 23-25

561 RTP **Old P3 Jul 2021** Q6 old scheme

The financial books of a company reveal the following data for the year ended 31st March, 2020:

	(₹)
Opening Stock:	
Finished goods 625 units	1,06,250
Work-in-process	92,000
01.04.2019 to 31.03.2020	
Raw materials consumed	16,80,000

	(₹)
Direct Labour	12,20,000
Factory overheads	8,44,000
Administration overheads (production related)	3,96,000
Dividend paid	2,44,000
Bad Debts	36,000
Selling and Distribution Overheads	1,44,000
Interest received	76,000
Rent received	92,000
Sales 12,615 units	45,60,000
Closing Stock: Finished goods 415 units	91,300
Work-in-process	82,400

The cost records provide as under:

- Factory overheads are absorbed at 70% of direct wages.
- Administration overheads are recovered at 15% of factory cost.
- Selling and distribution overheads are charged at ₹ 6 per unit sold.
- Opening Stock of finished goods is valued at ₹ 240 per unit.
- The company values work-in-process at factory cost for both Financial and Cost Profit Reporting.

Required:

(i) PREPARE statements for the year ended 31st March, 2020 showing:

- the profit as per financial records
- the profit as per costing records.

(ii) PREPARE a statement reconciling the profit as per costing records with the profit as per financial records.

ICAI'S ANSWER

(i) Statement of Profit as per financial records

(for the year ended March 31, 2020)

	(₹)		(₹)
To Opening stock of Finished Goods	1,06,250	By Sales	45,60,000
To Work-in-process	92,000	By Closing stock of finished Goods	91,300
To Raw materials consumed	16,80,000	By Work-in-Process	82,400

	(₹)		(₹)
To Direct labour	12,20,000	By Rent received	92,000
To Factory overheads	8,44,000	By Interest received	76,000
To Administration overheads	3,96,000		
To Selling & distribution overheads	1,44,000		
To Dividend paid	2,44,000		
To Bad debts	36,000		
To Profit	1,39,450		
	49,01,700		49,01,700

Statement of Profit as per costing records

(for the year ended March 31,2020)

	(₹)
Sales revenue (A) (12,615 units)	45,60,000
Cost of sales:	
Opening stock (625 units × ₹ 240)	1,50,000
Add: Cost of production of 12,405 units (Refer to working note 2)	43,28,140
Less: Closing stock (₹43,28,140 × 415 units/12,405 units)	(1,44,795)
Production cost of goods sold (12,615 units)	43,33,345
Selling & distribution overheads (12,615 units × ₹ 6)	75,690
Cost of sales: (B)	44,09,035
Profit: {(A) – (B)}	1,50,965

(ii) Statement of Reconciliation

(Reconciling the profit as per costing records with the profit as per financial records)

	(₹)	(₹)
Profit as per Cost Accounts		1,50,965
Add: Administration overheads over absorbed (₹ 5,64,540 – ₹ 3,96,000)	1,68,540	
Opening stock overvalued (₹1,50,000 – ₹ 1,06,250)	43,750	
Interest received	76,000	
Rent received	92,000	

	(₹)	(₹)
Factory overheads over recovered (₹ 8,54,000 – ₹ 8,44,000)	10,000	3,90,290
		5,41,255
Less: Selling & distribution overheads under recovery (₹ 1,44,000 – ₹ 75,690)	68,310	
Closing stock overvalued (₹1,44,795 – ₹ 91,300)	53,495	
Dividend	2,44,000	
Bad debts	36,000	(4,01,805)
Profit as per financial accounts		1,39,450

Working notes:

1. Number of units produced

	Units
Sales	12,615
Add: Closing stock	415
Total	13,030
Less: Opening stock	(625)
Number of units produced	12,405

2. Cost Sheet

	(₹)
Raw materials consumed	16,80,000
Direct labour	12,20,000
Prime cost	29,00,000
Factory overheads (70% of direct wages)	8,54,000
Factory cost	37,54,000
Add: Opening work-in-process	92,000
Less: Closing work-in-process	(82,400)
Factory cost of goods produced	37,63,600
Administration overheads (15% of factory cost)	5,64,540
Cost of production of 12,405 units (Refer to working note 1) Cost of production per unit: = Total Cost of Production/No. of units produced = ₹43,28,140/12,405 units = ₹348.90	43,28,140

562 RTP **Old P3 Nov 2020** Q6 old scheme

A manufacturing company disclosed a net loss of ₹6,94,000 as per their cost accounts for the year ended March 31,2020. The financial accounts however disclosed a net loss of ₹10,20,000 for the same period. The following information was revealed as a result of scrutiny of the figures of both the sets of accounts.

		(₹)
(i)	Factory Overheads under-absorbed	80,000
(ii)	Administration Overheads over-absorbed	1,20,000
(iii)	Depreciation charged in Financial Accounts	6,50,000
(iv)	Depreciation charged in Cost Accounts	5,50,000
(v)	Interest on investments not included in Cost Accounts	1,92,000
(vi)	Income-tax provided	1,08,000
(vii)	Interest on loan funds in Financial Accounts	4,90,000
(viii)	Transfer fees (credit in financial books)	48,000
(ix)	Stores adjustment (credit in financial books)	28,000
(x)	Dividend received	64,000

PREPARE a memorandum Reconciliation Account.

ICAI'S ANSWER

Memorandum Reconciliation Accounts

Dr.	(₹)	Cr.	(₹)
To Net Loss as per Costing books	6,94,000	By Administration overheads over recovered in cost accounts	1,20,000
To Factory overheads under absorbed in Cost Accounts	80,000	By Interest on investment not included in Cost Accounts	1,92,000
To Depreciation under charged in Cost Accounts	1,00,000	By Transfer fees in Financial books	48,000
To Income-Tax not provided in Cost Accounts	1,08,000	By Stores adjustment (Credit in financial books)	28,000
To Interest on Loan Funds in Financial Accounts	4,90,000	By Dividend received in financial books	64,000
		By Net loss as per Financial books	10,20,000
	14,72,000		14,72,000

ICAI · Old P3 · RTP Nov 2020 · question p. 5 · answer p. 18-19

563 MTP Sep 2026 S-I Q3(b) 5 marks

A manufacturing company has disclosed net loss of ₹ 48,700 as per their cost accounting records for the year ended 31st March, 2026. However, their financial accounting records disclosed net profit of ₹ 35,400 for the same period. A scrutiny of data of both the sets of books of accounts revealed the following information:

		(₹)
(i)	Factory overheads under absorbed	30,500
(ii)	Administrative overheads over absorbed	65,000
(iii)	Depreciation charged in financial accounts	2,25,000
(iv)	Depreciation charged in cost accounts	2,70,000
(v)	Income-tax provision	52,400
(vi)	Transfer fee (credited in financial accounts)	10,200
(vii)	Obsolescence loss charged in financial accounts	20,700
(viii)	Notional rent of own premises charged in cost accounts	54,000
(ix)	Value of opening stock:	
	(a) in cost accounts	1,38,000
	(b) in financial accounts	1,15,000
(x)	Value of closing stock:	
	(a) in cost accounts	1,22,000
	(b) in financial accounts	1,12,500

Prepare a Memorandum Reconciliation Account by taking costing loss as base.

ICAI'S ANSWER

Memorandum Reconciliation Accounts

Dr. Cr.

Particulars	Amount (₹)	Particulars	Amount (₹)
To Net Loss as per Cost Accounts	48,700	By Administration overheads over recovered in Cost Accounts	65,000
To Factory overheads under absorbed in Cost Accounts	30,500	By Depreciation overcharged in Cost Accounts (₹ 2,70,000 – ₹ 2,25,000)	45,000
To Provision for Income tax	52,400	By Transfer fees in Financial Accounts	10,200
To Obsolescence loss	20,700	By Notional Rent of own premises	54,000

Dr.			Cr.
To Overvaluation of closing stock in Cost Accounts**	9,500	By Overvaluation of Opening stock in Cost Accounts*	23,000
To Net Profit (as per Financial Accounts)	35,400		
	1,97,200		1,97,200

* Overvaluation of Opening Stock as per Cost Accounts

= Value in Cost Accounts – Value in Financial Accounts

= ₹ 1,38,000 – ₹ 1,15,000 = ₹ 23,000.

** Overvaluation of Closing Stock as per Cost Accounts

= Value in Cost Accounts – Value in Financial Accounts

= ₹ 1,22,000 – ₹ 1,12,500 = ₹ 9,500.

ICAI · P4 · MTP Sep 2026 · Series I · question p. 12 · answer p. 15

564 MTP **Sep 2026 S-I** Q6(a) 5 marks

Also asked in MTP May 2024 S-II Q6(a)

Why is it necessary to reconcile the Profits between the Cost Accounts and Financial Accounts?

ICAI'S ANSWER

When the cost and financial accounts are kept separately, It is imperative that these should be reconciled, otherwise the cost accounts would not be reliable. The reconciliation of two set of accounts can be made, if both the sets contain sufficient detail as would enable the causes of differences to be located. It is therefore, important that in the financial accounts, the expenses should be analysed in the same way as in cost accounts. It is important to know the causes which generally give rise to differences in the costs & financial accounts. These are:

(i) Items included in financial accounts but not in cost accounts

- Income-tax
- Transfer to reserve
- Dividends paid
- Goodwill / preliminary expenses written off
- Pure financial items
- Interest, dividends
- Losses on sale of investments
- Expenses of Co's share transfer office
- Damages & penalties

(ii) Items included in cost accounts but not in financial accounts

- Opportunity cost of capital
- Notional rent

(iii) Under / Over absorption of expenses in cost accounts

(iv) Different bases of inventory valuation

Motivation for reconciliation is:

- To ensure reliability of cost data
- To ensure ascertainment of correct product cost
- To ensure correct decision making by the management based on Cost & Financial data
- To report fruitful financial / cost data.

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565 MTP **Sep 2026 S-II** Q3(a) 10 marks

The financial books of a company reveal the following data for the year ended 31st March, 2026:

	₹
Opening Stock:	
Finished goods 625 units	53,125
Work-in-process	46,000
01.04.2025 to 31.03.2026	
Raw materials consumed	8,40,000
Direct Labour	6,10,000
Factory overheads	4,22,000
Administration overheads	1,98,000
Dividend paid	1,22,000
Bad Debts	18,000
Selling and Distribution Overheads	72,000
Interest received	38,000
Rent received	46,000
Sales 12,615 units	22,80,000
Closing Stock: Finished goods 415 units	45,650
Work-in-process	41,200

The cost records provide as under:

- Factory overheads are absorbed at 60% of direct wages.
- Administration overheads are recovered at 20% of factory cost.
- Selling and distribution overheads are charged at ₹ 2 per unit sold.

- Opening Stock of finished goods is valued at ₹ 120 per unit.
- The company values work-in-process at factory cost for both Financial and Cost Profit Reporting.

(i) Prepare statements for the year ended 31st March, 2026 show

The profit as per financial records

The profit as per costing records.

(ii) Present a statement reconciling the profit as per costing records with the profit as per Financial Records.

ICAI'S ANSWER

(i) **Profit as per financial records**

(For the year ended March 31, 2026)

	₹		₹
To Opening stock of Finished Goods	53,125	By Sales	22,80,000
To Work-in-process	46,000	By Closing stock of finished Goods	45,650
To Raw materials consumed	8,40,000	By Work-in-Process	41,200
To Direct labour	6,10,000	By Rent received	46,000
To Factory overheads	4,22,000	By Interest received	38,000
To Administration overheads	1,98,000		
To Selling & distribution overheads	72,000		
To Bad debts	18,000		
To Dividend paid	1,22,000		
To Profit	69,725		
	24,50,850		24,50,850

Profit as per costing records

(For the year ended March 31, 2026)

	(₹)
Sales revenue (12,615 units)	22,80,000
Cost of sales:	
Opening stock (625 units x ₹ 120)	75,000
Add: Cost of production of 12,405 units (Refer to working note 2)	21,84,960
Less: Closing stock (₹ 21,84,960×415 units/12,405 units)	73,096
Cost of production of goods sold (12,615 units)	21,86,864

	(₹)
Add: Selling & distribution overheads (12,615 units x ₹ 2)	25,230
Cost of sales	22,12,094
Profit	67,906

(ii) **Statement of Reconciliation**

(Reconciling the profit as per costing records with the profit as per financial records)

	(₹)	(₹)
Profit as per Cost Accounts		67,906
Add: Administration overheads over absorbed (₹ 3,64,160 – ₹ 1,98,000)	1,66,160	
Opening stock overvalued (₹ 75,000 – ₹ 53,125)	21 ,875	
Interest received	38,000	
Rent received	46,000	
		3,39,941
Less:		
Factory overheads under recovered (₹ 3,66,000 – ₹ 4,22,000)	56,000	
Selling & distribution overheads under recovery (₹ 72,000 – ₹ 25,230)	46,770	
Closing stock overvalued (₹ 73,096 – ₹ 45,650)	27,446	
Dividend	1,22,000	
Bad debts	18,000	(2,70,216)
Profit as per financial accounts		69,725

Working notes:

1. Number of units produced

	Units
Sales	12,615
Add: Closing stock	415
Total	13,030
less: Opening stock	625
Number of units produced	12,405

2. Cost Sheet

	(₹)
Raw materials consumed	8,40,000
Direct labour	6,10,000
Prime cost	14,50,000
Factory overheads (60% of direct wages)	3,66,000
Factory cost	18,16,000
Add: Opening work-in-process	46,000
Less: Closing work-in-process	41,200
Factory cost of goods produced	18,20,800
Administration overheads (20% of factory cost)	3,64,160
Cost of production of 12,405 units (Refer to working note 1)	21,84,960

3. Cost of production per unit:

$$\frac{\text{Total cost of Production}}{\text{No. of units produced}} = \frac{₹21,84,960}{12,405 \text{ units}} = ₹176.14$$

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566 MTP **May 2026 S-I** Q1(b) 5 marks

Also asked in MTP Jan 2026 S-II Q1(a)

A manufacturing company disclosed a net loss of ₹ 6,40,000 as per their cost accounts for the year ended 31st March, 2025. However, the financial accounts showed a net profit of ₹ 45,000 for the same period.

Upon scrutiny of both sets of accounts, the following differences were identified:

Particulars	Amount (in ₹)
(i) Transfer fee received	23,000
(ii) Interest on investment received	3,00,000
(iii) Goodwill written-off	38,000
(iv) Depreciation	
– Charged in Financial books	5,00,000
– Recovered in Cost books	6,00,000
(v) Opening stock of raw material	
– Cost books	4,50,000
– Financial books	4,82,000

Particulars	Amount (in ₹)
(vi) Closing stock of finished goods	
– Cost books	12,80,000
– Financial books	13,56,000
(vii) Notional rent of own premises recorded in Cost accounts	3,20,000
(viii) Factory overheads	
– Cost books	2,50,000
– Financial books	1,94,000
(ix) Income-tax provided	1,20,000

You are required to PREPARE Memorandum Reconciliation Account.

ICAI'S ANSWER

Memorandum Reconciliation Account

Debit	Amount (₹)	Credit	Amount (₹)
Net loss as per costing books	6,40,000	Transfer fee recorded in financial books	23,000
Goodwill written off not recorded in cost books	38,000	Interest on investment not included in costs	3,00,000
Under valuation of opening stock in cost accounts	32,000	Depreciation overcharge in cost books	1,00,000
Income tax not provided in costs	1,20,000	Under valuation of closing stock (finished goods) in cost accounts	76,000
Net profit as per financial books	45,000	Notional rent charges in cost books	3,20,000
		Factory overheads over recovered in cost books	56,000
Total	**8,75,000**	**Total**	**8,75,000**

ICAI · P4 · MTP May 2026 · Series I · question p. 8-9 · answer p. 7-8

567 MTP **May 2026 S-II** Q3(b) 8 marks

Stellar Equipments Pvt. Ltd. manufactures advanced renewable-energy control modules used in solar and wind-power systems. During the financial year ending 31st March 2025, the company upgraded its testing laboratory, strengthened quality assurance, and revised its overhead absorption methods. The following financial data has been extracted from the company's financial books:

Opening Stock

- Finished Goods: 1,050 units — ₹ 1,31,250
- Work-in-Process — ₹ 46,500

Expenses for the Year (01.04.2024 to 31.03.2025)

Particulars	Amount (₹)
Raw Materials Consumed	9,60,000
Direct Labour	6,10,000
Factory Overheads	3,45,000
Goodwill Written Off	1,20,000
Administrative Overheads	4,10,000
Interest Paid	96,000
Bad Debts	35,000
Selling & Distribution Overheads	88,000
Loss on Obsolete Inventory	42,000
Interest Received	27,000
Rent Received	96,000

Sales

- Units Sold: 18,700 units
- Total Sales: ₹ 26,18,000

Closing Stock

- Finished Goods: 620 units — ₹ 82,160
- Work-in-Process — ₹ 59,400

The cost records provide as under:

1. Factory overheads are absorbed at 50% of direct wages.
2. Administration overheads are recovered at 18% of factory cost.
3. Selling & Distribution overheads are charged at ₹ 6 per unit sold.
4. Opening Stock of Finished Goods is valued at ₹ 125 per unit under cost accounts.
5. Work-in-Process is valued at factory cost in both financial and cost records.
6. Loss on obsolete inventory is treated as abnormal loss and excluded from cost accounts.

You are required to:

(i) PREPARE statements for the year ended 31st March 2025 showing:

- Profit as per Financial Records
- Profit as per Costing Records

(ii) PREPARE a Reconciliation Statement between the profit as per Cost Accounts and the profit as per Financial Accounts.

ICAI'S ANSWER

(i) Statement of Profit as per Financial Records

(for the year ended March 31, 2025)

Particulars	₹	Particulars	₹
To Opening Stock:		By Sales (18,700 units)	26,18,000
Finished Goods	1,31,250	By Closing Stock:	
Work-in-Process	46,500	Finished Goods	82,160
To Raw Materials Consumed	9,60,000	Work-in-Process	59,400
To Direct Labour	6,10,000	By Rent Received	96,000
To Factory Overheads	3,45,000	By Interest Received	27,000
To Goodwill Written Off	1,20,000		
To Administration Overheads	4,10,000		
To Selling & Distribution OH	88,000		
To Interest Paid	96,000		
To Bad Debts	35,000		
To Loss on Obsolete Inventory	42,000		
To Profit / (Loss)	1,190 (Loss)		
Total	**28,82,560**	**Total**	**28,82,560**

Statement of Profit as per Costing Records

(for the year ended March 31, 2025)

Particulars	₹	₹
Sales Revenue (18,700 units) (A)		26,18,000
Cost of Sales:		
Opening stock: Finished Goods (1,050 × 125)	1,31,250	
Add: Cost of Production of 18,270 units (Working Note 1 & 2)	21,97,278	
Less: Closing Stock: 620 units × cost/unit	74,567	
Production Cost of Goods Sold	22,53,961	
Selling & Distribution Overheads (18,700 × 6)	1,12,200	
Cost of Sales (B)	23,66,161	23,66,161

Particulars	₹	₹
Profit {(A) – (B)}	**2,51,839**	

(ii) Reconciliation Statement

(Reconciling Profit as per Cost Accounts with Profit as per Financial Accounts)

Particulars	₹	₹
Profit as per Cost Accounts		2,51,839
Add: Over absorption of selling and distribution overhead (1,12,200-88,000)	24,200	
Add: Under valuation of closing stock (82,160 – 74,567)	7,593	
Add: Rent Received	96,000	
Add: Interest Received	27,000	1,54,793
Less: Under absorption of administrative overhead (4,10,000 - 3,35,178)	74,822	
Less: Under absorption of factory overhead (3,45,000 – 3,05,000)	40,000	
Less: Goodwill Written Off	1,20,000	
Less: Interest Paid	96,000	
Less: Bad Debts	35,000	
Less: Loss on obsolete inventory	42,000	(4,07,822)
Profit/(Loss) as per Financial Accounts		**(1,190)**

Working Notes

1. Number of Units Produced

Particulars	Units
Sales	18,700
Add: Closing Stock	620
Total	19,320
Less: Opening Stock	1,050
Number of units produced	**18,270**

2. Cost Sheet

Particulars	₹
Raw Materials Consumed	9,60,000
Direct Labour	6,10,000

Particulars	₹
Prime Cost	**15,70,000**
Factory Overheads (50% of Direct Wages)	3,05,000
Factory Cost	**18,75,000**
Add: Opening Work-in-Process	46,500
Less: Closing Work-in-Process	(59,400)
Factory Cost of Goods Produced	**18,62,100**
Administration Overheads (18% of Factory Cost)	3,35,178
Cost of Production of 18,270 units	21,97,278
Cost per unit	21,97,278/18,270 = ₹ 120.27

ICAI · P4 · MTP May 2026 · Series II · question p. 12-13 · answer p. 13-15

568 MTP **Jan 2026 S-I** Q3(b) 5 marks

A manufacturing company has disclosed a net loss of ₹ 3,15,000 as per their cost accounting records for the year ended March 31, 2025. However, their financial accounting records disclosed a net loss of ₹ 3,70,000 for the same period.

A scrutiny of both sets of books revealed the following information:

Particulars	Amount (₹)
(i) Factory overheads under-absorbed	8,000
(ii) Administration overheads over-absorbed	5,000
(iii) Depreciation charged in financial accounts	75,000
(iv) Depreciation charged in cost accounts	90,000
(v) Interest on investments (not included in cost accounts)	25,000
(vi) Income-tax provided in financial accounts	80,000
(vii) Transfer fees (credit in financial accounts)	3,000
(viii) Preliminary expenses written off (financial accounts only)	5,000
(ix) Overvaluation of closing stock of finished goods in cost accounts	10,000

PREPARE a Memorandum Reconciliation Account to reconcile the results shown by the cost and financial accounts.

ICAI'S ANSWER

Memorandum Reconciliation Account

Particulars	₹	Particulars	₹
To Net Loss as per Costing Books	3,15,000	By Administration Overheads over-absorbed in Cost Accounts	5,000
To Factory Overheads under-absorbed	8,000	By Depreciation over-charged in Cost Accounts (₹ 90,000 – ₹ 75,000)	15,000
To Income-tax provided in Financial Accounts	80,000	By Interest on Investments (not included in Cost Accounts)	25,000
To Preliminary Expenses written off (in Financial Accounts)	5,000	By Transfer Fees (income in Financial Accounts)	3,000
To Overvaluation of Closing Stock of Finished Goods in Cost Accounts	10,000	By Net Loss as per Financial Accounts	3,70,000
Total	**4,18,000**	**Total**	**4,18,000**

ICAI · P4 · MTP Jan 2026 · Series I · question p. 11-12 · answer p. 12-13

569 MTP **Sep 2025 S-I** Case scenario · Q3(b)(i) to Q3(b)(ii) 5 marks

CASE SCENARIO

Following Profit & Loss Account is provided from the financial accounts of XYZ Pvt. Ltd. for the year ended 31st March:

Particulars	(₹)	Particulars	(₹)
To Opening Stock	-	By Sales (50,000 units)	1,87,50,000
" Materials	75,00,000	" Closing Stock:	
" Wages	37,50,000	Finished goods (3,075 units)	11,25,000
" Factory Overheads	33,75,000	Work-in-Process:	
" Admn. Overheads (production related)	19,50,000	Materials ₹ 2,25,000 Labour ₹ 1,50,000 Factory overheads ₹ 1,50,000	5,25,000
" Selling and Distribution Overheads	13,50,000		
" Goodwill written off	15,00,000		
" Interest on loan	1,50,000		
" **Net Profit**	8,25,000		
	2,04,00,000		2,04,00,000

In the costing records, factory overhead is charged at 120% of wages, administrative overhead at 20% of the factory cost, and selling and distribution overhead at ₹ 15 per unit sold.

You are required to PREPARE a:

Q3(b)(i) 3 marks

Statement to find out profit as per cost records.

Q3(b)(ii) 2 marks

Statement reconciling the profit as per cost records with the profit as per financial records.

ICAI'S ANSWERS

Q3(b)(i)

Statement calculating profit as per cost records

Particulars	(₹)
Materials	75,00,000
Wages	37,50,000
Direct Expenses	Nil
Prime Cost	**1,12,50,000**
Add: Factory Overhead @ 120% of wages	45,00,000
Gross Factory Cost	**1,57,50,000**
Less: Closing WIP	(5,25,000)
Factory Cost of (50,000 + 3,075) units	1,52,25,000
Add: Admn. Overhead @ 20% of Factory cost	30,45,000
	1,82,70,000
Less: Closing Stock of finished goods (3,075 units)	(10,58,507)*
Cost of Goods sold (50,000 units)	**1,72,11,493**
Add: Selling & Dist. Overhead @ ₹ 15 per unit	7,50,000
Cost of sales (50,000 units)	**1,79,61,493**
Sales of 50,000 units	1,87,50,000
Profit	**7,88,507**

*$\left(\frac{3{,}075 \text{ units}}{53{,}075 \text{ units}} \text{ x } ₹\ 1{,}82{,}70{,}000\right)$

Q3(b)(ii)

Reconciliation Statement

	(₹)	(₹)
Profit as per Cost Accounts		**7,88,507**
Add: Factory overheads over-absorbed (₹ 45,00,000 - ₹ 33,75,000)	11,25,000	
Admn. overhead over-absorbed (₹ 30,45,000 – ₹ 19,50,000)	10,95,000	

	(₹)	(₹)
Difference in the valuation of closing stock of finished goods (₹ 11,25,000 – ₹ 10,58,507)	66,493	22,86,493
		30,75,000
Less: Selling & Dist. Overhead under-absorbed (₹ 13,50,000 – ₹ 7,50,000)	6,00,000	
Goodwill written off	15,00,000	
Interest on loan	1,50,000	22,50,000
Profit as per financial accounts		**8,25,000**

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570 MTP **Sep 2025 S-II** Q4(b) 7 marks

The financial books of a company reveal the following data for the year ended 31st March, 2025:

	(₹)
Opening Stock:	
Finished goods 875 units	76,525
Work-in-process	33,000
01.04.2024 to 31.3.2025	
Raw materials consumed	7,84,000
Direct Labour	4,65,000
Factory overheads	2,65,000
Goodwill written off	95,000
Administration overheads	3,15,000
Interest paid	72,000
Bad Debts	21,000
Selling and Distribution Overheads	65,000
Interest received	18,500
Rent received	72,000
Sales 14,500 units	20,80,000
Closing Stock: Finished goods 375 units	43,250
Work-in-process	48,200

The cost records provide as under:

Factory overheads are absorbed at 60% of direct wages.

Administration overheads are recovered at 20% of factory cost.

Selling and distribution overheads are charged at ₹ 5 per unit sold.

Opening Stock of finished goods is valued at ₹ 105 per unit.

The company values work-in-process at factory cost for both Financial and Cost Profit Reporting.

Required:

(i) PREPARE statements for the year ended 31st March, 2025 show

- the profit as per financial records
- the profit as per costing records.

(ii) PRESENT a statement reconciling the profit as per costing records with the profit as per Financial Records.

ICAI'S ANSWER

(i) **Statement of Profit as per financial records**

(for the year ended March 31, 2025)

	(₹)		(₹)
To Opening stock:		By Sales	20,80,000
Finished Goods	76,525	By Closing stock:	
Work-in-process	33,000	Finished Goods	43,250
To Raw materials consumed	7,84,000	Work-in-Process	48,200
To Direct labour	4,65,000	By Rent received	72,000
To Factory overheads	2,65,000	By Interest received	18,500
To Goodwill written off	95,000		
To Administration overheads	3,15,000		
To Selling & distribution overheads	65,000		
To Interest paid	72,000		
To Bad debts	21,000		
To Profit	70,425		
	22,61,950		22,61,950

Statement of Profit as per costing records

(for the year ended March 31,2025)

	(₹)	(₹)
Sales revenue (14,500 units) (A)		20,80,000
Cost of Sales:		
Opening stock (875 units x ₹ 105)	91,875	
Add: Cost of production of 14,000 units (Refer to Working Note 1& 2)	18,15,360	
Less: Closing stock $\left(\frac{₹18,15,360 \times 375 \text{ units}}{14,000 \text{ units}}\right)$	(48,626)	
Production cost of goods sold (14,500 units)	18,58,609	
Selling & distribution overheads (14,500 units x ₹ 5)	72,500	
Cost of sales: (B)	19,31,109	19,31,109
Profit: {(A) – (B)}		1,48,891

(ii) **Statement of Reconciliation**

(Reconciling the profit as per costing records with the profit as per financial records)

	(₹)	(₹)
Profit as per Cost Accounts		1,48,891
Add: Factory overheads over absorbed (₹ 2,79,000 – ₹ 2,65,000)	14,000	
S & D overheads over absorbed (₹ 72,500 – ₹ 65,000)	7,500	
Opening stock overvalued (₹ 91,875 – ₹ 76,525)	15,350	
Interest received	18,500	
Rent received	72,000	1,27,350
		2,76,241
Less: Administration overheads under recovery (₹ 3,15,000 – ₹ 3,02,560)	12,440	
Closing stock overvalued (₹ 48,626 – ₹ 43,250)	5,376	
Goodwill written off	95,000	
Interest paid	72,000	
Bad debts	21,000	2,05,816
Profit as per financial accounts		70,425

Working Notes:

		Units
1.	**Number of units produced**	
	Sales	14,500
	Add: Closing stock	375
	Total	14,875
	Less: Opening stock	875
	Number of units produced	14,000
2.	**Cost Sheet**	

	(₹)	(₹)
Raw materials consumed		7,84,000
Direct labour		4,65,000
Prime cost		12,49,000
Factory overheads (60% of direct wages)		2,79,000
Work cost		15,28,000
Add: Opening work-in-process		33,000
Less: Closing work-in-process		(48,200)
Factory cost of goods produced		15,12,800
Administration overheads (20% of factory cost)		3,02,560
Cost of production of 14,000 units		18,15,360

Cost of production per unit: $= \frac{\text{Total Cost of Production}}{\text{No. of units produced}} = \frac{₹\ 18,15,360}{14,000\ \text{units}} = ₹129.67$

ICAI · P4 · MTP Sep 2025 · Series II · question p. 13-14 · answer p. 19-21

571 MTP **May 2025 S-I** Q3(b) 4 marks

AeBee Ltd. disclosed a net loss of ₹ 9,02,200 as per their cost accounts for the year ending 31st March. However, the financial accounts disclosed a net loss of ₹ 13,26,000 for the same period. On scrutiny of both the sets of accounts, following information was revealed-

Particulars	(₹)
(i) Depreciation charged in Financial Accounts	8,45,000
(ii) Factory Overheads under-absorbed in Cost Accounts	7,41,000
(iii) Depreciation charged in Cost Accounts	7,15,000
(iv) Interest on investments not included in Cost Accounts	3,12,000
(v) Administration Overheads over-absorbed in cost accounts	1,56,000
(vi) Income-tax provided	1,40,400
(vii) Dividend received	83,200
(viii) Stores adjustment (credit in financial books)	36,400

PREPARE a memorandum Reconciliation Account.

ICAI'S ANSWER

Memorandum Reconciliation Accounts

Dr.	(₹)	Cr.	(₹)
To Net Loss as per Costing books	9,02,200	By Administration overheads over recovered in cost accounts	1,56,000
To Factory overheads under absorbed in Cost Accounts	7,41,000	By Interest on investment not included in Cost Accounts	3,12,000
To Depreciation under charged in Cost Accounts	1,30,000	By Stores adjustment (Credit in financial books)	36,400
To Income-Tax not provided in Cost Accounts	1,40,400	By Dividend received in financial books	83,200
		By Net loss as per Financial books	13,26,000
	19,13,600		19,13,600

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572 MTP **May 2025 S-II** Q2(b) 8 marks

The following is the summarised Trading and Profit and Loss Account of XYZ Ltd. for the year ended 31st

March 2024:

Particulars	Amount (₹)	Particulars	Amount (₹)
Direct Material	14,16,000	Sales (30,000 units)	30,00,000
Direct wages	7,42,000	Finished stock (2,000 units)	1,67,500
Works overheads	4,26,000	Work-in-progress:	
Administration overheads	1,50,000	- Materials 34,000 - Wages 16,000	
Selling and distribution overheads	1,65,000	- Works overhead 4,000	54,000
Net profit for the year	3,22,500		
	32,21,500		32,21,500

The company's cost records show that in course of manufacturing a standard unit (i) works overheads have been charged @ 20% on prime cost, (ii) administration overheads are related with production activities and are recovered at ₹5 per finished unit, and (iii) selling and distribution overheads are recovered at ₹6 per unit sold.

You are required to PREPARE:

(i) Costing Profit and Loss Account indicating the net profits,

(ii) A Statement showing reconciliation between profit as disclosed by the Cost Accounts and Financial Accounts.

ICAI'S ANSWER

(i) **Costing Profit and Loss Account for the year ended 31st March 2024:**

Particulars	Amount (₹)	Particulars	Amount (₹)
Material consumed	14,16,000	Sales (30,000 units)	30,00,000
Direct wages	7,42,000		
Prime Cost	21,58,000		
Works overheads (20% of Prime cost)	4,31,600		
	25,89,600		
Less: Work in progress	(54,000)		
Factory cost	25,35,600		
Administration overheads (₹ 5 × 32,000 units)	1,60,000		
Cost of production of goods produced	26,95,600		
Less: Finished stock	(1,68,475)		
Cost of production of goods sold	25,27,125		

Particulars	Amount (₹)	Particulars	Amount (₹)
Selling and distribution overheads (₹ 6 × 30,000 unit)	1,80,000		
Cost of sales	27,07,125		
Profit (balancing figure)	2,92,875		
	30,00,000		30,00,000

(ii) Statement reconciling the profit as per costing profit and loss account with the profit as per financial accounts

Particulars	Amount (₹)	Amount (₹)
Profit as per cost records		2,92,875
Add: Overheads over-absorbed:		
- Works overheads (₹ 4,31,600 – ₹ 4,26,000)	5,600	
- Administration OH (₹ 1,60,000 – ₹ 1,50,000)	10,000	
- Selling and Distribution (₹ 1,80,000 – ₹ 1,65,000)	15,000	30,600
Less: Closing stock overvalued (₹ 1,68,475 – ₹ 1,67,500)		(975)
Profit as per financial accounts		3,22,500

*It is assumed that there is no opening stock

No. of units produced = Number of units sold + Finished stock

= 30,000 + 2,000 = 32,000 units.

ICAI · P4 · MTP May 2025 · Series II · question p. 9-10 · answer p. 10-11

573 MTP Jan 2025 S-I MCQ 12 2 marks MCQ

WHICH of the following item is not the cause of differences in Financial and Cost Accounts?

A Income tax not treated in Cost Accounts

B Dividends credited in Financial Accounts

C Losses on the sale of investments not treated in Financial Accounts

D Cost Accounts showing notional depreciation on the assets fully depreciated for which book value is nil

ICAI'S ANSWER

(C) Losses on the sale of investments not treated in Financial Accounts

ICAI · P4 · MTP Jan 2025 · Series I · question p. 5 · answer p. 4

574 MTP **Jan 2025 S-II** Q4(c) 4 marks

Following information is available as per the cost accounts of a company for the year ended 31st March:

Particulars	Amount (₹)
Profit	7,77,150
Factory expenses under-charged	2,35,500
Administrative expenses under-charged	1,17,750
Selling & distribution expenses under-charged	31,400
Income from interest and dividends (not adjusted in cost statement)	2,35,500

You are required to PREPARE a reconciliation statement to ascertain Profit as per Financial Accounts.

ICAI'S ANSWER

Statement of Reconciliation

(to ascertain Profit as per Financial Accounts)

Particulars	(₹)	(₹)
Profit as per Cost Account		7,77,150
Add: Income from interest and dividends		2,35,500
		10,12,650
Less: Factory expenses under-charged in Cost Accounts	2,35,500	
Administrative expenses under-charged in Cost Accounts	1,17,750	
Selling & distribution expenses under-charged in Cost Accounts	31,400	(3,84,650)
Profit as per Financial Accounts		**6,28,000**

ICAI · P4 · MTP Jan 2025 · Series II · question p. 11 · answer p. 15

575 MTP **Sep 2024 S-I** Q1(b) 5 marks

A manufacturing company has disclosed net loss of ₹ 48,700 as per their cost accounting records for the year ended 31st March, 2024. However their financial accounting records disclosed net profit of ₹ 30,400 for the same period. A scrutiny of data of both the sets of books of accounts revealed the following informations:

		₹
(i)	Factory overheads under absorbed	30,500
(ii)	Administrative overheads over absorbed	65,000
(iii)	Depreciation charged in financial accounts	2,25,000
(iv)	Depreciation charged in cost accounts	2,70,000

		₹
(v)	Income-tax provision	52,400
(vi)	Transfer fee (credited in financial accounts)	10,200
(vii)	Obsolescence loss charged in financial accounts	20,700
(viii)	Notional rent of own premises charged in cost accounts	49,000
(ix)	Value of opening stock:	
	(a) in cost accounts	1,38,000
	(b) in financial accounts	1,15,000
(x)	Value of closing stock:	
	(a) in cost accounts	1,22,000
	(b) in financial accounts	1,12,500

PREPARE a Memorandum Reconciliation Account by taking costing loss as base.

ICAI'S ANSWER

Memorandum Reconciliation Accounts

Dr.			Cr.
Particulars	**Amount (₹)**	**Particulars**	**Amount (₹)**
To Net Loss as per Cost Accounts	48,700	By Administration overheads over recovered in Cost Accounts	65,000
To Factory overheads under absorbed in Cost Accounts	30,500	By Depreciation overcharged in Cost Accounts (₹ 2,70,000 –₹ 2,25,000)	45,000
To Provision for Income tax	52,400	By Transfer fees in Financial Accounts	10,200
To Obsolescence loss	20,700	By Notional Rent of own premises	49,000
To Overvaluation of closing stock in Cost Accounts**	9,500	By Overvaluation of Opening stock in Cost Accounts*	23,000
To Net Profit (as per Financial Accounts)	30,400		
	1,92,200		1,92,200

* Overvaluation of Opening Stock as per Cost Accounts

= Value in Cost Accounts – Value in Financial Accounts

= ₹ 1,38,000 – ₹ 1,15,000 = ₹ 23,000.

** Overvaluation of Closing Stock as per Cost Accounts

= Value in Cost Accounts – Value in Financial Accounts

= ₹ 1,22,000 – ₹ 1,12,500 = ₹ 9,500.

ICAI · P4 · MTP Sep 2024 · Series I · question p. 6-7 · answer p. 6

576 MTP **Sep 2024 S-II** Q5(a) 7 marks

The financial books of a company reveal the following data for the year ended 31st March, 2024:

	(₹)
Opening Stock:	
Finished goods 545 units	48,250
Work-in-process	38,000
01.04.2023 to 31.03.2024	
Raw materials consumed	5,00,000
Direct Labour	4,20,000
Factory overheads	3,56,000
Administration overheads	2,10,000
Stores Adjustment debited in financial Account	50,000
Dividend paid	98,000
Bad Debts	16,000
Selling and Distribution Overheads	84,000
Income tax paid	34,000
Interest received	42,000
Sales 14,250 units	13,96,500
Closing Stock: Finished goods 460 units	44,500
Work-in-process	36,200

The cost records provide as under:

- Factory overheads are absorbed at 60% of direct wages.
- Administration overheads are recovered at 20% of factory cost.
- Selling and distribution overheads are charged at ₹ 6 per unit sold.
- Opening Stock of finished goods is valued at ₹ 90 per unit.
- The company values work-in-process at factory cost for both Financial and Cost Profit Reporting.

Required:

(i) Prepare statements for the year ended 31st March, 2024 show

- the profit as per financial records
- the profit as per costing records.

(ii) Present a statement reconciling the profit as per costing records with the profit as per Financial Records

ICAI'S ANSWER

(i) **Statement of Profit as per financial records**

(for the year ended March 31, 2024)

	(₹)		(₹)
To Opening stock of Finished Goods	48,250	By Sales	13,96,500
To Work-in-process	38,000	By Closing stock of finished Goods	44,500
To Raw materials consumed	5,00,000	By Work-in-Process	36,200
To Direct labour	4,20,000	By Interest received	42,000
To Factory overheads	3,56,000	By Loss	3,35,050
To Administration overheads	2,10,000		
To Selling & distribution overheads	84,000		
To Dividend paid	98,000		
To Bad debts	16,000		
To Stores adjustment	50,000		
To Income tax	34,000		
	18,54,250		**18,54,250**

Statement of Profit as per costing records

(for the year ended March 31,2024)

	(₹)
Sales revenue (A) *(14,250 units)*	13,96,500
Cost of sales:	
Opening stock *(545 units x ₹ 90)*	49,050
Add: Cost of production of 14,165 units *(Refer to working note 2)*	14,08,560
Less: Closing stock (₹ 99.44 × 460 units)	45,742
Production cost of goods sold (14,250 units)	14,11,868
Selling & distribution overheads (14,250 units x ₹ 6)	85,500

	(₹)
Cost of sales: (B)	14,97,368
Profit/Loss: {(A) – (B)}	(1,00,868)

(ii) **Statement of Reconciliation**

(Reconciling the profit as per costing records with the profit as per financial records)

	(₹)	(₹)
Loss as per Cost Accounts		(1,00,868)
Add: Administration overheads over absorbed (₹ 2,34,760 – ₹ 2,10,000)	24,760	
Opening stock overvalued (₹ 49,050 – ₹ 48,250)	800	
Interest received	42,000	
Selling & distribution overheads over recovered (₹ 85,500 – ₹ 84,000)	1,500	69,060
		(31,808)
Less: Factory overheads over recovered (₹ 3,56,000 - ₹2,52,000)	1,04,000	
Closing stock overvalued (₹ 45,742 – ₹ 44,500)	1,242	
Stores adjustment	50,000	
Income tax	34,000	
Dividend	98,000	
Bad debts	16,000	(3,03,242)
Loss as per financial accounts		(3,35,050)

Working notes:

1. Number of units produced

	Units
Sales	14,250
Add: Closing stock	460
Total	14,710
Less: Opening stock	545
Number of units produced	14,165

2. Cost Sheet

	(₹)
Raw materials consumed	5,00,000
Direct labour	4,20,000
Prime cost	9,20,000
Factory overheads (60% of direct wages)	2,52,000
Factory cost	11,72,000
Add: Opening work-in-process	38,000
Less: Closing work-in-process	36,200
Factory cost of goods produced	11,73,800
Administration overheads (20% of factory cost)	2,34,760
Cost of production of 14,165 units (Refer to working note 1)	14,08,560
Cost of production per unit: $\frac{₹\ 14,08,560}{14,165}$	99.44

ICAI · P4 · MTP Sep 2024 · Series II · question p. 11 · answer p. 15-17

577 MTP **May 2024 S-II** Q4(c) 2 marks

WHEN is the reconciliation statement of Cost and Financial accounts not required?

ICAI'S ANSWER

When the Cost and Financial Accounts are integrated - there is no need to have a separate reconciliation statement between the two sets of accounts. Integration means that the same set of accounts fulfil the requirement of both i.e., Cost and Financial Accounts.

ICAI · P4 · MTP May 2024 · Series II · question p. 9 · answer p. 12

578 MTP **Old P3 May 2023 S-II** Q1(c) 5 marks old scheme

A manufacturing company disclosed a net profit ₹10,20,000 as per their cost accounts for the year ended 31st March, 2023. The financial accounts however disclosed a net profit of ₹ 6,94,000 for the same period. The following information was revealed as a result of scrutiny of the figures of both the sets of accounts.

	(₹)
(i) Factory Overheads under-absorbed	80,000
(ii) Administration Overheads over-absorbed	1,20,000
(iii) Depreciation charged in Financial Accounts	6,50,000
(iv) Depreciation charged in Cost Accounts	5,50,000
(v) Interest on investments not included in Cost Accounts	1,92,000

	(₹)
(vi) Income-tax provided	1,08,000
(vii) Interest on loan funds in Financial Accounts	4,90,000
(viii) Transfer fees (credit in financial books)	48,000
(ix) Stores adjustment (credit in financial books)	28,000
(x) Dividend received	64,000

PREPARE a Reconciliation statement.

ICAI'S ANSWER

Statement of Reconciliation

Particulars	Amount (₹)	Amount (₹)
Net profit as per Cost accounts		10,20,000
Add:		
Administration Overheads over-absorbed	1,20,000	
Interest on investments	1,92,000	
Transfer fees	48,000	
Stores adjustment	28,000	
Dividend received	64,000	4,52,000
Less:		
Factory Overheads under-absorbed	80,000	
Depreciation under charged	1,00,000	
Income-tax provided	1,08,000	
Interest on loan funds	4,90,000	(7,78,000)
Net profit as per Financial accounts		6,94,000

ICAI · Old P3 · MTP May 2023 · Series II · question p. 1-2 · answer p. 2

579 MTP **Old P3 Jul 2021 S-I** Q1(b) 5 marks old scheme

A manufacturing company disclosed a net profit Rs. 10,20,000 as per their cost accounts for the year ended 31st March 2021. The financial accounts however disclosed a net profit of Rs. 6,94,000 for the same period. The following information was revealed as a result of scrutiny of the figures of both the sets of accounts.

	(Rs.)
(i) Factory Overheads under-absorbed	80,000
(ii) Administration Overheads over-absorbed	1,20,000
(iii) Depreciation charged in Financial Accounts	6,50,000
(iv) Depreciation charged in Cost Accounts	5,50,000
(v) Interest on investments not included in Cost Accounts	1,92,000
(vi) Income-tax provided	1,08,000
(vii) Interest on loan funds in Financial Accounts	4,90,000
(viii) Transfer fees (credit in financial books)	48,000
(ix) Stores adjustment (credit in financial books)	28,000
(x) Dividend received	64,000

PREPARE a Reconciliation statement.

ICAI'S ANSWER

Statement of Reconciliation

Particulars	Amount (Rs.)	Amount (Rs.)
Net profit as per Cost accounts		10,20,000
Add:		
Administration Overheads over-absorbed	1,20,000	
Interest on investments	1,92,000	
Transfer fees	48,000	
Stores adjustment	28,000	
Dividend received	64,000	4,52,000
Less:		
Factory Overheads under-absorbed	80,000	
Depreciation under charged	1,00,000	
Income-tax provided	1,08,000	
Interest on loan funds	4,90,000	(7,78,000)
Net profit as per Financial accounts		6,94,000

ICAI · Old P3 · MTP Jul 2021 · Series I · question p. 1 · answer p. 1

580 MTP Old P3 Jul 2021 S-II Q3(a) 10 marks old scheme

The following figures have been taken from the financial accounts of a manufacturing firm for the year ended 31st March, 2021:

	(Rs.)
Direct material consumption	20,00,000
Direct wages	12,00,000
Factory overheads	6,40,000
Administrative overheads	2,80,000
Selling and distribution overheads	3,84,000
Bad debts	32,000
Preliminary expenses written off	16,000
Legal charges	4,000
Dividend received	40,000
Interest on fixed deposit	8,000
Sales - 48,000 units	48,00,000
Closing stock:	
- Finished stock - 4,000 units	3,20,000
- Work-in-process	96,000

The cost accounts for the same period reveal that the Direct Material consumption was Rs. 22,40,000; Factory overhead is recovered at 20% on prime cost; Administration overhead is recovered @ Rs. 4.8 per unit of production; and Selling and Distribution overheads are recovered at Rs. 6.40 per unit sold.

Required:

PREPARE Costing and Financial Profit & Loss Accounts and RECONCILE the difference in the profit as arrived at in the two sets of accounts.

ICAI'S ANSWER

Costing Profit and Loss Account

Particulars	Amount (Rs.)	Particulars	Amount (Rs.)
To Direct Material consumed	22,40,000	By Sales	48,00,000
To Direct Wages	12,00,000	By Closing Work-in-process	96,000
Prime Cost	34,40,000	By Closing Finished stock $\left(\frac{\text{Rs. } 41,28,000 - \text{Rs. } 96,000}{52,000\text{units}} \times 4,000\right)$	3,10,154

Particulars	Amount (Rs.)	Particulars	Amount (Rs.)
To Factory overheads (20% of prime cost)	6,88,000		
	41,28,000		
To Administrative overheads (Rs. 4.80 × 52,000* units)	2,49,600		
To Selling & distribution overheads (Rs.6.40 × 48,000 units)	3,07,200		
To Net profit (balancing figure)	5,21,354		
	52,06,154		52,06,154

* Units produced = Units sold + Closing stock - Opening stock

= 48,000 + 4,000 - 0 = 52,000 units

Financial Profit and Loss Account

Particulars	Amount (Rs.)	Particulars	Amount (Rs.)
To Direct Material consumed	20,00,000	By Sales	48,00,000
To Direct Wages	12,00,000	By Dividend received	40,000
To Factory overheads	6,40,000	By Interest on fixed deposit	8,000
To Administrative overheads	2,80,000	By Closing Work-in-process	96,000
To Selling & distribution overheads	3,84,000	By Closing Finished stock	3,20,000
To Bad debts	32,000		
To Preliminary expenses	16,000		
To Legal charges	4,000		
To Net profit (balancing figure)	7,08,000		
	52,64,000		52,64,000

Reconciliation Statement

	Particulars	Amount (Rs.)	Amount (Rs.)
	Net profit as per Financial Profit & Loss A/c		7,08,000
Add:	Administrative overheads (2,80,000 - 2,49,600)	30,400	
	Selling & Distribution overheads (3,84,000 - 3,07,200)	76,800	

	Particulars	Amount (Rs.)	Amount (Rs.)
	Bad debts	32,000	
	Preliminary expenses	16,000	
	Legal charges	4,000	1,59,200
			8,67,200
Less:	Difference in value of materials consumed (22,40,000 - 20,00,000)	2,40,000	
	Factory overheads (6,88,000 - 6,40,000)	48,000	
	Dividend received	40,000	
	Interest on fixed deposit	8,000	
	Closing stock (3,20,000 - 3,10,154)	9,846	(3,45,846)
	Profit as per Costing Profit & Loss A/c		5,21,354

ICAI · Old P3 · MTP Jul 2021 · Series II · question p. 4 · answer p. 5-6

581 MTP **Old P3 Nov 2020 S-I** Q6(c) 5 marks old scheme

LIST five financial expenses that causes differences in Financial and Cost Accounts.

ICAI'S ANSWER

Financial expenses causing differences in Financial and Cost Accounts:

(i) Interest on loans or bank mortgages.

(ii) Expenses and discounts on issue of shares, debentures etc.

(iii) Other capital losses i.e., loss by fire not covered by insurance etc.

(iv) Losses on the sales of fixed assets and investments.

(v) Goodwill written off.

(vi) Preliminary expenses written off.

(vii) Income tax, donations, subscriptions.

(viii) Expenses of the company's share transfer office, if any.

ICAI · Old P3 · MTP Nov 2020 · Series I · question p. 8 · answer p. 15

582 MTP **Old P3 May 2020 S-I** Q4(a) 10 marks old scheme

The Trading and Profit and Loss Account of a company for the year ended 31-03-2020 is as under:

Trading and Profit and Loss Account

Particulars	Rs.	Particulars	Rs.
To Materials	26,80,000	By Sales (50,000 units)	62,00,000
To Wages	17,80,000	By Closing stock (2,000 units)	1,50,000
To Factory expenses	9,50,000	By Dividend received	80,000
To General administrative expenses	4,80,200		
To Selling Expenses	2,50,000		
To Preliminary expenses written off	70,000		
To Net profit	2,19,800		
	64,30,000		64,30,000

In the Cost Accounts:

(i) Factory expenses have been allocated to production at 20% of Prime Cost.

(ii) General administrative expenses absorbed at 10% of factory cost.

(iii) Selling expenses charged at Rs.10 per unit sold.

Required:

PREPARE the Costing Profit and Loss Account of the company and RECONCILE the Profit/Loss with the profit as shown in the Financial Accounts.

ICAI'S ANSWER

Workings:

Preparation of Cost Sheet/ Cost Statement

Particulars	Amount (Rs.)
Materials	26,80,000
Wages	17,80,000
Prime Cost	44,60,000
Add: Factory expenses (20% of Rs. 44,60,000)	8,92,000
Factory cost/ Cost of Production	53,52,000
Less: Closing Stock $\left(\frac{\text{Rs.}53,52,000}{52,000\text{units}} \times 2,000\text{units}\right)$	(2,05,846)
Cost of Goods Sold	51,46,154
Add: General administrative expenses (10% of Rs.53,52,000)	5,35,200
Add: Selling expenses (Rs.10 × 50,000 units)	5,00,000
Cost of Sales	61,81,354

Particulars	Amount (Rs.)
Profit (Balancing figure)	18,646
Sales Value	62,00,000

Costing Profit and Loss Account

Particulars	Amount (Rs.)	Particulars	Amount (Rs.)
To Materials	26,80,000	By Sales	62,00,000
To Wages	17,80,000	By Closing stock	2,05,846
To Factory expenses	8,92,000		
To General administrative expenses	5,35,200		
To Selling expenses	5,00,000		
To Profit (Balancing figure)	18,646		
	64,05,846		64,05,846

Reconciliation of profit as per Cost Accounts and as per Financial Accounts

Particulars	Amount (Rs.)
Profit as per Cost Accounts	18,646
Additions:	
General administrative expenses (Over-absorbed) (Rs. 5,35,200 – Rs.4,80,200)	55,000
Selling expenses (Overcharged) (Rs. 5,00,000 – Rs. 2,50,000)	2,50,000
Dividend received	80,000
	4,03,646
Deductions:	
Factory expenses (Under -absorbed) (Rs.9,50,000 – 8,92,000)	58,000
Closing stock (Over-valued) (Rs.2,05,846 – Rs. 1,50,000)	55,846
Preliminary expenses written off	70,000
	1,83,846
Profit as per Financial Accounts	2,19,800

ICAI · Old P3 · MTP May 2020 · Series I · question p. 4-5 · answer p. 8

583 MTP Old P3 Nov 2019 S-I Q1(d) 5 marks old scheme

A manufacturing company has disclosed a net loss of Rs 2,25,000 as per their cost accounting records for the year ended March 31, 2019. However, their financial accounting records disclosed a net loss of Rs 2,70,000 for the same period. A scrutiny of data of both the sets of books of accounts revealed the following information:

		(Rs)
(i)	Factory overheads under-absorbed	5,000
(ii)	Administration overheads over-absorbed	3,000
(iii)	Depreciation charged in financial accounts	70,000
(iv)	Depreciation charged in cost accounts	80,000
(v)	Interest on investments not included in cost accounts	20,000
(vi)	Income-tax provided in financial accounts	65,000
(vii)	Transfer fees (credit in financial accounts)	2,000
(viii)	Preliminary expenses written off	3,000
(ix)	Over-valuation of closing stock of finished goods in cost accounts	7,000

Required:

PREPARE a Memorandum Reconciliation Account.

ICAI'S ANSWER

Memorandum Reconciliation Account

	Particulars	(Rs.)		Particulars	(Rs.)
To	Net loss as per Costing books	2,25,000	By	Administrative overhead over absorbed in costs	3,000
To	Factory overheads under absorbed	5,000	By	Depreciation over charged in Cost books (Rs. 80,000 – Rs.70,000)	10,000
To	Income tax not provided in Cost books	65,000	By	Interest on investments not included in Cost books	20,000
To	Preliminary expenses written off in Financial books	3,000	By	Transfer fees not considered in Cost books	2,000
To	Over-valuation of Closing Stock of finished goods in Cost books	7,000	By	Net loss as per Financial books	2,70,000
		3,05,000			3,05,000

ICAI · Old P3 · MTP Nov 2019 · Series I · question p. 2 · answer p. 3

CHAPTER 8

Unit and Batch Costing

8.1 Unit costing

1 question · asked in 1 ICAI document · 1 exam paper

584 EXAM May 2024 Q6(c) 4 marks

Describe Unit Costing and Batch Costing. Give three examples of industries for each method where these are used.

ICAI'S ANSWER

Unit costing is that method of costing where the output produced is identical and each unit of output requires identical cost. Under this method costs, are collected and analysed element wise and then total cost per unit is ascertained by dividing the total cost with the number of units produced. Such a method of costing is used in the industries like paper, cement, steel works, mining, breweries etc.

Batch Costing: Batch Costing is a type of specific order costing where articles are manufactured in predetermined lots, known as batch. Cost per unit in a batch is ascertained by dividing the total cost of a batch by the number of units produced in that batch.

Such a method of costing is used in the case of pharmaceutical or drug industries, readymade garment industries, industries, manufacturing electronic parts of T.V. radio sets etc.

ICAI · P4 · May 2024 exam · question p. 25 · answer p. 26-27

8.2 Batch costing and the economic batch quantity

43 questions · asked in 38 ICAI documents · 7 exam papers

585 EXAM May 2026 Q1(a) 5 marks

XYZ Limited specializes in manufacture of high end office furniture. They use a batch costing system to track production costs. The company received an order for a special production run, Batch A - 01, consisting of 20 Ergonomic Executive Chairs. The estimated costs for this specific: batch are as follows:

Direct Material	₹ 55,500
Direct Labour :	
Cutting & Shaping	300 hours @ ₹ 50 per hour
Polishing & Finishing	200 hours @ ₹ 40 per hour
Factory Overheads:	
Cutting & shaping	300 hours @ ₹ 8 per hour
Polishing & Shaping	200 hours @ ₹ 8 per hour
Administrative & Selling Overheads	20% of Works Cost
Profit	25% on Selling Price

You are required to calculate:

(i) Total Cost of the Batch.

(ii) Selling Price per Chair.

ICAI'S ANSWER

(i) Statement showing Total Cost of the Batch A – 01

Particulars	Amount (in ₹)
Direct Material Cost	55,500
Direct Labour:	
Cutting & Shaping (300 hours @ ₹ 50 per hour)	15,000
Polishing & Finishing (200 hours @ ₹ 40 per hour)	8,000
Prime Cost	**78,500**
Add: Factory Overheads:	
Cutting & Shaping (300 hours @ ₹ 8 per hour)	2,400
Polishing & Finishing (200 hours @ ₹ 8 per hour)	1,600
Factory Cost/Works Cost	**82,500**
Add: Administrative & Selling Overheads (20% of Works Cost)	16,500
Total Cost	**99,000**

(ii) Selling Price per Chair:

Units produced & sold = 20 chairs

Sales = Total cost + 25% profit on sales or 33.3% (1/3rd) profit on cost

= ₹99,000 + 33.33% (1/3rd) of ₹ 99,000

= ₹ 1,32,000

Selling price per chair = Total sales/no. of chairs sold

= ₹ 1,32,000/20= ₹ 6,600 per chair

ICAI · P4 · May 2026 exam · question p. 8 · answer p. 10

586 EXAM May 2025 Q5(b) 3 marks

Distinguish between "job costing and batch costing".

ICAI'S ANSWER

Difference between job and batch costing

Sr. No	Job Costing	Batch Costing
1	Method of costing used for non-standard and non-repetitive products produced as per customer specifications and against specific orders.	Homogeneous products produced in a continuous production flow in lots.
2	Cost determined for each Job	Cost determined in aggregate for the entire Batch and then arrived at on per unit basis.
3	Jobs are different from each other and independent of each other. Each Job is unique.	Products produced in a batch are homogeneous and lack of individuality

ICAI · P4 · May 2025 exam · question p. 26 · answer p. 28

587 **EXAM** **Jan 2025** Case scenario · Q1(b)(i) to Q1(b)(ii) 5 marks

CASE SCENARIO

PQR Ltd. manufactures a product in batches of 2,000 units.

The following costs are incurred for each batch

Particulars	Amount (in₹)
Direct Material Cost per Batch	2,40,000
Direct Labour Cost per Batch	1,65,000
Overhead Absorption Rate (variable)	120 per machine hour
Expected Rejection Rate	3%
Scrap Value per Rejected Unit	75

Other Information:

Particulars	Details
Selling Price per Good Unit	₹ 250
Total Available Machine Hours per month	3,000 hours
Fixed Overheads per Month	₹ 1,25,000
Batches Manufactured per Month	10 batches

Required:

Q1(b)(i) 3 marks

Calculate contribution per unit of good units after adjusting rejected units.

Q1(b)(ii) 2 marks

Calculate the company's total monthly profit.

ICAI'S ANSWERS

Q1(b)(i)

Calculation of Contribution per unit of good units

	Total Amount (₹)	Per Unit (₹)
Selling Price	48,50,000	250.00
Direct Material Cost	24,00,000	
Direct Labour Cost	16,50,000	
Variable Overhead (₹120 × 3,000 hours)	3,60,000	
Total Variable Cost	44,10,000	220.50
Less: Scrap Value of Rejected Units (₹ 75 × 20,000 × 3%)	45,000	
Net Variable Cost	43,65,000	225.00
Contribution per good units	4,85,000	25.00

Working Note:

Number of units manufactured in a month = Number of batches x number of units in a batch

= 10 x 2,000 = 20,000 units

Number of good units sold = 20,000 units x 97% = 19,400 units

Q1(b)(ii)

Calculation of Company's total monthly profit

	Total Amount (₹)
Contribution	4,85,000
Less: Fixed Cost	1,25,000
Total Monthly Profit	3,60,000

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588 EXAM Old P3 May 2023 Q1(a) 5 marks old scheme

TSK Limited manufactures a variety of products. The annual demand for one of its products- Product 'X' is estimated as ₹1,35,000 units. Product 'X' is to be manufactured done in batches. Set up cost of each batch is ₹ 3,375 and inventory holding cost is ₹ 5 per unit. It is expected that demand of Product 'X' would be uniform throughout the year.

Required:

(i) Calculate the Economic Batch (EBQ) for Product 'X'.

(ii) Assuming that the company has a policy of manufacturing 7,500 units of Product 'X' per batch, calculate the additional cost incurred as compared to the cost incurred as per Economic Batch Quantity (EBQ) as computed in (i) above.

ICAI'S ANSWER

(i) Economic Batch Quantity (EBQ) = $\sqrt{\frac{2DS}{C}}$

where,

D = Annual demand for the product

S = Set-up cost per batch

C = Carrying cost per unit per annum.

$\sqrt{\frac{2DS}{C}} = \sqrt{\frac{2 \times 1{,}35{,}000 \times 3{,}375}{5}}$ = **13,500 units.**

(ii) Total Cost (of maintaining the inventories) when batch size (Q) are 13,500 and 7,500 units respectively

Total cost = Total set-up cost + Total carrying cost.

	When batch size is 13,500 units	When batch size is 7,500 units
Total set up cost	= 1,35,000/13,500 × ₹ 3,375 = ₹ 33,750 Or, No. of setups = 10 = 10 x ₹ 3,375 = ₹ 33,750	= 1,35,000/7,500 × ₹ 3,375 = ₹ 60,750
Total Carrying cost	1/2 × 13,500 × 5 = ₹ 33,750	1/2 × 7,500 × 5 = ₹ 18,750
Total Cost	₹ 67,500	₹ 79,500

₹ 12,000 is the excess cost borne by the company due to batch size not being economic batch quantity.

Alternative presentation

	EOQ 13,500	Batch size 7500	Extra cost	Saving
No of setup	10	18	8 × 3375 = 27,000	
Carrying cost	13,500 – 7500 = 6000/ 2 @ 5			15,000

Net extra cost = (27,000- 15,000) = ₹ 12,000

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589 **EXAM** **Old P3 Nov 2022** Q1(a) 5 marks old scheme

A Ltd. is a pharmaceutical company which produces vaccines for diseases like Monkey Pox, Covid-19 and Chickenpox. A distributor had given an order for 1,600 Monkey Pox Vaccines. The company can

produce 80 vaccines at a time. To process a batch of 80 Monkey Pox vaccines, the following costs would be incurred:

	₹
Direct Materials	4,250
Direct wages	500
Lab set-up cost	1,400

The Production Overheads are absorbed at a rate of 20% of direct wages and 20% of total production cost is charged in each batch for Selling, distribution and administration Overheads. The company is willing to earn profit of 25% on sales value.

You are required to determine:

(i) Total Sales value for 1,600 Monkey Pox Vaccines

(ii) Selling price per unit of the Vaccine.

ICAI'S ANSWER

(i) & (ii) Calculation of Sales value and Selling price per unit of Monkey Pox vaccine

Particulars	Amount (₹) per Batch	Amount (₹) for 1600 units or 20 batches	Amount (₹) per unit
Direct materials	4,250	85,000	53.125
Direct wages	500	10,000	6.250
Lab set-up cost	1,400	28,000	17.500
Production overheads (20% of direct wages)	100	2,000	1.250
Production Cost	6,250	1,25,000	78.125
Selling, distribution and administration cost (20% of Production cost)	1,250	25,000	15.625
Total Cost	7,500	1,50,000	93.75
Add: Profit (1/3rd of Total cost or 25% of Sales value)	2,500	50,000	31.25
Sales value	10,000	2,00,000	125.00

ICAI · Old P3 · Nov 2022 exam · question p. 1 · answer p. 3

590 EXAM **Old P3 Jul 2021** Q1(c) 5 marks old scheme

AUX Ltd. has an Annual demand from a single customer for 60,000 Covid-19 vaccines. The customer prefers to order in the lot of 15,000 vaccines per order. The production cost of vaccine is ₹ 5,000 per

vaccine. The set-up cost per production run of Covid-19 vaccines is ₹ 4,800. The carrying cost is ₹ 12 per vaccine per month.

You are required to:

(i) Find the most Economical Production Run.

(ii) Calculate the extra cost that company incurs due to production of 15,000 vaccines in a batch.

ICAI'S ANSWER

(i) Calculation of most Economical Production Run

$$= \sqrt{\frac{2 \times 60{,}000 \times ₹\ 4{,}800}{12 \times 12}} = 2{,}000 \text{ Vaccine}$$

(ii) Calculation of Extra Cost due to processing of 15,000 vaccines in a batch

	When run size is 2,000 vaccines	When run size is 15,000 vaccines
Total set up cost	= 60,000/2,000 × ₹ 4,800 = ₹ 1,44,000	= 60,000/15,000 × ₹ 4,800 = ₹ 19,200
Total Carrying cost	½ × 2,000 × ₹ 144 = ₹ 1,44,000	½ × 15,000 × ₹ 144 = ₹ 10,80,000
Total Cost	₹ 2,88,000	₹ 10,99,200

Thus, extra cost = ₹ 10,99,200 – ₹ 2,88,000 = ₹ **8,11,200**

ICAI · Old P3 · Jul 2021 exam · question p. 54 · answer p. 56

591 **EXAM** **Old P3 Jan 2021** Q1(d) **5 marks** **old scheme**

GHI Ltd. manufactures 'Stent' that is used by hospitals in heart surgery. As per the estimates provided by Pharmaceutical Industry Bureau, there will be a demand of 40 Million 'Stents' in the coming year. GHI Ltd. is expected to have a market share of 2.5% of the total market demand of the Stents in the coming year. It is estimated that it costs ₹ 1.50 as inventory holding cost per stent per month and that the set-up cost per run of stent manufacture is ₹ 225.

Required:

(i) What would be the optimum run size for Stent manufacture?

(ii) What is the minimum inventory holding cost?

(iii) Assuming that the company has a policy of manufacturing 4,000 stents per run, how much extra costs the company would be incurring as compared to the optimum run suggested in (i) above?

ICAI'S ANSWER

(i) Computation of Optimum Run size of 'Stents' or Economic Batch Quantity (EBQ)

Economic Batch Quantity (EBQ) = $\sqrt{\frac{2DS}{C}}$

Where,

D = Annual demand for the Stents

= 4,00,00,000 × 2.5% = 10,00,000 units

S = Set- up cost per run

= ₹ 225

C = Carrying cost per unit per annum

= ₹ 1.50 × 12 = ₹ 18

$$\text{EBQ} = \sqrt{\frac{2 \times 10{,}00{,}000 \times ₹\ 225}{₹\ 18}}$$

= 5,000 units of Stents

(ii) Minimum inventory holding cost

Minimum Inventory Cost = Average Inventory × Inventory Carrying Cost per unit per annum

= (5,000 ÷ 2) × ₹ 18

= ₹ 45,000

(iii) Calculation of the extra cost due to manufacturing policy

	When run size is 4,000 units	When run size is 5,000 units i.e. at EBQ
Total set up cost	= 10,00,000/4,000 × ₹ 225 = ₹ 56,250	10,00,000/5,000 × ₹ 225 = ₹ 45,000
Total Carrying cost	½ × 4,000 × ₹ 18 = ₹ 36,000	½ × 5,000 × ₹ 18 = ₹ 45,000
Total Cost	**₹ 92,250**	**₹ 90,000**

Extra cost= ₹ 92,250 - ₹ 90,000 = **₹ 2,250**

ICAI · Old P3 · Jan 2021 exam · question p. 2-3 · answer p. 5-6

592 **RTP** **Sep 2026** Q10

Pintu Ltd. operates in beverages industry where it manufactures soft-drink in three sizes of Large (3 litres), Medium (1.5 litres) and Small (600 ml) bottles. The products are processed in batches. The 5,000 litres capacity processing plant consumes electricity of 90 Kilowatts per hour and a batch takes 1 hour 45 minutes to complete. Only symmetric size of products can be processed at a time. The machine set-up takes 15 minutes to get ready for next batch processing. During the set-up power consumption is only 20%.

(i) The current price of Large, Medium and Small are ₹ 150, ₹ 90 and ₹ 50 respectively.

(ii) To produce a litre of beverage, 14 litres of raw material-W and 25 ml of Material-C are required which costs ₹ 0.50 and ₹ 1,000 per litre respectively.

(iii) 20 direct workers are required. The workers are paid ₹ 880 for 8 hours shift of work.

(iv) The average packing cost per bottle is ₹ 3

(v) Power cost is ₹ 7 per Kilowatt -hour (Kwh)

(vi) Other variable cost is ₹ 30,000 per batch.

(vii) Fixed cost (Administration and marketing) is ₹ 4,90,00,000.

(viii) The holding cost is ₹ 1 per bottle per annum.

The marketing team has surveyed the following demand (bottle) of the product:

Large	Medium	Small
3,00,000	7,50,000	20,00,000

You are required to CALCULATE profit/ loss per batch and also COMPUTE Economic Batch Quantity (EBQ).

ICAI'S ANSWER

Workings:

1. Maximum number of bottles that can be processed in a batch:

2. $= \frac{5{,}000 \text{ ltrs}}{\text{Bottle volume}}$

Large		Medium		Small	
Qty (ltr)	**Max bottles**	**Qty (ltr)**	**Max bottles**	**Qty (ltr)**	**Max bottles**
3	1,666	1.5	3,333	0.6	8,333

*For simplicity of calculation small fractions has been ignored.

Number of batches to be run:

		Large	Medium	Small	Total
A	Demand	3,00,000	7,50,000	20,00,000	
B	Bottles per batch (Refer WN-1)	1,666	3,333	8,333	
C	No. of batches [A÷B]	180	225	240	645

*For simplicity of calculation small fractions has been ignored.

Quantity of Material-W and Material C required to meet demand:

	Particulars	Large	Medium	Small	Total
A	Demand (bottle)	3,00,000	7,50,000	20,00,000	
B	Qty per bottle (Litre)	3	1.5	0.6	
C	Output (Litre) [A×B]	9,00,000	11,25,000	12,00,000	32,25,000
D	Material-W per litre of output (Litre)	14	14	14	
E	Material-W required (Litre) [C×D]	**1,26,00,000**	**1,57,50,000**	**1,68,00,000**	**4,51,50,000**
F	Material-C required per litre of output (ml)	25	25	25	
G	Material-C required (Litre) [(C×F) ÷1000]	**22,500**	**28,125**	**30,000**	**80,625**

3. No. of Man-shift required:

		Large	Medium	Small	Total
A	No. of batches	180	225	240	645
B	Hours required per batch (Hours)	2	2	2	
C	Total hours required (Hours) [A×B]	360	450	480	1,290
D	No. of shifts required [C÷8]	45	57	60	162
E	**Total manshift [D×20 workers]**	**900**	**1,140**	**1,200**	**3,240**

4. Power consumption in Kwh

		Large	Medium	Small	Total
For processing					
A	No. of batches	180	225	240	645
B	Hours required per batch (Hours)	1.75	1.75	1.75	21.75
C	Total hours required (Hours) [A×B]	315	393.75	420	1,128.75
D	Power consumption per hour	90	90	90	90
E	**Power consumption in Kwh [C×D]**	**28,350**	**35,437.5**	**37,800**	**1,01,587.5**
F	**Per batch consumption (Kwh) [E÷A]**	**157.5**	**157.5**	**157.5**	**157.5**
For set-up					
G	Hours required per batch (Hours)	0.25	0.25	0.25	0.25
H	Total hours required (Hours) [A×G]	45	56.25	60	161.25
I	Power consumption per hour [20% × 90]	18	18	18	18
J	**Power consumption in Kwh [H×I]**	**810**	**1,012.5**	**1,080**	**2,902.5**
K	**Per batch consumption (Kwh) [J÷A]**	**4.5**	**4.5**	**4.5**	**4.5**

Calculation of Profit/ loss per batch:

	Particulars	Large	Medium	Small	Total
A	Demand (bottle)	3,00,000	7,50,000	20,00,000	30,50,000
B	Price per bottle (₹)	150	90	50	
C	**Sales value (₹) [A×B]**	**4,50,00,000**	**6,75,00,000**	**10,00,00,000**	**21,25,00,000**

	Particulars	Large	Medium	Small	Total
	Direct Material cost:				
E	Material-W (₹) [Qty in WN - 3 × ₹ 0.50]	63,00,000	78,75,000	84,00,000	2,25,75,000
F	Material-C (₹) [Qty in WN-3 × ₹ 1,000]	2,25,00,000	2,81,25,000	3,00,00,000	8,06,25,000
G	[E+F]	**2,88,00,000**	**3,60,00,000**	**3,84,00,000**	**10,32,00,000**
H	Direct Wages (₹) [Man-shift in WN-4 × ₹ 880]	7,92,000	10,03,200	10,56,000	28,51,200
I	Packing cost (₹) [A ×₹ 3]	9,00,000	22,50,000	60,00,000	91,50,000
	Power cost (₹)				
J	For processing (₹) [WN-5 × ₹ 7]	1,98,450	2,48,062.5	2,64,600	7,11,112.5
K	For set-up time (₹) [WN-5 × ₹ 7]	5,670	7,087.5	7,560	20,317.5
L	[J+K]	**2,04,120**	**2,55,150**	**2,72,160**	**7,31,430**
M	Other variable cost (₹) [No. of batch in WN - 2 × ₹ 30,000]	54,00,000	67,50,000	72,00,000	1,93,50,000
N	**Total Variable cost per batch [G+H+I+L+M]**	**3,60,96,120**	**4,62,58,350**	**5,29,28,160**	**13,52,82,630**
O	**Profit/ loss before fixed cost [C-N]**	**89,03,880**	**2,12,41,650**	**4,70,71,840**	**7,72,17,370**
P	Fixed Cost				4,90,00,000
Q	**Total Profit [O-P]**				**2,82,17,370**

Computation of Economic Batch Quantity (EBQ):

$$\text{EBQ} = \sqrt{\frac{2 \times D \times S}{C}}$$

D = Annual Demand for the Product = Refer A below

S = Set-up cost per batch = Refer D below

C = Carrying cost per unit per annum =Refer E below

	Particulars	Large	Medium	Small
A	Annual Demand (bottle)	3,00,000	7,50,000	20,00,000
Set-up Cost:				
B	Power cost for set-up time (₹) [Consumption per batch in WN-5 × ₹ 7]	31.50	31.50	31.50
C	Other variable cost (₹) *	30,000	30,000	30,000
D	Total Set-up cost [B+C]	30,031.50	30,031.50	30,031.50
E	Holding cost:	1.00	1.00	1.00
F	**EBQ (Bottle)**	**1,34,234**	**2,12,243**	**3,46,592**

* Other variable cost is assumed to be part of set-up cost.

ICAI · P4 · RTP Sep 2026 · question p. 12-13 · answer p. 35-38

593 RTP **Sep 2025** Q12

Pinku Confectioners (PC) owns a bakery which is used to make bakery items like pastries, cakes and muffins. PC use to bake atleast 50 units of any item at a time. A customer has given an order for 600 cakes. To process a batch of 50 cakes, the following cost would be incurred:

Direct materials- ₹ 5,000

Direct wages- ₹ 500

Oven set- up cost ₹ 750

PC absorbs production overheads at a rate of 20% of direct wages cost. 10% is added to the total production cost of each batch to allow for selling, distribution and administration overheads.

PC requires a profit margin of 25% of sales value.

Required:

(i) Determine the price to be charged for 600 cakes.

(ii) Calculate cost and selling price per cake.

(iii) What would be selling price per unit If the order is for 605 cakes.

ICAI'S ANSWER

Statement of cost per batch and per order

No. of batch = 600 units ÷ 50 units = 12 batches

	Particulars	Cost per batch (₹)	Total Cost (₹)
	Direct Material Cost	5,000.00	60,000
	Direct Wages	500.00	6,000
	Oven set-up cost	750.00	9,000

	Particulars	Cost per batch (₹)	Total Cost (₹)
	Add: Production Overheads (20% of Direct wages)	100.00	1,200
	Total Production cost	6,350.00	76,200
	Add: S&D and Administration overheads (10% of Total production cost)	635.00	7,620
	Total Cost	6,985.00	83,820
	Add: Profit (1/3rd of total cost)	2,328.33	27,940
(i)	**Sales price**	**9,313.33**	**1,11,760**
	No. of units in batch	50 units	
(ii)	**Cost per unit** (₹6,985 ÷ 50 units)	**139.70**	
	Selling price per unit (9,313.33 ÷ 50 units)	**186.27**	

(iii) If the order is for 605 cakes, then selling price per cake would be as below:

Particulars	Total Cost (₹)
Direct Material Cost	60,500
Direct Wages	6,050
Oven set-up cost	9,750
Add: Production Overheads (20% of Direct wages)	1,210
Total Production cost	77,510
Add: S&D and Administration overheads (10% of Total production cost)	7,751
Total Cost	85,261
Add: Profit (1/3rd of total cost)	28,420
Sales price	**1,13,681**
No. of units	605 units
Selling price per unit (₹ 1,13,681 ÷ 605 units)	**187.90**

ICAI · P4 · RTP Sep 2025 · question p. 16 · answer p. 41-42

594 RTP **Jan 2025** Q13

Phonick Ltd. accepted an order to supply 2,000 units per month of Product 'E' for the third quarter of the year. Each monthly batch order records the actual costs of materials and labour. Overheads are charged at a rate per labour hour. The selling price is established at ₹ 15 per unit.

Information relating to Material, Labour and Overheads is provided below:

Month	Batch Output (Numbers)	Material Cost (₹)	Labour Cost (₹)	Overheads (₹)	Total Labour Hours
October	2,500	12,500	5,000	24,000	8,000
November	3,000	18,000	6,000	18,000	9,000
December	2,000	10,000	4,000	30,000	10,000

Labour is paid at the rate of ₹ 2 per hour.

CALCULATE the cost and profit per unit of each batch order along with the overall position of the order for 6,000 units.

ICAI'S ANSWER

Statement of Cost and Profit per unit of each batch order

	October	November	December	Total
a) Batch Output (Nos.)	2,500	3,000	2,000	7,500
b) Sales Value (@ ₹ 15 per unit)	(₹) 37,500	(₹) 45,000	(₹) 30,000	(₹) 1,12,500
Cost				
Material	12,500	18,000	10,000	40,500
Wages	5,000	6,000	4,000	15,000
Overheads (working note)	7,500	6,000	6,000	19,500
c) Total	25,000	30,000	20,000	75,000
d) Profit per batch (b) – (c)	12,500	15,000	10,000	37,500
e) Cost per unit (c) ÷ (a)	**10**	**10**	**10**	
f) Profit per unit (d) ÷ (a)	**5**	**5**	**5**	

Overall Position of the Order for 6,000 Units

Particulars	Amount (₹)
Sales value (6,000 units × ₹ 15)	90,000
Less: Total cost (6,000 units × ₹ 10)	60,000
Profit	30,000

Working Note:

Calculation of overhead per hour

Particulars	October	November	December
i. Labour hours:			
$= \frac{\text{Labour cost}}{\text{Labour rates per hour}}$	$\frac{₹5,000}{2}$ = 2,500 hrs.	$\frac{₹6,000}{2}$ = 3,000 hrs.	$\frac{₹4,000}{2}$ = 2,000 hrs.
ii. Overhead per hour:			
$= \frac{\text{Total Overheads}}{\text{Totallabour hour}}$	$\frac{₹24,000}{8,000 \text{ hrs.}}$ = ₹ 3	$\frac{₹18,000}{9,000 \text{ hrs.}}$ = ₹ 2	$\frac{₹30,000}{10,000 \text{ hrs.}}$ = ₹ 3
iii. Overhead for the batch (i) × (ii)	₹ 7,500	₹ 6,000	₹ 6,000

ICAI · P4 · RTP Jan 2025 · question p. 13-14 · answer p. 35-36

595 RTP **Sep 2024** Q13

Same question, different ICAI answer: #624 Old P3 MTP Nov 2020 S-I

A jobbing factory has undertaken to supply 300 pieces of a component per month for the ensuing six months. Every month a batch order is opened against which materials and labour hours are booked at actual. Overheads are levied at a rate per labour hour. The selling price contracted for is ₹ 8 per piece. From the following data CALCULATE the cost and profit per piece of each batch order and overall position of the order for 1,800 pieces.

Month	Batch Output	Material cost (₹)	Direct wages (₹)	Direct labour hours
January	310	1150	120	240
February	300	1140	140	280
March	320	1180	150	280
April	280	1130	140	270
May	300	1200	150	300
June	320	1220	160	320

The other details are:

Month	Chargeable expenses (₹)	Direct labour (Hours)
January	12,000	4,800
February	10,560	4,400
March	12,000	5,000
April	10,580	4,600
May	13,000	5,000
June	12,000	4,800

ICAI'S ANSWER

Statement of Cost and Profit per batch

Particulars	Jan.	Feb.	March	April	May	June	Total
Batch output (in units)	310	300	320	280	300	320	1,830
Sale value (₹)	2,480	2,400	2,560	2,240	2,400	2,560	14,640
Material cost (₹)	1,150	1,140	1,180	1,130	1,200	1,220	7,020
Direct wages (₹)	120	140	150	140	150	160	860
Chargeable expenses* (₹)	600	672	672	621	780	800	4,145
Total cost (₹)	1,870	1,952	2,002	1,891	2,130	2,180	12,025
Profit per batch (₹)	610	448	558	349	270	380	2,615
Total cost per unit (₹)	6.03	6.51	6.26	6.75	7.10	6.81	6.57
Profit per unit (₹)	1.97	1.49	1.74	1.25	0.90	1.19	1.43

Overall position of the order for 1,800 units

Sales value of 1,800 units @ ₹ 8 per unit	₹ 14,400
Total cost of 1,800 units @ ₹ 6.57 per unit	₹ 11,826
Profit	₹ 2,574

$$* \frac{\text{Chargeable expenses}}{\text{Direct labour hour for the month}} \times \text{Direct labour hours for batch}$$

ICAI · P4 · RTP Sep 2024 · question p. 11 · answer p. 28

596 **RTP** **May 2024** Q7

Arnav Ltd. operates in beverages industry where it manufactures soft-drink in three sizes of Large (3 litres), Medium (1.5 litres) and Small (600 ml) bottles. The products are processed in batches. The 5,000 litres capacity processing plant consumes electricity of 90 Kilowatts per hour and a batch takes 1 hour 45

minutes to complete. Only symmetric size of products can be processed at a time. The machine set-up takes 15 minutes to get ready for next batch processing. During the set-up power consumption is only 20%.

(i) The current price of Large, Medium and Small are ₹ 150, ₹ 90 and ₹ 50 respectively.

(ii) To produce a litre of beverage, 14 litres of raw material-W and 25 ml of Material-C are required which costs ₹ 0.50 and ₹ 1,000 per litre respectively.

(iii) 20 direct workers are required. The workers are paid ₹ 880 for 8 hours shift of work.

(iv) The average packing cost per bottle is ₹ 3

(v) Power cost is ₹ 7 per Kilowatt -hour (Kwh)

(vi) Other variable cost is ₹ 30,000 per batch.

(vii) Fixed cost (Administration and marketing) is ₹ 4,90,00,000.

(viii) The holding cost is ₹ 1 per bottle per annum.

The marketing team has surveyed the following demand (bottle) of the product:

Large	Medium	Small
3,00,000	7,50,000	20,00,000

You are required to CALCULATE profit/ loss per batch and also COMPUTE Economic Batch Quantity (EBQ).

ICAI'S ANSWER

Workings:

1. Maximum number of bottles that can be processed in a batch: $= \frac{5{,}000 \text{ ltrs}}{\text{Bottle volume}}$

Large		Medium		Small	
Qty (ltr)	**Max bottles**	**Qty (ltr)**	**Max bottles**	**Qty (ltr)**	**Max bottles**
3	1,666	1.5	3,333	0.6	8,333

*For simplicity of calculation small fractions has been ignored.

2. Number of batches to be run:

		Large	Medium	Small	Total
A	Demand	3,00,000	7,50,000	20,00,000	
B	Bottles per batch (Refer WN-1)	1,666	3,333	8,333	
C	No. of batches [A÷B]	180	225	240	645

*For simplicity of calculation small fractions has been ignored.

Quantity of Material-W and Material C required to meet demand:

	Particulars	Large	Medium	Small	Total
A	Demand (bottle)	3,00,000	7,50,000	20,00,000	
B	Qty per bottle (Litre)	3	1.5	0.6	
C	Output (Litre) [A×B]	9,00,000	11,25,000	12,00,000	32,25,000
D	Material-W per litre of output (Litre)	14	14	14	
E	Material-W required (Litre) [C×D]	**1,26,00,000**	**1,57,50,000**	**1,68,00,000**	**4,51,50,000**
F	Material-C required per litre of output (ml)	25	25	25	
G	Material-C required (Litre) [(C×F)÷1000]	**22,500**	**28,125**	**30,000**	**80,625**

3. No. of Man-shift required:

		Large	Medium	Small	Total
A	No. of batches	180	225	240	645
B	Hours required per batch (Hours)	2	2	2	
C	Total hours required (Hours) [A×B]	360	450	480	1,290
D	No. of shifts required [C÷8]	45	57	60	162
E	**Total manshift [D×20 workers]**	**900**	**1,140**	**1,200**	**3,240**

4. Power consumption in Kwh

		Large	Medium	Small	Total
For processing					
A	No. of batches	180	225	240	645
B	Hours required per batch (Hours)	1.75	1.75	1.75	1.75
C	Total hours required (Hours) [A×B]	315	393.75	420	1,128.75
D	Power consumption per hour	90	90	90	90
E	**Power consumption in Kwh [C×D]**	**28,350**	**35,437.5**	**37,800**	**1,01,587.5**
F	**Per batch consumption (Kwh) [E÷A]**	**157.5**	**157.5**	**157.5**	**157.5**
For set-up					

		Large	Medium	Small	Total
G	Hours required per batch (Hours)	0.25	0.25	0.25	0.25
H	Total hours required (Hours) [A×G]	45	56.25	60	161.25
I	Power consumption per hour [20%×90]	18	18	18	18
J	**Power consumption in Kwh [H×I]**	**810**	**1,012.5**	**1,080**	**2,902.5**
K	**Per batch consumption (Kwh) [J÷A]**	**4.5**	**4.5**	**4.5**	**4.5**

Calculation of Profit/ loss per batch:

	Particulars	Large	Medium	Small	Total
A	Demand (bottle)	3,00,000	7,50,000	20,00,000	30,50,000
B	Price per bottle (₹)	150	90	50	
C	**Sales value (₹) [A×B]**	**4,50,00,000**	**6,75,00,000**	**10,00,00,000**	**21,25,00,000**
	Direct Material cost:				
E	Material-W (₹) [Qty in WN-3 × ₹ 0.50]	63,00,000	78,75,000	84,00,000	2,25,75,000
F	Material-C (₹) [Qty in WN-3 × ₹1,000]	2,25,00,000	2,81,25,000	3,00,00,000	8,06,25,000
G	[E+F]	**2,88,00,000**	**3,60,00,000**	**3,84,00,000**	**10,32,00,000**
H	Direct Wages (₹) [Man-shift in WN-4 × × ₹ 880]	7,92,000	10,03,200	10,56,000	28,51,200
I	Packing cost (₹) [A×₹3]	9,00,000	22,50,000	60,00,000	91,50,000
	Power cost (₹)				
J	For processing (₹) [WN-5 × ₹7]	1,98,450	2,48,062.5	2,64,600	7,11,112.5
K	For set-up time (₹) [WN-5 × ₹7]	5,670	7,087.5	7,560	20,317.5
L	[J+K]	**2,04,120**	**2,55,150**	**2,72,160**	**7,31,430**
M	Other variable cost (₹) [No. of batch in WN-2 × ₹ 30,000]	54,00,000	67,50,000	72,00,000	1,93,50,000
N	**Total Variable cost per batch [G+H+I+L+M]**	**3,60,96,120**	**4,62,58,350**	**5,29,28,160**	**13,52,82,630**
O	**Profit/ loss before fixed cost [C-N]**	**89,03,880**	**2,12,41,650**	**4,70,71,840**	**7,72,17,370**

	Particulars	Large	Medium	Small	Total
P	Fixed Cost				4,90,00,000
Q	**Total Cost [O-P]**				**2,82,17,370**

Computation of Economic Batch Quantity (EBQ):

$$\text{EBQ} = \sqrt{\frac{2 \times D \times S}{C}}$$

D = Annual Demand for the Product = Refer A below

S = Set-up cost per batch = Refer D below

C = Carrying cost per unit per annum =Refer E below

	Particulars	Large	Medium	Small
A	Annual Demand (bottle)	3,00,000	7,50,000	20,00,000
Set-up Cost:				
B	Power cost for set-up time (₹) [Consumption per batch in WN-5 × ₹7]	31.50	31.50	31.50
C	Other variable cost (₹) *	30,000	30,000	30,000
D	Total Set-up cost [B+C]	30,031.50	30,031.50	30,031.50
E	Holding cost:	1.00	1.00	1.00
F	**EBQ (Bottle)**	**1,34,234**	**2,12,243**	**3,46,592**

* Other variable cost is assumed to be part of set-up cost.

ICAI · P4 · RTP May 2024 · question p. 11 · answer p. 28-32

597 RTP **Old P3 May 2023** Q7 old scheme

A jobbing factory has undertaken to supply 200 pieces of a component per month for the ensuing six months. Every month a batch order is opened against which materials and labour hours are booked at actual. Overheads are levied at a rate per labour hour. The selling price contracted for is ₹ 80 per piece. From the following data.

COMPUTE the cost and profit per piece of each batch order and overall position of the order for 1,200 pieces.

Month	Batch Output (Pieces)	Material cost (₹)	Direct wages (₹)	Direct labour (Hours)
January	210	6,500	1,200	240
February	200	6,400	1,400	280
March	220	6,800	1,500	280
April	180	6,300	1,400	270
May	200	7,000	1,500	300
June	220	7,200	1,600	320

The other details are:

Month	Chargeable expenses (₹)	Direct labour Hours
January	1,20,000	4,800
February	1,05,600	4,400
March	1,20,000	5,000
April	1,05,800	4,600
May	1,30,000	5,000
June	1,20,000	4,800

ICAI'S ANSWER

Particulars	Jan. (₹)	Feb. (₹)	March (₹)	April (₹)	May (₹)	June (₹)	Total (₹)
Batch output (in pieces)	210	200	220	180	200	220	1,230
Sale value @ ₹80	16,800	16,000	17,600	14,400	16,000	17,600	98,400
Material cost	6,500	6,400	6,800	6,300	7,000	7,200	40,200
Direct wages	1,200	1,400	1,500	1,400	1,500	1,600	8,600
Chargeable expenses*	6,000	6,720	6,720	6,210	7,800	8,000	41,450
Total cost	13,700	14,520	15,020	13,910	16,300	16,800	90,250
Profit per batch	3,100	1,480	2,580	490	(300)	800	8,150
Total cost per piece	65.2	72.6	68.3	77.3	81.5	76.4	73.4
Profit per piece	14.8	7.4	11.7	2.7	(1.5)	3.6	6.6

Overall position of the order for 1,200 pieces

Sales value of 1,200 pieces @ ₹ 80 per piece	₹ 96,000
Total cost of 1,200 pieces @ ₹ 73.4 per piece	₹ 88,080
Profit	₹ 7,920

* $\frac{\text{Chargeable expenses}}{\text{Direct labour hour for the month}} \times$ Direct labour hours for batch

ICAI · Old P3 · RTP May 2023 · question p. 4-5 · answer p. 17

598 RTP **Old P3 May 2023** Q15(b) old scheme

DISCUSS the difference between Job costing and Batch costing.

ICAI'S ANSWER

Differences between Job costing and Batch costing:

Sr. No	Job Costing	Batch Costing
1.	Method of costing used for non- standard and non-repetitive products produced as per customer specifications and against specific orders.	Homogeneous products produced in a continuous production flow in lots.
2.	Cost determined for each Job.	Cost determined in aggregate for the entire Batch and then arrived at on per unit basis.
3.	Jobs are different from each other and independent of each other. Each Job is unique.	Products produced in a batch are homogeneous and lack of individuality.

ICAI · Old P3 · RTP May 2023 · question p. 11 · answer p. 26-27

599 RTP **Old P3 Nov 2022** Q7 old scheme

PS Ltd. manufactures articles in predetermined lots simultaneously. The following costs have been incurred for Batch No. 'PS143' in the month of March, 2022:

Units produced 1,000 units

Direct materials cost ₹ 2,00,000

Direct Labour -

Department A 800 labour hours @ ₹ 100 per hour.

Department B 1,400 labour hours @ ₹ 120 per hour.

Factory overheads are absorbed on labour hour basis and the rates are:

Department A @ ₹ 140 per hour.

Department B @ ₹ 80 per hour.

Administrative overheads are absorbed at 10% of selling price.

The firm expects 25% gross profit (sales value minus factory cost) for determining the selling price.

You are required to CALCULATE the selling price per unit of Batch No. 'PS143'.

ICAI'S ANSWER

Statement showing selling price per unit of Batch number 'PS143'

Particulars	Amount (₹)	Amount (₹)
Direct Materials		2,00,000
Direct Labour		
Department A 800 labour hours @ ₹100 per hour	80,000	
Department B 1400 labour hours @ ₹120 per hour	1,68,000	**2,48,000**
Factory overheads		
Department A 800 labour hours @ ₹140 per hour	1,12,000	
Department B 1400 labour hours @ ₹80 per hour	1,12,000	2,24,000
Factory Cost		**6,72,000**
Add: Administrative overheads (10% of selling price) (6,72,000/75% x 10%)		89,600
Cost of production		7,61,600
Add: Profit (15% of selling price) (6,72,000/75% x 15%)		1,34,400
Selling price of batch no 'PS143'		**8,96,000**
Selling price per unit (8,96,000 / 1000 units)		**896**

Alternatively, selling price calculation:- Selling price assume X

25% = (X – factory cost) / X

or 0.25 X = X- 6,72,000

or 0.75 X = 6,72,000

hence X = ₹ 8,96,000

ICAI · Old P3 · RTP Nov 2022 · question p. 7 · answer p. 25-26

600 RTP **Old P3 May 2022** Q7 old scheme

Brostom Ltd. manufactures 'Stent' that is used by hospitals in angioplasty, a procedure used to open blocked coronary arteries without open-heart surgery. As per the estimates provided by Pharmaceutical Industry Bureau, there will be a demand of 1 crore 'Stents' in the coming year. Brostom Ltd. is having a market share of 10% of the total market demand of the Stents. It is estimated that it costs ₹ 3.00 as inventory holding cost per stent per month and that the set-up cost per run of stent manufacture is ₹ 450.

Required:

(i) WHAT would be the optimum run size for Stent manufacture?

(ii) WHAT is the minimum inventory holding cost?

ICAI'S ANSWER

(i) Computation of Optimum Run size of 'Stents' or Economic Batch Quantity (EBQ)

Economic Batch Quantity (EBQ) = $\sqrt{\frac{2DS}{C}}$

Where, D = Annual demand for the Stents

= 1,00,00,000 × 10% = 10,00,000 units

S = Set- up cost per run

= ₹ 450

C = Carrying cost per unit per annum

= ₹ 3 × 12 = ₹ 36

EBQ = $\sqrt{\frac{2 \times 10{,}00{,}000 \times ₹\ 450}{₹\ 36}}$

= 5,000 units of Stents

(ii) Minimum inventory holding cost

Minimum Inventory Cost = Average Inventory × Inventory Carrying Cost per unit per annum

= (5,000 ÷ 2) × ₹ 36

= ₹ 90,000

(iii) Calculation of the extra cost due to manufacturing policy

	When run size is 6,000 units	When run size is 5,000 units i.e. at EBQ
Total set up cost	= 10,00,000/6,000 × ₹ 450 = ₹ 75,000	10,00,000/5,000 × ₹ 450 = ₹ 90,000
Total Carrying cost	½ × 6,000 × ₹ 36 = ₹ 1,08,000	½ × 5,000 × ₹ 36 = ₹ 90,000
Total Cost	**₹ 1,83,000**	**₹ 1,80,000**

Extra cost = ₹ 1,83,000 - ₹ 1,80,000 = **₹ 3,000**

ICAI · Old P3 · RTP May 2022 · question p. 6 · answer p. 22

601 RTP **Old P3 Dec 2021 Q7** old scheme

Rollon Ltd. is committed to supply 96,800 bearings per annum to Racing Ltd. on steady basis. It is estimated that it costs 25 paise as inventory carrying cost per bearing per month and the set-up cost per run of bearing manufacture is ₹ 588.

(a) COMPUTE what would be the optimum run size for bearing manufacture?

(b) Assuming that the company has a policy of manufacturing 8,800 bearings per run, CALCULATE how much extra costs the company would be incurring as compared to the optimum run suggested in (a) above?

ICAI'S ANSWER

(a) **Optimum production run size (Q)**

= $\sqrt{\frac{2DS}{C}}$ = $\sqrt{\frac{2 \times 96{,}800 \times ₹\ 588}{0.25 \times 12}}$ = 6,160 bearings.

(b) **Calculation of Extra Cost**

Total Cost (of maintaining the inventories) when production run size (Q) are 6,160 and 8,800 bearings respectively.

Total cost = Total set-up cost + Total carrying cost.

Particulars	When run size is 6,160 bearings	When run size is 8,800 bearings
Total set up cost	= 96,800/6,160 × ₹ 588 = ₹ 9,240 Or, No. of setups = 15.71 (16 setups) = 16 x ₹ 588 = ₹ 9,408	= 96,800/8,800 x ₹ 588 = ₹ 6,468
Total Carrying cost	½ × 6,160 × 0.25 × 12 = ₹ 9,240	½ × 8,800 × 0.25 × 12 = ₹ 13,200
Total Cost	**₹ 18,480/ ₹ 18,648**	**₹ 19,668**

₹ 1,188/ ₹ 1,020 is the extra cost incurred by the company due to run size not being optimum run size.

ICAI · Old P3 · RTP Dec 2021 · question p. 7 · answer p. 24

602 RTP **Old P3 Dec 2021** Q15(c) old scheme

DIFFERENTIATE between Job Costing and Batch Costing.

ICAI'S ANSWER

S. No.	Job Costing	Batch Costing
1	Method of costing used for non- standard and non-repetitive products produced as per customer specifications and against specific orders.	Homogeneous products produced in a continuous production flow in lots.
2	Cost determined for each Job.	Cost determined in aggregate for the entire Batch and then arrived at on per unit basis.
3	Jobs are different from each other and independent of each other. Each Job is unique.	Products produced in a batch are homogeneous and lack of individuality.

ICAI · Old P3 · RTP Dec 2021 · question p. 13 · answer p. 34

603 RTP **Old P3 Nov 2020** Q7 old scheme

A Ltd. manufactures mother boards used in smart phones. A smart phone requires one mother board. As per the study conducted by the Indian Cellular Association, there will be a demand of 180 million smart phones in the coming year. A Ltd. is expected to have a market share of 5.5% of the total market demand of the mother boards in the coming year. It is estimated that it costs ₹6.25 as inventory holding cost per board per month and that the set-up cost per run of board manufacture is ₹33,500.

(i) COMPUTE the optimum run size for board manufacturing?

(ii) Assuming that the company has a policy of manufacturing 80,000 boards per run, CALCULATE how much extra costs the company would be incurring as compared to the optimum run suggested in (i) above?

ICAI'S ANSWER

(i) Computation of optimum run size

Optimum run size or Economic Batch Quantity (EBQ) = $\sqrt{\frac{2 \times D \times S}{C}}$

Where, D = Annual demand i.e. 5.5% of 18,00,00,000 = 99,00,000 units

S = Set-up cost per run = ₹33,500

C = Inventory holding cost per unit per annum

= ₹6.25 × 12 months = ₹75

EBQ = $\sqrt{\frac{2 \times 99{,}00{,}000\text{units} \times ₹33{,}500}{₹\ 75}}$ = 94,042.5 units or 94,043 units

(ii) Calculation of Total Cost of set-up and inventory holding

	Batch size	No. of set-ups	Set-up Cost (₹)	Inventory holding cost (₹)	Total Cost (₹)
A	80,000 units	124 $\left(\frac{99{,}00{,}000}{80{,}000}\right)$	41,54,000 (124 × ₹33,500)	30,00,000 $\left(\frac{80{,}000 \times ₹75}{2}\right)$	71,54,000
B	94,043 units	106 $\left(\frac{99{,}00{,}000}{94{,}043}\right)$	35,51,000 (106 × ₹33,500)	35,26,612.5 $\left(\frac{94{,}043 \times ₹75}{2}\right)$	70,77,612.50
	Extra Cost (A – B)				76,387.50

ICAI · Old P3 · RTP Nov 2020 · question p. 5-6 · answer p. 19

604 **MTP** **Sep 2026 S-I** Q6(c) **4 marks**

Pinky Ltd. Produces product A in batches; management of Pinky Ltd. wants to know the number of batches of product A to be produced where the cost incurred on batch setup and carrying cost of production is at optimum level. EXPLAIN.

ICAI'S ANSWER

Economic batch quantity in Batch Costing: In batch costing the most important problem is the determination of 'Economic Batch Quantity'. The determination of economic batch quantity involves two types of costs viz, (i) set up cost and (ii) carrying cost. With the increase in the batch size, there is an increase in the carrying cost but the set up cost per unit of product is reduced. This situation is reversed when the batch size is reduced. Thus there is one particular batch size for which both set up and carrying costs are minimum. This size of a batch is known as economic or optimum batch quantity.

Economic batch quantity can be determined with the help of table, graph or mathematical formula. The mathematical formula usually used for its determination is as follows:

E.B.Q = $\sqrt{\frac{2DS}{C}}$

Where, D= Annual demand for the product

S = Setting up cost per batch

C = Carrying cost per unit of production per annum

ICAI · P4 · MTP Sep 2026 · Series I · question p. 15 · answer p. 21-22

605 MTP **May 2026 S-I** MCQ 15 2 marks MCQ

A company manufactures eye-glass frames at the rate of 1,200 frames per month. The company wants to determine the most economical production batch size to minimize total setup and holding costs.

The following information is available:

- Setup cost per batch = ₹ 7,200
- Cost per unit = ₹ 3,240
- Carrying cost = 10% per annum (including storage, cost of capital, and obsolescence)

What is the Economic Batch Quantity (EBQ)?

A 230 frames

B 566 frames

C 800 frames

D 1,200 frames

ICAI'S ANSWER

(C) 800 frames

EBQ For frames manufacturing

$$\text{EBQ} = \sqrt{\frac{2DS}{C}}$$

Where, D = Annual demand for the product = 1,200 X 12 = 14,400 frames

S = Setting up cost per batch = ₹ 7,200

C = Carrying cost per unit of production = 10% of ₹ 3,240 = ₹ 324

$$\text{EBQ} = \sqrt{\frac{2 \times 14{,}400 \times 7{,}200}{324}} = 800 \text{ frames}$$

ICAI · P4 · MTP May 2026 · Series I · question p. 7 · answer p. 6

606 MTP **May 2026 S-I** Q6(c) 4 marks

ABC Manufacturing produces a specialized fastener in batches to meet an annual demand of 24,000 units. The accountant is considering whether to increase the batch size. Currently, the setup cost per batch is significant, but the warehouse manager is concerned about rising inventory holding costs if larger batches are produced. After a meeting, the management asks the Cost Accountant to advise on the best approach.

(a) EXPLAIN, with reasons, how batch size affects both setup costs and inventory holding costs.

(b) Briefly DESCRIBE how the company can determine the Economic Batch Quantity (EBQ) and its importance in cost control.

ICAI'S ANSWER

(i) Explanation of the impact of batch size on setup and holding costs: If the size is higher, the set up cost may decline due to lesser number of set ups required; but units in inventory will go up leading to higher holding costs. If the lot size is lower, lower inventory holding costs are accomplished but only with higher set up costs.

(ii) Determination and importance of EBQ: Economic batch quantity is the size of a batch where total cost of set-up and holding costs are at minimum. This relationship is such that with the higher batch size, holding cost shows a tendency to increase whereas set-up costs show a declining trend. The point where both the cost lines intersect each other represents the lowest cost combination. The economic batch size or Economic Batch Quantity may be determined by calculating the total cost for a series of possible batch sizes and checking which batch size gives the minimum cost. Alternatively, a formula can be derived which is similar to determination of Economic Order Quantity (EOQ). The objective here being to determine the production lot (batch size) that optimizes on both set up and inventory holding costs.

ICAI · P4 · MTP May 2026 · Series I · question p. 15 · answer p. 20

607 MTP May 2026 S-II MCQ 14 2 marks MCQ

Also asked in MTP Sep 2025 S-I MCQ 14

A company manufactures a product with a monthly demand of 2,000 units. Each unit costs ₹ 50 to produce, and the setup cost per batch is ₹ 3,000. The company incurs a storage cost of ₹ 2 per unit per month and applies an annual interest rate of 18% on capital tied up in inventory. The finished goods are stored for an average of 2 months before being sold.

Considering both storage cost and opportunity cost based on the average inventory holding period, what is the Economic Batch Quantity (EBQ)?

A 2,376 units

B 1,095 units

C 1,200 units

D 1,400 units

ICAI'S ANSWER

(A) 2,376 units

$$\text{EBQ} = \sqrt{\frac{2DS}{C}}$$

Where, D = Annual demand for the product = 2,000 × 12 = 24,000 units

S = Setting up cost per batch = ₹ 3,000

C = Carrying cost per unit of production = ₹ 24 (storage) + ₹ 1.50 (interest) = ₹ 25.50

Storage cost per year = ₹ 2 × 12 = ₹ 24 per unit per year

Opportunity cost = 18% of ₹ 50 = ₹ 9 per unit per year

Since the average inventory is held for 2 months, the effective interest cost is:

= 9 x (2/12) = ₹ 1.5 per unit per year

$$\text{EBQ} = \sqrt{\frac{2 \text{ x } 24{,}000 \text{ x } 3{,}000}{25.50}} = 2{,}376 \text{ units}$$

ICAI · P4 · MTP May 2026 · Series II · question p. 6 · answer p. 5

608 MTP May 2026 S-II Q6(b) 5 marks

DESCRIBE Unit Costing and Batch Costing. GIVE three examples of industries for each method where these are used. Also, EXPLAIN the costing procedure in batch costing.

ICAI'S ANSWER

Unit costing is that method of costing where the output produced is identical and each unit of output requires identical cost. Under this method costs, are collected and analysed element wise and then total cost per unit is ascertained by dividing the total cost with the number of units produced.

Such a method of costing is used in the industries like paper, cement, steel works, mining, breweries etc.

Batch Costing: Batch Costing is a type of specific order costing where articles are manufactured in predetermined lots, known as batch. Cost per unit in a batch is ascertained by dividing the total cost of a batch by the number of units produced in that batch.

Such a method of costing is used in the case of pharmaceutical or drug industries, readymade garment industries, industries, manufacturing electronic parts of T.V. radio sets etc.

Costing Procedure in Batch Costing: To facilitate convenient cost determination, one number is allotted for each batch. Material cost for the batch is arrived at on the basis of material requisitions for the batch and labour cost is arrived at by multiplying the time spent on the batch by direct workers as ascertained from time cards or job tickets. Overheads are absorbed on some suitable basis like machine hours, direct labour hours etc.

ICAI · P4 · MTP May 2026 · Series II · question p. 16 · answer p. 20-21

609 MTP **Jan 2026 S-I** MCQ 14 2 marks MCQ

Consider the following information:

Market demand per month	: 600 units
Setting-up cost per batch	: ₹ 120
Cost of manufacture per unit	: ₹ 30
Rate of interest	: 10% p.a.

What is the EBQ?

A 758.95 units

B 848.53 units

C 268.33 units

D 1,200 units

ICAI'S ANSWER

(A) 758.95 units

$$\text{EBQ} = \sqrt{\frac{2DS}{C}} = \sqrt{\frac{2 \times 12 \times 600 \times 120}{30 \times 0.10}} = 758.95 \text{ units.}$$

ICAI · P4 · MTP Jan 2026 · Series I · question p. 6 · answer p. 4

610 MTP **Sep 2025 S-II** Q2(b) 2 marks

A juice company receives an annual order for 2,00,000 liters from a single customer. The order includes various flavors, each produced in a continuous run, with machinery reconfigured between flavors. Finished batches are stored and delivered in a single shipment before production of the next flavor begins.

The setup cost is ₹ 600 for each setup, and the required services are provided by an external company as needed.

Holding costs for rented storage are ₹ 600 per square meter annually, with each square meter holding 500 liters of juice.

You are required to CALCULATE the batch size which will minimise the total cost.

ICAI'S ANSWER

Economic Batch Quantity (EBQ):

$$EBQ = \sqrt{\frac{2DS}{C}}$$

Where, D = Annual demand for the product = 2,00,000 liters

S = Setting up cost per batch = ₹ 600 per set-up

C = Carrying cost per unit of production

$$= \frac{₹\ 600}{500\ \text{liters}} = 1.20 \text{ per liter per annum}$$

Therefore, $EBQ = \sqrt{\frac{2x2,00,000x600}{1.20}}$

= 14,142 Liters

ICAI · P4 · MTP Sep 2025 · Series II · question p. 11 · answer p. 14

611 MTP **Jan 2025 S-I** Q2(b) 2 marks

Luxz Ltd. is into luxury pens business manufacturing 120 pens in a batch. To process a single batch of 120 pens, company needs to incur following expenditure:

Particulars	(₹)
Direct Materials	57,375
Direct wages	6,750
Batch Set-up cost	18,900

For each batch, the company absorbs the Production Overheads at a rate of 20% of direct wages and 15% of the total production cost is allocated to cover selling, distribution, and administrative overheads.

During the month of March, Luxz Ltd. received an order for 2,400 pens and the company aims to achieve a profit margin of 25% on its sales value.

You are required to DETERMINE the total sales value for 2,400 pens.

ICAI'S ANSWER

Determination of total sales value of Luxury pens

Particulars	Amount per Batch (₹)	Amount for 2,400 units or 20 batches (₹)
Direct materials	57,375	11,47,500
Direct wages	6,750	1,35,000
Batch set-up cost	18,900	3,78,000

Particulars	Amount per Batch (₹)	Amount for 2,400 units or 20 batches (₹)
Production overheads (20% of direct wages)	1,350	27,000
Total Production Cost	84,375	**16,87,500**
Selling, distribution and administration cost (15% of Total Production cost)	12,656	2,53,125
Total Cost	97,031	**19,40,625**
Add: Profit (25% of Sales value or 1/3rd of Total cost)	32,344	6,46,875
Total Sales value	**1,29,375**	**25,87,500**

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612 MTP **Jan 2025 S-II** MCQ 14 2 marks MCQ

Due to technical and economical reasons, F8 Ltd. manufactures in batch. The latest contract requires the company to supply 9,000 bushings per month to G4 Ltd. The company has estimated that each set up for manufacturing the bushings will cost ₹ 16,002.25 and the inventory holding cost per bushing per annum will come to ₹ 60.

HOW many runs the company need to make throughout the year to complete the demand?

A 5 runs

B 10 runs

C 15 runs

D 20 runs

ICAI'S ANSWER

(C) Annual demand = 9,000 x 12 = 1,08,000

Economic Batch Quantity (EBQ):

$$\text{EBQ} = \sqrt{\frac{2DS}{C}}$$

$$= \sqrt{\frac{2 \text{ x } 1{,}08{,}000 \text{ x } 16{,}002.25}{60}}$$

= 7,590 bushings

Number of runs $= \frac{1{,}08{,}000}{7{,}590} = 14.23 =$ **15 runs**

ICAI · P4 · MTP Jan 2025 · Series II · question p. 6 · answer p. 6

613 MTP **Sep 2024 S-I** MCQ 11 2 marks MCQ

A FMCG company has an annual demand of 50,000 units for its specific product whose setting up cost per batch is ₹ 10,000 and carrying cost per unit per month is ₹ 1. What is the Economic Batch Quantity?

A 7,071 units

B 10,000 units

C 12,641 units

D 9,129 units

ICAI'S ANSWER

(D) 9,129 units

Annual demand (D) = 50,000 units

Setup cost per batch (S) = ₹ 10,000

Carrying cost per unit per month (C) = ₹ 1

EBQ = $\sqrt{\frac{2 \times D \times S}{C}}$

= 9,129 units

ICAI · P4 · MTP Sep 2024 · Series I · question p. 4-5 · answer p. 4

614 MTP Sep 2024 S-II MCQ 15 2 marks MCQ

A company uses batch costing and incurs a setup cost of ₹ 20,000 for a batch of 300 units. If direct materials cost ₹ 20 per unit and direct labor costs ₹ 10 per unit, what is the total cost of the batch?

A ₹ 25,000

B ₹ 29,000

C ₹ 32,000

D ₹ 7,000

ICAI'S ANSWER

(B)

Total cost ₹ 20,000 + (300 units × (₹ 20 + ₹10)) = ₹ 29,000

ICAI · P4 · MTP Sep 2024 · Series II · question p. 5 · answer p. 5

615 MTP Sep 2024 S-II Q1(c) 4 marks

Alpha Ltd. has an Annual demand from a single customer for 60,000 Covid-19 vaccines. The customer prefers to order in the lot of 15,000 vaccines per order. The production cost of vaccine is ₹ 5,000 per vaccine. The set-up cost per production run of Covid-19 vaccines is ₹ 4,800. The carrying cost is ₹ 12 per vaccine per month.

You are required to:

(i) FIND the most Economical Production Run.

(ii) CALCULATE the extra cost that company incurs due to production of 15,000 vaccines in a batch.

ICAI'S ANSWER

(i) Calculation of most Economical Production Run

= $\sqrt{\frac{2 \times 60{,}000 \times ₹\ 4{,}800}{12 \times 12}}$ = 2,000 Vaccine

(ii) Calculation of Extra Cost due to processing of 15,000 vaccines in a batch

	When run size is 2,000 vaccines	When run size is 15,000 vaccines
Total set up cost	$= \frac{60,000}{2,000} \times$ ₹ 4,800 = ₹ 1,44,000	$= \frac{60,000}{15,000} \times$ ₹ 4,800 = ₹ 19,200
Total Carrying cost	½ × 2,000 × ₹ 144 = ₹ 1,44,000	½ × 15,000 × ₹ 144 = ₹ 10,80,000
Total Cost	₹ 2,88,000	₹ 10,99,200

Thus, extra cost = ₹ 10,99,200 – ₹ 2,88,000 = ₹ 8,11,200

ICAI · P4 · MTP Sep 2024 · Series II · question p. 6 · answer p. 7

616 MTP **May 2024 S-I** MCQ 7 2 marks MCQ

A customer has been ordering 80,000 caps during the year. It is estimated that it costs ₹ 1 as inventory holding cost per cap per month and that the set up cost per run of cap manufacture is ₹ 3,500

What is optimum run size of cap manufacture?

A 12 runs

B 10 runs

C 15 runs

D 7 runs

ICAI'S ANSWER

Answer key: **(A)**

A Optimum batch size or Economic Batch Quantity (EBQ):

$$\text{EBQ} = \sqrt{\frac{2DS}{C}} = \sqrt{\frac{2 \times 80,000 \times 3,500}{12}} = 6,832 \text{ units.}$$

Number of Optimum runs = 80,000 ÷ 6,832 = 11.70 or 12 run

ICAI · P4 · MTP May 2024 · Series I · question p. 5 · answer p. 4

617 MTP **May 2024 S-II** Q2(b) 4 marks

Happi Ltd. Produces product RP in batches, management of the Happi Ltd. wants to know the number of batches of product RP to be produced where the cost incurred on batch setup and carrying cost of production is at optimum level.

ICAI'S ANSWER

Economic batch quantity in Batch Costing: In batch costing the most important problem is the determination of 'Economic Batch Quantity'. The determination of economic batch quantity involves two types of costs viz, (i) set up cost and (ii) carrying cost. With the increase in the batch size, there is an increase in the carrying cost but the set up cost per unit of product is reduced. This situation is reversed when the batch size is reduced. Thus there is one particular batch size for which both set up and carrying costs are minimum. This size of a batch is known as economic or optimum batch quantity.

Economic batch quantity can be determined with the help of table, graph or mathematical formula. The mathematical formula usually used for its determination is as follows:

$$\text{E.B.Q} = \sqrt{\frac{2DS}{C}}$$

Where, D= Annual demand for the product

S = Setting up cost per batch

C = Carrying cost per unit of production per annum

ICAI · P4 · MTP May 2024 · Series II · question p. 7 · answer p. 8

618 MTP **Old P3 Nov 2023 S-I** Q1(b) 5 marks old scheme

AT Ltd. manufactures machine parts used in industrial plants. As per market research it is expected that the annual demand for the parts will be 9,20,000 units. It is estimated that it costs ₹1.50 as inventory holding cost per unit per month and that the set-up cost per run is ₹ 3,500.

(i) DETERMINE the optimum run size for parts manufacturing?

(ii) Assuming that the company has a policy of manufacturing 40,000 parts per run, CALCULATE how much extra costs the company would be incurring as compared to the optimum run suggested in (i) above?

ICAI'S ANSWER

(i) Optimum run size or Economic Batch Quantity (EBQ) = $\sqrt{\frac{2 \times D \times S}{C}}$

Where, D = Annual demand = 9,20,000 units

S = Set-up cost per run = ₹ 3,500

C = Inventory holding cost per unit per annum

= ₹ 1.5 × 12 months = ₹ 18

EBQ = $\sqrt{\frac{2 \times 9{,}20{,}000 \text{ units} \times \text{Rs.}3{,}500}{\text{Rs.}18}}$ = 18,915 units

(ii) Calculation of Total Cost of set-up and inventory holding

	Batch size	No. of set-ups	Set-up Cost (₹)	Inventory holding cost (₹)	Total Cost (₹)
A	40,000 units	23 (9,20,000/40,000)	80,500 (23 × ₹ 3,500)	3,60,000 (40,000 × ₹ 18/2)	4,40,500
B	18,915 units	49 (9,20,000/18,915)	1,71,500 (49 × ₹ 3,500)	1,70,235 (18,915 × ₹ 18/2)	3,41,735
	Extra Cost (A – B)				98,765

ICAI · Old P3 · MTP Nov 2023 · Series I · question p. 1 · answer p. 1-2

619 MTP **Old P3 May 2023 S-I** Q1(c) 5 marks old scheme

Mithi Treat (MT) owns a confectionary store which sells items like sweets, cake, chocolates. MT use to produce at most 40 units of any item at a time. It has received an order for 800 chocolates from a customer. To process a batch of 40 chocolates, the following cost would be incurred:

Direct materials- ₹ 600

Direct wages- ₹ 55

Oven set- up cost ₹ 175

MT absorbs production overheads at a rate of 25% of direct wages cost. 15% is added to the total production cost of each batch to allow for selling, distribution and administration overheads.

MT requires a profit margin of 25% of cost.

DETERMINE the selling price for 800 Chocolates.

ICAI'S ANSWER

Statement of cost per batch and per order

No. of batch = 800 units ÷ 40 units = 20 batches

Particulars	*Cost per batch (₹)*	*Total Cost (₹)*
Direct Material Cost	600	12,000
Direct Wages	55	1100
Oven set-up cost	175	3500
Add: Production Overheads (25% of Direct wages)	13.75	275
Total Production cost	843.75	16875
Add: S&D and Administration overheads	126.56	2531.25
Total Cost	970.31	19406.25
Add: Profit (25% of total cost)	242.58	4851.56
Selling price	1,212.89	24,257.81
Selling Price per unit = 1,212.89÷40[0r 24,257.81÷800]	30.32	30.32

ICAI · Old P3 · MTP May 2023 · Series I · question p. 1 · answer p. 2

620 MTP Old P3 May 2022 S-I Q6(c) 5 marks old scheme

In Batch Costing, STATE how is Economic Batch Quantity determined?

ICAI'S ANSWER

The economic batch size or Economic Batch Quantity may be determined by calculating the total cost for a series of possible batch sizes and checking which batch size gives the minimum cost. The objective here being to determine the production lot (Batch size) that optimizes on both set up and inventory holding cots formula. The mathematical formula usually used for its determination is as follows:

$$\text{EBQ} = \sqrt{\frac{2DS}{C}}$$

Where,

D= Annual demand for the product

S =Setting up cost per batch

C=Carrying cost per unit of production

ICAI · Old P3 · MTP May 2022 · Series I · question p. 7 · answer p. 11

621 MTP Old P3 May 2022 S-II Q4(a) 10 marks old scheme

Arnav Ltd. operates in beverages industry where it manufactures soft-drink in three sizes of Large (3 litres), Medium (1.5 litres) and Small (600 ml) bottles. The products are processed in batches. The 5,000 litres capacity processing plant consumes electricity of 90 Kilowatts per hour and a batch takes 1 hour 45 minutes to complete. Only symmetric size of products can be processed at a time. The machine set-up takes 15 minutes to get ready for next batch processing. During the set-up, power consumption is only 20%.

(I) The current price of Large, Medium and Small are ₹ 150, ₹ 90 and ₹ 50 respectively.

(II) To produce a litre of beverage, 14 litres of raw material-W and 25 ml of Material-C are required which costs ₹ 0.50 and ₹1,000 per litre respectively.

(III) 20 direct workers are required. The workers are paid ₹ 880 for 8 hours shift of work.

(IV) The average packing cost per bottle is ₹3

(V) Power cost is ₹ 7 per Kilowatt -hour (Kwh)

(VI) Other variable cost is ₹ 30,000 per batch.

(VII) Fixed cost (Administration and marketing) is ₹ 4,90,00,000.

(VIII) The holding cost is ₹ 1 per bottle per annum.

The marketing team has surveyed the following demand (bottle) of products:

Large	Medium	Small
3,00,000	7,50,000	20,00,000

Required:

CALCULATE net profit/ loss of the organisation and also COMPUTE Economic Batch Quantity (EBQ).

ICAI'S ANSWER

Workings:

1. Maximum number of bottles that can be processed in a batch:

$$= \frac{5{,}000 \text{ ltrs}}{\text{Bottle volume}}$$

Large		Medium		Small	
Qty (ltr)	**Max bottles**	**Qty (ltr)**	**Max bottles**	**Qty (ml)**	**Max bottles**
3	1,666	1.5	3,333	600	8,333

For simplicity of calculation small fractions has been ignored.

2. Number of batches to be run:

		Large	Medium	Small	Total
A	Demand	3,00,000	7,50,000	20,00,000	
B	Bottles per batch (Refer WN-1)	1,666	3,333	8,333	
C	No. of batches [A÷B]	180	225	240	645

For simplicity of calculation small fractions has been ignored.

3. Quantity of Material-W and Material C required to meet demand:

	Particulars	Large	Medium	Small	Total
A	Demand (bottle)	3,00,000	7,50,000	20,00,000	
B	Qty per bottle (Litre)	3	1.5	0.6	
C	Output (Litre) [A×B]	9,00,000	11,25,000	12,00,000	32,25,000
D	Material-W per litre of output (Litre)	14	14	14	
E	Material-W required (Litre) [C×D]	**1,26,00,000**	**1,57,50,000**	**1,68,00,000**	**4,51,50,000**
F	Material-C required per litre of output (ml)	25	25	25	
G	Material-C required (Litre) [(C×F)÷1000]	**22,500**	**28,125**	**30,000**	**80,625**

4. No. of Man-shift required:

		Large	Medium	Small	Total
A	No. of batches	180	225	240	645
B	Hours required per batch (Hours)	2	2	2	
C	Total hours required (Hours) [A×B]	360	450	480	1,290
D	No. of shifts required [C÷8]	45	57	60	162
E	**Total manshift [D×20 workers]**	**900**	**1,140**	**1,200**	**3,240**

For simplicity of calculation small fractions has been ignored.

5. Power consumption in Kwh

		Large	Medium	Small	Total
For processing					
A	No. of batches	180	225	240	645
B	Hours required per batch (Hours)	1.75	1.75	1.75	1.75
C	Total hours required (Hours) [A×B]	315	393.75	420	1,128.75
D	Power consumption per hour (Kwh)	90	90	90	90
E	**Total Power consumption (Kwh) [C×D]**	**28,350**	**35,437.5**	**37,800**	**1,01,587**
F	**Per batch consumption* (Kwh) [E÷A]**	**157.5**	**157.5**	**157.5**	**157.5**

		Large	Medium	Small	Total
For set-up					
G	Hours required per batch (Hours)	0.25	0.25	0.25	0.25
H	Total hours required (Hours) [A×G]	45	56.25	60	161.25
I	Power consumption per hour (Kwh) [20%×90]	18	18	18	18
J	**Total Power consumption (Kwh) [H×I]**	**810**	**1,012.5**	**1,080**	**2,902.5**
K	**Per batch consumption* (Kwh) [J÷A]**	**4.5**	**4.5**	**4.5**	**4.5**

* Per batch consumption can be directly calculated as [Hours required per batch x Power consumption per hour]

Calculation of Profit/ loss per batch:

	Particulars	Large	Medium	Small	Total
A	Demand (bottle)	3,00,000	7,50,000	20,00,000	30,50,000
B	Price per bottle (₹)	150	90	50	
C	**Sales value (₹) [A×B]**	**4,50,00,000**	**6,75,00,000**	**10,00,00,000**	**21,25,00,000**
	Direct Material cost:				
E	Material-W (₹) [Qty in WN-3 × ₹0.50]	63,00,000	78,75,000	84,00,000	2,25,75,000
F	Material-C (₹) [Qty in WN-3 × ₹1,000]	2,25,00,000	2,81,25,000	3,00,00,000	8,06,25,000
G	[E+F]	**2,88,00,000**	**3,60,00,000**	**3,84,00,000**	**10,32,00,000**
H	Direct Wages (₹) [Man-shift in WN-4 × × ₹880]	7,92,000	10,03,200	10,56,000	28,51,200
I	Packing cost (₹) [A×₹3]	9,00,000	22,50,000	60,00,000	91,50,000
	Power cost (₹)				
J	For processing (₹) [WN-5 × ₹7]	1,98,450	2,48,062.5	2,64,600	7,11,112.5
K	For set-up time (₹) [WN-5 × ₹7]	5,670	7,087.5	7,560	20,317.5
L	[J+K]	**2,04,120**	**2,55,150**	**2,72,160**	**7,31,430**
M	Other variable cost (₹) [No. of batch in WN-2 × ₹30,000]	54,00,000	67,50,000	72,00,000	1,93,50,000

	Particulars	Large	Medium	Small	Total
N	**Total Variable cost per batch [G+H+I+L+M]**	**3,60,96,120**	**4,62,58,350**	**5,29,28,160**	**13,52,82,630**
O	**Profit/ loss before fixed cost [C-N]**	**89,03,880**	**2,12,41,650**	**4,70,71,840**	**7,72,17,370**
P	Fixed Cost				4,90,00,000
Q	**Net Profit [O-P]**				**2,82,17,370**

Computation of Economic Batch Quantity (EBQ):

$$EBQ = \sqrt{\frac{2 \times D \times S}{C}}$$

D = Annual Demand for the Product = Refer A below

S = Set-up cost per batch = Refer D below

C = Carrying cost per unit per annum =Refer E below

	Particulars	Large	Medium	Small
A	**Annual Demand (bottle)**	**3,00,000**	**7,50,000**	**20,00,000**
B	Power cost for set-up time (₹) [Consumption per batch in WN-5 × ₹7]	31.50	31.50	31.50
C	Other variable cost (₹)	30,000	30,000	30,000
D	**Total Set-up cost [B+C]**	**30,031.50**	**30,031.50**	**30,031.50**
E	**Holding cost:**	**1.00**	**1.00**	**1.00**
F	**EBQ (Bottle)**	**1,34,234**	**2,12,243**	**3,46,592**

ICAI · Old P3 · MTP May 2022 · Series II · question p. 5 · answer p. 7-9

622 MTP **Old P3 Dec 2021 S-I** Q1(b) 5 marks old scheme

Zee Ltd. manufactures pistons used in car engines. As per the study conducted by the Auto Parts Manufacturers Association, there will be a demand of 80 million pistons in the coming year. A Ltd. is expected to have a market share of 2.15% of the total market demand of the pistons in the coming year. It is estimated that it costs ₹ 2.50 as inventory holding cost per piston per month and that the set-up cost per run of piston manufacture is ₹ 4,500.

(i) COMPUTE the optimum run size for piston manufacturing?

(ii) Assuming that the company has a policy of manufacturing 20,000 pistons per run, CALCULATE how much extra costs the company would be incurring as compared to the optimum run suggested in (i) above?

ICAI'S ANSWER

(i) Optimum run size or Economic Batch Quantity (EBQ) = $\sqrt{\frac{2 \times D \times S}{C}}$

Where, D = Annual demand i.e. 2.15% of 8,00,00,000 = 17,20,000 units

S = Set-up cost per run = ₹ 4,500

C = Inventory holding cost per unit per annum

= ₹ 2.5 × 12 months = ₹ 30

$$\text{EBQ} = \sqrt{\frac{2 \times 17{,}20{,}000 \text{ units} \times ₹\ 4{,}500}{₹\ 30}} = 22{,}716 \text{ units}$$

(ii) Calculation of Total Cost of set-up and inventory holding

	Batch size	No. of set-ups	Set-up Cost (₹)	Inventory holding cost (₹)	Total Cost (₹)
A	20,000 units	86 (17,20,000/20,000)	3,87,000 (86 × ₹ 4,500)	3,00,000 (20,000 × ₹ 30/2)	6,87,000
B	22,716 units	76 (17,20,000/22,716)	3,42,000 (76 × ₹ 4,500)	3,40,740 (22,716 × ₹ 30/2)	6,82,740
	Extra Cost (A – B)				4,260

ICAI · Old P3 · MTP Dec 2021 · Series I · question p. 1 · answer p. 2

623 MTP **Old P3 Jul 2021 S-I** Q5(b)(ii) 5 marks old scheme

A jobbing factory has undertaken to supply 200 pieces of a component per month for the ensuing six months. Every month a batch order is opened against which materials and labour hours are booked at actual. Overheads are levied at a rate per labour hour. The selling price contracted for is Rs. 80 per piece. From the following data COMPUTE the cost and profit per piece of each batch order and overall position of the order for 1,200 pieces.

Month	Batch Output (Pieces)	Material cost (Rs.)	Direct wages (Rs.)	Direct labour (Hours)
January	210	6,500	1,200	240
February	200	6,400	1,400	280
March	220	6,800	1,500	280
April	180	6,300	1,400	270
May	200	7,000	1,500	300
June	220	7,200	1,600	320

The other details are:

Month	Chargeable expenses (Rs.)	Direct labour hours
January	1,20,000	4,800
February	1,05,600	4,400
March	1,20,000	5,000
April	1,05,800	4,600
May	1,30,000	5,000
June	1,20,000	4,800

ICAI'S ANSWER

Particulars	Jan. (Rs.)	Feb. (Rs.)	March (Rs.)	April (Rs.)	May (Rs.)	June (Rs.)	Total (Rs.)
Batch output (in pieces)	210	200	220	180	200	220	1,230
Sale value @ Rs.80	16,800	16,000	17,600	14,400	16,000	17,600	98,400
Material cost	6,500	6,400	6,800	6,300	7,000	7,200	40,200
Direct wages	1,200	1,400	1,500	1,400	1,500	1,600	8,600
Chargeable expenses*	6,000	6,720	6,720	6,210	7,800	8,000	41,450
Total cost	13,700	14,520	15,020	13,910	16,300	16,800	90,250
Profit per batch	3,100	1,480	2,580	490	(300)	800	8,150
Total cost per piece	65.2	72.6	68.3	77.3	81.5	76.4	73.4
Profit per piece	14.8	7.4	11.7	2.7	(1.5)	3.6	6.6

Overall position of the order for 1,200 pieces

Sales value of 1,200 pieces @ Rs. 80 per piece	Rs. 96,000
Total cost of 1,200 pieces @ Rs. 73.4 per piece	Rs. 88,080
Profit	Rs. 7,920

$$* \frac{\text{Chargeable expenses}}{\text{Direct labour hour for the month}} \times \text{Direct labour hours for batch}$$

ICAI · Old P3 · MTP Jul 2021 · Series I · question p. 7-8 · answer p. 13

624 MTP **Old P3 Nov 2020 S-I** Q1(a) 5 marks old scheme

Same question, different ICAI answer: #595 RTP Sep 2024

A jobbing factory has undertaken to supply 300 pieces of a component per month for the ensuing six months. Every month a batch order is opened against which materials and labour hours are booked at actual. Overheads are levied at a rate per labour hour. The selling price contracted for is ₹ 8 per piece. From the following data CALCULATE the cost and profit per piece of each batch order and overall position of the order for 1,800 pieces.

Month	Batch Output	Material cost (₹)	Direct wages (₹)	Direct labour hours
January	310	1150	120	240
February	300	1140	140	280
March	320	1180	150	280
April	280	1130	140	270
May	300	1200	150	300
June	320	1220	160	320

The other details are:

Month	Chargeable expenses (₹)	Direct labour (Hours)
January	12,000	4,800
February	10,560	4,400
March	12,000	5,000
April	10,580	4,600
May	13,000	5,000
June	12,000	4,800

ICAI'S ANSWER

Statement of Cost and Profit per batch

Particulars	Jan.	Feb.	March	April	May	June	Total
Batch output (in units)	310	300	320	280	300	320	1,830
Sale value (₹)	2,480	2,400	2,560	2,240	2,400	2,560	14,640
Material cost (₹)	1,150	1,140	1,180	1,130	1,200	1,220	7,020
Direct wages (₹)	120	140	150	140	150	160	860
Chargeable expenses* (₹)	600	672	672	621	780	800	4,145
Total cost (₹)	1,870	1,952	2,002	1,891	2,130	2,180	12,025
Profit per batch (₹)	610	448	558	349	270	380	2,615

Particulars	Jan.	Feb.	March	April	May	June	Total
Total cost per unit (₹)	6.03	6.51	6.26	6.75	7.10	6.81	6.57
Profit per unit (₹)	1.97	1.49	1.74	1.25	0.90	1.19	1.43

Overall position of the order for 1,200 units

Sales value of 1,800 units @ ₹ 8 per unit	₹ 14,400
Total cost of 1,800 units @ ₹ 6.57 per unit	₹ 11,826
Profit	₹ 2,574

* $\frac{\text{Chargeable expenses}}{\text{Direct labour hour for the month}} \times \text{Direct labour hours for batch}$

ICAI · Old P3 · MTP Nov 2020 · Series I · question p. 1 · answer p. 1

625 MTP **Old P3 May 2020 S-I** Q6(c) 5 marks old scheme

STATE the differences between Job costing and Batch costing.

ICAI'S ANSWER

Differences between Job costing and Batch costing:

Sr. No	Job Costing	Batch Costing
1	Method of costing used for non- standard and non-repetitive products produced as per customer specifications and against specific orders.	Homogeneous products produced in a continuous production flow in lots.
2	Cost determined for each Job.	Cost determined in aggregate for the entire Batch and then arrived at on per unit basis.
3	Jobs are different from each other and independent of each other. Each Job is unique.	Products produced in a batch are homogeneous and lack of individuality.

ICAI · Old P3 · MTP May 2020 · Series I · question p. 6 · answer p. 13

626 MTP **Old P3 Nov 2019 S-I** Q6(d) 5 marks old scheme

DISTINGUISH between Job and Batch costing.

ICAI'S ANSWER

Sr. No	Job Costing	Batch Costing
1	Method of costing used for non- standard and non-repetitive products produced as per customer specifications and against specific orders.	Homogeneous products produced in a continuous production flow in lots.
2	Cost determined for each Job.	Cost determined in aggregate for the entire Batch and then arrived at on per unit basis.
3	Jobs are different from each other and independent of each other. Each Job is unique.	Products produced in a batch are homogeneous and lack of individuality.

ICAI · Old P3 · MTP Nov 2019 · Series I · question p. 8 · answer p. 12

CHAPTER 9

Job Costing

9.1 Job costing

19 questions · asked in 18 ICAI documents · 4 exam papers

627 EXAM **May 2026** MCQ 13 2 marks MCQ

The following information pertains to Job Number 501 undertaken by PCM Limited:

Direct Materials	₹ 60,000
Direct Wages	₹ 50,000
Factory Overheads	60% of Direct Wages
Administrative Overheads	25% of Factory Cost
Profit percentage on total cost	25%

What is the selling price of Job Number 501?

A ₹ 1,89,550

B ₹ 2,24,675

C ₹ 2,18,750

D ₹ 2,33,640

ICAI'S ANSWER

Answer key: **(C)**

ICAI · P4 · May 2026 exam · question p. 6 · answer p. 7

628 EXAM **Sep 2025** Q6(b) 5 marks

List the advantages of Job Costing.

ICAI'S ANSWER

Some of the advantages of Job costing are summarised as below:

1. The details of Cost of material, labour and overhead for all job is available to control.
2. Profitability of each job can be derived.
3. It facilitates production planning.
4. Budgetary control and Standard Costing can be applied in job costing.
5. Spoilage and defective can be identified and responsibilities can be fixed accordingly.

ICAI · P4 · Sep 2025 exam · question p. 26 · answer p. 27-28

629 EXAM **Sep 2024** Q6(c) 4 marks

Define Job costing and explain differences between job and batch Costing.

ICAI'S ANSWER

Job Costing is defined as "the category of basic costing methods which is applicable where the work consists of separate contracts, jobs or batches, each of which is authorised by specific order or

contract." According to this method, costs are collected and accumulated according to jobs, contracts, products or work orders. Each job or unit of production is treated as a separate entity for the purpose of costing.

Difference between job and batch costing

Sr. No.	Job Costing	Batch Costing
1.	Method of costing used for non- standard and non-repetitive products produced as per customer specifications and against specific orders.	Homogeneous products produced in a continuous production flow in lots.
2.	Cost determined for each Job	Cost determined in aggregate for the entire Batch and then arrived at on per unit basis.
3.	Jobs are different from each other and independent of each other. Each Job is unique.	Products produced in a batch are homogeneous and lack of individuality

ICAI · P4 · Sep 2024 exam · question p. 36 · answer p. 39-40

630 EXAM Old P3 Nov 2019 Q1(b) 5 marks old scheme

The following data is presented by the supervisor of a factory for a Job:

	₹ per unit
Direct Material	120
Direct Wages @ ₹ 4 per hour (Departments A-4 hrs, B-7 hrs, C-2 hrs & D-2 hrs)	60
Chargeable Expenses	20
Total	200

Analysis of the Profit and Loss Account for the year ended 31st March, 2019

Material		2,00,000	Sales	4,30,000
Direct Wages				
Dept. A	12,000			
Dept. B	8,000			
Dept. C	10,000			
Dept. D	20,000	50,000		
Special Store items		6,000		
Overheads				
Dept. A	12,000			

Dept. B	6,000			
Dept. C	9,000			
Dept. D	17,000	44,000		
Gross Profit c/d		1,30,000		
		4,30,000		4,30,000
Selling Expenses		90,000	Gross Profit b/d	1,30,000
Net Profit		40,000		
		1,30,000		1,30,000

It is also to be noted that average hourly rates for all the four departments are similar.

Required:

(i) Prepare a Job Cost Sheet.

(ii) Calculate the entire revised cost using the above figures as the base.

(iii) Add 20% profit on selling price to determine the selling price.

ICAI'S ANSWER

Job Cost Sheet

Customer Details ——— Job No._____

Date of commencement —— Date of completion _____

Particulars		Amount (₹)
Direct materials		120
Direct wages:		
Deptt. A ₹ 4.00 × 4 hrs.	₹ 16.00	
Deptt. B ₹ 4.00 × 7 hrs.	₹ 28.00	
Deptt. C ₹ 4.00 × 2 hrs.	₹ 8.00	
Deptt. D ₹ 4.00 × 2 hrs.	₹ 8.00	60
Chargeable expenses		20
Prime cost		**200**
Overheads		
Deptt. A = ₹ 12,000/₹ 12,000 × 100 =100% of ₹ 16	₹ 16	
Deptt. B = ₹ 6,000/₹ 8,000 × 100 = 75% of ₹ 28	₹ 21	

Particulars		Amount (₹)
Deptt. C = ₹ 9,000/₹ 10,000 × 100 = 90% of ₹ 8 = ₹ 9,000/₹ 10,000 × 100 = 90% of ₹ 8 = ₹ 7.20	₹ 7.20	
Deptt. D = ₹ 17,000/₹ 20,000 × 100 = 85% of ₹ 8	₹ 6.80	**51.00**
Works cost		251.00
Selling expenses = ₹ 90,000/₹ 3,00,000 ×100 = 30% of work cost		**75.30**
Total cost		326.30
Profit (20% profit on selling price i.e 25% of total cost)		**81.58**
Selling price		**407.88**

ICAI · Old P3 · Nov 2019 exam · question p. 1-2 · answer p. 3-4

631 **RTP** **Jan 2026** **Q13**

A manufacturing company follows a job costing system. The following data is extracted from its books for the year ended 31st March, 2025:

Particulars	Amount (₹)
Direct Materials	12,50,000
Direct Wages	9,80,000
Selling & Distribution Overheads (Variable: 60%, Fixed: 40%)	6,30,000
Administration Overheads (Fixed)	5,40,000
Factory Overheads (70% Fixed, 30% Variable)	5,75,000
Profit	8,25,000

Additional Information:

1. **Overhead Recovery:**
   - Factory overheads are absorbed as a % of Direct Wages.
   - Selling & Distribution and Administration overheads are recovered as a % of Cost of Production.
2. In 2025-26, the company received a new job order. It is estimated that direct materials required will be ₹ 3,20,000 and direct labour will cost ₹ 2,10,000

Additional Information:

- Selling & Distribution overheads have increased by 20% (with variable portion now being 65%).
- Factory overheads have decreased by 10% due to efficiency improvements.
- The company wants to maintain the same profit % on sales as in the previous year.

You are required to prepare:

(a) Prepare a Job Cost Sheet showing Prime Cost, Cost of Production, Cost of Sales, Sales Value

(b) Determine the price to be quoted for the new job in 2025-26, considering Revised overhead structure and the company's profit objective.

(Assume relevant cost rates are based on previous year's data unless specified otherwise.)

ICAI'S ANSWER

Production Statement

For the Year Ended 31st March 2025

Particulars	Amount (₹)
Direct Materials	12,50,000
Direct Wages	9,80,000
Prime Cost	**22,30,000**
Factory Overheads	5,75,000
Cost of Production (COP)	**28,05,000**
Administration Overheads	5,40,000
Selling & Distribution Overheads	6,30,000
Cost of Sales	**39,75,000**
Profit	8,25,000
Sales Value	**48,00,000**

Calculation of Rates (2025-26)

1. Percentage of Administration Overheads to Cost of Production (COP) = $\frac{₹5,40,000}{₹28,05,000} \times 100 = 19.251\%$

2. Percentage of Profit to cost = 8,25,000 / 39,75,000 = 20.755%

3. Selling & Distribution Overheads (20% Increase, Variable 65%)

new S&D = 1.20 × 6,30,000 = **₹ 7,56,000**

S&D rate (revised) = 7,56,000 / 28,05,000 = **26.952%**

Note: variable portion changed to 65% - that affects fixed/variable split but not the total used for recovery here

4. Factory Overheads decreased by 10%:

new Factory OH = 0.90 × 5,75,000 = **₹5,17,500**

Factory OH rate (revised) = 5,17,500 / 9,80,000 = **52.806%**

Job Cost Sheet

Particulars	Amount (₹)
Direct Materials	3,20,000
Direct Wages	2,10,000

Particulars	Amount (₹)
Prime Cost	**5,30,000**
Add: Factory Overhead applied (52.806% of DL)	1,10,893
Cost of Production	**6,40,893**
Add: Administration overhead (19.251% of CP)	1,23,378
Cost of Sales (CP + Admin)	**7,64,271**
Add: Selling & Distribution overhead (26.952% of CP)	1,72,733
Total Cost (before profit)	9,37,004
Add: Profit (20.755% on cost)	1,94,475
Sales Value (Quoted Price)	**11,31,479**

ICAI · P4 · RTP Jan 2026 · question p. 17-18 · answer p. 39-41

632 RTP **May 2025** Q13

X Ltd. provides the following cost details for calculating the selling price of Job No. X2X:

Particulars	Per unit (₹)
Materials	1,330
Direct wages 18 hours @ ₹ 47.50 (Deptt. X: 8 hours; Deptt. Y: 6 hours; Deptt. Z: 4 hours)	855
Chargeable expenses	95
	2,280
Add: 1/3rd for expenses cost	760
	3,040

Analysis of the Profit/Loss Account

(for the current financial year)

		(₹)		(₹)
Materials used		28,50,000	Sales	47,50,000
Direct wages:				
Deptt. X	1,90,000			
Deptt. Y	2,28,000			
Deptt. Z	1,52,000	5,70,000		

			(₹)		(₹)
Special stores items			76,000		
Overheads:					
	Deptt. X	95,000			
	Deptt. Y	1,71,000			
	Deptt. Z	38,000	3,04,000		
Works cost			38,00,000		
Gross profit c/d			9,50,000		
			47,50,000		47,50,000
Selling expenses			3,80,000	Gross profit b/d	9,50,000
Net profit			5,70,000		
			9,50,000		9,50,000

It is also noted that average hourly rates for the three Departments X, Y and Z are similar.

You are required to prepare a job cost sheet to DETERMINE the selling price by calculating the entire revised cost using above figures as the base and adding 20% to the total cost.

ICAI'S ANSWER

Job cost Sheet

Particulars	Amount (₹)
Direct materials	1,330.00
Direct wages:	
Deptt. X ₹ 47.50 × 8 hrs. = ₹ 380	
Deptt. Y ₹ 47.50 × 6 hrs. = ₹ 285	
Deptt. Z ₹ 47.50 × 4 hrs. = ₹ 190	855.00
Chargeable expenses	95.00
Prime cost	**2,280.00**
Overheads:	
Deptt. X = (₹ 95,000/₹ 1,90,000 × 100) = 50% of ₹ 380 = ₹ 190.00	
Deptt. Y = (₹ 1,71,000/₹ 2,28,000 × 100) = 75% of ₹ 285 = ₹ 213.75	
Deptt. Z = (₹ 38,000/₹ 1,52,000 × 100) = 25% of ₹ 190 = ₹ 47.50	451.25

Particulars	Amount (₹)
Works cost	**2,731.25**
Selling expenses= (₹ 3,80,000/₹ 38,00,000 × 100) = 10% of work cost	273.13
Total cost	3,004.38
Profit (20% of total cost)	600.88
Selling price	**3,605.26**

ICAI · P4 · RTP May 2025 · question p. 18-19 · answer p. 42-43

633 RTP **Old P3 Nov 2023** Q7 old scheme

SM Motors Ltd. is a manufacturer of auto components. Following are the details of expenses for the year 2022-23:

		(₹)
(i)	Opening Stock of Material	15,00,000
(ii)	Closing Stock of Material	20,00,000
(iii)	Purchase of Material	1,80,50,000
(iv)	Direct Labour	90,50,000
(v)	Factory Overhead	30,80,000
(vi)	Administrative Overhead	20,50,400

During the FY 2023-24, the company has received an order from a car manufacturer where it estimates that the cost of material and labour will be ₹ 80,00,000 and ₹ 40,50,000 respectively. The company charges factory overhead as a percentage of direct labour and administrative overheads as a percentage of factory cost based on previous year's cost. Cost of delivery of the components at customer's premises is estimated at ₹ 4,50,000.

You are required to:

(i) Calculate the overhead recovery rates based on actual costs for 2022-23.

(ii) Prepare a Job cost sheet for the order received and the price to be quoted if the desired profit is 25% on sales.

ICAI'S ANSWER

(i) Calculation of Overhead Recovery Rate:

$$\text{Factory Overhead Recovery Rate} = \frac{₹\ 30,80,000}{₹\ 90,50,000} \times 100 = 34\% \text{ of Direct labour}$$

$$\text{Administrative Overhead Recovery Rate} = \frac{₹\ 20,50,400}{₹\ 2,96,80,000} \times 100 = 6.91\% \text{ of Factory Cost}$$

Working Note: Calculation of Factory Cost in 2022-23

Particulars	Amount (₹)
Opening Stock of Material	15,00,000
Add: Purchase of Material	1,80,50,000
Less: Closing Stock of Material	(20,00,000)
Material Consumed	1,75,50,000
Direct Labour	90,50,000
Prime Cost	2,66,00,000
Factory Overhead	30,80,000
Factory Cost	2,96,80,000

(ii) Job Cost Sheet for the order received in 2023-24

Particulars	Amount (₹)
Material	80,00,000
Labour	40,50,000
Factory Overhead (34% of ₹ 40,50,000)	13,77,000
Factory Cost	1,34,27,000
Administrative Overhead (6.91% of ₹1,34,27,000)	9,27,806
Cost of delivery	4,50,000
Total Cost	1,48,04,806
Add: Profit @ 25% of Sales or 33.33% of cost	49,34,935
Sales value (Price to be quoted for the order)	1,97,39,741

Hence the price to be quoted is ₹1,97,39,741.

ICAI · Old P3 · RTP Nov 2023 · question p. 4-5 · answer p. 16-17

634 RTP **Old P3 May 2022** Q8 old scheme

KJ Motors Ltd. is a manufacturer of auto components. Following are the details of expenses for the year

2020-21:

		(₹)
(i)	Opening Stock of Material	15,00,000
(ii)	Closing Stock of Material	20,00,000
(iii)	Purchase of Material	1,80,50,000
(iv)	Direct Labour	90,50,000
(v)	Factory Overhead	30,80,000
(vi)	Administrative Overhead	20,50,400

During the FY 2021-22, the company has received an order from a car manufacturer where it estimates that the cost of material and labour will be ₹ 80,00,000 and ₹ 40,50,000 respectively. The company charges factory overhead as a percentage of direct labour and administrative overheads as a percentage of factory cost based on previous year's cost. Cost of delivery of the components at customer's premises is estimated at ₹ 9,50,000.

You are required to:

(i) CALCULATE the overhead recovery rates based on actual costs for 2020-21.

(ii) PREPARE a Job cost sheet for the order received and the price to be quoted if the desired profit is 25% on sales.

ICAI'S ANSWER

(i) Calculation of Overhead Recovery Rate:

$$\text{Factory Overhead Recovery Rate} = \frac{\text{Factory Overhead in 2020-21}}{\text{Direct labour cost in 2020-21}} \times 100$$

$$= \frac{₹\ 30,80,000}{₹90,50,000} \times 100 = 34\% \text{ of Direct labour}$$

$$\text{Administrative Overhead Recovery Rate} = \frac{\text{Administrative Overhead in 2020-21}}{\text{Factory cost in 2020-21 (W.N)}} \times 100$$

$$= \frac{₹\ 20,50,400}{₹\ 2,96,80,000} \times 100 = 6.91\% \text{ of Factory Cost}$$

Working Note: Calculation of Factory Cost in 2020-21

Particulars	Amount (₹)
Opening Stock of Material	15,00,000
Add: Purchase of Material	1,80,50,000
Less: Closing Stock of Material	(20,00,000)
Material Consumed	1,75,50,000
Direct Labour	90,50,000
Prime Cost	2,66,00,000
Factory Overhead	30,80,000
Factory Cost	2,96,80,000

(ii) Job Cost Sheet for the order received in 2021-22

Particulars	Amount (₹)
Material	80,00,000
Labour	40,50,000
Factory Overhead (34% of ₹ 40,50,000)	13,77,000
Factory Cost	1,34,27,000
Administrative Overhead (6.91% of ₹ 1,34,27,000)	9,27,806
Cost of delivery	9,50,000
Total Cost	1,53,04,806
Add: Profit @ 25% of Sales or 33.33% of cost	51,01,602
Sales value (Price to be quoted for the order)	2,04,06,408

Hence the price to be quoted is ₹ 2,04,06,408.

ICAI · Old P3 · RTP May 2022 · question p. 6-7 · answer p. 23

635 RTP **Old P3 Jul 2021** Q7 old scheme

SM Motors Ltd. is a manufacturer of auto components. Following are the details of expenses for the year

2019-20:

		(₹)
(i)	Opening Stock of Material	15,00,000
(ii)	Closing Stock of Material	20,00,000
(iii)	Purchase of Material	1,80,50,000
(iv)	Direct Labour	90,50,000
(v)	Factory Overhead	30,80,000
(vi)	Administrative Overhead	20,50,400

During the FY 2020-21, the company has received an order from a car manufacturer where it estimates that the cost of material and labour will be ₹ 80,00,000 and ₹ 40,50,000 respectively. The company charges factory overhead as a percentage of direct labour and administrative overheads as a percentage of factory cost based on previous year's cost. Cost of delivery of the components at customer's premises is estimated at ₹ 4,50,000.

You are required to:

(i) CALCULATE the overhead recovery rates based on actual costs for 2019-20.

(ii) PREPARE a Job cost sheet for the order received and the price to be quoted if the desired profit is 25% on sales.

ICAI'S ANSWER

(i) Calculation of Overhead Recovery Rate:

$$\text{Factory Overhead Recovery Rate} = \frac{\text{Factory Overhead in } 2019-20}{\text{Direct Labour Costs in } 2019-20} \times 100$$

$$= \frac{₹\ 30,80,000}{₹\ 90,50,000} \times 100 = 34\% \text{ of Direct labour}$$

Administrative Overhead Recovery Rate

$$= \frac{\text{Administrative Overhead in } 2019-20}{\text{Factory Costs in } 2019-20 \text{ (W.N.)}} \times 100$$

$$= \frac{₹\ 20,50,400}{₹\ 2,96,80,000} \times 100 = 6.91\% \text{ of Factory Cost}$$

Working Note:

Calculation of Factory Cost in 2019-20

Particulars	Amount (₹)
Opening Stock of Material	15,00,000
Add: Purchase of Material	1,80,50,000
Less: Closing Stock of Material	(20,00,000)
Material Consumed	1,75,50,000
Direct Labour	90,50,000
Prime Cost	2,66,00,000
Factory Overhead	30,80,000
Factory Cost	2,96,80,000

(ii) Job Cost Sheet for the order received in 2020-21

Particulars	Amount (₹)
Material	80,00,000
Labour	40,50,000
Factory Overhead (34% of ₹ 40,50,000)	13,77,000
Factory Cost	1,34,27,000
Administrative Overhead (6.91% of ₹ 1,34,27,000)	9,27,806
Cost of delivery	4,50,000
Total Cost	1,48,04,806
Add: Profit @ 25% of Sales or 33.33% of cost	49,34,935
Sales value (Price to be quoted for the order)	1,97,39,741

Hence the price to be quoted is ₹ 1,97,39,741

ICAI · Old P3 · RTP Jul 2021 · question p. 6-7 · answer p. 22-23

636 RTP **Old P3 Nov 2020** Q8 old scheme

AP Ltd. received a job order for supply and fitting of plumbing materials. Following are the details related with the job work:

Direct Materials

AP Ltd. uses a weighted average method for the pricing of materials issues.

Opening stock of materials as on 12th August 2020:

- 15mm GI Pipe, 12 units of (15 feet size) @ ₹600 each
- 20mm GI Pipe, 10 units of (15 feet size) @ ₹ 660 each

- Other fitting materials, 60 units @ ₹ 26 each
- Stainless Steel Faucet, 6 units @ ₹ 204 each
- Valve, 8 units @ ₹ 404 each

Purchases:

On 16th August 2020:

- 20mm GI Pipe, 30 units of (15 feet size) @ ₹ 610 each
- 10 units of Valve @ ₹ 402 each

On 18th August 2020:

- Other fitting materials, 150 units @ ₹ 28 each
- Stainless Steel Faucet, 15 units @ ₹ 209 each

On 27th August 2020:

- 15mm GI Pipe, 35 units of (15 feet size) @ ₹ 628 each
- 20mm GI Pipe, 20 units of (15 feet size) @ ₹ 660 each
- Valve, 14 units @ ₹ 424 each

Issues for the hostel job:

On 12th August 2020:

- 20mm GI Pipe, 2 units of (15 feet size)
- Other fitting materials, 18 units

On 17th August 2020:

- 15mm GI Pipe, 8 units of (15 feet size)
- Other fitting materials, 30 units

On 28th August 2020:

- 20mm GI Pipe, 2 units of (15 feet size)
- 15mm GI Pipe, 10 units of (15 feet size)
- Other fitting materials, 34 units
- Valve, 6 units

On 30th August 2020:

- Other fitting materials, 60 units
- Stainless Steel Faucet, 15 units

Direct Labour:

Plumber: 180 hours @ ₹100 per hour (includes 12 hours overtime)

Helper: 192 hours @ ₹70 per hour (includes 24 hours overtime)

Overtimes are paid at 1.5 times of the normal wage rate.

Overheads:

Overheads are applied @ ₹26 per labour hour.

Pricing policy:

It is company's policy to price all orders based on achieving a profit margin of 25% on sales price.

You are required to

(a) CALCULATE the total cost of the job.

(b) CALCULATE the price to be charged from the customer.

ICAI'S ANSWER

(a) Calculation of Total Cost for the Job:

Particulars	Amount (₹)	Amount (₹)
Direct Material Cost:		
- 15mm GI Pipe (Working Note- 1)	11,051.28	
- 20mm GI Pipe (Working Note- 2)	2,588.28	
- Other fitting materials (Working Note- 3)	3,866.07	
- Stainless steel faucet 15 units × $\left(\frac{6 \times ₹\ 204 + 15 \times ₹\ 209}{21\text{units}}\right)$	3,113.57	
- Valve 6 units × $\left(\frac{8 \times ₹\ 404 + 10 \times ₹\ 402 + 14 \times ₹\ 424}{32\text{units}}\right)$	2,472.75	23,091.95
Direct Labour:		
- Plumber [(180 hours × ₹100) + (12 hours × ₹50)]	18,600.00	
- Helper [(192 hours × ₹70) + (24 hours × ₹35)]	14,280.00	32,880.00
- Overheads[₹26 × (180 + 192) hours]		9,672.00
Total Cost		**65,643.95**

(b) Price to be charged for the job work:

	Amount (₹)
Total Cost incurred on the job	65,643.95
Add: 25% Profit on Job Price $\left(\frac{65{,}643.95}{75\%} \times 25\%\right)$	21,881.32
	87,525.27

Working Note:

1. **Cost of 15mm GI Pipe**

Date		Amount (₹)
17-08-2020	8 units × ₹ 600	4,800.00
28-08-2020	10 units × $\left(\frac{4 \times ₹\ 600 + 35 \times ₹\ 628}{39\text{units}}\right)$	6,251.28
		11,051.28

2. **Cost of 20mm GI Pipe**

Date		Amount (₹)
12-08-2020	2 units × ₹ 660	1,320.00
28-08-2020	2 units × $\left(\frac{8 \times ₹\ 660 + 30 \times ₹\ 610 + 20 \times ₹\ 660}{58\text{units}}\right)$	1,268.28
		2,588.28

3. **Cost of Other fitting materials**

Date		Amount (₹)
12-08-2020	18 units × ₹ 26	468.00
17-08-2020	30 units × ₹ 26	780.00
28-08-2020	34 units × $\left(\frac{12 \times ₹\ 26 + 150 \times ₹\ 28}{162\text{units}}\right)$	946.96
30-08-2020	60 units × $\left(\frac{12 \times ₹\ 26 + 150 \times ₹\ 28}{162\text{units}}\right)$	1,671.11
		3,866.07

ICAI · Old P3 · RTP Nov 2020 · question p. 6-7 · answer p. 20-21

637 RTP **Old P3 May 2020** Q7 old scheme

A factory uses job costing system. The following data are obtained from its books for the year ended 31st March, 2020:

	Amount (₹)
Direct materials	18,00,000
Direct wages	15,00,000
Selling and distribution overheads	10,50,000
Administration overheads	8,40,000
Factory overheads	9,00,000
Profit	12,18,000

(i) PREPARE a Job Cost sheet indicating the Prime cost, Cost of Production, Cost of sales and the Sales value.

(ii) In 2019-20, the factory received an order for a job. It is estimated that direct materials required will be ₹4,80,000 and direct labour will cost ₹3,00,000. DETERMINE what should be the price for the job if factory intends to earn the same rate of profit on sales assuming that the selling and distribution overheads have gone up by 15%. The factory overheads is recovered as percentage of wages paid, whereas, other overheads as a percentage of cost of production, based on cost rates prevailing in the previous year.

ICAI'S ANSWER

(i) Production Statement

For the year ended 31st March, 2020

	Amount (₹)
Direct materials	18,00,000
Direct wages	15,00,000
Prime Cost	33,00,000
Factory overheads	9,00,000
Cost of Production	42,00,000
Administration overheads	8,40,000
Selling and distribution overheads	10,50,000
Cost of Sales	60,90,000
Profit	12,18,000
Sales value	73,08,000

Calculation of Rates:

1. Percentage of factory overheads to direct wages = $\frac{₹9,00,000}{₹15,00,000} \times 100 = 60\%$
2. Percentage of administration overheads to Cost of production

$= \frac{₹8,40,000}{₹\ 42,00,000} \times 100 = 20\%$

3. Selling and distribution overheads = ₹10,50,000 × 115% = ₹12,07,500

Selling and distribution overhead % to Cost of production

$= \frac{₹12,07,500}{₹\ 42,00,000} \times 100 = 28.75\%$

4. Percentage of profit to sales = $\frac{₹12,18,000}{₹73,08,000} \times 100 = 16.67\%$ or, 1/6

(ii) Calculation of price for the job received in 2019-20

	Amount (₹)
Direct materials	4,80,000
Direct wages	3,00,000
Prime Cost	7,80,000
Factory overheads (60% of ₹3,00,000)	1,80,000
Cost of Production	9,60,000
Administration overheads (20% of ₹9,60,000)	1,92,000
Selling and distribution overheads (28.75% of ₹9,60,000)	2,76,000
Cost of Sales	14,28,000
Profit (1/5 of ₹14,28,000)	2,85,600
Sales value	17,13,600

ICAI · Old P3 · RTP May 2020 · question p. 5-6 · answer p. 18-19

638 MTP **Sep 2026 S-I** Q2(a) 7 marks

Riya Engineering Works undertakes job work on a contract basis. The following information is available in respect of Job No. 707, which was completed during the month of June:

Particulars	Details
Direct Materials issued	₹ 45,000
Direct Materials returned to stores	₹ 3,000
Direct Labour: 250 hours @ ₹ 80 per hour	
Direct Expenses	₹ 6,000
Factory Overhead	60% of Direct Labour cost
Administration Overhead	10% of Works Cost
Selling & Distribution Overhead	8% of Cost of Production
Profit margin desired by the company	20% on Selling Price

Additional Information:

Rectification of defective work (additional direct labour of 15 hours @ ₹ 80 per hour, plus direct material of ₹ 1,200, required to rework units rejected at final inspection — defect arose due to a normal, recurring tolerance issue in this type of job)

You are required to:

(i) PREPARE a Job Cost Sheet for Job No. 707, showing Prime Cost, Works Cost (Factory Cost), Cost of Production, and Total Cost.

(ii) CALCULATE the price to be quoted to the customer for Job No. 707, given that profit is desired at 20% on selling price.

(iii) Briefly STATE how your treatment in (i) would differ if the defect had instead arisen due to an abnormal cause (e.g., a one-off machine malfunction).

ICAI'S ANSWER

Job Cost Sheet for Job No. 707

Particulars	Amount (₹)	Amount (₹)
Direct Materials issued	45,000	
Less: Direct Materials returned to stores	(3,000)	
Direct Materials consumed		42,000
Direct Labour (250 hours × ₹ 80)		20,000
Direct Expenses		6,000
Prime Cost		68,000
Add: Factory Overhead (60% of Direct Labour cost)		12,000
Add: Rectification cost of defective work (normal):		
— Direct material	1,200	
— Direct labour (15 hours × ₹ 80)	1,200	2,400
(ii) Works Cost (Factory Cost)		82,400
Add: Administration Overhead (10% of Works Cost)		8,240
(iii) Cost of Production		90,640
Add: Selling & Distribution Overhead (8% of Cost of Production)		7,251.20
(iv) Total Cost		97,891.20
Add: Profit (20% of SP or 25% of Cost		24,472.80
Selling Price		1,22,364

Treatment of rectification cost: Since the defect is stated to be normal for this type of job (a recurring tolerance issue inherent to the work), the rectification cost (₹ 2,400) is charged directly to the job itself, by adding it to the Works Cost / Factory Cost of Job No. 707. It is treated as a normal manufacturing cost specific to this job, since the customer's job benefits from — and should bear the cost of — bringing the units up to the required specification.

(ii) Price to be quoted to the customer for Job No. 707 = ₹ 1,22,364

(iii) Treatment if the Defect Were Abnormal

If the defect had arisen due to an abnormal cause (e.g., a one-off machine malfunction, not expected to recur in the normal course of operations), the treatment would differ as follows:

- The rectification cost of ₹ 2,400 (direct material ₹ 1,200 + direct labour ₹ 1,200) would not be charged to Job No. 707 or included in its Works Cost.
- Instead, it would be treated as an abnormal loss and charged directly to the Costing Profit & Loss Account, since it represents an avoidable, non-recurring inefficiency rather than a cost the customer should bear.

ICAI · P4 · MTP Sep 2026 · Series I · question p. 9-10 · answer p. 10-11

639 MTP **May 2026 S-I** Q1(a) 5 marks

A factory operates 8 hours per day with 6 working days in a week. The worker is paid @ ₹ 500 per day as basic plus 40% dearness allowance. Overtime rate is payable at double the normal hourly rate. The worker is allowed half an hour break for lunch and two tea breaks of 15 minutes each per day. During last week, worker X was present for 5 days. To meet out demand, X has done overtime for 4 hours. X has spent 60% time on Job A and 40% time on Job B. The overtime worked was for Job A only.

Direct material cost incurred are :

Job A : ₹ 5,000

Job B : ₹ 7,500

Overheads are applied @ ₹ 15 per labour hour.

You are required to CALCULATE the total cost of Job A and Job B.

ICAI'S ANSWER

Calculation of total cost of Job A and Job B

	Job A	Job B
Particulars	**Amount (in ₹)**	**Amount (in ₹)**
Direct Material Cost (Given)	5,000	7,500
Labour Cost (see working note)	2,900	1,400
Overheads	375 (21 hrs + 4 hrs x ₹15 per hour)	210 (14 hrs x ₹ 15 per hour)
Total	8,275	9,110

Working Note:

Normal time worked = (8 hrs – ½ hr – ½ hr) x 5day = 35 Hrs

Normal hourly wages earned

= (₹ 500 Basic + ₹ 200 D.A) x 5 days /35 Hrs

= ₹ 100 per hour

Overtime Wages = 2 x 100 per hour = ₹ 200 per hour

Cost of wages for Job A: = (35 hrs x 100 x 60%) + 4 hrs x 200

= ₹ 2,100 + ₹ 800

= ₹ 2,900

Cost of Wages of Job B: = (35 hrs x 100 x 40%)

= ₹ 1,400

ICAI · P4 · MTP May 2026 · Series I · question p. 8 · answer p. 7

640 MTP **Jan 2026 S-I** Q2(b) 4 marks

Distinguish between Job Costing and Process Costing?

ICAI'S ANSWER

The main points which distinguish Job Costing and Process Costing are as below:

Job Costing	Process Costing
(i) A Job is carried out or a product is produced by specific orders.	The process of producing the product has a continuous flow and the product produced is homogeneous.
(ii) Costs are determined for each job.	Costs are compiled on time basis i.e., for production of a given accounting period for each process or department.
(iii) Each job is separate and independent of other jobs.	Products lose their individual identity as they are manufactured in a continuous flow.
(iv) Each job or order has a number and costs are collected against the same job number.	The unit cost of process is an average cost for the period.
(v) Costs are computed when a job is completed. The cost of a job may be determined by adding all costs against the job.	Costs are calculated at the end of the cost period. The unit cost of a process may be computed by dividing the total cost for the period by the output of the process during that period.
(vi) As production is not continuous and each job may be different, so more managerial attention is required for effective control.	Process of production is usually standardized and is therefore, quite stable. Hence control here is comparatively easier.

ICAI · P4 · MTP Jan 2026 · Series I · question p. 10 · answer p. 10

641 MTP **Jan 2026 S-II** MCQ 15 2 marks MCQ

The following details are available for a job work:

Material issued ₹ 60,000; return to stores ₹ 8,000

Dept A labour: 80 hrs @ ₹ 180/hr; OH rate = 120% of labour

Dept B labour: 50 hrs @ ₹ 150/hr; OH rate = ₹200 per labour hour

Total job cost is:

A ₹ 1,04,600

B ₹ 1,06,800

C ₹ 1,12,400

D ₹ 1,01,180

ICAI'S ANSWER

(D) ₹ 1,01,180

Direct Material = Material issued – Material returned

= 60,000 - 8,000 = ₹ 52,000

Department A Labour = 80 hrs × ₹ 180 = ₹ 14,400

Dept A Overhead = 14,400 × 120% = ₹ 17,280

Department B Labour = 50 hrs × ₹ 150 = ₹ 7,500

Dept B Overhead = 50 hrs × ₹ 200 = ₹ 10,000

Total Job Cost = 52,000 + 14,400 + 17,280 + 7,500 + 10,000 = ₹ 1,01,180

ICAI · P4 · MTP Jan 2026 · Series II · question p. 7 · answer p. 7

642 MTP **May 2025 S-II** Q4(a) 6 marks

A factory uses job costing system. The following data are obtained from its books for the year ended 31st March, 2025:

	Amount (₹)
Direct materials	18,00,000
Direct wages	15,00,000
Selling and distribution overheads	10,50,000
Administration overheads	8,40,000
Factory overheads	9,00,000
Profit	12,18,000

(i) PREPARE a Job Cost sheet indicating the Prime cost, Cost of Production, Cost of sales and the Sales value.

(ii) In 2025-26, the factory received an order for a job. It is estimated that direct materials required will be ₹ 4,80,000 and direct labour will cost ₹ 3,00,000. DETERMINE what should be the price for the job if factory intends to earn the same rate of profit on sales assuming that the selling and distribution overheads have gone up by 15%. The factory overheads is recovered as percentage of wages paid, whereas, other overheads as a percentage of cost of production, based on cost rates prevailing in the previous year.

ICAI'S ANSWER

(i) **Production Statement**

For the year ended 31st March, 2025

	Amount (₹)
Direct materials	18,00,000

	Amount (₹)
Direct wages	15,00,000
Prime Cost	33,00,000
Factory overheads	9,00,000
Cost of Production	42,00,000
Administration overheads	8,40,000
Selling and distribution overheads	10,50,000
Cost of Sales	60,90,000
Profit	12,18,000
Sales value	73,08,000

Calculation of Rates:

1. Percentage of factory overheads to direct wages = $\frac{₹9,00,000}{₹15,00,000} \times 100 = 60\%$

2. Percentage of administration overheads to Cost of production

$= \frac{₹8,40,000}{₹\ 42,00,000} \times 100 = 20\%$

3. Selling and distribution overheads = ₹ 10,50,000 × 115% = ₹ 12,07,500

Selling and distribution overhead % to Cost of production

$= \frac{₹12,07,500}{₹\ 42,00,000} \times 100 = 28.75\%$

4. Percentage of profit to sales = $\frac{₹12,18,000}{₹73,08,000} \times 100 = 16.67\%$ or, 1/6

(ii) **Calculation of price for the job received in 2025-26**

	Amount (₹)
Direct materials	4,80,000
Direct wages	3,00,000
Prime Cost	7,80,000
Factory overheads (60% of ₹ 3,00,000)	1,80,000
Cost of Production	9,60,000
Administration overheads (20% of ₹ 9,60,000)	1,92,000
Selling and distribution overheads (28.75% of ₹ 9,60,000)	2,76,000
Cost of Sales	14,28,000
Profit (1/5 of ₹ 14,28,000)	2,85,600

	Amount (₹)
Sales value	17,13,600

ICAI · P4 · MTP May 2025 · Series II · question p. 11 · answer p. 12-14

643 MTP Old P3 Nov 2023 S-II Q4(b) 10 marks old scheme

In an engineering company, the factory overheads are recovered on a fixed percentage basis on direct wages and the administrative overheads are absorbed on a fixed percentage basis on factory cost.

The company has furnished the following data relating to two jobs undertaken by it in a period:

	Job 101	Job 102
	(₹)	**(₹)**
Direct materials	54,000	37,500
Direct wages	42,000	30,000
Selling price	1,66,650	1,28,250
Profit percentage on Total Cost	10%	20%

Required:

(i) Computation of percentage recovery rates of factory overheads and administrative overheads.

(ii) Calculation of the amount of factory overheads, administrative overheads and profit for each of the two jobs.

(iii) Using the above recovery rates determine the selling price of job 103. The additional data being:

Direct materials	₹ 24,000
Direct wages	₹ 20,000
Profit percentage on selling price	12-½%

ICAI'S ANSWER

(i) Computation of percentage recovery rates of factory overheads and administrative overheads.

Let the factory overhead recovery rate as percentage of direct wages be F and administrative overheads recovery rate as percentage of factory cost be A.

Factory Cost of Jobs:

Direct materials + Direct wages + Factory overhead

For Job 101 = ₹ 54,000 +₹ 42,000 + ₹ 42,000F

For Job 102 = ₹ 37,500 +₹ 30,000 + ₹ 30,000F

Total Cost of Jobs:

Factory cost + Administrative overhead

For Job 101 = (₹ 96,000 + ₹ 42,000F) + (₹ 96,000+ ₹ 42,000F) A = ₹ 1,51,500*

For Job-102 = (₹ 67,500 + ₹ 30,000F) + (₹ 67,500+ ₹ 30,000F) A = ₹ 1,06,875**

The value of F & A can be found using following equations

96,000 + 42,000F + 96,000A + 42,000AF	=	1,51,500	.....eqn (i)
67,500 + 30,000F + 67,500A + 30,000AF	=	1,06,875	.....eqn (ii)

Multiply equation (i) by 5 and equation (ii) by 7

4,80,000 + 2,10,000F + 4,80,000A + 2,10,000AF	=	7,57,500	.....eqn (iii)
4,72,500 + 2,10,000F + 4,72,500A + 2,10,000AF	=	7,48,125	.....eqn (iv)
- - - -		-	
7,500 + 7,500A	=	9,325	

7,500 A = 9,325 – 7,500

A = 0.25

Now put the value of A in equation (i) to find the value of F

96,000 + 42,000F + 24,000 + 10,500F = 1,51,500

52,500F = 1,51,500 – 1,20,000

F = 0.6

On solving the above relations: F = 0.60 and A = 0.25

Hence, percentage recovery rates of:

Factory overheads = 60% of wages and

Administrative overheads = 25% of factory cost.

Working note:

Total Cost = $\frac{\text{Selling price}}{(100\% + \text{Percentage of profit})}$

*For Job 101 = $\frac{₹1,66,650}{(100\% + 10\%)}$ = ₹ 1,51,500

**For Job 102 = $\frac{₹1,28,250}{(100\% + 20\%)}$ = ₹ 1,06,875

(ii) Statement of jobs, showing amount of factory overheads, administrative overheads and profit:

	Job 101	Job 102
	(₹)	**(₹)**
Direct materials	54,000	37,500
Direct wages	42,000	30,000
Prime cost	96,000	67,500
Factory overheads		

	Job 101	Job 102
60% of direct wages	25,200	18,000
Factory cost	1,21,200	85,500
Administrative overheads		
25% of factory cost	30,300	21,375
Total cost	1,51,500	1,06,875
Profit (10% & 20% respectively)	15,150	21,375
Selling price	1,66,650	1,28,250

(iii) Selling price of Job 103

	(₹)
Direct materials	24,000
Direct wages	20,000
Prime cost	44,000
Factory overheads (60% of Direct Wages)	12,000
Factory cost	56,000
Administrative overheads (25% of factory cost)	14,000
Total cost	70,000
Profit margin (balancing figure)	10,000
Selling price $\left[\frac{\text{Total Cost}}{87.5\%}\right]$	80,000

ICAI · Old P3 · MTP Nov 2023 · Series II · question p. 5-6 · answer p. 9-11

644 MTP **Old P3 Nov 2023 S-II** Q6(c) 5 marks old scheme

State the difference between job costing and process costing.

ICAI'S ANSWER

The main points which distinguish job costing and process costing are as below:

Job Costing	Process Costing
(i) A Job is carried out or a product is produced by specific orders.	The process of producing the product has a continuous flow and the product produced is homogeneous.
(ii) Costs are determined for each job.	Costs are compiled on time basis i.e., for production of a given accounting period for each process or department.

Job Costing	Process Costing
(iii) Each job is separate and independent of other jobs.	Products lose their individual identity as they are manufactured in a continuous flow.
(iv) Each job or order has a number and costs are collected against the same job number.	The unit cost of process is an average cost for the period.
(v) Costs are computed when a job is completed. The cost of a job may be determined by adding all costs against the job.	Costs are calculated at the end of the cost period. The unit cost of a process may be computed by dividing the total cost for the period by the output of the process during that period.
(vi) As production is not continuous and each job may be different, so more managerial attention is required for effective control.	Process of production is usually standardized and is therefore, quite stable. Hence control here is comparatively easier.

ICAI · Old P3 · MTP Nov 2023 · Series II · question p. 7 · answer p. 14

645 MTP **Old P3 Nov 2019 S-I** Q2(b) 10 marks old scheme

APFL Ltd. deals in plumbing materials and also provides plumbing services to its customers. On 12th August, 2019, APFL received a job order for a students' hostel to supply and fitting of plumbing materials. The work is to be done on the basis of specification provided by the hostel owner. Hostel will be inaugurated on 5th September, 2019 and the work is to be completed by 3rd September, 2019. Following are the details related with the job work:

Direct Materials

APFL uses a weighted average method for the pricing of materials issues.

Opening stock of materials as on 12th August 2019:

- 15mm GI Pipe, 12 units of 15 feet size @ Rs.600 each
- 20mm GI Pipe, 10 units of 15 feet size @ Rs.660 each
- Other fitting materials, 60 units @ Rs. 26 each
- Stainless Steel Faucet, 6 units @ Rs. 204 each
- Valve, 8 units @ Rs. 404 each

Purchases:

On 16th August 2019:

- 20mm GI Pipe, 30 units of 15 feet size @ Rs. 610 each
- 10 units of Valve @ Rs. 402 each

On 18th August 2019:

- Other fitting materials, 150 units @ Rs. 28 each
- Stainless Steel Faucet, 15 units @ Rs. 209 each

On 27th August 2019:

- 15mm GI Pipe, 35 units of 15 feet size @ Rs.628 each

- 20mm GI Pipe, 20 units of 15 feet size @ Rs.660 each
- Valve, 14 units @ Rs. 424 each

Issues for the hostel job:

On 12th August 2019:

- 20mm GI Pipe, 2 units of 15 feet size
- Other fitting materials, 18 units

On 17th August 2019:

- 15mm GI Pipe, 8 units of 15 feet size
- Other fitting materials, 30 units

On 28th August 2019:

- 20mm GI Pipe, 2 units of 15 feet size
- 15mm GI Pipe, 10 units of 15 feet size
- Other fitting materials, 34 units
- Valve, 6 units

On 30th August:

- Other fitting materials, 60 units
- Stainless Steel Faucet, 15 units

Direct Labour:

Plumber: 180 hours @ Rs. 50 per hour (includes 12 hours overtime)

Helper: 192 hours @ Rs.35 per hour (includes 24 hours overtime)

Overtimes are paid at 1.5 times of the normal wage rate.

Overheads:

Overheads are applied @ Rs. 13 per labour hour.

Pricing policy:

It is company's policy to price all orders based on achieving a profit margin of 25% on sales price.

You are required to

(a) CALCULATE the total cost of the job.

(b) CALCULATE the price to be charged from the customer.

ICAI'S ANSWER

(a) **Calculation of Total Cost for the Hostel Job**

Particulars	Amount (Rs.)	Amount (Rs.)
Direct Material Cost:		
- 15mm GI Pipe (Working Note- 1)	11,051.28	
- 20mm GI Pipe (Working Note- 2)	2,588.28	

Particulars	Amount (Rs.)	Amount (Rs.)
- Other fitting materials (Working Note- 3)	3,866.07	
- Stainless steel faucet 15 units × $\left(\frac{6 \times ₹\ 204 + 15 \times ₹\ 209}{21\text{units}}\right)$	3,113.57	
- Valve 6 units × $\left(\frac{8 \times ₹\ 404 + 10 \times ₹\ 402 + 14 \times ₹\ 424}{32\text{units}}\right)$	2,472.75	23,091.95
Direct Labour:		
Plumber [(180 hours × Rs. 50) + (12 hours × Rs. 25)]	9,300.00	
Helper [(192 hours × Rs. 35) + (24 hours × Rs. 17.5)]	7,140.00	16,440.00
- Overheads [Rs. 13 × (180 + 192) hours]		4,836.00
Total Cost		44,367.95

(b) Price to be charged for the job work:

	Amount (Rs.)
Total Cost incurred on the job	44,367.95
Add: 25% Profit on Job Price $\left(\frac{44{,}367.95}{75\%} \times 25\%\right)$	14,789.32
	59,157.27

Working Note:

1. Cost of 15mm GI Pipe

Date		Amount (Rs.)
17-08-2019	8 units × Rs. 600	4,800.00
28-08-2019	10 units × $\left(\frac{4 \times \text{Rs.}600 + 35 \times \text{Rs.}628}{39\text{units}}\right)$	6,251.28
		11,051.28

2. Cost of 20mm GI Pipe

Date		Amount (Rs.)
12-08-2019	2 units × Rs. 660	1,320.00
28-08-2019	2 units × $\left(\frac{8 \times \text{Rs.}660 + 30 \times \text{Rs.}610 + 20 \times \text{Rs.}660}{58\text{units}}\right)$	1,268.28
		2,588.28

3. Cost of Other fitting materials

Date		Amount (Rs.)
12-08-2019	18 units × Rs. 26	468.00
17-08-2019	30 units × Rs. 26	780.00
28-08-2019	34 units × $\left(\frac{12 \times \text{Rs.}26 + 150 \times \text{Rs.}28}{162\text{units}}\right)$	946.96
30-08-2019	60 units × $\left(\frac{12 \times \text{Rs.}26 + 150 \times \text{Rs.}28}{162\text{units}}\right)$	1,671.11
		3,866.07

ICAI · Old P3 · MTP Nov 2019 · Series I · question p. 3-4 · answer p. 4-5

CHAPTER 10

Process and Operation Costing

10.1 Process accounts, normal loss, abnormal loss and abnormal gain

35 questions · asked in 33 ICAI documents · 14 exam papers

646 EXAM **Sep 2026** Case scenario · Q4(b)(i) to Q4(b)(iii) 5 marks

CASE SCENARIO

New Pharma Chemicals Ltd. manufactures pharmaceutical-grade solvents for the drug formulation industry. Production passes through a series of processes and the company operates a FIFO process costing system to value work-in-process and finished goods.

The following information is given :

- Opening work-in-process at the beginning of the month was 800 litres which was 70% complete for labour and 60% complete for overheads.
- Closing work-in-process at the end of the month was 160 litres which was 30% complete for labour and 20% complete for overheads.
- Normal loss is 10% of input; total losses during the month were 1,800 litres, partly attributable to flood damage. Abnormal losses occurred in the production shop floor just before despatch to the warehouse.
- Output transferred to the finished goods warehouse was 4,200 litres.
- All raw materials are introduced at the commencement of the process.

Required :

Q4(b)(i) 2 marks

Calculate the quantity (in litres) of raw material input during the month.

Q4(b)(ii) 1 mark

Calculate the quantity (in litres) of normal loss and abnormal loss.

Q4(b)(iii) 2 marks

Calculate the equivalent units of labour and overheads completed in the process.

ICAI'S ANSWERS

Q4(b)(i)

ICAI has not published its answers to this paper yet.

Q4(b)(ii)

ICAI has not published its answers to this paper yet.

Q4(b)(iii)

ICAI has not published its answers to this paper yet.

ICAI · P4 · Sep 2026 exam · question p. 18-19

647 EXAM **Jan 2026** Q2(b) 5 marks

A producer of oils purchases 400 litres of oil at ₹ 75 per litre in containers of 40 litres each. Other expenses at the process were ₹ 37,350 including an abnormal loss of ₹ 12,500 due to an accident. Normal waste at the process is 10%. Waste is sold at ₹ 10 per litre and each empty container is sold for ₹ 50. Generally, 10% of containers gets damaged so badly that they became unsalable. However, in the given

situation, more containers were damaged and only 5 containers were in saleable condition. Actual output was 375 litres which is sold at a profit of 20% on cost.

You are required to prepare Process Account and Income Statement.

ICAI'S ANSWER

Process Account

Particulars	Container	Oil Litre	(₹)	Particulars	Container	Oil Litre	(₹)
To Material @ ₹ 75 per litre	10	400	30,000	By Normal loss oil (10% of 400 litres) × ₹ 10 per litre		40	400
To other Expenses (37,350-12,500)	-	-	24,850	By Normal loss Container	1	-	-
To Abnormal gain (₹ 150 × 15 litres)	-	15	2,250	By Abnormal loss (Container)	4	-	200
				By Sale of Container @ ₹ 50 per container	5	-	250
				By Output (₹150 × 375 litres)	-	375	56,250
	10	**415**	**57,100**		**10**	**415**	**57,100**

Cost per unit of completed units and abnormal gain:

$$= \frac{\text{Total Cost-Realisable value from normal loss-Realisable value of Container}}{\text{Inputs units-Normal loss units}}$$

$$= \frac{₹54,850 - ₹400 - ₹450}{400 \text{ litres-}40 \text{ litres}} = \frac{54,000}{360 \text{ litress}} = ₹ 150 \text{ per litre}$$

Income Statement

Particulars	(₹)	Particulars	(₹)
To Cost of sales (₹ 150 × 375 litres)	56,250	By Sales	67,500
To Abnormal loss (Container)	200	By Abnormal gain {15 litres × (₹ 150 – ₹ 10)}	2,100
To Abnormal loss (accidental loss-given)	12,500		
To Net Profit	650		
	69,600		**69,600**

ICAI · P4 · Jan 2026 exam · question p. 14 · answer p. 16-17

648 EXAM **Sep 2025** MCQ 14 2 marks MCQ

A product passes through Process-I and Process-II. Materials issued to Process-1 amounted to ₹ 1,60,000. Wages ₹ 70,000 and Manufacturing Overheads ₹ 58,000 were charged to Process 1. Anticipated normal loss was 6% of input. 9,200 units of output were produced and transferred to Process-II. There was no opening stock. Input of raw material issued to Process-1 was 10,000 units. Scrap has a realizable value of ₹ 10 per unit. What is the value of units transferred to Process-II?

A ₹ 2,76,000

B ₹ 2,82,000

C ₹ 2,88,000

D ₹ 2,78,000

ICAI'S ANSWER

Answer key: **(A)**

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649 EXAM **May 2025** MCQ 15 MCQ

A spice is passed through two processes and the output of Process I-Grinding, transferred to Process II-Packaging. The input units in Process 1 are 7,500 kgs and the output units are 7,275 kgs, abnormal gain is 150 kgs.

You are required to calculate the normal loss percentage and value of abnormal gain, if the total expenses incurred in Process I are ₹ 1,50,750 and scrap has realisable value of ₹ 3 per unit.

A 4% and ₹ 3,174

B 5% and ₹ 3,200

C 5% and ₹ 3,150

D 5.10% and ₹ 3,015

ICAI'S ANSWER

Answer key: **(C)**

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650 EXAM **Jan 2025** Q3(a) 8 marks

A chemical compound manufactured through two processes namely Process X and Process Y. Process Y is dependent on the output generated by Process X and the semi-finished chemical compound received

from Process X shall be mixed up with further materials in Process Y. The details of costs and other particulars for each process are given as follows:

	Process X	Process Y
Direct Material	1,000 kgs @ ₹ 50 per kg	700 kgs @ ₹ 90 per kg
Direct Labour ₹ 35,000	₹ 35,000	₹ 25,000
Process Plant time	200 hrs @ ₹ 60/hr	120 hrs @ ₹ 80/hr
Expected output	75% of input	80% of input
Actual output kgs	700	1150
Realizable value of Normal Loss	₹ 8 per kg	₹ 5per kg

Notes:

(i) The departmental overhead for the period was 30,000 and is absorbed in each process on direct labour cost.

(ii) Process plant time represents the attributable plant run time with respect to each process and is a part of direct process cost.

(iii) Assume no finished stock and work in progress either at the beginning and end of the period.

Required:

Prepare Process X Account, Process Y Account, Normal Loss Account and Abnormal Gain Account.

ICAI'S ANSWER

Process X Account

Particulars	Kg	Amount (₹)	Particulars	Kg	Amount (₹)
To Material	1,000	50,000	By Normal Loss (250 kg × ₹8 per kg)	250	2,000
To Direct Labour		35,000	By Abnormal loss A/c (50 kg × ₹ 150 per kg)	50	7,500
To Process Plant Time 200 hrs @ ₹ 60/hr		12,000	By Process Y (700 kg × ₹ 150 per kg)	700	1,05,000
To Departmental Overhead		17,500			
	1,000	1,14,500		1,000	1,14,500

$$\text{Cost per kg} = \frac{₹1,14,500 - ₹2,000}{1,000\ \text{kg} - 250\ \text{kg}} = ₹150.00\ \text{per kg}$$

Process Y Account

Particulars	Kg	Amount (₹)	Particulars	Kg	Amount (₹)
To Process X	700	1,05,000	By Normal Loss (280 kg × ₹ 5 per kg)	280	1,400

Particulars	Kg	Amount (₹)	Particulars	Kg	Amount (₹)
To Material	700	63,000	By Finished stock (1,150 kg × ₹190.803 per kg)	1,150	2,19,424
To Direct Labour		25,000			
To Process Plant Time 120 hrs @₹ 80/hr		9,600			
To Departmental Overheads		12,500			
To Abnormal Gain A/c (30 kg × ₹ 190.803 per kg)	30	5,724			
	1,430	2,20,824		1,430	2,20,824

$$\text{Cost per kg} = \frac{₹2,15,100 - ₹1,400}{1,400\ \text{kg} - 280\ \text{kg}} = ₹190.803\ \text{per kg}$$

Normal Loss Account

Particulars	Units	Amount (₹)	Particulars	Units	Amount (₹)
To Process X A/c	250	2,000	By Cash (Sales) (250 kg × ₹8 per kg)	250	2,000
To Process Y A/c	280	1,400	By Cash (Sales) (250 kg × ₹5 per kg)	250	1,250
			By Abnormal Gain A/c (30 kg × ₹5 per kg)	30	150
	430	3,400		430	3,400

Abnormal Gain Account

Particulars	Units	Amount (₹)	Particulars	Units	Amount (₹)
To Normal Loss (30 kg × ₹ 5 per kg)	30	150	By Process Y A/c	30	5,724
To Costing Profit and Loss		5,574			
	30	5,724		30	5,724

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651 EXAM **Sep 2024** MCQ 15 MCQ

A Chemical is passed through three processes and the output of Process 1 Account is transferred to Process 2 Account. The input units in Process 1 are 58,500 units and the output units are 55,200 units, normal loss is 2% and rest is abnormal loss.

You are required to calculate the per unit cost of output units in Process 1 Account, if the total expenses incurred in Process 1 are, 6,87,960.

A ₹ 11.76

B ₹ 12.00

C ₹ 12.20

D ₹ 12.46

ICAI'S ANSWER

Answer key: **(B)**

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652 EXAM **May 2024** Q4(a) 8 marks

Meta Company Ltd. is engaged in the production of product 'Trio' which passes through two different processes Process P and Process Q. Other information obtained from books of account for the year is as follows:

Particulars	Process P	Process Q
Raw material used	10,000	---
Raw material cost per unit	₹ 80	---
Direct wages	₹ 52,000	₹ 78,000
Direct Expenses	₹ 8,600	₹ 11,100
Selling price per unit of output	₹ 130	₹ 190

Production overheads of ₹ 3,00,000 are recovered as percentage of direct wages.

Actual output of the two processes was:

P-9,200 units and Q-6,400 units. 3/4th of the output of Process P was passed on to the Process Q and the balance was sold. The entire output of process Q was sold.

Management & Selling expenses during the year were ₹ 1,70,000.

These are not allocable to the processes.

The normal loss of the two processes, calculated on the input of every process was:

Process P- 6% and Process Q-10%

The Loss of Process P was sold at ₹ 5 per unit and that of Q at ₹ 8 per unit. Assume that Process P and Process Q are not the responsibility centres.

You are required to prepare:

(i) Process P Account

(ii) Process Q Account

(iii) Abnormal Loss and Abnormal Gain Account

(iv) Costing Profit & Loss Account.

ICAI'S ANSWER

Process- P Account

Particulars	Units	Amount (₹)	Particulars	Units	Amount (₹)
To Material	10,000	8,00,000	By Normal Loss	600	3,000
To Wages		52,000	By Process Q		
To Direct Exp.		8,600	(9,200 × 3/4)	6,900	7,17,600
To Production Overheads (3,00,000 × 2/5)		1,20,000	By Costing Profit and Loss (P&L) (9,200 × 1/4)	2,300	2,39,200
			By Abnormal Loss	200	20,800
	10,000	9,80,600		10,000	9,80,600

$$\text{Cost per unit} = \frac{9,80,600 - 3,000}{10,000 - 600} = ₹\ 104 \text{ per unit}$$

Process- Q Account

Particulars	Units	Amount (₹)	Particulars	Units	Amount (₹)
To Process P	6,900	7,17,600	By Normal Loss	690	5,520
To Wages		78,000			
To Direct Exp.		11,100	By Costing P&L	6,400	10,11,200
To Production Overheads (3,00,000 × 3/5)		180,000			
To Abnormal Gain	190	30,020			
	7,090	10,16,720		7,090	10,16,720

$$\text{Cost per unit} = \frac{9,86,700 - 5,520}{6900 - 690} = ₹158 \text{ per unit}$$

Abnormal Loss Account

Particulars	Units	Amount (₹)	Particulars	Units	Amount (₹)
To Process- P	200	20,800	By Bank	200	1,000
			By Costing P&L		19,800
	200	20,800		200	20,800

Abnormal Gain Account

Particulars	Units	Amount (₹)	Particulars	Units	Amount (₹)
To Normal Loss	190	1,520	By Process Q	190	30,020
To Costing Profit and Loss		28,500			
	190	30,020		190	30,020

Costing Profit & Loss Account for the year

Dr. Cr.

Particulars	Amount (₹)	Particulars	Amount (₹)
To Cost of Sales P - 2,39,200 Q - 10,11,200	12,50,400	By Sales P 2300 @ 130 Q 6400 @ 190	15,15,000
To Abnormal Loss	19,800	By Abnormal Gain	28,500
To Selling Expense	1,70,000		
To Net Profit	1,03,300		
	15,43,500		15,43,500

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653 EXAM Old P3 Nov 2023 Q4(c) 5 marks old scheme

A product passes through two processes; Process A and Process B.

The output of Process A is treated as input of Process B.

The following information has been furnished:

	Process A	Process B
Input Material 78,000 Kg.@ ₹ 5	₹ 3,90,000	-
Indirect Material	-	₹34,320
Wages	₹ 2,85,000	₹ 3,30,000
Overhead	₹ 1,67,400	₹ 1,11,600
Output transferred to Process B	68,640 kgs	
Transfer to Finished Stock	-	69,000 kgs
Normal loss of input material (weight in kgs.)	7,800 kgs	240 kgs

There is no realisable value for normal loss. No stock of raw materials on work -in-process was left at the end.

You are required to prepare the Process account for each Process.

ICAI'S ANSWER

Process A Account

Particulars	Units	₹	Particulars	Units	₹
To Material	78,000	3,90,000	By Normal Loss	7,800	-
To Wages		2,85,000	By Abnormal Loss	1,560	18,720
To Overheads		1,67,400	By Process B A/c	68,640	8,23,680
Total	**78,000**	**8,42,400**	**Total**	**78,000**	**8,42,400**

Cost per unit of completed units and abnormal loss = $\frac{8,42,400}{78,000 \text{ units} - 7,800 \text{ units}}$ = ₹ 12 unit

Process B Account

Particulars	Units	₹	Particulars	Units	₹
To Process A A/c	68,640	8,23,680	By Normal loss	240	-
To Indirect Material		34,320	By Finished stock	69,000	13,11,000
To Wages		3,30,000			
To Overheads		1,11,600			
To Abnormal gain	600	11,400			
Total	**69,240**	**13,11,000**	**Total**	**69,240**	**13,11,000**

Cost per unit of completed units and abnormal gains:

$$\frac{\text{Total cost}}{\text{Inputs - Normal loss}} = \frac{₹12,99,600}{68,640 \text{ units - } 240 \text{ units}} = ₹19$$

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654 EXAM **Old P3 May 2023** Q6(e) 5 marks old scheme

How will you treat normal loss, abnormal loss and abnormal gain in process costing? Explain

ICAI'S ANSWER

Treatment of normal loss, abnormal loss and abnormal gain in process costing

Treatment of Normal loss in Cost Accounts: The cost of normal process loss in practice is **absorbed by good units produced under the process.** The amount realised by the sale of normal process loss units should be credited to the process account.

Treatment of Abnormal loss in Cost Accounts: The cost of an abnormal process loss unit is equal to the cost of a good unit. The total cost of abnormal process loss is credited to the process account from which it arises. Cost of abnormal process loss is not treated as a part of the cost of the product. In fact, the total cost of abnormal process loss is debited to costing profit and loss account.

Treatment of Abnormal Gain in Cost Accounts: The **process account** under which abnormal gain arises is **debited with the abnormal gain and credited to abnormal gain account** which will be

closed by transferring to the Costing Profit and Loss account. The cost of **abnormal gain is computed on the basis of normal production.**

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655 EXAM **Old P3 Nov 2022** Q3(b) 10 marks old scheme

N Ltd. produces a product which passes through two processes – Process – I and Process-II. The company has provided following information related to the Financial Year 2021-22:

	Process-I	Process -II
Raw Material @ ₹ 65 per unit	6,500 units	-
Direct Wages	₹ 1,40,000	₹ 1,30,000
Direct Expenses	30% of Direct Wages	35% of Direct Wages
Manufacturing Overheads	₹ 21,500	₹ 24,500
Realisable value of scrap per unit	₹ 4.00	₹ 16.00
Normal Loss	250 units	500 units
Units transferred to Process-II / finished stock	6,000 units	5,500 units
Sales	-	5,000 units

There was no opening or closing stock of work-in progress.

You are required to prepare:

(i) Process-I Account

(ii) Process -II Account

(iii) Finished Stock Account

ICAI'S ANSWER

Process-I A/c

Particulars	Units	(₹)	Particulars	Units	(₹)
To Raw material used (₹ 65 × 6,500 units)	6,500	4,22,500	By Normal loss (250 units × ₹ 4)	250	1,000
To Direct wages	--	1,40,000	By Process- II A/c (₹ 100 × 6,000 units)	6,000	6,00,000
To Direct expenses (30% of ₹ 1,40,000)	--	42,000	By Abnormal loss (₹ 100 × 250 units)	250	25,000
To Manufacturing overhead		21,500			
	6,500	6,26,000		6,500	6,26,000

Cost per unit of completed units and abnormal loss: $\frac{\text{Total Cost-Realisable value from normal loss}}{\text{Inputs Units-Normal loss units}}$

$$= \frac{₹\ 6{,}26{,}000 - ₹\ 1{,}000}{6{,}500\text{units} - 250\ \text{units}} = \frac{₹\ 6{,}25{,}000}{6{,}250\ \text{units}} = ₹\ 100$$

Process- II A/c

Particulars	Units	(₹)	Particulars	Units	(₹)
To Process - I A/c	6,000	6,00,000	By Normal loss (500 units × ₹16)	500	**8,000**
To Direct wages	--	1,30,000	By Finished Stock A/c (₹144 × 5,500 units)	5,500	7,92,000
To Direct expenses (35% of ₹ 1,30,000)	--	45,500			
To Manufacturing overhead	--	24,500			
	6,000	8,00,000		6,000	8,00,000

Cost per unit of completed units and abnormal loss:

$$\frac{\text{TotalCost - Realisable value from normal loss}}{\text{Inputsunits - Normallossunits}}$$

$$= \frac{₹\ 8{,}00{,}000 - ₹\ 8{,}000}{6{,}000\ \text{units} - 500\ \text{units}} = \frac{₹\ 7{,}92{,}000}{5{,}500\ \text{units}} = ₹144$$

Finished Goods Stock A/c

Particulars	Units	(₹)	Particulars	Units	(₹)
To Process II A/c	5,500	7,92,000	By Cost of Sales (₹144 × 5,000 units)	5,000	7,20,000
			By Balance c/d	500	72,000
	5,500	7,92,000		5,500	7,92,000

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656 EXAM **Old P3 May 2022** Q2(c) 5 marks old scheme

Distinguish between Job costing and Process Costing. (Any five points of differences)

ICAI'S ANSWER

	Job Costing	Process Costing
(i)	A Job is carried out or a product is produced by specific orders.	The process of producing the product has a continuous flow and the product produced is homogeneous.
(ii)	Costs are determined for each job.	Costs are compiled on time basis i.e., for production of a given accounting period for each process or department.
(iii)	Each job is separate and independent of other jobs.	Products lose their individual identity as they are manufactured in a continuous flow.

	Job Costing	Process Costing
(iv)	Each job or order has a number and costs are collected against the same job number.	The unit cost of process is an average cost for the period.
(v)	Costs are computed when a job is completed. The cost of a job may be determined by adding all costs against the job.	Costs are calculated at the end of the cost period. The unit cost of a process may be computed by dividing the total cost for the period by the output of the process during that period.
(vi)	As production is not continuous and each job may be different, so more managerial attention is required for effective control.	Process of production is usually standardized and is therefore, quite stable. Hence control here is comparatively easier.

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657 EXAM **Old P3 Dec 2021** Q1(d) 5 marks old scheme

A product passes through Process-I and Process-II.

Particulars pertaining to the Process-I are:

Materials issued to Process-I amounted to ₹ 80,000, Wages ₹ 60,000 and manufacturing overheads were ₹ 52,500. Normal Loss anticipated was 5% of input, 9,650 units of output were produced and transferred out from Process-I to Process-II. Input raw materials issued to Process-I were 10,000 units.

There were no opening stocks.

Scrap has realizable value of ₹ 5 per unit.

You are required to prepare:

(i) Process-I Account

(ii) Abnormal Gain/Loss Account

ICAI'S ANSWER

(i) Process - I Account

Particulars	Units	(₹)	Particulars	Units	(₹)
To Materials	10,000	80,000	By Normal loss (5%of 10,000)	500	2,500
To Wages	-	60,000	By Process-II A/c (₹20*×9,650units)	9,650	1,93,000
To Manufacturing OH		52,500			
To Abnormal Gain A/c (₹20*×150units)	150	3,000			
	10,150	1,95,500		10,150	1,95,500

* $\frac{(80{,}000 + 60{,}000 + 52{,}500) - 2{,}500}{10{,}000 - 500} = ₹\ 20$

(ii) Abnormal Gain - Account

Particulars	Units	(₹)	Particulars	Units	(₹)
To Normal loss A/c	150	750	By Process-I A/c	150	3,000
To Costing P&L A/c	-	2,250			
	150	3,000		150	3,000

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658 EXAM **Old P3 Jul 2021** Q4(a) 10 marks old scheme

A Manufacturing unit manufactures a product 'XYZ' which passes through three distinct Processes - X, Y and Z. The following data is given:

	Process X	Process Y	Process Z
Material consumed (in ₹)	2,600	2,250	2,000
Direct wages (in ₹)	4,000	3,500	3,000

- The total Production Overhead of ₹ 15,750 was recovered @ 150% of Direct wages.
- 15,000 units at ₹ 2 each were introduced to Process 'X'.
- The output of each process passes to the next process and finally, 12,000 units were transferred to Finished Stock Account from Process 'Z'.
- No stock of materials or work in progress was left at the end.

The following additional information is given:

Process	% of wastage to normal input	Value of Scrap per unit (₹)
X	6%	1.10
Y	?	2.00
Z	5%	1.00

You are required to:

(i) Find out the percentage of wastage in process 'Y', given that the output of Process 'Y' is transferred to Process 'Z' at ₹ 4 per unit.

(ii) Prepare Process accounts for all the three processes X, Y and Z.

ICAI'S ANSWER

Dr. Process-X Account Cr.

Particulars	Units	(₹)	Particulars	Units	(₹)
To Material introduced	15,000	30,000	By Normal Loss A/c [(6% of 15,000 units) x ₹ 1.1]	900	990
" Additional material	--	2,600	" Process-Y A/c (₹ 2.951* × 14,100 units)	14,100	41,610
" Direct wages	--	4,000			
" Production OH	--	6,000			
	15,000	42,600		15,000	42,600

*Cost per unit of completed units

$$= \frac{\text{Total Cost} - \text{Realisable value from normal loss}}{\text{Inputs units} - \text{Normal loss units}} = \frac{₹\ 42,600 - ₹\ 990}{15,000\ \text{units} - 900\ \text{units}} = ₹\ 2.951$$

Dr. Process-Y Account Cr.

Particulars	Units	(₹)	Particulars	Units	(₹)
To Process-X A/c	14,100	41,610	By Normal Loss A/c [(#13.44% of 14,100 units) x ₹ 2]	1,895	3,790
" Additional material	--	2,250	" Process-Z A/c (₹ 4 × 12,205 units)	12,205	48,820
" Direct wages	--	3,500			
" Production OH	--	5,250			
	14,100	52,610		14,100	52,610

#Calculation for % of wastage in process 'Y':

Let's consider number of units lost under process 'Y' = A

Now, $\frac{\text{Total Cost - Realisable value from normal loss}}{\text{Inputs units - Normal loss units}} = 4$

$\frac{₹\ 52,610 - ₹\ 2A}{14,100\ \text{units - A}} = ₹\ 4$

₹ 52,610 - ₹ 2A = ₹ 56,400 - ₹ 4A

2A = ₹ 3,790 => A = 1,895 units

% of wastage = $\frac{1,895\ \text{units}}{14,100\ \text{units}}$ = **13.44%**

Dr. Process-Z Account Cr.

Particulars	Units	(₹)	Particulars	Units	(₹)
To Process-Y A/c	12,205	48,820	By Normal Loss A/c [(5% of 12,205 units) x ₹ 1]	610	610
″ Additional material	--	2,000	″ Finished Stock A/c (₹ 4.9771$ × 12,000 units)	12,000	59,726
″ Direct wages	--	3,000			
″ Production OH	--	4,500			
″ Abnormal gain (₹ 4.9771$ × 405 units)	405	2,016			
	12,610	60,336		12,610	60,336

$Cost per unit of completed units

$$= \frac{\text{Total Cost} - \text{Realisable value from normal loss}}{\text{Inputs units} - \text{Normal loss units}} = \frac{₹\ 58,320 - ₹\ 610}{12,205 \text{ units} - 610 \text{ units}} = ₹\ 4.9771$$

Alternative Solution

Dr. Process-X Account Cr.

Particulars	Units	(₹)	Particulars	Units	(₹)
To Material introduced	15,000	30,000	By Normal Loss A/c [(6% of 15,000 units) x ₹ 1.1]	900	990
″ Additional material	--	2,600	″ Process-Y A/c (₹ 2.951* × 14,100 units)	14,100	41,610
″ Direct wages	--	4,000			
″ Production OH	--	6,000			
	15,000	**42,600**		**15,000**	**42,600**

*Cost per unit of completed units

$$= \frac{\text{Total Cost} - \text{Realisable value from normal loss}}{\text{Inputs units} - \text{Normal loss units}} = \frac{₹\ 42,600 - ₹\ 990}{15,000 \text{ units} - 900 \text{ units}} = ₹\ 2.951$$

Dr. Process-Y Account Cr.

Particulars	Units	(₹)	Particulars	Units	(₹)
To Process-X A/c	14,100	41,610	By Normal Loss A/c [(#13.44% of 14,100 units) x ₹ 2]	1,895	3,790
" Additional material	--	2,250	" Process-Z A/c (₹ 4 × 12,631@ units)	12,631	50,524
" Direct wages	--	3,500			
" Production OH	--	5,250			
" Abnormal gain (₹ 4 × 426 units)	426	1,704			
	14,526	54,314		14,526	54,314

Working Notes:

@1. Units Transferred from Process Z Account to Finished Stock = 12,000 Units i.e 95% of Inputs.

So, Input of Z or Output of Y is 12,000 x 100/95 = 12,631 Units and Normal Loss (5%) is 631 units.

2. Let's consider number of units lost under process 'Y' as:

For Normal loss = A

For Abnormal loss = B

Now, A + B = 1,469 [i.e. 14,100 – 12,631] ...(I)

(A x ₹ 2 per unit) + (B x ₹ 4 per unit) = [52,610 – 50,524]

2A + 4B = 2,086 ...(II)

Now, putting the values of (I) in (II), we get,

2(1,469 – B) + 4B = 2,086

2938 – 2B + 4B = 2,086

2B = - 852 => B = - 426 units

Since, the figure of B is in negative, it is an abnormal gain of 426 units.

Further, A (i.e. normal loss) = 1,469 + 426 = 1,895 units

#3. % of wastage in Process Y Account = $\frac{1,895 \text{ units}}{14,100 \text{ units}}$ = **13.44%**

Dr. Process-Z Account Cr.

Particulars	Units	(₹)	Particulars	Units	(₹)
To Process-Y A/c	12,631	50,524	By Normal Loss A/c [(5% of 12,631 units) x ₹ 1]	631	631
" Additional material	--	2,000			
" Direct wages	--	3,000			
" Production OH	--	4,500	" Finished Stock A/c (₹ 4.9494**$** × 12,000 units)	12,000	59,393
	12,631	60,024		12,631	60,024

$Cost per unit of completed units

$$= \frac{\text{Total Cost - Realisable value from normal loss}}{\text{Inputs units - Normal loss units}} = \frac{₹\ 60{,}024 - ₹\ 631}{12{,}631 \text{ units} - 631 \text{ units}} = ₹\ 4.9494$$

ICAI · Old P3 · Jul 2021 exam · question p. 68-69 · answer p. 70-73

659 EXAM **Old P3 Nov 2019** Q4(b) 10 marks old scheme

A product passes through two distinct processes before completion.

Following information are available in this respect :

	Process-1	Process-2
Raw materials used	10,000 units	-
Raw material cost (per unit)	₹ 75	-
Transfer to next process/Finished good	9,000 units	8,200 units
Normal loss (on inputs)	5%	10%
Direct wages	₹ 3,00,000	₹ 5,60,000
Direct expenses	50% of direct wages	65% of direct wages
Manufacturing overheads	25% of direct wages	15% of direct wages
Realisable value of scrap (per unit)	₹ 13.50	₹ 145

8,000 units of finished goods were sold at a profit of 15% on cost. There was no opening and closing stock of work-in-progress.

Prepare:

(i) Process-1 and Process-2 Account

(ii) Finished goods Account

(iii) Normal Loss Account

(iv) Abnormal Loss Account

(v) Abnormal Gain Account.

ICAI'S ANSWER

(i)

Dr. **Process-1 Account** Cr.

	Particulars	Units	Total (₹)		Particulars	Units	Total (₹)
To	Raw Material Consumed	10,000	7,50,000	By	Normal Loss A/c @ 13.5	500	6,750
"	Direct Wages	--	3,00,000	"	Process 2 @ 133.5	9,000	12,01,500
"	Direct Expenses	--	1,50,000	"	By Abnormal Loss @ 133.5	500	66,750
"	Manufacturing Overheads		75,000				
		10,000	12,75,000			10,000	12,75,000

Cost per unit of completed units and abnormal loss:

$$= \frac{₹\ 12{,}75{,}000 - ₹\ 6{,}750}{10{,}000\text{units} - 500\text{units}} = \textbf{₹ 133.5}$$

(ii)

Dr. **Process-2 Account** Cr.

	Particulars	Units	Total (₹)		Particulars	Units	Total (₹)
To	Process-I A/c	9,000	12,01500	By	Normal Loss A/c @ 145	900	1,30,500
"	To Direct Wages	--	5,60,000	"	By Finished Stock A/c [bal fig]	8,200	21,04,667
"	Direct Expenses	--	3,64,000				
"	Manufacturing Overheads	--	84,000				
"	To Abnormal gain (₹ 256.67 × 100 units)	100	25,667				
		9,100	22,35,167			9,100	22,35,167

Cost per unit of completed units and abnormal gain:

$$= \frac{₹\ 22{,}09{,}500 - ₹\ 130500}{8{,}100\text{units}} = \textbf{₹ 256.67}$$

Dr. **Finished Goods A/c** Cr.

	Particulars	Units	Total (₹)		Particulars	Units	Total (₹)
To	Process II A/c	8,200	21,04,667	By	By Cost of Sales	8,000	20,53,333
				"	By Balance c/d	200	51,334
		8,200	**21,04,667**			**8,200**	**21,04,667**

(iii) Normal Loss A/c

Dr. Cr.

	Particulars	Units	Total (₹)		Particulars	Units	Total (₹)
To	Process I	500	6,750	By	By abnormal Gain II	100	14,500
	Process II	900	1,30,500		By Cash	500	6,750
					By Cash	800	1,16,000
		1400	1,37,250			1400	1,37,250

(iv) Abnormal Loss A/c

Dr. Cr.

	Particulars	Units	Total (₹)		Particulars	Units	Total (₹)
To	Process I	500	66,750	By	By Cost Ledger Control A/c	500	6,750
					By Costing P& L A/C (Abnormal Loss)		60,000
			66,750				66,750

(v) Abnormal Gain A/c

Dr. Cr.

	Particulars	Units	Total (₹)		Particulars	Units	Total (₹)
To	Normal Loss A/c @ 145	100	14,500	By	Process II	100	25,667
To	Costing P & L A/C		11,167				
		100	25,667			100	25,667

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660 RTP **Sep 2025** Q13

A product passes through three processes – A, B and C. The details of expenses incurred on the three

processes during the year 2025 were as under:

Process	A	B	C
Units issued / introduced cost per unit ₹ 100	10,000		
	₹	₹	₹
Sundry Materials	10,000	15,000	5,000
Labour	30,000	80,000	65,000
Direct Expenses	6,000	18,150	27,200
Selling price per unit of output	120	165	250

Management expenses during the year were ₹ 80,000 and selling expenses were ₹ 50,000 these are not allocable to the processes.

Actual output of the three processes was:

A – 9,300 units, B-5,400 units and C-2,100 units. Two third of the output of Process A and one half of the output of Process B was passed on to the next process and the balance was sold. The entire output of process C was sold.

The normal loss of the three processes, calculated on the input of every process was:

Process A-5%; B-15% and C-20%

The Loss of Process A was sold at ₹ 2 per unit, that of B at ₹ 5 per unit and of Process C at ₹ 10 per unit.

Prepare the Three Processes Accounts and the Profit and Loss Account.

ICAI'S ANSWER

Process A Account

Dr.					Cr.
Particulars	**Units**	**₹**	**Particulars**	**Units**	**₹**
To Units brought in (₹ 100 × 10,000)	10,000	10,00,000	By Normal Loss (5% of 10,000 units @ ₹ 2/- p.u.)	500	1,000
To Sundry Materials		10,000	By Abnormal loss (Working note 1)	200	22,000
To Labour		30,000	Process B A/c (Output to be transferred ₹ 110 × 6,200) (W. Note 1)	6,200	6,82,000
To Direct expenses		6,000	By Profit & Loss A/c (₹ 110 × 3,100 units) (W. Note 1)	3,100	3,41,000
	10,000	10,46,000		10,000	10,46,000

Process B Account

Dr. Cr.

Particulars	Units	₹	Particulars	Units	₹
To Process A A/c	6,200	6,82,000	By Normal Loss (15% of 6,200 Units = 930 units @ ₹ 5 p.u.)	930	4,650
To Sundry Materials		15,000	By Process C A/c (Output to be transferred) ₹ 150 × 2,700 (Working Note 2)	2,700	4,05,000
To Labour		80,000	By Profit & Loss A/c (₹ 150 × 2,700)	2,700	4,05,000
To Direct expenses		18,150			
To Abnormal gain (Working Note 2)	130	19,500			
	6,330	8,14,650		6,330	8,14,650

Process C Account

Dr. Cr.

Particulars	Units	₹	Particulars	Units	₹
To Process B A/c	2,700	4,05,000	By Normal Loss (20% of 2,700 units = 540 units @ ₹ 10/- p.u.)	540	5,400
To Sundry Materials		5,000	By Abnormal Loss (Working Note 3)	60	13,800
To Labour		65,000	By Profit & Loss A/c (₹ 230 × 2,100 units) (Working Note 3)	2,100	4,83,000
To Direct expenses		27,200			
	2,700	5,02,200		2,700	5,02,200

Profit & Loss Account

Dr. Cr.

Particulars	Units	₹	Particulars	Units	₹
To Process A A/c	3,100	3,41,000	By Sale (Process A's Output @ ₹ 120/- p.m.)	3,100	3,72,000
To Process B A/c	2,700	4,05,000	By Sale (Process B's Output @ ₹ 165/- p.u.)	2,700	4,45,500
To Process C A/c	2,100	4,83,000	By Sale (Process C's Output @ ₹ 250/- p.u.)	2,100	5,25,000

Dr.				Cr.
To Management Expenses	80,000	By Abnormal gain A/c (Working Note 5)		18,850
To Selling Expenses	50,000	By Net Loss		32,450
To Abnormal Loss A/c (Working Note 4)	34,800			
	7,900 13,93,800		7,900	13,93,800

Working Notes

1. (i) Per unit cost of normal production under process A:

$$= \frac{\text{Normal cost of normal output}}{\text{Normal production output}}$$

$$= \frac{₹\ 10{,}46{,}000 - ₹\ 1{,}000}{9{,}500\ \text{units}} = ₹\ 110$$

(ii) Value of Abnormal loss under process A:

Abnormal loss units = Normal production – Actual production

= 9,500 – 9,300 = 200 units

Value of Abnormal Loss

= Per unit cost of normal production × Abnormal loss units

= ₹ 110 × 200 – ₹ 22,000.

2. (i) Per unit cost of normal production under process B:

$$= \frac{(₹\ 7{,}95{,}150 - ₹\ 4{,}650)}{5{,}270} = \frac{₹\ 7{,}90{,}500}{5{,}270} = ₹\ 150$$

(ii) Value of Abnormal gain under process B:

Abnormal gain units = Normal loss – Actual loss

= 930 – 800 = 130 units

= Per unit cost of normal production × Abnormal gain units

= ₹ 150 × 130 units = ₹ 19,500.

3. (i) Per unit cost of normal production under process C:

$$= \frac{(₹\ 5{,}02{,}200 - ₹\ 5{,}400)}{2{,}160\text{units}} = \frac{₹\ 4{,}96{,}800}{2{,}160\ \text{units}} = ₹\ 230$$

(ii) Value of Abnormal loss under process C:

Abnormal loss units

= Normal production – Actual production

= 2,160 units – 2,100 units = 60 units

= ₹ 230 × 60 units = ₹ 13,800

4. **Abnormal Loss Account**

Dr.						Cr.
	Units	**Cost p.u. ₹**	**Amount ₹**	**Particulars**	**Units**	**Cost p.u. ₹**	**Amount ₹**
To Process A A/c	200	110	22,000	By Sale proceeds of Process A Loss	200	2	400
To Process C A/c	60	230	13,800	By Sale proceeds of Process C loss	60	10	600
				By Profit & Loss A/c			34,800
	260		**35,800**		**260**		**35,800**

5. **Abnormal Gain Account**

Dr.						Cr.
	Units	**Cost p.u. ₹**	**Amount ₹**	**Particulars**	**Units**	**Cost p.u. ₹**	**Amount ₹**
To Normal loss shortfall	130	5	650	By Process B	130	150	19,500
To Profit & Loss A/c			18,850				
			19,500				19,500

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661 RTP **May 2025** Q5 MCQ

A product passes through two processes. The output of Process-I is treated as the raw material of Process-II. The cost incurred at Process- II is as follows:

Particulars	Process-II (₹)
Transferred from Process-I A/c	27,85,700
Materials issued	10,00,000
Labour	2,00,000
Manufacturing overhead	5,00,000

The output of each process is as under:

Process	Output	Normal Loss
Process-I	48,750 units	2%
Process-II	47,000 units	5%

No stock of materials or of work-in-process was left at the end.

You are required to CALCULATE the value of Abnormal Gain/Loss in Process-II A/c.

A Abnormal Gain of ₹ 66,626

B Abnormal Loss of ₹ 66,626

C Abnormal Gain of ₹ 72,776

D Abnormal Loss of ₹ 72,776

ICAI'S ANSWER

(A)

Dr. **Process-II Account** Cr.

Particulars	Units	Total (₹)	Particulars	Units	Total (₹)
To Process-I A/c	48,750	27,85,700	By Normal Loss A/c (5% of 48,750 units)	2,438	--
" Material	--	10,00,000	" Output (₹ 96.858 × 47,000 units)	47,000	45,52,326
" Labour	--	2,00,000			
" Manufacturing OH	--	5,00,000			
" Abnormal Gain A/c (₹ 96.858 × 688 units)	688	**66,626** (round off)			
	49,438	45,52,326		49,438	45,52,326

Cost per unit of completed units and abnormal gain:

$$= \left(\frac{\text{Total Cost}}{\text{Input units - Normal Loss}}\right)$$

$$= \left(\frac{₹\ 44,85,700}{48,750 \text{ units - } 2,438 \text{ units}}\right)$$

= ₹ 96.858

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662 RTP **Jan 2025** Q19(d)

DISCUSS normal and abnormal Process Loss and ENUMERATE their treatment in Cost Accounts.

ICAI'S ANSWER

There are two types of material losses viz. (i) Normal loss and (ii) Abnormal loss.

(i) Normal Process Loss: It is also known as normal wastage. It is defined as **the loss of material which is inherent in the nature** of work. Such a loss can be reasonably anticipated from the nature of the material, nature of operation, the experience and technical data. It is unavoidable because of nature of the material or the process. It also includes units withdrawn from the process for test or sampling.

Treatment in Cost Accounts: The cost of **normal process loss in practice is absorbed by good units produced** under the process. The amount realised by the sale of normal process loss units should be credited to the process account.

(ii) Abnormal Process Loss: It is also known as abnormal wastage. It is defined as the **loss in excess of the pre-determined loss** (Normal process loss). This type of loss may occur due to the carelessness of workers, a bad plant design or operation, sabotage etc. Such a loss cannot obviously be estimated in advance. But it can be kept under control by taking suitable measures.

Treatment in Cost Accounts: The cost of an abnormal process loss unit is equal to the cost of a good unit. The total cost of abnormal process loss is credited to the process account from which it arises. Cost of abnormal process loss is not treated as a part of the cost of the product. In fact, **the total cost of abnormal process loss is debited to costing profit and loss account.**

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663 RTP Sep 2024 Q4 MCQ

The following information is available in respect of Process I: Raw material purchased and introduced 10,000 units @ 5 per unit Raw Material received from store 4000 units @ 6 per unit Direct Labour 40,000 Overheads 28,000 Output of Process is 13,500 units, Normal wastage 5% of inputs Scrap value of wastage 4 per unit The value of Abnormal Gain is:

A ₹ 2062.68

B ₹ 2135.34

C ₹ 2103.70

D ₹ 2093.2

ICAI'S ANSWER

(D) ₹ 2093.2

Process a/c

Particulars	Units	Amount	Particulars	units	Amount
Raw material	10,000	50,000	Normal loss	700	2,800
Stores	4,000	24,000	Units transferred	13,500	1,41,293.2
Direct Wages		40,000			
Production overheads		28,000			
Abnormal gain	200	2,093.2			
		1,44,093.2			1,44,093.2

Cost per unit= $\frac{1,42,000 - 2,800}{14,000 - 700}$ = 10.466 per unit

ICAI · P4 · RTP Sep 2024 · question p. 5 · answer p. 20

664 RTP Old P3 Nov 2022 Q9 old scheme

SM Pvt. Ltd. manufactures their products in three consecutive processes. The details are as below:

	Process A	Process B	Process C
Transferred to next Process	60%	50%	
Transferred to warehouse for sale	40%	50%	100%

In each process, there is a weight loss of 2% and scrap of 8% of input of each process. The realizable value of scrap of each process is as below:

Process A @ ₹ 2 per ton

Process B @ ₹ 4 per ton

Process C @ ₹ 6 per ton.

The following particulars relate to April, 2022:

	Process A	Process B	Process C
Materials used (in Tons)	1,000	260	140
Rate per ton	₹ 20	₹ 15	₹ 10
Direct Wages	₹ 4,000	₹ 3,000	₹ 2,000
Direct Expenses	₹ 3,160	₹ 2,356	₹ 1,340

PREPARE Process Accounts- A, B and C & calculate cost per ton at each process.

ICAI'S ANSWER

Process A Account

Particulars	Tones	Amount (₹)	Particulars	Tones	Amount (₹)
To Materials	1,000	20,000	By Weight Loss	20	---
To Wages		4,000	By Scrap	80	160
To Direct Expenses		3,160	By Process B	540	16,200
			By Warehouse	360	10,800
Total	1,000	27,160	Total	1,000	27,160

$$\text{Cost per Tonne} = \frac{27{,}160 - 160}{1{,}000 - 20 - 80}$$

$$= \frac{27{,}000}{900}$$

= ₹ 30 per ton

Process B Account

Particulars	Tones	Amount (₹)	Particulars	Tones	Amount (₹)
To Process A	540	16,200	By Weight Loss	16	---
To Materials	260	3,900	By Scrap	64	256
To Wages		3,000	By Process C	360	12,600
To Direct Expenses		2,356	By Warehouse	360	12,600
Total	800	25,456	Total	800	25,456

$$\text{Cost per Tonne} = \frac{25{,}456 - 256}{800 - 16 - 64}$$

$$= \frac{25{,}200}{720}$$

$$= ₹35 \text{ per ton}$$

Process C Account

Particulars	Tones	Amount (₹)	Particulars	Tones	Amount (₹)
To Process B	360	12,600	By Weight Loss	10	---
To Materials	140	1,400	By Scrap	40	240
To Wages		2,000	By Warehouse	450	17,100
To Direct Expenses		1,340			
Total	500	17,340	Total	500	17,340

$$\text{Cost per Tonne} = \frac{17{,}340 - 240}{500 - 10 - 40}$$

$$= \frac{17{,}100}{450}$$

$$= ₹\ 38 \text{ per ton}$$

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665 RTP **Old P3 Nov 2020** Q9 old scheme

M Ltd. produces a product-X, which passes through three processes, I, II and III. In Process-III a by-product arises, which after further processing at a cost of ₹85 per unit, product Z is produced. The information related for the month of August 2020 is as follows:

	Process-I	Process-II	Process-III
Normal loss	5%	10%	5%
Materials introduced (7,000 units)	1,40,000	-	-
Other materials added	62,000	1,36,000	84,200
Direct wages	42,000	54,000	48,000
Direct expenses	14,000	16,000	14,000

Production overhead for the month is ₹2,88,000, which is absorbed as a percentage of direct wages.

The scrapes are sold at ₹10 per unit

Product-Z can be sold at ₹135 per unit with a selling cost of ₹15 per unit

No. of units produced:

Process-I- 6,600; Process-II- 5,200, Process-III- 4,800 and Product-Z- 600

There is not stock at the beginning and end of the month.

You are required to PREPARE accounts for:

(i) Process-I, II and III

(ii) By-product process.

ICAI'S ANSWER

(i) Process-I A/c

Particulars	Units	Amt.(₹)	Particulars	Units	Amt.(₹)
To Materials	7,000	1,40,000	By Normal loss (5% of 7,000)	350	3,500
To Other materials	-	62,000	By Process-II*	6,600	3,35,955
To Direct wages	-	42,000	By Abnormal loss*	50	2,545
To Direct expenses	-	14,000			
To Production OH (200% of ₹42,000)	-	84,000			
	7,000	3,42,000		7,000	3,42,000

$* \frac{₹(3,42,000 - 3,500)}{(7,000 - 350)\text{units}} = ₹50.9022$

Process-II A/c

Particulars	Units	Amt.(₹)	Particulars	Units	Amt.(₹)
To Process-I A/c	6,600	3,35,955	By Normal loss (10% of 6,600)	660	6,600
To Other materials	-	1,36,000	By Process-III**	5,200	5,63,206
To Direct wages	-	54,000	By Abnormal loss**	740	80,149
To Direct expenses	-	16,000			
To Production OH (200% of ₹54,000)	-	1,08,000			
	6,600	6,49,955		6,600	6,49,955

$** \frac{₹(6,49,955 - 6,600)}{(6,600 - 660)\text{units}} = ₹108.3089$

Process-III A/c

Particulars	Units	Amt.(₹)	Particulars	Units	Amt.(₹)
To Process-I A/c	5,200	5,63,206	By Normal loss (5% of 5,200)	260	2,600
To Other materials	-	84,200	By Product-X***	4,800	8,64,670
To Direct wages	-	48,000			
To Direct expenses	-	14,000	By Product-Z# (₹35×600)	600	21,000
To Production OH (200% of ₹48,000)	-	96,000			
To Abnormal gain***	460	82,864			
	5,660	8,88,270		5,660	8,88,270

*** $\frac{₹(8,05,406 - 2,600 - 21,000)}{(5,200 - 260 - 600)\text{units}} = ₹180.1396$

Realisable value = ₹135 – (85+15) = ₹35

(ii) By-Product Process A/c

Particulars	Units	Amt.(₹)	Particulars	Units	Amt.(₹)
To Process-III A/c	600	21,000	By Product-Z	600	81,000
To Processing cost	-	51,000			
To Selling expenses	-	9,000			
	600	81,000		600	81,000

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666 RTP **Old P3 May 2020** Q8 old scheme

Star Ltd. manufactures chemical solutions for the food processing industry. The manufacturing takes place in a number of processes and the company uses FIFO method to value work-in-process and finished goods. At the end of the last month, a fire occurred in the factory and destroyed some of papers containing records of the process operations for the month.

Star Ltd. needs your help to prepare the process accounts for the month during which the fire occurred. You have been able to gather some information about the month's operating activities but some of the information could not be retrieved due to the damage. The following information was salvaged:

- Opening work-in-process at the beginning of the month was 1,600 litres, 70% complete for labour and 60% complete for overheads. Opening work-in-process was valued at ₹ 1,06,560.
- Closing work-in-process at the end of the month was 320 litres, 30% complete for labour and 20% complete for overheads.

- Normal loss is 10% of input and total losses during the month were 1,200 litres partly due to the fire damage.
- Output sent to finished goods warehouse was 8,400 litres.
- Losses have a scrap value of ₹15 per litre.
- All raw materials are added at the commencement of the process.
- The cost per equivalent unit (litre) is ₹78 for the month made up as follows:

	(₹)
Raw Material	46
Labour	14
Overheads	18
	78

Required:

(i) CALCULATE the quantity (in litres) of raw material inputs during the month.

(ii) CALCULATE the quantity (in litres) of normal loss expected from the process and the quantity (in litres) of abnormal loss / gain experienced in the month.

(iii) CALCULATE the values of raw material, labour and overheads added to the process during the month.

(iv) PREPARE the process account for the month.

ICAI'S ANSWER

(i) Calculation of Raw Material inputs during the month:

Quantities Entering Process	Litres	Quantities Leaving Process	Litres
Opening WIP	1,600	Transfer to Finished Goods	8,400
Raw material input (balancing figure)	8,320	Process Losses	1,200
		Closing WIP	320
	9,920		9,920

(ii) Calculation of Normal Loss and Abnormal Loss/Gain

	Litres
Total process losses for month	1,200
Normal Loss (10% input)	832
Abnormal Loss (balancing figure)	368

(iii) Calculation of values of Raw Material, Labour and Overheads added to the process:

	Material	Labour	Overheads
Cost per equivalent unit	₹46.00	₹14.00	₹18.00
Equivalent units (litre) (refer the working note)	7,488	7,744	7,872
Cost of equivalent units	₹3,44,448	₹1,08,416	₹1,41,696
Add: Scrap value of normal loss (832 units × ₹15)	₹12,480	--	--
Total value added	₹3,56,928	₹1,08,416	₹1,41,696

Workings:

Statement of Equivalent Units (litre):

Input Details	Units	Output details	Units	Equivalent Production					
				Material		Labour		Overheads	
				Units	(%)	Units	(%)	Units	(%)
Opening WIP	1,600	Units completed:							
Units introduced	8,320	- Opening WIP	1,600	--	--	480	30	640	40
		- Fresh inputs	6,800	6,800	100	6,800	100	6,800	100
		Normal loss	832	--	--	--	--	--	--
		Abnormal loss	368	368	100	368	100	368	100
		Closing WIP	320	320	100	96	30	64	20
	9,920		9,920	7,488		7,744		7,872	

(iv) Process Account for the month

	Litres	Amount (₹)		Litres	Amount (₹)
To Opening WIP	1,600	1,06,560	By Finished goods [8400 x ₹ 78]	8,400	6,55,200
To Raw Materials	8,320	3,56,928	By Normal loss [832 x ₹ 15]	832	12,480
To Wages	--	1,08,416	By Abnormal loss [368 x ₹ 78]	368	28,704
To Overheads	--	1,41,696	By Closing WIP [(320 x ₹ 46) + (320 x .30 x ₹ 14) + (320 x .20 x ₹ 18)]	320	17,216
	9,920	7,13,600		9,920	7,13,600

667 MTP **May 2026 S-I Q4(b)** 7 marks

AquaPaints Ltd. is engaged in the manufacturing of water-based paints for the construction industry. The production is carried out in a continuous process, and the company uses the FIFO method to value work-in-process and finished goods.

Due to a system failure during the previous month, part of the production data was lost. However, some key information was recovered, and the management has approached you to help prepare the process account for the affected month.

The following details were retrieved:

- Opening work-in-process: 1,200 litres, 50% complete for mixing labour, and 40% complete for utilities overheads. This WIP was valued at ₹ 36,000.
- Closing work-in-process: 250 litres, 30% complete for mixing labour, and 20% complete for utilities overheads.
- Normal process loss: 6% of total input. Actual loss during the month was 1,500 litres, partly due to equipment malfunction.
- Finished output transferred: 5,000 litres.
- Scrap value of normal loss: ₹ 12 per litre.
- All color pigments and base chemicals are added at the start of the process.
- The cost per equivalent litre for the month was:

Cost Element	Rate (₹)
Base Chemicals	21.00
Mixing Labour	6.00
Utilities Overheads	8.00
Total	35.00

(a) DETERMINE the quantity (in litres) of base chemicals input during the month.

(b) CALCULATE the normal loss and abnormal loss or gain for the period.

(c) COMPUTE the total cost of base chemicals, mixing labour, and utilities overheads added during the month.

(d) PREPARE the Process Account for the month.

ICAI'S ANSWER

(a) Calculation of Base Chemicals Input During the Month:

Quantities Entering Process	Litres	Quantities Leaving Process	Litres
Opening WIP	1,200	Transfer to Finished Goods	5,000
Base Chemicals Input (Bal. Fig.)	5,550	Process Losses	1,500
		Closing WIP	250
Total Input	6,750	Total Output	6,750

(b) Calculation of Normal Loss and Abnormal Loss/Gain

- **Normal Loss** = 6% of Total Input
- = 6% of 6,750 = 405 litres
- **Actual Loss** = 1,500 litres
- **Abnormal Loss** = 1,500 – 405 = 1,095 litres (Abnormal Loss)

(c) Calculation of Values of Base Chemicals, Labour, and Overheads Added

Cost Element	Rate per Litre (₹)	Equivalent Units (Ltr)	Total Cost (₹)
Base Chemicals	21	5,145	1,08,045
Mixing Labour	6	5,570	33,420
Utilities Overheads	8	5,665	45,320
Subtotal			1,86,785
Add: Scrap value of Normal Loss (405 × 12)			4,860
Total Cost Added to Process			1,91,645

Workings: Statement of Equivalent Units (Litres)

Input Details	Units	Output details	Units	Equivalent Production					
				Material		Labour		Overheads	
				Units	(%)	Units	(%)	Units	(%)
Opening WIP	1,200	Units completed:							
Units introduced	5,550	- Opening WIP	1,200	--	--	600	50	720	60
		- Fresh inputs	3,800	3,800	100	3,800	100	3,800	100
		Normal loss	405	--	--	--	--	--	--
		Abnormal loss	1,095	1,095	100	1,095	100	1,095	100

Input Details	Units	Output details	Units	Equivalent Production					
		Closing WIP	250	250	100	75	30	50	20
	6,750		6,750	5,145		5,570		5,665	

(d) Working Note:

Particulars	Equivalent Units (EU)	Cost per EU (₹)	Amount (₹)	Amount (₹)
(i) Opening W-I-P completed during the period				
Labour	600	6	3,600	
Overheads	720	8	5,760	9,360
Add: Cost of W-I-P at beginning	--	--	--	36,000
Complete cost of 1,200 litres of opening W-I-P	1,200			45,360
(ii) Cost of units introduced and completely processed units	3,800	35	1,33,000	1,33,000
Cost of 5,000 completed units (i) + (ii)	5,000			1,78,360
(iii) Abnormal Loss	1095	35	38,325	38,325
(iv) Closing W-I-P				
Material	250	21	5,250	
Labour	75	6	450	
Overheads	50	8	400	6,100

Calculate of Cost of material added:

Base chemical:	1,08,045
Add: Realisable value of Scrap:	4,200
Cost of material added	1,12,905

Process Account for the Month

Particulars	Litres	Amount (₹)	Particulars	Litres	Amount (₹)
To Opening WIP	1,200	36,000	By Finished Goods (Output)	5,000	1,78,360
To Base Chemicals	5,550	1,12,905	By Normal Loss (405 × 12)	405	4,860
To Mixing Labour	—	33,420	By Abnormal Loss (1,095 × 35)	1,095	38,325
To Utilities Overheads	—	45,320	By Closing WIP	250	6,100
Total	**6,750**	**2,27,645**	**Total**	**6,750**	**2,27,645**

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668 MTP Jan 2026 S-I MCQ 13 2 marks MCQ

For an input of 4,000 kg of material introduced in the process, the normal loss is 6%. If the actual production from the process is 3,700 kg, the abnormal loss is:

A 100 kg

B 140 kg

C 80 kg

D 60 kg

ICAI'S ANSWER

(D) 60 kg

Normal Loss = 6% of 4,000 = 240 kg

Normal Output = Input - Normal Loss

= 4,000 – 240 = 3,760 kg

Actual Output = 3,700 kg

Difference = 3,760-3,700 = 60 kg

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669 MTP Sep 2025 S-I MCQ 11 2 marks MCQ

The cost of production of 40 units in Process consisting of materials Rs 1,500; Labour ₹ 1,300 and Overhead ₹ 164. The normal waste is 5% of input.

A 40 units are transferred to next process @ ₹ 70 each

B 40 units are transferred to next process @ ₹ 74.10 each

C 38 units are transferred to next process @ ₹ 78 each

D 40 units are transferred to next process @ ₹ 78 each

ICAI'S ANSWER

Answer key: **(C)**

(c) Material cost = ₹ 1,500

Labour cost = ₹ 1,300

Overhead = ₹ 164

Total production cost = ₹ 1,500 + ₹ 1,300 + ₹ 164 = ₹ 2,964

Unit transferred = 40 – 5% of 40 = **38 units**

Value of units transferred to :

$$= \frac{\text{Total Cost - Realisable value of normal loss}}{\text{Total input units - Normal loss units}}$$

$$= \frac{₹\ 2{,}964 - ₹\ 0}{40\text{units} - 2\text{units}} = ₹\ 78$$

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670 MTP **May 2025 S-I** Q3(a) 10 marks

PR Ltd., a leading chemical company specializing in the production of specialty chemicals and industrial products, manufactures Product X through two distinct processes – Process-I and Process-II. Known for its innovation in chemical manufacturing, the company uses advanced technologies and industry best practices to ensure high-quality production. The following information is provided from the books of account for the year:

Particulars	Process- I	Process- II
Raw materials used	1,50,000 units	--
Raw materials cost per unit	₹ 400	--
Actual Output	1,38,000 units	96,000 units
Output transferred to next process	3/4th	--
Output sold	1/4th	96,000 units
Normal loss (on inputs)	6%	10%
Direct wages	₹ 39,00,000	₹ 58,50,000
Direct Expenses	20% of Direct wages	15% of Direct wages
Manufacturing overheads	₹ 2,25,00,000 to be recovered as a percentage of direct wages	
Realisable value of scrap per unit	₹ 25	₹ 40
Selling price per unit of output	₹ 650	₹ 950

Management & Selling expenses of ₹ 1,27,50,000 are directly attributable to the Costing Profit & Loss Account.

You are required to PREPARE the following:

(i) Process-I Account

(ii) Process-II Account

(iii) Abnormal Loss

(iv) Abnormal Gain Account

(v) Costing Profit & Loss Account

ICAI'S ANSWER

Process- I A/c

Particulars	Units	(₹)	Particulars	Units	(₹)
To Raw material used (₹ 400 × 1,50,000 units)	1,50,000	6,00,00,000	By Normal loss (6% of 1,50,000 units) × ₹ 25	9,000	2,25,000
To Direct wages	--	39,00,000	By Process- II A/c (₹ 520.96 × 1,03,500 units)	1,03,500	5,39,19,360
To Direct expenses (20% of Direct wages)	--	7,80,000	By Abnormal loss (₹ 520.96 × 3,000 units)	3,000	15,62,880
To Manufacturing overhead [₹ 2,25,00,000 × (390/975)]		90,00,000	By Costing Profit and Loss (P&L)	34,500	1,79,72,760
	1,50,000	7,36,80,000		1,50,000	7,36,80,000

Cost per unit:

$$\left(\frac{\text{Total Cost - Realisable value of normal loss}}{\text{Total input units - Normal Loss units}}\right)$$

$$= \left(\frac{₹7,36,80,000 - ₹2,25,000}{1,50,000 \text{ units} - 9,000 \text{ units}}\right) = ₹ 520.96 \text{ per unit}$$

Process- II A/c

Particulars	Units	(₹)	Particulars	Units	(₹)
To Process- I A/c	1,03,500	5,39,19,360	By Normal loss (10% of 1,03,500 units) × ₹ 40	10,350	4,14,000
To Direct wages	--	58,50,000	By Costing P & L A/c	96,000	7,59,88,778
To Direct expenses (15% of Direct wages)	--	8,77,500			
To Manufacturing overhead [₹ 2,25,00,000 × (585/975)]	--	1,35,00,000			
To Abnormal gain (₹ 791.55 × 2,850 units)	2,850	22,55,918			
	1,06,350	7,64,02,778		1,06,350	7,64,02,778

Cost per unit:

$$\left(\frac{\text{Total Cost - Realisable value of normal loss}}{\text{Total input units - Normal Loss units}}\right)$$

$$= \left(\frac{₹7,41,46,860 - ₹4,14,000}{1,03,500 \text{ units} - 10,350 \text{ units}}\right) = ₹\ 791.55 \text{ per unit}$$

Abnormal Loss A/c

Particulars	Units	(₹)	Particulars	Units	(₹)
To Process- I A/c	3,000	15,62,880	By Bank	3,000	75,000
			By Costing P & L A/c		14,87,880
	3,000	15,62,880		3,000	15,62,880

Abnormal Gain A/c

Particulars	Units	(₹)	Particulars	Units	(₹)
To Normal Loss	2,850	1,14,000	By Process- II A/c	2,850	22,55,918
To Costing P & L A/c		21,41,918			
	2,850	22,55,918		2,850	22,55,918

Costing Profit & Loss Account

Particulars	(₹)	Particulars	(₹)
To Cost of sales Process I- 1,79,72,760 Process II- 7,59,88,778	9,39,61,538	By Sales Process I- 34,500 @ 650 Process II- 96,000 @ 950	11,36,25,000
To Abnormal loss	14,87,880	By Abnormal gain	21,41,918
To Selling Expenses	1,27,50,000		
To Net Profit	75,67,500		
	11,57,66,918		11,57,66,918

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671 MTP **Sep 2024 S-II** MCQ 12 2 marks MCQ

1200 Kg of a material were input to a process in a period. The normal loss is 8% of input

There is no opening or closing work-in-progress. Output in the period was 1100 Kg. What was the abnormal gain/loss in the period?

A Abnormal gain of 12 Kg

B Abnormal loss of 12 kg

C Abnormal gain of 108 Kg

D Abnormal loss of 4 kg

ICAI'S ANSWER

(D)

Expected Output = Input Material–Normal Loss

Expected Output = 1,200 Kg–96 Kg=1,104 kg

Abnormal loss = 1,104 kg – 1,100 kg = 4 kg

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672 MTP **May 2024 S-II** MCQ 6 2 marks MCQ

A product passes through Process-I. Input raw material issued were 8,000 units. Normal loss anticipated was 10% of input with realisable value of ₹ 5 per unit. 7,600 units of output were produced and transferred to next process. If the total cost incurred under Process-I was ₹ 40,000, then amount of abnormal gain/(loss) is:

A ₹ 2,000

B (₹ 5,000)

C (₹ 2,500)

D ₹ 3,000

ICAI'S ANSWER

Answer key: **(A)**

A

Abnormal gain units = 7600 - [8000 - 800] = 400 Abnormal gain = [40,000 - (800 x 5)]/ 7200 units x 400 units = 2,000

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673 MTP **Old P3 Nov 2023 S-I** Q2(a) 10 marks old scheme

A Ltd. mixes powdered ingredients in two different processes to produce one product. The output of Process- I becomes the input of Process-II and the output of Process-II is transferred to the Packing department.

From the information given below, you are required to PREPARE accounts for Process-I, Process-II and Abnormal loss/ gain A/c to record the transactions for the month of August 2023.

Process-I

Input:	
Material A	6,000 kilograms at ₹ 50 per kilogram
Material B	4,000 kilograms at ₹ 100 per kilogram
Labour	430 hours at ₹ 50 per hour
Normal loss	5% of inputs. Scrap is disposed off at ₹16 per kilogram
Output	9,200 kilograms.

There is no work-in-process at the beginning or end of the month.

Process-II

Input:	
Material C	6,600 kilograms at ₹ 125 per kilogram
Material D	4,200 kilograms at ₹ 75 per kilogram
Flavouring Essence	₹ 3,300
Labour	370 hours at ₹50 per hour
Normal loss	5% of inputs with no disposal value
Output	18,000 kilograms.

There is no work-in-process at the beginning of the month but 1,000 kilograms in process at the end of the month and estimated to be only 50% complete so far as labour and overhead were concerned.

Overhead of ₹ 92,000 incurred to be absorbed on the basis of labour hours.

ICAI'S ANSWER

Process-I A/c

Particulars	Qty. (kgs)	Amount)	Particulars	Qty. (kgs)	Amount (₹)
To Material A	6,000	3,00,000	By Normal loss	500	8,000
To Material B	4,000	4,00,000	By Process-II A/c	9,200	7,38,857
To Labour	--	21,500	By Abnormal loss A/c	300	24,093
To Overhead (₹ 92,000 × 430 hrs/800 hrs)	--	49,450			
	10,000	7,70,950		10,000	7,70,950

$$* \frac{\{(₹3,00,000 + ₹4,00,000 + ₹21,500 + ₹49,450) - ₹8,000\}}{(10,000 - 500)\text{ units}} = \frac{₹7,70,950 - ₹8,000}{9,500\text{ units}} = ₹80.3105$$

Process-II A/c

Particulars	Qty. (kgs)	Amount (₹)	Particulars	Qty. (kgs)	Amount (₹)
To Process-I A/c	9,200	7,38,857	By Normal loss	1,000	--
To Material C	6,600	8,25,000	By Packing Dept. A/c (See the working notes)	18,000	18,42,496
To Material D	4,200	3,15,000	By WIP A/c (See the working notes)	1,000	1,00,711
To Flavouring essence	--	3,300			
To Labour	--	18,500			
To Overheads (₹ 92,000 × 370 hrs/800 hrs)	--	42,550			
	20,000	19,43,207		20,000	19,43,207

Abnormal loss A/c

Particulars	Qty. (kgs)	Amount (₹)	Particulars	Qty. (kgs)	Amount (₹)
To Process-I A/c	300	24,093	By Bank	300	4,800
			By Costing Profit & Loss A/c	--	19,293
	300	24,093		300	24,093

Working Notes:

Calculation of Equivalent Production units

Input	Units	Output	Units	Process-I		Mat-C & D		Labour & OH	
				(%)	Units	(%)	Units	(%)	Units
	9,200	Transferred to Packing.	18,000	100	18,000	100	18,000	100	18,000
Mat-C	6,600	Closing WIP	1,000	100	1,000	100	1,000	50	500
Mat-D	4,200	Normal loss	1,000	--	--	--	--	--	--
	20,000		20,000		19,000		19,000		18,500

Calculation of Unit cost

Cost component	Amount (₹)	Equivalent units	Cost per unit (₹)
Transferred-in	7,38,857	19,000	38.8872
Material-C	8,25,000	19,000	43.4211
Material-D	3,15,000	19,000	16.5789
Flavouring essence	3,300	19,000	0.1737
Total Material Cost	18,82,157	19,000	99.0609
Labour	18,500	18,500	1.0000
Overheads	42,550	18,500	2.3000
Total Cost	19,43,207		102.3609

Value of Materials transferred to Packing Department

= 18,000 unit × ₹102.3609 = 18,42,496

Value of WIP: For Materials- 1,000 units × ₹99.0609	= ₹99,061
For Labour & Overheads 500 units × ₹3.30	= ₹1,650
	₹1,00,711

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674 MTP **Old P3 Nov 2023 S-II** Q2(b) 10 marks old scheme

G K Ltd. produces a product "XYZ" which passes through two processes, viz. Process-A and Process-B. The details for the year ending 31st March, 2023 are as follows:

	Process- A	Process - B
40,000 units of input introduced at a cost of	₹ 3,60,000	-
Material consumed	₹ 2,42,000	2,25,000
Direct wages	₹ 2,58,000	1,90,000
Manufacturing expenses	₹ 1,96,000	1,23,720
Output in units	37,000	27,000
Normal wastage of inputs	5%	10%
Scrap value (per unit)	₹ 15	20
Selling price (per unit)	₹ 37	61

Additional Information:

(a) 80% of the output of Process-A, was passed on to the next process and the balance was sold. The entire output of Process- B was sold.

(b) Indirect expenses for the year were ₹ 4,48,080.

(c) It is assumed that Process-A and Process-B are not responsibility centre.

Required:

(i) Prepare Process-A and Process-B Account.

(ii) Prepare Costing Profit & Loss Account showing the net profit I net loss for the year.

ICAI'S ANSWER

(i) Process- A Account

Particulars	Units	Amount (₹)	Particulars	Units	Amount (₹)
To Inputs	40,000	3,60,000	By Normal wastage (2,000 units × ₹15)	2,000	30,000
To Material	---	2,42,000	By Abnormal loss A/c (1,000 units × ₹27)	1,000	27,000
To Direct wages	---	2,58,000	By Process- B (29,600 units × ₹27)	29,600	7,99,200
To Manufacturing Exp.	---	1,96,000	By Profit & Loss A/c (7,400 units × ₹27)	7,400	1,99,800
	40,000	10,56,000		40,000	10,56,000

Cost per unit = $\frac{₹10,56,000 - ₹30,000}{40,000 \text{ units} - 2,000 \text{ units}}$ = ₹ 27 per unit

Normal wastage = 40,000 units × 5% = 2,000 units

Abnormal loss = 40,000 units – (37,000 units + 2,000 units) = 1,000 units

Transfer to Process- B = 37,000 units × 80% = 29,600 units

Sale = 37,000 units × 20% = 7,400 units

Process- B Account

Particulars	Units	Amount (₹)	Particulars	Units	Amount (₹)
To Process- A A/c	29,600	7,99,200	By Normal wastage (2,960 units × ₹ 20)	2,960	59,200
To Material	---	2,25,000	By Profit & Loss A/c (27,000 units × ₹ 48)	27,000	12,96,000
To Direct Wages	---	1,90,000			
To Manufacturing Exp.	---	1,23,720			
To Abnormal Gain A/c (360 units × ₹ 48)	360	17,280			
	29,960	13,55,200		29,960	13,55,200

$$\text{Cost per unit} = \frac{₹13,37,920 - ₹59,200}{29,600 \text{ units} - 2,960 \text{ units}} = ₹ 48 \text{ per unit}$$

Normal wastage = 29,600 units × 10% = 2,960 units

Abnormal gain = (27,000 units + 2,960 units) – 29,600 units = 360 units

(ii) Costing Profit & Loss Account

Particulars	Amount (₹)	Particulars	Amount (₹)
To Process- A A/c	1,99,800	By Sales:	
To Process- B A/c	12,96,000	- Process-A (7,400 units × ₹ 37)	2,73,800
To Abnormal loss A/c	12,000	- Process- B (27,000 units × ₹ 61)	16,47,000
To Indirect Expenses	4,48,080	By Abnormal gain	10,080
		By Net loss	25,000
	19,55,880		19,55,880

Working Notes:

Normal wastage (Loss) Account

Particulars	Units	Amount (₹)	Particulars	Units	Amount (₹)
To Process- A A/c	2,000	30,000	By Abnormal Gain A/c (360 units × ₹ 20)	360	7,200
To Process- B A/c	2,960	59,200	By Bank (Sales)	4,600	82,000
	4,960	89,200		4,960	89,200

Abnormal Loss Account

Particulars	Units	Amount (₹)	Particulars	Units	Amount (₹)
To Process- A A/c	1,000	27,000	By Bank A/c (1,000 units × ₹ 15)	1,000	15,000
			By Profit & Loss A/c	---	12,000
	1,000	27,000		1,000	27,000

Abnormal Gain Account

Particulars	Units	Amount (₹)	Particulars	Units	Amount (₹)
To Normal loss A/c (360 units × ₹ 20)	360	7,200	By Process- B A/c	360	17,280
To Profit & Loss A/c		10,080			
	360	17,280		360	17,280

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675 MTP **Old P3 May 2023 S-I** Q4(a) 10 marks old scheme

SM Pvt. Ltd. manufactures their products in three consecutive processes. The details are as below:

	Process X	Process Y	Process Z
Transferred to next Process	60%	50%	
Transferred to warehouse for sale	40%	50%	100%

In each process, there is a weight loss of 2% and scrap of 4% of input of each process. The realizable value of scrap of each process is as below:

Process X @ ₹ 3 per ton

Process Y @ ₹ 5 per ton

Process Z @ ₹ 7 per ton.

The following particulars relate to January 2023:

	Process X	Process Y	Process Z
Materials used (in Tons)	1,500	454	189
Rate per ton	₹ 21.5	₹ 14	₹ 12
Direct Wages	₹ 5,000	₹ 3,260	₹ 2,540
Direct Expenses	₹ 3,820	₹ 2,775	₹ 1,900

PREPARE Process Accounts- X, Y and Z & calculate cost per ton at each process.

ICAI'S ANSWER

Process X Account

Particulars	Tones	Amount (₹)	Particulars	Tones	Amount (₹)
To Materials	1,500	32,250	By Weight Loss	30	---
To Wages		5,000	By Scrap	60	180
To Direct Expenses		3,820	By Process Y	846	24,534
			By Warehouse	564	16,356
Total	**1,500**	**41,070**	**Total**	**1,500**	**41,070**

Cost per Ton = (41,070 - 180)/(1,500-30-60) = ₹ 29 per ton

Process Y account

Particulars	Tones	Amount (₹)	Particulars	Tones	Amount (₹)
To Process X	846	24,534	By Weight Loss	26	---
To Materials	454	6,356	By Scrap	52	260
To Wages		3,260	By Process Z	611	18,332.5
To Direct Expenses		2,775	By Warehouse	611	18,332.5
Total	**1300**	**36,925**	**Total**	**1300**	**36,925**

Cost per Ton = (36,925-260)/(1,300-26-52)= ₹30 per ton

Process Z Accounts

Particulars	Tones	Amount (₹)	Particulars	Tones	Amount (₹)
To Process Y	611	18332.5	By Weight Loss	16	---
To Materials	189	2,268	By Scrap	32	224
To Wages		2,540	By Warehouse	752	24,817
To Direct Expenses		1,900			
Total	**800**	**25,041**	**Total**	**800**	**25041**

Cost per Ton = (25,041-224)/(800-16-32) = ₹ 33 per ton

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676 MTP **Old P3 May 2022 S-II** Q5(a) 10 marks old scheme

A Manufacturing unit manufactures a product which passes through three distinct Processes - A, B and

C. The following data is given:

	Process A	Process B	Process C
Material consumed (in ₹)	36,400	31,500	28,000
Direct wages (in ₹)	56,000	49,000	42,000

- The total Production Overhead of ₹ 2,20,500 was recovered @ 150% of Direct wages.
- 15,000 units at ₹ 28 each were introduced to Process 'A'.
- The output of each process passes to the next process and finally, 12,000 units were transferred to Finished Stock Account from Process 'C'.
- No stock of materials or work in progress was left at the end.

The following additional information is given:

Process	% of wastage to normal input	Value of Scrap per unit (₹)
A	6%	15.40
B	?	28.00
C	5%	14.00

You are required to:

(i) FIND OUT the percentage of wastage in process 'B', given that the output of Process 'B' is transferred to Process 'C' at ₹ 56 per unit.

(ii) PREPARE Process accounts for all the three processes A, B and C.

ICAI'S ANSWER

Dr. Process-A Account Cr.

Particulars	Units	(₹)	Particulars	Units	(₹)
To Material introduced	15,000	4,20,000	By Normal Loss A/c [(6% of 15,000 units) x ₹ 15.40]	900	13,860
" Additional material	--	36,400	" Process-B A/c (₹ 41.31* × 14,100 units)	14,100	5,82,540
" Direct wages	--	56,000			
" Production OH	--	84,000			
	15,000	5,96,400		15,000	5,96,400

*Cost per unit of completed units

$$= \frac{\text{Total Cost} - \text{Realisable value from normal loss}}{\text{Inputs units} - \text{Normal loss units}} = \frac{₹\ 5,96,400 - ₹\ 13,860}{15,000 \text{ units} - 900 \text{ units}} = ₹\ 41.31$$

Dr. Process-B Account Cr.

Particulars	Units	(₹)	Particulars	Units	(₹)
To Process-A A/c	14,100	5,82,540	By Normal Loss A/c [(#13.44% of 14,100 units) x ₹ 28]	1,895	53,060
" Additional material	--	31,500	" Process-C A/c (₹ 56 × 12,205 units)	12,205	6,83,480
" Direct wages	--	49,000			
" Production OH	--	73,500			
	14,100	7,36,540		14,100	7,36,540

#Calculation for % of wastage in process 'B':

Let's consider number of units lost under process 'B' = b

Now, $\frac{\text{Total Cost - Realisable value from normal loss}}{\text{Inputs units - Normal loss units}} = 56$

$\frac{₹\ 7,36,540 - ₹\ 28b}{14,100\ \text{units - b}} = ₹\ 56$

₹ 7,36,540 - ₹ 28b = ₹ 7,89,600 - ₹ 56b

28b = ₹ 53,060 => b = 1,895 units

% of wastage = $\frac{1,895\ \text{units}}{14,100\ \text{units}}$ = **13.44%**

Dr. Process-C Account Cr.

Particulars	Units	(₹)	Particulars	Units	(₹)
To Process-B A/c	12,205	6,83,480	By Normal Loss A/c [(5% of 12,205 units) x ₹ 14]	610	8,540
" Additional material	--	28,000	" Finished Stock A/c (₹ 69.68$ × 12,000 units)	12,000	8,36,160
" Direct wages	--	42,000			
" Production OH	--	63,000			
" Abnormal gain (₹ 69.68$ × 405 units)	405	28,220			
	12,610	8,44,700		12,610	8,44,700

$Cost per unit of completed units

$= \frac{\text{Total Cost} - \text{Realisable value from normal loss}}{\text{Inputs units} - \text{Normal loss units}} = \frac{₹\ 8,16,480 - ₹\ 8,540}{12,205\ \text{units - 610 units}} = ₹\ 69.68$

ICAI · Old P3 · MTP May 2022 · Series II · question p. 6 · answer p. 10-11

677 MTP **Old P3 Jul 2021 S-I** Q2(a) 10 marks old scheme

MP Ltd. produces a Product-X, which passes through three processes, I, II and III. In Process-III a by-product arises, which after further processing at a cost of Rs. 85 per unit, product Z is produced. The

information related for the month of September 2020 is as follows:

	Process-I	Process-II	Process-III
Normal loss	5%	10%	5%
Materials introduced (7,000 units)	1,40,000	-	-
Materials added	62,000	1,36,000	84,200
Direct wages	42,000	54,000	48,000
Direct expenses	14,000	16,000	14,000

Production overhead for the month is Rs. 2,88,000, which is absorbed as a percentage of direct wages.

The scraps are sold at Rs. 10 per unit

Product-Z can be sold at Rs. 135 per unit with a selling cost of Rs. 15 per unit

No. of units produced:

Process-I- 6,600; Process-II- 5,200, Process-III- 4,800 and Product-Z- 600

There is no stock at the beginning and end of the month.

You are required to PREPARE accounts for:

(i) Process-I, II and III

(ii) By-product-Z

ICAI'S ANSWER

Total direct wages

= Rs. 42,000 + Rs. 54,000 + Rs. 48,000 = Rs. 1,44,000

Percentage absorption of production overhead on the basis of direct wages

$= \frac{2,88,000}{1,44,000} \times 100 = 200\%$

(i) Process-I A/c

Particulars	Units	Amt. (Rs.)	Particulars	Units	Amt. (Rs.)
To Materials	7,000	1,40,000	By Normal loss (5% of 7,000 units)	350	3,500
To Other materials	-	62,000	By Process-II*	6,600	3,35,955
To Direct wages	-	42,000	By Abnormal loss*	50	2,545
To Direct expenses	-	14,000			
To Production OH (200% of Rs.42,000)	-	84,000			
	7,000	3,42,000		7,000	3,42,000

* Cost per unit $= \frac{\text{Rs.}(3,42,000 - 3,500)}{(7,000 - 350)\text{units}} = \text{Rs. } 50.9022$

Process-II A/c

Particulars	Units	Amt.(Rs.)	Particulars	Units	Amt.(Rs.)
To Process-I A/c	6,600	3,35,955	By Normal loss (10% of 6,600 units)	660	6,600
To Other materials	-	1,36,000	By Process-III**	5,200	5,63,206
To Direct wages	-	54,000	By Abnormal loss**	740	80,149
To Direct expenses	-	16,000			
To Production OH (200% of Rs.54,000)	-	1,08,000			
	6,600	6,49,955		6,600	6,49,955

** Cost per unit = $\frac{\text{Rs.}(6{,}49{,}955 - 6{,}600)}{(6{,}600 - 660)\text{units}}$ = Rs. 108.3089

Process-III A/c

Particulars	Units	Amt. (Rs.)	Particulars	Units	Amt. (Rs.)
To Process-I A/c	5,200	5,63,206	By Normal loss (5% of 5,200 units)	260	2,600
To Other materials	-	84,200	By Product-X***	4,800	8,64,670
To Direct wages	-	48,000			
To Direct expenses	-	14,000	By Product-Z# (Rs.35 × 600 units)	600	21,000
To Production OH (200% of Rs.48,000)	-	96,000			
To Abnormal gain***	460	82,864			
	5,660	8,88,270		5,660	8,88,270

*** Cost per unit = $\frac{\text{Rs.}(8{,}05{,}406 - 2{,}600 - 21{,}000)}{(5{,}200 - 260 - 600)\text{units}}$ = Rs. 180.1396

Realisable value = Rs. 135 – (85+15) = Rs. 35

(ii) By-Product Process A/c

Particulars	Units	Amt. (Rs.)	Particulars	Units	Amt. (Rs.)
To Process-III A/c	600	21,000	By Product-Z	600	81,000
To Processing cost	-	51,000			
To Selling expenses	-	9,000			
	600	81,000		600	81,000

678 MTP **Old P3 May 2020 S-I** Q3(b) 10 marks old scheme

G K Ltd. produces a product "XYZ" which passes through two processes, viz. Process-A and Process-B. The details for the year ending 31st March, 2020 are as follows:

	Process A	Process - B
40,000 units introduced at a cost of	Rs. 3,60,000	-
Material consumed	Rs. 2,42,000	2,25,000
Direct wages	Rs. 2,58,000	1,90,000
Manufacturing expenses	Rs. 1,96,000	1,23,720
Output in units	37,000	27,000
Normal wastage of inputs	5%	10%
Scrap value (per unit)	Rs. 15	20
Selling price (per unit)	Rs. 37	61

Additional Information:

(a) 80% of the output of Process-A, was passed on to the next process and the balance was sold. The entire output of Process- B was sold.

(b) Indirect expenses for the year was Rs. 4,48,080.

(c) It is assumed that Process-A and Process-B are not responsibility centre.

Required:

(i) PREPARE Process-A and Process-B Account.

(ii) PREPARE Costing Profit & Loss Account showing the net profit/ net loss for the year.

ICAI'S ANSWER

(i) Process- A Account

Particulars	Units	Amount (Rs.)	Particulars	Units	Amount (Rs.)
To Inputs	40,000	3,60,000	By Normal wastage (2,000 units × Rs.15)	2,000	30,000
To Material	---	2,42,000	By Abnormal loss A/c (1,000 units × Rs.27)	1,000	27,000
To Direct wages	---	2,58,000	By Process- B (29,600 units × Rs.27)	29,600	7,99,200
To Manufacturing Exp.	---	1,96,000	By Profit & Loss A/c (7,400 units × Rs.27)	7,400	1,99,800
	40,000	10,56,000		40,000	10,56,000

$$\text{Cost per unit} = \frac{\text{Rs.}10,56,000 - \text{Rs.}30,000}{40,000\text{units} - 2,000\text{units}} = \text{Rs. } 27 \text{ per unit}$$

Normal wastage = 40,000 units × 5% = 2,000 units

Abnormal loss = 40,000 units – (37,000 units + 2,000 units) = 1,000 units

Transfer to Process- B = 37,000 units × 80% = 29,600 units

Sale = 37,000 units × 20% = 7,400 units

Process- B Account

Particulars	Units	Amount (Rs.)	Particulars	Units	Amount (Rs.)
To Process- A A/c	29,600	7,99,200	By Normal wastage (2,960 units × Rs. 20)	2,960	59,200
To Material	---	2,25,000	By Profit & Loss A/c (27,000 units × Rs. 48)	27,000	12,96,000
To Direct Wages	---	1,90,000			
To Manufacturing Exp.	---	1,23,720			
To Abnormal Gain A/c (360 units × Rs. 48)	360	17,280			
	29,960	13,55,200		29,960	13,55,200

$$\text{Cost per unit} = \frac{\text{Rs.}13,37,920 - \text{Rs.}59,200}{29,600\text{units} - 2,960\text{units}} = \text{Rs. } 48 \text{ per unit}$$

Normal wastage = 29,600 units × 10% = 2,960 units

Abnormal gain = (27,000 units + 2,960 units) – 29,600 units = 360 units

(ii) Costing Profit & Loss Account

Particulars	Amount (Rs.)	Particulars	Amount (Rs.)
To Process- A A/c	1,99,800	By Sales:	
To Process- B A/c	12,96,000	- Process-A (7,400 units × Rs. 37)	2,73,800
To Abnormal loss A/c	12,000	- Process- B (27,000 units × Rs. 61)	16,47,000
To Indirect Expenses	4,48,080	By Abnormal gain	10,080
		By Net loss	25,000
	19,55,880		19,55,880

Working Notes:

Normal wastage (Loss) Account

Particulars	Units	Amount (Rs.)	Particulars	Units	Amount (Rs.)
To Process-A A/c	2,000	30,000	By Abnormal Gain A/c (360 units × Rs. 20)	360	7,200
To Process-B A/c	2,960	59,200	By Bank (Sales)	4,600	82,000
	4,960	89,200		4,960	89,200

Abnormal Loss Account

Particulars	Units	Amount (Rs.)	Particulars	Units	Amount (Rs.)
To Process-A A/c	1,000	27,000	By Bank A/c (1,000 units × Rs. 15)	1,000	15,000
			By Profit & Loss A/c	---	12,000
	1,000	27,000		1,000	27,000

Abnormal Gain Account

Particulars	Units	Amount (Rs.)	Particulars	Units	Amount (Rs.)
To Normal loss A/c (360 units × Rs. 20)	360	7,200	By Process- B A/c	360	17,280
To Profit & Loss A/c		10,080			
	360	17,280		360	17,280

ICAI · Old P3 · MTP May 2020 · Series I · question p. 4 · answer p. 6-7

10.2 Equivalent units: FIFO and weighted average

56 questions · asked in 37 ICAI documents · 6 exam papers

679 EXAM Sep 2026 MCQ 11 2 marks MCQ

J Ltd. uses process costing and has the following information for the month of August 2026 :

Particulars	Units
Beginning work in progress, (WIP) August 1	8,000
Started fresh units in production during August (besides opening WIP)	50,000
Completed production during August	46,000
Ending work in progress, August 31	12,000

The beginning work in progress was 60% complete for materials and 20% complete for conversion costs. The ending inventory was 80% complete for material and 40% complete for conversion costs. Using the FIFO method, what will be the equivalent units of production for conversion costs ?

A 46,000 units

B 49,200 units

C 50,800 units

D 51,400 units

ICAI'S ANSWER

Answer key: **ICAI has not published the key yet.**

ICAI · P4 · Sep 2026 exam · question p. 7

680 **EXAM** **May 2025** Q5(a) **4 marks**

Furniture Wala Ltd., a manufacturer of dining tables, procures wood as its direct material. The dining tables are initially processed in the Moulding department and subsequently transferred to the Laminating department, where a plastic layer is applied.

The Moulding department began manufacturing 35,000 initial dining tables during the month of March 2025 for the first time and their cost is as follows:

Direct material:	₹ 1,15,500
Moulding costs:	₹ 59,500
Total	₹ 1,75,000

A total of 28,000 dining tables were completed and transferred to the Laminating department, the rest 7,000 were still in the Moulding process at the end of the month. All of the Moulding department's direct materials were placed but on average, only 25% of the conversion costs were applied to the ending work in progress inventory.

Required to calculate:

(i) Equivalent units of production for each cost.

(ii) The Moulding cost per Equivalent units.

(iii) Cost of closing work-in-process (WIP) and finished products.

ICAI'S ANSWER

(i) Calculation of equivalent units of production:

Input Details	Units	Output Particulars	Units	Equivalent Units			
				Material		Conversion cost	
				%	Units	%	Units
Unit Introduced	35,000	Finished output	28,000	100	28,000	100	28,000
		Closing W-I-P	7,000	100	7,000	25	1,750
Total	35,000	Total	35,000		35,000		29,750

(ii) Calculation of Moulding cost per equivalent unit:

	Direct Material	Conversion costs
Total cost (₹)	1,15,500	59,500
Equivalent units	35,000	29,750
Cost per equivalent unit (₹)	3.30	2.00

(iii) The cost of closing work in process (WIP):

Costs	Equivalent units	Rate (₹)	Total Cost (₹)
Direct Material	7,000	3.30	23,100
Conversion Costs	1,750	2.00	3,500
Total			26,600

The cost of finished products:

Costs	Equivalent units	Rate (₹)	Total Cost (₹)
Direct Material	28,000	3.30	92,400
Conversion Costs	28,000	2.00	56,000
Total			1,48,400

ICAI · P4 · May 2025 exam · question p. 25-26 · answer p. 27-28

681 EXAM **May 2025** Q6(c) 4 marks

Describe briefly the methods for valuation of work-in-process followed in Process Costing.

ICAI'S ANSWER

Mainly two methods for valuation of work-in-process are followed:

(i) First-in-first-out (FIFO) method: Under this method the units completed and transferred are taken from both opening work-in-process (WIP) and freshly introduced materials/inputs. The cost to complete the opening WIP and other completed units are calculated separately. The cost of opening WIP is added to cost incurred on completing the incomplete (WIP) units into complete one. The total cost of units completed and transferred is calculated by adding opening WIP cost to cost on freshly introduced inputs. In this method the closing stock of work in process is valued at current cost.

(ii) Weighted Average (Average) Method: Under this method, the cost of opening work-in-process and cost of the current period are aggregated and the aggregate cost is divided by output in terms of completed units. The equivalent production in this case consists of work-load already contained in opening work-in-process and work-load of current period.

The main difference between FIFO method and average method is that units of opening work in process and their cost are taken in full under average method while under FIFO method only the remaining work done now is considered.

ICAI · P4 · May 2025 exam · question p. 31 · answer p. 32-33

682 EXAM Sep 2024 Case Scenario I · MCQ 1 to MCQ 5

CASE SCENARIO I

Sagar Limited, an oil refinery uses Process Costing for determining the cost of each process. Management of Sagar Limited is confused about method of valuation of WIP. They have FIFO and Weighted Average Cost methods under consideration.

Finance manager Mr. Sahil has put forward that Weighted Average Cost method is suitable when there are significant fluctuations in. price and quantity. In this method, calculation has to be done at every purchase and it is a complex and time-consuming method.

He also stated that price and quantity of input and output material of Sagar Limited is almost same for whole year; hence FIFO method would be more suitable for the company. He also revealed that in oil refinery industry; FIFO method is preferred over Weighted Average Cost method and switching to FIFO method will save time and money.

He further stated that by using FIFO method closing WIP is valued at current cost and provided the following information:

Opening WIP : 12,000 Units, Total cost ₹ 1,66,200.

Degree of Completion :	Material - 100%
	Labour and Overhead - 80%

Material introduced: (74,500 Units)	₹ 4,76,465
Direct Labour	₹ 3,70,395
Direct Overhead	₹ 2,96,316

Units Scrapped : 1,900 units Degree of Completion :

Material	100%
Labour and Overhead	70%

Closing WIP : 2,600 units Degree of Completion:

Material	100%
Labour and Overhead	60%

Rest of the units were transferred to next process.

Normal Loss is 2% of total input unit including opening work-in-process. Realizable value of normal loss is ₹ 2 per unit deducted from cost of material introduced.

You are required to calculate the following using FIFO method (MCQs 1 to 5) :

MCQ 1

Equivalent units of Material and Material cost per unit

A 86,500 units and ₹ 5.50 per unit

B 74,500 units and ₹ 6.39 per unit

C 72,770 units and ₹ 6.50 per unit

D 72,600 units and ₹ 6.56 per unit

MCQ 2

Equivalent units of labour and overheads and total cost per unit

A 82,490 units and ₹ 8.08 per unit

B 74,079 units and ₹ 9.00 per unit

C 75,290 units and ₹ 8.85 per unit

D 79,790 units and ₹ 8.35 per unit

MCQ 3

Value of abnormal loss to be shown in process account

A ₹ 2,176.00

B ₹ 2,182.00

C ₹ 2,168.35

D ₹ 1,896.52

MCQ 4

Value of units transferred to next process

A ₹ 11,10,660

B ₹ 12,75,600

C ₹ 12,51,200

D ₹ 12,72,800

MCQ 5

Value of closing WIP

A ₹ 31,096

B ₹ 31,044

C ₹ 30,940

D ₹ 28,340

ICAI'S ANSWERS

MCQ 1

Answer key: **(C)**

MCQ 2

Answer key: **(B)**

MCQ 3

Answer key: **(A)**

MCQ 4

Answer key: **(D)**

MCQ 5

Answer key: **(C)**

ICAI · P4 · Sep 2024 exam · question p. 1-2 · answer p. 7

683 EXAM Old P3 May 2022 Q4(a) 10 marks old scheme

STG Limited is a manufacturer of Chemical 'GK', which is required for industrial use. The complete production operation requires two processes. The raw material first passes through Process I, where Chemical 'G' is produced. Following data is furnished for the month April 2022:

Particulars	(in kgs.)
Opening work-in-progress quantity (Material 100% and conversion 50% complete)	9,500
Material input quantity	1,05,000
Work Completed quantity	83,000
Closing work-in-progress quantity (Material 100% and conversion 60% complete)	16,500

You are further provided that:

Particulars	(in ₹)
Opening work-in-progress cost	
Material cost	29,500
Processing cost	14,750
Material input cost	3,34,500
Processing cost	2,53,100

Normal process loss may be estimated to be 10% of material input. It has no realizable value. Any loss over and above normal loss is considered to be 100% complete in material and processing.

The Company transfers 60,000 kgs. of output (Chemical G) from Process I to Process II for producing Chemical 'GK'. Further materials are added in Process II which yield 1.20 kg. of Chemical 'GK' for every kg. of Chemical 'G' introduced. The chemicals transferred to Process II for further processing are then sold as Chemical 'GK' for ₹ 10 per kg. Any quantity of output completed in Process I, are sold as Chemical 'G' @ ₹ 9 per kg.

The monthly costs incurred in Process II (other than the cost of Chemical 'G') are:

Input 60,000 kg. of Chemical 'G'

Materials Cost ₹ 85,000

Processing Costs ₹ 50,000

You are required:

(i) Prepare Statement of Equivalent production and determine the cost per kg. of Chemical 'G' in Process I using the weighted average cost method.

(ii) Prepare a statement showing cost of Chemical 'G' transferred to Process II, cost of abnormal loss and cost of closing work-in progress.

(iii) STG is considering the option to sell 60,000 kg. of Chemical 'G' of Process I without processing it further in Process-II. Will it be beneficial for the company over the current pattern of processing 60,000

kg in process-II?

(Note: You are not required to prepare Process Accounts)

ICAI'S ANSWER

(i) Statement of Equivalent Production

Particulars	Input quantity	Particulars	Total	Material		Processing Cost	
				%	Units	%	Units
Opening WIP	9,500	Units completed	83,000	100%	83,000	100%	83,000
Material Input	1,05,000	Normal loss (10% of 1,05,000)	10,500	-	-	-	-
		Abnormal loss (Bal. fig.)	4,500	100%	4,500	100%	4,500
		Closing WIP	16,500	100%	16,500	60%	9,900
	1,14,500		1,14,500		1,04,000		97,400

Statement of Cost for each element

Particulars	Material (₹)	Processing (₹)	Total cost (₹)
Cost of opening WIP	29,500	14,750	44,250
Cost incurred during the month	3,34,500	2,53,100	5,87,600
Total cost (A)	3,64,000	2,67,850	6,31,850
Equivalent production (B)	1,04,000	97,400	
Cost per kg of Chemical 'G' (A/B)	**3.5**	**2.75**	6.25

Alternative Presentation

Statement showing cost per kg of each statement

	(₹)	(₹)
Material	(29,500 + 3,34,500)/1,04,000	3.5
Processing cost	(14,750 + 2,53,100)/97,400	2.75
Total Cost per kg		6.25

(ii) Statement showing cost of Chemical 'G' transferred to Process II, cost of abnormal loss and cost of closing work-in- progress

	(₹)
Units transferred (60,000 × 6.25)	**3,75,000**
Abnormal loss (4,500 × 6.25)	**28,125**
Closing work in progress:	
Material (16,500 × 3.5)	**57,750**
Processing cost (9,900 × 2.75)	**27,225**
	84,975

(iii) Calculation of Incremental Profit / Loss after further processing

Particulars	(₹)	(₹)
Sales if further processed (A) (60,000 × 1.20 x ₹ 10)	7,20,000	
Calculation of cost in Process II		
Chemical transferred from Process I	3,75,000	
Add: Material cost	85,000	
Add: Process cost	50,000	
Total cost of finished stock (B)	5,10,000	
Profit, if further processed (C = A – B)		2,10,000
If sold without further processing then,		
Sales (60,000 x ₹ 9)	5,40,000	
Less: Cost of input without further processing	3,75,000	
Profit without further processing (D)		1,65,000
Incremental Profit after further processing (C – D)		45,000

Additional net profit on further processing in Process II is 45,000. Therefore, it is advisable to process further chemical 'G'.

Alternative Presentation

Calculation of Incremental Profit / Loss after further processing

	(₹)
If 60,000 units are sold @ ₹ 9	5,40,000
If 60,000 units are processed in process II (60,000 × 1.2 × ₹ 10)	7,20,000
Incremental Revenue (A)	1,80,000
Incremental Cost: (B)	
Material Cost	85,000
Processing Cost	50,000
	1,35,000
Incremental Profit (A-B)	45,000

Additional net profit on further processing in Process II is 45,000. Therefore, it is advisable to process further chemical 'G'.

ICAI · Old P3 · May 2022 exam · question p. 15-17 · answer p. 17-19

684 EXAM **Old P3 Jan 2021** Q1(c) 5 marks old scheme

MNO Ltd has provided following details:

- Opening work in progress is 10,000 units at ₹ 50,000 (Material 100%, Labour and overheads 70% complete).
- Input of materials is 55,000 units at ₹ 2,20,000. Amount spent on Labour and Overheads is ₹ 26,500 and ₹ 61,500 respectively.
- 9,500 units were scrapped; degree of completion for material 100% and for labour & overheads 60%.
- Closing work in progress is 12,000 units; degree of completion for material 100% and for labour & overheads 90%.
- Finished units transferred to next process are 43,500 units.
  Normal loss is 5% of total input including opening work in progress. Scrapped units would fetch ₹ 8.50 per unit.

You are required to prepare using FIFO method:

(i) Statement of Equivalent production

(ii) Abnormal Loss Account

ICAI'S ANSWER

(i) Statement of Equivalent Production (Using FIFO method)

Particulars	Input Units	Particulars	Output Units	Equivalent Production			
				Material		Labour & O.H.	
				%	Units	%	Units

Particulars	Input Units	Particulars	Output Units	Equivalent Production			
Opening WIP	10,000	Completed and transferred to Process-II					
Units introduced	55,000	- From opening WIP	10,000	-		30	3,000
		- From fresh inputs	33,500	100	33,500	100	33,500
			43,500		33,500		36,500
		Normal Loss {5% (10,000 + 55,000 units)}	3,250	-			-
		Abnormal loss (9,500 – 3,250)	6,250	100	6,250	60	3,750
		Closing WIP	12,000	100	12,000	90	10,800
	65,000		**65,000**		**51,750**		**51,050**

(ii) Abnormal Loss A/c

Particulars	Units	(₹)	Particulars	Units	(₹)
To Process-I A/c (Refer Working Note-2)	6,250	29,698	By Cost Ledger Control A/c (6,250 units × ₹ 8.5)	6,250	53,125
To Costing Profit & Loss A/c	-	23,427			
	6,250	**53,125**		**6,250**	**53,125**

Working Notes:

1. Computation of Cost per unit

Particulars	Materials (₹)	Labour (₹)	Overhead (₹)
Input costs	2,20,000	26,500	61,500
Less: Realisable value of normal scrap (3,250 units x ₹ 8.5)	(27,625)	--	--
Net cost	1,92,375	26,500	61,500
Equivalent Units	51,750	51,050	51,050
Cost Per Unit	**3.7174**	**0.5191**	**1.2047**

Total cost per unit = ₹ (3.7174 + 0.5191 + 1.2047) = ₹ 5.4412

2. Valuation of Abnormal Loss

	(₹)
Materials (6,250 units × ₹ 3.7174)	23,233.75
Labour (3,750 units × ₹ 0.5191)	1,946.63
Overheads (3,750 units × ₹ 1.2047)	4,517.62
	29,698

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685 EXAM **Old P3 Nov 2020** Q4(a) 10 marks old scheme

Following details are related to the work done in Process-I by ABC Ltd. during the month of May 2019 :

	(₹)
Opening work in process (3,000 units)	
Materials	1,80,500
Labour	32,400
Overheads	90,000
Materials introduced in Process-I (42,000 units)	36,04,000
Labour	4,50,000
Overheads	15,18,000

Units Scrapped	: 4,800 units
Degree of completion	:
Materials	: 100%
Labour & overhead	: 70%
Closing Work-in-process	: 4,200 units
Degree of completion	:
Materials	: 100%
Labour & overhead	: 50%
Units finished and transferred to Process-II	: 36,000 units

Normal loss:

4% of total input including opening work-in-process

Scrapped units fetch ₹ 62.50 per piece.

Prepare:

(i) Statement of equivalent production.

(ii) Statement of cost per equivalent unit.

(iii) Process-I A/c

(iv) Normal Loss Account and

(v) Abnormal Loss Account

ICAI'S ANSWER

(i) Statement of Equivalent Production (Weighted Average method)

Particulars	Input Units	Particulars	Output Units	Equivalent Production			
				Material		Labour & O.H.	
				%	Units	%	Units
Opening WIP	3,000	Completed and transferred to Process-II	36,000	100	36,000	100	36,000
Units introduced	42,000	Normal Loss (4% of 45,000 units)	1,800	--	--	--	--

Particulars	Input Units	Particulars	Output Units	Equivalent Production			
		Abnormal loss (Balancing figure)	3,000	100	3,000	70	2,100
		Closing WIP	4,200	100	4,200	50	2,100
	45,000		45,000		**43,200**		**40,200**

(ii) Statement showing cost for each element

Particulars	Materials (₹)	Labour (₹)	Overhead (₹)	Total (₹)
Cost of opening work-in-process	1,80,500	32,400	90,000	3,02,900
Cost incurred during the month	36,04,000	4,50,000	15,18,000	55,72,000
Less: Realisable Value of normal scrap (₹ 62.50 × 1,800 units)	(1,12,500)	--	--	(1,12,500)
Total cost: (A)	**36,72,000**	**4,82,400**	**16,08,000**	57,62,400
Equivalent units: (B)	43,200	40,200	40,200	
Cost per equivalent unit: (C) = (A ÷ B)	**85.00**	**12.00**	**40.00**	137.00

Statement of Distribution of cost

	Particulars	Amount (₹)	Amount (₹)
1.	Value of units completed and transferred: (36,000 units × ₹ 137)		49,32,000
2.	Value of Abnormal Loss:		
	- Materials (3,000 units × ₹ 85)	2,55,000	
	- Labour (2,100 units × ₹ 12)	25,200	
	- Overheads (2,100 units × ₹ 40)	84,000	3,64,200
3.	Value of Closing W-I-P:		
	- Materials (4,200 units × ₹ 85)	3,57,000	
	- Labour (2,100 units × ₹ 12)	25,200	
	- Overheads (2,100 units × ₹ 40)	84,000	4,66,200

(iii) Process-I A/c

Particulars	Units	(₹)	Particulars	Units	(₹)
To Opening W.I.P:					

Particulars	Units	(₹)	Particulars	Units	(₹)
– Materials	3,000	1,80,500	By Normal Loss (₹ 62.5 × 1,800 units)	1,800	1,12,500
– Labour	--	32,400			
– Overheads	--	90,000			
To Materials introduced	42,000	36,04,000	By Abnormal loss	3,000	**3,64,200**
To Labour		4,50,000	By Process-I A/c	36,000	49,32,000
To Overheads		15,18,000	By Closing WIP	4,200	**4,66,200**
	45,000	**58,74,900**		45,000	**58,74,900**

(iv) Normal Loss A/c

Particulars	Units	(₹)	Particulars	Units	(₹)
To Process-I A/c	1,800	1,12,500	By Cost Ledger Control A/c	1,800	1,12,500
	1,800	**1,12,500**		1,800	**1,12,500**

(v) Abnormal Loss A/c

Particulars	Units	(₹)	Particulars	Units	(₹)
To Process-I A/c	3,000	3,64,200	By Cost Ledger Control A/c (₹ 62.5 × 3,000 units)	3,000	**1,87,500**
			By Costing Profit & Loss A/c (Bal. Figure)		**1,76,700**
	3,000	**3,64,200**		3,000	**3,64,200**

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686 RTP **Sep 2026** Q3 MCQ

Kanika Ltd. has furnished the following information pertaining to its process account for the month of March 2026:

Value of opening work-in-process	₹ 10,020
Cost incurred during the month	₹ 1,14,180
Finished goods units	1665
Closing work-in-process	150

Closing work-in-process is completed to the extent of 40%. Costs are incurred evenly throughout the month. The Company uses weighted average method for flow of costs.

What is the value of finished goods for the month of March 2026?

A ₹ 1,19,880

B ₹ 1,25,000

C ₹ 1,24,200

D ₹ 98,500

ICAI'S ANSWER

(A) ₹ 1,19,880

Equivalent Units = 100% of 1665 + 40% of 150 units

= 1665 + 60 = 1,725 units

Cost per unit = (₹ 10,020+ ₹ 1,14,180) ÷ 1,725

= ₹ 72 per unit

Cost of Completed units in the process:

= 1665 × 72 = ₹ 1,19,880

ICAI · P4 · RTP Sep 2026 · question p. 6-7 · answer p. 25

687 RTP **Sep 2026** Q11

A company produces a component, which passes through two processes. During the month of November, 2025, materials for 40,000 components were put into Process- I of which 30,000 were completed and transferred to Process- II. Those not transferred to Process- II were 100% complete as to materials cost and 50% complete as to labour and overheads cost. The Process- I costs incurred were as follows:

Direct Materials	₹ 3,00,000
Direct Wages	₹ 3,50,000
Factory Overheads	₹ 2,45,000

Of those transferred to Process II, 28,000 units were completed and transferred to finished goods stores. There was a normal loss with no salvage value of 200 units in Process II. There were 1,800 units, remained unfinished in the process with 100% complete as to materials and 25% complete as regard to wages and overheads.

Costs incurred in Process-II are as follows:

Packing Materials	₹ 80,000
Direct Wages	₹ 71,125
Factory Overheads	₹ 85,350

Packing material cost is incurred at the end of the second process as protective packing to the completed units of production.

Required:

(i) PREPARE Statement of Equivalent Production, Cost per unit and Process I A/c.

(ii) PREPARE statement of Equivalent Production, Cost per unit and Process II A/c.

ICAI'S ANSWER

Process I

Statement of Equivalent Production and Cost

Input (Units)	Particulars	Output Units	Equivalent Production					
			Materials		Labour		Overheads	
			(%)	Units	(%)	Units	(%)	Units
40,000	Completed	30,000	100	30,000	100	30,000	100	30,000
	Closing WIP	10,000	100	10,000	50	5,000	50	5,000
40,000		40,000		40,000		35,000		35,000

Particulars	Materials	Labour	Overhead	Total
Cost incurred (₹)	3,00,000	3,50,000	2,45,000	8,95,000
Equivalent units	40,000	35,000	35,000	
Cost per equivalent unit (₹)	7.50	10.00	7.00	24.50

Process-I Account

Particulars	Units	(₹)	Particulars	Units	(₹)
To Materials	40,000	3,00,000	By Process-II A/c (30,000 units × ₹ 24.5)	30,000	7,35,000
To Labour		3,50,000	By Closing WIP*	10,000	1,60,000
To Overhead		2,45,000			
	40,000	8,95,000		40,000	8,95,000

* (Material 10,000 units × ₹ 7.5) + (Labour 5,000 units × ₹ 10) + (Overheads 5,000 units × ₹7)

= ₹ 75,000 + ₹ 50,000 + ₹ 35,000 = ₹ 1,60,000

Process II

Statement of Equivalent Production and Cost

Input (Units)	Particulars	Output Units	Equivalent Production					
			Materials		Labour		Overheads	
			(%)	Units	(%)	Units	(%)	Units
30,000	Completed	28,000	100	28,000	100	28,000	100	28,000
	Normal loss	200		--		--		--
	Closing WIP	1,800	100	1,800	25	450	25	450
30,000		30,000		29,800		28,450		28,450

Particulars	Materials	Labour	Overhead	Total
Process-I Cost	7,35,000	--	--	7,35,000
Cost incurred (₹)	--	71,125	85,350	1,56,475
Equivalent units	29,800	28,450	28,450	--
Cost per equivalent unit (₹)	24.6644	2.5000	3.0000	30.1644

Process-II Account

Particulars	Units	(₹)	Particulars	Units	(₹)
To Process-I A/c	30,000	7,35,000	By Normal loss A/c	200	--
To Packing Material	--	80,000	By Finished Goods Stock A/c	28,000*	9,24,604
To Direct Wages	--	71,125	By Closing WIP	1,800**	46,871
To Factory Overhead	--	85,350			
	30,000	9,71,475		30,000	9,71,475

* 28,000 × ₹ 30.1644 = ₹ 8,44,603 + ₹ 80,000 (Packing Material Cost) = ₹ 9,24,604

** 1,800 units × ₹ 24.6644 + 450 units × (₹ 2.5 + ₹3) = ₹ 46,871

ICAI · P4 · RTP Sep 2026 · question p. 13-14 · answer p. 39-40

688 RTP May 2026 Q4 MCQ

In a chemical processing unit, materials are added at the beginning of the process. During the month, the department reported 9,200 equivalent units of production (EUP) under the FIFO method.

The process had an opening WIP of 800 units, which were 50% complete, and a closing WIP of 900 units, which were 70% complete with respect to conversion costs.

During the month, the plant also faced a minor machine breakdown, but it did not affect the material input, only the processing speed.

Based on the above information, the number of units completed during the month was:

A 9,100

B 8,970

C 8,450

D 8,650

ICAI'S ANSWER

(B) 8,970

Statement of Equivalent Production Units (Under FIFO Method)

Particulars	Output units	Equivalent Production	
		(%)	Equivalent units
From opening W-I-P	800	50	400
Units Started & Completed	X-800	100	X-800
Units completed (Transferred to next process)	X		-
Closing W-I-P	900	70	630
	Y		9,200

X - 800 + 1030 = 9,200

X + 230 = 9,200

X = 8,970

ICAI · P4 · RTP May 2026 · question p. 6 · answer p. 27-28

689 RTP **May 2026** Q13

A Company produces a component, which passes through two processes. During the month of April, 2025, materials for 40,000 components were put into Process I of which 30,000 were completed and transferred to Process II. Those not transferred to Process II were 100% complete as to materials cost and 50% complete as to labour and overheads cost. The Process I costs incurred were as follows:

Direct Materials	₹ 15,000
Direct Wages	₹ 18,000
Factory Overheads	₹ 12,000

Of those transferred to Process II, 28,000 units were completed and transferred to finished goods stores. There was a normal loss with no salvage value of 200 units in Process II. There were 1,800 units, remained unfinished in the process with 100% complete as to materials and 25% complete as regard to wages and overheads.

No further process material costs occur after introduction at the first process until the end of the second process, when protective packing is applied to the completed components. The process and packing costs incurred at the end of the Process II were:

Packing Materials	₹ 4,000
Direct Wages	₹ 3,500
Factory Overheads	₹ 4,500

Required:

(i) PREPARE Statement of Equivalent Production, Cost per unit and Process I A/c.

(ii) PREPARE statement of Equivalent Production, Cost per unit and Process II A/c.

ICAI'S ANSWER

Process I

Statement of Equivalent Production and Cost

Input (Units)	Particulars	Output Units	Equivalent Production					
			Materials		Labour		Overheads	
			(%)	Units	(%)	Units	(%)	Units
40,000	Completed	30,000	100	30,000	100	30,000	100	30,000
	Closing WIP	10,000	100	10,000	50	5,000	50	5,000
40,000		40,000		40,000		35,000		35,000

Particulars	Materials	Labour	Overhead	Total
Cost incurred (₹)	15,000	18,000	12,000	45,000
Equivalent units	40,000	35,000	35,000	
Cost per equivalent unit (₹)	0.3750	0.5143	0.3428	1.2321

Process- I Account

Particulars	Units	(₹)	Particulars	Units	(₹)
To Materials	40,000	15,000	By Process-II A/c (30,000 units × ₹ 1.2321)	30,000	36,964
To Labour		18,000	By Closing WIP	10,000	8,036*
To Overhead		12,000			
	40,000	45,000		40,000	45,000

* (Material 10,000 units × ₹ 0.3750) + (Labour 5,000 units × ₹ 0.5143) + (Overheads 5,000 units × ₹0.3428) = ₹ 3,750 + ₹ 2,572 + ₹ 1,714 = ₹ 8,036

Process II

Statement of Equivalent Production and Cost

Input (Units)	Particulars	Output Units	Equivalent Production					
			Materials		Labour		Overheads	
			(%)	Units	(%)	Units	(%)	Units
30,000	Completed	28,000	100	28,000	100	28,000	100	28,000
	Normal loss	200		--		--		--
	WIP	1,800	100	1,800	25	450	25	450
30,000		30,000		29,800		28,450		28,450

Particulars	Materials	Labour	Overhead	Total
Process-I Cost	36,964	--	--	36,964
Cost incurred (₹)	--	3,500	4,500	8,000
Equivalent units	29,800	28,450	28,450	
Cost per equivalent unit (₹)	1.2404	0.1230	0.1582	1.5216

Process- II Account

Particulars	Units	(₹)	Particulars	Units	(₹)
To Process-I A/c	30,000	36,964	By Normal loss A/c	200	--
To Packing Material	--	4,000	By Finished Goods Stock A/c	28,000*	46,605
To Direct Wages	--	3,500	By Closing WIP	1,800**	2,359
To Factory Overhead	--	4,500			
	30,000	48,964		30,000	48,964

*28,000 × ₹ 1.5216 = ₹ 42,605 + ₹ 4,000 (Packing Material Cost) = ₹46,605

**1,800 units × ₹ 1.2404 + 450 units × (₹ 0.1230 + ₹ 0.1582) = ₹ 2,359

ICAI · P4 · RTP May 2026 · question p. 17-18 · answer p. 43-45

690 RTP Jan 2026 Q14

Following data are available for a product for the month of April, 2025:

Particulars	Process- I (₹)	Process- II (₹)
Opening work-in- progress	Nil	Nil
Costs incurred during the month:		
- Direct materials	6,00,000	
- Labour	1,20,000	1,60,000
- Factory overheads	2,40,000	2,00,000
Units of production:		
Received in process	40,000	36,000
Completed and transferred	36,000	32,000
Closing work-in-progress	2,000	?
Normal loss in process	2,000	1,500

Production remaining in process has to be valued as follows:

Materials 100% Labour 50% Overheads 50%

There has been no abnormal loss in Process- II.

The company follows weighted average method for valuing inventory.

Prepare Process Accounts after working out the missing figures and with detailed workings.

ICAI'S ANSWER

Statement of Equivalent Units (Process- I)

Input (Units)	Particulars	Output (Units)	Equivalent Production			
			Materials		Labour and Overheads	
			Units	(%)	Units	(%)
40,000	Introduced and completed	36,000	36,000	100	36,000	100
	Normal loss	2,000	-	-	-	-
	Closing stock	2,000	2,000	100	1,000	50
40,000		40,000	38,000		37,000	

Computation of cost per Equivalent Unit for each element of cost (Process- I)

Elements of Cost	Total Cost (₹)	Equivalent units	Cost per Equivalent units (₹)
Direct Materials	6,00,000	38,000	15.7895
Labour	1,20,000	37,000	3.2432
Factory Overheads	2,40,000	37,000	6.4865

Statement of Apportionment of Cost

Items	Elements	Equivalent units	Cost per unit (₹)	Cost (₹)	Total (₹)
Units introduced and completed	Materials	36,000	15.7895	5,68,422.00	
	Labour	36,000	3.2432	1,16,755.20	
	Overheads	36,000	6.4865	2,33,514.00	9,18,691.20
Closing stock	Materials	2,000	15.7895	31,579.00	
	Labour	1,000	3.2432	3,243.20	
	Overheads	1,000	6.4865	6,486.50	41,308.70

Process- I Account

Particulars	Units	Amount (₹)	Particulars	Units	Amount (₹)
To Materials	40,000	6,00,000	By Normal loss	2,000	-
To Labour		1,20,000	By Process II	36,000	9,18,691
To Overheads		2,40,000	By Closing stock	2,000	41,309
	40,000	9,60,000		40,000	9,60,000

Statement of Equivalent Units (Process- II)

Input (Units)	Particulars	Output (Units)	Equivalent Production			
			Materials		**Labour and Overheads**	
			Units	**(%)**	**Units**	**(%)**
36,000	Units transferred from Process- I					
	Normal loss	1,500	-	-	-	-
	Completed	32,000	32,000	100	32,000	100
	Closing stock (balancing figure)	2,500	2,500	100	1,250	50
36,000		36,000	34,500		33,250	

Computation of cost per Equivalent Unit for each element of cost (Process- I)

Elements of Cost	Total Cost (₹)	Equivalent units	Cost per Equivalent units (₹)
Cost of 36,000 units transferred from Process- I	9,18,691	34,500	26.6287
Labour	1,60,000	33,250	4.8120
Factory Overheads	2,00,000	33,250	6.0150

Statement of Apportionment of Cost

Items	Elements	Equivalent units	Cost per unit (₹)	Cost (₹)	Total (₹)
Units introduced and completed	Materials	32,000	26.6287	8,52,118.40	
	Labour	32,000	4.8120	1,53,984.00	
	Overheads	32,000	6.0150	1,92,480.00	11,98,582.40
Closing stock	Materials	2,500	26.6287	66,571.75	
	Labour	1,250	4.8120	6,015.00	
	Overheads	1,250	6.0150	7,518.75	80,105.50

Process- II Account

Particulars	Units	Amount (₹)	Particulars	Units	Amount (₹)
To Units introduced	36,000	9,18,691	By Normal loss	1,500	-
To Labour		1,60,000	By Finished stock	32,000	11,98,582
To Overheads		2,00,000	By Closing stock	2,500	80,109*
	36,000	12,78,691		36,000	12,78,691

*Difference arose due to rounding-off has been adjusted.

ICAI · P4 · RTP Jan 2026 · question p. 18-19 · answer p. 41-43

691 RTP **May 2025** Q14

Following data are available for a product for the month of July, 2024:

Particulars	Process- I (₹)	Process- II (₹)
Opening work-in- progress	Nil	Nil
Costs incurred during the month:		
- Direct materials	6,00,000	
- Labour	1,20,000	1,60,000
- Factory overheads	2,40,000	2,00,000
Units of production:		
Received in process	40,000	36,000
Completed and transferred	36,000	32,000
Closing work-in-progress	2,000	?
Normal loss in process	2,000	1,500

Production remaining in process has to be valued as follows:

Materials 100% Labour 50% Overheads 50%

There has been no abnormal loss in Process- II.

The company follows weighted average method for valuing inventory.

PREPARE Process Accounts after working out the missing figures and with detailed workings.

ICAI'S ANSWER

Statement of Equivalent Units (Process- I)

Input (Units)	Particulars	Output (Units)	Equivalent Production			
			Materials		Labour and Overheads	
			Units	(%)	Units	(%)
40,000	Introduced and completed	36,000	36,000	100	36,000	100
	Normal loss	2,000	-	-	-	-
	Closing stock	2,000	2,000	100	1,000	50
40,000		40,000	38,000		37,000	

Computation of cost per Equivalent Unit for each element of cost (Process- I)

Elements of Cost	Total Cost (₹)	Equivalent units	Cost per Equivalent units (₹)
Direct Materials	6,00,000	38,000	15.7895
Labour	1,20,000	37,000	3.2432
Factory Overheads	2,40,000	37,000	6.4865

Statement of Apportionment of Cost

Items	Elements	Equivalent units	Cost per unit (₹)	Cost (₹)	Total (₹)
Units introduced and completed	Materials	36,000	15.7895	5,68,422.00	
	Labour	36,000	3.2432	1,16,755.20	
	Overheads	36,000	6.4865	2,33,514.00	9,18,691.20
Closing stock	Materials	2,000	15.7895	31,579.00	
	Labour	1,000	3.2432	3,243.20	
	Overheads	1,000	6.4865	6,486.50	41,308.70

Process- I Account

Particulars	Units	Amount (₹)	Particulars	Units	Amount (₹)
To Materials	40,000	6,00,000	By Normal loss	2,000	-
To Labour		1,20,000	By Process II	36,000	9,18,691
To Overheads		2,40,000	By Closing stock	2,000	41,309
	40,000	9,60,000		40,000	9,60,000

Statement of Equivalent Units (Process- II)

Input (Units)	Particulars	Output (Units)	Equivalent Production			
			Materials		**Labour and Overheads**	
			Units	**(%)**	**Units**	**(%)**
36,000	Units transferred from Process- I					
	Normal loss	1,500	-	-	-	-
	Completed	32,000	32,000	100	32,000	100
	Closing stock (balancing figure)	2,500	2,500	100	1,250	50
36,000		36,000	34,500		33,250	

Computation of cost per Equivalent Unit for each element of cost (Process- I)

Elements of Cost	Total Cost (₹)	Equivalent units	Cost per Equivalent units (₹)
Cost of 36,000 units transferred from Process- I	9,18,691	34,500	26.6287
Labour	1,60,000	33,250	4.8120
Factory Overheads	2,00,000	33,250	6.0150

Statement of Apportionment of Cost

Items	Elements	Equivalent units	Cost per unit (₹)	Cost (₹)	Total (₹)
Units introduced and completed	Materials	32,000	26.6287	8,52,118.40	
	Labour	32,000	4.8120	1,53,984.00	
	Overheads	32,000	6.0150	1,92,480.00	11,98,582.40
Closing stock	Materials	2,500	26.6287	66,571.75	
	Labour	1,250	4.8120	6,015.00	
	Overheads	1,250	6.0150	7,518.75	80,105.50

Process- II Account

Particulars	Units	Amount (₹)	Particulars	Units	Amount (₹)
To Units introduced	36,000	9,18,691	By Normal loss	1,500	-
To Labour		1,60,000	By Finished stock	32,000	11,98,582
To Overheads		2,00,000	By Closing stock	2,500	80,109*
	36,000	12,78,691		36,000	12,78,691

*Difference arose due to rounding-off has been adjusted.

ICAI · P4 · RTP May 2025 · question p. 19 · answer p. 43-45

692 RTP Sep 2024 Q14

The following data are available in respect of Process-I for June 2024:

(1) Opening stock of work in process: 600 units at a total cost of ₹ 4,20,000.

(2) Degree of completion of opening work in process:

Material	100%
Labour	60%
Overheads	60%

(3) Input of materials at a total cost of ₹ 55,20,000 for 9,200 units.

(4) Direct wages incurred ₹ 18,60,000

(5) Production overhead ₹ 8,63,000.

(6) Units scrapped 200 units. The stage of completion of these units was:

Materials	100%
Labour	80%
Overheads	80%

(7) Closing work in process; 700 units. The stage of completion of these units was:

Material	100%
Labour	70%
Overheads	70%

(8) 8,900 units were completed and transferred to the next process.

(9) Normal loss is 4% of the total input (opening stock plus units put in)

(10) Scrap value is ₹ 60 per unit.

You are required to:

(i) COMPUTE equivalent production,

(ii) CALCULATE the cost per equivalent unit for each element.

(iii) CALCULATE the cost of abnormal loss (or gain), closing work in process and the units transferred to the next process using the FIFO method.

ICAI'S ANSWER

(i) Statement of Equivalent Production (FIFO Method)

Input		Output		Equivalent Production					
				Materials		Labour		Production Overhead	
Details	Units	Details	Units	%	Units	%	Units	%	Units
Opening Stock	600	From opening stock	600	-	-	40	240	40	240
		- From fresh materials	8,300	100	8,300	100	8,300	100	8,300
		Closing W-I-P	700	100	700	70	490	70	490
Fresh inputs	9,200	Normal loss	392	-	-	-	-	-	-
			9,992		9,000		9,030		9,030
		Less: Abnormal Gain	(192)	100	(192)	100	(192)	100	(192)
	9,800		9,800		8,808		8,838		8,838

(ii) Statement of Cost per equivalent units

Elements	(₹)	Cost (₹)	Equivalent units (EU)	Cost per EU (₹)
Material Cost	55,20,000			
Less: Scrap realisation 392 units @ ₹ 60/- p.u.	(23,520)	54,96,480	8,808	624.03
Labour cost		18,60,000	8,838	210.45
Production OH Cost		8,63,000	8,838	97.65
Total Cost		82,19,480		932.13

(iii) Cost of Abnormal Gain – 192 Units

	(₹)	(₹)
Material cost of 192 units @ ₹ 624.03 p.u.	1,19,813.76	
Labour cost of 192 units @ ₹ 210.45 p.u.	40,406.40	
Production OH cost of 192 units @ ₹ 97.65 p.u.	18,748.80	1,78,968.96
Cost of closing WIP – 700 Units		
Material cost of 700 equivalent units @ ₹ 624.03 p.u.	4,36,821.00	
Labour cost of 490 equivalent units @ ₹ 210.45 p.u.	1,03,120.50	
Production OH cost of 490 equivalent @ ₹ 97.65 p.u.	47,848.50	5,87,790.00

Cost of 8,900 units transferred to next process

		₹
(i)	Cost of opening W-I-P Stock b/f – 600 units	4,20,000.00
(ii)	Cost incurred on opening W-I-P stock	
	Material cost	—
	Labour cost 240 equivalent units @ ₹ 210.45 p.u.	50,508.00
	Production OH cost 240 equivalent units @ ₹ 97.65 p.u.	23,436.00
		4,93,944.00
(iii)	Cost of 8,300 completed units	
	8,300 units @ ₹ 932.13 p.u.	77,36,679.00
	Total cost [(i) + (ii) + (iii))]	82,30,623.00

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The following data are available in respect of Process-I for January 2024:

(1) Opening stock of work in process: 600 units at a total cost of ₹ 4,200.

(2) Degree of completion of opening work in process:

Material	100%
Labour	60%
Overheads	60%

(3) Input of materials at a total cost of ₹ 55,200 for 9,200 units.

(4) Direct wages incurred ₹ 18,600

(5) Overheads ₹ 8,630.

(6) Units scrapped 200 units. The stage of completion of these units was:

Materials	100%
Labour	80%
Overheads	80%

(7) Closing work in process; 700 units. The stage of completion of these units was:

Material	100%
Labour	70%
Overheads	70%

(8) 8,900 units were completed and transferred to the next process.

(9) Normal loss is 4% of the total input (opening stock plus units put in)

(10) Scrap value is ₹ 6 per unit.

You are required to:

(i) PREPARE using FIFO method, Statement of equivalent production,

(ii) PREPARE Statement of cost,

(iii) CALCULATE cost of closing WIP,

(iv) CALCULATE the cost of the units to be transferred to the next process.

ICAI'S ANSWER

(i) Statement of Equivalent Production (FIFO Method)

Input		Output		Equivalent Production					
				Materials		Labour		Overheads	
Details	Units	Details	Units	%	Units	%	Units	%	Units

Input		Output		Equivalent Production					
Opening Stock	600	Finished goods transferred to next process:							
		-From opening stock	600	-	-	40	240	40	240
		-From fresh materials	8,300	100	8,300	100	8,300	100	8,300
		Closing W-I-P	700	100	700	70	490	70	490
Fresh inputs	9,200	Normal loss	392	-	-	-	-	-	-
			9,992		9,000		9,030		9,030
		Less: Abnormal Gain	(192)	100	(192)	100	(192)	100	(192)
	9,800		9,800		8,808		8,838		8,838

(ii) Statement of Cost per equivalent units

Elements	(₹)	Cost (₹)	Equivalent units	Cost per equivalent Unit (₹)
Material Cost	55,200			
Less: Scrap realisation 392 units @ ₹ 6/- p.u.	2,352	52,848	8,808	6.00
Labour cost		18,600	8,838	2.10
Overheads		8,630	8,838	0.98
Total Cost		80,078		9.08

Cost of Abnormal Gain – 192 Units

	(₹)	(₹)
Material cost of 192 units @ ₹ 6.00/- p.u.	1,152.00	
Labour cost of 192 units @ ₹ 2.10/- p.u.	403.20	
Overheads of 192 units @ ₹ 0.98/- p.u.	188.16	1,743.36

(iii) Cost of closing WIP – 700 Units

Material cost of 700 equivalent units @ ₹ 6.00/- p.u.	4,200.00	
Labour cost of 490 equivalent units @ ₹2.10/- p.u.	1,029.00	
Overheads of 490 equivalent @ ₹ 0.98/- p.u.	480.20	5709.20

(iv) Calculation of cost of 8,900 units transferred to next process

		(₹)
(i) Cost of opening W-I-P Stock b/f – 600 units		4,200.00
(ii) Cost incurred on opening W-I-P stock		
Material cost	—	
Labour cost 240 equivalent units @ ₹ 2.10 p.u.	504.00	
Overheads 240 equivalent units @ ₹ 0.98/- p.u.	235.20	
		739.20
(iii) Cost of 8,300 completed units		
8,300 units @ ₹9.08 p.u.		75,364.00
Total cost [(i) + (ii) + (iii))]		80,303.20

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The following information is furnished by ABC Company for Process - II of its manufacturing activity for the month of April 2023:

(i) Opening Work-in-Progress – Nil

(ii) Units transferred from Process I – 55,000 units at ₹ 3,27,800

(iii) Expenditure debited to Process – II:

Consumables	₹ 1,57,200
Labour	₹ 1,04,000
Overhead	₹ 52,000

(iv) Units transferred to Process III – 51,000 units

(v) Closing WIP – 2,000 units (Degree of completion):

Consumables	80%
Labour	60%
Overhead	60%

(vi) Units scrapped - 2,000 units, scrapped units were sold at ₹ 5 per unit

(vii) Normal loss – 4% of units introduced

You are required to:

(i) Prepare a Statement of Equivalent Production.

(ii) Determine the cost per unit

(iii) Determine the value of Work-in-Process and units transferred to Process – III

ICAI'S ANSWER

(i) Statement of Equivalent Production

Input Details	Units	Output Particulars	Units	Equivalent Production					
				Material- A*		Consumables		Labour & Overheads	
				%	Units	%	Units	%	Units
Units transferred from Process-I	55,000	Units transferred to Process-III	51,000	100	51,000	100	51,000	100	51,000
		Normal loss (4% of 55,000)	2,200	-	-	-	-	-	-
		Closing W-I-P	2,000	100	2,000	80	1,600	60	1,200
		Abnormal Gain	(200)	100	(200)	100	(200)	100	(200)
	55,000		55,000		52,800		52,400		52,000

*Material A represent transferred-in units from process-I

(ii) Determination of Cost per Unit

Particulars	Amount (₹)	Units	Per Unit (₹)
(i) Direct Material (Consumables) :			
Value of units transferred from Process-I	3,27,800		
Less: Value of normal loss (2,200 units × ₹ 5)	(11,000)		
	3,16,800	52,800	6.00
(ii) Consumables added in Process-II	1,57,200	52,400	3.00
(iii) Labour	1,04,000	52,000	2.00
(iii) Overhead	52,000	52,000	1.00
Total Cost per equivalent unit			12.00

(iii) Determination of value of Work-in-Process and units transferred to Process-III

Particulars	Units	Rate (₹)	Amount (₹)
Value of Closing W-I-P:			
Material from Process-I	2,000	6.00	12,000
Consumables	1,600	3.00	4,800
Labour	1,200	2.00	2,400
Overhead	1,200	1.00	1,200
			20,400
Value of units transferred to Process-III	51,000	12.00	6,12,000

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695 RTP **Old P3 May 2023** Q9 old scheme

'Dairy Wala Private limited' is engaged in the production of flavoured milk. Its process involve filtration and boiling of milk after that some sugar, flavour, colour is added and then letting it cool to fill the product into clean and sterile bottles. For Producing 10 litre of flavour milk, 100 litre of Raw milk is required, which extracts only 45 litres of standardized milk.

Following information regarding Process – I has been obtained from the manufacturing department of Dairy Wala Private limited for the month of December 2022:

Items	(₹)
Opening work-in process (13,500 litre)	
Milk	1,50,000
Labour	45,000
Overheads	1,35,000
Milk introduced for filtration and boiling (3,00,000 litre)	15,00,000
Direct Labour	6,00,000
Overheads	18,00,000
Abnormal Loss: 3,000 litres	
Degree of completion:	
Milk	100%
Labour and overheads	80%
Closing work-in process: 27,000 litres	
Degree of completion:	

Items	(₹)
Milk	100%
Labour and overheads	80%
Milk transferred for Packing: 1,18,500 litres	

You are required to PREPARE using average method:

(i) Statement of equivalent production,

(ii) Statement of cost,

(iii) Statement of distribution cost, and

(iv) Process-I Account.

ICAI'S ANSWER

(i) Statement of Equivalent Production

Particulars	Input Units	Particulars	Output Units	Equivalent Production			
				Milk		Labour & O.H.	
				%	Units	%	Units
Opening WIP	13,500	Completed and transferred to Process-II	1,18,500	100	1,18,500	100	1,18,500
Units introduced	3,00,000	Normal Loss (55%* of 3,00,000)	1,65,000	--	--	--	--
		Abnormal loss	3,000	100	3,000	80	2400
		Closing WIP	27,000	100	27,000	80	21,600
	3,13,500		3,13,500		1,48,500		1,42,500

* 100 litre of milk extracts only 45 litre of standardized milk. Thus, normal loss = 100 – 45 = 55%

(ii) Statement showing cost for each element

Particulars	Milk (₹)	Labour (₹)	Overhead (₹)	Total (₹)
Cost of opening work-in-process	1,50,000	45,000	1,35,000	3,30,000
Cost incurred during the month	15,00,000	6,00,000	18,00,000	39,00,000
Total cost: (A)	16,50,000	6,45,000	19,35,000	42,30,000
Equivalent units: (B)	1,48,500	1,42,500	1,42,500	
Cost per equivalent unit: (C) = (A ÷ B)	11.111	4.526	13.578	29.216

(iii) Statement of Distribution of cost

	(₹)	(₹)
1. Value of units completed and transferred (1,18,500 units × ₹ 29.216)		34,62,096
2. Value of Abnormal Loss: -		
Milk (3,000 units × ₹ 11.111)	33,333	
Labour (2400 units × ₹ 4.526)	10,863	
Overheads (2400 units × ₹ 13.579)	32,590	76,786
3. Value of Closing W-I-P:		
Milk (27,000 units × ₹ 11.111)	299997	
Labour (21,600 units × ₹ 4.526)	97,762	
Overheads (21,600 units × ₹ 13.579)	2,93,306	6,91,065

(iv) Process-I A/c

Particulars	Units	Amount (₹)	Particulars	Units	Amount (₹)
To Opening W.I.P:			By Normal Loss	1,65,000	--
Milk	13,500	1,50,000	By Abnormal Loss (₹.44 difference due to approximation)	3,000	76,839
Labour	--	45,000	By Process-II A/c	1,18,500	34,62,096
Overheads	--	1,35,000	By Closing WIP	27,000	6,91,065
To Milk introduced	3,00,000	15,00,000			
To Direct Labour		6,00,000			
To Overheads		18,00,000			

Particulars	Units	Amount (₹)	Particulars	Units	Amount (₹)
	3,13,500	42,30,000		3,13,500	42,30,000

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696 RTP Old P3 May 2022 Q9 old scheme

A company produces a component, which passes through two processes. During the month of December, 2021, materials for 40,000 components were put into Process-I of which 30,000 were completed and transferred to Process-II. Those not transferred to Process- II were 100% complete as to materials cost and 50% complete as to labour and overheads cost. The Process- I costs incurred were as follows:

Direct Materials	₹ 6,00,000
Direct Wages	₹ 7,00,000
Factory Overheads	₹ 4,90,000

Of those transferred to Process II, 28,000 units were completed and transferred to finished goods stores. There was a normal loss with no salvage value of 200 units in Process II. There were 1,800 units, remained unfinished in the process with 100% complete as to materials and 25% complete as regard to wages and overheads.

Costs incurred in Process-II are as follows:

Packing Materials	₹ 1,60,000
Direct Wages	₹ 1,42,250
Factory Overheads	₹ 1,70,700

Packing material cost is incurred at the end of the second process as protective packing to the completed units of production.

Required:

(i) PREPARE Statement of Equivalent Production, Cost per unit and Process I A/c.

(ii) PREPARE statement of Equivalent Production, Cost per unit and Process II A/c.

ICAI'S ANSWER

(i) Process I

Statement of Equivalent Production and Cost

Input (Units)	Particulars	Output Units	Equivalent Production					
			Materials		Labour		Overheads	
			(%)	Units	(%)	Units	(%)	Units
40,000	Completed	30,000	100	30,000	100	30,000	100	30,000
	Closing WIP	10,000	100	10,000	50	5,000	50	5,000
40,000		40,000		40,000		35,000		35,000

Particulars	Materials	Labour	Overhead	Total
Cost incurred (₹)	6,00,000	7,00,000	4,90,000	17,90,000
Equivalent units	40,000	35,000	35,000	
Cost per equivalent unit (₹)	15	20	14	49

Process-I Account

Particulars	Units	(₹)	Particulars	Units	(₹)
To Materials	40,000	6,00,000	By Process-II A/c (30,000 units × ₹49)	30,000	14,70,000
To Labour		7,00,000	By Closing WIP*	10,000	3,20,000
To Overhead		4,90,000			
	40,000	17,90,000		40,000	17,90,000

* (Material 10,000 units × ₹ 15) + (Labour 5,000 units × ₹ 20) + (Overheads 5,000 units × ₹ 14)

= ₹ 1,50,000 + ₹ 1,00,000 + ₹ 70,000 = ₹ 3,20,000

(ii) Process II

Statement of Equivalent Production and Cost

Input (Units)	Particulars	Output Units	Equivalent Production					
			Materials		Labour		Overheads	
			(%)	Units	(%)	Units	(%)	Units
30,000	Completed	28,000	100	28,000	100	28,000	100	28,000
	Normal loss	200		--		--		--
	Closing WIP	1,800	100	1,800	25	450	25	450
30,000		30,000		29,800		28,450		28,450

Particulars	Materials	Labour	Overhead	Total
Process-I Cost	14,70,000	--	--	14,70,000
Cost incurred (₹)	--	1,42,250	1,70,700	3,12,950
Equivalent units	29,800	28,450	28,450	--
Cost per equivalent unit (₹)	49.3289	5.00	6.00	60.3289

Process-II Account

Particulars	Units	(₹)	Particulars	Units	(₹)
To Process-I A/c	30,000	14,70,000	By Normal loss A/c	200	--
To Packing Material	--	1,60,000	By Finished Goods Stock A/c	28,000*	18,49,209
To Direct Wages	--	1,42,250	By Closing WIP	1,800**	93,741
To Factory Overhead	--	1,70,700			
	30,000	19,42,950		30,000	19,42,950

* 28,000 × ₹ 60.3289 = ₹ 16,89,209 + ₹1,60,000 (Packing Material Cost) = ₹ 18,49,209

** 1,800 units × ₹ 49.3289 + 450 units × (₹ 5 + ₹6) = ₹ 93,741

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697 RTP Old P3 Dec 2021 Q9 old scheme

Following information is available regarding Process-I of a manufacturing company for the month of February:

Production Record:

Units in process as on 1st February (All materials used, 1/4th complete for labour and overhead)	8,000
New units introduced	32,000
Units completed	28,000
Units in process as on 28th February (All materials used, 1/3rd complete for labour and overhead)	12,000

Cost Records:

	(₹)
Work-in-process as on 1st February	
Materials	1,20,000
Labour	20,000
Overhead	20,000

	(₹)
	1,60,000
Cost during the month:	
Materials	5,12,000
Labour	3,00,000
Overhead	3,00,000
	11,12,000

Presuming that average method of inventory is used, PREPARE the following:

(i) Statement of equivalent production.

(ii) Statement showing cost for each element.

(iii) Statement of apportionment of cost.

(iv) Process cost account for Process-I.

ICAI'S ANSWER

(i) **Statement of equivalent production (Average cost method)**

Particulars	Input Units	Particulars	Output Units	Equivalent Production			
				Material		Labour & O.H.	
				%	Units	%	Units
Opening WIP	8,000	Completed and transferred	28,000	100	28,000	100	28,000
Units introduced	32,000	Closing WIP	12,000	100	12,000	1/3rd	4,000
	40,000		40,000		40,000		32,000

(ii) **Statement showing cost for each element**

Particulars	Materials (₹)	Labour (₹)	Overhead (₹)	Total (₹)
Cost of opening work-in-process	1,20,000	20,000	20,000	1,60,000
Cost incurred during the month	5,12,000	3,00,000	3,00,000	11,12,000
Total cost: (A)	6,32,000	3,20,000	3,20,000	12,72,000
Equivalent units: (B)	40,000	32,000	32,000	
Cost per equivalent unit: (C) = (A ÷ B)	15.8	10	10	35.8

(iii) **Statement of apportionment of cost**

	Particulars	Amount (₹)	Amount (₹)
1.	Value of units completed and transferred (28,000 units × ₹ 35.8)		10,02,400
2.	Value of Closing W-I-P:		
	- Materials (12,000 units × ₹ 15.8)	1,89,600	
	- Labour (4,000 units × ₹ 10)	40,000	
	- Overheads (4,000 units × ₹ 10)	40,000	2,69,600

(iv) **Process-I Cost Account**

Particulars	Units	(₹)	Particulars	Units	(₹)
To Opening W-I-P	8,000	1,60,000	By Completed units	28,000	10,02,400
To Materials	32,000	5,12,000	By Closing W-I-P	12,000	2,69,600
To Labour	--	3,00,000			
To Overhead	--	3,00,000			
	40,000	12,72,000		40,000	12,72,000

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698 RTP **Old P3 Jul 2021** Q8 old scheme

A company produces a component, which passes through two processes. During the month of November, 2020, materials for 40,000 components were put into Process- I of which 30,000 were completed and transferred to Process- II. Those not transferred to Process- II were 100% complete as to materials cost and 50% complete as to labour and overheads cost. The Process- I costs incurred were as follows:

Direct Materials	₹ 3,00,000
Direct Wages	₹ 3,50,000
Factory Overheads	₹ 2,45,000

Of those transferred to Process II, 28,000 units were completed and transferred to finished goods stores. There was a normal loss with no salvage value of 200 units in Process II. There were 1,800 units, remained unfinished in the process with 100% complete as to materials and 25% complete as regard to wages and overheads.

Costs incurred in Process-II are as follows:

Packing Materials	₹ 80,000
Direct Wages	₹ 71,125
Factory Overheads	₹ 85,350

Packing material cost is incurred at the end of the second process as protective packing to the completed units of production.

Required:

(i) PREPARE Statement of Equivalent Production, Cost per unit and Process I A/c.

(ii) PREPARE statement of Equivalent Production, Cost per unit and Process II A/c.

ICAI'S ANSWER

Process I

Statement of Equivalent Production and Cost

Input (Units)	Particulars	Output Units	Equivalent Production					
			Materials		Labour		Overheads	
			(%)	Units	(%)	Units	(%)	Units
40,000	Completed	30,000	100	30,000	100	30,000	100	30,000
	Closing WIP	10,000	100	10,000	50	5,000	50	5,000
40,000		40,000		40,000		35,000		35,000

Particulars	Materials	Labour	Overhead	Total
Cost incurred (₹)	3,00,000	3,50,000	2,45,000	8,95,000
Equivalent units	40,000	35,000	35,000	
Cost per equivalent unit (₹)	7.50	10.00	7.00	24.50

Process-I Account

Particulars	Units	(₹)	Particulars	Units	(₹)
To Materials	40,000	3,00,000	By Process-II A/c (30,000 units × ₹24.5)	30,000	7,35,000
To Labour		3,50,000	By Closing WIP*	10,000	1,60,000
To Overhead		2,45,000			
	40,000	8,95,000		40,000	8,95,000

* (Material 10,000 units × ₹ 7.5) + (Labour 5,000 units × ₹ 10) + (Overheads 5,000 units × ₹7)

= ₹ 75,000 + ₹ 50,000 + ₹ 35,000 = ₹ 1,60,000

Process II

Statement of Equivalent Production and Cost

Input (Units)	Particulars	Output Units	Equivalent Production					
			Materials		Labour		Overheads	
			(%)	Units	(%)	Units	(%)	Units
30,000	Completed	28,000	100	28,000	100	28,000	100	28,000
	Normal loss	200		--		--		--
	Closing WIP	1,800	100	1,800	25	450	25	450
30,000		30,000		29,800		28,450		28,450

Particulars	Materials	Labour	Overhead	Total
Process-I Cost	7,35,000	--	--	7,35,000
Cost incurred (₹)	--	71,125	85,350	1,56,475
Equivalent units	29,800	28,450	28,450	--
Cost per equivalent unit (₹)	24.6644	2.5000	3.0000	30.1644

Process-II Account

Particulars	Units	(₹)	Particulars	Units	(₹)
To Process-I A/c	30,000	7,35,000	By Normal loss A/c	200	--
To Packing Material	--	80,000	By Finished Goods Stock A/c	28,000*	9,24,604
To Direct Wages	--	71,125	By Closing WIP	1,800**	46,871
To Factory Overhead	--	85,350			
	30,000	9,71,475		30,000	9,71,475

* 28,000 × ₹ 30.1644 = ₹ 8,44,603 + ₹ 80,000 (Packing Material Cost) = ₹ 9,24,604

** 1,800 units × ₹ 24.6644 + 450 units × (₹ 2.5 + ₹3) = ₹ 46,871

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699 MTP **Sep 2026 S-I** Case Scenario 1. · MCQ 1 to MCQ 5 10 marks

CASE SCENARIO 1.

Aarav Industrial Resins Pvt. Ltd., headquartered in Ghaziabad, is engaged in manufacturing specialty chemicals used in the paints, laminates, and adhesives industry. The company operates a continuous two-stage production system.

The company produces a base intermediate resin called “Phenolic Resin Base” in the first stage. This intermediate product can either:

- Be sold directly to small-scale manufacturers, or
- Be further processed into a higher-grade “Laminated Resin Compound”, which commands a better market price.

Following stage 1 production data and cost data are given for the month just ended:

Raw material input	: 2,20,000 kg
Opening Work-in-Progress	: 50,000 kg
(Materials 100% complete, Conversion 60% complete)	
Output transferred/sold	: 1,70,000 kg
Closing Work-in-Progress	: 40,000 kg
(Materials 100% complete, Conversion 70% complete)	
Normal process loss: 20% of Current input	
(Loss occurs due to evaporation and chemical reactions; no scrap value)	

Particulars	Amount (₹)
Raw Material Cost (Current Month)	88,000
Conversion Costs (Current Month)	1,20,000
Opening WIP	
– Material	25,000
– Conversion	15,000

The intermediate product can be sold in the open market at ₹1.80 per kg. Out of the total production, 50,000 kg were sold directly, while 1,20,000 kg were transferred to the second stage for further processing into a higher-grade product.

The further processing details in the Finishing and Compounding Unit are as follows:

In this stage, additional materials such as fillers and additives are introduced, resulting in an output where each kilogram of Phenolic Resin Base yields 2 kilograms of Laminated Resin Compound. The finished product is sold at ₹ 2.40 per kg in the market.

The monthly costs incurred in Process II (other than the cost of Product 'Phenolic Resin Base') are:

Input Quantity (Base Resin)	Material Cost (₹)	Processing Cost (₹)
1,20,000 kg	1,50,000	1,30,000
1,60,000 kg (Full Capacity)	2,00,000	1,55,000

The management is currently evaluating operational efficiency and profitability decisions based on the above information:

They are particularly interested in determining the cost per unit of the intermediate product, valuation of completed output and work-in-progress, profitability of further processing, and the feasibility of utilizing idle capacity in the second stage.

The finance team has been asked to compute the following requirements (MCQs 1 to 5):

MCQ 1 2 marks

What is the cost per kg of Phenolic Resin Base in Stage 1 using the weighted average cost method?

- A ₹ 0.90
- B ₹ 1.13
- C ₹ 1.20
- D ₹ 1.35

MCQ 2 2 marks

What is the total value of output completed during the month? (Round off intermittent calculations to three decimal places)

- A ₹ 1,92,270
- B ₹ 1,78,500
- C ₹ 2,04,000
- D ₹ 2,10,000

MCQ 3 2 marks

What is the value of closing work-in-progress for the month? (Round off intermittent calculations to three decimal places)

- A ₹ 32,000
- B ₹ 36,500
- C ₹ 40,000
- D ₹ 37,668

MCQ 4 2 marks

What is the incremental profit or loss on further processing 1,20,000 kg of Phenolic Resin Base into Laminated Resin Compound?

- A Profit ₹ 80,000
- B Profit ₹ 48,000
- C Loss ₹ 24,000
- D Loss ₹ 12,000

MCQ 5 2 marks

What is the minimum acceptable selling price per kg of Laminated Resin Compound for additional production using spare capacity?

- A ₹ 1.60
- B ₹ 1.84
- C ₹ 2.00
- D ₹ 2.20

ICAI'S ANSWERS

MCQ 1

(B) ₹ 1.13

Process – I

Statement of Equivalent Production

Particulars	Input (Kg.)	Particulars	Output (Kg.)	Material (%)	Material (Kg.)	Conversion (%)	Conversion (Kg.)
Opening WIP	50,000	Normal Loss	44,000	–	–	–	–
New Material Introduced	2,20,000	Units Completed	1,70,000	100	1,70,000	100	1,70,000
		Abnormal Loss	16,000	100	16,000	100	16,000
		Closing WIP	40,000	100	40,000	70	28,000
Total	**2,70,000**	**Total**	**2,70,000**		**2,26,000**		**2,14,000**

Statement of Cost for Each Element

Elements of Cost	Opening WIP (₹)	Current Cost (₹)	Total Cost (₹)	Equivalent Units (Kg.)	Cost per Kg. (₹)
Material	25,000	88,000	1,13,000	2,26,000	0.500
Conversion Cost	15,000	1,20,000	1,35,000	2,14,000	0.631
Total	**40,000**	**2,08,000**	**2,48,000**		**1.131**

MCQ 2

(A) ₹ 1,92,270

MCQ 3

(D) ₹ 37,668

Statement of Apportionment of Cost

Particulars	Elements	Equivalent Units (Kg.)	Cost per Unit (₹)	Cost (₹)	Total (₹)
Work Completed	Material	1,70,000	0.500	85,000	
	Conversion	1,70,000	0.631	1,07,270	1,92,270
Closing WIP	Material	40,000	0.500	20,000	
	Conversion	28,000	0.631	17,668	37,668

MCQ 4

(A) Profit ₹ 80,000

Statement Showing Comparative Data

Alternative I – Sell after Process I

Particulars	Amount (₹)
Sales (1,20,000 kg × ₹ 1.80)	2,16,000
Less: Cost (1,20,000 kg × ₹ 1.131)	1,35,720
Profit	**80,280**

Alternative II – Further Processing

Particulars	Amount (₹)
Sales (2,40,000 kg × ₹ 2.40)	5,76,000
Less: Cost from Process I	1,35,720
Material in Process II	1,50,000
Processing Cost in Process II	1,30,000
Total Cost	**4,15,720**
Profit	**1,60,280**

Conclusion:

Increase in Profit = ₹ 1,60,280 – ₹ 80,280 = ₹ 80,000

MCQ 5

(B) ₹ 1.84

Calculation of Minimum Selling Price

Cost of processing remaining 40,000 kg. further

Particulars	Amount (₹)
Additional Material Cost	50,000
Additional Processing Cost	25,000
Cost from Process I (40,000 × ₹ 1.131)	45,240
Opportunity Cost if 40,000 kg. are further processed (40,000 × ₹ 0.669)	26,760
Total Cost	**1,47,000**

Additional Output (40,000 × 2) = 80,000 kg

Minimum Selling Price per kg = ₹ $\frac{1,47,000}{80,000}$ = ₹ 1.84 per kg

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700 MTP **Sep 2026 S-II** Q2(a) 8 marks

The following data are available in respect of process I for February:

i. Opening stock of work-in-progress: 2,400 units at a total cost of ₹ 12000

ii. Degree of completion of opening work-in-progress:

Materials	100%
Labor	60%
Overhead	60%

i. Input of materials at a total cost of ₹ 1,10,400 for 27,600 units.

ii. Direct wages incurred ₹ 50,220

iii. Production overheads ₹ 25,110

iv. Units scrapped 3,600 units. The stage of completion of these units.

Materials	100%
Labor	80%
overheads	80%

v. Closing work-in-progress: 2,700 units. The stage of completion of these units was:

Materials	100%
Labor	70%
overheads	70%

vi. 23,700 units were completed and transferred to the next process.

vii. Normal loss is 8% of the total input (opening stock plus units put in)

viii. Scrap value is ₹ 4 per unit.

You are required to:

a. Compute equivalent production,

b. Calculate the cost per equivalent for each element,

c. Calculate the cost of abnormal loss (or gain), closing work-in-process and the units transferred to the next process using the FIFO method,

d. Show the process Account for February.

ICAI'S ANSWER

a. **Statement of Equivalent Production**

Process I

Input Details	Units	Output Particulars	Units	Equivalent Production					
				Material- A		Labour		Overhead	
				%	Units	%	Units	%	Units
Opening WIP	2,400	Work on Op. WIP	2,400	-	-	40	960	40	960
Introduced	27,600	Introduced & completed during the month	21,300	100	21,300	100	21,300	100	21,300
		Normal loss (8% of 30,000 units)	2,400	-	-	-	-	-	-
		Closing WIP	2,700	100	2,700	70	1,890	70	1,890
		Abnormal Loss	1,200	100	1,200	80	960	80	960
	30,000		30,000		25,200		25,110		25,110

Working note:

Production units = Opening units + Units Introduced - Closing Units - Units Scrapped

= 2,400 units + 27,600 units – 2,700 units - 3600 units

= 23,700 units (Opening WIP 2,400 + Introduced and completed 21,300)

b. **Statement of Cost**

	Cost (₹)	Equivalent units	Cost per equivalent units (₹)
Material Cost	1,104,00		
Less: Scrap value of normal loss (2,400 units x ₹ 4)	(9,600)		
	1,00,800	25,200	4.00
Labour	50,220	25,110	2.00
Overheads	25,110	25,110	1.00

c. **Statement of apportionment of process cost**

		Amount (₹)
Opening WIP (2400 units)		12,000
Completed opening WIP (960 units× ₹ 3)	Labour + Overheads (2 + 1)	2,880
Introduced & Completed- 21,300 units × ₹ 7	Material + Labour + Overheads (4 + 2 + 1)	**1,49,100**
Total cost of 23,700 finished goods units		**163,980**
Closing WIP units- 2700	Material (2,700 units × ₹ 4) Labour (1890 units x Rs 2.00) Overheads (1,890 units x ₹ 1.00)	10,800 3,780 1,890 **16,470**

		Amount (₹)
Abnormal Loss- 1,200 Units	Material (1,200 units × ₹ 4)	4,800
	Labour (960 units x ₹ 2.00)	1,920
	Overheads (960 units x ₹ 1.00)	960
		7,680

d. **Process I A/c**

Particulars	Units	Amount ₹	Particulars	Units	Amount ₹
To Balance b/d	2400	12,000	By Normal loss	2,400	9,600
To Units Introduced	27,600	1,10,400	Finished By goods	23,700	1,63,980
To Wages		50,220	By Closing WIP	2,700	16,470
To Overheads		25,110	By Abnormal Loss	1,200	7,680
	30,000	**1,97,730**		**30,000**	**1,97,730**

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701 MTP **May 2026 S-I** Case Scenario 1 · MCQ 1 to MCQ 5 10 marks

CASE SCENARIO 1

Mr. Sagar, the CFO of Super high Ltd. was shocked to see the plummeting margins in the second quarterly results of the company as presented in the committee meeting. He wondered how can things go so wrong all of a sudden, while he had been awarded for exemplary performance just three months back.

In order to set things right and ensure a positive turnaround, he appointed you to analyze and present a report giving narrative for the downfall.

You begin with seeking information which is presented as under:

There was no change in the selling price of the product from last quarter.

There was no opening or closing stocks of raw material or finished units in the beginning or end of the quarter. However, there were stocks of WIP (work in progress) units.

Opening WIP was 1,000 units which was 60% complete in respect of labour and overheads. Super high Ltd. was engaged in a process where material was fed immediately in the pipe line. The opening WIP was valued at standard cost which turns out to be ₹ 1,04,000 as shown later.

Cost of Material used: ₹ 3,00,000 (6000 units @ 50 ₹ per Kg)

The Cost of Labour and overheads incurred during the quarter was ₹ 3,15,900 and ₹ 1,74,960 respectively.

4,500 completed units were produced and 1,600 units were closing WIP which was 60% complete in respect of labour and overheads. The normal loss in process was 900 units which was 15% of the current input. It was sold as scrap for ₹ 4,200.

The standard cost card showed that 1 unit of finished product would cost ₹ 50 after considering the normal standard loss which is 15% and the labour hours should be 3 hrs. @ ₹ 20 per hour, the cost of

overheads per unit should be ₹ 30 per unit.

Opening WIP value:

- Material: 50 × 1,000 = ₹ 50,000
- Labour: 60% × (3 hrs × ₹ 20 × 1,000) = ₹ 36,000
- OH: 60% × ₹ 30 × 1,000 = ₹ 18,000
- Total Opening WIP = ₹ 1,04,000

MCQ 1 2 marks

Equivalent units for material (FIFO) are:

A 5,100

B 6,000

C 6,100

D 4,500

MCQ 2 2 marks

Cost per equivalent unit of labour:

A ₹ 65

B ₹ 61.94

C ₹ 70.20

D ₹ 90.26

MCQ 3 2 marks

Material Cost Variance and Labour Cost Variance for equivalent units is:

A ₹ 40,800 F and ₹ 40,800 A respectively

B ₹ 28,000 A and ₹ 40,800 F respectively

C ₹ 40,800 A and ₹ 24,300 A respectively

D ₹ 28,000 F and ₹ 25,000 F respectively

MCQ 4 2 marks

Standard cost of 4,500 units:

A ₹ 7,00,900

B ₹ 7,90,860

C ₹ 5,15,600

D ₹ 6,30,000

MCQ 5 2 marks

Actual cost of 4,500 units:

A ₹ 5,86,500

B ₹ 7,15,500

C ₹ 7,00,900

D ₹ 6,30,000

ICAI'S ANSWERS

MCQ 1

(A) 5,100

Equivalent Production (FIFO Method)

Input Details	Units	Output Particulars	Units	Equivalent Production					
				Material		Labour		Overheads	
				%	Units	%	Units	%	Units
Opening WIP	1,000	From Opening WIP	1,000	-	-	40	400	40	400
Fresh inputs	6,000	From fresh units	3,500	100	3,500	100	3,500	100	3,500
		Normal loss	900	-		-		-	
		Closing WIP	1,600	100	1,600	60	960	60	960
	7,000		7,000		5,100		4,860		4,860

MCQ 2

(A) ₹ 65

Cost per Equivalent Units (EU):

Element	Total cost (₹)	Equivalent units (EU)	Cost per EU (₹)
Value of raw material introduced	3,00,000		
Less: Scrap value of normal loss	(4,200)		
Total value of raw material	2,95,800	5,100	2,95,800/5,100 = **₹ 58**
Labour	3,15,900	4,860	3,15,900/4,860 = **₹ 65**
Overheads	1,74,960	4,860	1,74,960/4,860 = **₹ 36**

MCQ 3

(C) ₹ 40,800 A and ₹24,300 A respectively

Material Cost Variance for Equivalent units

Actual Material Cost = 2,95,800 (3,00,000 – 4,200)

Standard Price = ₹ 50

Equivalent units of Material = 5,100

Standard cost = 5,100 × 5 = ₹ 2,55,000

Material Cost Variance = Standard Cost – Actual Cost

= (2,55,000 – 2,95,800)

= ₹40,800 Adverse

Labour Cost Variance for Equivalent units

Actual cost of labour on completed units = ₹ 3,15,900

Standard rate per unit = ₹ 60

Standard cost = 4,860 × 60 = ₹ 2,91,600

Labour Cost Variance = Standard Cost – Actual Cost

= (₹ 2,91,600 - ₹ 3,15,900)

= ₹ 24,300 (A)

MCQ 4

(D) ₹ 6,30,000

Standard Cost per Unit

Material: = 50

Labour: 3 hrs × 20 = 60

Overhead Costs = 30

Total Standard Cost = ₹140 per unit

Standard Cost of 4,500 Units

= 4,500 × 140

= ₹6,30,000

MCQ 5

(C) ₹7,00,900

Completion of Opening WIP

Labour: 400 × 65 = 26,000

OH: 400 × 36 = 14,400

Add: opening cost: = 1,04,000

= **₹ 1,44,400**

Started & Completed (3,500 units)

Material = 3,500 × 58 = 2,03,000

Labour = 3,500 × 65 = 2,27,500

OH = 3,500 × 36 = 1,26,000

= **₹ 5,56,500**

Total Cost of 4,500 Units

= 1,44,400 + 5,56,500

= **₹ 7,00,900**

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702 MTP May 2026 S-II Q1(b) 5 marks

The following details are available of Process X for August 2025:

(1)	Opening work-in-progress	8,000 units
	Degree of completion and cost:	

	Material (100%)	₹ 63,900
	Labour (60%)	₹ 10,800
	Overheads (60%)	₹ 5,400
(2)	Input 1,82,000 units at	₹ 7,56,900
(3)	Labour paid	₹ 3,28,000
(4)	Over heads incurred	₹ 1,64,000
(5)	Units scrapped	14,000
	Degree of completion:	
	Material	100%
	Labour and overhead	80%
(6)	Closing work-in-process	18000 units
	Degree of completion:	
	Material	100%
	Labour and overhead	70%
(7)	1,58,000 units were completed and transferred to next process.	
(8)	Normal loss is 8% of total input including opening work-in-process	
(9)	Scrap value is ₹ 8 per unit to be adjusted in direct material cost	

You are required to COMPUTE, assuming that average method of inventory is used:

(i) Equivalent production, and

(ii) Cost per unit

ICAI'S ANSWER

(i) Statement of Equivalent Production

Particulars	Units	Material		Labour and Overhead	
		(%)	Units	(%)	Units
Production units completed	1,58,000	100	1,58,000	100	1,58,000
Normal Loss: 8% of Input (1,82,000 + 8,000)	15,200	--	--	--	--
Closing WIP	18,000	100	18,000	70	12,600
	1,91,200	--	1,76,000	--	1,70,600
Less: Abnormal Gain	1,200	100	1,200	100	1,200
Total	1,90,000		1,74,800		1,69,400

(ii) Statement of cost

Particulars	Materials (₹)	Labour (₹)	Overhead (₹)
Opening WIP	63,900	10,800	5,400
Input of Materials	7,56,900	-	-
Expenses	-	3,28,000	1,64,000
Total	8,20,800	3,38,800	1,69,400
Less: Sale of Scrap (15,200 x ₹ 8)	1,21,600	-	-
Net cost	6,99,200	3,38,800	1,69,400
Equivalent Units	1,74,800	1,69,400	1,69,400
Cost Per Unit	₹ 4.00	₹ 2.00	₹ 1.00

Total cost per unit = ₹ (4+2+1) = ₹ 7.00

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703 MTP **Jan 2026 S-I** Q4(a) 8 marks

Golden Harvest Foods Pvt. Ltd. is engaged in the production of organic honey blocks. Its process involves crushing of honeycombs for nectar extraction, followed by filtration and heating of the nectar along with certain natural stabilizers, and finally cooling and cutting the solidified honey into blocks.

The main process of nectar extraction (Process–I) is carried out in mechanical crushers, after which the nectar is filtered and boiled (Process–II) in steel vessels. The solidified honey blocks are then cut, packed, and dispatched.

For producing 10 kg of honey blocks, 100 kg of honeycombs are required, from which only 45 litres of nectar can be extracted.

Information for Process – I (Nectar Extraction) for January 2020

Particulars	Amount (₹)
Opening Work-in-Process (4,500 litres)	
Honeycombs	50,000
Labour	15,000
Overheads	45,000
Honeycombs introduced for nectar extraction (1,00,000 kg)	5,00,000
Direct Labour	2,00,000
Overheads	6,00,000

Additional Information:

- Abnormal Loss: 1,000 kg
- Degree of completion for abnormal loss:
  - Honeycombs – 100%
  - Labour & Overheads – 80%
- Closing Work-in-Process: 9,000 litres
  - Degree of completion:
    - Honeycombs – 100%
    - Labour & Overheads – 80%
- Extracted nectar transferred to Process–II: 39,500 litres
- (Assume 1 litre of nectar = 1 kg)

You are required to PREPARE (using Average Method):

(i) Statement of Equivalent Production

(ii) Statement of Cost

(iii) Statement of Distribution of Cost

(iv) Process–I Account

ICAI'S ANSWER

(i) Statement of Equivalent Production

Particulars	Input Units	Particulars	Output Units	Equivalent Production			
				Honeycomb		Labour & O.H.	
				%	Units	%	Units
Opening WIP	4,500	Completed and transferred to Process - II	39,500	100	39,500	100	39,500

Particulars	Input Units	Particulars	Output Units	Equivalent Production			
Units introduced	1,00,000	Normal Loss (55%* of 1,00,000)	55,000	--	--	--	--
		Abnormal loss	1,000	100	1,000	80	800
		Closing WIP	9,000	100	9,000	80	7,200
	1,04,500		1,04,500		49,500		47,500

* For every 100 kg of honeycombs, only 45 litres of nectar are extracted. Thus, normal loss = 100 – 45 = 55%

(ii) Statement showing cost for each element

Particulars	Honeycomb (₹)	Labour (₹)	Overhead (₹)	Total (₹)
Cost of opening work-in-process	50,000	15,000	45,000	1,10,000
Cost incurred during the month	5,00,000	2,00,000	6,00,000	13,00,000
Total cost: (A)	5,50,000	2,15,000	6,45,000	14,10,000
Equivalent units: (B)	49,500	47,500	47,500	
Cost per equivalent unit: (C) = (A ÷ B)	11.111	4.526	13.579	29.216

(iii) Statement of Distribution of cost

	Amount (₹)	Amount (₹)
1. Value of units completed and transferred (39,500 units × ₹ 29.216)		11,54,032
2. Value of Abnormal Loss:		
- Honeycomb (1,000 units × ₹ 11.111)	11,111	
- Labour (800 units × ₹ 4.526)	3,621	
- Overheads (800 units × ₹ 13.579)	10,863	25,595
3. Value of Closing W-I-P:		
- Honeycomb (9,000 units × ₹ 11.111)	99,999	
- Labour (7,200 units × ₹ 4.526)	32,587	
- Overheads (7,200 units × ₹ 13.579)	97,769	2,30,355

(iv) Process-I A/c (Nectar Extraction)

Particulars	Units	(₹)	Particulars	Units	(₹)
To Opening W.I.P:			By Normal Loss	55,000	--
- Honeycomb	4,500	50,000	By Abnormal loss (₹ 25,595 + ₹ 18 (difference due to approximation))	1,000	25,613
- Labour	--	15,000	By Process-II A/c	39,500	11,54,032
- Overheads	--	45,000	By Closing WIP	9,000	2,30,355
To Honeycomb introduced	100,000	5,00,000			
To Direct Labour		2,00,000			
To Overheads		6,00,000			
	104,500	14,10,000		104,500	14,10,000

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704 MTP **Jan 2026 S-II** Q2(a) 8 marks

Seema Ltd. is known for its two-stage production process, which is as follows:

Process A: Raw wool is converted into premium woollen thread.

Process B: Woollen threads are further refined to create luxurious woollen garments.

Following data is given for the quarter ended March 2025:

Particulars	Process A	Process B
Opening work-in-progress	2,000 units	Nil
– Material	₹ 4,50,000	—
– Labour & Overheads	₹ 1,44,000	—
Costs incurred during the quarter:		
– Material	₹ 18,00,000	₹ 6,87,500
– Labour & Overheads	₹ 5,70,000	—
– Other material	—	₹ 30,25,000
Units of production:		
Received in process	8,000 units	5,500 units
Completed units	7,000 units	5,500 units
Completed units transferred to next process	5,500 units	—

Particulars	Process A	Process B
Completed units sold in the market	1,500 units	5,500 units
Closing work-in-progress	1,500 units	Nil
Normal loss in process	10% of the total input (including opening WIP)	Nil

Additional Information:

- Normal loss units are sold at a realizable value of ₹ 90 per unit.
- Any loss beyond normal loss is considered 100% complete in respect of material and labour & overheads.
- In the market, woollen garments and remaining woollen threads were sold at 20% markup on the cost price.

Degree of Completion: Process A

Particulars	Material	Labour & Overheads
Opening work-in-progress	100%	50%
Closing work-in-progress	100%	60%

Using Weighted Average Method, you are required to PREPARE:

(i) Statement of equivalent production of Process A.

(ii) Statement showing cost for each element of Process A.

(iii) Statement of distribution of cost of Process A.

(iv) Statement showing sale proceeds of each of the product woollen thread and woollen garments along-with total sale proceeds of Seema Ltd. during the specified period.

ICAI'S ANSWER

(i) Statement of equivalent production : Process A

Particulars	Input Units	Particulars	Output Units	Equivalent Production			
				Material		Labour & OH	
				%	Units	%	Units
Opening WIP	2,000	Completed Units	7,000	100	7,000	100	7,000
Units Introduced	8,000	Normal Loss [(2,000+8,000) x 10%]	1,000	---	---	---	---
		Abnormal Loss (Bal. Fig.)	500	100	500	100	500
		Closing WIP	1,500	100	1,500	60	900
Total	10,000	Total	10,000		9,000		8,400

(ii) Statement showing cost for each element: Process A

Particulars	Material	Labour & OH	Total
Cost of Opening WIP	4,50,000	1,44,000	5,94,000
Cost of Completed Units	18,00,000	5,70,000	23,70,000
Less: Realizable Value (1,000 × 90)	(90,000)	----	(90,000)
Total Cost	21,60,000	7,14,000	28,74,000
Equivalent Units	9,000	8,400	
Cost per EQ Unit	240	85	325

(iii) Statement of distribution of cost: Process A

Particulars	EQ Units	Cost per EQ Unit	Amount
Completed Units	7,000	325	22,75,000
Abnormal Loss	500	325	1,62,500
Closing WIP:			
Material	1,500	240	3,60,000
Labour & Overheads	900	85	76,500
			4,36,500

(iv) Sale proceeds of WR fabrics:

Out of 7,000 units of Process A (woolen threads), only 5,500 units were further processed in Process B and sold as woolen garments. Thus balance 1,500 units of woolen threads were sold in the market.

	Units	Selling price p.u. (Refer Working note)	Amount in Rs.
Process A	1,500	390	5,85,000
Woolen Threads:			
Process B:			
Woolen Garments	5,500	1,200	66,00,000
		Total Sales	71,85,000

Working Notes:

Further Processing: Process B

Particulars	Units	Computation	Amount
Transferred from Process A	5,500	5,500 × 325	17,87,500
Material			30,25,000
Labour & Overheads			6,87,500
Total	5,500		55,00,000
Woolen Garment Cost per Unit		55,00,000/5,500	1,000

Sale Price of Woolen Thread = ₹ 325 per unit + 20% = ₹ 390 per unit

Sale Price of Woolen Garment = ₹ 1,000 per unit + 20% = ₹ 1,200 per unit

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705 MTP **Sep 2025 S-II** MCQ 13 2 marks MCQ

Consider the following data for a production process:

- Opening Work-in-Progress: 3,000 units (Degree of completion: 60%)
- Units introduced and completed during the period: 6,000 units
- Closing Work-in-Progress: 4,000 units (Degree of completion: 70%)

What is the equivalent production under the FIFO method in process costing?

A 9,600 units

B 9,800 units

C 10,000 units

D 8,600 units

ICAI'S ANSWER

(C) 10,000 units

1. Completion of Opening WIP
   = 3,000 units x (100% – 60%)
   = 3,000 x 40%
   = 1200 units
2. Units started and completed during the period
   These are new units that were completed in the period (excluding the opening WIP):
   = 6,000 units
3. Equivalent units in Closing WIP
   = 4,000 units x 70%
   = 2,800 units

   Total Equivalent Units (FIFO)
   = Completion of Opening WIP + Units started & completed + Closing WIP
   = 1,200 + 6,000 + 2,800

= 10,000 units

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706 MTP **May 2025 S-II** MCQ 14 2 marks MCQ

A firm introduced 3,000 units of material in the manufacturing process. During the period 2,500 units were completed and transferred to next process. However, the degree of completion on remaining 500 units was 100%, 60%, and 30% for materials, labour and overheads respectively. Which one of the following is the equivalent complete units with regard to labour?

A 2,500

B 2,800

C 2,650

D 2,500

ICAI'S ANSWER

(B) 2,800 units

Equivalent Units = Units Completed + (Units in Process x Degree of Completion)

Equivalent Units for Labour = 2,500 + (500 x 0.60)

= 2,800 units

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707 MTP **Jan 2025 S-I** Q5(a) 10 marks

Hawtt Veel is a renowned brand of HV Ltd. which manufactures toy car. The manufacturing process of the toy cars at first involve designing the parts, creating the mold and then simultaneously melting the plastic. As the mold created last year is being used as it is for the current year, the first process involves only melting the plastic (Process I). The next process is about injecting the plastic into the mold and assembling the parts formed (Process II).

During the month of April, the materials for 1,20,000 toy cars were put through Process I of which melting process were completed for 90,000 toy cars only before transferring to Process II.

The costs incurred in Process I are as follows:

Direct material	₹ 22,50,000
Direct wages	₹ 27,00,000
Factory overheads	₹ 18,00,000

Degree of completion for those not transferred to Process II is as follows:

Materials	100%
Labour and overheads	50%

Out of those transferred to Process II for injecting and assembling, 84,000 units of toy car were completed and transferred to finished goods store for protective packing. The process of protective packing is done at the end of the Process II and the costs incurred are as follows:

Packing materials	₹ 6,00,000
Direct wages	₹ 5,25,000
Factory overheads	₹ 6,75,000

There was a normal loss of 600 units in Process II with no salvage value.

Some units were still in progress under Process II and thus, shifted for the next month process. The degree of completion for those not transferred to finished goods store is as follows:

Materials	100%
Labour and overheads	25%

You are required to PREPARE-

(i) Statement of Equivalent Production, Cost per unit and Process I A/c.

(ii) Statement of Equivalent Production, Cost per unit and Process II A/c.

ICAI'S ANSWER

(i) Process I – Statement of Equivalent Production

Particulars	Completed Units	Closing stock of WIP			Equivalent Production units
		Units	% of Completion	Equivalent Units	
	(1)			(2)	(1) + (2)
Material	90,000	30,000	100%	30,000	1,20,000
Wages	90,000	30,000	50%	15,000	1,05,000
Overhead	90,000	30,000	50%	15,000	1,05,000

Process I

Particulars	Process Cost (₹)	Equivalent Production (units)	Process Cost p.u. (₹) (2)/(3)	WIP stock Equivalent units	Cost of WIP Stock (₹) (4) x (5)	Transfer to Process II (₹) (2)-(6)
(1)	(2)	(3)	(4)	(5)	(6)	(7)
Material	22,50,000	1,20,000	18.750	30,000	5,62,500	16,87,500
Wages	27,00,000	1,05,000	25.714	15,000	3,85,714	23,14,286
Overhead	18,00,000	1,05,000	17.143	15,000	2,57,143	15,42,857
	67,50,000				12,05,357	55,44,643

Process I A/c

	Particulars	Unit	(₹)		Particulars	Units	(₹)
To	Direct material	1,20,000	22,50,000	By	Process II A/c	90,000	55,44,643
To	Direct wages	--	27,00,000	By	Closing W-I-P	30,000	12,05,357
To	Factory overhead	--	18,00,000				
		1,20,000	67,50,000			1,20,000	67,50,000

(ii) Process II – Statement of Equivalent Production

Particulars	Completed Units	Closing stock of WIP			Equivalent Production units
		Units	% of Completion	Equivalent Units	
	(1)			(2)	(1) + (2)
Material	84,000	5,400*	100%	5,400	89,400
Wages	84,000	5,400	25%	1,350	85,350
Overhead	84,000	5,400	25%	1,350	85,350

*(90,000 - 84,000 - 600) units = 5,400 units

Process II

Particulars	Process Cost (₹)	Equivalent Production (units)	Process Cost p.u. (₹) (2)/(3)	WIP stock Equivalent units	Cost of WIP Stock (₹) (4) x (5)	Transfer to Finished Stock (₹) (2)-(6)
(1)	(2)	(3)	(4)	(5)	(6)	(7)
Material	55,44,643	89,400	62.021	5,400	3,34,911	52,09,732
Wages	5,25,000	85,350	6.151	1,350	8,304	5,16,696
Overhead	6,75,000	85,350	7.909	1,350	10,677	6,64,323
	67,44,643				3,53,892	63,90,751
Add: Packing Material Cost						6,00,000
Cost of Finished Stock						69,90,751

Process II A/c

	Particulars	Units	(₹)		Particulars	Units	(₹)
To	Process I	90,000	55,44,643	By	Finished Stock	84,000	69,90,751
To	Direct wages	--	5,25,000	By	Normal loss	600	--
To	Factory overhead	--	6,75,000	By	WIP stock	5,400	3,53,892
To	Packing charges	--	6,00,000				
		90,000	73,44,643			90,000	73,44,643

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708 MTP Sep 2024 S-I Q2(a) 10 marks

PQR Company Ltd. provides the following information relating to Process-P:

(i)	Opening Work-in-progress	-	NIL
(ii)	Units Introduced	-	45,000 units @ ₹ 10 per unit
(iii)	Expenses debited to the process:		
	Direct material		₹ 65,500
	Labour		₹ 90,800
	Overhead		₹ 1,80,700
(iv)	Normal loss in the process	-	2% of Input
(v)	Work-in progress	-	1800 units
	Degree of completion		
	Materials	-	100%
	Labour	-	50%
	Overhead	-	40%
(vi)	Finished output	-	42,000 units
(vii)	Degree of completion of abnormal loss:		
	Materials	-	100%
	Labour	-	80%
	Overhead	-	60%
(viii)	Units scrapped as normal loss were sold at ₹ 5 per unit.		
(ix)	All the units of abnormal loss were sold at ₹ 2 per unit.		

You are required to PREPARE:

- Statement of equivalent production.
- Statement showing the cost of finished goods, abnormal loss and closing balance of work-in-progress.
- Process-P account and abnormal loss account.

ICAI'S ANSWER

Statement of Equivalent Production

Input Details	Units	Output Particulars	Units	Equivalent Production					
				Material		Labour		Overhead	
				%	Units	%	Units	%	Units
Unit Introduced	45,000	Finished output	42,000	100	42,000	100	42,000	100	42,000
		Normal loss (2% of 45,000)	900	-	-	-	-	-	-
		Abnormal loss	300	100	300	80	240	60	180
		Closing W-I-P	1,800	100	1,800	50	900	40	720
	45,000		45,000		44,100		43,140		42,900

Statement of Cost

Particulars	Units	Rate (₹)	Amount (₹)	Amount (₹)
(i) Finished goods	42,000	17.9042		7,51,976.40
(ii) Abnormal Loss				
Material	300	11.5873	3,476.19	
Labour	240	2.1048	505.15	
Overhead	180	4.2121	758.18	4,739.52
(iii) Closing W-I-P:				
Material	1,800	11.5873	20,857.14	
Labour	900	2.1048	1,894.32	
Overhead	720	4.2121	3,032.71	25,784.17

Cost per Unit

Particulars	Amount (₹)	Units	Per Unit (₹)
(i) Direct Material :			
Unit Introduced	4,50,000		
Add: Material	65,500		
	5,15,500		
Less: Value of normal loss (900 units × ₹ 5)	(4,500)		
	5,11,000	44,100	11.5873
(ii) Labour	90,800	43,140	2.1048
(iii) Overhead	1,80,700	42,900	4.2121

Particulars	Amount (₹)	Units	Per Unit (₹)
			17.9042

Process – P A/c

Particulars	Units	Amount (₹)	Particulars	Units	Amount (₹)
To Input	45,000	4,50,000	By Normal loss	900	4,500
To Direct Material	-	65,500	By Abnormal loss	300	4,740
To Labour	-	90,800	By Finished goods	42,000	7,51,976
To Overhead		1,80,700	By Closing W-I-P	1,800	25,784
	45,000	7,87,000		45,000	7,87,000

Abnormal Loss A/c

Particulars	Units	Amount (₹)	Particulars	Units	Amount (₹)
To Process-B A/c	300	4,740	By Cost ledger control A/c or Bank A/c	300	600
			By Costing Profit & loss A/c	-	4,140
	300	4,740		300	4,740

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709 MTP **Sep 2024 S-II** Q2(a) 10 marks

As demand for LED light increases, more entrepreneurs are coming into its manufacturing process. eLED Pvt. Ltd. is also one of the recently formed company whose main business is related to LED lights.

The company has extended its hand into various LED products like COB (Chip On Board) LEDs, SMD (Surface Mounted Device) LEDs, RGB LEDs, Flashing LEDs, Miniature LEDs, OLEDs, Filament Bulbs, etc. However, at the beginning stage, the company has decided to only assemble the products and enter into manufacturing stage at later years.

The details relating to the first process of mounting for the month of August are given below:

Opening Work-in-Process:	31,000 units
Material	₹ 12,40,000
Labour	₹ 2,32,500
Overheads	₹ 6,97,500
Introduction during the process:	5,89,000 units
Material	₹ 2,29,40,000
Labour	₹ 55,64,500
Overheads	₹ 1,66,93,500

The process involve some wastage as well. The management estimated a normal loss of 5% of total input including opening work-in-process which can be sold out for ₹ 20 per unit. However, the workers reported 46,500 units as scrapped in which 100% material was used along with 80% of Labour and overheads.

5,42,500 units were transferred for next process of soldering.

Some units were still in process and thus, shifted for the next month process of mounting. With 100% material used along with 80% labour and overheads, 31,000 units were shifted.

Following the average method of inventory, you are required to PREPARE:

(i) Statement of cost showing cost per equivalent unit

(ii) Statement of distribution cost

(iii) Process Account (Mounting)

(iv) Normal Loss Account and Abnormal Loss Account.

ICAI'S ANSWER

(i) **Statement of Equivalent Production**

Particulars	Input Units	Particulars	Output Units	Equivalent Production			
				Material		Labour & O.H.	
				%	Units	%	Units
Opening WIP	31,000	Completed and transferred to Process (Soldering)	5,42,500	100	5,42,500	100	5,42,500
Units introduced	5,89,000	Normal Loss (5% of 6,20,000)	31,000	--	--	--	--

Particulars	Input Units	Particulars	Output Units	Equivalent Production			
		Abnormal loss (Balancing figure)	15,500	100	15,500	80	12,400
		Closing WIP	31,000	100	31,000	80	24,800
	6,20,000		6,20,000		5,89,000		5,79,700

Statement showing cost for each element

Particulars	Materials (₹)	Labour (₹)	Overhead (₹)	Total (₹)
Cost of opening work-in-process	12,40,000	2,32,500	6,97,500	21,70,000
Cost incurred during the month	2,29,40,000	55,64,500	1,66,93,500	4,51,98,000
Less: Realisable Value of normal scrap (₹ 20 × 31,000 units)	(6,20,000)	--	--	(6,20,000)
Total cost: (A)	2,35,60,000	57,97,000	1,73,91,000	4,67,48,000
Equivalent units: (B)	5,89,000	5,79,700	5,79,700	
Cost per equivalent unit: (C) = (A ÷ B)	**40.00**	**10.00**	**30.00**	**80.00**

(ii) **Statement of Distribution of cost**

	Amount (₹)	Amount (₹)
1. Value of units completed and transferred (5,42,500 units × ₹ 80)		4,34,00,000
2. Value of Abnormal Loss:		
- Materials (15,500 units × ₹ 40)	6,20,000	
- Labour (12,400 units × ₹ 10)	1,24,000	
- Overheads (12,400 units × ₹ 30)	3,72,000	11,16,000
3. Value of Closing W-I-P:		
- Materials (31,000 units × ₹ 40)	12,40,000	
- Labour (24,800 units × ₹ 10)	2,48,000	
- Overheads (24,800 units × ₹ 30)	7,44,000	22,32,000
Total		**4,67,48,000**

(iii) **Process Account (Mounting)**

Particulars	Units	(₹)	Particulars	Units	(₹)
To Opening W.I.P:			By Normal Loss (₹ 20 × 31,000 units)	31,000	6,20,000
- Materials	31,000	12,40,000	By Abnormal loss	15,500	11,16,000
- Labour	--	2,32,500	By Process A/c (Soldering)	5,42,500	4,34,00,000
- Overheads	--	6,97,500	By Closing WIP	31,000	22,32,000
To Materials introduced	5,89,000	2,29,40,000			
To Direct Labour		55,64,500			
To Overheads		1,66,93,500			
	6,20,000	4,73,68,000		6,20,000	4,73,68,000

(iv) **Normal Loss A/c**

Particulars	Units	(₹)	Particulars	Units	(₹)
To Process Account (Mounting)	31,000	6,20,000	By Cost Ledger Control A/c	31,000	6,20,000
	31,000	6,20,000		31,000	6,20,000

Abnormal Loss A/c

Particulars	Units	(₹)	Particulars	Units	(₹)
To Process Account (Mounting)	15,500	11,16,000	By Cost Ledger Control A/c	15,500	3,10,000
			By Costing Profit & Loss A/c		8,06,000
	15,500	11,16,000		15,500	11,16,000

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710 MTP **May 2024 S-I** Case scenario · MCQ 1(i) to MCQ 1(v) 10 marks

CASE SCENARIO

Arnav Ltd. manufactures chemical solutions used in paint and adhesive products. Chemical solutions are produced in different processes. Some of the processes are hazardous in nature which may results in fire accidents. At the end of the last month, one fire accident occurred in the factory. The fire destroyed some of the paper files containing records of the process operations for the month.

You being an associate to the Chief Manager (Finance), are assigned to prepare the process accounts for the month during which the fire occurred. From the documents and files of other sources, following information could be retrieved:

Opening work-in-process at the beginning of the month was 500 litres, 80% complete for labour and 60% complete for overheads. Opening work-in-process was valued at ₹ 2,78,000.

Closing work-in-process at the end of the month was 100 litres, 20% complete for labour and 10% complete for overheads.

Normal loss is 10% of input (fresh) and total losses during the month were 800 litres partly due to the fire damage.

Output transferred to finished goods was 3,400 litres.

Losses have a scrap value of ₹ 20 per litre.

All raw materials are added at the commencement of the process.

The cost per equivalent unit is ₹ 660 for the month made up as follows:

Raw Material ₹ 300 Labour ₹ 200 Overheads ₹ 160

The company uses FIFO method to value work-in-process and finished goods.

The following information are required for managerial decisions:

MCQ 1(i) 2 marks

How much quantity of raw material introduced during the month?

A 4,300 Litres

B 3,500 Litres

C 4,200 Litres

D 3,800 Litres

MCQ 1(ii) 2 marks

The Quantity of normal loss and abnormal loss are:

A Normal loss- 380 litres & Abnormal loss- 420 litres

B Normal loss- 350 litres & Abnormal loss – 450 litres

C Normal loss- 430 litres & Abnormal loss – 370 litres

D Normal loss- 420 litres & Abnormal loss – 380 litres.

MCQ 1(iii) 2 marks

Value of raw material added to the process during the month is:

A ₹ 10,10,000

B ₹ 10,33,600

C ₹ 10,18,400

D ₹ 10,20,000

MCQ 1(iv) 2 marks

Value of labour and overhead in closing Work-in-process are:

A ₹ 4,000 & ₹ 1,600 respectively

B ₹ 20,000 & ₹ 16,000 respectively

C ₹ 16,000 & ₹ 9,000 respectively

D ₹ 13,200 & ₹ 6,600 respectively

MCQ 1(v) 2 marks

Value of output transferred to finished goods is:

A ₹ 22,57,200

B ₹ 20,06,400

C ₹ 22,44,000

D ₹ 19,27,200

ICAI'S ANSWERS

MCQ 1(i)

Answer key: **(D)**

D

Inflow into process	Litres	Outflow from process	Litres
Opening WIP	500	Transferred to finished goods	3,400
Quantity introduced (Balancing figure)	3,800	Total loss	800
		Closing WIP	100
	4,300		4,300

MCQ 1(ii)

Answer key: **(A)**

A

Total loss	800 litres
Normal loss (10% of fresh input i.e. 3,800)	380 litres
Abnormal loss	420 litres

MCQ 1(iii)

Answer key: **(B)**

B

Calculation of Equivalent production units

Input Details	Units	Output Particulars	Units	Equivalent Production					
				Material		Labour		Overheads	
				%	Units	%	Units	%	Units
Opening WIP	500	From Opening WIP	500	-	-	20	100	40	200
Fresh inputs	3,800	From fresh units	2900	100	2900	100	2900	100	2900
		Normal loss	380	-		-		-	
		Closing WIP	100	100	100	20	20	10	10
		Abnormal loss	420	100	420	100	420	100	420
	4,300		4,300		3,420		3,440		3,530

Value of raw materials introduced during the month

	Equivalent units	Cost per EU (₹)	Total cost (₹)
Total value of raw material	3420	300	10,26,000
Add: Scrap value of normal loss	380	20	7,600
Value of raw material introduced			**10,33,600**

MCQ 1(iv)

Answer key: **(A)**

A

Value of labour and overhead in closing Work in process

Cost elements	Equivalent units	Cost per EU (₹)	Total cost (₹)
Labour	20	200	**4,000**
Overheads	10	160	**1,600**

MCQ 1(v)

Answer key: **(C)**

C

Value of output transferred to finished goods

Output transferred (Units) × Equivalent cost per unit

3,400 Litres × ₹660 = ₹22,44,000

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711 MTP **Old P3 Nov 2023 S-II** Q4(a) 10 marks old scheme

X Associates undertake to prepare income tax returns for individuals for a fee. They use the weighted average method and actual costs for the financial reporting purposes. However, for internal reporting,

they use a standard costs system. The standards, based on equivalent performance, have been established as follows:

Labour per return	5 hrs @ ₹ 40 per hour
Overhead per return	5 hrs @ ₹ 20 per hour

For July 2023 performance, budgeted overhead is ₹98,000 for standard labour hours allowed.

The following additional information pertains to the month of July 2023:

July 1	Return-in-process (25% complete)	200 No.
	Return started in July	825 Nos
July 31	Return-in-process (80% complete)	125 Nos
Cost Data:		
July 1	Return-in-process labour	₹ 12,000
	- Overheads	₹ 5,000
July 1 to 31	Labour : 4,000 hours	₹ 1,78,000
	Overheads	₹ 90,000

You are required to compute:

(i) For each element, equivalent units of performance and the actual cost per equivalent unit.

(ii) Actual cost of return-in-process on July 31.

(iii) The standard cost per return.

(iv) The labour rate and labour efficiency variance as well as overhead volume and overhead expenditure variance.

ICAI'S ANSWER

(i) Statement Showing Cost Elements Equivalent Units of Performance and the Actual Cost per Equivalent Unit

Detail of Returns	Detail of Input Units	Details	Equivalent Units				
			Output	Labour		Overheads	
			Units	Units	%	Units	%
Returns in Process at Start	200	Returns Completed in July	900	900	100	900	100
Returns Started in July	825	Returns in Process at the end of July	125	100	80	100	80
	1,025		1,025	1,000		1,000	

Costs:	Labour (₹)	Overhead (₹)
From previous month	12,000	5,000
During the month	1,78,000	90,000
Total Cost	1,90,000	95,000
Cost per Equivalent Unit	190.00	95.00

(ii) Actual cost of returns in process on July 31:

	Numbers	Stage of Completion	Rate per Return (₹)	Total (₹)
Labour	125 returns	0.80	190.00	19,000
Overhead	125 returns	0.80	95.00	9,500
				28,500

(iii) Standard Cost per Return:

Labour	5 Hrs × ₹ 40 per hour = ₹200
Overhead	5 Hrs × ₹ 20 per hour = ₹100
	₹ 300
Budgeted volume for July	= ₹ 98,000 / 1000 = 980 Returns
Actual labour rate	= ₹ 178000 / 4000 = ₹44.50

(iv) Computation of Variances:

Statement Showing Output (July only) Element Wise	Labour	Overhead
Actual performance in July in terms of equivalent units as Calculated above	1,000	1,000
Less: Returns in process at the beginning of July in terms of equivalent units i.e. 25% of returns (200)	50	50
	950	950

Variance Analysis:

Labour Rate Variance

= Actual Time × (Standard Rate – Actual Rate)

= Standard Rate × Actual Time – Actual Rate × Actual Time

= ₹ 40 × 4,000 hrs. – ₹ 1,78,000 = ₹ 18,000(A)

Labour Efficiency Variance

= Standard Rate × (Standard Time – Actual Time)

= Standard Rate × Standard Time – Standard Rate × Actual Time

= ₹ 40 × (950 units × 5 hrs.) – ₹ 40 × 4,000 hrs.

= ₹ 30,000(F)

Overhead Expenditure or Budgeted Variance

= Budgeted Overhead – Actual Overhead

= ₹ 98,000 – ₹ 90,000

= ₹ 8,000(F)

Overhead Volume Variance

= Recovered/Absorbed Overhead – Budgeted Overhead

= 950 Units × 5 hrs. × ₹20 – ₹ 98,000 = ₹ 3,000(A)

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712 MTP Old P3 Nov 2022 S-I Q3(a) 10 marks old scheme

XYZ Ltd. is manufacturer of medicines. It carries on production operation in two processes. The material first passes through Process I, where Medicine 'X' is produced. Following data are given for the month October, 2022:

Opening work-in-progress quantity (Material 100% and conversion 50% complete)	(in Liter)	12,000
Material input quantity	(in Liter)	60,000
Work completed quantity	(in Liter)	40,000
Closing work-in-progress quantity (Material 100% and conversion 80% complete)	(in Liter)	15,000
Opening work-in-progress cost		
Material cost	(in ₹)	1,75,000
Processing cost	(in ₹)	1,40,000
Material input cost	(in ₹)	7,70,000
Processing cost	(in ₹)	8,35,000

Normal process loss is 15% of material input. It has no realizable value.

Any quantity of Medicine 'X' can be sold for ₹ 42.50 per Liter. Alternatively, it can be transferred to Process II for further processing and then sold as Medicine ' XYZ' for ₹ 50 per Liter. Further materials are added in Process II, which yield 1.25 Liter of Medicine 'XYZ' for every Liter of Medicine 'X' of Process I. Out of the 40,000 Liter of work completed in Process I, 10,000 Liter are sold as Medicine 'X' and 30,000 Liter are passed through Process II for sale as Medicine 'XYZ'.

The monthly costs incurred in Process II (other than the cost of Medicine ' X') are:

Input	30,000 Liter of Medicine 'X'
Materials Cost	₹ 2,75,000
Processing Costs	₹ 2,50,000

You are required to:

(i) PREPARE Statement of Equivalent production and determine the cost per Liter of Medicine 'X' in Process I, using the weighted average cost method.

(ii) Company is mulling over the option to sell the 30,000 Liter of Medicine 'X' at Process-I without processing it further in Process-II. WILL IT BE beneficial for the company over the current pattern of processing 30,000 Liter in process-II?

ICAI'S ANSWER

(i) Process I Statement of Equivalent Production (Under Weighted Average Method)

Particulars	Input units (in Liter)	Particulars	Output units (in Liter)	Equivalent Production			
				Material		Conversion	
				(%)	Equivalent units (in Liter)	(%)	Equivalent units (in Liter)
Opening WIP	12,000	Units introduced and completed	40,000	100	40,000	100	40,000
New Material Introduced	60,000	Normal Loss (15% of 60,000 liters)	9,000	-	-	-	-
		Closing WIP	15,000	100	15,000	80	12,000
		Abnormal Loss (Bal. fig.)	8,000	100	8,000	100	8,000
	72,000		72,000		**63,000**		**60,000**

Statement of Cost for Each Element

Elements of Costs	Material (₹)	Conversion Cost (₹)
Costs of Opening WIP	1,75,000	1,40,000
Cost of the Process (for the period)	7,70,000	8,35,000
Total Cost	**9,45,000**	**9,75,000**
Equivalent Units (in liter)	63,000	60,000
Cost Per equivalent Units (in liter)	**₹ 15**	**₹ 16.25**

Therefore, Cost of Medicine 'X' is ₹ 31.25 per liter (₹ 15 + ₹ 16.25)

(ii) Statement showing comparative data to decide whether 30,000 Liters of Medicine 'X' should be further processed into 'XYZ'

	Alternative 1	Alternative 2
	Sell medicine 'X' after Process I (₹)	**Process further into 'XYZ' (₹)**

	Alternative 1	Alternative 2
Sales	**12,75,000 (30,000 liters x ₹ 42.50)**	**18,75,000 (37,500 liters x ₹ 50)**
Less: Costs:		
Process I - Costs (30,000 liters x ₹ 31.25)	9,37,500	9,37,500
Material in Process II	-	2,75,000
Conversion cost in Process II	-	2,50,000
Total Cost	**9,37,500**	**14,62,500**
Profit	**3,37,500**	**4,12,500**

Hence, company should process further as it will increase profit further by ₹ 75,000 (₹ 4,12,500 – ₹ 3,37,500)

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The following data are available in respect of Process-I for January 2022:

(1) Opening stock of work in process: 600 units at a total cost of ₹ 8,40,000.

(2) Degree of completion of opening work in process:

Material	100%
Labour	60%
Overheads	60%

(3) Input of materials at a total cost of ₹1,10,40,000 for 9,200 units.

(4) Direct wages incurred ₹37,20,000

(5) Production overhead ₹17,26,000.

(6) Units scrapped 200 units. The stage of completion of these units was:

Materials	100%
Labour	80%
Overheads	80%

(7) Closing work in process; 700 units. The stage of completion of these units was:

Material	100%
Labour	70%
Overheads	70%

(8) 8,900 units were completed and transferred to the next process.

(9) Normal loss is 4% of the total input (opening stock plus units put in)

(10) Scrap value is ₹600 per unit.

You are required to:

(i) COMPUTE equivalent production,

(ii) CALCULATE the cost per equivalent unit for each element.

(iii) CALCULATE the cost of abnormal loss (or gain), closing work in process and the units transferred to the next process using the FIFO method.

ICAI'S ANSWER

(i) Statement of Equivalent Production (FIFO Method)

Input		Output		Equivalent Production					
				Materials		Labour		Production Overhead	
Details	**Units**	**Details**	**Units**	**%**	**Units**	**%**	**Units**	**%**	**Units**
Opening Stock	600	From opening stock	600	-	-	40	240	40	240
		- From fresh materials	8,300	100	8,300	100	8,300	100	8,300
		Closing W-I-P	700	100	700	70	490	70	490
Fresh inputs	9,200	Normal loss	392	-	-	-	-	-	-
			9,992		9,000		9,030		9,030
		Less: Abnormal Gain	(192)	100	(192)	100	(192)	100	(192)
	9,800		9,800		8,808		8,838		8,838

(ii) Statement of Cost per equivalent units

Elements		Cost	Equivalent units (EU)	Cost per EU
	(Rs.)	(Rs.)		(Rs.)
Material Cost	1,10,40,000			
Less: Scrap realisation 392 units @ Rs. 600/- p.u.	(2,35,200)	1,08,04,800	8,808	1,226.70
Labour cost		37,20,000	8,838	420.91
Production OH Cost		17,26,000	8,838	195.29
Total Cost		1,62,50,800		1,842.90

(iii) Cost of Abnormal Gain – 192 Units

	(Rs.)	(Rs.)
Material cost of 192 units @ Rs. 1,226.70 p.u.	2,35,526.40	
Labour cost of 192 units @ Rs. 420.91 p.u.	80,814.72	
Production OH cost of 192 units @ Rs. 195.29 p.u.	37,495.68	3,53,836.80

Cost of closing WIP – 700 Units

Material cost of 700 equivalent units @ Rs. 1,226.70 p.u.	8,58,690.00	
Labour cost of 490 equivalent units @ Rs. 420.91 p.u.	2,06,245.90	
Production OH cost of 490 equivalent @ Rs. 195.29 p.u.	95,692.10	11,60,628.00

Cost of 8,900 units transferred to next process

(i)	Cost of opening W-I-P Stock b/f – 600 units	8,40,000.00
(ii)	Cost incurred on opening W-I-P stock	
	Material cost	—
	Labour cost 240 equivalent units @ Rs. 420.91 p.u.	1,01,018.40
	Production OH cost 240 equivalent units @ Rs197.29 p.u.	47,349.60
		1,48,368.00
(iii)	Cost of 8,300 completed units	
	8,300 units @ Rs. 1,842.90 p.u.	1,52,96,070.00
	Total cost [(i) + (ii) + (iii))]	1,62,84,438.00

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714 MTP Old P3 May 2022 S-I Q1(d) 5 marks old scheme

Chill Ltd. uses process costing to manufacture water density sensor for hydro sector. The following information pertains to operations for the month of February:

Particulars	Units
Beginning WIP, February 1	22,400
Started in production during February	1,40,000
Completed production during February	1,28,800
Ending work in progress, February 28	33,600

The beginning work in progress was 50% complete for materials and 30% complete for conversion costs. The ending inventory was 80% complete for material and 30% complete for conversion costs.

Costs pertaining to the month of February are as follows:

Beginning inventory costs are material ₹ 1,38,350, direct labour ₹ 1,50,600 and factory overhead ₹ 63,600

Cost incurred during February are material ₹ 23,95,000, direct labour ₹ 9,14,400, factory overheads ₹ 19,55,800.

CALCULATE:

(i) Using the FIFO method, the equivalent units of production for material.

(ii) Cost per equivalent unit for conversion cost.

ICAI'S ANSWER

(i) Calculation of equivalent units of production:

Input Details	Units	Output Particulars	Units	Equivalent Units			
				Material		Conversion cost	
				%	Units	%	Units
Beginning WIP	22,400	From beginning WIP	22,400	50	11,200	70	15,680
Unit Introduced	1,40,000	Completed output	1,06,400	100	1,06,400	100	1,06,400
		Closing W-I-P	33,600	80	26,880	30	10,080
Total	1,62,400	Total	1,62,400		1,44,480		1,32,160

(ii) Calculation of cost per equivalent unit for conversion costs

Particulars	
Direct labour	₹ 9,14,400
Factory overheads	₹ 19,55,800
Total	₹ 28,70,200
Equivalent units	1,32,160 units
Cost per equivalent unit	**₹ 21.72**

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715 MTP **Old P3 Dec 2021 S-I** Q3(a) 10 marks old scheme

Navyug Ltd. manufactures chemical solutions for the food processing industry. The manufacturing takes place in a number of processes and the company uses a FIFO process costing system to value work-in-process and finished goods. At the end of the last month, a fire occurred in the factory and destroyed some of the paper files containing records of the process operations for the month.

Navyug Ltd. needs your help to prepare the process accounts for the month during which the fire occurred. You have been able to gather some information about the month's operating activities but some of the information could not be retrieved due to the damage. The following information was salvaged:

- Opening work-in-process at the beginning of the month was 900 litres, 70% complete for labour and 60% complete for overheads. Opening work-in-process was valued at ₹ 29,970.
- Closing work-in-process at the end of the month was 160 litres, 30% complete for labour and 20% complete for overheads.
- Normal loss is 10% of input and total losses during the month were 1,800 litres partly due to the fire damage.
- Output sent to finished goods warehouse was 4,200 litres.
- Losses have a scrap value of ₹ 20 per litre.
- All raw materials are added at the commencement of the process.
- The cost per equivalent unit (litre) is ₹39 for the month made up as follows:

	(₹)
Raw Material	23
Labour	7
Overheads	9
	39

REQUIRED:

(i) Calculate the quantity (in litres) of raw material inputs during the month.

(ii) Calculate the quantity (in litres) of normal loss expected from the process and the quantity (in litres) of abnormal loss / gain experienced in the month.

(iii) Calculate the values of raw material, labour and overheads added to the process during the month.

(iv) Prepare the process account for the month.

ICAI'S ANSWER

(i) **Calculation of Raw Material inputs during the month:**

Quantities Entering Process	Litres	Quantities Leaving Process	Litres
Opening WIP	900	Transfer to Finished Goods	4,200
Raw material input (balancing figure)	5,260	Process Losses	1,800
		Closing WIP	160
	6,160		6,160

(ii) **Calculation of Normal Loss and Abnormal Loss/Gain**

Particulars	Litres
Total process losses for month	1,800
Normal Loss (10% input)	526
Abnormal Loss (balancing figure)	1,274

(iii) **Calculation of values of Raw Material, Labour and Overheads added to the process:**

	Material	Labour	Overheads
Cost per equivalent unit	₹ 23.00	₹ 7.00	₹ 9.00
Equivalent units (litre) (refer the working note)	4,734	4,892	4,966
Cost of equivalent units	₹ 1,08,882	₹ 34,244	₹ 44,694
Add: Scrap value of normal loss (526 units × ₹ 20)	₹ 10,520	--	--
Total value added	₹ 1,19,402	₹ 34,244	₹ 44,694

Workings:

Statement of Equivalent Units (litre):

Input Details	Units	Output details	Units	Equivalent Production					
				Material		Labour		Overheads	
				Units	(%)	Units	(%)	Units	(%)
Opening WIP	900	Units completed:							
Units introduced	5,260	- Opening WIP	900	--	--	270	30	360	40
		- Fresh inputs	3,300	3,300	100	3,300	100	3,300	100

Input Details	Units	Output details	Units	Equivalent Production					
		Normal loss	526	--	--	--	--	--	--
		Abnormal loss	1,274	1,274	100	1,274	100	1,274	100
		Closing WIP	160	160	100	48	30	32	20
	6,160		6,160	4,734		4,892		4,966	

(iv) **Process Account for Month**

	Litres	Amount (₹)		Litres	Amount (₹)
To Opening WIP	900	29,970	By Finished goods	4,200	1,63,800
To Raw Materials	5,260	1,19,402	By Normal loss	526	10,520
To Wages	--	34,244	By Abnormal loss	1,274	49,686
To Overheads	--	44,694	By Closing WIP	160	4,304
	6,160	2,28,310		6,160	2,28,310

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716 MTP Old P3 Dec 2021 S-II Q2(a) 10 marks old scheme

Following information is available regarding process A for the month of October, 2021:

Production Record:

Units in process as on 01.10.2021	8,000
(All materials used, 25% complete for labour and overhead)	
New units introduced	32,000
Units completed	28,000
Units in process as on 31.10.2021	12,000
(All materials used, 33-1/3% complete for labour and overhead)	

Cost Records:

Work-in-process as on 01.10.2021	(₹)
Materials	12,00,000
Labour	2,00,000
Overhead	2,00,000
	16,00,000

Cost during the month	
Materials	51,20,000
Labour	30,00,000
Overhead	30,00,000
	1,11,20,000

Presuming that average method of inventory is used, PREPARE:

(i) Statement of Equivalent Production.

(ii) Statement showing Cost for each element.

(iii) Statement of Apportionment of cost.

(iv) Process Cost Account for Process A.

ICAI'S ANSWER

(i) Statement of Equivalent Production (Average cost method)

Input (Units)	Particulars	Output Units	Equivalent Production					
			Materials		Labour		Overheads	
			(%*)	Units**	(%)*	Units**	(%)*	Units**
40,000	Completed	28,000	100	28,000	100	28,000	100	28,000
	WIP	12,000	100	12,000	33-1/3	4,000	33-1/3	4,000
40,000		40,000		40,000		32,000		32,000

*Percentage of completion ** Equivalent units

(ii) Statement showing Cost for each element

Particulars	Materials	Labour	Overhead	Total
Cost of opening work-in-progress (₹)	12,00,000	2,00,000	2,00,000	16,00,000
Cost incurred during the month (₹)	51,20,000	30,00,000	30,00,000	1,11,20,000
Total cost (₹) : (a)	63,20,000	32,00,000	32,00,000	1,27,20,000
Equivalent units : (B)	40,000	32,000	32,000	
Cost per equivalent unit (₹) : C= (A ÷ B)	158	100	100	358

(iii) Statement of Apportionment of cost

	(₹)	(₹)
Value of output transferred: (A) (28,000 units × ₹ 358)		1,00,24,000
Value of closing work-in-progress: (B)		
Material (12,000 units × ₹158)	18,96,000	
Labour (4,000 units × ₹ 100)	4,00,000	
Overhead (4,000 units × ₹ 100)	4,00,000	26,96,000
Total cost : (A + B)		1,27,20,000

(iv) Process- A Account

Particulars	Units	(₹)	Particulars	Units	(₹)
To Opening WIP	8,000	16,00,000	By Completed units	28,000	1,00,24,000
To Materials	32,000	51,20,000	By Closing WIP	12,000	26,96,000
To Labour		30,00,000			
To Overhead		30,00,000			
	40,000	1,27,20,000		40,000	1,27,20,000

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717 MTP **Old P3 Jul 2021 S-II** Q4(b) 10 marks old scheme

The following information relates to Process Q:

(i)	Opening Work-in-Progress	16,000 units at Rs.1,50,000
	Degree of Completion: Material	100%
	Labour and Overhead	60%
(ii)	Input - 3,64,000 units	Rs. 14,75,000
(iii)	Wages paid	Rs. 6,81,200
(iv)	Overheads paid	Rs. 3,40,600
(v)	Units scrapped	28,000
	Degree of Completion: Material	100%
	Labour and Overhead	80%
(vi)	Closing Work - in- Progress	36,000 units
	Degree of Completion: Material	100%

	Labour and Overhead	70%
(vii)	Units completed and transferred to next process	3,16,000
(viii)	Normal loss is 5% of total input including opening WIP	
(ix)	Scrap value is Rs. 5 per unit to be adjusted out of direct material cost	

You are required to COMPUTE on the basis of FIFO:

(i) Equivalent production

(ii) Cost per unit

(iii) Value of units transferred to next process

ICAI'S ANSWER

(i) Statement of Equivalent Production (FIFO Method)

Input		Output		Equivalent Production			
Particulars	**Units**	**Particulars**	**Units**	**Material**		**Labour & Overhead**	
				(%)	**Units**	**(%)**	**Units**
Opening WIP	16,000	Transfer to next Process:					
Introduced	3,64,000	Opening WIP completed	16,000	--	--	40	6,400
		Introduced & completed	3,00,000	100	3,00,000	100	3,00,000
		Normal loss 5% (16,000 + 3,64,000)	19,000	--	--	--	--
		Abnormal loss	9,000	100	9,000	80	7,200
		Closing WIP	36,000	100	36,000	70	25,200
	3,80,000		3,80,000		3,45,000		3,38,800

(ii) Computation of Cost per unit

Particulars	Material (Rs.)	Labour (Rs.)	Overhead (Rs.)
Input of Materials	14,75,000	--	--
Expenses	--	6,81,200	3,40,600
Total	14,75,000	6,81,200	3,40,600
Less: Sale of Scrap (19,000 units x Rs. 5)	(95,000)	--	--
Net cost (A)	13,80,000	6,81,200	3,40,600
Equivalent Units (B)	3,45,000	3,38,800	3,38,800
Cost Per Unit (A/B)	4.0000	2.0106	1.0053

Total cost per unit = Rs. (4.0000 + 2.0106 + 1.0053) = Rs. 7.0159

(iii) Value of units transferred to next process:

	Amount (Rs.)	Amount (Rs.)
Opening W-I-P	1,50,000	
Add: Labour (6,400 units × Rs. 2.0106)	12,868	
Overhead (6,400 units × Rs. 1.0053)	6,434	1,69,302
New introduced (3,00,000 units × Rs. 7.0159)		21,04,770
		22,74,072

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718 MTP **Old P3 Nov 2020 S-I** Q3(a) 10 marks old scheme

'Healthy Sweets' is engaged in the manufacturing of jaggery. Its process involve sugarcane crushing for juice extraction, then filtration and boiling of juice along with some chemicals and then letting it cool to cut solidified jaggery blocks.

The main process of juice extraction (Process – I) is done in conventional crusher, which is then filtered and boiled (Process – II) in iron pots. The solidified jaggery blocks are then cut, packed and dispatched. For manufacturing 10 kg of jaggery, 100 kg of sugarcane is required, which extracts only 45 litre of juice.

Following information regarding Process – I has been obtained from the manufacturing department of *Healthy Sweets* for the month of January, 2020:

	(₹)
Opening work-in process (4,500 litre)	
Sugarcane	50,000
Labour	15,000
Overheads	45,000
Sugarcane introduced for juice extraction (1,00,000 kg)	5,00,000
Direct Labour	2,00,000
Overheads	6,00,000

Abnormal Loss: 1,000 kg

Degree of completion:

Sugarcane	100%
Labour and overheads	80%

Closing work-in process: 9,000 litre

Degree of completion:

Sugarcane	100%
Labour and overheads	80%

Extracted juice transferred for filtering and boiling: 39,500 litre

(Consider mass of 1 litre of juice equivalent to 1 kg)

You are required to PREPARE using average method:

(i) Statement of equivalent production,

(ii) Statement of cost,

(iii) Statement of distribution cost, and

(iv) Process-I Account.

ICAI'S ANSWER

(i) **Statement of Equivalent Production**

Particulars	Input Units	Particulars	Output Units	Equivalent Production			
				Sugarcane		Labour & O.H.	
				%	Units	%	Units

Particulars	Input Units	Particulars	Output Units	Equivalent Production			
Opening WIP	4,500	Completed and transferred to Process - II	39,500	100	39,500	100	39,500
Units introduced	1,00,000	Normal Loss (55%* of 1,00,000)	55,000	--	--	--	--
		Abnormal loss	1,000	100	1,000	80	800
		Closing WIP	9,000	100	9,000	80	7,200
	1,04,500		1,04,500		49,500		47,500

* 100 kg of sugarcane extracts only 45 litre of juice. Thus, normal loss = 100 – 45 = 55%

(ii) **Statement showing cost for each element**

Particulars	Sugarcane (₹)	Labour (₹)	Overhead (₹)	Total (₹)
Cost of opening work-in-process	50,000	15,000	45,000	1,10,000
Cost incurred during the month	5,00,000	2,00,000	6,00,000	13,00,000
Total cost: (A)	5,50,000	2,15,000	6,45,000	14,10,000
Equivalent units: (B)	49,500	47,500	47,500	
Cost per equivalent unit: (C) = (A ÷ B)	11.111	4.526	13.579	29.216

(iii) **Statement of Distribution of cost**

		Amount (₹)	Amount (₹)
1.	Value of units completed and transferred (39,500 units × ₹ 29.216)		11,54,032
2.	Value of Abnormal Loss:		
	- Sugarcane (1,000 units × ₹ 11.111)	11,111	
	- Labour (800 units × ₹ 4.526)	3,621	
	- Overheads (800 units × ₹ 13.579)	10,863	25,595
3.	Value of Closing W-I-P:		
	- Sugarcane (9,000 units × ₹ 11.111)	99,999	
	- Labour (7,200 units × ₹ 4.526)	32,587	

	Amount (₹)	Amount (₹)
- Overheads (7,200 units × ₹ 13.579)	97,769	2,30,355

(iv) **Process-I A/c**

Particulars	Units	(₹)	Particulars	Units	(₹)
To Opening W.I.P:			By Normal Loss	55,000	--
- Sugarcane	4,500	50,000	By Abnormal loss (₹25,595 + ₹18 (difference due to approximation))	1,000	25,613
- Labour	--	15,000	By Process-II A/c	39,500	11,54,032
- Overheads	--	45,000	By Closing WIP	9,000	2,30,355
To Sugarcane introduced	100,000	5,00,000			
To Direct Labour		2,00,000			
To Overheads		6,00,000			
	104,500	14,10,000		104,500	14,10,000

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10.3 Inter-process profit and operation costing

16 questions · asked in 8 ICAI documents · 1 exam paper

719 EXAM **May 2026** Case Scenario - I · MCQ 1 to MCQ 5 10 marks

CASE SCENARIO - I

Graphite Core Industries established in 1995 is globally recognized for producing reliable, high quality wooden pencils. It begins its production process by sourcing raw graphite, clay, and cedar wood slats, holding them as raw materials.

The following are details about the manufacturing processes of the company:

- Core Manufacturing Process - In this process, the graphite and clay are mixed, extruded into rods and fired to create hardened pencil cores known as lead. Simultaneously, the cedar wood is cut into grooved slats. The output of this process i.e. lead and grooved slat is transferred to the Casing & Shaping Process at 30% profit on the transfer price.
- Casing & Shaping Process In this process, the leads are glued between the grooved slats, shaped into individual pencils and painted. The output of this process i.e. pencils, are then transferred to the Finished Stock at 10% profit on the transfer price. Stock in process is valued at prime cost.

The Production manager has provided the following information for the month of April 2026:

Particulars	Core Manufacturing Process (₹)	Casing & Shaping Process (₹)
Direct Materials	4,90,0000	8,95,000
Direct Wages	3,60,000	9,50,000
Direct Expenses	3,40,000	3,80,000
Factory Overheads	3,00,000	5,00,000

Additional Information:

Particulars	Opening Stock (₹)	Closing Stock (₹)
Core Manufacturing Process	5,10,000	2,50,000
Casing & Shaping Process	7,75,000	6,00,000
Finished Stock of Pencils	14,00,000	10,00,000

Finished stock is valued at the price at which it is received from the Casing & Shaping Process. During the month of April 2026, the company sold pencils worth ₹ 75,00,000. Inter process profits included in opening stock for Casing & Shaping Process and Finished Stock are ₹ 75,000 and ₹ 3,20,000 respectively.

On the basis of above case scenario for the month of April 2026, you are required to answer the following MCQs 1 to 5:

MCQ 1 2 marks

What is the Profit of the Core Manufacturing Process?

A ₹ 7,50,000

B ₹ 5,25,000

C ₹ 6,00,000

D ₹ 6,75,000

MCQ 2 2 marks

What is the Prime Cost (including Inter Process Profit) of Casing & Shaping Process?

A ₹ 53,25,000

B ₹ 44,00,000

C ₹ 46,75,000

D ₹ 55,00,000

MCQ 3 2 marks

What is the amount of profit included in the closing stock of Casing & Shaping Process?

A ₹ 84,000

B ₹ 90,000

C ₹ 87,000

D ₹ 93,000

MCQ 4 2 marks

What is the transfer price of pencils (including inter process profit) transferred to Finished Stock account from the Casing & Shaping Process?

A ₹ 57,50,000

B ₹ 62,00,000

C ₹ 60,00,000

D ₹ 64,00,000

MCQ 5 2 marks

What is the amount of total profit realized by Graphite Core Industries on sale of finished stock of pencils which is transferred to the Costing Profit and Loss Account?

A ₹ 25,40,000

B ₹ 25,25,000

C ₹ 25,17,500

D ₹ 25,32,500

ICAI'S ANSWERS

MCQ 1

Answer key: **(A)**

MCQ 2

Answer key: **(D)**

MCQ 3

Answer key: **(B)**

MCQ 4

Answer key: **(C)**

MCQ 5

Answer key: **(D)**

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720 RTP **Jan 2025** Case scenario · Q2(i) to Q2(v)

CASE SCENARIO

Knowing the hectic schedule of a student preparing for the examination, a homemaker managing work from home or a new parent busy in neonatal care, a freshly qualified professional (Mr. Rishi) entered into a start-up business of manufacturing frozen foods.

The process majorly involve washing and cutting the vegetables (Process I), blanching, cooling and mixing of ingredients with spices (Process II), forming, frying and freezing the final product (Process III).

In Accounts, Mr. Rishi normally transfers the output of one process to another process at cost but, being a young entrepreneur, he is interested in knowing the profit made at each and every process. Thus, it was decided to transfer the output of Process I and II to the next process at cost plus 25%. Further, the output of Process III is also transferred to finished stock at cost plus 33 1/3%.

Following information is extracted from the books of Mr. Rishi for the current year:

	Process I (₹)	Process II (₹)	Process III (₹)	Finished Stock (₹)
Opening stock	8,02,500	14,44,500	21,40,000	24,07,500
Direct materials	42,80,000	34,77,500	26,75,000	--
Direct wages	66,87,500	57,78,000	49,22,000	--
Factory overheads	51,36,000	38,52,000	35,57,750	--
Closing stock	10,70,000	17,12,000	20,86,500	26,75,000
Inter-process profit included in opening stock	NIL	2,14,000	5,35,000	10,70,000

Stock in processes is valued at prime cost. The finished stock is valued at the price at which it is received from Process III.

Mr. Rishi wants you to FIGURE OUT the following to analyse the profit generated at each process:

Q2(i)

What is the transfer price value at which the output of Process I is transferred to Process II?

A ₹ 1,97,95,000

B ₹ 39,59,000

C ₹ 1,58,36,000

D ₹ 1,69,06,000

Q2(ii)

What is the transfer price value at which the output of Process II is transferred to Process III?

A ₹ 1,20,97,476

B ₹ 4,07,93,750

C ₹ 2,86,96,274

D ₹ 3,43,47,000

Q2(iii)

What is the transfer price value at which the output of Process III is transferred to Finished Stock?

A ₹ 5,40,88,500

B ₹ 3,98,91,140

C ₹ 2,94,44,860

D ₹ 6,93,36,000

Q2(iv)

What is the cost value at which the output of Process III is transferred to Finished Stock?

A ₹ 5,40,88,500

B ₹ 3,98,91,140

C ₹ 2,94,44,860

D ₹ 6,93,36,000

Q2(v)

What is the cost value of closing stock of Process III A/c?

A ₹ 20,86,500

B ₹ 15,64,884

C ₹ 3,98,91,140

D ₹ 5,21,616

ICAI'S ANSWERS

Q2(i)

(A) Process I Account

Particulars	Cost (₹)	Profit (₹)	Total (₹)	Particulars	Cost (₹)	Profit (₹)	Total (₹)
Opening Stock	8,02,500	–	8,02,500	Process II A/c (Transfer)*	1,58,36,000	39,59,000	1,97,95,000
Direct Material	42,80,000	–	42,80,000	Closing stock	10,70,000	–	10,70,000
Direct Wages	66,87,500	–	66,87,500				
Prime Cost	1,17,70,000	–	1,17,70,000				
Manufacturing Overheads	51,36,000	–	51,36,000				
Total cost	1,69,06,000	–	1,69,06,000				
Costing Profit and Loss A/c**		39,59,000	39,59,000				
	1,69,06,000	39,59,000	2,08,65,000		1,69,06,000	39,59,000	2,08,65,000

*Transfer price = (Total Cost - Closing Stock) (1 + 25%)

= (1,69,06,000 - 10,70,000) x 1.25

= ₹ 1,97,95,000

**Profit on transfer = (1,69,06,000 - 10,70,000) x .25 = ₹ 39,59,000

Q2(ii)

(B) Process II Account

Particulars	Cost (₹)	Profit (₹)	Total (₹)	Particulars	Cost (₹)	Profit (₹)	Total (₹)
Opening Stock	12,30,500	2,14,000	14,44,500	By Process III A/c (Transfer)**	2,86,96,274	1,20,97,476	4,07,93,750
Process I A/c	1,58,36,000	39,59,000	1,97,95,000	Closing stock*	14,77,726	2,34,274	17,12,000
Direct Material	34,77,500	-	34,77,500				
Direct Wages	57,78,000	-	57,78,000				

Particulars	Cost (₹)	Profit (₹)	Total (₹)	Particulars	Cost (₹)	Profit (₹)	Total (₹)
Prime Cost	2,63,22,000	41,73,000	3,04,95,000				
Manufacturing Overheads	38,52,000	-	38,52,000				
Total cost	3,01,74,000	41,73,000	3,43,47,000				
Costing Profit and Loss A/c***	-	81,58,750	81,58,750				
	3,01,74,000	1,23,31,750	4,25,05,750		3,01,74,000	1,23,31,750	4,25,05,750

* Cost of Closing Stock = $\left(\frac{₹2,63,22,000}{₹3,04,95,000}\right)$ x ₹ 17,12,000 = ₹ 14,77,726

**Transfer price = (Total Cost - Closing Stock) (1 + 25%)

= (3,43,47,000 - 17,12,000) x 1.25 = ₹ 4,07,93,750

***Profit on transfer = (3,43,47,000 - 17,12,000) x .25 = ₹ 81,58,750

Q2(iii)

(D) Process III Account

Particulars	Cost (₹)	Profit (₹)	Total (₹)	Particulars	Cost (₹)	Profit (₹)	Total (₹)
Opening Stock	16,05,000	5,35,000	21,40,000	By Finished Stock A/c** (Transfer)	3,98,91,140	2,94,44,860	6,93,36,000
Process II A/c	2,86,96,274	1,20,97,476	4,07,93,750	Closing stock*	15,64,884	5,21,616	20,86,500
Direct Material	26,75,000	--	26,75,000				
Direct Wages	49,22,000	--	49,22,000				
Prime Cost	3,78,98,274	1,26,32,476	5,05,30,750				
Manufacturing Overheads	35,57,750	--	35,57,750				
Total cost	4,14,56,024	1,26,32,476	5,40,88,500				
Costing Profit and Loss A/c***	-	1,73,34,000	1,73,34,000				
	4,14,56,024	2,99,66,476	7,14,22,500		4,14,56,024	2,99,66,476	7,14,22,500

* Cost of Closing Stock = $\left(\frac{₹3,78,98,274}{₹5,05,30,750}\right)$ X ₹ 20,86,500

= ₹ 15,64,884

**Transfer price = (Total Cost - Closing Stock) (1 + 33 1/3%)

= (5,40,88,500 - 20,86,500) x (1 + 33 1/3%)

= ₹ 6,93,36,000

***Profit on transfer = (5,40,88,500 - 20,86,500) x 33 1/3%

= ₹ 1,73,34,000

Q2(iv)

(B) Refer part (iii) above.

Q2(v)

(B) Refer part (iii) above.

ICAI · P4 · RTP Jan 2025 · question p. 3-4 · answer p. 22-23, 23-24, 24-25, 25

721 RTP **Old P3 Nov 2020** Q15(d) old scheme

DESCRIBE Operation costing with two examples of industries where operation costing is applied.

ICAI'S ANSWER

This product costing system is used when an entity produces more than one variant of final product using different materials but with similar conversion activities. This means conversion activities are similar for all the product variants but materials differ significantly. Operation Costing method is also known as Hybrid product costing system as materials costs are accumulated by job order or batch wise but conversion costs i.e. labour and overheads costs are accumulated by department, and process costing methods are used to assign these costs to products. Moreover, under operation costing, conversion costs are applied to products using a predetermined application rate. This predetermined rate is based on budgeted conversion costs.

The two examples of industries are Ready made garments and Jewellery making.

ICAI · Old P3 · RTP Nov 2020 · question p. 12 · answer p. 31

722 MTP **Sep 2025 S-I** Q2 14 marks

ColdD Pvt. Ltd. is a well-established cold drink manufacturing company specializing in producing a variety of refreshing beverages. The company operates a modern production facility where advanced techniques and efficient processes are employed to ensure consistency, taste, and customer satisfaction. For this, the company's renowned product 'Pinaco' go through a meticulous two major process system, wherein, Process-I involves water purification, mixing & blending of essential ingredients and Process-II involves carbonation, chilling & filling the bottles. Later on, the product is transferred to the finished stock.

Following data is provided relating to the manufacturing process of product 'Pinaco' for the month of September:

Particulars	Process - I (₹)	Process - II (₹)
Opening stock	15,15,000	18,18,000
Direct materials	30,30,000	31,81,500
Direct wages	22,62,400	22,72,500
Factory overheads	21,21,000	9,09,000
Closing stock	7,47,400	9,09,000
Inter-process profit included in opening stock	--	3,03,000

The profit ratio on which output is transferred to next process is as follows:

Process	Transferred to	Profit
Process - I	Process - II	1/3rd of cost
Process - II	Finished Stock	1/4th of cost

The opening and closing stock of finished stock are ₹ 45,45,000 (which includes ₹ 16,66,500 as Inter-process profit) and ₹ 22,72,500 respectively.

Stock in processes is valued at prime cost. Finished stock is valued at the price at which it is received from process II. Sales during the period are ₹ 2,82,80,000.

You are required to PREPARE the following accounts showing the profit element at each stage:

(a) Process- I Account

(b) Process- II Account

(c) Finished Stock Account

ICAI'S ANSWER

Process- I Account

Particulars	Total (₹)	Cost (₹)	Profit (₹)	Particulars	Total (₹)	Cost (₹)	Profit (₹)
Opening stock	15,15,000	15,15,000	--	Process-II A/c*	1,09,08,000	81,81,000	27,27,000
Direct materials	30,30,000	30,30,000	--	Closing Stock	7,47,400	7,47,400	--
Direct wages	22,62,400	22,62,400	--				
Prime Cost	68,07,400	68,07,400					
Overheads	21,21,000	21,21,000	--				
Total Cost	89,28,400	89,28,400					
Profit**	27,27,000	--	27,27,000				
	1,16,55,400	89,28,400	27,27,000		1,16,55,400	89,28,400	27,27,000

*Transfer price = (Total Cost - Closing stock) $\left(1+\frac{1}{3}\right)$

= (₹ 89,28,400 - ₹ 7,47,400) $\left(1+\frac{1}{3}\right)$

= ₹ 1,09,08,000

**Profit on transfer = ₹ 1,09,08,000 × 1/4th

= ₹ 27,27,000

Process- II Account

Particulars	Total (₹)	Cost (₹)	Profit (₹)	Particulars	Total (₹)	Cost (₹)	Profit (₹)
Opening stock	18,18,000	15,15,000	3,03,000	Finished Stock A/c**	2,27,25,000	1,53,01,500	74,23,500
Transferred from Process- I	1,09,08,000	81,81,000	27,27,000	Closing stock*	9,09,000	7,57,500	1,51,500
Direct materials	31,81,500	31,81,500	--				
Direct wages	22,72,500	22,72,500	--				
Prime cost	1,81,80,000	1,51,50,000	30,30,000				
Overheads	9,09,000	9,09,000	--				
Total cost	1,90,89,000	1,60,59,000	30,30,000				
Profit***	45,45,000	--	45,45,000				
	2,36,34,000	1,60,59,000	75,75,000		2,36,34,000	1,60,59,000	75,75,000

* Cost of Closing Stock = $\frac{₹\ 1,51,50,000}{₹\ 1,81,80,000}$ x ₹ 9,09,000

= ₹ 7,57,500

**Transfer price = (Total Cost - Closing stock) $\left(1+\frac{1}{4}\right)$

= (₹ 1,90,89,000 - ₹ 9,09,000) $\left(1+\frac{1}{4}\right)$

= ₹ 2,27,25,000

***Profit on transfer = ₹ 2,27,25,000 × 1/5th

= ₹ 45,45,000

Finished Stock Account

Particulars	Total (₹)	Cost (₹)	Profit (₹)	Particulars	Total (₹)	Cost (₹)	Profit (₹)
Opening stock	45,45,000	28,78,500	16,66,500	Costing P&L A/c	2,82,80,000	1,66,49,850	1,16,30,150
Process-II	2,27,25,000	1,53,01,500	74,23,500	Closing stock*	22,72,500	15,30,150	7,42,350
Profit	32,82,500	--	32,82,500				
	3,05,52,500	1,81,80,000	1,23,72,500		3,05,52,500	1,81,80,000	1,23,72,500

*Cost of Closing Stock = $\frac{\text{Cost of transfer from Process-II}}{\text{Transfer price from Process-II}}$ × Value of closing stock

= $\frac{₹\ 1,53,01,500}{₹\ 2,27,25,000}$ x ₹ 22,72,500

= ₹ 15,30,150

ICAI · P4 · MTP Sep 2025 · Series I · question p. 9-10 · answer p. 8-10

723 MTP Jan 2025 S-II Q3(a) 6 marks

The following are the details in respect of Process A and Process B of a processing factory:

	Process A (₹)	Process B (₹)
Materials	40,000	--
Labour	40,000	56,000
Overheads	16,000	40,000

The output of Process A is transferred to Process B at a price calculated to give a profit of 20% on the transfer price and the output of Process B is charged to finished stock at a profit of 25% on the transfer price. The finished stock department realized ₹ 4,00,000 for the finished goods received from Process B.

You are asked to SHOW process accounts and total profit, assuming that there was no opening or closing work-in-progress.

ICAI'S ANSWER

Dr. **Process A Account** Cr.

	₹		₹
To Materials	40,000	By Transfer to Process B A/c	1,20,000
To Labour	40,000		
To Overheads	16,000		
	96,000		
To Profit (20% of transfer price, i.e., 25% of cost)	24,000		
	1,20,000		1,20,000

Dr. **Process B Account** Cr.

	₹		₹
To Transferred from Process A A/c	1,20,000	By Transfer to Finished Stock A/c	2,88,000
To Labour	56,000		
To Overhead	40,000		
	2,16,000		
To Profit (25% of transfer price i.e., 33.33% of cost)	72,000		
	2,88,000		2,88,000

Statement of Total Profit

	₹
Profit from Process A	24,000
Profit from Process B	72,000
Profit on Sales (₹ 4,00,000 – ₹ 2,88,000)	1,12,000
Total Profit	2,08,000

ICAI · P4 · MTP Jan 2025 · Series II · question p. 9-10 · answer p. 12

724 **MTP** **May 2024 S-II** **Q1(a)** **5 marks**

The product of a manufacturing concern passes through two processes A and B and then to finished stock. The details of expenses incurred on the two processes during the year were as under:

	Process A (₹)	Process B (₹)
Materials	40,000	--
Labour	40,000	56,000
Overheads	16,000	40,000

On completion, the output of Process A is transferred to Process B at a price calculated to give a profit of 20% on the transfer price and the output of Process B is charged to finished stock at a profit of 25% on the transfer price. The finished stock department realized ₹ 4,00,000 for the finished goods received from Process B.

You are asked to SHOW process accounts and total profit, assuming that there was no opening or closing work-in-progress.

ICAI'S ANSWER

Process A Account

Dr			Cr.
	₹		₹
To Materials	40,000	By Transfer to Process B A/c	1,20,000
To Labour	40,000		
To Overheads	16,000		
	96,000		
To Profit (20% of transfer price, i.e., 25% of cost)	24,000		
	1,20,000		1,20,000

Process B Account

Dr			Cr.
	₹		₹
To Transferred from Process A A/c	1,20,000	By Transfer to Finished Stock A/c	2,88,000
To Labour	56,000		
To Overhead	40,000		
	2,16,000		
To Profit (25% of transfer price i.e., 33.33% of cost)	72,000		
	2,88,000		2,88,000

Statement of Total Profit

	₹
Profit from Process A	24,000
Profit from Process B	72,000
Profit on Sales (₹ 4,00,000 – ₹ 2,88,000)	1,12,000
Total Profit	2,08,000

ICAI · P4 · MTP May 2024 · Series II · question p. 5 · answer p. 4-5

725 MTP **Old P3 May 2022 S-I** Q6(e) 5 marks old scheme

WHAT is inter-process profit? STATE its advantages and disadvantages.

ICAI'S ANSWER

Inter-Process Profit: To control cost and to measure performance, different processes within an organization are designated as separate profit centres. In this type of organizational structure, the output of one process is transferred to the next process not at cost but at market value or cost plus a percentage of profit. The difference between cost and the transfer price is known as inter-process profits.

The advantages and disadvantages of using inter-process profit, in the case of process type industries are as follows:

Advantages:

1. Comparison between the cost of output and its market price at the stage of completion is facilitated.
2. Each process is made to stand by itself as to the profitability.

Disadvantages:

1. The use of inter-process profits involves complication.

2. The system shows profits which are not realised because of stock not sold out.

ICAI · Old P3 · MTP May 2022 · Series I · question p. 7 · answer p. 11

726 MTP Old P3 Jul 2021 S-II Q1(d) 5 marks old scheme

An article passes through three successive operations from raw materials stage to the finished product stage. The following data are available from the production records for the month of March, 2021:

Operation	No. of pieces (Input)	No. of pieces (Rejected)	No. of pieces (Output)
1	1,80,000	60,000	1,20,000
2	1,98,000	18,000	1,80,000
3	1,44,000	24,000	1,20,000

(i) DETERMINE the input required to be introduced in the first operation in no. of pieces in order to obtain finished output of 500 pieces after the last operation.

(ii) CALCULATE the cost of raw material required to produce one piece of finished product, if the weight of the finished piece is 0.5 kg. and the price of raw material is Rs. 80 per kg.

ICAI'S ANSWER

Statement of production

Operation	Input	Rejections		Output
		Total	% of output	
1	1,80,000	60,000	50	1,20,000
2	1,98,000	18,000	10	1,80,000
3	1,44,000	24,000	20	1,20,000

(i) Determination of input required to obtain 500 pieces of finished output:

Particulars	No. of pieces
Output required after operation 3	500
Add: Rejection in operation 3 (20%)	100
Output required after operation 2	600
Add: Rejection in operation 2 (10%)	60
Output required after operation 1	660
Add: Rejection in operation 1 (50%)	330
Input required in operation 1	**990**

(ii) Calculation of cost of raw material:

To produce 500 pieces of final output, 990 pieces of inputs are required at operation 1. Thus, to get a finished piece of 0.5 kg. of output, the weight of input required is:

$= \frac{0.5}{500} \times 990 = 0.99$ kg.

The cost of raw material would be Rs. 80 × 0.99 kg. = Rs.79.20

ICAI · Old P3 · MTP Jul 2021 · Series II · question p. 2 · answer p. 2-3

CHAPTER 11

Joint Products and By-Products

11.1 Apportioning joint costs: physical units, sales value and net realisable value

46 questions · asked in 39 ICAI documents · 12 exam papers

727 EXAM **Sep 2026** Q6(b)(ii) 1 mark

Products A and B are produced from the same raw material but Product A requires more refining before it can be sold. How would you apportion the joint costs between Products A and B using net realizable method ?

ICAI'S ANSWER

ICAI has not published its answers to this paper yet.

ICAI · P4 · Sep 2026 exam · question p. 22

728 EXAM **May 2026** Q3(a) 8 marks

Apex Chemicals Ltd. processes a basic chemical raw material in 8 Department I, which yields two joint products, A and B, at the split-off point in the ratio 60 40. The input-output ratio of Department I is 100: 90.

Product A can either be sold at split-off or further processed in Department II to produce A-Prime. Department II is used exclusively for Product A, and its input-output ratio is 100: 80.

Production & Cost Data

Particulars	Department I	Department II
Raw Material Input	1,20,000 kg @ ₹ 7 per kg	-
Conversion Costs:	₹ 4,50,000	
Other Direct Material	₹ 4,50,000	₹ 1,00,000
Direct Labour	₹ 4,00,000	₹ 1,40,000
Variable Overhead	₹ 6,00,000	₹ 2,60,000
Fixed Overhead	₹ 5,50,000	₹ 40,000
Total Conversion Cost	**₹ 20,00,000**	**₹ 5,40,000**

Other Data:

Product	Selling Price (₹ per kg)	Selling Expense (₹)
A (split – off)	45	30,000
B	45	26,000
A - Prime	75	22,000

Required:

(i) Prepare a joint cost apportionment statement for Products A and B on the basis of sales value at split-off point.

(ii) Advise whether Product A should be sold at split-off or processed into A-Prime, showing incremental revenue, incremental cost, and profit.

ICAI'S ANSWER

Calculation of Total Joint Cost of Department I

Particulars	Amount (₹)
Raw material (1,20,000 kg × ₹ 7)	8,40,000.00
Conversion Cost:	
Other direct material	4,50,000.00
Direct labour	4,00,000.00
Variable overhead	6,00,000.00
Fixed Overheads	5,50,000.00
Total Cost	**28,40,000.00**
(i) Joint Cost Apportionment Based on Sales Value at Split-off	
Output total (1,20,000 × 90%)	1,08,000 units
Product A **(60%)**	64,800 units
Product B **(40%)**	43,200 units
Sales Value at split off point:	
Product A (64,800 x ₹ 45)	29,16,000.00
Product B (43,200 x ₹ 45)	19,44,000.00
Total Sales value	48,60,000.00
Joint Cost (28,40,000) Apportioned:	
Product A (28,40,000 × 29,16,000)/48,60,000	17,04,000.00
Product B (28,40,000 ×19,44,000)/48,60,000	11,36,000.00

(ii) If product A is processed further into A prime

Particulars	Amount (₹)
Output (64,800 × 80%)	51,840
(A) Incremental Revenue {(₹ 75 × 51,840) - (₹ 45 × 64,800)}	9,72,000.00
Increment Cost:	
Conversion cost in Department II	5,40,000.00
Less: Savings on Selling Expenses (₹ 30,000 - ₹ 22,000)	8,000.00
(B) Total Incremental Cost	5,32,000.00
(C) Incremental profit (A-B)	4,40,000.00

Apex Chemicals Ltd. should further process Product A into A-Prime. This decision results in an incremental profit of ₹ 4,40,000.

ICAI · P4 · May 2026 exam · question p. 16-17 · answer p. 18-19

729 EXAM Jan 2026 Q1(b) 5 marks

ECO Cotton Limited manufactures two joint products Cotton Fiber, Cotton Seed and a by-product Cotton Linters by processing a raw cotton,

The joint processing costs are ₹ 1,46,100.

Joint products can be further processed and sold at a higher market price, with some sales promotion efforts whereas, by-product Cotton Linters can be sold only after further processing.

The relevant details of three products are as follows:

Particulars	Cotton Fiber	Cotton Seed	Cotton Linters
Output (in kgs.)	2,800	1,800	300
Selling price at the split -off point (per kg.)	₹ 90	₹ 60	-
Further processing cost (per kg.)	₹ 9	₹ 6	₹ 2
Further marketing cost (per kg.)	₹ 5	₹ 2	₹ 1
Selling price after further processing (per kg.)	₹ 110	₹ 67	₹ 10

You are required to: -

(i) Show how joint cost would be apportioned between Cotton Fiber and Cotton Seed using the sales value at split-off, after crediting net realisable value of by-product Cotton Linters.

(ii) Determine the total profit or loss if the joint products are sold without further processing.

(iii) Which of the joint products can be processed further for maximizing profits using incremental analysis of profit?

ICAI'S ANSWER

Working Note:

Calculation of Net Realizable value of By-Product:

Particulars	
Selling Price after further processing (₹)	10
Less: Further Marketing Cost (₹)	1
Less: Further Processing Cost (₹)	2
Net Realisable Value (₹ 10 - ₹ 3)	7
Quantity (Kgs)	300
Net Realisable Value of By-Product (300 kg@₹7) (₹)	2,100

Calculation of Net joint costs to be apportioned:

Particulars	Amount (₹)
Joint Costs	1,46,100
Less: Net Realizable value of by-product	2,100
Net joint costs to be apportioned	1,44,000

(i) Statement showing apportionment of Joint Cost between Cotton fiber and Cotton Seed using sales value at split-off point

Product	Cotton Fibre	Cotton Seeds
Quantity (in Kgs)	2,800	1800
Selling Price at Split-off point (per kg) (₹)	90	60
Sales Value at Split off point (₹)	2,52,000	1,08,000
Joint Cost Apportioned (₹ 1,44,000)	1,00,800	43,200

(ii) Statement showing Profit or loss if the Joint Products are Sold Without further Processing

Product	Cotton Fibre	Cotton Seeds	Total
Sales Value at Split off point (₹)	2,52,000	1,08,000	3,60,000
Less: Joint Cost Apportioned (₹1,44,000)	1,00,800	43,200	1,44,000
Profit	1,51,200	64,800	2,16,000

(iii) Statement of increment profit/loss to decide further processing

Product	Cotton Fibre	Cotton Seeds
Incremental Revenue	56,000 [(110-90) × 2,800]	12,600 [(67-60) × 1,800]
Less: Further processing cost	39,200 (9+5) x 2800	14,400 (6+2) x 1800
Incremental profit /(loss)	**16,800**	**(1,800)**

Increment Profit/(loos) can be presented on unit basis also.

It is advisable to further process only product Cotton Fibre and to sale product Cotton Seeds at the point of separation.

ICAI · P4 · Jan 2026 exam · question p. 8-9 · answer p. 10-12

730 EXAM **Sep 2025** MCQ 12 2 marks MCQ

A company produces two joint products X and Y, using the same type of material. The cost data to produce 200 units of product X and 400 units of product Y are as under:

Material	₹ 80,000
Labour	₹ 40,000
Fixed overheads	₹ 20,000

Sales of product X was 200 units @₹ 350 per unit and product Y was 400 units @ ₹ 250 per unit.

Using the Contribution margin method, in what ratio fixed overheads will be apportioned between Product X and Product Y?

A 4:8

B 3:2

C 2:3

D 8:4

ICAI'S ANSWER

Answer key: **(B)**

ICAI · P4 · Sep 2025 exam · question p. 5-6 · answer p. 7

731 EXAM **May 2025** Case scenario · Q1(b)(i) to Q1(b)(ii) 5 marks

CASE SCENARIO

Aroma Park Ltd. Produces two perfumes named Floral, Oriental, and one Cologne, all created through

a joint production process. Below are the data from the most recent month of production:

	Products		
	Floral	**Oriental**	**Cologne**
Sales Price	₹ 80	₹200	₹ 300
Quantity (in units)	5,000	3,000	2,000
Joint Cost	₹ 60	₹ 60	₹ 60
Cost after split off	₹ 40	₹ 80	₹ 100
Total cost	₹ 100	₹ 140	₹ 160

The management on reviewing the above cost data is of the opinion that either they are selling the largest – volume product at a loss or the product cost data is flawed.

Required:

Q1(b)(i) 2 marks

Prepare statement showing profit / loss for each product based on the given data.

Q1(b)(ii) 3 marks

Respond to the management perception by showing joint cost apportionment under Net Realisable Value method.

ICAI'S ANSWERS

Q1(b)(i)

Statement showing profit/loss by each product after further processing products

	Floral (in ₹)	Oriental (in ₹)	Cologne (in ₹)	Total (in ₹)
Sales price after further processing	80	200	300	
Less: Cost after split off	40	80	100	
Less: Joint Cost (already apportioned)	60	60	60	
Profit/(loss) per unit	(20)	60	140	
Quantity (in units)	5,000	3,000	2,000	
Profit/(loss)	**(1,00,000)**	**1,80,000**	**2,80,000**	**3,60,000**

Q1(b)(ii)

Apportionment of joint costs on the basis of Net Realisable Value method

	Floral (in ₹)	Oriental (in ₹)	Cologne (in ₹)	Total (in ₹)
Sales price	80	200	300	
Less: Post split-off cost (Further processing cost)	40	80	100	
Net Realisable Value per unit	40	120	200	
Net Realisable Value	2,00,000 (₹ 40 × 5,000)	3,60,000 (₹ 120 × 3,000)	4,00,000 (₹ 200 × 2,000)	
Apportionment of Joint Cost of ₹ 6,00,000* in ratio of 5:9:10	1,25,000	2,25,000	2,50,000	
Total Profit	**75,000**	**1,35,000**	**1,50,000**	**3,60,000**

*Total Joint Cost = 60 x 5,000 + 60 x 3,000 + 60 x 2,000 = 6,00,000

Note: Profitability statement prepared after distributing joint cost based on NRV clearly shows that they are not selling its largest volume product, i.e floral at a loss. Actually, it yields a profit of ₹ 75,000.

ICAI · P4 · May 2025 exam · question p. 8 · answer p. 11, 11-12

732 EXAM **Jan 2025** MCQ 12 MCQ

Two products Y and Z are obtained in a crude form and require further processing at a cost of ₹ 6 for Y and ₹ 5 for Z per litre before the products can be sold in the market. The final prices of product Y and Z are ₹ 15 and ₹ 8.75 per litre respectively. The company earns a net margin of 25% on Cost. The following data is available for output of both the products for the year

Y 8,000 Litres

Z 6,000 Litres

A joint cost of ₹ 60,000 was incurred for the year and company apportions the joint costs on the basis of net realisable value after further processing.

Calculate the joint cost per unit of product Y.

A ₹ 4.74

B ₹ 5.00

C ₹ 5.625

D ₹ 6.00

ICAI'S ANSWER

Answer key: **(D)**

ICAI · P4 · Jan 2025 exam · question p. 5 · answer p. 6

733 EXAM **May 2024** Q1(b) 5 marks

A company produces two products, A and B, through a joint production process. The total joint

production cost incurred is as under:

Material	₹ 20,000
Labour	₹ 10,000
Variable overheads	₹ 6,000
Fixed Overheads	₹ 24,000

Product A and B can be sold for ₹ 20 per unit and ₹ 15 per unit respectively at split off point. The produced quantities are Product A-2,000 units and Product B – 4,000 units.

(i) You are required to calculate the joint production cost allocation for each product using the:

(a) Physical unit method.

(b) Contribution margin method.

(ii) Product B can be further processed by incurring expenditure of ₹ 12,000. Loss in further processing is 2%. It can be sold @ ₹ 18 per unit. Explain the impact on profitability if Product B is further processed.

ICAI'S ANSWER

Working

Calculation of joint cost

Description	Amount (₹)
Material	20,000
Labour	10,000
Variable overheads	6,000
Total variable cost	**36,000**
Fixed overheads	24,000
Total joint cost	**60,000**

(i) (a) Allocation of joint cost using physical unit method:

Product A = ₹60,000 × 2,000/6,000 = ₹ 20,000

Product B = ₹60,000 × 4,000/6,000 = ₹ 40,000

(b) Allocation of joint cost using contribution margin method:

	Description	Product-A	Product-B
	Units produced	2,000	4,000
	Selling price per unit (₹)	20	15
A.	Sales value (₹)	40,000	60,000
B.	**Allocation of joint variable cost on the basis of physical unit**		

	Description	Product-A	Product-B
	₹36,000 × 2,000/6,000	(12,000)	
	₹36,000 × 4,000/6,000		(24,000)
C= A-B	**Contribution**	**28,000**	**36,000**
D	Allocation of fixed joint cost on the basis of contribution margin		
	₹24,000 × 28,000/64,000	(10,500)	
	₹24,000 × 36,000/64,000		(13,500)
C-D	Profit at split off point	**17,500**	**22,500**

Allocation of Joint Cost on the basis of Contribution Margin Method:

Particulars	Product A	Product B
Allocation of Variable Cost	₹ 12,000	₹ 24,000
Allocation of Fixed Cost	₹ 10,500	₹ 13,500
Total Joint Cost	**₹ 22,500**	**₹ 37,500**

(ii) Profitability after further processing of Product B

Description	Amount (₹)
Units produced and sold 98% of 4,000 units	3,920
Selling price per unit (₹)	18
Sales value (₹)	70,560
Joint cost upto split off point	(37,500)
Further processing cost	(12,000)
Profit after further processing	**21,060**

Calculation of the profitability after further processing of product B can also be done in the following manner:

Profitability after further processing of Product B

Description	Amount (₹)
Incremental revenue on further processing (3,920 x ₹18)- (4,000 x ₹15)	10,560
Further processing cost	(12,000)
Incremental loss after further processing	**1,440**

Impact on profitability on Product B

If Product B is sold at split off point it earns profit of ₹ 22,500, but after further processing the profit is reduced to ₹ 21,060/- i.e. an opportunity loss of ₹ 1,440/-.

ICAI · P4 · May 2024 exam · question p. 1-2 · answer p. 4-5

734 EXAM Old P3 Nov 2023 Q1(c) 5 marks old scheme

XYZ Limited manufactures three joint products A, B and C from a joint process. Product B is sold at split off point whereas product A and C are sold after further processing. 10% of the quantity of product A is lost in further processing. Data regarding these products for the year ending 31st March,2023 are as follows:

	A	B	C
Number of units produced and sold	3,60,000	2,10,000	4,50,000
Selling price per unit at split off point	-	₹ 6	-
Selling price per unit after further processing	₹ 9.50	-	₹ 12
Further processing costs	₹ 8,60,000	-	₹10,40,000

The joint production cost upto the split off point at which A, B and C become separable products is ₹ 57,26,000.

Required:

(i) Prepare a statement showing apportionment of joint cost to the products using Net realizable value method.

(ii) Assume XYZ Limited has received an offer from D Limited to purchase product 'A' at the split off point at ₹ 7 per unit and another company PQR Limited has offered to purchase product 'C' at split off point at 9 per unit.

Advise whether these offers should be accepted or not?

ICAI'S ANSWER

(i) Statement showing apportionment of joint cost to the products using NRV method

Particulars	Product A (₹)	Product B (₹)	Product C (₹)
Sales value	34,20,000 (3,60,000 x ₹ 9.5)	12,60,000 (2,10,000 x ₹ 6)	54,00,000 (4,50,000 x ₹ 12)
Less: Further processing cost	8,60,000	-	10,40,000
Net Realisable Value	25,60,000	12,60,000	43,60,000
Apportionment of Joint cost of ₹ 57,26,000 in the ratio of 256:126:436	17,92,000	8,82,000	30,52,000

(ii) Decision whether to Process further or not

Particulars	Product A (₹)	Product C (₹)
Incremental Revenue	9,00,000 (₹ 9.5-₹ 7) x 3,60,000	13,50,000 (₹ 12- ₹ 9) x 4,50,000
Less: Further processing cost	8,60,000	10,40,000
Less: wastage if further processed	2,80,000 ₹ 7 x (3,60,000*10%/90%)	-
Incremental profit/(loss)	**(2,40,000)**	**3,10,000**

On comparing incremental sales revenue with further processing cost, there is net loss of ₹ 2,40,000 in case of product A and profit of ₹ 3,10,000 in case of product C. Hence **offer of D Ltd should be accepted and Product A should be sold at split off point** Whereas product C should be **sold after further processing.**

The solution can also be presented in following way:

Profit from further processing

Particulars	Product A (₹)	Product C (₹)
Sales Revenue	34,20,000 (3,60,000 × 9.5)	54,00,000 (4,50,000 × 12)
Less: Joint cost	17,92,000	30,52,000
Less: Further processing cost	8,60,0000	10,40,000
(i) Profit/(loss)	**7,68,000**	**13,08,000**

Profit from Accepting offer (Sale at separation point)

Particulars	Product A (₹) D Limited offer accepted	Product C (₹) PQR Limited offer accepted
Sales Revenue	28,00,000 (3,60,000/0.90) x 7	40,50,000 (4,50,000 × 9)
Less: Joint cost	17,92,000	30,52,000
(ii) Profit/(loss)	**10,08,000**	**9,98,000**
Incremental profit (loss) (i)-(ii)	**(2,40,000)**	**3,10,000**

On comparing profit at separation point with further processing profit, there is net loss of ₹ 2,40,000 in case of product A and profit of ₹ 3,10,000 in case of product C. Hence **offer of D Ltd should be accepted and Product A should be sold at split off point** Whereas product C should be **sold after further processing.**

ICAI · Old P3 · Nov 2023 exam · question p. 2 · answer p. 5-6

735 EXAM **Old P3 May 2023** Q4(a) 10 marks old scheme

ABC Company produces a Product 'X' that passes through three processes: R, S and T. Three types of raw materials, viz., J, K, and L are used in the ratio of 40:40:20 in process R. The output of each process is transferred to next process. Process loss is 10% of total input in each process. At the stage of output in

process T, a by-product 'Z' is emerging and the ratio of the main product 'X' to the by-product 'Z' is 80:20. The selling price of product 'X' is ₹60 per kg.

The company produced 14,580 kgs of product 'X'

Material price : Material J @ ₹ 15 per kg; Material K @ ₹ 9 per kg.

Material L@ ₹ 7 per kg Process costs are as follows:

Process	Variable cost per kg (₹)	Fixed cost of Input (₹)
R	5.00	42,000
S	4.50	5,000
T	3.40	4,800

The by-product 'Z' cannot be processed further and can be sold at ₹ 30 per kg at the split-off stage. There is no realizable value of process losses at any stage.

Required:

Present a statement showing the apportionment of joint costs on the basis of the sales value of product 'X' and by-product 'Z' at the split- off point and the profitability of product 'X' and by-product 'Z.

ICAI'S ANSWER

Working Notes:

1. Calculation of Input of Raw Material

Let assume total raw material in Process R be 100%

∴ Output of Process T will be equal to:

Input R	100%
- 10% Normal Loss	₹ 10
Input S	₹ 90%
- 10% Normal loss	₹ 9
Input T	81%
- 10% Normal loss	₹ 8.1
Output of T	72.9
Actual output of X	14,580 units

Which is 80% of the total output

∴ Output of Process T

$= \frac{14580}{80\%} = 18{,}225$

∴ Input of Process R $= \frac{18225}{72.9\%} = 25{,}000$ kgs

Alternative presentation for Calculation of Input in Process R, S and T

Working notes:

Process T (Kg.)			
To Input (Transfer from process S)	20,250	By Normal loss	2,025
		By Output Product X	14,580
		By output of by-product Z	3,645
	20,250		20,250

Process S (kg.)			
To Input (Transfer from process S)	22,500	By Normal loss (10%)	2,250
		By Transfer to process T	20,250
	22,500		22,500

Process R (kg.)			
To Input	25,000	By Normal loss (10%)	2,500
		By Transfer to process S	22,500
	25,000		25,000

2. Calculation of Joint Cost

Process	Inputs	Variable cost per kg	Variable cost	Fixed Cost	Total Cost
		₹	₹	₹	₹
R	25,000	5	1,25,000	42,000	1,67,000
S	22,500	4.5	1,01,250	5,000	1,06,250
T	20,250	3.4	68,850	4,800	73,650
					3,46,900

Raw material	J	10000 × 15	₹ 1,50,000
	K	10000 × 9	₹ 90,000
	L	5000 × 7	₹ 35,000
			2,75,000
Add: Processing cost (as above)			₹ 3,46,900
Total Joint Cost			6,21,900

(i) Statement showing apportionment of Joint Cost

Particulars	Product X	By-Product Z	Total
Units	14,580	3,645	
Selling price (₹)	60	30	
Sales Value (₹)	8,74,800	1,09,350	9,84,150
(₹ 6,21,900 to apportioned in ratio of sales value at split off point)	5,52,800	69,100	6,21,900

(ii) Statement of Profitability

Particulars	Product X	By-Product Z	Total
Sales Value	8,74,800	1,09,350	9,84,150
Joint Cost (As apportioned above)	(5,52,800)	(69,100)	(6,21,900)
Profit	3,22,000	40,250	3,62,250

ICAI · Old P3 · May 2023 exam · question p. 18 · answer p. 20-21

736 EXAM **Old P3 Jan 2021** Q4(a) 10 marks old scheme

Mayura Chemicals Ltd buys a particular raw material at ₹ 8 per litre. At the end of the processing in Department- I, this raw material splits-off into products X, Y and Z. Product X is sold at the split-off point, with no further processing. Products Y and Z require further processing before they can be sold. Product Y is processed in Department-2, and Product Z is processed in Department-3. Following is a summary of the costs and other related data for the year 2019-20:

Particulars	Department		
	1	**2**	**3**
Cost of Raw Material	₹ 4,80,000	-	-
Direct Labour	₹ 70,000	₹ 4,50,000	₹ 6,50,000
Manufacturing Overhead	₹ 48,000	₹ 2,10,000	₹ 4,50,000
	Products		
	X	**Y**	**Z**
Sales (litres)	10,000	15,000	22,500
Closing inventory (litres)	5,000	-	7,500
Sale price per litre (₹)	30	64	50

There were no opening and closing inventories of basic raw materials at the beginning as well as at the end of the year. All finished goods inventory in litres was complete as to processing. The company uses the Net-realisable value method of allocating joint costs.

You are required to prepare:

(i) Schedule showing the allocation of joint costs.

(ii) Calculate the Cost of goods sold of each product and the cost of each item in Inventory.

(iii) A comparative statement of Gross profit.

ICAI'S ANSWER

(i) Statement of Joint Cost allocation of inventories of X, Y and Z

	Products			Total (₹)
	X (₹)	**Y (₹)**	**Z (₹)**	
Final sales value of total production (Working Note 1)	4,50,000 (15,000 x ₹ 30)	9,60,000 (15,000 x ₹ 64)	15,00,000 (30,000 x ₹ 50)	29,10,000
Less: Additional cost	--	6,60,000	11,00,000	17,60,000
Net realisable value (at split-off point)	4,50,000	3,00,000	4,00,000	11,50,000
Joint cost allocated (Working Note 2)	2,34,000	1,56,000	2,08,000	5,98,000

(ii) Calculation of Cost of goods sold and Closing inventory

	Products			Total (₹)
	X (₹)	**Y (₹)**	**Z (₹)**	
Allocated joint cost	2,34,000	1,56,000	2,08,000	5,98,000
Add: Additional costs	--	6,60,000	11,00,000	17,60,000
Cost of goods sold (COGS)	2,34,000	8,16,000	13,08,000	23,58,000
Less: Cost of closing inventory (Working Note 1)	78,000 (COGS × 100/3%)	--	3,27,000 (COGS × 25%)	4,05,000
Cost of goods sold	1,56,000	8,16,000	9,81,000	19,53,000

(iii) Comparative Statement of Gross Profit

	Products			Total (₹)
	X (₹)	**Y (₹)**	**Z (₹)**	
Sales revenue	3,00,000 (10,000 x ₹ 30)	9,60,000 (15,000 x ₹ 64)	11,25,000 (22,500 x ₹ 50)	23,85,000
Less: Cost of goods sold	1,56,000	8,16,000	9,81,000	19,53,000
Gross Profit	**1,44,000**	**1,44,000**	**1,44,000**	4,32,000

Working Notes:

1. Total production of three products for the year 2019-2020

Products	Quantity sold in litres	Quantity of closing inventory in litres	Total production	Closing inventory percentage (%)
(1)	**(2)**	**(3)**	**(4) = [(2) + (3)}**	**(5) = (3)/ (4)**
X	10,000	5,000	15,000	100/3
Y	15,000	--	15,000	--
Z	22,500	7,500	30,000	25

2. Joint cost apportioned to each product:

$$= \frac{\text{Total Joint cost}}{\text{Total Net Realisable Value}} \text{ x Net Realisable Value of each product}$$

$$\text{Joint cost of product X} = \frac{₹\ 5,98,000}{₹\ 11,50,000} \text{ x } ₹\ 4,50,000 = ₹\ 2,34,000$$

$$\text{Joint cost of product Y} = \frac{₹\ 5,98,000}{₹\ 11,50,000} \text{ x } ₹\ 3,00,000 = ₹\ 1,56,000$$

$$\text{Joint cost of product Z} = \frac{₹\ 5,98,000}{₹\ 11,50,000} \text{ x } ₹\ 4,00,000 = ₹\ 2,08,000$$

ICAI · Old P3 · Jan 2021 exam · question p. 15 · answer p. 16-18

737 EXAM **Old P3 Nov 2020** Q1(c) 5 marks old scheme

A company's plant processes 6,750 units of a raw material in a month to produce two products 'M' and 'N'.

The process yield is as under:

Product M	80%
Product N	12%
Process Loss	8%

The cost of raw material is ₹ 80 per unit.

Processing cost is ₹ 2,25,000 of which labour cost is accounted for 66%. Labour is chargeable to products 'M' and 'N' in the ratio of 100:80.

Prepare a Comprehensive Cost Statement for each product showing:

(i) Apportionment of joint cost among products 'M' and 'N' and

(ii) Total cost of the products 'M' and 'N'.

ICAI'S ANSWER

Comprehensive Cost Statement

Particulars	Total Cost (₹)	Product-M (₹)	Product-N (₹)
No. of units produced *		5,400 units	810 units
Cost of raw material (₹ 80 × 6,750 units)	5,40,000		

Particulars	Total Cost (₹)	Product-M (₹)	Product-N (₹)
Processing cost:			
- Labour cost (₹ 2,25,000 × 66%)	1,48,500		
- Other costs (₹ 2,25,000 - 1,48,500)	76,500		
Total joint cost	**7,65,000**		
(i) Apportionment of joint costs between the joint products			
Labour cost in the ratio of 100:80	1,48,500	**82,500** $\left(\frac{1,48,500 \times 100}{180}\right)$	**66,000** $\left(\frac{1,48,500 \times 80}{180}\right)$
Other joint costs (including material) in the ratio of output (5,400:810)	6,16,500	**5,36,087** $\left(\frac{6,16,500 \times 5,400}{6,210}\right)$	**80,413** $\left(\frac{6,16,500 \times 810}{6,210}\right)$
(ii) Total product cost	7,65,000	**6,18,587**	**1,46,413**

* No. of units produced of Product M = 6750 units x 80% = 5400 units

No. of units produced of Product N = 6750 units x 12% = 810 units

ICAI · Old P3 · Nov 2020 exam · question p. 2 · answer p. 4-5

738 EXAM Old P3 Nov 2019 Q1(c) 5 marks old scheme

A Factory produces two products, 'A' and 'B' from a single process. The joint processing costs during a particular month are :

Direct Material ₹30,000

Direct Labour ₹ 9,600

Variable Overheads ₹ 12,000

Fixed Overheads ₹ 32,000

Sales: A- 100 units@ ₹ 600 per unit; B – 120 units @ ₹ 200 per unit.

I. Apportion joints costs on the basis of:

(i) Physical Quantity of each product.

(ii) Contribution Margin method, and

II. Determine Profit or Loss under both the methods.

ICAI'S ANSWER

Total Joint Cost

	Amount (₹)
Direct Material	30,000
Direct Labour	9,600
Variable Overheads	12,000
Total Variable Cost	51,600
Fixed Overheads	32,000
Total joint cost	83,600

Apportionment of Joint Costs:

			Product-A	Product-B
I.	(i)	Apportionment of Joint Cost on the basis of 'Physical Quantity'	**₹ 38,000** $\left(\frac{₹\ 83,600}{100+120\ \text{units}} \times 100\right)$	**₹ 45,600** $\left(\frac{₹\ 83,600}{100+120\ \text{units}} \times 120\right)$
	(ii)	Apportionment of Joint Cost on the basis of 'Contribution Margin Method':		
		- Variable Costs (on basis of physical units)	**₹ 23,455** $\left(\frac{₹\ 51,600}{100+120\ \text{units}} \times 100\right)$	**₹ 28,145** $\left(\frac{₹\ 51,600}{100+120\ \text{units}} \times 120\right)$
		Contribution Margin	**36,545** (₹600×100 – 23,455)	**-4,145** (₹200×120 – 28,145)
		Fixed Costs*	₹ 32,000	
		Total apportioned cost	₹ 55,455	₹ 28,145
II.	**(iii)**	**Profit or Loss:**		
	When Joint cost apportioned on basis of physical units			
	A.	Sales Value	₹ 60,000	₹ 24,000
	B.	Apportioned joint cost on basis of 'Physical Quantity':	₹ 38,000	₹ 45,600
	A-B	**Profit or (Loss)**	**22,000**	**(21,600)**

			Product-A	Product-B
	When Joint cost apportioned on basis of 'Contribution Margin Method'			
	C	Apportioned joint cost on basis of 'Contribution Margin Method'	₹ 55,455	₹ 28,145
	A-C	**Profit or (Loss)**	₹ 4,545	₹ (4,145)

* The fixed cost of ₹ 32,000 is to be apportioned over the joint products A and B in the ratio of their contribution margin but contribution margin of Product B is Negative so fixed cost will be charged to Product A only.

ICAI · Old P3 · Nov 2019 exam · question p. 2 · answer p. 4-5

739 RTP **May 2026** Q5 MCQ

In a manufacturing enterprise, two joint products, X and Y, emerge simultaneously from a common production process. The company incurs joint costs up to the split-off point, after which the products become separately identifiable. For the previous month, the following details relating to production and inventory are available:

Product	Sales (units)	Finished Goods – Opening Stock (units)	Finished Goods – Closing Stock (units)
X	96,000	2,400	6,000
Y	64,000	4,800	2,400

The total joint production cost incurred during the month amounted to ₹ 18,60,000. These joint costs are apportioned between products X and Y in proportion to the number of units produced.

Required:

Determine the amount of joint production cost to be allocated to Product X for the month.

A ₹ 10,23,000

B ₹ 11,49,230

C ₹ 10,71,000

D ₹ 11,16,000

ICAI'S ANSWER

(B) ₹ 11,49,230

Joint Production Cost = ₹ 18,60,000

Product	Sales (units) (A)	Finished Goods (units) Opening stock (B)	Finished Goods (units) Closing stock (C)	Production (units) (D) = (A) + (C) - (B)
X	96,000	2,400	6,000	99,600
Y	64,000	4,800	2,400	61,600
Total				1,61,200

Apportionment of Joint Production Cost to Product X:

$$= \left(\frac{99{,}600 \text{ units}}{1{,}61{,}200 \text{ units}} \times ₹18{,}60{,}000 \right)$$

= ₹ 11,49,230

ICAI · P4 · RTP May 2026 · question p. 7 · answer p. 28

740 **RTP** **May 2026** Q14

ABC Ltd. operates a simple chemical process to convert a single material into three separate items, referred to here as X, Y and Z. All three end products are separated simultaneously at a single split-off point.

Product X and Y are ready for sale immediately upon split off without further processing or any other additional costs. Product Z, however, is processed further before being sold. There is no available market price for Z at the split-off point.

The selling prices quoted here are expected to remain the same in the coming year. During 2024-25, the selling prices of the items and the total amounts sold were:

X – 186 tons sold for ₹ 3,000 per ton

Y – 527 tons sold for ₹ 2,250 per ton

Z – 736 tons sold for ₹ 1,500 per ton

The total joint manufacturing costs for the year were ₹ 12,50,000. An additional ₹ 6,20,000 was spent to finish product Z.

There were no opening inventories of X, Y or Z at the end of the year. The following inventories of complete units were on hand:

X 180 tons

Y 60 Tons

Z 25 tons

There was no opening or closing work-in-progress.

Required:

COMPUTE the cost of inventories of X, Y and Z and cost of goods sold for year ended March 31, 2025, using Net realizable value (NRV) method of joint cost allocation.

ICAI'S ANSWER

(i) (a) Statement of Joint Cost allocation of inventories of X, Y and Z

(By using Net Realisable Value Method)

	Products			Total
	X	Y	Z	
	(₹)	(₹)	(₹)	(₹)
Final sales value of total production (Working Note 1)	10,98,000 (366 × ₹3,000)	13,20,750 (587 × ₹2,250)	11,41,500 (761 × ₹1,500)	35,60,250
Less: Additional cost	--	--	(6,20,000)	(6,20,000)
Net realisable value (at split-off point)	10,98,000	13,20,750	5,21,500	29,40,250
Joint cost allocated (Working Note 2)	4,66,797	5,61,496	2,21,707	12,50,000

Cost of goods sold as on March 31, 2025

(By using Net Realisable Value Method)

	Products			Total
	X	Y	Z	
	(₹)	(₹)	(₹)	(₹)
Allocated joint cost	4,66,797	5,61,496	2,21,707	12,50,000
Additional costs	--	--	6,20,000	6,20,000
Cost of goods available for sale (CGAS)	4,66,797	5,61,496	8,41,707	18,70,000
Less: Cost of ending inventory (Working Note 1)	2,29,571 (CGAS× 49.18%)	57,385 (CGAS × 10.22%)	27,692 (CGAS × 3.29%)	3,14,648
Cost of goods sold	2,37,226	5,04,111	8,14,015	15,55,352

Working Notes

1. **Total production of three products for the year 2024-25**

Products	Quantity sold in tones	Quantity of ending inventory in tons	Total production	Ending inventory percentage (%)
(1)	**(2)**	**(3)**	**(4) = [(2) + (3)}**	**(5) = (3)/ (4)**
X	186	180	366	49.18
Y	527	60	587	10.22
Z	736	25	761	3.29

2. **Joint cost apportioned to each product:**

$$\frac{\text{Total Joint cost}}{\text{Total Net Realisable Value}} \times \text{Net Realisable Value of each product}$$

$$\text{Total cost of Product X} = \frac{₹12,50,000}{₹29,40,250} \times ₹10,98,000 = ₹\ 4,66,797$$

$$\text{Total cost of Product Y} = \frac{₹12,50,000}{₹29,40,250} \times ₹13,20,750 = ₹\ 5,61,496$$

$$\text{Total cost of Product Z} = \frac{₹12,50,000}{₹29,40,250} \times ₹5,21,500 = ₹\ 2,21,707$$

ICAI · P4 · RTP May 2026 · question p. 18-19 · answer p. 45-46

741 RTP **Jan 2026** Q15

A company processes a raw material in its Department 1 to produce three products, viz. P, Q and C at the same split-off stage. During a period 1,80,000 kgs of raw materials were processed in Department 1 at a total cost of ₹ 12,88,000 and the resultant output of P, Q and C were 18,000 kgs, 10,000 kgs and 54,000 kgs respectively. P and Q were further processed in Department 2 at a cost of ₹1,80,000 and ₹1,50,000 respectively.

C was further processed in Department 3 at a cost of ₹1,08,000. There is no waste in further processing. The details of sales affected during the period were as under:

	P	Q	C
Quantity Sold (kgs.)	17,000	5,000	44,000
Sales Value (₹)	12,24,000	2,50,000	7,92,000

There were no opening stocks. If these products were sold at split-off stage, the selling prices of P, Q and C would have been ₹ 50, ₹ 40 and ₹ 10 per kg respectively.

Required:

(i) Prepare a statement showing the apportionment of joint costs to P, Q and C.

(ii) Present a statement showing the cost per kg of each product indicating joint cost and further processing cost and total cost separately.

(iii) Prepare a statement showing the product wise and total profit for the period.

(iv) State with supporting calculations as to whether any or all the products should be further processed or not

ICAI'S ANSWER

(i) Statement showing the apportionment of joint costs to P, Q and C

Products	P	Q	C	Total
Output (kg)	18,000	10,000	54,000	
Sales value at the point of split off (₹)	9,00,000 (₹ 50 x 18,000)	4,00,000 (₹ 40 x 10,000)	5,40,000 (₹ 10 x 54,000)	18,40,000
Joint cost apportionment on the basis of sales value at the point of split off (₹)	6,30,000 (₹ 12,88,000/₹ 18,40,000 x ₹ 9,00,000)	2,80,000 (₹ 12,88,000/₹ 18,40,000 x ₹ 4,00,000)	3,78,000 (₹ 12,88,000/₹ 18,40,000 x ₹ 5,40,000)	12,88,000

(ii) Statement showing the cost per kg. of each product (indicating joint cost; further processing cost and total cost separately)

Products	P	Q	C
Joint costs apportioned (₹): (I)	6,30,000	2,80,000	3,78,000
Production (kg): (II)	18,000	10,000	54,000
Joint cost per kg (₹): (I ÷ II)	35	28	7
Further processing Cost per kg. (₹)	10 (₹ 1,80,000/18,000kg)	15 (₹ 1,50,000/10,000kg)	2 (₹ 1,08,000/54,000kg)
Total cost per kg (₹)	45	43	9

(iii) Statement showing the product wise and total profit for the period

Products	P	Q	C	Total
Sales value (₹)	12,24,000	2,50,000	7,92,000	
Add: Closing stock value (₹) (Refer to Working note 2)	45,000	2,15,000	90,000	
Value of production (₹)	12,69,000	4,65,000	8,82,000	26,16,000
Apportionment of joint cost (₹)	6,30,000	2,80,000	3,78,000	
Add: Further processing cost (₹)	1,80,000	1,50,000	1,08,000	
Total cost (₹)	8,10,000	4,30,000	4,86,000	17,26,000
Profit (₹)	4,59,000	35,000	3,96,000	8,90,000

Working Notes

1.

Products	P	Q	C
Sales value (₹)	12,24,000	2,50,000	7,92,000
Quantity sold (Kgs.)	17,000	5,000	44,000
Selling price ₹/kg	72 (₹ 12,24,000/17,000kg)	50 (₹ 2,50,000/5,000kg)	18 (₹ 7,92,000/44,000kg)

2. Valuation of closing stock:

Since the selling price per kg of products P, Q and C is more than their total costs, therefore closing stock will be valued at cost.

Products	P	Q	C	Total
Closing stock (kgs.)	1,000	5,000	10,000	
Cost per kg (₹)	45	43	9	
Closing stock value (₹)	45,000 (₹ 45 x 1,000 kg)	2,15,000 (₹ 43 x 5,000 kg)	90,000 (₹ 9 x 10,000 kg)	3,50,000

(iv) Calculations for processing decision

Products	P	Q	C
Selling price per kg at the point of split off (₹)	50	40	10
Selling price per kg after further processing (₹) (Refer to working Note 1)	72	50	18
Incremental selling price per kg (₹)	22	10	8
Less: Further processing cost per kg (₹)	(10)	(15)	(2)
Incremental profit (loss) per kg (₹)	12	(5)	6

Product P and C has an incremental profit per unit after further processing, hence, these two products may be further processed.

However, further processing of product Q is not profitable hence, product Q shall be sold at split off point.

ICAI · P4 · RTP Jan 2026 · question p. 19 · answer p. 43-46

742 RTP **Sep 2025** Q5 MCQ

In a company's production process, two joint products, A and B, are created simultaneously. The company operates in the manufacturing sector, focusing on producing goods that require multiple stages of production, often resulting in the creation of joint products that share common production costs up until a certain split-off point. For the last month, the following information is provided relating to the inventory and sales:

Product	Sales (units)	Finished Goods (units)	
		Opening stock	**Closing stock**
A	1,14,000	1,900	5,700
B	76,000	7,600	3,800

Joint production costs for the last month amounted to ₹ 20,90,000. These costs were allocated between both the joint products, A and B, based on the number of units produced. This method of apportioning costs ensures that each product is fairly charged for its share of the overall production expenses.

CALCULATE the joint production costs apportioned to product A for last month?

A ₹ 12,12,200

B ₹ 12,54,000

C ₹ 12,95,800

D ₹ 13,58,500

ICAI'S ANSWER

(c) ₹ 12,95,800

Product	Sales (units) (A)	Finished Goods (units)		Production (units) (D) = (A) + (C) - (B)
		Opening stock (B)	Closing stock (C)	
A	1,14,000	1,900	5,700	1,17,800
B	76,000	7,600	3,800	72,200
Total				1,90,000

Apportioned to A = $\left(\frac{1,17,800 \text{ units}}{1,90,000 \text{ units}} \text{ x } ₹\ 20,90,000\right)$

= **₹ 12,95,800**

ICAI · P4 · RTP Sep 2025 · question p. 8-9 · answer p. 30

743 RTP **Sep 2025** Q14

Same question, different ICAI answer: #760 Old P3 MTP Nov 2023 S-I

A company processes a raw material in its Department 1 to produce three products, viz. A, B and X at the same split-off stage. During a period 1,80,000 kgs of raw materials were processed in Department 1 at a total cost of ₹ 12,88,000 and the resultant output of A, B and X were 18,000 kgs, 10,000 kgs and 54,000 kgs respectively. A and B were further processed in Department 2 at a cost of ₹ 1,80,000 and ₹ 1,50,000 respectively.

X was further processed in Department 3 at a cost of ₹1,08,000. There is no waste in further processing. The details of sales affected during the period were as under:

	A	B	X
Quantity Sold (kgs.)	17,000	5,000	44,000
Sales Value (₹)	12,24,000	2,50,000	7,92,000

There were no opening stocks. If these products were sold at split-off stage, the selling prices of A, B and X would have been ₹ 50, ₹ 40 and ₹ 10 per kg respectively. Required:

(i) Prepare a statement showing the apportionment of joint costs to A, B and X.

(ii) Present a statement showing the cost per kg of each product indicating joint cost and further processing cost and total cost separately.

(iii) Prepare a statement showing the product wise and total profit for the period.

(iv) State with supporting calculations as to whether any or all the products should be further processed or not

ICAI'S ANSWER

(i) Statement showing the apportionment of joint costs to A, B and X

Products	A	B	X	Total
Output (kg)	18,000	10,000	54,000	
Sales value at the point of split off (₹)	9,00,000 (₹ 50 × 18,000)	4,00,000 (₹ 40 × 10,000)	5,40,000 (₹ 10 × 54,000)	18,40,000
Joint cost apportionment on the basis of sales value at the point of split off (₹)	6,30,000 (₹12,88,000/₹ 18,40,000 x ₹ 9,00,000)	2,80,000 (₹ 12,88,000/₹ 18,40,000 x ₹ 4,00,000)	3,78,000 (₹ 12,88,000/₹ 18,40,000 x ₹ 5,40,000)	12,88,000

(ii) Statement showing the cost per kg. of each product (indicating joint cost; further processing cost and total cost separately)

Products	A	B	X
Joint costs apportioned (₹): (I)	6,30,000	2,80,000	3,78,000
Production (kg): (II)	18,000	10,000	54,000
Joint cost per kg (₹): (I ÷ II)	35	28	7
Further processing Cost per kg. (₹)	10 (₹ 1,80,000/18,000kg)	15 (₹ 1,50,000/10,000kg)	2 (₹ 1,08,000/54,000kg)
Total cost per kg (₹)	45	43	9

(iii) Statement showing the product wise and total profit for the period

Products	A	B	X	Total
Sales value (₹)	12,24,000	2,50,000	7,92,000	
Add: Closing stock value (₹) (Refer to Working note 2)	45,000	2,15,000	90,000	
Value of production (₹)	12,69,000	4,65,000	8,82,000	26,16,000
Apportionment of joint cost (₹)	6,30,000	2,80,000	3,78,000	
Add: Further processing cost (₹)	1,80,000	1,50,000	1,08,000	
Total cost (₹)	8,10,000	4,30,000	4,86,000	17,26,000
Profit (₹)	4,59,000	35,000	3,96,000	8,90,000

Working Notes

1.

Products	A	B	X
Sales value (₹)	12,24,000	2,50,000	7,92,000
Quantity sold (Kgs.)	17,000	5,000	44,000
Selling price ₹/kg	72 (₹ 12,24,000/17,000kg)	50 (₹ 2,50,000/5,000kg)	18 (₹ 7,92,000/44,000kg)

2. Valuation of closing stock:

Since the selling price per kg of products A, B and X is more than their total costs, therefore closing stock will be valued at cost.

Products	A	B	X	Total
Closing stock (kgs.)	1,000	5,000	10,000	
Cost per kg (₹)	45	43	9	
Closing stock value (₹)	45,000 (₹ 45 × 1,000 kg)	2,15,000 (₹ 43 × 5,000 kg)	90,000 (₹ 9 × 10,000 kg)	3,50,000

(iv) Calculations for processing decision

Products	A	B	X
Selling price per kg at the point of split off (₹)	50	40	10
Selling price per kg after further processing (₹) (Refer to working Note 1)	72	50	18
Incremental selling price per kg (₹)	22	10	8
Less: Further processing cost per kg (₹)	(10)	(15)	(2)
Incremental profit (loss) per kg (₹)	12	(5)	6

Product A and X has an incremental profit per unit after further processing, hence, these two products may be further processed. However, further processing of product B is not profitable hence, product B shall be sold at split off point.

ICAI · P4 · RTP Sep 2025 · question p. 17-18 · answer p. 46-48

744 RTP **May 2025** Q6 MCQ

Sterling Industries manages various manufacturing processes. In process I, joint products P1 and P2 are produced in the ratio of 6:4 in units from the raw material input. A normal loss of 2% of the raw material input is expected in this process, with losses having a realizable value of ₹ 12.5 per kg. The company has no work in progress. The joint costs are apportioned between the joint products using the physical measure basis.

The following information relates to process I for last month:

Raw materials input	75,000 kg (at a cost of ₹ 4,76,250)
Direct labour	₹ 2,25,000
Direct expenses	₹ 67,500
Production Overheads	110% of direct labour cost
Abnormal gain	1,250 kg

You are required to CALCULATE the number of unit and its value of the Normal loss, Abnormal gain and joint products P1 & P2.

A Normal loss: 1,500 units ₹ 16,964; Abnormal gain: 1,250 units ₹ 18,750; Product P1: 44,850 units ₹ 6,08,678 and Product P2: 29,900 units ₹ 4,05,786.

B Normal loss: 1,250 units ₹ 18,750; Abnormal gain: 1,500 units ₹ 16,964; Product P1: 44,850 units ₹ 6,08,678 and Product P2: 29,900 units ₹ 4,05,786.

C Normal loss: 1,500 units ₹ 18,750; Abnormal gain: 1,250 units ₹ 16,964; Product P1: 44,850 units ₹ 6,08,678 and Product P2: 29,900 units ₹ 4,05,786.

D Normal loss: 1,500 units ₹ 18,750; Abnormal gain: 1,250 units ₹ 16,964; Product P1: 44,850 units ₹ 4,05,786 and Product P2: 29,900 units ₹ 6,08,678.

ICAI'S ANSWER

(C)

Process- I Account

Particulars	Units	(₹)	Particulars	Units	(₹)
To Raw material	75,000	4,76,250	By Normal loss (W.N. (ii))	**1,500**	**18,750**
To Direct labour		2,25,000	By Output (W.N. (iii))		
To Direct expenses		67,500	P1 (W.N. (iv))	**44,850**	**6,08,678**
To Production overheads (W.N.(i))		2,47,500	P2 (W.N. (iv))	**29,900**	**4,05,786**
To Abnormal gain (W.N. (iv))	**1,250**	**16,964**			
	76,250	10,33,214		76,250	10,33,214

Working Note Working Notes (W.N.):

(i) Production overheads = 110% x 2,25,000

= ₹ 2,47,500

(ii) Normal loss = 2% x 75,000 = 1,500 kg at ₹ 12.5

= ₹ 18,750

(iii) Total output = 75,000 input + 1,250 abnormal gain - 1,500 normal loss

= 74,750 kg

P1 and P2 is produced in the ratio 6:4

P1 = $\left(\frac{6}{10}\right)$ x 74,750 = 44,850 kg

P2 = $\left(\frac{4}{10}\right)$ x 74,750 = 29,900 kg

(iv) Value of Abnormal Gain:

$= (\frac{\text{Total Cost - Realisable value of normal loss}}{\text{Total input units - Normal Loss units}} \text{ x Abnormal Gain units})$

$= (\frac{₹\ 4{,}76{,}250 + ₹\ 2{,}25{,}000 + ₹\ 67{,}500 + ₹\ 2{,}47{,}500 - ₹\ 18{,}750}{75{,}000 - 1{,}500 \text{ units}} \text{ x } 1{,}250 \text{ units})$

= ₹ 16,964

Value of Joint Products:

$= (\frac{\text{Total Cost - Realisable value of normal loss}}{\text{Total input units - Normal Loss units}} \text{ x Output units})$

$\text{P1}= (\frac{₹\ 4{,}76{,}250 + ₹\ 2{,}25{,}000 + ₹\ 67{,}500 + ₹\ 2{,}47{,}500 - ₹\ 18{,}750}{75{,}000 - 1{,}500 \text{ units}} \text{ x } 44{,}850 \text{ units})$

= ₹ 6,08,679

$\text{P2}= (\frac{₹\ 4{,}76{,}250 + ₹\ 2{,}25{,}000 + ₹\ 67{,}500 + ₹\ 2{,}47{,}500 - ₹\ 18{,}750}{75{,}000 - 1{,}500 \text{ units}} \text{ x } 29{,}900 \text{ units})$

= ₹ 4,05,786

ICAI · P4 · RTP May 2025 · question p. 9-10 · answer p. 30-32

745 RTP **May 2025** Q15

JB Ltd. manufactures three chemical products, XR, YS, and ZT. It processes input material in common plant facility to generate two intermediate products, X and Y, in 4:1 proportion after a normal loss of 1/12th of the input material. There is no market for these two intermediate products. Thus, X is processed further through separate finishing process R to yield the product XR and Y is converted into product YS by a process S. The S finishing process also produces a waste material, Z, which has no market value. Thus, the company converts Z, after additional processing through process T, into a saleable by-product, ZT.

11,40,000 kg of the common input material are processed each month in common plant facility. And after the separate finishing processes (all losses are normal losses), proportions of XR, YS and ZT emerge as follows:

Product	Quantity (kg)	Market price per kg (₹)
XR	7,60,000	38.80
YS	1,90,000	72.00
ZT	19,000	24.00

The material and processing costs are as follows:

Particulars	Common plant facility (₹)	Separate finishing processes		
		R (₹)	S (₹)	T (₹)
Direct material	97,28,000	33,44,000	4,56,000	30,400
Direct labour	45,60,000	68,40,000	27,36,000	1,67,200
Factory overhead	24,32,000	22,80,000	9,12,000	1,06,400
Total	1,67,20,000	1,24,64,000	41,04,000	3,04,000

You are required to CALCULATE the cost per unit and total operating profit/loss attributed to both the products XR and YS considering all joint costs are allocated based on net realisable value method.

ICAI'S ANSWER

Statement showing operating profit/loss by each product after further processing

	Product XR (7,60,000 kg)		Product YS (1,90,000 kg)	
	Total (₹)	Cost per unit (₹)	Total (₹)	Cost per unit (₹)
Joint costs (W.N.)	1,06,40,000	14.0	60,80,000	32.0
Further processing costs	1,24,64,000	16.40	41,04,000	21.60
By-product net revenues			(1,52,000)	(0.80)
Total cost	2,31,04,000	**30.40**	1,00,32,000	**52.80**
Sales	2,94,88,000	38.80	1,36,80,000	72.00
Operating profit	**63,84,000**	8.40	**36,48,000**	19.20

Working Note:

Calculation of joint costs using Net realisable value method:

Particulars	Product XR (₹)	Product YS (₹)
Sales Value	2,94,88,000 (₹ 38.80 × 7,60,000 kg)	1,36,80,000 (₹ 72.00 × 1,90,000 kg)
Add: By-product net revenue	-	1,52,000 [(₹ 24.00 × 19,000 kg) - ₹ 3,04,000]
Less: Post split-off cost (Further processing cost)	(1,24,64,000)	(41,04,000)
Net Realisable Value	1,70,24,000	97,28,000
Apportionment of Joint Cost of ₹ 1,67,20,000 in ratio of 1,70,24:97,28	**1,06,40,000**	**60,80,000**

ICAI · P4 · RTP May 2025 · question p. 19-20 · answer p. 46

746 RTP **May 2025** Q20(c)

Also asked in RTP May 2024 Q13(d) · Old P3 RTP Jul 2021 Q13(b) · MTP May 2024 S-I Q6(d)

How apportionment of joint costs upto the point of separation amongst the joint products using market value at the point of separation and net realizable value method is done? DISCUSS.

ICAI'S ANSWER

Apportionment of Joint Cost amongst Joint Products using:

Market value at the point of separation

This method is used for apportionment of joint costs to joint products upto the split off point. It is difficult to apply if the market value of the product at the point of separation is not available. It is useful method where further processing costs are incurred disproportionately.

Net realizable value Method

From the sales value of joint products (at finished stage) the followings are deducted:

- Estimated profit margins
- Selling & distribution expenses, if any
- Post split off costs.

The resultant figure so obtained is known as net realizable value of joint products. Joint costs are apportioned in the ratio of net realizable value.

ICAI · P4 · RTP May 2025 · question p. 25 · answer p. 57

747 RTP **Jan 2025** Q6 MCQ

Same question, different ICAI answer: #752 MTP May 2026 S-I

ICT Ltd. belongs to pharmaceutical industries. The chemical process that ICT Ltd. operates convert one compound into three category of medicines viz. BetaTab, Folick and TegriCap. Though BetaTab and

Folick are already converted to final product at split-off point, Tegricap needs further processing along with addition of new compound with it.

The market for BetaTab and Folick is highly active, thus the production is sold at split-off point, however, Tegricap can be sold only after further processing.

Following information is provided for the current year:

Products	Quantity sold (tons)	Selling price per ton (₹)
BetaTab	372	7,500
Folick	1,054	5,625
TegriCap	1,472	3,750

The selling price is expected to remain the same for coming years.

The total joint manufacturing costs till split-off point is ₹ 62,50,000 and the amount spent for further processing w.r.t. Tegricap is ₹ 31,00,000

The details regarding closing inventories are as follows:

Products	Completed units (tons)
BetaTab	360
Folick	120
TegriCap	50

You are required to COMPUTE the joint cost allocated to BetaTab, Folick and TegriCap using Net realizable value (NRV) method.

A BetaTab- ₹ 15,65,481, Folick - ₹ 33,26,647 and TegriCap - ₹ 13,57,872

B BetaTab - ₹ 23,33,985, Folick - ₹ 28,07,478 and TegriCap - ₹ 11,08,537

C BetaTab - ₹ 19,27,533, Folick - ₹ 23,18,570 and TegriCap - ₹ 20,03,897

D BetaTab - ₹ 11,08,537, Folick - ₹ 28,07,478 and TegriCap - ₹ 23,33,985

ICAI'S ANSWER

(B) Calculation of total production of BetaTab, Folick and TegriCap

Products	Quantity sold (tons)	Quantity of closing inventories (tons)	Total production
(1)	(2)	(3)	(4) = [(2) + (3)]
BetaTab	372	360	732
Folick	1,054	120	1,174
TegriCap	1,472	50	1,522

Calculation of Net Realisable Value (at split-off point)

	Products			Total (₹)
	BetaTab	**Folick**	**TegriCap**	
Total Production (tons) (A)	732	1,174	1,522	
Selling price per ton (₹) (B)	7,500	5,625	3,750	
Final sales value of total production (₹) [(A) x (B)]	54,90,000	66,03,750	57,07,500	1,78,01,250
Less: Additional cost (₹)	-	-	(31,00,000)	(31,00,000)
Net realisable value (₹) (at split-off point)	54,90,000	66,03,750	26,07,500	1,47,01,250

Joint cost allocated using Net Realisable Value (at split-off point):

$$\frac{\text{Total Jointcost}}{\text{Total Net Realisable Value}} \times \text{NetRealisable Valueof each product}$$

$$\text{BetaTab} = \left(\frac{₹62,50,000}{₹1,47,01,250}\right) \text{ x } ₹\ 54,90,000$$

= ₹ 23,33,985

$$\text{Folick} = \left(\frac{₹62,50,000}{₹1,47,01,250}\right) \text{ x } ₹\ 66,03,750$$

= ₹ 28,07,478

$$\text{TegriCap} = \left(\frac{₹62,50,000}{₹1,47,01,250}\right) \text{ x } ₹\ 26,07,500$$

= ₹ 11,08,537

ICAI · P4 · RTP Jan 2025 · question p. 8-9 · answer p. 27-28

748 RTP **Sep 2024** Q15

Three products X, Y and Z alongwith a byproduct B are obtained again in a crude state which require further processing at a cost of ₹ 5 for X; ₹ 4 for Y; and ₹ 2.50 for Z per unit before sale. The byproduct is however saleable as such to a nearby factory. The selling prices for the three main products and byproduct, assuming they should yield a net margin of 25 percent of cost, are fixed at ₹ 13.75 ₹ 8.75 and ₹ 7.50 and ₹ 1.00 respectively – all per unit quantity sold.

During a period, the joint input cost including the material cost was ₹ 90,800 and the respective outputs were:

X	8,000 units
Y	6,000 units
Z	4,000 units
B	1,000 units

By product should be credited to the joint cost and only the net joint costs are to be allocated to the main products.

CALCULATE the joint cost per unit of each product and the margin available as a percentage on cost.

ICAI'S ANSWER

Working Notes:

(i) Computation of Allocation Ratio for Joint Costs

	Products		
	X ₹	Y ₹	Z. ₹
Selling Price	13.75	8.75	7.50
Less: anticipated margin@ 25% on cost of 20% on sales	2.75	1.75	1.50
Cost of sales	11.00	7.00	6.00
Less: post split off cost	5.00	4.00	2.50
Joint cost per unit	6.00	3.00	3.50
Output (units)	8,000	6,000	4,000
Total output cost	48,000	18,000	14,000
Allocation ratio for joint costs	24	9	7

(ii) Computation of net allocable joint costs

	₹	₹
Joint input cost including material cost		90,800
Less: Credit for realization from by-product B:		
Sales revenue (1,000 × Re. 1)	1,000	
Less: profit @ 25% on cost or 20% on sales	200	800
Net joint costs to be allocated		90,000

Determination of joint cost per unit of each product

Product	Net joint costs allocation ₹	Output (units) ₹	Joint cost per unit ₹
X	54,000 (Note: 1)	8,000	6.75
Y	20,250	6,000	3.38
Z	15,750	4,000	3.94
	90,000		

Profit margin available on each product as a percentage on cost

Product	Joint Cost ₹	Post spilt-off cost ₹	Total Cost ₹	Selling Price ₹	Margin ₹	Margin % on cost ₹
X	6.75	5.00	11.75	13.75	2.00	17.02
Y	3.38	4.00	7.38	8.75	1.37	18.56
Z	3.94	2.50	6.44	7.50	1.06	16.46

Note: 1

$$X = \frac{24}{40} \times 90{,}000 = 54{,}000$$

$$Y = \frac{9}{40} \times 90{,}000 = 20{,}250$$

$$Z = \frac{7}{40} \times 90{,}000 = 15{,}750$$

90,000

ICAI · P4 · RTP Sep 2024 · question p. 13 · answer p. 30-32

749 RTP **Old P3 Nov 2022 Q10** old scheme

JP Ltd. uses joint production process that produces three products at the split-off point. Joint production costs during the month of July, 2022 were ₹ 33,60,000.

Product information for the month of July is as follows:

Particulars	Product A	Product B	Product C
Units produced	3,000	6,000	9,000
Sales prices:			
At the split-off	₹ 200		
After further processing	₹ 300	₹ 350	₹ 100
Costs to process after split-off	₹ 6,00,000	₹ 6,00,000	₹ 6,00,000

Other information is as follows:

Product C is a by-product and the company accounts for the by-product at net realizable value as a reduction of joint cost. Further, Product B & C must be processed further before they can be sold. FIND OUT the joint cost allocated to Product A in the month of July if joint cost allocation is based on Net Realizable Value.

ICAI'S ANSWER

Product A

As the question says that "Products B and C must be processed further before they can be sold", it means Product A can be sold at the split-off point.

Cost to process Product A after the split-off point = ₹ 6,00,000

Additional revenue to be earned by processing further = ₹ 3,00,000 (₹ 100 increase in selling price per unit x 3,000 units)

Therefore, Product A will not be processed further, and the sales value at split-off for A will be used for allocating the joint costs.

Sales value at the split-off for A = ₹ 6,00,000 (₹ 200 × 3,000 units)

Product B

Since Product B must be processed further, we use its net realizable value for the joint cost allocation.

Net realizable value of Product B = ₹ 15,00,000 [(₹ 350 × 6,000 units) – ₹ 6,00,000 further processing costs]

Product C

Product C, the by-product, must also be processed further to be sold.

Net realizable value of Product C = ₹ 3,00,000 [(₹ 100 × 9,000 units) – ₹ 6,00,000 in further processing costs]

Joint Cost Allocation

Joint production cost = ₹ 33,60,000

Since, by-product C is accounted for as a reduction to the joint costs, the joint costs to be allocated = ₹ 30,60,000 (₹ 33,60,000 - ₹ 3,00,000 NRV of Product C)

Allocation of joint costs between Product A and B will be on the basis of ₹ 6,00,000: ₹ 15,00,000

Joint Cost allocated to Product A = ₹ 30,60,000 x $\frac{₹\ 6,00,000}{₹\ 21,00,000}$ = ₹ 8,74,286

ICAI · Old P3 · RTP Nov 2022 · question p. 8-9 · answer p. 28-29

750 RTP Old P3 Dec 2021 Q10 old scheme

A company produces two joint products A and B from the same basic materials. The processing is completed in three departments.

Materials are mixed in Department I. At the end of this process, A and B get separated. After separation, A is completed in the Department II and B in Department III. During a period, 4,00,000 kg of raw material was processed in Department I at a total cost of ₹ 17,50,000, and the resultant 50% becomes A and 40% becomes B and 10% normally lost in processing.

In Department II, 1/5th of the quantity received from Department I is lost in processing. A is further processed in Department II at a cost of ₹ 2,60,000.

In Department III, further new material is added to the material received from Department I and weight mixture is doubled, there is no quantity loss in the department III. Further processing cost (with material cost) in Department III is ₹ 3,00,000.

The details of sales during the said period are:

	Product A	Product B
Quantity sold (kg)	1,50,000	3,00,000
Sales price per kg (₹)	10	4

There were no opening stocks. If these products sold at split-off-point, the selling price of A and B would be ₹ 8 and ₹ 4 per kg respectively.

Required:

(i) PREPARE a statement showing the apportionment of joint cost to A and B in proportion of sales value at split off point.

(ii) PREPARE a statement showing the cost per kg of each product indicating joint cost, processing cost and total cost separately.

(iii) PREPARE a statement showing the product wise profit for the year.

(iv) On the basis of profits before and after further processing of product A and B, give your COMMENT that products should be further processed or not.

ICAI'S ANSWER

Calculation of quantity produced

	Dept I (kg)	Dept II (kg)	Dept III (kg)
Input	4,00,000	2,00,000 (50% of 4,00,000 kg.)	1,60,000 (40% of 4,00,000 kg.)
Weight (lost) or added	(40,000) (10% of 4,00,000 kg.)	(40,000) (1/5th of 2,00,000 kg.)	1,60,000
	3,60,000	1,60,000	3,20,000
Production of A	2,00,000	1,60,000	--
Production of B	1,60,000	--	3,20,000

(i) **Statement of apportionment of joint cost of dept I**

	Product A	Product B
Output (kg)	2,00,000	1,60,000
Selling price per kg (₹)	8	4
Sales value (₹)	16,00,000	6,40,000
Share in Joint cost (5:2)	12,50,000 (₹ 17,50,000 × 5 ÷ 7)	5,00,000 (₹ 17,50,000 × 2 ÷ 7)

(ii) **Statement of cost per kg**

	Product A	Product B
Output (kg)	1,60,000	3,20,000
Share in joint cost (₹)	12,50,000	5,00,000
Joint Cost per kg (₹) (A)	7.8125	1.5625
Further processing cost (₹)	2,60,000	3,00,000
Further processing cost per kg (₹) (B)	1.625	0.9375
Total cost per kg (₹) {(A)+(B)}	9.4375	2.5000

(iii) **Statement of profit**

	Product A	Product B
Output (kg)	1,60,000	3,20,000
Sales (kg)	(1,50,000)	(3,00,000)
Closing stock (kg)	10,000	20,000
	(₹)	**(₹)**
Sales	15,00,000 (1,50,000 kg × ₹ 10)	12,00,000 (3,00,000 kg × ₹ 4)
Add: closing stock (at full cost)	94,375 (10,000 kg × ₹ 9.4375)	50,000 (20,000 kg × ₹ 2.5)
Value of production	15,94,375	12,50,000
Less: Share in joint cost	12,50,000	5,00,000
Further processing cost	2,60,000	3,00,000
Profit	84,375	4,50,000

(iv) **Profitability statement before and after processing**

	Product A		Product B	
	Before (₹)	**After (₹)**	**Before (₹)**	**After (₹)**
Sales Value	16,00,000		6,40,000	
Share in joint costs	12,50,000		5,00,000	
Profit	3,50,000	84,375 (as per iii above)	1,40,000	4,50,000 (as per iii above)

Product A should be sold at split off point and product B after processing because of higher profitability.

ICAI · Old P3 · RTP Dec 2021 · question p. 9 · answer p. 27-28

751 RTP **Old P3 Nov 2020** Q10 old scheme

ABC Ltd. operates a simple chemical process to convert a single material into three separate items, referred to here as X, Y and Z. All three end products are separated simultaneously at a single split-off point.

Product X and Y are ready for sale immediately upon split off without further processing or any other additional costs. Product Z, however, is processed further before being sold. There is no available market price for Z at the split-off point.

The selling prices quoted here are expected to remain the same in the coming year. During 2019-20, the selling prices of the items and the total amounts sold were:

X – 186 tons sold for ₹3,000 per ton

Y – 527 tons sold for ₹2,250 per ton

Z – 736 tons sold for ₹1,500 per ton

The total joint manufacturing costs for the year were ₹12,50,000. An additional ₹ 6,20,000 was spent to finish product Z.

There were no opening inventories of X, Y or Z at the end of the year. The following inventories of complete units were on hand:

X	180 tons
Y	60 Tons
Z	25 tons

There was no opening or closing work-in-progress.

Required:

COMPUTE the cost of inventories of X, Y and Z and cost of goods sold for year ended March 31, 2020, using Net realizable value (NRV) method of joint cost allocation.

ICAI'S ANSWER

(i) (a) Statement of Joint Cost allocation of inventories of X, Y and Z

(By using Net Realisable Value Method)

	Products			Total
	X	**Y**	**Z**	
	(₹)	**(₹)**	**(₹)**	**(₹)**
Final sales value of total production (Working Note 1)	10,98,000 (366 × ₹3,000)	13,20,750 (587 × ₹2,250)	11,41,500 (761 × ₹1,500)	35,60,250
Less: Additional cost	--	--	(6,20,000)	(6,20,000)
Net realisable value (at split-off point)	10,98,000	13,20,750	5,21,500	29,40,250
Joint cost allocated (Working Note 2)	4,66,797	5,61,496	2,21,707	12,50,000

Cost of goods sold as on March 31, 2020

(By using Net Realisable Value Method)

	Products			Total
	X	Y	Z	
	(₹)	(₹)	(₹)	(₹)
Allocated joint cost	4,66,797	5,61,496	2,21,707	12,50,000
Additional costs	--	--	6,20,000	6,20,000
Cost of goods available for sale (CGAS)	4,66,797	5,61,496	8,41,707	18,70,000
Less: Cost of ending inventory (Working Note 1)	2,29,571 (CGAS×49.18%)	57,385 (CGAS × 10.22%)	27,692 (CGAS × 3.29%)	3,14,648
Cost of goods sold	2,37,226	5,04,111	8,14,015	15,55,352

Working Notes

1. **Total production of three products for the year 2019-2020**

Products	Quantity sold in tones	Quantity of ending inventory in tons	Total production	Ending inventory percentage (%)
(1)	(2)	(3)	(4) = [(2) + (3)}	(5) = (3)/ (4)
X	186	180	366	49.18
Y	527	60	587	10.22
Z	736	25	761	3.29

2. **Joint cost apportioned to each product:**

$$\frac{\text{Total Joint cost}}{\text{Total Net Realisable Value}} \times \text{Net Realisable Value of each product}$$

$$\text{Total cost of Product X} = \frac{₹\ 12,50,000}{₹\ 29,40,250} \times ₹10,98,000 = ₹\ 4,66,797$$

$$\text{Total cost of Product Y} = \frac{₹\ 12,50,000}{₹\ 29,40,250} \times ₹13,20,750 = ₹\ 5,61,496$$

$$\text{Total cost of Product Z} = \frac{₹\ 12,50,000}{₹\ 29,40,250} \times ₹\ 5,21,500 = ₹\ 2,21,707$$

ICAI · Old P3 · RTP Nov 2020 · question p. 8-9 · answer p. 23-24

752 MTP May 2026 S-I MCQ 14 2 marks MCQ

Same question, different ICAI answer: #747 RTP Jan 2025

ICT Ltd. belongs to pharmaceutical industries. The chemical process that ICT Ltd. operates convert one compound into three category of medicines viz. BetaTab, Folick and TegriCap. Though BetaTab and Folick are already converted to final product at split-off point, Tegricap needs further processing along with addition of new compound with it. The market for BetaTab and Folick is highly active, thus the production is sold at split-off point, however, Tegricap can be sold only after further processing.

Following information is provided for the current year:

Products	Quantity sold (tons)	Selling price per ton (₹)
BetaTab	372	7,500
Folick	1,054	5,625
TegriCap	1,472	3,750

The selling price is expected to remain the same for coming years.

The total joint manufacturing costs till split-off point is ₹ 62,50,000 and the amount spent for further processing w.r.t. Tegricap is ₹ 31,00,000

The details regarding closing inventories are as follows:

Products	Completed units (tons)
BetaTab	360
Folick	120
TegriCap	50

You are required to COMPUTE the joint cost allocated to BetaTab, Folick and TegriCap using Net realizable value (NRV) method.

A BetaTab- ₹ 15,65,481, Folick - ₹ 33,26,647 and TegriCap - ₹ 13,57,872

B BetaTab - ₹ 23,33,985, Folick - ₹ 28,07,478 and TegriCap - ₹ 11,08,537

C BetaTab - ₹ 19,27,533, Folick - ₹ 23,18,570 and TegriCap - ₹ 20,03,897

D BetaTab - ₹ 11,08,537, Folick - ₹ 28,07,478 and TegriCap - ₹ 23,33,985

ICAI'S ANSWER

(B) BetaTab - ₹ 23,33,985, Folick - ₹ 28,07,478 and TegriCap - ₹ 11,08,537

Calculation of total production of BetaTab, Folick and TegriCap

Products	Quantity sold (tons)	Quantity of closing inventories (tons)	Total production
(1)	(2)	(3)	(4) = [(2) + (3)]
BetaTab	372	360	732
Folick	1,054	120	1,174
TegriCap	1,472	50	1,522

Calculation of Net Realisable Value (at split-off point)

	Products			Total (₹)
	BetaTab	**Folick**	**TegriCap**	
Total Production (tons) (A)	732	1,174	1,522	
Selling price per ton (₹) (B)	7,500	5,625	3,750	
Final sales value of total production (₹) [(A) x (B)]	54,90,000	66,03,750	57,07,500	1,78,01,250
Less: Additional cost (₹)	-	-	(31,00,000)	(31,00,000)
Net realisable value (₹) (at split-off point)	54,90,000	66,03,750	26,07,500	1,47,01,250

Joint cost allocated using Net Realisable Value (at split-off point):

$$\frac{\text{Total Joint cost}}{\text{Total Net Realisable Value}} \times \text{Net Realisable Value of each product}$$

$$\text{BetaTab} = \left(\frac{₹\ 62,50,000}{₹\ 1,47,01,250}\right) \text{x } ₹\ 54,90,000$$

= ₹ 23,33,985

$$\text{Folick} = \left(\frac{₹\ 62,50,000}{₹\ 1,47,01,250}\right) \text{x } ₹\ 66,03,750$$

= ₹ 28,07,478

$$\text{TegriCap} = \left(\frac{₹\ 62,50,000}{₹\ 1,47,01,250}\right) \text{x } ₹\ 26,07,500$$

= ₹ 11,08,537

ICAI · P4 · MTP May 2026 · Series I · question p. 6-7 · answer p. 5-6

753 MTP **Jan 2026 S-I** Q1(c) 4 marks

Creamy Delights Ltd. is engaged in the production of Flavored Milk, Cheese, and Clarified Cheese Oil. It purchases processed milk cream and subjects it to a churning process until it separates into flavored milk and cheese.

For the month of January 2025, the company purchased 50 kilolitres of processed cream @ ₹100 per 1,000 ml.

Conversion costs of ₹1,00,000 were incurred up to the split-off point, where two saleable products were obtained — flavored milk and cheese.

Cheese can be further processed into clarified cheese oil.

Production and Sales Information for January 2025

Product	Production (KL / tonne)	Sales Quantity (KL / tonne)	Selling Price per Litre / Kg (₹)
Flavored Milk	28	28	30
Cheese	20	—	—
Clarified Cheese Oil	16	16	480

All 20 tonnes of cheese were further processed at an incremental cost of ₹ 1,20,000 to yield 16 kilolitres of clarified cheese oil.

There was no opening or closing inventory of flavored milk, cheese, or clarified cheese oil during January 2025.

Required:

(i) SHOW how the joint cost would be apportioned between Flavored Milk and Cheese under the Estimated Net Realisable Value (NRV) method.

(ii) In February, "Dairy Fresh Foods" offers to purchase 20 tonnes of cheese at ₹ 360 per kg.

If Creamy Delights Ltd. accepts this offer, no clarified cheese oil would be produced in February.

SUGGEST whether Creamy Delights Ltd. should accept the offer or continue further processing the cheese into clarified cheese oil, considering its effect on operating income.

ICAI'S ANSWER

(i) Estimated Net Realisable Value Method:

	Flavoured Milk Amount (₹)	Cheese Amount (₹)
Sales Value	8,40,000 (₹ 30 × 28 x 1000)	76,80,000 (₹ 480 × 16 x 1000)
Less: Post split-off cost (Further processing cost)	-	(1,20,000)
Net Realisable Value	8,40,000	75,60,000
Apportionment of Joint Cost of ₹ 51,00,000* in ratio of 1:9	5,10,000	45,90,000

* [(₹ 100 × 50 × 1000) + ₹ 1,00,000] = ₹ 51,00,000

(ii) Incremental revenue from further processing of Cheese into Clarified Cheese Oil

(₹ 480 × 16 × 1000 - ₹ 360 × 20 × 1000)	₹ 4,80,000
Less: Incremental cost of further processing of Cheese into Clarified Cheese Oil	₹ 1,20,000
Incremental operating income from further processing	₹ 3,60,000

The operating income of Creamy Delights Ltd. will decrease by ₹ 3,60,000 in February if it sells 20 tonnes of Cheese to Dairy Fresh Foods, instead of further processing the cheese into Clarified Cheese Oil for sale.

Therefore, Creamy Delights Ltd. is advised not to accept the offer and should continue further processing the cheese into Clarified Cheese Oil itself.

ICAI · P4 · MTP Jan 2026 · Series I · question p. 7-8 · answer p. 7

754 MTP **Jan 2026 S-II** Q6(c) 4 marks

MENTION the circumstances under which following types of method of apportioning joint costs can be used.

1. Net Realisable Value (NRV) at Split-off Point Method

2. Technical Estimates
3. Market value at the point of separation
4. Market value after further processing

ICAI'S ANSWER

Method	Circumstances
Net Realisable Value (NRV) at Split-off Point Method	When the realisable value of joint products at split-off is not known.
Technical Estimates	When the result obtained by Net Realisable Value (NRV) at Split-off Point Method does not match with the resources consumed by joint products.
Market value at the point of separation	Where further processing costs are incurred disproportionately.
Market value after further processing	Where further processing costs after the point of separation are proportionate and all the joint products are subject to further processing.

ICAI · P4 · MTP Jan 2026 · Series II · question p. 18 · answer p. 22

755 MTP **Sep 2025 S-II Q4(a)** 7 marks

Pintoo Pvt. Ltd. runs a straightforward chemical process that transforms a single material into three distinct products, named Ae, Bee, and Cee. These three end products are separated simultaneously at a single split-off point.

Products Ae and Bee are ready for sale immediately after the split-off, with no additional processing or costs. However, Product Cee undergoes further processing before it can be sold. There is no market price available for Cee at the split-off point.

For the current year, the selling price of the products and the units sold were as follows:

Product	Units sold (tons)	Selling price (₹)
Ae	558	4,500
Bee	1,581	3,375
Cee	2,208	2,250

Cost related information for the same year is as follows:

Particulars	Amount (₹)
Total joint manufacturing costs	56,25,000
Additional cost spent to finish product Cee	27,90,000

There were no opening or closing work-in-progress, however, the following inventories of completed units were on hand at the end of the year:

Product	Units (tons)
Ae	540
Bee	180
Cee	75

You are required to:

(i) COMPUTE the cost of inventories of Ae, Bee and Cee for Balance Sheet purpose and cost of goods sold for income statement purpose using Net realizable value (NRV) method of joint cost allocation.

(ii) PREPARE Income Statement showing gross margin percentage of Ae, Bee and Cee.

ICAI'S ANSWER

(i) **Statement of Joint Cost allocation of Ae, Bee and Cee using net realisable value method**

Particulars	Product			
	Ae (₹)	**Bee (₹)**	**Cee (₹)**	**Total (₹)**
Sales Value of total production (Working Note 1)	49,41,000 (1,098 tons x ₹ 4,500)	59,43,375 (1,761 tons x ₹ 3,375)	51,36,750 (2,283 tons x ₹ 2,250)	1,60,21,125
Less: Additional cost	---------	---------	27,90,000	27,90,000
Net realisable value (at split-off point)	49,41,000	59,43,375	23,46,750	1,32,31,125
Joint cost allocated (Working Note 2)	21,00,587	25,26,730	9,97,683	56,25,000

Calculation of Cost of ending inventory and Cost of goods sold:

	Product			
	Ae (₹)	**Bee (₹)**	**Cee (₹)**	**Total (₹)**
Allocated joint cost	21,00,587	25,26,730	9,97,683	56,25,000
Additional costs	-------	--------	27,90,000	27,90,000
Cost of goods available for sale (CGAS)	21,00,587	25,26,730	37,87,683	84,15,000
Less: **Cost of ending inventory** (Working Note 3)	**(10,33,069)**	**(2,58,232)**	**(1,24,615)**	**(14,15,915)**
Cost of goods sold	**10,67,518**	**22,68,498**	**36,63,068**	**69,99,085**

(ii) **Income statement showing gross margin percentage**

	Product			
	Ae (₹)	**Bee (₹)**	**Cee (₹)**	**Total (₹)**
Sales Value of goods sold (A)	25,11,000 (558 tons x ₹ 4,500)	53,35,875 (1,581 tons x ₹ 3,375)	49,68,000 (2,208 tons x ₹ 2,250)	1,28,14,875
Less: Cost of goods sold	10,67,518	22,68,498	36,63,068	69,99,085
Gross margin (₹) (B)	14,43,482	30,67,377	13,04,932	58,15,790
Gross margin (%) (B/A) x 100	**57.49%**	**57.49%**	**26.27%**	

Working notes:

1. **Total production of three products for the year**

Product	Quantity sold (tons)	Ending inventory (tons)	Total production	Ending inventory %
(1)	(2)	(3)	(4) = (2) + (3)	(5) = (3)/(4)
Ae	558	540	1,098	49.18
Bee	1,581	180	1,761	10.22
Cee	2,208	75	2,283	3.29

2. **Joint cost apportioned to each product:**

$$\left(\frac{\text{Total joint cost}}{\text{Total net realisable value}} \text{ x Net realisable value of each product}\right)$$

$$\text{Ae} = \left(\frac{₹\ 56,25,000}{₹\ 1,32,31,125} \text{ x } ₹\ 49,41,000\right)$$

= ₹ 21,00,587

$$\text{Bee} = \left(\frac{₹\ 56,25,000}{₹\ 1,32,31,125} \text{ x } ₹\ 59,43,375\right)$$

= ₹ 25,26,730

$$\text{Cee} = \left(\frac{₹\ 56,25,000}{₹\ 1,32,31,125} \text{ x } ₹\ 23,46,750\right)$$

= ₹ 9,97,683

3. **Cost of ending inventory**

Product	CGAS (₹)	Ending inventory %	Cost of ending inventory (₹)
(1)	(2)	(3)	(4) = (2) x (3)
Ae	21,00,587	49.18	10,33,069
Bee	25,26,730	10.22	2,58,232
Cee	37,87,683	3.29	1,24,615

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756 MTP **May 2025 S-I** Q2(c) 2 marks

DESCRIBE briefly, how joint costs upto the point of separation may be apportioned amongst the joint products under the following methods:

(i) Market value after further processing

(ii) Net realizable value method

ICAI'S ANSWER

(i) Market Value after further Processing: Here the basis of apportionment of joint costs is the total sales value of finished products at the further processing. The use of this method is unfair where further processing costs after the point of separation are disproportionate or when all the joint products are not subjected to further processing.

(ii) Net Realisable Value Method: Here joint costs is apportioned on the basis of net realisable value of the joint products,

Net Realisable Value	=	Sale value of joint products (at finished stage)
	(-)	estimated profit margin
	(-)	selling & distribution expenses, if any
	(-)	post split off cost

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757 MTP **Jan 2025 S-I** MCQ 15 2 marks MCQ

RN Ltd. manufactures two primary products, P1 and P2, through a joint process and a by-product, R12, is produced spontaneously. The relationship between output quantities to the direct material input stays stable.

To allocate joint production costs to the primary products, the company utilizes the physical volume method.

During the month of March, company incurred joint production costs of ₹ 1,30,00,000. As the primary products are not freely marketable at the split-off point, they are processed further.

The net realizable value of the by-product is treated as deductions from the joint production costs before the joint costs are allocated to the primary products.

The information regarding company's production and its cost during the month of March is provided below:

Particulars	P1	P2	R 12
Output (kg.)	1,95,000	3,90,000	81,250
Selling price per kg.	₹ 200	₹ 120	₹ 40
Further processing costs	₹ 26,00,000	₹ 39,00,000	-

FIND OUT the amount of joint product cost to be allocated to P2 by using the physical volume method.

A ₹ 65,00,000

B ₹ 97,50,000

C ₹ 39,00,000

D ₹ 32,50,000

ICAI'S ANSWER

(A) ₹ 65,00,000

Calculation of Net joint costs to be allocated:

Particulars	Amount (₹)
Joint Costs	1,30,00,000
Less: Net Realizable value of by-product R 12 (81,250 kg. × ₹ 40)	32,50,000
Net joint costs to be allocated	97,50,000

Therefore, the amount of joint product cost to be allocated to P2 by using the physical volume method

$$= \left(\frac{\text{Physical quantity of P2}}{\text{Total quantity}}\right) \text{x Net joint costs to be allocated}$$

$$= \left(\frac{3{,}90{,}000\ \text{kg}}{5{,}85{,}000\ \text{kg}}\right) \text{x ₹ } 97{,}50{,}000 = \text{₹ } 65{,}00{,}000$$

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758 MTP **Jan 2025 S-I** Q1(a) 5 marks

Petro Ltd. is a petroleum refining company which uses cracking process for producing gasoline, diesel and Heavy fuel oil (HFO). All three final products are extracted simultaneously at one common split-off point.

Gasoline and diesel are immediately available for sale upon separation, requiring no further processing. In contrast, heavy fuel oil (HFO) undergoes additional processing before it can be sold, as there is no market for it at the split-off point.

Throughout the year, the selling prices and total quantities sold for each item were as follows:

Product	Quantity sold (Gallons)	Selling Price per gallon (₹)
Gasoline	1,674	400
Diesel	4,743	300
Heavy fuel oil (HFO)	6,624	200

The selling prices listed above are projected to remain unchanged in the upcoming year.

The total joint manufacturing costs for the year amounted to ₹ 15,00,000, with an additional cost of ₹ 7,44,000 incurred for finishing Heavy fuel oil (HFO).

There were no opening inventories of gasoline, diesel and Heavy fuel oil (HFO). Though, at the end of the period, the following inventories of complete units were available: 1,620 gallons of gasoline, 540 gallons of diesel, and 225 gallons of Heavy fuel oil (HFO).

You are required to COMPUTE the following for gasoline, diesel and Heavy fuel oil (HFO)-

(i) joint cost allocated, and

(ii) cost of goods sold

using Net Realisable Value Method of joint cost allocation.

ICAI'S ANSWER

(i) Statement of Joint Cost allocation of inventories of gasoline, diesel and Heavy fuel oil (HFO)

(By using Net Realisable Value Method)

	Products			Total
	Gasoline	**Diesel**	**Heavy fuel oil (HFO)**	
	(₹)	**(₹)**	**(₹)**	**(₹)**
Final sales value of total production (Working Note 1)	13,17,600 (3,294 × ₹ 400)	15,84,900 (5,283 × ₹ 300)	13,69,800 (6,849 × ₹ 200)	42,72,300
Less: Additional cost	-	-	(7,44,000)	(7,44,000)
Net realisable value (at split-off point)	13,17,600	15,84,900	6,25,800	35,28,300
Joint cost allocated (Working Note 2)	**5,60,156**	**6,73,795**	**2,66,049**	**15,00,000**

(ii) Cost of goods sold

(By using Net Realisable Value Method)

	Products			Total
	Gasoline	**Diesel**	**Heavy fuel oil (HFO)**	

	Products			Total
	(₹)	(₹)	(₹)	(₹)
Allocated joint cost (from (i))	5,60,156	6,73,795	2,66,049	15,00,000
Additional costs	--	--	7,44,000	7,44,000
Cost of goods available for sale (CGAS)	5,60,156	6,73,795	10,10,049	22,44,000
Less: Cost of ending inventory (Working Note 1)	2,75,485 (CGAS × 49.18%)	68,862 (CGAS × 10.22%)	33,231 (CGAS × 3.29%)	3,77,578
Cost of goods sold	**2,84,671**	**6,04,933**	**9,76,818**	**18,66,422**

Working Notes

1. Total production of three products for the year

Products	Quantity sold (in gallon)	Quantity of ending inventory (in gallon)	Total production	Ending inventory percentage (%)
(1)	(2)	(3)	(4) = [(2) + (3)}	(5) = (3)/ (4)
Gasoline	1,674	1,620	3,294	49.18
Diesel	4,743	540	5,283	10.22
Heavy fuel oil (HFO)	6,624	225	6,849	3.29

2. Joint cost apportioned to each product

$$\frac{\text{Total Joint cost}}{\text{Total Net Realisable Value}} \times \text{Net Realisable Value of each product}$$

Total cost of Gasoline	$\frac{₹\ 15,00,000}{₹\ 35,28,300}$ x ₹ 13,17,600	₹ 5,60,156
Total cost of Diesel	$\frac{₹\ 15,00,000}{₹\ 35,28,300}$ x ₹ 15,84,900	₹ 6,73,795
Total cost of Heavy fuel oil (HFO)	$\frac{₹\ 15,00,000}{₹\ 35,28,300}$ x ₹ 6,25,800	₹ 2,66,049

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759 MTP **Jan 2025 S-II** Case Scenario II · MCQ 6 to MCQ 10 10 marks

CASE SCENARIO II

eSalt is the biggest producer of sodium hydroxide in India. This main product of the company has a strong reactivity with other organic compounds. It is highly versatile and is alkaline in nature. However, the basic material required for the production of this product is salt along with the electricity.

The manufacturing process involve electrolysis which produces Halogen as co-product. Modern use of Halogen is widespread. However, the common use is in disinfection like for purifying drinking

water or swimming pool water. It is also an important ingredient of toothpaste. Thus, the company's management affirmed the simultaneous production of Halogen.

During the previous financial year, the company purchased the base material of ₹ 5,34,000. For the current year, company decided to increase the production by 2 times. Due to increased production, the total conversion cost hiked to 3 times. Last year, the conversion cost accounted to ₹ 8,01,000 up to the point at which two products i.e. sodium hydroxide and Halogen are separated.

The production and sales information for current year is provided as below:

	Sodium hydroxide	Halogen
Production/ Sales(in tonne)	24,030	16,020
Selling price per tonne (₹)	100	150

During the current year, the management of the company pointed the extensive use of Vinyl which can be produced by further processing Halogen. Having selling price of ₹ 250 per tonne higher than that of the Halogen, it was decided not to sell Halogen and further process it into Vinyl. The incremental processing cost took ₹ 8,01,000 producing 10,012.50 tonnes of Vinyl.

You are required to FIGURE OUT the following for managerial decision (MCQs 6 to 10):

MCQ 6 2 marks

For the current year, the amount of base material purchased and the conversion cost up to the point at which two products i.e. Sodium hydroxide and Halogen are separated would be:

A base material ₹ 10,68,000 and conversion cost ₹ 24,03,000

B base material ₹ 10,68,000 and conversion cost ₹ 16,02,000

C base material ₹ 16,02,000 and conversion cost ₹ 24,03,000

D base material ₹ 24,03,000 and conversion cost ₹ 16,02,000

MCQ 7 2 marks

Joint cost to be apportioned between Sodium hydroxide and Halogen as per the physical unit method would be:

A Sodium hydroxide ₹ 24,03,000 and Halogen ₹ 10,68,000

B Sodium hydroxide ₹ 10,68,000 and Halogen ₹ 16,02,000

C Sodium hydroxide ₹ 16,02,000 and Halogen ₹ 24,03,000

D Sodium hydroxide ₹ 24,03,000 and Halogen ₹ 16,02,000

MCQ 8 2 marks

Joint cost to be apportioned between Sodium hydroxide and Halogen as per the sales value at split- off point method would be:

A Sodium hydroxide ₹ 20,02,500 and Halogen ₹ 20,02,500

B Sodium hydroxide ₹ 16,02,000 and Halogen ₹ 24,03,000

C Sodium hydroxide ₹ 24,03,000 and Halogen ₹ 16,02,000

D Sodium hydroxide ₹ 10,68,000 and Halogen ₹ 20,02,500

MCQ 9 2 marks

Joint cost to be apportioned between Sodium hydroxide and Halogen as per the estimated net realisable value method would be:

A Sodium hydroxide ₹ 23,44,390 and Halogen ₹ 16,60,610

B Sodium hydroxide ₹ 17,16,429 and Halogen ₹ 22,88,571

C Sodium hydroxide ₹ 22,88,571 and Halogen ₹ 17,16,429

D Sodium hydroxide ₹ 16,60,610 and Halogen ₹ 23,44,390

MCQ 10 2 marks

Considering that the decision relating to further processing Halogen is not approved, suggest whether this would be in favour of the management by calculating incremental revenue /loss from further processing Halogen into Vinyl.

A Incremental loss would be ₹ 16,02,000, thus the decision of not further processing Halogen is correct.

B Incremental loss would be ₹ 8,01,000, thus the decision of not further processing Halogen is correct.

C Incremental revenue would be ₹ 8,01,000, thus the decision relating to further processing Halogen needs to be approved.

D Incremental revenue would be ₹ 16,02,000, thus the decision relating to further processing Halogen needs to be approved.

ICAI'S ANSWERS

MCQ 6

(C)

Particulars	Base Material	Conversion cost
Previous year cost (₹)	5,34,000	8,01,000
Increased by	2 times	-
Increased to		3 times
Current year cost (₹)	5,34,000 + (5,34,000 × 2) = 16,02,000	8,01,000 × 3 = 24,03,000

MCQ 7

(D)

Products	Production/ Sales(in tonne)	Joint Cost Apportioned (₹)
Sodium hydroxide	24,030	24,03,000
Halogen	16,020	16,02,000
Total	**40,050**	**40,05,000**

Joint cost = base material + conversion cost

= 16,02,000 + 24,03,000

= 40,05,000

$$\text{Apportioned joint cost} = \frac{\text{Total joint cost}}{\text{Total physical value}} \text{ x Physical units of each product}$$

For Sodium hydroxide = $\frac{₹40,05,000}{40,050 \text{ tonnes}}$ x 24,030 tonnes

= ₹ 24,03,000

For Halogen = $\frac{₹40,05,000}{40,050 \text{ tonnes}}$ x 16,020 tonnes

= ₹ 16,02,000

MCQ 8

(A)

Products	Sales (in Tonne)	Selling Price per Tonne (₹)	Sales Revenue (₹)	Joint Cost Apportioned (₹)
Sodium hydroxide	24,030	100	24,03,000	20,02,500
Halogen	16,020	150	24,03,000	20,02,500
Total	**40,050**		**48,06,000**	**40,05,000**

Apportioned joint cost = $\frac{\text{Total joint cost}}{\text{Total sale revenue}}$ x Sale revenue of each product

For Sodium hydroxide = $\frac{₹40,05,000}{₹48,06,000}$ x 24,03,000 = ₹ 20,02,500

For Halogen = $\frac{₹40,05,000}{₹48,06,000}$ x 24,03,000 = ₹ 20,02,500

MCQ 9

(B)

Products	Sales (in Tonne)	Selling Price per Tonne (₹)	Sales Value (₹)	Post split-off cost (₹)	Net Realisable Value (₹)	Joint Cost Apportioned (₹)
Sodium hydroxide	24,030	100	24,03,000	-	24,03,000	17,16,429
Halogen (Vinyl after further processing)	10,012.50	150 + 250 = 400	40,05,000	8,01,000	32,04,000	22,88,571
Total					**56,07,000**	**40,05,000**

Apportioned joint cost = $\frac{\text{Total joint cost}}{\text{Total Net Realisable Value}}$ x Net Realisable Value of each product

For Sodium hydroxide = $\frac{₹40,05,000}{₹56,07,000}$ x 24,03,000

= ₹ 17,16,429

For Halogen = $\frac{₹40,05,000}{₹56,07,000}$ x 32,04,000

= ₹ 22,88,571

MCQ 10

(C)

Particulars	Amount (in ₹)
Revenue from sales of Vinyl if Halogen further processed (10,012.50 tonnes × ₹ 400) (A)	40,05,000
Revenue from sales of Halogen if no further processing done (16,020 tonnes × ₹ 150)(B)	24,03,000
Incremental revenue from further processing of Halogen into Vinyl (A-B)	**16,02,000**
Incremental cost of further processing Halogen into Vinyl	8,01,000
Incremental operating income from further processing	**8,01,000**

Incremental revenue would be ₹ 8,01,000, thus the decision relating to further processing Halogen needs to be approved.

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760 MTP **Old P3 Nov 2023 S-I** Q3(a) 10 marks old scheme

Same question, different ICAI answer: #743 RTP Sep 2025

A company processes a raw material in its Department 1 to produce three products, viz. A, B and X at the same split-off stage. During a period 1,80,000 kgs of raw materials were processed in Department 1 at a total cost of ₹ 12,88,000 and the resultant output of A, B and X were 18,000 kgs, 10,000 kgs and 54,000 kgs respectively. A and B were further processed in Department 2 at a cost of ₹ 1,80,000 and ₹ 1,50,000 respectively.

X was further processed in Department 3 at a cost of ₹1,08,000. There is no waste in further processing. The details of sales affected during the period were as under:

	A	B	X
Quantity Sold (kgs.)	17,000	5,000	44,000
Sales Value (₹)	12,24,000	2,50,000	7,92,000

There were no opening stocks. If these products were sold at split-off stage, the selling prices of A, B and X would have been ₹ 50, ₹ 40 and ₹ 10 per kg respectively. Required:

(i) Prepare a statement showing the apportionment of joint costs to A, B and X.

(ii) Present a statement showing the cost per kg of each product indicating joint cost and further processing cost and total cost separately.

(iii) Prepare a statement showing the product wise and total profit for the period.

(iv) State with supporting calculations as to whether any or all the products should be further processed or not.

ICAI'S ANSWER

(i) Statement showing the apportionment of joint costs to A, B and X

Products	A	B	X	Total
Output (kg)	18,000	10,000	54,000	
Sales value at the point of split off (₹)	9,00,000 (₹ 50 × 18,000)	4,00,000 (₹ 40 × 10,000)	5,40,000 (₹ 10 × 54,000)	18,40,000
Joint cost apportionment on the basis of sales value at the point of split off (₹)	6,30,000 (₹ 12,88,000/₹ 18,40,000 x ₹ 9,00,000)	2,80,000 (₹ 12,88,000/₹ 18,40,000 x ₹ 4,00,000)	3,78,000 (₹ 12,88,000/₹ 18,40,000 x ₹ 5,40,000)	12,88,000

(ii) Statement showing the cost per kg. of each product

(indicating joint cost; further processing cost and total cost separately)

Products	A	B	X
Joint costs apportioned (₹) : (I)	6,30,000	2,80,000	3,78,000
Production (kg) : (II)	18,000	10,000	54,000
Joint cost per kg (₹): (I ÷ II)	35	28	7
Further processing Cost per kg. (₹)	10 (₹ 1,80,000/18,000 kg)	15 (₹ 1,50,000/10,000 kg)	2 (₹ 1,08,000/54,000 kg)
Total cost per kg (₹)	45	43	9

(iii) Statement showing the product wise and total profit for the period

Products	A	B	X	Total
Sales value (₹)	12,24,000	2,50,000	7,92,000	
Add: Closing stock value (₹) (Refer to Working note 2)	45,000	2,15,000	90,000	
Value of production (₹)	12,69,000	4,65,000	8,82,000	26,16,000
Apportionment of joint cost (₹)	6,30,000	2,80,000	3,78,000	
Add: Further processing cost (₹)	1,80,000	1,50,000	1,08,000	
Total cost (₹)	8,10,000	4,30,000	4,86,000	17,26,000
Profit (₹)	4,59,000	35,000	3,96,000	8,90,000

Working Notes

1.

Products	A	B	X
Sales value (₹)	12,24,000	2,50,000	7,92,000
Quantity sold (Kgs.)	17,000	5,000	44,000
Selling price ₹/kg	72 (₹ 12,24,000/17,000 kg)	50 (₹ 2,50,000/5,000 kg)	18 (₹ 7,92,000/44,000 kg)

2. Valuation of closing stock:

Since the selling price per kg of products A, B and X is more than their total costs, therefore closing stock will be valued at cost.

Products	A	B	X	Total
Closing stock (kgs.)	1,000	5,000	10,000	
Cost per kg (₹)	45	43	9	
Closing stock value (₹)	45,000 (₹ 45 × 1,000 kg)	2,15,000 (₹ 43 × 5,000 kg)	90,000 (₹9×10,000 kg)	3,50,000

(iv) Calculations for processing decision

Products	A	B	X
Selling price per kg at the point of split off (₹)	50	40	10
Selling price per kg after further processing (₹) (Refer to working Note 1)	72	50	18
Incremental selling price per kg (₹)	22	10	8
Less: Further processing cost per kg (₹)	(10)	(15)	(2)
Incremental profit (loss) per kg (₹)	12	(5)	6

Product A and X has an incremental profit per unit after further processing, hence, these two products may be further processed. However, further processing of product B is not profitable hence, product B shall be sold at split off point.

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761 MTP **Old P3 May 2023 S-I** Q5(b) 10 marks old scheme

Nero Chemicals Ltd. operates a simple chemical process to convert material RV into three separate items, such as T, U and V. All three end products are separated simultaneously at a single split-off point, at which time Product T and Product U are ready for sale without additional costs. Product V, however, is processed further before being sold. There is no available market price for V at the split-off point.

The selling prices quoted here are expected to remain the same in the coming year.

During 2021-22, the selling prices of the items and the total units sold were:

T – 1,000 tons sold for ₹ 6,000 per ton

U – 2500 tons sold for ₹ 5,000 per ton

V – 3000 tons sold for ₹ 6,500 per ton

The total joint manufacturing costs for the year were ₹62,50,000. An additional ₹9,00,000 was spent to finish product V.

There were no opening inventories of T, U or V at the end of the year. The following inventories of complete units were on hand.

T - 900 tons

U - 300 Tons

V - 125 tons

There was no opening or closing work-in-progress.

Required:

COMPUTE the cost of inventories of T, U and V and cost of goods sold for year ended March 31,2022, using Net realizable value (NRV) method of joint cost allocation.

ICAI'S ANSWER

(i) (a) Statement of Joint Cost allocation of inventories of X, Y and Z

(By using Net Realisable Value Method)

	Products			Total
	T	**U**	**V**	
	(₹)	**(₹)**	**(₹)**	**(₹)**
Final sales value of total production (Working Note 1)	1,14,00,000 (1,900 × ₹ 6,000)	1,40,00,000 (2,800 × ₹5,000)	2,03,12,500 (3,125 × ₹6,500)	4,57,12,500
Less: Additional cost	--	--	(9,00,000)	(9,00,000)
Net realisable value (at split-off point)	1,14,00,000	1,40,00,000	1,94,12,500	4,48,12,500
Joint cost allocated (Working Note 2)	15,89,958	19,52,580	27,07,462	62,50,000

Cost of goods sold as on March 31, 2022

(By using Net Realisable Value Method)

	Products			Total
	T	U	V	
	(₹)	(₹)	(₹)	(₹)
Allocated joint cost	15,89,958	19,52,580	27,07,462	62,50,000
Additional costs	--	--	9,00,000	9,00,000
Cost of goods available for sale (CGAS)	15,89,958	19,52,580	36,07,462	71,50,000
Less: Cost of ending inventory (Working Note 1)	7,53,138 (CGAS×47.37%)	2,09,205 (CGAS × 10.71%)	1,44,298 (CGAS × 4%)	11,06,642
Cost of goods sold	8,36,820	17,43,375	34,63,163	60,43,358

Working Note:

1. Total production of three products for the year 2021-2022

Products	Quantity sold in tones	Quantity of ending inventory in tons	Total production	Ending inventory percentage (%)
1	**2**	**3**	**(4) = [(2) + (3)}**	**(5) = (3)/ (4)**
T	1000	900	1900	47.37
U	2500	300	2800	10.71
V	3000	125	3125	4.00

2. Joint cost apportioned to each product:

$$\frac{\text{Total Joint cost}}{\text{Total Net RealisableValue}} \times \text{Net Realisable Value of each product}$$

$$\text{Total cost of Product T} = \frac{62,50,000}{4,48,12,500} \times 1,14,00,000 = 15,89,958$$

$$\text{Total cost of Product U} = \frac{62,50,000}{4,48,12,500} \times 1,40,00,000 = 19,52,580$$

$$\text{Total cost of Product V} = \frac{62,50,000}{4,48,12,500} \times 1,94,12,500 = 27,07,462$$

ICAI · Old P3 · MTP May 2023 · Series I · question p. 6 · answer p. 9-10

762 MTP **Old P3 May 2023 S-II** Q3(b) 10 marks old scheme

ABC Ltd. operates a simple chemical process to convert a single material into three separate items, referred to here as X, Y and Z. All three end products are separated simultaneously at a single split-off point.

Product X and Y are ready for sale immediately upon split off without further processing or any other additional costs. Product Z, however, is processed further before being sold. There is no available market price for Z at the split-off point.

The selling prices quoted here are expected to remain the same in the coming year. During 2022-23, the selling prices of the items and the total amounts sold were:

X – 186 tons sold for ₹3,000 per ton

Y – 527 tons sold for ₹2,250 per ton

Z – 736 tons sold for ₹1,500 per ton

The total joint manufacturing costs for the year were ₹12,50,000. An additional ₹6,20,000 was spent to finish product Z.

There were no opening inventories of X, Y or Z at the end of the year. The following inventories of complete units were on hand:

X 180 tons

Y 60 Tons

Z 25 tons

There was no opening or closing work-in-progress.

Required:

COMPUTE the cost of inventories of X, Y and Z and cost of goods sold for year ended March 31, 2023, using Net realizable value (NRV) method of joint cost allocation.

ICAI'S ANSWER

(i) (a) Statement of Joint Cost allocation of inventories of X, Y and Z

(By using Net Realisable Value Method)

	Products			Total
	X	Y	Z	
	(₹)	(₹)	(₹)	(₹)
Final sales value of total production (Working Note 1)	10,98,000 (366 × ₹3,000)	13,20,750 (587 × ₹2,250)	11,41,500 (761 × ₹1,500)	35,60,250
Less: Additional cost	--	--	(6,20,000)	(6,20,000)
Net realisable value (at split-off point)	10,98,000	13,20,750	5,21,500	29,40,250
Joint cost allocated (Working Note 2)	4,66,797	5,61,496	2,21,707	12,50,000

Cost of goods sold as on March 31, 2023

(By using Net Realisable Value Method)

	Products			Total
	X	Y	Z	
	(₹)	(₹)	(₹)	(₹)
Allocated joint cost	4,66,797	5,61,496	2,21,707	12,50,000
Additional costs	--	--	6,20,000	6,20,000
Cost of goods available for sale (CGAS)	4,66,797	5,61,496	8,41,707	18,70,000
Less: Cost of ending inventory (Working Note 1)	2,29,571 (CGAS×49.18%)	57,385 (CGAS × 10.22%)	27,692 (CGAS × 3.29%)	3,14,648
Cost of goods sold	2,37,226	5,04,111	8,14,015	15,55,352

Working Notes

1. Total production of three products for the year 2022-2023

Products	Quantity sold in tones	Quantity of ending inventory in tons	Total production	Ending inventory percentage (%)
(1)	(2)	(3)	(4) = [(2) + (3)}	(5) = (3)/ (4)
X	186	180	366	49.18
Y	527	60	587	10.22
Z	736	25	761	3.29

2. Joint cost apportioned to each product:

$$\frac{\text{Total Joint cost}}{\text{Total Net Realisable Value}} \times \text{Net Realisable Value of each product}$$

$$\text{Total cost of Product X} = \frac{₹\ 12,50,000}{₹\ 29,40,250} \times ₹10,98,000 = ₹4,66,797$$

$$\text{Total cost of Product Y} = \frac{₹\ 12,50,000}{₹\ 29,40,250} \times ₹13,20,750 = ₹5,61,496$$

$$\text{Total cost of Product Z} = \frac{₹\ 12,50,000}{₹\ 29,40,250} \times ₹5,21,500 = ₹2,21,707$$

ICAI · Old P3 · MTP May 2023 · Series II · question p. 4-5 · answer p. 7-8

763 MTP **Old P3 May 2022 S-I** Q1(c) 5 marks old scheme

Mili Ltd., a manufacturing company, produces two main products and a by-product out of a joint process. The ratio of output quantities to input quantities of direct material used in the joint process remains consistent on yearly basis.

Company has employed the physical volume method to allocate joint production costs to the main products. The net realizable value of the by-product is used to reduce the joint production costs before the joint costs are allocated to the main products.

During a month, company incurred joint production costs of ₹ 15,00,000. The main products are not marketable at the split off point and thus have to be processed further. Details of company's operation are given in the table below.

Particulars	Product-Q	Product-R	By product
Monthly output in kg.	90,000	1,80,000	75,000
Selling price per kg.	₹ 50	₹ 30	₹ 5
Process costs	₹ 3,00,000	₹ 4,50,000	

FIND OUT the amount of joint product cost that Mili Ltd. would allocate to product-R by using the physical volume method to allocate joint production costs?

ICAI'S ANSWER

Calculation of Net joint costs to be allocated:

Particulars	Amount (₹)
Joint Costs	15,00,000
Less: Net Realizable value of by-product (75,000 × 5)	3,75,000
Net joint costs to be allocated	11,25,000

Therefore, amount of joint product cost that Mili Ltd. would allocate to the product-R by using the physical volume method to allocate joint production costs:

$$= \frac{\text{Physical quantity of Product-R}}{\text{Total Quantity}} \times \text{Net joint costs to be allocated}$$

$$= \frac{1,80,000 \text{ units}}{2,70,000 \text{ units}} \times 11,25,000 = ₹\ 7,50,000$$

ICAI · Old P3 · MTP May 2022 · Series I · question p. 2 · answer p. 2

764 MTP **Old P3 May 2022 S-II** Q1(c) 5 marks old scheme

Following information is available for A Ltd.:

Sales-

P: 200 kg @ ₹ 120 per kg.

Q: 240 kg @ ₹ 60 per kg.

Joint costs-

Marginal cost	₹ 17,600
Fixed cost	₹ 15,600

You are required to FIND OUT the cost of joint products P and Q using contribution margin method.

ICAI'S ANSWER

The marginal cost (variable cost) of ₹ 17,600 is apportioned over the joint products P and Q in the ratio of their physical quantity i.e. 200 : 240

Marginal cost for Product P : ₹ 17,600 × $\frac{200}{440}$ = ₹ 8,000

Marginal cost for Product Q : ₹ 17,600 × $\frac{240}{440}$ = ₹ 9,600

The fixed cost of ₹ 15,600 is apportioned over the joint products P and Q in the ratio of their contribution margin *i.e.* 160 : 48 *(Refer to working note)*

Product P : ₹ 15,600 × 160/208 = ₹ 12,000

Product Q : ₹ 15,600 × 48/208 = ₹ 3,600

Working Note:

Computation of contribution margin ratio

Products	Sales revenue (₹)	Marginal cost (₹)	Contribution (₹)
P	24,000	8,000	16,000
Q	14,400	9,600 (Refer to above)	4,800

Contribution ratio is 160 : 48

ICAI · Old P3 · MTP May 2022 · Series II · question p. 2 · answer p. 2

765 MTP **Old P3 Dec 2021 S-I** Q1(a) 5 marks old scheme

A factory produces two products, 'Ghee' and 'Cream' from a single process. The joint processing costs during a particular month are:

Direct Material	₹ 60,000
Direct Labour	₹ 19,200
Variable Overheads	₹ 24,000
Fixed Overheads	₹ 64,000

Sales: Ghee - 200 litre @ ₹ 600 per litre; Cream – 240 litre @ ₹ 200 per litre.

REQUIRED:

I. Apportion joints costs on the basis of:

(i) Physical Quantity of each product.

(ii) Contribution Margin method, and

II. Determine Profit or Loss under both the methods.

ICAI'S ANSWER

Total Joint Cost

Particulars	Amount (₹)
Direct Material	60,000
Direct Labour	19,200
Variable Overheads	24,000
Total Variable Cost	1,03,200
Fixed Overheads	64,000
Total joint cost	1,67,200

Apportionment of Joint Costs:

			Product-Ghee	Product-Cream
I.	(i)	Apportionment of Joint Cost on the basis of 'Physical Quantity'	**₹ 76,000** $\left(\frac{₹\ 1{,}67{,}200}{200+240\ \text{litre}} \times 200\right)$	**₹ 91,200** $\left(\frac{₹\ 1{,}67{,}200}{200+240\ \text{litre}} \times 240\right)$
	(ii)	Apportionment of Joint Cost on the basis of 'Contribution Margin Method':		
		- Variable Costs (on basis of physical units)	**₹ 46,909** $\left(\frac{₹\ 1{,}03{,}200}{200+240\ \text{litre}} \times 200\right)$	**₹ 56,291** $\left(\frac{₹\ 1{,}03{,}200}{200+240\ \text{litre}} \times 240\right)$
		Contribution Margin	**73,091** (₹600×200 – 46,909)	**- 8,291** (₹200×240 – 56,291)
		Fixed Costs*	₹ 64,000	
		Total apportioned cost	₹ 1,10,909	₹ 56,291
II.	**(iii)**	**Profit or Loss:**		
	When Joint cost apportioned on basis of physical units			
	A.	Sales Value	₹ 1,20,000	₹ 48,000
	B.	Apportioned joint cost on basis of 'Physical Quantity':	₹ 76,000	₹ 91,200
	A-B	**Profit or (Loss)**	**44,000**	**(43,200)**

			Product-Ghee	Product-Cream
	When Joint cost apportioned on basis of 'Contribution Margin Method'			
	C	Apportioned joint cost on basis of 'Contribution Margin Method'	₹ 1,10,909	₹ 56,291
	A-C	**Profit or (Loss)**	₹ 9,091	₹ (8,291)

* The fixed cost of ₹ 64,000 is to be apportioned over the joint products- Ghee and Cream in the ratio of their contribution margin but contribution margin of Product- Cream is Negative so fixed cost will be charged to Product- Ghee only.

ICAI · Old P3 · MTP Dec 2021 · Series I · question p. 1 · answer p. 1-2

766 MTP **Old P3 Dec 2021 S-II** Q6(a) 5 marks old scheme

How apportionment of joint costs up-to the point of separation amongst the joint products using market value at the point of separation and net realizable value method is done? DISCUSS.

ICAI'S ANSWER

Apportionment of Joint Cost amongst Joint Products using:

Market value at the point of separation: This method is used for apportionment of joint costs to joint products upto the split off point. It is difficult to apply if the market value of the product at the point of separation is not available. It is useful method where further processing costs are incurred disproportionately.

Net realizable value Method: From the sales value of joint products (at finished stage) the followings are deducted:

- Estimated profit margins
- Selling & distribution expenses, if any
- Post- split off costs.

The resultant figure so obtained is known as net realizable value of joint products. Joint costs are apportioned in the ratio of net realizable value.

ICAI · Old P3 · MTP Dec 2021 · Series II · question p. 9 · answer p. 19

767 MTP **Old P3 Nov 2020 S-I** Q1(d) 5 marks old scheme

'Buttery Butter' is engaged in the production of Buttermilk, Butter and Ghee. It purchases processed cream and let it through the process of churning until it separates into buttermilk and butter. For the month of January, 2020, *'Buttery Butter'* purchased 50 Kilolitre processed cream @ ₹ 100 per 1000 ml. Conversion cost of ₹ 1,00,000 were incurred up-to the split off point, where two saleable products were produced i.e. buttermilk and butter. Butter can be further processed into Ghee.

The January, 2020 production and sales information is as follows:

Products	Production (in Kilolitre/tonne)	Sales Quantity (in Kilolitre/tonne)	Selling price per Litre/Kg (₹)
Buttermilk	28	28	30
Butter	20	—	—
Ghee	16	16	480

All 20 tonne of butter were further processed at an incremental cost of ₹ 1,20,000 to yield 16 Kilolitre of Ghee. There was no opening or closing inventories of buttermilk, butter or ghee in January, 2020.

Required:

(i) SHOW how joint cost would be apportioned between Buttermilk and Butter under Estimated Net Realisable Value method.

(ii) *'Healthy Bones'* offers to purchase 20 tonne of butter in February at ₹ 360 per kg. In case *'Buttery Butter'* accepts this offer, no Ghee would be produced in February. SUGGEST whether *'Buttery Butter'* shall accept the offer affecting its operating income or further process butter to make Ghee itself?

ICAI'S ANSWER

(i) **Estimated Net Realisable Value Method:**

	Buttermilk Amount (₹)	Butter Amount (₹)
Sales Value	8,40,000 (₹ 30 × 28 × 1000)	76,80,000 (₹ 480 × 16 × 1000)
Less: Post split-off cost (Further processing cost)	-	(1,20,000)
Net Realisable Value	8,40,000	75,60,000
Apportionment of Joint Cost of ₹ 51,00,000* in ratio of 1:9	5,10,000	45,90,000

* [(₹ 100 × 50 × 1000) + ₹ 1,00,000] = ₹ 51,00,000

(ii) Incremental revenue from further processing of Butter into Ghee

(₹ 480 × 16 × 1000 - ₹ 360 × 20 × 1000)	₹ 4,80,000
Less: Incremental cost of further processing of Butter into Ghee	₹ 1,20,000
Incremental operating income from further processing	₹ 3,60,000

The operating income of 'Buttery Butter' will be reduced by ₹ 3,60,000 in February if it sells 20 tonne of Butter to 'Healthy Bones', instead of further processing of Butter into Ghee for sale. Thus, 'Buttery Butter' is advised not to accept the offer and further process butter to make Ghee itself.

ICAI · Old P3 · MTP Nov 2020 · Series I · question p. 2 · answer p. 2-3

11.2 Sell at split-off or process further

11 questions · asked in 7 ICAI documents · 2 exam papers

768 EXAM Old P3 May 2022 Q5(c) 5 marks old scheme

RST Limited produces three joint products X, Y and Z. The products are processed further. Pre-separation costs are apportioned on the basis of weight of output of each joint product. The following data are provided for the month of April, 2022.

Cost incurred up to separation point: ₹ 10,000

	Product X	Product Y	Product Z
Output (in Litre)	100	70	80
	₹	₹	₹
Cost incurred after separation point	2,000	1,200	800
Selling Price per Litre:			
After further processing	50	80	60
At pre-separation point (estimated)	25	70	45

You are required to:

(i) Prepare a statement showing profit or loss made by each product after further processing using the presently adopted method of apportionment of pre-separation cost.

(ii) Advise the management whether, on purely financial consideration, the three products are to be processed further or not.

ICAI'S ANSWER

(i) Statement showing profit/loss by each product after further processing products

	Product X (in ₹)	Product Y (in ₹)	Product Z (in ₹)
Sales value after further processing	5,000	5,600	4,800
Less: Further processing cost	2,000	1,200	800
Less: Joint Cost* (as apportioned)	4,000	2,800	3,200
Profit/(loss)	(1,000)	1,600	800

* Statement showing apportionment of joint cost on the basis of physical units

	Product X (in ₹)	Product Y (in ₹)	Product Z (in ₹)	Total (₹)
Output (in litre)	100	70	80	250
Weight	0.4 (100/250)	0.28 (70/250)	0.32 (80/250)	
Joint cost apportioned	4,000	2,800	3,200	

(ii) Decision whether to process further or not

	Product X (in ₹)	Product Y (in ₹)	Product Z (in ₹)
Incremental Revenue	2,500 [(50-25) × 100]	700 [(80-70) × 70]	1,200 [(60-45) × 80]
Less: Further processing cost	2,000	1,200	800
Incremental profit /(loss)	500	(500)	400

	Product X (in ₹)	Product Y (in ₹)	Product Z (in ₹)	Total
Sales	2500	4900	3600	11000
Pre separation costs	4000	2800	3200	10000
Profit/(Loss)	(1500)	2100	400	1000

It is advisable to further process only product X and Z and to sale product Y at the point of separation.

ICAI · Old P3 · May 2022 exam · question p. 23 · answer p. 25-26

769 EXAM **Old P3 Jul 2021** Q2(b) 5 marks old scheme

OPR Ltd. purchases crude vegetable oil. It does refining of the same. The refining process results in four products at the spilt-off point - S, P, N and A. Product 'A' is fully processed at the split-off point. Product S, P and N can be individually further refined into SK, PM, and NL respectively. The joint cost of purchasing the crude vegetable oil and processing it were ₹ 40,000. Other details are as follows:

Product	Further processing costs (₹)	Sales at split-off point (₹)	Sales after further processing (₹)
S	80,000	20,000	1,20,000
P	32,000	12,000	40,000
N	36,000	28,000	48,000
A	-	20,000	-

You are required to identify the products which can be further processed for maximizing profits and make suitable suggestions.

ICAI'S ANSWER

Statement of Comparison of Profits before and after further processing

		S (₹)	P (₹)	N (₹)	A (₹)	Total (₹)
A.	Sales at split off point	20,000	12,000	28,000	20,000	80,000
B.	Apportioned Joint Costs (Refer Working Note)	**10,000**	**6,000**	**14,000**	**10,000**	40,000
C.	**Profit at split-off point**	10,000	6,000	14,000	10,000	40,000
D.	Sales after further processing	1,20,000	40,000	48,000	-	2,08,000

		S (₹)	P (₹)	N (₹)	A (₹)	Total (₹)
E.	Further processing cost	80,000	32,000	36,000	-	1,48,000
F.	Apportioned Joint Costs (Refer Working Note)	10,000	6,000	14,000	-	-
G.	**Profit if further processing (D – E + F)**	**30000**	**2,000**	**(-) 2,000**	-	-
H.	Increase/ decrease in profit after further processing (G-C)	20,000	- 4000	- 16,000	-	-

Suggested Product to be further processed for maximising profits:

On comparing the figures of "Profit if no further processing" and "Profits if further processing", one observes that OPR Ltd. is earning more after further processing of Product S only i.e. ₹ 20,000. Hence, for maximizing profits, only Product S should be further processed and Product P, N and A should be sold at split-off point.

Working Note:

Apportionment of joint costs on the basis of Sales Value at split-off point

$$\text{Apportioned joint cost} = \frac{\text{Total joint cost}}{\text{Total Sales value at split-off point}} \times \text{Sales value of each product}$$

Where,

Total Joint cost = ₹ 40,000

Total sales at split off point (S, P, N and A) = 20,000 + 12,000 + 28,000 + 20,000 = ₹ 80,000

$$\text{Share of S in joint cost} = \frac{₹\ 40,000}{₹\ 80,000} \text{ x } ₹\ 20,000 = ₹\ 10,000$$

$$\text{Share of P in joint cost} = \frac{₹\ 40,000}{₹\ 80,000} \text{ x } ₹\ 12,000 = ₹\ 6,000$$

$$\text{Share of N in joint cost} = \frac{₹\ 40,000}{₹\ 80,000} \text{ x } ₹\ 28,000 = ₹\ 14,000$$

$$\text{Share of A in joint cost} = \frac{₹\ 40,000}{₹\ 80,000} \text{ x } ₹\ 20,000 = ₹\ 10,000$$

Alternative Solution

Decision for further processing of Product S, P and N

Products	S (₹)	P (₹)	N (₹)
Sales revenue after further processing	1,20,000	40,000	48,000
Less: sales value at split-off point	20,000	12,000	28,000
Incremental Sales Revenue	**1,00,000**	**28,000**	**20,000**
Less: Further Processing cost	80,000	32,000	36,000
Profit/ loss arising due to further processing	**20,000**	**(-)4,000**	**(-)16,000**

Suggested Product to be further processed for maximising profits:

On comparing the figures of "Profit if no further processing" and "Profits if further processing", one observes that OPR Ltd. is earning more after further processing of Product S only i.e. ₹ 20,000. Hence, for maximizing profits, only Product S should be further processed and Product P, N and A should be sold at split-off point.

ICAI · Old P3 · Jul 2021 exam · question p. 58-59 · answer p. 62-63

770 RTP Jan 2025 Q14

JPBP Ltd. manufactures two joint products A and B simultaneously from the same process. The process produces another product C which is recovered incidentally from the material used in the manufacture of A and B.

The expenditures incurred up to the point of separation i.e. split-off point are ₹ 14,82,000. As the joint products are capable of being measured in the same units, joint costs are allocated on the basis of physical unit.

Though the joint products A and B are saleable at split-off point, these can also be further processed and sold at a higher market price, with some sales promotion efforts. However, product C can be sold only after further processing.

The management is of the view that, as the net realisable value of the product C at split off point is too small, the value may be deducted from the joint production cost.

The relevant details of the products are as follows:

Particulars	Product A	Product B	Product C
Output (kg.)	16,250	8,125	1,625
Selling price at the split-off point (per kg.) (₹)	72	80	-
Further processing cost (per kg.) (₹)	16	20	8
Further marketing cost (per kg.) (₹)	8	8	4
Selling price after further processing (per kg.) (₹)	112	104	24

You are required the following:

(i) DETERMINE the profit/ (loss) of each joint product if these are sold without further processing.

(ii) WHETHER joint products be processed further? Decide on the basis of incremental profit/ (loss).

ICAI'S ANSWER

Workings -

1. Product C is produced incidentally from the material used in the manufacture of A and B, thus, Product C is a By-product.

	Per unit (₹)
Selling price after further processing (per kg.) (₹)	24
Less: Further Processing Cost (per kg)	8
Further Marketing Cost (per kg)	4
	12

Calculation of Joint Cost to be borne by By-product C

Joint Costs to be borne by By-product C = Output (kg.) x ₹ **12**

= 1,625 kg. × ₹ 12

= ₹ 19,500

2. Allocation of joint cost among joint products (on the basis of physical units) (given)

Product A: (₹ 14,82,000 - ₹ 19,500) x $\left(\frac{16,250}{24,375}\right)$ = **₹ 9,75,000**

Product B: (₹ 14,82,000 - ₹ 19,500) x $\left(\frac{8,125}{24,375}\right)$ = **₹ 4,87,500**

(i) Statement of Profit/ (Loss) if joint products are sold without processing

Particulars	Product A	Product B	Total
(a) Output (kg.)	16,250	8,125	
(b) Selling price at the split-off point (per kg.) (₹)	72	80	
(c) Sales Value (a) x (b)	11,70,000	6,50,000	18,20,000
(d) Allocation of joint costs	9,75,000	4,87,500	14,62,500
(e) Profit at the point of separation (c)-(d)	**1,95,000**	**1,62,500**	**3,57,500**

(ii) Further processing decision

Particulars	Product A (₹)	Product B (₹)
(a) Selling price at split off	72	80
(b) Selling price after further processing	112	104
(c) Incremental revenue (b)-(a)	40	24
(d) Further processing cost	16	20
(e) Further Marketing Cost	8	8
(f) Incremental cost (d)+(e)	24	28
(g) Incremental profit/ (loss) per kg (c)-(f)	**16**	**(4)**
(h) Total Incremental profit/ (loss)	**₹ 16 × 16,250 kg ₹ 2,60,000**	**(₹ 4) x 8,125 kg (₹ 32,500)**

Therefore, Product A should be processed further as they give incremental profit. On the other hand, Product B should be sold at split-off point as they suffer incremental losses after further processing.

ICAI · P4 · RTP Jan 2025 · question p. 14-15 · answer p. 36-38

771 RTP **Old P3 May 2023** Q10 old scheme

Key Pee Limited produces and sells the following products:

Products	Units	Selling price at split-off point (₹)	Selling price after further processing (₹)
A	500000	42.5	62.5
B	75000	32.5	42.5
C	62500	20	30
D	50000	25	-
E	187500	35	50

Cost of raw material ₹ 89,75,000 and other manufacturing ex-penses cost ₹13,67,500 in the manufacturing process which are absorbed on the products on the basis of their 'Net realisable value'. The further processing costs of A, B, C and E are ₹31,25,000; ₹ 3,75,000; ₹1,25,000 and ₹3,75,000 respectively. Fixed costs are ₹11,82,500.

You are required to PREPARE the following in respect of the coming year:

(a) Statement showing income forecast of the company assuming that none of its products are to be further processed.

(b) Statement showing income forecast of the company assuming that products A, B, C and E are to be processed further.

ICAI'S ANSWER

Working Note:

Apportionment of joint costs on the basis of Net Realisable Value method

Products	Sales Value (₹)	Post separation Cost (₹)	Net Realisable Value (₹)	Apportioned Cost (₹)
A	3,12,50,000 (5,00,000 units x ₹ 62.50)	31,25,000	2,81,25,000	67,74,563
B	31,87,500 (75,000 units x ₹ 42.5)	3,75,000	28,12,500	6,77,456
C	18,75,000 (62,500 units x ₹ 30)	1,25,000	17,50,000	4,21,528
D	12,50,000 (50,000 units x ₹ 25)	---	12,50,000	3,01,092
E	93,75,000 (1,87,500 units x ₹ 50)	3,75,000	90,00,000	21,67,860
			4,29,37,500	1,03,42,500

Total joint cost =Raw material costs + Manufacturing expenses = ₹89,75,000 + ₹13,67,500 = ₹1,03,42,500

Apportioned joint cost = (Total Joint Cost/ Total Net Realisable value of each X Net Realisable value of each product)

Apportioned joint cost for Product A = (1,03,42,500 / 4,29,37,500 X 2,81,25,000) = ₹67,74,563.32

Similarly, the apportioned joint cost for products B, C, D and E are ₹6,77,456, ₹4,21,528, ₹3,01,092 and ₹21,67,860 respectively.

(a) Statement showing income forecast of the company assum-ing that none of its products are further processed.

	Products					
	A (₹)	**B (₹)**	**C (₹)**	**D (₹)**	**E (₹)**	**Total (₹)**
Sales revenue	2,12,50,000 (₹42.5 × 5,00,000)	24,37,500 (₹ 32.5 × 75,000)	12,50,000 (₹ 20 × 62,500)	12,50,000 (₹ 25 × 50,000)	65,62,500 (₹ 35 × 1,87,500)	3,27,50,000
Less: Apportioned Costs (Refer Working note)	67,74,563	6,77,456	4,21,528	3,01,092	21,67,860	1,03,42,500
	1,44,75,437	17,60,044	8,28,472	9,48,908	43,94,640	2,24,07,500
Less: Fixed Cost						11,82,500
Profit						2,12,25,000

(b) Statement showing income forecast of the company: assuming that products A, B, C and E are further processed (Refer to working note)

	Products					
	A (₹)	B (₹)	C (₹)	D (₹)	E (₹)	Total (₹)
A. Sales revenue	3,12,50,000	31,87,500	18,75,000	12,50,000	93,75,000	4,69,37,500
B. Apportioned Costs	67,74,563	6,77,456	4,21,528	3,01,092	21,67,860	1,03,42,500
C. Further processing cost	31,25,000	3,75,000	1,25,000	-	3,75,000	40,00,000
D. Total processing cost (B+ C)	98,99,563	10,52,456	5,46,528	3,01,092	25,42,860	1,43,42,500
E. Excess of sales revenue (A-D)	2,13,50,437	21,35,044	13,28,472	9,48,908	68,32,140	3,25,95,000
F. Fixed Cost						11,82,500
G. Profit (E - F)						3,14,12,500

ICAI · Old P3 · RTP May 2023 · question p. 7 · answer p. 19-21

772 MTP **Sep 2026 S-II** Case Scenario - 1 · MCQ 1 to MCQ 5 10 marks

CASE SCENARIO - 1

Sunrise Oil Mills Ltd. is engaged in refining crude vegetable oil into multiple marketable products. During the refining process, a single manufacturing operation results in the simultaneous production of four joint products—AOXE, BOXE, COXE, and DOXE. Up to the point where these products become separately identifiable (known as the split-off point), the company incurs common manufacturing costs that cannot be traced to any individual product. These costs are therefore treated as joint costs.

For the quarter ending 31 March 2026, the company incurred a total joint processing cost of ₹ 1,48,000 up to the split-off point. After separation, management has the option of either selling the products immediately at the split-off point or carrying out additional processing to enhance their quality, increase their market value, and sell them as finished products.

The production, additional processing costs, and final sales values after further processing for the quarter are summarized below:

Product	Output (Liters)	Additional Processing Cost after split off (₹)	Final Sales Value (₹)
AOXE	8,000	43,000	1,72,500
BOXE	4,000	9,000	15,000
COXE	2,000	–	6,000
DOXE	4,000	1,500	45,000

The production manager has informed the finance department that all four products could alternatively be sold immediately at the split-off point, thereby avoiding any further processing costs.

Based on prevailing market conditions, the estimated selling prices at the split-off point are as follows:

Product	Selling Price at Split-off (₹ per Liter)
AOXE	15.00
BOXE	6.00
COXE	3.00
DOXE	7.50

The finance manager has been asked to evaluate the profitability of the products under different joint cost allocation methods and to determine whether the additional processing undertaken after the split-off point is economically justified. Management is particularly interested in comparing allocations based on physical output, sales value at split-off, and net realizable value (NRV), as well as assessing the incremental revenues and incremental costs associated with further processing.

Assume that all units produced during the quarter were sold and that there are no opening or closing inventories. Unless otherwise stated, ignore taxes and other indirect costs.

Based on the above information, answer the following questions (MCQ 1 to 5):

MCQ 1 2 marks

Which product is loss making if they are sold after further processing

A AOXE

B BOXE

C COXE

D DOXE.

MCQ 2 2 marks

If all the products are sold after further processing, if possible, what is the total profit or loss?

A ₹ 37,000 Profit

B ₹ 50,733 Profit

C ₹ 37,000 Loss

D ₹ 50,733 Loss

MCQ 3 2 marks

What is the ratio of share in the joint costs, if Joint costs are allocated on the basis of Sales Value at split off point

A 115:10:4:30.

B 4:3:2:1

C 20:4:1:5

D 10:4:2:5.

MCQ 4 2 marks

If product BOXE is sold at split off point, what is the total profit or loss on BOXE?

A ₹ 1,067 Profit

B ₹ 5,333 Profit

C ₹ 4,267 Loss

D ₹ 4,267 Profit

MCQ 5 2 marks

If optimum profit is achieved, what is the total profit?

A ₹ 37,000 Profit

B ₹ 32,000 Profit

C ₹ 55,000 Profit

D ₹ 50,733 Profit

ICAI'S ANSWERS

MCQ 1

(B) BOXE

MCQ 2

(A) ₹ 37,000 Profit

MCQ 3

(C) 20:4:1:5

MCQ 4

(D) ₹ 4,267 Profit

MCQ 5

(C) ₹ 55,000 Profit

(1) **Statement of Profitability of an Oil Mill**

(after Carrying out further processing)

(for the quarter ending March 2026)

Product's name	Sales Value after further processing	Share of joint cost	Further processing cost	Total cost after split off	Profit (Loss)
(a)	(b)	(c)	(d)	(e) = (c) + (d)	(f = (b) – (e)
	₹	₹	₹	₹	₹
AOXE	1,72,500	98,667	43,000	1,41,667	30,833
BOXE	15,000	19,733	9,000	28,733	(13,733)
COXE	6,000	4,933	-	4,933	1,067
DOXE	45,000	24,667	1,500	26,167	18,833
Total	2,38,500	1,48,000	53,500	2,01,500	37,000

Working Notes

1. Sales Value at split off point for:

AOXE = 8,000 x ₹ 15 = 1,20,000

BOXE = 4,000 x ₹ 6 = 24,000

COXE = 2,000 x ₹ 3 = 6,000

DOXE = 4,000 x ₹ 7.50 = 30,000

The ratio between the sale values of AOXE: BOXE: COXE: DOXE: 20: 4:1: 5.

2. Share of joint cost of AOXE, BOXE, COXE and DOXE has been determined by dividing the total joint cost viz ₹ 1,48,000 in the ratio 20:4:1: 5

AOXE = ₹ $\frac{1,48,000}{30}$ x 20 = ₹ 98,667

BOXE = ₹ $\frac{1,48,000}{30}$ x 4 = ₹ 19,733

COXE = ₹ $\frac{1,48,000}{30}$ x 1 = ₹ 4,933

DOXE = ₹ $\frac{1,48,000}{30}$ x 5 = ₹ 24,667

(2) **Statement of Profitability of an Oil Mill**

(At the split off point)

(for the quarter ending March 2026)

Product's name	Selling price at split off point	Output In units	Sales value at split off point	Share of joint cost	Profit at split off point
(0)	(1)	(2)	(3) = (1) x (2)	(4)	(5) = (3) – (4)
	₹	₹	₹	₹	₹
AOXE	15	8,000	1,20,000	98,667	21,333
BOXE	6	4,000	24,000	19,733	4,267
COXE	3	2,000	6,000	4,933	1,067
DOXE	7.50	4,000	30,000	24,667	5,333
	Total		1,80,000	1,48,000	32,000

(3) Product BOXE should be sold without further processing. This will maximize the profit.

ICAI · P4 · MTP Sep 2026 · Series II · question p. 1-2 · answer p. 1, 1-2

773 MTP **May 2025 S-II** Q5(a) 8 marks

In a chemical manufacturing company, three products A, B and C emerge at a single split off stage in department P. Product A is further processed in department Q, product B in department R and product C in department S. There is no loss in further Processing of any of the three products. The cost data for a month are as under:

Cost of raw materials introduced in department P	₹ 12,68,800
Direct Wages Department	(₹)
P	3,84,000
Q	96,000
R	64,000
S	36,000

Factory overheads of ₹ 4,64,000 are to be apportioned to the departments on direct wage basis.

During the month under reference, the company sold all three products after processing them further as under:

Products	A	B	C
Output sold (kg.)	44,000	40,000	20,000
Selling Price per kg. (₹)	32	24	16

There is no opening or closing stocks. If these products were sold at the split off stage, that is, without further processing, the selling prices would have been ₹ 20, ₹ 22 and ₹ 10 each per kg respectively for A, B and C.

Required:

(i) PREPARE a statement showing the apportionment of joint costs to joint products.

(ii) PRESENT a statement showing product-wise and total profit for the month under reference as per the company's current processing policy.

(iii) WHAT processing decision should have been taken to improve the profitability of the company?

(iv) CALCULATE the product-wise and total profit arising from your recommendation in (iii) above.

ICAI'S ANSWER

(i) **Statement showing the apportionment of joint costs to joint products**

	Products			
	A	**B**	**C**	**Total**
Output sold Kg.: (I)	44,000	40,000	20,000	
Selling price per kg. at split off (₹): (II)	20	22	10	
Sales value at split off (₹): (I) x (II)	8,80,000	8,80,000	2,00,000	19,60,000
Joint costs (costs incurred in department P (₹) (apportioned on the basis of sales value at the point of split off) i.e. (22:22:5) (Working Note 1)	8,80,000	8,80,000	2,00,000	19,60,000

(ii) **Statement showing product-wise and total profit for the month under reference (as per the company's current processing policy)**

	Products			
	A	**B**	**C**	**Total**
Output (kg.) : (a)	44,000	40,000	20,000	
Selling price per kg. after further processing (₹): (b)	32	24	16	
Sales value after further processing (₹).:(c) = {(a) x (b)}	14,08,000	9,60,000	3,20,000	26,88,000
Joint costs (₹): (d)	8,80,000	8,80,000	2,00,000	19,60,000
Further processing costs (₹): (e) (Working Note 2)	1,72,800	1,15,200	64,800	3,52,800
Total costs (₹): (f) = [(d) + (e)}	10,52,800	9,95,200	2,64,800	23,12,800
Profit/ (Loss) (₹): [(c))– (f)}	3,55,200	(35,200)	55,200	3,75,200

(iii) **Processing decision to improve the profitability of the company.**

44,000 units of product A and 20,000 units of product C should be further processed because the incremental sales revenue generated after further processing is more than the further processing costs incurred. 40,000 units of product B should be sold at the point of-split off because the incremental revenue generated after further processing is less than the further processing costs.

(iv) **The product wise and total profit arising from the recommendation in (iii) above is as follows:**

Product	A	B	C	Total
Profit (₹)	3,55,200	-	55,200	4,10,400

Working Notes:

1. **Statement of department-wise costs**

	P (₹)	Q (₹)	R (₹)	S (₹)
Raw materials	12,68,800			
Wages	3,84,000	96,000	64,000	36,000
Overheads (Apportioned on the basis of departmental direct wages i.e. 96:24:16:9)	3,07,200	76,800	51,200	28,800
Total Cost	19,60,000	1,72,800	1,15,200	64,800

2. **Joint costs and further processing costs**

(i) Costs incurred in the department P are joint costs of products A, B and C and are equal to ₹ 19,60,000.

(ii) Costs incurred in the departments Q, R and S are further processing costs of products A, B and C respectively. Further processing costs of products A, B and C thus are ₹ 1,72,800; ₹ 1,15,200 and ₹ 64,800 respectively.

ICAI · P4 · MTP May 2025 · Series II · question p. 12-13 · answer p. 15-16

774 MTP Old P3 Nov 2019 S-I Q5(b) 10 marks old scheme

In an Oil Mill, four products emerge from a refining process. The total cost of input during the quarter ending March 2019 is Rs.22,20,000. The output, sales and additional processing costs are as under:

Products	Output in Litres	Additional processing cost after split off (Rs.)	Sales value (Rs.)
A	8,000	6,45,000	25,87,500
B	4,000	1,35,000	2,25,000
C	2,000	–	90,000
D	4,000	22,500	6,75,000

In case these products were disposed-off at the split off point that is before further processing, the selling price per litre would have been:

A (Rs.)	B (Rs.)	C (Rs.)	D (Rs.)
225.00	90.00	45.00	112.50

PREPARE a statement of profitability based on:

(i) If the products are sold after further processing is carried out in the mill.

(ii) If they are sold at the split off point.

ICAI'S ANSWER

(i) Statement of profitability of an Oil Mill (after carrying out further processing) for the quarter ending 31st March 2019.

Products	Sales Value after further processing	Share of Joint cost	Additional processing cost	Total cost after processing	Profit (loss)
A	25,87,500	14,80,000	6,45,000	21,25,000	4,62,500
B	2,25,000	2,96,000	1,35,000	4,31,000	(2,06,000)
C	90,000	74,000	–	74,000	16,000
D	6,75,000	3,70,000	22,500	3,92,500	2,82,500
	35,77,500	22,20,000	8,02,500	30,22,500	5,55,000

(ii) Statement of profitability at the split off point

Products	Selling price of split off	Output in units	Sales value at split off point	Share of joint cost	Profit at split off point
A	225.00	8,000	18,00,000	14,80,000	3,20,000
B	90.00	4,000	3,60,000	2,96,000	64,000
C	45.00	2,000	90,000	74,000	16,000
D	112.50	4,000	4,50,000	3,70,000	80,000
			27,00,000	22,20,000	4,80,000

Note: Share of Joint Cost has been arrived at by considering the sales value at split off point.

ICAI · Old P3 · MTP Nov 2019 · Series I · question p. 8 · answer p. 11

11.3 Treatment of by-products

17 questions · asked in 19 ICAI documents · 3 exam papers

775 EXAM **Sep 2026** Q6(b)(i) 2 marks

How would you account for the amount realized on sale of molasses, a by-product in a sugar manufacturing process ?

ICAI'S ANSWER

ICAI has not published its answers to this paper yet.

ICAI · P4 · Sep 2026 exam · question p. 22

776 EXAM **Sep 2026** Q6(b)(iii) 2 marks

Suppose that a by-product requires further processing before being sold. How would you determine and treat the net realizable value at the point of split-off ?

ICAI'S ANSWER

ICAI has not published its answers to this paper yet.

ICAI · P4 · Sep 2026 exam · question p. 23

777 EXAM **Sep 2024** Q6(c) 4 marks

Discuss the treatment of By-products in cost accounting.

ICAI'S ANSWER

By-product cost can be dealt in cost accounting in the following ways:

(i) When they are of small total value: When the by-products are of small total value, the amount realised from their sale may be dealt in any one the following two ways:

1. The sales value of the by-products may be credited to the Costing Profit and Loss Account and no credit be given in the Cost Accounts. The credit to the Costing Profit and Loss Account here is treated either as miscellaneous income or as additional sales revenue.
2. The sale proceeds of the by-product may be treated as deductions from the total costs. The sale proceeds in fact should be deducted either from the production cost or from the cost of sales.

(ii) When the by-products are of considerable total value: Where by-products are of considerable total value, they may be regarded as joint products rather than as by-products. To determine exact cost of by-products the costs incurred upto the point of separation, should be apportioned over by-products and joint products by using a logical basis. In this case, the joint costs may be divided over joint products and by-products by using relative market values; physical output method (at the point of split off) or ultimate selling prices (if sold).

(iii) Where they require further processing: In this case, the net realisable value of the by-product at the split-off point may be arrived at by subtracting the further processing cost from the realisable value of by-products.

If total sales value of by-products at split-off point is small, it may be treated as per the provisions discussed above under (*i*).

In the contrary case, the amount realised from the sale of by-products will be considerable and thus it may be treated as discussed under (ii).

ICAI · P4 · Sep 2024 exam · question p. 36 · answer p. 39

778 EXAM Old P3 Nov 2022 Q5(c) 5 marks old scheme

ASR Ltd mainly produces Product 'L' and gets a by-Product 'M' out of a joint process. The net realizable value of the by-product is used to reduce the joint production costs before the joint costs are allocated to the main product. During the month of October 2022, company incurred joint production costs of ₹ 4,00,000. The main Product 'L' is not marketable at the split off point. Thus, it has to be processed further. Details of company's operation are as under:

Particulars	Product L	By- Product M
Production (units)	10,000	200
Selling price per kg	₹ 45	₹5
Further processing cost	₹ 1,01,000	-

You are required to find out:

(i) Profit earned from Product 'L'.

(ii) Selling price per kg of product 'L', if the company wishes to earn a profit of ₹ 1,00,000 from the above production.

ICAI'S ANSWER

(i) Calculation of profit on product 'L'

Particular	₹
Sales	4,50,000
Less: Further processing cost	(1,01,000)
	3,49,000
Less: Joint Production Cost*	(3,99,000)
loss	(50,000)

*Joint Production Cost = [4,00,000 – (200 × 5)] = 3,99,000

(ii) Calculation of desired selling price of product 'L'

$$\text{Desired selling price} = \frac{\text{Desired Profit} + \text{Total Cost}}{\text{units measured}}$$

$$= \frac{1,00,000 + 1,01,000 + 3,99,000}{10,000} = ₹\ 60 \text{ per kg.}$$

ICAI · Old P3 · Nov 2022 exam · question p. 19 · answer p. 21-22

779 RTP Sep 2026 Q12

A manufacturing unit producing Product X also generates a by-product Y, which undergoes additional processing to obtain the finished product. The details of the joint manufacturing costs are provided

below:

Material	₹ 5,000
Labour	₹ 3,000
Overhead	₹ 2,000
	₹ 10,000

Subsequent cost in ₹ are given below:

	X	Y
Material	3,000	1,500
Labour	1,400	1,000
Overhead	600	500
	5,000	3,000

Selling prices are X ₹ 16,000

Y ₹ 8,000

Estimated profit on selling prices is 25% for X and 20% for Y.

On the assumption that selling and distribution expenses are incurred in proportion to the respective sales values, distribute the joint manufacturing costs and prepare a statement indicating the cost of production of X and Y.

ICAI'S ANSWER

Apportionment of Joint Costs

Particulars	X (₹)	Y (₹)
Selling Price	16,000	8,000
Less: Estimated profit	4,000 (25% of ₹ 16,000)	1,600 (20% of ₹ 8,000)
Cost of sales	12,000	6,400
Less: Selling & Distribution exp. (Refer working note)	267 (₹ 400 × 2/3)	133 (₹ 400 × 1/3)
Less: Subsequent cost	5,000	3,000
Share of Joint cost	6,733	3,267

So, Joint cost of manufacture is to be distributed to X & Y in the ratio of 6733:3267.

Statement showing Cost of Production of X and Y

Elements of cost	Joint Cost		Subsequent Cost		Total Cost	
	X	Y	X	Y	X	Y
Material	3,367	1,633	3,000	1,500	6,367	3,133
Labour	2,020	980	1,400	1,000	3,420	1,980
Overheads	1,346	654	600	500	1,946	1,154
				Cost of production	11,733	6,267

Working Note:

Calculation of Selling and Distribution Expenses

Particulars	(₹)
Total Sales Revenue (₹ 16,000 + ₹ 8,000)	24,000
Less: Estimated Profit (₹ 4,000 + ₹ 1,600)	(5,600)
Cost of Sales	18,400
Less: Cost of production:	
- Joint Costs	(10,000)
- Subsequent costs (₹ 5,000 + ₹ 3,000)	(8,000)
Selling and Distribution expenses (Balancing figure)	400

ICAI · P4 · RTP Sep 2026 · question p. 14-15 · answer p. 40-41

780 RTP **Jan 2025** Q19(c)

DISCUSS the treatment of By-product cost in joint cost accounting when they are of small total value.

ICAI'S ANSWER

By-product cost, when they are of small total value, can be dealt in cost accounting in the following ways:

When the by-products are of small total value, the amount realised from their sale may be dealt in any one the following two ways:

1. The sales value of the by-products may be **credited to the Costing Profit and Loss Account** and no credit be given in the Cost Accounts. The credit to the Costing Profit and Loss Account here is treated either as miscellaneous income or as additional sales revenue.
2. The sale proceeds of the by-product may be **treated as deductions from the total costs**. The sale proceeds in fact should be deducted either from the production cost or from the cost of sales.

ICAI · P4 · RTP Jan 2025 · question p. 20 · answer p. 46

781 RTP **Sep 2024** Q19(a)

Also asked in Old P3 MTP Jul 2021 S-I Q6(a)

DISCUSS the Net Realisable Value (NRV) method of apportioning joint costs to by-products.

ICAI'S ANSWER

Net Realisable Value method: The realisation on the disposal of the by-product may be deducted from the total cost of production so as to arrive at the cost of the main product. For example, the amount realised by the sale of molasses in a sugar factory goes to reduce the cost of sugar produced in the factory.

When the by-product requires some additional processing and expenses are incurred in making it saleable to the best advantage of the concern, the expenses so incurred should be deducted from the total value realised from the sale of the by-product and only the net realisations should be deducted from the total cost of production to arrive at the cost of production of the main product. Separate accounts should be maintained for collecting additional expenses incurred on:

(i) further processing of the by-product, and

(ii) selling, distribution and administration expenses attributable to the by-product.

ICAI · P4 · RTP Sep 2024 · question p. 16 · answer p. 36-37

782 RTP **Old P3 Nov 2023** Q9 old scheme

Also asked in Old P3 MTP Nov 2023 S-II Q1(a)

A factory producing article A also produces a by-product B which is further processed into finished product. The joint cost of manufacture is given below:

Material	₹ 5,000
Labour	₹ 3,000
Overhead	₹ 2,000
	₹ 10,000

Subsequent cost in ₹ are given below:

	A	B
Material	3,000	1,500
Labour	1,400	1,000
Overhead	600	500
	5,000	3,000

Selling prices are A ₹ 16,000

B ₹ 8,000

Estimated profit on selling prices is 25% for A and 20% for B.

Assume that selling and distribution expenses are in proportion of sales prices. Show how you would apportion joint costs of manufacture and prepare a statement showing cost of production of A and B.

ICAI'S ANSWER

Apportionment of Joint Costs

Particulars	A (₹)	B (₹)
Selling Price	16,000	8,000
Less: Estimated profit	4,000 (25% of ₹16,000)	1,600 (20% of ₹ 8,000)
Cost of sales	12,000	6,400
Less: Selling & Distribution exp. (Refer working note)	267 (₹ 400 × 2/3)	133 (₹ 400 × 1/3)
Less: Subsequent cost	5,000	3,000
Share of Joint cost	6,733	3,267

So, Joint cost of manufacture is to be distributed to A & B in the ratio of 6733 : 3267

Statement showing Cost of Production of A and B

Elements of cost	Joint Cost		Subsequent Cost		Total Cost	
	A	B	A	B	A	B
Material	3,367	1,633	3,000	1,500	6,367	3,133
Labour	2,020	980	1,400	1,000	3,420	1,980
Overheads	1,346	654	600	500	1,946	1,154
				Cost of production	11,733	6,267

Working Note:

Calculation of Selling and Distribution Expenses

Particulars	(₹)
Total Sales Revenue (₹ 16,000 + ₹ 8,000)	24,000
Less: Estimated Profit (₹ 4,000 + ₹ 1,600)	(5,600)
Cost of Sales	18,400
Less: Cost of production:	
- Joint Costs	(10,000)
- Subsequent costs (₹ 5,000 + ₹ 3,000)	(8,000)
Selling and Distribution expenses (Balancing figure)	400

ICAI · Old P3 · RTP Nov 2023 · question p. 6 · answer p. 18-19

783 RTP **Old P3 Dec 2021** Q15(d) old scheme

DISCUSS the treatment of by-product cost in Cost Accounting when they are of small total value.

ICAI'S ANSWER

When the by-products are of small total value, the amount realised from their sale may be dealt in any one the following two ways:

(i) The sales value of the by-products may be **credited to the Costing Profit and Loss Account** and no credit be given in the Cost Accounts. The credit to the Costing Profit and Loss Account here is treated either as miscellaneous income or as additional sales revenue.

(ii) The sale proceeds of the by-product may be **treated as deductions from the total costs**. The sale proceeds in fact should be deducted either from the production cost or from the cost of sales.

ICAI · Old P3 · RTP Dec 2021 · question p. 13 · answer p. 34

784 RTP **Old P3 May 2020** Q13(d) old scheme

Same question, different ICAI answer: #785 MTP Sep 2026 S-II

DISCUSS the treatment of by-product cost in cost accounting.

ICAI'S ANSWER

By-product cost can be dealt in cost accounting in the following ways:

(i) **When they are of small total value:** When the by-products are of small total value, the amount realised from their sale may be dealt in any one the following two ways:

1. The sales value of the by-products may be credited to the Costing Profit and Loss Account and no credit be given in the Cost Accounts. The credit to the Costing Profit and Loss Account here is treated either as miscellaneous income or as additional sales revenue.
2. The sale proceeds of the by-product may be treated as deductions from the total costs. The sale proceeds in fact should be deducted either from the production cost or from the cost of sales.

(ii) **When the by-products are of considerable total value:** Where by-products are of considerable total value, they may be regarded as joint products rather than as by-products. To determine exact cost of by-products the costs incurred upto the point of separation, should be apportioned over by-products and joint products by using a logical basis. In this case, the joint costs may be divided over joint products and by-products by using relative market values; physical output method (at the point of split off) or ultimate selling prices (if sold).

(iii) **Where they require further processing:** In this case, the net realisable value of the by-product at the split-off point may be arrived at by subtracting the further processing cost from the realisable value of by-products.

If total sales value of by-products at split-off point is small, it may be treated as per the provisions discussed above under (i).

In the contrary case, the amount realised from the sale of by-products will be considerable and thus it may be treated as discussed under (ii).

ICAI · Old P3 · RTP May 2020 · question p. 10 · answer p. 29

785 MTP **Sep 2026 S-II** Q6(a) 5 marks

Same question, different ICAI answer: #784 Old P3 RTP May 2020

Discuss the treatment of by-product Cost in Cost Accounting.

ICAI'S ANSWER

Treatment of by-product Cost in Cost Accounting:

By-Product cost can be dealt in Cost Accounting in the following ways:

(a) **When they are of small total value:** When the by-products are of small total value, the amount realized from their sale **may be dealt in any one the following two ways:**

1. The sales value of the by-products may be **credited to the Costing Profit and Loss Account** and no credit be given in the Cost Accounts. The credit to the Costing Profit and Loss Account here is treated either as miscellaneous income or as additional sales revenue.

2. The sale proceeds of the by-product may be **treated as deductions from the total costs**. The sale proceeds in fact should be deducted either from the production cost or from the cost of sales.

(b) **When the by-products are of considerable total value:** Where by-products are of considerable total value, **they may be regarded as joint products rather than as by-products**. To determine exact cost of by-products the costs incurred up to the point of separation, should be apportioned over by-products and joint products by using a logical basis. In this case, the joint costs may be divided over joint products and by-products by using relative market values; physical output method (at the point of split off or ultimate selling prices (if sold).

(c) **Where they require further processing:** In this case, the net realisable value of the by-product at the split-off point may be arrived at by subtracting the further processing cost from the realisable value of by-products.

If total sales value of by-products at split-off point is small, it may be treated as per the provisions discussed above under (a).

In the contrary case, the amount realised from the sale of by-products will be considerable and thus it may be treated as discussed under (b).

ICAI · P4 · MTP Sep 2026 · Series II · question p. 14 · answer p. 20-21

786 MTP **May 2026 S-II** MCQ 13 2 marks MCQ

Also asked in MTP May 2025 S-I MCQ 14

Product Y yields two by-products A and B. The joint cost of manufacture is ₹ 2,00,000. Sales of A and B are ₹ 80,000 and ₹ 50,000 respectively. Manufacturing expenses after separation of A and B are ₹ 10,000 and ₹ 8,000 respectively while their respective estimated selling expenses on sales is 20% for both products. The estimated profit on sales of A and B is 25% and 30% respectively. What is the cost of product Y after adjusting joint cost apportioned to by-products A and B?

A ₹ 1,50,000

B ₹ 1,49,000

C ₹ 1,82,000

D ₹ 1,20,000

ICAI'S ANSWER

(B) ₹ 1,49,000

Total Cost = Sales Revenue−Profit

For A:

Total Cost of A = 80,000−20,000=60,000

For B:

Total Cost of B = 50,000−15,000=35,000

For A:

Joint Cost Allocated to A = 60,000 − (Manufacturing Expenses + Selling Expenses

= 60,000−(10,000+16,000) = 34,000

For B:

Joint Cost Allocated to B = 35,000 − (Manufacturing Expenses + Selling Expenses

= 35,000 − (8,000 + 10,000) = 17,000

Cost of Product Y = Total Joint Cost−Joint Cost Allocated to A and B

= 2,00,000 − 51,000 = 1,49,000

ICAI · P4 · MTP May 2026 · Series II · question p. 6 · answer p. 4-5

787 MTP **Sep 2025 S-I** Q3(a) 5 marks

GreenChem Organics Ltd. is a company that processes agricultural waste to extract valuable chemical compounds used in the cosmetics and pharmaceutical industries.

The process yields:

- One main product (Active Bio-Extract - ABE)
- Two by-products:
  - Bio-Wax (BWX) – used in lip balms and creams
  - Bio-Fertiliser (BFT) – sold to organic farms

For the month of April 2025, the following production data and financial estimates are provided:

- Total joint processing cost up to separation point: ₹ 3,60,000

Particulars	ABE (Main)	BWX (By-Product 1)	BFT (By-Product 2)
Cost incurred after separation	–	₹ 62,000	₹ 45,000
Quantity produced (units)	6,000	2,200	3,000
Selling price per unit	₹ 150	₹ 55	₹ 40
Estimated net profit (% of sales)	–	25%	20%
Estimated selling expenses (% of sales)	15%	10%	10%

- There is no opening or closing inventory.
- The company allocates joint costs to by-products based on net realizable value (NRV).

PREPARE the following for the month of April 2025:

(i) Statement showing allocation of joint cost to BWX and BFT

(ii) Statement showing product-wise and overall profitability as well as profit per unit.

ICAI'S ANSWER

(i) Statement Showing Allocation of Joint Cost

Particulars	BWX (Bio-Wax) (₹)	BFT (Bio-Fertiliser) (₹)
No. of Units Produced	2,200	3,000
Selling Price per Unit (₹)	55	40
Sales Value	**1,21,000**	**1,20,000**
Less: Estimated Profit (25% of sales for BWX, 20% for BFT)	(30,250)	(24,000)
Cost of Sales	**90,750**	**96,000**
Less: Selling Expenses (10% of sales for both)	(12,100)	(12,000)
Cost of Production	**78,650**	**84,000**
Less: Cost after Separation	(62,000)	(45,000)
Joint Cost Allocated	**16,650**	**39,000**

(ii) Statement of Product-wise and Overall Profitability for April 2025

Particulars	ABE (₹)	BWX (₹)	BFT (₹)
Sales Value (A)	9,00,000 (6,000 units x ₹ 150)	1,21,000	1,20,000
Less:			
Joint Cost	3,04,350 (3,60,000 -16,650 - 39,000)	16,650	39,000
Cost after Separation	–	62,000	45,000
Selling Expenses (15%, 10%, 10%)	1,35,000	12,100	12,000
Total Cost (B)	4,39,350	90,750	96,000
Profit (A – B)	**4,60,650**	**30,250**	**24,000**
Profit per Unit	**76.775**	**13.750**	**8.00**

Overall Profit = ₹4,60,650 + ₹30,250 + ₹24,000 = ₹5,14,900

ICAI · P4 · MTP Sep 2025 · Series I · question p. 10-11 · answer p. 10-11

788 MTP **Sep 2024 S-I** Q3(a) 6 marks

A company manufactures one main product (MN) and two by-products AB and PQ. For the month of January 2024, following details are available:

Total Cost upto separation Point ₹ 2,12,400

	MN	AB	PQ
Cost after separation	-	₹ 35,000	₹ 24,000
No. of units produced	4,000	1,800	3,000
Selling price per unit	₹ 100	₹ 40	₹ 30
Estimated net profit as percentage to sales value	-	20%	30%
Estimated selling expenses as percentage to sales value	30%	15%	15%

There are no beginning or closing inventories.

PREPARE statement showing:

(i) Allocation of joint cost; and

(ii) Product-wise and overall profitability of the company for January 2024.

ICAI'S ANSWER

(i) Statement showing allocation of Joint Cost

Particulars	AB	PQ
No. of units Produced	1,800	3,000
Selling Price Per unit (₹)	40	30
Sales Value (₹)	72,000	90,000
Less: Estimated Profit (AB -20% & PQ -30%)	(14,400)	(27,000)
Cost of Sales	57,600	63,000
Less: Estimated Selling Expenses (AB -15% & PQ -15%)	(10,800)	(13,500)
Cost of Production	46,800	49,500
Less: Cost after separation	(35,000)	(24,000)
Joint Cost allocated	11,800	25,500

(ii) Statement of Profitability

Particulars	MA (₹)	AB (₹)	PQ (₹)
Sales Value (A)	4,00,000 (4,000x ₹ 100)	72,000	90,000
Less:- Joint Cost	1,75,100 (2,12,400 -11,800 - 25,500)	11,800	25,500
Cost after separation	-	35,000	24,000
Selling Expenses (MA- 30%, AB-15% & PQ-15%)	1,20,000	10,800	13,500
(B)	2,95,100	57,600	63,000
Profit (A –B)	1,04,900	14,400	27,000
Overall Profit = 1,04,900 + 14,400 + 27,000 = ₹ 1,46,300			

ICAI · P4 · MTP Sep 2024 · Series I · question p. 8-9 · answer p. 9-10

789 MTP **Sep 2024 S-II** Q6(b) 5 marks

Anju Ltd. is engaged in production of butter. While producing butter buttermilk is also produced. Buttermilk is identified as by-product of butter. What is the TREATMENT of buttermilk in the cost accounts of Anju Ltd.

ICAI'S ANSWER

Buttermilk is a by-product of butter and treatment of by-product in cost accounting is as follows.

(i) When they are of small total value, the amount realized from their sale may be dealt as follows:

- Sales value of the by-product may be credited to Profit and Loss Account and no credit be given in Cost Accounting. The credit to Profit and Loss Account here is treated either as a miscellaneous income or as additional sales revenue.
- The sale proceeds of the by-product may be treated as deduction from the total costs. The sales proceeds should be deducted either from production cost or cost of sales.

(ii) When the by-products are of considerable total value: Where by-products are of considerable total value, they may be regarded as joint products rather than as by- products. To determine exact cost of by-products the costs incurred upto the point of separation, should be apportioned over by-products and joint products by using a logical basis.

(iii) When they require further processing: In this case, the net realisable value of the by-product at the split-off point may be arrived at by subtracting the further processing cost from realisable value of by-product. If the value is small, it may be treated as discussed in (i) above.

ICAI · P4 · MTP Sep 2024 · Series II · question p. 12 · answer p. 20

790 MTP **Old P3 Nov 2022 S-I** Q1(b) 5 marks old scheme

A manufacturing process yields the following products out of the raw materials introduced in the

process:

Main Product X	60% of Raw Materials
By-Product Y	15% of Raw Materials
By Product Z	20% of Raw Materials
Wastage	5% of Raw Materials

Other information is as follows:

a. Total Cost: Raw Materials 1,000 units of ₹ 9,200; Labour ₹ 8,200; Overheads ₹ 12,000

b. One unit of product z requires ½ the raw materials required for one unit of product Y, one unit of product X requires1½ times the raw materials required for product Y.

c. Product X required double the time needed for production of one unit of Y and one unit of Z.

d. Product Z requires ½ the time required for the production of one unit of product Y.

e. Overheads are to be apportioned in the ratio of 6:1:1.

You are required to CALCULATE the total and per unit of cost of each of the products.

ICAI'S ANSWER

Statement of Distribution of Costs

Cost Elements	Basis	Total Cost	Main Product X (600 Units)		By-Product Y (150 Units)		By-Product Z (200 Units)	
			Total	Per Unit	Total	Per Unit	Total	Per Unit
Raw Materials	18:3:2	9,200	7,200	12	1,200	8	800	4
Labour	36:3:2	8,200	7,200	12	600	4	400	2
Overheads	6:1:1	12,000	9,000	15	1,500	10	1,500	7.50
Total		**29,400**	**23,400**	**39**	**3,300**	**22**	**2,700**	**13.50**

Working Notes:

1. Calculation of Units produced:

Main Product X	60% of Raw Materials	600 Units
By-Product Y	15% of Raw Materials	150 Units
By Product Z	20% of Raw Materials	200 Units
Wastage 5% of Raw Materials		50 Units
		1000 Units

2. Cost Allocation

Raw Materials

Let Product Z requires 1 unit of raw materials then, Product Y will require 2 units of raw materials and Product X will require 3 units of raw materials.

Product	X		Y		Z
Individual Unit ratio (a)	3	:	2	:	1
Units (b)	600		150		200
Ratio for Cost Allocation (a*b)	1800	:	300	:	200
Ratio	18	:	3	:	2

Labour:

Let Product Z requires 1 hour of Labour then, Product Y will require 2 hours of Labour and Product X will require 6 hours of Labour.

Product	X		Y		Z
Individual hour ratio (a)	6	:	2	:	1
Units (b)	600		150		200
Ratio for Cost Allocation (a*b)	3600	:	300	:	200
Ratio	36	:	3	:	2

ICAI · Old P3 · MTP Nov 2022 · Series I · question p. 1-2 · answer p. 1-2

791 MTP **Old P3 May 2020 S-I** Q6(d) 5 marks old scheme

EXPLAIN the treatment of by-product cost in cost accounting.

ICAI'S ANSWER

By-product cost can be dealt in cost accounting in the following ways:

(i) When they are of small total value: When the by-products are of small total value, the amount realised from their sale may be dealt in any one the following two ways:

1. The sales value of the by-products may be credited to the Costing Profit and Loss Account and no credit be given in the Cost Accounts. The credit to the Costing Profit and Loss Account here is treated either as miscellaneous income or as additional sales revenue.
2. The sale proceeds of the by-product may be treated as deductions from the total costs. The sale proceeds in fact should be deducted either from the production cost or from the cost of sales.

(ii) When the by-products are of considerable total value: Where by-products are of considerable total value, they may be regarded as joint products rather than as by-products. To determine exact cost of by-products the costs incurred upto the point of separation, should be apportioned over by-products and joint products by using a logical basis. In this case, the joint costs may be divided over joint products and by-products by using relative market values; physical output method (at the point of split off) or ultimate selling prices (if sold).

(iii) Where they require further processing: In this case, the net realisable value of the by-product at the split-off point may be arrived at by subtracting the further processing cost from the realisable value of by-products.

If total sales value of by-products at split-off point is small, it may be treated as per the provisions discussed above under (*i*).

In the contrary case, the amount realised from the sale of by-products will be considerable and thus it may be treated as discussed under (ii).

ICAI · Old P3 · MTP May 2020 · Series I · question p. 6 · answer p. 13

CHAPTER 12

Service Costing

12.1 Service costing and composite cost units

11 questions · asked in 7 ICAI documents · 1 exam paper

792 EXAM **Old P3 Nov 2019** Q6(a) 5 marks old scheme

Describe Composite Cost unit as used in Service Costing and discuss the ways of computing it .

ICAI'S ANSWER

Composite Cost Unit: Sometime two measurement units are combined together to know the cost of service or operation. These are called composite cost units. For example, a public transportation undertaking would measure the operating cost per passenger per kilometre.

Examples of Composite units are Ton- km., Quintal- km, Passenger-km., Patient-day etc. **Composite unit may be computed in two ways**:

(i) Absolute (Weighted Average) basis.

(ii) Commercial (Simple Average) basis.

In both bases of computation of service cost unit, weightage is also given to qualitative factors rather quantitative (which are directly related with variable cost elements) factors alone.

(i) Weighted Average or Absolute basis – It is summation of the products of qualitative and quantitative factors. For example, to calculate absolute Ton-Km for a goods transport is calculated as follows.:

$\sum$ (Weight Carried × Distance)$_1$ + (Weight Carried × Distance)$_2$ +....+ (Weight Carried × Distance)$_n$

Similarly, in case of Cinema theatres, price for various classes of seats are fixed differently. For example–

First class seat may be provided with higher quality service and hence charged at a higher rate, whereas Second Class seat may be priced less. In this case, appropriate weight to be given effect for First Class seat and Second Class seat – to ensure proper cost per composite unit.

(ii) Simple Average or Commercial basis – It is the product of average qualitative and total quantitative factors. For example, in case of goods transport, Commercial Ton -Km is arrived at by multiplying total distance km., by average load quantity.

$$\sum (\text{Distance}_1 + \text{Distance}_2 + \text{.....} + \text{Distance}_n) \times \left(\frac{W_1 + W_2 + ... + W_n}{n}\right)$$

In both the example, variable cost is dependent of distance and is a quantitative factor. Since, the weight carried does not affect the variable cost hence and is a qualitative factor.

ICAI · Old P3 · Nov 2019 exam · question p. 23 · answer p. 24-25

793 RTP **May 2025** Q20(b)

WHAT do you understand by Operating Costs? DESCRIBE its essential features and state where it can be usefully implemented?

ICAI'S ANSWER

Operating Costs or Service Costing are the costs incurred by undertakings which do not manufacture any product but provide a service. Such undertakings for example are — Transport concerns, Gas

agencies; Electricity Undertakings; Hospitals; Theatres etc. Because of the varied nature of activities carried out by the service undertakings, the cost system used is obviously different from that followed in manufacturing concerns.

The essential features of operating costs are as follows:

(1) The operating costs can be classified under three categories. For example in the case of transport undertaking these three categories are as follows:

(a) Operating and running charges: It includes expenses of variable nature. For example expenses on petrol, diesel, lubricating oil, and grease etc.

(b) Maintenance charges: These expenses are of semi-variable nature and includes the cost of tyres and tubes, repairs and maintenance, spares and accessories, overhaul, etc.

(c) Fixed or standing charges: These includes garage rent, insurance, road licence, depreciation, interest on capital, salary of operating manager, etc.

(2) The cost unit used is composite like passenger-mile; Kilowatt-hour, etc.

It can be implemented in all firms of transport, airlines, bus-service, etc., and by all firms of distribution undertakings.

ICAI · P4 · RTP May 2025 · question p. 25 · answer p. 56

794 RTP **Sep 2024** Q19(b)

DIFFERENTIATE between Service costing and Product costing.

ICAI'S ANSWER

Service costing differs from product costing (such as job or process costing) in the following ways due to some basic and peculiar nature.

(i) Unlike products, services are intangible and cannot be stored, hence, there is no inventory for the services.

(ii) Use of Composite cost units for cost measurement and to express the volume of outputs.

(iii) Unlike a product manufacturing, employee (labour) cost constitutes a major cost element than material cost.

(iv) Indirect costs like administration overheads are generally have a significant proportion in total cost of a service as unlike manufacturing sector, service sector heavily depends on support services and traceability of costs to a service may not economically feasible

ICAI · P4 · RTP Sep 2024 · question p. 16 · answer p. 37

795 MTP **Jan 2025 S-I** Q6(b) 5 marks

EXPLAIN the Methods for ascertaining Service Cost Unit.

ICAI'S ANSWER

Methods for ascertaining Service Cost Unit

Composite Cost Unit

Sometime two measurement units are combined together to know the cost of service or operation. These are called composite cost units. For example, a public transportation undertaking would measure the operating cost per passenger per kilometer.

Examples of Composite units are Tonne- km., Quintal- km, Passenger-km., Patient-day etc.
Composite unit may be computed in two ways.

(i) Absolute (Weighted Average) basis.

(ii) Commercial (Simple Average) basis.

In both bases of computation of service cost unit, weightage is also given to qualitative factors rather quantitative (which are directly related with variable cost elements) factors alone.

(i) Weighted Average or Absolute basis – It is a summation of the products of qualitative and quantitative factors. For example, to calculate absolute Tonne-Km for a goods transport is calculated as follows.:

$$\sum(\text{Weight Carried} \times \text{Distance})_1 + (\text{Weight Carried} \times \text{Distance})_2 + \ldots. + (\text{Weight Carried} \times \text{Distance})_n$$

Similarly, in case of Cinema theatres, price for various classes of seats is fixed differently. For example–

First class seat may be provided with higher quality service and hence charged at a higher rate, whereas Second Class seat may be priced less. In this case, appropriate weight to be given effect for First Class seat and Second Class seat – to ensure proper cost per composite unit.

(ii) Simple Average or Commercial Basis – It is the product of average qualitative and total quantitative factors. For example, in case of goods transport, Commercial Tonne-Km is arrived at by multiplying total distance km., by average load quantity.

$$\sum(\text{Distance}_1 + \text{Distance}_2 + \ldots.\ldots.\ldots.\ldots + \text{Distance}_n) \times \left(\frac{W_1 + W_2 + ... + W_n}{n}\right)$$

In both the example, variable cost is dependent of distance and is a quantitative factor. Since, the weight carried does not affect the variable cost hence and is a qualitative factor.

Equivalent Cost Unit/ Equivalent Service Unit:

To calculate cost or pricing of two more different grade of services which uses common resources, **each grade of service is assigned a weight and converted into equivalent units**. Converting services into equivalent units make different grade of services equivalent and comparable.

ICAI · P4 · MTP Jan 2025 · Series I · question p. 16 · answer p. 20-21

796 MTP **Old P3 Nov 2022 S-I** Question 6 · Q6(a) to Q6(e) 25 marks old scheme

QUESTION 6

Answer **any four** of the following:

Q6(a) 5 marks

LIST OUT cost unit examples of following service industry:

Hospital, Electricity Supply service, Cinema, Canteen, Hotels

Q6(b) 5 marks

LIST OUT five purely financial expenses that are included only in Financial Accounts.

Q6(c) 5 marks

DESCRIBE Unit Costing. WHAT kind of industries follow this method of costing?

Q6(d) 5 marks

WRITE DOWN the corresponding cost drivers related to the following activity cost pools:

Inspecting and testing costs, Setting-up machines cost, Machining costs, Supervising Costs, Ordering and Receiving Materials cost

Q6(e) 5 marks

BRIEF the treatment of following while calculating purchase cost of material:

Trade Discount, Cash Discount, Penalty, Insurance charges, Commission paid.

ICAI'S ANSWERS

Q6(a)

Service industry	Unit of cost (examples)
Hospital	Patient per day, room per day or per bed, per operation etc.
Electricity Supply service	Kilowatt- hour (kWh)
Cinema	Per ticket
Canteen	Per item, per meal etc.
Hotels	Guest Days or Room Days

Q6(b)

Purely Financial Expenses included in Financial Accounts only:

(i) Interest on loans or bank mortgages.

(ii) Expenses and discounts on issue of shares, debentures etc.

(iii) Other capital losses i.e., loss by fire not covered by insurance etc.

(iv) Losses on the sales of fixed assets and investments

(v) Income tax, donations, subscriptions

(vi) Expenses of the company's share transfer office, if any. (Any five)

Q6(c)

Unit costing: It is that method of costing where the output produced is identical and each unit of output requires identical cost. Unit costing is synonymously known as single or output costing, but these are sub-division of unit costing method.

This method of costing is followed by industries which produce single output or few variants of a single output, therefore, this method of costing, finds its application in industries like paper, cement, steel works, mining, breweries etc. These types of industries produce identical products and therefore have identical costs.

Q6(d)

Activity Cost Pools	Related Cost Drivers
Inspecting and testing costs	Number of tests
Setting up machines cost	Number of set-ups
Machining costs	Machine hours
Supervising Costs	Direct labour hours
Ordering and Receiving Materials cost	Number of purchase orders

Q6(e)

Trade Discount	Trade discount is deducted from the purchase price if it is not shown as deduction in the invoice.
Cash Discount	Cash discount is not deducted from the purchase price. It is treated as interest and finance charges. It is ignored.
Penalty	Penalty of any type is not included with the cost of purchase
Insurance charges	Insurance charges are paid for protecting goods during transit. It is added with the cost of purchase.
Commission paid	Commission or brokerage paid is added with the cost of purchase.

ICAI · Old P3 · MTP Nov 2022 · Series I · question p. 7 · answer p. 11, 12

797 MTP **Old P3 May 2022 S-II** Q6(c) 5 marks old scheme

STATE the unit of cost for the following service industries:

(i) Electricity Supply service

(ii) Hospital

(iii) Cinema

(iv) Hotels

ICAI'S ANSWER

S. No.	Service industry	Unit of cost
(i)	Electricity Supply service	Kilowatt- hour (kWh)
(ii)	Hospital	Patient per day, room per day or per bed, per operation etc.
(iii)	Cinema	Per ticket.
(iv)	Hotels	Guest Days or Room Days

ICAI · Old P3 · MTP May 2022 · Series II · question p. 7 · answer p. 13

798 MTP Old P3 Jul 2021 S-I Q6(b) 5 marks old scheme

DIFFERENCIATE between Service costing and Product costing.

ICAI'S ANSWER

Service costing differs from product costing (such as job or process costing) in the following ways due to some basic and peculiar nature.

(i) Unlike products, services are intangible and cannot be stored, hence, there is no inventory for the services.

(ii) Use of Composite cost units for cost measurement and to express the volume of outputs.

(iii) Unlike a product manufacturing, employee (labour) cost constitutes a major cost element than material cost.

(iv) Indirect costs like administration overheads are generally have a significant proportion in total cost of a service as unlike manufacturing sector, service sector heavily depends on support services and traceability of costs to a service may not economically feasible.

ICAI · Old P3 · MTP Jul 2021 · Series I · question p. 8 · answer p. 13

12.2 Transport costing: passenger-km and tonne-km

53 questions · asked in 37 ICAI documents · 7 exam papers

799 EXAM Sep 2026 Q3(b) 7 marks

BH Transit Services Ltd. has been awarded a contract to operate a 72-kilometre intercity route using 8 deluxe buses during the next year. Each bus has a seating capacity of 60 passengers. The buses are expected to make 4 round trips per day, carrying on an average, 80% of their seating capacity. The buses will run for 26 days in a month. Assume that the buses operate as a point to point service with no stops intervening during a single trip of 72 kilometres.

The following information relates to the operating cost of the service :

Garage rent : ₹ 48,000 per month (Total for all buses)

Annual repairs and maintenance : ₹ 1,92,000 per bus

Total salaries of all drivers, conductors, cleaners and managers for the year (Total for all buses) ₹ 60,28,800

Road tax, permit fee, etc. : ₹ 48,000 for a quarter (Total for all buses)

Total Office expenses : ₹ 20,000 per month

Cost of diesel per litre : ₹ 108, Diesel Usage : 6 kilometres per litre

Annual depreciation and Insurance : 24% of cost

Engine oils and lubricants : ₹ 56,30,000 (Total for all buses for the whole year).

The cost of each bus is ₹ 60,00,000.

However, during the year, two buses are expected to be under major servicing for one month each and therefore those buses will run only for 11 months, while the remaining 6 buses will operate for the whole year.

The company targets a profit of one third of revenue.

Present the following figures :

(i) Kilometres run by all vehicles for the whole year.

(ii) Number of passengers in all the trips put together for the whole year for all the buses.

(iii) Diesel Cost in total for the whole year.

Calculate the one-way bus fare to be charged from each passenger, rounding off to the next higher rupee value.

ICAI'S ANSWER

ICAI has not published its answers to this paper yet.

ICAI · P4 · Sep 2026 exam · question p. 16-17

800 EXAM **May 2026** MCQ 14 2 marks MCQ

AMC Travels is running 4 luxury buses between two cities which are 60 kms. apart. The seating capacity of each bus is 50 passengers. The actual passengers carried were 70% of the seating capacity. All the four buses run for 30 days for the month. Each bus made one round trip per day. During the month of April 2026, AMC incurred total cost of ₹ 25,05,000.

What is the cost per passenger kilometer for the month of April 2026?

A ₹ 7.55

B ₹ 3.75

C ₹ 6.84

D ₹ 4.97

ICAI'S ANSWER

Answer key: **(D)**

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801 EXAM **May 2025** MCQ 13 MCQ

A Lorry starts with a load of 15 tons of goods from Station 'X'. It unloads 5 tons in Station 'Y' and balance goods in Station 'Z'. On return trip, it reaches Station 'X' with a load of 8 tons, loaded at Station 'Z'. The distance between X to Y, Y to Z and Z to X are 50 kms, 60 kms and 80 kms, respectively.

Compute "Absolute Tons-Kilometre" and "Commercial Tons-Kilometre".

A 1,690 & 2,000

B 1,990 & 2,090

C 2,090 & 1,990

D 2,100 & 1,980

ICAI'S ANSWER

Answer key: **(B)**

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802 EXAM **Jan 2025** Q4(a) 8 marks

XYZ Transport is running a bus between town A and town B which are 25 kms apart. The bus will make 4 round trips every day carrying on an average 30 passengers on each trip. The bus costs the company a

sum of ₹ 5,00,000. It has been insured at 2% per annum and the annual tax will amount to ₹ 2,000 and the garage rent is ₹ 500 per month. Annual repairs will be ₹ 8,000 and the bus is likely to last for 5 years. The driver's salary will be ₹ 15,000 per month and the conductor's salary will be ₹ 12,000 per month in addition to 10% of the takings as commission (to be shared by the driver and conductor equally). Cost of stationery will be ₹ 800 per month. Manager-cum-accountant's salary is ₹ 35,000 per month. Petrol and Oil will be ₹ 1,000 per 100 km. Assuming 15% profit on takings. Depreciation will be charged at straight line method.

You are required to calculate the bus fare to be charged for per passenger kilometer. The bus will run on an average 25 days in a month.

ICAI'S ANSWER

Statement of Cost per Passenger – Km

Particulars	Cost per annum (₹)	Per month (₹)
Standing Charges (A):		
Taxes	2,000	
Insurance (5,00,000 × 0.02)	10,000	
Garage Rent (500 × 12)	6,000	
Salary of Driver (15,000× 12)	1,80,000	
Salary of Conductor (12,000× 12)	1,44,000	
Stationery (800 × 12)	9,600	
Salary of Manager cum Accountant (35,000 × 12)	4,20,000	
Total	7,71,600	
Monthly Standing Charges: (7,71,600/12)		64,300.00
Running Charges (B):		
Depreciation [(5,00,000/5/12]		8,333.33
Diesel and oil [(4 × 2 × 25 × 25) × 10]		50,000.00
Monthly Running Charges		58,333.33
Maintenance Charges (C):		
Repairs	8,000	666.67
Total cost before commission		1,23,300.00
Commission: (D)		16,440.00
Total cost (A+B+C+D)		1,39,740.00
Profit		24,660.00

Particulars	Cost per annum (₹)	Per month (₹)
Total takings (Total Cost + Profit)		**1,64,400.00**

Total passenger-km = 4 × 2 × 25 × 25 × 30 = 1,50,000 passenger km

Hence, Fare per passenger-km = 1,64,400.00/ 1,50,000 = ₹ 1.09

Working Note:

Calculation of Commission and Profit:

Let total takings = ₹ x

Commission = 0.10 x

Profit = 0.15 x

Takings = Cost before commission and profit + Commission + Profit

Therefore, x = 1,23,300 + 0.10 x + 0.15 x

0.75x = 1,23,300

x = ₹ 1,64,400

Commission = 0.10 × ₹ 1,64,400 = ₹ 16,440

Profit = 0.15 × ₹ 1,64,400 = ₹ 24,660

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803 **EXAM** **May 2024** Q2(b) **6 marks**

Star Airlines operates a single aircraft of 180 seats capacity between city 'ND' and 'GA'. The average normal occupancy is estimated at 70% per flight. The average one-way fare is ₹ 12,500 from city 'ND" to 'GA'. The costs of operation of the flight as collected by an expert analyst are:

Fuel cost (Variable) per flight from 'ND' to 'GA'	₹ 2,28,000 per flight
Food served on flight from 'ND' to 'GA' (no charge to passenger)	₹ 270 per passenger
Commission paid to Travel Agents (All ticket booking through agents)	7.5% of fare
Fixed costs:	
Lease & landing charges per flight 'ND' to 'GA'	₹ 9,12,000
Salaries of flight crew per flight 'ND' to 'GA'	₹ 90,000

Note: Assume that fuel costs are unaffected by the actual number of passengers on a flight.

You are required to:

(i) Calculate the net operating income that Star Airlines makes per flight from 'ND' to 'GA'.

(ii) Star Airlines expects that its occupancy will increase to 144 passengers per flight if the fare is reduced to ₹ 11,670. Advise whether this proposal should be implemented or not.

ICAI'S ANSWER

(i) No. of passengers 180 seats × 70% = 126

	(₹)	(₹)
Fare collection (126 passengers × ₹12,500)		15,75,000
Variable costs:		
Fuel	2,28,000	
Food (126 passengers × ₹270)	34,020	
Commission (7.5 % of ₹15,75,000)	1,18,125	3,80,145
Contribution per flight		11,94,855
Fixed costs:		
Lease and Landing Charges	9,12,000	
Salaries of flight Crew	90,000	10,02,000
Net income per flight		1,92,855

(ii)

Fare collection (144 passengers × ₹11,670)		16,80,480
Variable costs:		
Fuel	2,28,000	
Food (144 passengers × ₹270)	38,880	
Commission (7.5% of ₹16,80,480)	1,26,036	3,92,916
Contribution		12,87,564
Fixed costs:		
Lease and Landing Charges	9,12,000	
Salaries of flight Crew	90,000	10,02,000
Net income per flight		2,85,564

There is an increase in contribution by ₹ 92,709. Hence the proposal is acceptable.

ICAI · P4 · May 2024 exam · question p. 7-8 · answer p. 10-11

804 EXAM **Old P3 May 2022** Q1(d) 5 marks old scheme

Coal is transported from two mines X & Y and unloaded at plots in a railway station. X is at distance of 15 kms and Y is at a distance of 20 kms from the rail head plots. A fleet of lorries having carrying capacity of 4 tonnes is used to transport coal from the mines. Records reveal that average speed of the lorries is 40 kms per hour when running and regularly take 15 minutes to unload at the rail head.

At Mine X average loading time is 30 minutes per load, while at mine Y average loading time is 25 minutes per load.

Additional Information:

Drivers' wages, depreciation, insurance and taxes, etc. ₹ 12 per hour

Operated Fuel, oil tyres, repairs and maintenance, etc. ₹ 1.60 per km

You are required to prepare a statement showing the cost per tonne kilometre of carrying coal from each mine 'X' and 'Y'.

ICAI'S ANSWER

Statement showing the cost per tonne-kilometre of carrying mineral from each mine

	Mine X (₹)	Mine Y (₹)
Fixed cost per trip: (Refer to working note 1)		
(Driver's wages, depreciation, insurance and taxes)		
X: 1 hour 30 minutes @ ₹ 12 per hour	18.00	
Y: 1 hour 40 minutes @ ₹ 12 per hour		20.00
Running and maintenance cost:		
(Fuel, oil, tyres, repairs and maintenance)		
X: 30 km. ₹ 1.60 per km.	48.00	
Y: 40 km. ₹ 1.60 per km.		64.00
Total cost per trip (₹)	66.00	84.00
Cost per tonne – km (Refer to working note 2)	1.1 (₹ 66/60 tonne - km)	1.05 (₹ 84/80 tonne - km)

Working notes:

		Mine- X	Mine- Y
(1)	Total operated time taken per trip		
	Running time to & fro	**45 minutes** (30 km. × 60 minutes/40 km.)	**60 minutes** (40 km. × 60 minutes/40 km.)
	Un-loading time	15 minutes	15 minutes
	Loading time	30 minutes	25 minutes
	Total operated time	**90 minutes or** 1 hour 30 minutes	**100 minutes or** 1 hour 40 minutes
(2)	Effective tones – km.	60 (4 tonnes × 15 km.)	80 (4 tonnes × 20 km.)

ICAI · Old P3 · May 2022 exam · question p. 2 · answer p. 5-6

805 EXAM **Old P3 Dec 2021** Q3(a) 10 marks old scheme

Paras Travels provides mini buses to an IT company for carrying its employees from home to office and dropping back after office hours. It runs a fleet of 8 mini buses for this purpose. The buses are parked in a garage adjoining the company's premises. Company is operating in two shifts (one shift in the morning and one shift in the afternoon). The distance travelled by each mini bus one way is 30 kms. The company works for 20 days in a month.

The seating capacity of each mini bus is 30 persons. The seating capacity is normally 80% occupied during the year. The details of expenses incurred for a year are as under:

Particulars	
Driver's salary	₹ 20,000 per driver per month
Lady attendant's salary (mandatorily required for each mini bus)	₹ 10,000 per attendant per month
Cleaner's salary (One cleaner for 2 mini buses)	₹ 15,000 per cleaner per month
Diesel (Avg. 8 kms per litre)	₹ 80 per litre
Insurance charges (per annum)	2% of Purchase Price
License fees and taxes	₹ 5,080 per mini bus per month
Garage rent paid	₹ 24,000 per month
Repair & maintenance including engine oil and lubricants (for every 5,760 kms)	₹ 2,856 per mini bus
Purchase Price of mini bus	₹ 15,00,000 each

Particulars	
Residual life of mini bus	8 Years
Scrap value per mini bus at the end of residual life	₹ 3,00,000

Paras Travels charges two types of fare from the employees. Employees coming from a distance of beyond 15 kms away from the office are charged double the fare which is charged from employees coming from a distance of up-to 15 kms. away from the office. 50% of employees travelling in each trip are coming from a distance beyond 15 kms. from the office. The charges are to be based on average cost.

You are required to:

(i) Prepare a statement showing expenses of operating a single mini bus for a year,

(ii) Calculate the average cost per employee per month in respect of:

(a) Employees coming from a distance upto 15 kms. from the office.

(b) Employees coming from a distance beyond 15 kms. from the office.

ICAI'S ANSWER

(i) Statement of Expenses of operating a mini bus in a year

Particulars		Rate (₹)	Per Bus per annum (₹)
(A)	**Standing Charges:**		
	Driver's salary	20,000 p.m	2,40,000
	Lady attendant's salary	10,000 p.m	1,20,000
	Average Cleaner's salary (50%)	15,000 p.m	90,000
	Insurance charge	30,000 p.a.	30,000
	License fee, taxes etc.	5,080 p.m.	60,960
	Average Garage Rent	24,000 p.m	36,000
	Depreciation {(15,00,000 – 3,00,000) ÷ 8}	1,50,000 p.a.	1,50,000
(B)	**Maintenance Charges:**		
	Repairs & maintenance including engine oil and lubricants (Working Note 1)	28,560 p.a.	
(C)	**Operating Charges:**		
	Diesel (Working Note 2)		5,76,000
	Total Cost (A + B + C)		13,31,520

Particulars	Rate (₹)	Per Bus per annum (₹)
Cost per month		1,10,960

(ii) Average cost per employee per month:

A. Employee coming from distance of upto 15 km

$$= \frac{\text{Total cost per month}}{\text{Total no.of equivalent employee}} = \frac{1,10,960}{72^*} = ₹\ 1,541.11$$

B. Employee coming from a distance beyond 15 km

= 1541.11 × 2 = ₹ 3,082.22

*** Considering half fare employees as a base**

Full fare employees (12 × 2)	24 employees
Add: Half fare employees (Working Note 3)	12 employees
Total Equivalent number of employees per month	36 employees
Total Equivalent number of employees per month (morning + afternoon shift of company)	72 employees

Working Notes:

1. Calculation of Repairs and maintenance cost of a bus :

Distance travelled in a year:

(4 trip × 2 shifts × 30 km. × 20 days × 12 months)

Distance travelled p.a.: 57,600 km.

Repairs and maintenance cost per Bus per annum:

$$= \frac{57,600 \text{ km.}}{5,760 \text{ km}} \times ₹\ 2,856 \text{ per bus}$$

= ₹ 28,560 per annum

2. Calculation of diesel cost per bus per annum:

Distance travelled in a year = 57,600 km

Diesel cost per Bus per annum:

$$= \frac{57,600 \text{ km.}}{8 \text{ Km}} \times ₹\ 80$$

= 5,76,000

3. Calculation of equivalent number of employees per bus:

Seating capacity of a bus	30 employees
Occupancy (80% of capacity)	24 employees
Half fare employees (50% of 24 employees)	12 employees
Full fare employees (50% of 24 employees)	12 employee

[**Note**: Total Equivalent number of employees per month (morning + afternoon shift of company can also be calculated considering full fare employees as a base. In that case the number will be 36. Then fare for employees coming from distance beyond 15km will be $\frac{1,10,960}{36}$ = ₹ 3,082.22 and employees coming from distance upto 15 km will be 3,082.22 / 2 = ₹ 1,541.11]

ICAI · Old P3 · Dec 2021 exam · question p. 9-10 · answer p. 11-12

806 RTP **Sep 2026** Q13

Royal Transport Services runs fleet of buses within the limits of Jaipur city. The following are the details which were incurred by the company during October, 2025:

	(₹)
Cost of each Bus	24,00,000
Garage Rent	1,00,000
Insurance	25,000
Road tax	20,000
Manager's Salary	60,000
Assistant's Salary (Two)	32,000 each
Supervisor's Salary (Three)	24,000 each
Driver's Salary (Twenty-Five)	20,000 each
Cleaner's Salary (Twenty)	5,000 each
Office Staff's Salary	1,00,000
Consumables	1,20,000
Repairs & Maintenance	90,000
Other Fixed Expenses	72,000
Diesel (10 Kms per Litre)	80 per litre
Oils & Lubricants	1,45,000
Tyres and tubes	35,000

	(₹)
Depreciation	10% p.a. on Cost

Other details are as below:

Capacity

12 Buses 60 Passengers

13 Buses 50 Passengers

Each bus makes 4 round trips a day covering a distance of 10 Kilometers in each trip (One Way) on an average. During the trips 80% of the seats are occupied. The annual records show that 5 buses are generally required to be kept away from roods each day for repairs.

You are required to CALCULATE cost per passenger-km.

Cost sheet to be prepared on the basis of 25 buses.

ICAI'S ANSWER

Operating Cost Sheet

Particulars	Amount (₹)	Amount (₹)
Standing Charges:		
Depreciation (₹ 24,00,000 × 10% x 1/12 × 25)	5,00,000	
Garage Rent	1,00,000	
Insurance	25,000	
Road Tax	20,000	
Manager's Salary	60,000	
Assistant's Salary (₹ 32,000 × 2)	64,000	
Supervisor's Salary (₹ 24,000 × 3)	72,000	
Driver's Salary (₹ 20,000 × 25)	5,00,000	
Cleaner's Salary (₹ 5,000 × 20)	1,00,000	
Office Staff's Salary	1,00,000	
Consumables	1,20,000	
Repairs & Maintenance	90,000	
Other Fixed Expenses	72,000	18,23,000
Running Charges		
Diesel (49,600 Kms / 10 Kms x ₹ 80 per unit)	3,96,800	
Oils & Lubricants	1,45,000	

Particulars	Amount (₹)	Amount (₹)
Tyres and tubes	35,000	5,76,800
Total Operating Cost		**23,99,800**

$$\text{Cost per passenger-km} = \frac{\text{Total Operating Cost}}{\text{Passenger –kms}}$$

$$= \frac{23,99,800}{27,18,080}$$

$$= 0.883$$

Working Note:

Calculation of Total Kilometers and Passenger Kilometers

Specification	Total Km.	Passenger–Km.
12 Buses (60 Passengers)	29,760 Kms (10 Kms × 4 × 2 trips × 31 days x 12 Buses)	14,28,480 (29760 Kms x 60 Pass. x 80%)
13 Buses (50 Passengers)	32,240 Kms (10 Kms × 4 × 2 trips × 31 days x 13 Buses)	12,89,600 (32240 Kms x 50 Pass. x 80%)
Total	**62,000**	**27,18,080**

Since 5 buses out of 25 buses are kept for repairs every day

Actual total Km. 62,000 × 20/25 = 49,600

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807 RTP **Jan 2026** Case Scenario 1 · Q1(i) to Q1(v)

CASE SCENARIO 1

EcoTrans Logistics Pvt. Ltd. is a regional goods transport company based in the industrial hub of LogiPort, providing daily freight services to and from GreenCity, a fast-growing metropolitan area located 120 kilometers away. The company has been operating for over a decade and has built a reputation for timely deliveries, especially for fast-moving consumer goods (FMCGs), packaged food items, and light industrial components.

To meet increasing demand, EcoTrans has recently expanded its fleet and now operates 24 transport vehicles of varying capacities. The company promises reliable daily service, and each vehicle typically completes one round trip per day, carrying cargo in both directions. EcoTrans emphasizes sustainability and efficiency, with systems in place for fuel monitoring, vehicle maintenance, and cost control.

However, to remain competitive and ensure profitability in a cost-sensitive industry, management has decided to undertake a detailed cost analysis for the month of April 2025. This analysis will guide pricing strategy and assess whether current operations are yielding the desired profit margin of 20% on freight revenue.

The following data has been compiled by the Finance and Operations departments for cost analysis and pricing decision-making.

Fleet Composition:

No. of Vehicles	Capacity per Vehicle (MT)
6	10 MT
8	14 MT
6	18 MT
4	22 MT

Each vehicle operates one round trip per day on average. Vehicles are loaded to 85% capacity on the forward journey and 65% on the return. On average, 12% of the fleet is under maintenance at any time.

The company operated for 26 days in April 2025.

Monthly Expenditure (April 2025)

Description	Amount (₹)
Salary of Transport Manager	75,000
Salary of 32 Drivers @ ₹22,000 each	7,04,000
Wages of 28 Helpers @ ₹13,000 each	3,64,000
Loading & Unloading Charges per Trip	950
Consumable Stores (Fuel Additives, Tools)	1,50,000
Toll Charges	250 per Trip
Insurance (Annual Premium)	9,60,000
Road Licence Fee (Annual)	5,40,000
Diesel Cost per Litre	82
Mileage per Vehicle	4.5 km/litre
Lubricants & Oils	1,30,000
Tyres, Tubes, Spare Parts (Running basis)	4,80,000
GPS & Fleet Tracking Subscription (Monthly Basis)	2,000 per vehicle
Driver Safety & Training Programmes	80,000
Garage Rent (Annual)	10,80,000
Workshop Repairs (Routine & Emergency)	3,40,000
Electricity & Utility Expenses	62,000

Description	Amount (₹)
Depreciation on Vehicles (Monthly)	6,80,000

The company operates an in-house workshop that handles routine and emergency repairs for both its own fleet and external vehicles. As part of the internal cost allocation, 45% of the Transport Manager's salary is charged to the workshop to reflect time and resources dedicated to overseeing maintenance operations. In return, the transport department has been apportioned ₹95,000 by the workshop for repair and service support during the month.

Being a finance manager of the company, you are required to answer the followings (5 MCQs):

Q1(i)

What is the total fixed cost for April 2025?

A ₹21,95,000

B ₹20,89,750

C ₹22,89,250

D ₹19,98,250

Q1(ii)

Calculate the total variable cost for April 2025

A ₹44,71,885

B ₹31,50,000

C ₹29,65,920

D ₹48,45,000

Q1(iii)

What should be the freight rate per ton-km to achieve a 20% profit margin?

A ₹4.461

B ₹5.576

C ₹4.372

D ₹3.845

Q1(iv)

Calculate the total wages paid to loading and unloading labour in April 2025.

A ₹10,43,520

B ₹10,42,560

C ₹9,85,000

D ₹10,37,400

Q1(v)

What is the total ton-kilometers (ton-km) generated by EcoTrans Logistics in April 2025?

A 1,31,789 ton-km

B 5,49,120 ton-km

C 15,15,571 ton-km

D 26,00,000 ton-km

ICAI'S ANSWERS

Q1(i)

(C) ₹22,89,250

Q1(ii)

(A) ₹44,71,885

Q1(iii)

(B) ₹5.576

Q1(iv)

(D) ₹10,37,400

Q1(v)

(C) 15,15,571 ton-km

Operating Cost Sheet for the month

Particulars	Amount (₹)
A. Fixed Charges:	
Transport Manager's Salary (55% x ₹75,000)	41,250
Driver Salaries (32 x ₹22,000)	7,04,000
Helper Wages (28 x ₹13,000)	3,64,000
GPS Subscription (24 x ₹2,000)	48,000
Driver Safety & Training Programmes	80,000
Insurance (₹9,60,000 ÷ 12)	80,000
Road Licence Fee (₹5,40,000 ÷ 12)	45,000
Garage Rent (₹10,80,000 ÷ 12)	90,000
Electricity & Utility Expenses	62,000
Depreciation on Vehicles	6,80,000
Workshop Repairs (Apportioned)	95,000
Total Fixed Costs	**22,89,250**
B. Variable Charges:	
Loading & Unloading Charges (WN.1)	10,37,400
Toll Charges (W.N.2)	2,73,000
Diesel Cost (W.N.3)	24,01,485
Lubricants & Oils	1,30,000

Particulars	Amount (₹)
Tyres, Tubes, Spare Parts	4,80,000
Consumable Stores	1,50,000
Total Variable Costs	**44,71,885**
C. Total Cost (A + B)	**67,61,135**
D. Total Ton-Kms. (W. N. 4)	15,15,571
E. Cost per ton-km. (C ÷ D)	**4.461**

Calculation of Chargeable Freight

Cost per ton-km.	₹ 4.461
Add: Profit @ 20% on freight or 25% on cost	₹ 1.115
Chargeable freight per ton-km.	**₹ 5.576**

Working Notes:

1. Wages paid to loading and unloading labours

Numbers of vehicles available per day x No. of days x trips x wages per trip

(24 vehicles x 88%) x 26 days x 2 trips x ₹950

21 x 2 x 26x 950 = ₹10,37,400

2. Toll Charges

Numbers of vehicles available per day x No. of days x trips x toll charges per trip

21 x 2 x 26x 250 = ₹2,73,000

3. Cost of Diesel:

Distance covered by each vehicle during the month

= 120 k.m. × 2 × 26 days × 88% = 5,491.2 km.

Consumption of diesel = $\frac{5{,}491.2 \text{ k.m.} \times 24 \text{ vehicles}}{4.5 \text{ k.m.}}$ = 29,286.4 litres

Cost of diesel = 29,286.4 litres × ₹ 82 = ₹24,01,485

4. Calculation of total ton-km:

Total Ton-Km. = Total Capacity × Distance covered by each vehicle × Average Capacity Utilisation ratio.

$= [(6 \times 10 \text{ MT}) + (8 \times 14 \text{ MT}) + (6 \times 18 \text{ MT}) + (4 \times 22 \text{ MT})] \times 5{,}491.2 \text{ k.m.} \times \frac{(85\% + 65\%)}{2}$

= (60 + 112 + 108 + 88) x 5,491.2 x 75%

= 368 x 5,491.2 x 75%

= 15,15,571 ton-km.

ICAI · P4 · RTP Jan 2026 · question p. 1-3 · answer p. 23, 23-25

808 RTP Jan 2025 Q15

Roshan Travels provide bus facility to a College for carrying its students from home to College and dropping them back at home after study hours. The travel company runs a fleet of 6 buses for this purpose and park them in the college premises.

The information regarding bus running is as follows:

(I) The College operates in two shifts (one in the morning and one in the afternoon).

(II) The distance travelled by each bus one way is 20 kms.

(III) The students need to attend the college for 30 days in a month.

(IV) The seating capacity of each bus is 30 persons.

(V) The seating capacity is normally 80% occupied during the year.

The information regarding expenses incurred for a year is as follows:

Particulars	Amount
Driver and attendant salary	₹ 60,000 per bus per month
Cleaner's salary (One cleaner for 2 buses)	₹ 30,000 per cleaner per month
Diesel (Avg. 8 kms per litre)	₹ 160 per litre
Insurance charges (per annum)	2% of Purchase Price
License fees and taxes	₹ 10,160 per bus per month
Parking charges paid	₹ 36,000 per month
Repair & maintenance including engine oil and lubricants (for every 5,760 kms)	₹ 5,712 per bus
Purchase Price of each bus	₹ 30,00,000
Residual life of each bus	8 Years
Scrap value per bus at the end of residual life	₹ 6,00,000

Students coming from a distance of beyond 10 kms away from the College are charged double the fare than that from students coming from a distance of up-to 10 kms. away from the College. 50% of students travelling in each trip are coming from a distance beyond 10 kms. from the College. The charges are to be based on average cost.

You are required to:

(i) PREPARE a statement showing expenses of operating a single bus for a year.

(ii) CALCULATE the average cost per student per month in respect of:

(a) Students coming from a distance up-to 10 kms. from the College.

(b) Students coming from a distance beyond 10 kms. from the College.

ICAI'S ANSWER

(i) Statement of Expenses of operating a single bus for a year

	Particulars	Rate (₹)	Per Bus per annum (₹)
(A)	**Standing Charges:**		
	Driver and attendant salary	60,000 p.m	7,20,000
	Average Cleaner's salary (50%)	30,000 p.m	1,80,000
	Insurance charge	60,000 p.a.	60,000
	License fee, taxes etc.	10,160 p.m.	121,920
	Average Parking Charges	36,000 p.m	72,000
	Depreciation {(30,00,000 – 6,00,000) ÷ 8}	3,00,000 p.a.	3,00,000
(B)	**Maintenance Charges:**		
	Repairs & maintenance including engine oil and lubricants (Working Note 1)	5,7120 p.a.	5,7120
(C)	**Operating Charges:**		
	Diesel (Working Note 2)		11,52,000
	Total Cost (A + B + C)		26,63,040
	Cost per month		2,21,920

(ii) Average cost per students per month:

A. Student coming from distance of up-to 10 km

$$= \frac{\text{Total cost per month}}{\text{Total no. of equivalent student}} = \frac{₹2,21,920}{72^*} = ₹\ 3,082.22$$

B. Student coming from a distance beyond 10 km

= ₹ 3,082.22 × 2 = ₹ 6,164.44

* Considering half fare students as a base

Full fare students (12 × 2)	24 students
Add: Half fare students (Working Note 3)	12 students
Total Equivalent number of students per month	36 students
Total Equivalent number of students per month (morning + afternoon shift)	72 students

Working Notes:

1. Calculation of Repairs and maintenance cost of a bus:

Distance travelled in a year:

(4 trips × 2 × 20 km. × 30 days × 12 months)

Distance travelled p.a.: 57,600 km.

Repairs and maintenance cost per Bus per annum:

$= \frac{57,600 \text{ km.}}{5,760 \text{ km}} \times ₹ 5,712 \text{ per bus}$

= ₹ 57,120 per annum

2. Calculation of diesel cost per bus per annum:

Distance travelled in a year = 57,600 km

Diesel cost per Bus per annum:

$= \frac{57,600 \text{ km.}}{8 \text{ Km}} \times ₹ 160$

= ₹ 11,52,000

3. Calculation of equivalent number of students per bus:

Seating capacity of a bus	30 students
Occupancy (80% of capacity)	24 students
Half fare students (50% of 24 students)	12 students
Full fare students (50% of 24 students)	12 students

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809 **RTP** **May 2024** Q9

A LMV Pvt. Ltd, operates cab/ car rental service in Delhi/NCR. It provides its service to the offices of Noida, Gurugram and Faridabad. At present it operates CNG fuelled cars but it is also considering to upgrade these into Electric vehicle (EV). The following details related with the owning of CNG & EV propelled cars are as tabulated below:

Particulars	CNG Car	EV Car
Car purchase price (₹)	9,20,000	15,20,000
Govt. subsidy on purchase of car (₹)	--	1,50,000
Life of the car	15 years	10 years
Residual value (₹)	95,000	1,70,000
Mileage	20 km/kg	240 km per charge
Electricity consumption per full charge	--	30 Kwh
CNG cost per Kg (₹)	60	--
Power cost per Kwh (₹)	--	7.60
Annual Maintenance cost (₹)	8,000	5,200
Annual insurance cost (₹)	7,600	14,600

Particulars	CNG Car	EV Car
Tyre replacement cost in every 5 - year (₹)	16,000	16,000
Battery replacement cost in every 8- year (₹)	12,000	5,40,000

Apart from the above, the following are the additional information:

Particulars	
Average distance covered by a car in a month	1,500 km
Driver's salary (₹)	20,000 p.m
Garage rent per car (₹)	4,500 p.m
Share of Office & Administration cost per car (₹)	1,500 p.m

You have been approached by the management of A LMV Pvt. Ltd. for consultation on the two options of operating the cab service.

CALCULATE the operating cost of vehicle per month per car for both CNG & EV options.

ICAI'S ANSWER

Workings:

1. Calculation of Depreciation per month:

	Particulars	CNG Car	EV Car
A	Car purchase price (₹)	9,20,000	15,20,000
B	Less: Govt. subsidy (₹)	--	(1,50,000)
C	Less: Residual value (₹)	(95,000)	(1,70,000)
D	Depreciable value of car (₹) [A-B-C]	8,25,000	12,00,000
E	Life of the car	15 years	10 years
F	Annual depreciation (₹) [D÷E]	55,000	1,20,000
G	**Depreciation per month (₹) [F÷12]**	**4,583.33**	**10,000**

2. Fuel/ Electricity consumption cost per month:

	Particulars	CNG Car	EV Car
A	Average distance covered in a month (KM)	1,500	1,500
B	Mileage (KM)	20	240
C	Qty. of CNG/ Full charge required [A÷B]	75 kg.	6.25
D	Electricity Consumption [C×30kwh]	-	187.5
E	Cost of CNG per kg (₹)	60	-
F	Power cost per Kwh (₹)	-	7.60
G	**CNG Cost per month (₹) [C×E]**	**4,500**	**-**
H	**Power cost per month (₹) [D×F]**	**-**	**1,425**

3. Amortised cost of Tyre replacement:

	Particulars	CNG Car	EV Car
A	Life of vehicle	15 years	10 years
B	Replacement interval	5 years	5 years
C	No. of time replacement required	2 times	1 time
D	Cost of tyres for each replacement (₹)	16,000	16,000
E	Total replacement cost (₹) [C×D]	32,000	16,000
F	Amortised cost per year (₹) [E÷A]	2,133.33	1,600
E	**Cost per month (₹) [F÷12]**	**177.78**	**133.33**

4. Amortised cost of Battery replacement:

	Particulars	CNG Car	EV Car
A	Life of vehicle	15 years	10 years
B	Replacement interval	8 years	8 years
C	No. of time replacement required	1 time	1 time
D	Cost of battery for each replacement (₹)	12,000	5,40,000
E	Total replacement cost (₹) [C×D]	12,000	5,40,000
F	Amortised cost per year (₹) [E÷A]	800	54,000
E	**Cost per month (₹) [F÷12]**	**66.67**	**4,500**

Calculation of Operating cost per month

	Particulars	CNG Car (₹)	EV Car (₹)
A	**Running cost:**		
	Fuel cost/ Power consumption cost [Refer WN-2]	4,500	1,425
B	**Maintenance cost:**		
	Annual Maintenance cost [Annual cost ÷12]	666.67	433.33
	Annual Insurance cost [Annual cost ÷12]	633.33	1,216.67
	Amortised cost of Tyre replacement [Refer WN-3]	177.78	133.33
	Amortised cost of Battery replacement [Refer WN-4]	66.67	4,500
		1,544.45	6,283.33
C	**Fixed cost:**		
	Depreciation [Refer WN-1]	4,583.33	10,000
	Driver's salary	20,000	20,000
	Garage rent	4,500	4,500
	Share of Office & Administration cost	1,500	1,500
		30,583.33	36,000
D	**Operating cost per month [A+B+C]**	**36,627.78**	**43,708.33**

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810 RTP **Old P3 Nov 2022** Q11 old scheme

Royal Transport Services runs fleet of buses within the limits of Jaipur city. The following are the details which were incurred by the company during October, 2021:

	(₹)
Cost of each Bus	24,00,000
Garage Rent	1,00,000
Insurance	25,000
Road tax	20,000
Manager's Salary	60,000
Assistant's Salary (Two)	32,000 each
Supervisor's Salary (Three)	24,000 each

	(₹)
Driver's Salary (Twenty-Five)	20,000 each
Cleaner's Salary (Twenty)	5,000 each
Office Staff's Salary	1,00,000
Consumables	1,20,000
Repairs & Maintenance	90,000
Other Fixed Expenses	72,000
Diesel (10 Kms per Litre)	80 per litre
Oils & Lubricants	1,45,000
Tyres and tubes	35,000
Depreciation	10% p.a. on Cost

Other details are as below:

	Capacity
12 Buses	60 Passengers
13 Buses	50 Passengers

Each bus makes 4 round trips a day covering a distance of 10 Kilometers in each trip (One Way) on an average. During the trips 80% of the seats are occupied. The annual records show that 5 buses are generally required to be kept away from roods each day for repairs.

You are required to CALCULATE cost per passenger-km.

Cost sheet to be prepared on the basis of 25 buses.

ICAI'S ANSWER

Operating Cost Sheet

Particulars	Amount (₹)	Amount (₹)
Standing Charges:		
Depreciation (₹ 24,00,000 X 10% X 1/12 X 25)	5,00,000	
Garage Rent	1,00,000	
Insurance	25,000	
Road Tax	20,000	
Manager's Salary	60,000	
Assistant's Salary (₹ 32,000 X 2)	64,000	

Particulars	Amount (₹)	Amount (₹)
Supervisor's Salary (₹ 24,000 X 3)	72,000	
Driver's Salary (₹ 20,000 X 25)	5,00,000	
Cleaner's Salary (₹ 5,000 X 20)	1,00,000	
Office Staff's Salary	1,00,000	
Consumables	1,20,000	
Repairs & Maintenance	90,000	
Other Fixed Expenses	72,000	18,23,000
Running Charges		
Diesel (49,600 Kms / 10 Kms X ₹ 80 per unit)	3,96,800	
Oils & Lubricants	1,45,000	
Tyres and tubes	35,000	5,76,800
Total Operating Cost		23,99,800

$$\text{Cost per passenger-km} = \frac{\text{Total Operating Cost}}{\text{Passenger} - \text{kms}}$$

$$= \frac{23,99,800}{27,18,080}$$

$$= 0.883$$

Working Note:

Calculation of Total Kilometers and Passenger Kilometers

Specification	Total Km.	Passenger–Km.
12 Buses (60 Passengers)	29,760 Kms (10 Kms × 4 X 2 trips × 31 days x 12 Buses)	14,28,480 (29760 Kms x 60 Pass. x 80%)
13 Buses (50 Passengers)	32,240 Kms (10 Kms × 4 X 2 trips × 31 days x 13 Buses)	12,89,600 (32240 Kms x 50 Pass. x 80%)
Total	**62,000**	**27,18,080**

Since 5 buses out of 25 buses are kept for repairs every day

Actual total Km. $62,000 \times 20/25 = 49,600$

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811 RTP **Old P3 May 2022** Q10 old scheme

Navya LMV Pvt. Ltd, operates cab/ car rental service in Delhi/NCR. It provides its service to the offices of Noida, Gurugram and Faridabad. At present it operates CNG fuelled cars but it is also considering to

upgrade these into Electric vehicle (EV). The details related with the owning of CNG & EV propelled cars are as tabulated below:

Particulars	CNG Car	EV Car
Car purchase price (₹)	9,20,000	15,20,000
Govt. subsidy on purchase of car (₹)	--	1,50,000
Life of the car	15 years	10 years
Residual value (₹)	95,000	1,70,000
Mileage	20 km/kg	240 km per charge
Electricity consumption per full charge	--	30 Kwh
CNG cost per Kg (₹)	60	--
Power cost per Kwh (₹)	--	7.60
Annual Maintenance cost (₹)	8,000	5,200
Annual insurance cost (₹)	7,600	14,600
Tyre replacement cost in every 5 -year (₹)	16,000	16,000
Battery replacement cost in every 8- year (₹)	12,000	5,40,000

Apart from the above, the following are the additional information:

Particulars	
Average distance covered by a car in a month	1,500 km
Driver's salary (₹)	20,000 p.m
Garage rent per car (₹)	4,500 p.m
Share of Office & Administration cost per car (₹)	1,500 p.m

Required:

CALCULATE the operating cost of vehicle per month per car for both CNG & EV options.

ICAI'S ANSWER

Working Notes:

1. Calculation of Depreciation per month:

	Particulars	CNG Car	EV Car
A	Car purchase price (₹)	9,20,000	15,20,000
B	*Less:* Govt. subsidy (₹)	--	(1,50,000)
C	*Less:* Residual value (₹)	(95,000)	(1,70,000)
D	Depreciable value of car (₹) [A-B-C]	8,25,000	12,00,000
E	Life of the car	15 years	10 years
F	Annual depreciation (₹) [D÷E]	55,000	1,20,000
G	**Depreciation per month (₹) [F÷12]**	**4,583.33**	**10,000**

2. Fuel/ Electricity consumption cost per month:

	Particulars	CNG Car	EV Car
A	Average distance covered in a month (KM)	1,500	1,500
B	Mileage (KM)	20	240
C	Qty. of CNG/ Full charge required [A÷B]	75 kg.	6.25
D	Electricity Consumption [C×30kwh]	-	187.5
E	Cost of CNG per kg (₹)	60	-
F	Power cost per Kwh (₹)	-	7.60
G	**CNG Cost per month (₹) [C×E]**	**4,500**	**-**
H	**Power cost per month (₹) [D×F]**	**-**	**1,425**

3. Amortised cost of Tyre replacement:

	Particulars	CNG Car	EV Car
A	Life of vehicle	15 years	10 years
B	Replacement interval	5 years	5 years
C	No. of time replacement required	2 times	1 time
D	Cost of tyres for each replacement (₹)	16,000	16,000
E	Total replacement cost (₹) [C×D]	32,000	16,000
F	Amortised cost per year (₹) [E÷A]	2,133.33	1,600
E	**Cost per month (₹) [F÷12]**	**177.78**	**133.33**

4. Amortised cost of Battery replacement:

	Particulars	CNG Car	EV Car
A	Life of vehicle	15 years	10 years
B	Replacement interval	8 years	8 years
C	No. of time replacement required	1 time	1 time
D	Cost of battery for each replacement (₹)	12,000	5,40,000
E	Total replacement cost (₹) [C×D]	12,000	5,40,000
F	Amortised cost per year (₹) [E÷A]	800	54,000
E	**Cost per month (₹) [F÷12]**	**66.67**	**4,500**

Calculation of Operating cost per month:

	Particulars	CNG Car (₹)	EV Car (₹)
A	**Running cost:**		
	Fuel cost/ Power consumption cost [Refer WN-2]	4,500	1,425
B	**Maintenance cost:**		
	Annual Maintenance cost [Annual cost ÷12]	666.67	433.33
	Annual Insurance cost [Annual cost ÷12]	633.33	1,216.67
	Amortised cost of Tyre replacement [Refer WN-3]	177.78	133.33
	Amortised cost of Battery replacement [Refer WN-4]	66.67	4,500
		1,544.45	6,283.33

	Particulars	CNG Car (₹)	EV Car (₹)
C	**Fixed cost:**		
	Depreciation [Refer WN-1]	4,583.33	10,000
	Driver's salary	20,000	20,000
	Garage rent	4,500	4,500
	Share of Office & Administration cost	1,500	1,500
		30,583.33	36,000
D	**Operating cost per month [A+B+C]**	**36,627.78**	**43,708.33**

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812 RTP **Old P3 Dec 2021** Q11 old scheme

Mr. PS owns a bus which runs according to the following schedule:

(i)	Delhi to Hisar and back, the same day	
	Distance covered:	160 km. one way
	Number of days run each month:	9
	Seating capacity occupied	90%.
(ii)	Delhi to Aligarh and back, the same day	
	Distance covered:	160 km. one way
	Number of days run each month:	12
	Seating capacity occupied	95%
(iii)	Delhi to Alwar and back, the same day	
	Distance covered:	170 km. one way
	Number of days run each month:	6
	Seating capacity occupied	100%
(iv)	Following are the other details:	
	Cost of the bus	₹ 15,00,000
	Salary of the Driver	₹ 30,000 p.m.
	Salary of the Conductor	₹ 26,000 p.m.
	Salary of the part-time Accountant	₹ 7,000 p.m.

	Insurance of the bus	₹ 6,000 p.a.
	Diesel consumption 5 km. per litre at	₹ 90 per litre
	Road tax	₹ 21,912 p.a.
	Lubricant oil	₹ 30 per 100 km.
	Permit fee	₹ 500 p.m.
	Repairs and maintenance	₹ 5,000 p.m.
	Depreciation of the bus	@ 30% p.a.
	Seating capacity of the bus	50 persons

Passenger tax is 20% of the total takings.

CALCULATE the bus fare to be charged from each passenger to earn a profit of 30% on total takings.

The fares are to be indicated per passenger for the journeys: (i) Delhi to Hisar (ii) Delhi to Aligarh and (iii) Delhi to Alwar.

ICAI'S ANSWER

Working Notes:

1. Total Distance (in km.) covered per month

Bus route	Km. per trip	Trips per day	Days per month	Km. per month
Delhi to Hisar	160	2	9	2,880
Delhi to Aligarh	160	2	12	3,840
Delhi to Alwar	170	2	6	2,040
Total				8,760

2. Passenger- km. per month

	Total seats available per month (at 100% capacity)	Capacity utilised		Km. per trip	Passenger-Km. per month
		(%)	Seats		
Delhi to Hisar & Back	900 (50 seats × 2 trips × 9 days)	90	810	160	1,29,600 (810 seats × 160 km.)
Delhi to Aligarh & Back	1,200 (50 seats × 2 trips × 12 days)	95	1,140	160	1,82,400 (1,140 seats × 160 km.)
Delhi to Alwar & Back	600 (50 seats × 2 trips × 6 days)	100	600	170	1,02,000 (600 seats × 170 km.)
Total					4,14,000

Monthly Operating Cost Statement

	Particulars	(₹)	(₹)
(i)	**Running Costs**		
	Diesel {(8,760 km ÷ 5 km) × ₹ 90}	1,57,680.00	
	Lubricant oil {(8,760 km ÷ 100) × ₹ 30}	2,628.00	1,60,308.00
(ii)	**Maintenance Costs**		
	Repairs & Maintenance		5,000.00
(iii)	**Standing charges**		
	Salary to driver	30,000.00	
	Salary to conductor	26,000.00	
	Salary of part-time accountant	7,000.00	
	Insurance (₹ 6,000 ÷12)	500.00	
	Road tax (₹ 21,912 ÷12)	1,826.00	
	Permit fee	500.00	
	Depreciation {(₹ 15,00,000 × 30%) ÷ 12}	37,500.00	1,03,326.00
	Total costs per month before Passenger Tax (i)+(ii)+(iii)		2,68,634.00
	Passenger Tax*		1,07,453.60
	Total Cost		3,76,087.60
	Add: Profit*		1,61,180.40
	Total takings per month		5,37,268.00

*Let total takings be X then,

X = Total costs per month before passenger tax + 0.2 X (passenger tax) + 0.3 X (profit)

X = ₹ 2,68,634 + 0.2 X + 0.3 X

0.5 X = ₹ 2,68,634 or, X = ₹ 5,37,268

Passenger Tax = 20% of ₹ 5,37,268 = ₹ 1,07,453.60

Profit = 30% of ₹ 5,37,268 = ₹ 1,61,180.40

Calculation of Rate per passenger km. and fares to be charged for different routes

$$\textbf{Rate per Passenger-Km.} = \frac{\text{Total takings per month}}{\text{Total Passenger - Km. per month}}$$

$$= \frac{₹\ 5,37,268}{4,14,000\ \text{Passenger-Km.}} = ₹\ 1.30\ \text{(approx.)}$$

Bus fare to be charged per passenger:

Delhi to Hisar	=	₹ 1.30 × 160 km	=	₹ 208.00
Delhi to Aligarh	=	₹ 1.30 × 160 km	=	₹ 208.00
Delhi to Alwar	=	₹ 1.30 × 170 km	=	₹ 221.00

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813 RTP **Old P3 Jul 2021** Q9 old scheme

VPS is a public school having 25 buses each plying in different directions for the transport of its school students. In view of large number of students availing of the bus service, the buses work two shifts daily both in the morning and in the afternoon. The buses are garaged in the school. The workload of the students has been so arranged that in the morning, the first trip picks up senior students and the second trip plying an hour later picks up junior students. Similarly, in the afternoon, the first trip takes the junior students and an hour later the second trip takes the senior students home.

The distance travelled by each bus, one way is 8 km. The school works 22 days in a month and remains closed for vacation in May and June. The bus fee, however, is payable by the students for all the 12 months in a year.

The details of expenses for a year are as under:

Driver's salary – payable for all the 12 in months	₹ 12,000 per month per driver
Cleaner's salary payable for all the 12 months	₹ 8,000 per month per cleaner
License fees, taxes etc.	₹ 8,400 per bus per annum
Insurance Premium	₹ 15,600 per bus per annum
Repairs and Maintenance	₹ 20,500 per bus per annum
Purchase price of the bus	₹ 20,00,000 each
Life of the bus	16 years
Scrap value	₹ 1,60,000
Diesel Cost	₹ 78.50 per litre

Each bus gives an average of 5 km. per litre of diesel. The seating capacity of each bus is 40 students.

The school follows differential transportation fees based on distance travelled as under:

Students picked up and dropped within the range of distance from the school	Transportation fee	Percentage of students availing this facility
2 km.	25% of Full	15%
4 km.	50% of Full	30%
8 km.	Full	55%

Due to a pandemic, lockdown imposed on schools and the school remained closed from April 2020 to December 2020. Drivers and cleaners were paid 75% of their salary during the lockdown period.

Repairing cost reduced to 75% for the year 2020.

Ignore the interest cost.

Required:

(i) PREPARE a statement showing the expenses of operating a single bus and the fleet of 25 buses for a year.

(ii) FIND OUT transportation fee per student per month in respect of:

(a) Students coming from a distance of upto 2 km. from the school.

(b) Students coming from a distance of upto 4 km. from the school; and

(c) Students coming from a distance of upto 8 km. from the school.

(iii) CALCULATE the minimum bus fare that has to be recovered from the students for the year 2020.

ICAI'S ANSWER

(i) Statement showing the expenses of operating a single bus and the fleet of 25 buses for a year

Particulars	Per bus per annum (₹)	Fleet of 25 buses per annum (₹)
Running costs : (A)		
Diesel (Refer to working note 1)	2,21,056	55,26,400
Repairs & maintenance costs: (B)	20,500	5,12,500
Fixed charges:		
Driver's salary (₹ 12,000 × 12 months)	1,44,000	36,00,000
Cleaners salary (₹ 8,000 × 12 months)	96,000	24,00,000
Licence fee, taxes etc.	8,400	2,10,000
Insurance	15,600	3,90,000
Depreciation ((₹ 20,00,000 – ₹1,60,000)/16 years)	1,15,000	28,75,000
Total fixed charges: (C)	3,79,000	94,75,000
Total expenses: (A+B+C)	6,20,556	1,55,13,900

(ii) Average cost per student per month in respect of students coming from a distance of:

(a)	2 km. from the school {₹ 6,20,556 / (236 students × 12 months)} (Refer to Working Note 2)	₹ 219.12
(b)	4 km. from the school (₹ 219.12 × 2)	₹ 438.24
(c)	8 km. from the school (₹ 219.12 × 4)	₹ 876.48

(iii) Calculation of minimum bus fare to be recovered from the students during the year 2020:

Statement showing the expenses of operating a single bus in year 2020

Particulars	Per bus per annum (₹)
Running costs : (A)	
Diesel (Refer to working note 3)	66,316.80
Repairs & maintenance costs: (B) (₹ 20,500 × 0.75)	15,375
Fixed charges:	
Driver's salary {₹ 12,000 × 3 months + (75% of ₹ 12,000 × 9 months)}	1,17,000
Cleaners salary {₹ 8,000 × 3 months + (75% of ₹ 8,000 × 9 months)}	78,000
Licence fee, taxes etc.	8,400
Insurance	15,600
Depreciation ((₹ 20,00,000 – ₹1,60,000)/16 years)	1,15,000
Total fixed charges: (C)	3,34,000
Total expenses: (A+B+C)	4,15,691.80

Minimum bus fare to be recovered:

(a)	2 km. from the school {₹ 4,15,691.8 / (236 students × 12 months)} (Refer to Working Note 2)	₹ 146.78
(b)	4 km. from the school (₹ 146.78 × 2)	₹ 293.56
(c)	8 km. from the school (₹146.78 × 4)	₹ 587.12

Working Notes:

1. Calculation of diesel cost per bus:

No. of trips made by a bus each day	4
Distance travelled in one trip both ways (8 km. × 2 trips)	16 km.
Distance travelled per day by a bus (16 km. × 4 shifts)	64 km.
Distance travelled during a month (64 km. × 22 days)	1,408 km.
Distance travelled per year (1,408 × 10 months)	14,080 km.
No. of litres of diesel required per bus per year (14,080 km. ÷ 5 km.)	2,816 litres
Cost of diesel per bus per year (2,816 litres × ₹ 78.50)	₹ 2,21,056

2. Calculation of equivalent number of students per bus:

Bus capacity of 2 trips (40 students × 2 trips)	80 students
1/4th fare students (15% × 80 students)	12 students
½ fare students (30% × 80 students × 2) (equivalent to 1/4th fare students)	48 students
Full fare students (55% × 80 students × 4) (equivalent to 1/4th fare students)	176 students
Total students equivalent to 1/4th fare students	236 students

3. Calculation of diesel cost per bus in Year 2020:

Distance travelled during a month (64 km. × 22 days)	1,408 km.
Distance travelled during the year 2020 (1,408 × 3 months)	4,224 km.
No. of litres of diesel required per bus per year (4,224 km. ÷ 5 km.)	844.8 litres
Cost of diesel per bus per year (844.8 litres × ₹ 78.50)	₹ 66,316.80

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814 RTP **Old P3 Nov 2020** Q11 old scheme

A transport company has 20 vehicles, the capacities are as follows:

No. of Vehicles	Capacity per vehicle
5	9 MT
6	12 MT
7	15 MT
2	20 MT

The company provides the goods transport service between stations 'A' to station 'B'. Distance between these stations is 100 kilometers. Each vehicle makes one round trip per day on an average. Vehicles are loaded with an average of 90 per cent of capacity at the time of departure from station 'A' to station 'B' and at the time of return back loaded with 70 per cent of capacity. 10 per cent of vehicles are laid up for repairs every day. The following information is related to the month of August, 2020:

Salary of Transport Manager	₹ 60,000
Salary of 30 drivers	₹ 20,000 each driver
Wages of 25 Helpers	₹ 12,000 each helper
Loading and unloading charges	₹ 850 each trip
Consumable stores (depends on running of vehicles)	₹ 1,35,000
Insurance (Annual)	₹ 8,40,000

Road Licence (Annual)	₹ 6,00,000
Cost of Diesel per litre	₹ 78
Kilometres run per litre each vehicle	5 Km.
Lubricant, Oil etc.	₹ 1,15,000
Cost of replacement of Tyres, Tubes, other parts etc. (on running basis)	₹ 4,25,000
Garage rent (Annual)	₹ 9,00,000
Routine mechanical services	₹ 3,00,000
Electricity charges (for office, garage and washing station)	₹ 55,000
Depreciation of vehicles (on time basis)	₹ 6,00,000

There is a workshop attached to transport department which repairs these vehicles and other vehicles also. 40 per cent of transport manager's salary is debited to the workshop. The transport department has been apportioned ₹88,000 by the workshop during the month. During the month operation was for 25 days.

You are required:

(i) CALCULATE per ton-km operating cost.

(ii) DETERMINE the freight to be charged per ton-km, if the company earned a profit of 25 per cent on freight.

ICAI'S ANSWER

(i) Operating Cost Sheet for the month of August, 2020

	Particulars	Amount (₹)
A.	**Fixed Charges:**	
	Manager's salary (₹60,000 × 60%)	36,000
	Drivers' Salary (₹20,000 × 30 drivers)	6,00,000
	Helpers' wages (₹12,000 × 25 helpers)	3,00,000
	Insurance (₹8,40,000 ÷ 12 months)	70,000
	Road licence (₹6,00,000 ÷ 12 months)	50,000
	Garage rent (₹9,00,000 ÷ 12 months)	75,000
	Routine mechanical services	3,00,000
	Electricity charges (for office, garage and washing station)	55,000
	Depreciation of vehicles	6,00,000
	Apportioned workshop expenses	88,000

	Particulars	Amount (₹)
	Total (A)	21,74,000
B.	**Variable Charges:**	
	Loading and unloading charges (Working Note 1)	7,65,000
	Consumable Stores	1,35,000
	Cost of diesel (Working Note 2)	14,04,000
	Lubricant, Oil etc.	1,15,000
	Replacement of Tyres, Tubes & other parts	4,25,000
	Total (B)	28,44,000
C.	**Total Cost (A + B)**	**50,18,000**
D.	Total Ton-Kms. (Working Note 3)	9,43,200
E.	**Cost per ton-km. (C ÷ D)**	**5.32**

(ii) Calculation of Chargeable Freight

Cost per ton-km.	₹ 5.32
Add: Profit @ 25% on freight or 33⅓% on cost	₹ 1.77
Chargeable freight per ton-km.	**₹ 7.09**

Working Notes:

1. **Wages paid to loading and unloading labours**

Numbers of vehicles available per day × No. of days × trips × wages per trip

(20 vehicles × 90%) × 25 days × 2 trips × ₹850

18 × 25 × 2 × 850 = ₹7,65,000

2. **Cost of Diesel:**

Distance covered by each vehicle during August, 2020

= 100 k.m. × 2 × 25 days × 90% = 4,500 km.

Consumption of diesel = $\frac{4{,}500\text{k.m.} \times 20\text{vehicles}}{5\text{k.m.}}$ = 18,000 litres.

Cost of diesel = 18,000 litres × ₹ 78 = ₹14,04,000.

3. **Calculation of total ton-km:**

Total Ton-Km. = Total Capacity × Distance covered by each vehicle × Average Capacity Utilisation ratio.

$= [(5 \times 9\ \text{MT}) + (6 \times 12\text{MT}) + (7 \times 15\ \text{MT}) + (2 \times 20\ \text{MT})] \times 4{,}500\text{k.m.} \times \frac{(90\% + 70\%)}{2}$

= $(45 + 72 + 105 + 40) \times 4{,}500$ k.m. $\times 80\%$

= 262 × 4,500 × 80%.

= 9,43,200 ton-km.

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815 MTP **Sep 2026 S-I** Q2(b) 7 marks

In a bid to promote regional tourism, the Civil Aviation Ministry has granted Sky Connect Airways Pvt. Ltd. a permit to operate four flights a week between cities P and Q (both directions), as part of its regional connectivity scheme. The airline plans to deploy a single aircraft with a seating capacity of 180 seats. Based on historical data for similar routes, the airline's revenue team estimates a normal occupancy of 65% throughout the year of 52 weeks.

The airline's costing department has compiled the following data to assess the route's profitability and to evaluate a proposed fare-reduction scheme:

Particulars	Details
One-way fare	₹ 7,800
Fuel cost (variable)	₹ 1,05,000 per flight
Food served on board (non-chargeable basis)	₹ 140 per passenger
Commission	6% of fare, applicable on all bookings
Passenger tax	₹ 150 per passenger, payable to the airport authority
Aircraft lease	₹ 3,80,000 per flight
Landing charges	₹ 78,000 per flight
Ground handling and parking charges	₹ 45,000 per flight

You are required to:

(i) CALCULATE the net operating income per flight.

(ii) The airline expects that occupancy will increase to 130 passengers per flight if the fare is reduced to ₹ 7,150. Advise, with supporting calculations, WHETHER this proposal should be implemented.

(iii) CALCULATE the Total Takings per flight for the current bookings (i.e., Net Takings plus Passenger Tax collected), which the finance team needs to report separately to the airport regulatory authority.

ICAI'S ANSWER

(i) Calculation of Net Operating Income per Flight

No. of passengers = 180 seats × 65% = 117

	(₹)	(₹)
Fare collection (117 passengers × ₹ 7,800)		9,12,600
Variable costs:		

	(₹)	(₹)
Fuel	1,05,000	
Food (117 passengers × ₹ 140)	16,380	
Commission (6% of ₹ 9,12,600)	54,756	
Passenger tax (117 passengers × ₹ 150)	17,550	1,93,686
Contribution per flight		**7,18,914**
Fixed costs:		
Aircraft Lease	3,80,000	
Landing charges	78,000	
Ground handling and parking charges	45,000	5,03,000
Net income per flight		**2,15,914**

(ii) Evaluation of Proposal (Fare Reduction to ₹7,150 with 130 Passengers)

	(₹)	(₹)
Fare collection (130 passengers × ₹ 7,150)		9,29,500
Variable costs:		
Fuel	1,05,000	
Food (130 passengers × ₹ 140)	18,200	
Commission (6% of ₹ 9,29,500)	55,770	
Passenger tax (130 passengers × ₹ 150)	19,500	1,98,470
Contribution		**7,31,030**

There is an increase in contribution by ₹ 12,116 (i.e., ₹ 7,31,030 – ₹ 7,18,914). **Hence the proposal is acceptable.**

(iii) Calculation of Total Takings per Flight

Particulars	Amount (₹)
Total operating Cost (Variable cost excl. Passenger tax + Fixed cost)	
— Variable cost (excl. passenger tax) = 1,93,686 – 17,550	1,76,136
— Add: Fixed cost	5,03,000
Total Operating Cost	**6,79,136**
Add: Profit (Net income per flight)	2,15,914
Net Takings	**8,95,050**
Add: Passenger Tax	17,550
Total Takings	**9,12,600**

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816 MTP **Sep 2026 S-II** Q2(b) 6 marks

M/s XY Travels has been given a 25 km. long route to run an air-conditioned Mini Bus. The cost of bus is ₹ 20,00,000. It has been insured @3% premium per annum while annual road tax amounts to ₹ 36,000. Annual repairs will be ₹ 50,000 and the bus is likely to last for 5 years. The driver's salary will be ₹2,40,000 per annum and the conductor's salary will be ₹ 1,80,000 per annum in addition to 10% of the takings as commission (to be shared by the driver and the conductor equally). Office and administration overheads will be ₹ 18,000 per annum. Diesel and oil will be ₹ 1,500 per 100 km. The bus will make 4 round trips carrying on an average 40 passengers on each trip.

Assuming 25% profit on takings and considering that the bus will run on an average 25 days in a month, you are required to:

(i) Prepare operating cost sheet (for the month).

(ii) Calculate fare to be charged per passenger km.

ICAI'S ANSWER

(i) **Statement showing the Operating Cost per Passenger-km.**

		Yearly (₹)	Monthly (₹)
(A)	***Standing Charges:***		
	Insurance Charge ₹. 20,00,000 × 3%	60,000	5,000
	Road Tax	36,000	3,000
	Depreciation (20,00,000/5)	4,00,000	33,333.33
	Total	4,96,000	41,333.33
(B)	***Maintenance Charges:***		
	Annual Repairs	50,000	4166.67

		Yearly (₹)	Monthly (₹)
	Office and administration overheads	3,18,000	26,500
	Total	3,68,000	30666.67
(C)	***Running Cost/Charges:***		
	Driver's Salary	2,40,000	20,000
	Conductor's Salary	1,80,000	15,000
	Diesel & Oil $\left(60{,}000 \times \frac{1{,}500}{100}\right)$	9,00,000	75,000
	Total	13,20,000	41,333.33
	Total (A+B+C) Cost before commission and profit	21,84,000	1,82,000
	Commission (33,60,000 × 10%) (working note 2)	3,36,000	28,000
	Profit (33,60,000 × 25%) (working note 2)	8,40,000	70,000
	Takings (working note 1)	33,60,000	2,80,000

(ii) Fare per Passenger-km. = $\frac{\text{Total Collection/Takings}}{\text{Total Passenger-km (Working note 3)}}$

= $\frac{33{,}60{,}000}{24{,}00{,}000}$ = ₹ 1.40

OR

Fare per Passenger-km. (monthly) = $\frac{2{,}80{,}000}{2{,}00{,}000}$ = ₹ 1.40

Working note:

1. Cost before commission (10%) and profit (25%) is 21,84,000 which is 65% of total takings. So total takings is (21,84000÷65) ×100=₹ 33,60,000

2. Commission is 10% of ₹ 33,60,000=₹ 3,36,000 and Profit is 25% of ₹ 33,60,000 = ₹ 8,40,000

3. Total Km is (4 Round Trips × Days in a month × Month = (4×2×25 ×25×12) = 60,000 km

Passenger km is 60,000 km×40 passenger= 24,00,000

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817 MTP **May 2026 S-I** Q1(c) 4 marks

A transport truck starts with a load of 30 MT of goods from Delhi. It unloads 12 MT at Agra and the remaining goods at Jaipur.

On its return journey, the truck carries 20 MT of goods from Jaipur back to Delhi. The distances between Delhi to Agra, Agra to Jaipur, and Jaipur to Delhi are 230 kms, 240 kms, and 280 kms respectively.

COMPUTE "Absolute MT-Kilometer" and "Commercial MT – Kilometer".

(MT = Metric Ton or Ton).

ICAI'S ANSWER

Weighted Average or Absolute basis – MT – Kilometer:

= (30 MT × 230 Kms) + (18 MT × 240 Kms) + (20 MT × 280 Kms)

= 6,900 + 4,320 + 5,600 = **16,820 MT - Kilometer**

Simple Average or Commercial basis – MT – Kilometer:

= [{(30 + 18 + 20) / 3} MT × {(230 + 240 + 280) Kms}]

= 22.667 MT × 750 Kms = **17,000 MT – Kilometer (approx.)**

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818 MTP **May 2026 S-II** Q2(b) 7 marks

Skyline Travels Pvt. Ltd., a New Delhi-based cab rental company, offers transportation services to tourists across major destinations including Delhi, Lucknow, and Ayodhya. With the objective of enhancing its customer base and attracting more tourists, the company has recently introduced an exclusive four-day tour package covering the route Delhi-Lucknow-Ayodhya-Delhi. This package is designed to provide a seamless and comfortable travel experience, combining convenience with cost-effectiveness. The management has compiled relevant operational and cost-related information pertaining to this package to evaluate its feasibility, pricing strategy, and profitability.

Distance Delhi to Lucknow (Km.)	555
Distance between Lucknow to Ayodhya (Km.)	135
Distance between Ayodhya to Delhi (Km.)	690
Price of diesel in Delhi	₹ 89 per litre
Price of diesel in Lucknow	₹ 92 per litre
Price of diesel in Ayodhya	₹ 93 per litre
Mileage of cab per litre of diesel (Km.)	18
Chauffeur's salary	₹ 18,000 per month
Cost of the cab	₹ 14,50,000
Expected life of the cab	3,00,000 kms.
Servicing cost	₹ 45,000 after every 40,000 kilometres run.
Chauffeur's meal allowance	₹ 120 for every 200 kilometres of completed journey
Other set up and office cost	₹ 6,000 per month.

Skyline Travels Pvt. Ltd. has entered tie-ups with fuel service centres at Delhi, Lucknow, and Ayodhya to facilitate diesel refilling for its cabs through a fuel passbook system. As per company policy, fuel is filled only to the extent required to reach the next destination.

You are required to COMPUTE the package price (inclusive of GST @ 5%) to be quoted for the four-day Delhi-Lucknow-Ayodhya-Delhi tour, assuming that the company desires to earn a profit of 25% on its net

takings (i.e., excluding GST). Take 30 days in a month.

ICAI'S ANSWER

Calculation of Price of Delhi-Lucknow-Ayodhya-Delhi tour package

Particulars	Amount (₹)	Amount (₹)
Diesel Cost (Working Note-2)		6,999.00
Servicing Cost $\left(\frac{₹\ 45{,}000}{40{,}000\ \text{kms}} \times 1380\ \text{kms.}\right)$		1,553.00
Chauffeur's meal cost (six 200 km. completed journey × ₹ 120)		720.00
Other Allocable costs:		
Depreciation $\left(\frac{₹\ 14{,}50{,}000}{3{,}00{,}000\ \text{kms}} \times 1380\ \text{kms.}\right)$	6,670.00	
Other set-up and office cost $\left(\frac{₹\ 6{,}000}{30\ \text{days}} \times 4\ \text{days}\right)$	800.00	
Chauffeur's salary $\left(\frac{₹\ 18{,}000}{30\ \text{days}} \times 4\ \text{days}\right)$	2,400.00	9,870.00
Total Cost		19,142.00
Add: Profit (25% of net takings or 1/3rd of total cost)		6,381.00
Net Takings		25,223.00
Add: GST @5%		1,276.15
Price of the package (inclusive of GST)		**26,799.15**

Working Notes

(1) Total distance of journey

From	To	Distance (in Km.)
Delhi	Lucknow	555
Lucknow	Ayodhya	135
Ayodhya	Delhi	690
Total Distance		1,380

(2) Cost of Diesel

From	To	Distance (in Km.)	Price of diesel per litre (₹)	Total diesel Cost (₹)
I	**II**	**III**	**IV**	**V = (III ÷ 18 km) × IV**
Delhi	Lucknow	555	89	2,744
Lucknow	Ayodhya	135	92	690
Ayodhya	Delhi	690	93	3,565
			Total cost	6,999

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819 MTP **Sep 2025 S-I** Q5(b) 6 marks

You are required to DETERMINE a suggested fare per passenger-km from the following information for a Mini Bus:

(i) Length of route: 30 km

(ii) Purchase price ₹ 4,00,000

(iii) Part of above cost met by loan, annual interest of which is ₹ 10,000 p.a.

(iv) Other annual charges: Insurance ₹ 15,000, Garage rent ₹ 9,000, Road tax ₹ 3,000, Repairs & maintenance ₹ 15,000, Administrative charges ₹ 5,000.

(v) Running Expenses: Driver & Conductor ₹ 5,000 p.m., Repairs/ Replacement of tyre-tube ₹ 3,600 p.a., Diesel and oil cost per km ₹ 5.

(vi) Effective life of vehicle is estimated at 5 years at the end of which it will have a scrap value of ₹ 10,000.

(vii) Mini Bus has 20 seats and is planned to make Six no. two way trips for 25 days/p.m.

(viii) Provide profit @ 20% of total revenue.

ICAI'S ANSWER

Working Notes:

1. Depreciation per annum: $= \frac{\text{Purchase price - Scrap value}}{\text{Estimated life}}$

$= \frac{₹\ 4,00,000 - ₹\ 10,000}{5 \text{ years}} = ₹\ 78,000$

2. Total distance travelled by mini-bus in 25 days:

= Length of the route (two -sides) × No. of trips per day × No. of days

= 60 km × 6 trips × 25 days = 9,000 km

3. Total Passenger-Km:

= Total distance travelled by mini-bus in 25 days × No. of seats

= 9,000 km × 20 seats = 1,80,000 passenger-km

Statement suggesting fare per passenger-km

	Cost per annum ₹	Cost per month ₹
Fixed expenses:		
Insurance	15,000	
Garage rent	9,000	
Road tax	3,000	
Administrative charges	5,000	
Depreciation	78,000	
Interest on loan	10,000	
	1,20,000	10,000
Running expenses:		
Repair and maintenance	15,000	1,250
Replacement of tyre-tube	3,600	300
Diesel and oil cost (9,000 km × ₹ 5)	-	45,000
Driver and conductor's salary	-	5,000
Total cost (per month)		61,550.00
Add: Profit 20% of total revenue or 25% of total cost		15,387.50
Total revenue		76,937.50

Rate per passenger-km ₹ 76,937.50/1,80,000 passenger km = 0.42743 i.e., = 0.43 i.e., 43 paise

ICAI · P4 · MTP Sep 2025 · Series I · question p. 15 · answer p. 17-18

820 MTP **Sep 2025 S-II** Q5(a) 9 marks

BPS is a public school that operates a comprehensive transportation system with 25 buses, each serving different routes to transport students to and from school. Given the large student population requiring bus services, the school has implemented a dual-shift system where each bus operates twice daily during both morning and afternoon sessions. The morning schedule is strategically designed with the first trip collecting senior students, followed by a second trip one hour later for junior students, while the afternoon reverses this pattern with junior students departing first and senior students following an hour later. Each bus travels a one-way distance of 8 kilometers per route, and all vehicles are housed in the school's on-site garage facility. The school works 22 days per month and observes vacation closures during May and June, yet the bus fee structure requires students to pay for all 12 months of the year regardless of the vacation periods.

The details of expenses for a year are as under:

Driver's salary – payable for all the 12 in months	₹ 12,000 per month per driver
Cleaner's salary payable for all the 12 months	₹ 8,000 per month per cleaner
License fees, taxes etc.	₹ 8,400 per bus per annum
Insurance Premium	₹ 15,600 per bus per annum
Repairs and Maintenance	₹ 20,500 per bus per annum
Purchase price of the bus	₹ 20,00,000 each
Life of the bus	16 years
Scrap value	₹ 1,60,000
Diesel Cost	₹ 84.50 per litre

Each bus gives an average of 5 km. per litre of diesel. The seating capacity of each bus is 40 students.

The school follows differential transportation fees based on distance travelled as under:

Students picked up and dropped within the range of distance from the school	Transportation fee	Percentage of students availing this facility
2 km.	25% of Full	15%
4 km.	50% of Full	30%
8 km.	Full	55%

Due to excess pollution, the school remained closed from December 2024 to January 2025. Drivers and cleaners were paid 75% of their salary during these period. Repairing cost reduced to 90% for the year 2024.

Ignore the interest cost.

Required:

(i) PREPARE a statement showing the expenses of operating a single bus and the fleet of 25 buses for a year.

(ii) FIND OUT transportation fee per student per month in respect of:

(a) Students coming from a distance of upto 2 km. from the school.

(b) Students coming from a distance of upto 4 km. from the school; and

(c) Students coming from a distance of upto 8 km. from the school.

(iii) CALCULATE the minimum bus fare that has to be recovered from the students for the year 2025.

ICAI'S ANSWER

(i) **Statement showing the expenses of operating a single bus and the fleet of 25 buses for a year**

Particulars	Per bus per annum (₹)	Fleet of 25 buses per annum (₹)
Running costs : (A)		

Particulars	Per bus per annum (₹)	Fleet of 25 buses per annum (₹)
Diesel (Refer to working note 1)	2,37,952	59,48,800
Repairs & maintenance costs: (B)	20,500	5,12,500
Fixed charges:		
Driver's salary (₹ 12,000 × 12 months)	1,44,000	36,00,000
Cleaners salary (₹ 8,000 × 12 months)	96,000	24,00,000
Licence fee, taxes etc.	8,400	2,10,000
Insurance	15,600	3,90,000
Depreciation $\left(\frac{₹\ 20{,}00{,}000 - ₹1{,}60{,}000}{16\ \text{years}}\right)$	1,15,000	28,75,000
Total fixed charges: (C)	3,79,000	94,75,000
Total expenses: (A+B+C)	6,37,452	1,59,36,300

(ii) **Average cost per student per month in respect of students coming from a distance of:**

(a) 2 km. from the school {₹ 6,37,452 / (236 students × 12 months)} (Refer to Working Note 2)	₹ 225.09
(b) 4 km. from the school (₹ 225.09 × 2)	₹ 450.18
(c) 8 km. from the school (₹ 225.09 × 4)	₹ 900.36

(iii) **Calculation of minimum bus fare to be recovered from the students during the year 2025:**

Statement showing the expenses of operating a single bus in year 2025

Particulars	Per bus per annum (₹)
Running costs : (A)	
Diesel (Refer to working note 3)	1,90,362
Repairs & maintenance costs: (B) (₹ 20,500 × 0.90)	18,450
Fixed charges:	
Driver's salary {₹ 12,000 × 10 months + (75% of ₹ 12,000 × 2 months)}	1,38,000
Cleaners' salary {₹ 8,000 × 10 months + (75% of ₹ 8,000 × 2 months)}	92,000
Licence fee, taxes etc.	8,400
Insurance	15,600
Depreciation $\left(\frac{₹\ 20{,}00{,}000 - ₹1{,}60{,}000}{16\ \text{years}}\right)$	1,15,000

Particulars	Per bus per annum (₹)
Total fixed charges: (C)	3,69,000
Total expenses: (A+B+C)	5,77,812

Minimum bus fare to be recovered:

(a) 2 km. from the school {₹ 5,77,812 / (236 students × 12 months)} (Refer to Working Note 2)	₹ 204.03
(b) 4 km. from the school (₹ 204.03 × 2)	₹ 408.06
(c) 8 km. from the school (₹ 204.03 × 4)	₹ 816.12

Working Notes:

1. **Calculation of diesel cost per bus**

Distance travelled per year (8 km x 2 trips x 4 shifts x 22 days x 10 months)	14,080 km.
No. of litres of diesel required per bus per year (14,080 km. ÷ 5 km.)	2,816 litres
Cost of diesel per bus per year (2,816 litres × ₹ 84.50)	₹ 2,37,952

2. **Calculation of equivalent number of students per bus**

Bus capacity of 2 trips (40 students × 2 trips)	80 students
1/4th fare students (15% × 80 students)	12 students
½ fare students (30% × 80 students × 2) (equivalent to 1/4th fare students)	48 students
Full fare students (55% × 80 students × 4) (equivalent to 1/4th fare students)	176 students
Total students equivalent to 1/4th fare students	236 students

3. **Calculation of diesel cost per bus in Year 2025:**

Distance travelled during a month (64 km. × 22 days)	1,408 km.
Distance travelled during the year 2025 (1,408 × 8 months)	11,264 km.
No. of litres of diesel required per bus per year (11,264 km. ÷ 5 km.)	2252.8 litres
Cost of diesel per bus per year (2252.8 litres × ₹ 84.50)	₹ 1,90,362

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821 MTP **May 2025 S-I** MCQ 12 2 marks MCQ

A truck having a capacity of 5 tonnes of goods normally carries 80% of the load on the outward journey and 40% of the load on inward journey. The distance is 300 km for one side. It takes 2 days to complete the round trip. The truck is on the road for 310 days in a year. Which one of the following is the total tonne-km in a year?

A 3,70,000

B 3,20,000

C 2,79,000

D 2,50,000

ICAI'S ANSWER

(C) 2,79,000

Tonne-Km for Outward Journey = 4 tonnes × 300 km = 1,200 tonne-km

Tonne-Km for Inward Journey = 2 tonnes × 300 km = 600 tonne-km

Total Tonne-Km per Round Trip = 1,200 + 600 = 1,800 tonne-km

Total Tonne-Km in a year = 1,800 × 155 = 2,79,000 tonne-km

ICAI · P4 · MTP May 2025 · Series I · question p. 5 · answer p. 6

822 MTP **May 2025 S-I** Q1(b) 5 marks

S Travels has been promised a contract to run a tourist car on a 20 km. long route for a multinational firm. He buys a car costing ₹ 4,50,000. The annual cost of insurance and taxes are ₹ 7,500 and ₹ 1,800 respectively. He has to pay ₹ 2,500 per month for a garage where he keeps the car when it is not in use. The annual repair costs are estimated at ₹ 12,000. The car is estimated to have a life of 10 years at the end of which the scrap value is likely to be ₹ 50,000.

He hires a driver who is to be paid ₹ 3,000 per month plus 10% of the takings as commission. Other incidental expenses are estimated at ₹ 2,000 per month.

Petrol and oil will cost ₹ 220 per 100 kms. The car will make 4 round trips each day. Assuming that a profit of 15% on takings is desired and that the car will be on the road for 25 days on an average per month, WHAT should he charge per round-trip?

ICAI'S ANSWER

Statement of Operating cost.

	Per Annum (₹)	Per Month (₹)
Standing charges:		
Depreciation [(4,50,000- 50,000)/10]	40,000	3,333.33
Insurance	7,500	625.00
Taxes	1,800	150.00
Garage (₹ 2,500 × 12)	30,000	2,500.00
Annual repairs	12,000	1,000.00
Driver's Salary (₹ 3,000× 12)	36,000	3,000.00
Incidental expenses (₹ 2,000 × 12)	24,000	2,000.00
	1,51,300	12,608.33

	Per Annum (₹)	Per Month (₹)
Variable expenses:		
Petrol and Oil (4,000* kms × 1/100 kms. × Rs. 220)		8,800.00
Total Cost (without commission)		21,408.33

[* 20 km. × 2 × 4 round trips × 25 days = 4,000 km.]

Let X be the total takings per month

Driver's Commission = 10% of X = $\frac{X}{10}$

Profit = 15% of X = $\frac{15}{100}X = \frac{3X}{20}$

Total takings per month = Total cost + Driver's Commission + Profit

or X = ₹ 21,408.33 + $\frac{X}{10} + \frac{3X}{20}$

or $X - \frac{3X}{20} - \frac{X}{10}$ = ₹ 21,408.33

or $\frac{20X - 3X - 2X}{20}$ = ₹ 21,408.33

or $\frac{15X}{20}$ = ₹ 21,408.33

or X = $\frac{\text{Rs. } 21{,}408.33 \times 4}{3}$

X = ₹ 28,544.44

Total number of round trips per month: 25 days × 4 round trips per day = 100 trips

Hence the charge per round trip = $\frac{₹28{,}544.44}{100}$

= ₹ 285.44

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823 MTP **May 2025 S-II** Case Scenario 2 · MCQ 6 to MCQ 10 10 marks

CASE SCENARIO 2

ABC Transport Services Pvt. Ltd. is a private bus company renowned for providing reliable and comfortable intercity passenger services. The company operates a fleet of buses that connect two major cities, Mumbai and Pune, which are 150 kilometers apart. By ensuring timely and efficient services, ABC Transport Services has become a preferred choice for travelers commuting between these two bustling cities.

Fleet and Operations

The company operates a total of 10 buses, each designed for optimal comfort and safety. Every bus in the fleet has a seating capacity of 50 passengers, equipped with modern amenities to enhance the travel experience. The buses adhere to strict maintenance schedules to ensure safety and reliability on the road. The company operates multiple trips daily to accommodate the high demand for travel between Mumbai and Pune. The buses normally operate at 80% capacity.

Cost Data:

Cost Category	Amount (₹)
Fixed Costs (per month)	
Insurance	2,00,000
License Fees	50,000
Salaries to Driver and Conductor	5,00,000
Garage Rent	1,00,000
Depreciation	3,00,000
Administration Expenses	1,50,000
Variable Costs (per kilometer)	
Fuel	₹ 35 per km
Lubricants and Oils	₹ 5 per km
Wages per bus (additional per trip)	₹ 10,000 per trip
Operational Data	
Number of round trips per bus in this month	20 trips
Average occupancy rate	80%

Additional Info:

In the past few months, the repairs and maintenance costs for ABC Transport Services Pvt. Ltd. have shown some variability due to fluctuating operational conditions. For instance, in April, the total repairs & maintenance costs amounted to ₹ 1,40,000, with the company reporting 18 trips per bus. In May, these costs increased to ₹ 1,60,000 due to additional maintenance activities and 22 trips per bus.

You are required to answer the following requirements (MCQs 6 to 10)

MCQ 6 2 marks

Calculate the cost per trip per bus.

A ₹ 21,750

B ₹ 29,250

C ₹ 23,450

D ₹ 28,250

MCQ 7 2 marks

Determine the total cost of operating one bus for a month.

A ₹ 5,20,000

B ₹ 4,45,000

C ₹ 6,10,000

D ₹ 5,85,000

MCQ 8 2 marks

What is the monthly revenue if each ticket is priced at ₹1,000 per trip?

A ₹ 90,00,000

B ₹ 1,00,00,000

C ₹ 80,00,000

D ₹ 75,00,000

MCQ 9 2 marks

Calculate the break-even number of passengers per trip if the ticket price is ₹635.

A 44 passengers

B 49 passengers

C 47 passengers

D 50 passengers

MCQ 10 2 marks

Calculate the cost per passenger-kilometer.

A ₹ 2.438

B ₹ 4.88

C ₹ 3.75

D ₹ 5.25

ICAI'S ANSWERS

MCQ 6

(B) ₹ 29,250

Fixed Costs per Trip per Bus

$$= \frac{\text{Total Fixed Costs}}{\text{Number of Buses x Number of Trips}}$$

= ₹ 13,00,000 / (10 x 20) = ₹ 6,500

Semi-Variable Costs (Repairs & Maintenance) per Trip per Bus

$$= \frac{\text{Total Semi} - \text{Variable Costs}}{\text{Number of Buses x Number of Trips}}$$

= ₹ 1,50,000 / (10 x 20) = ₹ 750

Variable Costs per Trip per Bus

= (Fuel + Lubricants and Oils) x Distance per Trip + Wages

= (₹ 35 + ₹ 5) x (150 kms x 2) + ₹ 10,000 = ₹ 22,000

Total Cost per Trip per Bus = ₹ 6,500 + ₹ 750 + ₹ 22,000 = ₹ 29,250

Total Fixed Costs:

- Insurance: ₹ 2,00,000
- License Fees: ₹ 50,000
- Salaries to Driver and Conductor: ₹ 5,00,000

- Garage Rent: ₹ 1,00,000
- Depreciation: ₹ 3,00,000
- Administration Expenses: ₹ 1,50,000
- **Total Fixed Costs: ₹ 13,00,000**

Repairs & Maintenance calculation

- Let x be the fixed portion of the semi-variable costs.
- Let y be the variable cost per trip.

Formulate Equations from Given Data:

- April: x + 18y = 1,40,000
- May: x + 22y = 1,60,000

Solve for y:

- Subtract the April equation from the May equation:
- (x+22y) − (x+18y) = 1,60,000 – 1,40,000
- y = 5,000
- Using the April equation:
- x + 18 x (5,000) = 1,40,000
- x = 50,000

Calculate Semi-Variable Costs for 20 Trips:

- Semi-variable costs = x + 20y
- Semi-variable costs = 50,000 + 20 x 5,000 = 1,50,000

MCQ 7

(D) ₹ 5,85,000

Fixed Costs per Bus per Month = ₹ 13,00,000 / 10 = ₹ 1,30,000

Semi-Variable Costs per Bus per Month = ₹ 1,50,000 / 10 = ₹ 15,000

Variable Costs per Trip = (Fuel + Lubricants and Oils) x Distance per Trip + Wages

= (₹ 35 + ₹ 5) x 300 + ₹ 10,000 = ₹ 22,000

Variable Costs for 20 trips = ₹ 4,40,000

Total Cost per Bus per Month = ₹ 1,30,000 + ₹ 15,000 + ₹ 4,40,000 = ₹ 5,85,000

MCQ 8

(C) ₹ 80,00,000

Monthly Revenue = Number of Buses x Number of Trips x Average Occupancy Rate x Ticket Price

= 10 x 20 x 50 x 80% x ₹ 1,000 = ₹ 80,00,000

MCQ 9

(C) 47 passengers

No. of Passengers per trip to recover total cost = Total Cost per Trip / Ticket Price

= ₹ 29,250 / ₹ 635 = 46.03 passengers per trip ≈ 47 passengers per trip

MCQ 10

(A) ₹ 2.438

Total Passenger-Kilometers = 10 buses x 20 trips x 40 passengers (50 x 80%) x 150 km x 2

= 24,00,000 passenger-kms

Cost per Passenger-Kilometer = Total Monthly Cost / Total Passenger-Kilometers

= ₹ 29,250 x (10x20) / 24,00,000

= ₹ 58,50,000/24,00,000

= ₹ 2.438 per passenger-kilometer

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824 MTP **May 2025 S-II** MCQ 13 2 marks MCQ

PG Ferry services Pvt Ltd. provide ferry services between two towns. Distance one way is 18.52 nautical miles. Seating capacity of a ferry is 125 passengers. Actual passengers carried in each trip is 80% of seating capacity. Ferry run on all days of month (30 days). Ferry makes a round trips in a day. company is expecting a monthly revenue of ₹ 55,56,000. Calculate fare to be charged from a passenger for round trip.

A 100

B 926

C 1852

D 50.95

ICAI'S ANSWER

(C) 1,852

Calculation of fare per passenger nautical mile:

$$\frac{55,56,000}{18.52 \text{ x } 100 \text{ x } 2 \text{ x } 30}$$

= 50 per passenger nautical mile

Fare for round trip = 50 x 18.52 x 2 = 1,852

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825 MTP **Jan 2025 S-II** Case Scenario I · MCQ 1 to MCQ 5 10 marks

CASE SCENARIO I

A truck driver, named Raju, owns a truck which can carry 5 tonne of material at a time. Raju has no other truck and he has listed himself with various carriage services agencies, to offer his services. He gets his work from these agencies and they pay him as per the load and the distance. Raju has one condition that he must be paid for at least 75% of his total capacity. Raju charges freight at ₹ 10 per tonne-km.

He received a work contract, from one of these agencies, where he has to take 4 tonne from Delhi in the morning and drop it off at Chandigarh. After that he will move to Ludhiana, where he again loads 3 tonne and come back to Delhi by evening. This contract is for nearly 3 months.

Raju is excited to accept the order but it is not physically possible for Raju to complete this project alone. He decides to hire a helper cum driver who will assist him in this work contract and will also drive in turns with Raju. Thus, such a long contract will be managed comfortably. This helper will take ₹ 15,000 per month.

The contract will start from 15th June, 2024 and will run till 14th September, 2024. Throughout this time period there are only 2 days holidays, both falling in August (1 for Independence Day and 1 for Raksha Bandhan).

Some information about the Truck and its associated costs:

- Truck was purchased on 1st April, 2021 by taking a loan of ₹ 20,00,000 @ 10% p.a. from Punjab national bank for 5 years. Raju mortgaged jewellery of his wife to get this loan.
- Every year-end he has to pay ₹ 5,27,595 as instalment.
- Scrap value after 10 years is expected to be ₹ 500,000.
- Depreciation is charged on straight-line method.
- Services and maintenance charges each month is ₹ 80,000.
- Truck runs on diesel and its running average is 8kms/ litre.
- Diesel cost per litre:

June	80.30
July	80.50
August	81.25
September	80.90

Yearly interest amount of loan and yearly depreciation is charged to a work contract on the basis of days worked in a year in the contract.

Distance between these places:

(1) Delhi to Chandigarh = 250 kms

(2) Chandigarh to Ludhiana = 100 kms

(3) Ludhiana to Delhi = 150 kms

Answer the following questions (MCQs 1 to 5):

MCQ 1 2 marks

What would be the amount of profit Raju would have earned if he had no minimum charges limit of 75% of total capacity on absolute Tonne-km basis? (If the vehicle runs empty then he would only charge for Diesel expenses).

A 3,34,249

B 4,43,249

C 5,96,977

D 4,34,249

MCQ 2 2 marks

If payment was made on commercial Tonne-km basis and Raju had no minimum charges limit of 75%, how much he would have lost due to no minimum requirement?

A ₹ 6,37,500

B ₹ 5,93,750

C ₹ 4,92,438

D ₹ 3,91,126

MCQ 3 2 marks

What should be the minimum amount charged on basis of absolute Tonne-km if Raju wants to earn ₹ 2,70,000?

A ₹ 4.58

B ₹ 6.13

C ₹ 8.39

D ₹ 3.21

MCQ 4 2 marks

Choose the correct amount of depreciation and interest that should be charged to this work contract.

A 56,983 & 22,588

B 36,986 & 22,578

C 63,963 & 12,568

D 63,953 & 12,558

MCQ 5 2 marks

What is the profit as per current rate charged by Raju? (Use absolute Tonne-Km).

A 7,34,249

B 9,44,863

C 5,96,977

D 4,34,249

ICAI'S ANSWERS

MCQ 1

(C) Profit if no minimum charges are there, on absolute tonne basis, but he will charge for diesel petrol when running empty

Absolute tonne-kms: (250 kms x 4 tonnes + 150 kms x 3 tonnes) x 90 days

= 1,30,500 tonne-kms

Vacant moving (Chandigarh to Ludhiana) = 100kms x 90 days = 9,000 kms

Charges for vacant running:

	(₹)
June (80.30 × 16 × 100)/8	16,060
July (80.50 × 31 × 100) /8	31,194
August (81.25 × 29 × 100) /8	29,453
September (80.90 × 14 × 100) /8	14,158
Total Charges	**90,864**

	(₹)
Total revenue (1,30,500 × 10)	13,05,000
Add: diesel recovery for vacant running	90,864
Less: service & maintenance (80,000 × 3)	(2,40,000)
Less: salary (15,000 × 3)	(45,000)
Less: diesel cost	(4,54,323)
Less: interest	(22,578)
Less: depreciation	(36,986)
Profit	**5,96,977**

Bifurcation of principal and interest

Years	Calculation of interest (₹)	Interest (₹)	Principal repayment (₹)	Loan balance (₹)
0	-	-	-	20,00,000
1	20,00,000 × 10%	2,00,000	3,27,595	16,72,405
2	16,72,405 × 10%	1,67,241	3,60,354	13,12,051
3	13,12,051 × 10%	1,31,205	3,96,390	9,15,661
4	9,15,661 × 10%	91,566	4,36,029	4,79,632
5	4,79,632 × 10%	47,963	4,79,632	-

Interest allocated to this job = 91,566 x 90 / 365 = 22,578

Depreciation = $\frac{20,00,000 - 5,00,000}{10}$ x $\frac{90}{365}$ = 36,986

Diesel expenses:

	(₹)
June (80.30 × 16 × 500)/8	80,300
July (80.50 × 31 × 500)/8	1,55,969
August (81.25 × 29 × 500)/8	1,47,266
September (80.90 × 14 × 500)/8	70,788
Total diesel expenses	**4,54,322**

MCQ 2

(A)

	With minimum limit (₹)	Without minimum limit (₹)
Commercial tonne kms	3.75 × 500 × 90 = 1,68,750	((4+0+3)/3) x 500 × 90 = 1,05,000
revenue	1,68,750 × 10 = 16,87,500	1,05,000 × 10 = 10,50,000
Less: costs	(7,98,887)	(7,98,887)
Profit/(loss)	8,88,613	2,51,113

Loss arising due to no minimum limit = 8,88,613-2,51,113 = 6,37,500

MCQ 3

(B) Total Revenue = Cost + Profit = 7,98,887+ 2,70,000 = ₹ 10,68,887

Absolute Tonne-Kms = 1,74,375

Rate = 10,68,887/1,74,375 = ₹ 6.13

MCQ 4

(B)

MCQ 5

(B) Profit at current rate (based on minimum charges of 75%)

Absolute tonne-kms: (250 kms x 4 tonnes + 100 kms x 3.75 tonnes + 150 kms x 3.75 tonnes) x 90 days = 1,74,375 tonne-kms

	(₹)
Total revenue (1,74,375 × 10)	17,43,750
Less: service & maintenance (80,000 × 3)	(2,40,000)
Less: salary (15,000 × 3)	(45,000)
Less: diesel cost	(4,54,323)
Less: interest	(22,578)
Less: depreciation	(36,986)
Profit	**9,44,863**

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826 MTP **Sep 2024 S-I** MCQ 14 2 marks MCQ

A truck carrying 10 tons of goods over 200 kilometres per day for 26 days in a month. The ton kms applicable is -

A 52,000

B 20,000

C 5200

D 260

ICAI'S ANSWER

(A) 52,000

Ton-kilometers = 10 tons x 200 kilometers x 26 days

= 52,000

ICAI · P4 · MTP Sep 2024 · Series I · question p. 5 · answer p. 5

827 MTP **Sep 2024 S-I** Q3(b) 8 marks

A mini-bus, having a capacity of 32 passengers, operates between two places - 'A' and 'B'. The distance between the place 'A' and place 'B' is 30 km. The bus makes 10 round trips in a day for 25 days in a month. On an average, the occupancy ratio is 70% and is expected throughout the year.

The details of other expenses are as under:

	Amount (₹)	
Insurance	15,600	Per annum
Garage Rent	2,400	Per quarter
Road Tax	5,000	Per annum
Repairs	4,800	Per quarter

	Amount (₹)	
Salary of operating staff	7,200	Per month
Tyres and Tubes	3,600	Per quarter
Diesel: (one litre is consumed for every 5 km)	13	Per litre
Oil and Sundries	22	Per 100 km run
Depreciation	68,000	Per annum

Passenger tax @ 22% on total taking is to be levied and bus operator requires a profit of 25% on total taking.

PREPARE operating cost statement on the annual basis and find out the cost per passenger kilometer and one way fare per passenger.

ICAI'S ANSWER

Operating Cost Statement

	Particulars	Total Cost Per annum (₹)
A.	Fixed Charges:	
	Insurance	15,600
	Garage rent (₹ 2,400 × 4 quarters)	9,600
	Road Tax	5,000
	Salary of operating staff (₹ 7,200 × 12 months)	86,400
	Depreciation	68,000
	Total (A)	1,84,600
B.	Variable Charges:	
	Repairs (₹ 4,800 × 4 quarters)	19,200
	Tyres and Tubes (₹ 3,600 × 4 quarters)	14,400
	Diesel {(1,80,000 km. ÷ 5 km.) × ₹ 13}	4,68,000
	Oil and Sundries {(1,80,000 km. ÷ 100 km.) × ₹ 22}	39,600
	Total (B)	5,41,200
	Total Operating Cost (A+B)	7,25,800
	Add: Passenger tax (Refer to WN-1)	3,01,275
	Add: Profit (Refer to WN-1)	3,42,359
	Total takings	13,69,434

Calculation of Cost per passenger kilometre and one way fare per passenger:

$$\text{Cost per Passenger-Km.} = \frac{\text{Total Operating Cost}}{\text{Total Passenger - Km.}}$$

$$= \frac{₹\ 7{,}25{,}800}{40{,}32{,}000\ \text{Passenger} - \text{Km.}} = ₹\ 0.18$$

$$\text{One way fare per passenger} = \frac{\text{Total Takings}}{\text{Total Passenger} - \text{Km.}} \times 30\text{Km.}$$

$$= \frac{₹\ 13{,}69{,}434}{40{,}32{,}000\ \text{Passenger -Km.}} \times 30\text{km} = ₹\ 10.20$$

Working Notes:

1. Let total taking be X then Passenger tax and profit will be as follows:

$X = ₹\ 7{,}25{,}800 + 0.22\ X + 0.25X$

$X - 0.47\ X = ₹\ 7{,}25{,}800$

$$X = \frac{₹7{,}25{,}800}{0.53} = ₹\ 13{,}69{,}434$$

Passenger tax = ₹ 13,69,434 × 0.22 = ₹ 3,01,275

Profit = ₹ 13,69,434 × 0.25 = ₹ 3,42,359

2. Total Kilometres to be run during the year

= 30 km.× 2 sides × 10 trips × 25 days × 12 months = 1,80,000 Kilometres

3. Total passenger Kilometres

= 1,80,000 km. × 32 passengers × 70% = 40,32,000 Passenger-km.

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828 MTP **Sep 2024 S-II** Q1(b) 5 marks

SpeedEx Logistics, established in 2010 and headquartered in Mumbai, India, operates within the transportation and logistics industry as a third-party logistics (3PL) provider. The company's fleet consists of 10 trucks, 15 vans, and 5 trailer, each serving distinct purposes. The records of Truck R-40 reveal the following information for July 2024.

Days Maintained	30
Days Operated	25
Total Hours Operated	300
Total Kilometres Covered	2,500
Total Tonnage Carried (4 tonne-load per trip, return journey empty 2 round trips per day)	

The following further information is made available:

A. Operating Costs for the month: Petrol ₹ 400, oil ₹170, Grease ₹ 90, Wages to driver ₹ 550, Wages to Worker ₹ 350.

B. Maintenance Costs for the month: Repair ₹ 170, Overhaul ₹ 60, Tyres ₹ 150, Garage charges ₹ 100.

C. Fixed Costs for the month based on the estimates for the year: Insurance ₹ 50, Licence, tax etc. ₹ 80, Interest ₹ 40, Other Overheads ₹ 190

D. Capital costs: Cost of acquisition ₹ 54,000; Residual Value at the end of 5 years life ₹ 36,000.

You are required to CALCULATE:

(i) cost per days maintained

(ii) cost per days operated

(iii) cost per hours operated

(iv) cost per kilometres covered

(v) cost per commercial tonne km

ICAI'S ANSWER

	Particulars	Amount in ₹
A	**Operating costs:**	
	Petrol	400
	Oil	170
	Grease	90
	Wages to Driver	550
	Wages to Worker	350
	(A)	1,560
B	**Maintenance Costs:**	
	Repairs	170
	Overhead	60
	Tyres	150
	Garage Charges	100
	(B)	480
C	**Fixed Cost:**	
	Insurance	50
	License, Tax etc	80
	Interest	40
	Other Overheads	190
	Depreciation $\frac{(54{,}000 - 36{,}000)}{5 \times 12}$	300
	(C)	660
	Total Cost (A + B + C)	2,700

(i) Cost per days maintained = ₹ 2700/30 days = ₹ 90

(ii) Cost per days operated = ₹ 2700/25 days = ₹ 108

(iii) Cost per hours operated = ₹ 2700/300 hours = ₹ 9

(iv) Cost per kilometres covered = ₹ 2700/2500 kms = ₹ 1.08

(v) Cost per commercial tonne kms= ₹ 2700/5000 tonne kms = ₹ 0.54

*Commercial tonne kms = Total distance travelled x Average load

$= \frac{(4 \text{ tonnes} + 0 \text{ tonnes})}{2}$ x 2500 kms

= 5000 tonne kms

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829 MTP **May 2024 S-I** Case scenario · Q2(a)(i) to Q2(a)(ii) 9 marks

CASE SCENARIO

Chiku Transport Service is a Delhi based national goods transport service provider, owning four trucks for this purpose. The cost of running and maintaining these trucks are as follows:

Particulars	Amount
Diesel cost	₹ 19.20 per km.
Engine oil	₹ 4,200 for every 13,000 km.
Repair and maintenance	₹ 36,000 for every 10,000 km.
Driver's salary	₹ 24,000 per truck per month
Cleaner's salary	₹ 15,000 per truck per month
Supervision and other general expenses	₹ 14,000 per month
Cost of loading of goods	₹ 180 per Metric Ton (MT)

All four trucks were purchased for ₹ 30 lakhs with an estimated life of 7,20,000 km each.

During the next month, it is expecting 6 bookings, the details are as follows:

Sl. No.	Journey	Distance in km	Weight-Up (in MT)	Weight- Down (in MT)
1.	Delhi to Kochi	2,700	14	6
2.	Delhi to Guwahati	1,890	12	0
3.	Delhi to Vijayawada	1,840	15	0
4.	Delhi to Varanasi	815	10	0
5.	Delhi to Asansol	1,280	12	4
6.	Delhi to Chennai	2,185	10	8
	Total	10,710	73	18

Required

Q2(a)(i) 3 marks

Calculate the total absolute Ton-km for the vehicles.

Q2(a)(ii) 6 marks

Calculate the cost per ton-km.

ICAI'S ANSWERS

Q2(a)(i)

Calculation of Absolute Ton-km for the next month:

Journey	Distance in km	Weight-Up (in MT)	Ton-km	Weight-Down (in MT)	Ton-km	Total
	(a)	(b)	(c)= (a)×(b)	(d)	(e)= (a)×(d)	(c)+(e)
Delhi to Kochi	2,700	14	37,800	6	16,200	54,000
Delhi to Guwahati	1,890	12	22,680	0	0	22,680
Delhi to Vijayawada	1,840	15	27,600	0	0	27,600
Delhi to Varanasi	815	10	8,150	0	0	8,150
Delhi to Asansol	1,280	12	15,360	4	5,120	20,480
Delhi to Chennai	2,185	10	21,850	8	17,480	39,330
Total	10,710	73	1,33,440	18	38,800	1,72,240

Total Ton-Km = 1,72,240 ton-km

Q2(a)(ii)

Calculation of cost per ton-km:

Particulars	Amount (₹)	Amount (₹)
A. Running cost:		
- Diesel Cost {₹19.20 × (10,710 × 2)}	4,11,264.00	
- Engine oil cost (₹4,200/13,000km × 21,420km)	6,920.31	
- Cost of loading of goods {₹180 × (73+18)}	16,380.00	
- Depreciation {(30,00,000/720,000×21,420 km)×4}	3,57,000.00	7,91,564.31
B. Repairs & Maintenance Cost (36,000/10,000×21,420)		77,112.00
C. Standing Charges		

Particulars	Amount (₹)	Amount (₹)
- Drivers' salary (₹24,000 × 4 trucks)	96,000.00	
- Cleaners' salary (₹15,000 × 4 trucks)	60,000.00	
- Supervision and other general exp.	14,000.00	1,70,000.00
Total Cost (A + B + C)		10,38,676.31
Total ton-km		1,72,240
Cost per ton-km		6.03

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830 MTP **May 2024 S-II** MCQ 7 2 marks MCQ

Find out the most appropriate unit cost from the following information of ZMD Transport Services Ltd. dealing in goods carriage:

Total cost = ₹ 5,25,000

Kms. Travelled = 8,75,000

Tonnes carries = 4,000

No. of Drivers = 25

No. of trucks = 20

Tonnes Km carried = 6,55,000

A ₹ 0.6

B ₹ 0.8

C ₹ 21,000

D ₹ 131.25

ICAI'S ANSWER

Answer key: **(B)**

B

Total cost = ₹ 5,25,000

Tonnes Km carried = 6,55,000

Unit Cost = ₹ 525000/655000 Km = ₹ 0.801

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831 MTP **Old P3 Nov 2023 S-I** Q5(a) 10 marks old scheme

A transport company has a fleet of three trucks of 10 tonnes capacity each plying in different directions for transport of customer's goods. The trucks run loaded with goods and return empty. The distance

travelled, number of trips made and the load carried per day by each truck are as under:

Truck No.	One way Distance Km	No. of trips per day	Load carried per trip / day tonnes
1	16	4	6
2	40	2	9
3	30	3	12

The analysis of maintenance cost and the total distance travelled during the last two years is as under

Year	Total distance travelled	Maintenance Cost (₹)
1	1,60,200	46,050
2	1,56,700	45,175

The following are the details of expenses for the year under review:

Diesel	₹ 65 per litre. Each litre gives 4 km per litre of diesel on an average.
Driver's salary	₹ 24,000 per month
Licence and taxes	₹ 25,000 per annum per truck
Insurance	₹ 45,000 per annum for all the three vehicles
Purchase Price per truck	₹ 30,00,000, Life 10 years. Scrap value at the end of life is ₹ 1,00,000.
Oil and sundries	₹ 250 per 100 km run.
General Overhead	₹ 1,15,600 per annum

The vehicles operate 24 days per month on an average.

On the basis of commercial tone-km, you are required to:

(i) PREPARE an Annual Cost Statement covering the fleet of three vehicles.

(ii) CALCULATE the cost per km. run.

(iii) DETERMINE the freight rate per tonne km. to yield a profit of 10% on freight.

ICAI'S ANSWER

(i) Annual Cost Statement of three vehicles

	(₹)
Diesel {(1,34,784 km. ÷ 4 km) × ₹ 65) (Refer to Working Note 1)	21,90,240
Oil & sundries {(1,34,784 km. ÷ 100 km.) × ₹ 250}	3,36,960
Maintenance {(1,34,784 km. × ₹ 0.25) + ₹ 6,000} (Refer to Working Note 2)	39,696
Drivers' salary {(₹24,000 × 12 months) × 3 trucks}	8,64,000

	(₹)
Licence and taxes (₹ 25,000 × 3 trucks)	75,000
Insurance	45,000
Depreciation {(₹ 29,00,000 ÷ 10 years) × 3 trucks}	8,70,000
General overhead	1,15,600
Total annual cost	45,36,496

(ii) Cost per km. run

$$\text{Cost per kilometer run} = \frac{\text{Total annual cost of vehicles}}{\text{Total kilometre travelled annually}} \text{ (Refer to Working Note 1)}$$

$$= \frac{₹\ 45{,}36{,}496}{1{,}34{,}784\ \text{Kms}} = ₹\ 33.66$$

(iii) Freight rate per tonne km (to yield a profit of 10% on freight)

$$\text{Cost per tonne km.} = \frac{\text{Total annual cost of three vehicles}}{\text{Total effective tonnes kms. per annum}} \text{ (Refer to Working Note 1)}$$

$$= \frac{₹\ 45{,}36{,}496}{6{,}06{,}528\ \text{kms}} = ₹\ 7.48$$

$$\text{Freight rate per tonne km.} \left(\frac{₹\ 7.48}{0.9}\right) \times 1 = ₹\ 8.31$$

Working Notes:

1. Total kilometer travelled and Commercial tonnes kilometer (load carried) by three trucks in one year

Truck	One way distance in kms	No. of trips	Total distance covered in km per day (with load)	Total distance covered in km per day (up & down)	Load carried per trip / day in tonnes	Total effective tonnes km
	a	b	c = a × b	d = c × 2	e	f = 27/3 × c
1	16	4	64	128	6	576
2	40	2	80	160	9	720
3	30	3	90	180	12	810
Total			234	468	27	2,106

Total kilometre travelled by three trucks in one year

(468 km. × 24 days × 12 months) = 1,34,784

Total effective tonnes kilometre of load carried by three trucks during one year

(2,106 tonnes km. × 24 days × 12 months) = 6,06,528 tonne-km

2. Fixed and variable component of maintenance cost:

$$\text{Variable maintenance cost per km.} = \frac{\text{Difference in maintenance cost}}{\text{Difference in distance travelled}}$$

$$= \frac{₹\ 46{,}050 - ₹\ 45{,}175}{1{,}60{,}200\ \text{kms} - 1{,}56{,}700\ \text{kms}} = ₹\ 0.25$$

Fixed maintenance cost = Total maintenance cost–Variable maintenance cost

= ₹ 46,050 – 1,60,200 kms × ₹ 0.25= ₹ 6,000

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832 MTP Old P3 May 2023 S-II Q5(b) 10 marks old scheme

Arnav LMV Pvt. Ltd, operates cab/ car rental service in Delhi/NCR. It provides its service to the offices of Noida, Gurugram and Faridabad. At present it operates CNG fueled cars but it is also considering to upgrade these into Electric vehicle (EV). The following details related with the owning of CNG & EV propelled cars are as tabulated below:

Particulars	CNG Car	EV Car
Car purchase price (₹)	9,20,000	15,20,000
Govt. subsidy on purchase of car (₹)	--	1,50,000
Life of the car	15 years	10 years
Residual value (₹)	95,000	1,70,000
Mileage	20 km/kg	240 km per charge
Electricity consumption per full charge	--	30 Kwh
CNG cost per Kg (₹)	90	--
Power cost per Kwh (₹)	--	7.60
Annual Maintenance cost (₹)	8,000	5,200
Annual insurance cost (₹)	7,600	14,600
Tyre replacement cost in every 5 -year (₹)	16,000	16,000
Battery replacement cost in every 8- year (₹)	12,000	5,40,000

Apart from the above, the following are the additional information:

Particulars	
Average distance covered by a car in a month	1,500 km
Driver's salary (₹)	20,000 p.m
Garage rent per car (₹)	4,500 p.m
Share of Office & Administration cost per car (₹)	1,500 p.m

Required:

(i) CALCULATE the operating cost of vehicle per month per car for both CNG & EV options.

ICAI'S ANSWER

Working Notes:

1. Calculation of Depreciation per month:

	Particulars	CNG Car	EV Car
A	Car purchase price (₹)	9,20,000	15,20,000
B	Less: Govt. subsidy (₹)	--	(1,50,000)
C	Less: Residual value (₹)	(95,000)	(1,70,000)
D	Depreciable value of car (₹) [A-B-C]	8,25,000	12,00,000
E	Life of the car	15 years	10 years
F	Annual depreciation (₹) [D÷E]	55,000	1,20,000
G	**Depreciation per month (₹) [F÷12]**	**4,583.33**	**10,000**

2. Fuel/ Electricity consumption cost per month:

	Particulars	CNG Car	EV Car
A	Average distance covered in a month (KM)	1,500	1,500
B	Mileage (KM)	20	240
C	Qty. of CNG/ Full charge required [A÷B]	75 kg.	6.25
D	Electricity Consumption [C×30kwh]	-	187.5
E	Cost of CNG per kg (₹)	90	-
F	Power cost per Kwh (₹)	-	7.60
G	**CNG Cost per month (₹) [C×E]**	**6,750**	**-**
H	**Power cost per month (₹) [D×F]**	**-**	**1,425**

3. Amortised cost of Tyre replacement:

	Particulars	CNG Car	EV Car
A	Life of vehicle	15 years	10 years
B	Replacement interval	5 years	5 years
C	No. of time replacement required	2 times	1 time
D	Cost of tyres for each replacement (₹)	16,000	16,000
E	Total replacement cost (₹) [C×D]	32,000	16,000
F	Amortised cost per year (₹) [E÷A]	2,133.33	1,600
E	**Cost per month (₹) [F÷12]**	**177.78**	**133.33**

4. Amortised cost of Battery replacement:

	Particulars	CNG Car	EV Car
A	Life of vehicle	15 years	10 years
B	Replacement interval	8 years	8 years
C	No. of time replacement required	1 time	1 time
D	Cost of battery for each replacement (₹)	12,000	5,40,000
E	Total replacement cost (₹) [C×D]	12,000	5,40,000
F	Amortised cost per year (₹) [E÷A]	800	54,000
E	**Cost per month (₹) [F÷12]**	**66.67**	**4,500**

Calculation of Operating cost per month:

	Particulars	CNG Car (₹)	EV Car (₹)
A	**Running cost:**		
	Fuel cost/ Power consumption cost [Refer WN-2]	6,750	1,425
B	**Maintenance cost:**		
	Annual Maintenance cost [Annual cost ÷12]	666.67	433.33
	Annual Insurance cost [Annual cost ÷12]	633.33	1,216.67
	Amortised cost of Tyre replacement [Refer WN-3]	177.78	133.33
	Amortised cost of Battery replacement [Refer WN-4]	66.67	4,500
		1,544.45	6,283.33

	Particulars	CNG Car (₹)	EV Car (₹)
C	**Fixed cost:**		
	Depreciation [Refer WN-1]	4,583.33	10,000
	Driver's salary	20,000	20,000
	Garage rent	4,500	4,500
	Share of Office & Administration cost	1,500	1,500
		30,583.33	36,000
D	**Operating cost per month [A+B+C]**	**38,877.78**	**43,708.33**

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833 MTP **Old P3 Nov 2022 S-I** Q1(a) 5 marks old scheme

A company has the following three alternative proposals for conveyance facilities for its sales personnel who has to do substantial traveling, approximately 20,000 kilometers yearly:

(i) Purchasing and maintaining its own fleet of cars. The average cost of a car is ₹ 7,20,000

(ii) Allow the Executive to use their own car and reimburse the expenses @ ₹ 12 per kilometer and also bear insurance costs.

(iii) Hire cars from an agency at ₹ 2,16,000 per year per car. The company will have to bear costs of petrol, taxes and tyres.

The following further details are available:

Petrol	₹ 7.20 per km.
Tyre	₹ 0.144 per km.
Taxes	₹ 960 per car per annum
Repairs and maintenance	₹ 0.24 per km.
Insurance	₹ 1,440 per car per annum
Life of the car	5 years with annual mileage of 20,000 km.
Resale value	₹ 96,000 at the end of the fifth year.

WORK OUT the relative costs of three proposals and rank them.

ICAI'S ANSWER

Calculation of relative costs of three proposals and their ranking

	I- Use of company's car per km. (₹)	II- Use of own car per km. (₹)	III- Use of hired car per km. (₹)
Reimbursement	--	12.00	--
Hire Charges	--	--	10.80*

	I- Use of company's car per km. (₹)	II- Use of own car per km. (₹)	III- Use of hired car per km. (₹)
Fixed cost:			
Insurance	0.072	0.072	--
Taxes	0.048	--	0.048
Depreciation	6.24#	--	--
Running and Maintenance Cost:			
Petrol	7.20	--	7.20
Repairs and Maintenance	0.24	--	--
Tyre	0.144	--	0.144
Total cost per km.	13.944	12.072	18.192
Cost for 20,000 km.	2,78,880	2,41,440	3,63,840
Ranking of proposals	II	I	III

*(₹ 2,16,000 ÷ 20,000 km.) = ₹ 10.80

#[(₹ 7,20,000 - ₹ 96,000) ÷ 5 years] ÷ 20,000 km. = ₹ 6.24

The Second alternative i.e., use of own car by the executive and reimbursement of expenses by the company is the best alternative from company's point of view.

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834 MTP **Old P3 Nov 2022 S-II** Q5(a) 10 marks old scheme

Royal transport company has been given a 50-kilometre-long route to run 6 buses. The cost of each bus is ₹ 75,00,000. The buses will make 3 round trips per day carrying on an average 75 percent passengers of their seating capacity. The seating capacity of each bus is 48 passengers. The buses will run on an average 25 days in a month. The other information for the year 2021-22 is given below:

Garage Rent	₹ 60,000 per month
Annual Repairs & Maintenance	₹ 2,40,000 each bus
Salaries of 6 drivers	₹ 20,000 each per month
Wages of 6 conductors	₹ 16,000 each per month
Wages of 6 cleaners	₹ 10,000 each per month
Manager's salary	₹ 50,000 per month

Road Tax, Permit fee, etc.	₹ 60,000 for a quarter
Office expenses	₹ 25,000 per month
Cost of diesel per litre	₹92
Kilometer run per litre for each bus	6 kilometres
Annual Depreciation	20% of cost
Annual Insurance	4% of cost
Engine oils & lubricants (for 1,000 kilometres)	₹ 20,000

You are required to calculate the bus fare to be charged from each passenger per kilometer (upto four decimal points), if the company wants to earn profit of $33\frac{1}{3}$ percent on taking (total receipts from passengers).

ICAI'S ANSWER

Working Notes:

1. Total Kilometres to be run during the year 2021-22

= 50 km.× 2 sides × 3 trips × 25 days × 12 months × 6 buses = 5,40,000 Kilometres

2. Total passenger Kilometres

= 5,40,000 km. × 48 passengers × 75% = 1,94,40,000 Passenger- km.

Operating Cost Sheet for the year 2021- 22

	Particulars	Total Cost (Rs.)
A.	**Fixed Charges:**	
	Garage rent (Rs. 60,000 × 12 months)	7,20,000
	Salary of drivers (Rs. 20,000 × 6 drivers ×12 months)	14,40,000
	Wages of Conductors (Rs. 16,000 × 6 conductors × 12 months)	11,52,000
	Wages of Cleaners (Rs. 10,000 × 6 cleaners × 12 months)	7,20,000
	Manager's salary (Rs.50,000 × 12 months)	6,00,000
	Road Tax, Permit fee, etc. (Rs. 60,000 × 4 quarters)	2,40,000
	Office expenses (Rs. 25,000 × 12 months)	3,00,000
	Depreciation (Rs. 75,00,000 × 6 buses × 20%)	90,00,000
	Insurance (Rs. 75,00,000 × 6 buses × 4%)	18,00,000
	Total (A)	1,59,72,000
B.	**Variable Charges:**	

	Particulars	Total Cost (Rs.)
	Repairs and Maintenance (Rs. 2,40,000 × 6 buses)	14,40,000
	Diesel {(5,40,000 km. ÷ 6 km.) × Rs.92}	82,80,000
	Engine oils & lubricants {(Rs. 20,000. ÷ 1000 km.) × 5,40,000 km}	1,08,00,000
	Total (B)	2,05,20,000
	Total Cost (A+B)	3,64,92,000
	Add: 33 1/3 % Profit on takings or 50% on cost	1,82,46,000
C.	Total Takings (Total bus fare collection)	5,47,38,000
D.	Total Passenger-km. (Working Note 2)	1,94,40,000
E.	Bus fare to be charged from each passenger per km. (C ÷ D)	2.82

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835 MTP **Old P3 May 2022 S-I** Q4(a) 10 marks old scheme

YSPP Transport Company is running local city buses. It has a fleet of 20 Buses. Each bus can carry average 40 passengers per day and cover distance of 112.50 kms per day. Due to Covid-19 pandemic, the company is running 90% buses on average.

Below are the operational expenses worked out for the month of November, 2021:

Original cost per bus	₹ 48,00,000
Insurance for 20 buses	₹ 63,36,000 per annum
Diesel & Oil	₹ 10 per km.
Salary of drivers per bus	₹ 25,000
Salary of cleaners per bus	₹ 15,000
Tyres and tubes	₹ 12,58,040
Lubricants	₹ 10,70,000
Repairs	₹ 24,70,000
Road tax per bus	₹ 1,50,000
Administrative overhead	₹ 50,88,000 per annum

Depreciation on buses is computed @ 20% using Straight Line Method.

Passenger tax is 15% on total taking.

Based on abovementioned information, you are required to COMPUTE the fare to be charged from each passenger per kilometer assuming 25% margin on total taking (Total receipts from passengers.)

ICAI'S ANSWER

Operating Cost Statement

Particulars	Total Cost Per Month (in ₹)
Fixed Charges:	
Salary of Drivers (₹ 25,000 × 20 buses)	5,00,000
Salary of Cleaners (₹ 15,000 × 20 buses)	3,00,000
Road Tax (₹ 1,50,000 × 20 buses)	30,00,000
Insurance (₹ 63,36,000/12 months)	5,28,000
Depreciation (48,00,000 × 20% × 20 buses/12 months)	16,00,000
Administrative Overheads (₹ 50,88,000/12 months)	4,24,000
Total (A)	**63,52,000**
Variable Charges:	
Diesel (60,750 km. × ₹10)	6,07,500
Tyres and Tubes	12,58,040
Lubricants	10,70,000
Repairs	24,70,000
Total (B)	**54,05,540**
Total Operating Cost (A+B)	1,17,57,540
Add: Passenger tax (Refer to WN-1)	29,39,385
Add: Profit (Refer to WN-1)	48,98,975
Total takings (C)	**1,95,95,900**
No. of passengers kms. in a month (D)	24,30,000
Cost per passenger km. (C/D)	**8.06**

Working Notes:

1. Let total takings be X then Passenger tax and profit will be as follows:

X	= ₹ 1,17,57,540 + 0.15X + 0.25X
X – 0.40X	= ₹ 1,17,57,540
X	= $\frac{1,17,57,540}{0.60}$ = ₹ 1,95,95,900
Passenger tax	= ₹ 1,95,95,900 × 0.15 = ₹ 29,39,385
Profit	= ₹ 1,95,95,900 × 0.25 = ₹ 48,98,975

2. Total Kilometres to run during the month of November, 2021
   = (112.50 km. × 30 days × 20 Buses) x 90% = 60,750 Kilometres
3. Total passenger Kilometres during the month of November, 2021
   = 60,750 km. × 40 passengers = 24,30,000 Passenger- km.

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836 MTP **Old P3 Jul 2021 S-I** Q3(b) 10 marks old scheme

GMCS Ltd. collects raw milk from the farmers of Ramgarh, Pratapgarh and Devgarh panchayats and processes this milk to make various dairy products. GMCS Ltd. has its own vehicles (tankers) to collect and bring the milk to the processing plant. Vehicles are parked in the GMCS Ltd.'s garage situated within the plant compound. Following are the information related with the vehicles:

	Ramgarh	Pratapgarh	Devgarh
No. of vehicles assigned	4	3	5
No. of trips a day	3	2	4
One way distance from the processing plant	24 k.m.	34 k.m.	16 k.m.
Fess & taxes per month (Rs.)	5,600	6,400	---

All the 5 vehicles assigned to Devgarh panchayat, were purchased five years back at a cost of Rs. 9,25,000 each. The 4 vehicles assigned to Ramgarh panchayat, were purchased two years back at a cost of Rs. 11,02,000 each and the remaining vehicles assigned to Pratapgarh were purchased last year at a cost of Rs. 13,12,000 each. With the purchase of each vehicle a two years free servicing warranty is provided. A vehicle gives 10 kmpl mileage in the first two year of purchase, 8 kmpl in next two years and 6 kmpl afterwards. The vehicles are subject to depreciation of 10% p.a. on straight line basis irrespective of usage. A vehicle has the capacity to carry 10,000 litres of milk but on an average only 70% of the total capacity is utilized.

The following expenditures are related with the vehicles:

Salary of Driver (a driver for each vehicle)	Rs. 24,000 p.m.
Salary to Cleaner (a cleaner for each vehicle)	Rs. 12,000 p.m.
Allocated garage parking fee	Rs. 4,200 per vehicle per month
Servicing cost	Rs. 15,000 for every complete 5,000 k.m. run.
Price of diesel per litre	Rs. 78.00

From the above information you are required to CALCULATE

(i) Total operating cost per month for each vehicle. (Take 30 days for the month)

(ii) Vehicle operating cost per litre of milk.

ICAI'S ANSWER

(i) Calculation of Operating Cost per month for each vehicle

	Ramgarh (Rs.)	Pratapgarh (Rs.)	Devgarh (Rs.)	Total (Rs.)
A. Running Costs:				
- Cost of diesel (Working Note- 2)	1,68,480	95,472	2,49,600	5,13,552
- Servicing cost (Working Note- 3)	45,000	-	45,000	90,000
	2,13,480	95,472	2,94,600	6,03,552
B. Fixed Costs:				
- Salary to drivers	96,000 (4 drivers × Rs. 24,000)	72,000 (3 drivers × Rs. 24,000)	1,20,000 (5 drivers × Rs. 24,000)	2,88,000
- Salary to cleaners	48,000 (4 cleaners × Rs. 12,000)	36,000 (3 cleaners × Rs. 12,000)	60,000 (5 cleaners × Rs. 12,000)	1,44,000
- Allocated garage parking fee	16,800 (4 vehicles × Rs.4,200)	12,600 (3 vehicles × Rs.4,200)	21,000 (5 vehicles × Rs.4,200)	50,400
- Depreciation (Working Note- 4)	36,733	32,800	38,542	1,08,075
- Fess & taxes	5,600	6,400	---	12,000
	2,03,133	1,59,800	2,39,542	6,02,475
Total [A + B]	4,16,613	2,55,272	5,34,142	12,06,027

	Ramgarh (Rs.)	Pratapgarh (Rs.)	Devgarh (Rs.)	Total (Rs.)
Operating Cost per vehicle	1,04,153 (Rs.4,16,613 ÷ 4 vehicles)	85,091 (Rs.2,55,272 ÷ 3 vehicles)	1,06,828 (Rs.5,34,142 ÷ 5 vehicles)	1,00,502 (Rs.12,06,027 ÷ 12 vehicles)

(ii) Vehicle operating cost per litre of milk

$$\frac{\text{Total Operating Cost per month}}{\text{Total milk carried a month}} = \frac{\text{Rs.12,06,027}}{\text{79,80,000 Litres (Working Note - 5)}} = \text{Rs. 0.15}$$

Working Notes:

1. **Distance covered by the vehicles in a month**

Route		Total Distance (in K.M.)
Ramgarh	(4 vehicles × 3 trips × 2 × 24 km. × 30 days)	17,280
Pratapgarh	(3 vehicles × 2 trips × 2 × 34 km. × 30 days)	12,240
Devgarh	(5 vehicles × 4 trips × 2 × 16 km. × 30 days)	19,200

2. **Cost of diesel consumption**

	Ramgarh	Pratapgarh	Devgarh
Total distance travelled (K.M.)	17,280	12,240	19,200
Mileage per litre of diesel	8 kmpl	10 kmpl	6 kmpl
Diesel consumption (Litre)	2,160 (17,280 ÷ 8)	1,224 (12,240 ÷ 10)	3,200 (19,200 ÷ 6)
Cost of diesel consumption @ Rs. 78 per litre (Rs.)	1,68,480	95,472	2,49,600

3. **Servicing Cost**

	Ramgarh	Pratapgarh	Devgarh
Total distance travelled (K.M.)	17,280	12,240	19,200
Covered under free service warranty	No	Yes	No
No. of services required	3 (17,280 k.m. ÷ 5,000 k.m.)	2 (12,240 k.m. ÷ 5,000 k.m.)	3 (19,200 k.m. ÷ 5,000 k.m.)
Total Service Cost (Rs.)	45,000 (Rs. 15,000 × 3)	---	45,000 (Rs. 15,000 × 3)

4. **Calculation of Depreciation**

	Ramgarh	Pratapgarh	Devgarh
No. of vehicles	4	3	5
Cost of a vehicle (Rs.)	11,02,000	13,12,000	9,25,000
Total Cost of vehicles (Rs.)	44,08,000	39,36,000	46,25,000
Depreciation per month (Rs.)	36,733 (Rs. 44,08,000×10% / 12months)	32,800 (Rs.39,36,000×10% / 12months)	38,542 (Rs.46,25,000×10% / 12months)

5. **Total volume of Milk Carried**

Route		Milk Qty. (Litre)
Ramgarh	(10,000 ltr. × 0.7 × 4 vehicles × 3 trips × 30 days)	25,20,000
Pratapgarh	(10,000 ltr. × 0.7 × 3 vehicles × 2 trips × 30 days)	12,60,000
Devgarh	(10,000 ltr. × 0.7 × 5 vehicles × 4 trips × 30 days)	42,00,000
		79,80,000

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837 MTP **Old P3 Jul 2021 S-II** Q4(a) 10 marks old scheme

Harry Transport Service is a Delhi based national goods transport service provider, owning five trucks for this purpose. The cost of running and maintaining these trucks are as follows:

Particulars	Amount
Diesel cost	Rs.15 per km.
Engine oil	Rs. 4,200 for every 14,000 km.
Repair and maintenance	Rs.12,000 for every 10,000 km.
Driver's salary	Rs. 20,000 per truck per month
Cleaner's salary	Rs. 7,000 per truck per month
Supervision and other general expenses	Rs.15,000 per month
Cost of loading of goods	Rs. 200 per Metric Ton (MT)

Each truck was purchased for Rs. 20 lakhs with an estimated life of 7,20,000 km.

During the next month, it is expecting 6 bookings, the details of which are as follows:

Sl. No.	Journey	Distance (in km)	Weight - Up (in MT)	Weight - Down (in MT)
1.	Delhi to Kochi	2,700	15	7
2.	Delhi to Guwahati	1,890	13	0
3.	Delhi to Vijayawada	1,840	16	0
4.	Delhi to Varanasi	815	11	0
5.	Delhi to Asansol	1,280	13	5
6.	Delhi to Chennai	2,185	11	9
	Total	10,710	79	21

Required:

(i) CALCULATE the total absolute Ton-km for the next month.

(ii) CALCULATE the cost per ton-km.

ICAI'S ANSWER

(i) Calculation of Absolute Ton-km for the next month:

Journey	Distance (in km)	Weight-Up (in MT)	Ton-km	Weight-Down (in MT)	Ton-km	Total
	(a)	**(b)**	**(c) = (a)×(b)**	**(d)**	**(e) = (a)×(d)**	**(f) = (c)+(e)**
Delhi to Kochi	2,700	15	40,500	7	18,900	59,400
Delhi to Guwahati	1,890	13	24,570	0	0	24,570
Delhi to Vijayawada	1,840	16	29,440	0	0	29,440
Delhi to Varanasi	815	11	8,965	0	0	8,965
Delhi to Asansol	1,280	13	16,640	5	6,400	23,040
Delhi to Chennai	2,185	11	24,035	9	19,665	43,700
Total	10,710	79	1,44,150	21	44,965	1,89,115

Total absolute Ton-Km = 1,89,115 ton-km

(ii) Calculation of cost per ton-km:

Particulars	Amount (Rs.)	Amount (Rs.)
A. Running cost:		
- Diesel Cost {Rs.15 × (10,710 × 2)}	3,21,300	
- Engine oil cost (Rs. 4,200/14,000 km × 21,420 km)	6,426	
- Cost of loading of goods {Rs.200 × (79 + 21)}	20,000	
- Depreciation (Rs.20,00,000/7,20,000km × 21,420km)	59,500	4,07,226
B. Repair & Maintenance Cost (Rs.12,000/10,000km × 21,420km)		25,704
C. Standing Charges		
- Drivers' salary (Rs.20,000 × 5 trucks)	1,00,000	
- Cleaners' salary (Rs.7,000 × 5 trucks)	35,000	
- Supervision and other general expenses	15,000	1,50,000
Total Cost (A + B + C)		5,82,930
Total absolute ton-km		1,89,115
Cost per ton-km		3.08

ICAI · Old P3 · MTP Jul 2021 · Series II · question p. 5 · answer p. 7-8

838 MTP **Old P3 Nov 2020 S-I** Q6(d) 5 marks old scheme

EXPLAIN standing charges and running charges in the case of transport organisations. LIST three examples of both.

ICAI'S ANSWER

Standing Charges: These are the fixed costs that remain constant irrespective of the distance travelled. These costs include the following-

- Insurance
- License fees
- Salary to Driver, Conductor, Cleaners, etc. if paid on monthly basis
- Garage costs, including garage rent
- Depreciation (if related to efflux of time)
- Taxes
- Administration expenses, etc.

Running Charges: These costs are generally associated with the distance travelled. These costs include the following-

- Petrol and Diesel

- Lubricant oils,
- Wages to Driver, Conductor, Cleaners, etc. if it is related to operations
- Depreciation (if related to activity)
- Any other variable costs identified.

ICAI · Old P3 · MTP Nov 2020 · Series I · question p. 8 · answer p. 15

12.3 Hotels, hospitals, power, IT and other services

41 questions · asked in 32 ICAI documents · 11 exam papers

839 EXAM Jan 2026 Q3(a) 8 marks

Gold Drive bank is having a branch which is engaged in processing of 'Gold Loan' and 'Car Loan' applications in addition to other services to customers.

The bank has hired three direct professional workers for processing 'Gold Loan' applications and four direct professional workers for processing 'Car Loan' applications.

In addition to above, following expenses are incurred by the branch:

(1) Branch manager who supervises all the activities of branch, is paid at ₹ 1,10,000 per month.

(2) Legal charges, printing & stationery and advertising expenses are incurred at ₹ 28,000, ₹ 15,000 and ₹ 20,000 respectively for a month.

(3) Other expenses are ₹ 51,000 per quarter.

(4) Depreciation on building is ₹ 4,20,000 per annum.

(5) Overhead costs of the branch allocable to processing of 'Gold Loan' applications and to 'Car Loan' applications is 20% & 15% respectively of total overhead costs of the branch.

Number of applications processed:

- 'Gold Loan' - 400 applications per month; and
- 'Car Loan' - 500 applications per month.

You are required to:

(i) Compute the total salary of direct professionals for processing of 'Gold Loan' and 'Car Loan' applications respectively, by assuming that cost of processing a 'Car Loan' application is same as 'Gold Loan' application.

(ii) Find out the total cost of processing a 'Car Loan' application.

ICAI'S ANSWER

Let the Salary of one Direct Professional Be "x"

Particulars	Total Monthly Cost (₹)	Gold loan Applications (₹)	Car loan Application (₹)
Employee Cost		3x (₹ x × 3 Professionals)	4x (₹ x × 4 Professionals)
Apportionment of Branch manager's salary	110000	22,000	16,500

Particulars	Total Monthly Cost (₹)	Gold loan Applications (₹)	Car loan Application (₹)
Legal charges, Printing & stationery and Advertising expenses	63000	12,600	9,450
Other expenses (51,000/3 quarters)	17000	3,400	2,550
Depreciation (4,20,000/12 months)	35000	7,000	5,250
Total cost per month		**45,000 + 3x**	**33,750 + 4x**
Number of Applications per month		400	500

(i) Computation of Salary of Direct Professionals

Cost per Car Loan application = Cost per Gold Loan application

$$\frac{3x + 45{,}000}{400} = \frac{4x + 33{,}750}{500}$$

1,500x + 2,25,00,000 = 1,600x + 1,35,00,000

Salary of Direct Professionals "x"= ₹ 90,000

Salary of Direct Professionals for Gold Loan applications = 3x = ₹ 2,70,000

Salary of Direct Professionals for Car Loan applications = 4x = ₹ 3,60,000

(ii) Computation of cost of processing a Car loan application:

Total Cost = 33,750 + 4x

= 33,750 + 3,60,000 = ₹ 3,93,750

Cost per Car Loan Application

$= \frac{3{,}93{,}750}{500}$ = ₹ 787.50 per application

ICAI · P4 · Jan 2026 exam · question p. 17-18 · answer p. 18-19

840 EXAM **Jan 2026** Q3(b) 6 marks

Wings Institute of Air-hostess training has reserved a spacious training hall for ₹ 25,000 on a weekly basis with a seating capacity of 320 trainees. The institute has designed an intensive training schedule comprising 3 training session per day, 5 days a week for 25 weeks, on aviation and hospitability.

Training delivered by the lead trainer to whom a honorarium of ₹ 3,600 per training session is paid. In addition, she receives travel reimbursement of ₹ 500 per day and refreshments costing ₹ 1,200 per week to maintain comfort and focus during sessions.

She also takes 2 lectures per week on safety and customer services during weekends only at ₹ 5,000 per lecture (inclusive of travelling and refreshment expenses), ensuring complete coverage of aviation education. Trainee support services covering utilities and essential training aids amount to ₹ 2,020 per week.

You are required to:

(i) Calculate the total cost per training batch.

(ii) Determine the minimum training fee per trainee in a batch to cover costs, if the batch is fully occupied.

(iii) If the institute targets 20% profit margin on the training fee and decides to charge ₹ 14,800 per trainee, then calculate minimum percentage of batch size to be filled.

ICAI'S ANSWER

(i) Calculation of Total cost per batch

Particulars	Amount (₹)
Hall Charges (₹ 25,000 × 25)	6,25,000
Honorarium of instructor (₹ 3,600 × 3 × 5 × 25)	13,50,000
Reimbursement of travel expenses (₹ 500 × 5 × 25)	62,500
Refreshment (₹ 1,200 × 25)	30,000
Lectures on Weekends (₹ 5,000 × 2 × 25)	250000
Administrative and miscellaneous expenses (₹ 2,020 × 25)	50,500
Total Cost	23,68,000
No. of Batches	1
Total cost per batch	23,68,000

(ii) Minimum fee per student in a batch to cover costs

$$= \frac{\text{Total cost per batch}}{\text{Students per batch}} = \frac{23,68,000}{320} = ₹\ 7,400$$

(iii) Total Fee to be recovered to achieve 20% profit margin on the fee

= ₹ 23,68,000 + (₹ 23,68,000 x 1/5th of sales or 1/4rd of the cost)

= ₹ 29,60,000

$$\frac{\text{Total Fee per batch}}{\text{Students per batch}} = ₹\ 14,800$$

$$\frac{29,60,000}{\text{Students per batch}} = ₹\ 14,800$$

Students per batch = 200

Minimum % of batch size to be filled to charge ₹ 14,800 per trainee

= 200/320 = 62.5%

ICAI · P4 · Jan 2026 exam · question p. 18 · answer p. 19-20

841 EXAM **Sep 2025** Q2(a) 10 marks

XYZ Highway Toll Plaza Limited operates a toll plaza on a 100 km highway and collects tolls from vehicles passing through the plaza. The company has estimated that every year a total of ₹ 60 lakh

vehicles (60% Passenger vehicles, 15% Heavy Commercial Vehicles and rest are Buses) will be using the highway during the 15 years toll collection tenure.

Toll Operating and Maintenance cost for the month (30 days in a month) are as follows:

(1) Personnel Costs (Salaries):

Collection Personnel:

- Number of shifts: 3
- Number of toll collection personnel per shift: 10
- Salary per day per person: ₹ 800

Supervisors:

- Number of shifts: 2
- Number of supervisors per shift: 3
- Salary per day per supervisor: ₹ 1,200

Security Personnel:

- Number of shifts: 3
- Number of security personnel per shift: 10
- Salary per day per security person: ₹ 500

Toll Plaza Manager:

- Number of shifts: 2
- Number of managers per shift: 1
- Salary per day per manager: ₹ 2,000

(2) Other Annual costs:

Electricity	₹ 14,40,000
Telephone & Communication Cost	₹ 2,40,000
Maintenance Cost	₹ 60,00,000
Depreciation and amortization	₹ 12,00,00,000
Insurance and safety cost	₹ 15,00,000
Interest expense incurred for servicing term loans	₹ 7,83,48,000

The toll rate per vehicle is to be fixed as under:

Heavy commercial vehicles	500% of toll rate for Passenger vehicle
Bus	400% of toll rate for Passenger vehicle

Required:

(i) Calculate the total cost per month for the toll plaza.

(ii) The company aims to achieve a 20% profit margin over total takings. Calculate the toll rate to be charged for each type of vehicle. (Assume a 360 days year)

ICAI'S ANSWER

(i) **Statement of Cost**

Particulars	Amount (₹)
Personnel Cost:	
Collection Personnel (3 × 10 × 800 × 30)	7,20,000
Supervisor (2 × 3 × 1,200 × 30)	2,16,000
Security Personnel (3 × 10 × 500 × 30)	4,50,000
Toll Plaza manager (2 × 1 × 2,000 × 30)	1,20,000
Other Cost:	
Electricity	1,20,000
Telephone and Communication	20,000
Maintenance Cost	5,00,000
Depreciation and Amortization	1,00,00,000
Insurance and Safety Cost	1,25,000
Interest Expense	65,29,000
Total Cost	**1,88,00,000**

(ii)

Total cost	₹ 1,88,00000
Profit	₹ 47,00,000
Takings (1,88,00,000/80%)	₹ 2,35,00,000
No. of vehicles (monthly)	
Passenger vehicles (5,00,000 × 60%)	3,00,000
Heavy Commercial vehicles (5,00,000 × 15%)	75,000
Buses (5,00,000 × 25%)	1,25,000
Let the toll rate for Passenger vehicles be	X
Then toll rate for Heavy Commercial vehicles be	5X
Then toll rate for Buses be	4X

Therefore

3,00,000*X +75,000*5X +1,25,000*4X = 11,75,000X = ₹ 2,35,00,000

X = ₹ 2,35,00,000/11,75,000 = ₹ 20

Toll Rate (per vehicle)

Passenger vehicles	20
Heavy Commercial vehicles	100
Buses	80

ICAI · P4 · Sep 2025 exam · question p. 12-13 · answer p. 14-15

842 EXAM May 2025 Q3(b) 6 marks

Max Cinemas has three types of seats – Classic, Prime & Recliner where the total capacity is 306 seats, which are divided in the ratio of 12: 4: 1 respectively. The ticket price of Prime is twice of Classic ticket price and that of Recliner is thrice of Prime ticket price. Following information is given:

Types of seats	Occupancy percentage
Classic	75%
Prime	50%
Recliner	50%

On a daily basis, 4 movie shows are run throughout the year. The total cost per day is estimated to be ₹ 77,760. Assume 25% profit on total revenue.

Required:

(i) Calculate Equivalent Classic seats per day.

(ii) Calculate Ticket prices of all three types of seats.

ICAI'S ANSWER

(i) Total Equivalent Classic seats per day:

Type of Seats	Occupied seats per day	Equivalent Classic seats per day
Classic	648 (216 seats x 75% x 4 shows)	648 (648 × 1)
Prime	144 (72 seats x 50% x 4 shows)	288 (144 × 2)
Recliner	36 (18 seats x 50% x 4 shows)	216 (36 × 6)
		1,152

(ii) Calculation of Ticket Prices for all types of seats:

Total cost per day = ₹ 77,760

Profit is 25% on total revenue

$\therefore$ Total Revenue = $\frac{₹77,760}{75}$ x 100 = ₹ 1,03,680

Calculation of Ticket Price for Classic seat per show:

Total Revenue / Equivalent Classic Seats per day = $\frac{₹1,03,680}{1,152}$ = ₹ 90

Ticket Prices for all three types of seats:

Ticket price per show for Classic Seat	= ₹ 90
Ticket price per show for Prime Seat	= ₹ 90 × 2 = ₹ 180
Ticket price per show for Recliner Seat	= ₹ 90 × 6 = ₹ 540

ICAI · P4 · May 2025 exam · question p. 19 · answer p. 21

843 EXAM **Sep 2024** Q4(b) 5 marks

Explain Build-Operate-Transfer (BOT) approach and classify the following expenses in Capital Cost or Operating and Maintenance Cost for Toll Roads:

(i) Land acquisition

(ii) Interest expenses incurred for servicing term loans

(iii) Material and Labour

(iv) Toll Collection Expenses

(v) Contingency Allowance

(vi) Periodic painting cost of railings etc.

ICAI'S ANSWER

Build-Operate-Transfer (BOT) Approach: BOT is an option for the Government to outsource public projects to the private sector.

With BOT, the private sector designs, finances, constructs and operates the facility and eventually, after specified concession period, the ownership is transferred to the Government. Therefore, BOT can be seen as a developing technique for infrastructure projects by making them amenable to private sector participation.

Expenses	Classification
Land acquisition	Capital Cost
Interest expenses incurred for servicing term loans	Operating and Maintenance Cost
Material and Labour	Capital Cost
Toll Collection Expenses	Operating and Maintenance Cost
Contingency Allowance	Capital Cost
Periodic painting cost of railings etc.	Operating and Maintenance Cost

ICAI · P4 · Sep 2024 exam · question p. 25-26 · answer p. 27-28

844 EXAM Old P3 Nov 2023 Q3(b) 10 marks old scheme

Royal Hotel offers three types of rooms to its guests - Deluxe Room, Executive Room and Suite Room. Other information is as follows:-

	Deluxe Room	Executive Room	Suite Room
Room Tariff per day	₹ 1,500	₹ 2,400	₹ 3,800
No. of rooms	20	10	4
Average occupancy during the year	80%	60%	75%
Housekeeping expenses per day	₹280	₹320	₹425

The hotel provides complimentary breakfast facility to its executive room and suite room guests while swimming pool facility is provided free of cost only to suite room guests. The restaurant and swimming pool is run by a contractor. The contractor recovers charges of ₹ 150 per person for breakfast and ₹ 200 per person for using swimming pool facility from Royal Hotel.

Besides the above-mentioned charges, annual fixed expenses are as follows:

Salaries to staff	₹ 57,60,000
Electricity Expenses	₹ 24,00,000

Salaries to staff are apportioned to Deluxe Room. Executive Room and Suite Room in the ratio of 25:35:40 and electricity expenses are to be apportioned in proportion to occupancy.

You are required to calculate the total profit of each room type on annual basis.

Note: Assume 360 days in a year and double occupancy in each category of room.

ICAI'S ANSWER

Calculation of room days:

Nature of Room	Occupancy (Room-days)
Deluxe room	5760 (20 × 80% x 360)
Executive room	2160 (10 × 60% x 360)
Suite room	1080 (4 × 75% x 360)

Statement showing Total Profit for each room type

Elements	Deluxe room (₹)	Executive room (₹)	Suite room (₹)	Total (₹)
Room Days	**5760**	**2160**	**1080**	
Revenue	86,40,000	51,84,000	41,04,000	1,79,28,000
Cost				
Housekeeping @ ₹ 280 per room day	16,12,800	6,91,200	4,59,000	27,63,000

Elements	Deluxe room (₹)	Executive room (₹)	Suite room (₹)	Total (₹)
Breakfast @ ₹ 150 per person	-	6,48,000	3,24,000	9,72,000
Swimming pool @ ₹ 200 per person	-	-	4,32,000	4,32,000
Salaries to staff (25:35:40)	14,40,000	20,16,000	23,04,000	57,60,000
Electricity expenses (occupancy)	15,36,000	5,76,000	2,88,000	24,00,000
Total cost	45,88,800	39,31,200	38,07,000	1,23,27,000
Profit	**40,51,200**	**12,52,800**	**2,97,000**	**56,01,000**

The solution can also be presented in following way:

Calculation of room days

Particulars	Occupancy during the year		
	Deluxe Room	**Executive Room**	**Suite Room**
(i) No. of Rooms	20	10	4
(ii) Occupancy in %	80%	60%	75%
No. of rooms occupied per day	16	6	3
No. of rooms occupied per year	5,760	2,160	1,080

Statement showing Total Profit for each room type

Annual Room Rent	Deluxe Room	Executive Room	Suite Room
Room Rent per day per room	₹ 1,500	₹ 2,400	₹ 3,800
Annual Room Rent (A)	₹ 86,40,000	₹ 51,84,000	₹ 41,04,000
Annual Fixed Expenses			
Staff Salary (25:35:40)	₹ 14,40,000	₹ 20,16,000	₹ 23,04,000
Electricity Expenses (Occupancy)	₹ 15,36,000	₹ 5,76,000	₹ 2,88,000
Total (B)	₹ 29,76,000	₹ 25,92,000	₹ 25,92,000
Housekeeping Expenses	₹ 16,12,800	₹ 6,91,200	₹ 4,59,000
Breakfast Charges		₹ 6,48,000 (2,160 × 2 x 150)	₹ 3,24,000 (1,080 × 2 x 150)
Swimming Pool Charges			₹ 4,32,000 (1,080 × 2 x 200)
Total (C)	₹ 16,12,800	₹ 13,39,200	₹12,15,000

Annual Room Rent	Deluxe Room	Executive Room	Suite Room
Total Cost (B+C)	₹ 45,88,800	₹ 39,31,200	₹ 38,07,000
Profit	**₹ 40,51,200**	**₹ 12,52,800**	**₹ 2,97,000**

ICAI · Old P3 · Nov 2023 exam · question p. 14-15 · answer p. 18-19

845 EXAM **Old P3 May 2023** Q1(d) 5 marks old scheme

RST Toll Plaza Limited built an 80-kilometre-long highway between two cities and operates a toll plaza to collect tolls from passing vehicles using the highway. The company has estimated that 50,000 light weight, 12,000 medium weight and 10,000 heavy weight vehicles will be using the highway in one month in outward journey and the same number for return journey.

As per government notification, vehicles used for medical emergencies, Members of Parliament, and essential services are exempt from toll charges. It is estimated that 10% of light weight vehicles will pass the highway for such use.

It is the policy of the company that if vehicles return within 24 hours of their outward journey, the toll fare will be reduced by 25 percent automatically. It is estimated that 30% of chargeable light weight vehicles return within the specified time frame.

The toll charges for medium weight vehicles is to be fixed as 2.5 times of the light weight vehicles and that of heavy weight vehicles as 2 times of the medium weight vehicles.

The toll and maintenance cost for a month is ₹ 59,09,090, The company requires a profit of 10% over the total cost to cover interest and other costs.

Required:

(i) Calculate the toll rate for each type of vehicle if concession facilities are not available on the return journey.

(ii) Calculate the toll rate that will be charged from light weight vehicles if a return journey concession facility is available, assuming that the revenue earned from light weight vehicles calculated in option (i) remains the same.

ICAI'S ANSWER

Working Notes:

(1) Calculation of equivalent numbers of Light weight vehicles (when no concession is provided on return journey)

Type of vehicle	Monthly traffic (A)	Return traffic (B)	Ratio (C)	Equivalent light weight [(A + B) × C]
Light weight	45,000*	45,000	1	90,000
Medium weight	12,000	12,000	2.5	60,000
Heavy weight	10,000	10,000	5	1,00,000
				2,50,000

*50,000 light vehicles less 10% exempted vehicles

(2) Calculation of equivalent numbers of Light weight vehicles (when concession is provided on return journey)

Type of vehicle	Monthly traffic (A)	Return traffic (B)	Ratio (C)	Equivalent light weight [(A + B) × C]
Light weight	45,000*	41,625 [45,000- (45,000 × 30% × 25%)]	1	86,625
Medium weight	12,000	12,000	2.5	60,000
Heavy weight	10,000	10,000	5	1,00,000
				2,46,625

(i) Calculation of toll rate for each type of vehicle:

Total cost to cover ÷ Equivalent type of vehicles

(₹ 59,09,090 + 10% of ₹ 59,09,090) ÷ 2,50,000 equivalent vehicles (Refer working note 1)

= 65,00,000 ÷ 2,50,000 = ₹ 26

Toll rate for:

Light weight vehicle = ₹ 26

Medium weight vehicle = ₹ 26 × 2.5 = ₹ 65

Heavy weight vehicle = ₹ 26 × 5 = ₹ 130

(ii) Calculation of toll rate for each type of vehicle:

Revenue earned from Light weight vehicle in (i) above

= 90,000 vehicles × ₹ 26 = ₹ 23,40,000

New toll rate to maintain the same revenue from Light weight vehicle

= ₹ 23,40,000 ÷ 86,625 (Refer working note-2) = ₹ 27.01

Light weight vehicle = ₹ 27.01

Rate to be charged from 13,500 light weight vehicles = 27.01 × 0.75 = 20.26

Alternative presentation

(ii) Toll rate to be charged from light weight vehicles if concession applicable

Revenue share in light vehicles = 90,000 × 26 = ₹ 23,40,000

Suppose rate is x, then outward journey 45,000 x; return journey (45,000 - 30% of 45,000) + 13,500 (x - 0.25)

45,000x + 31,500x + 13500 (0.75x) = ₹ 23,40,000

Toll rate to be charged from light weight vehicles : 86,625x = ₹ 23,40,000 = ₹ 27.01

Rate to be charged from 76,500 light weight vehicles @ 27.01; revenue will be ₹ 20,66,494

Rate to be charged from 13,500 light weight vehicles = 27.01 × 0.75 = 20.26 revenue will be ₹ 2,73,506

ICAI · Old P3 · May 2023 exam · question p. 2-3 · answer p. 6-7

846 EXAM Old P3 Nov 2022 Q1(b) 5 marks old scheme

ABC Bank is having a branch which is engaged in processing of 'Vehicle Loan' and 'Education Loan' applications in addition to other services to customers. 30% of the overhead costs for the branch are estimated to be applicable to the processing of 'Vehicle Loan' applications and 'Education Loan' applications each.

Branch is having four employees at a monthly salary of ₹ 50,000 each, exclusively for processing of Vehicle Loan applications and two employees at a monthly salary of ₹ 70,000 each, exclusively for processing of Education Loan applications.

In addition to above, following expense are incurred by the Branch:

- Branch Manager who supervises all the activities of branch, is paid at ₹ 90,000 per month.
- Legal charges, Printing & stationery and Advertising Expenses are incurred at ₹ 30,000, ₹ 12,000 and ₹ 18,000 respectively for a month.
- Other expenses are ₹ 10,000 per month.

You are required to:

(i) Compute the cost of processing a Vehicle Loan application on the assumption that 496 Vehicle Loan applications are processed each month.

(ii) Find out the number of Education Loan Applications processed, if the total processing cost per Education Loan Application is same as in the Vehicle Loan Application as computed in (i) above.

ICAI'S ANSWER

Particulars	Vehicle loan Applications (₹)	Education loan Application (₹)	Total (₹)
Employee Cost	2,00,000 (₹ 50,000 × 4)	1,40,000 (₹ 70,000 × 2)	3,40,000
Apportionment of Branch manager's salary	27,000	27,000	54,000
Legal charges, Printing & stationery and Advertising expenses	18,000	18,000	36,000
Other expenses	3,000	3,000	6,000
Total cost	2,48,000	1,88,000	4,36,000

(i) Computation of cost of processing a vehicle loan application:

Total Cost ÷ No. of applications

₹ 2,48,000 ÷ 496 = ₹ 500

(ii) Computation of no. of Education loan Processed

Total Cost = No. of applications × Processing cost per application

₹ 1,88,000 = No. of applications × ₹ 500

No. of education loan applications = ₹1,88,000 ÷ ₹500 = 376 applications

ICAI · Old P3 · Nov 2022 exam · question p. 1-2 · answer p. 3-4

847 EXAM Old P3 Jul 2021 Q4(b) 5 marks old scheme

MRSL Healthcare Ltd. has incurred the following expenditure during the last year for its newly launched 'COVID-19' Insurance policy:

Office administration cost	48,00,000
Claim management cost	3,80,000
Employees cost	16,20,000
Postage and logistics	32,40,000
Policy issuance cost	29,50,000
Facilities cost	46,75,000
Cost of marketing of the policy	1,38,90,000
Policy development cost	35,00,000
Policy servicing cost	96,45,000
Sales support expenses	32,00,000
I.T. Cost	?

Number of Policy sold: 2,800

Total insured value of policies - ₹ 3,500 Crores

Cost per rupee of insured value - ₹ 0.002

You are required to:

(i) Calculate Total Cost for "COVID-19" Insurance policy segregating the costs into four main activities namely (a) Marketing and Sales support (b) Operations (c) I.T. Cost and (d) Support functions.

(ii) Calculate Cost Per Policy.

ICAI'S ANSWER

(i) Calculation of total cost for 'COVID-19' Insurance policy

	Particulars	Amount (₹)	Amount (₹)
a.	**Marketing and Sales support:**		
	- Policy development cost	35,00,000	
	- Cost of marketing	1,38,90,000	
	- Sales support expenses	32,00,000	**2,05,90,000**
b.	**Operations:**		
	- Policy issuance cost	29,50,000	
	- Policy servicing cost	96,45,000	

	Particulars	Amount (₹)	Amount (₹)
	- Claim management cost	3,80,000	**1,29,75,000**
c.	**IT Cost***		**2,21,00,000**
d.	**Support functions**		
	- Postage and logistics	32,40,000	
	- Facilities cost	46,75,000	
	- Employees cost	16,20,000	
	- Office administration cost	48,00,000	**1,43,35,000**
	Total Cost		**7,00,00,000**

*IT cost

= (₹ 3,500 crores x 0.002) – ₹ 4,79,00,000 = ₹ 2,21,00,000

(ii) Calculation of cost per policy = $\frac{\text{Total cost}}{\text{No.of policies}} = \frac{₹\ 7,00,00,000}{2,800} = ₹\ 25,000$

ICAI · Old P3 · Jul 2021 exam · question p. 69 · answer p. 73-74

848 EXAM Old P3 Jul 2021 Q6(d) 5 marks old scheme

What do you understand by Build-Operate-Transfer (BOT) approach in Service Costing? How is the Toll rate computed?

ICAI'S ANSWER

Build-Operate-Transfer (BOT) Approach: In recent years a growing trend emerged among Governments in many countries to solicit investments for public projects from the private sector under BOT scheme. **BOT is an option for the Government to outsource public projects to the private sector.**

With BOT, the private sector designs, finances, constructs and operate the facility and eventually, after specified concession period, the ownership is transferred to the Government. Therefore, BOT can be seen as a developing technique for infrastructure projects by making them amenable to private sector participation.

Toll Rate: In general, the toll rate should have a direct relation with the benefits that the road users would gain from its improvements. The benefits to road users are likely to be in terms of fuel savings, improvement in travel time and good riding quality.

To compute the toll rate, following formula may be used

$= \frac{\text{Total Cost} + \text{Profit}}{\text{Number of Vehicles}}$

Or, to compute the toll rate following formula with rounding off to nearest multiple of five has been adopted: User fee = Total distance x Toll rate per km.

ICAI · Old P3 · Jul 2021 exam · question p. 78 · answer p. 80

849 EXAM Old P3 Jan 2021 Q5(a) 10 marks old scheme

ABC Health care runs an Intensive Medical Care Unit. For this purpose, it has hired a building at a rent of ₹ 50,000 per month with the agreement to bear the repairs and maintenance charges also.

The unit consists of 100 beds and 5 more beds can comfortably be accommodated when the situation demands. Though the unit is open for patients all the 365 days in a year, scrutiny of accounts for the year 2020 reveals that only for 120 days in the year, the unit had the full capacity of 100 patients per day and for another 80 days, it had, on an average only 40 beds occupied per day. But, there were occasions when the beds were full, extra beds were hired at a charge of ₹ 50 per bed per day. This did not come to more than 5 beds above the normal capacity on any one day. The total hire charges for the extra beds incurred for the whole year amounted to ₹ 20,000.

The unit engaged expert doctors from outside to attend on the patients and the fees were paid on the basis of the number of patients attended and time spent by them which on an average worked out to ₹ 30,000 per month in the year 2020.

The permanent staff expenses and other expenses of the unit were as follows:

	₹
2 Supervisors each at a per month salary of	5,000
4 Nurses each at a per month salary of	3,000
2 Ward boys each at a per month salary of	1,500
Other Expenses for the year were as under:	
Repairs and Maintenance	28,000
Food supplied to patients	4,40,000
Caretaker and Other services for patients	1,25,000
Laundry charges for bed linen	1,40,000
Medicines supplied	2,80,000
Cost of Oxygen etc. other than directly borne for treatment of patients	75,000
General Administration Charges allocated to the unit	71,000

Required:

(i) What is the profit per patient day made by the unit in the year 2020, if the unit recovered an overall amount of ₹ 200 per day on an average from each patient.

(ii) The unit wants to work on a budget for the year 2021, but the number of patients requiring medical care is a very uncertain factor. Assuming that same revenue and expenses prevail in the year 2021 in the first instance, work out the number of patient days required by the unit to break even.

ICAI'S ANSWER

Workings:

Calculation of number of Patient days

100 Beds × 120 days	=	12000
40 Beds × 80 days	=	3,200
Extra beds	=	400
Total	=	15,600

(i) Statement of Profitability

Particulars	Amount (₹)	Amount (₹)
Income for the year (₹ 200 per patient per day × 15,600 patient days)		31,20,000
Variable Costs:		
Doctor Fees (₹ 30,000 per month × 12)	3,60,000	
Food to Patients (Variable)	4,40,000	
Caretaker Other services to patients (Variable)	1,25,000	
Laundry charges (Variable)	1,40,000	
Medicines (Variable)	2,80,000	
Bed Hire Charges (₹ 50 × 400 Beds)	20,000	
Total Variable costs		(13,65,000)
Contribution		17,55,000
Fixed Costs:		
Rent (₹ 50,000 per month × 12)	6,00,000	
Supervisor (2 persons × ₹ 5,000 × 12)	1,20,000	
Nurses (4 persons × ₹ 3,000 × 12)	1,44,000	
Ward Boys (2 persons x ₹ 1500 ×12)	36,000	
Repairs (Fixed)	28,000	
Cost of Oxygen	75,000	
Administration expenses allocated	71,000	
Total Fixed Costs		(10,74,000)
Profit		**6,81,000**

Calculation of Contribution and profit per Patient day

Total Contribution	= ₹ 17,55,000
Total Patient days	= 15,600 days
Contribution per Patient day	= ₹ 17,55,000 / 15,600 days = ₹ 112.50
Total Profit	= ₹ 6,81,000
Total Patient days	= 15,600 days
Profit per Patient day	= ₹ 43.65

(ii) Breakeven Point = Fixed Cost / Contribution per Patient day

= ₹ 10,74,000 / ₹ 112.50

= 9,547 patient days

ICAI · Old P3 · Jan 2021 exam · question p. 20-21 · answer p. 22-23

850 EXAM **Old P3 Nov 2020** Q5(a) 10 marks old scheme

SEZ Ltd. built a 120 km. long highway and now operates a toll road to collect tolls. The company has invested ₹ 900 crore to build the road and has estimated that a total of 120 crore vehicles will be using the highway during the 10 years toll collection tenure. The other costs for the month of "June 2020" are as follows:

(i) Salary:

- Collection personnel (3 shifts and 5 persons per shift) - ₹ 200 per day per person.
- Supervisor (3 shifts and 2 persons per shift) - ₹ 350 per day per person.
- Security personnel (2 shifts and 2 persons per shift) - ₹ 200 per day per person.
- Toll Booth Manager (3 shifts and 1 person per shift) - ₹ 500 per day per person.

(ii) Electricity - ₹ 1,50,000

(iii) Telephone - ₹ 1,00,000

(iv) Maintenance cost - ₹ 50 lakhs

(v) The company needs 30% profit over total cost.

Required:

(1) Calculate cost per kilometre.

(2) Calculate the toll rate per vehicle.

ICAI'S ANSWER

Statement of Cost

Particulars		(₹)
A. Apportionment of capital cost	$\left(\frac{₹\ 900\text{crore}}{10\text{years}} \times \frac{1}{12\text{months}}\right)$	**7,50,00,000**
B. Other Costs		

Particulars		(₹)
Salary to Collection Personnel	(3 Shifts × 5 persons per shift × 30 days × ₹ 200 per day)	**90,000**
Salary to Supervisor	(3 Shifts × 2 persons per shift × 30 days × ₹ 350 per day)	**63,000**
Salary to Security Personnel	(2 Shifts × 2 persons per shift × 30 days × ₹ 200 per day)	**24,000**
Salary to Toll Booth Manager	(3 Shifts × 1 person per shift × 30 days × ₹ 500 per day)	**45,000**
Electricity		1,50,000
Telephone		1,00,000
		4,72,000
C. Maintenance cost		50,00,000
Total (A + B + C)		**8,04,72,000**

(1) Calculation of cost per kilometre:

$$= \frac{\text{Total Cost}}{\text{Total km.}} = \frac{₹\ 8{,}04{,}72{,}000}{120\text{km.}} = ₹\ 6{,}70{,}600$$

(2) Calculation of toll rate per vehicle:

$$= \frac{\text{Total Cost} + 25\%\ \text{profit}}{\text{Vehicles per month}} = \frac{₹\ 8{,}04{,}72{,}000 + ₹\ 2{,}41{,}41{,}600}{1{,}00{,}00{,}000\ \text{vehicles}} = \mathbf{₹\ 10.46}$$

Working:

$$\text{Vehicles per month} = \frac{\text{Total estimated vehicles}}{10\ \text{years}} \times \frac{1\ \text{month}}{12\ \text{months}}$$

$$= \frac{120\text{crore}}{10\text{years}} \times \frac{1\ \text{month}}{12\text{months}} = 1\ \text{Crore vehicles}$$

ICAI · Old P3 · Nov 2020 exam · question p. 18-19 · answer p. 20

851 EXAM **Old P3 Nov 2019** Q3(a) 10 marks old scheme

A hotel is being run in a Hill station with 200 single rooms. The hotel offers concessional rates during six off-season months in a year.

During this period, half of the full room rent is charged. The management's profit margin is targeted at 20% of the room rent. The following are the cost estimates and other details for the year ending 31st March ,2019:

(i) Occupancy during the season is 80% while in the off-season it is 40%.

(ii) Total investment in the hotel is ₹ 300 lakhs of which 80% relates to Buildings and the balance to Furniture and other Equipment.

(iii) Room attendants are paid ₹ 15 per room per day on the basis of occupancy of rooms in a month.

(iv) Expenses:

- Staff salary (excluding that of room attendants) ₹ 8,00,000
- Repairs to Buildings ₹ 3,00,000
- Laundry Charges ₹ 1,40,000
- Interior Charges ₹ 2,50,000
- Miscellaneous Expenses ₹ 2,00,200

(v) Annual Depreciation is to be provided on Buildings @ 5% and 15% on Furniture and other Equipments on straight line method.

(vi) Monthly lighting charges are ₹ 110, except in four months in winter when it is ₹ 30 per room and this cost is on the basis of full occupancy for a month.

You are required to workout the room rent chargeable per day both during the season and the off-season months using the foregoing information.

(Assume a month to be of 30 days and winter season to be considered as part of off-season).

ICAI'S ANSWER

Working Notes:

(i) Total Room days in a year

Season	Occupancy (Room-days)	Equivalent Full Room charge days
Season – 80% Occupancy	200 Rooms × 80% × 6 months × 30 days in a month = 28,800 Room Days	28,800 Room Days × 100% = 28,800
Off-season – 40% Occupancy	200 Rooms × 40% × 6 months × 30 days in a month = 14,400 Room Days	14,400 Room Days × 50% = 7,200
Total Room Days	28,800 + 14,400 = 43,200 Room Days	36,000 Full Room days

(ii) Lighting Charges:

It is given in the question that lighting charges for 8 months is ₹110 per month and during winter season of 4 months it is ₹30 per month. Further it is also given that peak season is 6 months and off season is 6 months.

It should be noted that – being Hill station, winter season is to be considered as part of Off season. Hence, the non-winter season of 8 months include – Peak season of 6 months and Off season of 2 months.

Accordingly, the lighting charges are calculated as follows:

Season	Occupancy (Room-days)
Season & Non-winter – 80% Occupancy	200 Rooms × 80% × 6 months × ₹ 110 per month = ₹ **1,05,600**
Off- season & Non-winter – 40% Occupancy (8 – 6 months)	200 Rooms × 40% × 2 months × ₹110 per month = ₹ **17,600**
Off- season & -winter – 40% Occupancy months)	200 Rooms × 40% × 4 months × ₹ 30 per month = ₹ **9,600**
Total Lighting charges	₹ 1,05,600+ ₹ 17,600 + ₹ 9,600 = ₹ **132,800**

Statement of total cost:

	(₹)
Staff salary	8,00,000
Repairs to building	3,00,000
Laundry	1,40,000
Interior	2,50,000
Miscellaneous Expenses	2,00,200
Depreciation on Building (₹ 300 Lakhs × 80% × 5%)	**12,00,000**
Depreciation on Furniture & Equipment (₹ 300 Lakhs × 20% × 15%)	**9,00,000**
Room attendant's wages (₹ 15 per Room Day for 43,200 Room Days)	**6,48,000**
Lighting charges	1,32,800
Total cost	45,71,000
Add: Profit Margin (20% on Room rent or 25% on Cost)	11,42,750
Total Rent to be charged	57,13,750

Calculation of Room Rent per day:

Total Rent / Equivalent Full Room days = ₹ 57,13,750/ 36,000 = ₹ 158.72

Room Rent during Season – ₹ 158.72

Room Rent during Off season = ₹ 158.72 × 50% = ₹ 79.36

ICAI · Old P3 · Nov 2019 exam · question p. 11 · answer p. 12-14

852 RTP **May 2026** Q12

Analysis of costs by behaviour for decision-making. The Northshire Hospital Trust operates two types of

specialist X-ray scanning machine, XR1 and XR50. Details for the next period are estimated as follows:

Machine	XR1	XR50
Running hours	1,100	2,000
	(₹)	(₹)
Variable running costs (excluding plates)	27,500	64,000
Fixed costs	20,000	97,500

A brain scan is normally carried out on machine type XR1: this task uses special X-ray plates costing ₹40 each and takes four hours of machine time. Because of the nature of the process, around 10 per cent of the scans produce blurred and therefore useless results.

Required:

(a) Calculate the cost of a satisfactory brain scan on machine type XR1.

(b) Brain scans can also be done on machine type XR50 and would take only 1.8 hours per scan with a reduced reject rate of 6 per cent. However, the cost of the X-ray plates would be ₹ 55 per scan.

Required:

ADVISE which type should be used, assuming sufficient capacity is available on both types of machines.

ICAI'S ANSWER

(a) Variable Cost per Running Hour of Machine XR1

Cost Component	Calculation	₹
Variable cost per running hour	₹ 27,500 ÷ 1,100 hours	25.00
Fixed cost per running hour	₹ 20,000 ÷ 1,100 hours	18.18

Cost of Brain Scan on Machine XR1

Cost Component	Calculation	₹
Variable machine cost	4 hours × ₹ 25	100.00
X-ray plates		40.00
Total variable cost		140.00
Fixed machine cost	4 hours × ₹ 18.182	72.73
Total cost of a scan		212.73
Total cost of a satisfactory scan	₹ 212.73 ÷ 0.9	236.37

(b) Relevant Costs for an Additional Scan

Machine	Cost Component	Calculation	₹
XR1	Variable cost per scan		140.00
	Variable cost per satisfactory scan	₹ 140 ÷ 0.9	155.56
XR50	Variable machine cost per scan	(₹ 64,000 ÷ 2,000 hours) × 1.8 hours = ₹ 32 × 1.8	57.60
	X-ray plates		55.00
	Variable cost per scan		112.60
	Variable cost per satisfactory scan	₹ 112.60 ÷ 0.94	119.79

The relevant costs per satisfactory scan are cheaper on Machine XR50 and therefore brain scans should be undertaken on this machine.

ICAI · P4 · RTP May 2026 · question p. 16-17 · answer p. 42-43

853 RTP **May 2026** Q16

Fit'n'fine Ltd. runs a fitness club in Delhi. It offers three types of membership plans: Bronze, Silver, and Gold.

The club is using a traditional costing system with the following information:

Cost Information

Description	Amount (in ₹)
Rent	1,00,000 per month
Gym equipments	30,00,000 (service life 5 years)
Licenses & Permits	86,000 (renewed in every six month)
Insurance	12,000 per annum
Salary to the instructors	7,48,000 per annum
Utilities	75,000 per quarter
Cleaning supplies	1,44,000 per annum
AMC of equipments & machinery	1,20,000 per month
Marketing expenses	3,00,000 per annum
Electricity expenses @ ₹ 14 per unit	April to July – 60 units per hour August to March – 25 units per hour

Membership Details

Type of Members	Club Access time	Occupancy % (April to July)	Occupancy % (August to March)
Gold	5.00 AM to 9.00 AM	100%	80%
Silver	6.00 PM to 9.00 PM	70%	50%
Bronze	9.00 AM to 6.00 PM	50%	30%

The maximum capacity of the fitness club is 200 persons in any time slot. The membership fee per month for Gold members is to be fixed at 1.5 times of the Silver membership and that of Silver membership as twice of the Bronze membership fee after providing profit @ 25% on membership fees. The club opens 25 days in a month.

You are required to calculate the membership fee per month to be charged by the club for each member type.

ICAI'S ANSWER

Statement showing Total Cost per annum

Particulars	Amount (₹)
Rent (1,00,000 × 12)	12,00,000
Depreciation on Gym Equipment (30,00,000 / 5)	6,00,000
Licenses & Permits (86,000 × 2)	1,72,000
Insurance	12,000
Salary to the instructors	7,48,000
Utilities (75,000 × 4)	3,00,000
Cleaning supplies	1,44,000
AMC of Equipment & Machinery (1,20,000 × 12)	14,40,000
Marketing expenses	3,00,000
Electricity Expenses April to July (25 days × 16 Hrs × 4 Month × 60 units × 14)	13,44,000
Electricity Expenses August to March (25 days × 16 Hrs × 8 Month × 25 units × 14)	11,20,000
Total cost per annum	73,80,000
Add: Profit 25% on sale or 1/3 on cost	24,60,000
Total amount to be charged	98,40,000

Calculation of per member per month Subscription charges:

Total amount to be charged = ₹ 98,40,000

Equivalent Bronze membership = 9,840

Rate per Bronze membership per month

= 98,40,000 / 9,840 = ₹ 1,000

Rate per Silver membership per month

= 1,000 × 2 = ₹ 2,000

Rate per Gold membership per month

= 2,000 × 1.5 = ₹ 3,000

Working note:

Calculation of equivalent Bronze membership:

Type of Membership	April to July	August to March
Gold	200 × 100% = 200	200 × 80% = 160
Silver	200 × 70% = 140	200 × 50% = 100
Bronze	200 × 50% = 100	200 × 30% = 60

Equivalent membership:

Type of Membership	April to July	August to March
Gold	3 × 200 = 600	3 × 160 = 480
Silver	2 × 140 = 280	2 × 100 = 200
Bronze	1 × 100 = 100	1 × 60 = 60
Total	980 × 4 = 3,920	740 × 8 = 5,920

Total = 3,920 + 5,920 = 9,840

ICAI · P4 · RTP May 2026 · question p. 19-20 · answer p. 48-49

854 **RTP** **Sep 2025** Q15

Due to frequent power cuts and inconsistent water supply from the Electricity Board, Anju Manufacturing Pvt. Ltd. — a mid-sized engineering goods manufacturer based in Pune — decided to set up its own captive power generation plant. The plant is critical for ensuring uninterrupted production across its two major units:

- Unit 1: Produces industrial machinery components (highly power-intensive operations)
- Unit 2: Manufactures agricultural equipment parts (moderate power load)

The factory operations heavily depend on continuous electric and water supply for running CNC machines, foundry operations, welding units, and assembly lines. Any electricity failure leads to major production delays, idle labour costs, and penalties from clients due to late deliveries.

After analysing the risk of depending entirely on the electricity board, the company's Board of Directors approved a proposal to invest in a dedicated steam boiler and power generation plant within the factory premises.

The following information has been taken from the records in connection with the generation of power during the month:

- Coal used 1,800 Tonnes @ ₹ 230 per tonne
- Freight and handling charges: 7% of the value of coal used.
- Oil: 15 Tonnes @ ₹ 4,500 per tonne
- Water: 80,000 litres Re. 0.07 per litre
- Steam boiler cost ₹ 50,00,000 with a residual value of ₹ 2 lakh after a life of 10 years.
- Salaries and wages of the boiler house:
  - 15 skilled workers @ ₹ 1,300 per month each
  - 30 semi-skilled workers @ 1100 per month each
  - 50 unskilled workers @ ₹ 900 per month each
- Recovery on account of sale of ashes: 150 tonnes ₹ 60 per tonne.
- Salary and wages of generating station:
  - 60 skilled workers @ ₹ 1,200 per month each
  - 20 semi-skilled workers @ 1000 per month each
  - 30 unskilled workers @ ₹ 850 per month each
- Repairs and maintenance of generating equipment: ₹ 30,000
- Depreciation of generating equipment: ₹ 75,000
- Insurance premium on generating plant: ₹ 15,000
- Fuel handling and storage expenses: ₹ 8,000
- Rent of generating station premises: ₹ 12,000
- Share of administration charges: ₹ 65,000
- Total number of units generated (before losses): 2,00,000 units
- Normal losses in the process: 3,000 units generated.
- Electricity purchased from grid to meet emergency requirements: 5,000 units @ ₹ 6 per unit.
- 10% of the effective units generated are used internally by the generator department itself

Additional Information:

- The purchased electricity units are not subject to any loss adjustment.
- Emergency purchase units should be added to the effective units available for use.

You are required to CALCULATE the total cost of electricity generated and the cost of generating electricity per unit.

ICAI'S ANSWER

Operating Cost Sheet

Particulars	Amount (₹)
Boiler House Cost	
Coal consumed (1,800 tonnes @ ₹ 230)	4,14,000
Freight, loading and unloading charges (7% of coal cost)	28,980

Particulars		Amount (₹)
Oil consumed (15 tonnes @ ₹ 4,500)	67,500	
Water consumed (80,000 litres @ Re. 0.07)	5,600	
Depreciation of steam boiler (50,00,000-2,00,000/10 × 12)	40,000	
Salaries and wages of boiler house workers:		
Skilled workers (15 @ ₹ 1,300)	19,500	
Semi-skilled workers (30 @ 1,100)	33,000	
Unskilled workers (50 @ ₹ 900)	45,000	
Less: Recovery from sale of ash (150 tonnes @ ₹ 60)	(9,000)	
Total Cost (I)		**6,44,580**
Generating Station Costs		
Salaries and wages of generating station staff:		
Skilled workers (60 @ ₹ 1,200)	72,000	
Semi-skilled workers (20 @ 1,000)	20,000	
Unskilled workers (30 @ ₹ 850)	25,500	
Repairs and maintenance of generating equipment	30,000	
Depreciation of generating equipment	75,000	
Insurance premium on generating plant	15,000	
Fuel handling and storage expenses	8,000	
Rent of generating station premises	12,000	
Share of administration and general expenses	65,000	
Total Cost (II)		**3,22,500**
Other Costs		
Electricity purchased from grid (5,000 units @ ₹ 6)	30,000	
Total Cost (III)		**30,000**
Total Cost of Electricity Generated and Purchased (I+II+III)		**9,97,080**
Units Available (W.N.)		**1,81,800 units**
Cost per unit		**5.48 per unit**

Working Note: Units Available

Particulars	Units
Total Units Generated	2,00,000
Less: Normal loss in generation	(3,000)
Net Own Units Generated	1,97,000
Add: Units purchased from grid	5,000
Total Effective Units Available	**2,02,000**
Less: 10% Units used by generating department	(20,200)
Units Available for Factory Operations	**1,81,800**

ICAI · P4 · RTP Sep 2025 · question p. 18-20 · answer p. 48-50

855 RTP May 2025 Q16

Mr. Intell newly sets up a Home Stay in Hanle, Ladakh offering two types of room, single and double. The expected occupancy percentage for the rooms is provided below:

Type of rooms	Number	Occupancy percentage
Single	20	100%
Double	10	80%

The details of the expenses as forecasted are provided below:

Particulars	(₹)
Staff salaries	30,20,000
Food and beverage costs	20,16,000
Lighting and power	8,60,000
Repairs and renovation	4,94,000
Laundry charges	3,22,000
Building rent	14,40,000
Miscellaneous expenses	6,12,000

Room attendants to be paid @ ₹ 125 per room day.

Mr. Intell knowing about the power cut during nights at Hanle to preserve the natural light for astronomical research, sets up emergency power backup, being charged from the customers separately @ ₹ 200 per room, if requested.

The rent of the double room is to be fixed at 1.5 times the single room.

In the month of June, Mr. Matrix along with his family visited Hanle Dark Sky Reserve to experience the breathtaking view of the stars. For one night stay at the place, he approached Mr. Intell to book a double bedroom and also requested for emergency power backup at night.

You are required to CALCULATE the rent to be charged from Mr. Matrix, considering the profit @ 25% on total taking.

(Assume 360 days in a year for calculation purpose.)

ICAI'S ANSWER

Working Notes:

Total equivalent single rooms

Nature of room	Occupancy (Room-days)	Equivalent single rooms (Room-days)
Single room	7,200 (20 rooms × 360 days × 100%)	7,200 (7,200 × 1)
Double rooms	2,880 (10 rooms × 360 days × 80%)	4,320 (2,880 × 1.5)
	10,080	11,520

Statement of total cost

Particulars	(₹)
Staff salaries	30,20,000
Room attendant's wages (₹ 125 per Room Day for 10,080 Room Days)	12,60,000
Food and beverage costs	20,16,000
Lighting and power	8,60,000
Repairs and renovation	4,94,000
Laundry charges	3,22,000
Building rent	14,40,000
Miscellaneous expenses	6,12,000
Total cost	**1,00,24,000**

Profit is 25% on total taking

∴ Total taking = ₹ 1,00,24,000 + 25% of total taking

Let R be rent for single room

Then 11,520 R	= 1,00,24,000 + (0.25 × 11,520 R)
Or, 8,640 R	= 1,00,24,000
Or, R	= ₹ 1,160 (approx.)
Rent to be charged for single room	= ₹ 1,160

Rent for double room ₹ 1,160 × 1.5 = ₹ 1,740

Rent to be charged from Mr. Matrix

Double room rent	₹ 1,740
Add: Power backup charges	₹ 200
	₹ 1,940

ICAI · P4 · RTP May 2025 · question p. 21 · answer p. 47-48

856 RTP **Sep 2024** Q5 MCQ

Also asked in MTP Sep 2026 S-II MCQ 12

A hotel has 200 rooms (120 Deluxe rooms and 80 Premium rooms). The normal occupancy in summer is 80% and winter 60%. The period of summer and winter is taken as 8 months and 4 months respectively. Assume 30 days in each month. Room rent of Premium room will be double of Deluxe room. Hotel is expecting a profit of 20% on total revenue, total cost for the year is 2,66,11,200. Calculate the room rent to be charged for Premium room.

A ₹ 450 per room day

B ₹ 900 per room day

C ₹ 380 per room day

D ₹ 760 per room day

ICAI'S ANSWER

(B) ₹ 900 per room day

Total Revenue (2,66,11,200/80%) = 3,32,64,000

Calculation of Room Days:

	Deluxe	Premium
Summer	120 rooms x 80% x 30 days x 8 months = 23,040	80 rooms x 80% x 30 days x 8 months = 15,360
Winter	120 rooms x 60% x 30 days x 4 months = 8,640	80 rooms x 60% x 30 days x 4 months = 5,760
Total room days	31,680	21,120

Let's assume the room rent of Deluxe room be 'x'

Then rent of Premium room will be '2x'

Therefore: 31,680x + 42,240x = 3,32,64,000

X = 450

Rent of Premium room will be 450 x 2 = ₹ 900 per room day

ICAI · P4 · RTP Sep 2024 · question p. 6 · answer p. 20-21

857 RTP Sep 2024 Q6 MCQ

ALC Ltd. is a insurance company. It launched a new term insurance policy Names as Protection Plus. The total cost for the policy during the year is ₹ 1,60,00,000. Total number of policies sold is 410 and total insured value of policies is ₹ 920 crore.

What is the cost per rupee of insured value?

A ₹ 0.0017

B ₹ 0.18

C ₹ 575

D ₹ 2.24

ICAI'S ANSWER

(A) ₹ 0.0017

Cost per rupee of insured value

= Total Cost/ Total Insured Value

= 1.6 cr/920 cr = ₹ 0.0017

ICAI · P4 · RTP Sep 2024 · question p. 6 · answer p. 21

858 RTP Sep 2024 Q16

BK Infra Ltd. built and operates a 110 k.m. long highway on the basis of Built-Operate-Transfer (BOT) model for a period of 25 year. A traffic assessment has been carried out to estimate the traffic flow per day. The details are as below:

Sl. No.	Type of vehicle	Daily traffic volume
1.	Two wheelers	44,500
2.	Car and SUVs	3,450
3.	Bus and LCV	1,800
4.	Heavy commercial vehicles	816

The following is the estimated cost of the project:

Sl. No.	Activities	Amount (₹ in lakh)
1	Site clearance	170.70
2	Land development and filling work	9,080.35
3	Sub base and base courses	10,260.70
4	Bituminous work	35,070.80
5	Bridge, flyovers, underpasses, Pedestrian subway, footbridge, etc.	29,055.60
6	Drainage and protection work	9,040.50
7	Traffic sign, marking and road appurtenance	8,405.00

Sl. No.	Activities	Amount (₹ in lakh)
8	Maintenance, repairing and rehabilitation	12,429.60
9	Environmental management	982.00
	Total Project cost	114,495.25

An average cost of ₹ 1,120 lakh has to be incurred on administration and toll plaza operation.

On the basis of the vehicle specifications (i.e. weight, size, time saving etc.), the following weights has been assigned to the passing vehicles:

Sl. No.	Type of vehicle	
1.	Two wheelers	5%
2.	Car and SUVs	20%
3.	Bus and LCV	30%
4.	Heavy commercial vehicles	45%

Required:

(i) CALCULATE the total project cost per day of concession period.

(ii) COMPUTE toll fee to be charged for per vehicle of each type, if the company wants to earn a profit of 15% on total cost.

[Note: Concession period is a period for which an infrastructure is allowed to operate and recovers its investment]

ICAI'S ANSWER

(i) Calculation of total project cost per day of concession period:

Activities	Amount (₹ in lakh)
Site clearance	170.70
Land development and filling work	9,080.35
Sub base and base courses	10,260.70
Bituminous work	35,070.80
Bridge, flyovers, underpasses, Pedestrian subway, footbridge, etc.	29,055.60
Drainage and protection work	9,040.50
Traffic sign, marking and road appurtenance	8,405.00
Maintenance, repairing and rehabilitation	12,429.60
Environmental management	982.00
Total Project cost	**114,495.25**

Activities	Amount (₹ in lakh)
Administration and toll plaza operation cost	1,120.00
Total Cost	115,615.25
Concession period in days (25 years × 365 days)	9,125
Cost per day of concession period (₹ in lakh)	**12.67**

(ii) Computation of toll fee:

Cost to be recovered per day = Cost per day of concession period + 15% profit on cost

= ₹ 12,67,000 + ₹ 1,90,050

= ₹ 14,57,050

$$\text{Cost per equivalent vehicle} = \frac{₹\ 14{,}57{,}050}{76{,}444 \text{ units (Refer working note)}}$$

= ₹ 19.06 per equivalent vehicle

Vehicle type-wise toll fee:

Sl. No.	Type of vehicle	Equivalent cost [A]	Weight [B]	Toll fee per vehicle [A×B]
1.	Two wheelers	₹ 19.06	1	19.06
2.	Car and SUVs	₹ 19.06	4	76.24
3.	Bus and LCV	₹ 19.06	6	114.36
4.	Heavy commercial vehicles	₹ 19.06	9	171.54

Working Note:

The cost per day has to be recovered from the daily traffic. Each type of vehicle is to be converted into equivalent unit. Let's convert all vehicle types equivalent to Two-wheeler

Sl. No.	Type of vehicle	Daily traffic volume [A]	Weight	Ratio [B]	Equivalent Two-wheeler [A×B]
1.	Two wheelers	44,500	0.05	1	44,500
2.	Car and SUVs	3,450	0.20	4	13,800
3.	Bus and LCV	1,800	0.30	6	10,800
4.	Heavy commercial vehicles	816	0.45	9	7,344
	Total				76,444

ICAI · P4 · RTP Sep 2024 · question p. 13-14 · answer p. 32-33

859 RTP Old P3 Nov 2023 Q10 old scheme

P Holiday Resorts offers three types of rooms to its guests, viz deluxe room, super deluxe room and luxury suite. You are required to ascertain the tariff to be charged to the customers for different types of rooms on the basis of following information:

Types of Room	Number of Rooms	Occupancy
Deluxe Room	100	90%
Super Deluxe Room	60	75%
Luxury Suite	40	60%

Rent of 'super deluxe' room is to be fixed at 2 times of 'deluxe room' and that of 'luxury suite' is 3 times of 'deluxe room'. Annual expenses are as follows:

Particulars	Amount (₹ lakhs)
Staff salaries	680.00
Lighting, Heating and Power	300.00
Repairs, Maintenance and Renovation	180.00
Linen	30.00
Laundry charges	24.00
Interior decoration	75.00
Sundries	30.28

An attendant for each room was provided when the room was occupied and he was paid ₹ 500 per day towards wages. Further, depreciation is to be provided on building @ 5% on ₹ 900 lakhs, furniture and fixtures @ 10% on ₹ 90 lakhs and air conditioners @ 10% on ₹ 75 lakhs.

Profit is to be provided @ 25% on total taking and assume 360 days in a year.

ICAI'S ANSWER

Operating cost statement of P Holiday Resorts

Particulars	Cost per annum (₹ In lakhs)
Staff Salaries	680.00
Room Attendant's Wages (Refer WN-3)	286.20
Lighting, Heating & Power	300.00
Repairs, Maintenance & Renovation	180.00
Linen	30.00
Laundry charges	24.00
Interior Decoration	75.00

Particulars	Cost per annum (₹ In lakhs)
Sundries	30.28
Depreciation: (Refer WN-4)	
Building	45.00
Furniture & Fixture	9.00
Air Conditioners	7.50
Total cost for the year	**1666.98**

Computation of profit:

Let ₹ x be the rent for deluxe from.

Equivalent deluxe room days are 90,720 (Refer WN-2)

Total takings = ₹ 90,720x

Profit is 25% of total takings.

Profit = 25% of ₹ 90,720x = ₹ 22,680x

Total takings = Total Cost + Profit

₹ 90,720x = ₹ 16,66,98,000 + ₹ 22,680x

₹ 90,720x - ₹ 22,680x = ₹ 16,66,98,000

₹ 68,040x = ₹ 16,66,98,000

$X = \frac{₹\ 16,66,98,000}{₹\ 68,040} = ₹\ 2,450$

Rent to be charged for deluxe room	₹ 2,450
Rent to be charged for super deluxe room = Rent of deluxe room x 2 = ₹ 2,450 × 2	₹ 4,900
Rent to be charged for luxury suite = Rent of Deluxe room x 3 = ₹ 2,450 × 3	₹ 7,350

Working Notes:

1. Computation of Room Occupancy

Type of Room	No. of rooms x no. of days x occupancy %	Room days
Deluxe Room	100 rooms x 360 days x 90% occupancy	32,400
Super Deluxe Room	60 rooms x 360 days x 75% occupancy	16,200
Luxury Suite	40 × 360 days x 60% occupancy	8,640
	Total	57,240

2. Computation of equivalent deluxe room days

Rent of 'super deluxe' room is to be fixed at 2 times of 'deluxe room' and luxury suite' is 3 times of 'deluxe room'. Therefore, equivalent room days would be:

Type of Room	Room days	Equivalent deluxe room days
Deluxe Room	32,400 × 1	32,400
Super Deluxe Room	16,200 × 2	32,400
Luxury Suite	8,640 × 3	25,920
	Total	90,720

3. Computation of room attendant's wages:

Room occupancy days @ ₹ 500 per day

= 57,240 days × ₹ 500 per day = ₹ 2,86,20,000

4. Computation of Depreciation per annum:

Particulars	Cost (₹)	Rate of Depreciation	Depreciation (₹)
Building	900,00,000	5%	45,00,000
Furniture & Fixtures	90,00,000	10%	9,00,000
Air Conditioners	75,00,000	10%	7,50,000

ICAI · Old P3 · RTP Nov 2023 · question p. 6-7 · answer p. 19-21

860 RTP **Old P3 May 2023** Q11 old scheme

PREPARE cost statement of Panipat Thermal Power Station showing the cost of electricity generated per kwh, from the following data.

Total units generated	16,50,000 kWh
	(₹)
Operating labour	21,75,000
Repairs & maintenance	7,25,000
Lubricants, spares and stores	5,80,000
Plant supervision	4,35,000
Administration overheads	29,00,000
Insurance Charges	15,00,000
Fuel Charges	8,00,000

7 kWh. of electricity generated per kg. of coal consumed @ ₹4.75 per kg. Depreciation charges @ 5% on capital cost of ₹3,10,00,000.

ICAI'S ANSWER

Total units generated 16,50,000 kWh.

Cost Statement of Panipat Thermal Power Station

	Per annum (₹)	Per kWh. (₹)
Fixed costs:		
Plant supervision	4,35,000	
Administration overheads	29,00,000	
Insurance Charges	15,00,000	
Depreciation (5% of ₹ 3,10,00,000 p.a.)	15,50,000	
Total fixed cost: (A)	63,85,000	3.87
Variable costs:		
Operating labour	21,75,000	
Fuel Charges	8,00,000	
Lubricants, spares and stores	5,80,000	
Repairs & maintenance	7,25,000	
Coal cost (Refer to working note)	11,19,643	
Total variable cost: (B)	53,99,643	3.27
Total cost [(A) + (B)]	1,17,84,643	7.14

Working Note:

Coal cost (16,50,000 kWh. ÷ 7 kWh) × ₹4.75 per kg. = ₹11,19,643

ICAI · Old P3 · RTP May 2023 · question p. 8 · answer p. 21

861 RTP **Old P3 May 2020** Q9 old scheme

AD Higher Secondary School (AHSS) offers courses for 11th & 12th standard in three streams i.e. Arts, Commerce and Science. AHSS runs higher secondary classes alongwith primary and secondary classes but for accounting purpose it treats higher secondary as a separate responsibility centre. The Managing committee of the school wants to revise its fee structure for higher secondary students. The accountant of the school has provided the following details for a year:

	Amount (₹)
Teachers' salary (15 teachers × ₹35,000 × 12 months)	63,00,000
Principal's salary	14,40,000
Lab attendants' salary (2 attendants × ₹15,000 × 12 months)	3,60,000

	Amount (₹)
Salary to library staff	1,44,000
Salary to peons (4 peons × ₹10,000 × 12 months)	4,80,000
Salary to other staffs	4,80,000
Examinations expenditure	10,80,000
Office & Administration cost	15,20,000
Annual day expenses	4,50,000
Sports expenses	1,20,000

Other information:

(i)

	Standard 11 & 12			Primary & Secondary
	Arts	**Commerce**	**Science**	
No. of students	120	360	180	840
Lab classes in a year	0	0	144	156
No. of examinations in a year	2	2	2	2
Time spent at library per student per year	180 hours	120 hours	240 hours	60 hours
Time spent by principal for administration	208 hours	312 hours	480 hours	1,400 hours
Teachers for 11 & 12 standard	4	5	6	-

(ii) One teacher who teaches economics for Arts stream students also teaches commerce stream students. The teacher takes 1,040 classes in a year, it includes 208 classes for commerce students.

(iii) There is another teacher who teaches mathematics for Science stream students also teaches business mathematics to commerce stream students. She takes 1,100 classes a year, it includes 160 classes for commerce students.

(iv) One peon is fully dedicated for higher secondary section. Other peons dedicate their 15% time for higher secondary section.

(v) All school students irrespective of section and age participate in annual functions and sports activities.

Requirement:

(a) CALCULATE cost per student per annum for all three streams.

(b) If the management decides to take uniform fee of ₹ 1,000 per month from all higher secondary students, CALCULATE stream wise profitability.

(c) If management decides to take 10% profit on cost, COMPUTE fee to be charged from the students of all three streams respectively.

ICAI'S ANSWER

Calculation of Cost per annum

Particulars	Arts (₹)	Commerce (₹)	Science (₹)	Total (₹)
Teachers' salary (W.N-1)	16,80,000	21,00,000	25,20,000	63,00,000
Re-apportionment of Economics & Mathematics teachers' salary (W.N- 2)	(84,000)	1,45,091	(61,091)	-
Principal's salary (W.N-3)	1,24,800	1,87,200	2,88,000	6,00,000
Lab assistants' salary (W.N-4)	-	-	1,72,800	1,72,800
Salary to library staff (W.N-5)	43,200	28,800	57,600	1,29,600
Salary to peons (W.N-6)	31,636	94,909	47,455	1,74,000
Salary to other staffs (W.N-7)	38,400	1,15,200	57,600	2,11,200
Examination expenses (W.N- 8)	86,400	2,59,200	1,29,600	4,75,200
Office & Administration expenses (W.N- 7)	1,21,600	3,64,800	1,82,400	6,68,800
Annual Day expenses (W.N-7)	36,000	1,08,000	54,000	1,98,000
Sports expenses (W.N- 7)	9,600	28,800	14,400	52,800
Total Cost per annum	20,87,636	34,32,000	34,62,764	89,82,400

(a) Calculation of cost per student per annum

Particulars	Arts (₹)	Commerce (₹)	Science (₹)	Total (₹)
Total Cost per annum	20,87,636	34,32,000	34,62,764	89,82,400
No. of students	120	360	180	660
Cost per student per annum	17,397	9,533	19,238	13,610

(b) Calculation of profitability

Particulars	Arts (₹)	Commerce (₹)	Science (₹)	Total (₹)
Total Fees per annum	12,000	12,000	12,000	
Cost per student per annum	17,397	9,533	19,238	
Profit/ (Loss) per student per annum	(5,397)	2,467	(7,238)	
No. of students	120	360	180	
Total Profit/ (Loss)	(6,47,640)	8,88,120	(13,02,840)	(10,62,360)

(c) Computation of fees to be charged to earn a 10% profit on cost

Particulars	Arts (₹)	Commerce (₹)	Science (₹)
Cost per student per annum	17,397	9,533	19,238
Add: Profit @10%	1,740	953	1,924
Fees per annum	19,137	10,486	21,162
Fees per month	1,595	874	1,764

Working Notes:

(1) Teachers' salary

Particulars	Arts	Commerce	Science
No. of teachers	4	5	6
Salary per annum (₹)	4,20,000	4,20,000	4,20,000
Total salary	16,80,000	21,00,000	25,20,000

(2) Re-apportionment of Economics and Mathematics teachers' salary

	Economics		Mathematics	
Particulars	**Arts**	**Commerce**	**Science**	**Commerce**
No. of classes	832	208	940	160
Salary re-apportionment (₹)	(84,000)	84,000	(61,091)	61,091
		(₹4,20,000/1,040 × 208)		(₹4,20,000/1,100 × 160)

Total addition to Commerce stream = ₹ *84,000* + ₹ *61,091* = ₹ *1,45,091*

(3) Principal's salary has been apportioned on the basis of time spent by him for administration of classes.

(4) Lab attendants' salary has been apportioned on the basis of lab classes attended by the students.

(5) Salary of library staffs are apportioned on the basis of time spent by the students in library.

(6) Salary of Peons are apportioned on the basis of number of students. The peons' salary allocable to higher secondary classes is calculated as below:

	Amount (₹)
Peon dedicated for higher secondary (1 peon × ₹10,000 × 12 months)	1,20,000
Add: 15% of other peons' salary {15% of (3 peons × ₹10,000 × 12 months)}	54,000
	1,74,000

(7) Salary to other staffs, office & administration cost, Annual day expenses and sports expenses are apportioned on the basis of number of students.

(8) Examination Expenses has been apportioned taking number of students and number of examinations into account.

ICAI · Old P3 · RTP May 2020 · question p. 7-8 · answer p. 21-23

862 MTP **May 2026 S-I** MCQ 13 2 marks MCQ

BCIC Ltd. is a insurance company. It launched a new term insurance policy Names as Protection Plus. The total cost for the policy during the year is ₹ 1,60,00,000. Total number of policies sold is 410 and total insured value of policies is ₹ 920 crore.

What is the cost per rupee of insured value?

A ₹ 0.0017

B ₹ 0.18

C ₹ 575

D ₹ 2.24

ICAI'S ANSWER

(A) ₹ 0.0017

Cost per rupee of insured value

= Total Cost/ Total Insured Value

= 1.6 cr/920 cr = ₹ 0.0017

ICAI · P4 · MTP May 2026 · Series I · question p. 6 · answer p. 5

863 MTP **Jan 2026 S-I** Q5(b) 7 marks

A hotel is being run in a Hill station with 200 single rooms. The hotel offers concessional rates during six off-season (winter) months in a year. During this period, half of the full room rent is charged. The management's profit margin is targeted at 20% of the room rent. The following are the cost estimates and other details for the year ending 31st March, 2025:

(i) Occupancy during the season is 80% while in the off-season it is 40%.

(ii) Total investment in the hotel is ₹ 300 lakhs of which 80% relates to Buildings and the balance to Furniture and other Equipment.

(iii) Room attendants are paid ₹ 15 per room per day on the basis of occupancy of rooms in a month.

(iv) Expenses:

- Staff salary (excluding that of room attendants) ₹ 8,00,000
- Repairs to Buildings ₹ 3,00,000
- Laundry Charges ₹ 1,40,000
- Interior Charges ₹ 2,50,000
- Miscellaneous Expenses ₹ 2,00,200

(v) Annual Depreciation is to be provided on Buildings @ 5% and 15% on Furniture and other Equipments on straight line method.

(vi) Monthly lighting charges are ₹ 110 per room, except in four months in winter when it is ₹ 30 per room and this cost is on the basis of full occupancy for a month.

You are required to WORKOUT the room rent chargeable per day both during the season and the off-season months using the foregoing information.

(Assume a month to be of 30 days and winter season to be considered as part of off-season).

ICAI'S ANSWER

Working Notes:

(i) Total Room days in a year

Season	Occupancy (Room-days)	Equivalent Full Room charge days
Season – 80% Occupancy	200 Rooms × 80% × 6 months × 30 days in a month = **28,800** Room Days	28,800 Room Days × 100% = 28,800
Off-season – 40% Occupancy	200 Rooms × 40% × 6 months × 30 days in a month = **14,400** Room Days	14,400 Room Days × 50% = 7,200
Total Room Days	28,800 + 14,400 = 43,200 Room Days	36,000 Full Room days

(ii) Lighting Charges:

It is given in the question that lighting charges for 8 months is ₹ 110 per month and during winter season of 4 months it is ₹ 30 per month. Further it is also given that peak season is 6 months and off season is 6 months.

It should be noted that – being Hill station, winter season is to be considered as part of Off season. Hence, the non-winter season of 8 months include – Peak season of 6 months and Off season of 2 months.

Accordingly, the lighting charges are calculated as follows:

Season	Occupancy (Room-days)
Season & Non-winter – 80% Occupancy	200 Rooms × 80% × 6 months × ₹ 110 per month = ₹ **1,05,600**
Off- season & Non-winter – 40% Occupancy (8 – 6 months)	200 Rooms × 40% × 2 months × ₹ 110 per month = ₹ **17,600**
Off- season & -winter – 40% Occupancy months)	200 Rooms × 40% × 4 months × ₹ 30 per month = ₹ **9,600**
Total Lighting charges	₹ 1,05,600+ ₹ 17,600 + ₹ 9,600 = ₹ **132,800**

Statement of total cost:

	(₹)
Staff salary	8,00,000
Repairs to building	3,00,000
Laundry	1,40,000

	(₹)
Interior	2,50,000
Miscellaneous Expenses	2,00,200
Depreciation on Building (₹ 300 Lakhs × 80% × 5%)	**12,00,000**
Depreciation on Furniture & Equipment (₹ 300 Lakhs × 20% × 15%)	**9,00,000**
Room attendant's wages (₹ 15 per Room Day for 43,200 Room Days)	**6,48,000**
Lighting charges	1,32,800
Total cost	**45,71,000**
Add: Profit Margin (20% on Room rent or 25% on Cost)	11,42,750
Total Rent to be charged	**57,13,750**

Calculation of Room Rent per day:

Total Rent / Equivalent Full Room days = ₹ 57,13,750/ 36,000 = ₹ 158.72

Room Rent during Season – ₹ 158.72

Room Rent during Off season = ₹ 158.72 × 50% = ₹ 79.36

ICAI · P4 · MTP Jan 2026 · Series I · question p. 15 · answer p. 17-18

864 MTP **Jan 2026 S-II** Case Scenario II · MCQ 6 to MCQ 10 10 marks

CASE SCENARIO II

The Greenfield Recreation Club, a well-known community hub in the city, has been operating for over 25 years. Besides sports facilities and hobby classes, the club runs a dedicated public library for its members. Over the years, the library has grown into a quiet learning space where adults, students, and retirees gather for reading, research, and social interaction.

Recently, the club noticed a steady rise in library usage, especially after it introduced digital reading corners and weekend storytelling sessions for kids. As a result, the committee has decided to review the financial performance of the library to ensure it remains sustainable while continuing to serve the community.

To support the library, the club follows a long-standing policy of offering an annual subsidy of up to ₹ 6 per club member, drawn from the general funds. This subsidy can cover maintenance, utilities, and book purchases when necessary. However, with rising costs and increasing expectations, the club now wants to re-evaluate whether this subsidy is sufficient and whether the library is being managed efficiently.

The management provides the following operational details for the current year:

- The club has 4,800 active members, out of which 1,200 have opted for the library membership.
- Each library member pays a monthly fee of ₹ 110, which supports routine expenses.
- To encourage timely book returns, the library imposes a fine of ₹ 2 per book per day.
- On average, 450 books are returned late per month, each delayed by around 6 days.

- Due to increasing member engagement, late returns have slowly increased, indicating heavy circulation of books.

The library's book collection includes:

- 60,000 old books, requiring routine repairs, dusting, rebinding, and pest control.
- These activities cost about ₹ 15 per book per year.
- To stay updated with current releases and replace damaged copies, the library purchases 1,000 new books annually at a cost of ₹ 320 per book.

To manage daily operations smoothly, the library employs:

- 1 Librarian earning ₹ 12,000 per month
- 3 Assistant Librarians, each earning ₹ 8,000 per month
- 1 Clerk earning ₹ 5,000 per month

These employees handle catalogue management, member services, issuing and returning books, event coordination, and stock upkeep.

You are required to answer the following questions (MCQ 6 to 10):

MCQ 6 2 marks

What is the Cost incurred per library member per month (excluding cost of new books)

A ₹ 96.67

B ₹ 55.00

C ₹ 61.25

D ₹ 73.75

MCQ 7 2 marks

What is the Cost incurred per club member per month (excluding cost of new books)

A ₹ 96.67

B ₹ 24.17

C ₹ 52.25

D ₹ 13.75

MCQ 8 2 marks

Calculate the Net income earned by the library per year.

A ₹ 7,65,200

B ₹ 6,42,600

C ₹ 4,86,200

D ₹ 2,56,800

MCQ 9 2 marks

How many extra books are currently being purchased per year?

A 0 books (no excess; purchase is within limit)

B 350 books

C 198 books

D 220 books

MCQ 10 2 marks

Calculate the amount of more subsidy required:

A ₹ 63,360

B ₹ 42,600

C ₹ 86,200

D ₹ 21,500

ICAI'S ANSWERS

MCQ 6

(A) ₹ 96.67

Particular		Amount (₹)
Librarian	₹ 12,000 × 12 × 1 person	1,44,000
Assistant librarians	₹ 8,000 × 12 × 3 persons	2,88,000
Clerk	₹ 5,000 × 12 × 1 person	60,000
Total Staff Cost		4,92,000
Old Books Maintenance	60,000 old books × ₹ 15 per year	9,00,000
Total Annual Cost (excluding new books)		13,92,000

Cost per member per year

= 13,92,000 / 1,200

= ₹ 1,160

Cost per library member per month

= 1,160 / 12

= ₹ 96.67

MCQ 7

(B) ₹ 24.17

Cost per member per year

= 13,92,000 / 4,800

= ₹ 290

Cost per library member per month

= 290 / 12

= ₹ 24.17

MCQ 8

(D) ₹ 2,56,800

Total Revenue

Particulars	Amount (₹)
Library fees per month (1,200 members × ₹ 110)	1,32,000
Late fees per month (450 books × 6 days × ₹ 2)	5,400
Total Revenue per month	1,37,400
Total Revenue per year (1,37,400×12)	16,48,800

Net Income per Year

= Total revenue – Total cost

= 16,48,800 – 13,92,000

= ₹ 2,56,800

MCQ 9

(C) 198 books

Cost per new book	320
Maximum number of new books per annum (₹ 2,56,800 ÷ ₹ 320)	802 nos.
Number of books purchased	1,000 nos.
Excess books purchased (1,000 nos. – 802 nos.)	198 nos.

MCQ 10

(A) ₹ 63,360

Additional Subsidy Required

= Extra books × cost per book

= 198 × 320

= ₹ 63,360

ICAI · P4 · MTP Jan 2026 · Series II · question p. 3-4 · answer p. 4, 4-5, 5

865 MTP **Jan 2026 S-II** Q5(b) 3 marks

Star study centre provides coaching classes to school students. The study centre has taken an auditorium of 250 seat capacity on rent of ₹ 3,75,000 per month. It has also hired some renowned teachers for taking classes. A teacher takes ₹ 3,000 per hour. The study centre has decided to conduct a batch of 2-hour per day for 3 days a week for 4 months.

(i) CALCULATE the total cost per batch.

(ii) COMPUTE the minimum fee to be charged per student in a batch, if the centre operates at 60% capacity.

(iii) DETERMINE the fee per student if the study centre desires to earn a profit of 50% and study centre operates at 50% capacity.

ICAI'S ANSWER

(i) Calculation of total cost per batch:

Particulars	Amount (₹)
(i) Auditorium hire charges (₹ 3,75,000 × 4 months)	15,00,000
(ii) Teachers' remuneration (₹ 3,000 × 2 hours × 3 days × 4 weeks × 4 months)	2,88,000
Total cost	17,88,000

(ii) Computation of minimum fee per student per batch:

$$\text{Minimum fee to be charged} = \frac{\text{Total Cost}}{\text{No. of students}} = \frac{₹17,88,000}{150 \text{ students}} = ₹\ 11,920/-$$

(iii) Fee to be charged per student $= \frac{\text{Total Cost} + \text{Profit}}{\text{No. of students}}$

$$= \frac{₹17,88,000 + ₹8,94,000}{125 \text{ students}}$$

$$= \frac{₹26,82,000}{125 \text{ students}} \quad ₹\ 21,456$$

ICAI · P4 · MTP Jan 2026 · Series II · question p. 17 · answer p. 20

866 MTP Sep 2025 S-I MCQ 12 2 marks MCQ

The loan department of a bank is responsible for several functions, including processing home loan applications, managing customer inquiries, and overseeing loan approvals. After careful analysis, it has been estimated that 30% of the department's total overhead costs such as legal advice, telephone expenses, staff salaries, etc. are directly related to the processing of home loan applications, though only 480 home loan applications are processed each month. The following information is given concerning the processing of a loan application:

Particulars	(₹)
Home Loan processors monthly salary: (16 employees @ ₹ 94,500 each)	15,12,000
Loan department overhead costs (monthly)	
Chief loan officer's salary	4,72,500
Telephone expenses	47,250
Depreciation Building	1,76,400
Legal advice	1,51,200
Advertising	2,52,000
Miscellaneous	40,950
Total overhead costs	11,40,300

COMPUTE processing cost per home loan application.

A ₹ 2,375.63

B ₹ 3,150.00

C ₹ 3,862.69

D ₹ 5,525.63

ICAI'S ANSWER

Answer key: **(C)**

(c) Statement showing computation of the cost of processing a home loan application

Particulars	(₹)
Direct professional labour cost (16 employees @ ₹ 94,500 each)	15,12,000
Service overhead cost (30% of ₹ 11,40,300)	3,42,090
Total processing cost per month	18,54,090
No. of applications processed per month	480
Total processing cost per home loan application	3,862.69

ICAI · P4 · MTP Sep 2025 · Series I · question p. 5-6 · answer p. 4

867 MTP **Jan 2025 S-I** MCQ 14 2 marks MCQ

Parth Ltd. operates in insurance business. Previous Year, the company launched a new term insurance policy called 'Max Jivan' and incurred the following expenditure throughout the year:

Particulars	Amount (₹)
Claim management cost	52,82,000
Facilities cost	6,49,82,500
Employees cost	2,25,18,000
Cost of marketing of the policy	19,30,71,000
Policy development cost	4,86,50,000
Policy issuance cost	4,10,05,000
Policy servicing cost	13,40,65,500
Sales support expenses	4,44,80,000
Office administration cost	6,67,20,000
I.T. Cost	30,71,90,000
Postage and logistics	4,50,36,000

You are required to ASCERTAIN the cost of the policy 'Max Jivan' segregated into four main activities namely (a) Marketing and Sales support (b) Operations (c) I.T. Cost and (d) Support functions.

A Marketing and Sales support- ₹ 23,75,51,000, Operations - ₹ 22,90,02,500, I.T. Cost- ₹ 30,71,90,000 and Support functions- ₹ 19,92,56,500

B Marketing and Sales support- ₹ 28,62,01,000, Operations- ₹ 22,53,88,500, I.T. Cost- ₹ 30,71,90,000 and Support functions- ₹ 15,42,20,500

C Marketing and Sales support- ₹ 28,62,01,000, Operations- ₹ 18,03,52,500, I.T. Cost- ₹ 30,71,90,000 and Support functions- ₹ 19,92,56,500

D Marketing and Sales support- ₹ 24,17,21,000, Operations- ₹ 22,48,32,500, I.T. Cost- ₹ 30,71,90,000 and Support functions- ₹ 19,92,56,500

ICAI'S ANSWER

(C) Marketing and Sales support- ₹ 28,62,01,000, Operations- ₹ 18,03,52,500, I.T. Cost- ₹ 30,71,90,000 and Support functions- ₹ 19,92,56,500

Calculation of total cost for 'Max Jivan' Insurance policy

	Particulars	Amount (₹)	Amount (₹)
a.	**Marketing and Sales support:**		
	- Policy development cost	4,86,50,000	
	- Cost of marketing	19,30,71,000	
	- Sales support expenses	4,44,80,000	28,62,01,000
b.	**Operations:**		
	- Policy issuance cost	4,10,05,000	
	- Policy servicing cost	13,40,65,500	
	- Claim management cost	52,82,000	18,03,52,500
c.	**IT Cost**		30,71,90,000
d.	**Support functions**		
	- Postage and logistics	4,50,36,000	
	- Facilities cost	6,49,82,500	
	- Employees cost	2,25,18,000	
	- Office administration cost	6,67,20,000	19,92,56,500
	Total Cost		**97,30,00,000**

ICAI · P4 · MTP Jan 2025 · Series I · question p. 6 · answer p. 5

868 MTP Jan 2025 S-II Q3(c) 2 marks

STI is majorly providing education loan in its loan department. For the month of August, salary paid to the education loan processors is ₹ 21,60,000. W.r.t. overhead cost, 30% is applicable to the processing of education loan out of the total overhead cost of loan department.

The total overhead cost for the month of August is ₹ 16,40,000 which includes payment of ₹ 11,000 w.r.t. legal advice relating to one of the education loan processing.

The education loan applications processed during this month are 500. You are required to COMPUTE the cost of processing per education loan application.

ICAI'S ANSWER

Statement showing computation of the cost of processing an education loan application

Particulars	(₹)
Salary paid to the education loan processors	21,60,000
Legal advice cost relating to education loan	11,000
Overhead cost (30% of (₹ 16,40,000 - ₹ 11,000)]	4,88,700
Total processing cost per month	26,59,700
No. of applications processed per month	500
Total processing cost per education loan application	**5,319.40**

ICAI · P4 · MTP Jan 2025 · Series II · question p. 10 · answer p. 13

869 MTP **May 2024 S-II** Q5(b) 6 marks

A hotel having 20 single rooms is having 80% occupancy in normal season (8 months) and 50% in off-season (4 months) in a year (take 30 days month).

Annual fixed expenses	Amount in ₹
Salary of the staff (excluding room attendant)	15,00,000
Repair & maintenance	12,60,000
Depreciation on building & furniture	12,40,000
Other fixed expenses like dusting, sweeping etc.	13,25,000
	53,25,000
Variable expenses (per guest per day)	
Linen, laundry & security support	80.00
Electricity & other facilities	120.00
Misc. expenses like attendant etc.	300.00
	500.00

Management wishes to make a margin of 25% of total cost.

Required

(a) CALCULATE the Tariff per room per day.

(b) CALCULATE the break-even occupancy in normal season (in percentage also) assuming there is 50% occupancy in off-season.

ICAI'S ANSWER

Workings:

Total occupancy = Occupancy in normal season + Occupancy in off-season

= (20 rooms × 80% × 8 months × 30 days) + (20 rooms × 50% × 4 months × 30 days)

= 3,840 + 1,200 = 5,040 room-days

Total Cost = Variable cost + Fixed cost

= (₹ 500 × 5,040 room-days) + ₹ 53,25,000

= ₹ 25,20,000 + ₹ 53,25,000

= 78,45,000

(a) **Calculation of tariff rate per room**

Tariff per room per day = (Total cost + 25% Margin on total cost) ÷ Total occupancy

= (₹ 78,45,000 + 19,61,250) ÷ 5,040 = ₹ 1,945.68

(b) **Calculation of break-even occupancy**

Contribution per day = Tariff – Variable cost

= ₹ 1,945.68 – 500 = ₹ 1445.68

Break-even occupancy = ₹ 53,25,000 ÷ 1445.68

= 3683

Occupancy in normal season = Break-even occupancy – Occupancy in off-season

= 3683 – (20 rooms × 50% × 4 months × 30 days)

= 3683 – 1200 = 2483 room-days

In Percentage = 2483 ÷ 4800 = 51.73%

ICAI · P4 · MTP May 2024 · Series II · question p. 10 · answer p. 13

870 MTP **Old P3 Nov 2023 S-II** Q3(b) 10 marks old scheme

ABC Hospital runs a Critical Care Unit (CCU) in a hired building. CCU consists of 35 beds and 5 more beds can be added, if required.

Rent per month - ₹ 75,000

Supervisors – 2 persons – ₹ 25,000 per month – each

Nurses – 4 persons – ₹ 20,000 per month – each

Ward Boys – 4 persons – ₹ 5,000 per month – each

Doctors paid ₹ 2,50,000 per month – paid on the basis of number of patients attended and the time spent by them

Other expenses for the year are as follows:

Repairs (Fixed) – ₹ 81,000

Food to Patients (Variable) – ₹ 8,80,000

Other services to patients (Variable) – ₹ 3,00,000

Laundry charges (Variable) – ₹ 6,00,000

Medicines (Variable) – ₹ 7,50,000

Other fixed expenses – ₹ 10,80,000

Administration expenses allocated – ₹ 10,00,000

It was estimated that for 150 days in a year 35 beds are occupied and for 80 days only 25 beds are occupied.

The hospital hired 750 beds at a charge of ₹ 100 per bed per day, to accommodate the flow of patients. However, this does not exceed more than 5 extra beds over and above the normal capacity of 35 beds on any day.

You are required to –

(i) Calculate profit per patient day, if the hospital recovers on an average ₹ 2,000 per day from each patient.

(ii) Find out breakeven point for the hospital.

ICAI'S ANSWER

Working Notes:

(1) Calculation of number of patient days

35 Beds × 150 days	=	5,250
25 Beds × 80 days	=	2,000
Extra beds	=	750
Total	=	8,000

Statement of Profitability

Particulars	Amount	Amount
Income for the year (₹ 2,000 per patient per day × 8,000 patient days)		1,60,00,000
Variable Costs:		
Doctor Fees (₹ 2,50,000 per month × 12)	30,00,000	
Food to Patients (Variable)	8,80,000	
Other services to patients (Variable)	3,00,000	
Laundry charges (Variable) – (₹)	6,00,000	
Medicines (Variable) – (₹)	7,50,000	
Bed Hire Charges (₹100 × 750 Beds)	75,000	
Total Variable costs		56,05,000
Contribution		1,03,95,000
Fixed Costs:		

Particulars	Amount	Amount
Rent (₹ 75,000 per month × 12)	9,00,000	
Supervisor (2 persons × ₹25,000 × 12)	6,00,000	
Nurses (4 persons × ₹ 20,000 × 12)	9,60,000	
Ward Boys (4 persons × ₹ 5,000 × 12)	2,40,000	
Repairs (Fixed)	81,000	
Other fixed expenses – (₹)	10,80,000	
Administration expenses allocated – (₹)	10,00,000	
Total Fixed Costs		48,61,000
Profit		55,34,000

(i) Calculation of Contribution per Patient- day

Total Contribution – ₹ 1,03,95,000

Total Patient days – 8,000

Contribution per Patient -day – ₹ 1,03,95,000 / 8,000 = ₹ 1,299.375

(ii) Breakeven Point = Fixed Cost / Contribution per Patient- day

= ₹ 48,61,000 / ₹1,299.375

= 3,741 patient days

ICAI · Old P3 · MTP Nov 2023 · Series II · question p. 4-5 · answer p. 7-8

871 MTP **Old P3 May 2023 S-I** Q1(d) 5 marks old scheme

Secure lifeline Ltd. operates in life insurance business. It launched a new insurance policy 'Total secure'. The company has incurred the following expenditures during the last year for the policy:

	₹
Cost of marketing of the policy	74,58,000
Sales support expenses	18,89,250
Policy issuance cost	16,59,735
Claims management cost	2,07,240
Policy development cost	18,56,250
Postage and logistics	16,91,250
Facilities cost	25,14,600
Policy servicing cost	58,09,155

	₹
Employees cost	9,24,000
IT cost	1,22,62,800
Office administration cost	26,73,660

Number of policies sold- 844.

Total insured value of policies - ₹ 1,640 crore.

Required:

(i) CALCULATE total cost for Professionals Protection Plus' policy segregating the costs into four main activities namely (a) Marketing and Sales support, (b) Operations, (c) IT and (d) Support functions.

(ii) CALCULATE cost per policy.

(iii) CALCULATE cost per rupee of insured value.

ICAI'S ANSWER

(i) Calculation of total cost for 'Professionals Protection Plus' policy

	Particulars	Amount (₹)	Amount (₹)
1	**Marketing and Sales support:**		
	- Policy development cost	18,56,250	
	- Cost of marketing	74,58,000	
	- Sales support expenses	18,89,250	1,12,03,500
2	**Operations:**		
	- Policy issuance cost	16,59,735	
	- Policy servicing cost	58,09,155	
	- Claims management cost	2,07,240	76,76,130
3	**IT Cost**		1,22,62,800
4	**Support functions**		
	- Postage and logistics	16,91,250	
	- Facilities cost	25,14,600	
	- Employees cost	9,24,000	
	- Office administration cost	26,73,660	78,03,510
	Total Cost		**3,89,45,940**

(ii) Calculation of cost per policy $\frac{\text{Total cost}}{\text{Number of policies}} = \frac{3,89,45,940}{844}$

= ₹ 46,144.48

(iii) Cost per rupee of insured value $\frac{\text{Total cost}}{\text{Total insured value}} = \frac{3,89,45,940}{1,640 \text{ crore}}$

= ₹ 0.0024

ICAI · Old P3 · MTP May 2023 · Series I · question p. 2 · answer p. 2

872 MTP **Old P3 Dec 2021 S-I** Q4(a) 10 marks old scheme

A hotel is being run in a Hill station with 200 single rooms. The hotel offers concessional rates during six off-season (winter) months in a year.

During this period, half of the full room rent is charged. The management's profit margin is targeted at 20% of the room rent. The following are the cost estimates and other details for the year ending 31st March, 2021:

(i) Occupancy during the season is 80% while in the off-season it is 40%.

(ii) Total investment in the hotel is ₹ 300 lakhs of which 80% relates to Buildings and the balance to Furniture and other Equipment.

(iii) Room attendants are paid ₹ 15 per room per day on the basis of occupancy of rooms in a month.

(iv) Expenses :

- Staff salary (excluding that of room attendants) ₹ 8,00,000
- Repairs to Buildings ₹ 3,00,000
- Laundry Charges ₹ 1,40,000
- Interior Charges ₹ 2,50,000
- Miscellaneous Expenses ₹ 2,00,200

(v) Annual Depreciation is to be provided on Buildings @ 5% and 15% on Furniture and other Equipments on straight line method.

(vi) Monthly lighting charges are ₹ 110 per room, except in four months in winter when it is ₹ 30 per room and this cost is on the basis of full occupancy for a month.

You are REQUIRED to workout the room rent chargeable per day both during the season and the off-season months using the foregoing information.

(Assume a month to be of 30 days and winter season to be considered as part of off-season).

ICAI'S ANSWER

Working Notes:

(i) **Total Room days in a year**

Season	Occupancy (Room-days)	Equivalent Full Room charge days
Season – 80% Occupancy	200 Rooms × 80% × 6 months × 30 days in a month = **28,800** Room Days	28,800 Room Days × 100% = 28,800
Off-season – 40% Occupancy	200 Rooms × 40% × 6 months × 30 days in a month = **14,400** Room Days	14,400 Room Days × 50% = 7,200
Total Room Days	28,800 + 14,400 = 43,200 Room Days	36,000 Full Room days

(ii) **Lighting Charges:**

It is given in the question that lighting charges for 8 months is ₹110 per month and during winter season of 4 months it is ₹30 per month. Further it is also given that peak season is 6 months and off season is 6 months.

It should be noted that – being Hill station, winter season is to be considered as part of Off season. Hence, the non-winter season of 8 months include – Peak season of 6 months and Off season of 2 months.

Accordingly, the lighting charges are calculated as follows:

Season	Occupancy (Room-days)
Season & Non-winter – 80% Occupancy	200 Rooms × 80% × 6 months × ₹ 110 per month = ₹ **1,05,600**
Off- season & Non-winter – 40% Occupancy (8 – 6 months)	200 Rooms × 40% × 2 months × ₹110 per month = ₹ **17,600**
Off- season & -winter – 40% Occupancy months)	200 Rooms × 40% × 4 months × ₹ 30 per month = ₹ **9,600**
Total Lighting charges	₹ 1,05,600+ ₹ 17,600 + ₹ 9,600 = ₹ **132,800**

Statement of total cost:

	(₹)
Staff salary	8,00,000
Repairs to building	3,00,000
Laundry	1,40,000
Interior	2,50,000
Miscellaneous Expenses	2,00,200
Depreciation on Building (₹ 300 Lakhs × 80% × 5%)	**12,00,000**
Depreciation on Furniture & Equipment (₹ 300 Lakhs × 20% × 15%)	**9,00,000**
Room attendant's wages (₹ 15 per Room Day for 43,200 Room Days)	**6,48,000**
Lighting charges	1,32,800
Total cost	**45,71,000**
Add: Profit Margin (20% on Room rent or 25% on Cost)	11,42,750
Total Rent to be charged	**57,13,750**

Calculation of Room Rent per day:

Total Rent / Equivalent Full Room days = ₹ 57,13,750/ 36,000 = ₹ 158.72

Room Rent during Season – ₹ 158.72

Room Rent during Off season = ₹ 158.72 × 50% = ₹ 79.36

ICAI · Old P3 · MTP Dec 2021 · Series I · question p. 4-5 · answer p. 7-8

873 MTP **Old P3 Dec 2021 S-II** Q3(a) 10 marks old scheme

MKL Infrastructure built and operates 110 k.m. highway on the basis of Built-Operate-Transfer (BOT) for a period of 21 years. A traffic assessment has been carried out to estimate the traffic flow per day which shows the following figures:

Sl. No.	Type of vehicle	Daily traffic volume
1.	Two wheelers	44,500
2.	Car and SUVs	3,450
3.	Bus and LCV	1,800
4.	Heavy commercial vehicles	816

The following is the estimated cost of the project:

Sl. no.	Activities	Amount (₹ in lakh)
1	Site clearance	341.00
2	Land development and filling work	9,160.00
3	Sub base and base courses	10,520.00
4	Bituminous work	32,140.00
5	Bridge, flyovers, underpasses, Pedestrian subway, footbridge, etc	28,110.00
6	Drainage and protection work	9,080.00
7	Traffic sign, marking and road appurtenance	8,810.00
8	Maintenance, repairing and rehabilitation	12,850.00
9	Environmental management	1,964.00
	Total Project cost	1,12,975.00

An average cost of ₹1,200 lakh has to be incurred on administration and toll plaza operation.

On the basis of the vehicle specifications (i.e. weight, size, time saving etc.), the following weights has been assigned to the passing vehicles:

Sl. No.	Type of vehicle	
1.	Two wheelers	5%
2.	Car and SUVs	20%
3.	Bus and LCV	30%
4.	Heavy commercial vehicles	45%

Required:

(i) CACULATE the total project cost per day of concession period.

(ii) COMPUTE toll fee to be charged for per vehicle of each type, if the company wants earn a profit of 15% on total cost.

[Note: Concession period is a period for which an infrastructure is allowed to operate and recover its investment]

ICAI'S ANSWER

(i) Calculation of total project cost per day of concession period:

Activities	Amount (₹ in lakh)
Site clearance	341.00
Land development and filling work	9,160.00
Sub base and base courses	10,520.00
Bituminous work	32,140.00
Bridge, flyovers, underpasses, Pedestrian subway, footbridge, etc	28,110.00
Drainage and protection work	9,080.00
Traffic sign, marking and road appurtenance	8,810.00
Maintenance, repairing and rehabilitation	12,850.00
Environmental management	1,964.00
Total Project cost	**1,12,975.00**
Administration and toll plaza operation cost	1,200.00
Total Cost	1,14,175.00
Concession period in days (21 years × 365 days)	7,665
Cost per day of concession period (₹ in lakh)	**14.90**

(ii) Computation of toll fee:

Cost to be recovered per day = Cost per day of concession period + 15% profit on cost

= ₹ 14,90,000 + ₹ 2,23,500 = ₹ 17,13,500

$$\text{Cost per equivalent vehicle} = \frac{₹\ 17{,}13{,}500}{76{,}444 \text{ units (Refer working note)}}$$

= ₹ 22.42 per equivalent vehicle

Vehicle type-wise toll fee:

Sl. No.	Type of vehicle	Equivalent cost [A]	Weight [B]	Toll fee per vehicle [A×B]
1.	Two wheelers	₹22.42	1	**22.42**
2.	Car and SUVs	₹22.42	4	**89.68**
3.	Bus and LCV	₹22.42	6	**134.52**
4.	Heavy commercial vehicles	₹22.42	9	**201.78**

Working Note:

The cost per day has to be recovered from the daily traffic. The each type of vehicle is to be converted into equivalent unit. Let's convert all vehicle types equivalent to Two-wheelers..

Sl. No.	Type of vehicle	Daily traffic volume [A]	Weight	Ratio [B]	Equivalent Two-wheeler [A×B]
1.	Two wheelers	44,500	0.05	1	44,500
2.	Car and SUVs	3,450	0.20	4	13,800
3.	Bus and LCV	1,800	0.30	6	10,800
4.	Heavy commercial vehicles	816	0.45	9	7,344
	Total				**76,444**

ICAI · Old P3 · MTP Dec 2021 · Series II · question p. 4-5 · answer p. 7-8

874 MTP **Old P3 May 2020 S-I** Q5(a) 10 marks old scheme

KR Resorts (P) Ltd. offers three types of rooms to its guests, viz deluxe room, super deluxe room and luxury suite. You are required to DETERMINE the tariff to be charged to the customers for different types of rooms on the basis of following information:

Types of Room	Number of Rooms	Occupancy
Deluxe Room	100	90%
Super Deluxe Room	60	75%
Luxury Suite	40	60%

Rent of 'super deluxe' room is to be fixed at 2 times of 'deluxe room' and that of 'luxury suite' is 3 times of 'deluxe room'. Annual expenses are as follows:

Particulars	Amount (Rs. lakhs)
Staff salaries	780.00
Lighting, Heating and Power	350.00
Repairs, Maintenance and Renovation	220.00
Linen	60.00
Laundry charges	34.00
Interior decoration	85.00
Sundries	36.28

An attendant for each room was provided when the room was occupied and he was paid Rs. 500 per day towards wages. Further, depreciation is to be provided on building @ 5% on Rs. 900 lakhs, furniture and fixtures @ 10% on Rs. 90 lakhs and air conditioners @ 10% on Rs. 75 lakhs.

Profit is to be provided @ 25% on total taking and assume 360 days in a year.

ICAI'S ANSWER

Total cost statement of KR Resort (P) Limited

Particulars	Cost per annum (Rs. in lakhs)
Staff Salaries	780.00
Room Attendant's Wages (Refer working note 3)	286.20
Lighting, Heating & Power	350.00
Repairs, Maintenance & Renovation	220.00
Linen	60.00
Laundry charges	34.00
Interior Decoration	85.00
Sundries	36.28
Depreciation: (Refer working note 4)	
- Building	45.00
- Furniture & Fixture	9.00
- Air Conditioners	7.50
Total cost for the year	1912.98

Computation of profit:

Let Rs. x be the rent for deluxe from.

Equivalent deluxe room days are 90,720 (Refer working note 2)

Total takings = Rs. 90,720x

Profit is 25% of total takings.

Profit = 25% of Rs. 90,720x = Rs. 22,680x

Total takings = Total Cost + Profit

Rs. 90,720x = Rs. 19,12,98,000 + Rs. 22,680x

Rs. 90,720x -Rs. 22,680x = Rs. 19,12,98,000

Rs. 68,040x = Rs. 19,12,98,000

$$X = \frac{Rs.19,12,98,000}{Rs.68,040} = Rs.\ 2,811.55$$

Rent to be charged for deluxe room	Rs. 2,811.55
Rent to be charged for super deluxe room = Rent of deluxe room x 2 = Rs. 2,811.55 × 2	Rs. 5,623.10
Rent to be charged for luxury suite = Rent of Super Deluxe room x 1.5 = Rs. 5,623.10 × 1.5	Rs. 8,434.65

Working Notes:

(1) Computation of Room Occupancy

Type of Room	No. of rooms x no. of days x occupancy %	Room days
Deluxe Room	100 rooms x 360 days x 90% occupancy	32,400
Super Deluxe Room	60 rooms x 360 days x 75% occupancy	16,200
Luxury Suite	40 rooms x 360 days x 60% occupancy	8,640
	Total	57,240

(2) Computation of equivalent deluxe room days

Rent of 'super deluxe' room is to be fixed at 2 times of 'deluxe room' and luxury suite' is 3 times of 'deluxe room'. Therefore, equivalent room days would be:

Type of Room	Room days	Equivalent deluxe room days
Deluxe Room	32,400 × 1	32,400
Super Deluxe Room	16,200 × 2	32,400
Luxury Suite	8,640 × 3	25,920
	Total	90,720

(3) Computation of room attendant's wages

Room occupancy days @ Rs. 500 per day = 286.2 lakhs (i.e. 57,240 days ×Rs. 500)

(4) Computation of Depreciation per annum

Particulars	Cost (Rs.)	Rate of Depreciation	Depreciation (Rs.)
Building	9,00,00,000	5%	45,00,000
Furniture & Fixtures	90,00,000	10%	9,00,000
Air Conditioners	75,00,000	10%	7,50,000

ICAI · Old P3 · MTP May 2020 · Series I · question p. 5-6 · answer p. 10-11

875 MTP Old P3 Nov 2019 S-I Q5(a) 10 marks old scheme

SLS Infrastructure builts and operates a 110 k.m. long highway on the basis of Built-Operate-Transfer (BOT) model for a period of 25 years. A traffic assessment has been carried out to estimate the traffic flow per day. The details are as below:

Sl. No.	Type of vehicle	Daily traffic volume
1.	Two wheelers	44,500
2.	Car and SUVs	3,450
3.	Bus and LCV	1,800
4.	Heavy commercial vehicles	816

The following is the estimated cost of the project:

Sl. no.	Activities	Amount (Rs. in lakh)
1	Site clearance	170.70
2	Land development and filling work	9,080.35
3	Sub base and base courses	10,260.70
4	Bituminous work	35,070.80
5	Bridge, flyovers, underpasses, Pedestrian subway, footbridge, etc.	29,055.60
6	Drainage and protection work	9,040.50
7	Traffic sign, marking and road appurtenance	8,405.00
8	Maintenance, repairing and rehabilitation	12,429.60
9	Environmental management	982.00
	Total Project cost	1,14,495.25

An average cost of Rs.1,120 lakh has to be incurred on administration and toll plaza operation.

On the basis of the vehicle specifications (i.e. weight, size, time saving etc.), the following weights has been assigned to the passing vehicles:

Sl. No.	Type of vehicle	
1.	Two wheelers	5%
2.	Car and SUVs	20%
3.	Bus and LCV	30%
4.	Heavy commercial vehicles	45%

Required:

(i) CACULATE the total project cost per day of concession period.

(ii) COMPUTE toll fee to be charged for per vehicle of each type, if the company wants to earn a profit of 15% on total cost.

[**Note:** Concession period is a period for which an infrastructure is allowed to operate and recovers its investment]

ICAI'S ANSWER

(i) Calculation of total project cost per day of concession period:

Activities	Amount (Rs. in lakh)
Site clearance	170.70
Land development and filling work	9,080.35
Sub base and base courses	10,260.70
Bituminous work	35,070.80
Bridge, flyovers, underpasses, Pedestrian subway, footbridge, etc	29,055.60
Drainage and protection work	9,040.50
Traffic sign, marking and road appurtenance	8,405.00
Maintenance, repairing and rehabilitation	12,429.60
Environmental management	982.00
Total Project cost	**1,14,495.25**
Administration and toll plaza operation cost	1,120.00
Total Cost	1,15,615.25
Concession period in days (25 years × 365 days)	9,125
Cost per day of concession period (Rs. in lakh)	**12.67**

(ii) Computation of toll fee:

Cost to be recovered per day = Cost per day of concession period + 15% profit on cost

= Rs.12,67,000 + Rs.1,90,050 = Rs.14,57,050

$$\text{Cost per equivalent vehicle} = \frac{₹14,57,050}{76,444 \text{ units (Refer working note)}}$$

= Rs.19.06 per equivalent vehicle

Vehicle type-wise toll fee:

Sl. No.	Type of vehicle	Equivalent cost [A]	Weight [B]	Toll fee per vehicle [A×B]
1.	Two wheelers	Rs.19.06	1	19.06
2.	Car and SUVs	Rs.19.06	4	76.24
3.	Bus and LCV	Rs.19.06	6	114.36
4.	Heavy commercial vehicles	Rs.19.06	9	171.54

Working Note:

The cost per day has to be recovered from the daily traffic. The each type of vehicle is to be converted into equivalent unit. Let's convert all vehicle types equivalent to Two-wheelers.

Sl. No.	Type of vehicle	Daily traffic volume [A]	Weight	Ratio [B]	Equivalent Two-wheeler [A×B]
1.	Two wheelers	44,500	0.05	1	44,500
2.	Car and SUVs	3,450	0.20	4	13,800
3.	Bus and LCV	1,800	0.30	6	10,800
4.	Heavy commercial vehicles	816	0.45	9	7,344
	Total				76,444

ICAI · Old P3 · MTP Nov 2019 · Series I · question p. 7-8 · answer p. 9-10

CHAPTER 13

Standard Costing

13.1 Standards, their types and standard costing

3 questions · asked in 6 ICAI documents · 1 exam paper

876 EXAM Old P3 May 2022 Q6(d) 5 marks old scheme

Discuss briefly some of the criticism which may be levelled against the Standard Costing System.

ICAI'S ANSWER

Criticism of Standard Costing

(i) Variation in price: One of the chief problem faced in the operation of the standard costing system is the precise estimation of likely prices or rate to be paid. The variability of prices is so great that even actual prices are not necessarily adequately representative of cost. But the use of sophisticated forecasting techniques should be able to cover the price fluctuation to some extent. Besides this, the system provides for isolating uncontrollable variances arising from variations to be dealt with separately.

(ii) Varying levels of output: If the standard level of output set for pre-determination of standard costs is not achieved, the standard costs are said to be not realised. However, the statement that the capacity utilisation cannot be precisely estimated for absorption of overheads may be true only in some industries of jobbing type. In vast majority of industries, use of forecasting techniques, market research, etc., help to estimate the output with reasonable accuracy and thus the variation is unlikely to be very large. Prime cost will not be affected by such variation and, moreover, variance analysis helps to measure the effects of idle time.

(iii) Changing standard of technology: In case of industries that have frequent technological changes affecting the conditions of production, standard costing may not be suitable. This criticism does not affect the system of standard costing. Cost reduction and cost control is a cardinal feature of standard costing because standards once set do not always remain stable. They have to be revised.

(iv) Attitude of technical people: Technical people are accustomed to think of standards as physical standards and, therefore, they will be misled by standard costs. Since technical people can be educated to adopt themselves to the system through orientation courses, it is not an insurmountable difficulty.

(v) Mix of products: Standard costing presupposes a pre-determined combination of products both in variety and quantity. The mixture of materials used to manufacture the products may vary in the long run but since standard costs are set normally for a short period, such changes can be taken care of by revision of standards.

(vi) Level of Performance: Standards may be either too strict or too liberal because they may be based on (a) theoretical maximum efficiency, (b) attainable good performance or (c) average past performance. To overcome this difficulty, the management should give thought to the selection of a suitable type of standard. The type of standard most effective in the control of costs is one which represents an attainable level of good performance.

(vii) Standard costs cannot possibly reflect the true value in exchange: If previous historical costs are amended roughly to arrive at estimates for ad hoc purposes, they are not standard costs in the strict sense of the term and hence they cannot also reflect true value in exchange. In arriving at standard costs, however, the economic and technical factors, internal and external, are brought

together and analysed to arrive at quantities and prices which reflect optimum operations. The resulting costs, therefore, become realistic measures of the sacrifices involved.

(viii) Fixation of standards may be costly: It may require high order of skill and competency. Small concerns, therefore, feel difficulty in the operation of such system.

ICAI · Old P3 · May 2022 exam · question p. 27 · answer p. 29-30

877 RTP **Sep 2024** Q19(c)

Also asked in Old P3 MTP Jul 2021 S-I Q6(c)

DISCUSS the Controllable and un-controllable variances.

ICAI'S ANSWER

Controllable and un-controllable variances: The purpose of the standard costing reports is to investigate the reasons for significant variances so as to identify the problems and take corrective action.

Variances are broadly of two types, namely, controllable and uncontrollable. Controllable variances are those which can be controlled by the departmental heads whereas uncontrollable variances are those which are beyond their control. Responsibility centres are answerable for all adverse variances which are controllable and are appreciated for favourable variances. Controllability is a subjective matter and varies from situation to situation. If the uncontrollable variances are of significant nature and are persistent, the standard may need revision.

ICAI · P4 · RTP Sep 2024 · question p. 16 · answer p. 37

878 MTP **May 2026 S-II** Q4(b) 5 marks

Also asked in MTP Sep 2025 S-I Q6(c) · MTP Jan 2025 S-II Q2(b)

"Calculation of variances in standard costing is not an end in itself, but a means to an end." DISCUSS.

ICAI'S ANSWER

The crux of standard costing lies in variance analysis. Standard costing is the technique whereby standard costs are predetermined and subsequently compared with the recorded actual costs. It is a technique of cost ascertainment and cost control. It establishes predetermined estimates of the cost of products and services based on management's standards of efficient operation. It thus lays emphasis on "what the cost should be". These should be costs are when compared with the actual costs. The difference between standard cost and actual cost of actual output is defined as the variance.

The variance in other words in the difference between the actual performance and the standard performance. The calculations of variances are simple. A variance may be favourable or unfavourable. If the actual cost is less than the standard cost, the variance is favourable but if the actual cost is more than the standard cost, the variance will be unfavourable. They are easily expressible and do not provide detailed analysis to enable management of exercise control over them. It is not enough to know the figures of these variances from month to month. We in fact are required to trace their origin and causes of occurrence for taking necessary remedial steps to reduce / eliminate them.

A detailed probe into the variance particularly the controllable variances helps the management to ascertain:

(i) the amount of variance

(ii) the factors or causes of their occurrence

(iii) the responsibility to be laid on executives and departments and

(iv) corrective actions which should be taken to obviate or reduce the variances.

Mere calculation and analysis of variances is of no use. The success of variance analysis depends upon how quickly and effectively the corrective actions can be taken on the analysed variances. In fact variance gives information. The manager needs to act on the information provided for taking corrective action. Information is the means and action taken on it is the end. In other words, the calculation of variances in standard costing is not an end in itself, but a means to an end.

ICAI · P4 · MTP May 2026 · Series II · question p. 14 · answer p. 17

13.2 Material variances

46 questions · asked in 38 ICAI documents · 7 exam papers

879 EXAM **May 2026** Q1(b) 5 marks

EXA Ltd. operates a standard costing system for controlling its material costs and has furnished the following information for the month of April, 2026:

Standard Price of raw material	₹ 30 per Kg
Standard Quantity of material required per unit of output	5 kgs.
Standard labour rate	₹ 50 per hour
Standard labour requirement	2.5 hours per unit
Actual Output	2000 units
Material Price Variance	₹ 1,75,000 (Adverse)
Actual Quantity of raw material purchased and consumed	14,000 kgs.
Actual wages paid for 6,000 hours	₹ 3,24,000

You are required to calculate:

(i) Actual price per kg of raw material

(ii) Material usage variance

(iii) Material cost variance

(iv) Labour rate variance

(v) Labour efficiency variance

(Indicate the nature of variance Favourable (F) or Adverse (A))

ICAI'S ANSWER

Workings:

1. Standard Quantity for actual output = 2,000 x 5 kg = 10,000 kg
2. Standard Labour Hours for actual output = 2,000 x 2.5 hours per unit = 5,000 hours
3. Actual Labour Rate = ₹ 3,24,000/6,000 hours = ₹ 54

(i) Actual Price Per Kg

Material Price Variance = (SP - AP) x AQ

Material Price Variance = ₹ 1,75,000 (A)

Therefore:

(30 - AP) x 14,000 = -1,75,000

Actual Price = ₹ 42.5

(ii) Material Usage Variance = (SQ - AQ) x SP

= (10,000 – 14,000) x ₹ 30

= ₹ 1,20,000 (A)

(iii) Material Cost Variance = SQ x SP - AQ X AP

= 10,000 x ₹ 30 – 14,000 x ₹ 42.5

= ₹ 2,95,000 (A)

(iv) Labour Rate Variance = (SR - AR) X AH

= (₹ 50 - ₹ 54) x 6000

= ₹ 24,000 (A)

(v) Labour Efficiency Variance = (SH - AH) X SR

= (5,000 – 6,000) x ₹ 50

= ₹ 50,000 (A)

ICAI · P4 · May 2026 exam · question p. 8-9 · answer p. 10-11

880 EXAM Jan 2026 Q4(a) 7 marks

The following data pertaining to a company engaged in manufacturing and sale of a single product during the year:

Particulars	Budget	Actual
Sales (in units)	60,000	66,000
Sales (₹)	1,80,00,000	2,14,50,000
Direct Materials (₹)	28,80,000	36,30,000
Direct labour (₹)	43,20,000	52,80,000
Variable Overheads (₹)	72,00,000	81,84,000

Additional information is given below:

	Standard (₹)	Actual (₹)
Direct Material price per kg	10	8
Direct labour rate per hour	10	12

You are required to calculate:

(i) Direct material usage variance

(ii) Direct material price variance

(iii) Direct labour efficiency variance

(iv) Direct labour rate variance

(v) Variable overhead cost variance

ICAI'S ANSWER

	Budget			Revised standard			Actual		
	Qty	Price (₹)	Amount (₹)	Qty	Price (₹)	Amount (₹)	Qty	Price (₹)	Amount (₹)
Material	2,88,000	10	28,80,000	3,16,800	10	31,68,000	4,53,750	8	36,30,000

(i) Direct Material Usage Variance = (SQ – AQ) X SP

(3,16,800 – 4,53,750) x 10 = ₹ 13,69,500 A

(ii) Direct Material Price Variance = (SP – AP) X AQ

(10 - 8) x 4,53,750 = ₹ 9,07,500F

	Budget			Revised standard			Actual		
	Hrs	Rate (₹)	Amount (₹)	Hrs	Rate (₹)	Amount (₹)	Hrs	Rate (₹)	Amount (₹)
Labour	4,32,000	10	43,20,000	4,75,200	10	47,52,000	4,40,000	12	52,80,000

(iii) Direct Labor Efficiency Variance = (SH – AH) X SR

(4,75,200 – 4,40,000) x 10 = ₹ 3,52,000 F

(iv) Direct Labor rate Variance = (SR – AR) X AH

(10 – 12) x 4,40,000 = ₹ 8,80,000 A

(v) Variable overhead cost Variance = (Variable overhead for actual output – Actual Variable overheads)

$(72,00,000 \times \frac{66,000}{60,000} - 81,84,000) = ₹\ 2,64,000\ A$

ICAI · P4 · Jan 2026 exam · question p. 20-21 · answer p. 22

881 EXAM **Sep 2025** MCQ 11 2 marks MCQ

The Cost Accountant of AQ Limited has provided the following information for investigation of variances:

Material Cost Variance	₹ 4,220 (F)
Material Usage Variance	₹ 18,540 (A)
Material Yield Variance	₹ 8,390 (F)

The Management Accountant is not able to comment on the reason of variances as information is not sufficient and seeks your help to find out the correct amount of Material Price Variance and Material Mix Variance. What is the correct amount of Material Price Variance (MPV) and Material Mix Variance (MMV)?

A MPV ₹ 24,880 (A) and MMV ₹ 27,440 (F)

B MPV ₹ 14,320 (A) and MMV ₹ 10,150 (F)

C MPV ₹ 14,320 (F) and MMV ₹ 10,150 (A)

D MPV ₹ 22,760 (F) and MMV ₹ 26,930 (A)

ICAI'S ANSWER

Answer key: **(D)**

ICAI · P4 · Sep 2025 exam · question p. 5 · answer p. 7

882 **EXAM** **Sep 2024** Q4(a) 9 marks

BG company produces a standard product and sold in a packet of 10 kg. The standard cost card per pack is as follows:

Direct Material:

A - 4 kg @ ₹ 50 per kg

B - 8 kg @ ₹ 40 per kg

Direct Labour:

6 hours @ ₹ 20 per hour

The company manufactured and sold 1,600 packets during the month. Actual data for material and labour recorded as under.

Direct Material:

A - 7,000 kg @ ₹ 40

B - 12,500 kg @ ₹ 45

Labour hours paid for two different categories of workers:

Skilled	6,000 hours @ ₹ 25
Semi-skilled	4,000 hours @ ₹ 20

5% of the time paid was lost due to an abnormal reason.

Calculate the following variances indicating their nature (Favourable or Adverse):

(i) Material cost variances

(ii) Material price variances

(iii) Material usage variances

(iv) Material mix variances

(v) Material yield variances

(vi) Labour cost variances

(vii) Labour rate variances

(viii) Labour efficiency variances

(ix) Labour Idle time variances

ICAI'S ANSWER

	Budget			Actual		
	Qty. (Kg.) [SQ]	Price (₹) [SP]	Amount (₹) [SQ x SP]	Qty. (Kg.) [AQ]	Price (₹) [AP]	Amount (₹) [AQ x AP]
A	6,400	50	3,20,000	7,000	40	2,80,000
B	12,800	40	5,12,000	12,500	45	5,62,500
	19,200		**8,32,000**	**19,500**		**8,42,500**

Material Cost Variance = (SQ X SP – AQ X AP)

= ₹ 8,32,000 – ₹ 8,42,500 = **₹ 10,500 (A)**

Material Price Variance = (SP – AP) x AQ

A (₹ 50 – ₹ 40) × 7,000 Kg = **₹ 70,000 (F)**

B (₹ 40 – ₹ 45) × 12,500 Kg = **₹ 62,500 (A)**

₹ 7,500 (F)

Material Usage Variance = SP x (SQ – AQ)

A ₹ 50 × (6,400 Kg – 7000 Kg) = **₹ 30,000 (A)**

B ₹ 40 × (12,800 Kg – 12,500 Kg) = **₹ 12,000 (F)**

₹ 18,000 (A)

Material Mix Variance = (RSQ - AQ) X SP

A = (6,500 Kg – 7,000 Kg) x ₹ 50 = **₹ 25,000 (A)**

B = (13,000 Kg – 12,500 Kg) x ₹ 40 = **₹ 20,000 (F)**

₹ 5,000 (A)

Material Yield Variance = (SQ – RSQ) X SP

A = (6,400 Kg – 6,500 Kg) x ₹ 50 = **₹ 5,000 (A)**

B = (12,800 Kg – 13,000 Kg) x ₹ 40 = **₹ 8,000 (A)**

₹ 13,000 (A)

Labour

Standard Hours for actual Production = 6 Hours X 1,600 Units = 9,600 Hours

Labour Cost Variance = (SH X SR – AH X AR)

= 9,600 hrs x ₹ 20 – {(6,000 hrs x ₹ 25) + (4,000 hrs x ₹ 20)}

= ₹ 1,92,000 – ₹ 2,30,000 = **₹ 38,000 (A)**

Labour Rate Variance = (SR – AR) X AH

= (₹ 20 -₹ 25) x 6,000 hrs = **₹ 30,000 (A)**

Efficiency Variance = (SH – AH worked) x SR

= (9,600 hrs – 9,500 hrs) x ₹ 20 = **₹ 2,000 (F)**

Idle time Variance = Idle Hours X SR

= (AH – AH#) × SR

= (10,000 hours – 9,500 hours) x ₹ 20 = **₹ 10,000 (A)**

AH# refers to Actual Hours Worked

ICAI · P4 · Sep 2024 exam · question p. 24-25 · answer p. 26-27

883 EXAM **Old P3 May 2023** Q5(a) 10 marks old scheme

NC Limited uses a standard costing system for the manufacturing of its product 'X'. The following information is available for the last week of the month:

- 25,000 kg of raw material were actually purchased for ₹ 3,12,500. The expected output is 8 units of product 'X' from each one kg of raw material. There is no opening and closing inventories. The material price variance and material cost variance, as per cost records, are ₹ 12,500 (F) and ₹ 1800 (A), respectively.
- The standard time to produce a batch of 10 units of product 'X' is 15 minutes. The standard wage rate per labour hour is 50. The company employs 125 workers in two categories, skilled and semi-skilled, in a ratio of 60:40. The hourly wages actually paid were ₹ 50 per hour for skilled workers and ₹ 40 per hour for semi-skilled workers. The weekly working hours are 40 hours per worker. Standard wage rate is the same for skilled and semi- skilled workers.
- The monthly fixed overheads are budgeted at ₹ 76,480 Overheads are evenly distributed throughout the month and assume 4 weeks in a month. In the last week of the month, the actual fixed overhead expenses were ₹ 19,500.

Required:

(i) Calculate the standard price per kg and the standard quantity of raw material.

(ii) Calculate the material usage variance, labour cost variance, and labour efficiency variance.

(iii) Calculate the fixed overhead cost variance, the fixed overhead expenditure variance and the fixed overhead volume variance.

Note: Indicate the nature of variance i.e Favourable or Adverse.

ICAI'S ANSWER

(i) Calculation of Standard price per kg and the standard quantity of raw material:

Standard Price

(a) Material Price Variance = Standard Cost of Actual Quantity – Actual Cost

12,500 (F) = (SP × AQ) – ₹ 3,12,500

12,500 (F) = (SP × 25,000) – ₹ 3,12,500

SP = ₹ 13

Standard Quantity

(b) Material Cost Variance = Standard Cost – Actual Cost

1,800 (A) = SQ × ₹13 – ₹ 3,12,500

SQ = 23,900 kg.

(ii) Calculation of Material Usage Variance, Labour Cost Variance and Labour Efficiency Variance

(a) Material Usage Variance = Standard Cost of Standard Quantity for Actual Output – Standard Cost of Actual Quantity

= SQ × SP – AQ × SP

Or

= SP × (SQ – AQ)

= ₹ 13 × (23,900 kg. – 25,000 kg.)

= **₹ 14,300 (A)**

(b) Labour Cost Variance = Standard Cost – Actual Cost

= (SH × SR) – (AH × AR)

= ₹ 2,39,000 – ₹ 2,30,000

= **₹ 9,000 (F)**

(c) Labour Efficiency Variance = Standard Cost of Standard Time for Actual Production – Standard Cost of Actual Time

= (SH × SR) – (AH × SR)

Or

= (SH – AH) × SR

= ₹ 50 × [4,780 hrs. – 5,000 hrs.]

= **₹ 11,000 (A)**

(iii) Calculation of Fixed Overhead Cost Variance, Fixed Overhead Expenditure Variance and Fixed Overhead Volume Variance:

(a) Fixed overhead cost variance = Standard Fixed Overheads – Actual Fixed Overheads

= 18,279 – 19,500

= ₹ 1,221(A)

(b) Fixed Overhead Expenditure Variance = Budgeted Fixed Overheads – Actual Fixed Overheads

= ₹ 19,120 – ₹ 19,500

= **₹ 380 (A)**

(c) Fixed overhead volume variance = (Budgeted output – Actual Output) X Budgeted rate per unit

= (2,00,000 – 1,91,200) 0.0956

= ₹ 8,800 x 0.0956

= **₹ 841 (A)**

Alternative presentation to part (iii) (a) and (b)

(i) Fixed Overhead Cost Variance:

= Overhead absorbed for actual production – Actual overhead incurred

= $\frac{₹19,120}{2,00,000}$ x 1,91,200 – 19,500 = **₹ 1,221(A)**

(iii) Fixed Overhead Volume Variance:

= Absorbed overhead – Budgeted overhead

= $\frac{₹19,120}{2,00,000}$ x 1,91,200 – 19,120 = **₹ 841(A)**

Working Notes:

1. Standard time to produce 10 units of product X is 15 minutes. Therefore we can manufacture 40 units in an hour.

Hours available in a week

125 Workers x 40 Hours = 5,000 hours

Therefore budgeted output = 5,000 x 40 units per hour = 2,00,000 units

Alternatively

Budgeted time per unit = $\frac{15 \text{ units}}{10 \text{ units}}$ = 1.5 minutes

So, Budgeted output = $\frac{5{,}000 \text{ Hours} \times 60 \text{ Minutes}}{1.5 \text{ Minutes}}$ = 2,00,000 units

Actual output = 23,900 x 8 units = 1,91,200 units

Standard hour for actual output = $1{,}91{,}200 \times \frac{0.25 \text{ Hrs}}{10 \text{ units}} = 4{,}780 \text{ Hrs}$

2.

Labour

Budget			Revised standard			Actual			
Hours	Rate	₹	Hours	Rate	₹		Hours	Rate	₹
5,000	50	2,50,000	4,780	50	2,39,000	Skilled	3000	50	1,50,000
						Semi-Skilled	2000	40	80,000
							5000		2,30,000

3.

	Budget	Actual
Units	2,00,000	1,91,200
Fixed Overheads	19,120	19,500

4. Standard Fixed overheads:

$\frac{19,120}{2,00,000} \times 1{,}91{,}200$ = ₹18,279

Budgeted rate per unit:

$$\frac{19,120}{2,00,000} = ₹\ 0.0956$$

ICAI · Old P3 · May 2023 exam · question p. 23-24 · answer p. 24-27

884 EXAM Old P3 Nov 2022 Q5(a) 10 marks old scheme

Y Lid manufactures "Product M" which requires three types of raw materials - "A", "B" & "C". Following information related to 1st quarter of the F.Y. 2022-23 has been collected from its books of accounts. The standard material input required for 1,000 kg of finished product 'M' are as under:

Material	Quantity (Kg.)	Std. Rate per Kg. (₹)
A	500	25
B	350	45
C	250	55
	1100	
Standard Loss	100	
Standard Output	1000	

During the period, the company produced 20,000 kg of product "M" for which the actual quantity of materials consumed and purchase prices are as under:

Material	Quantity (Kg.)	Purchase price per Kg. (₹)
A	11,000	23
B	7,500	48
C	4,500	60

You are required to calculate:

(i) Material Cost Variance

(ii) Material Price Variance for each raw material and Product 'M'

(iii) Material Usage Variance for each raw material and Product 'M'

(iv) Material Yield Variance

Note: Indicate the nature of variance i.e. Favourable or Adverse.

ICAI'S ANSWER

Basic Calculations:

	Standard for 20,000 kg.			Actual for 20,000 kg.		
	Qty.	Rate	Amount	Qty.	Rate	Amount
	Kg.	(₹)	(₹)	Kg.	(₹)	(₹)
A	10,000	25	2,50,000	11,000	23	2,53,000
B	7,000	45	3,15,000	7,500	48	3,60,000
C	5,000	55	2,75,000	4,500	60	2,70,000
Total	22,000		8,40,000	23,000		8,83,000

Calculation of Variances:

(i) Material Cost Variance = Std. Cost for actual output–Actual cost

MCV =8,40,000– 8,83,000 = ₹ 43,000(A)

(ii) Material Price Variance = (SP–AP) × AQ

A	= (25 - 23) x 11,000	= 22,000 (F)
B	= (45 – 48) x 7,500	= 22,500 (A)
C	= (55 – 60) x 4,500	= 22,500 (A)
		23000 (A)

(iii) Material Usages Variance = (SQ–AQ) × SP

A	= (10,000 – 11,000) x 25	= 25,000 (A)
B	= (7,000 – 7,500) x 45	= 22,500 (A)
C	= (5,000 – 4,500) x 55	= 27,500 (F)
		20,000 (A)

(iv) Material Yield Variance = (SQ–RSQ*) × SP

A	= (10,000 – 10,454.54) x 25	= 11,363.5(A)
B	= (7,000 – 7,318.18) x 45	= 14,318.1(A)
C	= (5,000 – 5,227.27) x 55	= 12,500(A)
		38,181.6(A)

*Revised Standard Quantity (RSQ)

$$A = \frac{10,000}{22,000} \times 23,000 = 10,454.54$$

$B = \frac{7,000}{22,000} \times 23,000 = 7,318.18$

$C = \frac{5,000}{22,000} \times 23,000 = 5,227.27$

Material Yield Variance can also be Calculated as below

Material yield variance = Standard cost per unit (Actual yield – Standard yield)

Standard cost per unit = $\frac{₹\ 8,40,000}{20,000} = ₹\ 42$

New Standard Yield = $\frac{20,000}{22,000} \times 23,000 = 20,909$

Material yield variance = ₹ 42 (20,000 – 20,909)

= ₹ 38,178 (A)

ICAI · Old P3 · Nov 2022 exam · question p. 18 · answer p. 20-21

885 EXAM **Old P3 Nov 2019** Q5(b) 10 marks old scheme

The standard cost of a chemical mixture is as follows:

60% of Material A @ ₹ 50 per kg

40% Material B @ ₹ 60 per kg

A standard loss of 25% on output is expected in production. The cost records for a period has shown the following usage.

540 kg of Material A @ ₹ 60 per kg

260 kg of Material B @ ₹ 50 per kg

The quantity processed was 680 kilograms of good product.

From the above given information

Calculate:

(i) Material Cost Variance

(ii) Material Price Variance

(iii) Material Usage Variance

(iv) Material Mix Variance

(v) Material Yield Variance.

ICAI'S ANSWER

Basic Calculation

Material	Standard for 640 kg. output			Actual for 680 kg. output		
	Qty. Kg.	Rate (₹)	Amount (₹)	Qty Kg.	Rate (₹)	Amount (₹)
A	480	50	24,000	540	60	32,400
B	320	60	19,200	260	50	13,000
Total	800		43,200	800		45,400
Less: Loss	160	–	–	120	–	–
	640		43,200	680		45,400

Std. cost of actual output = ₹ 43,200 × 680/640 = ₹ 45,900

Calculation of Variances

(i) **Material Cost Variance** = (Std. cost of actual output – Actual cost)

= (45,900– 45,400)

= ₹ **500 (F)**

(ii) **Material Price Variance** = (SP – AP) × AQ

Material A = (50 – 60) × 540 = ₹ 5400 (A)

Material B = (60 – 50)) × 260 = ₹ 2600 (F)

MPV = **₹ 2800 (A)**

(iii) **Material Usage Variance (MUV)** = (Std. Quantity for actual output – Actual Quantity) × Std. Price

Material A = $\left(\frac{480 \times 680}{640} - 540\right) \times 50$ = ₹ 1,500 (A)

Material B = $\left(\frac{320 \times 680}{640} - 260\right) \times 60$ = ₹ 4,800 (F)

MUV = **₹ 3,300 (F)**

(iv) **Material Mix Variance** = SP × (RAQ – AQ)

A = ₹ 50× (480 Kg – 540 Kg) = ₹ 3,000 (A)

B = ₹ 60 × (320 Kg. – 260 Kg.) = ₹ 3,600 (F)

Total = ₹ 3,000 (A) + ₹ 3,600 (F) = **₹ 600 (F)**

(v) **Material Yield Variance** = SP × (SQ – RAQ)

A = ₹ 50 × (510 Kg. – 480 Kg) = ₹ 1,500 (F)

B = ₹ 60 × (340 Kg. – 320 Kg.) = ₹ 1,200 (F)

Total = ₹ 1,500 (F) + ₹ 1,200 (F) = ₹ **2,700 (F)**

ICAI · Old P3 · Nov 2019 exam · question p. 21 · answer p. 22-23

886 RTP **May 2026** Q15

Also asked in Old P3 RTP Dec 2021 Q12

BabyMoon Ltd. uses standard costing system in manufacturing one of its product 'Baby Cap'. The details are as follows:

Direct Material 1 Meter @ ₹ 60 per meter	₹ 60
Direct Labour 2 hour @ ₹ 20 per hour	₹ 40
Variable overhead 2 hour @ ₹ 10 per hour	₹ 20
Total	₹ 120

During the month of August, 10,000 units of 'Baby Cap' were manufactured. Details are as follows:

Direct material consumed	11,400 meters @ ₹ 58 per meter		
Direct labour Hours	?	@ ?	₹ 4,48,800
Variable overhead incurred			₹ 2,24,400

Variable overhead efficiency variance is ₹ 4,000 A. Variable overheads are based on Direct Labour Hours.

You are required to CALCULATE the following Variances:

(a) Material Variances- Material Cost Variance, Material Price Variance and Material Usage Variance.

(b) Variable Overheads variances- Variable overhead Cost Variance, Variable overhead Efficiency Variance and Variable overhead Expenditure Variance.

(c) Labour variances- Labour Cost Variance, Labour Rate Variance and Labour Efficiency Variance.

ICAI'S ANSWER

(i) Material Variances

Budget			Std. for actual			Actual		
Quantity (Meter)	Price (₹)	Amount (₹)	Quantity (Meter)	Price (₹)	Amount (₹)	Quantity (Meter)	Price (₹)	Amount (₹)
1	60	60	10,000	60	6,00,000	11,400	58	6,61,200

Material Cost Variance = (SQ × SP – AQ × AP)

= 6,00,000 – 6,61,200 = ₹ 61,200 (A)

Material Price Variance = (SP – AP) AQ

= (60 - 58) 11,400 = ₹ 22,800 (F)

Material Usage Variance = (SQ – AQ) SP

= (10,000 – 11,400) 60 = ₹ 84,000 (A)

(ii) Variable Overheads variances

Variable overhead cost Variance

= Standard variable overhead – Actual Variable Overhead

= (10,000 units × 2 hours × ₹ 10) – 2,24,400 = ₹ 24,400 (A)

Variable overhead Efficiency Variance

= (Standard Hours – Actual Hours) × Standard Rate per Hour

Let Actual Hours be 'X', then:

(20,000 – X) × 10 = 4,000 (A)

2,00,000 – 10X = - 4,000

X = 2,04,000 ÷ 10

Therefore, Actual Hours (X) = 20,400

Variable overhead Expenditure Variance

= Variable Overhead at Actual Hours - Actual Variable Overheads

= 20,400 × ₹ 10 – 2,24,400 = ₹ 20,400 (A)

(iii) Labour variances

Budget			Std. for actual			Actual		
Hours	Rate (₹)	Amount (₹)	Hours	Rate (₹)	Amount (₹)	Hours	Rate (₹)	Amount (₹)
2	20	40	20,000	20	4,00,000	20,400	22*	4,48,800

*Actual Rate = ₹ 4,48,800 ÷ 20,400 hours = ₹ 22

Labour Cost Variance = (SH × SR) – (AH × AR)

= 4,00,000 – 4,48,800 = ₹ 48,800 (A)

Labour Rate Variance = (SR – AR) × AH

= (20 – 22) × 20,400 = ₹ 40,800 (A)

Labour Efficiency Variance = (SH – AH) × SR

= (20,000 – 20,400) × 20 = ₹ 8,000 (A)

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887 RTP **Jan 2026** Q5 MCQ

Product AB has a standard cost of ₹ 32, ₹ 6 of which relates to direct materials. Budgeted production for the month was 1600 units. During the month, 1500 units of AB were produced and materials worth ₹ 9,000 were purchased. There was no opening stock of materials but closing stock, which was valued at standard cost, amounted to ₹ 1,500. What is the total variance for materials?

A ₹ 500 (F)

B ₹ 500 (A)

C ₹ 1500 (F)

D ₹ 1500 (A)

ICAI'S ANSWER

(C) ₹ 1500 (F)

Material Cost Variance

= Standard cost of actual output – Actual cost of materials consumed

= (1,500 x 6) – (9,000 - 1,500)

= 1,500 (F)

ICAI · P4 · RTP Jan 2026 · question p. 7-8 · answer p. 28

888 RTP **Jan 2026** Q16

A company produces a product X, using raw materials A and B. The standard mix of A and B is 1: 1 and the standard loss is 10% of input. You are required to COMPUTE the MISSING INFORMATION indicated by "?" based on the data given below:

	A	B	Total
Standard price of raw material (₹ / kg.)	24	30	
Actual input (kg.)	?	70	
Actual output (kg.)			?
Actual price ₹ / kg.	30	?	
Standard input quantity (kg.)	?	?	
Yield variance (sub usage)	?	?	270(A)
Mix variance	?	?	?
Usage variance	?	?	?
Price variance	?	?	?
Cost variance	0	?	1,300(A)

ICAI'S ANSWER

Working for Finding – Missing Figures

Cost Variance $_A$	=	0
Cost Variance $_{(A+B)}$	=	₹ 1,300 (A)
Yield Variance $_{(A+B)}$	=	₹ 270 (A)

Standard Cost and Actual Cost (Incomplete Information)

Raw Material	Standard Data			Actual Data		
	Qty. (Kg.) [SQ]	Price (₹) [SP]	Amount (₹) [SQ x SP]	Qty. (Kg.) [AQ]	Price (₹) [AP]	Amount (₹) [AQ x AP]
A	? ??	24	? ? ?	? ? ?	30	? ? ?
B	? ??	30	? ? ?	70	? ? ?	? ? ?
Total	? ? ?		? ??	? ??		? ? ?

Material Cost Variance $_A$	=	Standard Cost – Actual Cost
⇒ 0	=	(SQ_A × ₹ 24 – AQ_A × ₹ 30)
⇒ SQ_A	=	1.25 AQ_A
Material Yield Variance $_{(A+B)}$	=	Average Standard Price per unit of Standard Mix × [Total Standard Quantity (units) – Total Actual Quantity (units)]
⇒ ₹ 270 (A)	=	$\left(\frac{₹24 \text{ x } SQ_A + ₹30 \text{ x } SQ_B}{SQ_A + SQ_B}\right)$ × [(SQ_A + SQ_B) – (AQ_A+70)]
SQ_A	=	*SQ_B as Standard Mix is in ratio 1:1*
⇒₹ 270 (A)	=	$\left(\frac{₹24 \text{ x } SQ_A + ₹30 \text{ x } SQ_A}{SQ_A + SQ_A}\right)$ × [(SQ_A + SQ_A) – (AQ_A+70)]
⇒₹ 270 (A)	=	27 × [2 x SQ_A – (AQ_A+70)]
⇒₹ 270 (A)	=	27 × [2 × 1.25 AQ_A– (AQ_A+70)]
⇒AQ_A	=	40 Kg.
As SQ_A	=	1.25 AQ_A
	=	1.25 × 40 Kg.
	=	50 Kg.
As SQ_B	=	SQ_A
	=	50 Kg.
Cost Variance $_{(A+B)}$	=	Standard Cost – Actual Cost
⇒ 1,300 (A)	=	(50 Kg. × ₹ 24 + 50 Kg. × ₹ 30) – (40Kg. ×₹ 30 +70 Kg. × AP_A)
⇒ AP_A	=	₹ 40

Standard Cost and Actual Cost (Complete Information)

Raw Material	Standard Data			Actual Data			Std. Cost of Actual Qty. (₹) [AQ x SP]
	Qty. (Kg.) [SQ]	Price (₹) [SP]	Amount (₹) [SQ x SP]	Qty. (Kg.) [AQ]	Price (₹) [AP]	Amount (₹) [AQ x AP]	
A	50	24	1,200	40	30	1,200	960
B	50	30	1,500	70	40	2,800	2,100
Total	100		2,700	110		4,000	3,060

Computation of Variances

Material Cost Variance	=	Standard Cost – Actual Cost
	=	SQ × SP – AQ × AP
(A)	=	₹1,200 – ₹1,200
	=	₹0
(B)	=	₹1,500 – ₹2,800
	=	₹1,300 (A)
Total	=	₹0 + ₹1,300 (A)
	=	₹1,300 (A)
Material Price Variance	=	Standard Cost of Actual Quantity – Actual Cost
	=	AQ × SP – AQ × AP
		Or
	=	AQ × (SP – AP)
(A)	=	40 Kg. × (₹24.00 – ₹30.00)
	=	₹240 (A)
(B)	=	70 Kg. × (₹30.00 – ₹40.00)
	=	₹700 (A)
Total	=	₹240 (A) + ₹700 (A)
	=	₹940 (A)
Material Usage Variance	=	Standard Cost of Standard Quantity for Actual Output – Standard Cost of Actual Quantity
	=	SQ × SP – AQ × SP
		Or
	=	SP × (SQ – AQ)
(A)	=	₹24 × (50 Kg. – 40 Kg.)
	=	₹240 (F)
(B)	=	₹30 × (50 Kg. – 70 Kg.)
	=	₹600 (A)
Total	=	₹240 (F) + ₹600 (A)

	=	₹360 (A)
Material Mix Variance	=	Total Actual Quantity (units) × (Average Standard Price Per unit of Standard Mix - Average Standard Price per unit of Actual Mix)
	=	110 Kg. × $\left(\frac{₹2,700}{100\ \text{Kg.}} - \frac{₹3,060}{110\ \text{Kg.}}\right)$
	=	₹90 (A)
Material Yield Variance	=	Average Standard Price per unit of Standard Mix × [Total Standard Quantity (units) – Total Actual Quantity (units)]
	=	$\left(\frac{₹2,700}{100\ \text{Kg.}}\right)$ × (100 Kg. – 110 Kg.)
	=	₹ 270 (A)
Standard Output	=	Standard Input – Standard Loss
	=	100 Kg. – 10 Kg.
	=	90Kg.
Actual Output	=	90 Kg.

(Actual Output and Standard Output are always equal numerically in any Material Variance Analysis)

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889 RTP **May 2025** Q17

Same question, different ICAI answer: #899 MTP May 2026 S-I

COMPUTE the missing data indicated by the question marks from the following:

Particulars	A	B
Standard Price/ unit	₹ 12	₹ 15
Actual Price/ unit	₹ 15	₹ 20
Standard Input (kgs.)	50	?
Actual Input (kgs.)	?	70
Material Price Variance	?	?
Material Usage Variance	?	₹ 300 Adverse
Material Cost Variance	?	?

Material mix variance for both products together was ₹ 45 Adverse.

ICAI'S ANSWER

(i) Standard input (kgs.) of Material- B:

Material usage variance = Std. Rate (Std. Quantity – Actual Quantity)

₹ 300 Adverse = ₹ 15 (SQ – 70)

Or, -300 = 15 SQ – 1,050

Or, SQ = 50 kgs.

(ii) Actual Input (kgs) of Material- A:

Let the actual input in for Material-A is X kgs.

Material Mix Variance = Std. Price (Actual Quantity in Std. mix – Actual Quantity)

Or, Material Mix Variance (A+B) = Material Mix Variance for Material - A + Material Mix Variance for Material -B

Or, - 45 = $[₹12\{(\frac{X+70}{2})-X\}]+[₹15\{(\frac{X+70}{2})-70\}]$

Or, - 45 = $[₹12\{\frac{X+70-2X}{2}\}]+[₹15\{\frac{X+70-140}{2}\}]$

Or, - 45 = $[₹12\{\frac{70-X}{2}\}]+[₹15\{\frac{X-70}{2}\}]$

Or, - 45 = $[-6X+420]+[\frac{15X-1,050}{2}]$

Or, - 45 = $[\frac{-12X+840+15X-1,050}{2}]$

Or, - 90 = 3X−210

Or, X = $\frac{120}{3}$ = 40kgs.

(iii) (a) Material Price Variance of A = Actual Quantity (Std. Rate – Actual Rate)

= 40 kg. (12–15) = ₹ 120 Adverse

(b) Material Price Variance of B = 70 kg. (15 – 20) = ₹ 350 Adverse

(iv) Material usage variance of A = Std. Rate (Std. Quantity – Actual Quantity)

= 12 (50 – 40) = ₹ 120 Favourable

(v) (a) Material Cost variance of A = Std. Cost – Actual Cost

= (50 kgs. @ ₹ 12) – (40 kgs. @ ₹ 15)

= 600 – 600 = Nil

(b) Material Cost variance of B = (50 kgs. @ ₹ 15) – (70 kgs. @ ₹ 20)

= 750 – 1,400 = ₹ 650 Adverse

ICAI · P4 · RTP May 2025 · question p. 22 · answer p. 48-49

890 RTP **Jan 2025** Q16

Banku manufacturing Ltd. is engaged in producing a item named 'ABC'. It produces 'ABC' in a batch of 100 kgs. Standard material inputs required for 100 kgs. of 'ABC' are as below:

Material	Quantity (in kgs.)	Rate per kg. (in ₹)
A	50	110
B	30	320
C	30	460

During the month of April, 2024, actual production was 50,000 kgs. of 'ABC' for which the actual quantities of material used for a batch and the prices paid thereof are as under:

Material	Quantity (in kgs.)	Rate per kg. (in ₹)
A	60	115
B	25	330
C	20	405

You are required to CALCULATE the following variances based on the above given information for the month of April, 2024 for Banku manufacturing Ltd.:

(i) Material Cost Variance;

(ii) Material Price Variance;

(iii) Material Usage Variance;

(iv) Material Mix Variance;

(v) Material Yield Variance.

ICAI'S ANSWER

(i)

Material	SQ* × SP (₹)	AQ** × SP (₹)	AQ** × AP (₹)	RSQ*** × SP (₹)
A	27,50,000	33,00,000	34,50,000	26,24,600
	(25,000 kg. × ₹ 110)	(30,000 kg. × ₹ 110)	(30,000 kg. × ₹ 115)	(23,860 kg. × ₹ 110)
B	48,00,000	40,00,000	41,25,000	45,82,400
	(15,000 kg. × ₹ 320)	(12,500 kg. × ₹ 320)	(12,500 kg. × ₹ 320)	(14,320 kg. × ₹ 320)
C	69,00,000	46,00,000	40,50,000	65,87,200
	(15,000 kg. × ₹ 460)	(10,000 kg. × ₹ 460)	(10,000 kg. × ₹ 405)	(14,320 kg. × ₹ 460)
Total	1,44,50,000	1,19,00,000	1,16,25,000	1,37,94,200

* Standard Quantity of materials for actual output :

A	$= \frac{50 \text{ kgs.}}{100 \text{ kgs}} \times 50{,}000 \text{ kgs.} = 25{,}000 \text{ kgs.}$
B	$= \frac{30 \text{ kgs.}}{100 \text{ kgs}} \times 50{,}000 \text{ kgs.} = 15{,}000 \text{ kgs.}$
C	$= \frac{30 \text{ kgs.}}{100 \text{ kgs}} \times 50{,}000 \text{ kgs.} = 15{,}000 \text{ kgs.}$

** Actual Quantity of Material used for actual output:

A	$= \frac{60 \text{ kgs.}}{100 \text{ kgs}} \times 50{,}000 \text{ kgs.} = 30{,}000 \text{ kgs.}$
B	$= \frac{25 \text{ kgs.}}{100 \text{ kgs}} \times 50{,}000 \text{ kgs.} = 12{,}500 \text{ kgs.}$
C	$= \frac{20 \text{ kgs.}}{100 \text{ kgs}} \times 50{,}000 \text{ kgs.} = 10{,}000 \text{ kgs.}$

***Revised Standard Quantity (RSQ):

A	$= \frac{50 \text{ kgs.}}{110 \text{ kgs}} \times 52{,}500 \text{ kgs.} = 23{,}860 \text{ kgs.}$
B	$= \frac{30 \text{ kgs.}}{110 \text{ kgs}} \times 52{,}500 \text{ kgs.} = 14{,}320 \text{ kgs.}$
C	$= \frac{30 \text{ kgs.}}{110 \text{ kgs}} \times 52{,}500 \text{ kgs.} = 14{,}320 \text{ kgs.}$

(i) Material Cost Variance = (Std. Qty. × Std. Price) – (Actual Qty. × Actual Price)

Or = (SQ × SP) – (AQ × AP)

A	= ₹ 27,50,000 - ₹ 34,50,000	= ₹ 7,00,000 (A)
B	= ₹ 48,00,000 - ₹ 41,25,000	= ₹ 6,75,000 (F)
C	= ₹ 69,00,000 - ₹ 40,50,000	= ₹ 28,50,000 (F)
		= ₹ 28,25,000 (F)

(ii) Material Price Variance = Actual Quantity (Std. Price – Actual Price)

= (AQ × SP) – (AQ × AP)

A	= ₹ 33,00,000 - ₹ 34,50,000	= ₹ 1,50,000 (A)
B	= ₹ 40,00,000 - ₹ 41,25,000	= ₹ 1,25,000 (A)
C	= ₹ 46,00,000 - ₹ 40,50,000	= ₹ 5,50,000 (F)
		= ₹ 2,75,000 (F)

(iii) Material Usage Variance = Std. Price (Std. Qty. – Actual Qty.)

Or = (SQ × SP) – (AQ × SP)

A	= ₹ 27,50,000 - ₹ 33,00,000	= ₹ 5,50,000 (A)
B	= ₹ 48,00,000 - ₹ 40,00,000	= ₹ 8,00,000 (F)
C	= ₹ 69,00,000 - ₹ 46,00,000	= ₹ 23,00,000 (F)
		= ₹ 25,50,000 (F)

(iv) Material Mix Variance = Std. Price (Revised Std. Qty. – Actual Qty.)

Or = (RSQ × SP) – (AQ × SP)

A	= ₹ 26,24,600 - ₹ 33,00,000	= ₹ 6,75,400 (A)
B	= ₹ 45,82,400 - ₹ 40,00,000	= ₹ 5,82,400 (F)
C	= ₹ 65,87,200 - ₹ 46,00,000	= ₹ 19,87,200 (F)
		= ₹ 18,94,200 (F)

(v) Material Yield Variance = Std. Price (Std. Qty. – Revised Std. Qty.)

Or = (SQ × SP) – (RSQ × SP)

A	= ₹ 27,50,000 - ₹ 26,24,600	= ₹ 1,25,400 (F)
B	= ₹ 48,00,000 - ₹ 45,82,400	= ₹ 2,17,600 (F)
C	= ₹ 69,00,000 - ₹ 65,87,200	= ₹ 3,12,800 (F)
		= ₹ 6,55,800 (F)

ICAI · P4 · RTP Jan 2025 · question p. 16-17 · answer p. 40-42

891 RTP **Sep 2024** Case scenario · Q2(i) to Q2(v)

CASE SCENARIO

ABC Pvt Ltd is engaged in the manufacture of a Product Q. The product has the following standard production requirements determined by the technical team of the company post satisfactory completion of test run.

Raw Material Z – 2 units @ ₹ 2 per unit

Skilled labour of – 2.5 hours@ ₹ 5 per hour

Fixed Overheads – ₹ 7.5 per unit

The input of Raw material Z has a yield of 80% everytime when infused into production. The actual quantity of Raw material Z consumed for production during the year was 24,000 units. The Usage variance of Material Z was 2,000 Favourable. Further the actual amount of material cost for the material consumed amounted to ₹ 45,000.

During the said year, the actual working hours were 30,000 for which the labour cost paid by the company amounted to ₹1,20,000. The idle time variance amounted to 10,000 Adverse.

The actual fixed overheads incurred for the year amounted to ₹ 1,50,000 and the expenditure variance was ₹25,000 Favourable.

In the context of the above, the following needs to be determined:

Q2(i)

The Actual output of Product Q produced during the year is:

A 10,000 units

B 12,500 units

C 25,000 units

D 15,000 units

Q2(ii)

The Material price and material cost variance are:

A Price variance – 3,000 Adverse, Cost Variance – 5,000 Adverse

B Price variance – 3,000 Favourable, Cost Variance – 5,000 Favourable

C Price variance – 3,000 Favourable, Cost Variance – 8,000 Adverse

D Price variance – 5,000 Adverse, Cost Variance – 3,000 Favourable

Q2(iii)

The Standard Hours, Net Actual hours and the idle time are:

A Standard Hours – 27,500 Net Actual Hours – 28,000 hours Idle Time – 2,000 hours

B Standard Hours – 22,500 Net Actual Hours – 28,500 hours Idle Time – 1,500 hours

C Standard Hours – 24,000 Net Actual Hours – 29,000 hours Idle Time – 1,000 hours

D Standard Hours – 25,000 hours Net Actual Hours –28,000 hours Idle Time – 2,000 hours

Q2(iv)

Labour Efficiency variance and Labour rate variance are:

A Labour Efficiency Variance – 30,000 Favourable Labour rate Variance – 25,000 Adverse

B Labour Efficiency Variance – 25,000 Favourable, Labour rate Variance – 30,000 Adverse

C Labour Efficiency Variance – 25,000 Adverse, Labour rate Variance – 30,000 Favourable

D Labour Efficiency Variance – 30,000 Adverse Labour rate Variance – 25,000 Favourable

Q2(v)

Fixed Overhead volume variance is:

A Fixed Overhead volume variance – 1,00,000 Favourable

B Fixed Overhead volume variance – 50,000 Adverse

C Fixed Overhead volume variance – 1,00,000 Adverse

D Fixed Overhead volume variance – 50,000 Favourable

ICAI'S ANSWERS

Q2(i)

(A) 10,000 units

Usage variance of Material Z = 2,000 F

Usage Variance = SQ x SP – AQ x SP

SP =₹2

AQ = 24,000 units

2 x (SQ – 24,000) = 2,000

2SQ = 50,000

Therefore SQ = 25,000

No of units of Input required per output =2

Yield of input = 80%

= (25000/2) x 80% = 10,000 units.

Q2(ii)

(B) Price variance – 3,000 Favourable, Cost Variance – 5,000 Favourable

Price variance = AQ x (SP-AP)

24,000 x (2-1.875) = 3,000 Favourable.

Cost variance = SQ x SP – AQ x AP

= 50,000–45,000=5,000 Favourable.

Q2(iii)

(D) Standard Hours – 25,000 hours Net Actual Hours –28,000 hours Idle Time – 2,000 hours

Actual output = 10,000 units

Standard hours per unit = 2.5

Therefore standard hours = 10,000 x 2.5 = 25,000 hours.

Idle time variance = SR x (Net AH – AH)

5 x (Net AH – 30,000) = 10,000 Adverse

5 Net AH – 1,50,000 = -10,000

5 Net AH = 1,40,000

Net AH = 28,000 hours

Idle time = 2,000 hours

Q2(iv)

(C) Labour Efficiency Variance – 25,000 Adverse, Labour rate Variance – 30,000 Favourable

Efficiency Variance = SR x (SH-AH)

= 5 x (25,000 – 30,000)

= 25,000 Adverse

Rate Variance = AH x (SR – AR)

= 30,000 (5 – 4) [1,20,000/30,000]

= 30,000 Favourable.

Q2(v)

(C) Fixed Overhead Volume variance – 1,00,000 Adverse

Overhead Volume variance = Actual OutputxSR per unit – Budgeted FOH

Budgeted FOH = Actual FOH (+/-) Expenditure variance

1,50,000 + 25,000 = 1,75,000

AO x SR = 10,000 x 7.5 = 75,000

Therefore volume variance = 75,000 – 1,75,000

=1,00,000 Adverse.

ICAI · P4 · RTP Sep 2024 · question p. 3 · answer p. 18, 18-19, 19

892 RTP May 2024 Q10

EML operates in coal mining through open cast mining method. Explosives and detonators are used for excavation of coal from the mines. The following are the details of standard quantity of explosives materials used for mining:

Particulars	Rate (₹)	Standard Qty. for Iron ore	Standard Qty. for Overburden (OB)
SME	40.00 per kg.	2.4 kg per tonne	1.9 kg per cubic- meter
Detonators	20.00 per piece	2 pcs per tonne	2 pcs per cubic- meter

The standard stripping ratio is 3:1 (means 3 cubic- meter of overburden soil to be removed to get one tonne of coal).

During the month of December 2023, the company produces 20,000 tonnes of coal and 58,000 cubic-meter of OB. The quantity of explosive materials used and paid for the month is as below:

Material	Quantity	Amount (₹)
SME	1,67,200 kg.	63,53,600
Detonators	1,18,400 pcs	24,27,200

Explosive suppliers are paid for the explosive materials on the basis of performance of the explosives which is termed as powder factor. One of the suppliers has presented their bill for explosive supplied for the month of December 2023. You being a bill passing officer of EML is required to COMPUTE the material price variance, material quantity variance and material cost variance.

ICAI'S ANSWER

Workings:

1. Calculation of Standard Qty. of Explosives and Detonators for actual output:

	Particulars	Coal	Overburden (OB)	Total
SME:				
A	Actual Output	20,000 tonne	58,000 M^3	
B	Standard Qty per unit	2.4 kg./ tonne	1.9 kg./M^3	
C	**Standard Qty. for actual production [A×B]**	**48,000 kg.**	**1,10,200 kg.**	**1,58,200 kg.**
Detonators:				
D	Standard Qty per unit	2 pcs/ tonne	2 pcs/ M^3	
E	**Standard Qty. for actual production [A × D]**	**40,000 pcs.**	**1,16,000 pcs**	**1,56,000 pcs**

2. Calculation of Actual Price per unit of materials:

Material	Quantity [A]	Amount (₹) [B]	Rate (₹) [C = B÷A]
SME	1,67,200 kg.	63,53,600	38.00
Detonators	1,18,400 pcs	24,27,200	20.50

Computation of material price variance:

Material Price Variance = Actual Qty. × (Std. Price - Actual Price)

SME = 1,67,200 kg. × (₹40 – ₹38) = ₹3,34,400 (F)

Detonators = 1,18,400 pcs × (₹20 – ₹20.5) = ₹59,200 (A)

Total = ₹2,75,200 (F)

Computation of material quantity variance:

Material Qty. Variance = Std. Price × (Std. Qty for actual output - Actual Qty.)

SME = ₹40 × (1,58,200 kg. - 1,67,200 kg.) = ₹3,60,000 (A)

Detonators = ₹20 × (1,56,000 pcs -1,18,400 pcs) = ₹7,52,000 (F)

Total = ₹3,92,000 (F)

Computation of material cost variance:

Material cost variance = Std. cost – Actual Cost

Or, (Std. Price × Std. Qty) – (Actual Price × Actual Qty.)

SME = (₹ 40 × 1,58,200 kg) – (₹ 38 × 1,67,200 kg.)

= ₹ 63,28,000 – ₹ 63,53,600 = ₹ 25,600 (A)

Detonators = (₹ 20 × 1,56,000 pcs) – (₹ 20.50 × 1,18,400 pcs)

= ₹31,20,000 – ₹24,27,200 = 6,92,800 (F)

Total = ₹6,67,200 (F)

ICAI · P4 · RTP May 2024 · question p. 14 · answer p. 36-38

893 RTP **Old P3 May 2023** Q12 old scheme

XYZ Manufacturing Ltd. had prepared the following estimation for the month of January:

	Quantity	Rate (₹)	(₹)
Raw Material-DF	1,600 kg.	50	80,000
Raw Material-CE	1,200 kg.	35	42,000
Skilled labour	2,000 hours	40	80,000
Semiskilled labour	1,600 hours	25	40,000

Standard loss in the process was expected to be 10% of total input materials and an idle labour time of 5% of expected labour hours was also estimated.

At the end of the month the following information has been collected from the cost accounting department:

The company has produced 2,960 kg. finished product by using the followings:

	Quantity	Rate (₹)	(₹)
Raw Material-DF	1,800 kg.	40	72,000
Raw Material-CE	1,300 kg.	30	39,000
Skilled labour	2,400 hours	35	84,000
Semiskilled labour	1,720 hours	20	34,400

You are required to CALCULATE:

(a) Material Cost Variance;

(b) Material Price Variance;

(c) Material Mix Variance;

(d) Material Yield Variance;

(e) Labour Cost Variance;

(f) Labour Efficiency Variance and

(g) Labour Yield Variance

ICAI'S ANSWER

Material Variance

Raw Material	SQ (kg.)	SP	SQ × SP	RSQ (WN-2) (kg.)	RSQ × SP	AQ	AQ × SP	AP	AQ × AP
	(WN-1)	**(₹)**	**(₹)**		**(₹)**		**(₹)**	**(₹)**	**(₹)**
DF	1879	50	93,950	1771	88,550	1800	90,000	40	72,000
CE	1410	35	49,350	1329	46,515	1300	45,500	30	39,000
	3289		1,43,300	3,100	1,35,065	3100	1,35,500		1,11,000

WN-1: Standard Quantity (SQ):

1879.365 or 1879 kg. = $\left(\frac{1{,}600 \text{ kg.}}{0.9 \times 2{,}800 \text{ kg.}} \times 2{,}960 \text{ kg.}\right)$

Raw Material DF = $\left(\frac{1{,}200 \text{ kg.}}{0.9 \times 2{,}800 \text{ kg.}} \times 2{,}960 \text{ kg.}\right)$

Raw Material CE = 1409.52 or 1410 kg. $\left(\frac{1{,}200 \text{ kg.}}{2{,}800 \text{ kg.}} \times 3{,}100 \text{ kg.}\right)$

WN- 2: Revised Standard Quantity (RSQ):

Raw Material DF = 1,771.43 or 1,771 kg. $\left(\frac{1{,}200 \text{ kg.}}{2{,}800 \text{ kg.}} \times 3{,}100 \text{ kg.}\right)$

Raw Material CE = 1,328.57 or 1,329 kg.

(a) Material Cost Variance (A + B) = {(SQ × SP) – (AQ × AP)}

{1,43,300 –1,11,000} = 32,300(F) (F)

(b) Material Price Variance (A + B) = {(AQ × SP) – (AQ ×AP)

{1,35,500 – 1,11,000} = 24,500(F)

(c) Material Mix Variance (A + B) = {(RSQ × SP) – (AQ × SP)}

{1,35,065 – 1,35,500} = 435 (A)

(d) Material Yield Variance (A + B) = {(SQ × SP) – (RSQ × SP)}

{1,43,300 – 1,35,065} = 8,235 (F)

Labour Variances:

Labour	SH (WN-3)	SR (₹)	SH × SR (₹)	RSH (WN-4)	RSH × SR (₹)	AH	AH × SR (₹)	AR (₹)	AH × AR (₹)
Skilled	2232	40	89,280	2289	91,560	2,400	96,000	35	84,000
Semiskilled	1785	25	44,625	1831	45,775	1720	43,000	20	34,400
	4,017 hrs		1,33,905	4,120	1,37,335	4,120	1,39,000		1,18,400

WN- 3: Standard Hours (SH):

Skilled labour = 2,231.746 or 2,232 hrs $\left(\frac{0.95 \text{ x } 2{,}000 \text{ hr}}{0.90 \text{ x } 2{,}800 \text{ kg}} \text{ x } 2{,}960 \text{ kg}\right)$

Semiskilled labour = 1785.397 or 1785 hrs $\left(\frac{0.95 \text{ x } 1600 \text{ hr}}{0.90 \text{ x } 2{,}800 \text{ kg}} \text{ x } 2{,}960 \text{ kg}\right)$

WN- 4: Revised Standard Hours (RSH):

Skilled labour = 2,288.889 or 2,289 hrs. = $\left(\frac{2{,}000 \text{ hrs}}{3{,}600 \text{ kg}} \text{ x } 4{,}120 \text{ hrs}\right)$

Semiskilled labour = 1831.11 or 1831 hrs. = $\left(\frac{1{,}600 \text{ hrs}}{3{,}600 \text{ kg}} \text{ x } 4{,}120 \text{ hrs}\right)$

(e) Labour Cost Variance (Skilled + Semiskilled) = {(SH × SR) – (AH × AR)}

{1,33,905 – 1,18,400} =15,505 (F)

(f) Labour Efficiency Variance (Skilled + Semiskilled) = {(SH × SR) – (AH × SR)}

{1,33,905 – 1,39,000} = 5,095 (A)

(g) Labour Yield Variance (Skilled + Semiskilled) = {(SH × SR) – (RSH × SR)}

= {1,33,905 – 1,37,335} = 3,430 (A)

ICAI · Old P3 · RTP May 2023 · question p. 8-9 · answer p. 21-23

894 RTP **Old P3 Nov 2022** Q12 old scheme

Ahaan Limited operates a system of standard costing in respect of one of its products 'AH1' which is manufactured within a single cost centre. Details of standard per unit are as follows:

- The standard material input is 20 kilograms at a standard price of ₹ 24 per kilogram.
- The standard wage rate is ₹ 72 per hour and 5 hours are allowed to produce one unit.

- Fixed production overhead is absorbed at the rate of 100% of wages cost.

During the month of April 2022, the following was incurred:

- Actual price paid for material purchased @ ₹ 22 per kilogram.
- Total direct wages cost was ₹ 43,92,000
- Fixed production overhead cost incurred was ₹ 45,00,000

Analysis of variances was as follows:

Variances	Favourable	Adverse
Direct material price	₹ 4,80,000	-
Direct material usage	₹ 48,000	
Direct labour rate	-	₹ 69,120
Direct labour efficiency	₹ 33,120	-
Fixed production overhead expenditure		₹ 1,80,000

You are required to CALCULATE the following for the month of April, 2022

(i) Material cost variance

(ii) Budgeted output (in units)

(iii) Quantity of raw materials purchased (in kilograms)

(iv) Actual output (in units)

(v) Actual hours worked

(vi) Actual wage rate per labour hour

(vii) Labour cost variance

(viii) Production overhead cost variance

ICAI'S ANSWER

(i) Direct Material Cost Variance = Direct Material Price Variance + Direct Material Usage Variance

= ₹ 4,80,000 F + ₹ 48,000 F = ₹ **5,28,000 F**

(ii) Budgeted Output (units)

Fixed Production Overhead Expenditure Variance

= Budgeted Fixed Overhead - Actual Fixed Overheads

= Budgeted Output x Standard Overhead Rate - Actual Fixed Overheads

₹ 1,80,000 A = Budgeted Output x ₹ 360 (5 hrs @₹ 72) - ₹ 45,00,000

Budgeted Output = $\frac{₹45,00,000 - ₹\ 1,80,000}{₹360}$ = **12,000 units**

(iii) Quantity of Materials purchased (in kilograms)

Material Price Variance = Actual Usage (Standard Price per kg - Actual price per kg)

₹ 4,80,000 F = Actual Usage (₹ 24 -₹ 22)

Actual usage in kgs = $\frac{₹4,80,000 - ₹\ 1,80,000}{₹2}$ = **2,40,000 kgs**

(iv) Actual Output (units)

Actual Direct Wages	₹ 43,92,000
Direct labour rate variance	₹ 69,120 A
Direct labour efficiency variance	₹ 33,120 F
Standard labour cost for actual output	₹ 43,56,000

Actual Output = $\frac{\text{Standard labour cost for actual output}}{\text{Standard wage rate per unit}}$

= $\frac{₹\ 43,56,000}{₹\ 360\ (72\ x\ 5)}$ = **12,100 units**

Alternatively, let X be the actual quantity of output

Then, Standard Quantity of input for actual output 'X'

20X = SQ

Material cost variance = (SQ x SP) - (AQ x AP)

₹ 5,28,000 = (20 X x ₹ 24) - (2,40,000 kgs x ₹ 22)

480X = ₹ 52,80,000 + ₹ 5,28,000

480X = ₹ 58,08,000

X = $\frac{₹\ 58,08,000}{480}$ = **12,100 units**

(v) Actual hours worked

Labour Efficiency Variance = Standard Labour Rate (Standard time for actual output - Actual time)

₹ 33,120 F = ₹ 72 (5 hours x 12100 units - Actual time)

460 hours = 60,500 hours - Actual time

Actual time = 60,500 - 460 = **60,040 hours**

(vi) Actual wage rate per hour

Actual Wages paid = ₹ 43,92,000

Actual hours worked = 60,040 hours

Actual Wage rate per hour = $\frac{₹\ 43,92,000}{60,040\ \text{hours}}$ = **₹ 73.15 per hour**

(vii) Labour cost variance

= Labour rate variance + Labour efficiency variance

=₹ 69,120 A + ₹ 33,120 F

= ₹ **36,000 A**

(viii) Production Overhead Cost Variance

= Actual Output x Standard overhead rate - Actual Overheads Incurred

= 12,100 units x₹ 360 - ₹ 45,00,000

= ₹ 43,56,000 - ₹ 45,00,000

= ₹ 1,44,000 A

ICAI · Old P3 · RTP Nov 2022 · question p. 10-11 · answer p. 30-32

895 RTP **Old P3 Jul 2021** Q10 old scheme

LM Limited produces a product 'SX4' which is sold in a 10 Kg. packet. The standard cost card per packet of 'SX4' is as follows:

	(₹)
Direct materials 10 kg @ ₹ 90 per kg	900
Direct labour 8 hours @ ₹ 80 per hour	640
Variable Overhead 8 hours @ ₹ 20 per hour	160
Fixed Overhead	250
	1,950

Budgeted output for a quarter of a year was 10,000 Kg. Actual output is 9,000 Kg.

Actual costs for this quarter are as follows:

	(₹)
Direct Materials 8,900 Kg @ ₹ 92 per Kg.	8,18,800
Direct Labour 7,000 hours @ ₹ 84 per hour	5,88,000
Variable Overhead incurred	1,40,000
Fixed Overhead incurred	2,60,000

You are required to CALCULATE:

(i) Material Usage Variance

(ii) Material Price Variance

(iii) Material Cost Variance

(iv) Labour Efficiency Variance

(v) Labour Rate Variance

(vi) Labour Cost Variance

(vii) Variable Overhead Cost Variance

(viii) Fixed Overhead Cost Variance

ICAI'S ANSWER

(i) Material Usage Variance = Std. Price (Std. Quantity – Actual Quantity)

= ₹ 90 (9,000 kg. – 8,900 kg.)

= ₹ 9,000 (Favourable)

(ii) Material Price Variance = Actual Quantity (Std. Price – Actual Price)

= 8,900 kg. (₹ 90 – ₹ 92) = ₹ 17,800 (Adverse)

(iii) Material Cost Variance = Std. Material Cost – Actual Material Cost

= (SQ × SP) – (AQ × AP)

= (9,000 kg. × ₹ 90) – (8,900 kg. × ₹ 92)

= ₹ 8,10,000 – ₹ 8,18,800

= ₹ 8,800 (Adverse)

(iv) Labour Efficiency Variance = Std. Rate (Std. Hours – Actual Hours)

$= ₹\ 80\ (\frac{9,000}{10} \times 8 \text{ hours} - 7,000 \text{ hrs.})$

= ₹ 80 (7,200 hrs. – 7,000 hrs.)

= ₹ 16,000 (Favourable)

(v) Labour Rate Variance = Actual Hours (Std. Rate – Actual Rate)

= 7,000 hrs. (₹ 80 – ₹ 84)

= ₹ 28,000 (Adverse)

(vi) Labour Cost Variance = Std. Labour Cost – Actual Labour Cost

= (SH × SR) – (AH × AR)

= (7,200 hrs. × ₹ 80) – (7,000 hrs. × ₹ 84)

= ₹ 5,76,000 – ₹ 5,88,000

= ₹ 12,000 (Adverse)

(vii) Variable Cost Variance = Std. Variable Cost – Actual Variable Cost

= (7,200 hrs. × ₹ 20) – ₹ 1,40,000

= ₹ 4,000 (Adverse)

(viii) Fixed Overhead Cost Variance = Absorbed Fixed Overhead – Actual Fixed Overhead

$= \frac{₹\ 250}{10 \text{ kgs.}} \times 9,000 \text{ kgs.} - ₹\ 2,60,000$

= ₹ 2,25,000 – ₹ 2,60,000 = ₹ 35,000 (Adverse)

ICAI · Old P3 · RTP Jul 2021 · question p. 9 · answer p. 27-28

896 RTP **Old P3 Nov 2020** Q12 old scheme

Also asked in Old P3 MTP May 2023 S-II Q4(b)

Following are the standard cost for a product-X:

	(₹)
Direct materials 10 kg @ ₹ 90 per kg	900
Direct labour 8 hours @ ₹100 per hour	800
Variable Overhead 8 hours @ ₹15 per hour	120
Fixed Overhead	400
	2,220

Budgeted output for the year was 2,000 units. Actual output is 1,800 units.

Actual cost for year is as follows:

	(₹)
Direct Materials 17,800 Kg @ ₹ 92 per Kg.	16,37,600
Direct Labour 14,000 hours @ ₹ 104 per hour	14,56,000
Variable Overhead incurred	2,17,500
Fixed Overhead incurred	7,68,000

You are required to CALCULATE:

(i) Material Usage Variance

(ii) Material Price Variance

(iii) Material Cost Variance

(iv) Labour Efficiency Variance

(v) Labour Rate Variance

(vi) Labour Cost Variance

(vii) Variable Overhead Cost Variance

(viii) Fixed Overhead Cost Variance.

ICAI'S ANSWER

(i) Material Usage Variance = Std. Price (Std. Quantity – Actual Quantity)

= ₹ 90 (18,000 kg. – 17,800 kg.)

= ₹ 18,000 (Favourable)

(ii) Material Price Variance = Actual Quantity (Std. Price – Actual Price)

= 17,800 kg. (₹ 90 – ₹ 92) = ₹ 35,600 (Adverse)

(iii) Material Cost Variance = Std. Material Cost – Actual Material Cost

= (SQ × SP) – (AQ × AP)

= (18,000 kg. × ₹ 90) – (17,800 kg. × ₹ 92)

= ₹ 16,20,000 – ₹ 16,37,600

= ₹17,600 (Adverse)

(iv) Labour Efficiency Variance = Std. Rate (Std. Hours – Actual Hours)

= ₹ 100 (1,800 units × 8 – 14,000 hrs.)

= ₹ 100 (14,400 hrs. – 14,000 hrs.)

= ₹ 40,000 (Favourable)

(v) Labour Rate Variance = Actual Hours (Std. Rate – Actual Rate)

= 14,000 hrs. (₹ 100 – ₹104)

= ₹ 56,000 (Adverse)

(vi) Labour Cost Variance = Std. Labour Cost – Actual Labour Cost

= (SH × SR) – (AH × AR)

= (14,400 hrs. × ₹ 100) – (14,000 hrs. × ₹ 104)

= ₹ 14,40,000 – ₹ 14,56,000

= ₹16,000 (Adverse)

(vii) Variable Cost Variance = Std. Variable Cost – Actual Variable Cost

= (14,400 hrs. × ₹ 15) – ₹ 2,17,500

= ₹ 1,500 (Adverse)

(viii) Fixed Overhead Cost Variance = Absorbed Fixed Overhead – Actual Fixed Overhead

= (1,800 units × ₹400) - ₹ 7,68,000

= ₹ 7,20,000 – ₹ 7,68,000 = ₹ 48,000 (Adverse)

ICAI · Old P3 · RTP Nov 2020 · question p. 10-11 · answer p. 26-27

897 RTP **Old P3 May 2020** Q10 old scheme

ABC Ltd. had prepared the following estimation for the month of January:

	Quantity	Rate (₹)	Amount (₹)
Material-A	800 kg.	90.00	72,000
Material-B	600 kg.	60.00	36,000
Skilled labour	1,000 hours	75.00	75,000
Unskilled labour	800 hours	44.00	35,200

Normal loss was expected to be 10% of total input materials and an idle labour time of 5% of expected labour hours was also estimated.

At the end of the month the following information has been collected from the cost accounting department:

The company has produced 1,480 kg. finished product by using the followings:

	Quantity	Rate (₹)	Amount (₹)
Material-A	900 kg.	86.00	77,400
Material-B	650 kg.	65.00	42,250
Skilled labour	1,200 hours	71.00	85,200
Unskilled labour	860 hours	46.00	39,560

You are required to CALCULATE:

(a) Material Cost Variance;

(b) Material Price Variance;

(c) Material Mix Variance;

(d) Material Yield Variance;

(e) Labour Cost Variance;

(f) Labour Efficiency Variance and

(g) Labour Yield Variance.

ICAI'S ANSWER

Material Variances:

Material	SQ (WN-1)	SP (₹)	SQ × SP (₹)	RSQ (WN-2)	RSQ × SP (₹)	AQ	AQ × SP (₹)	AP (₹)	AQ × AP (₹)
A	940 kg.	90.00	84,600	886 kg.	79,740	900 kg.	81,000	86.00	77,400
B	705 kg.	60.00	42,300	664 kg.	39,840	650 kg.	39,000	65.00	42,250
	1645 kg		1,26,900	1550 kg	1,19,580	1550 kg	1,20,000		1,19,650

WN-1: Standard Quantity (SQ):

Material A- $\left(\frac{800\text{kg.}}{0.9 \times 1{,}400\text{kg.}} \times 1{,}480\text{kg.}\right)$ = 939.68 or 940 kg.

Material B- $\left(\frac{600\text{kg.}}{0.9 \times 1{,}400\text{kg.}} \times 1{,}480\text{kg.}\right)$ = 704.76 or 705 kg.

WN- 2: Revised Standard Quantity (RSQ):

Material A- $\left(\frac{800\text{kg.}}{1{,}400\text{kg.}} \times 1{,}550\text{kg.}\right)$ = 885.71 or 886 kg.

Material B- $\left(\frac{600\text{kg.}}{1{,}400\text{kg.}} \times 1{,}550\text{kg.}\right)$ = 664.28 or 664 kg.

(a) Material Cost Variance (A + B) = {(SQ × SP) – (AQ × AP)}

= {1,26,900 – 1,19,650} = 7,250 (F)

(b) Material Price Variance (A + B) = {(AQ × SP) – (AQ × AP)

= {1,20,000 – 1,19,650} = 350 (F)

(c) Material Mix Variance (A + B) = {(RSQ × SP) – (AQ × SP)}

= {1,19,580 – 1,20,000} = 420 (A)

(d) Material Yield Variance (A + B) = {(SQ × SP) – (RSQ × SP)}

= {1,26,900 – 1,19,580} = 7,320 (F)

Labour Variances:

Labour	SH (WN-3)	SR (₹)	SH × SR (₹)	RSH (WN-4)	RSH × SR (₹)	AH	AH × SR (₹)	AR (₹)	AH × AR (₹)
Skilled	1,116 hrs	75.00	83,700	1144	85,800	1,200	90,000	71.00	85,200
Unskilled	893 hrs	44.00	39,292	916	40,304	860	37,840	46.00	39,560
	2,009 hrs		1,22,992	2,060	1,26,104	2,060	1,27,840		1,24,760

WN- 3: Standard Hours (SH):

Skilled labour- $\left(\frac{0.95 \times 1{,}000\text{hr.}}{0.90 \times 1{,}400\text{kg.}} \times 1{,}480\text{kg.}\right)$ =1,115.87 or 1,116 hrs.

Unskilled labour- $\left(\frac{0.95 \times 800\text{hr.}}{0.90 \times 1{,}400\text{kg.}} \times 1{,}480\text{kg.}\right)$ = 892.69 or 893 hrs.

WN- 4: Revised Standard Hours (RSH):

Skilled labour- $\left(\frac{1{,}000\text{hr.}}{1{,}800\text{hr.}} \times 2{,}060\text{hr.}\right)$ = 1,144.44 or 1,144 hrs.

Unskilled labour- $\left(\frac{800\text{hr.}}{1{,}800\text{hr.}} \times 2{,}060\text{hr.}\right)$ = 915.56 or 916 hrs.

(e) Labour Cost Variance (Skilled + Unskilled) = {(SH × SR) – (AH × AR)}

= {1,22,992 – 1,24,760} = 1,768 (A)

(f) Labour Efficiency Variance (Skilled + Unskilled) = {(SH × SR) – (AH × SR)}

= {1,22,992 – 1,27,840} = 4,848 (A)

(g) Labour Yield Variance (Skilled + Unskilled) = {(SH × SR) – (RSH × SR)}

= {1,22,992 – 1,26,104} = 3,112 (A)

ICAI · Old P3 · RTP May 2020 · question p. 8-9 · answer p. 23-25

898 MTP **Sep 2026 S-I** Q4 14 marks

Astra Components Pvt. Ltd. is a Tier-2 supplier to the electrical vehicle (EV) industry. The company manufactures a single machined housing component in a single cost centre. Management uses standard costing for control and performance measurement.

The Standard Cost Card for the component is as follows:

Standard — Unit cost (₹)

- Direct material: 6.00 kg @ 5.00 per kg = 30.00
- Direct labour: 2.50 hours @ 4.50 per hour = 11.25
- Factory overhead: 2.00 per labour hour = 5.00

- Total standard manufacturing cost per unit = ₹ 46.25

Budget & actual production activity for the month of June 2025:

- Production scheduled (required completion) for the month: 50,000 units.
- Actual completed units during the month: 51,200 units (no opening or closing WIP).
- Purchases of direct material during the month: 3,20,000 kg at ₹ 5.20 per kg.
- Actual material used in production: 2,96,000 kg.
- Actual direct labour hours worked: 125,000 hours; total labour cost incurred: ₹ 6,40,000.
- Actual total factory overhead incurred during the month: ₹ 2,80,000.
- Sales during the month: 50,000 units.

Selling price policy: Selling price is to be fixed so as to allow a mark-up of 25% on the selling price (i.e., profit = 25% of selling price and standard manufacturing cost per unit should be considered for the same.)

Incentive scheme: The company operates a labour bonus scheme — employees are paid a bonus equal to 60% of the standard direct labour rate for each standard direct labour hour saved (i.e., for hours saved compared to standard hours allowed for actual output).

Required: Using the above data, prepare calculations and reports for June 2025 as follows:

(i) CALCULATE all material variances based on actual material consumption.

(ii) CALCULATE the labour variances (rate and efficiency) and compute the total factory overhead variance.

(iii) PREPARE an Income Statement for June 2025 showing the actual gross margin

(iv) COMPUTE if any bonus is payable under the incentive scheme.

ICAI'S ANSWER

(i) Material variances:

(a) Direct Material Cost Variance = Standard Cost – Actual Cost

= (Actual output × Std qty per unit × Std price) – (Actual qty used × Actual price)

= (51,200 units × 6 kg × ₹ 5.00) – (2,96,000 kg × ₹ 5.20)

= ₹ 15,36,000 – ₹ 15,39,200 = ₹ 3,200 (A)

(b) Material Price Variance = Actual Qty used × (Std Price – Actual Price)

= 2,96,000 kg × (₹ 5.00 – ₹ 5.20)

= 2,96,000 × (–₹ 0.20) = ₹ 59,200 (A)

(Material variances are calculated on the basis of consumption)

(c) Material Usage (Quantity) Variance = Std Price × (Std Qty for actual output – Actual Qty used)

= ₹ 5.00 × (51,200 × 6 – 2,96,000)

= ₹ 5.00 × (3,07,200 – 2,96,000)

= ₹ 5.00 × 11,200 = ₹ 56,000 (F)

(ii) Labour variances and Factory Overhead variance:

(a) Labour Cost Variance = Standard Cost – Actual Cost

= (Std hrs allowed for actual output × Std rate) – Actual labour cost

Std hrs allowed = 51,200 units × 2.5 hrs = 1,28,000 hrs

= (1,28,000 × ₹ 4.50) – ₹ 6,40,000

= ₹ 5,76,000 – ₹ 6,40,000 = ₹ 64,000 (A)

(b) Labour Rate Variance = Actual Hours × (Std Rate – Actual Rate)

Actual rate = Actual labour cost / Actual hours

= ₹ 6,40,000 / 1,25,000 = ₹ 5.12 / hr

= 1,25,000 × (₹ 4.50 – ₹ 5.12)

= 1,25,000 × (–₹ 0.62) = ₹ 77,500 (A)

(c) Labour Efficiency Variance = Std Rate × (Std Hours – Actual Hours)

= ₹ 4.50 × (128,000 – 125,000)

= ₹ 4.50 × 3,000 = ₹ 13,500 (F)

(d) Total Factory Overhead Variance = Factory Overhead Absorbed – Actual Factory Overhead

Factory overhead absorbed = Actual hours × Std FOH rate

= 1,25,000 × ₹ 2.00 = ₹ 2,50,000

Actual FOH incurred = ₹ 2,80,000

FOH variance = ₹ 2,50,000 – ₹ 2,80,000 = ₹ 30,000 (A)

(iii) Preparation of Income Statement

Calculation of unit selling price (markup 25% on selling price ⇒ Selling price = Cost ÷ 0.75)

Standard manufacturing cost per unit = ₹ 46.25 (given)

Selling price per unit = ₹ 46.25 ÷ 0.75 = ₹ 61.67

Income Statement

Particulars	Amount (₹)
Sales (50,000 units × ₹61.6667)	30,83,333
Less: Standard cost of goods sold (50,000 × ₹ 46.25)	23,12,500
Standard gross margin	7,70,833
Less: Adverse variances:	
Material Price variance	59,200
Labour Rate variance	77,500
Factory Overhead variance	30,000
Total Adverse variances	1,66,700
Add: Favourable variances:	

Particulars	Amount (₹)
Material Usage variance	56,000
Labour Efficiency variance	13,500
Total Favourable variances	69,500
Actual gross margin = 7,70,833 – 1,66,700 + 69,500	6,73,633

(iv) Bonus under incentive scheme

Standard labour hours for actual output = 51,200 × 2.5 = 128,00 hrs

Actual labour hours worked = 125,000 hrs

Labour hours saved = 128,000 – 125,000 = 3,000 hrs

Bonus rate = 60% of standard direct labour rate per hour

= 60% × ₹ 4.50 = ₹2.70 / hr

Bonus payable = Hours saved × ₹2.70

= 3,000 × 2.70 = ₹8,100

ICAI · P4 · MTP Sep 2026 · Series I · question p. 13 · answer p. 15-17

899 MTP **May 2026 S-I** Q2(a) 8 marks

Same question, different ICAI answer: #889 RTP May 2025

COMPUTE the missing data indicated by the question marks from the following:

Particulars	A	B
Standard Price/ unit	₹ 12	₹ 15
Actual Price/ unit	₹ 15	₹ 20
Standard Input (kgs.)	50	?
Actual Input (kgs.)	?	70
Material Price Variance	?	?
Material Usage Variance	?	₹ 300 Adverse
Material Cost Variance	?	?

Material mix variance for both products together was ₹ 45 Adverse.

ICAI'S ANSWER

(i) Standard input (kgs.) of Material- B:

Material usage variance = Std. Rate (Std. Quantity – Actual Quantity)

₹ 300 Adverse = ₹ 15 (SQ – 70)

Or, -300 = 15 SQ – 1,050

Or, SQ = 50 kgs.

(ii) Actual Input (kgs) of Material- A:

Let the actual input in for Material-A is X kgs.

Material Mix Variance = Std. Price (Actual Quantity in Std. mix – Actual Quantity)

Or, Material Mix Variance (A+B) = Material Mix Variance for Material - A + Material Mix Variance for Material -B

Or, - 45 = $[₹\ 12\{(\frac{X+70}{2}) - X\}] + [₹\ 15\{(\frac{X+70}{2}) - 70\}]$

Or, - 45 = $[₹\ 12\{\frac{X+70-2X}{2}\}] + [₹\ 15\{\frac{X+70-140}{2}\}]$

Or, - 45 = $[₹\ 12\{\frac{70-X}{2}\}] + [₹\ 15\{\frac{X-70}{2}\}]$

Or, - 45 = $[-6X + 420] + [\frac{15X-1,050}{2}]$

Or, - 45 = $[\frac{-12X+840+15X-1,050}{2}]$

Or, - 90 = 3X – 210

Or, X = $\frac{120}{3}$ = 40 kgs.

(iii) (a) Material Price Variance of A = Actual Quantity (Std. Rate – Actual Rate)

= 40 kg. (12 – 15) = ₹ 120 Adverse

(b) Material Price Variance of B = 70 kg. (15 – 20) = ₹ 350 Adverse

(iv) Material usage variance of A

= Std. Rate (Std. Quantity – Actual Quantity)

= 12 (50 – 40) = ₹ 120 Favourable

(v) (a) Material Cost variance of A = Std. Cost – Actual Cost

= (50 kgs. × ₹ 12) – (40 kgs. × ₹ 15)

= 600 – 600 = Nil

(b) Material Cost variance of B

= (50 kgs. × ₹ 15) – (70 kgs. × ₹ 20)

= 750 – 1,400 = ₹ 650 Adverse

ICAI · P4 · MTP May 2026 · Series I · question p. 9 · answer p. 8-9

900 **MTP** **May 2026 S-II** **Q1(c)** **4 marks**

Fruit Ltd. produces "Mango Burst", a popular beverage. It is reviewing their material costs for the month of March, 2025.

Following data is available :

(i) Standard cost: Each bottle of "Mango Burst" should use 0.5 kg. of mangoes, costing ₹ 30 per kg.

(ii) Actual production: 50,000 bottles of "Mango Burst".

(iii) Actual material usage: 26,000 kgs of mangoes.

(iv) Actual material cost: The total cost of the mangoes used was ₹ 7,67,000.

You are required to CALCULATE:

(i) Material Price Variance

(ii) Material Usage Variance

(iii) Material Cost Variance

ICAI'S ANSWER

(i) Material Price Variance (MPV):

MPV= (Std Price – Actual Price) x Actual Quantity

Actual Price per Kg = Actual Cost / Actual Qty

= 7,67,000 / 26,000 = ₹ 29.50

MPV = (30 -29.50) x 26,000 = ₹ 13,000 (Fav)

(ii) Material Usage Variance (MUV):

Std qty of Mangoes for Actual Output : Actual output x Std Qty per Unit

Std Qty of Mangoes for Actual Output: 50,000 bottles x 0.5 kg/ bottle

= 25, 000 kg

MUV = (Std Qty – Actual Qty) x Std Price

MUV = (25,000 -26,000) x 30 = ₹ 30,000 (Adv)

(iii) Material Cost Variance (MCV):

MCV = MPV + MUV

MCV = 13,000 +(-30,000) =₹ 17,000 (Adv)

ICAI · P4 · MTP May 2026 · Series II · question p. 8-9 · answer p. 7-8

901 MTP **Jan 2026 S-I** Q3(a) 9 marks

You are working as a Cost Analyst at TechMech Ltd., a company that manufactures Product-Y, a specialized industrial tool. The company uses standard costing to monitor cost performance.

Below is the standard cost structure for one unit of Product-Y:

Component	Amount (₹)
Direct Materials 15 kg @ ₹ 80 per kg	1,200
Direct Labour 5 hours @ ₹ 120 per hour	600
Variable Overhead 5 hours @ ₹ 25 per hour	125
Fixed Overhead	300
Total	**2,225**

Production Details:

- Budgeted Output: 3,000 units
- Actual Output: 2,750 units
- Abnormal Material Loss: 400 kg

- Idle Time: 300 hours

Actual Costs:

Component	Amount (₹)
Direct Materials 41,600 kg @ ₹ 84 per kg	34,94,400
Direct Labour 14,500 hours @ ₹ 125 per hour	18,12,500
Variable Overhead	3,62,500
Fixed Overhead	8,50,000

You are required to CALCULATE:

(i) Material Usage Variance (adjust for abnormal loss)

(ii) Material Price Variance

(iii) Material Cost Variance

(iv) Labour Efficiency Variance (exclude idle time)

(v) Labour Rate Variance (include idle time)

(vi) Labour Idle Time Variance

(vii) Labour Cost Variance

(viii) Variable Overhead Cost Variance

(ix) Fixed Overhead Cost Variance.

ICAI'S ANSWER

(i) Material Usage Variance

= Std. Price x (Std. Quantity – Actual Quantity Adjusted)

= ₹ 80 x [(2,750 x 15 kg) – (41,600 – 400)]

= ₹ 80 x (41,250 – 41,200)

= ₹ 4,000 **Favourable**

(ii) Material Price Variance

= Actual Quantity x (Std. Price – Actual Price)

= 41,600 kg x (₹ 80 – ₹ 84)

= 41,600 x (–4)

= ₹ 1,66,400 **Adverse**

(iii) Material Cost Variance

= (SQ x SP) – (AQ x AP)

= (41,250 x ₹ 80) – (41,600 x ₹ 84)

= ₹ 33,00,000 – ₹ 34,94,400

= ₹ 1,94,400 **Adverse**

(iv) Labour Efficiency Variance (Exclude idle time)

= Std. Rate x (Std. Hours – Actual Hours worked)

= ₹ 120 x [(2,750 x 5 hrs) – (14,500 – 300)]

= ₹ 120 x (13,750 – 14,200)

= ₹ 54,000 **Adverse**

(v) Labour Rate Variance

= Actual Hours x (Std. Rate – Actual Rate)

= 14,500 x (₹120 – ₹ 125)

= 14,500 x (–5)

= ₹ 72,500 **Adverse**

(vi) Labour Idle Time Variance

= Idle Hours x Std. Rate

= 300 hrs x ₹ 120

= ₹ 36,000 **Adverse**

(vii) Labour Cost Variance

= (SH x SR) – (AH x AR)

= (13,750 x ₹120) – (14,500 x ₹ 125)

= ₹ 16,50,000 – ₹ 18,12,500

= ₹ 1,62,500 **Adverse**

(viii) Variable Overhead Cost Variance

= (SH x SR) – Actual Overhead

= (13,750 hrs x ₹ 25) – ₹ 3,62,500

= ₹ 3,43,750 – ₹ 3,62,500

= ₹ 18,750 **Adverse**

(ix) Fixed Overhead Cost Variance

= Absorbed Fixed Overhead – Actual Fixed Overhead

= (2,750 units x ₹ 300) – ₹ 8,50,000

= ₹ 8,25,000 – ₹ 8,50,000

= ₹ 25,000 **Adverse**

ICAI · P4 · MTP Jan 2026 · Series I · question p. 10-11 · answer p. 11-12

902 MTP **Jan 2026 S-II** Q5(a) 11 marks

Astra Components Pvt. Ltd. is a Tier-2 supplier to the electrical vehicle (EV) industry. The company manufactures a single machined housing component in a single cost centre. Management uses standard costing for control and performance measurement.

The Standard Cost Card for the component is as follows:

Standard — Unit cost (₹)

- Direct material: 6.00 kg @ 5.00 per kg = 30.00
- Direct labour: 2.50 hours @ 4.50 per hour = 11.25
- Factory overhead: 2.00 per labour hour = 5.00
- Total standard manufacturing cost per unit = ₹ 46.25

Budget & actual production activity for the month of June 2025:

- Production scheduled (required completion) for the month: 50,000 units.
- Actual completed units during the month: 51,200 units (no opening or closing WIP).
- Purchases of direct material during the month: 3,20,000 kg at ₹ 5.20 per kg.
- Actual material used in production: 2,96,000 kg.
- Actual direct labour hours worked: 125,000 hours; total labour cost incurred: ₹ 6,40,000.
- Actual total factory overhead incurred during the month: ₹ 2,80,000.
- Sales during the month: 50,000 units.

Selling price policy: Selling price is to be fixed so as to allow a mark-up of 25% on the selling price (i.e., profit = 25% of selling price).

Incentive scheme: The company operates a labour bonus scheme — employees are paid a bonus equal to 60% of the standard direct labour rate for each standard direct labour hour saved (i.e., for hours saved compared to standard hours allowed for actual output).

Required: Using the above data, prepare calculations and reports for June, 2025 as follows:

(i) CALCULATE all material variances based on actual material consumption.

(ii) CALCULATE the labour variances (rate and efficiency) and compute the total factory overhead variance.

(iii) PREPARE an Income Statement for June 2025 showing the actual gross margin

(iv) COMPUTE if any bonus is payable under the incentive scheme

ICAI'S ANSWER

(i) Material variances:

(a) Direct Material Cost Variance = Standard Cost – Actual Cost

= (Actual output × Std qty per unit × Std price) – (Actual qty used × Actual price)

= (51,200 units × 6 kg × ₹ 5.00) – (296,000 kg × ₹ 5.20)

= ₹ 15,36,000 – ₹ 15,39,200 = ₹ 3,200 (A)

(b) Material Price Variance = Actual Qty used × (Std Price – Actual Price)

= 2,96,000 kg × (₹ 5.00 – ₹ 5.20)

= 2,96,000 × (–₹ 0.20) = ₹ 59,200 (A)

(Material variances are calculated on the basis of consumption)

(c) Material Usage (Quantity) Variance = Std Price × (Std Qty for actual output – Actual Qty used)

= ₹ 5.00 × (51,200 × 6 – 2,96,000)

= ₹ 5.00 × (3,07,200 – 2,96,000)

= ₹ 5.00 × 11,200 = ₹ 56,000 (F)

(ii) Labour variances and Factory Overhead variance:

(a) Labour Cost Variance = Standard Cost – Actual Cost

= (Std hrs allowed for actual output × Std rate) – Actual labour cost

Std hrs allowed = 51,200 units × 2.5 hrs = 1,28,000 hrs

= (1,28,000 × ₹ 4.50) – ₹ 6,40,000

= ₹ 5,76,000–₹ 6,40,000 = ₹ 64,000 (A)

(b) Labour Rate Variance = Actual Hours × (Std Rate – Actual Rate)

Actual rate = Actual labour cost / Actual hours

= ₹ 6,40,000 / 1,25,000 = ₹ 5.12 / hr

= 1,25,000 × (₹ 4.50 – ₹ 5.12) = 125,000 × (–₹ 0.62) = ₹ 77,500 (A)

(c) Labour Efficiency Variance = Std Rate × (Std Hours – Actual Hours)

= ₹ 4.50 × (1,28,000 – 1,25,000) = ₹ 4.50 × 3,000 = ₹ 13,500 (F)

(d) Total Factory Overhead Variance = Factory Overhead Absorbed – Actual Factory Overhead

= 51,200 units x 2.5 Hrs. x ₹ 2 - ₹ 2,80,000

FOH variance = ₹ 2,56,000 – ₹ 2,80,000 = ₹ 24,000 (A)

(iii) Preparation of Income Statement

Calculation of unit selling price

Standard manufacturing cost per unit = ₹ 46.25 (given)

Selling price per unit = ₹ 46.25 ÷ 0.75 = ₹ 61.6667

Income Statement

Particulars	Amount (₹)
Sales (50,000 units × ₹ 61.6667)	30,83,333
Less: Standard cost of goods sold (50,000 × ₹ 46.25)	23,12,500
Standard gross margin	7,70,833
Less: Adverse variances:	
Material Price variance	59,200
Labour Rate variance	77,500
Factory Overhead variance	24,000
Total Adverse variances	1,60,700
Add: Favourable variances:	
Material Usage variance	56,000
Labour Efficiency variance	13,500

Particulars	Amount (₹)
Total Favourable variances	69,500
Actual gross margin = 7,70,833 – 1,60,700 + 69,500	**6,79,633**

(iv) Bonus under incentive scheme

Standard labour hours for actual output = 51,200 × 2.5 = 1,28,000 hrs

Actual labour hours worked = 1,25,000 hrs

Labour hours saved = 1,28,000 – 1,25,000 = 3,000 hrs

Bonus rate = 60% of standard direct labour rate per hour

= 60% × ₹ 4.50

= ₹ 2.70 / hr

Bonus payable = Hours saved × ₹ 2.70 = 3,000 × 2.70 = ₹ 8,100

ICAI · P4 · MTP Jan 2026 · Series II · question p. 16-17 · answer p. 18-20

903 MTP **Sep 2025 S-I** MCQ 13 2 marks MCQ

A company provides the following information:

- Standard price per unit of raw material = ₹ 5.50
- Actual quantity of raw materials purchased = 3,000 kg
- Standard quantity for actual production = 3,200 kg
- Material price variance (favourable) = ₹ 600

What is the actual price per unit of raw materials?

A ₹ 5.70

B ₹ 5.30

C ₹ 5.50

D ₹ 5.00

ICAI'S ANSWER

Answer key: **(B)**

(b) ₹ 5.30

Material Price Variance = (Standard Price - Actual Price) x Actual Quantity

600 = (5.50 - AP) x 3,000

AP = 5.50 - 0.20 = ₹5.30

ICAI · P4 · MTP Sep 2025 · Series I · question p. 6 · answer p. 4

904 MTP **Sep 2025 S-I** Q1(b) 5 marks

Also asked in MTP Jan 2025 S-II Q1(b)

Following standards have been set for manufacturing a product 'XYZ':

Direct Material:	(₹)
4 units of X @ ₹ 8 per unit	32.00
6 units of Y @ ₹ 6 per unit	36.00
30 units of Z @ ₹ 2 per unit	60.00
	128.00
Direct Labour:	
6 hrs @ ₹ 16 per hour	96.00
Total standard prime cost	224.00

The company actually manufactured and sold 12,000 units of the product 'XYZ' during the year.

Direct material costs were as follows:

50,000 units of X at ₹ 8.80 per unit

72,000 units of Y at ₹ 5.60 per unit

354,000 units of Z at ₹ 2.40 per unit

The company worked 70,000 direct labour hours during the year. For 10,000 of these hours, the company paid at ₹ 24 per hour while for the remaining, the wages were paid at standard rate.

You are required to CALCULATE the following:

(i) Material Price Variance

(ii) Material Usage Variance

(iii) Labour Rate Variance

(iv) Labour Efficiency Variance

ICAI'S ANSWER

For Material Cost Variances:

	SQ × SP	AQ × AP	AQ × SP
X	12,000 × 4 x ₹ 8 = ₹ 3,84,000	50,000 x ₹ 8.80 = ₹ 4,40,000	50,000 x ₹ 8 = ₹ 4,00,000
Y	12,000 × 6 x ₹ 6 = ₹ 4,32,000	72,000 x ₹ 5.60 = ₹ 4,03,200	72,000 x ₹ 6 = ₹ 4,32,000
Z	12,000 × 30 x ₹ 2 = ₹ 7,20,000	3,54,000 x ₹ 2.40 = ₹ 8,49,600	3,54,000 x ₹ 2 = ₹ 7,08,000
Total	₹ 15,36,000	₹ 16,92,800	₹ 15,40,000

Material Price Variance = Actual quantity (Std. price – Actual price)

= (AQ x SP) – (AQ x AP)

= ₹ 15,40,000 – ₹ 16,92,800

= ₹ 1,52,800 (A)

Material Usage Variance = Standard Price (Std. Quantity – Actual Quantity)

= (SP x SQ) – (SP x AQ)

= ₹ 15,36,000 – ₹ 15,40,000

= ₹ 4,000 (A)

For Labour Cost Variance:

	SH × SR	AH × AR	AH × SR
Labour	(12,000 × 6) x ₹ 16 = ₹ 11,52,000	10,000 x ₹ 24 = ₹ 2,40,000 60,000 x ₹ 16 = ₹ 9,60,000	70,000 x ₹ 16 = ₹ 11,20,000
Total	₹ 11,52,000	₹ 12,00,000	₹ 11,20,000

Labour Rate Variance = Actual Hours (Std. Rate – Actual Rate)

= (AH x SR) – (AH x AR)

= ₹ 11,20,000 – ₹ 12,00,000

= ₹ 80,000 (A)

Labour Efficiency Variance = Standard Rate (Std. Hours – Actual Hours)

= (SR x SH) – (SR x AH)

= ₹ 11,52,000 – ₹ 11,20,000

= ₹ 32,000 (F)

ICAI · P4 · MTP Sep 2025 · Series I · question p. 8 · answer p. 6-7

905 MTP **May 2025 S-II** Q3(a) 10 marks

The details corresponding to the manufacture of a product X is as follows:

Materials 20 units @ ₹ 3 per unit – ₹ 60

Wages 10 hours @ 16 per hour – ₹ 160

Production OH 10 hours @ 20 – ₹ 200

Actual Material cost amounts to ₹ 25,740 and labour cost amounts to ₹ 65,368.

The variances have been analysed and the following information is being made available:

1. Material Price Variance - 1170A
2. Material Usage Variance - 750F
3. Wages Rate Variance - 1352F
4. Wages Efficiency - 800A
5. POH Expenditure - 900F
6. POH Volume Variance - 1400F

From the above DETERMINE:

i. Actual output in Units

ii. Actual price of material per unit

iii. Actual quantity of materials consumed

iv. Actual wage rate

v. Actual hours worked and Standard Hours

vi. Amount of overhead absorbed

vii. Amount of Overhead incurred

viii. Production Overhead capacity variance

ix. Production Overhead efficiency variance

x. Budgeted output in units.

ICAI'S ANSWER

Actual Cost of Material (Actual Quantity x Actual Price) = 25,740

Material Cost Variance = Price Variance + Usage Variance

= 1,170A + 750F = 420A

Standard Cost of Materials (Standard Quantity x Standard Price)

= Actual cost + Cost variance

= 25,740 – 1170 + 750 = 25,320

SQ x SP = 25,320, SP = 3 per unit **hence SQ = 8,440**

AQ x SP = (SQ x SP) + Usage Variance = 25,320 – 750 = 24,570

Hence AQ of Material = 8,190

AQ x AP = 25,740

Hence AP of Material = ₹ 3.142

Actual Wages paid (AH x AR) = 65,368

Wage cost variance = Wage efficiency variance + Wage rate variance

= 1352 F – 800 A = 552 F

SH x SR = Actual Wages paid + Wage cost variance

= 65,368 + 552 = 65,920

SH = 4,120 hours

AH x SR = SH x SR + Efficiency variance = 66,720

AH thereon = 4,170 hours

AH x AR = 65,368

AR = 15.676

SH x SR (Absorbed Overheads) = 4,120 x 20 = 82,400

Actual Output = 4,120/10 = 412 units

OH Cost variance = Volume Variance + Expenditure Variance

= 1,400 F + 900 F

= 2,300 F

Actual FOH = 82,400 – 2,300 = 80,100

Budgeted FOH = Actual FOH + Expenditure Variance

Budgeted FOH = 80,100 + 900 = 81,000

BO x SR = BFOH = 81,000/200 = 405 units

Budgeted Units = 405 units

(SH x SR – AH x SR) = 82,400 – (4,170 x 20) = 1,000A

OH Efficiency Variance = 1,000A

AH x SR – BFOH = 83,400 – 81,000

OH Capacity Variance = 2,400 F

ICAI · P4 · MTP May 2025 · Series II · question p. 10-11 · answer p. 11-12

906 MTP **Sep 2024 S-I** MCQ 12 2 marks MCQ

A furniture company uses premium wood for sofa. Standard quantity of premium wood per sofa is 5 sq. ft. Standard price per sq. ft. of premium wood is ₹ 10. Actual production of sofa is 1,000. Premium wood actually used is 5,300 sq. ft. Actual purchase price of premium wood per sq. ft. is ₹ 10. What is material cost variance?

A ₹ 3,000 (A)

B ₹ 4,300 (A)

C ₹ 7,300 (A)

D ₹ 5,300 (F)

ICAI'S ANSWER

(A) ₹ 3000(A)

Standard quantity = Standard quantity per sofa × Actual production

= 5 sq. ft x 1000 = 5,000 sq. ft.

Standard material cost = Standard quantity× Standard price per sq. ft.

= 5,000 sq. ft.×₹ 10/sq. ft.= ₹ 50,000

Actual material cost = Actual quantity used × Actual purchase price per sq. ft.

= 5,300 sq. ft.×₹ 10/sq. ft.= ₹ 53,000

Material cost variance = Standard material cost–Actual Material cost

= ₹ 50,000–₹ 53,000 = –₹ 3,000

ICAI · P4 · MTP Sep 2024 · Series I · question p. 5 · answer p. 4-5

907 MTP **Sep 2024 S-I** Q4(b) 8 marks

Anju Limited produces a product 'Pect' which is sold in a 10 Kg. packet. The standard cost card per

packet of 'Pect' are as follows:

	₹
Direct materials 10 kg @ ₹ 45 per kg	450
Direct labour 8 hours @ ₹ 50 per hour	400
Variable Overhead 8 hours @ ₹ 10 per hour	80
Fixed Overhead	200
	1,130

Budgeted output for the third quarter of a year was 10,000 Kg. Actual output is 9,000 Kg.

Actual cost for this quarter are as follows :

	₹
Direct Materials 8,900 Kg @ ₹ 46 per Kg.	4,09,400
Direct Labour 7,000 hours @ ₹ 52 per hour	3,64,000
Variable Overhead incurred	72,500
Fixed Overhead incurred	1,92,000

You are required to CALCULATE:

(i) Material Usage Variance

(ii) Material Price Variance

(iii) Material Cost Variance

(iv) Labour Efficiency Variance

(v) Labour Rate Variance

(vi) Labour Cost Variance

(vii) Variable Overhead Cost Variance

(viii) Fixed Overhead Cost Variance

ICAI'S ANSWER

(i) Material Usage Variance = Std. Price (Std. Quantity – Actual Quantity)

= ₹ 45 (9,000 kgs. – 8,900 kgs.)

= ₹ 4,500 (Favourable)

(ii) Material Price Variance = Actual Quantity (Std. Price – Actual Price)

= 8,900 kgs. (₹ 45 – ₹ 46)

= ₹ 8,900 (Adverse)

(iii) Material Cost Variance = Std. Material Cost – Actual Material Cost

= (SQ × SP) – (AQ × AP)

= (9,000 kgs. × ₹ 45) – (8,900 kgs. × ₹ 46)

= ₹ 4,05,000 – ₹ 4,09,400

= ₹ 4,400 (Adverse)

(iv) Labour Efficiency Variance = Std. Rate (Std. Hours – Actual Hours)

= ₹ 50 ($\frac{9,000}{10}$ ×8hours – 7,000 hrs.)

= ₹ 50 (7,200 hrs. – 7,000 hrs.)

= ₹ 10,000 (Favourable)

(v) Labour Rate Variance = Actual Hours (Std. Rate – Actual Rate)

= 7,000 hrs. (₹ 50 – ₹ 52)

= ₹ 14,000 (Adverse)

(vi) Labour Cost Variance = Std. Labour Cost – Actual Labour Cost

= (SH × SR) – (AH × AR)

= (7,200 hrs. × ₹ 50) – (7,000 hrs. × ₹ 52)

= ₹ 3,60,000 – ₹ 3,64,000

= ₹ 4,000 (Adverse)

(vii) Variable Overhead Cost Variance = Std. Overhead for Actual Production – Actual Variable Overhead Cost

= (7,200 hrs. × ₹ 10) – ₹ 72,500

= ₹ 500 (Adverse)

(viii) Fixed Overhead Cost Variance = Absorbed Fixed Overhead – Actual Fixed Overhead

= $\frac{₹\ 200}{10\ \text{kgs.}}$ ×9,000kgs.- ₹ 1,92,000

= ₹ 1,80,000 – ₹ 1,92,000

= ₹ 12,000 (Adverse)

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908 MTP **Sep 2024 S-II** Case scenario · MCQ 6 to MCQ 10 10 marks

CASE SCENARIO

Hilfy textiles Ltd. has been a major player in the textile industry, producing high-quality polyester mix cotton fabric. The production process is complex and involves multiple stages, including spinning, weaving, quality control, and packaging. The company has been facing challenges in controlling costs and maintaining profitability, mainly due to fluctuating material costs and labor inefficiencies.

To address these challenges, the company's management has decided to implement a **standard costing** system to better manage costs, set benchmarks, and identify variances. The goal is to gain better control over production costs, improve budgeting accuracy, and enhance decision-making.

Hilfy textiles Ltd. had prepared the following estimation for the month of April:

	Quantity/Time	Rate (₹)	Amount (₹)
Cotton	8,000 m	50.00	4,00,000
Polyester	6,000 m	40.00	2,40,000
Skilled labour	1,000 hours	37.50	37,500
Unskilled labour	800 hours	22.00	17,600

Normal loss was expected to be 10% of total input materials and an idle labour time of 5% of expected labour hours was also estimated.

At the end of the month the following information has been collected from the cost accounting department:

The company has produced 14,800 m finished product by using the followings:

	Quantity/Time	Rate (₹)	Amount (₹)
Cotton	9,000 m	48.00	4,32,000
Polyester	6,500 m	37.00	2,40,500
Skilled labour	1,200 hours	35.50	42,600
Unskilled labour	860 hours	23.00	19,780

On the basis of analysis of standard costing system, company's management wants to take actions like supplier negotiation, process optimisation, employee training, etc.

Being the cost manager of the company, you are required to answer the following five requirements of the management:

MCQ 6 2 marks

Compute Material mix variance and Material Yield Variance

A ₹ 1430 (A) & 43,200 (F)

B ₹ 1430 (F) & 43,200 (F)

C ₹ 24,000 (A) & 37,500 (F)

D ₹ 19,300 (A) & 37,500 (F)

MCQ 7 2 marks

Compute Material Price Variance for supplier negotiation

A ₹ 18,000 (A)

B ₹ 43,200 (F)

C ₹ 37,500 (A)

D ₹ 37,500 (F)

MCQ 8 2 marks

Compute Material Cost Variance

A ₹ 32,500 (F)

B ₹ 24,500 (A)

C ₹ 79,270 (F)

D ₹ 79,270 (A)

MCQ 9 2 marks

Compute Labour Efficiency Variance and Labour Yield Variance.

A ₹ 940 (A) & 1,140 (A)

B ₹ 2,424 (A) & 1,556 (A)

C ₹ 2,424 (A) & 1,556 (A)

D ₹ 940 (A) & 1,140 (F)

MCQ 10 2 marks

Compute Labour Cost Variance.

A ₹ 884 (A)

B ₹ 1,556 (F)

C ₹ 884 (F)

D ₹ 1,556 (A)

ICAI'S ANSWERS

MCQ 6

(A)

Material Mix Variance (Cotton + Polyester) = {(RSQ × SP) – (AQ × SP)}

= {7,08,570- 7,10,000}

= 1,430 (A)

Material Yield Variance (Cotton + Polyester) = {(SQ × SP) – (RSQ × SP)}

= {7,51,770 – 7,08,570}

= 43,200 (F)

MCQ 7

(D)

Material Price Variance (Cotton + Polyester) = {(AQ × SP) – (AQ × AP)

= {7,10,000 – 6,72,500}

= 37,500 (F)

MCQ 8

(C)

Material Cost Variance (Cotton + Polyester) = {(SQ × SP) – (AQ × AP)}

= {7,51,770 – 6,72,500}

= 79,270 (F)

Working Note

Material Variances:

Material	SQ (WN-1)	SP (₹)	SQ × SP (₹)	RSQ (WN-2)	RSQ × SP (₹)	AQ	AQ × SP (₹)	AP (₹)	AQ × AP (₹)
Cotton	9,397 m	50	4,69,850	8,857 m	4,42,850	9,000 m	4,50,000	48	4,32,000
Polyester	7,048 m	40	2,81,920	6,643 m	2,65,720	6,500 m	2,60,000	37	2,40,500
	16,445 m		**7,51,770**	**15,500 m**	**7,08,570**	**15,500 m**	**7,10,000**		**6,72,500**

WN-1: Standard Quantity (SQ):

Cotton - $\left(\frac{8{,}000\text{m}}{0.9 \times 14{,}000\text{m}} \times 14{,}800\text{m}\right)$ = 9,396.8 or 9,397 m

Polyester- $\left(\frac{6{,}000\text{m}}{0.9 \times 14{,}000\text{m}} \times 14{,}800\text{m}\right)$ = 7,047.6 or 7048 m

WN- 2: Revised Standard Quantity (RSQ):

Cotton - $\left(\frac{8{,}000\text{m}}{14{,}000\text{m}} \times 15{,}500\text{m}\right)$ = 8,857.1 or 8857 m

Polyester - $\left(\frac{6{,}000\text{m}}{14{,}000\text{m}} \times 15{,}500\text{m}\right)$ = 6,642.8 or 6643 m

MCQ 9

(B)

Labour Efficiency Variance (Skilled + Unskilled) = {(SH × SR) – (AH × SR)}

= {61,496 – 63,920}

= 2,424 (A)

Labour Yield Variance (Skilled + Unskilled) = {(SH × SR) – (RSH × SR)}

= {61,496 – 63,052}

= 1,556 (A)

MCQ 10

(A)

Labour Cost Variance (Skilled + Unskilled) = {(SH × SR) – (AH × AR)}

= {61,496 – 62,380}

= 884 (A)

Working Note

Labour Variances:

Labour	SH (WN-3)	SR (₹)	SH × SR (₹)	RSH (WN-4)	RSH × SR (₹)	AH	AH × SR (₹)	AR (₹)	AH × AR (₹)
Skilled	1,116 hrs	37.50	41,850	1144	42,900	1,200	45,000	35.50	42,600
Unskilled	893 hrs	22.00	19,646	916	20,152	860	18,920	23.00	19,780
	2,009 hrs		61,496	2,060	63,052	2,060	63,920		62,380

WN- 3: Standard Hours (SH):

Skilled labour- $\left(\frac{0.95 \times 1{,}000\text{hr.}}{0.90 \times 14{,}000\text{m.}} \times 14{,}800\text{m.}\right)$ =1,115.87 or 1,116 hrs.

Unskilled labour- $\left(\frac{0.95 \times 800\text{hr.}}{0.90 \times 14{,}000\text{m.}} \times 14{,}800\text{m.}\right)$ = 892.69 or 893 hrs.

WN- 4: Revised Standard Hours (RSH):

Skilled labour- $\left(\frac{1{,}000\text{hr.}}{1{,}800\text{hr.}} \times 2{,}060\text{hr.}\right)$ = 1,144.44 or 1,144 hrs.

Unskilled labour- $\left(\frac{800\text{hr.}}{1{,}800\text{hr.}} \times 2{,}060\text{hr.}\right)$ = 915.56 or 916 hrs.

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909 MTP **May 2024 S-I** Q5(a) 6 marks

SARA Ltd. has furnished the following standard cost data per' unit of production:

Material 15 kg @ ₹ 15 per kg.

Labour 6 hours @ ₹ 5 per hour

Variable overhead 6 hours @ ₹ 12 per hour.

Fixed overhead ₹ 4,50,000 per month (Based on a normal volume of 30,000 labour hours.)

The actual cost data for the month of August 2023 are as follows:

Material used 65,000 kg at a cost of ₹ 9,85,000.

Labour paid ₹ 1,40,000 for 31,500 hours worked.

Variable overheads ₹ 3,60,200

Fixed overheads ₹ 4,70,000

Actual production 4,800 units.

CALCULATE:

(i) Material Cost Variance.

(ii) Labour Cost Variance.

(iii) Fixed Overhead Cost Variance.

(iv) Variable Overhead Cost Variance.

ICAI'S ANSWER

Budgeted Production 30,000 hours ÷ 6 hours per unit = 5,000 units

Budgeted Fixed Overhead Rate = ₹ 4,50,000 ÷ 5,000 units = ₹ 90 per unit Or

= ₹ 4,50,000 ÷ 30,000 hours = ₹ 15 per hour.

(i) Material Cost Variance	= (Std. Qty. × Std. Price) – (Actual Qty. × Actual Price)
	= (4,800 units × 15 kg. × ₹15) - ₹ 9,85,000
	= ₹ 10,80,000 – ₹ 9,85,000
	= ₹ 95,000 (F)
(ii) Labour Cost Variance	= (Std. Hours × Std. Rate) – (Actual Hours × Actual rate)
	= (4,800 units × 6 hours × ₹ 5) – ₹1,40,000

	= ₹ 1,44,000 – ₹ 1,40,000
	= ₹ 4,000 (F)
(iii) Fixed Overhead Cost Variance	= (Budgeted Rate × Actual Qty) – Actual Overhead
	= (₹ 90 × 4,800 units) – ₹ 4,70,000
	= ₹ 38,000 (A)
OR	
	= (Budgeted Rate × Std. Hours) – Actual Overhead
	= (₹ 15 × 4,800 units × 6 hours) – ₹ 4,70,000
	= ₹ 38,000 (A)
(iv) Variable Overhead Cost Variance	= (Std. Rate × Std. Hours) – Actual Overhead
	= (4,800 units × 6 hours × ₹ 12) - ₹ 3,60,200
	= ₹ 3,45,600 - ₹ 3,60,200
	= ₹ 14,600 (A)

ICAI · P4 · MTP May 2024 · Series I · question p. 10 · answer p. 11-12

910 MTP **Old P3 Nov 2023 S-I** Q5(b) 10 marks old scheme

Essel Minerals Ltd. operates in iron ore mining through open cast mining method. Explosives and detonators are used for excavation of iron ores from the mines. The following are the details of standard quantity of explosives materials used for mining:

Particulars	Rate (₹)	Standard Qty. for Iron ore	Standard Qty. for Overburden (OB)
SME	40.00 per kg.	2.4 kg per tonne	1.9 kg per cubic- meter
Detonators	20.00 per piece	2 pcs per tonne	2 pcs per cubic- meter

The standard stripping ratio is 3:1 (means 3 cubic- meter of overburden soil to be removed to get one tonne of iron ore).

During the month of December 2021, the company produces 20,000 tonnes of iron ore and 58,000 cubic-meter of OB. The quantity of explosive materials used and paid for the month is as below:

Material	Quantity	Amount (₹)
SME	1,67,200 kg.	63,53,600
Detonators	1,18,400 pcs	24,27,200

Required:

(i) COMPUTE the material price variance

(ii) COMPUTE the material quantity variance

(iii) COMPUTE the material cost variance

ICAI'S ANSWER

Workings:

1. Calculation of Standard Qty. of Explosives and Detonators for actual output:

	Particulars	Iron ore	Overburden (OB)	Total
SME:				
A	Actual Output	20,000 tonne	58,000 M^3	
B	Standard Qty per unit	2.4 kg./tonne	1.9 kg./M^3	
C	**Standard Qty. for actual production [A×B]**	**48,000 kg.**	**1,10,200 kg.**	**1,58,200 kg.**
Detonators:				
D	Standard Qty per unit	2 pcs/ tonne	2 pcs/ M^3	
E	**Standard Qty. for actual production [A×D]**	**40,000 pcs.**	**1,16,000 pcs**	**1,56,000 pcs**

2. Calculation of Actual Price per unit of materials:

Material	Quantity [A]	Amount (₹) [B]	Rate (₹) [C = B÷A]
SME	1,67,200 kg.	63,53,600	38.00
Detonators	1,18,400 pcs	24,27,200	20.50

(i) Computation of material price variance:

Material Price Variance	= Actual Qty. × (Std. Price - Actual Price)
SME	= 1,67,200 kg. × (₹ 40 – ₹ 38) = ₹ 3,34,400 (F)
Detonators	= 1,18,400 pcs × (₹ 20 – ₹ 20.5) = ₹ 59,200 (A)
Total	= ₹ 2,75,200 (F)

(ii) Computation of material quantity variance:

Material Qty. Variance	= Std. Price × (Std. Qty for actual output - Actual Qty.)
SME	= ₹ 40 × (1,58,200 kg. - 1,67,200 kg.) = ₹ 3,60,000 (A)
Detonators	= ₹ 20 × (1,56,000 pcs -1,18,400 pcs) = ₹ 7,52,000 (F)
Total	**= ₹3,92,000 (F)**

(iii) Computation of material cost variance:

Material cost variance	= Std. cost – Actual Cost
Or, (Std. Price × Std. Qty) – (Actual Price × Actual Qty.)	
SME	= (₹ 40 × 1,58,200 kg) – (₹ 38 × 1,67,200 kg.) = ₹ 63,28,000 – ₹ 63,53,600 = ₹ 25,600 (A)
Detonators	= (₹ 20 × 1,56,000 pcs) – (₹ 20.50 × 1,18,400 pcs) = ₹ 31,20,000 – ₹ 24,27,200 = 6,92,800 (F)
Total	**= ₹ 6,67,200 (F)**

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911 MTP **Old P3 May 2023 S-I** Q2(b) 10 marks old scheme

Following data is extracted from the books of RAMZY Ltd. for the month of March:

(i) Estimation-

Particulars	Quantity (kg.)	Price (₹)	Amount (₹)
Material-A	1320	?	--
Material-B	990	50	49500
			--

Normal loss was expected to be 5% of total input materials.

(ii) Actuals- 2,500 kg of output produced.

Particulars	Quantity (kg.)	Price (₹)	Amount (₹)
Material-A	1500	?	--
Material-B	?	53	--
			98,000

(iii) Other Information-

Material Cost Variance = ₹ 5,500 (F)

Material Price Variance = ₹ 300 (F)

You are required to CALCULATE:

(i) Standard Price of Material-A;

(ii) Actual Quantity of Material-B;

(iii) Actual Price of Material-A;

(iv) Revised standard quantity of Material-A and Material-B; and

(v) Material Mix Variance.

ICAI'S ANSWER

(i)	Material Cost Variance (A + B)	= {(SQ × SP) – (AQ × AP)}
	Or ₹5,500	= (SQ × SP) – ₹98,000
	Or (SQ × SP)	= ₹1,03,500
	Or (SQA × SPA) + (SQB × SPB)	= ₹ 1,03,500
	Or (1,503.8 kg × SPA) + (1,127.8 kg ×₹50)	= ₹ 1,03,500
	Or (1,503.8 kg × SPA) + ₹56,390	= ₹1,03,500
	Or (1,503.8 kg × SPA)	= ₹ 47,110
	Or SP_A	= $\frac{47,110}{1503.80\ \text{kg}}$ = ₹31.33
(ii)	Material Price Variance (A + B)	= {(AQ × SP) – (AQ × AP)}
	Or ₹300	= (AQ × SP) – ₹ 98,000
	Or (AQ × SP)	= ₹ 98,300
	Or (AQA × SPA) + (AQB × SPB)	= ₹98,300
	Or (1,500 kg × ₹31.33 (from (i) above)) + AQB x ₹50	= ₹98,300
	Or ₹ 46,995 + (AQB × ₹ 50)	= ₹ 98,300
	Or (AQB × ₹ 50)	= ₹ 51,305
	Or AQB	= 1,026kg
	Actual Quantity of Material B	**= 1,026 kg.**
(iii)	(AQ × AP)	= ₹98,000
	Or (AQ_A × AP_A) + (AQ_B × AP_B)	= ₹ 98,000
	Or (1,500 kg × AP_A) + (1,026 kg (from (ii) above) × ₹53)	= ₹ 98,000
	Or (1,500 kg × AP_A) + ₹ 54,378	= ₹ 98,000
	Or (1,500 kg AP_A)	= ₹ 43,622
	Or AP_A = $\frac{43,622}{1,500}$	= ₹ 29.10
	Actual Price of Material A	**= ₹ 29.10**
(iv)	Total Actual Quantity of Material-A and Material-B	= AQA + AQB
	Or 1,500 kg + 1,026 kg (from (ii) above)	= 2,526 kg

Revised SQ_A = $\frac{1320\ kg}{(1{,}320+990)}$ x 2,526 kg	= 1,443 kg
Revised SQ_B = $\frac{990\ kg}{(1{,}320+990)}$ x 2,526 kg	= 1,083 kg

(v) Material Mix Variance (A + B) ={(RSQ × SP) – (AQ × SP)}

= {(RSQA × SPA) + (RSQB × SPB) – ₹98,300} = (1,443 kg (from (iv) above) × ₹ 31.33 (from (i) above)) + (1,083 kg (from (iv) above) × ₹50) - ₹98,300

= (₹45,209 + ₹54,150) – ₹98,300 = ₹ 1059 (F)

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912 MTP **Old P3 Nov 2022 S-I** Q2(a) 10 marks old scheme

The details regarding a product manufactured by the company for the last one week are as follows:

Standard cost (per unit)	
Direct materials 10 units @ ₹ 22.50	₹ 225
Direct wages 5 hours @ ₹ 120	₹ 600
Total:	₹ 825
Actual (for whole activity):	
Direct materials	₹ 96,525
Direct wages	₹ 2,44,860
Analysis of variances:	
Direct materials:	
Price	₹ 8,775 (Adverse)
Usage	₹ 5,625 (Favourable)
Direct wages (labour):	
Efficiency	₹ 5,400 (Adverse)

You are required to CALCULATE:

(i) Material Cost variance

(ii) Actual output units

(iii) Actual price of material per unit

(iv) Actual Wages rate per labour hour

(v) Labour rate variance

(vi) Labour Cost variance

ICAI'S ANSWER

(i) Material Cost Variance = Material Price Variance+ Material Usage Variance

= ₹ 8,775 A + ₹ 5,625 F= ₹ 3,150 Adverse

(ii) Actual output units

Let x be the actual quantity of output

Then Standard Quantity of input for actual output 'x'

SQ = 10x

Material cost variance = (SQ x SP) - (AQ x AP)

-3,150 = (10x x ₹ 22.50) - ₹ 96,525

-3,150 = 225x - ₹ 96,525

225x = ₹ 96,525 – 3,150 = ₹ 93,375

x = ₹ 93,375/225 = 415 Units

(iii) Actual Price of Material per unit

Material Usage variance = (SQ - AQ) x SP

5,625 = (10x - AQ) x ₹ 22.50

5,625 = (10 x 415 units - AQ) x ₹ 22.50

5,625/22.50 = 4,150 - AQ

AQ = 4,150 - 250 = 3,900 units

Now, AQ x AP = ₹ 96,525 (given)

AP = ₹ 96,525/AQ

= ₹ 96,525/3,900 units = ₹ 24.75

(iv) Actual wages rate per labour hour

Labour efficiency variance = 5,400 Adverse (given)

Standard rate per hour (Standard time – Actual time) = -5,400

₹ 120 [(Actual output units x Number of hours per output) – Actual time] = -5,400

₹ 120 [(415 units x 5 hrs) – Actual time] = -5,400

2,075 hrs – Actual time = -5,400/120

Actual time = 2,075 + 45

= 2,120 hrs

Now Direct wages = ₹ 2,44,860 (given)

Actual time x Actual rate per hour = ₹ 2,44,860

Actual rate per hour = ₹ 2,44,860 / 2,120 hrs

= ₹ 115.50

(v) Labour rate variance

= Actual time (Standard Rate – Actual Rate)

= 2,120 hrs (₹ 120 - ₹ 115.50)

= 2,120 hrs x ₹ 4.50 = **9,540 Favourable**

(vi) Labour Cost variance

= Labour rate variance+ Labour efficiency variance

= 9,540 F + 5,400 A = 4,140 Favourable

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913 MTP **Old P3 May 2022 S-II** Q2(b) 10 marks old scheme

Rounak Minerals Ltd. operates in iron ore mining through open cast mining method. Explosives and detonators are used for excavation of iron ores from the mines. The following are the details of standard quantity of explosives materials used for mining:

Particulars	Rate (₹)	Standard Qty. for Iron ore	Standard Qty. for Overburden (OB)
SME	40.00 per kg.	2.4 kg per tonne	1.9 kg per cubic- meter
Detonators	20.00 per piece	2 pcs per tonne	2 pcs per cubic-meter

The standard stripping ratio is 3:1 (means 3 cubic- meter of overburden soil to be removed to get one tonne of iron ore).

During the month of December 2021, the company produced 20,000 tonnes of iron ore and removed 58,000 cubic- meter of OB. The quantity of explosive materials used and paid for the month is as below:

Material	Quantity	Amount (₹)
SME	1,67,200 kg.	63,53,600
Detonators	1,18,400 pcs	24,27,200

You are required to COMPUTE:

(i) Material price variance

(ii) Material quantity variance

(iii) Material cost variance.

ICAI'S ANSWER

Workings:

1. Calculation of Standard Qty. of Explosives and Detonators for actual output:

	Particulars	Iron ore	Overburden (OB)	Total
SME:				
A	Actual Output	20,000 tonne	58,000 M3	
B	Standard Qty per unit	2.4 kg./tonne	1.9 kg./M3	

	Particulars	Iron ore	Overburden (OB)	Total
C	**Standard Qty. for actual production [A×B]**	**48,000 kg.**	**1,10,200 kg.**	**1,58,200 kg.**
Detonators:				
D	Standard Qty per unit	2 pcs/ tonne	2 pcs/ M3	
E	**Standard Qty. for actual production [A×D]**	**40,000 pcs.**	**1,16,000 pcs**	**1,56,000 pcs**

2. Calculation of Actual Price per unit of materials:

Material	Quantity [A]	Amount (₹) [B]	Rate (₹) [C = B÷A]
SME	1,67,200 kg.	63,53,600	38.00
Detonators	1,18,400 pcs	24,27,200	20.50

(i) Computation of material price variance:

Material Price Variance	= Actual Qty. × (Std. Price - Actual Price)	
SME	= 1,67,200 kg. × (₹40 – ₹38)	= ₹ 3,34,400 (F)
Detonators	= 1,18,400 pcs × (₹20 – ₹20.5)	= ₹ 59,200 (A)
Total		**= ₹ 2,75,200 (F)**

(ii) Computation of material quantity variance:

Material Qty. Variance	= Std. Price × (Std. Qty for actual output - Actual Qty.)	
SME	= ₹40 × (1,58,200 kg. - 1,67,200 kg.)	= ₹ 3,60,000 (A)
Detonators	= ₹20 × (1,56,000 pcs -1,18,400 pcs)	= ₹ 7,52,000 (F)
Total		**= ₹ 3,92,000 (F)**

(iii) Computation of material cost variance:

Material cost variance	= Std. cost – Actual Cost	
Or, (Std. Price × Std. Qty) – (Actual Price × Actual Qty.)		
SME	= (₹40 × 1,58,200 kg) – (₹38 × 1,67,200 kg.)	
	= ₹63,28,000 – ₹63,53,600	= ₹ 25,600 (A)
Detonators	= (₹20 × 1,56,000 pcs) – (₹20.50 × 1,18,400 pcs)	
	= ₹31,20,000 – ₹24,27,200	= ₹ 6,92,800 (F)
Total		**= ₹ 6,67,200 (F)**

ICAI · Old P3 · MTP May 2022 · Series II · question p. 3 · answer p. 4-5

914 MTP **Old P3 Dec 2021 S-II** Q1(d) 5 marks old scheme

JK Ltd. has furnished the following standard cost data per unit of production:

Material 10 kg @ ₹ 200 per kg.

Labour 6 hours @ ₹ 110 per hour

Variable overhead 6 hours @ ₹ 200 per hour.

Fixed overhead ₹ 90,00,000 per month (Based on a normal volume of 30,000 labour hours.)

The actual cost data for the month of September 2021 are as follows:

Material used 50,000 kg at a cost of ₹ 1,05,00,000.

Labour paid ₹ 31,00,000 for 31,000 hours

Variable overheads ₹ 58,60,000

Fixed overheads ₹ 94,00,000

Actual production 4,800 units.

CALCULATE:

(i) Material Cost Variance.

(ii) Labour Cost Variance.

(iii) Fixed Overhead Cost Variance.

(iv) Variable Overhead Cost Variance.

ICAI'S ANSWER

Budgeted Production 30,000 hours ÷ 6 hours per unit = 5,000 units

Budgeted Fixed Overhead Rate = ₹ 90,00,000 ÷ 5,000 units = ₹ 1,800 per unit

= ₹ 90,00,000 ÷ 30,000 hours = ₹ 300 per hour.

(i) Material Cost Variance = (Std. Qty. × Std. Price) – (Actual Qty. × Actual Price)

= (4,800 units × 10 kg. × ₹200) - ₹1,05,00,000

= ₹ 96,00,000 – ₹ 1,05,00,000

= ₹ 9,00,000 (A)

(ii) Labour Cost Variance = (Std. Hours × Std. Rate) – (Actual Hours × Actual rate)

= (4,800 units × 6 hours × ₹ 110) – ₹31,00,000

= ₹ 31,68,000 – ₹ 31,00,000

= ₹ 68,000 (F)

(iii) Fixed Overhead Cost Variance= (Budgeted Rate × Actual Qty) – Actual Overhead

= (₹ 1,800 × 4,800 units) – ₹ 94,00,000

= ₹ 7,60,000 (A)

OR = (Budgeted Rate × Std. Hours) – Actual Overhead

= (₹ 300 × 4,800 units × 6 hours) – ₹ 94,00,000

= ₹ 7,60,000 (A)

(iv) Variable Overhead Cost Variance= (Std. Rate × Std. Hours) – Actual Overhead

= (4,800 units × 6 hours × ₹ 200) - ₹ 58,60,000

= ₹ 57,60,000 - ₹ 58,60,000

= ₹ 1,00,000 (A)

ICAI · Old P3 · MTP Dec 2021 · Series II · question p. 2 · answer p. 3-4

915 MTP **Old P3 Jul 2021 S-I** Q1(d) 5 marks old scheme

AK Ltd. has furnished the following standard cost data per unit of production:

Material 10 kg @ Rs. 100 per kg.

Labour 6 hours @ Rs. 55 per hour

Variable overhead 6 hours @ Rs. 100 per hour.

Fixed overhead Rs.45,00,000 per month (Based on a normal volume of 30,000 labour hrs)

The actual cost data for the month of September 2020 are as follows:

Material used 50,000 kg at a cost of Rs. 52,50,000.

Labour paid Rs. 15,50,000 for 31,000 hours

Variable overheads Rs. 29,30,000

Fixed overheads Rs. 47,00,000

Actual production 4,800 units.

CALCULATE:

(i) Material Cost Variance.

(ii) Labour Cost Variance.

(iii) Fixed Overhead Cost Variance.

(iv) Variable Overhead Cost Variance.

ICAI'S ANSWER

Budgeted Production 30,000 hours ÷ 6 hours per unit = 5,000 units

Budgeted Fixed Overhead Rate = Rs. 45,00,000 ÷ 5,000 units = Rs. 900 per unit Or

= Rs. 45,00,000 ÷ 30,000 hours = Rs. 150 per hour.

(i) Material Cost Variance = (Std. Qty. × Std. Price) – (Actual Qty. × Actual Price)

= (4,800 units × 10 kg. × Rs.100) - Rs. 52,50,000

= Rs. 48,00,000 – Rs. 52,50,000

= Rs. 4,50,000 (A)

(ii) Labour Cost Variance = (Std. Hours × Std. Rate) – (Actual Hours × Actual rate)

= (4,800 units × 6 hours × Rs. 55) – Rs. 15,50,000

= Rs. 15,84,000 – Rs. 15,50,000

= Rs. 34,000 (F)

(iii) Fixed Overhead Cost Variance = (Budgeted Rate × Actual Qty) – Actual Overhead

= (Rs. 900 × 4,800 units) – Rs.47,00,000

= Rs. 3,80,000 (A)

OR

= (Budgeted Rate × Std. Hours) – Actual Overhead

= (Rs. 150 × 4,800 units × 6 hours) – Rs. 47,00,000

= Rs. 3,80,000 (A)

(iv) Variable Overhead Cost Variance = (Std. Rate × Std. Hours) – Actual Overhead

= (4,800 units × 6 hours × Rs. 100) - Rs. 29,30,000

= Rs. 28,80,000 - Rs. 29,30,000

= Rs. 50,000 (A)

ICAI · Old P3 · MTP Jul 2021 · Series I · question p. 2 · answer p. 2-3

916 MTP **Old P3 Nov 2020 S-I** Q2(a) 10 marks old scheme

Following data is extracted from the books of XYZ Ltd. for the month of January, 2020:

(i) Estimation-

Particulars	Quantity (kg.)	Price (₹)	Amount (₹)
Material-A	800	?	--
Material-B	600	30.00	18,000
			--

Normal loss was expected to be 10% of total input materials.

(ii) Actuals-

1480 kg of output produced.

Particulars	Quantity (kg.)	Price (₹)	Amount (₹)
Material-A	900	?	--
Material-B	?	32.50	--
			59,825

(iii) Other Information-

Material Cost Variance = ₹ 3,625 (F)

Material Price Variance = ₹ 175 (F)

You are required to CALCULATE:

(i) Standard Price of Material-A;

(ii) Actual Quantity of Material-B;

(iii) Actual Price of Material-A;

(iv) Revised standard quantity of Material-A and Material-B; and

(v) Material Mix Variance;

ICAI'S ANSWER

(i) Material Cost Variance (A + B) = {(SQ × SP) – (AQ × AP)}

₹ 3,625 = (SQ × SP) – ₹ 59,825

(SQ × SP) = ₹ 63,450

(SQA × SPA) + (SQB × SPB) = ₹ 63,450

(940 kg × SPA) + (705 kg × ₹ 30) = ₹ 63,450

(940 kg × SPA) + ₹ 21,150 = ₹ 63,450

(940 kg × SPA) = ₹ 42,300

$$\text{SPA} = \frac{\text{Rs.42,300}}{\text{940kg}}$$

Standard Price of Material-A = ₹ 45

Working Note:

SQ i.e. quantity of inputs to be used to produce actual output

$$= \frac{\text{1,480kg}}{90\%} = \textbf{1,645 kg}$$

$$\text{SQA} = \frac{\text{800kg}}{(800 + 600)} \times \text{1,645kg.} = 940 \text{ kg}$$

$$\text{SQB} = \frac{\text{600kg}}{(800 + 600)} \times \text{1,645kg.} = 705 \text{ kg}$$

(ii) Material Price Variance (A + B) = {(AQ × SP) – (AQ × AP)}

₹ 175 = (AQ × SP) – ₹ 59,825

(AQ × SP) = ₹ 60,000

(AQA × SPA) + (AQB × SPB) = ₹ 60,000

(900 kg × ₹ 45 (from (i) above)) + (AQB × ₹30) = ₹ 60,000

₹ 40,500 + (AQB × ₹ 30) = ₹ 60,000

(AQB × ₹ 30) = ₹ 19,500

$$AQB = \frac{19,500}{30} = 650 \text{ kg}$$

Actual Quantity of Material B = 650 kg.

(iii) (AQ × AP) = ₹ 59,825

(AQA × APA) + (AQB × APB) = ₹ 59,825

(900 kg × APA) + (650 kg (from (ii) above) × ₹ 32.5) = ₹ 59,825

(900 kg × APA) + ₹ 21,125 = ₹ 59,825

(900 kg × APA) = ₹ 38,700

$$APA = \frac{38,700}{900} = 43$$

Actual Price of Material-A = ₹ 43

(iv) Total Actual Quantity of Material-A and Material-B

= AQA + AQB

= 900 kg + 650 kg (from (ii) above)

= 1,550 kg

Now,

$$\text{Revised SQA} = \frac{800\text{kg}}{(800+600)} \times 1,550\text{kg.} = 886 \text{ kg}$$

$$\text{Revised SQB} = \frac{600\text{kg}}{(800+600)} \times 1,550\text{kg.} = 664 \text{ kg}$$

(v) Material Mix Variance (A + B) = {(RSQ × SP) – (AQ × SP)}

= {(RSQA × SPA) + (RSQB × SPB) – 60,000}

= (886 kg (from (iv) above) × ₹ 45 (from (i) above)) + (664 kg (from (iv) above) × ₹ 30) - ₹60,000

= (39,870 + 19,920) – 60,000 = ₹ 210 (A)

ICAI · Old P3 · MTP Nov 2020 · Series I · question p. 3 · answer p. 3-4

13.3 Labour variances, including idle time

24 questions · asked in 12 ICAI documents · 5 exam papers

917 **EXAM** **Sep 2026** Q4(a) **9 marks**

XYZ Ltd. employs workforce classified into Skilled, Semi-Skilled and Unskilled categories for its manufacture of precision-engineered automobile components.

In the month of August 2026, the Production Department completed the manufacture of 5000 units of Product X.

The following information is available :

Budgeted Labour Requirement for manufacturing **2** units of Product X.

Labour Category	Standard Hours	Standard Rate (₹ per hour)
Skilled	2	6
Semi-skilled	3	5
Unskilled	5	2

Actual Labour Details :

Labour Category	Actual Hours paid (including abnormal idle hours)	Abnormal Idle time (hours)	Actual Rate (₹)	Amount (₹)
Skilled	6,000	200	7	42,000
Semi-skilled	8,000	300	4.5	36,000
Unskilled	11,000	400	2.5	27,500
Total	**25,000**	**900**		**1,05,500**

Required :

Compute category wise labour cost variances analysed into mix, efficiency, yield and rate variances, indicating whether they are Adverse (A) or Favourable (F).

ICAI'S ANSWER

ICAI has not published its answers to this paper yet.

ICAI · P4 · Sep 2026 exam · question p. 17-18

918 **EXAM** **Jan 2026** MCQ 11 **2 marks** **MCQ**

A company's wage budget for the last month was based on a standard production time of 30 minutes per unit and a standard wage rate of ₹ 50 per hour.

During the last month, it produced 30,000 units. The labour rate variance was ₹ 7,500 adverse and the labour efficiency variance was nil.

What is the actual wage rate per unit during the last month?

A ₹ 50.00

B ₹ 25.50

C ₹ 50.50

D ₹ 25.25

ICAI'S ANSWER

Answer key: **(D)**

ICAI · P4 · Jan 2026 exam · question p. 5 · answer p. 7

919 EXAM Jan 2025 Case scenario · Q5(a)(i) to Q5(a)(v) 6 marks

CASE SCENARIO

The following information has been provided by a company:

Number of units produced and sold	7,000
Standard labour rate per hour	₹ 9
Actual hours worked	17,820 hours
Labour efficiency	106.8%
Labour rate variance	71,280 (A)

You are required to calculate :

Q5(a)(i) 1 mark

Actual labour rate per hour

Q5(a)(ii) 2 marks

Standard hours required for 7,000 units

Q5(a)(iii) 1 mark

Labour Efficiency variance

Q5(a)(iv) 1 mark

Standard labour cost per unit

Q5(a)(v) 1 mark

Actual labour cost per unit

ICAI'S ANSWERS

Q5(a)(i)

Actual Hours Worked	17,820
Standard Rate Per hour	9
Labour Rate Variance	71280 (A)

Labour Rate Variance = (Standard Rate – Actual Rate) Actual Hours

(9 – Actual Rate) 17,820 = -71,280

(9 – Actual Rate) = - 4

Actual Rate = 13

Q5(a)(ii)

$$\text{Labour Efficiency} = \frac{\text{Standard Hours x 100}}{\text{Actual Hours}}$$

$$106.8\% = \frac{\text{Standard Hours}}{17{,}820}$$

Standard Hours = 19,031.76

Q5(a)(iii)

Labour Efficiency Variance = (Standard Hours – Actual Hours) x Standard Rate

= (19,031.76 – 17,820) x 9

= 10,905.84 (F)

Q5(a)(iv)

Standard Labour Cost per unit= $\frac{19{,}031.76 \text{ x } 9}{7{,}000}$ = 24.47

Q5(a)(v)

Actual Labour cost per unit = $\frac{17{,}820 \text{ x } 13}{7{,}000}$ = 33.09

ICAI · P4 · Jan 2025 exam · question p. 23-24 · answer p. 24-25, 25

920 EXAM **Old P3 May 2022** Q5(b) 5 marks old scheme

A manufacturing department of a company has employed 120 workers. The standard output of product "NPX" is 20 units per hour and the standard wage rate is ₹ 25 per labour hour.

In a 48 hours week, the department produced 1,000 units of 'NPX' despite 5% of the time paid being lost due to an abnormal reason. The hourly wages actually paid were ₹ 25.70 per hour.

Calculate:

(i) Labour Cost Variance

(ii) Labour Rate Variance

(iii) Labour Efficiency Variance

(iv) Labour Idle time Variance

ICAI'S ANSWER

Working Notes:

1. Calculation of standard man hours

When 120 worker works for 1 hr., then the std. output is 20 units.

Std. man hour per unit = $\frac{120 \text{ hrs.}}{20 \text{ units}}$ = 6 hrs.

2. Calculation of std. man hours for actual output

Total std. man hours = 1,000 units × 6 hrs. = 6,000 hrs.

Standard for actual			Actual				
Hours	**Rate (₹)**	**Amount (₹)**	**Actual hrs. paid**	**Idle time hrs.**	**Production hrs.**	**Rate (₹)**	**Amount paid (₹)**
6,000	25	1,50,000	5,760 (48 hrs. x 120 workers)	288	5,472	25.70	1,48,032

(i) Labour cost variance

= Std. labour cost – Actual labour cost

= 1,50,000 – 1,48,032 = ₹ 1,968 F

(ii) Labour rate variance

= (SR – AR) × AH_{Paid}

= (25 - 25.70) × 5,760 = ₹ 4,032 A

(iii) Labour efficiency variance

= (SH – AH) × SR

= (6,000 – 5,472) × 25 = ₹ 13,200 F

(iv) Labour Idle time variance

= Idle Hours × SR

= 288 × 25 = ₹ 7,200 A

Note: Variances can also be calculated for one worker instead of 120.

ICAI · Old P3 · May 2022 exam · question p. 22 · answer p. 24-25

921 EXAM **Old P3 Jul 2021** Q5(a) 10 marks old scheme

The standard output of a Product 'DJ' is 25 units per hour in manufacturing department of a Company employing 100 workers. In a 40 hours week, the department produced 960 units of product 'DJ' despite 5% of the time paid was lost due to an abnormal reason. The hourly wage rates actually paid were ₹ 6.20, ₹ 6.00 and ₹ 5.70 respectively to Group 'A' consisting 10 workers, Group 'B' consisting 30 workers and Group 'C' consisting 60 workers. The standard wage rate per labour is same for all the workers. Labour Efficiency Variance is given ₹ 240 (F).

You are required to compute:

(i) Total Labour Cost Variance.

(ii) Total Labour Rate Variance.

(iii) Total Labour Gang Variance.

(iv) Total Labour Yield Variance, and

(v) Total Labour Idle Time Variance.

ICAI'S ANSWER

Working Notes:

1. Calculation of Standard Man hours

When 100 workers work for 1 hour, the standard output is 25 units.

Standard man hours per unit = $\frac{100 \text{ hours}}{25 \text{ units}}$ **= 4 hours per unit**

2. Calculation of standard man hours for actual output:

= 960 units x 4 hours **= 3,840 hours.**

3. Calculation of actual cost

Type of Workers	No of Workers	Actual Hours Paid	Rate (₹)	Amount (₹)	Idle Hours (5% of hours paid)	Actual hours Worked
Group 'A'	10	400	6.2	2,480	20	380
Group 'B'	30	1,200	6	7,200	60	1,140
Group 'C'	60	2,400	5.7	13,680	120	2,280
	100	**4,000**		**23,360**	**200**	**3,800**

4. Calculation of Standard wage Rate:

Labour Efficiency Variance	= 240F
(Standard hours for Actual production – Actual Hours) x SR	= 240F
(3,840 – 3,800) x SR	= 240
Standard Rate (SR)	**= ₹ 6 per hour**

(i) Total Labour Cost Variance

= (Standard hours x Standard Rate) – (Actual Hours x Actual rate)

= (3,840 x 6) – 23,360 **= 320A**

(ii) Total Labour Rate Variance

= (Standard Rate – Actual Rate) x Actual Hours

Group 'A' = (6- 6.2) 400	=	80A
Group 'B' = (6- 6) 1,200	=	0
Group 'C' = (6 – 5.7) 2,400 =		720F
		640F

(iii) Total Labour Gang Variance

= Total Actual Time Worked (hours) × {Average Standard Rate per hour of Standard Gang -Average Standard Rate per hour of Actual Gang@}

@ on the basis of hours worked

$$= 3{,}800 \times \left(6 - \frac{3{,}840 \times 6}{3{,}800}\right)$$

= 0

(iv) Total Labour Yield Variance

= Average Standard Rate per hour of Standard Gang × {Total Standard Time (hours) - Total Actual Time worked (hours)}

= 6 x (3,840 – 3,800)

= 240F

(v) Total Labour idle time variance

= Total Idle hours x standard rate per hour

= 200 hours x 6

= 1,200A

ICAI · Old P3 · Jul 2021 exam · question p. 74-75 · answer p. 75-77

922 RTP Sep 2025 Case scenario · Q1(i) to Q1(v)

CASE SCENARIO

ALZO Toys Ltd. is an exciting new player in the toy manufacturing industry, founded with a passion for creating high-quality, engaging, and educational toys. The company aims to make a positive impact on the industry and also on the development of young minds through imaginative play.

The following statement provides a comprehensive analysis of the various cost variances for a particular period, outlining the differences between the expected costs and the actual expenditures incurred.

Cost variances	(₹)
Direct material price	25,000F
Direct material usage	3,750A
Direct labour rate	5,000A
Direct labour efficiency	3,750A
Variable overhead expenditure	15,000A
Variable overhead efficiency	1,875A
Fixed overhead expenditure	62,500F

The budget for the same period reflected the following data:

Production volume	7,500 units
Direct materials purchased	3,750kg
Direct materials used	3,750kg
Direct material cost	₹ 1,12,500
Direct labour hours	5,625 hours
Direct labour cost	₹ 1,12,500
Variable overhead cost	₹ 56,250
Fixed overhead cost	₹ 1,12,500

Some other information relating to the same period is provided below:

(i) Stocks of raw materials and finished goods are valued at a predetermined standard cost for easier cost comparison and reporting.

(ii) The actual number of units produced was 7,750.

(iii) The direct materials purchased were 5,000 kg.

From the information given above, you are required to FIGURE OUT the following in actual:

Q1(i)

Quantity of materials used and direct material cost-

A 3,875 kg and ₹ 1,50,000

B 3,875 kg and ₹ 1,25,000

C 4,000 kg and ₹ 1,25,000

D 4,000 kg and ₹ 1,50,000

Q1(ii)

Direct labour hours-

A 7,937.50 hours

B 6,000 hours

C 5,812.50 hours

D 5,000 hours

Q1(iii)

Direct labour cost-

A ₹ 1,16,250

B ₹ 1,25,000

C ₹ 1,55,000

D ₹ 1,63,750

Q1(iv)

Variable overhead cost-

A ₹ 75,000

B ₹ 73,125

C ₹ 60,000

D ₹ 58,125

Q1(v)

Fixed overhead cost-

A ₹ 1,75,000

B ₹ 1,12,500

C ₹ 62,500

D ₹ 50,000

ICAI'S ANSWERS

Q1(i)

Answer key: **(C)**

(c) Calculation of actual quantity of materials used:

Standard quantity of material used per units of output

$$= \left(\frac{\text{budgeted usage}}{\text{budgeted production}}\right)$$

$$= \left(\frac{3{,}750 \text{ kg}}{7{,}500 \text{ units}}\right) = 0.5 \text{ kg}$$

$$\text{Standard price} = \left(\frac{\text{budgeted material cost}}{\text{budgeted usage}}\right)$$

$= \left(\frac{₹\ 1{,}12{,}500}{3{,}750\ \text{kg}}\right) = ₹\ 30$

Material usage variance = (Std. qty. for actual output - Actual qty.) × Std. price

₹ 3,750A = [(7,750 x 0.5kg) - AQ] x ₹ 30

- ₹ 3,750 = ₹ 1,16,250 - 30AQ

30AQ = ₹ 1,20,000

AQ = 4,000 kg

Actual quantity of materials used = **4,000 kg**

Calculation of actual direct material cost:

Material Price Variance = Actual Quantity purchased (AQ) x {Std. Price (SP) – Actual Price(AP)}

₹ 25,000F = 5,000 kg x (₹ 30 - AP)

₹ 25,000 = ₹ 1,50,000 - 5,000 AP

5,000 AP = ₹ 1,25,000

AP = ₹ 25

Actual direct material cost = 5,000 kg x ₹ 25 = ₹ **1,25,000**

PassBook's note: ICAI's key is irregular: the answer to (i) is headed "(i) (c) Calculation of actual quantity of materials used" (p.23) and works out 4,000 kg and ₹ 1,25,000, which is option (c).

Q1(ii)

Answer key: **(B)**

(b) Standard hours per unit of output $= \left(\frac{\text{Budgeted hours}}{\text{Budgeted output}}\right)$

$= \left(\frac{5{,}625\ \text{hours}}{7{,}500\ \text{units}}\right) = 0.75\ \text{hours}$

Standard wage rate $= \left(\frac{\text{budgeted labour cost}}{\text{budgeted hours}}\right)$

$= \left(\frac{₹\ 1{,}12{,}500}{5{,}625\ \text{hours}}\right) = ₹\ 20$

Labour Efficiency Variance = Std. Rate (SR) x {Std. Hours (SH) – Actual Hours (AH)}

₹ 3,750A = ₹ 20 x [(7,750 x 0.75 hours) - AH]

- ₹ 3,750 = ₹ 1,16,250 - 20AH

20AH = ₹ 1,20,000

AH = 6,000

Actual Hours = **6,000**

Q1(iii)

Answer key: **(B)**

(b) Labour Cost Variance = [Standard Labour Cost – Actual Labour Cost]

Labour Rate Variance + Labour Efficiency Variance = [Standard Labour Cost – Actual Labour Cost]

₹ 5,000A + ₹ 3,750A = (7,750 x 0.75 hours x ₹ 20) - Actual Labour Cost

- ₹ 5,000 - ₹ 3,750 = ₹ 1,16,250 - Actual Cost

Actual Labour Cost = ₹ 1,16,250 + ₹ 8,750

Actual Labour Cost = ₹ **1,25,000**

Q1(iv)

Answer key: **(A)**

(a) Standard variable overhead per unit = $\left(\frac{\text{Budgeted variable overheads cost}}{\text{Budgeted output}}\right)$

= $\left(\frac{₹\ 56{,}250}{7{,}500\ \text{units}}\right)$ = ₹ 7.50

Total Variable Overhead Variance = [Standard Variable Overhead – Actual Variable Overhead]

Variable Overhead Expenditure Variance + Variable Overhead Efficiency Variance = [Standard Variable Overhead – Actual Variable Overhead]

₹ 15,000A + ₹ 1,875A = (7,750 x ₹ 7.50) - Actual Variable Overhead

- ₹ 15,000 - ₹ 1,875 = ₹ 58,125 - Actual Variable Overhead

Actual Variable Overhead = ₹ 58,125 + ₹ 16,875

Actual Variable Overhead = ₹ **75,000**

Q1(v)

Answer key: **(D)**

(d) Fixed Overhead Expenditure Variance = Budgeted Fixed Overheads - Actual Fixed Overheads

₹ 62,500 F = ₹ 1,12,500 - Actual Fixed Overheads

Actual Fixed Overheads = ₹ **50,000**

Summary

(i) (d) 4,000 kg and ₹ 1,25,000

(ii) (b) 6,000 hours

(iii) (b) ₹ 1,25,000

(iv) (a) ₹ 75,000

(v) (d) ₹ 50,000

ICAI · P4 · RTP Sep 2025 · question p. 1-2 · answer p. 23-24, 24-25, 25, 26

923 RTP **Old P3 Nov 2023** Q11 old scheme

The following information has been provided by a company:

Number of units produced and sold	6,000
Standard labour rate per hour	₹8
Standard hours required for 6,000 units	-
Actual hours required	17094 hours
Labour efficiency	105.3%
Labour rate variance	₹ 68,376 (A)

You are required to calculate:

(i) Actual labour rate per hour

(ii) Standard hours required for 6,000 units

(iii) Labour Efficiency variance

(iv) Standard labour cost per unit

(v) Actual labour cost per unit.

ICAI'S ANSWER

SR – Standard labour Rate per Hour

AR – Actual labour rate per hour

SH – Standard Hours

AH – Actual hours

(i) Labour rate Variance = AH (SR – AR)

Or 17,094 (8 – AR) = 68,376(A)

Or 17,094 (8 – AR) = - 68,376

Or 8 – AR = -4

Or AR = ₹12

(ii) Labour Efficiency = $\frac{SH}{AH}$ x 100 = 105.3

= SH = $\frac{AH \times 105.3}{100} = \frac{17{,}094 \times 105.3}{100}$

= 17,999.982

= SH = 18,000 hours

(iii) Labour Efficiency Variance = SR (SH – AH)

= 8(18,000 – 17,094)

= 8 X 906

= ₹ 7,248(F)

(iv) Standard Labour Cost per Unit = $\frac{18{,}000 \times 8}{6{,}000}$ = ₹ 24

(v) Actual Labour Cost Per Unit = $\frac{17{,}094 \times 12}{6{,}000}$ = ₹ 34.19

ICAI · Old P3 · RTP Nov 2023 · question p. 7 · answer p. 21

924 RTP **Old P3 May 2022** Q11 old scheme

The standard output of a Product 'D' is 50 units per hour in manufacturing department of a Company employing 100 workers. In a 40 hours week, the department produced 1,920 units of product 'D' despite 5% of the time paid was lost due to an abnormal reason. The hourly wage rates actually paid were ₹ 12.40, ₹ 12.00 and ₹ 11.40 respectively to Group 'A' consisting 10 workers, Group 'B' consisting 30 workers and Group 'C' consisting 60 workers. The standard wage rate per labour is same for all the workers. Labour Efficiency Variance is given ₹ 480 (F).

You are required to COMPUTE:

(i) Total Labour Cost Variance.

(ii) Total Labour Rate Variance.

(iii) Total Labour Gang Variance.

(iv) Total Labour Yield Variance, and

(v) Total Labour Idle Time Variance.

ICAI'S ANSWER

Working Notes:

1. Calculation of Standard Man hours

When 100 workers work for 1 hour, the standard output is 50 units.

Standard man hours per unit = $\frac{100 \text{ hours}}{50 \text{ units}}$ **= 2 hours per unit**

2. Calculation of standard man hours for actual output:

= 1,920units x 2 hours **= 3,840 hours.**

3. Calculation of actual cost

Type of Workers	No of Workers	Actual Hours Paid	Rate (₹)	Amount (₹)	Idle Hours (5% of hours paid)	Actual hours Worked
Group 'A'	10	400	12.40	4,960	20	380
Group 'B'	30	1,200	12	14,400	60	1,140
Group 'C'	60	2,400	11.40	27,360	120	2,280
	100	**4,000**		**46,720**	**200**	**3,800**

4. Calculation of Standard wage Rate:

Labour Efficiency Variance	= 480F
(Standard hours for Actual production – Actual Hours) x SR	= 480F
(3,840 – 3,800) x SR	= 480
Standard Rate (SR)	**= ₹ 12 per hour**

(i) Total Labour Cost Variance

= (Standard hours x Standard Rate) – (Actual Hours x Actual rate)

= (3,840 x 12) – 46,720 **= 640A**

(ii) Total Labour Rate Variance

= (Standard Rate – Actual Rate) x Actual Hours

Group 'A' = (12 - 12.40) 400	=	160A
Group 'B' = (12 - 12) 1,200	=	0
Group 'C' = (12 – 11.40) 2,400	=	1,440F
		1,280F

(iii) Total Labour Gang Variance

= Total Actual Time Worked (hours) × {Average Standard Rate per hour of Standard Gang -Average Standard Rate per hour of Actual Gang@}

@ on the basis of hours worked

$$= 3{,}800 \times \left(12 - \frac{3{,}840 \times 12}{3{,}800}\right)$$

= 0

[Note: As the number of workers in standard and actual is the same, there is no difference in mix ratio, so labour gang variance will be NIL]

(iv) Total Labour Yield Variance

= Average Standard Rate per hour of Standard Gang × {Total Standard Time (hours) - Total Actual Time worked (hours)}

= 12 x (3,840 – 3,800)

= 480F

(v) Total Labour idle time variance

= Total Idle hours x standard rate per hour

= 200 hours x 12

= 2,400A

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925 MTP **May 2026 S-II** MCQ 12 2 marks MCQ

In a workshop, there is a standard requirement of 30 labour hours for producing a wooden toy. The standard wage rate is set at ₹ 20 per hour. In January 2025, the workshop successfully produced 1,000 wooden toys. The payroll records show that 22,800 labour hours were paid including 500 idle hours caused by a temporary power outage. The labour rate variance for the month was ₹ 15,960 (Favourable).

Calculate total actual labour cost incurred in the month of January 2025.

A ₹ 4,40,040

B ₹ 4,40,140

C ₹ 4,40,240

D ₹ 4,30,040

ICAI'S ANSWER

(A) ₹4,40,040

Labour Rate Variance = ₹ 15,960 (Favourable)

Labour Rate Variance = Actual Hours Paid x (Standard Rate – Actual Rate)

₹ 15,960 = 22,800 hrs x (₹ 20 – Actual Rate)

(15,960/22,800) = 20 – Actual Rate

0.70 = 20 - Actual Rate

Actual Rate = ₹ 19.30 per hour

Actual Wages Paid = Actual Hours Paid x Actual Rate

= 22,800 hrs. x ₹ 19.30

= ₹ 4,40,040

ICAI · P4 · MTP May 2026 · Series II · question p. 5 · answer p. 4

926 MTP May 2025 S-I Case Scenario II · MCQ 6 to MCQ 10 10 marks

CASE SCENARIO II

A company is working in manufacturing sector and uses labour-force, which consists of skilled, semi-skilled, and unskilled workers.

The rate of Labour per week for skilled worker is ₹ 120, for semi-skilled workers is ₹ 80 and for unskilled worker is ₹ 60.

It had planned the labour for a job that would take 60 weeks to be completed, ideally. Ratio of workers employed in skilled, semi-skilled, and unskilled would be 5:3:4. The management consultant who was employed to handle this job died in an accident. Board hired a new consultant to take over this job.

This new consultant had a different approach for doing this job. He thought that workers actually required would be in the ratio of 7:3:8. He changed the composition of labour-force and employed the workers in the new ratio.

Project was carried out successfully and it turned out that this change in Mix of labour-force saved the company ₹ 76,800 in the job but at the same time company also lost ₹ 1,29,600 due to poor productivity of the labourers.

As a result, management consultant was promoted and labour force was fired & was replaced with more experienced labour-force.

As the new management consultant was promoted, company found a new one to replace him. This 3rd management consultant was asked to understand and analyse the previous job done by the 2nd management consultant. Data in addition to above-given information was provided as follows:-

- Total time period taken for the job was 64 weeks.
- Net extra cost spent, in comparison to planned cost, was ₹ 1,04,000.
- Total number of weeks worked by all the type of labourers was 23,040 weeks.
- Skilled and semi-skilled labourers in reality charged ₹ 20 extra per week, whereas unskilled labourers were negotiated to charge ₹ 20 less per week, than what they had charged earlier.

Let say, you are the 3rd management consultant, and you have been given with all this information. Work out calculations and answer the following questions (MCQs 6 to 10) based on the information given above.

MCQ 6 2 marks

If the total of labour efficiency variance of skilled and semiskilled labourers is ₹ 1,29,600 (Favourable), what is the labour efficiency variance of unskilled labour?

- A 1,82,400 (Adverse)
- B 52,800 (Adverse)
- C 1,29,600 (Adverse)
- D 76,800 (Favourable)

MCQ 7 2 marks

What are the amounts that company saved & paid extra to labourers, respectively?

- A 2,56,000 & 2,04,800
- B 1,79,200 & 204,800
- C 2,04,800 & 2,56,000
- D 204,800 & 1,79,200

MCQ 8 2 marks

If in total 1,440 weeks were worked in addition to what was planned, how many extra/less workers were used in actual, compared to standard? (Answer in sequence of skilled, semi-skilled, & unskilled).

- A -10, -30, -40
- B -10, -30, +40
- C +10, +30, -40
- D +10, +30, +40

MCQ 9 2 marks

Calculate revised standard weeks for all 3 types of labour forces? (Answer in sequence of skilled, semi-skilled, & unskilled).

- A 9,400, 5,650, & 7,990
- B 9,550, 6,280, & 7,210
- C 8,520, 6,850, & 7,670
- D 9,600, 5,760, & 7,680

MCQ 10 2 marks

If standard rates charged by skilled, semi-skilled, and unskilled were ₹ 120, ₹ 80, & ₹ 60, respectively, then which labour force performed worst, better, & best, due to change in labour composition?

- A Semi-skilled, skilled, & unskilled
- B Skilled, semi-skilled, unskilled
- C Unskilled, skilled, & semi-skilled
- D Unskilled, semi-skilled, & skilled

ICAI'S ANSWERS

MCQ 6

(A) 1,82,400 (Adverse)

Working Note

Total actual weeks = 23,040

Ratio of actual workers = 7:3:8

Let's assume that total number of workers is 'X'

(7/18 x X x64) + (3/18 x X x 64) + (8/18 x X x 64) = 23,040

X = 360

	Standard		Actual	
	Workers	Rate per week	Workers	Rate per week
Skilled	150	120	140	140
Semi-skilled	90	80	60	100
unskilled	120	60	160	40

Standard time = 60 weeks

Actual time = 64 weeks

	Standard			Actual		
	Weeks	Rate	Amount	Weeks	Rate	Amount
Skilled	9,000	120	10,80,000	8,960	140	12,54,400
Semi-skilled	5,400	80	4,32,000	3,840	100	3,84,000
unskilled	7,200	60	4,32,000	10,240	40	4,09,600
total	21,600		19,44,000	23,040		20,48,000

1. Cost Variance = 19,44,000 – 20,48,000 = 1,04,000(A)

2. Rate Variance =

Skilled	8,960x(120-140)	1,79,200 (A)
Semi-skilled	3,840x(80-100)	76,800 (A)
unskilled	10,240x(60-40)	2,04,800 (F)
		51,200 (A)

3. Efficiency Variance =

Skilled	120x(9,000-8,960)	4,800(F)
Semi-skilled	80x(5,400-3,840)	1,24,800(F)
unskilled	60x(7,200-10,240)	1,82,400(A)
		52,800(A)

4. Revised Labour Efficiency Rate=

Skilled	120x(9,000-9,600)	72,000(A)
Semi-skilled	80x(5,400-5,760)	28,800(A)
unskilled	60x(7,200-7,680)	28,800(A)
		1,29,600(A)

5. Labour Mix Variance=

Skilled	120x(9,600-8,960)	76,800(F)
Semi-skilled	80x(5,760-3,840)	1,53,600(F)
unskilled	60x(7,680-10,240)	1,53,600(A)
		76,800(F)

Labour Mix Variance (76,800 F) + Revised Labour Efficiency Variance (1,29,600 A)

= Total Labour Efficiency (52,800 A)

Total Lab Efficiency Variance – Labour Efficiency Variance of Skilled And Semi Skilled

= Labour Efficiency Variance of Unskilled

52,800A – 1,29,600A = 1,82,400A,

So, Labour Efficiency Variance of Unskilled = 1,82,400A

	No. of workers	Weeks worked
Skilled	140	140×64= 8,960
Semi-skilled	60	60×64= 3,840
unskilled	160	160×64= 10,240

MCQ 7

(C) 2,04,800 & 2,56,000

Money saved or spent:

	Weeks worked	Saved/spent extra, per week	Total
Skilled	8,960	20 spent extra	-1,79,200
Semi-skilled	3,840	20 spent extra	-76,800
unskilled	10,240	20 saved	+2,04,800

Saved = 2,04,800

Spent = 2,56,000

MCQ 8

(B) -10, -30, +40

Actual weeks = 23,040

(-) extra = (1,440)

Planned weeks =21,600

Original standard ratio = 5:3:4

Standard weeks chart:

	Weeks	Workers
Skilled	21,600×5/12 = 9,000	9,000/60 weeks = 150
Semi-skilled	21,600×3/12 = 5,400	5,400/60 weeks = 90
unskilled	21,600×4/12 = 7,200	7,200/60 weeks = 120

	Planned workers	Actual workers	Actual compared to standard
Skilled	150	140	-10
Semi-skilled	90	60	-30
unskilled	120	160	+40

MCQ 9

(D) 9,600, 5,760, & 7,680

Actual weeks is divided in standard ratio of workers

23,040 in 5:3:4

	Revised Standard Weeks
Skilled	9,600
Semi-skilled	5,760
unskilled	7,680

MCQ 10

(C) Unskilled, skilled, & semi-skilled

Change in performance of workers, due to change in labour composition can be evaluated through labour mix variance. We have already calculated labour mix variance above. Answers are:

	Labour mix variance
Skilled	76,800 F
Semi-skilled	1,53,600 F
unskilled	1,53,600 A

This means unskilled labour performed the worst, skilled labour performed better than unskilled, and semi skilled performed the best.

ICAI · P4 · MTP May 2025 · Series I · question p. 3-4 · answer p. 2-4, 4, 4-5, 5

927 MTP **May 2024 S-I** MCQ 3 2 marks MCQ

The wages budget for the last period was based on a standard repair time of 30 minutes per unit and a standard wage rate of ₹ 50 per hour. The actual data for the last period are as follows:

Number of units = 30,000

Labour rate variance = 7,500 (A)

Labour efficiency variance = Nil

From the information find out the actual rate of wages per unit

A ₹ 50

B ₹ 25.50

C ₹ 50.50

D ₹ 25.25

ICAI'S ANSWER

Answer key: **(D)**

D Labour rate variance = Standard time for actual production (SR- AR)

7,500 (A) = (30,000 × 30 minutes/60 minutes) × (50-AR)

AR = (7,50,000 + 7,500)/15,000 = ₹50.50 per hour

Actual wages per unit = 50.50/2 = ₹25.25

ICAI · P4 · MTP May 2024 · Series I · question p. 4 · answer p. 3-4

928 MTP **Old P3 Dec 2021 S-I** Q5(b) 10 marks old scheme

Following information has been provided by a company:

Number of units produced and sold	9,000
Standard labour rate per hour	₹ 12
Standard hours required for 9,000 units	-
Actual hours required	25,641 hours
Labour efficiency	105.3%
Labour rate variance	₹ 1,53,846 (A)

You are required to CALCULATE:

(i) Actual labour rate per hour

(ii) Standard hours required for 9,000 units

(iii) Labour Efficiency variance

(iv) Standard labour cost per unit

(v) Actual labour cost per unit.

ICAI'S ANSWER

SR – Standard labour Rate per Hour

AR – Actual labour rate per hour

SH – Standard Hours

AH – Actual hours

(i) Labour rate Variance = AH (SR – AR)

- 1,53,846 = 25,641 (12 – AR)

- 6 = 12 – AR

AR = ₹ 18

(ii) Labour Efficiency = $\frac{SH}{AH}$ x 100 = 105.3

$SH = \frac{AH \times 105.3}{100} = \frac{25{,}641 \times 105.3}{100}$

SH = 26,999.973

SH = 27,000 hours

(iii) Labour Efficiency Variance = SR (SH – AH)

= 12 (27,000 – 25,641)

= ₹ 16,308 (F)

(iv) Standard Labour Cost per Unit = $\frac{27{,}000 \times 12}{9{,}000}$ = ₹ 36

(v) Actual Labour Cost Per Unit = $\frac{25{,}641 \times 18}{9{,}000}$ = ₹ 51.282

ICAI · Old P3 · MTP Dec 2021 · Series I · question p. 6 · answer p. 10-11

13.4 Variable overhead variances

7 questions · asked in 8 ICAI documents · 4 exam papers

929 EXAM Sep 2025 Q1(c) 4 marks

MD Limited has furnished following information for the month of August, 2025:

Standard Variable Overhead rate	₹ 3 per hour
Standard Hours for per unit of production	5 hours
Actual Output	15,560 units
Variable Overhead Efficiency Variance	₹ 11,400 (F)
Variable Overhead Expenditure Variance	₹ 37,000 (A)
Standard Fixed Overhead rate	₹ 2 per hour
Actual Fixed Overheads	₹ 1,85,000

You are required to calculate:

(i) Actual Hours

(ii) Actual Variable Overhead rate per hour

(iii) Variable Overhead Cost Variance

(iv) Fixed Overhead Cost Variance

ICAI'S ANSWER

(i) Actual Hours

Variable overhead efficiency variance:

(Standard hours for Actual Production – Actual Hours) x Standard Rate

{(15,560 x 5) - Actual Hours} x 3 = ₹ 11,400 F

Actual Hours = 74,000

(ii) Actual Variable Overhead rate per hour

Variable overhead expenditure Variance:

Std. overhead for Actual hours – Actual Variable Overhead

(74,000 x 3) – Actual Variable overheads = ₹ 37,000 (A)

Actual Variable overhead = ₹ 2,59,000

Actual Variable Rate per hour = $\frac{2,59,000}{74,000}$ = ₹ 3.5 per hour

(iii) Variable overhead Cost Variance

Variable overhead Efficiency Variance + Variable overhead Expenditure Variance

11,400 (F) + 37,000 (A) = ₹ 25,600(A)

(iv) Fixed Overhead cost Variance

Fixed overhead Cost Variance:

(Absorbed Fixed Overheads) - (Actual Fixed Overheads)

(SH × SR) – (AH × AR)

(77,800 x 2) – 1,85,000 = ₹29,400 (A)

ICAI · P4 · Sep 2025 exam · question p. 9 · answer p. 11-12

930 **EXAM** **Jan 2025** MCQ 14 MCQ

During a certain period, 4,000 labour hours were utilized, and the standard hours for actual production were 5,500 hours. The Variable Overhead Efficiency Variance amounted to ₹ 15,000 (Favourable).

Calculate the Standard Variable Overhead Rate per hour.

A ₹ 15 per hour

B ₹ 20 per hour

C ₹ 10 per hour

D ₹ 25 per hour

ICAI'S ANSWER

Answer key: **(C)**

ICAI · P4 · Jan 2025 exam · question p. 5-6 · answer p. 6

931 EXAM Old P3 Nov 2023 Q5(a) 10 marks old scheme

PQR Alloys Ltd. uses a standard costing system.

Budgeted information for the year:

Budgeted output	84,000 units
Variable Factory Overhead per unit	₹ 16
Standard time for one unit of output	0.80 machine hour
Fixed factory overheads	₹ 6,72,000

Actual results for the year:

Actual output	87,600 units
Variable Overhead efficiency variance	₹ 67,200 (A)
Actual Fixed factory overheads	₹ 7,05,000
Actual variable factory overheads	₹ 14,37,000

Required:

Calculate the following variances clearly indicating Adverse(A) or Favourable (F):

(i) Variable factory overhead expenditure variance.

(ii) Fixed factory overhead expenditure variance.

(iii) Fixed factory overhead efficiency variance.

(iv) Fixed factory overhead capacity variance.

ICAI'S ANSWER

Calculation of actual hours

Standard rate per hour = $\frac{\text{Variable factory overhead per unit}}{\text{Standard time for one unit of output}} = \frac{₹16}{0.8} = ₹\ 20$

Variable Overhead Efficiency Variance:

(Standard hours for actual production – Actual hours) × Standard rate per hour

Let actual hours be x

[(87,600 × 0.8) – x] ×20 = -67,200

(70,080 – x) × 20 = -67,200

x = 73,440

(i) Variable Factory Overhead Expenditure Variance:

(Variable overhead at actual hours – Actual variable overheads)

$$\left[\left(\frac{13,44,000}{67,200} \times 73,440\right) - 14,37,000\right]$$

= 31,800 F

(ii) Fixed Factory Overhead Expenditure Variance:

Budgeted fixed overhead – Actual fixed overhead.

(6,72,000 – 7,05,000)

= 33,000 A

(iii) Fixed Factory Overhead Efficiency Variance:

(Standard hours for actual production – Actual hours) × Standard rate per hour

(70,080 – 73,440) × 10 = 33,600 A

(iv) Fixed Overhead Capacity Variance:

(Actual hours - Budgeted hours) × Standard rate per hour

(73,440 - 67,200) ×10 = 62,400 F

The solution can also be presented in following way based on Quantity (units)

Calculation of standard quantity for actual hours:

Variable standard rate per unit (SR) = ₹ 16

Variable Overhead Efficiency Variance:

(SR x AQ) – (SR x standard quantity for Actual hours worked)

-67,200 = (16 x 87,600) – 16 x

-67200 = 14,01,600- 16 x

x = 14,68,800 / 16 = **91,800** (SQ for actual hours worked)

(i) Variable Factory Overhead Expenditure Variance:

(SR x SQ for actual hour worked – Actual variable overheads)

16 x 91,800 – 14,37,000 or 14,68,800 – 14,37,000

= 31,800 F

(ii) Fixed Factory Overhead Expenditure Variance:

Budgeted fixed overhead – Actual fixed overhead.

(6,72,000 – 7,05,000)

= 33,000 A

(iii) Fixed Factory Overhead Efficiency Variance:

Standard rate per unit (SR) = 6,72,000 / 84,000 = ₹ 8 per unit

(SR x AQ) – (SR x standard quantity for Actual hours)

(8 x 87,600) – (8 x 91,800)

(7,00,800 – 7,34,400) = 33,600 A

(iv) Fixed Overhead Capacity Variance:

(SR x standard quantity for Actual hours - Budgeted fixed overheads)

(8 x 91800) – (6,72,000)

(7,34,400 – 6,72,000) = 62,400 F

932 EXAM **Old P3 Jan 2021** Q5(b) 10 marks old scheme

Premier Industries has a small factory where 52 workers are employed on an average for 25 days a month and they work 8 hours per day. The normal down time is 15%. The firm has introduced standard costing for cost control. Its monthly budget for November, 2020 shows that the budgeted variable and fixed overhead are ₹ 1,06,080 and ₹ 2,21,000 respectively.

The firm reports the following details of actual performance for November, 2020, after the end of the month:

Actual hours worked	8,100 hrs.
Actual production expressed in standard hours	8,800 hrs.
Actual Variable Overheads	₹ 1,02,000
Actual Fixed Overheads	₹ 2,00,000

You are required to calculate:

(i) Variable Overhead Variances:

(a) Variable overhead expenditure variance.

(b) Variable overhead efficiency variance.

(ii) Fixed Overhead Variances:

(a) Fixed overhead budget variance.

(b) Fixed overhead capacity variance.

(c) Fixed overhead efficiency variance.

(iii) Control Ratios:

(a) Capacity ratio.

(b) Efficiency ratio.

(c) Activity ratio.

ICAI'S ANSWER

Workings:

Calculation of budgeted hours

Budgeted hours = (52 x 25 x 8) x 85% = 8,840 hours

(i) Variable overheads variance

(a) Variable overhead expenditure variance

= Std. overhead for Actual hours – Actual variable Overhead

$$= \left(\frac{₹\ 1,06,080}{8,840} \text{ x } 8,100\right) - ₹\ 1,02,000$$

= **4800 A**

(b) Variable overhead efficiency variance

Std. rate per hour × (Std. hours for actual production – Actual hours)

$= \frac{₹\ 1,06,080}{8,840}$ (8,800 hours – 8,100 hours)

= **8400 F**

(ii) Fixed overhead variances

(a) Fixed overhead budget variance

= Budgeted overhead – Actual overhead

= ₹ 2,21,000 – ₹ 2,00,000

= 21,000 F

(b) Fixed overhead capacity variance

= Std rate x (Actual hours – budgeted hours)

$= \frac{₹\ 2,21,000}{8,840}$ x (8,100 – 8,840)

= **18,500 A**

(c) Fixed overhead efficiency variance

= Std rate x (Std hours for actual production – Actual hours)

$= \frac{₹\ 2,21,000}{8,840}$ x (8,800 – 8,100)

= **17,500 F**

(iii) Control Ratios

(a) Capacity Ratio

$= \frac{\text{Actual hours}}{\text{Budgeted hours}}$ x 100

$= \frac{8,100}{8,840}$ x 100 = **91.63%**

(b) Efficiency Ratio

$= \frac{\text{Standard hours}}{\text{Actual hours}}$ x 100

$= \frac{8,800}{8,100}$ x 100 = **108.64 %**

(c) Activity Ratio

$= \frac{\text{Standard hours}}{\text{Budgted hours}}$ x 100

$= \frac{8,800}{8,840}$ x 100 = **99.55%**

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933 MTP **Sep 2026 S-II** MCQ 14 2 marks MCQ

Also asked in MTP May 2025 S-II MCQ 11

The following figures are extracted from the books of a company:

Budgeted overheads ₹ 20,000 (Fixed ₹ 12,000, Variable ₹ 8,000)

Budgeted output 2,500 units

Actual Overheads ₹ 21,800 (Fixed ₹ 11,800, Variable ₹ 10,000)

Actual output 3000

Variable Overheads and fixed overheads cost variance will be:

A 400 (A) and 2600 (F)

B 400 (A) and 200 (F)

C 2000 (A) and 200 (F)

D 2000 (F) and 200 (A)

ICAI'S ANSWER

(A) 400 (A) and 2,600 (F)

Variable overhead cost variance: Standard Variable overheads - Actual variable overheads

8,000/2,500 x 3,000 – 10,000 = 400A

Fixed overhead cost variance: Standard fixed overheads - Actual fixed overheads

12,000/2,500 x 3,000 – 11,800 = 2,600F

ICAI · P4 · MTP Sep 2026 · Series II · question p. 7 · answer p. 6

934 MTP Jan 2026 S-I MCQ 12 2 marks MCQ

The standard variable overhead cost of a product is ₹ 12, i.e. 6 hours at the rate of ₹ 2 per hour.

In a certain month, it took 2,400 hours at a cost of ₹ 5,800 to manufacture 450 units.

The variable overhead expenditure and efficiency variances, respectively, are:

A ₹ 1,000 (F) and ₹ 600 (A)

B ₹ 1,000 (A) and ₹ 600 (F)

C ₹ 1,000 (F) and ₹ 600 (F)

D ₹ 1,000 (A) and ₹ 600 (A)

ICAI'S ANSWER

(B) ₹ 1,000 (A) and ₹ 600 (F)

Standard Hours = 450 units x 6 hours per unit = 2,700 hours

Expenditure Variance = (Standard Rate x Actual Hours) – Actual Overheads

= (₹ 2 x 2,400) – ₹ 5,800 = ₹ 4,800– ₹ 5,800

= ₹ 1,000 (A)

Efficiency Variance = (Standard Hours – Actual Hours) x Standard Rate

= (2,700–2,400) x ₹ 2 = 300 x 2 = ₹ 600(F)

ICAI · P4 · MTP Jan 2026 · Series I · question p. 5 · answer p. 4

935 MTP Sep 2025 S-II Q1(c) 4 marks

Premier Industries has a small factory where 52 workers are employed on an average for 25 days a month and they work 8 hours per day. The normal down time is 15%. The firm has introduced standard costing for cost control. Its monthly budget for March, 2025 shows that the budgeted variable and fixed overhead are ₹ 1,06,080 and ₹ 2,21,000 respectively.

The firm reports the following details of actual performance for March,2025, after the end of the month:

Actual hours worked	8,100 hrs.
Actual production expressed in standard hours	8,800 hrs.
Actual Variable Overheads	₹ 1,02,000
Actual Fixed Overheads	₹ 2,00,000

You are required to CALCULATE:

(i) Variable Overhead Variances:

(a) Variable overhead expenditure variance.

(b) Variable overhead efficiency variance.

(ii) Fixed Overhead Variances:

(a) Fixed overhead capacity variance.

(b) Fixed overhead efficiency variance.

ICAI'S ANSWER

Workings:

Calculation of budgeted hours

Budgeted hours = (52 x 25 x 8) x 85% = 8,840 hours

(i) Variable overheads variance

(a) Variable overhead expenditure variance

= Std. overhead for Actual hours – Actual variable Overhead

$= \left(\frac{₹\ 1,06,080}{8,840} \text{ x } 8,100\right) - ₹\ 1,02,000$

= **4800 A**

(b) Variable overhead efficiency variance

Std. rate per hour × (Std. hours for actual production – Actual hours)

$= \frac{₹\ 1,06,080}{8,840}$ (8,800 hours – 8,100 hours)

= **8400 F**

(ii) Fixed overhead variances

(a) Fixed overhead capacity variance

= Std rate x (Actual hours – budgeted hours)

$= \frac{₹\ 2,21,000}{8,840}$ x (8,100 – 8,840)

= **18,500 A**

(b) Fixed overhead efficiency variance

= Std rate x (Std hours for actual production – Actual hours)

$= \frac{₹\ 2,21,000}{8,840}$ x (8,800 – 8,100)

= **17,500 F**

ICAI · P4 · MTP Sep 2025 · Series II · question p. 9-10 · answer p. 10-11

13.5 Fixed overhead variances

22 questions · asked in 14 ICAI documents · 5 exam papers

936 EXAM May 2026 Q3(b) 6 marks

Alpha Limited uses Standard Costing System for manufacturing its single product 'STAR'. The following data has been gathered from the cost records of the company to compute various fixed overhead variances for a specific period:

Number of budgeted working days	25
Budgeted man-hours per day	12,000
Output (budgeted) per man-hour (in units)	1
Fixed overhead cost as budgeted	₹ 6,00,000
Actual number of working days	28
Actual man-hours per day	12,500
Actual output per man-hour (in units)	0.9
Actual fixed overhead incurred	₹ 6,24,000

You are required to ascertain the following:

(i) Fixed Overhead Expenditure Variance

(ii) Fixed Overhead Volume Variance

(iii) Fixed Overhead Calendar Variance

(iv) Fixed Overhead Cost Variance

(Indicate the nature of variance Favourable (F) or Adverse (A))

ICAI'S ANSWER

Particulars	Budget	Actual
Days	25	28
Man hours per day	12,000	12,500
Hours	3,00,000	3,50,000
Units per man hour	1	0.9
Units	3,00,000	3,15,000
Fixed Overheads (₹)	6,00,000	6,24,000

(i) Fixed Overhead Expenditure Variance

= Budgeted Fixed Overhead - Actual Fixed Overhead

= (₹ 6,00,000 – ₹ 6,24,000) = ₹ 24,000 (A)

(ii) Fixed Overhead Volume Variance

= Absorbed Overhead – Budgeted Overhead

= (₹ 2 x 3,15,000) – ₹ 6,00,000 = ₹ 30,000 (F)

(iii) Fixed Overhead Calendar Variance

= (Actual Days - Budgeted Days) x Budgeted Rate per day

= (28 - 25) x ₹ 24,000 = ₹ 72,000 (F)

(iv) Fixed Overhead Cost Variance

= Absorbed Overhead - Actual Fixed Overheads

= (₹ 2 x 3,15,000) – ₹ 6,24,000 = ₹ 6,000 (F)

Working Note:

$$\text{Standard fixed overhead rate per unit} = \frac{₹6,00,000}{3,00,000 \text{ units}} = ₹ 2 \text{ per unit}$$

$$\text{Standard fixed overhead per day} = \frac{₹6,00,000}{25 \text{ days}} = ₹ 24,000$$

ICAI · P4 · May 2026 exam · question p. 17 · answer p. 19-20

937 EXAM **May 2025** Q4(b) 6 marks

A manufacturing unit using Standard costing system and the following information was obtained from its records:

	Standard	Actual
Production	4,800 units	4,560 units
Working days	25	27
Fixed overhead	₹ 48,000	₹ 46,800
Variable overhead	₹ 14,400	₹ 14,400

Required:

Calculate the following overhead variances:

(i) Variable overhead variance

(ii) Fixed overhead variance

(iii) Fixed overhead Expenditure variance

(iv) Fixed overhead Volume variance

(v) Fixed overhead Calendar variance

ICAI'S ANSWER

(i) Variable Overhead Variance:

= Standard Overhead for Actual Production - Actual Variable Overhead

$= \left(\frac{₹14,400}{4,800 \text{ units}} \times 4,560 \text{ units} \right)$ - ₹ 14,400

= ₹ 13,680 – ₹ 14,400 = ₹ 720 (A)

(ii) Fixed Overhead Variance:

= Absorbed Fixed Overhead – Actual Fixed Overhead

$= \left(\frac{₹48,000}{4,800 \text{ units}} \times 4,560 \text{ units} \right)$ - ₹ 46,800

= ₹ 45,600 - ₹ 46,800 = ₹ 1,200 (A)

(iii) Fixed Overhead Expenditure Variance:

= Budgeted Fixed Overhead – Actual Fixed Overhead

= ₹ 48,000 - ₹ 46,800 = ₹ 1,200 (F)

(iv) Fixed Overhead Volume Variance:

= Absorbed Fixed Overhead – Budgeted Fixed Overhead

$= \left(\frac{₹48,000}{4,800 \text{ units}} \times 4,560 \text{ units} \right)$ - ₹ 48,000

= ₹ 45,600 - ₹ 48,000 = ₹ 2,400 (A)

(v) Fixed Overhead Calendar Variance:

= (Actual Days - Budgeted Days) x Standard rate per day

$= (27 \text{ days} – 25 \text{ days}) \text{ x } \frac{₹48,000}{25\text{days}}$ = ₹ 3,840 (F)

ICAI · P4 · May 2025 exam · question p. 22-23 · answer p. 24-25

938 EXAM Jan 2025 MCQ 11 MCQ

AB limited has furnished the following data:

	Budget	Actual (for the month of March)
Production in units	40,000	48,000
Fixed overheads (₹)	78,000	84,000

Calculate fixed overhead volume variance.

A ₹ 15,600 F

B ₹ 15,600 A

C ₹ 6,000 A

D ₹ 14,000 A

ICAI'S ANSWER

Answer key: **(A)**

ICAI · P4 · Jan 2025 exam · question p. 4 · answer p. 6

939 EXAM Old P3 Dec 2021 Q5(a) 10 marks old scheme

In a manufacturing company the standard units of production for the year were fixed at 1,20,000 units and overhead expenditures were estimated to be as follows:

Particulars	Amount (₹)
Fixed	12,00,000
Semi-variable (60% expenses are of fixed nature and 40% are of variable nature)	1,80,000
Variable	6,00,000

Actual production during the month of April, 2021 was 8,000 units. Each month has 20 working days. During the month there was one public holiday. The actual overheads were as follows:

Particulars	Amount (₹)
Fixed	1,10,000
Semi-variable (60% expenses are of fixed nature and 40% are of variable)	19,200
Variable	48,000

You are required to calculate the following variances for the month of April 2021:

i. Overhead Cost variance

ii. Fixed Overhead Cost variance

iii. Variable Overhead Cost variance

iv. Fixed Overhead Volume variance

v. Fixed Overhead Expenditure Variance

vi. Calendar Variance

ICAI'S ANSWER

Working Notes

Fixed Overheads = $\frac{\text{Budgeted Fixed Overheads}}{\text{Budgeted Output}} = \frac{₹12,00,000}{1,20,000 \text{ units}}$	₹ 10
Fixed Overheads element in Semi-Variable Overheads i.e. 60% of ₹1,80,000	₹ 1,08,000
Fixed Overheads = $\frac{\text{Budgeted Fixed Overheads}}{\text{Budgeted Output}} = \frac{₹1,08,000}{1,20,000 \text{units}}$	₹ 0.90
Standard Rate of Absorption of Fixed Overheads *per unit* (₹10 + ₹0.90)	₹ 10.90
Fixed Overheads Absorbed on 8,000 units @ ₹ 10.90	₹ 87,200
Budgeted Variable Overheads	₹ 6,00,000
Add: Variable element in Semi-Variable Overheads 40% of ₹ 1,80,000	₹ 72,000
Total Budgeted Variable Overheads	₹ 6,72,000

Standard Variable Cost *per unit* = $\frac{\text{Budgeted Variable Overheads}}{\text{Budgeted Output}} = \frac{₹6,72,000}{1,20,000 \text{ units}}$	₹5.60
Standard Variable Overheads for 8,000 units @ ₹5.60	₹ 44,800
Budgeted Annual Fixed Overheads (₹ 12,00,000 + 60% of ₹ 1,80,000)	₹ 13,08,000
Possible Fixed Overheads = $\frac{\text{Budgeted Fixed Overheads}}{\text{Budgeted Days}} \times \text{Actual Days} = \left[\frac{₹1,09,000}{20 \text{ Days}} \times 19 \text{ Days}\right]$	₹ 1,03,550
Actual Fixed Overheads (₹1,10,000 + 60% of ₹ 19,200)	₹ 1,21,520
Actual Variable Overheads (₹48,000 + 40% of ₹19,200)	₹ 55,680

COMPUTATION OF VARIANCES

i. Overhead Cost Variance = Absorbed Overheads – Actual Overheads

= (₹ 87,200 + ₹ 44,800) – (₹ 1,21,520 + ₹ 55,680)

= ₹ 45,200 (A)

ii. Fixed Overhead Cost Variance = Absorbed Fixed Overheads – Actual Fixed Overheads

= ₹ 87,200 – ₹ 1,21,520

= ₹ 34,320 (A)

iii. Variable Overhead Cost Variance = Standard Variable Overheads for Production– Actual Variable Overheads

= ₹ 44,800 – ₹ 55,680

= ₹ 10,880 (A)

iv. Fixed Overhead Volume Variance = Absorbed Fixed Overheads – Budgeted Fixed Overheads

= ₹ 87,200 – ₹1,09,000

= ₹ 21,800 (A)

v. Fixed Overhead Expenditure Variance = Budgeted Fixed Overheads – Actual Fixed Overheads

= ₹ 10.90 × 10,000 units – ₹ 1,21,520

= ₹ 12,520 (A)

vi. Calendar Variance = Possible Fixed Overheads – Budgeted Fixed Overheads

= ₹ 1,03,550 – ₹ 1,09,000

= ₹ 5,450 (A)

OR

Calendar Variance = (Actual days – Budgeted days) x Standard fixed overhead rate per day

Standard fixed overhead rate per day = 1308000/20*12 = ₹ 5450

Fixed Overhead Calendar Variance = (19-20) x 5450 = 5450(A)

ICAI · Old P3 · Dec 2021 exam · question p. 19 · answer p. 20-21

940 EXAM Old P3 Nov 2020 Q3(a) 10 marks old scheme

ABC Ltd. has furnished the following information regarding the overheads for the month of June 2020 :

(i)	Fixed Overhead Cost Variance	₹ 2,800 (Adverse)
(ii)	Fixed Overhead Volume Variance	₹ 2,000 (Adverse)
(iii)	Budgeted Hours for June, 2020	2,400 hours
(iv)	Budgeted Overheads for June,2020	₹ 12,000
(v)	Actual rate of recovery of overheads	₹ 8 Per Hour

From the above given information

Calculate:

(1) Fixed Overhead Expenditure Variance

(2) Actual Overheads Incurred

(3) Actual Hours for Actual Production

(4) Fixed Overhead Capacity Variance

(5) Standard hours for Actual Production

(6) Fixed Overhead Efficiency Variance

ICAI'S ANSWER

(1) Fixed Overhead Expenditure Variance

= Budgeted Fixed Overheads – Actual Fixed Overheads

= ₹ 12,000 – ₹ 12,800 (as calculated below) **= ₹ 800 (A)**

(2) Fixed Overhead Cost Variance= Absorbed Fixed Overheads – Actual Fixed Overheads

2,800 (A) = ₹ 10,000 – Actual Overheads

Actual Overheads **= ₹ 12,800**

(3) Actual Hours for Actual Production = ₹ 12,800/ ₹8 **= 1,600 hrs.**

(4) Fixed Overhead capacity Variance

= Budgeted Fixed Overheads for Actual Hours– Budgeted Fixed Overheads

= ₹ 5 x 1600 hrs. – ₹ 12,000 **= ₹ 4,000 (A)**

(5) Standard Hours for Actual Production

= Absorbed Overheads/ Std. Rate

= ₹ 10,000/ ₹ 5 **= 2,000 hrs.**

(6) Fixed Overhead Efficiency Variance

= Absorbed Fixed Overheads – Budgeted Fixed Overheads for Actual Hours

= ₹ 10,000 – ₹ 5 x 1,600 hrs. **= ₹ 2,000 (F)**

Working Note:

(i) Fixed Overhead Volume Variance = Absorbed Fixed Overheads – Budgeted Fixed Overheads

2,000 (A) = Absorbed Fixed Overheads – ₹12,000

Absorbed Fixed Overheads = ₹ 10,000

(ii) Standard Rate/ Hour = ₹ 5 (₹ 12,000/2,400 hrs.)

ICAI · Old P3 · Nov 2020 exam · question p. 11 · answer p. 12

941 RTP Jan 2026 Q3 MCQ

A manufacturing company has set the standard cost for producing one unit of its product as follows:

- Direct Material: 10 kg @ ₹ 5/kg
- Direct Labour: 4 hours @ ₹ 20/hour
- Variable Overhead: 4 hours @ ₹ 5/hour
- Fixed Overhead: ₹ 10 per unit (based on normal output of 5,000 units)

During a particular month, the company produced 4,800 units. The actual results were:

- Direct Material Used: 49,000 kg costing ₹ 2,45,000
- Direct Labour: 19,200 hours costing ₹ 3,84,000
- Variable Overhead Incurred: ₹ 95,000
- Fixed Overhead Incurred: ₹ 52,000

Based on this information, which of the following statements is most accurate?

A The material usage variance is ₹ 5,000 (Favourable)

B The labour rate variance is ₹ 19,200 (Adverse)

C The labour rate variance is ₹ 19,200 (Favourable)

D The fixed overhead volume variance is ₹ 2,000 (Adverse)

ICAI'S ANSWER

(D) The fixed overhead volume variance is ₹ 2,000 (Adverse)

Material Usage Variance (MUV)

= (Standard Quantity - Actual Quantity) × Standard Price

= (48,000-49,000) × ₹ 5 = (-1,000) × ₹ 5 = ₹ 5,000 (Adverse)

Labour Rate Variance (LRV)

= (Standard Rate - Actual Rate) × Actual Hours

= (₹ 20 - ₹ 20) × 19,200 = ₹ 0

Fixed Overhead Volume Variance

= (Actual Output – Budgeted Output) × Standard FOH rate per unit = (4,800 – 5,000) × ₹ 10 = ₹ 2,000 Adverse

ICAI · P4 · RTP Jan 2026 · question p. 6-7 · answer p. 27-28

942 MTP Sep 2026 S-II Q4(a) 8 marks

Following is the standard cost card of a component:

Martials	2 Units at ₹ 15	₹ 30
Labour3 Hours at	₹ 20	₹ 60
Total overheads	3 Hours at ₹ 10	₹ 30

During a particular month 10,000 units of the component were produced and the same was found to be at 60% capacity of the budget. In preparing the variances report for the month, the cost accountant gathered the following information:

Labour	₹ 6,50,000
Variable overheads	₹ 2,00,000
Fixed overheads	₹ 3,00,000
Material price variance	₹70,000 (A)
Material cost variance	₹ 50,000 (A)
Labour rate variance	₹ 50,000 (F)
Fixed overhead expenditure variance	₹ 50,000 (A)

You are required to prepare from the above details:

(1) Actual material cost incurred

(2) Standard cost of materials actually consumed

(3) Labour efficiency variance

(4) Variable OH efficiency variance

(5) Variable OH expenditure variance

(6) Fixed OH efficiency variance

(7) Fixed OH capacity variance

(8) Fixed OH volume variance.

ICAI'S ANSWER

Working Notes:

1. Standard hours = 10,000 × 3 = 30,000 hours

The capacity is 60%

So at100 % capacity the total hours would be = 30,000/ 60% = 50,000 hours

Budgeted hours = 50,000

2. Budgeted Fixed Overheads = Actual Fixed overheads – Expenditure Variance(A)

₹ 3,00,000 – ₹ 50,000 = ₹ 2,50,000

3. Standard Fixed overhead rate = 2,50,000 / 50,000 = ₹ 5 per hour or ₹ 15 per unit (As time taken is 3 hrs. per unit)

4. Total Overhead rate = ₹ 10 per hour

So, Variable overhead rate per hour = ₹ 10-₹5 = ₹ 5per hour or ₹ 15 per unit.

5. Labour Rate Variance = (Standard Rate per hour × Actual hours) – (Actual cost)

₹ 50,000 = (20 × Actual Hours – ₹ 6,50,000)

Actual Hours = 35,000 (₹7,00,000/20)

1. Actual Material cost incurred:

Material Cost Variance = Standard cost of Standard Quantity for Actual output – Actual cost

Actual cost = Standard cost of Standard Quantity for Actual output - Material Cost Variance

Material Cost Variance = ₹ 50,000(A)

(10,000 units × 2 units ×₹ 15) + ₹50,000

= ₹ 3,00,000 + ₹50,000 = ₹ 3,50,000

2. Standard Cost of material consumed:

Material price Variance = [Standard Price (S.P) – Actual Price (A.P)] × Actual Quantity

Standard Cost of material consumed = (Actual Price (A.P) × Actual Quantity) + Material price Variance

Material price Variance = ₹ 70,000(A)

= 3,50,000 – 70,000 = ₹ 2,80,000

3. Labour Efficiency Variance

[Standard Hours – Actual Hours (W. N -5)] × Standard Rate per hour

(10,000× 3 - 35000) × 20

(600,000 - 7,00,000) = ₹ 1,00,000(A)

4. Variable Overhead Efficiency Variance

Variable Overhead Efficiency Variance = (Standard hours for Actual Output – Actual Hours) × Standard variable rate per hour (W. N.4)

(30,000 – 35,000) × 5 = ₹25000 (A)

5. Variable Overhead Expenditure Variance:

Variable Overhead Expenditure Variance= (Budgeted Variable Overheads for Actual Hours) – (Actual Variable Overheads)

= (5 × 35000 – ₹ 2,00,000) = ₹ 25,000(A)

6. Fixed Overhead Efficiency Variance:

Fixed Overhead Efficiency Variance= (Standard hours for Actual output – Actual Hours) × Standard Fixed overhead rate per hour

(30,000 -35,000) ×5 = ₹25,000(A)

7. Fixed Overhead Capacity Variance:

Fixed Overhead Capacity Variance= (Actual Hours – Budgeted Hours) × Standard Fixed overhead rate per hour

(35,000 -50,000) ×5 = ₹ 75,000(A)

8. Fixed Overhead Volume Variance:

Fixed Overhead Volume Variance= (Actual output – Budgeted output) × Standard Fixed overhead rate per unit

(10,000 - 50,000/3 hrs.) ×15 = ₹ 1,00,000(A)

ICAI · P4 · MTP Sep 2026 · Series II · question p. 11-12 · answer p. 15-17

943 MTP Jan 2025 S-I Case Scenario I · MCQ 1 to MCQ 5 10 marks

CASE SCENARIO I

XYZ Manufacturing Ltd. is a mid-sized enterprise that has established a strong reputation in the field of precision engineering. The company specializes in producing high-quality engineering components that meet the stringent requirements of various industries including automotive, aerospace, medical devices, and industrial machinery. With a commitment to precision and excellence, XYZ Manufacturing Ltd. has positioned itself as a reliable supplier of critical components that demand the highest levels of accuracy and durability.

To maintain stringent control over its production costs and enhance cost efficiency, XYZ Manufacturing Ltd. operates under a standard costing system. This system plays a pivotal role in the company's financial and operational management. Standard costing involves setting predetermined costs for each production element, including materials, labor, and overheads. These predetermined costs, known as standard costs, serve as benchmarks against which actual production costs are measured.

Particulars	Budgeted Data	Actual Data
Units Produced	10,000 units	9,500 units
Fixed Overheads	₹ 20,00,000	₹ 19,50,000 + ₹ 1,00,000 (additional quality control cost for 1,000 units chosen on sample basis)
Hours Worked	15,000 hours	14,250 hours
Variable Overhead Rate	₹ 50 per hour	₹ 50 per hour (first 10,000 hours) ₹ 60 per hour (additional hours)

Based on the given information, you are being required to answer the following questions (MCQs 1 to 5):

MCQ 1 2 marks

What is the Fixed Overhead Cost Variance for XYZ Manufacturing Ltd. in May 2024?

A ₹ 50,000 (A)

B ₹ 1,00,000 (A)

C ₹ 1,50,000 (A)

D ₹ 2,00,000 (A)

MCQ 2 2 marks

What is the Fixed Overhead Volume Variance for XYZ Manufacturing Ltd. in May 2024?

A ₹ 50,000 (F)

B ₹ 50,000 (A)

C ₹ 1,00,000 (F)

D ₹ 1,00,000 (A)

MCQ 3 2 marks

What is the Variable Overhead Efficiency Variance for XYZ Manufacturing Ltd. in May 2024?

A ₹ 37,500 (A)

B ₹ 42,500 (A)

C ₹ 0

D ₹ 25,000 (A)

MCQ 4 2 marks

What is the Variable Overhead Expenditure Variance for XYZ Manufacturing Ltd. in May 2024?

A ₹ 40,000 (A)

B ₹ 42,500 (A)

C ₹ 45,000 (A)

D ₹ 45,030 (A)

MCQ 5 2 marks

What is the Fixed Overhead Expenditure Variance for XYZ Manufacturing Ltd. in May 2024?

A ₹ 50,000 (F)

B ₹ 50,000 (A)

C ₹ 1,00,000 (F)

D ₹ 1,00,000 (A)

ICAI'S ANSWERS

MCQ 1

(C) ₹ 1,50,000 (A)

Fixed Overhead Cost Variance = Absorbed Fixed Overheads - Actual Fixed Overheads

Absorbed Fixed Overheads = (Budgeted Fixed Overheads / Budgeted Production) x Actual Production

= (₹ 20,00,000 / 10,000 units) x 9,500 units

= ₹ 19,00,000

Adjusted Actual Fixed Overheads = ₹ 19,50,000 + ₹ 1,00,000

= ₹ 20,50,000

Fixed Overhead Cost Variance = ₹ 19,00,000 - ₹ 20,50,000

= ₹ 1,50,000 (Adverse)

MCQ 2

(D) ₹ 1,00,000 (A)

Fixed Overhead Volume Variance = (Actual Production - Budgeted Production) x Standard Fixed Overhead Rate per Unit

Standard Fixed Overhead Rate per Unit = ₹ 20,00,000 / 10,000 units

= ₹ 200 per unit

Fixed Overhead Volume Variance = (9,500 units - 10,000 units) x ₹ 200

= 500 units x ₹ 200

= ₹ 1,00,000 (Adverse)

MCQ 3

(C) 0

Variable Overhead Efficiency Variance = (Standard Hours for Actual Production - Actual Hours Worked) x Standard Variable Overhead Rate

Standard Hours for Actual Production = 9,500 units x 1.5 hours/unit

= 14,250 hours

Variable Overhead Efficiency Variance = (14,250 – 14,250) x ₹ 50 = 0

MCQ 4

(B) ₹ 42,500 (A)

Variable Overhead Expenditure Variance = (Standard Rate - Actual Rate) x Actual Hours Worked

Total Variable Overhead for Actual Hours: (10,000 x ₹ 50) + (4,250 x ₹ 60) = ₹ 5,00,000 + ₹ 2,55,000 = ₹ 7,55,000

Variable Overhead Expenditure Variance = (₹ 50 x 14,250 hours) - ₹ 7,55,000

= ₹ 42,500 (Adverse)

MCQ 5

(B) ₹ 50,000 (A)

Fixed Overhead Expenditure Variance = Budgeted Fixed Overheads - Actual Fixed Overheads

= ₹ 20,00,000 - ₹ 20,50,000

= ₹ 50,000 (Adverse)

ICAI · P4 · MTP Jan 2025 · Series I · question p. 1-2 · answer p. 1, 1-2, 2

944 MTP Jan 2025 S-II MCQ 15 2 marks MCQ

The Budgeted fixed overhead for the month of August was ₹ 75,00,000 with the units of production estimated at 15,000. However, the actual units produced is 15,600 with no Fixed overhead cost variance.

CALCULATE the actual fixed overhead incurred.

A ₹ 75,00,000

B ₹ 72,11,538

C ₹ 78,00,000

D ₹ 79,00,000

ICAI'S ANSWER

(C) Fixed Overhead Cost Variance = Absorbed Fixed Overheads - Actual Fixed Overheads

$0 = (\frac{₹75,00,000}{15,000} \times 15,600)$ - Actual Fixed Overheads

Actual Fixed Overheads = ₹ 78,00,000

ICAI · P4 · MTP Jan 2025 · Series II · question p. 6 · answer p. 6

945 MTP **May 2024 S-II** Case scenario · MCQ 2(i) to MCQ 2(v) 10 marks

CASE SCENARIO

K Ltd. is a manufacturer of a single product A. 8,000 units of the product A has been produced in the month of March 2024. At the beginning of the year a total 1,20,000 units of the product-A has been planned for production. The cost department has provided the following estimates of overheads:

Fixed	₹ 12,00,000	Variable	₹ 6,00,000
Semi-Variable	₹ 1,80,000		

Semi-variable charges are considered to include 60 per cent expenses of fixed nature and 40 per cent of variable character.

The records of the production department shows that the company could have operated for 20 days but there was a festival holiday during the month.

The actual cost data for the month of March 2024 are as follows:

Fixed	₹ 1,19,000	Variable	₹ 48,000
Semi-Variable	₹ 19,200		

The cost department of the company is now preparing a cost variance report for managerial information and action. You being an accounts officer of the company are asked to calculate the following information for preparation of the variance report:

MCQ 2(i) 2 marks

What is the amount of variable overhead cost variance for the month of March 2024:

A ₹ 10,200 (A)

B ₹ 10,400 (A)

C ₹ 10,800 (A)

D ₹ 10,880 (A)

MCQ 2(ii) 2 marks

What is the amount of fixed overhead volume variance for the month of March 2024:

A ₹ 9,000 (F)

B ₹ 9,000 (A)

C ₹ 21,800 (A)

D ₹ 11,000 (A)

MCQ 2(iii) 2 marks

What is the amount of fixed overhead expenditure variance for the month of March 2024:

A ₹ 21,520 (A)

B ₹ 21,500 (A)

C ₹ 21,400 (A)

D ₹ 21,480 (A)

MCQ 2(iv) 2 marks

What is the amount of fixed overhead calendar variance for the month of March 2024:

A ₹ 5,400 (A)

B ₹ 5,450 (A)

C ₹ 5,480 (A)

D ₹ 5.420 (A)

MCQ 2(v) 2 marks

What is the amount of fixed overhead cost variance for the month of March 2024:

A ₹ 43,320 (A)

B ₹ 43,300 (A)

C ₹ 43,200 (A)

D ₹ 43,380 (A)

ICAI'S ANSWERS

MCQ 2(i)

Answer key: **(D)**

D

Variable Overhead Cost Variance = Standard Variable Overheads for Production – Actual Variable Overheads

= ₹ 44,800 – ₹ 55,680

= ₹ 10,880 (A)

MCQ 2(ii)

Answer key: **(C)**

C

Fixed Overhead Volume Variance = Absorbed Fixed Overheads – Budgeted Fixed Overheads

= ₹ 87,200 – ₹ 1,09,000

=₹ 21,800 (A)

MCQ 2(iii)

A

Fixed Overhead Expenditure Variance = Budgeted Fixed Overheads – Actual Fixed Overheads

= ₹ 10.9 × 10,000 units – ₹ 1,30,520

= ₹ 21,520 (A)

MCQ 2(iv)

Answer key: **(B)**

B

Calendar Variance = Possible Fixed Overheads – Budgeted Fixed Overheads

= ₹ 1,03,550 – ₹ 1,09,000

= ₹ 5,450 (A)

MCQ 2(v)

A

Fixed Overhead Cost Variance = Absorbed Fixed Overheads – Actual Fixed Overheads

= ₹ 87,200 – ₹ 1,30,520

= ₹ 43,320 (A)

WORKING NOTE

Fixed Overheads = Budgeted Fixed Overheads/Budgeted Output = 12,00,000÷1,20,000	₹ 10.00
Fixed Overheads element in *Semi-Variable* Overheads i.e. 60% of ₹ 1,80,000	₹ 1,08,000
Fixed Overheads = Budgeted Fixed Overheads/Budgeted Output ₹ 1,08,000/120,000	₹ 0.90
Standard Rate of Absorption of Fixed Overheads *per unit* (₹ 10.00 + ₹ 0.90)	₹ 10.90
Fixed Overheads Absorbed on 8,000 units @ ₹10.90	₹ 87,200
Budgeted Variable Overheads	₹ 6,00,000
Add: Variable element in Semi-Variable Overheads 40% of ₹ 1,80,000	₹ 72,000
Total Budgeted Variable Overheads	₹ 6,72,000
Standard Variable Cost *per unit* = Budgeted Variable Overheads/Budgeted Output	₹5.60
Standard Variable Overheads for 8,000 units @ ₹5.60	₹ 44,800
Budgeted Annual Fixed Overheads (₹ 12,00,000 + 60% of ₹ 1,80,000)	₹ 13,08,000
Possible Fixed Overheads = Budgeted Fixed Overheads/Budgeted Days x Actual Days = 1,09,000/20 days ×19 days	₹ 1,03,550
Actual Fixed Overheads (₹ 1,19,000 + 60% of ₹ 19,200)	₹ 1,30,520
Actual Variable Overheads (₹ 48,000 + 40% of ₹ 19,200)	₹ 55,680

ICAI · P4 · MTP May 2024 · Series II · question p. 2-3 · answer p. 2, 2-3, 3

946 MTP **Old P3 Nov 2022 S-II** Q2(a) 10 marks old scheme

The following information is available from the cost records of a company for the month of July, 2022:

(1)	Material purchased	22,000 pieces	₹ 9,00,000
(2)	Material consumed	21,000 pieces	
(3)	Actual wages paid for	5,150 hours	₹ 2,57,500
(4)	Fixed Factory overhead incurred		₹ 4,60,000

(5)	Fixed Factory overhead budgeted		₹ 4,20,000
(6)	Units produced	1,900	
(7)	Standard rates and prices are:		
	Direct material	₹ 45 per piece	
	Standard input	10 pieces per unit	
	Direct labour rate	₹ 60 per hour	
	Standard requirement	2.5 hours per unit	
	Overheads	₹ 80 per labour hour	

You are required to CALCULATE the following variances:

(i) Material price variance

(ii) Material usage variance

(iii) Labour rate variance

(iv) Labour efficiency variance

(v) Fixed overhead expenditure variance

(vi) Fixed overhead efficiency variance

(vii) Fixed overhead capacity variance

ICAI'S ANSWER

(i) Material price variance (on the basis of Single plan):

= Actual $\text{Quantity}_{\text{Purchased}}$ (Std. Price – Actual Price)

$=22{,}000 \text{ pcs} \left(\text{Rs. } 45 - \frac{\text{Rs.}9{,}00{,}000}{22{,}000\text{pcs}}\right) = \text{Rs.}90{,}000^{*} \text{ (Favourable)}$

OR

Material price variance (on the basis of Partial plan):

= Actual $\text{Quantity}_{\text{consumed}}$ (Std. Price – Actual Price)

$= 21{,}000 \text{ pcs} \left(\text{Rs. } 45 - \frac{\text{Rs.}9{,}00{,}000}{22{,}000\text{pcs}}\right) = \text{Rs.}85{,}909^{*} \text{ (Favourable)}$

(*Figure may slightly differ due to rounding off the actual price per unit)

(ii) Material usage variance:

= Std. price per piece (Std. Quantity – Actual $\text{Quantity}_{\text{consumed}}$)

= Rs.45 (1,900 units × 10 – 21,000) = Rs. 90,000 (Adverse)

(iii) Labour rate variance:

= Actual hours paid (Std. rate – Actual rate)

$= 5{,}150 \text{ hours} \left(\text{Rs. } 60 - \frac{\text{Rs.}2{,}57{,}500}{5{,}150\text{hours}}\right) = \text{Rs. } 51{,}500 \text{ (Favourable)}$

(iv) Labour efficiency variance:

= Std. rate per hour (Std. hours – Actual $\text{hours}_{\text{worked}}$)

= Rs.60 (1,900 units × 2.5 hours – 5,150 hours) = Rs. 24,000 (Adverse)

(v) Fixed overhead expenditure variance:

= Budgeted Overhead – Actual Overhead

= Rs. 4,20,000 – Rs. 4,60,000 = Rs. 40,000 (Adverse)

(vi) Fixed overhead efficiency variance:

= Std. rate (Std. hours - Actual hours worked)

= Rs.80 (1,900 units × 2.5 hours - 5,150 hours) = Rs. 32,000 (Adverse)

Or,

Fixed overhead efficiency variance on basis of units

= Std. rate per unit (Actual output – Standard output for actual hours)

= Rs.200 (1,900 units - 5,150 / 2.5 hours) = Rs. 32,000 (Adverse)

(vii) Fixed overhead capacity variance:

= Std. rate (Actual hours worked – Budgeted hours)

$= \text{Rs. } 80 \left(5{,}150\text{hours} - \frac{\text{Rs.}4{,}20{,}000}{\text{Rs.}80}\right) = \text{Rs. } 8{,}000 \text{ (Adverse)}$

Or,

Fixed overhead capacity variances on basis of units

= Std. rate per unit (Standard output for actual hours – Budgeted output)

= Rs.200 (2,060 units - 4,20,000 / 200) = Rs. 8,000 (Adverse)

ICAI · Old P3 · MTP Nov 2022 · Series II · question p. 2-3 · answer p. 3-4

947 MTP **Old P3 May 2022 S-I** Q2(b) 10 marks old scheme

Following are the details given:

Budgeted Days	25
Budgeted Fixed Overheads	1,00,000
Budgeted Production	800 units per day
Actual Production	21,000 units

Fixed Overheads are absorbed @ ₹ 10 per hour.

Fixed overheads efficiency variance	10,000A
Fixed overheads calendar variance	8,000F
Fixed overheads cost variance	15,000A

You are required to CALCULATE:

(a) Actual Fixed Overheads

(b) Actual Days

(c) Actual Hours

(d) Fixed overheads Expenditure variance

(e) Fixed overheads volume variance

(f) Fixed overheads capacity variance

ICAI'S ANSWER

(i) Fixed Overhead Cost Variance = (Std Fixed Overheads – Actual Fixed Overheads)

= ($\frac{1,00,000}{20,000}$ × 21,000 units – Actual Fixed Overheads) = 15,000A

= (1,05,000 - Actual Fixed Overheads) = 15,000A

=> Actual Fixed Overheads = 1,20,000

(ii) Fixed Overhead Calendar Variance=(Actual Days – Budgeted Days) x Budgeted rate per day

= (Actual Days - 25)× $\frac{1,00,000}{25}$ = 8,000F

= (Actual Days - 25) = 2

=> Actual Days = 27

(iii) Fixed Overhead Efficiency Variance =(Standard Hours for Actual Production – Actual Hours) x Budgeted rate per hour

= $\left(\frac{10,000}{20,000} \times 21,000 \text{ - Actual Hours}\right) \times 10$ = 10,000A

= (10,500 – Actual Hours) = -1,000

=> Actual Hours = 11,500

(iv) Fixed overheads Expenditure variance=(Budgeted Fixed Overheads – Actual Fixed Overheads)

= (1,00,000 – 1,20,000) = **20,000A**

(v) Fixed overheads volume variance = (Budgeted units – Actual Units) x Budgeted Rate per unit

= $(20,000 - 21,000) \times \frac{1,00,000}{20,000}$ = **5,000F**

(vi) Fixed overheads capacity variance = (Budgeted Hours for Actual Days – Actual Hours) x Budgeted Rate per Hour

= $\left(\frac{10,000}{25} \times 27 - 11,500\right) \times 10$ = **7,000F**

ICAI · Old P3 · MTP May 2022 · Series I · question p. 3 · answer p. 3-4

948 MTP Old P3 May 2020 S-I Q1(b) 5 marks old scheme

ZX Ltd. has furnished the following information:

	Budgeted	Actual March 2020
Number of working days	25	27
Production (in units)	20,000	22,000
Fixed Overheads	Rs. 3,00,000	Rs. 3,10,000

Budgeted fixed overhead rate is Rs. 10.00 per hour. In March 2020, the actual hours worked were 31,500. In relation to fixed overheads, CALCULATE:

(i) Efficiency Variance

(ii) Capacity Variance

(iii) Calendar Variance

(iv) Volume Variance

(v) Expenditure Variance

ICAI'S ANSWER

Workings:

(1) Budgeted Hours = $\frac{\text{Rs.3,00,000}}{\text{Rs.10 per hour}}$ = 30,000 hours

(2) Standard Fixed Overhead rate per hour (Standard Rate):

= $\frac{\text{Budgeted fixed overheads}}{\text{Budgeted Hours}} = \frac{\text{Rs.3,00,000}}{\text{30,000hours}}$ = Rs.10.00

(3) Standard hour per unit of output = $\frac{\text{30,000hours}}{\text{20,000units}}$ = 1.5 hours

(4) Standard hours for Actual Output = 22,000 units × 1.5 hours = 33,000 Hours

(5) Budgeted Overhead per day for budgeted days= $\frac{\text{Rs.3,00,000}}{\text{25days}}$ = Rs.12, 000

(6) Budgeted Overhead for actual days worked = Rs.12,000 × 27 days = Rs.3,24,000

(7) Budgeted Hours for Actual days worked = $\frac{\text{30,000hours}}{\text{25days}} \times \text{27days}$ = 32,400 hours

Computation of Variances in relation to Fixed Overheads:

(i) **Efficiency Variance**

= Standard Rate × (Standard hours for actual output – Actual hours worked)

= Rs.10 (33,000 hours – 31,500 hours) = Rs.15,000 (Favourable)

(ii) **Capacity Variance**

= Standard Rate × (Actual Hours – Budgeted Hours for actual days worked)

= Rs.10 (31,500 hours – 32,400 hours) = Rs.9,000 (Adverse)

(iii) **Calendar Variance**

= Standard/Budgeted Fixed Overhead Rate per day × (Actual Working days – Budgeted working days)

= Rs.12,000 (27 days – 25 days) = Rs.24,000 (Favourable)

(iv) **Volume Variance**

= Standard Rate × (Standard hours – Budgeted hours)

= Rs.10 (33,000 hours – 30,000 hours) = Rs.30,000 (Favourable)

(v) **Expenditure Variance**

= Budgeted Overheads – Actual Overheads

= Rs.3,00,000 – Rs.3,10,000 = Rs.10,000 (Adverse)

Note: *Overhead Variances may also be calculated based on output.*

ICAI · Old P3 · MTP May 2020 · Series I · question p. 1 · answer p. 1-2

949 MTP **Old P3 Nov 2019 S-I** Q4(a) 10 marks old scheme

In a manufacturing company the standard units of production of the year were fixed at 1,20,000 units and overhead expenditures were estimated to be:

Fixed Rs. 12,00,000; Variable Rs. 6,00,000;

Semi-Variable Rs. 1,80,000

Actual production during the April, 2019 of the year was 8,000 units. Each month has 20 working days.

During the month there was one public holiday. The actual overheads amounted to:

Fixed Rs. 1,10,000; Variable Rs. 48,000

Semi-variable Rs. 19,200

Semi-variable charges are considered to include 60 per cent expenses of fixed nature and 40 per cent of variable character.

CALCULATE the followings:

(i) Overhead Cost Variance

(ii) Fixed Overhead Cost Variance

(iii) Variable Overhead Cost Variance

(iv) Fixed Overhead Volume Variance

(v) Fixed Overhead Expenditure Variance

(vi) Calendar Variance.

ICAI'S ANSWER

COMPUTATION OF VARIANCES

(i) Overhead Cost Variance = Absorbed Overheads – Actual Overheads

= (Rs.87,200 + Rs.44,800) – (Rs.1,21,520 + Rs.55,680)

= Rs. 45,200 (A)

(ii) Fixed Overhead Cost Variance = Absorbed Fixed Overheads – Actual Fixed Overheads

= Rs. 87,200 – Rs.1,21,520

= Rs.34,320 (A)

(iii) Variable Overhead Cost Variance = Standard Variable Overheads for Production – Actual Variable Overheads

= Rs. 44,800 – Rs. 55,680

= Rs. 10,880 (A)

(iv) Fixed Overhead Volume Variance = Absorbed Fixed Overheads – Budgeted Fixed Overheads

= Rs. 87,200 – Rs.1,09,000

= Rs. 21,800 (A)

(v) Fixed Overhead Expenditure Variance = Budgeted Fixed Overheads – Actual Fixed Overheads

= Rs.10.90 × 10,000 units – Rs.1,21,520

= Rs.12,520 (A)

(vi) Calendar Variance = Possible Fixed Overheads – Budgeted Fixed Overheads

= Rs.1,03,550 – Rs.1,09,000

= Rs. 5,450 (A)

WORKING NOTE

Fixed Overheads per Unit = $\frac{\text{Budgeted Fixed Overheads}}{\text{Budgeted Output}} = \frac{\text{Rs.12,00,000}}{\text{1,20,000units}}$	Rs. 10
Fixed Overheads element in *Semi-Variable* Overheads i.e. 60% of Rs.1,80,000	Rs. 1,08,000
Fixed Overheads per Unit = $\frac{\text{Budgeted Fixed Overheads}}{\text{Budgeted Output}} = \frac{\text{Rs.1,08,000}}{\text{1,20,000units}}$	Rs. 0.90
Standard Rate of Absorption of Fixed Overheads *per unit* (Rs.10 + Rs.0.90)	Rs.10.90
Fixed Overheads Absorbed on 8,000 units @ Rs10.90	Rs. 87,200
Budgeted Variable Overheads	Rs. 6,00,000
Add : Variable element in Semi-Variable Overheads 40% of Rs. 1,80,000	Rs. 72,000
Total Budgeted Variable Overheads	Rs. 6,72,000
Standard Variable Cost *per unit* = $\frac{\text{Budgeted Variable Overheads}}{\text{Budgeted Output}} = \frac{\text{Rs.6,72,000}}{\text{1,20,000units}}$	Rs.5.60
Standard Variable Overheads for 8,000 units @ Rs.5.60	Rs. 44,800
Budgeted Annual Fixed Overheads (Rs. 12,00,000 + 60% of Rs. 1,80,000)	Rs.13,08,000
Possible Fixed Overheads = $\frac{\text{Budgeted Fixed Overheads}}{\text{Budgeted Days}} \times \text{Actual Days}$ = $\left[\frac{\text{Rs.1,09,000}}{\text{20 Days}} \times \text{19 Days}\right]$	Rs.1,03,550
Actual Fixed Overheads (Rs.1,10,000 + 60% of Rs. 19,200)	Rs.1,21,520
Actual Variable Overheads (Rs.48,000 + 40% of Rs.19,200)	Rs. 55,680

ICAI · Old P3 · MTP Nov 2019 · Series I · question p. 6 · answer p. 8-9

CHAPTER 14

Marginal Costing

14.1 Marginal cost, contribution and the P/V ratio

20 questions · asked in 11 ICAI documents · 4 exam papers

950 EXAM Sep 2025 Case Scenario - II · MCQ 6 to MCQ 10 10 marks

CASE SCENARIO - II

JMS Limited, a soft drink company, is intending to introduce a new product viz. 'Herbs Infused Mineral Water to the market. Annual sales of this new product is estimated at 36,000 units with a selling price of ₹ 75 per unit. The cost estimates for this new product are as follows:

Elements of Cost	Amount (₹)
Direct material consumed	9,50,000
Direct labour cost	5,93,750
Manufacturing overheads (variable)	2,85,000
Manufacturing overheads (fixed)	1,90,000
General & Administration overheads (variable)	1,42,500
General & Administration overheads (fixed)	2,13,750
Selling and distribution overheads (variable)	80,750
Selling and distribution overheads (fixed)	64,250

There will be no closing stock of 'Herbs Infused Mineral Water'.

On the basis of above Case Scenario, you are required to answer the following MCQs 6 to 10:

MCQ 6 2 marks

What is the 'Cost of production' and 'Total Cost' as per Absorption costing?

A ₹ 18,28,750 and ₹ 27,00,000

B ₹ 19,71,250 and ₹ 25,00,000

C ₹ 20,18,750 and ₹ 25,20,000

D ₹ 17,67,000 and ₹ 26,20,000

MCQ 7 2 marks

What is the amount of total variable cost?

A ₹ 20,52,000

B ₹ 19,71,250

C ₹ 23,75,000

D ₹ 25,20,000

MCQ 8 2 marks

What is the amount of contribution per unit earned at the estimated level of sales?

A ₹ 21

B ₹ 18

C ₹ 23

D ₹ 15

MCQ 9 2 marks

What is the Break-even Sales (in Rupees)?

A ₹ 32,50,000

B ₹ 23,69,000

C ₹ 16,33,000

D ₹ 19,50,000

MCQ 10 2 marks

What will be the profit if the actual sales are 10% less than the estimated sales?

A ₹ 1,15,200

B ₹ 1,62,000

C ₹ 1,71,500

D ₹ 1,51,200

ICAI'S ANSWERS

MCQ 6

Answer key: **(C)**

MCQ 7

Answer key: **(A)**

MCQ 8

Answer key: **(B)**

MCQ 9

Answer key: **(D)**

MCQ 10

Answer key: **(A)**

ICAI · P4 · Sep 2025 exam · question p. 3-4 · answer p. 7

951 EXAM **May 2025** MCQ 14 MCQ

A company forecasts its labour costs and material cost to go up by 12% and 8% respectively per unit in the next financial year. If the ratio between material and labour is 5: 3, determine the increase in selling price as a percentage that the company shall keep to maintain its P/V of 12%, assuming variable overheads as nil.

A 7.45%

B 8.01%

C 9.95%

D 9.46%

ICAI'S ANSWER

Answer key: **(D)**

952 EXAM **May 2025** Q1(a) 5 marks

Axion Industries is a heavy industrial gear manufacturing company with a manufacturing setup based in Pune. Mr. Andrew, the CFO of the company furnished the following information to Mr. Joe who heads the Finance department.

For FY 2024-25:

Particulars	Amount ₹ (in crores)
Total Sales	1,00,000
Raw material cost	50,000
Direct wages	15,000
Fixed & variable overheads	25,000
Profit	10,000
Total Number of Units sold	40,000 units

The market being very competitive and with the raw materials rates rising, Mr. Andrew raises his concern with Mr. Joe where he expects in the next financial year 2025-26 workers' wages to rise by 20%, fixed costs component to decrease by ₹ 500 crores. Total fixed & variable overhead however is to be ₹ 28,500 crores. Total fixed & variable overhead however is to be ₹ 28,500 crores. The total number of units expected to be sold would be 50,000.

Required:

Calculate the minimum number of units to be sold to sustain the same per unit profit in the financial year 2025-26 also.

(Ignore further effects on Fixed costs.)

ICAI'S ANSWER

(i) Calculation of Variable Overhead Per Unit & Fixed Cost (F.Y. 2024-25)

$$\text{Variable cost per unit} = \frac{\text{Difference in Total Cost}}{\text{Difference in Units}}$$

$$= \frac{₹28{,}500 \text{ crores} - ₹25{,}000 \text{ crores}}{50{,}000 \text{ units} - 40{,}000 \text{ units}}$$

= ₹ 0.35 crore per unit

Therefore, Fixed cost = ₹ 25,000 crores - (40,000 x 0.35) = ₹ 11,000 crores

Revised Fixed cost (FY 2025-26) = 10,500 Crores

Desired Profit per unit (₹ 10,000 crores/40,000 units) = ₹ 0.25 crores per unit

Calculation of Contribution per unit

Particulars		Per Unit (₹ in crores)
Selling price per unit		2.50
Variable Cost:		
Material	1.25	
Labour (0.375 × 120%)	0.45	
Variable Overheads	0.35	2.05
Contribution per Unit		**0.45**

Let the units sold be X

Contribution	= 0.45X
Profit	= 0.25X
Profit	= Contribution – Fixed Cost
0.25X	= 0.45X – 10,500 Crores

Minimum no. of units to be sold to sustain the same per unit profit, X = 52,500 Units

Alternatively, solution can be done in following way:

Calculation of Variable Overhead Per Unit & Fixed Cost (F.Y. 2024-25)

$$\text{Variable cost per unit} = \frac{\text{Difference in Total Cost}}{\text{Difference in Units}}$$

$$= \frac{₹29{,}000 \text{ crores} - ₹25{,}000 \text{ crores}}{50{,}000 \text{ units} - 40{,}000 \text{ units}}$$

= ₹ 0.40 crore per unit

Therefore, Fixed cost = ₹ 25,000 crores - (40,000 x 0.40) = ₹ 9,000 crores

Revised Fixed cost (FY 2025-26) = 8,500 Crores

Desired Profit per unit (₹ 10,000 crores/40,000 units) = ₹ 0.25 crores per unit

Calculation of Contribution per unit (2025-26)

Particulars		Per Unit (₹ in crores)
Selling price per unit		2.50
Variable Cost:		
Material	1.25	
Labour (0.375 × 120%)	0.45	
Variable Overheads	0.40	2.10
Contribution per Unit		**0.40**

Let the units sold be X

Contribution	= 0.40X
Profit	= 0.25X
Profit	= Contribution – Fixed Cost
0.25X	= 0.40X – 8,500 Crores

Minimum no. of units to be sold to sustain the same per unit profit, X = 56,667 Units

ICAI · P4 · May 2025 exam · question p. 7 · answer p. 9-11

953 EXAM **Sep 2024** MCQ 11 MCQ

PS Limited is facing downfall in its demand. Marketing team has suggested to reduce the selling price by 5% to compete in the market. Variable cost is 76% of the current selling price.

You are required to find out the PN Ratio after reducing the price by 5%

A 20%

B 24%

C 25.26%

D 19%

ICAI'S ANSWER

Answer key: **(A)**

ICAI · P4 · Sep 2024 exam · question p. 5 · answer p. 7

954 EXAM **Old P3 Nov 2019** Q1(d) 5 marks old scheme

When volume is 4,000 units; average cost is ₹ 3.75 per unit. When volume is 5,000 units, average cost is ₹ 3.50 per unit. The Break-Even point is 6,000 units.

Calculate: (i) Variable Cost per unit (ii) Fixed Cost and (iii) Profit Volume Ratio.

ICAI'S ANSWER

(i) Variable cost per unit = $\frac{\text{Change in Total cost}}{\text{Change in units}}$

$= \frac{(₹\ 3.50 \times 5{,}000 \text{ units}) - (₹\ 3.75 \times 4{,}000 \text{ units})}{5{,}000 - 4{,}000}$

$= \frac{₹\ 17{,}500 - ₹\ 15{,}000}{1{,}000}$ = ₹ **2,500**/1000 = **₹2.5**

(ii) Fixed cost = Total Cost – Variable cost (at 5,000 units level)

= ₹17,500 – ₹2.5 × 5,000 = **₹5,000**

(iii) Contribution per unit = $\frac{\text{Fixed cost}}{\text{BEP (in units)}} = \frac{₹\ 5{,}000}{6{,}000 \text{ units}} = 0.833$

P/V Ratio = $\frac{\text{Contribution per unit}}{\text{Sale price per unit}} = \frac{0.833}{2.5 + 0.833}$ = **25%**

ICAI · Old P3 · Nov 2019 exam · question p. 2 · answer p. 6

955 RTP May 2026 Q3 MCQ

In two consecutive quarters, a company reported the following performance:

Quarter 1 sales were ₹ 2,10,000 with a profit of ₹ 12,000, while Quarter 2 sales rose to ₹ 2,40,000 with a profit of ₹ 24,000. Assume that the fixed cost remained constant across both quarters and that the company did not introduce any new product lines.

Based on the above information, the P/V ratio must be:

A 20%

B 30%

C 40%

D 50%

ICAI'S ANSWER

(C) 40%

$$\text{P/V Ratio} = \frac{\text{Contribution}}{\text{Sales}}$$

For Quarter I

Sales = ₹ 2,10,000

Profit = ₹ 12,000

F + 12,000 = C × 2,10,000

For Quarter II

Sales = ₹ 2,40,000

Profit = ₹ 24,000

F + 24,000 = C × 2,40,000

OR

Change in Profit / Change in Sales gives P/V Ratio

$$\text{P/V Ratio} = \frac{\text{Change in Profit}}{\text{Change in Sales}}$$

$$= \frac{24{,}000 - 12{,}000}{2{,}40{,}000 - 2{,}10{,}000}$$

$$= \frac{12{,}000}{30{,}000}$$

= 0.40

ICAI · P4 · RTP May 2026 · question p. 6 · answer p. 26-27

956 RTP Jan 2025 Q19(a)

DISCUSS advantages of Marginal Costing.

ICAI'S ANSWER

Advantages of Marginal Costing:

1. **Simplified Pricing Policy:** The marginal cost remains constant per unit of output whereas the fixed cost remains constant in total. Since marginal cost per unit is constant from period to period within a short span of time, firm decisions on pricing policy can be taken.

2. **Proper recovery of Overheads:** Overheads are recovered in costing on the basis of pre-determined rates. If fixed overheads are included on the basis of pre-determined rates, there will be under- recovery of overheads if production is less or if overheads are more. There will be over-recovery of overheads if production is more than the budget or actual expenses are less than the estimate. This creates the problem of treatment of such under or over-recovery of overheads. Marginal costing avoids such under or over recovery of overheads.
3. **Shows Realistic Profit:** Advocates of marginal costing argues that under the marginal costing technique, the stock of finished goods and work-in-progress are carried on marginal cost basis and the fixed expenses are written off to profit and loss account as period cost. This shows the true profit of the period.
4. **How much to produce:** Marginal costing helps in the preparation of break-even analysis which shows the effect of increasing or decreasing production activity on the profitability of the company.
5. **More control over expenditure:** Segregation of expenses as fixed and variable helps the management to exercise control over expenditure. The management can compare the actual variable expenses with the budgeted variable expenses and take corrective action through analysis of variances.
6. **Helps in Decision Making:** Marginal costing helps the management in taking a number of business decisions like make or buy, discontinuance of a particular product, replacement of machines, etc.
7. **Short term profit planning:** It helps in short term profit planning by B.E.P charts.

ICAI · P4 · RTP Jan 2025 · question p. 20 · answer p. 44-45

957 MTP **Sep 2026 S-II** Q5(b) 6 marks

Calculate Sales in rupees for desired profit on the basis of the following information:

(a) Fixed cost ₹ 25,000, Variable Cost ₹ 2,50,000; Sales ₹ 5,00,000; Desired profit ₹ 20,000

(b) Fixed cost ₹ 40,000; Desired profit ₹ 30,000; P/V Ratio 40%

(c) Fixed cost ₹ 60,000; Selling price ₹ 20 per unit. Variable cost ₹ 12 per unit., Desired profit ₹ 2 per unit.

(d) Fixed cost ₹ 80,000; Variable Cost ₹ 2,40,000; Sales Rs,.4,00,000, Desired profit 15% on sales.

ICAI'S ANSWER

(a)

Sales:	₹ 5,00,000
Variable Cost:	₹ 2,50,000
Contribution:	₹ 2,50,000
Contribution %	= 50%
Desired Contribution	= Fixed Cost + Desired Profit
	= ₹ 25,000 + ₹ 20,000
	= ₹ 45,000
Therefore, Sales	= ₹ 90,000 (45,000/50%)

(b)

P/V Ratio % = 40%	
Desired Contribution	= Fixed Cost + Desired Profit
	= ₹ 40,000 + ₹ 30,000
	= ₹ 70,000
Therefore, Sales	= ₹ 1,75,000 (70,000/40%)

(c)

Selling Price:	₹ 20
Variable Cost per unit:	₹ 12
Contribution:	₹8
Contribution %	= 40%
Desired Profit	=₹2
Contribution Per unit	= Fixed Cost per unit + Desired Profit per unit
₹8	=X+₹2
Fixed Cost per unit (X)	=₹6
Total Fixed Cost	= ₹ 60,000
No. of units sold	= 10,000 (60,000/6)
Therefore, Sales	= ₹ 2,00,000 (10,000 × 20)

(d)

Sales:	₹ 4,00,000

Variable Cost:	₹ 2,40,000
Contribution:	₹ 1,60,000
Contribution %	= 40%
Contribution	= Fixed Cost + Desired Profit
Contribution%	= Fixed Cost% + 15%
40%	= Fixed Cost% + 15%
Fixed Cost%	= 40% - 15%
	= 25%
Fixed cost ₹ 80,000	
Therefore, Sales	= ₹ 3,20,000 (80,000/25%)

ICAI · P4 · MTP Sep 2026 · Series II · question p. 13-14 · answer p. 19-20

958 MTP **May 2026 S-I** Q6(c) 4 marks

"Cost-Volume-Profit (CVP) analysis is a vital tool for profit planning and decision-making."

DISCUSS this statement and EXPLAIN the impact of various changes in cost and volume on profit.

ICAI'S ANSWER

Cost-Volume-Profit (CVP) analysis is a managerial tool showing the relationship between various ingredients of profit planning viz., cost, selling price and volume of activity. As the name suggests, cost volume profit (CVP) analysis is the analysis of three variables, cost, volume and profit. Such an analysis explores the relationship between costs, revenue, activity levels and the resulting profit. It aims at measuring variations in cost and volume.

An understanding of CVP analysis is extremely useful to management in budgeting and profit planning.

It elucidates the impact of the following on the net profit:

(i) Changes in selling prices,

(ii) Changes in volume of sales,

(iii) Changes in variable cost,

(iv) Changes in fixed cost.

ICAI · P4 · MTP May 2026 · Series I · question p. 15 · answer p. 20

959 MTP **May 2025 S-II** Case Scenario 1 · MCQ 1 to MCQ 5 10 marks

CASE SCENARIO 1

Company Rontomax maintains its accounts in Delhi head office. All the records of Rontomax are safely kept in this office only. In the 2nd quarter Delhi office went under repair. Thus, for the 2nd quarter records were maintained in Gurugram branch office. This branch's main work is to bring business to the company and thus generally no records are maintained in this branch office.

So for 2nd quarter all the records were recorded and maintained in this Gurugram office only. At the end of 2nd quarter, fire broke out in this branch and unfortunately all the records were burned. In the beginning of 3rd quarter a board meeting was going to be conducted and performance of 2nd quarter were to be discussed. Company secretary, Mr. Manoj, was responsible for preparing a report of performance to be presented to the board. Now he is under immense pressure as the records were burned and thus he was not able to prepare a performance report.

Manoj contacted the Delhi head office and received a copy of 1st quarter records. He also got some information through emails shared between head office and branch office. He somehow got a lot of information but this information doesn't make any sense as it is in parts and pieces. He called out for help of Finance and cost head, Miss Bharti, who is also a Chartered accountant. Now both of them are at task to work out this information and be able to present a summary performance report to be presented to the board in the board meeting. Data that Manoj was able to gather was:

- Rontomax garnered revenue of ₹ 80,00,000 in 1st quarter of 2023. Its tax provision expense was ₹ 4,50,000 calculated on earning before tax in the same period.
- Cost of Goods Sold (COGS) and Operating expenses in 1st quarter were Rs, 38,00,000 and ₹ 12,50,000 respectively.
- Quarterly interest expense was ₹ 1,50,000.
- Non-operating expense other than interest was ₹ 13,00,000 in the 1st quarter.
- Selling price was reduced by 8% & no. of units sold increased by 25% from 1st quarter to 2nd quarter.
- Variable cost per unit for maintaining the day-to-day business operations is 30% of variable cost per unit of producing the goods.
- EBIT per unit for 2nd quarter is ₹ 38.857 which has gone down by ₹ 8.285 from 1st quarter.

Manoj tells Bharti about the general format of questions that board asks from him, every quarter. So, they decide to find out the answers of such questions before-hand so that meeting can be conducted smoothly.

Following are those questions that they are seeking for solutions. Consider yourself as their assistant trainee and help to find these answers (MCQs 5 to 10).

MCQ 1 2 marks

Find out the sales amount of Quarter 2. Select the correct answer.

A ₹ 76,50,000

B ₹ 86,00,000

C ₹ 92,00,000

D ₹ 96,50,000

MCQ 2 2 marks

What is the total variable cost & fixed cost in quarter 1 as per marginal costing income statement, respectively?

A ₹ 49,40,000 & ₹ 1,10,000

B ₹ 3,26,000 & ₹ 17,90,000

C ₹ 17,90,000 & ₹ 3,26,000

D ₹ 4,94,000 & ₹ 11,10,000

MCQ 3 2 marks

If Fixed cost & total variable cost as per marginal costing doesn't change, what is the cost change in 2nd quarter?

A Operating expenses increased by ₹ 11,50,000

B Non-operating expense decreased by ₹ 11,50,000

C Operating expenses decreased by ₹ 11,50,000

D Non-operating expenses increased by ₹ 11,50,000

MCQ 4 2 marks

If operating fixed cost, total variable cost, & interest cost remains same in quarter 2, what is the tax provision for 2nd quarter?

A ₹ 4,65,000

B ₹ 4,75,000

C ₹ 4,85,000

D ₹ 4,45,000

MCQ 5 2 marks

What is the amount of profit excluding non-operating expenses in quarter 2?

A ₹ 38,50,000

B ₹ 36,50,000

C ₹ 41,50,000

D ₹ 29,50,000

ICAI'S ANSWERS

MCQ 1

(C) ₹ 92,00,000

Quarter 1:

	Amount (₹)
Sales	80,00,000
(-) COGS	(38,00,000)
(-) Operating expenses	(12,50,000)
(-) Non-operating expenses	(13,00,000)
EBIT of 1ST Quarter	16,50,000

Let's assume no. of units sold in 1st quarter = X

EBIT per unit of 1st quarter = 38.857 + 8.285 = ₹ 47.142

Then, 16,50,000/X = 47.142

X = 35,000 units

Number of units sold in 2nd Qtr. = 35,000 + 35,000x0.25 = 43,750 units

Selling price in Qtr 1 = 80,00,000/35,000 = ₹ 228.571

Selling price in Qtr 2 = ₹ 210.285

Sales in Qtr 2 = 210.285 x 43,750 = ₹ 92,00,000

MCQ 2

(A) ₹ 49,40,000 & ₹ 1,10,000

COGS is 100% variable

VC per unit of operating expenses = 30% of COGS per unit

COGS per unit = 38,00,000/35,000 = ₹ 108.571

VC per unit of operating expenses = 30% x 108.571 = ₹ 32.571

VC in operating expenses = 32.571 x 35,000 = 11,40,000

Fixed cost in operating expenses = 12,50,000 – 11,40,000 = ₹ 1,10,000

Total Variable Cost = 11,40,000 + 38,00,000 = ₹ 49,40,000

Total fixed cost = ₹ 1,10,000

MCQ 3

(D) Non-operating expenses increased by ₹ 11,50,000

EBIT in Qtr 2 = 38.857 x 43,750 = ₹ 17,00,000

Non operating expenses in 2nd Qtr = Revenue – VC – FC - EBIT

= 92,00,000 – 49,40,000 – 1,10,000 – 17,00,000

= ₹ 24,50,000

Non operating expenses in 1st Qtr = ₹ 13,00,000

NOE increased by 11,50,000

MCQ 4

(A) ₹ 4,65,000

EBIT of 1ST Qtr = ₹ 16,50,000

EBT of 1ST Qtr = EBIT – Int = 16,50,000 – 1,50,000 = ₹ 15,00,000

Tax Provision for 1st Qtr = ₹ 4,50,000

Tax rate = 4,50,000/15,00,000 = 30%

EBT for 2nd Qtr = 17,00,000 – 1,50,000 = ₹ 15,50,000

Tax provision for 2nd Qtr = 15,50,000 x 30% = ₹ 4,65,000

MCQ 5

(C) ₹ 41,50,000

Profit in 2nd Qtr as per Marginal Costing

= Sales – VC – FC

= 92,00,000 - 49,40,000 - 1,10,000 = ₹ 41,50,000

ICAI · P4 · MTP May 2025 · Series II · question p. 1-2 · answer p. 1, 1-2, 2

960 MTP Sep 2024 S-II Q3(a) 7 marks

A company manufactures and sells a product, the price of which is controlled by the Government. Raw material required for this product is also made available at a fixed controlled price. The following figures have been called for the previous two accounting years of the company:

	Year- I	Year- II
Quantity Sold (tones)	1,26,000	1,44,000
Price per tone	₹ 185	₹ 185
	(₹ In thousands)	
Sales Value	23,310	26,640
Raw Materials	11,340	12,960
Direct Labour	1,512	1,872
Factory, Administration and Selling Expenses	9,702	11,232
Profit	756	576

During the year II direct labour rates increased by 8 1/3%. Increases in factory, administration and selling expenses during the year were ₹ 8,10,000 on account of factors other than the increased quantities produced and sold. The managing director desires to know, what quantity if they had produced and sold would have given the company the same net profit per tonne in Year II as it earned during the Year I Advise him.

ICAI'S ANSWER

Contribution *per tonne*	(₹)
Sales Price	185.00
Variable Cost:	
Material (W.N.-1)	90.00
Labour (W.N.-2)	13.00
Variable Overhead (W.N.-3)	40.00
Contribution	42.00
Profit Required (₹7,56,000 /1,26,000 tonnes)	6.00
Balance Contribution *per tonne* for meeting Fixed Costs	36.00
Fixed Costs (W.N.-4)	54,72,000
Quantity Required (₹54,72,000 ÷ ₹36)	1,52,000 tonnes

Working Notes

1. Materials Cost per tonne in Year II ₹90

$\left(\frac{₹1,29,60,000}{1,44,000\text{tonnes}}\right)$

2. Labour Cost per tonne in Year II ₹13
$\left(\frac{₹18,72,000}{1,44,000\text{tonnes}}\right)$

3. Variable portion of Factory, Administration and Sell. Expenditure, etc	₹
Total in Year II	1,12,32,000
Less: Increase otherwise than on account of increased turnover	8,10,000
	1,04,22,000
Less: Amount Spent in Year I	97,02,000
Increase	7,20,000
Increase in Quantity Sold	18,000 tonnes
Variable Expenses per tonne $\left(\frac{₹7,20,000}{18,000\text{tonnes}}\right)$	₹40
4. Fixed portion of Factory, Administration and Selling Expenses (Yr. 2)	₹1,12,32,000
Variable Expenses @ ₹ 40 per tonne	₹57,60,000
Fixed Portion	₹54,72,000

ICAI · P4 · MTP Sep 2024 · Series II · question p. 7-8 · answer p. 9-10

961 MTP **Old P3 Jul 2021 S-II** Q5(a) 10 marks old scheme

Following data is available from the costing department of Aarya Ltd. which manufactures and markets a single product:

Material	Rs. 32 per unit	Fixed Cost (Rs.)	Rs. 10,00,000
Conversion Cost (Variable)	Rs. 24 per unit	Present Sales (units)	90,000
Dealer's Margin (10% of Sales)	Rs. 8 per unit	Capacity Utilization	60 %
Selling Price	Rs. 80 per unit		

There is acute competition in the market, thus extra efforts are necessary to enhance the sales. For this, following suggestions have been proposed:

(i) Reducing selling price by 5 per cent.

(ii) Increasing dealer's margin by 20 per cent over the existing rate.

Which of these two suggestions would you RECOMMEND, if the company desires to maintain the present profit? GIVE REASONS.

ICAI'S ANSWER

Workings:

Statement Showing Profit on Sale of 90,000 units

	(Rs.)	(Rs.)
Selling Price per unit		80
Less: Variable Cost per unit		
Material	32	
Conversion Cost	24	
Dealers' Margin	8	64
Contribution per unit		16
Total Contribution (90,000 units × Rs. 16)		14,40,000
Less: Fixed Cost		10,00,000
Profit		4,40,000

In both the proposed suggestions, the fixed costs remain unchanged. Therefore, the present profit of Rs. 4,40,000 can be maintained by maintaining the total contribution at the present level i.e. Rs. 14,40,000.

(i) Reducing Selling Price by 5%

New Selling Price (Rs. 80 − 5% of Rs. 80) = Rs. 76

New Dealer's Margin (10% of Rs. 76) = Rs. 7.60

New Variable Cost (Rs. 32 + Rs. 24 + Rs. 7.60) = Rs. 63.60

New Contribution per unit (Rs. 76 − Rs. 63.60) = Rs. 12.40

Level of sales required for present level of Profits = $\frac{\text{Total Contribution Required}}{\text{New Contribution per unit}}$

= $\frac{\text{Rs. } 14,40,000}{\text{Rs. } 12.40}$

= 1,16,129 units

(ii) Increasing Dealer's Margin by 20%

New Dealer's Margin after increasing it by 20% = Rs. 8 + (20% of Rs. 8)

= Rs. 9.60

New Variable Cost (Rs. 32 + Rs. 24 + Rs. 9.60) = Rs. 65.60

Contribution (Rs. 80 − Rs. 65.60) = Rs. 14.40

Level of sales required for present level of Profits = $\frac{\text{Total Contribution Required}}{\text{New Contribution per unit}}$

= $\frac{\text{Rs. } 14,40,000}{\text{Rs. } 14.40}$

= 1,00,000 units

Conclusion:

The second proposal, i.e., increasing the Dealer's Margin is recommended because:

1. The contribution per unit is higher which is Rs. 14.40 in comparison to Rs. 12.40 in the first proposal; and
2. The sales (in units) required to earn the same level of profit are lower. They are at 1,00,000 units as against 1,16,129 units in the first proposal. This means a lower sales effort and less finance would be required for implementing proposal (ii) as against proposal (i). Of course, under proposal (ii) the company can earn higher profits than at present level if it can increase its sales beyond 1,00,000 units.

ICAI · Old P3 · MTP Jul 2021 · Series II · question p. 6 · answer p. 9-10

14.2 Break-even point, margin of safety and angle of incidence

62 questions · asked in 43 ICAI documents · 9 exam papers

962 EXAM **May 2026** MCQ 15 2 marks MCQ

The following information has been extracted from the records of XYZ Ltd. for two consecutive years:

Particulars	Year 1	Year 2
Sales (₹)	1,80,00,000	2,40,00,000
Profit (₹)	24,00,000	42,00,000

What is the Break-even Sales Value?

A ₹ 1,40,00,000

B ₹ 1,00,00,000

C ₹ 1,25,00,000

D ₹ 90,00,000

ICAI'S ANSWER

Answer key: **(B)**

ICAI · P4 · May 2026 exam · question p. 7 · answer p. 7

963 EXAM **May 2026** Q2(b) 7 marks

PQR Ltd is a manufacturing company that operates at 80% capacity, generating a profit of ₹ 7,50,000 by

selling 36000 units. The cost details are as follows:

Particulars	Amount (in ₹)
Material Cost per unit	25
Labour Cost per unit	15
Semi Variable Cost (Variable Component = 10 per unit)	8,20,000
Contribution per unit	50
Margin of Safety Sales at the current level of sales	15,00,000
Fixed Cost upto 80% Capacity (Including depreciation of ₹ 1,90,000)	5,90,000
Additional Fixed Cost (Beyond 80% Capacity)	1,50,000

You are required to ascertain:

(i) Break Even Point in units at current Capacity utilisation.

(ii) Total Sales value and Cash Break Even Sales Quantity at current capacity utilization.

(iii) Capacity utilisation & Margin of safety sales, if the company wants to earn a profit of ₹ 9,37,500. (Assume selling price per unit and variable cost per unit remain unchanged.)

ICAI'S ANSWER

Workings:

1. No. of units at 80% level = 36,000 units.

No. of units at 100% level = $\frac{36,000}{80\%}$ = 45,000 units.

2. Contribution per unit = ₹ 50

Variable cost per unit = Direct Material + Direct wages +Variable cost in semi variable component

= (25 + 15 + 10) = ₹ 50 per unit

Selling Price (SP) = Variable cost per unit+ Contribution per unit

= ₹ 50 + ₹ 50 = ₹ 100 per unit.

3. Calculation of Total fixed cost:

= Fixed cost part of semi-variable cost + Fixed overheads

= (Total Semi-variable cost at 80% level – Variable cost part) + Fixed Cost (at 80% capacity)

= {₹ 8,20,000 – (₹ 10 × 36,000 units)} + ₹ 5,90,000

= ₹ 4,60,000 + ₹ 5,90,000 = ₹ 10,50,000

(i) Calculation of Break-even point at 80% Capacity:

$$= \frac{\text{Total fixed cost}}{\text{Contribution per unit}}$$

$$= \frac{₹10,50,000}{₹50} = 21,000 \text{ units}$$

(ii) Total Sales Value & Cash BEP at 80% Capacity

Total Sales Value = 36,000 units × ₹ 100 = ₹ 36,00,000

Now, Cash Fixed Cost = Total Fixed Cost - Depreciation

Or Cash Fixed Cost = ₹10,50,000 - ₹ 1,90,000 = ₹ 8,60,000

Cash BEP (units) = $\frac{\text{Total cash fixed cost}}{\text{Contribution per unit}} = \frac{₹8,60,000}{50}$

= 17,200 units

(iii) Capacity utilisation & Margin of safety (MOS) sales to earn a profit of ₹ 9,37,500

To earn a profit of ₹ 9,37,500, the company must exceed 80% capacity, requiring additional fixed costs.

So, New Fixed Cost (>80%) = ₹ 10,50,000 + ₹ 1,50,000 = ₹ 12,00,000.

Required Units = $\frac{\text{Fixed cost} + \text{Target profit}}{\text{Contribution per unit}}$

= $\frac{₹12,00,000 + ₹9,37,500}{₹50}$

= 42,750 units

So, Capacity Utilisation = (42,750 / 45,000) × 100 = 95%

New BEP (units) = $\frac{₹12,00,000}{₹50}$ = 24,000 units.

So, MOS (units) = 42,750 - 24,000 = 18,750 units

ICAI · P4 · May 2026 exam · question p. 13 · answer p. 14-16

964 EXAM May 2026 Q5(b) 4 marks

The following information has been furnished by the management accountant of CMC Limited:

Margin of Safety	5600 units
Break even sales	2400 units
Margin of Safety	₹ 9,80,000
Total Cost	₹ 11, 94, 200

You are required to calculate:

(i) Profit

(ii) Profit Volume (P/V) Ratio

(iii) Break even sales value

(iv) Fixed Cost

ICAI'S ANSWER

Working Notes:

Total Sales Units = Margin of Safety (Units) + Break-even Sales (Units)

= 5,600 + 2,400 = 8,000 units

Selling Price per Unit = $\frac{\text{Margin of Safety (MoS in Value)}}{\text{Margin of Safety (Units)}}$

= ₹ 9,80,000 / 5,600 = ₹ 175 per unit

Total Sales Value = Total Units × Selling Price = 8,000 × ₹ 175

= ₹ 14,00,000

(i) Profit

= Total Sales – Total Cost

= ₹ 14,00,000 – ₹ 11,94,200 = ₹ 2,05,800

(ii) Profit Volume (P/V) Ratio

$$= \frac{\text{Profit}}{\text{Margin of Safety (Value)}} \times 100$$

= (₹ 2,05,800 / ₹ 9,80,000) × 100 = 21%

(iii) Break-even Sales Value

= Break-even Units × Selling Price

= 2,400 units × ₹ 175 = ₹ 4,20,000

Alternatively,

= Total Sales × [100 – Margin of Safety %]

= ₹ 14,00,000 × 0.30 = ₹ 4,20,000

(iv) Fixed Cost

= Break-even Sales Value × P/V Ratio

= ₹ 4,20,000 × 21% = ₹ 88,200

ICAI · P4 · May 2026 exam · question p. 24 · answer p. 27-28

965 EXAM Jan 2025 Case scenario · Q1(a)(i) to Q1(a)(ii) 5 marks

CASE SCENARIO

XYZ Company has an option to buy any one of the two machines N or M to manufacture its unique industrial component P. Each of the machines have the capacity to produce same quality of component P and are almost identical except for the fact that they are being manufactured by a different manufacturers. The specifications for each Machine are:

Machine M: It has the capacity to produce 50,000 components of P per annum, the fixed costs being 1,50,000 and could generate a profit of 2,25,000 on the sale of all the components produced.

Machine N: It is also having the equal capacity to produce same number of components as that of Machine M per annum and all the components thus produced could be sold in the open market without any difficulty. Fixed cost of Machine N is 60,000 less than that of Machine M and yield a profit of 1,60,000 by selling all the components that are produced.

The selling price of each component of P is 100.

Required:

Q1(a)(i) 3 marks

Calculate break even sales in value for each machine.

Q1(a)(ii) 2 marks

Calculate sales levels in units where both the machines are equally profitable.

ICAI'S ANSWERS

Q1(a)(i)

	Machine M			Machine N		
	Quantity	**Price**	**Amount (₹)**	**Quantity**	**Price**	**Amount (₹)**
Sales	50,000	100	50,00,000	50,000	100	50,00,000
Variable cost	50,000	92.5	46,25,000	50,000	95	47,50,000
Contribution		7.5	3,75,000		5	2,50,000
Fixed Cost			1,50,000			90,000
Profit			2,25,000			1,60,000
P/V Ratio						
Contribution x 100/Sales			3,75,000 x100/50,00,000			2,50,000 x100/50,00,000
			7.5%			5%
Break even Sales						
FixedCost/P / V Ratio			20,00,000			18,00,000

Q1(a)(ii)

Let the units be x

7.5x – 1,50,000 = 5x – 90,000

2.5x = 60,000

X = 24,000 Units

ICAI · P4 · Jan 2025 exam · question p. 7 · answer p. 9

966 EXAM **May 2024** Q5(b) 4 marks

The following information is given by PQR Ltd:

Year	Sales (₹)	Profit (Loss (₹)
2022-23	1,80,00,000	(3,80,000)
2023-24	2,40,00,000	11,20,000

You are required to:

(i) Calculate the Break even sales.

(ii) In 2024-25, it is estimated that the variable cost will go up by 5% and fixed cost will reduce by ₹ 4,80,000. Selling price will remain same. Calculate the sales volume to earn a profit of ₹ 15,00,000.

ICAI'S ANSWER

(i) Break-even sales = $\frac{\text{Fixed Cost}}{\text{P/V Ratio}}$

P/V Ratio = $\frac{\text{Change in Profit}}{\text{Change in Sales}}$ ×100 or, $\frac{15,00,000}{2,40,00,000 - 1,80,00,000}$ ×100

Or, $\frac{15,00,00}{60,00,000}$ ×100 or, 25%

Fixed Cost = Contribution – Profit

= ₹ 2,40,00,000 × 25% - ₹ 11,20,000

= ₹ 60,00,000 – ₹ 11,20,000 = ₹ 48,80,000

Break-even sales = $\frac{48,80,000}{25\%}$ = ₹ 1,95,20,000

(iii) Desired Contribution in 2024-25 = Revised Fixed Cost + Target Profit

= (₹ 48,80,000- ₹ 4,80,000) + ₹ 15,00,000

= ₹ 59,00,000

Earlier P/V ratio = 25%. So Variable Cost ratio =75%.

Selling price remain the same.

Variable cost increased by 5% i.e. Variable Cost ratio will be 78.75% (75+5%of 75).

Now revised P/V ratio=21.25%

Sales Volume in 2024-25=$\frac{59,00,000}{21.25\%}$ = ₹ 2,77,64,706(approx.)

If it is assumed that variable cost will go up by 5% on total. So, it will be increased from 75% to 80% and solution can be done in following way:

(i) Desired Contribution in 2024-25 = Revised Fixed Cost + Target Profit

= (₹ 48,80,000-₹ 4,80,000) + ₹ 15,00,000

= ₹ 59,00,000

Earlier P/V ratio = 25%. So Variable Cost ratio =75%.

Selling price remain the same.

Variable cost increased by 5% i.e. Variable Cost ratio will be 80% (75%+5%).

Now revised P/V ratio=20%

Sales Volume in 2024-25==$\frac{59,00,000}{20\%}$ = ₹ 2,95,00,000

ICAI · P4 · May 2024 exam · question p. 20-21 · answer p. 23-24

967 **EXAM** **Old P3 May 2023** Q1(c) 5 marks old scheme

The following information pertains to ZB Limited for the year:

Profit volume ratio	30%
Margin of Safety (as % of total sales	25%
Fixed cost	₹ 12,60,000

You are required to calculate:

(i) Break even sales value (₹).

(ii) Total sales value (₹) at present,

(iii) Proposed sales value (₹) if company wants to earn the present profit after reduction of 10% in fixed cost,

(iv) Sales in value (₹) to be made to earn a profit of 20% on sales assuming fixed cost remains unchanged,

(v) New Margin of Safety if the sales value at present as computed in (ii) decreased by 12.5%.

ICAI'S ANSWER

(i) Calculation of Break-even sales in value:

= Fixed Cost ÷ P/V Ratio

= ₹ 12,60,000 ÷ 30% **= ₹ 42,00,000**

(ii) Calculation of Total Sales value:

Sales value (S) = Break-even Sales + Margin of Safety

Or, S = 42,00,000 + 0.25 S

Or, 0.75 S = 42,00,000

Or, S = 42,00,000 ÷ 0.75

Or, Sales = **₹ 56,00,000**

(iii) Calculation of proposed sales value to earn present profit:

Present profit = Sales – Variable cost – Fixed Cost

= ₹ 56,00,000 – 70% of 56,00,000 – ₹ 12,60,000

= ₹ 56,00,000 – ₹ 39,20,000 – ₹ 12,60,000

= ₹ 4,20,000

Proposed Sales value (S) = 0.7S + (90% of ₹ 12,60,000) + 4,20,000

S = 0.7S + 11,34,000 + 4,20,000

S = 15,54,000 ÷ 0.3 = **₹ 51,80,000**

(iv) Calculation of sales value to earn 20% on sales:

Sales Value (S) = 0.7 S + 12,60,000 + 0.2S

S = 12,60,0000 ÷ 0.10 = **₹ 1,26,00,000**

(v) New Margin of Safety:

= (Sales – BES) ÷ Sales

= (87.5% of 56,00,000 – 42,00,000) ÷ (87.5% of 56,00,000)

= (49,00,000 – 42,00,000) ÷ 49,00,000

= 7,00,000 ÷ 49,00,000 **= 14.29%**

Or

= (Sales – BES)

= (87.5% of 56,00,000 – 42,00,000)

= ₹ 7,00,000

ICAI · Old P3 · May 2023 exam · question p. 2 · answer p. 4-5

968 EXAM Old P3 Nov 2022 Q1(d) 5 marks old scheme

ABC Ltd sells its Product 'Y' at a price of ₹ 300 per unit and its variable cost is ₹ 180 per unit. The fixed costs are ₹ 16,80,000 per year uniformly incurred throughout the year. The Profit for the year is ₹ 7,20,000.

You are required to calculate:

(i) BEP in value (₹) and units.

(ii) Margin of Safety

(iii) Profits made when sales are 24,000 units.

(iv) Sales in value (₹) to be made to earn a net profit of ₹ 10,00,000 for the year.

ICAI'S ANSWER

(i) Calculation of BEP in value

$$\text{P / V ratio} = \frac{\text{Sales price - Variable Cost}}{\text{Sales}} = \frac{300 - 180}{300} = 40\%$$

$$\text{Break Even Point in Value (₹)} = \frac{\text{Fixed Cost}}{\text{P / V ratio}} = \frac{16,80,000}{40\%} = ₹\ 42,00,000$$

$$\text{Break Even Point in Units} = \frac{\text{Fixed Cost}}{\text{Contribution}} = \frac{16,80,000}{120} = 14,000 \text{ Units}$$

$$\text{(Alternatively, } \frac{₹\ 42,00,000}{300} = 14000 \text{ units)}$$

(ii) Margin of safety (In Amount) $= \frac{\text{Profit}}{\text{P / V ratio}} = \frac{7,20,000}{40\%} = ₹\ 18,00,000$

Margin of safety may also be calculated by deducting BEP sales from present sale. Present sale is ₹ 60,00,000 i.e. (16,80,000 + 7,20,000)/40%.

Margin of safety (In units) $= \frac{\text{Profit}}{\text{Contribution per unit}} = \frac{7,20,000}{120} = 6,000$ units

(iii) Profit when sales are 24,000 units

Particular	(₹)
Contribution (24,000 × 120)	28,80,000
Less: Fixed cost	16,80,000
Profit	12,00,000

(iv) Sales in value to earn a net profit of ₹10,00,000

$$\frac{\text{Fixed Cost + Desired profit}}{\text{P / V Ratio}} = \frac{16,80,000 + 10,00,000}{40\%} = ₹\ 67,00,000$$

ICAI · Old P3 · Nov 2022 exam · question p. 2 · answer p. 4-5

969 EXAM Old P3 May 2022 Q4(b) 5 marks old scheme

UV Limited started a manufacturing unit from 1st October 2021. It produces designer lamps and sells its lamps at ₹ 450 per unit.

During the quarter ending on 31st December, 2021, it produced and sold 12,000 units and suffered a loss of ₹ 35 per unit.

During the quarter ending on 31st March, 2022, it produced and sold 30,000 units and earned a profit of ₹ 40 per unit.

You are required to calculate:

(i) Total fixed cost incurred by UV ltd. per quarter.

(ii) Break Even sales value (in rupees)

(iii) Calculate Profit, if the sale volume reaches 50,000 units in the next quarter (i.e., quarter ending on 30th June, 2022).

ICAI'S ANSWER

	Quarter ending 31st December, 2021 (₹)	Quarter ending 31st March, 2022 (₹)
Sales (No. of units sold x ₹ 450 per unit)	54,00,000	1,35,00,000
Profit (Loss)	(4,20,000) [12,000 × 35]	12,00,000 [30,000 × 40]

$$\text{P/V Ratio} = \frac{\text{Change in profit}}{\text{Change in Sales}} \times 100$$

$$\therefore \frac{16,20,000}{81,00,000} \times 100 = \mathbf{20\%}$$

(i) Fixed Cost = Sales × P/V ratio – profit

= ₹ 1,35,00,000 × 20% – 12,00,000

= ₹ 15,00,000

Alternative Presentation for the calculation of Fixed cost

	Quarter ending 31st December, 2021 (₹)	Quarter ending 31st March, 2022 (₹)
Sales (No. of units sold x ₹ 450 per unit)	54,00,000	1,35,00,000
Profit (Loss)	(4,20,000) [12,000 × 35]	12,00,000 [30,000 × 40]
Total cost	58,20,000	1,23,00,000

VC per unit = (1,23,00,000 – 58,20,000) / (30,000 – 12,000)

= 64,80,000 / 18,000 =₹ 360 per unit

Fixed cost = TC – VC , 58,20,000 (360 x12,000 units) ₹15,00,000

(ii) Break even sales value (in Rupees) = $\frac{\text{Fixed cost}}{\text{P/V ratio}} \times 100$

$= \frac{15,00,000}{20\%} =$ **₹75,00,000**

(iii) Profit, if sales reach 50,000 units for the quarter ending 30th June, 2022

	(₹)
Sales (50,000 × ₹ 450)	2,25,00,000
Less: Variable cost	1,80,00,000
Contribution	45,00,000
Less: Fixed cost	15,00,000
Profit	30,00,000

ICAI · Old P3 · May 2022 exam · question p. 17 · answer p. 19-20

970 EXAM **Old P3 Dec 2021** Q2(b) 10 marks old scheme

AZ company has prepared its budget for the production of 2,00,000 units. The variable cost per unit is ₹ 16 and fixed cost is ₹ 4 per unit. The company fixes its selling price to fetch a profit of 20% on total cost.

You are required to calculate:

(i) Present break-even sales (in ₹ and in quantity).

(ii) Present profit-volume ratio.

(iii) Revised break-even sales in ₹ and the revised profit-volume ratio, if it reduces its selling price by 10%.

(iv) What would be revised sales- in quantity and the amount, if a company desires a profit increase of 20% more than the budgeted profit and selling price is reduced by 10% as above in point (iii).

ICAI'S ANSWER

Variable Cost per Unit=₹16

Fixed Cost per Unit =₹ 4, Total Fixed Cost= 2,00,000 units x ₹ 4 = ₹8,00,000

Total Cost per Unit =₹20

Selling Price per Unit=Total Cost+ Profit =₹ 20+₹ 4 =₹ 24

Contribution per Unit=₹ 24-₹16=₹ 8

(i) Present Break-even Sales (Quantity) = $\frac{\text{Fixed cost}}{\text{Contribution margin per unit}} = \frac{₹8,00,000}{₹8}$

= 1,00,000 units

Present Break-even Sales (₹) = 1,00,000 units × ₹ 24 = ₹ 24,00,000

(ii) Present P/V Ratio = $\frac{8}{24} \times 100 = 33.33\%$

(iii) Revised Selling Price per Unit = ₹ 24 – 10% of ₹ 24 = ₹ 21.60

Revised Contribution per Unit=₹ 21.60-₹ 16 = ₹ 5.60

Revised P/V Ratio = $\frac{5.60}{21.60} \times 100 = 25.926\%$

Revised Break-even point (₹) = $\frac{\text{Fixed cost}}{\text{P/V ratio}} = \frac{8,00,000}{25.926\%}$ = ₹ 30,85,705

Or

Revised Break-even point (units) = $\frac{\text{Fixed cost}}{\text{Contribution margin per unit}} = \frac{8,00,000}{5.60}$ = 1,42,857 units

Revised Break-even point (₹) = 1,42,857 units x ₹ 21.60 = ₹ 30,85,711

(iv) Present profit =₹ 8,00,000

Desired Profit = 120% of ₹ 8,00,000 =₹ 9,60,000

Sales to earn a profit of ₹ 9,60,000

Total contribution required = 8.00.000 + 9,60,000 = ₹ 17,60,000

$\frac{\text{Fixed cost} + \text{Desired profit}}{\text{Contribution per unit}} = \frac{8,00,000 + 9,60,000}{5.60}$ = 3,14,286 units

Revised sales (in ₹) = 3,14,286 units x ₹ 21.60 = ₹ 67,88,578

ICAI · Old P3 · Dec 2021 exam · question p. 6 · answer p. 8-9

971 EXAM **Old P3 Jul 2021** Q6(b) 5 marks old scheme

What is Margin of Safety? What does a large Margin of Safety indicates? How can you calculate Margin of Safety?

ICAI'S ANSWER

Margin of Safety: The margin of safety can be defined as the difference between the expected level of sale and the breakeven sales.

The larger the margin of safety, the higher is the chances of making profits.

The Margin of Safety can be **calculated by** identifying the difference between the projected sales and breakeven sales in units multiplied by the contribution per unit. This is possible because, at the breakeven point all the fixed costs are recovered and any further contribution goes into the making of profits.

Margin of Safety = (Projected sales – Breakeven sales) in units x contribution per unit

It also can be calculated as:

Margin of Safety = $\frac{\text{Profit}}{\text{P / V Ratio}}$

ICAI · Old P3 · Jul 2021 exam · question p. 78 · answer p. 79

972 EXAM **Old P3 Jan 2021** Q1(a) 5 marks old scheme

During a particular period ABC Ltd has furnished the following data:

Sales ₹ 10,00,000

Contribution to sales ratio 37% and

Margin of safety is 25% of sales.

A decrease in selling price and decrease in the fixed cost could change the "contribution to sales ratio" to 30% and "margin of safety" to 40% of the revised sales. Calculate:

(i) Revised Fixed Cost.

(ii) Revised Sales and

(iii) New Break-Even Point.

ICAI'S ANSWER

Contribution to sales ratio (P/V ratio) = 37%

Variable cost ratio = 100% - 37% = 63%

Variable cost = ₹ 10,00,000 x 63% = ₹ 6,30,000

After decrease in selling price and fixed cost, sales quantity has not changed. Thus, variable cost is ₹ 6,30,000.

Revised Contribution to sales = 30%

Thus, Variable cost ratio = 100% – 30% = 70%

Thus, Revised sales = $\frac{₹\ 6,30,000}{70\%}$ = ₹ 9,00,000

Revised, Break-even sales ratio = 100% – 40% (revised Margin of safety) = 60%

(i) Revised fixed cost = revised breakeven sales x revised contribution to sales ratio

= ₹ 5,40,000 (₹ 9,00,000 x 60%) x 30%

= ₹ 1,62,000

(ii) Revised sales = ₹ 9,00,000 (as calculated above)

(iii) Revised Break-even point = Revised sales x Revised break-even sales ratio

= ₹ 9,00,000 x 60%

= ₹ 5,40,000

ICAI · Old P3 · Jan 2021 exam · question p. 1 · answer p. 3

973 EXAM **Old P3 Jan 2021** Q3(a) 10 marks old scheme

Two manufacturing companies A and B are planning to merge. The details are as follows:

	A	B
Capacity utilisation (%)	90	60
Sales (₹)	63,00,000	48,00,000
Variable Cost (₹)	39,60,000	22,50,000
Fixed Cost (₹)	13,00,000	15,00,000

Assuming that the proposal is implemented, calculate:

(i) Break-Even sales of the merged plant and the capacity utilization at that stage.

(ii) Profitability of the merged plant at 80% capacity utilization.

(iii) Sales Turnover of the merged plant to earn a profit of ₹ 60,00,000.

(iv) When the merged plant is working at a capacity to earn a profit of ₹ 60,00,000, what percentage of increase in selling price is required to sustain an increase of 5% in fixed overheads.

ICAI'S ANSWER

Workings:

1. Statement showing computation of Breakeven of merged plant and other required information

S. No.	Particulars	Plan A		Plant B		Merged Plant (100%) (₹)
		Before (90%) (₹)	After (100%) (₹)	Before (60%) (₹)	After (100%) (₹)	
(i)	Sales	63,00,000	70,00,000	48,00,000	80,00,000	1,50,00,000
(ii)	Variable cost	39,60,000	44,00,000	22,50,000	37,50,000	81,50,000
(iii)	Contribution (i - ii)	23,40,000	26,00,000	25,50,000	42,50,000	68,50,000
(iv)	Fixed Cost	13,00,000	13,00,000	15,00,000	15,00,000	28,00,000
(v)	Profit (iii - iv)	10,40,000	13,00,000	10,50,000	27,50,000	40,50,000

2. PV ratio of merged plant = $\frac{\text{Contribution}}{\text{Sales}}$ x 100

= $\frac{₹\ 68,50,000}{₹\ 1,50,00,000}$ x 100 = 45.67 %

(i) Break even sales of merged plant = $\frac{\text{Fixed Cost}}{\text{P/V Ratio}}$

= $\frac{₹\ 28,00,000}{45.67\%}$

= ₹ 61,30,939.34 (approx.)

Capacity utilisation = $\frac{₹\ 61,30,939.34}{₹\ 1,50,00,000}$ x 100= 40.88%

(ii) Profitability of the merged plant at 80% capacity utilisation

= (₹ 1,50,00,000 x 80%) x P/v ratio – fixed cost

= ₹ 1,20,00,000 x 45.67% – ₹ 28,00,000

= ₹ 26,80,400

(iii) Sales to earn a profit of ₹ 60,00,000

Desired sales = $\frac{\text{Fixed Cost} + \text{desired profit}}{\text{P/V Ratio}}$

= $\frac{₹\ 28,00,000 + ₹\ 60,00,000}{45.67\%}$

= ₹ 1,92,68,666 (approx.)

(iv) Increase in fixed cost

= ₹ 28,00,000 x 5% = ₹ 1,40,000

Therefore, percentage increase in sales price

= $\frac{₹\ 1,40,000}{₹\ 1,92,68,666}$ x 100 = 0.726% (approx.)

ICAI · Old P3 · Jan 2021 exam · question p. 11-12 · answer p. 13-14

974 RTP Sep 2026 Q4 MCQ

The current selling price is ₹ 136 per unit. Variable costs are expected to increase from ₹ 76 to ₹ 82 per unit. Fixed costs or 5,32,000 will not change. How many additional units are required to be sold in the order to maintain an operating income of ₹ 3,45,500?

A 1,500 units

B 1,462 units

C 1,625 units

D 1,250 units

ICAI'S ANSWER

(C) 1,625 units

Projected Unit sales = (Fixed costs + Target operating income) ÷ unit Contribution margin.

= (₹ 5,32,000 + ₹ 3,45,500) ÷ (₹ 136 - ₹82)

= ₹ 8,77,500 ÷ 54

= 16,250 units

Existing Sales in units = (Fixed costs + operating income) ÷ unit Contribution margin.

= (₹ 5,32,000 + ₹ 3,45,500) ÷ (₹ 136 - ₹ 76)

= ₹ 8,77,500 ÷ 60

= 14,625 units

Increase in units: 16,250 – 14,625 = 1,625 units

ICAI · P4 · RTP Sep 2026 · question p. 7 · answer p. 25-26

975 RTP Sep 2025 Q6 MCQ

A company's sales could decrease by 50% before it starts incurring losses. And, for every Re. 1 of sales, it can contribute 40 paise towards fixed costs and generating profit. For the current year, the company's fixed cost amounts to ₹ 5,00,000.

CALCULATE the Projected sales.

A ₹ 10,00,000

B ₹ 12,50,000

C ₹ 20,83,333

D ₹ 25,00,000

ICAI'S ANSWER

Answer key: **(D)**

(d) ₹ 25,00,000

Margin of Safety = 50%

P/V ratio = 40%

Fixed Cost = ₹ 5,00,000

Break-even Sales (BES) = $\frac{\text{Fixed Cost}}{\text{P/V ratio}}$

= $\frac{₹\ 5,00,000}{0.40}$

BES = ₹ 12,50,000

Margin of Safety = Projected sales(S) – Break Even Sales(BES)

S = BES + Margin of Safety

S = ₹ 12,50,000 + (0.50 x S)

Or, S – 0.50S = ₹ 12,50,000

Or, S = **₹ 25,00,000**

ICAI · P4 · RTP Sep 2025 · question p. 9 · answer p. 30

976 RTP **May 2025** Q18

Gourmet Food Products is a new entrant in the market for chocolates. It has introduced a new product—Sweetee. This is a small rectangular chocolate bar. The bars are wrapped in aluminium foil and packed in attractive cartons containing 50 bars. A carton, is therefore, considered the basic sales unit. Although management had made detailed estimates of costs and volumes prior to undertaking this venture, new projections based on actual cost experience are now required.

Income Statements for the last two quarters are each thought to be representative of the costs and productive efficiency we can expected in the next few quarter. There were virtually no inventories on hand at the end of each quarter. The income statements reveal the following:–

	First Quarter (₹)	Second Quarter (₹)
Sales :		
50,000 × ₹ 24	12,00,000	—
70,000 × ₹ 24	—	16,80,000
Less: Cost of Goods Sold	7,00,000	8,80,000
Gross Margin	5,00,000	8,00,000
Less: Selling and Administration	6,50,000	6,90,000
Net Income / (Loss) before Taxes	(1,50,000)	1,10,000
Less: Tax	(60,000)	44,000
Net Income / (Loss)	(90,000)	66,000

The firm's overall marginal and average income tax rate is 40%. This 40% figure has been used to estimate the tax liability arising from the chocolate operations.

REQUIRED:

(a) Management would like to know the breakeven point in terms of quarterly carton sales for the chocolates.

(b) Management estimates that there is an investment of ₹ 30,00,000 in this product line. What quarterly carton sales and total revenue are required in each quarter to earn an after tax return of 20% per annum on investment?

(c) The firm's marketing people predict that if the selling price is reduced by ₹ 1.50 per carton (₹ 0.03 off per chocolate bar) and a ₹ 1,50,000 advertising campaign among school children is mounted, sales will increase by 20% over the second quarter sales. Should the plan be implemented?

ICAI'S ANSWER

(a) Estimation of the Fixed and Variable Costs.

Variable Manufacturing Cost per carton:

$$= \frac{\text{Change in Costs}}{\text{Change in Activity}}$$

$$= \frac{₹8,80,000 - ₹7,00,000}{70,000 - 50,000}$$

$$= \frac{₹\ 1,80,000}{20,000}$$

= ₹ 9 per carton

Fixed Manufacturing Costs:

Costs of Goods Sold = Fixed Manufacturing Cost + Variable Manufacturing Cost

₹ 7,00,000 = Fixed Manufacturing Cost + (50,000 Cartons × ₹9)

Fixed Manufacturing Cost = ₹7,00,000 – ₹4,50,000

= ₹2,50,000

Variable Selling and Administration Cost per unit:

$$= \frac{₹6,90,000 - ₹6,50,000}{70,000 - 50,000}$$

$$= \frac{₹\ 40,000}{20,000}$$

= ₹ 2 *per unit*

Fixed Selling & Administration Costs:

Total Selling & Admn. Costs = Fixed Selling & Admn. Cost + Variable Selling & Admn. Costs

₹6,50,000 = Fixed Selling & Admn. Costs + (50,000 Cartons × ₹ 2)

Fixed Selling & Admn. Cost = ₹6,50,000 – ₹1,00,000

= ₹5,50,000

So the Total Variable Costs *per unit* are ₹ 11 *per unit* (₹ 9 + ₹ 2).

Total Fixed Costs are ₹8,00,000 *per quarter* (₹ 2,50,000 + ₹ 5,50,000).

Given Sale Price of ₹24 *per carton* and Variable Costs of ₹ 11 *per carton*, the Contribution *per carton* is ₹ 13 (₹ 24 – ₹ 11).

Breakeven Point (in terms of carton units)

$$= \frac{\text{Fixed cost (per quarter)}}{\text{Contribution per Carton}}$$

$$= \frac{₹\ 8,00,000}{₹\ 13}$$

= 61,539 Cartons

(b) To earn an After Tax Return of 20% on ₹30,00,000, the Desired Annual After Tax Net Income is ₹6,00,000 (₹30,00,000 × 20%). The Quarterly After Tax Net Income will be ₹1,50,000. Given the Tax Rate of 40%, the Pre-tax Return will be ₹ 2,50,000 (₹ 1,50,000 × 100/60).

$$\text{Quarterly Sales (units)} = \frac{\text{Fixed Cost} + \text{Desired Return}}{\text{Contribution per unit}}$$

$$= \frac{₹\ (8{,}00{,}000 + 2{,}50{,}000)}{₹\ 13}$$

$$= \frac{₹\ 10{,}50{,}000}{₹\ 13}$$

= 80,769 Cartons

Quarterly Sales Revenue = ₹19,38,456 (80,769 Cartons × ₹ 24)

(c) The proposal involves reducing Selling Price from ₹24 *per carton* to ₹ 22.50 *per carton*. Hence the Contribution *per carton* will be ₹ 11.50 (₹22.50 – ₹11.00).

The increase in Advertising Costs will push Fixed Costs up by ₹1,50,000 to ₹9,50,000.

A 20% increase over second quarter's Sales would increase Sales form 70,000 cartons to 84,000 cartons.

The Expected Earnings Before Taxes will be ₹ 16,000 [(84,000 Cartons × ₹ 11.50) – ₹ 9,50,000].

After deducting Tax at 40%, the Net Income will be ₹9,600 (₹ 16,000 – ₹ 6,400).

Earning has reduced from ₹ 66,000 to ₹ 9,600, accordingly this plan should not be implemented.

ICAI · P4 · RTP May 2025 · question p. 22-23 · answer p. 49-51

977 RTP Jan 2025 Q7 MCQ

Ms. Gauri has the business of selling pens. She has setup this pen retailing for over 10 years with good profit volume ratio. Her average cost from the retailing is ₹ 11.25 per unit if she sells 16,000 units and is ₹ 11 per unit if she sells 20,000 units.

For the current month, she also charged ₹ 5,000 towards depreciation and the rental payment due.

The excess of sales revenue over the variable costs is ₹ 3.333 per unit.

You are required to CALCULATE Break-even Point (in units), Cash Break-even Point (in units) and Profit Volume Ratio.

A Break-even Point- 6,000 units, Cash Break-even Point- 6,000 units and Profit Volume Ratio- 33.33%

B Break-even Point- 6,000 units, Cash Break-even Point- 4,500 units and Profit Volume Ratio- 25%

C Break-even Point- 4,500 units, Cash Break-even Point- 4,500 units and Profit Volume Ratio- 33.33%

D Break-even Point- 4,500 units, Cash Break-even Point- 4,500 units and Profit Volume Ratio- 25%

ICAI'S ANSWER

(B) $$\text{Variable cost per unit} = \frac{\text{Change in Total cost}}{\text{Change in units}}$$

$$= \left(\frac{(₹11 \text{ x } 20{,}000 \text{ units}) - (₹11.25 \text{ x } 16{,}000 \text{ units})}{20{,}000 \text{ units} - 16{,}000 \text{ units}}\right)$$

$$= \left(\frac{₹2{,}20{,}000 - ₹1{,}80{,}000}{4{,}000 \text{ units}}\right) = ₹\ 10$$

Fixed cost = Total Cost – Variable cost (at 20,000 units level)

= (₹ 11 x 20,000 units) – (₹ 10 x 20,000 units)

= ₹ 20,000

(i) Break-even Point (in units) = $\left(\frac{\text{Fixed Costs}}{\text{Contribution per unit*}}\right)$

= $\left(\frac{₹20,000}{₹3.333}\right)$

= **6,000 units**

* Contribution is the excess of sales revenue over the variable costs.

(ii) Cash Break-even Point (in units) = $\left(\frac{\text{Cash Fixed Costs**}}{\text{Contribution per unit}}\right)$

= $\left(\frac{₹20,000 - ₹5,000}{₹3.333}\right)$

= **4,500 units**

** depreciation and other non-cash fixed costs are excluded from the fixed costs to compute cash break-even point.

(ii) P/V Ratio = $\frac{\text{Contributionperunit}}{\text{Salepriceperunit}}$

= $\left(\frac{₹3.333}{₹10 + ₹3.333}\right)$

= **25%**

ICAI · P4 · RTP Jan 2025 · question p. 9 · answer p. 28-29

978 **RTP** **Sep 2024** Q17

RS Ltd. manufactures and sells a single product X whose selling price is ₹ 100 per unit and the variable cost is ₹ 60 per unit.

(i) If the Fixed Costs for this year are ₹ 24,00,000 and the annual sales are at 60% margin of safety, CALCULATE the rate of net return on sales, assuming an income tax level of 40%

(ii) For the next year, it is proposed to add another product line Y whose selling price would be ₹ 150 per unit and the variable cost ₹ 100 per unit. The total fixed costs are estimated at ₹ 28,00,000. The sales mix of X : Y would be 5 : 3. COMPUTE the break-even sales in units for both the products.

ICAI'S ANSWER

(i) Contribution per unit = Selling price – Variable cost

= ₹ 100 – ₹ 60

= ₹ 40

Break-even Point = $\frac{₹\ 24,00,000}{₹\ 40}$

= 60,000 units

Percentage Margin of Safety = $\frac{\text{Actual Sales} - \text{Break - even Sales}}{\text{Actual Sales}}$

Or, 60% = $\frac{\text{Actual Sales} - 60,000\text{units}}{\text{Actual Sales}}$

∴ Actual Sales = 1,50,000 units

	(₹)
Sales Value (1,50,000 units × ₹ 100)	1,50,00,000
Less: Variable Cost (1,50,000 units × ₹ 60)	(90,00,000)
Contribution	60,00,000
Less: Fixed Cost	(24,00,000)
Profit	36,00,000
Less: Income Tax @ 40%	(14,40,000)
Net Return	21,60,000

Rate of Net Return on Sales = 14.40% $\left(\frac{₹\ 21,60,000}{₹\ 1,50,00,000} \times 100\right)$

(ii) **Products**

	X (₹)	Y (₹)
Selling Price per unit	100	150
Variable Cost per unit	60	100
Contribution per unit	40	50

Composite contribution will be as follows:

Contribution per unit = $\left(\frac{40}{8} \times 5\right) + \left(\frac{50}{8} \times 3\right)$

= 25 + 18.75 = ₹ 43.75

Break-even Sale = 64,000 units $\left(\frac{₹28,00,000}{₹43.75}\right)$

Break-even Sales Mix:

X (64,000 units × 5/8) = 40,000 units

Y (64,000 units × 3/8) = 24,000 units

ICAI · P4 · RTP Sep 2024 · question p. 15 · answer p. 33-34

979 RTP **May 2024** Q11

The analysis of cost sheet of A Ltd. for the last financial year has revealed the following information for

it's product R:

Elements of Cost	Variable Cost portion	Fixed Cost
Direct Material	30% of cost of goods sold	--
Direct Labour	15% of cost of goods sold	--
Factory Overhead	10% of cost of goods sold	₹ 2,30,000
General & Administration Overhead	2% of cost of goods sold	₹ 71,000
Selling & Distribution Overhead	4% of cost of sales	₹ 68,000

Last year 5,000 units were sold at ₹185 per unit.

You being an associate to cost controller of the A Ltd., CALCULATE :

(i) Break-even Sales (in rupees),

(ii) Profit earned during last year,

(iii) Margin of safety (in %) and

(iv) the profit if the sales were 10% less than the actual sales.

ICAI'S ANSWER

Workings:

Calculation of Cost of Goods Sold (COGS):

COGS = {(DM- 0.3 COGS) + (DL- 0.15 COGS) + (FOH- 0.10 COGS + ₹ 2,30,000) + (G&AOH- 0.02 COGS + ₹ 71,000)}

Or COGS = 0.57 COGS + ₹ 3,01,000

Or COGS = $\frac{₹3,01,000}{0.43}$ = ₹ 7,00,000

Calculation of Cost of Sales (COS):

COS = COGS + (S&DOH- 0.04 COS + ₹ 68,000)

Or COS = ₹ 7,00,000 + (0.04 COS + ₹ 68,000)

Or COS = $\frac{₹7,68,000}{0.96}$ = ₹ 8,00,000

Calculation of total Fixed Costs:

Factory Overhead	₹ 2,30,000
General & Administration OH	₹ 71,000
Selling & Distribution OH	₹ 68,000
	₹ 3,69,000

Calculation of Variable Costs:

Direct Material	(0.3 × ₹ 7,00,000)	₹ 2,10,000
Direct Labour	(0.15 × ₹ 7,00,000)	₹ 1,05,000
Factory Overhead	(0.10 × ₹ 7,00,000)	₹ 70,000
General & Administration OH	(0.02 × ₹ 7,00,000)	₹ 14,000
Selling & Distribution OH	(0.04 × ₹ 8,00,000)	₹ 32,000
		₹ 4,31,000

Calculation of P/V Ratio:

$$\text{P/V Ratio} = \frac{\text{Contribution}}{\text{Sales}} \times 100 = \frac{\text{Sales-Variable Costs}}{\text{Sales}} \times 100$$

$$= \frac{(₹185 \times 5{,}000 \text{ units}) - ₹4{,}31{,}000}{₹185 \times 5{,}000 \text{ units}} \times 100 = 53.41\%$$

$$\text{(i) Break-Even Sales} = \frac{\text{Fixed Costs}}{\text{P / V Ratio}} = \frac{₹3{,}69{,}000}{53.41\%} = ₹\ 6{,}90{,}882$$

(ii) Profit earned during the last year

= (Sales – Total Variable Costs) – Total Fixed Costs

= (₹ 9,25,000 - ₹ 4,31,000) - ₹ 3,69,000

= ₹ 1,25,000

$$\text{(iii) Margin of Safety (\%)} = \frac{\text{Sales-Breakeven sales}}{\text{Sales}} \times 100$$

$$= \frac{₹\ 9{,}25{,}000 - ₹\ 6{,}90{,}882}{₹\ 9{,}25{,}000} \times 100 = 25.31\%$$

(iv) Profit if the sales were 10% less than the actual sales:

Profit = 90% (₹ 9,25,000 - ₹ 4,31,000) - ₹ 3,69,000

= ₹ 4,44,600 - ₹ 3,69,000 = ₹ 75,600

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980 **RTP** **Old P3 Nov 2023** Q12(a) old scheme

A dairy product company manufacturing baby food with a shelf life of one year furnishes the following information:

(i) On 1st April, 2023, the company has an opening stock of 20,000 packets whose variable cost is ₹ 180 per packet.

(ii) In 2022-23, production was 1,20,000 packets and the expected production in 2023-24 is 1,50,000 packets. Expected sales for 2023-24 is 1,60,000 packets.

(iii) In 2022-23, fixed cost per unit was ₹ 60 and it is expected to increase by 10% in 2023-24. The variable cost is expected to increase by 25%. Selling price for 2023-24 has been fixed at ₹ 300 per packet.

You are required to calculate the Break-even volume in units for 2023-24.

ICAI'S ANSWER

Working Notes:

Particulars	2022-23 (₹)	2023-24 (₹)
Fixed Cost	72,00,000 (₹ 60 × 1,20,000 units)	79,20,000 (110% of ₹ 72,00,000)
Variable Cost	180	225 (125% of ₹ 180)

Calculation of Break-even Point (in units):

Since, shelf life of the product is one year only, hence, opening stock is to be sold first.

	(₹)
Total Contribution required to recover total fixed cost in 2023-24 and to reach break-even volume.	79,20,000
Less: Contribution from opening stock {20,000 units × (₹ 300 – ₹ 180)}	24,00,000
Balance Contribution to be recovered	55,20,000

Units to be produced to get balance contribution

$$= \frac{₹\ 55{,}20{,}000}{₹\ 300 - ₹\ 225} = 73{,}600 \text{ packets.}$$

Break-even volume in units for 2023-24

	Packets
From 2023-24 production	73,600
Add: Opening stock from 2022-23	20,000
	93,600

ICAI · Old P3 · RTP Nov 2023 · question p. 7-8 · answer p. 22

981 RTP **Old P3 May 2023** Q13 old scheme

The following data are available from the budget records of Finesign Women's Handbag Company for the forthcoming budget period.

	₹
Selling Price per unit	1000
Variable cost per unit:	
Cost of Material used	750.00
Sales commission	50.00
Total Variable Cost	800.00
Annual fixed expenses:	

	₹
Rent	7,00,000
Salaries	11,00,000
Other fixed expenses	5,00,000
Total Fixed Cost	23,00,000

Although the firm manufactures Bags with different styles, they have identical purchase costs and selling price.

Requirement:

(a) What is the annual break-even point both in terms of units and value?

(b) If the store manager is paid 1 per cent commission on sales, what would be the annual break-even point both in terms of units and value?

(c) If the firm decides to pay a fixed salary of ₹ 9,00,000 in lieu of sales commission, what would be the annual break-even point in terms of units and value.

Considering break-even point in requirement (a), If the stores manager is paid 2 per cent commission on each bag sold in excess of the break-even point, what would be the profit if 20000 bags were sold.

ICAI'S ANSWER

(a) P/V ratio: $\frac{\text{Sales per unit - Variable Cost per unit}}{\text{Selling price per unit}} \times 100$

$= \frac{1000 - 800}{1000} \times 100$

$= \frac{200}{1000} \times 100 = 20\%$

Annual BEP in units: $\frac{\text{Annual fixed cost}}{\text{Contribution per unit}}$

$= \frac{₹\ 23,00,000}{₹\ 200} = 11,500$ units

Annual BEP in value: $\frac{\text{Annual fixed cost}}{\text{P / V ratio}}$

$\frac{₹\ 23,00,000}{₹\ 20\%} =$ ₹1,15,00,000

(b) Revised P/V ratio and BEP :

commission on sales per unit= 1% of 1,000= ₹10

So, P/V ratio : $\frac{1000 - (750 + 50 + 10)}{1000}$

$= \frac{190}{1000} \times 100 = 19\%$

BEP in terms of units: $\frac{\text{Annual fixed cost}}{\text{Contribution per unit}}$

$= \frac{23,00,000}{190} = 12,106$ units

BEP in terms of value: $\frac{\text{Annual fixed cost}}{\text{P/V}}$

$= \frac{23,00,000}{19\%} =$ ₹1,21,05,263

(c) Break-even point under fixed salary plan:

P/V ratio = $\frac{\text{Contribution per unit}}{\text{Selling price per unit}} = \frac{1000 - 750}{1000} \times 100 = \frac{250}{1000} \times 100 = 25\%$

Revised fixed cost :

Original fixed cost	₹ 23,00,000
Proposed fixed salary	₹ 9,00,000
Total	₹ 32,00,000

BEP in terms of units: $\frac{\text{Annual fixed cost}}{\text{Contribution per unit}} = \frac{32,00,000}{250} = 12,800 \text{ units}$

BEP in terms of value: $\frac{\text{Annual fixed cost}}{\text{P / v ratio}} = \frac{32,00,000}{25\%} = 1,28,00,000$

(d) Annual break-even point under requirement (a) is 11,500 units.

Margin of safety at sales volume of 20,000 unit of bags (20,000 – 11,500) = 8500 units

Contribution on sales beyond break-even sales:

Revised contribution per unit: 200 – (2% of 1000) = 180

Profit = Margin of safety (in units) × Contribution per unit

= 8500 × 180 = ₹ 15,30,000

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982 RTP **Old P3 Dec 2021** Q13 old scheme

A company has three factories situated in North, East and South with its Head Office in Mumbai. The Management has received the following summary report on the operations of each factory for a period:

(₹ in '000)

Factory	Sales		Profit	
	Actual	**Over / (Under) Budget**	**Actual**	**Over / (Under) Budget**
North	1,100	(400)	135	(180)
East	1,450	150	210	90
South	1,200	(200)	330	(110)

CALCULATE the following for each factory and for the company as a whole for the period:

(i) Fixed Cost

(ii) Break-even Sales

ICAI'S ANSWER

Computation of Profit Volume Ratio

(₹ in '000)

Factory	Sales			Profit			P/V Ratio
	Actual	**Over / (Under) Budget**	**Budgeted Sales**	**Actual**	**Over / (Under) Budget**	**Budget Profit**	$\left(\frac{\text{Change in Profit}}{\text{Change in Sales}}\right)$
North	1,100	(400)	1,500	135	(180)	315	45%
East	1,450	150	1,300	210	90	120	60%
South	1,200	(200)	1,400	330	(110)	440	55%

(i) **Computation of Fixed Costs** (₹ in '000)

Factory	Actual Sales	P/V Ratio	Contribution	Actual Profit	Fixed Cost
	(1)	**(2)**	**(3) = (1) × (2)**	**(4)**	**(5) = (3) - (4)**
North	1,100	45%	495	135	**360**
East	1,450	60%	870	210	**660**
South	1,200	55%	660	330	**330**
Total	3,750		2,025	675	**1,350**

(ii) **Computation of Break-Even Sales**

Factory	Fixed Cost (a)	P/V Ratio (b)	Break-even Sales (a) / (b)
North	360	45%	**800**
East	660	60%	**1,100**
South	330	55%	**600**
			2,500

$$\text{Break-even Sales (Company as Whole)} = \frac{\text{Fixed Cost}}{\text{Composite P / V Ratio *}}$$

$$= \frac{₹\ 13,50,000}{54\%}$$

$$= ₹\ 25,00,000$$

$$\text{*Composite P/V Ratio} = \frac{\text{Total Contribution}}{\text{Total Actual sales}} = \frac{2,025}{3,750} = 54\%$$

ICAI · Old P3 · RTP Dec 2021 · question p. 11-12 · answer p. 31-32

983 RTP **Old P3 Nov 2020** Q13 old scheme

J Ltd. manufactures a Product-Y. Analysis of income statement indicated a profit of ₹ 250 lakhs on a sales volume of 5,00,000 units. Fixed costs are ₹1,000 lakhs which appears to be high. Existing selling price is ₹680 per unit. The company is considering revising the profit target to ₹ 700 lakhs. You are required to COMPUTE –

(i) Break- even point at existing levels in units and in rupees.

(ii) The number of units required to be sold to earn the target profit.

(iii) Profit with 10% increase in selling price and drop in sales volume by 10%.

(iv) Volume to be achieved to earn target profit at the revised selling price as calculated in (ii) above, if a reduction of 10% in the variable costs and ₹ 170 lakhs in the fixed cost is envisaged.

ICAI'S ANSWER

Sales Volume 5,00,000 Units

Computation of existing contribution

Particulars	Per unit (₹)	Total (₹ In lakhs)
Sales	680	3,400
Fixed Cost	200	1,000
Profit	50	250
Contribution	250	1,250
Variable Cost (Sales – Contribution)	430	2,150

(i) Break even sales in units = $\frac{\text{Fixed Cost}}{\text{Contribution per unit}} = \frac{₹10,00,00,000}{₹250}$ = 4,00,000 units

Break even sales in rupees = 4,00,000 units × ₹ 680 = ₹ 2,720 lakhs

OR

P/V Ratio = $\frac{250}{680} \times 100 = 36.76\%$

B.EP (Rupees) = $\frac{\text{Fixed Cost}}{\text{P / VRatio}} = \frac{10,00,00,000}{36.76\%}$ = ₹ 2,720 lakhs (approx.)

(ii) Number of units sold to achieve a target profit of ₹700 lakhs:

Desired Contribution = Fixed Cost + Target Profit

= 1,000 L + 700 L = 1,700 L

Number of units to be sold = $\frac{\text{Desired Contribution}}{\text{Contribution per unit}} = \frac{17,00,00,000}{250}$ = 6,80,000 units

(iii) Profit if selling price is increased by 10% and sales volume drops by 10%:

Existing Selling Price per unit = ₹ 680

Revised selling price per unit = ₹ 680 × 110% = ₹748

Existing Sales Volume = 5,00,000 units

Revised sales volume = 5,00,000 units – 10% of 5,00,000 = 4,50,000 units.

Statement of profit at sales volume of 4,50,000 units @ ₹ 748 per unit

Particulars	Per unit (₹)	Total (₹ In lakhs)
Sales	748	3,366
Less: Variable Costs	430	1,935
Contribution	318	1,431
Less: Fixed Cost		1,000
Profit		**431**

(iv) Volume to be achieved to earn target profit of ₹700 lakhs with revised selling price and reduction of 10% in variable costs and ₹170 lakhs in fixed cost:

Revised selling price per unit = ₹748

Variable costs per unit existing = ₹430

Revised Variable Costs

Reduction of 10% in variable costs = ₹ 430 – 10% of 430

= ₹ 430 – ₹43

= ₹387

Total Fixed Cost (existing) = ₹ 1,000 lakhs

Reduction in fixed cost = ₹ 170 lakhs

Revised fixed cost = ₹ 1,000 lakhs – ₹ 170 lakhs = ₹830 lakhs

Revised Contribution (unit) = Revised selling price per unit – Revised Variable Costs per units

Revised Contribution per unit = ₹ 748 – ₹ 387 = ₹ 361

Desired Contribution = Revised Fixed Cost + Target Profit

= ₹ 830 lakhs + ₹700 lakhs = ₹1,530 lakhs

No. of units to be sold = $\frac{\text{Desired Contribution}}{\text{Contribution per unit}} = \frac{₹15,30,00,000}{₹361}$ =4,23,823 units

ICAI · Old P3 · RTP Nov 2020 · question p. 11 · answer p. 27-28

984 RTP **Old P3 May 2020** Q11 old scheme

A Ltd. manufacture and sales its product R-9. The following figures have been collected from cost records of last year for the product R-9:

Elements of Cost	Variable Cost portion	Fixed Cost
Direct Material	30% of Cost of Goods Sold	--
Direct Labour	15% of Cost of Goods Sold	--
Factory Overhead	10% of Cost of Goods Sold	₹ 2,30,000
Administration Overhead	2% of Cost of Goods Sold	₹ 71,000
Selling & Distribution Overhead	4% of Cost of Sales	₹ 68,000

Last Year 5,000 units were sold at ₹185 per unit. From the given DETERMINE the followings:

(i) Break-even Sales (in rupees)

(ii) Profit earned during last year

(iii) Margin of safety (in %)

(iv) Profit if the sales were 10% less than the actual sales.

(Assume that Administration Overhead is related with production activity)

ICAI'S ANSWER

Working Notes:

(1) Calculation of Cost of Goods Sold (COGS):

COGS = DM + DL + FOH + AOH

COGS = {0.3 COGS + 0.15 COGS + (0.10 COGS + ₹ 2,30,000) + (0.02 COGS + ₹ 71,000)}

Or, COGS = 0.57 COGS + ₹ 3,01,000

Or, COGS = $\frac{₹\ 3,01,000}{0.43}$ = ₹ 7,00,000

(2) Calculation of Cost of Sales (COS):

COS = COGS + S&DOH

COS = COGS + (0.04 COS + ₹ 68,000)

Or, COS = ₹ 7,00,000 + (0.04 COS + ₹ 68,000)

Or, COS = $\frac{₹\ 7,68,000}{0.96}$ = ₹ 8,00,000

(3) Calculation of Variable Costs:

Direct Material-	(0.30 × ₹ 7,00,000)	₹ 2,10,000
Direct Labour-	(0.15 × ₹ 7,00,000)	₹ 1,05,000
Factory Overhead-	(0.10 × ₹ 7,00,000)	₹ 70,000
Administration OH-	(0.02 × ₹ 7,00,000)	₹ 14,000
Selling & Distribution OH	(0.04 × ₹ 8,00,000)	₹ 32,000
		₹ 4,31,000

(4) Calculation of total Fixed Costs:

Factory Overhead-	₹ 2,30,000
Administration OH-	₹ 71,000
Selling & Distribution OH	₹ 68,000
	₹ 3,69,000

(5) Calculation of P/V Ratio:

$$\text{P/V Ratio} = \frac{\text{Contribution}}{\text{Sales}} \times 100 = \frac{\text{Sales} - \text{Variable Costs}}{\text{Sales}} \times 100$$

$$= \frac{(₹185 \times 5{,}000 \text{ units}) - ₹\ 4{,}31{,}000}{₹185 \times 5{,}000 \text{ units}} \times 100 = 53.41\%$$

(i) Break-Even Sales

$$= \frac{\text{Fixed Costs}}{\text{P / V Ratio}} = \frac{₹\ 3{,}69{,}000}{53.41\%} = ₹6{,}90{,}882$$

(ii) Profit earned during the last year

= (Sales – Total Variable Costs) – Total Fixed Costs

= (₹9,25,000 - ₹4,31,000) - ₹3,69,000

= ₹1,25,000

(iii) Margin of Safety (%)

$$= \frac{\text{Sales} - \text{Breakeven sales}}{\text{Sales}} \times 100$$

$$= \frac{₹\ 9{,}25{,}000 - ₹\ 6{,}90{,}882}{₹\ 9{,}25{,}000} \times 100 = 25.31\%$$

(iv) Profit if the sales were 10% less than the actual sales:

Profit = 90% (₹9,25,000 - ₹4,31,000) - ₹3,69,000

= ₹4,44,600 - ₹3,69,000 = ₹75,600

ICAI · Old P3 · RTP May 2020 · question p. 9 · answer p. 25-26

985 MTP Sep 2026 S-I Q1(c) 4 marks

M Ltd. has an annual fixed cost of ₹ 28,50,000. In the year 2024-25, sales amounted to ₹ 7,80,60,000 as compared with ₹ 4,80,60,000 in the preceding year 2023-24. The profit in the year 2024-25 is ₹ 37,50,000 more than that in 2023-24.

Required:

(i) CALCULATE Break-even sales of the company;

(ii) DETERMINE profit/ loss on a forecasted sales volume of ₹12,00,00,000.

(iii) If there is a reduction in selling price by 10% in the financial year 2025-26 and company desires to earn the same amount of profit as in 2024-25, WHAT would be the required sales volume?

ICAI'S ANSWER

(i) Break-even sales = $\frac{\text{Fixed Cost}}{\text{P / V Ratio}}$

P/V Ratio = $\frac{\text{Change in Profit}}{\text{Change in Sales}} \times 100$ or,

$$\frac{₹\ 37{,}50{,}000}{₹\ 7{,}80{,}60{,}000 - ₹\ 4{,}80{,}60{,}000} \times 100$$

Or, $\frac{₹37{,}50{,}000}{₹3{,}00{,}00{,}000} \times 100$ or, 12.5%

Break-even sales = $\frac{₹\ 28{,}50{,}000}{12.5\%} = ₹\ 2{,}28{,}00{,}000$

(ii) Profit/ loss = Contribution – Fixed Cost

= ₹ 12,00,00,000 × 12.5% - ₹ 28,50,000

= ₹ 1,50,00,000 – ₹ 28,50,000 = ₹ 1,21,50,000

(iii) To earn same amount of profit in 2025-26 as was in 2024-25, it has to earn the same amount of contribution as of 2024-25.

Sales – Variable cost = Contribution equal to 2024-25 contribution

Contribution in 2024-25 = Sales in 2024-25 × P/V Ratio in 2024-25

= ₹ 7,80,60,000 × 12.5% = ₹ 97,57,500

Let the Selling price be ₹100 in 2024-25, then the Variable cost would be ₹ 87.5 and Contribution would be ₹12.5.

(As Contribution is 12.5%, the Variable cost would be 87.5%.)

If there is 10% reduction in Selling price,

New Selling Price = ₹ 90

Variable cost = ₹ 87.5

So, Contribution = ₹ 2.5

New Contribution % = 2.77778% (2.5/90) ×100

Therefore, Sales volume required to earn a profit equals to 2024-25 profit

= ₹ 97,57,500/ 2.77778% = ~₹ 35,12,69,719.

ICAI · P4 · MTP Sep 2026 · Series I · question p. 9 · answer p. 9

986 MTP **May 2026 S-II** MCQ 11 2 marks MCQ

A manufacturer of industrial fasteners produces high-strength bolts. Last year, they sold 35,000 bolts at a price of ₹ 50 each. Their total fixed cost was ₹ 7,50,000 (including ₹ 2,50,000 depreciation on machinery) incurred uniformly throughout the year. There was no inventory at the beginning and all bolts produced during the year were shipped out, resulting in no inventory at the end of the year. The company's cash break-even point is 20,000 bolts.

Calculate the amount of total variable costs incurred.

A ₹ 8,57,000

B ₹ 8,75,000

C ₹ 4,37,500

D ₹ 4,73,500

ICAI'S ANSWER

(B) ₹ 8,75,000

Cash Break-even Sales Quantity = $\frac{\text{Fixed Costs (excluding dep.)}}{\text{Contribution per unit}}$

20,000 units = $\frac{(₹\ 7,50,000 - ₹\ 2,50,000)}{\text{Contribution per unit}}$

Contribution per unit = ₹ 5,00,000 /20,000 = ₹ 25

Variable Cost per unit = Selling Price per unit – Contribution per unit

= ₹ 50 - ₹ 25 = ₹ 25

Total Variable cost = 25 x 35,000 = ₹ 8,75,000

ICAI · P4 · MTP May 2026 · Series II · question p. 5 · answer p. 4

987 MTP Jan 2026 S-I Case Scenario I · MCQ 1 to MCQ 5 10 marks

CASE SCENARIO I

A company deals in manufacturing and selling hand sanitizers since 1998. Due to it's reputation company never really required any innovative selling techniques to sell its sanitizers. It sells the same at a very competitive price of ₹ 150 per unit. The variable cost ratio of the company was 67% in the previous year. After the COVID-19 government had ease the policies on manufacturing of sanitizers, this resulted in sudden increase of manufacturers in this industry. The reason that lured people to enter was that the process is pretty simpler to manufacture the sanitizers and it has a very high demand post Coronavirus. Due to this company lost a fair market share and incurred a loss of ₹ 85,000 last year, first time in 25 years. Company is proactive and took immediate steps to control the situation. For this they appointed a modern age sales manager and fired the old one. This new manager used digital marketing, data science, and artificial intelligence to increase the market presence & sales and as a result this year company sold 19,500 units, 30% more than the previous year, without changing the selling price. Old marketing techniques were not required anymore, thus they were removed. So it balanced out the new cost of digital marketing and other new techniques.

Based on the above data, you are required to answer the following questions (MCQ 1 to 5):

MCQ 1 2 marks

Find out the fixed cost of the company?

A 10,50,250

B 8,27,500

C 10,60,000

D 8,35,000

MCQ 2 2 marks

What is MOS in value of current year?

A 4,42,500

B - 2,57,700

C 417,300

D - 2,32,500

MCQ 3 2 marks

If target profit for the next year is ₹ 160,000, how much percentage increase in sale is required in comparison to current year.

A 2.307%

B 1.282%

C 2.255%

D 1.265%

MCQ 4 2 marks

In current year, if selling price is increased by 25%, demand will reduce by 9%. What will be the new MOS in value?

A 15,43,687.5

B 24,21,000

C 20,91,937.5

D 18,72,750

MCQ 5 2 marks

BEP in percentage and value of current year?

A 85.75%, ₹ 25,07,750

B 84.87%, ₹ 29,25,500

C 84.90%, ₹ 29,25,000

D 85.73%, ₹ 25,07,700

ICAI'S ANSWERS

MCQ 1

(B)

Selling price pu = 150

Variable cost pu = 150 x 67% = 100.5

Contribution pu = 49.5

Total contribution = 49.5 x 15,000 = 7,42,500

Contribution + Loss = Fixed cost,

Fixed cost = 7,42,500 + 85,000 = 8,27,500

MCQ 2

(C)

$$\text{BEP} = \frac{\text{Fixed cost}}{\text{contribution per unit}}$$

$$= \frac{8,27,500}{49.5} = 16,718 \text{ units}$$

MOS in value = (Sales in units - BEP in units) x Selling price pu

= (19,500 – 16,718) x 150 = 4,17,300

MCQ 3

(A)

$$\text{Target sales in units} = \frac{(\text{Fixed cost} + \text{Target profit})}{\text{contribution per unit}}$$

$$= \frac{(8,27,500 + 1,60,000)}{49.5} = 19,950 \text{ units}$$

Increase = 19,950 – 19,500 = 450 units

$$\text{Increase in \%} = \frac{450}{19,500} = 2.307\%$$

MCQ 4

(A)

New selling price = 187.5

Variable cost will remain same = 100.5

New contribution pu = 87

$$\text{BEP in units} = \frac{\text{Fixed cost}}{\text{New contribution per unit}}$$

$= \frac{8,27,500}{87} = 9,512$ units

New sales units = 19,500(1 - 9%) = 17,745 units

MOS sales in ₹ = (17,745 – 9,512) x 187.5 = 15,43,687.5

MCQ 5

(D)

BEP as calculated in part 2 = 16,718 units

BEP in ₹ = 16,718 x 150 = 25,07,700

Total sales = 150 x 19,500 units = 29,25,000

BEP in % = BEP in ₹/ total sales = $\frac{25,07,700}{29,25,000} = 85.73\%$

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988 MTP Sep 2025 S-I Q1(c) 4 marks

During the first half of the year, the company recorded fixed expenses of ₹ 13,50,000, total sales of ₹ 45,00,000, and a profit of ₹ 9,00,000. However, in the second half, the company incurred a loss of ₹ 4,50,000.

From the information given above, you are required to CALCULATE the following:

(i) Profit-volume ratio, break-even point and margin of safety for the first half year.

(ii) Expected sales volume for the second half year assuming that selling price and fixed expenses remained unchanged during the second half year.

ICAI'S ANSWER

(i) In the First half year:

Contribution = Fixed cost + Profit

= ₹ 13,50,000 + ₹ 9,00,000

= ₹ 22,50,000

$\text{P/V ratio} = \frac{\text{Contribution}}{\text{Sales}} \times 100$

$= \frac{22,50,000}{45,00,000} \times 100$

= 50%

$\text{Break-even point} = \frac{\text{Fixed cost}}{\text{P/V ratio}}$

$= \frac{13,50,000}{50\%}$

= ₹ 27,00,000

Margin of safety = Actual sales – Break-even point

= ₹ 45,00,000 – ₹ 27,00,000

= ₹ 18,00,000

(ii) In the second half year:

Contribution = Fixed cost – Loss

= ₹ 13,50,000 – ₹ 4,50,000

= ₹ 9,00,000

$$\text{Expected sales volume} = \frac{\text{Contribution}}{\text{P/V ratio}}$$

$$= \frac{9,00,000}{50\%}$$

= ₹ 18,00,000

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989 **MTP** **Sep 2025 S-II Q1(a)** **5 marks**

C.T Ltd. manufactures and sells a single product X whose selling price is ₹ 100 per unit and the variable cost is ₹ 60 per unit.

(i) If the Fixed Costs for this year are ₹ 24,00,000 and the annual sales are at 60% margin of safety, CALCULATE the rate of net return on sales, assuming an income tax level of 40%

(ii) For the next year, it is proposed to add another product line Y whose selling price would be ₹ 150 per unit and the variable cost ₹ 100 per unit. The total fixed costs are estimated at ₹ 28,00,000. The sales mix of X : Y would be 5 : 3. COMPUTE the break- even sales in units for both the products.

ICAI'S ANSWER

(i) Contribution per unit = Selling price – Variable cost

= ₹ 100 – ₹ 60

= ₹ 40

$$\text{Break-even Point} = \frac{₹\ 24,00,000}{₹\ 40}$$

= 60,000 units

$$\text{Percentage Margin of Safety} = \frac{\text{Actual Sales} - \text{Break - even Sales}}{\text{Actual Sales}}$$

$$\textit{Or, } 60\% = \frac{\text{Actual Sales} - 60,000\ \textit{units}}{\text{Actual Sales}}$$

∴ Actual Sales = 1,50,000 units

	(₹)
Sales Value (1,50,000 units × ₹ 100)	1,50,00,000
Less: Variable Cost (1,50,000 units ×₹ 60)	90,00,000
Contribution	60,00,000
Less: Fixed Cost	24,00,000
Profit	36,00,000
Less: Income Tax @ 40%	14,40,000
Net Return	21,60,000

$$\text{Rate of Net Return on Sales} = 14.40\% \left(\frac{₹\ 21,60,000}{₹\ 1,50,00,000} \times 100\right)$$

(ii) **Products**

	X (₹)	Y (₹)
Selling Price *per unit*	100	150
Variable Cost *per unit*	60	100
Contribution *per unit*	40	50

Composite contribution will be as follows:

Contribution per unit = $\left(\frac{40}{8} \times 5\right) + \left(\frac{50}{8} \times 3\right)$

= 25 + 18.75 = ₹ 43.75

Break-even Sale = 64,000 units $\left(\frac{₹\ 28,00,000}{₹\ 43.75}\right)$

Break-even Sales Mix:

X (64,000 units × 5/8) = 40,000 units

Y (64,000 units × 3/8) = 24,000 units

ICAI · P4 · MTP Sep 2025 · Series II · question p. 8-9 · answer p. 8-9

990 MTP Jan 2025 S-I MCQ 13 2 marks MCQ

Mefttal Ltd. is currently operating at 60% of its total capacity which is 1.5 times than the previous year. The total capacity of the company is 2,00,000 units.

Other information relating to the production is provided below:

(i) The total cost of production for the current year is ₹ 59,28,000, and for the previous year, it was ₹ 44,72,000.

(iii) No changes are anticipated in the cost structure for the upcoming years.

Selling price is ₹ 52 per unit and is expected to remain the same in the coming years.

You are required to CALCULATE Break-Even Point (in units).

A 1,20,000 units

B 40,000 units

C 80,000 units

D 1,00,000 units

ICAI'S ANSWER

(D) 1,00,000 units

Current Year production = 60% of 2,00,000 units

= 1,20,000 units

Previous Year production = $\left(\frac{1,20,000 \text{ units}}{1.5 \text{ times}}\right)$

= 80,000 units

Particulars	Previous Year	Current Year	Difference
Sales (Units)	80,000	1,20,000	40,000
Total Cost (₹)	44,72,000	59,28,000	14,56,000

$$\text{Variable Cost per unit} = \frac{\text{Change in Total Cost}}{\text{Change in sales volume}}$$

$$= \left(\frac{₹\ 14{,}56{,}000}{40{,}000\ \text{units}}\right)$$

= ₹ 36.40 per unit

Total Fixed Cost (₹) = ₹ 59,28,000-(1,20,000 units × ₹ 36.40)

= ₹ 15,60,000

$$\text{Break- even point (in units)} = \frac{\text{Fixed Cost}}{\text{Contribution per unit}}$$

$$= \left(\frac{₹\ 15{,}60{,}000}{₹\ 52\text{- }₹\ 36.40}\right)$$

= 1,00,000 units

ICAI · P4 · MTP Jan 2025 · Series I · question p. 5-6 · answer p. 4-5

991 MTP **Jan 2025 S-II** MCQ 13 2 marks MCQ

Due to sudden rise in demand of the product, the sales of Arrow Ltd. for current year enhanced to 3 times the average of last 4 years. The Break even point and the variable cost of the company for the current year is ₹ 1,17,00,000 and 93,60,000 respectively.

The sales data relating to past years is given below:

Year	Sales (₹)
Year 1 (latest)	62,00,000
Year 2	50,00,000
Year 3	52,00,000
Year 4	44,00,000
Year 5	66,00,000

CALCULATE the fixed cost to the company for the current year.

A ₹ 64,35,000

B ₹ 48,12,453

C ₹ 65,34,340

D ₹ 46,80,000

ICAI'S ANSWER

(D) Sales for current year = 3 x $\left(\frac{62{,}00{,}000 + 50{,}00{,}000 + 52{,}00{,}000 + 44{,}00{,}000}{4}\right)$

= ₹ 1,56,00,000

$$\text{P/V ratio} = \frac{\text{Sales - Variable Cost}}{\text{Sales}}$$

$$= \frac{₹1,56,00,000 - 93,60,000}{₹1,56,00,000}$$

= 40%

$$\text{Now, Break even point} = \frac{\text{Fixed Cost}}{\text{P/V ratio}}$$

Therefore, **Fixed Cost** = Break even point x P/V ratio

= ₹ 1,17,00,000 x 40%

= ₹ 46,80,000

ICAI · P4 · MTP Jan 2025 · Series II · question p. 6 · answer p. 6

992 MTP **Jan 2025 S-II** Q5(a) 8 marks

A Korean beverage company plans to set up a subsidiary in India to manufacture fruit juice. Based on projected annual sales of 40,000 bottles, cost analysis has provided the following estimates for the Indian subsidiary:

	Total Annual Costs (₹)	Percentage of Total Annual Cost which is Variable
Material	3,15,000	100%
Labour	1,40,000	75%
Factory Overheads	1,35,000	50%
Administrative Overheads	50,000	35%

The fruit juice produced in India will be sold through manufacturer's representatives, who will earn a commission of 10% of the sales price. Expenses from the Korean office will not be allocated to the Indian subsidiary.

Required

(i) COMPUTE the sale price per bottle to enable the management to realise an estimated 10% profit on sale proceeds in India.

(ii) CALCULATE the break-even point in Rupee sales and also in number of bottles for the Indian subsidiary on the assumption that the sale price is ₹ 19 per bottle.

ICAI'S ANSWER

(i) **Computation of Sale Price *Per Bottle***

Output: 40,000 Bottles

	(₹)
Variable Cost:	
Material	3,15,000
Labour (₹ 1,40,000 × 75%)	1,05,000
Factory Overheads (₹ 1,35,000 × 50%)	67,500

	(₹)
Administrative Overheads (₹ 50,000 × 35%)	17,500
Commission (10% on ₹ 8,00,000) (W.N.-1)	80,000
Fixed Cost:	
Labour (₹ 1,40,000 × 25%)	35,000
Factory Overheads (₹ 1,35,000 × 50%)	67,500
Administrative Overheads (₹ 50,000 × 65%)	32,500
Total Cost	7,20,000
Profit (W.N.-1)	80,000
Sales Proceeds (W.N.-1)	8,00,000
Sales Price per bottle (₹ 8,00,000/40,000Bottles)	20

(ii) **Calculation of Break-even Point**

Sales Price per Bottle = ₹19

Variable Cost per Bottle = $\frac{₹5,85,000\ (W.N.-2)}{40,000\ Bottles}$

= ₹ 14.625

Contribution per Bottle = ₹ 19 – ₹14.625

= ₹ 4.375

Break -even Point (in number of Bottles) = $\frac{Fixed\ Costs}{Contribution\ per\ Bottle}$

= $\frac{₹1,35,000}{₹4.375}$ = 30,857 Bottles

Break- even Point (in Sales Value) = 30,857 Bottles × ₹ 19

= ₹ 5,86,285/-

Working Note

W.N.-1

Let the Sales Price be 'x'

Commission = $\frac{10x}{100}$

Profit = $\frac{10x}{100}$

x = 6,40,000 + $\frac{10x}{100} + \frac{10x}{100}$

100x - 10x - 10x = 6,40,00,000

80x = 6,40,00,000

x = 6,40,00,000 / 80

= ₹ 8,00,000

W.N.-2

Total Variable Cost

	(₹)
Material	3,15,000
Labour	1,05,000
Factory Overheads	67,500
Administrative Overheads	17,500
Commission [(40,000 Bottles x ₹20) x 10%]	80,000
Total	5,85,000

ICAI · P4 · MTP Jan 2025 · Series II · question p. 11-12 · answer p. 15-16

993 MTP **Sep 2024 S-I** Case scenario · MCQ 6 to MCQ 10 10 marks

CASE SCENARIO

XYZ Manufacturing Pvt. Ltd. is a prominent company in the electric appliances industry, known for producing a diverse range of high-quality products. The company has built a reputation for reliability and innovation in the manufacturing of household appliances, including fans, mixers, and heaters. XYZ Manufacturing Pvt. Ltd. is dedicated to delivering products that meet the needs of its customers while adhering to the highest standards of quality and performance.

The company operates a state-of-the-art factory that is fully equipped with advanced machinery and technology to ensure efficient and consistent production. The factory operates 25 days a month, running multiple shifts to meet the growing demand for its products. The company have spare capacity to additional orders. Each product type—fans, mixers, and heaters—undergoes a meticulous manufacturing process that includes assembly, quality testing, and packaging.

Cost Category	Amount (₹)
Fixed Costs (per month)	
Factory Rent	₹ 3,00,000
Depreciation	₹ 2,00,000
Administrative Expenses	₹ 1,00,000
Salaries	₹ 4,00,000
Total Fixed Costs	₹ 10,00,000
Number of units produced per month (Note: Last month there was an additional special order of 2000 units which resulted in higher production)	10,000 units

Cost Category	Amount (₹)
Selling price per unit	₹ 1,500

Additional Info: Raw Materials include Copper, Plastic, and Other Materials. The per unit cost of Copper is ₹ 80 more than the cost of Plastic, while the cost of Other Materials is twice that of Plastic. And the total Raw Material Cost per unit is ₹ 210 more than the combined cost of Copper & Plastic.

The Labour Hour Rate is ₹ 100 per hour. The total labour hours used in the last month were 36,000 Hours. The Utilities Cost per unit is ₹ 100, and the Packaging Cost per unit is ₹ 50. Being a finance manager of the company, you are required to answer the following:

MCQ 6 2 marks

Calculate the contribution margin per unit.

A ₹ 550

B ₹ 600

C ₹ 650

D ₹ 700

MCQ 7 2 marks

Determine the break-even point in sales revenue.

A ₹ 31,28,593

B ₹ 25,85,153

C ₹ 27,27,025

D ₹ 27,05,983

MCQ 8 2 marks

If the company wants to achieve a target profit of ₹ 5,00,000, what should be the sales volume (in units)?

A 2,000 units

B 2,727 units

C 2,750 units

D 3,000 units

MCQ 9 2 marks

What would be the impact on the break-even point if the variable cost per unit increases by 10%?

A 2,178 units

B 2,198 units

C 2,248 units

D 2,258 units

MCQ 10 2 marks

Calculate the margin of safety in percentage if the company sells 4,000 units if the variable cost per unit increases by 10%

A 44.85%

B 42.55%

C 45.05%

D 45.75%

ICAI'S ANSWERS

MCQ 6

(A) ₹ 550

Contribution Margin per Unit = Selling Price per Unit - Variable Cost per Unit

= Variable Cost per unit = ₹ 500*+ ₹ 300**+ ₹ 100+ ₹ 50

Contribution Margin per Unit = ₹ 1,500 - ₹ 950 = ₹ 550

*Raw Material Cost Calculation

Let the cost of Plastic be x

1. The cost of Copper is ₹ 80 more than the cost of Plastic: Cost of Copper = x + 80
2. The cost of Other Materials is twice that of Plastic: Cost of Other Materials = 2x
3. The total Raw Material Cost per unit is ₹ 210 more than the combined cost of Copper & Plastic: x + (x+80) + 2x = (x + (x+80)) + 210

Solving for X = 105

Now, calculate the total cost of Raw Materials:

105 + (105+80) +210 = 500

So, the total cost of Raw Materials is ₹ 500.

** Labour Cost Calculation

1. The Labour Hour Rate is ₹ 100 per hour.
2. The total labour hours used in the last month were 36,000 hours.
3. The production units last month were 12,000 units (10000 normal units plus 2000 special order).
   Total Labour Cost = Labour Hour Rate × Total Labour Hours

   Total Labour Cost= ₹ 100 /hour × 36,000 hours = ₹ 3,600,000

   Per Unit Labour Cost = Total Labour Cost/Production Units

   Per Unit Labour Cost = ₹ 3,600,000/12000

   Per Unit Labour Cost = ₹ 300

   So, the per unit labour cost is ₹ 300.

MCQ 7

(C) ₹ 27,27,025

- Break-even Point (Sales Revenue) = Total Fixed Costs / Contribution Margin Ratio
- Contribution Margin Ratio = Contribution Margin per Unit / Selling Price per Unit
- = ₹ 550 / ₹ 1,500 = 0.3667
- Break-even Point = ₹ 10,00,000 / 0.3667 ≈ ₹ 27,27,025

MCQ 8

(B) 2,727 units

- Required Sales Volume (Units) = (Total Fixed Costs + Target Profit) / Contribution Margin per Unit

- = (₹ 10,00,000 + ₹ 5,00,000) / ₹ 550 ≈ 2,727.27 units ≈ 2,727 units (rounded up)

MCQ 9

(B) 2,198 units

- New Variable Cost per Unit = ₹ 950 + 10% of ₹ 950 = ₹ 950 + ₹ 95 = ₹ 1,045
- New Contribution Margin per Unit = ₹ 1,500 - ₹ 1,045 = ₹ 455
- New Break-even Point (Units) = Total Fixed Costs / New Contribution Margin per Unit
- = ₹ 10,00,000 / ₹ 455 ≈ 2198 units

MCQ 10

(C) 45.05%

- Margin of Safety (Units) = Actual Sales - Break-even Sales
- = 4,000 - 2198 = 1,802 units
- Margin of Safety (%) = (Margin of Safety in Units / Actual Sales in Units) * 100
- = (1,802 / 4,000) * 100 ≈ 45.05%

ICAI · P4 · MTP Sep 2024 · Series I · question p. 3-4 · answer p. 3, 3-4, 4

994 MTP Sep 2024 S-II MCQ 11 2 marks MCQ

A company's fixed costs are ₹ 5,00,000, the selling price per unit is ₹ 200, and the variable cost per unit is ₹100. How many units must the company sell to earn the targeted profit of ₹ 2,00,000?

A 2,000 units

B 5,000 units

C 10,000 units

D 7,000 units

ICAI'S ANSWER

(D)

$$\text{Break-even point} = \frac{\text{Fixed Costs} + \text{Targeted Profit}}{(\text{Selling Price per Unit} - \text{Variable Cost per Unit})}$$

= (5,00,000 + 2,00,000)/100 = 7,000 units

ICAI · P4 · MTP Sep 2024 · Series II · question p. 4-5 · answer p. 4

995 MTP May 2024 S-I Q4(b) 4 marks

PQ Ltd. sells bottles and currently is trying to find out the profitability of opening another store which will have the following expenses and revenues:

	Amount per piece (₹)
Selling Price	600
Variable costs:	
Material cost	410
Salesmen's commission	60

	Amount per piece (₹)
Total variable cost	470
Annual fixed expenses are:	**(₹)**
- Rent	6,00,000
- Office and administrative expenses	20,00,000
- Advertising	8,00,000
- Other fixed expenses	2,00,000

Calculate the annual break-even point in units and in value. Also determine the profit or loss if 35,000 units of bottles are sold.

ICAI'S ANSWER

Total Fixed Cost = ₹ 6,00,000 + ₹20,00,000 + ₹8,00,000 + ₹ 2,00,000

= ₹ 36,00,000

Contribution per unit = ₹600 - ₹470 = ₹130

$$\text{P/V Ratio} = \frac{\text{Contribution per unit}}{\text{Selling Price}} \times 100 = \frac{₹130}{₹600} \times 100 = 21.67\%$$

$$\text{Break-even Point} = \frac{\text{Total Fixed Cost}}{\text{Contribution per unit}}$$

$$= \frac{₹36,00,000}{₹130} = 27,692.31 \text{ or } 27,693 \text{ units}$$

$$\text{Break-even Sales} = \frac{\text{Total Fixed Cost}}{\text{P / V Ratio}} = \frac{₹36,00,000}{21.67\%} = ₹1,66,12,829$$

Calculation of Profit/ (loss):

Total Contribution (₹130 × 35,000 units)	= ₹45,50,000
Less: Fixed Cost	= ₹36,00,000
Profit	= ₹ 9,50,000

ICAI · P4 · MTP May 2024 · Series I · question p. 9-10 · answer p. 11

996 MTP **May 2024 S-II** Case scenario · MCQ 1(i) to MCQ 1(v) 10 marks

CASE SCENARIO

A meeting of the heads of departments of the Arnav Ltd. has been called to review the operating performance of the company in the last financial year. The head of the production department appraised that during the last year the company could operate at 70% capacity level but in the coming financial year 95% capacity level can be achieved if an additional amount of ₹100 Crore on capex and working capital is incurred.

The head of the finance department has presented that during the last financial year the company had a P/V ratio of 40%, margin of safety and the break-even were ₹50 crore and ₹200 crore respectively.

To the reply to the proposal of increasing the production capacity level to 95%, the head of the finance department has informed that this could be achieved if the selling price and variable cost are reduced by 8% and 5% of sales respectively. Fixed cost will also increase by ₹20 crore due to increased depreciation on additional assets. The additional capital will be arranged at a cost of 15% p.a. from a bank.

In the coming financial year, it has been aimed to achieve an additional profit of ₹10 crore over and above the last year's profit after adjusting the interest cost on the additional capital.

The following points is required to be calculated on urgent basis to put the same in the meeting. You being an assistant to the head of finance, has been asked the followings:

MCQ 1(i) 2 marks

What will be the revised sales for the coming financial year?

A ₹ 322.22 Crore

B ₹ 311.11 Crore

C ₹ 300.00 Crore

D ₹ 324.24 Crore

MCQ 1(ii) 2 marks

What will be the revised break-even point for the coming financial year?

A ₹ 222.22 Crore

B ₹ 252.22 Crore

C ₹ 244.44 Crore

D ₹ 255.56 Crore

MCQ 1(iii) 2 marks

What will be the revised margin of safety for the coming financial year?

A ₹ 100 Crore

B ₹ 58.89 Crore

C ₹ 55.56 Crore

D ₹ 66.66 Crore

MCQ 1(iv) 2 marks

The profit of the last year and for the coming year are:

A ₹ 50 Crore & ₹95 Crore respectively

B ₹ 20 Crore & ₹ 65 Crore respectively

C ₹ 20 Crore & ₹ 30 Crore respectively

D ₹ 45 Crore & ₹ 66.66 Crore respectively

MCQ 1(v) 2 marks

The total cost of the last year and for the coming year are:

A ₹ 230 Crore & ₹292.22

B ₹ 230 Crore & ₹275 Crore

C ₹ 220 Crore & ₹282.22 Crore

D ₹ 220 Crore & ₹292.22 Crore

ICAI'S ANSWERS

MCQ 1(i)

Answer key: **(A)**

A

$$\text{Revised Sale} = \frac{\text{Revised Fixed Cost} + \text{Expected Profit}}{\text{P / V Ratio}}$$

= {₹115 + (20+10)} ÷ 45% **= ₹ 322.22 crores**

MCQ 1(ii)

Answer key: **(D)**

D

$$\text{Revised Break – even Point} = \frac{\text{Fixed Cost}}{\text{P / V Ratio}}$$

= ₹115 Crore ÷ 45% = ₹255.56 Crore (Refer working notes)

MCQ 1(iii)

Answer key: **(D)**

D

Revised Margin of Safety = Revised Sales – Revised Break–even Sales

= ₹ 322.22Crores – ₹ 255.56Crores = **₹ 66.66 Crores.**

MCQ 1(iv)

Answer key: **(C)**

C

₹ 20 Crore & ₹30 Crore respectively (Refer working note)

MCQ 1(v)

Answer key: **(A)**

A

Total cost in last year = ₹230 Crore

Total cost in coming year = Variable Cost + Fixed Cost

Revised sales × 55% + 115 Crore

= ₹ 322.22 Crore × 55% + ₹ 115 Crore = ₹ 292.22 Crore

Working Note

Present Sales and Profit

Total Sales	=	Break – even Sales + Margin of Safety
	=	₹ 200 Crores + ₹ 50 Crores
	=	₹ 250 Crores
P/V Ratio	=	40%
Variable Cost	=	60% of Sales
	=	₹ 250 Crores × 60%

	=	₹ 150 Crores
Fixed Cost	=	Break – even Sales × P/V Ratio
	=	₹ 200 Crores × 40%
	=	₹ 80 Crores
Total Cost	=	₹ 150 Crores + ₹ 80 Crores
	=	₹ 230 Crores
Profit	=	Total Sales – Total Cost
	=	₹ 250 Crores – ₹ 230 Crores
	=	₹ 20 Cores

Revised Sales (₹ in Crores)

Present Fixed Cost	80.00
Increase in Fixed Cost	20.00
Interest at 15 *per cent* on Additional Capital (₹100Crores × 15%)	15.00
Total Revised Fixed Cost (in crore)	115.00
Assuming that the Present Selling Price is ₹100	
Revised Selling Price will be (8% Less)	92.00
New Variable Cost (Reduced from 60% to 55%) of Sales (₹ 92 × 55%)	50.60
Contribution (₹92.00 – ₹ 50.60)	41.40

$$\text{New P / V Ratio} = \frac{₹41.40}{₹92.00} \text{ x } 100$$

= 45%

ICAI · P4 · MTP May 2024 · Series II · question p. 1-2 · answer p. 1, 1-2

997 MTP **May 2024 S-II** Q6(c) 4 marks

Same question, different ICAI answer: #1007 Old P3 MTP Jul 2021 S-II

DISCUSS basic assumptions of Cost Volume Profit analysis.

ICAI'S ANSWER

CVP Analysis:-Assumptions

(i) Changes in the levels of revenues and costs arise only because of changes in the number of products (or service) units produced and sold.

(ii) Total cost can be separated into two components: Fixed and variable

(iii) Graphically, the behaviour of total revenues and total cost are linear in relation to output level within a relevant range.

(iv) Selling price, variable cost per unit and total fixed costs are known and constant.

(v) All revenues and costs can be added, sub traded and compared without taking into account the time value of money.

ICAI · P4 · MTP May 2024 · Series II · question p. 10 · answer p. 15

998 MTP Old P3 Nov 2023 S-I Q1(a) 5 marks old scheme

LK Ltd. has an annual fixed cost of ₹ 98,50,000. In the year 2022-23, sales amounted to ₹7,80,60,000 as compared to ₹5,93,10,000 in the preceding year 2021-22. Profit in the year 2022-23 is ₹37,50,000 more than that in 2021-22.

Required:

(i) CALCULATE Break-even sales of the company;

(ii) DETERMINE profit/ loss on a forecasted sales volume of ₹8,20,00,000.

(iii) If there is a reduction in selling price by 10% in the financial year 2022-23 and company desires to earn the same amount of profit as in 2021-22, COMPUTE the required sales amount?

ICAI'S ANSWER

(i) Break-even sales = $\frac{\text{Fixed Cost}}{\text{P / V Ratio}}$

P/V Ratio = $\frac{\text{Change in Profit}}{\text{Change in Sales}} \times 100$ or, $\frac{₹\ 37,50,000}{₹\ 7,80,60,000 - ₹\ 5,93,10,000} \times 100$

Or, $\frac{₹\ 37,50,000}{₹\ 1,87,50,000} \times 100$ or, 20%

Break-even sales = $\frac{₹\ 98,50,000}{20\%}$ = ₹4,92,50,000

(ii) Profit/ loss = Contribution – Fixed Cost

= ₹8,20,00,000 × 20% - ₹98,50,000

= ₹1,64,00,000 – ₹98,50,000 = ₹65,50,000

(iii) To earn same amount of profit in 2022-23 as was in 2021-22, it has to earn the same amount of contribution as in 2021-22.

Sales – Variable cost = Contribution equal to 2021-22 contribution

Contribution in 2021-22 = Sales in 2021-22 × P/V Ratio in 2021-22

= ₹5,93,10,000 × 20% = ₹1,18,62,000

Let the number of units to be sold in 2022-23 = X

Sales in 2022-23 – Variable cost in 2022-23 = Desired Contribution

90 X – 80 X = ₹1,18,62,000

Or, 10 X = 1,18,62,000

Or, X = 11,86,200 units

Therefore, Sales amount required to earn a profit equal to 2021-22 profit

= ₹ 90 × 11,86,200 units = ₹ 10,67,58,000

999 MTP **Old P3 May 2023 S-I** Q3(b) 10 marks old scheme

LNP Ltd. and MNT Ltd. are engaged in manufacturing of identical products. Existing revenue and cost data is as follows:

	LNP Ltd. (₹)	MNT Ltd. (₹)
Sales	13,60,000	17,00,000
Variable Cost	10,88,000	10,20,000
Fixed Cost	1,72,000	5,80,000

You are required to calculate:

(i) Break-even point (in Value) for each company

Sales at which each company will earn a profit of ₹ 5,00,000.

Sales at which both companies will have same profits.

ICAI'S ANSWER

Income Statement

	LNP Ltd. (₹)	MNT Ltd. (₹)
Sales (Rs.)	13,60,000	17,00,000
Less: Variable Cost	10,88,000	10,20,000
Contribution	2,72,000	6,80,000
P.V. Ratio ($\frac{\text{Contribution}}{\text{Sales}} \times 100$)	20%	40%
Fixed Cost (₹)	1,72,000	5,80,000
Profit (₹)	1,00,000	1,00,000

(i) Break-Even Point = $\frac{\text{Fixed Cost}}{\text{P. V. Ratio}}$

LNP Ltd. = = $\frac{₹\ 1,72,000}{20\%}$ = ₹ 8,60,000

MNT Ltd. = = $\frac{₹\ 5,80,000}{40\%}$ = ₹ 14,50,000

(ii) Sales value to earn a profit of ₹ 5,00,000

Sales = $\frac{\text{Fixed Cost} + \text{Desired Profit}}{\text{P. V. Ratio}}$

LNP Ltd. = $\frac{1,72,000 + 5,00,000}{40\%}$ = ₹ 33,60,000

MNT Ltd. = $\frac{5,80,000 + 5,00,000}{40\%}$ = ₹ 27,00,000

(iii) Sales value at which both companies will earn same profit

Let S = Sales value and P = Profit

Sales – Variable cost = Fixed cost + Profit

or, Contribution = Fixed cost + Profit

LNP Ltd.:

20% S = ₹1,72,000 + P

or, 0.20S = ₹ 1,72,000 + P .....(i)

MNT Ltd.

40% S = ₹5,80,000 + P

or, 0.40S = ₹ 5,80,000 + P .....(ii)

By solving these equations, we will get the value of 'S' and 'P'

0.20S = 1,72,000 + P

0.40S = 5,80,000 + P

- - -

- 0.20S = -4,08,000

or, S = ₹ 20,40,000

Putting the value of 'S' in equation no. (i) we will get the value of 'P'

0.20 × 20,40,000 = 1,72,000 + P

or, P = ₹2,36,000

Therefore, at Sale value of ₹20,40,000 both the companies will earn same profit of ₹ 2,36,000

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1000 **MTP** **Old P3 May 2023 S-II** Q1(a) **5 marks** **old scheme**

NG Ltd. has an annual fixed cost of ₹ 98,50,000. In the year 2022-23, sales amounted to ₹7,80,60,000 as compared to ₹5,93,10,000 in the preceding year 2021-22. Profit in the year 2022-23 is ₹37,50,000 more than that in 2021-22.

Required:

(i) CALCULATE Break-even sales of the company.

(ii) DETERMINE profit/ loss on a forecasted sales volume of ₹8,20,00,000.

(iii) If there is a reduction in selling price by 10% in the financial year 2022-23 and company desires to earn the same amount of profit as in 2021-22, COMPUTE the required sales amount?

ICAI'S ANSWER

(i) Break-even sales = $\frac{\text{Fixed Cost}}{\text{P / V Ratio}}$

P/V Ratio = $\frac{\text{Change in Profit}}{\text{Change in Sales}} \times 100$ or, $\frac{₹\ 37,50,000}{₹\ 7,80,60,000 - ₹\ 5,93,10,000} \times 100$

Or, $\frac{₹\ 37,50,000}{₹\ 1,87,50,000} \times 100$ or, 20%

Break-even sales = $\frac{₹\ 98,50,000}{20\%}$ = ₹4,92,50,000

(ii) Profit/ loss = Contribution – Fixed Cost

= ₹8,20,00,000 × 20% - ₹98,50,000

= ₹1,64,00,000 – ₹98,50,000 = ₹65,50,000

(iii) To earn same amount of profit in 2022-23 as it was in 2021-22, the company has to earn the same amount of contribution as it had earned in 2021-22.

Sales – Variable cost = Contribution equal to 2021-22 contribution

Contribution in 2021-22 = Sales in 2021-22 × P/V Ratio in 2021-22

= ₹5,93,10,000 × 20% = ₹1,18,62,000

Let the number of units to be sold in 2022-23 = X

Sales in 2022-23 – Variable cost in 2022-23 = Desired Contribution

90 X – 80 X = ₹1,18,62,000

Or, 10 X = 1,18,62,000

Or, X = 11,86,200 units

Therefore, Sales amount required to earn a profit equal to 2021-22 profit

= ₹ 90 × 11,86,200 units = ₹ 10,67,58,000

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1001 MTP **Old P3 May 2023 S-II** Q3(a)(2) 5 marks old scheme

The following figures are related to KG Limited for the year ending 31st March, 2023:

Sales - 48,000 units @ ₹ 400 per unit;

P/V Ratio 25% and Break-even Point 50% of sales.

You are required to CALCULATE:

(i) Fixed cost for the year

(ii) Profit earned for the year

(iii) Units to be sold to earn a target net profit of ₹ 22,00,000 for a year.

(iv) Number of units to be sold to earn a net income of 25% on cost.

ICAI'S ANSWER

Break- even point (in units) is 50% of sales i.e. 24,000 units.

Hence, Break- even point (in sales value) is 24,000 units × ₹ 400 = ₹ 96,00,000

(i) Break even sales = $\frac{\text{Fixed Cost}}{\text{P / V ratio}}$

Or, ₹ 96,00,000 = $\frac{\text{Fixed Cost}}{25\%}$

Or, Fixed Cost = ₹ 96,00,000 × 25%

= ₹ 24,00,000

So Fixed Cost for the year is ₹ 24,00,000

(ii) Contribution for the year = (48,000 units × ₹ 400) × 25%

= ₹ 48,00,000

Profit for the year = Contribution – Fixed Cost

= ₹48,00,000 - ₹ 24,00,000

= ₹ 24,00,000

(iii) Target net profit is ₹22,00,000

Hence, Target contribution = Target Profit + Fixed Cost

= ₹ 22,00,000 + ₹ 24,00,000

= ₹ 46,00,000

Contribution per unit = 25% of ₹ 400 = ₹ 100 per unit

No. of units = $\frac{₹46,00,000}{₹100 \text{ per unit}}$ = 46,000 units

So, 46,000 units to be sold to earn a target net profit of ₹ 22,00,000 for a year.

(iv) Let desired total Sales (Number of units × Selling price) be x then desired profit is 25% on Cost or 20% on Sales i.e. 0.2 x

Desired Sales = $\frac{\text{Fixed Cost} + \text{Desired Profit}}{\text{P / V ratio}}$

$x = \frac{₹24,00,000 + 0.2x}{25\%}$

or, 0.25 x = ₹ 24,00,000 + 0.2 x

or, 0.05 x = ₹ 24,00,000

or, x = ₹ 4,80,00,000

No. of units to be sold = $\frac{₹4,80,00,000}{₹400}$ = 1,20,000 units

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1002 MTP **Old P3 Nov 2022 S-I** Q2(b) 10 marks old scheme

PS Limited is a manufacturing company and is operating at 75% capacity utilization. The PV ratio at this level of activity is 40%.

The flexible budget drafted by the company for two levels of activity is given below:

	Capacity utilization (75 %)	Capacity utilization (100 %)
	Amount in ₹ (Lakhs)	Amount in ₹ (Lakhs)
Direct materials	180	240
Direct wages	120	160
Power and fuel	12	16
Repairs and maintenance	18	21
Consumables	21	28
Supervision	20	20
Indirect labour	36	42

	Capacity utilization (75 %)	Capacity utilization (100 %)
Administrative expenses	21	21
Selling expenses	18	18
Depreciation	54	54

You are required to:

i. CALCULATE the profit earned by PS Limited at 75% level of activity.

ii. CALCULATE the break-even level of activity.

ICAI'S ANSWER

Calculation of Semi Variable component

	Repairs and Maintenance (₹)	Indirect labour (₹)
At 75% capacity	18,00,000	36,00,000
At 100% capacity	21,00,000	42,00,000
Variable component for 25%	3,00,000	6,00,000
Hence variable cost at 75%	3,00,000 × 75/25=**9,00,000**	6,00,000 × 75/25 = **18,00,000**
Fixed cost at 75% capacity	18,00,000 – 9,00,000 = **9,00,000**	36,00,000 – 18,00,000=**18,00,000**

Segregation of Fixed and Variable cost

	75%	100%	VC at 75%	FC at 75%
Direct Material	180	240	180	
Direct Labour	120	160	120	
Power and fuel	12	16	12	
Repairs and maintenance	18	21	**9**	**9**
Consumables	21	28	21	
Supervision	20	20		20
Indirect labour	36	42	**18**	**18**
Administrative expenses	21	21		21
Selling expenses	18	18		18
Depreciation	54	54		54
Total	500	620	**360**	**140**

(i) Calculation of profit earned at 75% capacity

Given PV ratio = 40%, Hence variable cost would be 60%

If variable cost is ₹ 360 lakhs then sales would be 360/ 0.60	= ₹ 600 lakhs
Less: Variable cost	= ₹ 360 lakhs
Less: Fixed cost	= ₹ 140 lakhs
Profit	**= ₹ 100 lakhs**

(ii) Break-even level of activity

BEP Sales = FC/ P/V ratio = 140 /0.40 = **₹ 350 lakhs**

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1003 MTP **Old P3 Nov 2022 S-II** Q1(a) 5 marks old scheme

A company makes 1,500 units of a product for which the profitability statement is given below:

	(₹)
Sales	1,20,000
Direct Materials	30,000
Direct Labour	35,000
Variable Overheads	15,000
Fixed Cost	16,800
Profit	22,200

After the first 500 units of production, the company has to pay a premium of ₹ 5 per unit towards overtime labour. The premium so paid has been included in the direct labour cost of ₹ 35,000 given above.

You are required to COMPUTE the Break-even point.

ICAI'S ANSWER

Data / Unit	1 – 500 (Rs.)	501 – 1,500 (Rs.)
Sales (Rs.1,20,000 / 1,500 units)	80	80
Direct Material (Rs.30,000 / 1,500 units)	20	20
Direct Labour*	20	25
Variable Overheads (Rs.15,000 / 1,500 units)	10	10
Contribution	30	25

Contribution at 500 units	=	Rs. 15,000
Fixed Cost	=	Rs. 16,800
Shortfall	=	Rs. 1,800
No. of units to recover shortfall	=	72 units (Rs. 1,800 / Rs.25)
Break Even Point	=	572 units (500 units + 72 units)

(*)

Let X be the Direct Labour per unit up to 500 units. Total Direct Labour-

500X + 1,000 × (X + 5)	=	35,000
1,500X + 5,000	=	35,000
X	=	20

Therefore, up to 500 units the Direct Labour is Rs. 20. After 500 units it is Rs. 25.

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1004 MTP **Old P3 May 2022 S-I** Q3(b) 10 marks old scheme

At budget activity of 80% of total capacity, a company earns a P/V ratio of 30% and a profit of 15% of total sales. Due to covid pandemic resulting in poor demand, the company has to reduce its selling price by 10%. The company was able to achieve a production and sales volume for the year equivalent to 50% of total capacity. The sales value at this level was ₹ 27,00,000 at a reduced price of ₹ 18 per unit. Due to reduction in production, the actual variable cost went up by 5% of the budget.

You are required to:

(i) PREPARE statement of profitability at budget and actual activity.

(ii) FIND P/V ratio and BES (in ₹ and unit of the actual sales activity).

ICAI'S ANSWER

Actual Sales	₹ 27,00,000
Actual Selling Price per unit	18
Actual units (50%) (27,00,000/18)	1,50,000
Therefore, budgeted units (80%) (1,50,000 × 80/50)	2,40,000
Budgeted Selling Price (18/90%)	20

$$\text{Budgeted Variable cost per unit} = \frac{(2,40,000 \text{ x } 20)\ (1 - .30)}{2,40,000 \text{ units}} = \frac{33,60,000}{2,40,000 \text{ units}} = ₹\ 14$$

(i) Statement of profitability at budget and actual activity

Particulars		Budget (80%)	Actual (50%)
Units		2,40,000	1,50,000
Sales (₹)	(a)	48,00,000	27,00,000
Variable cost (₹)	(b)	33,60,000	22,05,000
Contribution (₹)	(c = a - b)	14,40,000	4,95,000
Fixed cost (₹)	(d)	7,20,000	7,20,000
Profit (₹)	(e = c – d)	7,20,000	(2,25,000)

(ii) Calculation of P/V ratio and BES

$$\text{P/V ratio} = \frac{\text{Contribution}}{\text{Sales}} \times 100$$

$$= \frac{4,95,000}{27,00,000} \times 100 = 18.33\%$$

$$\text{Break Even Sales (in ₹)} = \frac{\text{Fixed Cost}}{\text{P/v Ratio}}$$

$$= \frac{7,20,000}{18.33\%} = ₹\ 39,27,987$$

$$\text{Break Even Sales (in Units)} = \frac{\text{Fixed Cost}}{\text{Contribution per unit}}$$

$$= \frac{7,20,000}{3.3^*} = 2,18,182 \text{ Units}$$

$$^*\text{Contribution per unit} = \frac{4,95,000}{1,50,000 \text{ units}} = 3.3 \text{ per unit}$$

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1005 MTP **Old P3 Dec 2021 S-I** Q5(a) 10 marks old scheme

Amy Ltd. manufacture and sales its product RM. The following figures have been collected from cost records of last year for the product RM:

Elements of Cost	Variable Cost portion	Fixed Cost
Direct Material	30% of Cost of Goods Sold	--
Direct Labour	15% of Cost of Goods Sold	--
Factory Overhead	10% of Cost of Goods Sold	₹ 3,45,000
Administration Overhead	2% of Cost of Goods Sold	₹ 1,06,500
Selling & Distribution Overhead	4% of Cost of Sales	₹ 1,02,000

Last Year, 7,500 units were sold at ₹ 185 per unit. From the given information, DETERMINE the followings:

(i) Break-even Sales (in rupees)

(ii) Profit earned during last year

(iii) Margin of safety (in %)

(iv) Profit if the sales were 10% less than the actual sales.

(Assume that Administration Overhead is related with production activity)

ICAI'S ANSWER

Working Notes:

(1) **Calculation of Cost of Goods Sold (COGS):**

COGS = DM + DL + FOH + AOH

COGS = {0.3 COGS + 0.15 COGS + (0.10 COGS + ₹ 3,45,000) + (0.02 COGS + ₹ 1,06,500)}

Or, COGS = 0.57 COGS + ₹ 4,51,500

Or COGS = $\frac{₹\ 4,51,500}{0.43}$ = ₹ 10,50,000

(2) **Calculation of Cost of Sales (COS):**

COS = COGS + S&DOH

COS = COGS + (0.04 COS + ₹ 1,02,000)

Or COS = ₹ 10,50,000 + (0.04 COS + ₹ 1,02,000)

Or, COS = $\frac{₹\ 11,52,000}{0.96}$ = ₹ 12,00,000

(3) **Calculation of Variable Costs:**

Direct Material-	(0.30 × ₹ 10,50,000)	₹ 3,15,000
Direct Labour-	(0.15 × ₹ 10,50,000)	₹ 1,57,500
Factory Overhead-	(0.10 × ₹ 10,50,000)	₹ 1,05,000
Administration OH-	(0.02 × ₹ 10,50,000)	₹ 21,000
Selling & Distribution OH	(0.04 × ₹ 12,00,000)	₹ 48,000
		₹ 6,46,500

(4) **Calculation of total Fixed Costs:**

Factory Overhead	₹ 3,45,000
Administration OH	₹ 1,06,500
Selling & Distribution OH	₹ 1,02,000
	₹ 5,53,500

(5) **Calculation of P/V Ratio:**

$$\text{P/V Ratio} = \frac{\text{Contribution}}{\text{Sales}} \times 100 = \frac{\text{Sales} - \text{Variable Costs}}{\text{Sales}} \times 100$$

$$= \frac{(₹\ 185 \times 7,500 \text{ units}) - ₹\ 6,46,500}{₹\ 185 \times 7,500 \text{ units}} \times 100$$

$$= \frac{₹\ 13,87,500 - ₹\ 6,46,500}{₹\ 13,87,500} \times 100 = 53.41\%$$

(i) Break-Even Sales = $\frac{\text{Fixed Costs}}{\text{P / V Ratio}} = \frac{₹\ 5,53,500}{53.41\%} = ₹\ 10,36,323$

(ii) Profit earned during the last year

= (Sales – Total Variable Costs) – Total Fixed Costs

= (₹ 13,87,500 - ₹ 6,46,500) - ₹ 5,53,500

= ₹ 1,87,500

(iii) Margin of Safety (%) = $\frac{\text{Sales} - \text{Breakeven sales}}{\text{Sales}} \times 100$

= $\frac{₹\ 13,87,500 - ₹\ 10,36,323}{₹\ 13,87,500} \times 100 = 25.31\%$

(iv) Profit if the sales were 10% less than the actual sales:

Profit = 90% (₹ 13,87,500 - ₹ 6,46,500) - ₹ 5,53,500

= ₹ 1,13,400

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1006 MTP **Old P3 Jul 2021 S-I** Q5(a) 10 marks old scheme

The following information has been obtained from the records of a manufacturing unit:

	Rs.	Rs.
Sales 80,000 units @ Rs. 50		40,00,000
Material consumed	16,00,000	
Variable Overheads	4,00,000	
Labour Charges	8,00,000	
Fixed Overheads	7,20,000	35,20,000
Net Profit		4,80,000

CALCULATE:

(i) The number of units by selling which the company will neither lose nor gain anything.

(ii) The sales needed to earn a profit of 20% on sales.

(iii) The extra units which should be sold to obtain the present profit if it is proposed to reduce the selling price by 20% and 25%.

(iv) The selling price to be fixed to bring down its Break-even Point to 10,000 units under present conditions.

ICAI'S ANSWER

Workings:

(1) Contribution per unit = Selling price per unit – Variable cost per unit

= Rs. 50 – {Rs. (16,00,000 + 4,00,000 + 8,00,000) ÷ 80,000 units}

= Rs. 50 – Rs. 35 = Rs. 15

(2) Profit-Volume (P/V) Ratio = $\frac{\text{Contribution per unit}}{\text{Selling price per unit}} \times 100 = \frac{\text{Rs.15}}{\text{Rs.50}} \times 100 = 30\%$

Calculations:

(i) The number of units to be sold for neither loss nor gain i.e. Break-even units:

$= \frac{\text{Fixed Overheads}}{\text{Contribution per unit}} = \frac{\text{Rs.7,20,000}}{\text{Rs.15}} = 48{,}000 \text{ units}$

(ii) The sales needed to earn a profit of 20% on sales:

As we know

S=V+F+P

(S = Sales; V = Variable Cost; F = Fixed Cost; P = Profit)

Suppose Sales units are x then

Rs. 50*x* = Rs. 35 x + Rs. 7,20,000 + Rs. 10*x*

Rs. 50*x* – Rs. 45*x* = Rs. 7,20,000

Or, $x = \frac{\text{Rs.7,20,000}}{\text{Rs.5}} = 1{,}44{,}000 \text{ units}$

Therefore, Sales needed = 1,44,000 units × Rs. 50 = Rs. 72,00,000 to earn a profit of 20% on sales.

(iii) Calculation of extra units to be sold to earn present profit of Rs.4,80,000 under the following proposed selling price:

		When selling price is reduced by	
		20% (Rs.)	**25% (Rs.)**
	Selling price per unit	40.00 (Rs. 50 × 80%)	37.50 (Rs. 50 × 75%)
	Less: Variable Cost per unit	35.00	35.00
	Contribution per unit	5.00	2.50
	Desired Contribution:		
	Fixed Overheads	7,20,000	7,20,000
	Desired Profit	4,80,000	4,80,000
		12,00,000	12,00,000
(a)	Sales unit for desired contribution (Desired Contribution / Contribution per unit)	2,40,000 units (Rs. 12,00,000 / Rs. 5)	4,80,000 units (Rs.12,00,000 / Rs. 2.5)
(b)	Units presently sold	80,000 units	80,000 units
(c)	Extra units to be sold {(a) – (b)}	1,60,000 units	4,00,000 units

(iv) Sales price to bring down BEP to 10,000 units:

B.E.P (Units) = $\frac{\text{Fixed Cost}}{\text{Contribution per unit}}$

Or, Contribution per unit = $\frac{\text{Rs.}7,20,000}{10,000 \text{ units}}$ = Rs. 72

So, Sales Price (per unit) = Variable Cost + Contribution

= Rs. 35 + Rs. 72 = Rs. 107

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1007 MTP **Old P3 Jul 2021 S-II** Q6(c) 5 marks old scheme

Same question, different ICAI answer: #997 MTP May 2024 S-II

DISCUSS basic assumptions of Cost Volume Profit analysis.

ICAI'S ANSWER

Assumptions of Cost Volume Profit analysis:

1. **Changes in the levels of revenues and costs arise only because of changes in the number of product (or service) units produced and sold –** for example, the number of television sets produced and sold by Sony Corporation or the number of packages delivered by Overnight Express. The number of output units is the only revenue driver and the only cost driver. Just as a cost driver is any factor that affects costs, a revenue driver is a variable, such as volume, that causally affects revenues.
2. **Total costs can be separated into two components**; a fixed component that does not vary with output level and a variable component that changes with respect to output level. Furthermore, variable costs include both direct variable costs and indirect variable costs of a product. Similarly, fixed costs include both direct fixed costs and indirect fixed costs of a product
3. When represented graphically, **the behaviours of total revenues and total costs are linear** (meaning they can be represented as a straight line) in relation to output level within a relevant range (and time period).
4. **Selling price, variable cost per unit, and total fixed costs (within a relevant range and time period) are known and constant**.
5. The analysis either covers a single product or assumes that **the proportion of different products when multiple products are sold will remain constant** as the level of total units sold changes.
6. All revenues and costs can be added, subtracted, and compared **without taking into account the time value of money.**

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1008 MTP **Old P3 Nov 2020 S-I** Q2(b) 10 marks old scheme

CanCola, a zero sugar cold drink manufacturing Indian company, is planning to establish a subsidiary company in Nepal to produce coconut flavoured juice. Based on the estimated annual sales of 60,000

bottles of the juice, cost studies produced the following estimates for the Nepalese subsidiary:

	Total Annual Costs (₹)	Percent of Total Annual Cost which is variable
Material	2,70,000	100%
Labour	1,97,000	80%
Factory Overheads	1,20,000	60%
Administration Expenses	52,000	35%

The Nepalese production will be sold by manufacturer's representatives who will receive a commission of 9% of the sale price. No portion of the Indian office expenses is to be allocated to the Nepalese subsidiary. You are required to-

(i) COMPUTE the sale price per bottle to enable the management to realize an estimated 20% profit on sale proceeds in Nepal.

(ii) CALCULATE the break-even point in rupees value sales and also in number of bottles for the Nepalese subsidiary on the assumption that the sale price is ₹ 14 per bottle.

ICAI'S ANSWER

(i) **Computation of Sale Price Per Bottle**

Output: 60,000 Bottles

	(₹)
Variable Cost:	
Material	2,70,000
Labour (₹ 1,97,000 × 80%)	1,57,600
Factory Overheads (₹1,20,000 × 60%)	72,000
Administrative Overheads (₹ 52,000 × 35%)	18,200
Commission (9% on ₹9,00,000 (Working Note -1))	81,000
Fixed Cost:	
Labour (₹ 1,97,000 × 20%)	39,400
Factory Overheads (₹ 1,20,000 × 40%)	48,000
Administrative Overheads (₹ 52,000 × 65%)	33,800
Total Cost	7,20,000
Profit (20% of ₹ 9,00,000)	1,80,000
Sales Proceeds	9,00,000
Sales Price per bottle $\left(\frac{\text{Rs.}9,00,000}{60,000}\right)$	15

(ii) **Calculation of Break-even Point**

Sales Price per Bottle = ₹ 14

Variable Cost per Bottle = $\frac{\text{Rs.5,93,400 (working note} - 2)}{\text{60,000 bottles}}$ = ₹ 9.89

Contribution per Bottle = ₹ 14 – ₹ 9.89 = ₹ 4.11

Break -even Point (in number of Bottles) = $\frac{\text{Fixed cost}}{\text{Contribution per bottle}}$

= $\frac{\text{Rs.1,21,200}}{\text{Rs.4.11}}$ = 29,489

Break- even Point (in Sales Value) = 29,489 Bottles × ₹14

= ₹ 4,12,846

Working Note

(1) Let the Sales Price be 'X'

Commission = $\frac{9X}{100}$

Profit = $\frac{20X}{100}$

X= ₹ 2,70,000 + ₹1,57,600 + ₹ 72,000 + ₹ 18,200 + ₹ 39,400 + ₹ 48,000 + ₹ 33,800 + $\frac{9X}{100} + \frac{20X}{100}$

X = ₹ 6,39,000 + $\frac{9X}{100} + \frac{20X}{100}$

100X – 9X – 20X = 6,39,00,000

71X = 6,39,00,000

X = $\frac{6,39,00,000}{71}$ = ₹ 9,00,000

(2)

Total Variable Cost	(₹)
Material	2,70,000
Labour	1,57,600
Factory Overheads	72,000
Administrative Overheads	18,200
Commission [(60,000 Bottles × ₹ 14) × 9%]	75,600
	5,93,400

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1009 MTP **Old P3 May 2020 S-I** Q1(a) 5 marks old scheme

A company gives the following information:

Margin of Safety	Rs.7,50,000
Total Cost	Rs.7,75,000
Margin of Safety (Qty.)	15,000 units
Break Even Sales in Units	5,000 units

You are required to CALCULATE:

(i) Selling price per unit

(ii) Profit

(iii) Profit/ Volume Ratio

(iv) Break Even Sales (in Rupees)

(v) Fixed Cost

ICAI'S ANSWER

(i) Selling Price per unit = $\frac{\text{Margin of Safety in Rupee value}}{\text{Margin of Safety in Quantity}}$

= $\frac{\text{Rs.7,50,000}}{\text{15,000units}}$ = Rs.50

(ii) Profit = Sales Value – Total Cost

= Selling price per unit × (BEP units + MoS units) – Total Cost

= Rs.50 × (5,000 + 15,000) units – Rs.7,75,000

= Rs.10,00,000 – Rs.7,75,000 = Rs.2,25,000

(iii) Profit/ Volume (P/V) Ratio = $\frac{\text{Profit}}{\text{Margin of Safety in Rupee value}} \times 100$

= $\frac{\text{Rs.2,25,000}}{\text{Rs.7,50,000}} \times 100 = 30\%$

(iv) Break Even Sales (in Rupees) = BEP units × Selling Price per unit

= 5,000 units × Rs.50 = Rs.2,50,000

(v) Fixed Cost = Contribution – Profit

= Sales Value × P/V Ratio – Profit

= (Rs.10,00,000 × 30%) – Rs.2,25,000

= Rs.3,00,000 – Rs.2,25,000 = Rs.75,000

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1010 MTP **Old P3 Nov 2019 S-I** Q1(a) 5 marks old scheme

C.T. Ltd. manufactures and sells a single product X whose selling price is Rs. 100 per unit and the variable cost is Rs. 60 per unit.

(i) If the Fixed Costs for this year are Rs. 24,00,000 and the annual sales are at 60% margin of safety, CALCULATE the rate of net return on sales, assuming an income tax level of 40%.

(ii) For the next year, it is proposed to add another product line Y whose selling price would be Rs. 150 per unit and the variable cost Rs. 100 per unit. The total fixed costs are estimated at Rs. 28,00,000. The sales mix of X : Y would be 5 : 3. COMPUTE the break-even sales in units for both the products.

ICAI'S ANSWER

(i) Contribution per unit = Selling price – Variable cost

= Rs.100 – Rs.60

= Rs.40

$$\text{Break-even Point} = \frac{\text{Rs.24,00,000}}{\text{Rs.40}}$$

= 60,000 units

$$\text{Percentage Margin of Safety} = \frac{\text{Actual Sales} - \text{Break - even Sales}}{\text{Actual Sales}}$$

$$\textit{Or, } 60\% = \frac{\text{Actual Sales} - 60{,}000\ \textit{units}}{\text{Actual Sales}}$$

∴ Actual Sales = 1,50,000 units

	(Rs.)
Sales Value (1,50,000 units × Rs.100)	1,50,00,000
Less: Variable Cost (1,50,000 units ×Rs.60)	90,00,000
Contribution	60,00,000
Less: Fixed Cost	24,00,000
Profit	36,00,000
Less: Income Tax @ 40%	14,40,000
Net Return	21,60,000

$$\text{Rate of Net Return on Sales} = 14.40\% \left(\frac{\text{Rs.21,60,000}}{\text{Rs.1,50,00,000}} \times 100\right)$$

(ii) Products

	X (Rs.)	Y (Rs.)
Selling Price *per unit*	100	150
Variable Cost *per unit*	60	100
Contribution *per unit*	40	50

Composite contribution will be as follows:

$$\text{Contribution per unit} = \left(\frac{40}{8} \times 5\right) + \left(\frac{50}{8} \times 3\right)$$

= 25 + 18.75 = Rs.43.75

$$\text{Break-even Sale} = 64{,}000 \text{ units } \left(\frac{\text{Rs.28,00,000}}{\text{Rs.43.75}}\right)$$

Break-even Sales Mix:

X (64,000 units × 5/8) = 40,000 units

Y (64,000 units × 3/8) = 24,000 units

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14.3 Marginal versus absorption costing and profit reconciliation

4 questions · asked in 3 ICAI documents · 1 exam paper

1011 EXAM **Old P3 Nov 2020** Q6(e) 5 marks old scheme

Differentiate between "Marginal and Absorption Costing".

ICAI'S ANSWER

Difference between Marginal costing and Absorption costing

S. No.	Marginal costing	Absorption costing
1.	Only variable costs are considered for product costing and inventory valuation.	Both fixed and variable costs are considered for product costing and inventory valuation.
2.	Fixed costs are regarded as period costs. The Profitability of different products is judged by their P/V ratio.	Fixed costs are charged to the cost of production. Each product bears a reasonable share of fixed cost and thus the profitability of a product is influenced by the apportionment of fixed costs.
3.	Cost data presented highlight the total contribution of each product.	Cost data are presented in conventional pattern. Net profit of each product is determined after subtracting fixed cost along with their variable costs.
4.	The difference in the magnitude of opening stock and closing stock does not affect the unit cost of production.	The difference in the magnitude of opening stock and closing stock affects the unit cost of production due to the impact of related fixed cost.
5.	In case of marginal costing the cost per unit remains the same, irrespective of the production as it is valued at variable cost	In case of absorption costing the cost per unit reduces, as the production increases as it is fixed cost which reduces, whereas, the variable cost remains the same per unit.

ICAI · Old P3 · Nov 2020 exam · question p. 22 · answer p. 24

1012 MTP **May 2025 S-I** Case scenario · Q5(a)(i) to Q5(a)(ii) 6 marks

CASE SCENARIO

RN Ltd. is a manufacturing company producing single product. The following information is available relating to current financial year:

Activity Level	100%
Sales and production (units)	4,400
	(₹)

Activity Level	100%
Sales	88,00,000
Production costs:	
- Variable	35,20,000
- Fixed	8,80,000
Selling and distribution costs:	
- Variable	17,60,000
- Fixed	13,20,000

Fixed costs are incurred evenly throughout the year, and actual fixed costs are the same as budgeted. Thus, at 50% activity level, fixed production cost and fixed selling and distribution costs remains the same and other data/figures which are variable changes proportionately.

The normal level of activity for the year is 4,400 units.

There were no stock at the beginning of the year.

In the first quarter, 1,210 units were produced and 880 units were sold.

You are required to COMPUTE the profit for the quarter using-

Q5(a)(i) 3 marks

Absorption costing

Q5(a)(ii) 3 marks

Marginal costing.

ICAI'S ANSWERS

Q5(a)(i)

Profit for the Quarter (Absorption Costing)

Particulars	(₹)	(₹)
Sales revenue (880 units × (₹ 88,00,000/4,400 units)): (A)		17,60,000
Less: Production costs:		
- Variable cost (1,210 units × (₹ 35,20,000/4,400 units))	9,68,000	
- Fixed overheads absorbed (1,210 units × ₹ 200*)	2,42,000	12,10,000
Add: Opening stock		--
Less: Closing Stock[(₹ 12,10,000/1,210 units) x 330 units]		(3,30,000)
Cost of Goods sold		8,80,000
Less: Adjustment for over-absorption of fixed production overheads **		(22,000)
Add: Selling & Distribution Overheads:		

Particulars	(₹)	(₹)
- Variable (880 units × (₹ 17,60,000/4,400 units))	3,52,000	
- Fixed (1/4th of ₹ 13,20,000)	3,30,000	6,82,000
Cost of Sales (B)		15,40,000
Profit {(A) – (B)}		**2,20,000**

* absorption rate: ₹ 8,80,000/4,400 = ₹ 200 per unit

** Actual fixed production overhead - Absorbed fixed production overhead

= ₹ 8,80,000/4 - 2,42,000 = ₹ (22,000)

Q5(a)(ii)

Profit for the Quarter (Marginal Costing)

	(₹)	(₹)
Sales revenue (880 units × (₹ 88,00,000/4,400 units)): (A)		17,60,000
Less: Production costs:		
- Variable cost (1,210 units × (₹ 35,20,000/4,400 units))		9,68,000
Add: Opening stock		--
Less: Closing Stock [(₹ 9,68,000/1,210 units) x 330 units]		(2,64,000)
Variable cost of goods sold		7,04,000
Add: Selling & Distribution Overheads:		
- Variable (880 units × (₹ 17,60,000/4,400 units))		3,52,000
Cost of Sales (B)		10,56,000
Contribution {(C) = (A) – (B)}		**7,04,000**
Less: Fixed Costs:		
- Production cost (₹ 8,80,000/4)	(2,20,000)	
- Selling & distribution cost (₹ 13,20,000/4)	(3,30,000)	(5,50,000)
Profit		**1,54,000**

ICAI · P4 · MTP May 2025 · Series I · question p. 13 · answer p. 19, 19-20

1013 MTP **May 2024 S-I** Q4(a) 10 marks

AB Ltd produces a single product V2 and sells it at a fixed price of ₹ 2,050 per unit. The production and

sales data for first quarter of the year 2023-24 are as follows:

	April	May	June
Sales in units	4,200	4,500	5,200
Production in units	4,600	4,400	5,500

Actual/budget information for each month was as follows:

Direct materials	4 kilograms at ₹ 120 per kilogram
Direct labour	6 hours at ₹ 60 per hour
Variable production overheads	150% of direct labour
Fixed production overheads	₹ 5,00,000
Fixed selling overheads	₹ 95,000

There was no opening inventory at the start of the quarter. Fixed production overheads are budgeted at ₹ 60,00,000 per annum and are absorbed into products based on a budgeted normal output of 60,000 units per annum.

Required:

(i) Prepare a profit statement for each of the three months using absorption costing principles.

(ii) Prepare a profit statement for each of the three months using marginal costing principles.

(iii) Present a reconciliation of the profit or loss figures given in your answer to (i) and (ii).

ICAI'S ANSWER

(i) Statement of Profit under Absorption Costing

Particulars	April (₹)	May (₹)	June (₹)
Sales (units)	4,200	4,500	5,200
Selling price per unit	2,050	2,050	2,050
Sales value (A)	86,10,000	92,25,000	1,06,60,000
Cost of Goods Sold:			
Opening Stock @ ₹1,480	0	5,92,000	4,44,000
Production cost @ ₹1,480	68,08,000	65,12,000	81,40,000
Closing Stock @ ₹1,480	(5,92,000)	(4,44,000)	(8,88,000)
Under/ (Over) absorption	40,000	60,000	(50,000)
Add: Fixed Selling Overheads	95,000	95,000	95,000
Cost of Sales (B)	63,51,000	68,15,000	77,41,000

Particulars	April (₹)	May (₹)	June (₹)
Profit (A – B)	22,59,000	24,10,000	29,19,000

Workings:

1. Calculation of full production cost

	(₹)
Direct Materials (4 kg. × ₹ 120)	480
Direct labour (6 hours × ₹ 60)	360
Variable production Overhead (150% of ₹ 360)	540
Total Variable cost	1,380
Fixed production overhead (₹60,00,000/60,000units)	100
	1,480

2. Calculation of Opening and Closing stock

	April	May	June
Opening Stock	0	400	300
Add: Production	4,600	4,400	5,500
Less: Sales	4,200	4,500	5,200
Closing Stock	400	300	600

3. Calculation of Under/Over absorption of fixed production overhead

	April (₹)	May (₹)	June (₹)
Actual Overhead	5,00,000	5,00,000	5,00,000
Overhead absorbed	4,60,000 (4,600 units × ₹100)	4,40,000 (4,400 units × ₹100)	5,50,000 (5,500 units × ₹100)
Under/(Over) absorption	40,000	60,000	(50,000)

(ii) Statement of Profit under Marginal Costing

Particulars	April (₹)	May (₹)	June (₹)
Sales (units)	4,200	4,500	5,200
Selling price per unit	2,050	2,050	2,050
Sales value	86,10,000	92,25,000	1,06,60,000
Less: Variable production cost @ ₹1,380	57,96,000	62,10,000	71,76,000
Contribution	28,14,000	30,15,000	34,84,000
Less: Fixed Production Overheads	5,00,000	5,00,000	5,00,000
Less: Fixed Selling Overheads	95,000	95,000	95,000
Profit	22,19,000	24,20,000	28,89,000

(iii) Reconciliation of profit under Absorption costing to Marginal Costing

Particulars	April (₹)	May (₹)	June (₹)
Profit under Absorption Costing	22,59,000	24,10,000	29,19,000
Add: Opening Stock	0	40,000 (400 × ₹100)	30,000 (300 × ₹100)
Less: Closing Stock	40,000 (400 × ₹100)	30,000 (300 × ₹100)	60,000 (600 × ₹100)
Profit under Marginal Costing	22,19,000	24,20,000	28,89,000

ICAI · P4 · MTP May 2024 · Series I · question p. 9 · answer p. 9-11

14.4 Limiting factor and product mix

10 questions · asked in 10 ICAI documents · 7 exam papers

1014 EXAM Sep 2026 Q1(a) 7 marks

A company manufactures two products – DSP and EM. It has adopted marginal costing for managerial decision-making and has provided the following information for the coming year :

	DSP	EM	Remarks
Selling Price per unit (₹)	15,000	10,000	–
Variable costs per unit (₹)	9,000	4,600	–
Maximum Demand for the year (in units)	700	720	Any quantity at or below this limit can be readily sold in the market.
Maximum Production Capacity	–	–	770 units of DSP and EM put together (any one or both can be produced but the overall limit is 770 units).
Fixed Expenses (₹)	6,00,000	–	Incurred only if DSP is manufactured.

	DSP	EM	Remarks
	–	2,16,000	Incurred only if EM is manufactured.
Other Fixed expenses (excluding the above)	–	–	₹ 18,00,000 incurred whether any product is made or not.

Assume no inventory is built up. Whatever is produced is sold. The company can choose to make any of the products or both the products without affecting its future prospects.

Requirement :

Calculate the optimum profit for the year in rupees giving the product mix (in units).

(Present only relevant figures. Detailed statements are not required.)

ICAI'S ANSWER

ICAI has not published its answers to this paper yet.

ICAI · P4 · Sep 2026 exam · question p. 10-11

1015 **EXAM** **Sep 2025** Q3(b) **6 marks**

SM Limited is the manufacturer of the two products A & B. The following particulars are extracted from the records of the company:

	A	B
Maximum Capacity	5,000 units	3,500 units
Selling price per unit	₹ 1,000	₹ 1,500
Cost per unit:		
Raw Material @ ₹ 20 per kg	₹ 200	₹ 400
Wages @ 10 per hour	₹ 150	₹ 100
Direct Expenses	₹ 200	₹ 300
Variable overhead	₹ 80	₹ 120

The total fixed overhead for product A is ₹ 2,50,000 and for product B is ₹ 3,50,000.

The company manufactures both the products using the same grade of material. The company is facing a constraint of raw material which is available in limited quantity of 1,10,000 kgs only.

Required:

Determine the optimum product mix, considering material as the limiting factor, to generate maximum profit and calculate the maximum profit.

ICAI'S ANSWER

	A	B
	(₹)	**(₹)**
Selling Price	1,000	1,500

	A	B
Less: Direct Materials	200	400
Direct Labour	150	100
Direct Expenses	200	300
Variable Overheads	80	120
Contribution per unit	370	580
Raw Material required per unit	10 kg	20 kg
Contribution per kg of raw material	37	29
Rank	I	II

Calculation of Product Mix:

Raw material available	1,10,000 kg	
Product A (5,000 × 10)	50,000 kg	5,000 units
Product B (3,000 × 20)	60,000 kg	3,000 units

Statement of Profit:

Particulars	A	B	Total
Units	5,000	3,000	
Total Contribution	18,50,000	17,40,000	35,90,000
Fixed Cost	2,50,000	3,50,000	6,00,000
Profit	16,00,000	13,90,000	29,90,000

ICAI · P4 · Sep 2025 exam · question p. 17-18 · answer p. 19-20

1016 EXAM **Sep 2024** Q5(b) 7 marks

ABC Ltd. is a well-known company for producing baby care products.

The company produces and sells two variants of organic shampoo for children: "Baby Rose" and "Baby Lily". The sales and cost data for both products are provided below:

Particulars	Baby Rose	Baby Lily
Current demand and Sales (Number of bottles)	4,000	3,000
Production Capacity (Number of bottles)	7,500	6,000
Selling Price per bottle (₹)	600	750
Variable Costs per bottle:		
- Direct Materials (₹ 20 Per litre)	160	200
- Other Variable Costs	270	350

The fixed costs amount to ₹ 5,00,000 and ₹ 4,50,000 for Baby Rose and Baby Lily respectively. The Production Manager has informed that 1,00,000 litres of material is available for production. A dealer has approached the company and proposed to purchase both products at the existing selling prices, which are to be produced by utilizing the remaining unused material. However, he has insisted that all the bottles must be packed with eco-friendly packaging, which will result in an additional cost of ₹ 10 per bottle for the company. Presently, the company is not using eco-friendly material for packing of bottles.

Required:

Prepare a detailed statement showing the overall contribution and profit of the company after acceptance of the dealer's proposal.

ICAI'S ANSWER

Statement showing the Overall contribution and profit of the company

Particulars	Baby Rose	Baby Lily	Total
	(₹)	**(₹)**	**(₹)**
Selling price per bottle	600	750	-
Less: Direct Materials	160	200	-
Other variable costs	270	350	-
Additional packaging	10	10	
Contribution per bottle	**160**	**190**	-
Material required per bottle	8 litres	10	
Contribution per litre of material	**20**	**19**	
Ranking on the basis of Contribution per litre of material	I	II	
	Baby Rose (4,000 + 3,500 bottles)	Baby Lily (3,000 +1,000 bottles)	

Particulars	Baby Rose	Baby Lily	Total
Selling price per bottle	600	750	-
Sales Value	45,00,000	30,00,000	
Variable Cost			
Direct Materials	7,500 units x ₹ 160 = 12,00,000	4,000 units x ₹ 200 = 8,00,000	
Other Variable Costs	7,500 units x ₹ 270 = 20,25,000	4,000 units x ₹ 350 = 14,00,000	
Eco-friendly pack cost	3,500 units x ₹ 10 = 35,000	1,000 units x ₹ 10 = 10,000	
Total Variable Costs	32,60,000	22,10,000	
Contribution	4,000 units x 170 3,500 units x 160 = 12,40,000	3,000 units x 200 1,000 units x 190 = 7,90,000	20,30,000
Less: Fixed Cost	5,00,000	4,50,000	9,50,000
Profit	**7,40,000**	**3,40,000**	**10,80,000**

WN1

	Baby Rose	Baby Lily
Raw Material used per unit of bottle **(a)**	8 litres (₹ 160/₹ 20)	10 litres (₹ 200/₹ 20)
Current Demand and Sales **(b)**	4,000 bottles	3,000 bottles
Total Raw Material used **(c = a x b)**	32,000 litres	30,000 litres

WN2

Raw Material available after current sales = 1,00,000 litres – 62,000 litres = 38,000 litres

Since the contribution per unit of Baby Rose is higher than Baby Lily, the company will produce and sale Baby Rose shampoo to the dealer.

Number of units that can be produced in 38,000 litres = 38,000 litres/8 litres

= 4,750 bottles

However, the Production capacity of Baby Rose is 7,500 bottles, only 3,500 bottles can be produced.

Raw materials used in 3,500 bottles = 8 litres x 3,500 bottles = 28,000 litres

Remaining material = 10,000 litres

Number of Baby Lily that can be produced in 10,000 litres = 10,000 litres/10 litres

= 1,000 bottles

Alternatively, Solution can also be presented in following way:

Statement showing the Overall contribution and profit of the company

Particulars	Baby Rose	Baby Lily	Total
	(₹)	(₹)	(₹)
Selling price per bottle	600	750	-
Less: Direct Materials	160	200	-
Other variable costs	270	350	-
Contribution per bottle Before additional packaging	170	200	-
Contribution per bottle per unit of raw material Before additional packaging	21.25	20	
Ranking on the basis of Contribution per bottle per unit of raw material	I	II	

Particulars	Current Sales (WN2)	Additional Sales of Baby Rose (3,500 bottles)	Additional Sales of Baby Lily (1,000 bottles)	Total
	(₹)	(₹)	(₹)	(₹)
Selling price per bottle	-	600	750	-
Less: Direct Materials	-	160	200	-
Other variable costs	-	270	350	-
Additional packaging	-	10	10	
Contribution per unit	-	160	190	-
Total Contribution	12,80,000	5,60,000	1,90,000	20,30,000
Less: Fixed Cost	9,50,000	-	-	9,50,000
Profit	3,30,000	5,60,000	1,90,000	10,80,000

WN1

	Baby Rose	Baby Lily
Raw Material used per unit of bottle **(a)**	8 litres (₹ 160/₹ 20)	10 litres (₹ 200/₹ 20)
Current Demand and Sales **(b)**	4,000 bottles	3,000 bottles
Total Raw Material used **(c = a x b)**	32,000 litres	30,000 litres

WN2

Statement showing the current contribution and profit of the company

Particulars	Baby Rose	Baby Lily	Total
	(₹)	(₹)	(₹)
Selling price per bottle	600	750	-
Less: Direct Materials	160	200	-
Other variable costs	270	350	-
Contribution per bottle Before additional packaging	170	200	-
Contribution per bottle per unit of raw material Before additional packaging	21.25	20	
Total Contribution Before additional packaging	6,80,000	6,00,000	12,80,000
Less: Fixed Cost	5,00,000	4,50,000	9,50,000
Profit	1,80,000	1,50,000	3,30,000

WN3

Raw Material available after current sales = 1,00,000 litres – 62,000 litres = 38,000 litres

Since the contribution per unit of Baby Rose is higher than Baby Lily, the company will produce and sale Baby Rose shampoo to the dealer.

Number of units that can be produced in 38,000 litres = 38,000 litres/8 litres

= 4,750 bottles

However, the Production capacity of Baby Rose is 7,500 bottles, only 3,500 bottles can be produced.

Raw materials used in 3,500 bottles = 8 litres x 3,500 bottles = 28,000 litres

Remaining material = 10,000 litres

Number of Baby Lily that can be produced in 10,000 litres = 10,000 litres/10 litres

= 1,000 bottles

ICAI · P4 · Sep 2024 exam · question p. 29-30 · answer p. 32-36

1017 EXAM **Old P3 May 2023** Q4(c) 5 marks old scheme

MNP Company Limited produces two products 'A' and 'B'. The relevant cost and sales data per unit of output is as follows.

Particulars	Product A	Product B
	(₹)	(₹)
Direct material	55	60
Direct labour	35	45
Variable factory overheads	40	20
Selling Price	180	175

The availability of machine hours is limited to 55,000 hours for the month. The monthly demand for product 'A' and product 'B' is 5,000 units and 6,000 units, respectively. The fixed expenses of the company are ₹1,40,000 per month. Variable factory overheads are ₹ 4 per machine hour. The company can produce both products according to the market demand.

Required:

Calculate the product mix that generates maximum profit for the company in the situation and also calculate profit of the company.

ICAI'S ANSWER

Particulars	Product A	Product B
	₹	₹
Selling Price	180	175
Variable cost:		
Direct Material	55	60
Direct labour	35	45
Variable factory overheads	40	20
	130	125
Contribution	50	50
Machine hour (p.u.)	10	5
Contribution per hour	**5**	**10**
Rank	**II**	**I**

Calculation of Product Mix

Hours available	55,000
Product B (6000 × 5)	**30,000**
Balance Hours	25,000
Product A (2500 × 10)	**25,000**
Balance Hours	0

Calculation of Profit

		₹
Contribution		
A	2500 units x 50	
B	6000 units x 50	4,25,000
Less:	Fixed cost	(1,40,000)
	Profit	**2,85,000**

ICAI · Old P3 · May 2023 exam · question p. 19 · answer p. 22-23

1018 **EXAM** **Old P3 Nov 2022** Q4(a) 10 marks old scheme

An agriculture based company having 210 hectares of land is engaged in growing three different cereals namely, wheat, rice and maize annually. The yield of the different crops and their selling prices are given below:

	Wheat	Rice	Maize
Yield (in kgs per hectare)	2,000	500	100
Selling Price (₹ per kg)	20	40	250

The variable cost data of different crops are given below:

(All figures in ₹ per kg)

Crop	Labour charges	Packing Materials	Other variable expenses
Wheat	8	2	4
Rice	10	2	1
Maize	120	10	20

The company has a policy to produce and sell all the three kinds of crops. The maximum and minimum area to be cultivated for each crop is as follows:

Crop	Maximum Area (in hectares)	Minimum Area (in hectares)
Wheat	160	100
Rice	50	40
Maize	60	10

You are required to:

(i) Rank the crops on the basis of contribution per hectare.

(ii) Determine the optimum product mix considering that all the three cereals are to be produced.

(iii) Calculate the maximum profit which can be achieved if the total fixed cost per annum is ₹ 21,45,000.

(Assume that there are no other constraints applicable to this company)

ICAI'S ANSWER

(i) Statement showing Ranking of crops on the basis of Contribution per hectare

Sl. No	Particulars	Wheat	Rice	Maize
(I)	Sales price per kg (₹)	20	40	250
(II)	Variable cost* per kg (₹)	14	13	150
(III)	Contribution per kg (₹)	6	27	100
(IV)	Yield (in kgs per hectare)	2,000	500	100
(V)	Contribution per hectare (₹)	12,000	13,500	10,000
(VI)	Ranking	II	I	III

*Variable cost = Labour Charges +Packing Material+ Other Variable Expenses

Therefore, to maximize profits, the order of priority of production would be Rice, Wheat and Maize.

(ii) & (iii) Statement showing optimum product mix considering that all the three cereals are to be produced and maximum profit thereof

Sl. No.	Particulars	Wheat	Rice	Maize	Total
(i)	Minimum Area (in hectare)	100	40	10	150
(ii)	Remaining area (in hectare)				60
(iii)	Distribution of remaining area based on ranking considering Maximum area	50	10	-	60
(iv)	Optimum mix (in hectare)	150	50	10	210
(v)	Contribution per hectare (₹)	12,000	13,500	10,000	
(vi)	Total contribution (₹)	18,00,000	6,75,000	1,00,000	25,75,000
(vii)	Fixed cost (₹)				21,45,000
(viii)	Maximum Profit (₹)				4,30,000

Optimum Product Mix and calculation of maximum profit earned by company can also be presented as below

(ii) Optimum Product Mix:

Particular	Area (in hectares)	Yield (kg per hectare)	Total Production (in kgs)
(a) Maximum of Rice	50	500	25000
(b) Minimum of Maize	10	100	1000
(c) Balance of Wheat	150	2000	300000
	210		326000

(iii) Calculation of maximum profit earned by the company:

	Production (in kgs)	Contribution (₹ per kg)	Total contribution (₹)
(a) Rice	25,000	24	6,75,000
(b) Maize	1,000	100	1,00,000
(c) Wheat	3,00,000	6	18,00,000
Total contribution			25,75,000
Less: Total Fixed Cost per annum			(21,45,000)
Maximum profits earned by the company			4,30,000

ICAI · Old P3 · Nov 2022 exam · question p. 14 · answer p. 15-17

1019 EXAM **Old P3 May 2022** Q3(a) 10 marks old scheme

SR Ltd. is a manufacturer of Garments. For the first three months of financial year 2022-23 commencing on 1st April 2022, production will be constrained by direct labour. It is estimated that only 12,000 hours of direct labour hours will be available in each month.

For market reasons, production of either of the two garments must be at least 25% of the production of the other. Estimated cost and revenue per garment are as follows:

	Shirt (₹)	Short (₹)
Sales price	60	44
Raw Materials		
Fabric @12 per metre	24	12
Dyes and cotton	6	4
Direct labour @ 8 per hour	8	4
Fixed Overhead @ 4 per hour	4	2
Profit	18	22

From the month of July 2022 direct labour will no longer be a constraint. The company expects to be able to sell 15,000 shirts and 20,000 shorts in July, 2022. There will be no opening stock at the beginning of July 2022.

Sales volumes are expected to grow at 10% per month cumulatively thereafter throughout the year. Following additional information is available:

- The company intends to carry stock of finished garments sufficient to meet 40% of the next month's sale from July 2022 onwards.
- The estimated selling price will be same as above.

Required:

I. Calculate the number of shirts and shorts to be produced per month in the first quarter of financial year 2022-2023 to maximize company's profit.

II. Prepare the following budgets on a monthly basis for July, August and September 2022:

(i) Sales budget showing sales units and sales revenue for each product.

(ii) Production budget (in units) for each product.

ICAI'S ANSWER

I. Calculation of number of shirts & shorts to be produced per month:

Contribution per labour hour:

		Shirts (₹)	Shorts (₹)
A	Sales Price per unit	60	44
B	Variable Cost:		
	- Raw materials	30	16
	- Direct labour	8	4
		38	20
C	Contribution per unit [A-B]	22	24
D	Labour hour per unit	1 hour	0.5 hour
E	Contribution per labour hour [C÷D]	**22**	**48**

Production plan for the first three months:

Since, Shorts has the higher Contribution per labour hour, it will be made first. Shirts will be 25% of Shorts. The quantity will be determined as below:

Let the Quantity of Shorts be X and Shirts will be 0.25 X, then

(Qty. of Shorts × labour hour per unit) + (Qty. of Shirts × labour hour per unit) = Total labour hours available

Or, (X × 0.5 hour) + (0.25X × 1 hour) = 12,000 hours

Or, 0.5X + 0.25X = 12,000

Or, 0.75X = 12,000

Or, X = 12,000÷0.75

= 16,000 units of Shorts

Therefore, for Shirts = 25% of 16,000 units

= 4,000 units

Production per month for the first quarter will be:

Shorts- 16,000 units &

Shirts- 4,000 units

II. (i) Sales Budget for the month of July, August & September 2022:

		July 2022		August 2022		September 2022	
		Shirts	**Shorts**	**Shirts**	**Shorts**	**Shirts**	**Shorts**
A	Sales demand	15,000	20,000	16,500	22,000	18,150	24,200
B	Selling price per unit (₹)	60	44	60	44	60	44
C	Sales Revenue (₹)	9,00,000	8,80,000	9,90,000	9,68,000	10,89,000	10,64,800

(ii) Production budget for the month of July, August & September 2022:

		July 2022		August 2022		September 2022		October 2022	
		Shirts	**Shorts**	**Shirts**	**Shorts**	**Shirts**	**Shorts**	**Shirts**	**Shorts**
A	Opening stock	0	0	6,600	8,800	7,260	9,680		
B	Sales demand	15,000	20,000	16,500	22,000	18,150	24,200	19,965	26,620
C	Closing stock	6,600	8,800	7,260	9,680	7,986	10,648		
D	Production [B+C-A]	21,600	28,800	17,160	22,880	18,876	25,168		

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1020 **EXAM** **Old P3 Nov 2020** Q1(b) 5 marks old scheme

Moon Ltd. produces products 'X', 'Y' and 'Z' and has decided to analyse it's production mix in respect of these three products - 'X', 'Y' and 'Z'.

You have the following information :

	X	Y	Z
Direct Materials ₹ (per unit)	160	120	80
Variable Overheads ₹ (per unit)	8	20	12

Direct labour :

Departments:	Rate per Hour (₹)	Hours per unit	Hours per unit	Hours per unit
		X	Y	Z
Department-A	4	6	10	5
Department-B	8	6	15	11

From the current budget, further details are as below :

	X	Y	Z
Annual Production at present (in units)	10,000	12,000	20,000
Estimated Selling Price per unit (₹)	312	400	240
Sales departments estimate of possible sales in the coming year (in units)	12,000	16,000	24,000

There is a constraint on supply of labour in Department-A and its manpower cannot be increased beyond its present level.

Required:

(i) Identify the best possible product mix of Moon Ltd.

(ii) Calculate the total contribution from the best possible product mix.

ICAI'S ANSWER

(i) Statement Showing "Calculation of Contribution/ unit"

Particulars	X (₹)	Y (₹)	Z (₹)
Selling Price (A)	312	400	240
Variable Cost:			
Direct Material	160	120	80
Direct Labour			
Dept. A (Rate x Hours)	24	40	20
Dept. B (Rate x Hours)	48	120	88
Variable Overheads	8	20	12
Total Variable Cost (B)	240	300	200
Contribution per unit (A - B)	**72**	**100**	**40**
Hours in Dept. A	6	10	5
Contribution per hour	12	10	8
Rank	I	II	III

Existing Hours = 10,000 x 6hrs. + 12,000 x 10 hrs. + 20,000 x 5 hrs. = 2,80,000 hrs.

Best possible product mix (Allocation of Hours on the basis of ranking)

Produce 'X'	= 12,000 units
Hours Required	= 72,000 hrs (12,000 units × 6 hrs.)
Balance Hours Available	= 2,08,000 hrs (2,80,000 hrs. – 72,000 hrs.)
Produce 'Y' (the Next Best)	= 16,000 units
Hours Required	= 1,60,000 hrs (16,000 units × 10 hrs.)
Balance Hours Available	= 48,000 hrs (2,08,000 hrs. – 1,60,000 hrs.)
Produce 'Z' (balance)	= 9,600 units (48,000 hrs./ 5 hrs.)

(ii) Statement Showing "Contribution"

Product	Units	Contribution/ Unit (₹)	Total Contribution (₹)
X	12,000	72	8,64,000
Y	16,000	100	16,00,000
Z	9,600	40	3,84,000
		Total	**28,48,000**

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1021 RTP **Sep 2025** Q16

N.P. Ltd. produces two products P and Q. The draft budget for the next month is as under:

	P	Q
Budgeted Production and Sales (units)	40,000	80,000
Selling Price ₹ / unit	25	50
Total Costs ₹ / unit	20	40
Machine Hours / unit	2	1
Maximum Sales Potential (units)	60,000	1,00,000

The fixed expenses are estimated at ₹ 9,60,000 per month. The company absorbs fixed overheads on the basis of machine hours which are fully utilised by the budgeted production and cannot be further increased.

When the budget was discussed, the Managing Director stated that the product mix should be altered to yield optimum profit.

The Marketing Director suggested that he would introduce a new Product-C, each unit of which will take 1.5 machine hours. However, a new machine involving a capital outlay of ₹ 2,00,000 is to be installed for

processing Product-C. The additional fixed overheads relating to the new machine was estimated at ₹ 60,000 per month. The variable cost of Product- C was estimated at ₹ 21 per Unit.

Required:

(i) Calculate the profit as per draft budget for the next month.

(ii) Revise the product mix based on data given for P and Q to yield optimum profit.

(iii) The company decides to discontinue either Product- P or Q whichever is giving lower profit and proposes to substitute Product- C instead. Fix the selling price of product- C in such a way as to yield 15% return on additional capital employed besides maintaining the same overall profit as envisaged in (ii) above.

ICAI'S ANSWER

(i) Profit as per Draft Budget for the Next Month:

	Products				
	P		**Q**		**Total**
Budgeted Production and Sales	40,000 units		80,000 units		1,20,000 units
	Per Unit (₹)	**Total (₹)**	**Per unit (₹)**	**Total (₹)**	**(₹)**
Sales	25	10,00,000	50	40,00,000	50,00,000
Less: Total Costs	20	8,00,000	40	32,00,000	40,00,000
Profit	5	2,00,000	10	8,00,000	10,00,000

(ii) Basic Calculations:

(a)	Machine Hrs.	P: 40,000 units x 2 hrs. Q: 80,000 units x 1 hr.	1,60,000 hrs.
(b)	Fixed Overhead Rate *per machine hr.*	₹ 9,60,000 / 1,60,000 hrs.	₹6
(c)	Fixed Overhead *per unit*	P: 2 hrs. x ₹ 6 Q: 1 hr. x ₹ 6	₹ 12 ₹6
(d)	Contribution *per unit*	P: ₹ 12 + ₹ 5 Q: ₹ 6 + ₹ 10	₹ 17 ₹ 16
(e)	Product-wise Contribution *per machine hr.*	P: ₹ 17/2hrs. Q: ₹ 16/1 hr.	₹ 8.50 ₹ 16.00

Revised Product Mix to Yield Optimum Profit:

Product Q has higher contribution per machine hour. Since machine hour is a limiting factor hence maximum units of product Q should be produced. However, maximum sales potentiality of Product Q is 1,00,000 units. This will take 1,00,000 machine hour. The balance 60,000 hours should be used to produce 30,000 units of P.

The Revised Product Mix to Yield Optimum Profit will be as follows:

Product Q	(1,00,000 units × ₹ 16)	₹ 16,00,000
Product P	(30,000 units × ₹ 17)	₹ 5,10,000
Total Contribution		₹ 21,10,000
Less: Fixed Expenses		₹ 9,60,000
Profit		₹ 11,50,000

(iii) Product - P gives lower contribution per machine hour, hence, it will be discontinued and Product - C will be manufactured in its place. The discontinuance of Product - P will make available 60,000 machine hours to produce 40,000 units of Product - C.

The computation of selling price of Product - C can be done as follows:

Computation of Selling Price of Product- C

	(₹)
Variable Cost (40,000 units × ₹ 21)	8,40,000
Additional Fixed Cost *per month*	60,000
Return on Capital (₹ 2,00,000 × 1.25%)	2,500
Present Contribution from Product P	5,10,000
Total Sales Value *to be recovered*	14,12,500
Selling Price *per unit* of C (₹ 14,12,500/40,000 units)	35.31

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1022 MTP **Old P3 Nov 2023 S-II** Q5(b) 10 marks old scheme

A company manufactures four products. The annual demand for products, selling prices and variable production costs are as follows:

Product	P	Q	R	S
Demand (Units)	1,20,000	1,86,000	1,71,000	99,000
	₹	₹	₹	₹
Selling price/unit	23.88	28.68	55.08	47.88
Direct Material/Unit	10.08	13.20	30.48	24.96
Direct Labour/unit	4.08	4.08	6.72	6.36
Variable overheads/unit	1.44	1.44	2.40	2.16

Other data:

(i) The variable overheads are absorbed on a machine hour basis at a rate of ₹ 1.20 per machine hour.

(ii) Fixed overheads total ₹ 46,84,000 per annum.

(iii) Production capacity available 8,15,000 machine hours per annum.

(iv) Products P, Q and R can be bought-in at ₹ 21.36 per unit, ₹ 24 per unit and ₹ 48 per unit respectively.

You are required to calculate Best product mix and Profitability statement for the year.

ICAI'S ANSWER

(i) Statement Showing "Calculation of Contribution/ unit"

	P (₹)	Q (₹)	R (₹)	S (₹)
Selling Price ...(A)	23.88	28.68	55.08	47.88
Variable Cost				
Direct Material	10.08	13.20	30.48	24.96
Direct Labour	4.08	4.08	6.72	6.36
Variable Overheads	1.44	1.44	2.40	2.16
Total Variable Cost ...(B)	15.60	18.72	39.60	33.48
Contribution per unit ...(A) - (B)	8.28	9.96	15.48	14.40

(ii) Calculation of Machine Hours/ unit

Machine Hours per unit	1.20	1.20	2.00	1.80

(iii) Machine Hours Required

Machine Hours per unit	1,44,000*	2,23,200%	3,42,000@	1,78,200#
			Total	8,87,400

* - (1,20,000 × 1.2); % - (1,86,000 × 1.2); @ - (1,71,000 × 2); # - (99,000 × 1.8)

(iv) Total Machine Hours Available 8,15,000. Hence, it is a key factor. Product 'S' is to be manufactured, since it is not available with sub-contractor/ market.

(v) **Statement Showing "Make or Buy for Products P, Q, R"**

	P (₹)	Q (₹)	R (₹)
Sub-Contractor/ Buy Price	21.36	24.00	48.00
Less: Variable Manufacturing Cost	15.60	18.72	39.60
Saving in Cost	5.76	5.28	8.40
Saving in Cost per machine hour	4.8	4.4	4.20
Ranking	I	II	III

(vi) **Statement Showing "Best Product Mix"**

Product	Units	Machine Hour/ Unit	Total Machine Hours
S	99,000	1.8	1,78,200
P	1,20,000	1.2	1,44,000
Q	1,86,000	1.2	2,23,200
R (Balance)	1,34,800	2.0	2,69,600
		Total	8,15,000

Balance quantity of R to be purchased 36,200 units (1,71,000 – 1,34,800).

(vii) **Profitability Statement**

Product	No of Units	Contribution/unit (₹)	Total Cont. (₹)
P (Mfg)	1,20,000	8.28	9,93,600
Q (Mfg)	1,86,000	9.96	18,52,560
R (Mfg)	1,34,800	15.48	20,86,704
R (Buy)	36,200	7.08 (₹55.08 - ₹48.00)	2,56,296
S (Mfg)	99,000	14.40	14,25,600
Total Contribution			66,14,760
Less: Fixed Overheads			46,84,000
Net Profit			19,30,760

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1023 MTP **Old P3 May 2022 S-II** Q2(a) 10 marks old scheme

Company manufacture and sell 3 types of mobile handset. It also manufactures wireless charger for mobile. The company has worked out following estimates for next year.

	Annual Demand (in units)	Selling Price (₹ per unit)	Material cost (₹ per unit)	Labour cost (₹ per unit)
X5	5,000	8,000	2,000	1,000
X6	4,000	9,000	2,500	1,500
X7	3,000	12,000	3,000	2,000
Wireless Charger	15,000	1,500	300	200

To encourage the sale of wireless charger a discount of 10% in its price is being offered if it were to be purchased along with mobile. It is expected that customer buying mobile will also buy the wireless

charger. The company factory has an effective capacity of 35,000 labour hours. The labour is paid @ ₹ 500 per hour. Overtime of labour has to be paid at double the normal rate. Other variable cost work out to be 50% of direct labour cost and fixed cost is ₹ 1,00,00,000. There will be no inventory at the end of the year.

PREPARE statement of profitability.

ICAI'S ANSWER

Calculation of Labour overtime hours

Total hours required for production

X5	(5,000 × 2 hrs)	10,000
X6	(4,000 × 3 hrs)	12,000
X7	(3,000 × 4 hrs)	12,000
Wireless Charger	(15,000 × 0.40 hrs)	6,000
		40,000
Hours available		(35,000)
Overtime		**5,000**

Statement of Profitability

Particulars	Amount (₹)	Amount (₹)
Sales		
X5 (5,000 × 8,000)	4,00,00,000	
X6 (4,000 × 9,000)	3,60,00,000	
X7 (3,000 × 12,000)	3,60,00,000	
Wireless Charger [(12,000 × 1,350) + (3,000 × 1,500)	2,07,00,000	**13,27,00,000**
Less: Variable cost		
Material:		
X5 (5,000 × 2,000)		
X6 (4,000 × 2,500)		
X7 (3,000 × 3,000)		
Wireless Charger (15,000 × 300)	3,35,00,000	
Labour:		
X5 (5,000 × 1,000)		
X6 (4,000 × 1,500)		

Particulars	Amount (₹)	Amount (₹)
X7 (3,000 × 2,000)		
Wireless Charger (15,000 × 200)		
Overtime (5,000 × 1,000)	2,50,00,000	
Other variable overheads	1,25,00,000	7,10,00,000
Contribution		**6,17,00,000**
Less: Fixed Cost		1,00,00,000
Profit		**5,17,00,000**

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14.5 Make or buy, shut down and other short-term decisions

39 questions · asked in 24 ICAI documents · 6 exam papers

1024 EXAM Jan 2025 Q3(b) 6 marks

SW Limited manufactures Lenin bed covers. The present cost data are as below:

Variable Cost of manufacturing per unit	₹ 200
Variable cost of selling and distribution per unit	₹ 100
Fixed costs	₹ 16,00,000
Selling price per unit	₹ 800
Expected Profit for the coming year	₹ 8,00,000

The management could sense a stage of stagnation/deterioration in future sales with the new entrant RK Enterprises. The SW limited has approached to one marketing consulting firm for the study of cost volume profit analysis. The firm suggested three alternatives to fuel the sales growth by tinkering with the selling price.

Alternatives	Reduce selling price %	Projected increase in sales (units) % (from the sales level that would generate ₹ 8,00,000 profit)
1	10.00	15
2	12.50	20
3	15.00	25

Required:

Calculate the effect on profits under each alternative and recommend which alternative is most likely to be adopted to get the maximum profit.

ICAI'S ANSWER

Calculation of Profit under each Alternative

Particulars	Present Situation (₹)	Alternative 1 (₹)	Alternative 2 (₹)	Alternative 3 (₹)
Selling Price per unit	800	720	700	680
Less: Variable Cost per unit				
Manufacturing Cost	200	200	200	200
Selling and Distribution Cost	100	100	100	100
Contribution per unit	**500**	**420**	**400**	**380**
No. of units sold	4,800	5,520	5,760	6,000
Total Contribution	24,00,000	23,18,400	23,04,000	22,80,000
Less: Fixed Cost	16,00,000	16,00,000	16,00,000	16,00,000
Profit	8,00,000	**7,18,400**	**7,04,000**	**6,80,000**
Change in profit		(81,600)	(96,000)	(1,20,000)
Percentage change in profit		-10.2%	-12%	-15%

Among the various alternatives, alternative 1 is giving maximum profit. However, in each of the alternative (alternative 1,2,3) maximum profit is reducing. Accordingly, SW limited should continue to sell at current selling price to get the maximum profit.

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1025 **EXAM** **Sep 2024** Q1(c) 4 marks

JC Ltd. has a production capacity of 80,000 units per year. Presently a company produces 60,000 units. Its cost structure is as under:

Material Cost	₹ 6 per unit
Labour Cost	₹ 4 per unit
Variable overheads	₹ 2 per unit

Total fixed cost ₹ 3,00,000 per annum. Present selling price ₹ 20 per unit in the month of January, 2024 company received an offer from a Japanese client to supply 20,000 units at a price of ₹ 14 per unit with the additional shipping cost of ₹ 8,000.

Required:

(i) On the basis of changes in the profit, advice to the company, whether the offer should be accepted or not?

(ii) Will your advice be different, if the customer is local one?

(iii) If Japanese client offer for supply of 30,000 units to a price of ₹ 14 (part supply of order not accepted) and shipping cost treated as variable cost, analyze the impact on the profit of JC Ltd., if order accepted.

ICAI'S ANSWER

(i) Statement Showing "Cost and Profit under Both Situation"

Particulars	Existing Production (60,000 units) (₹)	After Offer (80,000 units) (₹)
Sales		
Existing (60,000 x ₹ 20)	12,00,000	12,00,000
Offer (20,000 x ₹ 14)	-	2,80,000
Total Sales	12,00,000	14,80,000
Less: Direct Materials @ ₹ 6	3,60,000	4,80,000
Direct Labour @ ₹ 4	2,40,000	3,20,000
Variable Overheads @ ₹ 2	1,20,000	1,60,000
Contribution	4,80,000	5,20,000
Less: Additional Shipping cost	-	8,000
Less: Fixed Cost	3,00,000	3,00,000
Profit	**1,80,000**	**2,12,000**

Since the Profit has increased by ₹ 32,000, the proposal of the Japanese client should be accepted

(ii) Yes, the advice will be different, if the customer is local one since the company is currently selling at ₹ 20 in local market and therefore, selling at discounted price of ₹ 14 may impact its local market.

(iii) Statement Showing "Cost and Profit"

Particulars	After Offer (80,000 units) (₹)
Sales	
Existing (50,000 x ₹ 20)	10,00,000
Offer (30,000 x ₹ 14)	4,20,000
Total Sales	14,20,000
Less: Direct Materials @ ₹ 6	4,80,000
Direct Labour @ ₹ 4	3,20,000
Variable Overheads @ ₹ 2	1,60,000
Additional Shipping cost (₹ 8,000/20,000 units) x 30,000 units	12,000
Contribution	4,48,000
Less: Fixed Cost	3,00,000
Profit	1,48,000

If offer of Japanese client to supply 30,000 units at a price of ₹ 14 is accepted, **the Profit will decrease by ₹32,000 from the current level.**

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1026 **EXAM** **Old P3 Nov 2023** Q4(b) 5 marks old scheme

R Ltd. produces and sells 60,000 units of product 'AN', at its Noida Plant. The selling price of the product is ₹ 15 per unit. The variable cost is 80% of selling price per unit. Fixed cost during this period is ₹ 4,20,000. The company is continuously suffering losses, and management plans to shut down the Noida Plant.

The fixed cost is expected to be reduced by ₹ 2,50,000.

Additional costs of plant shut down are expected at ₹ 25,000.

You are required to comment on:

(i) Whether the Noida plant be shut down?

(ii) Find the shut-down point in units.

ICAI'S ANSWER

Statement of profit

Particulars	₹
Selling Price	15 per unit
Less : Variable cost	12 per unit
Contribution	3 per unit
Capacity	60,000 units
Total contribution (60,000 units × ₹ 3)	1,80,000
Less: Fixed Cost	4,20,000
Loss	**(2,40,000)**

Shut down cost

Particular	₹
Fixed cost	1,70,000
Additional cost	25,000
Shut down cost	**1,95,000**

(i) Since the loss of Noida plant exceeds shut down cost it is better to shut down the plant.

(ii) Shut down point: $\frac{\text{Total fixed cost - Shut down cost}}{\text{Contribution per unit}}$

$\frac{4,20,000 - 1,95,000}{3} = 75,000 \text{ units}$

The solution can also be presented in following way

Statement of profit

Particulars	If plant is continued ₹	If plant is shut down ₹
Selling Price	15 per unit	-
Less : Variable cost	12 per unit	-
Contribution	3 per unit	-
Capacity	60,000 units	-
Total contribution (60,000 units × ₹ 3)	1,80,000	
Less : Fixed Cost	4,20,000	1,70,000
Additional Fixed Cost	-	25,000
Loss	**2,40,000**	**1,95,000**

(i) Since the loss of Noida plant exceeds shut down cost it is better to shut down the plant.

(ii) Shut down point: $\frac{\text{Total fixed cost - Shut down cost}}{\text{Contribution per unit}}$

$\frac{4,20,000 - 1,95,000}{3} = 75,000 \text{ units}$

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1027 EXAM **Old P3 May 2022** Q1(c) 5 marks old scheme

Top-tech a manufacturing company is presently evaluating two possible machines for the manufacture of superior Pen-drives. The following information is available:

Particulars	Machine A	Machine B
Selling price per unit	₹ 400.00	₹ 400.00
Variable cost per unit	₹ 240.00	₹ 260.00
Total fixed costs per year	₹ 350 lakhs	₹ 200 lakhs
Capacity (in units)	8,00,000	10,00,000

Required:

(i) Recommend which machine should be chosen?

(ii) Would you change your answer, if you were informed that in near future demand will be unlimited and the capacities of the two machines are as follows?

Machine A - 12,00,000 units

Machine B - 12,00,000 units

Why?

ICAI'S ANSWER

		Machine-A	Machine-B	Total
A	Selling price per unit (₹)	400	400	
B	Variable cost per cost (₹)	240	260	
C	Contribution per unit (₹) [A-B]	160	140	
D	Units	8,00,000	10,00,000	
E	Total contribution (₹ [C×D]	12,80,00,000	14,00,00,000	26,80,00,000
F	Fixed Cost (₹)	3,50,00,000	2,00,00,000	5,50,00,000
G	Profit [E-F] (₹)	9,30,00,000	12,00,00,000	21,30,00,000
H	Profit per unit [G÷D] (₹)	116.25	120.00	

(i) Machine B has the higher profit of ₹2,70,00,000 than the Machine-A. Further, Machine-B's fixed cost is less than the fixed cost of Machine-A and higher capacity. Hence, **Machine B be recommended.**

Note: This question can also be solved as below:

Indifferent point = Difference in fixed cost / difference in variable cost per unit

= 1,50,00,000 / 20 = 7,50,000 units

At the level of demand 7,50,000 units both machine options equally profitable.

If demand below 7,50,000 units, select machine B (with lower FC).

If demand above 7,50,000 units, select machine A (with lower VC).

(ii) When the capacities of both the machines are same and demand for the product is unlimited, calculation of profit will be as follows:

		Machine-A	Machine-B	Total
A	Contribution per unit (₹)	160	140	
B	Units	12,00,000	12,00,000	
C	Total contribution (₹) [A×B]	19,20,00,000	16,80,00,000	36,00,00,000
D	Fixed Cost (₹)	3,50,00,000	2,00,00,000	5,50,00,000
E	Profit [C-E] (₹)	15,70,00,000	14,80,00,000	30,50,00,000
F	Profit per unit [E÷B] (₹)	130.83	123.33	

Yes, the preference for the machine would change because now, Machine A is having higher contribution and higher profit, hence recommended.

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1028 EXAM **Old P3 Jul 2021** Q1(d) 5 marks old scheme

LR Ltd. is considering two alternative methods to manufacture a new product it intends to market. The two methods have a maximum output of 50,000 units each and produce identical items with a selling price of ₹ 25 each. The costs are:

	Method-1 Semi-Automatic (₹)	Method-2 Fully-Automatic (₹)
Variable cost per unit	15	10
Fixed costs	1,00,000	3,00,000

You are required to calculate:

(1) Cost Indifference Point in units. Interpret your results.

(2) The Break-even Point of each method in terms of units.

ICAI'S ANSWER

(i) Cost Indifference Point

	Method-1 and Method-2 (₹)
Differential Fixed Cost (I)	₹ 2,00,000 (₹ 3,00,000 – ₹ 1,00,000)
Differential Variable Costs (II)	₹ 5 (₹ 15 – ₹ 10)
Cost Indifference Point (I/II) (Differential Fixed Cost / Differential Variable Costs per unit)	**40,000**

Interpretation of Results

At activity level below the indifference points, the alternative **with lower fixed costs and higher variable costs should be used.** At activity level above the indifference point, alternative with **higher fixed costs and lower variable costs should be used.**

No. of Product	Alternative to be Chosen
Product ≤ 40,000 units	Method-1, Semi-Automatic
Product ≥ 40,000 units	Method-2, Automatic

(ii) Break Even point (in units)

	Method-1	Method-2
BEP (in units) = Fixed cost/Contribution per unit	1,00,000/(25-15) = 10,000	3,00,000/(25-10) = 20,000

ICAI · Old P3 · Jul 2021 exam · question p. 54 · answer p. 57

1029 EXAM **Old P3 Nov 2019** Q5(a) 10 marks old scheme

PJ Ltd manufactures hockey sticks. It sells the products at ₹ 500 each and makes a profit of ₹ 125 on each stick. The Company is producing 5,000 sticks annually by using 50% of its machinery capacity.

The cost of each stick is as under:

Direct Material	₹ 150
Direct Wages	₹ 50
Works Overhead	₹ 125 (50% fixed)
Selling Expenses	₹ 50 (25% variable)

The anticipation for the next year is that cost will go up as under:

Fixed Charges	10%
Direct Wages	20%
Direct Material	5%

There will not be any change in selling price.

There is an additional order for 2,000 sticks in the next year.

Calculate the lowest price that can be quoted so that the Company can earn the same profit as it has earned in the current year?

ICAI'S ANSWER

Selling Price = ₹ 500

Profit = ₹ 125

No of Sticks = 5,000

Particular	Current Year (₹)	Next Year (₹)
Direct Material	150	**157.50** (150 + 5%)
Direct Wages	50	**60** (50+20%)
Works Overheads	62.50 (125 × 50%)	**62.5**
Selling Expenses	12.50 (50 × 25%)	**12.5**
Total Variable Cost	275	292.50
Fixed Cost (62.5 × 5,000) = 3,12,500; (37.5 × 5,000) = 1,87,500	**5,00,000**	**5,50,000**

Let: Lowest Price Quoted = K

Now, Sales = Target Profit (5,000 units × ₹ 125) + Variable Cost + Fixed Cost

Or, = (5,000 × 500) + (2,000 × K) = **6,25,000** + 20,47,500 + 5,50,000

Or, K = ₹ **361.25**

So, Lowest Price that can be quoted to earn the profit of ₹ 6,25,000 (same as current year) is ₹ 361.25

1030 RTP **Sep 2026** Q15

XYX Limited manufactures three different products and the following information has been collected from the books of accounts:

	Products		
	A	**B**	**C**
Sales Mix	35%	35%	30%
Selling Price	₹ 30	₹40	₹20
Variable Cost	₹ 15	₹ 20	₹ 12
Total Fixed Costs			₹ 1,80,000
Total Sales			₹ 6,00,000

The company has currently under discussion, a proposal to discontinue the manufacture of Product C and replace it with Product P, when the following results are anticipated:

	Products		
	A	**B**	**C**
Sales Mix	50%	25%	25%
Selling Price	₹ 30	₹ 40	₹ 30
Variable Cost	₹ 15	₹ 20	₹ 15
Total Fixed Costs			₹ 1,80,000
Total Sales			₹ 6,40,000

Will you advise the company to change over to production of P? Give reasons for your answer.

ICAI'S ANSWER

XYZ Limited

Evaluation of Change over to Product P from Product C

Existing Production

	Products			Total
	A	**B**	**C**	
Selling Price (₹)	30	40	20	
Less: Variable Cost (₹)	15	20	12	
Contribution per unit (₹)	15	20	8	

	Products			Total
P/V Ratio	50%	50%	40%	
Sales Mix	35%	35%	30%	
Contribution *per rupee of sales* (P/V Ratio × Sales Mix)	17½%	17½%	12%	47%
Present Total Contribution (₹ 6,00,000 × 47%)				₹ 2,82,000
Less: Fixed Costs				₹ 1,80,000
Present Profit				₹ 1,02,000
Present Break-Even Sales (₹ 1,80,000/0.47)				₹ 3,82,979

Proposed Production

	Products			Total
	A	**B**	**C**	
Selling Price (₹)	30	40	30	
Less: Variable Cost (₹)	15	20	15	
Contribution *per unit* (₹)	15	20	15	
P/V Ratio	50%	50%	50%	
Sales Mix	50%	25%	25%	
Contribution *per rupee of sales* (P/V Ratio x Sales Mix)	25%	12½%	12½%	50%
Proposed Total Contribution (₹ 6,40,000 × 50%)				₹ 3,20,000
Less: Fixed Costs				₹ 1,80,000
Proposed Profit				₹ 1,40,000
Proposed Break Even Sales (₹ 1,80,000/0.50)				₹ 3,60,000

ICAI · P4 · RTP Sep 2026 · question p. 18-19 · answer p. 46-47

1031 RTP **May 2026** Case scenario · Q2(i) to Q2(v)

CASE SCENARIO

White Kollars Pvt. Ltd. is a company engaged solely in the manufacture of shirts, which are bought mainly by office goers. Currently, the company sells its products directly to the retailers. In the recent years, sales has steadily declined due to rising competition. The forecast for next year suggests that the current decline in profits is expected to persist.

The management believes that a profit of ₹ 60,00,000 is necessary to ensure an adequate return on capital. Thus, it asked the marketing director to review the marketing policies.

Now, the marketing director submits the following profit and loss account for the last year along with 3 proposals available with him:

(Amount in '000)

Particulars	(₹)	(₹)	(₹)
Sales revenue (50,000 shirts at ₹ 1,500 each)			75,000
Factory cost of goods sold:			
Direct materials	7,500		
Direct labour	26,250		
Variable factory overheads	4,500		
Fixed factory overheads	16,500	54,750	
Fixed Administration overhead		10,500	
Selling and distribution overhead:			
Sales commission (2% of sales)	1,500		
Delivery costs (variable per unit sold)	3,750		
Fixed costs	3,000	8,250	73,500
Profit			1,500

Proposals for upcoming year:

(i) 10% reduction in selling price which may increase demand by 40%.

(ii) Selling of 25,000 units to eFlip, an online platform selling products directly to consumers, at a particular price. eFlip would transport the shirts from White Kollars Pvt. Ltd. to its own warehouse. Thus, abolishing the delivery costs and sales commission to be paid for each sale by White Kollars Pvt. Ltd. However, if a suitable price is negotiated, White Kollars Pvt. Ltd. would contribute ₹ 45,00,000 annually towards the online order catalogue cost and provide special packaging at ₹ 75 per shirt.

Remaining units of 25,000 would continue to be sold at ₹ 1,500 each as per existing business.

(iii) 10% price reduction together with an advertising campaign costing ₹ 22,50,000 which may boost sales by 60%.

From the information given above, you are required to FIGURE OUT the following for managerial decision:

Q2(i)

P/V ratio and Break-even sales (in value) based on the last year accounts would be-

A 40%, ₹ 7,14,28,571

B 42%, ₹ 7,14,28,571

C 40%, ₹ 7,50,00,000

D 42%, ₹ 7,50,00,000

Q2(ii)

Profit/ (Loss) from proposal (i) would be:

A ₹ 3,38,10,000

B ₹ 3,00,00,000

C ₹ 38,10,000

D ₹ 30,00,000

Q2(iii)

With respect to proposal (i), how many number of units White Kollars Pvt. Ltd. would need to sell at ₹ 1,350 each to earn a desired profit of ₹ 60,00,000?

A 75,000 units

B 74,534 units

C 62,112 units

D 47,619 units

Q2(iv)

With respect to proposal (ii), how much selling price per unit be offered to eFlip, that would at least break even on this proposal?

A ₹ 180

B ₹ 660

C ₹ 840

D ₹ 1,020

Q2(v)

Profit/ (Loss) from proposal (iii) would be:

A ₹ 63,90,000

B ₹ 86,40,000

C ₹ 81,00,000

D (₹ 81,00,000)

ICAI'S ANSWERS

Q2(i)

(B) 42%, ₹ 7,14,28,571

Calculation of Contribution: (Amount in '000)

Particulars	(₹)	(₹)
Sales revenue (50,000 shirts at ₹ 1,500 each)		75,000
Less: Variable costs-		
Direct materials	7,500	
Direct labour	26,250	
Variable factory overheads	4,500	
Sales commission (2% of sales)	1,500	
Delivery costs (variable per unit sold)	3,750	43,500
Contribution		31,500

Calculation of Fixed Costs: (Amount in '000)

Particulars	(₹)
Fixed factory overheads	16,500
Administration overhead	10,500
Selling and distribution overhead:	
Fixed costs	3,000
Total Fixed Costs	30,000

$$\text{P/V ratio} = \frac{\text{Contribution}}{\text{Sales}} \text{ x } 100$$

$$= \frac{₹31,500}{₹75,000} \text{ x } 100$$

$$= 42\%$$

$$\text{Break even sales value} = \frac{\text{Fixed Cost}}{\text{P/V ratio}}$$

$$= \frac{₹3,00,00,000}{42\%}$$

= ₹ 7,14,28,571

Q2(ii)

(C) ₹ 38,10,000

Particulars	(₹)	(₹)
Revised selling price (₹ 1,500 - 10%)		1,350
Less: variable costs-		
Direct materials	150	
Direct labour	525	

Particulars	(₹)	(₹)
Variable overhead	90	
Delivery expenses	75	
Sales commission (2% of selling price)	27	867
Contribution per unit		483
Number of units sold		70,000
Total contribution (70,000 units x 483) (A)		3,38,10,000
Fixed costs (B)		3,00,00,000
Profit from proposal (i) (A - B)		38,10,000

Q2(iii)

(B) 74,534 units

We know that,

Sales × P/V Ratio = Fixed Cost + Profit

Or, (₹ 1,350 x required no. of units) x $\frac{\text{Contribution}}{\text{Sales}}$

= Fixed Cost + Profit

Or, (₹ 1,350 x required no. of units) x $\frac{₹483 \text{ per unit}}{₹1,350 \text{ per unit}}$

= ₹ 3,00,00,000 + ₹ 60,00,000

Therefore, required no. of units = $\frac{₹3,60,00,000}{₹483}$

= 74,534 units

Q2(iv)

(D) ₹ 1,020

The variable cost of selling to eFlip is:

Particulars	(₹)
Direct material	150
Direct labour	525
Variable overhead	90
Delivery expenses	-
Sales commission	-
Additional package cost	75
	840

$$\text{Break-even Point} = \frac{\text{Fixed Cost}}{\text{Contribution per unit}}$$

$$25{,}000 \text{ units} = \frac{₹45{,}00{,}000}{\text{Contribution per unit}}$$

Contribution per unit = ₹ 180

Selling Price at Break-even Point = Variable cost + Contribution

= ₹ 840 + ₹ 180

= ₹ 1,020 per unit

Q2(v)

(A) ₹ 63,90,000

Particulars	(₹)
Contribution per unit (similar to Proposal (i))	483
Number of units sold	80,000
Total contribution (80,000 units x 483) (A)	3,86,40,000
Fixed costs (₹ 3,00,00,000 + ₹ 22,50,000) (B)	3,22,50,000
Profit from proposal (iii) (A - B)	63,90,000

ICAI · P4 · RTP May 2026 · question p. 3-5 · answer p. 24, 25, 26

1032 RTP Jan 2026 Q6 MCQ

Rinki Ltd. has two divisions, A and B. Target profit of A is ₹ 100 lakh. A has capacity to produce 20,000 units of product P but only 15,000 units are produced and sold in the market at ₹ 3,000 per unit. B received an order for which P would be required as input. B approaches A for purchase of 5000 units of P. What price should A charge from B per unit of P so as to meet its target profit? (Other details of A are: Fixed cost is ₹ 90 lakh; Variable cost per unit of P is ₹ 2,000)

A ₹ 3,000

B ₹ 2,800

C ₹ 2,500

D ₹ 2,000

ICAI'S ANSWER

(B) ₹ 2,800

Revenue = 15,000 × 3,000 = 450,00,000 (₹ 450 lakh)

Variable cost = 15,000 × 2,000 = 300,00,000 (₹ 300 lakh)

Contribution = 450 - 300 = ₹ 150 lakh

Profit = Contribution - Fixed cost = 150 - 90 = ₹ 60 lakh

Target Profit = 100 - 60 = ₹ 40 lakh

Required contribution per unit = 40,00,000/5,000 = ₹ 800/unit

Transfer price = Variable cost + 800 = 2,000 + 800 = ₹ 2,800/unit

ICAI · P4 · RTP Jan 2026 · question p. 8 · answer p. 29

1033 **RTP** **Jan 2026** **Q17**

Jupiter Ltd, a 'Fast-Moving Consumer Goods (FMCG)' company intends to diversify the product line to achieve full utilisation of its plant capacity. As a result of considerable research made, the company has been able to develop a new product called 'EXE'.

'EXE' is packed in cans of 100 ml capacity and is sold to the wholesalers in cartons of 24 cans at ₹120 per carton. Since the company uses its spare capacity for the manufacture of 'EXE', no additional fixed expenses will be incurred. However, accountant has allocated a share of ₹1,12,500 per month as fixed expenses to be absorbed by 'EXE' as a fair share of the company's present fixed costs to the new product for costing purposes.

The company estimates the production and sale of 'EXE' at 1,50,000 cans per month and on this basis the following cost estimates (per carton) have been developed:

	₹
Direct Materials	54
Direct Wages	36
All Overheads	27
Total Costs	117

After a detailed market survey the economy is confident that the production and sales of 'EXE' can be increased to 1,75,000 cans per month and ultimately to 2,25,000 cans per month.

The company at present has a capacity for the manufacture of 1,50,000 empty cans and the cost of the empty cans if purchased from outside will result in a saving of 20% in direct material, 10% in direct wages and 10% in variable overhead costs of 'EXE'. The price at which the outside firm is willing to supply the empty cans is ₹0.675 per empty can. If the company desires to manufacture empty cans in excess of 1,50,000 cans, a machine involving an additional fixed overhead of ₹7,500 per month will have to be installed.

Required:

(a) State by showing your workings whether the company should make or buy the empty cans at each of the three volumes of production of 'EXE' namely, 1,50,000, 1,75,000 and 2,25,000 cans.

(b) At what volume of sales will it be economical for the company to install the additional equipment for the manufacture of empty cans?

(c) Evaluate the profitability on the sale of 'EXE' at each of the aforesaid three levels of output based on your decision and showing the cost of empty cans as a separate element of cost.

ICAI'S ANSWER

Workings

(1) All Overheads for one carton or 24cans	₹ 27
Therefore, per can Overheads (₹27/24)	1.125
Fixed Overheads Allocated for 1,50,000 cans:	₹ 1,12,500
Per can Fixed Overheads (₹1,12,500 / 1,50,000cans)	₹ 0.75
Variable Overheads per can (₹1.125 – ₹0.75)	₹ 0.375
(2) Direct Wage per carton	₹ 36
Per can (₹36/24)	₹ 1.50
(3) Direct Materials per carton	₹ 54
Per can (₹54/24)	₹ 2.25

(4) Cost of making one empty can:

	Cost per can of 'EXE' (₹)	Cost % empty can	Cost empty can (₹)	Cost of per can of 'EXE' without empty can (₹)
Direct Material	2.250	20	0.4500	1.8000
Direct Wages	1.500	10	0.1500	1.3500
Variable Overheads	0.375	10	0.0375	0.3375
Total	4.125		0.6375	3.4875

(5) Cost of manufacturing/buying of 1,50,000 empty cans of 'EXE':

	Empty can Cost (₹)	If empty can made (₹)	If empty can purchased (₹)
Direct Material	0.4500	67,500.00	-----
Direct Wages	0.1500	22,500.00	-----
Variable Overheads	0.0375	5,625.00	-----
Purchase Price	0.6750	-----	1,01,250.00
Total		95,625.00	1,01,250.00

The company should manufacture the empty cans for a production volume of 1,50,000 'EXE' cans as capacity is available and cost of manufacture is lower.

(6) After the level of 1,50,000 empty cans, the company has to install a new machine involving a total additional Fixed Overheads of ₹ 7,500. The cost of making and buying the additional cans of 25,000 and 75,000 will be as follows:

	Cost per can (₹)	Make (₹)	Buy (₹)	Make (₹)	Buy (₹)
		25,000 cans		**75,000 cans**	
Direct Material	0.4500	11,250.00	-	33,750.00	-
Direct Wages	0.1500	3,750.00	-	11,250.00	-
Variable Overheads	0.0375	937.50	-	2,812.50	-
Additional Overheads		7,500.00	-	7,500.00	-
Purchase Price	0.6750	-	16,875.00	-	50,625.00
Total		23,437.50	16,875.00	55,312.50	50,625.00

The cost of buying additional empty containers at both the levels is lower than the cost of their manufacture.

(a) If the company increases production to 1,75,000 cans of 'EXE', 1,50,000 empty cans should be manufactured and additional 25,000 cans should be purchased at ₹16,875 [Refer W.N. 5&6]

If the company increases production to 2,25,000 cans of 'EXE', 1,50,000 empty cans should be manufactured and additional 75,000 cans should be purchased at a cost of ₹ 50,625. [Refer W.N. 5&6]

(b) Additional fixed overheads to be incurred on a new machine: ₹7,500 Savings per unit if empty cans are made instead of buying: ₹ 0.675 – ₹ 0.6375 = ₹ 0.0375

Minimum additional quantity of empty containers to be made to recover the additional fixed costs:

₹ 7,500/ ₹0.0375 = 2,00,000 empty cans

Installation of the new machine for the manufacture of empty cans will be economical at production level of 3,50,000 cans per month.

(c) Evaluation of the profitability on sale of "EXE" at the three levels.

	Per can (₹)	1,50,000 can (₹)	1,75,000 can (₹)	2,25,000 can (₹)
Sales	5.0000	7,50,000.00	8,75,000.00	11,25,000.00
Less: Direct Material	1.8000	2,70,000.00	3,15,000.00	4,05,000.00
Direct Wages	1.3500	2,02,500.00	2,36,250.00	3,03,750.00
Variable Overheads	0.3375	50,625.00	59,062.50	75,937.50
Empty can made	0.6375	95,625.00	95,625.00	95,625.00
Empty can purchases	0.6750		16,875.00	50,625.00
Net Gain		1,31,250.00	1,52,187.50	1,94,062.50

ICAI · P4 · RTP Jan 2026 · question p. 20-21 · answer p. 49-52

1034 RTP Jan 2025 Q17

XYZ Ltd. is a company involved in production and construction specialised equipment and machines on the demand of customers. The company received an order for construction of a specialised machine, it had nearly completed this job relating to construction of a specialised machine, when it discovered that the customer had gone out of business. At this stage, the position of the job was as under:

	(₹)
Original cost estimate	27,50,000
Costs incurred so far	24,80,000
Costs to be incurred	3,70,000
Progress payment received from original customer	15,50,000

After searches, a new customer for the machine has been found. He is interested to take the machine, if certain modifications are carried out. The new customer wanted the machine in its original condition, but without its AI device and with certain other modifications. The costs of these additions and modifications are estimated as under:

Direct Materials (at cost)	₹ 1,05,000
Direct Wages Dept.: X	35 men days
Dept.: Y	55 men days
Variable Overheads	30% of Direct Wages in each Dept.
Delivery Costs	₹ 15,500

Fixed overheads will be absorbed at 50% of direct wages in each department.

The following additional information is available:

(1) The direct materials required for the modification are in stock and if not used for modification of this order, they will be used in another job in place of materials that will now cost ₹ 1,50,000.

(2) Department X is working normally and hence any engagement of labour will have to be paid at the direct wage rate of ₹ 1,000 per man day.

(3) Department Y is extremely busy. Its direct wages rate is ₹ 1,200 per man day and it is currently yielding a contribution of ₹ 3 per rupee of direct wages.

(4) Additional supervisory required for the modification cost ₹ 80,000.

(5) The cost of the AI device that the new customer does not require is ₹ 1,35,000. If it is taken out, it can be used in another job in place of a different mechanism. The latter mechanism has otherwise to be bought for ₹ 1,05,000. The dismantling and removal of the control mechanism will take 5 man day in department X.

(6) If the conversion is not carried out, some of the materials in the original machine can be used in another contract in place of materials that would have cost ₹ 2,00,000. It would have taken 5 men days of work in department X to make them suitable for this purpose. The remaining materials will realize ₹ 1,50,000 as scrap. The drawings, which are included as part for the job can he sold for ₹ 45,000.

You are required to CALCULATE the minimum price, which the company can afford to quote for the new customer as staled above.

ICAI'S ANSWER

Statement of Minimum Price Which the Company Can Afford to Quote for the New Customer

	(₹)	(₹)
Cost to be incurred to bring the machine in its original condition		3,70,000
Direct Material (Replacement Value)		1,50,000
Direct Wages		
Dept. X: (35 men days × ₹ 1,000)	35,000	
Dept. Y: (55 men days × ₹ 1,200)	66,000	
Opportunity Cost of Contribution Lost by Dept. Y (₹66,000 × ₹3)	1,98,000	2,99,000
Variable Overheads [30%×(₹35,000+₹ 66,000)]		30,300
Delivery Costs		15,500
Additional Supervisory required for modification		80,000
Saving Due to Alternative Use of AI Device		
Bought Out Price	1,05,000	
Less: Dismantling & Removal Cost (5 men day × ₹1,000)	5,000	
Less: Variable Cost (30% × ₹ 5,000)	1,500	(98,500)
Net Loss on Material Cost Savings (W.N.)		1,93,500
Opportunity Cost of Remaining Materials which can be sold as scrap		1,50,000
Opportunity Cost of Sale of Drawings		45,000
Total Minimum Price which may be quoted		12,34,800

Working Note

	(₹)
Loss on Material Cost Saving of Machine	2,00,000
Less: Conversion Cost (5 men days × ₹1,000)	5,000
Less: Variable Cost (30% × ₹5,000)	1,500
Net Loss on Material Cost Saving of Machine	1,93,500

ICAI · P4 · RTP Jan 2025 · question p. 17-19 · answer p. 42-43

1035 RTP Old P3 Nov 2023 Q12(b) old scheme

The M-Tech Manufacturing Company is presently evaluating two possible processes for the manufacture of a toy. The following information is available:

Particulars	Process A (₹)	Process B (₹)
Variable cost per unit	12	14
Sales price per unit	20	20
Total fixed costs per year	30,00,000	21,00,000
Capacity (in units)	4,30,000	5,00,000
Anticipated sales (Next year, in units)	4,00,000	4,00,000

Suggest:

1. Identify the process which gives more profit.

2. Would you change your answer as given above, if you were informed that the capacities of the two processes are as follows:

A - 6,00,000 units; B - 5,00,000 units?

ICAI'S ANSWER

(1) Comparative Profitability Statements

Particulars	Process- A (₹)	Process- B (₹)
Selling Price per unit	20.00	20.00
Less: Variable Cost per unit	12.00	14.00
Contribution per unit	8.00	6.00
Total Contribution	32,00,000 (₹ 8 × 4,00,000)	24,00,000 (₹ 6 × 4,00,000)
Less: Total fixed costs	30,00,000	21,00,000
Profit	2,00,000	3,00,000
Capacity (units)	4,30,000	5,00,000
Total Contribution at full capacity	34,40,000 (₹ 8 × 4,30,000)	30,00,000 (₹ 6 × 5,00,000)
Fixed Cost	30,00,000	21,00,000
Profit	4,40,000	9,00,000

Process - B gives more profit.

(2)

Particulars	Process- A (₹)	Process- B (₹)
*Capacity (units)	6,00,000	5,00,000
Total contribution	48,00,000 (₹ 8 × 6,00,000)	30,00,000 (₹ 6 × 5,00,000)
Fixed Cost	30,00,000	21,00,000
Profit	18,00,000	9,00,000

Process-A be chosen.

***Note: It is assumed that capacity produced equals sales.**

ICAI · Old P3 · RTP Nov 2023 · question p. 8 · answer p. 22-23

1036 RTP **Old P3 Nov 2022** Q13(a) old scheme

RPP Manufacturers is approached by an international customer for one-time special order similar to one offered to its domestic customers. Per unit data for sales to regular customers is provided below:

Direct material	₹ 693
Direct labour	₹ 315
Variable manufacturing support	₹ 504
Fixed manufacturing support	₹ 1092
Total manufacturing costs	₹ 2604
Markup (50%)	₹ 1302
Targeted selling price	₹ 3906

It is provided that RPP Manufacturers has excess capacity.

Required:

(i) WHAT is the full cost of the product per unit?

(ii) WHAT is the contribution margin per unit?

(iii) WHICH costs are relevant for making the decision regarding this one-time special order? WHY?

(iv) For RPP Manufacturers, WHAT is the minimum acceptable price of this one-time-special order only

(v) For this one-time-only special order, SHOULD RPP Manufacturers consider a price of ₹ 2100 per unit? WHY or why not?

ICAI'S ANSWER

(i) Full cost of the product per unit

Direct material	₹ 693
Direct labour	₹ 315
Variable manufacturing support	₹ 504
Fixed manufacturing support	₹ 1092
Total manufacturing costs	₹ 2604

(ii) Contribution margin per unit

Selling price	₹ 3906
Less: Variable costs	
Direct material	₹ 693
Direct labour	₹ 315
Variable manufacturing support	₹ 504
Contribution margin per unit	₹ 2394

(iii) Costs for decision making are those costs that differ between alternatives, which in this situation are the incremental costs.

Direct material	₹ 693
Direct labour	₹ 315
Variable manufacturing support	₹ 504
Total incremental costs	₹ 1512

(iv) Minimum acceptable price would be the incremental costs in the short term i.e. ₹ 1512

(v) Yes, RPP Manufacturers may consider a price of ₹ 2100 per unit because this price is greater than the minimum acceptable price.

ICAI · Old P3 · RTP Nov 2022 · question p. 11 · answer p. 32

1037 RTP **Old P3 Nov 2022** Q13(b) old scheme

The lab corner of Newlife Hospital Trust operates two types of specialist MRI scanning machine- MR10 and MR59. Following details are estimated for the next period:

Machine	MR10	MR59
Running hours	1,100	2,000
	(₹)	(₹)
Variable running costs excluding special technology	68,750	1,60,000
Fixed Costs	50,000	2,43,750

A brain scan is normally carried out on machine type MR10. This task uses special technology costing ₹ 100 each and takes four hours of machine time. Because of the nature of the process, around 10% of the scans produce blurred and therefore useless results.

Required:

(i) CALCULATE the total cost of a satisfactory brain scan on machine type MR10.

(ii) Brain scans can also be done on machine type MR59 and would take only 1.8 hours per scan with a reduced reject rate of 6%. However, the cost of the special technology would be ₹ 137.50 per scan. ADVISE which type should be used, assuming sufficient capacity is available on both types of machines. Consider fixed costs will remain unchanged.

ICAI'S ANSWER

(i)

Particulars	(₹)
Variable cost per running hour of Machine MR10 (₹ 68,750/1100 hours)	62.50
Fixed cost (₹ 50,000/1100 hours)	45.46
Cost of brain scan on Machine MR10:	(₹)
Variable machine cost (4 hours × ₹ 62.50)	250.00
Special technology	100.00
Total variable cost	350.00
Fixed machine cost (4 hours × ₹ 45.46)	181.84
Total cost of a scan	531.84
Total cost of a satisfactory scan (₹ 531.84/0.9)	**590.93**

(ii) It is given that fixed cost will remain unchanged and thus they are not relevant for the decision. The relevant costs would be the incremental costs of an additional scan:

Machine MR10:	(₹)
Variable cost per scan	350.00
Variable cost per satisfactory scan (₹ 350/0.9)	**388.89**
Machine MR59:	(₹)
Variable machine cost per scan (₹ 1,60,000 / 2000 hours × 1.8 hours)	144.00
Special technology	137.50
Variable cost per scan	281.50
Variable cost per satisfactory scan (₹ 281.50/0.94)	**299.47**

The relevant costs per satisfactory scan are cheaper on Machine MR59 and therefore brain scans should be undertaken on said machine.

ICAI · Old P3 · RTP Nov 2022 · question p. 12 · answer p. 33

1038 MTP **Sep 2026 S-I** MCQ 11 2 marks MCQ

A company is preparing a quotation for a customised engineering project. The project requires 40 kg of Material P. The company has 200 kg of Material P in stock, originally bought at ₹ 10 per kg.

Material P, if processed further at a cost of ₹ 3 per kg, can be converted into Material Q, which the company currently purchases from the market at ₹ 9 per kg.

What is the relevant cost of using Material P for the new project?

A ₹40

B ₹160

C ₹ 240

D ₹ 200

ICAI'S ANSWER

(C) ₹ 240

Opportunity value of 1 kg of Material P:

Opportunity cost = Purchase Price of Material Q – Processing Cost of Material P

= ₹ 9 - ₹ 3 = ₹ 6 per kg

Relevant cost for 40 kg = 40 kg x ₹ 6 = ₹ 240

ICAI · P4 · MTP Sep 2026 · Series I · question p. 5-6 · answer p. 5

1039 MTP **Sep 2026 S-II** Q5(a) 8 marks

A laboratory carrying out various tests on products produced by various drug companies to ascertain whether drugs are fit for medical use or nol. At present, the laboratory carries out 10,000 tests each year and a survey carried out by the laboratory shows a rise in number of tests to 15,000 tests a year, to carrying out all these tests would require an additional shift to be worked.

The current cost of carrying out a full test is:

	₹ per test
Materials	1,800
Technicians' fees	180
Variable expenses	45
Fixed cost	200

Working the additional shift would

(i) require a shift premium of 50 per cent to be paid to the technicians on the additional shift;

(ii) enable a quantity discount of 10 per cent to be obtained for all materials if an order was placed to cover 15,000 tests;

(iii) Increase fixed costs by ₹ 5, 00,000 per year.

The current fee per test is ₹ 2500.

(1) Calculate the profit for the period at the current capacity of 10,000 tests.

(2) A profit statement if the additional shift was worked and 15,000 tests were carried out.

ICAI'S ANSWER

(1) Profit for the period at the current capacity of 10,000 tests:

Particulars	Amount (₹)
Fee per test	2,500
Less: Variable costs:	
Direct Materials	1,800
Technician fees	180
Other Variable expenses	45
Contribution per test	475
Total contribution (₹ 475 × 10,000 tests)	47,50,000
Less: Fixed Cost (₹ 200 × 10,000)	20,00,000
Profit	27,50,000

(2) Statement of Profit for expected 15,000 tests, with additional shift:

Particulars	Amount (₹)	Amount (₹)
Fees (15,000 tests x ₹ 2,500)		3,75,00,000
Less: Variable Costs:		
Direct materials [(90% of 1,800) x 15,000 tests]		2,43,00,000
Technician's fee		
10,000 tests @ ₹ 180	18,00,000	
5,000 tests @ ₹ (150% of 180)	13,50,000	31,50,000
Other Variable expenses (₹ 45 × 15,000 tests)		6,75,000
Contribution		93,75,000
Less: Fixed costs (20,00,000 + 5,00,000)		25,00,000
Profit		68,75,000

ICAI · P4 · MTP Sep 2026 · Series II · question p. 13 · answer p. 18-19

1040 MTP **Sep 2026 S-II** Q6(c) 4 marks

Elaborate the practical application of Marginal Costing.

ICAI'S ANSWER

Practical applications of Marginal costing:

(i) **Pricing Policy:** Since marginal cost per unit is constant from period to period, firm decisions on pricing policy can be taken particularly in short term.

(ii) **Decision Making:** Marginal costing helps the management in taking a number of business decisions like make or buy, discontinuance of a particular product, replacement of machines, etc.

(iii) **Ascertaining Realistic Profit:** Under the marginal costing technique, the stock of finished goods and work-in-progress are carried on marginal cost basis and the fixed expenses are written off to profit and loss account as period cost. This shows the true profit of the period.

(iv) **Determination of production level:** Marginal costing helps in the preparation of break-even analysis which shows the effect of increasing or decreasing production activity on the profitability of the company.

ICAI · P4 · MTP Sep 2026 · Series II · question p. 14 · answer p. 24-25

1041 MTP **May 2026 S-I** Q4(a) 7 marks

MintCraft Pvt. Ltd. specializes in producing custom-designed award medallions and corporate commemorative coins. The company operates at a monthly production capacity of 12,000 medallions, currently producing and selling 9,000 medallions per month in the domestic market at a price of ₹180 per unit.

The cost data for the month of March, 2025 is as under:

Cost Elements	Amount (₹)
Variable Costs (dependent on units produced)	
Direct Materials (for 9,000 units)	3,42,000
Direct Labour	3,96,000
Variable Batch Costs (180 batches x ₹ 550 per batch)	99,000
Fixed Costs	
Fixed Manufacturing Overhead	3,20,000
Fixed Marketing and Admin	2,10,000

The batch size for MintCraft Pvt. Ltd.'s regular production is 50 medallions per batch. However, the special order received from an international client involves the production of 3,000 medallions, which differs in batch configuration. For this special order, the company is required to manufacture the medallions in 30 batches, with each batch consisting of 100 medallions. The client has offered a price of ₹125 per medallion for this one-time order.

Required:

(i) SHOULD MintCraft Pvt. Ltd. accept the special order? Support your recommendation with relevant calculations and brief justification.

(ii) Assume the plant capacity is reduced to 10,000 units due to scheduled maintenance. The special order must be accepted in full or not at all. SHOULD MintCraft accept the special order under these new conditions? PROVIDE brief reasoning supported by contribution margin analysis.

ICAI'S ANSWER

Statement of Contribution Margin on Accepting the Special Order of 3,000 Medallions

Particulars	₹
Sales Revenue (3,000 medallions x ₹ 125 per medallion)	3,75,000
Less: Variable Costs	
• Direct Materials (3,000 medallions x ₹ 38 per medallion*)	1,14,000
• Direct Labour (3,000 medallions x ₹ 44 per medallion*)	1,32,000
• Setups, Quality Control, Handling (30 batches x ₹ 550 per batch)	16,500
Contribution Margin	**1,12,500**

*Rates derived from regular cost data:

→ Direct Materials = ₹3,42,000 / 9,000 units = ₹38/unit

→ Direct Labour = ₹3,96,000 / 9,000 units = ₹44/unit

Decision – When Capacity Is 12,000 Medallions:

The plant has 3,000 units of unused capacity (12,000 – 9,000), and the special order fits within this limit. Since fixed costs will not increase and the order provides a positive contribution margin of ₹ 1,12,500, MintCraft Pvt. Ltd. should accept the special order, as it will increase operating profit.

Statement of Contribution Margin Loss Due to Capacity Constraint (When Capacity = 10,000 Medallions):

Particulars	₹
Gain in Contribution Margin from Special Order	1,12,500
Less: Loss of Contribution Margin on 2,000 Medallions of Regular Sales (W.N.1 & W.N.2)	(1,74,000)
Net Loss in Contribution Margin	(61,500)

Decision – When Capacity Is 10,000 Medallions:

Accepting the special order means MintCraft must reduce 2,000 medallions from its regular, higher-priced domestic sales. The loss in contribution from regular sales exceeds the gain from the special order, resulting in a net contribution loss of ₹ 61,500. Therefore, the special order should not be accepted under capacity constraint.

Working Notes:

W.N. 1: Contribution Margin on Current 9,000 Medallions

Particulars	₹
Sales Revenue (9,000 x ₹ 180)	16,20,000
Less: Variable Costs	
• Direct Materials (9,000 x ₹38)	3,42,000
• Direct Labour (9,000 x ₹44)	3,96,000
• Setup, Handling, QC (180 batches x ₹550)	99,000
Contribution Margin	**7,83,000**

W.N. 2: Contribution Margin on 7,000 Medallions (after reducing 2,000 units)

Particulars	₹
Sales Revenue (7,000 x ₹180)	12,60,000
Less: Variable Costs	
• Direct Materials (7,000 x ₹38)	2,66,000
• Direct Labour (7,000 x ₹44)	3,08,000
• Setup, Handling, QC (140 batches x ₹550)	77,000
Contribution Margin	**6,09,000**

Loss of Contribution Margin = ₹ 7,83,000 – ₹ 6,09,000 = ₹ 1,74,000

ICAI · P4 · MTP May 2026 · Series I · question p. 11-12 · answer p. 13-15

1042 MTP **May 2026 S-II** Q2(a) 7 marks

GreenLeaf Furniture Ltd., a small but rapidly growing manufacturer of bespoke wooden furniture, has prepared its operating budget for the forthcoming year (figures in ₹000s). The company supplies premium handcrafted items to interior designers, boutique resorts, and high-end retail outlets. Due to rising demand for sustainable and custom-made furniture, GreenLeaf has recently expanded its workshop capacity, although it still operates close to full utilisation during peak months. The annual budget reflects management's efforts to control costs while maintaining craftsmanship standards, and it serves as the basis for setting selling prices, estimating profitability, and negotiating large contracts for the year.

Particulars	Amount (₹000)
Budgeted Sales revenue	2,000
Direct materials	360
Direct labour	640
Factory overheads — Variable	220

Particulars	Amount (₹000)
Factory overheads — Fixed	260
Administration overheads — Fixed	180
Sales commission (variable)	100
Fixed selling overheads	40
Budgeted Profit	200

Company pricing policy: all production and selling overheads other than prime costs (i.e., everything except direct materials and direct labour) are recovered as a percentage of direct wages. After total cost is computed, the company adds a markup equal to one-ninth (1/9) of total cost to arrive at the selling price (this markup is intended to give the budgeted profit).

Situation: A boutique lakeside hotel approaches GreenLeaf directly to place a one-off order for a set of custom dining tables. Because the client is buying directly (no dealer commission) and the order will be produced in addition to the budgeted output, GreenLeaf estimates the extra prime costs for the special order as follows:

- Direct materials: ₹ 45,000
- Direct labour: ₹ 75,000

Using its standard pricing policy, GreenLeaf prepared a quotation of ₹ 2,37,500 for the special order. The hotel, however, says it can only pay ₹ 1,80,000 and asks whether GreenLeaf will accept that price.

Management is reluctant to accept any order below the company's calculated total cost because they believe this will cause a loss.

Required:

1. STATE and EXPLAIN the arguments for and against accepting the hotel's offer of ₹ 180,000 for the special order. Consider:
2. Provide a clear RECOMMENDATION to the management (accept / reject / accept with conditions).

ICAI'S ANSWER

Analysis of Cost and Profit:

Budgeted Cost Sheet

Particulars	Amount (₹ 000)
Prime Cost	
Direct Material	360
Direct Labour	640
Total Prime Cost	**1,000**
Overheads	
Variable Factory Overhead	220

Particulars	Amount (₹ 000)
Fixed Factory Overhead	260
Administration Overheads	180
Selling Commission	100
Fixed Selling Overheads	40
Total Overheads	**800**
Total Cost	**1,800**
Profit	**200**

Rate of Profit on Cost = 2 / 18 = 1/9

Overhead Absorption Rate = 125% of Direct Wages (800 / 640 × 100)

Break up of New Order:

Particulars	Amount (₹)
Direct Materials	45,000
Direct Labour	75,000
Overheads @ 125% of Direct Wages	93,750
Total Cost (as per company policy)	**2,13,750**
Profit @ 1/9	23,750
Quoted Price	**2,37,500**

The following points emerge:

(i) Factory overheads only are to be recovered on the basis of direct wages.

(ii) The special order is a direct order. Hence commission is not payable.

(iii) The budgeted sales are achieved. Hence all fixed overheads are recovered. Hence, no fixed overheads will be chargeable to the special order.

Calculation of Factory Variable Overhead Rate

Particulars	Amount (₹ 000)
Total Variable Factory Overheads	220
Direct Wages	640
Variable Factory OH Rate (220 / 640 × 100)	**34.375%**

Applying this rate the Cost of the Special Order will be as under:

Particulars	Amount (₹)
Direct Materials	45,000
Direct Labour	75,000
Overheads @ 34.375% of Direct Wages	25,781
Total Relevant Cost	**1,45,781**
Price Offered	1,80,000
Margin	**34,219**

Since the offer price of 1,80,000 provides a positive contribution of 34,219, which is higher than the budgeted profit margin (1/9), the special order should be accepted.

ICAI · P4 · MTP May 2026 · Series II · question p. 9-10 · answer p. 8-9

1043 MTP **Jan 2026 S-II** Case Scenario I · MCQ 1 to MCQ 5 10 marks

CASE SCENARIO I

Bharat Pharma Ltd was established three years ago by a group of biotechnology researchers to commercialize a new therapeutic drug they had created. The production technology required for this drug is highly sophisticated and capital-intensive. As a result, the company incurs very high fixed manufacturing costs.

This situation is a major concern for Dr. Kavita Rao, the company's CEO. She recently organised a meeting of all senior managers to discuss profitability. Dr. Rao demonstrated how the average cost per unit decreases as production increases, due to the company's heavy fixed-cost base.

She explained:

"It is evident that as we move closer to the plant's maximum production capacity of 90,000 vials, the average cost per vial continues to fall. Producing and selling as near to that capacity as possible must improve our profitability."

The data she presented are shown below:

Production Volume (vials)	50,000	60,000	75,000	90,000
Average Cost per Unit (₹)	3,400	3,050	2,700	2,466.66

Current annual sales and production: 78,000 vials Selling price per vial: ₹ 3,900

(Average cost is defined as total fixed + variable cost divided by number of units produced.)

You are a member of the management accounting team. Soon after the meeting, you are contacted by Arun Mehta, the Marketing Director. He wants to understand how profitability changes with production.

Arun then informs you about a discussion he had with Dr. Rao. She again emphasized pushing production closer to the capacity of 90,000 vials.

Arun has the opportunity to secure an export order for 8,000 additional vials, but due to strong competition, the customer will pay only ₹ 2,550 per vial.

Dr. Rao believes the price is below cost and wants to reject it.

However, she says she would be willing to sell at ₹ 2,500 per vial, but only if the order is increased to 20,000 vials.

Based on the above facts, you are required to answer the following questions (MCQ 1 to 5):

MCQ 1 2 marks

What are the total fixed costs of Bharat Pharma Ltd?

A ₹ 6.5 crore

B ₹ 8.4 crore

C ₹ 10.5 crore

D ₹ 12.3 crore

MCQ 2 2 marks

What is the profit at the current sales volume of 78,000 vials?

A ₹ 4.2 crore

B ₹ 7.1 crore

C ₹ 9.78 crore

D ₹ 11.3 crore

MCQ 3 2 marks

What is the break-even point (units) and margin of safety (as a percentage)?

A 25,450 units and 52.25%

B 32,900 units and 60%

C 40,385 units and 48.22%

D 50,200 units and 40%

MCQ 4 2 marks

What is the change in profit if the company accepts the order for 8,000 vials at ₹ 2,550?

A Increase of ₹ 82 lakh

B Increase of ₹ 1 crore

C Increase of ₹ 2.1 crore

D No change in profit

MCQ 5 2 marks

What is the change in profit if the company accepts the order for 20,000 vials at ₹ 2,500?

A Increase of ₹ 1.68 crore

B Increase of ₹ 0.32 crore

C Increase of ₹ 2.8 crore

D Decrease of ₹ 1.1 crore

ICAI'S ANSWERS

MCQ 1

(C) ₹ 10.5 crore

Total cost at 50,000

= 50,000 × ₹ 3,400 = ₹ 17,00,00,000

Total cost at 60,000

= 60,000 × ₹ 3,050 = ₹ 18,30,00,000

Difference in total cost = 18.3 crore – 17 crore = ₹ 1,30,00,000

Difference in output = 10,000 units

Variable cost per unit = $\frac{1,30,00,000}{10,000}$

= ₹ 1,300

Total Fixed Costs

Use total cost at 50,000 units:

Total cost = 50,000 × 3,400 = ₹ 17,00,00,000

Variable cost = 50,000 × 1,300 = ₹ 6,50,00,000

Fixed cost = 17,00,00,000– 6,50,00,000

= ₹ 10,50,00,000

Fixed Costs = ₹ 10.5 crore

MCQ 2

(C) ₹ 9.78 crore

Profit at current sales of 78,000 units

Selling price = ₹ 3,900

Variable cost = ₹ 1,300

Contribution per unit = 3,900 – 1,300

= ₹ 2,600

Revenue = 78,000 × 3,900 = ₹ 30,42,00,000

Variable cost = 78,000 × 1,300 = ₹ 10,14,00,000

Contribution = 30.42 cr – 10.14 cr = ₹ 20,28,00,000

Profit = Contribution – Fixed cost

Profit = 20.28 cr – 10.5 cr

= ₹ 9,78,00,000

Profit = ₹ 9.78 crore

MCQ 3

(C) 40,385 units and 48.22%

Contribution per unit = ₹ 2,600

Break-even units = $\frac{\text{Fixed cost}}{\text{CPU}}$

$= \frac{10,50,00,000}{2,600}$

= 40,385 units

Break-even point = 40,385 units

Current sales = 78,000 units

Break-even = 40,385 unit

MOS = $\frac{78,000 - 40,385}{78,000} \times 100$

$= \frac{37,615}{78,000} \times 100$

= 48.22%

Margin of safety = 48.22%

MCQ 4

(B) Increase of ₹ 1 crore

Profit impact of accepting 8,000 vials at ₹ 2,550

Contribution per unit = Selling price – Variable cost

= 2,550 – 1,300

= ₹ 1,250

Additional contribution = 8,000 × 1,250

= ₹ 1,00,00,000

Profit increases by ₹ 1 crore

(Within plant capacity: 78,000 + 8,000 = 86,000 < 90,000)

MCQ 5

(B) Increase of ₹ 0.32 crore

Domestic sales contribution per unit

Selling price = ₹ 3,900

Variable cost = ₹ 1,300

Contribution = ₹ 2,600 per vial

Export order contribution per unit

Selling price = ₹ 2,500

Variable cost = ₹ 1,300

Contribution = ₹ 1,200 per vial

(i) Contribution gained from export order

20,000 × ₹ 1,200

= ₹ 2,40,00,000 (₹ 2.4 crore)

(ii) Contribution lost by reducing domestic sales

8,000 × ₹ 2,600

= ₹ 2,08,00,000 (₹ 2.08 crore)

Net Change = Gain – Loss

= ₹ 2,40,00,000 – ₹ 2,08,00,000

= ₹ 32,00,000

Net Profit Increase = ₹ 0.32 crore

ICAI · P4 · MTP Jan 2026 · Series II · question p. 1-2 · answer p. 1, 2, 3

1044 MTP **Sep 2025 S-I** Q4(a) 6 marks

Apex Electronics Ltd. is a leading Indian manufacturer of smart home devices. Until now, the company has produced all device components in-house. However, due to a temporary shortage of skilled labour, the company is exploring the possibility of outsourcing the production of some key components for its flagship product: the Smart Speaker.

The components under consideration are:

1. Voice Module
2. Control Panel
3. Power Adapter

The following data relates to the manufacturing of 5,000 units of each component:

Component	Voice Module	Control Panel	Power Adapter
Labour Hours per Unit	1.2	2	3.5
Machine Hours per Unit	0.8	0.6	0.3
Direct Labour (₹)	1,500	2,000	1,200
Direct Material (₹)	1,200	1,800	1,100
Variable Overhead (₹)	700	900	600
Fixed Overhead (₹)	600	750	550
Purchase Price from Supplier (per unit) (₹)	3,700	4,600	3,300

Additional details:

Labour availability for the upcoming month is limited to 4,000 hours.

The company must produce 4,000 units of each component in the next month.

Required

CALCULATE how many of each product should be made in-house and how many should be outsourced.

ICAI'S ANSWER

Calculation of Relevant In-House Cost and saving:

Component	Voice Module	Control Panel	Power Adapter
Direct Labour (₹)	1,500	2,000	1,200

Component	Voice Module	Control Panel	Power Adapter
Direct Material (₹)	1,200	1,800	1,100
Variable Overhead (₹)	700	900	600
Total In-House Cost (₹)	3,400	4,700	2,900
Outsource Price (₹)	3,700	4,600	3,300
Saving (₹)	300	(100)	400
Labour Hours per Unit	1.2	2	3.5
Savings per Labour hour (₹)	250	-	114.29
Rank	I	-	II

Note: As the internal cost of making the control panel exceeds the external purchase price of buying, it would be beneficial to outsource their production entirely.

Allocate Labour Efficiently

Make Voice Modules first:

Labour needed per unit = 1.2 hours

To make all 4,000 units: 4,000 × 1.2 = 4,800 hours (available 4,000 hours)

Max units company can make = 4,000 ÷ 1.2 = 3,333 units

Component	Make In-House	Outsource
Voice Module	3,333 units	667 units
Control Panel	0 units	4,000 units
Power Adapter	0 units	4,000 units

ICAI · P4 · MTP Sep 2025 · Series I · question p. 12-13 · answer p. 13

1045 MTP Sep 2025 S-II MCQ 11 2 marks MCQ

A machine which originally cost ₹ 25,000 has an estimated life of 10 years and is to be depreciated at ₹ 2,500 per year. It has remained idle recently due to a lack of orders. A special job has now been received that would require the use of the machine for three months. The current net realisable value of the machine is ₹ 18,000. If used for the order, its value is expected to fall to ₹ 15,000. The routine maintenance cost of the machine is ₹ 500 per month when idle. With usage, maintenance and repair costs are expected to increase to ₹ 700 per month. Additionally, a temporary machine operator would need to be hired at ₹ 800 per month for the duration of the job.

What would be the relevant cost of using the machine for the special order?

A ₹ 3,600

B ₹ 4,200

C ₹ 4,800

D ₹ 6,000

ICAI'S ANSWER

(D) ₹ 6,000

Relevant Cost :

- Opportunity Cost of machine = ₹ 18,000 – ₹ 15,000 = ₹ 3,000
- Additional Maintenance:
  Incremental maintenance = ₹ 200/month

  Duration = 3 months

  Total additional maintenance cost = ₹ 200 x 3 = ₹ 600
- Operator Cost = ₹ 800/month x 3 months = ₹ 2,400
  Total Relevant Cost = ₹ 3,000 + ₹ 600 + ₹ 2,400 = ₹ 6,000

ICAI · P4 · MTP Sep 2025 · Series II · question p. 6 · answer p. 5-6

1046 MTP May 2025 S-I MCQ 11 2 marks MCQ

A manufacturing firm is presently producing and selling 10,000 units of a product at ₹ 500 per unit in the domestic market. The fixed cost per unit at the current level of operation is ₹ 150 and variable cost is ₹ 300 per unit. The firm has received an export order for supply of 5,000 units of the product at ₹ 400 per unit. After meeting the domestic demand, if the firm accepts the export offer, the profit of the firm is expected to:

A decrease by ₹ 5,00,000.

B increase by ₹ 2,00,000.

C decrease by ₹ 3,00,000.

D increase by ₹ 5,00,000.

ICAI'S ANSWER

(D) increase by ₹ 5,00,000.

Total Revenue (Domestic) = 10,000 × 500 = 50,00,000

Total Cost (Domestic) = Fixed Cost + Variable Cost

= (10,000 × 150) + (10,000 × 300) = 15,00,000+30,00,000 = 45,00,000

Profit (Domestic) =Total Revenue−Total Cost = 50,00,000−45,00,000 = 5,00,000

Contribution per Unit (Export) = Export Price−Variable Cost = 400−300 = 100

Total Contribution from Export = 100 × 5,000 = 5,00,000

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1047 MTP Jan 2025 S-I Case Scenario II · MCQ 6 to MCQ 10 10 marks

CASE SCENARIO II

A garment manufacturer has been producing and selling T-shirts exclusively for Indian market. His T-shirts are made of a specific material which is eco-friendly. It means that T-shirts are bio-degradable in soil after it becomes unsuitable for use.

This invention has been applauded throughout the country. Owner, Vikas, registered for the patent rights for his invention so that no one else could use it.

Vikas feels that this invention will also be liked in foreign markets, and thus plans to expand his business outside India. He feels that US market is the first foreign market he should tap into.

Current cost structure (each T-shirt):

Direct material	90
Direct labour	60
Special service	80
(Used in T-shirt making, 50% fixed)	
Fixed overhead	50
Administration overhead (fixed)	20
Total cost per T-shirt	300
(+) Profit margin	200
Selling price in India	500

There is no limitation of any resources in India. Vikas is able to sell 80,000 T-shirts each year. He is currently working at 80% of his total capacity.

After searching for potential customers in US, Vikas received an inquiry for 30,000 units from a wholesale distributor in California. As per the inquiry, order will be placed if price per T-shirt is reasonable and the order has to be satisfied in full.

Vikas decided to send a quote and the order was placed by the foreign client, on the same day. Vikas, without a second thought accepted the order, but did not feel the need to extend the manufacturing capacity; therefore he decided forgo a few Indian clients.

This foreign order also required special packaging. It is spent at 20% of the total prime cost per T-shirt. The production was done quickly and foreign consignment was transported to custom port via services from a carriage agency. It charged ₹ 80,000 for 1 truck, whose capacity was 500 kg, to transport whole of the consignment. Truck was 20% vacant after loading the consignment.

Bill of lading was filed and a professional fee of ₹ 25,000 for filing this was paid to a Chartered accountant. Custom port also charged ₹ 80 per kg per day to handle the material, storing it in warehouse, and for loading the goods on ship.

The shipping company, which was booked by Vikas for taking the consignment to US, got delayed due to bad weather. Stock was held at port for 5 days and on 6th day it was loaded on ship. Shipping company charged ₹ 2,800/ 10kg of goods. Insurance was charged flat at ₹ 1,11,000.

There is no custom duty on such exports.

Answer the following questions (MCQs 6 to 10):

MCQ 6 2 marks

Vikas had sufficient funds in his hands but he still raised a short-term working capital loan @ 6.5% p.a. for the satisfaction of this foreign order because he found a one time investment opportunity which was

giving him 9.25% returns. Foreign order was accepted on 1st June and loan was taken on the same day. Repayment of the loan will be made on 1st September. Calculate net cash outflow due to this export order. Which of the following is correct?

A ₹ 73,91,000

B ₹ 75,47,750

C ₹ 74,76,500

D ₹ 71,06,000

MCQ 7 2 marks

What would have been the minimum price that Vikas could have quoted per T-shirt in US dollars? (exchange rate on 1st June, $1 = ₹ 83.86)

A $ 4.23

B $ 4.20

C $ 4.17

D $4.05

MCQ 8 2 marks

Payment from foreign client was received on 8th October when exchange rate was ₹ 86 for each US $. Calculate the profit earned from this export order if actual quoted price was $4.90 per T-shirt. Select the correct amongst following:

A ₹ 40,65,500

B ₹ 41,51,000

C ₹ 39,94,250

D ₹ 44,36,000

MCQ 9 2 marks

What is the net cash Inflow from this export order?

A ₹ 55,36,000

B ₹ 51,65,500

C ₹ 52,51,000

D ₹ 50,94,250

MCQ 10 2 marks

What is the Incremental benefit from this export order?

A ₹ 19,94,250

B ₹ 21,51,000

C ₹ 20,65,500

D ₹ 24,36,000

ICAI'S ANSWERS

MCQ 6

(B) ₹ 75,47,750

Funds required for foreign order:

Costs	Amounts
Direct material per unit	90
Add: Direct labour per unit	60
Add: special services per unit	40
	190
Add: packaging per unit (20% x prime cost, 20% x (90 + 60 + 80))	46
Variable cost per unit	236
Total variable cost (236×30,000)	**70,80,000**
Add: freight	80,000
Add: professional fees	25,000
Add: custom charges (500kg x 80% x 80 × 6)	1,92,000
	73,77,000
Add: shipping ((500×80%/10) x 2,800)	1,12,000
Add: insurance	1,11,000
Funds required	**76,00,000**

Net amount of interest earned (interest earned in 9.25% and paid is 6.50% for 3 months) = 76,00,000 x (9.25% - 6.50%) x 3/12 = 52,250

So, net cash outflow due to export order = 76,00,000 - 52,250 = 75,47,750

MCQ 7

(A) $ 4.23

Minimum price :-

Variable cost (net)	75,47,750
Add: fixed cost recovery (110 × 10,000 units)	11,00,000
Add: loss of profit (200 × 10,000 units)	20,00,000
Minimum price	1,06,47,750
Minimum price per unit 1,06,47,750/30,000	₹ 354.925
Minimum price is $ ($1 = ₹ 83.864)	$4.23

MCQ 8

(C) ₹ 39,94,250

PROFIT EARNED:

SALES ($4.90 × 30,000 x RS. 86)	**₹ 1,26,42,000**
(-) Variable cost (net)	(75,47,750)
(-) allotted fixed cost (10,000 units x110)	(11,00,000)
PROFIT	₹ 39,94,250

MCQ 9

(D) ₹ 50,94,250

CASH INFLOW:

SALES ($4.90 × 30,000 x RS. 86)	₹ 1,26,42,000
(-) Variable cost (net)	(75,47,750)
CASH INFLOW	**₹ 50,94,250**

MCQ 10

(A) ₹ 19,94,250

Incremental benefits:

SALES ($4.90 × 30,000 x RS. 86)	₹ 1,26,42,000
(-) Variable cost (net)	(75,47,750)
(-) allotted fixed cost (10,000 units x110)	(11,00,000)
(-) loss of profit (10,000×200)	(20,00,000)
Incremental benefits	**19,94,250**

ICAI · P4 · MTP Jan 2025 · Series I · question p. 3-4 · answer p. 2, 3

1048 MTP **Sep 2024 S-I** Q6(d) 4 marks

WHAT are the principles of estimation of costs and benefits?

ICAI'S ANSWER

After identification of the costs and benefits, it is now required to be quantified i.e., the cost and benefit should be measured and estimated. The estimation is done by following the two principles as discusses below:

(i) Variability: Variability means by how much a cost or benefit increased or decreased due to the choice of the option. Variable costs are the cost which differs under the different volume or activities. On the other hand, fixed costs remain same irrespective of volume and activities.

(ii) Traceability: Traceability of cost means degree of relationship between the cost and the choice of the option. Direct costs are directly assigned to the option on the other hand indirect costs needs to be apportioned to the option on some reasonable basis.

ICAI · P4 · MTP Sep 2024 · Series I · question p. 12 · answer p. 18

1049 MTP **Old P3 Nov 2023 S-II** Q1(b) 5 marks old scheme

T Ltd., produces and sells 95,000 units of 'X' in a year at its 80% production capacity. The selling price of product is ₹ 8 per unit. The variable cost is 75% of sales price per unit. The fixed cost is ₹ 3,50,000. The company is continuously incurring losses and management plans to shut-down the plant. The fixed cost is expected to be reduced to ₹ 1,30,000. Additional costs of plant shut-down are expected at ₹ 15,000.

Should the plant be shut-down? Find the shut-down point in units and also in percentage of capacity level of production.

ICAI'S ANSWER

Statement Showing "Operating Loss"

	If Plant is Continued	If Plant is Shutdown
Sales	7,60,000	---
Less: Variable Cost	5,70,000	---
Contribution	1,90,000	---
Less: Fixed Cost	3,50,000	1,30,000
Less: Additional Cost	---	15,000
Operating Loss	1,60,000	1,45,000

Decision on Shut Down

A comparison of loss figures (indicated as above) points out that loss is reduced by **₹15,000** (₹ 1,60,000 - ₹ 1,45,000) if plant is shut down.

→ Accordingly, plant should be Shut Down.

Shut Down Point = $\frac{₹3,50,000 - ₹1,45,000}{₹8 - ₹6}$ = 1,02,500 units

Capacity Level at Shut Down Point (%)

At 100% Level – Production Capacity

= 1,18,750 $\left(\frac{95,000 \text{ units}}{0.80}\right)$

Capacity Level at Shut Down Point

= 86.32% $\left(\frac{1,02,500 \text{ units}}{1,18,750 \text{ units}}\right)$

ICAI · Old P3 · MTP Nov 2023 · Series II · question p. 1 · answer p. 1-2

1050 MTP **Old P3 May 2020 S-I** Q5(b)(ii) 5 marks old scheme

A company can make any one of the 3 products X, Y or Z in a year. It can exercise its option only at the beginning of each year.

Relevant information about the products for the next year is given below.

	X	Y	Z
Selling Price (Rs. / unit)	100	120	120
Variable Costs (Rs. / unit)	60	90	70
Market Demand (unit)	3,000	2,000	1,000
Production Capacity (unit)	2,000	3,000	900
Fixed Costs (Rs.)	3,00,000		

Required

COMPUTE the opportunity costs for each of the products.

ICAI'S ANSWER

		X	Y	Z
I.	Contribution *per unit* (Rs.)	40	30	50
II.	Units (Lower of Production / Market Demand)	2,000	2,000	900
III.	Possible Contribution (Rs.) [I × II]	80,000	60,000	45,000
IV.	Opportunity Cost* (Rs.)	60,000	80,000	80,000

(*) Opportunity cost is the maximum possible contribution forgone by not producing alternative product i.e. if Product X is produced then opportunity cost will be maximum of (Rs. 60,000 from Y, Rs. 45,000 from Z).

ICAI · Old P3 · MTP May 2020 · Series I · question p. 6 · answer p. 12

14.6 Multi-product break-even analysis

14 questions · asked in 6 ICAI documents · 1 exam paper

1051 EXAM Jan 2026 Case Scenario - I · MCQ 1 to MCQ 5 10 marks

CASE SCENARIO - I

Zed Manufacturing Ltd. produces three different types of specialized products namely X, Y and Z. The following information has been collected from their books of accounts for the year ended March 2025:

Particulars	Products		
	X	Y	Z
Sales (in ₹)	40,00,000	23,80,000	16,20,000
Selling Price per unit (in ₹)	500	700	400
Variable Cost per unit (in ₹)	250	420	240

Total fixed costs: ₹ 18,00,000

Zed Manufacturing Ltd. is currently discussing proposals to maintain a sales mix ratio, to discontinue the manufacturing of Product Z and replace it with a new product - Product W in anticipation of following results:

	Products		
Particulars	**X**	**Y**	**W**
Sales Mix Ratio	55%	30%	15%
Selling Price per unit (in ₹)	500	700	500
Variable Cost per unit (in ₹)	250	420	250

Total fixed costs: ₹ 18,80,000

Total sales: ₹ 85,00,000

Based on the above, you are required to calculate the following (MCQ 1 to 5):

MCQ 1 2 marks

What is the combined contribution-to-sales (C/S) ratio at the exiting sales level?

A 37.5%

B 40%

C 32.5%

D 45%

MCQ 2 2 marks

What is the total profit at the existing sales level?

A ₹ 18,00,000

B ₹ 32,00,000

C ₹ 36,00,000

D ₹ 16,00,000

MCQ 3 2 marks

What is the new contribution-to-sales (C/S) ratio in the proposed situation (with Product W)?

A 38.5%

B 35%

C 47%

D 40.5%

MCQ 4 2 marks

What is the break-even sales at the level of proposed sales mix (with Product W)?

A ₹ 40,75,000

B ₹ 40,00,000

C ₹ 48,83,117

D ₹ 53,80,000

MCQ 5 2 marks

What is the total profit at the level of proposed sales mix (with Product W)?

A ₹ 10,95,000

B ₹ 21,15,000

C ₹ 15,62,500

D ₹ 19,60,000

ICAI'S ANSWERS

MCQ 1

Answer key: **(D)**

MCQ 2

Answer key: **(A)**

MCQ 3

Answer key: **(C)**

MCQ 4

Answer key: **(B)**

MCQ 5

Answer key: **(B)**

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1052 RTP **May 2025** Q7 MCQ

PR Ltd. sells two types of pen, ball pen and gel pen. Currently, the company is expecting to sell 6,000 units of ball pen along with 3,600 units of gel pen in the coming month. Other information as forecasted is provided below:

Particulars	Ball pen	Gel pen
Selling price (per unit)	₹ 150	₹ 100
Variable cost (per unit)	₹ 90	₹ 60
Contribution (per unit)	₹ 60	₹ 40
Fixed Costs		₹ 3,36,000

You are required to CALCULATE the Composite Break-even Batch and individual break-even of the pens (in units).

A Composite Break-even Batch- 6,400 batches, Break-even units of Ball pen- 4,000 units, Break-even units of Gel pen- 2,400 units.

B Composite Break-even Batch- 800 batches, Break-even units of Ball pen- 500 units, Break-even units of Gel pen- 300 units.

C Composite Break-even Batch- 800 batches, Break-even units of Ball pen- 4,000 units, Break-even units of Gel pen- 2,400 units.

D Composite Break-even Batch- 6,400 batches, Break-even units of Ball pen- 500 units, Break-even units of Gel pen- 300 units.

ICAI'S ANSWER

(C)

PR Ltd. is expecting to sell 6,000 units of ball pen along with 3,600 units of gel pen, resulting in a sales mix of 5:3 per batch.

Thus, composite contribution per batch = (₹ 60 x 5 ball pens) + (₹ 40 x 3 gel pens)

= ₹ 420

Composite Break-even Batch = $\left(\frac{\text{Commom fixed costs}}{\text{Composite contribution per batch}}\right)$

= $\left(\frac{₹\ 3,36,000}{₹\ 420}\right)$

= 800 batches

Break-even units of Ball pen = 800 x 5= 4,000 units

Break-even units of Gel pen = 800 x 3= 2,400 units

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1053 **RTP** **Jan 2025** Case scenario · Q1(i) to Q1(v)

CASE SCENARIO

Popular company produces various articles for student purposes. It has been in industry since last 25 years. Company had a very humble start but gained popularity over the years due to excellent quality products which were sold at very competitive prices. Company has huge reserves and feel that it is also obligated to give back to the society from which it has grown.

Last year management decided to produce and supply special quality school bags, water bottles, & geometry boxes to NGOs, at no price, as a social responsibility. These articles were simple looking but were more durable, that would not have wore-off easily and could have been used for long-term.

This year management wants to add another dimension to this social work. It approached charitable schools and government run schools and offered them the supply of the same articles, at cost. This will help students in these schools to get these things at a very low price compared to market.

The variable costs are ₹ 100, ₹ 80, and ₹ 40 for school bags, water bottles, and geometry boxes, respectively. These articles are made using a single machine. 0.20 hours of machine operation is required for manufacturing 1 unit of school bag. Similarly, machine hours required for each units of water bottle and geometry box is 0.15 hours and 0.10 hours, respectively. Fixed overhead related to machine is ₹ 7,40,000 per year. Machine can operate for 8,000 hours in a year.

Company has decided to sell its 80% capacity production in markets. Rest is divided amongst the 2 undergoing social works, equally.

All Schools requests these items in the ratio of 2:3:5, as per their demand by the school students.

Company wants to set a price for these articles to be offered to the schools. Management has few questions they need the answers to. They assigned the task to their team. Team made rough calculations but as there were too many people on the team, each came up with different answers. As a Chartered accountant, you have been approached. Understand the case closely, find the correct answers and help management to set a price.

Answer the following:

Q1(i)

What is allocated fixed cost per unit of School bags, water bottles, and geometry boxes?

- A 18.5, 13.875, 9.75
- B 18.5, 13.875, 9.25
- C 18.5, 13.785, 9.25
- D 18.5, 13.785, 9.50

Q1(ii)

If the prices were ₹ 200, ₹ 160, and ₹ 100, what would be the overall break-even point in units in relation to fixed cost allocated to these supplies?

- A 308.33 units
- B 500 units
- C 508.33 units
- D 1,000 units

Q1(iii)

Find out the maximum number of units of each article that can be given at the prices given in Part (ii).

- A 61, 92, 154
- B 200, 300, 500
- C 101, 152, 254
- D 100, 150, 250

Q1(iv)

What will be the maximum units that can be supplied to the schools of each article?

- A 1103, 1645, 2726
- B 1093, 1655, 2748
- C 1185, 1777, 2962
- D 1133, 1675, 2958

Q1(v)

What should be the correct price for each item as per the management's decision?

- A 118.50, 93.875, 49.75
- B 118.50, 93.785, 49.25
- C 118.50, 93.785, 49.50
- D 118.50, 93.875, 49.25

ICAI'S ANSWERS

Q1(i)

(B) Fixed overhead = 740000

Total machine hours = 8000 hours

Fixed OH per hour= ₹ 92.5

Fixed OH per unit of:

- School bag = 0.20 x 92.5 = ₹ 18.5

- Water bottle = 0.15 x 92.5 = ₹ 13.875
- Geometry box = 0.10 x 92.5 = ₹ 9.25

Q1(ii)

(D) Hours allocated = 8000 x 10% = 800 hours

Fixed overhead allocated = 800 x 92.5 = ₹ 74,000

Contribution:

- Bag = 200-100 = 100
- Bottle = 160-80=80
- Geometry = 100- 40 = 60

Composite contribution = 100 x 2/10 + 80 x 3 / 10 + 60 x 5/10 = ₹ 74

Overall breakeven point for this assignment is = fixed cost allocated/composite contribution = 74,000/74 = **1,000 units**

Q1(iii)

(B) 1000 units are to be distributed in the ratio of 2:3:5

Bag = 200 units, bottle = 300 units, geometry = 500 units

Q1(iv)

(C) Total hours = 800 hours

let total no of units = X

Supply: bag 2/10 x X; bottle 3/10 x X; geometry 5/10 x X

Hours: (2X/10) x 0.20 + (3X/10) x 0.15 + (5X/10) x 0.10 = 800 hours

X = 5925

Units of :

- Bag = 2/10 x 5925 = 1185
- Bottle = 3/10 x 5925 = 1777.5 or 1777
- Geometry = 5/10 x 5925 = 2962.5 or 2962

Q1(v)

(D) Correct price is AT COST.

COST = Marginal Cost Per Unit + Fixed Overhead Cost Allocated Per Unit

	Bag	Bottle	Geometry
Variable cost per unit	100	80	40
Fixed cost per unit	18.5	13.875	9.25
Total	**118.5**	**93.875**	**49.25**

ICAI · P4 · RTP Jan 2025 · question p. 1-2 · answer p. 20, 20-21, 21

1054 RTP **Old P3 May 2022** Q12 old scheme

Same question, different ICAI answer: #1056 Old P3 MTP Dec 2021 S-II

A Limited manufactures three different products and the following information has been collected from the books of accounts:

	Products		
	S	**T**	**U**
Sales Mix	25%	35%	40%
Selling Price	₹ 600	₹800	₹400
Variable Cost	₹ 300	₹400	₹240
Total Fixed Costs			₹ 36,00,000
Total Sales			₹ 1,20,00,000

The company has currently under discussion, a proposal to discontinue the manufacture of Product U and replace it with Product M, when the following results are anticipated:

	Products		
	S	**T**	**M**
Sales Mix	40%	35%	25%
Selling Price	₹ 600	₹ 800	₹ 600
Variable Cost	₹ 300	₹ 400	₹ 300
Total Fixed Costs			₹ 36,00,000
Total Sales			₹ 1,28,00,000

Required:

(i) COMPUTE the PV ratio, total contribution, profit and Break-even sales for the existing product mix.

(ii) COMPUTE the PV ratio, total contribution, profit and Break-even sales for the proposed product mix

ICAI'S ANSWER

(i) Computation of PV ratio, contribution, profit and break-even sales for existing product mix

	Products			Total
	S	**T**	**U**	
Selling Price (₹)	600	800	400	
Less: Variable Cost (₹)	300	400	240	
Contribution per unit (₹)	300	400	160	
P/V Ratio (Contribution/Selling price)	50%	50%	40%	
Sales Mix	25%	35%	40%	

	Products			Total
Contribution per rupee of sales (P/V Ratio × Sales Mix)	12.5%	17.5%	16%	46%
Present Total Contribution (₹1,20,00,000 × 46%)				₹ 55,20,000
Less: Fixed Costs				₹ 36,00,000
Present Profit				₹ 19,20,000
Present Break Even Sales (₹ 36,00,000/0.46)				₹ 78,26,087

(ii) Computation of PV ratio, contribution, profit and break-even sale for proposed product mix

	Products			
	S	**T**	**M**	**Total**
Selling Price (₹)	600	800	600	
Less: Variable Cost (₹)	300	400	300	
Contribution per unit (₹)	300	400	300	
P/V Ratio (Contribution/Selling price)	50%	50%	50%	
Sales Mix	40%	35%	25%	
Contribution per rupee of sales (P/V Ratio x Sales Mix)	20%	17.5%	12.5%	50%
Proposed Total Contribution (₹ 1,28,00,000 × 50%)				₹ 64,00,000
Less: Fixed Costs				₹ 36,00,000
Proposed Profit				₹ 28,00,000
Proposed Break- Even Sales (₹ 36,00,000/0.50)				₹ 72,00,000

ICAI · Old P3 · RTP May 2022 · question p. 9 · answer p. 29

1055 RTP **Old P3 Jul 2021** Q11 old scheme

Aditya Limited manufactures three different products and the following information has been collected

from the books of accounts:

	Products		
	S	T	U
Sales Mix	35%	35%	30%
Selling Price	₹ 300	₹ 400	₹ 200
Variable Cost	₹ 150	₹ 200	₹ 120
Total Fixed Costs			₹ 18,00,000
Total Sales			₹ 60,00,000

The company has currently under discussion, a proposal to discontinue the manufacture of Product U and replace it with Product M, when the following results are anticipated:

	Products		
	S	T	M
Sales Mix	50%	25%	25%
Selling Price	₹ 300	₹ 400	₹ 300
Variable Cost	₹ 150	₹ 200	₹ 150
Total Fixed Costs			₹ 18,00,000
Total Sales			₹ 64,00,000

Required

(i) COMPUTE the PV ratio, total contribution, profit and Break-even sales for the existing product mix.

(ii) COMPUTE the PV ratio, total contribution, profit and Break-even sales for the proposed product mix.

ICAI'S ANSWER

(i) Computation of PV ratio, contribution and break-even sales for existing product mix

	Products			Total
	S	T	U	
Selling Price (₹)	300	400	200	
Less: Variable Cost (₹)	150	200	120	
Contribution per unit (₹)	150	200	80	
P/V Ratio (Contribution/Selling price)	50%	50%	40%	
Sales Mix	35%	35%	30%	
Contribution per rupee of sales (P/V Ratio × Sales Mix)	17.5%	17.5%	12%	47%

	Products	Total
Present Total Contribution (₹60,00,000 × 47%)		₹ 28,20,000
Less: Fixed Costs		₹ 18,00,000
Present Profit		₹ 10,20,000
Present Break Even Sales (₹18,00,000/0.47)		₹ 38,29,787

(ii) Computation of PV ratio, contribution and break-even sale for proposed product mix

	Products			
	S	**T**	**M**	**Total**
Selling Price (₹)	300	400	300	
Less: Variable Cost (₹)	150	200	150	
Contribution per unit (₹)	150	200	150	
P/V Ratio (Contribution/Selling price)	50%	50%	50%	
Sales Mix	50%	25%	25%	
Contribution per rupee of sales (P/V Ratio x Sales Mix)	25%	12.5%	12.5%	50%
Proposed Total Contribution (₹64,00,000 × 50%)				₹ 32,00,000
Less: Fixed Costs				₹ 18,00,000
Proposed Profit				₹ 14,00,000
Proposed Break Even Sales (₹18,00,000/0.50)				₹ 36,00,000

ICAI · Old P3 · RTP Jul 2021 · question p. 9-10 · answer p. 28-29

1056 MTP **Old P3 Dec 2021 S-II** Q4(b) 10 marks old scheme

Same question, different ICAI answer: #1054 Old P3 RTP May 2022

A Limited manufactures three different products and the following information has been collected from

the books of accounts:

	Products		
	S	T	U
Sales Mix	25%	35%	40%
Selling Price	₹ 600	₹ 800	₹ 400
Variable Cost	₹ 300	₹ 400	₹ 240
Total Fixed Costs			₹ 36,00,000
Total Sales			₹ 1,20,00,000

The company has currently under discussion, a proposal to discontinue the manufacture of Product U and replace it with Product M, when the following results are anticipated:

	Products		
	S	T	M
Sales Mix	40%	35%	25%
Selling Price	₹ 600	₹800	₹600
Variable Cost	₹ 300	₹400	₹300
Total Fixed Costs			₹ 36,00,000
Total Sales			₹1,28,00,000

Required

(i) Compute the PV ratio, total contribution, profit and Break-even sales for the existing product mix.

(ii) Compute the PV ratio, total contribution, profit and Break-even sales for the proposed product mix.

ICAI'S ANSWER

(i) Computation of PV ratio, contribution and break-even sales for existing product mix

	Products			Total
	S	T	U	
Selling Price (₹)	600	800	400	
Less: Variable Cost (₹)	300	400	240	
Contribution per unit (₹)	300	400	160	
P/V Ratio (Contribution/Selling price)	50%	50%	40%	
Sales Mix	25%	35%	40%	
Contribution per rupee of sales (P/V Ratio × Sales Mix)	12.5%	17.5%	16%	46%

Present Total Contribution (₹ 1,20,00,000 × 46%)	₹55,20,000
Less: Fixed Costs	₹36,00,000
Present Profit	₹19,20,000
Present Break Even Sales (₹ 36,00,000/0.46)	₹ 78,26,087

(ii) Computation of PV ratio, contribution and break-even sale for proposed product mix

	Products			
	S	**T**	**M**	**Total**
Selling Price (₹)	600	800	600	
Less: Variable Cost (₹)	300	400	300	
Contribution per unit (₹)	300	400	300	
P/V Ratio (Contribution/Selling price)	50%	50%	50%	
Sales Mix	40%	35%	25%	
Contribution per rupee of sales (P/V Ratio x Sales Mix)	20%	17.5%	12.5%	50%
Proposed Total Contribution (₹1,28,00,000 × 50%)				₹64,00,000
Less: Fixed Costs				₹36,00,000
Proposed Profit				₹28,00,000
Proposed Break Even Sales (₹36,00,000/0.50)				₹72,00,000

ICAI · Old P3 · MTP Dec 2021 · Series II · question p. 7 · answer p. 13-14

CHAPTER 15

Budgets and Budgetary Control

15.1 Budgets, budgetary control, the budget manual and key factor

12 questions · asked in 14 ICAI documents · 3 exam papers

1057 EXAM Sep 2024 Q6(b) 5 marks

Define the following terms:

(i) Controllable Variance

(ii) Uncontrollable Variance

(iii) Budget Manual

(iii) Performance Budgeting

(iv) Budget Period

ICAI'S ANSWER

(i) Controllable variances: Controllable variances are those which can be controlled under the normal operating conditions if a responsibility centre takes preventive measures and acts prudently. Responsibility centres are answerable for all adverse variances which could have been controlled.

(ii) Uncontrollable variances: Uncontrollable variances are those which occurs due to conditions which are beyond the control of a responsibility centre and cannot be controlled even though all preventive measures are in place.

(iii) Budget Manual: The budget manual is a booklet specifying the objectives of an organisation in relation to its strategy. The budget is made to decide how much an organisation would earn and spend and in what manner. In the budget, the organisation sets its priorities too.

Budget manual is also defined as, "A document which sets out the responsibilities of the persons engaged in, the routine of, and the forms and records required for, budgetary control."

Effective budgetary planning relies on the provision of adequate information to the individuals involved in the planning process. Many of these information needs are contained in the budget manual. A budget manual is a collection of documents that contains key information for those involved in the planning process.

(iv) Performance budgeting: Performance budgeting (PB) involves evaluation of the performance of an organisation in the context of both specific as well as overall objectives of the organisation.

Performance Budgeting provide a meaningful relationship between estimated inputs and expected outputs as an integral part of the budgeting system. A performance budget is one which presents the purposes and objectives for which funds are required, the costs of the programmes proposed for achieving those objectives, and quantitative data measuring the accomplishments and work performed under each programme. Thus, PB is a technique of presenting budgets for costs and revenues in terms of functions. Programmes and activities correlate the physical and financial aspect of the individual items comprising the budget.

(v) Budget period: The period covered by a budget is known as budget period. There is no general rule governing the selection of the budget period. In practice the Budget Committee determines the length of the budget period suitable for the business. Normally, a calendar year or a period co-terminus with the financial year is adopted. The budget period for the calendar or financial year is

then divided into shorter periods; it may be monthly or quarterly or for such periods as coincide with period of trading activity of the business.

ICAI · P4 · Sep 2024 exam · question p. 36 · answer p. 37-38

1058 EXAM **Old P3 Nov 2022** Q6(d) 5 marks old scheme

Define Budget Manual. What are the salient features of Budget Manual?

ICAI'S ANSWER

Budget Manual: The budget manual is a booklet specifying the objectives of an organisation in relation to its strategy. The budget is made to decide how much an organisation would earn and spend and in what manner. In the budget, the organisation sets its priorities too.

Effective budgetary planning relies on the provision of adequate information to the individuals involved in the planning process. Many of these information needs are contained in the budget manual. A budget manual is a collection of documents that contains key information for those involved in the planning process.

CIMA London defines budget manual as, 'A document which sets out the responsibilities of the persons engaged in, the routines of, and the forms and records required for, budgetary control'.

Contents of a budget manual: Typical budget manual may include the following:

(i) A statement regarding the objectives of the organisation and how they can be achieved through budgetary control;

(ii) A statement about the functions and responsibilities of each executive, both regarding preparation and execution of budgets;

(iii) Procedures to be followed for obtaining the necessary approval of budgets. The authority of granting approval should be stated in explicit terms. Whether, one two or more signatures are required on each document should be clearly stated;

(iv) A form of organisation chart to show who are responsible for the preparation of each functional budget and the way in which the budgets are interrelated.

(v) A timetable for the preparation of each budget.

(vi) The manner of scrutiny and the personnel to carry it out;

(vii) Reports, statements, forms and other record to be maintained.

(viii) The accounts classification to be employed. It is necessary that the framework within which the costs, revenue and other financial accounts are classified must be identical both in the accounts and budget department.

(ix) The reporting of the remedial action.

(x) The manner in which budgets, after acceptance and issuance, are to be revised or amended, these are included in budgets and on which action can be taken only with the approval of top management

(xi) This will prevent the formation of a 'bottleneck' with the late preparation of one budget holding up the preparation of all others.

(xii) Copies of all forms to be completed by those responsible for preparing budgets, with explanations concerning their completion.

(xiii) A list of the organization's account codes, with full explanations of how to use them.

(xiv) Information concerning key assumptions to be made by managers in their budgets, for example the rate of inflation, key exchange rates, etc.

(Any four points)

ICAI · Old P3 · Nov 2022 exam · question p. 23 · answer p. 24-25

1059 EXAM **Old P3 Jan 2021** Q6(b) 5 marks old scheme

State the limitations of Budgetary Control System.

ICAI'S ANSWER

Limitations of Budgetary Control System

Points	Description
1. Based on Estimates	Budgets are based on a series of estimates, which are based on the conditions prevalent or expected at the time budget is established. It requires revision in plan if conditions change.
2. Time factor	Budgets cannot be executed automatically. Some preliminary steps are required to be accomplished before budgets are implemented. It requires proper attention and time of management. Management must not expect too much during the initial development period.
3. Co-operation Required	Staff co-operation is usually not available during the initial budgetary control exercise. In a decentralised organisation, each unit has its own objective and these units enjoy some degree of discretion. In this type of organisation structure, coordination among different units is required. The success of the budgetary control depends upon willing co-operation and teamwork,
4. Expensive	The implementation of budget is somewhat expensive. For successful implementation of the budgetary control, proper organisation structure with responsibility is prerequisite. Budgeting process start from the collection of information to for preparing the budget and performance analysis. It consumes valuable resources (in terms of qualified manpower, equipment, etc.) for this purpose; hence, it is an expensive process.
5. Not a substitute for management	Budget is only a managerial tool and must be intelligently applied for management to get benefited. Budgets are not a substitute for good management.
6. Rigid document	Budgets are sometime considered as rigid documents. But in reality, an organisation is exposed to various uncertain internal and external factors. Budget should be flexible enough to incorporate ongoing developments in the internal and external factors affecting the very purpose of the budget.

ICAI · Old P3 · Jan 2021 exam · question p. 25 · answer p. 26-27

1060 RTP **Sep 2026** Q16(c)

Describe the steps involved in the budgetary control technique.

ICAI'S ANSWER

There are certain steps involved in the budgetary control technique. They are as follows:

(i) Definition of objectives: A budget being a plan for the achievement of certain operational objectives, it is desirable that the same are defined precisely. The objectives should be written out; the areas of control demarcated; and items of revenue and expenditure to be covered by the budget stated.

(ii) Location of the key (or budget) factor: There is usually one factor (sometimes there may be more than one) which sets a limit to the total activity. Such a factor is known as key factor. For proper budgeting, it must be located and estimated properly.

(iii) Appointment of controller: Formulation of a budget usually required whole time services of a senior executive known as budget controller; he must be assisted in this work by a Budget Committee, consisting of all the heads of department along with the Managing Director as the Chairman.

(iv) Budget Manual: Effective budgetary planning relies on the provision of adequate information which are contained in the budget manual. A budget manual is a collection of documents that contains key information for those involved in the planning process.

(v) Budget period: The period covered by a budget is known as budget period. The Budget Committee determines the length of the budget period suitable for the business. It may be months or quarters or such periods as coincide with period of trading activity.

(vi) Standard of activity or output: For preparing budgets for the future, past statistics cannot be completely relied upon, for the past usually represents a combination of good and bad factors. Therefore, though results of the past should be studied but these should only be applied when there is a likelihood of similar conditions repeating in the future.

ICAI · P4 · RTP Sep 2026 · question p. 19 · answer p. 48-49

1061 RTP **Jan 2026** Q19(c)

Also asked in MTP May 2025 S-I Q6(c)

Besides having advantages of Budgetary Control System being a powerful instrument used by business entity for the control of their expenditure, it has certain limitations as well. ELABORATE any four limitations of Budgetary Control System.

ICAI'S ANSWER

Limitations of Budgetary Control System

Points	Description
1. Based on Estimates	Budgets are based on a series of estimates, which are based on the conditions prevalent or expected at the time budget is established. It requires revision in plan if conditions change.
2. Time factor	Budgets cannot be executed automatically. Some preliminary steps are required to be accomplished before budgets are implemented. It requires proper attention and time of management. Management must not expect too much during the initial development period.

Points	Description
3. Co-operation Required	Staff co-operation is usually not available during the initial budgetary control exercise. In a decentralised organisation, each unit has its own objective and these units enjoy some degree of discretion. In this type of organisation structure, coordination among different units is required. The success of the budgetary control depends upon willing co-operation and teamwork.
4. Expensive	The implementation of budget is somewhat expensive. For successful implementation of the budgetary control, proper organisation structure with responsibility is prerequisite. Budgeting process start from the collection of information to for preparing the budget and performance analysis. It consumes valuable resources (in terms of qualified manpower, equipment, etc.) for this purpose; hence, it is an expensive process.
5. Not a substitute for management	Budget is only a managerial tool and must be intelligently applied for management to get benefited. Budgets are not a substitute for good management.
6. Rigid document	Budgets are sometime considered as rigid documents. But in reality, an organisation is exposed to various uncertain internal and external factors. Budget should be flexible enough to incorporate ongoing developments in the internal and external factors affecting the very purpose of the budget.

ICAI · P4 · RTP Jan 2026 · question p. 23 · answer p. 57-58

1062 RTP **Old P3 Jul 2021** Q13(d) old scheme

Also asked in Old P3 MTP Dec 2021 S-II Q6(d)

DESCRIBE the salient features of budget manual.

ICAI'S ANSWER

Salient features of Budget Manual

- Budget manual contains much information which is required for effective budgetary planning.
- A budget manual is a collection of documents that contains key information for those involved in the planning process.
- An introductory explanation of the budgetary planning and control process, including a statement of the budgetary objective and desired results is included in Budget Manual.
- Budget Manual contains a form of organisation chart to show who is responsible for the preparation of each functional budget and the way in which the budgets are interrelated.
- In contains a timetable for the preparation of each budget.
- Copies of all forms to be completed by those responsible for preparing budgets, with explanations concerning their completion is included in Budget Manual.

ICAI · Old P3 · RTP Jul 2021 · question p. 11 · answer p. 33

1063 MTP **Sep 2026 S-II** Q6(c) 4 marks

Same question, different ICAI answer: #1064 MTP May 2025 S-II

ABC Ltd., a mid-sized manufacturing company, has been facing frequent cost overruns and profit fluctuations over the past year. The management is considering implementing a Budgetary Control

System to improve financial planning and operational efficiency.

As a financial consultant, you have been asked to IDENTIFY any EXPLAIN at least four key objectives of a Budgetary Control System that would benefit ABC Ltd.

ICAI'S ANSWER

Objectives of Budgetary Control System

1. **Portraying with precision the overall aims of the business** and determining targets of performance for each section or department of the business.
2. **Laying down the responsibilities** of each of the executives and other personnel so that everyone knows what is expected of him and how he will be judged. Budgetary control is one of the few ways in which an objective assessment of executives or department is possible.
3. **Providing a basis for the comparison** of actual performance with the predetermined targets and investigation of deviation, if any, of actual performance and expenses from the budgeted figures. This naturally helps in adopting corrective measures.
4. **Ensuring optimum use of available resources** to maximise profit or production, subject to the limiting facto₹ Since budgets cannot be properly drawn up without considering all aspects, usually there is good co-ordination when a system of budgetary control operates.
5. **Co-ordinating various activities** of the business, and centralising control and yet enabling management to decentralise responsibility and delegate authority in the overall interest of the business.
6. **Engendering a spirit of careful forethought**, assessment of what is possible and an attempt at it. It leads to dynamism without being reckless. Of course, much depends on the objectives of the firm and the dynamism r of its management.
7. **Providing a basis for revision** of current and future policies.
8. **Drawing up long range plans** with a fair measure of accuracy.
9. **Providing a yardstick** against which actual results can be compared.

*** Any Four Points**

ICAI · P4 · MTP Sep 2026 · Series II · question p. 14 · answer p. 24

1064 MTP **May 2025 S-II** Q6(c) 4 marks

Same question, different ICAI answer: #1063 MTP Sep 2026 S-II

ABC Ltd., a mid-sized manufacturing company, has been facing frequent cost overruns and profit fluctuations over the past year. The management is considering implementing a Budgetary Control System to improve financial planning and operational efficiency.

As a financial consultant, you have been asked to IDENTIFY any EXPLAIN at least four key objectives of a Budgetary Control System that would benefit ABC Ltd.

ICAI'S ANSWER

Objectives of Budgetary Control System

1. **Portraying with precision the overall aims of the business** and determining targets of performance for each section or department of the business.
2. **Laying down the responsibilities** of each of the executives and other personnel so that everyone knows what is expected of him and how he will be judged. Budgetary control is one of the few ways in which an objective assessment of executives or department is possible.

3. **Providing a basis for the comparison** of actual performance with the predetermined targets and investigation of deviation, if any, of actual performance and expenses from the budgeted figures. This naturally helps in adopting corrective measures.
4. **Ensuring optimum use of available resources** to maximise profit or production, subject to the limiting factors. Since budgets cannot be properly drawn up without considering all aspects, usually there is good co-ordination when a system of budgetary control operates.
5. **Co-ordinating various activities** of the business, and centralising control and yet enabling management to decentralise responsibility and delegate authority in the overall interest of the business.
6. **Engendering a spirit of careful forethought**, assessment of what is possible and an attempt at it. It leads to dynamism without being reckless. Of course, much depends on the objectives of the firm and the dynamism r of its management.
7. **Providing a basis for revision** of current and future policies.
8. **Drawing up long range plans** with a fair measure of accuracy.
9. **Providing a yardstick** against which actual results can be compared.

ICAI · P4 · MTP May 2025 · Series II · question p. 15 · answer p. 20

1065 **MTP** **Jan 2025 S-I** Q6(c) **4 marks**

Despite the many benefits of Budgetary Control System, it does have its own limitations. DISCUSS those limitations.

ICAI'S ANSWER

	Points	Description
1.	Based on Estimates	Budgets are based on a series of estimates, which are based on the conditions prevalent or expected at the time budget is established. It requires revision in plan if conditions change.
2.	Time factor	Budgets cannot be executed automatically. Some preliminary steps are required to be accomplished before budgets are implemented. It requires proper attention and time of management. Management must not expect too much during the initial development period.
3.	Co-operation Required	Staff co-operation is usually not available during the initial budgetary control exercise. In a decentralised organisation, each unit has its own objective and these units enjoy some degree of discretion. In this type of organisation structure, coordination among different units is required. The success of the budgetary control depends upon willing co-operation and teamwork,
4.	Expensive	The implementation of budget is somewhat expensive. For successful implementation of the budgetary control, proper organisation structure with responsibility is prerequisite. Budgeting process start from the collection of information to for preparing the budget and performance analysis. It consumes valuable resources (in terms of qualified manpower, equipment, etc.) for this purpose; hence, it is an expensive process.
5.	Not a substitute for management	Budget is only a managerial tool and must be intelligently applied for management to get benefited. Budgets are not a substitute for good management.

	Points	Description
6.	Rigid document	Budgets are sometime considered as rigid documents. But in reality, an organisation is exposed to various uncertain internal and external factors. Budget should be flexible enough to incorporate ongoing developments in the internal and external factors affecting the very purpose of the budget.

ICAI · P4 · MTP Jan 2025 · Series I · question p. 17 · answer p. 21-22

1066 MTP **Jan 2025 S-II** Q6(b) 5 marks

As a consultant, a client has approached you to set up a budgetary control system in their organization. WHAT sequential steps would you follow to design, implement, and monitor the system?

ICAI'S ANSWER

The following steps are necessary for establishing a good budgetary control system:

1. Determining the objectives to be achieved, over the budget period, and the policy or policies that might be adopted for the achievement of these objectives.
2. Determining the activities that should be undertaken for the achievement of the objectives.
3. Drawing up a plan or a scheme of operation in respect of each class of activity, in quantitative as well as monetary terms for the budget period.
4. Laying out a system of comparison of actual performance by each person, or department with the relevant budget and determination of causes for the variation, if any.
5. Ensuring that corrective action will be taken where the plan has not been achieved and, if that is not possible, for the revision of the plan.

ICAI · P4 · MTP Jan 2025 · Series II · question p. 13 · answer p. 19

1067 MTP **Old P3 Jul 2021 S-II** Q6(d) 5 marks old scheme

DESCRIBE the steps necessary for establishing a good budgetary control system.

ICAI'S ANSWER

The following steps are necessary for establishing a good budgetary control system:

1. Determining the objectives to be achieved, over the budget period, and the policy or policies that might be adopted for the achievement of these objectives.
2. Determining the activities that should be undertaken for the achievement of the objectives.
3. Drawing up a plan or a scheme of operation in respect of each class of activity, in quantitative as well as monetary terms for the budget period.
4. Laying out a system of comparison of actual performance by each person, or department with the relevant budget and determination of causes for the variation, if any.
5. Ensuring that corrective action will be taken where the plan has not been achieved and, if that is not possible, for the revision of the plan.

ICAI · Old P3 · MTP Jul 2021 · Series II · question p. 7 · answer p. 12

1068 MTP **Old P3 Nov 2020 S-I** Q6(e) 5 marks old scheme

DESCRIBE objectives of Budgetary Control System.

ICAI'S ANSWER

Objectives of Budgetary Control System

1. **Portraying with precision the overall aims of the business** and determining targets of performance for each section or department of the business.
2. **Laying down the responsibilities** of each of the executives and other personnel so that everyone knows what is expected of him and how he will be judged. Budgetary control is one of the few ways in which an objective assessment of executives or department is possible.
3. **Providing a basis for the comparison** of actual performance with the predetermined targets and investigation of deviation, if any, of actual performance and expenses from the budgeted figures. This naturally helps in adopting corrective measures.
4. **Ensuring the best use of all available resources** to maximise profit or production, subject to the limiting factors. Since budgets cannot be properly drawn up without considering all aspects usually there is good co-ordination when a system of budgetary control operates.
5. **Co-ordinating the various activities** of the business, and centralising control and yet enabling management to decentralise responsibility and delegate authority in the overall interest of the business.
6. **Engendering a spirit of careful forethought**, assessment of what is possible and an attempt at it. It leads to dynamism without recklessness. Of course, much depends on the objectives of the firm and the vigour of its management.
7. **Providing a basis for revision** of current and future policies.
8. **Drawing up long range plans** with a fair measure of accuracy.
9. **Providing a yardstick** against which actual results can be compared.

ICAI · Old P3 · MTP Nov 2020 · Series I · question p. 8 · answer p. 15-16

15.2 Functional budgets and the master budget

71 questions · asked in 37 ICAI documents · 9 exam papers

1069 EXAM **Sep 2026** Case Scenario – II · MCQ 6 to MCQ 10 10 marks

CASE SCENARIO – II

ABCD Ltd. manufactures two types of products - WM for women and KD for children. These are sold in containers of one kg each. Raw materials W and M are used for product WM and raw materials K and D are used for product KD. There is no process loss or scrap unit. During the second quarter (July to September 2026), the company expects a market demand of 262 containers of WM and 190 containers of KD. The following additional information is given :

Raw Material	Product WM % of Raw Material	Product KD % of Raw Material	Cost per kg (₹)
W	75%	–	225
M	25%	–	440
K	–	40%	315
D	–	60%	380

Stock	Actual Quantity as on 01-07-2026	Budgeted Quantity as on 30-09-2026
W (kgs)	22	16
M (kgs)	07	13
K (kgs)	24	21
D (kgs)	34	32
WM (containers)	20	18
KD (containers)	38	28

On the basis of the above Case Scenario, you are required to answer the following MCQs : **(6 to 10)**

MCQ 6 2 marks

How many containers of KD will be produced in the quarter according to the production budget ?

A 190

B 218

C 180

D 200

MCQ 7 2 marks

How many kilograms of raw material M will be consumed as per raw material budget ?

A 195 kgs

B 65 kgs

C 65.5 kgs

D 78 kgs

MCQ 8 2 marks

How many kilograms of raw material K will be purchased as per raw material budget ?

A 64 kgs

B 69 kgs

C 61 kgs

D 93 kgs

MCQ 9 2 marks

What is the value of budgeted raw materials purchase (₹) during the quarter for WM ?

A 65,112.50

B 72,807.50

C 72,475

D 73,765

MCQ 10 2 marks

How many kilograms of WM will be the budgeted production in the quarter ?

A 262 kgs

B 260 kgs

C 280 kgs

D 244 kgs

ICAI'S ANSWERS

MCQ 6

Answer key: **ICAI has not published the key yet.**

MCQ 7

Answer key: **ICAI has not published the key yet.**

MCQ 8

Answer key: **ICAI has not published the key yet.**

MCQ 9

Answer key: **ICAI has not published the key yet.**

MCQ 10

Answer key: **ICAI has not published the key yet.**

ICAI · P4 · Sep 2026 exam · question p. 5-6

1070 **EXAM** **Sep 2026** Q2(a) **7 marks**

Vivida Ltd. manufactures two products – Ace eyeglasses & Sharp eyeglasses using various types of materials and one grade of labour – general optical technicians who cut lenses, drill frames and align hinges. Shown below is an extract from the company's working papers for the next month's budget :

	Ace eyeglasses	Sharp eyeglasses
Budgeted sales (in units)	1,100	2,200
Standard labour hours allowed per unit of product at full productivity	3	4.5

Labour is paid at ₹ 30 per hour. Overtime premium is 60% and is payable only if a worker works for more than 45 hours a week. There are 110 direct workers. Assume that they can work on either product equally efficiently. Efficiency during normal working hours may be taken to be the same as in overtime hours also.

The target productivity for the productive hours worked by the direct workers in actually manufacturing the products is 75%. In addition, the non-productive down-time is budgeted at 15% of the productive hours worked.

There are four 5-day weeks in the budgeted period and it is anticipated that sales and production will occur evenly throughout the whole period. The following information is given :

	Ace eyeglasses	Sharp eyeglasses
Opening Stock (units)	125	140
Closing Stock	5 days' sales	4 days' sales

Calculate the Wages Budget for the direct workers, showing the quantities and values for the next month.

ICAI'S ANSWER

ICAI has not published its answers to this paper yet.

1071 **EXAM** **May 2026** **Q4(a)** **9 marks**

ZED Ltd. manufactures two categories of products, Omega and Theta. These products are sold through two primary distribution channels, Online Portals and Retail Stores. In the current financial year, Omega performed exceptionally well online due to a viral trend, while Theta saw a surge in retail stores. The following information is available for the year 2025-26:

Product	Channel	Budgeted Sales	Actual Sales	Selling Price per unit (For Budgeted and Actual Sales)
Omega	Online	1000 units	1200 units	₹ 15,000
	Retail	800 units	700 units	₹ 18,000
Theta	Online	500 units	450 units	₹40,000
	Retail	400 units	500 units	₹ 45,000

Market research indicates that Omega series is facing stiff competition offline, therefore, management has decided to reduce the price by 10% across both channels. Theta series has gained luxury status, so the management has decided to increase the price by ₹ 5,000 per unit for both Online and Retail sales. Based on these price changes, the Sales Manager has submitted the following estimates of increase in sales for the year 2026-27 over the original budgeted sales:

Product	Online Channel	Retail Channel
Omega	+ 15%	+5%
Theta	+ 10%	+20%

Furthermore, the company has signed a contract for an Influencer Marketing Campaign. It is estimated that for the year 2026-27 this campaign will generate the following additional sales:

Product	Online channel	Retail Channel
Omega	200 units	Nil
Theta	100 units	50 Units

You are required to prepare:

(i) Sales Budget for 2026-27 after incorporating the above estimates.

(ii) Statement showing Budgeted and Actual Sales for 2025-26.

ICAI'S ANSWER

(i) Statement Showing Sales Budget for 2026-27

Channel	Omega			Theta			Total
	Qty.	Rate (₹)	Amt. (₹)	Qty.	Rate (₹)	Amt. (₹)	Amt. (₹)
Online	1,350[1]	13,500	1,82,25,000	650[3]	45,000	2,92,50,000	4,74,75,000
Retail	840[2]	16,200	1,36,08,000	530[4]	50,000	2,65,00,000	4,01,08,000
Total	2,190		3,18,33,000	1,180		5,57,50,000	8,75,83,000

Workings

1. 1,000 × 115% + 200 = 1,350 units
2. 800 × 105% + Nil = 840 units
3. 500 × 110% + 100 = 650 units
4. 400 × 120% + 50 = 530 units

(ii) Statement Showing Budgeted Sales for 2025-26

Channel	Omega			Theta			Total
	Qty.	Rate (₹)	Amt. (₹)	Qty.	Rate (₹)	Amt. (₹)	Amt. (₹)
Online	1,000	15,000	1,50,00,000	500	40,000	2,00,00,000	3,50,00,000
Retail	800	18,000	1,44,00,000	400	45,000	1,80,00,000	3,24,00,000
Total	1,800		2,94,00,000	900		3,80,00,000	6,74,00,000

Statement Showing Actual Sales for 2025-26

Channel	Omega			Theta			Total
	Qty.	Rate (₹)	Amt. (₹)	Qty.	Rate (₹)	Amt. (₹)	Amt. (₹)
Online	1,200	15,000	1,80,00,000	450	40,000	1,80,00,000	3,60,00,000
Retail	700	18,000	1,26,00,000	500	45,000	2,25,00,000	3,51,00,000
Total	1,900		3,06,00,000	950		4,05,00,000	7,11,00,000

ICAI · P4 · May 2026 exam · question p. 20-21 · answer p. 21-22

1072 EXAM **Jan 2026** MCQ 13 2 marks MCQ

A garment factory stitches each shirt through a single hand sewing machine operator. The sewing time required to stitch each shirt is 15 minutes. Operator is paid at ₹ 65 per hour. The factory works for 40 hours per week and the production target is 480 dozens of shirts per week.

What is the number of hours and the number of operators required to meet the shirts target?

A 1,800 Hours & 45 Operators

B 1,440 Hours & 36 Operators

C 1,800 Hours & 36 Operators

D 1,440 Hours & 40 Operators

ICAI'S ANSWER

Answer key: **(B)**

ICAI · P4 · Jan 2026 exam · question p. 6 · answer p. 7

1073 **EXAM** **Jan 2026** MCQ 15 **2 marks** **MCQ**

The following information has been given regarding the two machines of a manufacturing department of X Ltd. for the month of September 2025.

Particulars	Total cost	Machine A	Machine B
Spare parts cost (in ₹)	1,00,000	40,000	60,000

Further estimates for the month of October 2025:

(i) There is an increase of 15% in the prices of spare parts of both machines.

(ii) There is an increase of 25% in the consumption of spare parts for machine B only.

What is the total spare parts cost for the month of October 2025?

A ₹ 1,25,000

B ₹ 1,43,750

C ₹ 1,26,500

D ₹ 1,32,250

ICAI'S ANSWER

Answer key: **(D)**

ICAI · P4 · Jan 2026 exam · question p. 6-7 · answer p. 7

1074 **EXAM** **Jan 2026** Q6(c) **4 marks**

Discuss the four components of Budgetary Control System.

ICAI'S ANSWER

These functional budgets are broadly grouped under the following heads:

1. **Physical budgets:** Those budgets which contain information in quantitative terms such as the physical units of sales, production etc. This may include quantity of sales, quantity of production, inventories, and manpower budgets are physical budgets.
2. **Cost budgets:** Budgets which provides cost information in respect of manufacturing, administration, selling and distribution, etc. for example, manufacturing costs, selling costs, administration cost, and research and development cost budgets are cost budgets.
3. **Profit budgets:** A budget which enables the ascertainment of profit. For example, sales budget, profit and loss budget, etc.
4. **Financial budgets:** A budget which facilitates in ascertaining the financial position of a concern, for example, cash budgets, capital expenditure budget, budgeted balance sheet etc.

ICAI · P4 · Jan 2026 exam · question p. 28 · answer p. 30

1075 EXAM Sep 2025 Q4(a) 9 marks

SVS Limited manufactures a single product 'Al'. The company has estimated its quarter-wise sales for the next year as follows:

Quarter	I	II	III	IV
Sales (Units)	72,000	90,000	99,000	1,08,000

In the beginning of the year, the opening stock of finished goods is 14,400 units and the company expects to maintain the closing stock of finished goods at 29,400 units at the end of the year. The production pattern in each quarter is based on 80% of the sales of the current quarter and 20% of the sales of the next quarter. The company maintains this 20% of sales of next quarter as closing stock of current quarter.

The opening stock of raw materials in the beginning of the year is 24,000 kgs and the closing stock at the end of the year is required to be maintained at 12,000 kgs. Each unit of finished output requires 2 kgs of raw material. The production time required to produce one unit of product 'A1' is 5 hours. During the production, the product uses two machines as under:

Product	Machine A	Machine B	Total
A1	2 hours	3 hours	5 hours

Machine A requires 100 hours of maintenance after a use of 5000 hours and Machine B requires 100 hours of maintenance after use of 3000 hours.

Required:

(i) Prepare quarter-wise Production Budget (in units) and the Raw Material Consumption budget (in quantity) for the next year.

(ii) Calculate total machine hours including maintenance time required during the year for Machine A and Machine B to manufacture product 'A1'.

ICAI'S ANSWER

(i) **Production Budget (in units)**

Quarters	I Units	II Units	III Units	IV Units	Total Units
Sales	72,000	90,000	99,000	1,08,000	3,69,000
Production in current quarter	57,600	72,000	79,200	86,400	
(80% of the sale of current quarter)					
Production for next quarter	18,000	19,800	21,600	29,400	
(20% of the sale of next quarter)					
Total production	75,600	91,800	1,00,800	1,15,800	3,84,000

Raw material consumption budget in quantity

Quarters	I	II	III	IV	Total
Units to be produced in each quarter: (A)	75,600	91,800	1,00,800	1,15,800	3,84,000
Raw material consumption p.u. (kg.): (B)	2	2	2	2	
Total raw material consumption (Kg.): (A × B)	1,51,200	1,83,600	2,01,600	2,31,600	7,68,000

(ii)

	Machine A	Machine B
Units to be produced	3,84,000	3,84,000
Machine hours per unit	2	3
Total Hours	7,68,000	11,52,000
Maintenance Hours	15,360	38,400
Total machine hours including maintenance	**7,83,360**	**11,90,400**

ICAI · P4 · Sep 2025 exam · question p. 20-21 · answer p. 21-22

1076 **EXAM** **May 2025** Case Scenario - II · MCQ 6 to MCQ 10

CASE SCENARIO - II

Allure Metallurgy Ltd. is a stainless-steel manufacturing company which manufactures two grades of stainless steel products namely SS304 & SS316 made of a common raw material iron procured at ₹ 52 per kg from the market. The usage of the raw material is expected to be at a constant rate over the entire period. The raw material supplier to the company charges ₹ 24,000 per order but its delivery is limited to 1200 tons per annum. There is no alternate source to procure the raw material. In consideration of the above limitations, the company decided to review its inventory management policies for the forthcoming year.

The following forecasted information has been extracted from departmental estimates for the budget year ending on 31 March 2025:

	SS304	SS316
Sales (units)	56,000	86,000
Finished Goods stock increase by year end (units)	1,614	1,215
Post Production rejection rate (%)	3	7
Iron usage in kg (per completed unit, net of wastage)	5.5	8
Iron wastage (%)	8	11

You are required to calculate the following (MCQ's 6 to 10):

MCQ 6

The minimum number of units of SS304 & SS316, the company shall produce to justify the sales forecast would be:

- A 56,000 & 86,000
- B 57,614 & 87,215
- C 59,396 & 93,780
- D 64,561 & 1,05,371

MCQ 7

The ratio in which the raw material utilized for SS304 & SS316 from the total quantity of raw material procured, to produce the number of units desired in Q-6 above?

- A 29.59% & 70.24%
- B 29.64% & 70.36%
- C 30.33% & 69.67%
- D 38.77% & 61.23%

MCQ 8

Assuming that all the available 1200 tons of raw material is procured per annum and would be utilized for production, what would be the raw material needed for production of SS 304 in order to maintain the same production mix arrived in Q-7 above?

- A 3,26,678 kg
- B 3,27,209 kg
- C 3,55,085 kg
- D 3,55,663 kg

MCQ 9

Assuming that all the available 1200 tons of raw material is procured per annum and would be utilized for production, what would be the raw material needed for production of SS 316 in order to maintain the same production mix arrived in Q-7 above?

- A 7,50,240 kg
- B 7,51,460 kg
- C 8,42,966 kg
- D 8,44,337 kg

MCQ 10

Keeping the management purchase policy & production quantity mix in consideration for SS304 & SS316, the maximum number of units of each product that company would produce (in units) respectively by utilizing 1200 tons of raw material:

- A 59,396 & 93,780
- B 59,493 & 93,933
- C 64,561 & 1,05,371
- D 64,666 & 1,05,542

ICAI'S ANSWERS

MCQ 6

Answer key: **(C)**

MCQ 7

Answer key: **(B)**

MCQ 8

Answer key: **(D)**

MCQ 9

Answer key: **(D)**

MCQ 10

Answer key: **(B)**

ICAI · P4 · May 2025 exam · question p. 2-3 · answer p. 6

1077 **EXAM** **May 2025** MCQ 11 MCQ

Healthy & Fit Ltd. manufactures & sells a single product captioned as 'Exercise bikes'. The estimated units to be sold in the last quarter of the year 2024-25 are as under:

Particulars	January 2025	February 2025	March 2025
Exercise bikes (in units)	1,500	1,800	1,000

The company's policy is to hold closing stock of finished goods at 20% of the expected sales volume of the succeeding month.

Each unit of exercise bike requires one unit of main body with resistance system & two units of pedals. Calculate the number of pedals required to be purchased for January 2025 production.

A 1,560 pedals

B 1,440 pedals

C 3,120 pedals

D 2,880 pedals

ICAI'S ANSWER

Answer key: **(C)**

ICAI · P4 · May 2025 exam · question p. 4-5 · answer p. 6

1078 **EXAM** **Jan 2025** Case Scenario - II · MCQ 6 to MCQ 10

CASE SCENARIO - II

XYZ Limited produces the product P. The cost accountant of the company has to prepare its budget for a particular year.

The following information are made available for this purpose:

The expected sales of the product P is 1,00,000 units during the year at a selling price of ₹ 50 per unit.

Each unit of product P requires 3 kgs of raw material Q and 4 kgs of raw material R.

The expected stock levels are as follows:

	Beginning of year	End of year
Finished product P in units	12,000	15,000
Raw material Q in kgs	26,000	20,000
Raw material R in kgs	36,000	42,000

Raw material Q costs ₹ 2 per kg and R costs ₹ 3 per kg.

It requires 10 minutes of direct labour time to produce one unit of product P.

Labour cost is ₹ 50 per hour.

Variable manufacturing overheads are ₹ 10 per unit.

Fixed manufacturing cost is ₹ 3,00,000 per year.

Fixed Administration and selling expenses are ₹ 25,000 per year.

On the basis of above Case Scenario, you are required to answer the following MCQs 6 to 10:

MCQ 6

The total number of units to be produced of product P is:

A 1,03,000 units

B 97,000 units

C 92,000 units

D 1,27,000 units

MCQ 7

The total quantity of raw material R to be purchased during the year -

A 4,06,000 kgs

B 4,18,000 kgs

C 3,82,000 kgs

D 3,75,000 kgs

MCQ 8

The total cost of purchase of Raw Material Q during the year is -

A ₹ 6,00,000

B ₹ 6,06,000

C ₹ 5,88,000

D ₹ 6,12,000

MCQ 9

The budgeted variable cost of production of one unit of product P is -

A ₹ 46.33

B ₹ 36.38

C ₹ 25.33

D ₹ 36.33

MCQ 10

What is the budgeted net income for the year?

A ₹ 10,41,667

B ₹ 13,66,650

C ₹ 10,67,000

D ₹ 10,37,000

ICAI'S ANSWERS

MCQ 6

Answer key: **(A)**

MCQ 7

Answer key: **(B)**

MCQ 8

Answer key: **(B)**

MCQ 9

Answer key: **(D)**

MCQ 10

Answer key: **(A)**

ICAI · P4 · Jan 2025 exam · question p. 3-4 · answer p. 6

1079 EXAM **Old P3 Nov 2023** Q2(b) 10 marks old scheme

HL Limited produces and sells four varieties of beverage. The past data shows different demand patterns for various quarters during the year. The sales quantity and selling price for the month of September 2023 is as follows:

	Sales Quantity	Selling Price per unit
Hot Coffee	1,40,000 Units	₹ 20/-
Cold Coffee	3,40,000 Units	₹ 40/-
Fruit Juice	4,20,000 Units	₹ 20/-
Carbonated Soft Drink	2,70,000 units	₹ 20/-

For the quarter October to December 2023, it is estimated that due to climate changes the demand for Hot Coffee would increase every month by 50% of the previous month and the demand for Cold Coffee would decrease every month by 30% of the previous month. The demand for Fruit Juice would decrease by 20% in the month of October 2023 and thereafter it will remain constant. HL Limited would be able to sell only 60,000 units, 50,000 units and 30,000 units of Carbonated Soft Drink respectively during the months of October, November and December 2023. There would be no change in the selling price of all the products during the next quarter.

Standard Quantity of closing stock for the period September 2023 to December 2023 is as follows: (in units)

	Hot Coffee	Cold Coffee	Fruit Juice	Carbonated Soft Drink
September 2023	12,000	13,000	11,000	7,500
October 2023	15,000	14,000	12,000	5,500
November 2023	13,000	15,000	10,000	6,000
December 2023	11,000	16,000	13,000	7,000

You are required to prepare a Production Budget (in units) and Sales Budget (in units and sales value) for the months of October, November and December 2023.

ICAI'S ANSWER

Production Budget (in units)

Particulars	Hot Coffee	Cold Coffee	Fruit Juice	Carbonated Soft Drink
October 2023				
Sales*	2,10,000	2,38,000	3,36,000	60,000
Add: Closing stock	15,000	14,000	12,000	5,500
Total Quantity Required	2,25,000	2,52,000	3,48,000	65,500
Less: Opening stock	12,000	13,000	11,000	7,500
Production	**2,13,000**	**2,39,000**	**3,37,000**	**58,000**
November 2023				
Sales*	3,15,000	1,66,600	3,36,000	50,000
Add: Closing stock	13,000	15,000	10,000	6,000
Total Quantity Required	3,28,000	1,81,600	3,46,000	56,000
Less: Opening stock	15,000	14,000	12,000	5,500
Production	**3,13,000**	**1,67,600**	**3,34,000**	**50,500**
December 2023				
Sales*	4,72,500	1,16,620	3,36,000	30,000
Add: Closing stock	11,000	16,000	13,000	7,000
Total Quantity Required	4,83,500	1,32,620	3,49,000	37,000
Less: Opening stock	13,000	15,000	10,000	6,000
Production	**4,70,500**	**1,17,620**	**3,39,000**	**31,000**

*sales units are taken from sales budget

Sales Budget (in Units and sales value)

Particulars	Hot Coffee	Cold Coffee	Fruit Juice	Carbonated Soft Drink
October 2023 (in units)	2,10,000 [1,40,000 + (1,40,000 x 50%)]	2,38,000 [3,40,000 - (3,40,000 x 30%)]	3,36,000 [420000 - (4,20,000×20%)]	60,000
October 2023 (Sales Value in ₹)	42,00,000 (2,10,000 x ₹ 20)	95,20,000 (2,38,000 x ₹ 40)	67,20,000 (3,36,000 x ₹20)	12,00,000 (60,000 x ₹ 20)
November 2023 (in units)	3,15,000 [2,10,000 + (2,10,000 x 50%)]	1,66,600 [2,38,000 - (2,38,000 x 30%)]	3,36,000	50,000
November 2023 (Sales Value in ₹)	63,00,000 (3,15,000 x ₹ 20)	66,64,000 (1,66,600 x ₹ 40)	67,20,000 (3,36,000 x ₹ 20)	10,00,000 (50,000 x ₹ 20)
December 2023 (in units)	4,72,500 [3,15,000 + (3,15,000 x 50%)]	1,16,620 [1,66,600 - (1,66,600 x 30%)]	3,36,000	30,000
December 2023 (Sales Value in ₹)	94,50,000 (4,72,500 x ₹ 20)	46,64,800 (1,16,620 x ₹ 40)	67,20,000 (3,36,000 x ₹ 20)	6,00,000 (30,000 x ₹ 20)

Sales Budget can also be presented in following way:

	Oct 2023		Nov 2023		Dec 2023	
	Quantity (units)	**Amount (₹)**	**Quantity (units)**	**Amount (₹)**	**Quantity (units)**	**Amount (₹)**
Hot Coffee @ ₹ 20 per unit	2, 10,000	42,00,000	3,15,000	63,00,000	4,72,500	94,50,000
Cold Coffee @ ₹ 40 per unit	2,38,000	95,20,000	1,66,600	66,64,000	1,16,620	46,64,800
Fruit Juice @ ₹ 20 per unit	3,36,000	67,20,000	3,36,000	67,20,000	3,36,000	67,20,000
Carbonated Soft Drink @ ₹ 20 per unit	60,000	12,00,000	50,000	10,00,000	30,000	6,00,000
		2,16,40,000		**2,06,84,000**		**2,14,34,800**

ICAI · Old P3 · Nov 2023 exam · question p. 9-10 · answer p. 11-13

1080 EXAM Old P3 May 2023 Q3(a) 10 marks old scheme

PQR Limited manufactures three products - Product X, Product Y and Product Z. The output for the current year is 2,50,000 units of Product X, 2,80,000 units of Product Y and 3,20,000 units of Product Z respectively.

Selling price of Product X is 1.25 times of Product Z whereas Product Y can be sold at double the price at which product Z can be sold. Product Z can be sold at a profit of 20% on its marginal cost.

Other information are as follows:

	Product X	Product Y	Product Z
Direct Material Cost (Per unit)	₹ 20	₹ 20	₹ 20
Direct Wages Cost (per unit)	₹ 16	₹ 24	₹ 16

Raw material used for manufacturing all the three products is the same. Direct Wages are paid @ ₹ 4 per labour hour,

Total overhead cost of the company is ₹ 52,80,000 for the year, out of which ₹ 1 per labour hour is variable and the rest is fixed.

In the next year it is expected that sales of product X and product Z will increase by 12% and 15% respectively and sale of product Y will decline by 5%. The total overhead cost of the company for the next year is estimated at ₹ 55,08,000. The variable cost of ₹ 1 per labour hour remains unchanged.

It is anticipated that all other costs will remain same for the next year and there is opening and closing stock. Selling Price per unit of each product will remain unchanged in the next year.

Required:

Prepare a budget showing the current position and the position for the next year clearly indicating the total product-wise contribution and profit for the company as a whole.

ICAI'S ANSWER

(i) Budget showing current position of total product wise contribution and profitability

	Particulars	Product X (₹)	Product Y (₹)	Product Z (₹)	Total (₹)
A	Direct material cost (per unit)	20	20	20	
B	Direct wages cost (per unit)	16	24	16	
C	Variable overhead per unit (Refer WN-1)	4	6	4	
D	Total variable cost/ Marginal cost per unit [A+B+C]	40	50	40	
E	Add: Profit [20% of D]	-	-	8	
F	Selling price unit [D+E]	-	-	48	
G	Price weight	1.25	2	1	
H	Selling price per unit [Selling price of Product Z × G]	60	96	48	

	Particulars	Product X (₹)	Product Y (₹)	Product Z (₹)	Total (₹)
I	Contribution per unit [H-D]	20	46	8	
J	Quantity to be sold	2,50,000	2,80,000	3,20,000	
K	Total Contribution [J×I]	50,00,000	1,28,80,000	25,60,000	2,04,40,000
L	Fixed Overheads [Refer WN-1]				13,20,000
M	Profit				1,91,20,000

Working Notes:

1. Segregation of Overheads into variable and fixed in current year

	Particulars	Product X (₹)	Product Y (₹)	Product Z (₹)	Total (₹)
A	Total overhead cost	-	-	-	52,80,000
B	Labour hour per unit [Direct wages Cost ÷ Re.1]	4	6	4	
C	Quantity produced	2,50,000	2,80,000	3,20,000	
D	Total variable overhead cost [B×C]	10,00,000	16,80,000	12,80,000	39,60,000
E	Fixed overhead cost [A-D]				13,20,000

(ii) Budget showing next year's position of total product wise contribution and profitability

	Particulars	Product X (₹)	Product Y (₹)	Product Z (₹)	Total (₹)
A	Selling price per unit	60	96	48	
B	Contribution per unit	20	46	8	
C	Quantity to be sold	2,80,000 [112% of 2,50,000]	2,66,000 [95% of 2,80,000]	3,68,000 [115% of 3,20,000]	
D	Total Contribution [B×C]	56,00,000	1,22,36,000	29,44,000	2,07,80,000
	Fixed Overheads [Refer WN-2]				13,20,000
	Profit				1,94,60,000

Working Notes:

2. Segregation of Overheads into variable and fixed in next year

	Particulars	Product X (₹)	Product Y (₹)	Product Z (₹)	Total (₹)
A	Total overhead cost	-	-	-	55,08,000
B	Labour hour per unit [Direct wages Cost ÷ Re.1]	4	6	4	
C	Quantity produced	2,80,000	2,66,000	3,68,000	
D	Total variable overhead cost [B×C]	**11,20,000**	**15,96,000**	**14,72,000**	**41,88,000**
E	Fixed overhead cost [A-D]				13,20,000

ICAI · Old P3 · May 2023 exam · question p. 11-12 · answer p. 13-15

1081 **EXAM** **Old P3 Jul 2021** Q5(b) 10 marks old scheme

PSV Ltd. manufactures and sells a single product and estimated the following related information for the period November, 2020 to March, 2021.

Particulars	November, 2020	December, 2020	January, 2021	February, 2021	March, 2021
Opening Stock of Finished Goods (in Units)	7,500	3,000	9,000	8,000	6,000
Sales (in Units)	30,000	35,000	38,000	25,000	40,000
Selling Price per unit (in ₹)	10	12	15	15	20

Additional Information:

- Closing stock of finished goods at the end of March, 2021 is 10,000 units.
- Each unit of finished output requires 2 kg of Raw Material 'A' and 3 kg of Raw Material 'B'.

You are required to prepare the following budgets for the period November, 2020 to March, 2021 on monthly basis:

(i) Sales Budget (in ₹)

(ii) Production budget (in units) and

(iii) Raw material Budget for Raw material 'A' and 'B' separately (in units)

ICAI'S ANSWER

(i) Sales Budget (in ₹)

Particulars	Nov, 20	Dec, 20	Jan, 21	Feb, 21	Mar, 21	Total
Sales (in Units)	30,000	35,000	38,000	25,000	40,000	1,68,000
Selling Price per unit (₹)	10	12	15	15	20	-
Total Sales (₹)	3,00,000	4,20,000	5,70,000	3,75,000	8,00,000	24,65,000

(ii) Production Budget (in units)

Particulars	Nov, 20	Dec, 20	Jan, 21	Feb, 21	Mar, 21	Total
Sales	30,000	35,000	38,000	25,000	40,000	1,68,000
Add: Closing stock of finished goods	3,000	9,000	8,000	6,000	10,000	36,000
Total quantity required	33,000	44,000	46,000	31,000	50,000	2,04,000
Less: Opening stock of finished goods	7,500	3,000	9,000	8,000	6,000	33,500
Units to be produced	**25,500**	**41,000**	**37,000**	**23,000**	**44,000**	1,70,500

(iii) Raw material budget (in units)

For Raw material 'A'

Particulars	Nov, 20	Dec, 20	Jan, 21	Feb, 21	Mar, 21	Total
Units to be produced: (a)	25,500	41,000	37,000	23,000	44,000	1,70,500
Raw material consumption p.u. (kg.): (b)	2	2	2	2	2	-
Total raw material consumption (Kg.): (a × b)	**51,000**	**82,000**	**74,000**	**46,000**	**88,000**	3,41,000

For Raw material 'B'

Particulars	Nov, 20	Dec, 20	Jan, 21	Feb, 21	Mar, 21	Total
Units to be produced: (a)	25,500	41,000	37,000	23,000	44,000	1,70,500
Raw material consumption p.u. (kg.): (b)	3	3	3	3	3	-
Total raw material consumption (Kg.): (a × b)	**76,500**	**1,23,000**	**1,11,000**	**69,000**	**1,32,000**	5,11,500

ICAI · Old P3 · Jul 2021 exam · question p. 75 · answer p. 77-78

1082 RTP **Sep 2026** Case scenario · Q14(A) to Q14(B)

CASE SCENARIO

ABC Manufacturing Ltd. is a medium-sized manufacturing company engaged in the production of a single industrial component used in engineering applications. In recent months, the company has experienced fluctuations in demand due to changing market conditions and increased competition.

In order to ensure better cost control, efficient utilization of resources, and improved planning, the management has decided to implement a detailed budgeting and standard costing system for the financial year 2025. The Cost Accountant of the company has been entrusted with the responsibility of preparing budgets and analyzing variances for performance evaluation.

You are provided with the following data relating to the first six months of the year 2025:

Sales Budget (Units)

Month	January	February	March	April	May	June
Units	12,000	13,500	15,000	16,500	17,000	18,500

The company follows a policy of maintaining finished goods inventory at the end of each month equal to 25% of the budgeted sales for the following month. The opening finished goods inventory as on 1st January 2025 was 3,000 units. There is no work-in-progress inventory at any point of time.

Each unit of finished product requires two types of materials as detailed below:

Material	Quantity per Unit	Rate per kg
Material A	5 kg	₹ 12
Material B	4 kg	₹ 18

The stock of raw materials on 1st January 2025 was:

- Material A: 25,000 kg
- Material B: 18,000 kg

The company maintains closing stock of raw materials at the end of each month equal to 60% of the next month's production requirement to avoid stock-outs and ensure uninterrupted production.

Direct Labour

Each unit of finished product requires 0.8 direct labour hours.

Based on past experience and wage agreements, the total budgeted direct labour cost for the first quarter of the year has been estimated at ₹ 15,36,000.

Actual data for the quarter one, ended on March 31, 2025 is as under:

Despite careful planning, actual results differed from estimates due to operational inefficiencies and changes in input prices.

Actual production: 42,000 units

Actual Material Consumption

Material	Quantity (kg)	Rate per kg
Material A	2,10,000	₹12.50
Material B	1,65,000	₹18.40

Actual Labour Data

- Actual hours worked: 35,000 hours
- Actual labour cost: ₹ 17,85,000

Required:

Q14(A)

Prepare the following budgets:

1. Monthly production budget for the quarter January to March, 2025.
2. Monthly raw material consumption quantity budget for January to April, 2025.
3. Materials purchase quantity budget for the first quarter of the year 2025.

Q14(B)

Calculate the following variances for the quarter ended 31st March, 2025:

1. Material Cost Variance
2. Material Price Variance
3. Material Usage Variance
4. Labour Cost Variance
5. Labour Rate Variance
6. Labour Efficiency Variance

ICAI'S ANSWERS

Q14(A)

(i) Production Budget for January to March 2025

(Quantitative)

Particulars	Jan	Feb	Mar	April
Budgeted Sales	12,000	13,500	15,000	16,500
Add: Closing Stock (25% of next month sales)	3,375	3,750	4,125	4,250
Total Requirement	**15,375**	**17,250**	**19,125**	**20,750**
Less: Opening Stock	3,000	3,375	3,750	4,125
Budgeted Production	**12,375**	**13,875**	**15,375**	**16,625**

Total Budgeted Production (Q1) = 12,375 + 13,875 + 15,375 = 41,625 units

(ii) Raw Material Consumption Budget (Quantity)

Month	Output (Units)	Material A (5 kg/unit)	Material B (4 kg/unit)
January	12,375	61,875	49,500
February	13,875	69,375	55,500
March	15,375	76,875	61,500
April	16,625	83,125	66,500
Total (Q1)		**2,08,125**	**1,66,500**

(iii) Raw Materials Purchase Quantity Budget (Quarter I)

Material A (kg)

Particulars	Quantity
Material required for production	2,08,125
Add: Closing Stock (60% of April requirement = 83,125 × 60%)	49,875
Total Requirement	**2,58,000**
Less: Opening Stock	25,000
Purchase Quantity	**2,33,000 kg**

Material B (kg)

Particulars	Quantity
Material required for production	1,66,500
Add: Closing Stock (60% of April requirement = 66,500 × 60%)	39,900
Total Requirement	**2,06,400**
Less: Opening Stock	18,000
Purchase Quantity	**1,88,400 kg**

Q14(B)

Calculation of Material Cost Variance

Working Data

- Standard Quantity for actual output (42,000 units):
  - Material A = 42,000 × 5 = 2,10,000 kg
  - Material B = 42,000 × 4 = 1,68,000 kg

Material Cost Variance Statement

	Particulars	Material A (₹)	Material B (₹)	Total (₹)
(a)	Std Price × Std Qty	₹ 12 × 2,10,000 = 25,20,000	₹ 18 × 1,68,000 = 30,24,000	55,44,000
(b)	Std Price × Actual Qty	₹ 12 × 2,10,000 = 25,20,000	₹ 18 × 1,65,000 = 29,70,000	54,90,000
(c)	Actual Cost	₹ 12.5 × 2,10,000 = 26,25,000	₹ 18.40 × 1,65,000 = 30,36,000	56,61,000

Material Usage Variance = Standard Rate x Std. Qty. – Standard Rate x Actual Quantity

- A = 25,20,000 – 25,20,000 = Nil
- B = 30,24,000 – 29,70,000 = 54,000 (F)

Total = 54,000 (F)

Material Price Variance = Actual Quantity x Standard rate – Actual Quantity x Actual Rate

- A = 25,20,000 – 26,25,000 = 1,05,000 (A)
- B = 29,70,000 – 30,36,000 = 66,000 (A)

Total = 1,71,000 (A)

Material Cost Variance = Standard Cost for Actual output – Actual cost

- A = 25,20,000 – 26,25,000 = 1,05,000 (A)
- B = 30,24,000 – 30,36,000 = 12,000 (A)

Total = 1,17,000 (A)

Calculation of Labour Cost Variances:

Working Notes

Budgeted output (Q1) = 41,625 units

Standard labour hours = 41,625 × 0.8 = 33,300 hours

Standard rate per hour = $= \frac{\text{Budgeted direct labour cost}}{\text{Budgeted direct labour hours}}$

= ₹ 15,36,000 / 33,300 = ₹ 46.13 (approx.)

Standard hours for actual output = 42,000 × 0.8 = 33,600 hours

Actual rate per hour = ₹ 17,85,000 / 35,000 = ₹ 51

Direct Labour Efficiency Variance = Std Rate × (Std Hours – Actual Hours)

= 46.13 × (33,600 – 35,000)

= 46.13 × (–1,400)

= ₹ 64,582 (A)

Direct Labour Rate Variance = Actual Hours × (Std Rate – Actual Rate)

= 35,000 × (46.13 – 51)

= 35,000 × (–4.87)

= ₹ 1,70,450 (A)

Direct Labour Cost Variance = (Std Rate × Std Hours) – Actual Cost

= (46.13 × 33,600) – 17,85,000

= 15,49,968 – 17,85,000

= ₹ 2,35,032 (A)

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1083 RTP **Jan 2026** Q7 MCQ

A company plans to produce 44,000 units during the budget period. The company requires 4 units of raw material to produce 1 unit of finished goods. The company currently has 1,00,000 units of raw material on hand and wishes to have an inventory of 1,10,000 units of raw material on hand at the end of the budget period. The number of raw material units to be purchased during the budget period is

A 1,76,000 units

B 1,67,000 units

C 1,86,000 units

D 1,68,000 units

ICAI'S ANSWER

(C) 1,86,000 units

Raw material required for production = 44,000 units x 4 units/unit

= 1,76,000

Raw material to be purchased = Closing stock + RM required – Opening stock

= 1,10,000 + 1,76,000 – 1,00,000

= 1,86,000 units

ICAI · P4 · RTP Jan 2026 · question p. 8 · answer p. 29

1084 RTP **Jan 2026** Q18

SIAM Ltd. manufactures two products using one type of material and one grade of labour. Shown below is an extract from the company's working papers for the next period's budget.

Particulars	Product A	Product B
Budgeted Sales (Units)	1,800	2,400
Budgeted Material Consumption per Product (Kg.)	5	3
[Budgeted Material Cost ₹12 per Kg.]		
Standard Hours Allowed per product	5	4
[Budgeted Wage Rate ₹8 per hour]		

Overtime premium is 50% and is payable, if a worker works for more than 40 hours a week. There are 45 direct workers.

The target productivity ratio (or efficiency ratio) for the productive hours worked by the direct workers in actually manufacturing the products is 80%; in addition the non-productive downtime is budgeted at 20% of the productive hours worked.

There are twelve 5-day weeks in the budget period and it is anticipated that sales and production will occur evenly throughout the whole period.

It is anticipated that stock at the beginning of the period will be:

Product A 510 units; Product B 1,200 units; Raw material 2,150 Kg.

The target closing stock, expressed in terms of anticipated activity during budget period are - Product A 15 days sales; Product B 20 days sales; Raw material 10 days consumption.

Required:

(a) the material purchases budget, and

(b) the wages budget for the direct workers, showing the quantities and values, for the next period.

ICAI'S ANSWER

Material Purchase Budget

(in quantities and value)

Particulars	Total
Material Consumption (Kg.)	
A (1,740 units x 5 Kg. = 8,700)	
B (2,000 units x 3 Kg. = 6,000)	14,700
Add: Closing Balance of Material (Kg.) (W.N. 3)	2,450
Less: Anticipated Opening Balance of Material (Kg.)	2,150
Total Quantity of Material (kg.) to be purchased	15,000
Total Value of Material to be purchased (₹) (15,000 Kg. x ₹12)	₹ 1,80,000

Direct Workers Wages Budget

(showing hours required and wages paid)

Particulars	Total
Standard Hours for Budgeted Production	16,700
A (1,740 units x 5 hrs.= 8,700)	
B (2,000 units x 4 hrs.= 8,000)	
Standard Hours for Budgeted Production at Targeted Efficiency Ratio (W.N. 4)	20,875
Add: Non Productive Downtime (20% x 20,875 hours)	4,175
Total Labour Hours Required	25,050
Less: Normal Labour Hours (45 workers x 12 weeks x 5 days x 8 hours)	21,600
Difference (Overtime hours)	3,450
Wages for normal hours (21,600 hours x ₹ 8)	₹1,72,800
Overtime Wages (3,450 x ₹12)	₹41,400
Total Wages	₹2,14,200

Working Notes:

1. Computation of Closing Stock Balance of Products A and B

Budgeted Period of Sales (In days) = 12 weeks x 5 days = 60 days

Closing Stock of Product A (Units) (15 days sales)

$$= \frac{1{,}800 \text{ units} \times 15 \text{ days}}{60 \text{ days}} = 450 \text{ units}$$

Closing stock of Product B (units) (20 days sales)

$= \frac{2{,}400 \text{ units} \times 20 \text{ days}}{60 \text{ days}} = 800 \text{ units}$

2. Production Budget (Units)

	Products	
	A	**B**
Sales in units (60 days)	1,800	2,400
Add: Closing Stock (W.N. 1)	450	800
	2,250	3,200
Less: Anticipated Opening Balance	510	1,200
Total Number of Units to be produced	1,740	2,000

3. Closing Balance of Material (Kg.)

Closing Balance of Material (10 days consumption)

$= \frac{14{,}700 \text{ kg}}{60 \text{ days}} \times 10 \text{ days} = 2{,}450 \text{ Kg.}$

4. Standard Hours for Budgeted Production at targeted 80% efficiency ratio $= \frac{16{,}700 \text{ Hrs.}}{80} \times 100 =$ 20,875 Hrs.

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1085 **RTP** **Sep 2025** **Q17**

X Ltd. has carefully analyzed market trends and consumer behaviour to project its sales figures for the upcoming year as follows:

Quarter	Sales (Units)
I	45,000
II	56,250
III	61,875
IV	67,500

Information regarding production and finished goods stock is as follows:	
Opening Stock of Finished Goods	11,250 Units
Expected Closing Stock of Finished Goods at the end of the year	18,375 Units
Production Pattern	75% of Current Quarter Sales Plus 25% of Next Quarter Sales
Closing Stock of current quarter to be maintained	25% of Next Quarter Sales

Further information regarding stock of raw materials is as follows:	
Opening Stock of Raw Materials	15,000 Units
Required Closing Stock of Raw Materials at the end of the year	7,500 Units

Each unit of finished output requires 4 kg. of raw materials.

The company plans to acquire its total annual requirement of raw materials during the first three quarters, in the specified proportions and at the prices detailed below:

Quarter	Purchase of raw materials % to total annual requirement in quantity	Price per kg. (₹)
I	25%	12
II	50%	13
III	25%	14

You are required to PREPARE the following for the next year, quarter wise:

(i) Production budget (in units).

(ii) Raw material consumption budget (in quantity).

(iii) Raw material purchase budget (in quantity).

(iv) Raw material purchase budget (in value).

ICAI'S ANSWER

Working Note:

Calculation of total annual production

Particulars	(Units)
Sales in 4 quarters	2,30,625
Add: Closing balance	18,375
	2,49,000
Less: Opening balance	(11,250)
Total number of units to be produced in the next year	**2,37,750**

(i) Production Budget (in units)

Quarters	I (Units)	II (Units)	III (Units)	IV (Units)	Total (Units)
Sales	45,000	56,250	61,875	67,500	2,30,625
Production in current quarter (75% of the sale of current quarter)	33,750	42,188	46,406	50,625	
Production for next quarter (25% of the sale of next quarter)	14,063	15,469	16,875	18,375	
Total production	**47,813**	**57,656**	**63,281**	**69,000**	**2,37,750**

(ii) Raw material consumption budget in quantity

Quarters	I	II	III	IV	Total
Units to be produced in each quarter: (A)	47,813	57,656	63,281	69,000	2,37,750
Raw material consumption p.u. (kg.): (B)	4	4	4	4	
Total raw material consumption (Kg.) : (A × B)	1,91,250	2,30,625	2,53,125	2,76,000	9,51,000

(iii) Raw material purchase budget (in quantity)

Particulars	Qty. (kg.)
Raw material required for production	9,51,000
Add: Closing balance of raw material	7,500
	9,58,500
Less: Opening balance	(15,000)
Material to be purchased	9,43,500

(iv) Raw material purchase budget (in value)

Quarters	% of annual requirement	Qty. of material	Rate per kg. (₹)	Amount (₹)
(1)	**(2)**	**(3)**	**(4)**	**(5) = (3×4)**
I	25	2,35,875 (9,43,500 kg. x 25%)	12	28,30,500
II	50	4,71,750 (9,43,500 kg. x 50%)	13	61,32,750
III	25	2,35,875 (9,43,500 kg. x 25%)	14	33,02,250
Total		**9,43,500**		**1,22,65,500**

ICAI · P4 · RTP Sep 2025 · question p. 21-22 · answer p. 52-53

1086 RTP **Sep 2024** Q7 MCQ

Also asked in MTP Sep 2026 S-II MCQ 13

A business manufactures a single product and is preparing its production budget for the year ahead. It is estimated that 2,00,000 units of the product can be sold in the year and the opening inventory is currently 25,000 units. The inventory level is to be reduced by 40% by the end of the year. What is production budget in units?

A 1,95,000 units

B 1,90,000 units

C 1,84,000 units

D 1,75,000 units

ICAI'S ANSWER

(B) 1,90,000 units

	Units
Sales budget	2,00,000
Add: Closing Inventory (25,000 × 0.6)	15,000
Less: Opening Inventory	(25,000)
Production Budget	1,90,000

ICAI · P4 · RTP Sep 2024 · question p. 6-7 · answer p. 21

1087 RTP **Sep 2024** Q18

Raja Ltd manufactures and sells a single product and has estimated sales revenue of ₹ 302.4 lakh during the year based on 20% profit on selling price. Each unit of product requires 6 kg of material A and 3 kg of material B and processing time of 4 hours in machine shop and 2 hours in assembly shop. Factory overheads are absorbed at a blanket rate of 20% of direct labour. Variable selling & distribution overheads are ₹ 60 per unit sold and fixed selling & distribution overheads are estimated to be ₹ 69,12,000.

The other relevant details are as under:

Purchase Price:	Material A	₹ 160 per kg
	Materials B	₹ 100 per kg
Labour Rate:	Machine Shop	₹ 140 per hour
	Assembly Shop	₹ 70 per hour

	Finished Stock	Material A	Material B
Opening Stock	2,500 units	7,500 kg	4,000 kg
Closing Stock	3,000 units	8,000 kg	5,500 kg

Required

(i) CALCULATE number of units of product proposed to be sold and selling price per unit,

(ii) PREPARE Production Budget in units and

(iii) PREPARE Material Purchase Budget in units.

ICAI'S ANSWER

Workings

Statement Showing "Total Variable Cost for the year"

Particulars	Amount (₹)
Estimated Sales Revenue	3,02,40,000
Less: Desired Profit Margin on Sale @ 20%	60,48,000
Estimated Total Cost	2,41,92,000
Less: Fixed Selling and Distribution Overheads	69,12,000
Total Variable Cost	1,72,80,000

Statement Showing "Variable Cost per unit"

Particulars	Variable Cost p.u. (₹)
Direct Materials:	
A: 6 Kg. @ ₹ 160 per kg.	960
B: 3 Kg. @ ₹ 100 per kg.	300
Labour Cost:	
Machine Shop: 4 hrs @ ₹ 140 per hour	560
Assembly Shop: 2 hrs @ ₹ 70 per hour	140
Factory Overheads: 20% of (₹ 560 + ₹ 140)	140
Variable Selling & Distribution Expenses	60
Total Variable Cost per unit	2,160

(i) Calculation of number of units of product proposed to be sold and selling price per unit:

Number of Units Sold = Total Variable Cost/Variable Cost per unit

= ₹ 1,72,80,000 / ₹ 2,160

= 8,000 units

Selling Price per unit = Total Sales Value / Number of Units Sold

= ₹ 3,02,40,000 / 8,000 units

= ₹ 3,780

(ii) Production Budget (units)

Particulars	Units
Budgeted Sales	8,000
Add: Closing Stock	3,000
Total Requirements	11,000
Less: Opening Stock	(2,500)
Required Production	8,500

(iii) Materials Purchase Budget (Kg.)

Particulars	Material A	Material B
Requirement for Production	51,000 (8,500 units × 6 Kg.)	25,500 (8,500 units × 3 Kg.)
Add: Desired Closing Stock	8,000	5,500
Total Requirements	59,000	31,000
Less: Opening Stock	(7,500)	(4,000)
Quantity to be purchased	51,500	27,000

ICAI · P4 · RTP Sep 2024 · question p. 15-16 · answer p. 35-36

1088 RTP **Old P3 Nov 2023** Q13 old scheme

XY Co. Ltd manufactures two products viz., X and Y and sells them through two divisions, East and West. For the purpose of Sales Budget to the Budget Committee, following information has been made available for the year 2022-23:

Product	Budgeted Sales		Actual Sales	
	East Division	**West Division**	**East Division**	**West Division**
X	400 units at ₹ 9	600 units at ₹ 9	500 units at ₹ 9	700 units at ₹ 9
Y	300 units at ₹ 21	500 units at ₹ 21	200 units at ₹ 21	400 units at ₹ 21

Adequate market studies reveal that product X is popular but underpriced. It is expected that if the price of X is increased by ₹ 1, it will, find a ready market. On the other hand, Y is overpriced and if the price of Y is reduced by ₹ 1 it will have more demand in the market. The company management has agreed for the aforesaid price changes. On the basis of these price changes and the reports of salesmen, following estimates have been prepared by the Divisional Managers:

Percentage increase in sales over budgeted sales

Product	East Division	West Division
X	+ 10%	+ 5%
Y	+ 20%	+ 10%

With the help of intensive advertisement campaign, following additional sales (over and above the above-mentioned estimated sales by Divisional Mangers) are possible:

Product	East Division	West Division
X	60 units	70 units
Y	40 units	50 units

You are required to prepare Sales Budget for 2023-24 after incorporating above estimates and also show the Budgeted Sales and Actual Sales of 2022-23.

ICAI'S ANSWER

Statement Showing Sales Budget for 2023-24

Division	Product X			Product Y			Total
	Qty.	Rate (₹)	Amt. (₹)	Qty.	Rate (₹)	Amt. (₹)	Amt. (₹)
East	500[1]	10	5,000	400[3]	20	8,000	13,000
West	700[2]	10	7,000	600[4]	20	12,000	19,000
Total	1,200		12,000	1,000		20,000	32,000

Workings

1. 400 × 110% + 60 = 500 units
2. 600 × 105% + 70 = 700 units
3. 300 × 120% + 40 = 400 units
4. 500 × 110% + 50 = 600 units

Statement Showing Sales Budget for 2022-23

Division	Product X			Product Y			Total
	Qty.	Rate (₹)	Amt. (₹)	Qty.	Rate (₹)	Amt. (₹)	Amt. (₹)
East	400	9	3,600	300	21	6,300	9,900
West	600	9	5,400	500	21	10,500	15,900
Total	1,000		9,000	800		16,800	25,800

Statement Showing Actual Sales for 2022-23

Division	Product X			Product Y			Total
	Qty.	Rate (₹)	Amt. (₹)	Qty.	Rate (₹)	Amt. (₹)	Amt. (₹)
East	500	9	4,500	200	21	4,200	8,700
West	700	9	6,300	400	21	8,400	14,700
Total	1,200		10,800	600		12,600	23,400

1089 RTP **Old P3 May 2023** Q14 old scheme

EDF Ltd. produces two products using Skilled labour and two types of materials. Shown below the information for the next month's budget:

	Product- A	Product-B
Budgeted sales (in units)	4,080	6,120
Budgeted material consumption per unit (in kg):		
Material-X	8.5	5.1
Material-Y	6.8	10.2
Standard labour hours allowed per unit of product	5.1	8.5

Material-X and Material-Y cost ₹8 and ₹10 per kg and labours are paid ₹30 per hour. Overtime premium is 75% and is payable, if a worker works for more than 45 hours a week. There are 400 direct workers.

The target efficiency ratio for the productive hours worked by the direct workers in actually manufacturing the products is 85%. In addition the non-productive down-time is budgeted at 15% of the productive hours worked.

There are four 6-days weeks in the budgeted period and it is anticipated that sales and production will occur evenly throughout the whole period.

It is anticipated that stock at the beginning of the period will be:

Product-A	550 units
Product-B	350 units
Material-X	1,200 kgs.
Material-Y	600 kgs.

The anticipated closing stocks for budget period are as below:

Product-A	5 days sales
Product-B	5 days sales
Material-X	10 days consumption
Material-Y	5 days consumption

Required:

CALCULATE the Material Purchase Budget and the Wages Budget for the direct workers, showing the quantities and values, for the next month.

ICAI'S ANSWER

Number of days in budget period = 4 weeks × 6 days = 24 days

Number of units to be produced

	Product-A (units)	Product-B (units)
Budgeted Sales	4,080	6,120
Add: Closing stock	850 (4,080 units/24 days × 5 days)	1275 (6,120 units/24 days × 5 days)
Less: Opening stock	550	350
	4,380	7,045
(i) Material Purchase Budget		
	Material-X (Kg.)	**Material-Y (Kg.)**
Material required:		
Product-A	37,230 (4,380 units × 8.5 kg.)	29,784 (4,380 units × 6.8 kg.)
Product-B	35,930 (7,045 units × 5.1 kg.)	71,859 (7,045 units × 10.2 kg.)
	73,160	1,01,643
Add: Closing stock	30,483 (73,160 kgs./24 days × 10 days)	21,176 (1,01,643 kgs./24 days × 5 days)
Less: Opening stock	1,200	600
Quantity to be purchased	1,02,443	1,22,219
Rate per kg. of Material	8	10
Total Cost	8,19,541	12,22,186
(ii) Wages Budget		
	Product-A (Hours)	**Product-B (Hours)**
Units to be produced	4,380	7,045
Standard hours allowed per unit	5.1	8.5
Total Standard Hours allowed	22,338	59,883
Productive hours required for production	22,338 hours/85% × 26,280	59,883 hours/85% × 70,450
Add: Non-Productive down time hours	3942 (15% of 26,280 hours)	10568 (15% of 70,450 hours)

	Product-A (units)	Product-B (units)
Hours to be paid	30,222	81,018
Total Hours to be paid =	1,11,240	
Hours to be paid at normal rate (4 weeks × 45 hours × 400 workers) =	72000	
Hours to be paid at premium rate	39,240	
Total wages to be paid = (72,000 hours × ₹30 + 39,240 hours × ₹ 52.5)	= ₹ 21,60,000 + ₹ 20,60,100 = ₹ 42,20,100	

ICAI · Old P3 · RTP May 2023 · question p. 10 · answer p. 25-26

1090 RTP **Old P3 Dec 2021 Q14** old scheme

The accountant of manufacturing company provides you the following details for year 2019-20:

Particulars	(₹)
Direct materials	28,00,000
Direct Wages	16,00,000
Fixed factory overheads	16,00,000
Variable factory overheads	16,00,000
Other variable costs	12,80,000
Other fixed costs	12,80,000
Profit	18,40,000
Sales	1,20,00,000

During the year, the company manufactured two products A and B and the output and costs were:

Particulars	A	B
Output (units)	2,00,000	1,00,000
Selling price per unit	₹ 32.00	₹ 56.00
Direct materials per unit	₹ 8.00	₹ 12.00
Direct wages per unit	₹ 4.00	₹ 8.00

Variable factory overhead is absorbed as a percentage of direct wages. Other variable costs have been computed as: Product A ₹ 4.00 per unit; and B ₹ 4.80 per unit.

During 2020-21, it is expected that the demand for product A will fall by 25% and for B by 50%. It is decided to manufacture a new product C, the cost for which is estimated as follows:

Particulars	Product C
Output (units)	2,00,000
Selling price per unit	₹ 28.00
Direct materials per unit	₹ 6.40
Direct wages per unit	₹ 4.00

It is anticipated that the other variable costs per unit of Product C will be same as for product A.

PREPARE a budget to present to the management, showing the current position and the position for 2020-21. COMMENT on the comparative results.

ICAI'S ANSWER

Budget Showing Current Position and Position for 2020-21

	Position for 2019-20			Position for 2020-21			
	A	**B**	**Total (A+B)**	**A**	**B**	**C**	**Total (A+B+C)**
Sales (units)	2,00,000	1,00,000	–	1,50,000	50,000	2,00,000	–
	(₹)	(₹)	(₹)	(₹)	(₹)	(₹)	(₹)
(A) Sales	**64,00,000**	**56,00,000**	**1,20,00,000**	**48,00,000**	**28,00,000**	**56,00,000**	**1,32,00,000**
Direct Material	16,00,000	12,00,000	28,00,000	12,00,000	6,00,000	12,80,000	30,80,000
Direct wages	8,00,000	8,00,000	16,00,000	6,00,000	4,00,000	8,00,000	18,00,000
Factory overhead (variable)	8,00,000	8,00,000	16,00,000	6,00,000	4,00,000	8,00,000	18,00,000
Other variable costs	800,000	4,80,000	12,80,000	6,00,000	240,000	8,00,000	16,40,000
(B) Marginal Cost	40,00,000	32,80,000	72,80,000	30,00,000	16,40,000	36,80,000	83,20,000
(C) Contribution (A-B)	**24,00,000**	**23,20,000**	**47,20,000**	**18,00,000**	**11,60,000**	**19,20,000**	**48,80,000**
Fixed costs							
– Factory			16,00,000				16,00,000
– Others			12,80,000				12,80,000
(D) Total fixed cost			28,80,000				28,80,000
Profit (C – D)			**18,40,000**				**20,00,000**

Comments: Introduction of Product C is likely to increase profit by ₹ 1,60,000 (i.e. from ₹ 18,40,000 to ₹ 20,00,000) in 2020-21 as compared to 2019-20 even if the demand for Product A & B falls. Therefore, introduction of product C is recommended.

1091 RTP **Old P3 Jul 2021** Q12 old scheme

RS Ltd manufactures and sells a single product and has estimated sales revenue of ₹ 302.4 lakh during the year based on 20% profit on selling price. Each unit of product requires 6 kg of material A and 3 kg of material B and processing time of 4 hours in machine shop and 2 hours in assembly shop. Factory overheads are absorbed at a blanket rate of 20% of direct labour. Variable selling & distribution overheads are ₹ 60 per unit sold and fixed selling & distribution overheads are estimated to be ₹ 69,12,000.

The other relevant details are as under:

Purchase Price:	Material A	₹ 160 per kg
	Materials B	₹ 100 per kg
Labour Rate:	Machine Shop	₹ 140 per hour
	Assembly Shop	₹ 70 per hour

	Finished Stock	Material A	Material B
Opening Stock	2,500 units	7,500 kg	4,000 kg
Closing Stock	3,000 units	8,000 kg	5,500 kg

Required:

(i) CALCULATE number of units of product proposed to be sold and selling price per unit,

(ii) PREPARE Production Budget in units, and

(iii) PREPARE Material Purchase Budget in units.

ICAI'S ANSWER

Workings:

Statement Showing "Total Variable Cost for the year"

Particulars	Amount (₹)
Estimated Sales Revenue	3,02,40,000
Less: Desired Profit Margin on Sale @ 20%	60,48,000
Estimated Total Cost	2,41,92,000
Less: Fixed Selling and Distribution Overheads	69,12,000
Total Variable Cost	1,72,80,000

Statement Showing "Variable Cost per unit"

Particulars	Variable Cost p.u. (₹)
Direct Materials:	

Particulars	Variable Cost p.u. (₹)
A: 6 Kg. @ ₹ 160 per kg.	960
B: 3 Kg. @ ₹ 100 per kg.	300
Labour Cost:	
Machine Shop: 4 hrs. @ ₹ 140 per hour	560
Assembly Shop: 2 hrs. @ ₹ 70 per hour	140
Factory Overheads: 20% of (₹ 560 + ₹ 140)	140
Variable Selling & Distribution Expenses	60
Total Variable Cost per unit	2,160

(i) Calculation of number of units of product proposed to be sold and selling price per unit:

Number of Units Sold = Total Variable Cost / Variable Cost per unit

= ₹ 1,72,80,000 / ₹ 2,160

= 8,000 units

Selling Price per unit = Total Sales Value / Number of Units Sold

= ₹ 3,02,40,000 / 8,000 units

= ₹ 3,780

(ii) Production Budget (units)

Particulars	Units
Budgeted Sales	8,000
Add: Closing Stock	3,000
Total Requirements	11,000
Less: Opening Stock	(2,500)
Required Production	8,500

(iii) Materials Purchase Budget (Kg.)

Particulars	Material A	Material B
Requirement for Production	51,000 (8,500 units × 6 Kg.)	25,500 (8,500 units × 3 Kg.)
Add: Desired Closing Stock	8,000	5,500
Total Requirements	59,000	31,000
Less: Opening Stock	(7,500)	(4,000)
Quantity to be purchased	51,500	27,000

ICAI · Old P3 · RTP Jul 2021 · question p. 10-11 · answer p. 29-31

1092 RTP **Old P3 May 2020** Q12 old scheme

A Vehicle manufacturer has prepared sales budget for the next few months, and the following draft figures are available:

Month	No. of vehicles
October	40,000
November	35,000
December	45,000
January	60,000
February	65,000

To manufacture a vehicle a standard cost of ₹11,42,800 is incurred and sold through dealers at a uniform selling price of ₹17,14,200 to customers. Dealers are paid 15% commission on selling price on sale of a vehicle.

Apart from other materials, four units of Part - X are required to manufacture a vehicle. It is a policy of the company to hold stocks of Part-X at the end of each month to cover 40% of next month's production. 48,000 units of Part-X are in stock as on 1st October.

There are 9,500 nos. of completed vehicles in stock as on 1st October and it is policy to have stocks at the end of each month to cover 20% of the next month's sales.

You are required to -

(i) PREPARE Production budget (in nos.) for the month of October, November, December and January.

(ii) PREPARE a Purchase budget for Part-X (in units) for the months of October, November and December.

(iii) CALCULATE the budgeted gross profit for the quarter October to December.

ICAI'S ANSWER

(i) Preparation of Production Budget (in units)

	October	November	December	January
Demand for the month (Nos.)	40,000	35,000	45,000	60,000
Add: 20% of next month's demand	7,000	9,000	12,000	13,000
Less: Opening Stock	(9,500)	(7,000)	(9,000)	(12,000)
Vehicles to be produced	37,500	37,000	48,000	61,000

(ii) Preparation of Purchase budget for Part-X

	October	November	December
Production for the month (Nos.)	37,500	37,000	48,000
Add: 40% of next month's production	14,800 (40% of 37,000)	19,200 (40% of 48,000)	24,400 (40% of 61,000)
	52,300	56,200	72,400
No. of units required for production	2,09,200 (52,300 × 4 units)	2,24,800 (56,200 × 4 units)	2,89,600 (72,400 × 4 units)
Less: Opening Stock	(48,000)	(59,200) (14,800 × 4 units)	(76,800) (19,200 × 4 units)
No. of units to be purchased	1,61,200	1,65,600	2,12,800

(iii) Budgeted Gross Profit for the Quarter October to December

	October	November	December	Total
Sales in nos.	40,000	35,000	45,000	1,20,000
Net Selling Price per unit* (₹)	14,57,070	14,57,070	14,57,070	
Sales Revenue (₹ in lakh)	5,82,828	5,09,974.50	6,55,681.50	17,48,484
Less: Cost of Sales (₹ in lakh) (Sales unit × Cost per unit)	4,57,120	3,99,980	5,14,260	13,71,360
Gross Profit (₹ in lakh)	1,25,708	1,09,994.50	1,41,421.50	3,77,124

* Net Selling price unit =₹17,14,200 – 15% commission on ₹17,14,200 = ₹14,57,070.

ICAI · Old P3 · RTP May 2020 · question p. 9-10 · answer p. 26-27

1093 MTP **Sep 2026 S-I** MCQ 12 2 marks MCQ

A company plans to produce 50,000 units of finished goods during the upcoming budget period. Each finished unit requires 3 units of raw material. At the beginning of the period, the company has 80,000 units of raw material in inventory. By the end of the period, the company wants to increase the raw material inventory to 90,000 units.

Based on this information, calculate how many units of raw material need to be purchased during the budget period.

A 1,40,000 units

B 1,50,000 units

C 1,60,000 units

D 1,70,000 units

ICAI'S ANSWER

(C) 1,60,000 units

Raw material required for production:

= 50,000 × 3

= **1,50,000 units**

Raw material to be purchased

= Required for production + Closing stock – Opening stock

= 1,50,000 + 90,000 – 80,000

= **1,60,000 units**

ICAI · P4 · MTP Sep 2026 · Series I · question p. 6 · answer p. 5

1094 **MTP** **Sep 2026 S-I** Q5(a) **10 marks**

Same question, different ICAI answer: #1096 MTP Jan 2026 S-II

Kantur Ltd. is a medium-sized manufacturer of precision components used in the automobile aftermarket. The factory runs a single production line where two product variants (Product A and Product B) are made from the same raw material and by the same grade of direct labour. Management is preparing the master budget for the next period and wants the material purchases and direct wages budgets prepared so that procurement and staffing decisions can be finalised.

Below is an extract from the company's budget working papers (all figures are for the whole budget period unless stated otherwise):

Production & material data

Particulars	Product A	Product B
Budgeted sales (units)	2,100	2,700
Budgeted material consumption per product (kg)	6.00	2.50
Budgeted material cost	₹ 14 per kg (for both products)	
Standard hours allowed per product	6.0	5.0
Budgeted basic wage rate	₹ 9 per hour (for all direct workers)	

Labour / working-time assumptions

- Overtime premium is 60% of the basic wage rate and is payable for any hours worked over 40 hours in a 5-day week.

- There are 48 direct workers available in the budget period.
- The target productivity (efficiency) ratio for productive hours actually used in manufacturing is 85%. In addition, non-productive downtime (set-up, maintenance, unavoidable delays) is budgeted at 15% of the productive hours worked.
- The budget period contains thirteen 5-day weeks and production and it is anticipated that sales and production will occur evenly throughout the whole period.

Opening and desired closing stocks

- Expected opening stocks at the start of the budget period: Product A 600 units; Product B 1,000 units; Raw material 2,800 kg.
- Target closing stocks (expressed in terms of anticipated activity during the budget period):
  - Product A — 18 days' sales
  - Product B — 22 days' sales
  - Raw material — 12 days' consumption

Other assumptions

- Production is planned to meet budgeted sales and to build up / run down stocks to the closing targets stated above.
- Sales and production occur evenly over the budget period (i.e., the daily demand pattern is smooth).

Required

PREPARE the following budgets for the next period showing quantities and values:

1. Material purchases budget.
2. Wages budget for direct workers

ICAI'S ANSWER

(i) Material Purchase Budget

(in quantities and value)

Particulars	Total
Material Consumption (Kg.)	
A (2,082 units × 6 kg = 12,492)	12,492
B (2,614 units × 2.5 kg = 6,535)	6,535
Total	19,027
Add: Closing Balance of Material (Kg.) (W.N. 3)	3,513
Less: Anticipated Opening Balance of Material (Kg.)	2,800
Total Quantity of Material (Kg.) to be purchased	19,740
Total Value of Material to be purchased (19,740 kg × ₹ 14)	₹ 2,76,360

(ii) Direct Workers Wages Budget

(Showing hours required and wages paid)

Particulars	Total
Standard Hours for Budgeted Production	
A (2,082 units × 6 hrs = 12,492)	12,492
B (2,614 units × 5 hrs = 13,070)	13,070
Total Standard Hours for Budgeted Production	25,562
Standard Hours for Budgeted Production at Targeted Efficiency Ratio (W.N. 4)	30,072.94
Add: Non-Productive Downtime (15% × 30,072.94 hours)	4,510.94
Total Labour Hours Required	34,583.88
Less: Normal Labour Hours (48 workers × 13 weeks × 5 days × 8 hours)	24,960
Difference (Overtime hours)	9,624
Wages for normal hours (24,960 hours × ₹ 9)	₹ 2,24,640.00
Overtime Wages (9,624 hrs × ₹ 14.40)	₹ 1,38,585.60
Total Wages	₹ 3,63,225.60

Working Notes:

1. Computation of Closing Stock Balance of Products A and B

Budgeted Period of Sales (In days) = 13 weeks × 5 days = 65 days

Closing stock of Product A (Units) (18 days' sales)

$$= \frac{2,100}{65} \times 18 = 582 \text{ units}$$

Closing stock of Product B (Units) (22 days' sales)

$$= \frac{2,700}{65} \times 22 = 914 \text{ units}$$

2. Production Budget (Units)

Particulars	Product A	Product B
Sales in units (65 days)	2,100	2,700
Add: Closing Stock (W.N. 1)	582	914
Total	2,682	3,614
Less: Anticipated Opening Balance	600	1,000
Total Number of Units to be produced	2,082	2,614

3. Closing Balance of Material (Kg.)

Total material consumption for production = 12,492 kg (A) + 6,535 kg (B) = 19,027 kg

Daily consumption = $\frac{19,027}{65}$ = 292.72 kg/day

Closing stock of material (12 days consumption) = 292.72 × 12 = 3,513 kg (rounded up)

4. Standard Hours for Budgeted Production at targeted 85% efficiency ratio

Total standard hours for production = (2,082 × 6) + (2,614 × 5) = 25,562 hrs

Productive hours required = Standard hours ÷ Efficiency

= $\frac{25,562}{0.85}$ = 30,072.94 hrs

Non-productive downtime = 15% of productive hours = 0.15 × 30,072.94 = 4,510.94 hrs

Total labour hours required = 30,072.94 + 4,510.94 = 34,583.88 hrs

ICAI · P4 · MTP Sep 2026 · Series I · question p. 14-15 · answer p. 18-19

1095 MTP **Jan 2026 S-I** Case Scenario II · MCQ 6 to MCQ 10 10 marks

CASE SCENARIO II

Aroma Beverages Ltd. is a growing beverage company that produces and sells four varieties of tea-based drinks. The company operates across major cities and experiences strong seasonal fluctuations in demand. The management closely monitors quarterly sales trends to plan production and inventory levels effectively.

The sales quantity and selling price for the month of September 2024 are given below:

Product	Sales Quantity	Selling Price per unit (₹)
Classic Tea	1,40,000 units	20
Iced Tea	3,40,000 units	40
Herbal Infusion	4,20,000 units	20
Sparkling Tea	2,70,000 units	20

During the quarter October to December 2024, the company expects significant changes in demand due to seasonal weather variations, consumer preferences, and a marketing campaign launched for its flagship product, Classic Tea.

Based on market forecasts and internal analysis, the following estimates are made:

- The demand for Classic Tea will increase by 50% every month compared to the previous month, owing to cooler weather and the ongoing marketing campaign.
- The demand for Iced Tea will decrease by 30% each month as the weather cools down.
- The demand for Herbal Infusion will fall by 20% in October 2024 due to new competition but will remain constant thereafter.
- Due to a limited production capacity and distributor commitments, sales of Sparkling Tea will be restricted to 60,000 units in October, 50,000 units in November, and 30,000 units in December 2024.

There will be no change in the selling prices of the products during the quarter.

The standard quantity of closing stock from September 2024 to December 2024 is as follows:

Month	Classic Tea	Iced Tea	Herbal Infusion	Sparkling Tea
September 2024	12,000	13,000	11,000	7,500
October 2024	15,000	14,000	12,000	5,500
November 2024	13,000	15,000	10,000	6,000
December 2024	11,000	16,000	13,000	7,000

Additionally, the company plans to maintain sufficient stock levels to meet holiday season demand in December while ensuring that excess production is minimized due to limited storage capacity.

Based on the above data, you are required to answer the following questions (MCQ 6 to 10):

MCQ 6 2 marks

What will be the required production units for Classic Tea for each month — October, November, and December 2024 respectively?

A 2,13,000; 3,13,000; 4,70,500

B 2,10,000; 3,15,000; 4,72,500

C 2,15,000; 3,12,000; 4,75,000

D 2,14,000; 3,13,000; 4,72,000

MCQ 7 2 marks

What will be the production quantity for Iced Tea for October, November, and December 2024 respectively?

A 2,38,000; 1,67,000; 1,17,000

B 2,39,000; 1,67,600; 1,17,620

C 2,40,800; 1,69,000; 1,19,500

D 2,36,000; 1,66,000; 1,15,000

MCQ 8 2 marks

Find the production quantities for Herbal Infusion for October, November, and December 2024 respectively.

A 3,37,000; 3,34,000; 3,39,000

B 3,36,000; 3,36,000; 3,36,000

C 3,40,000; 3,38,000; 3,40,000

D 3,34,000; 3,30,000; 3,35,000

MCQ 9 2 marks

Determine the production requirements for Sparkling Tea for the three months respectively.

A 59,000; 50,000; 31,000

B 60,000; 50,000; 32,000

C 58,000; 50,500; 31,000

D 61,000; 49,000; 29,000

MCQ 10 2 marks

What will be the total sales value for Classic Tea and Iced Tea combined for the month of December 2024?

A ₹ 1,48,20,000

B ₹ 94,40,000

C ₹ 46,80,000

D ₹ 1,41,14,800

ICAI'S ANSWERS

MCQ 6

(A) 2,13,000; 3,13,000; 4,70,500

MCQ 7

(B) 2,39,000; 1,67,600; 1,17,620

MCQ 8

(A) 3,37,000; 3,34,000; 3,39,000

MCQ 9

(C) 58,000; 50,500; 31,000

MCQ 10

(D) ₹ 1,41,14,800

Working Note:

Production Budget (in units)

Particulars	Classic Tea	Iced Tea	Herbal Infusion	Sparkling Tea
October 2024				
Sales*	2,10,000	2,38,000	3,36,000	60,000
Add: Closing stock	15,000	14,000	12,000	5,500
Total Quantity Required	2,25,000	2,52,000	3,48,000	65,500
Less: Opening stock	12,000	13,000	11,000	7,500
Production	**2,13,000**	**2,39,000**	**3,37,000**	**58,000**
November 2024				
Sales*	3,15,000	1,66,600	3,36,000	50,000
Add: Closing stock	13,000	15,000	10,000	6,000
Total Quantity Required	3,28,000	1,81,600	3,46,000	56,000
Less: Opening stock	15,000	14,000	12,000	5,500
Production	**3,13,000**	**1,67,600**	**3,34,000**	**50,500**
December 2024				

Particulars	Classic Tea	Iced Tea	Herbal Infusion	Sparkling Tea
Sales*	4,72,500	1,16,620	3,36,000	30,000
Add: Closing stock	11,000	16,000	13,000	7,000
Total Quantity Required	4,83,500	1,32,620	3,49,000	37,000
Less: Opening stock	13,000	15,000	10,000	6,000
Production	**4,70,500**	**1,17,620**	**3,39,000**	**31,000**

*sales units are taken from sales budget

Sales Budget (in Units and sales value)

Particulars	Classic Tea	Iced Tea	Herbal Infusion	Sparkling Tea
October 2024 (in units)	2,10,000 [1,40,000 + (1,40,000 x 50%)]	2,38,000 [3,40,000 - (3,40,000 x 30%)]	3,36,000 [4,20,000 - (4,20,000×20%)]	60,000
October 2024 (Sales Value in ₹)	42,00,000 (2,10,000 x ₹ 20)	95,20,000 (2,38,000 x ₹ 40)	67,20,000 (3,36,000 x ₹20)	12,00,000 (60,000 x ₹ 20)
November 2024 (in units)	3,15,000 [2,10,000 + (2,10,000 x 50%)]	1,66,600 [2,38,000 - (2,38,000 x 30%)]	3,36,000	50,000
November 2024 (Sales Value in ₹)	63,00,000 (3,15,000 x ₹ 20)	66,64,000 (1,66,600 x ₹ 40)	67,20,000 (3,36,000 x ₹ 20)	10,00,000 (50,000 x ₹ 20)
December 2024 (in units)	4,72,500 [3,15,000 + (3,15,000 x 50%)]	1,16,620 [1,66,600 -(1,66,600 x 30%)]	3,36,000	30,000
December 2024 (Sales Value in ₹)	94,50,000 (4,72,500 x ₹ 20)	46,64,800 (1,16,620 x ₹ 40)	67,20,000 (3,36,000 x ₹ 20)	6,00,000 (30,000 x ₹ 20)

ICAI · P4 · MTP Jan 2026 · Series I · question p. 3-4 · answer p. 2, 2-3

1096 MTP **Jan 2026 S-II** Q3(a) 9 marks

Same question, different ICAI answer: #1094 MTP Sep 2026 S-I

Kantur Ltd. is a medium-sized manufacturer of precision components used in the automobile aftermarket. The factory runs a single production line where two product variants (Product A and Product B) are made from the same raw material and by the same grade of direct labour. Management is preparing the master budget for the next period and wants the material purchases and direct wages budgets prepared so that procurement and staffing decisions can be finalised.

Below is an extract from the company's budget working papers (all figures are for the whole budget period unless stated otherwise):

Production & material data

Particulars	Product A	Product B
Budgeted sales (units)	2,100	2,700
Budgeted material consumption per product (kg)	6.00	2.50
Budgeted material cost	₹ 14 per kg (for both products)	
Standard hours allowed per product	6.0	5.0
Budgeted basic wage rate	₹ 9 per hour (for all direct workers)	

Labour / working-time assumptions

- Overtime premium is 60% of the basic wage rate and is payable for any hours worked over 40 hours in a 5-day week.
- There are 48 direct workers available in the budget period.
- The target productivity (efficiency) ratio for productive hours actually used in manufacturing is 85%. In addition, non-productive downtime (set-up, maintenance, unavoidable delays) is budgeted at 15% of the productive hours worked.
- The budget period contains thirteen 5-day weeks and production and it is anticipated that sales and production will occur evenly throughout the whole period.

Opening and desired closing stocks

- Expected opening stocks at the start of the budget period: Product A 600 units; Product B 1,000 units; Raw material 2,800 kg.
- Target closing stocks (expressed in terms of anticipated activity during the budget period):
  - Product A — 18 days' sales
  - Product B — 22 days' sales
  - Raw material — 12 days' consumption

Other assumptions

- Production is planned to meet budgeted sales and to build up / run down stocks to the closing targets stated above.
- Sales and production occur evenly over the budget period (i.e., the daily demand pattern is smooth).

Required

PREPARE the following budgets for the next period showing quantities and values:

1. Material purchases budget.
2. Wages budget for direct workers

ICAI'S ANSWER

(i) Material Purchase Budget

(in quantities and value)

Particulars	Total
Material Consumption (Kg.)	
A (2,082 units × 6 kg = 12,492)	12,492
B (2,614 units × 2.5 kg = 6,535)	6,535
Total	19,027
Add: Closing Balance of Material (Kg.) (W.N. 3)	3,513
Less: Anticipated Opening Balance of Material (Kg.)	2,800
Total Quantity of Material (Kg.) to be purchased	19,740
Total Value of Material to be purchased (19,740 kg × ₹ 14)	₹ 2,76,360

(ii) Direct Workers Wages Budget

(Showing hours required and wages paid)

Particulars	Total
Standard Hours for Budgeted Production	
A (2,082 units × 6 hrs = 12,492)	12,492
B (2,614 units × 5 hrs = 13,070)	13,070
Total Standard Hours for Budgeted Production	25,562
Standard Hours for Budgeted Production at Targeted Efficiency Ratio (W.N. 4)	30,072.94
Add: Non-Productive Downtime (15% × 30,072.94 hours)	4,510.94
Total Labour Hours Required	34,583.88
Less: Normal Labour Hours (48 workers × 13 weeks × 5 days × 8 hours)	24,960
Difference (Overtime hours)	9,624
Wages for normal hours (24,960 hours × ₹ 9)	₹ 2,24,640.00
Overtime Wages (9,624 hrs × ₹ 14.40)	₹ 1,38,585.60
Total Wages	₹ 3,63,225.60

Working Notes:

1. Computation of Closing Stock Balance of Products A and B

Budgeted Period of Sales (In days) = 13 weeks × 5 days = 65 days

Closing stock of Product A (Units) (18 days' sales)

$= \frac{2,100}{65} \times 18 = 582$ units (rounded up)

Closing stock of Product B (Units) (22 days' sales)

$= \frac{2,700}{65} \times 22 = 914$ units (rounded up)

2. Production Budget (Units)

Particulars	Product A	Product B
Sales in units (65 days)	2,100	2,700
Add: Closing Stock (W.N. 1)	582	914
Total	2,682	3,614
Less: Anticipated Opening Balance	600	1,000
Total Number of Units to be produced	2,082	2,614

3. Closing Balance of Material (Kg.)

Total material consumption for production = 12,492 kg (A) + 6,535 kg (B)

= 19,027 kg

Daily consumption = $\frac{19,027}{65}$ = 292.72 kg/day

Closing stock of material (12 days consumption) = 292.72 × 12 = 3,513 kg (rounded up)

4. Standard Hours for Budgeted Production at targeted 85% efficiency ratio

Total standard hours for production = (2,082 × 6) + (2,614 × 5) = 25,562 hrs

Productive hours required = Standard hours ÷ Efficiency

$= \frac{25,562}{0.85}$ = 30,072.94 hrs

Non-productive downtime = 15% of productive hours = 0.15 × 30,072.94

= 4,510.94 hrs

Total labour hours required = 30,072.94 + 4,510.94 = 34,583.88 hrs

ICAI · P4 · MTP Jan 2026 · Series II · question p. 12-13 · answer p. 13-15

1097 MTP **Sep 2025 S-I** Case Scenario 2 · MCQ 6 to MCQ 10 10 marks

CASE SCENARIO 2

Valley Ltd., a medium-sized manufacturing firm, is reviewing its operational strategy for Q2 of 2025 due to an anticipated rise in market demand for its signature product, 'X'—a pre-packaged consumer item. To maintain profitability and manage costs efficiently, the management team is preparing a detailed budget.

The following information are made available for this purpose:

(a) It expects to sell 50,000 bags of 'X' during the second quarter of 2025 at the selling price of ₹ 9 per bag.

(b) Each bag of 'X' requires 2.5 kgs. of a raw – material called 'Y' and 7.5 kgs. of raw – material called 'Z'.

(c) Stock levels are planned as follows:

Particulars	Beginning of Quarter	End of Quarter
Finished Bags of 'X' (Nos.)	15,000	11,000
Raw – Material 'Y' (Kgs.)	32,000	26,000
Raw – Material 'Z' (Kgs.)	57,000	47,000
Empty Bag (Nos.)	37,000	28,000

(d) 'Y' cost ₹ 1.20 per Kg., 'Z' costs 20 paise per Kg. and 'Empty Bag' costs 80 paise each.

(e) It requires 9 minutes of direct labour to produce and fill one bag of 'X'. Labour cost is ₹ 5 per hour.

(f) Variable manufacturing costs are ₹ 0.45 bag. Fixed manufacturing costs ₹ 30,000 per quarter.

(g) Variable selling and administration expenses are 5% of sales and fixed administration and selling expenses are ₹ 25,000 per quarter.

As part of the budgeting exercise, management is looking for clarity on production needs, raw material purchases, cost per unit, and profitability to support strategic decisions and investor confidence.

Based on above information, you are required to answer the following (MCQs 6 to 10):

MCQ 6 2 marks

The required production of "X" in second quarter will be:

A 45,000 bags

B 46,000 bags

C 61,000 bags

D 50,000 bags

MCQ 7 2 marks

What is the quantity to be purchased for 'Y', 'Z' and 'Empty bags'?

A 1,41,000, 3,92,000 and 74,000

B 1,15,000, 3,45,000 and 46,000

C 1,30,800, 67,000 and 26,600

D 1,09,000, 3,35,000 and 37,000

MCQ 8 2 marks

What is the cost of quantity purchased for 'Y', 'Z' and 'Empty bags'

A 1,30,800, 67,000 and 29,600

B 1,09,000, 3,35,000 and 37,000

C 1,41,000, 3,92,000 and 74,000

D 1,15,000, 3,45,000 and 46,000

MCQ 9 2 marks

What is the budgeted variable cost of producing one bag of 'X'.

A ₹ 5.50

B ₹ 4.75

C ₹ 6.50

D ₹ 6.05

MCQ 10 2 marks

What is the Budgeted Net Income for the Second Quarter

A ₹ 30,000

B ₹ 1,25,000

C ₹ 75,000

D ₹ 47,500

ICAI'S ANSWERS

MCQ 6

Answer key: **(B)**

(b) 46,000 bags

MCQ 7

Answer key: **(D)**

(d) 1,09,000, 3,35,000 and 37,000

MCQ 8

Answer key: **(A)**

(a) 1,30,800, 67,000 and 29,600

MCQ 9

Answer key: **(C)**

(c) ₹ 6.50

MCQ 10

Answer key: **(D)**

(d) ₹ 47,500

Working Note:

(i) Production Budget of 'X' for the Second Quarter

Particulars	Bags (Nos.)
Budgeted Sales	50,000
Add: Desired Closing stock	11,000
Total Requirements	61,000
Less: Opening stock	15,000
Required Production	46,000

(ii) Raw–Materials Purchase Budget in Quantity as well as in ₹ for 46,000 Bags of 'X'

Particulars	'Y' Kgs.	'Z' Kgs.	Empty Bags Nos.
Production Requirements Per bag of 'X'	2.5	7.5	1.0
Requirement for Production	1,15,000 (46,000 × 2.5)	3,45,000 (46,000 × 7.5)	46,000 (46,000 × 1)
Add: Desired Closing Stock	26,000	47,000	28,000
Total Requirements	1,41,000	3,92,000	74,000
Less: Opening Stock	32,000	57,000	37,000
Quantity *to be purchased*	1,09,000	3,35,000	37,000
Cost per Kg./Bag	₹ 1.20	₹ 0.20	₹ 0.80
Cost of Purchase (₹)	1,30,800	67,000	29,600

(iii) Computation of Budgeted Variable Cost of Production of 1 Bag of 'X'

Particulars	(₹)
Raw – Material	
Y 2.5 Kg @1.20	3.00
Z 7.5 Kg. @0.20	1.50
Empty Bag	0.80
Direct Labour (₹ 5 × 9 minutes / 60 minutes)	0.75
Variable Manufacturing Overheads	0.45
Variable Cost of Production *per bag*	6.50

(iv) Budgeted Net Income for the Second Quarter

Particulars	Per Bag (₹)	Total (₹)
Sales Value (50,000 Bags)	9.00	4,50,000
Less: Variable Cost:		
Production Cost	6.50	3,25,000
Admn. & Selling Expenses (5% of Sales Price)	0.45	22,500
Budgeted Contribution	2.05	1,02,500
Less: Fixed Expenses:		
Manufacturing		30,000

Particulars	Per Bag (₹)	Total (₹)
Admn. & Selling		25,000
Budgeted Net Income		47,500

ICAI · P4 · MTP Sep 2025 · Series I · question p. 3-4 · answer p. 2, 2-4

1098 MTP **Sep 2025 S-I** MCQ 15 2 marks MCQ

A company prepares its monthly production budget using the following policy:

- 70% of the current month's sales are to be produced in the same month.
- The company maintains an opening finished goods inventory of 6,000 units for June.
- The desired closing inventory for June is 30% of July's forecasted sales.

The sales forecasts are as follows:

Month	Sales Forecast (Units)
May	20,000
June	24,000
July	30,000
August	18,000

What is the production budget for June?

A 25,200

B 28,200

C 27,000

D 19,800

ICAI'S ANSWER

Answer key: **(D)**

(d) 19,800

Production = 70% of June sales + 30% of July sales (Closing inventory) – Opening inventory

= 24,000 x 70% + 30,000 x 30% - 6,000

= 19,800 units

ICAI · P4 · MTP Sep 2025 · Series I · question p. 6-7 · answer p. 5

1099 MTP **Sep 2025 S-II** MCQ 12 2 marks MCQ

A company plans to produce 50,000 units during the budget period. To produce 1 unit of finished goods, the company requires 5 units of raw material, including an allowance for 5% raw material wastage during production. The company currently has 90,000 units of raw material in stock and wishes to maintain an ending inventory of 1,20,000 units of raw material at the end of the budget period.

How many units of raw material should the company plan to purchase during the budget period?

A 3,45,000 units

B 3,50,000 units

C 2,80,000 units

D 3,70,000 units

ICAI'S ANSWER

(C) 2,80,000 units

Raw material required for production (including 5% wastage)

= 50,000 units x 5 = 2,50,000 units

Raw material to be purchased

= Required for production + Desired closing stock – Opening stock

= 2,50,000 + 1,20,000– 90,000 = 2,80,000 units

ICAI · P4 · MTP Sep 2025 · Series II · question p. 6-7 · answer p. 6

1100 MTP **May 2025 S-I** Case scenario · Q4(b)(i) to Q4(b)(iv) 8 marks

CASE SCENARIO

A company manufactures two products, X and Y, which are sold to retailers. These products are part of the company's core offerings and are crucial to meeting the demands of its retail partners. The company is anticipating a specific sales volume for each product in the upcoming quarter, which is expected to play a significant role in production planning, inventory management, and overall financial forecasting.

The budgeted sales volumes for the next quarter are as follows:

Product	Units
X	64,000
Y	1,12,000

The finished goods inventory is projected to rise by 2,000 units of Product X and fall by 4,000 units of Product Y by the end of the quarter.

Raw materials A and B are essential for the production of both products. The quantities needed of each raw material to produce one unit of the finished product, along with their purchase prices, are outlined in the table below:

	Raw Material A	Raw Material B
Product X	24 kg	12 kg
Product Y	12 kg	9 kg
Purchase price per kg	₹ 18.75	₹ 27.00
Budgeted opening inventory	1,80,000 kg	1,20,000 kg

The company intends to maintain a raw material inventory at the end of the quarter equivalent to 5% of the quarter's total raw material usage budget.

Additional information is provided relating to production of the products, X and Y:

(i) Product X requires 16 minutes of direct labour, while Product Y requires 20 minutes of direct labour. Labour cost is ₹ 540 for 1 day of 9 hours.

(ii) Variable manufacturing overhead of ₹ 8 per unit is same for both the products.

(iii) Variable selling and administration expenses are 10% of sales.

You are required to:

Q4(b)(i) 2 marks

PREPARE Production budget for the quarter(in units)

Q4(b)(ii) 2 marks

PREPARE Raw material usage budget for the quarter (in kg)

Q4(b)(iii) 2 marks

PREPARE Raw material purchase budget for the quarter(in kg and ₹)

Q4(b)(iv) 2 marks

COMPUTE the budgeted variable cost to produce one unit of Product Y.

ICAI'S ANSWERS

Q4(b)(i)

Production budget (in units)

Particulars	Product X	Product Y
Budgeted Sales (units)	64,000	1,12,000
Increase/(decrease) in inventory	2,000	(4,000)
Production budget (units)	66,000	108,000

Q4(b)(ii)

Raw material usage budget (in kg)

	Raw material A			Raw material B		
	X	**Y**	**Total**	**X**	**Y**	**Total**
Production budget (units) (a)	66,000	108,000	1,74,000	66,000	108,000	1,74,000
Kg per unit (b)	24	12	--	12	9	--
Material usage (kg) (a x b)	15,84,000	12,96,000	28,80,000	7,92,000	9,72,000	17,64,000

Q4(b)(iii)

Raw material purchase budget (in kg and ₹)

Particulars	Raw Material A	Raw material B
Material usage (kg) (Total from (ii))	28,80,000	17,64,000
Add: Desired Closing Stock (5% of the quarter's material usage Budget)	1,44,000	88,200
Total Requirements	30,24,000	18,52,200
Less: Opening Stock	(1,80,000)	(1,20,000)
Quantity to be purchased (kg)	28,44,000	17,32,200
Price per kg	₹ 18.75	₹ 27.00
Cost of Purchase (₹)	5,33,25,000	4,67,69,400

Q4(b)(iv)

Computation of the budgeted variable cost to produce one unit of Product Y

Particulars	(₹)
Raw – Material	
A: 12 kg @ ₹ 18.75	225
B: 9 kg @ ₹ 27.00	243
Direct Labour (₹ 540/9 hours x 60 minutes) x 20 minutes	20
Variable Manufacturing Overheads	8
Variable Cost of Production per unit of Product Y	**496**

ICAI · P4 · MTP May 2025 · Series I · question p. 12-13 · answer p. 17, 18

1101 MTP **Jan 2025 S-II** Q5(b) 6 marks

C Ltd. manufactures two products using two types of materials and one grade of labour. Shown below is an extract from the company's working papers for the next month's budget:

	Product-A	Product-B
Budgeted sales (in units)	2,400	3,600
Budgeted material consumption per unit (in kg):		
Material-X	5	3
Material-Y	4	6
Standard labour hours allowed per unit of product	3	5

Material-X and Material-Y cost ₹ 4 and ₹ 6 per kg and labours are paid ₹ 25 per hour. Overtime premium is 50% and is paid, if a worker works for more than 40 hours a week. There are 180 direct workers.

The target productivity ratio (or efficiency ratio) for the productive hours worked by the direct workers in actually manufacturing the products is 80%. In addition, the non-productive down-time is budgeted at 20% of the hours worked.

There are four 5-days weeks in the budgeted period and it is anticipated that sales and production will occur evenly throughout the whole period.

It is anticipated that stock at the beginning of the period will be:

Product-A	400 units
Product-B	200 units
Material-X	1,000 kg.
Material-Y	500 kg.

The anticipated closing stocks for budget period are as below:

Product-A	4 days sales
Product-B	5 days sales
Material-X	10 days consumption
Material-Y	6 days consumption

Required:

CALCULATE the Material Purchase Budget and the Wages Budget for the direct workers, showing the quantities and values, for the next month.

ICAI'S ANSWER

Number of days in budget period = 4 weeks × 5 days = 20 days

Number of units to be produced

	Product-A (units)	Product-B (units)
Budgeted Sales	2,400	3,600
Add: Closing stock	480 (2,400 units/20 days × 4 days)	900 (3,600 units/20 days × 5 days)
Less: Opening stock	(400)	(200)
	2,480	4,300

(i) **Material Purchase Budget**

	Material-X (Kg.)	Material-Y (Kg.)
Material required:		
- Product-A	12,400 (2,480 units × 5 kg.)	9,920 (2,480 units × 4 kg.)
- Product-B	12,900 (4,300 units × 3 kg.)	25,800 (4,300 units × 6 kg.)

	Material-X (Kg.)	Material-Y (Kg.)
	25,300	35,720
Add: Closing stock	12,650 (25,300kgs./20days × 10days)	10,716 (35,720kgs./20days × 6days)
Less: Opening stock	(1,000)	(500)
Quantity to be purchased	36,950	45,936
Rate per kg. of Material	₹ 4	₹ 6
Total Cost	₹ 1,47,800	₹ 2,75,616

(ii) **Wages Budget**

	Product-A (Hours)	Product-B (Hours)
Units to be produced	2,480 units	4,300 units
Standard hours allowed per unit	3	5
Total Standard Hours allowed	7,440	21,500
Productive hours required for production	7,440hours/80% =9,300	21,500hours/80% =26,875
Add: Non-Productive down time	1,860 hours. (20% of 9,300 hours)	5,375 hours. (20% of 26,875 hours)
Hours to be paid	11,160	32,250

Total Hours to be paid = 43,410 hours (11,160 + 32,250)

Hours to be paid at normal rate = 4 weeks × 40 hours × 180 workers

= 28,800 hours

Hours to be paid at premium rate = 43,410 hours – 28,800 hours

= 14,610 hours

Total wages to be paid = 28,800 hours × ₹ 25 + 14,610 hours × ₹ 37.5

= ₹ 7,20,000 + ₹ 5,47,875

= ₹ 12,67,875

ICAI · P4 · MTP Jan 2025 · Series II · question p. 12-13 · answer p. 16-17

1102 MTP **Sep 2024 S-I** Q5(a) 7 marks

Bicon Ltd. manufactures two products using two types of materials and one grade of labour. Shown

below is an extract from the company's working papers for the next month's budget:

	Product - A	Product-B
Budgeted sales (in units)	2,400	3,600
Budgeted material consumption per unit (in kg):		
Material-X	5	3
Material-Y	4	6
Standard labour hours allowed per unit of product	3	5

Material-X and Material-Y cost ₹ 4 and ₹ 6 per kg and labours are paid ₹ 25 per hour. Overtime premium is 50% and is payable, if a worker works for more than 40 hours a week. There are 180 direct workers.

The target productivity ratio (or efficiency ratio) for the productive hours worked by the direct workers in actually manufacturing the products is 80%. In addition the non-productive down-time is budgeted at 20% of the productive hours worked.

There are four 5-days weeks in the budgeted period and it is anticipated that sales and production will occur evenly throughout the whole period.

It is anticipated that stock at the beginning of the period will be:

Product-A	400 units
Product-B	200 units
Material-X	1,000 kgs.
Material-Y	500 kgs.

The anticipated closing stocks for budget period are as below:

Product-A	4 days sales
Product-B	5 days sales
Material-X	10 days consumption
Material-Y	6 days consumption

Required:

CALCULATE the Material Purchase Budget and the Wages Budget for the direct workers, showing the quantities and values, for the next month.

ICAI'S ANSWER

Number of days in budget period = 4 weeks × 5 days = 20 days

Number of units to be produced

	Product-A (units)	Product-B (units)
Budgeted Sales	2,400	3,600
Add: Closing stock (2,400units/20days × 4days) (3,600units/20days × 5days)	480	900
Less: Opening stock	400	200
Production (units)	2,480	4,300

(i) Material Purchase Budget

	Material-X (Kg.)	Material-Y (Kg.)
Material required:		
Product-A	12,400 (2,480 units × 5 kg.)	9,920 (2,480 units × 4 kg.)
Product-B	12,900 (4,300 units × 3 kg.)	25,800 (4,300 units × 6 kg.)
	25,300	35,720
Add: Closing stock (25,300kgs./20days × 10days) (35,720kgs./20days × 6days)	12,650	10,716
Less: Opening stock	1,000	500
Quantity to be purchased	36,950	45,936
Rate per kg. of Material	₹ 4	₹ 6
Total Cost	₹ 1,47,800	₹ 2,75,616

(ii) Wages Budget

	Product-A (Hours)	Product-B (Hours)
Units to be produced	2,480 units	4,300 units
Standard hours allowed per unit	3	5
Total Standard Hours allowed	7,440	21,500
Productive hours required for production	7,440hours/80% =9,300	21,500hours/80% =26,875
Add: Non-Productive down time	1,860 hours. (20% of 9,300 hours)	5,375 hours. (20% of 26,875 hours)
Hours to be paid	11,160	32,250

Total Hours to be paid	= 43,410 hours (11,160 + 32,250)
Hours to be paid at normal rate	= 4 weeks × 40 hours × 180 workers = 28,800 hours
Hours to be paid at premium rate	= 43,410 hours – 28,800 hours = 14,610 hours
Total wages to be paid	= 28,800 hours × ₹ 25 + 14,610 hours × ₹ 37.5 = ₹ 7,20,000 + ₹ 5,47,875 = ₹ 12,67,875

ICAI · P4 · MTP Sep 2024 · Series I · question p. 11 · answer p. 13-14

1103 MTP **May 2024 S-I** Case scenario · Q1(a)(i) to Q1(c) 14 marks

CASE SCENARIO

P Ltd. manufactures two products called 'X' and 'Y'. Both products use a common raw material Z. The raw material Z is purchased @ ₹ 72 per kg from the market. The company has decided to review inventory management policies for the forthcoming year.

The following forecast information has been extracted from departmental estimates for the year ended 31st March 2025 (the budget period):

	Product X	Product Y
Sales (units)	28,000	13,000
Finished goods stock increase by year-end	320	160
Post-production rejection rate (%)	4	6
Material Z usage (per completed unit, net of wastage)	5 kg	6 kg
Material Z wastage (%)	10	5

Additional information:

- Usage of raw material Z is expected to be at a constant rate over the period.
- Annual cost of holding one unit of raw material in stock is 11% of the material cost.
- The cost of placing an order is ₹ 15,600 per order.
- The management of P Ltd. has decided that there should not be more than 40 orders in a year for the raw material Z.

Required:

Q1(a)(i) 3 marks

Prepare Production budget for Products X and Y (in units) for the year ended 31st March 2025.

Q1(a)(ii) 2 marks

Calculate the Economic Order Quantity for Material Z (in kgs).

Q1(b) 5 marks

Prepare Purchases budget for Material Z (in kgs and value) for the year ended 31st March 2025.

Q1(c) 4 marks

If there is a sole supplier for the raw material Z in the market and the supplier do not sale more than 4,000 kg. of material Z at a time. Keeping the management purchase policy and production quantity mix into consideration, calculate the maximum number of units of Product X and Y that could be produced.

ICAI'S ANSWERS

Q1(a)(i)

Production Budget (in units) for the year ended 31st March 2025

	Product X	Product Y
Budgeted sales (units)	28,000	13,000
Add: Increase in closing stock	320	160
No. good units to be produced	28,320	13,160
Post production rejection rate	4%	6%
No. of units to be produced	29,500 (28,320/0.96)	14,000 (13,160/0.94)

Q1(a)(ii)

Calculation of Economic Order Quantity for Material Z

$$\text{EOQ} = \sqrt{\frac{2 \times 2{,}52{,}310 \times 15{,}600}{72 \times 11\%}} = \sqrt{\frac{5{,}04{,}620 \times 15{,}600}{72 \times 11\%}} = 31{,}526.95 \text{ kg.}$$

Q1(b)

Purchase budget (in kgs and value) for Material Z

	Product X	Product Y
No. of units to be produced	29,500	14,000
Usage of Material Z per unit of production	5 kg.	6 kg.
Material needed for production	1,47,500 kg.	84,000 kg.
Materials to be purchased	1,63,889 kg. (1,47,500/0.90)	88,421 kg. (84,000/0.95)
Total quantity to be purchased	2,52,310 kg.	
Rate per kg. of Material Z	₹72	
Total purchase price	₹1,81,66,320	

Q1(c)

Since, the maximum number of orders per year cannot be more than 40 orders and the maximum quantity per order that can be purchased is 4,000 kg. Hence, the total quantity of Material Z that can be available for production:

= 4,000 kg. × 40 orders =1,60,000 kg.

	Product X	Product Y
Material needed for production to maintain the same production mix	1,03,929 kg. (1,60,000 × 1,63,889/2,52,310)	56,071 kg. (1,60,000 × 88,421/2,52,310)
Less: Process wastage	10,393 kg.	2,804 kg.
Net Material available for production	93,536 kg.	53,267 kg.
Units to be produced	18,707 units (93,536kg./5kg.)	8,878 units (53,267kg./6kg.)

ICAI · P4 · MTP May 2024 · Series I · question p. 6 · answer p. 4, 4-5, 5

1104 MTP **Old P3 Nov 2023 S-I** Q3(b) 10 marks old scheme

G Ltd. manufactures two products called 'M' and 'N'. Both products use a common raw material Z. The raw material Z is purchased @ ₹ 36 per kg from the market. The company has decided to review inventory management policies for the forthcoming year.

The following information has been extracted from departmental estimates for the year ended 31st March 2023 (the budget period):

	Product M	Product N
Sales (units)	28,000	13,000
Finished goods stock increase by year-end	320	160
Post-production rejection rate (%)	4	6
Material Z usage (per completed unit, net of wastage)	5 kg	6 kg
Material Z wastage (%)	10	5

Additional information:

- Usage of raw material Z is expected to be at a constant rate over the period.
- Annual cost of holding one unit of raw material in stock is 11% of the material cost.
- The cost of placing an order is ₹ 320 per order.
- The management of G Ltd. has decided that there should not be more than 40 orders in a year for the raw material Z.

Required:

(a) Prepare functional budgets for the year ended 31st March 2023 under the following headings:

(i) Production budget for Products M and N (in units).

(ii) Purchases budget for Material Z (in kgs and value).

(b) Calculate the Economic Order Quantity for Material Z (in kgs).

ICAI'S ANSWER

(a) (i) Production Budget (in units) for the year ended 31st March 2023

	Product M	Product N
Budgeted sales (units)	28,000	13,000
Add: Increase in closing stock	320	160
No. good units to be produced	28,320	13,160
Post production rejection rate	4%	6%
No. of units to be produced	29,500 (28,320/0.96)	14,000 (13,160/0.94)

(ii) Purchase budget (in kgs and value) for Material Z

	Product M	Product N
No. of units to be produced	29,500	14,000
Usage of Material Z per unit of production	5 kg.	6 kg.
Material needed for production	1,47,500 kg.	84,000 kg.
Materials to be purchased	1,63,889 kg. (1,47,500/0.90)	88,421 kg. (84,000/0.95)
Total quantity to be purchased	2,52,310 kg.	
Rate per kg. of Material Z	₹36	
Total purchase price	₹90,83,160	

(b) Calculation of Economic Order Quantity for Material Z

$$\text{EOQ} = \sqrt{\frac{2 \times 2{,}52{,}310 \text{ kg.} \times ₹320}{₹36 \times 11\%}} = \sqrt{\frac{16{,}14{,}78{,}400}{₹3.96}} = 6{,}385.72 \text{ kg.}$$

ICAI · Old P3 · MTP Nov 2023 · Series I · question p. 4-5 · answer p. 7-8

1105 MTP **Old P3 Nov 2023 S-II** Q5(a) 10 marks old scheme

Same question, different ICAI answer: #1107 Old P3 MTP Nov 2022 S-II

N Ltd., a vehicle manufacturer, has prepared sales budget for the next few months, and the following draft figures are available:

Month	No. of vehicles
October	40,000
November	35,000
December	45,000
January	60,000
February	65,000

To manufacture a vehicle a standard cost of ₹5,71,400 is incurred and sold through dealers at a uniform selling price of ₹8,57,100 to customers. Dealers are paid 15% commission on selling price on sale of a vehicle.

Apart from other materials four units of Part - X are required to manufacture a vehicle. It is a policy of the company to hold stocks of Part-X at the end of each month to cover 40% of next month's production. 48,000 units of Part-X are in stock as on 1st October.

There are 9,500 nos. of completed vehicles are in stock as on 1st October and it is policy to have stocks at the end of each month to cover 20% of the next month's sales.

You are required to -

(i) Prepare Production budget (in nos.) for the month of October, November, December and January.

(ii) Prepare a Purchase budget for Part-X (in units) for the months of October, November and December.

(iii) Calculate the budgeted gross profit for the quarter October to December.

ICAI'S ANSWER

(i) Preparation of Production Budget (in units)

	October	November	December	January
Demand for the month (Nos.)	40,000	35,000	45,000	60,000
Add: 20% of next month's demand	7,000	9,000	12,000	13,000
Less: Opening Stock	(9,500)	(7,000)	(9,000)	(12,000)
Vehicles to be produced	37,500	37,000	48,000	61,000

(ii) Preparation of Purchase budget for Part-X

	October	November	December
Production for the month (Nos.)	37,500	37,000	48,000
Add: 40% of next month's production	14,800 (40% of 37,000)	19,200 (40% of 48,000)	24,400 (40% of 61,000)
	52,300	56,200	72,400
No. of units required for production	2,09,200 (52300 × 4 units)	2,24,800 (56200 × 4 units)	2,89,600 (72,400 × 4 units)
Less: Opening Stock	(48,000)	(59,200) (14800 × 4 units)	(76,800) (19200 × 4 units)
No. of units to be purchased	1,61,200	1,65,600	2,12,800

(iii) Budgeted Gross Profit for the Quarter October to December

	October	November	December	Total
Sales in nos.	40,000	35,000	45,000	1,20,000
Net Selling Price per unit*	7,28,535	7,28,535	7,28,535	
Sales Revenue (₹ in lakh)	2,91,414	2,54,987.25	3,27,840.75	8,74,242
Less: Cost of Sales (₹ in lakh) (Sales unit × Cost per unit)	2,28,560	1,99,990.00	2,57,130.00	6,85,680
Gross Profit (₹ in lakh)	62,854	54,997.25	70,710.75	1,88,562

* Net Selling price unit = ₹ 8,57,100 – 15% commission on ₹ 8,57,100 = ₹7,28,535.

ICAI · Old P3 · MTP Nov 2023 · Series II · question p. 6 · answer p. 11-12

1106 MTP **Old P3 May 2023 S-II** Q2(a) 10 marks old scheme

JD Ltd. produces and markets a very popular product called 'X'. The company is interested in presenting its budget for the first quarter of 2023.

The following information are made available for this purpose:

(i) It expects to sell 50,000 bags of 'X' during the first quarter of 2023 at the selling price of ₹ 900 per bag.

(ii) Each bag of 'X' requires 2.5 kgs. of a raw – material called 'Y' and 7.5 kgs. of raw – material called 'Z'.

(iii) Stock levels are planned as follows:

Particulars	Beginning of Quarter	End of Quarter
Finished Bags of 'X' (Nos.)	15,000	11,000
Raw – Material 'Y' (Kgs.)	32,000	26,000
Raw – Material 'Z' (Kgs.)	57,000	47,000
Empty Bag (Nos.)	37,000	28,000

(iv) 'Y' cost ₹120 per Kg., 'Z' costs ₹20 per Kg. and 'Empty Bag' costs ₹80 each.

(v) It requires 9 minutes of direct labour to produce and fill one bag of 'X'. Labour cost is ₹50 per hour.

(vi) Variable manufacturing costs are ₹45 bag. Fixed manufacturing costs ₹30,00,000 per quarter.

(vii) Variable selling and administration expenses are 5% of sales and fixed administration and selling expenses are ₹20,50,000 per quarter.

Required

(i) PREPARE a production budget for the said quarter.

(ii) PREPARE a raw – material purchase budget for 'Y', 'Z' and 'Empty Bags' for the said quarter in quantity as well as in rupees.

(iii) COMPUTE the budgeted variable cost to produce one bag of 'X'.

(iv) PREPARE a statement of budgeted net income for the said quarter and show both per unit and total cost data.

ICAI'S ANSWER

(i) Production Budget of 'X' for the first Quarter

Particulars	Bags (Nos.)
Budgeted Sales	50,000
Add: Desired Closing stock	11,000
Total Requirements	61,000
Less: Opening stock	15,000
Required Production	46,000

(ii) Raw–Materials Purchase Budget in Quantity as well as in ₹ for 46,000 Bags of 'X'

Particulars	'Y'	'Z'	Empty Bags
	Kgs.	**Kgs.**	**Nos.**
Production Requirements Per bag of 'X'	2.5	7.5	1.0
Requirement for Production	1,15,000 (46,000 × 2.5)	3,45,000 (46,000 × 7.5)	46,000 (46,000 × 1)
Add: Desired Closing Stock	26,000	47,000	28,000
Total Requirements	1,41,000	3,92,000	74,000
Less: Opening Stock	32,000	57,000	37,000
Quantity *to be purchased*	1,09,000	3,35,000	37,000
Cost per Kg./Bag	₹120	₹20	₹80
Cost of Purchase (₹)	1,30,80,000	67,00,000	29,60,000

(iii) Computation of Budgeted Variable Cost of Production of 1 Bag of 'X'

Particulars	(₹)
Raw – Material	
Y 2.5 Kg @120	300.00
Z 7.5 Kg. @20	150.00
Empty Bag	80.00
Direct Labour(₹50× 9 minutes / 60 minutes)	7.50
Variable Manufacturing Overheads	45.00
Variable Cost of Production *per bag*	582.50

(iv) Budgeted Net Income for the first Quarter

Particulars	Per Bag (₹)	Total (₹)
Sales Value (50,000 Bags)	900.00	4,50,00,000
Less: Variable Cost:		
Production Cost	582.50	2,91,25,000
Admn. & Selling Expenses (5% of Sales Price)	45.00	22,50,000
Budgeted Contribution	272.50	1,36,25,000
Less: Fixed Expenses:		
Manufacturing		30,00,000
Admn. & Selling		20,50,000
Budgeted Net Income		85,75,000

ICAI · Old P3 · MTP May 2023 · Series II · question p. 2-3 · answer p. 3-4

1107 MTP **Old P3 Nov 2022 S-II** Q4(a) 10 marks old scheme

Same question, different ICAI answer: #1105 Old P3 MTP Nov 2023 S-II

N Ltd a vehicle manufacturer has prepared sales budget for the next few months, and the following draft figures are available:

Month	No. of vehicles
October	40,000
November	35,000
December	45,000
January	60,000
February	65,000

To manufacture a vehicle a standard cost of ₹5,71,400 is incurred and sold through dealers at a uniform selling price of ₹8,57,100 to customers. Dealers are paid 15% commission on selling price on sale of a vehicle.

Apart from other materials four units of Part - X are required to manufacture a vehicle. It is a policy of the company to hold stocks of Part-X at the end of each month to cover 40% of next month's production. 48,000 units of Part-X are in stock as on 1st October.

There are 9,500 nos. of completed vehicles are in stock as on 1st October and it is policy to have stocks at the end of each month to cover 20% of the next month's sales.

You are required to

(i) PREPARE Production budget (in nos.) for the month of October, November, December and January.

(ii) PREPARE a Purchase budget for Part-X (in units) for the months of October, November and December.

(iii) CALCULATE the budgeted gross profit for the quarter October to December.

ICAI'S ANSWER

(i) Preparation of Production Budget (in units)

	October	November	December	January
Demand for the month (Nos.)	40,000	35,000	45,000	60,000
Add: 20% of next month's demand	7,000	9,000	12,000	13,000
Less: Opening Stock	(9,500)	(7,000)	(9,000)	(12,000)
Vehicles to be produced	37,500	37,000	48,000	61,000

(ii) Preparation of Purchase budget for Part-X

	October	November	December
Production for the month (Nos.)	37,500	37,000	48,000
Add: 40% of next month's production	14,800 (40% of 37,000)	19,200 (40% of 48,000)	24,400 (40% of 61,000)
	52,300	56,200	72,400
No. of units required for production	2,09,200 (52300 × 4 units)	2,24,800 (56200 × 4 units)	2,89,600 (72,400 × 4 units)
Less: Opening Stock	(48,000)	(59,200) (14800 × 4 units)	(76,800) (19200 × 4 units)
No. of units to be purchased	1,61,200	1,65,600	2,12,800

(iii) Budgeted Gross Profit for the Quarter October to December

	October	November	December	Total
Sales in nos.	40,000	35,000	45,000	1,20,000
Net Selling Price per unit*	7,28,535	7,28,535	7,28,535	
Sales Revenue (Rs. in lakh)	2,91,414	2,54,987.25	3,27,840.75	8,74,242
Less: Cost of Sales (Rs. in lakh) (Sales unit × Cost per unit)	2,28,560	1,99,990.00	2,57,130.00	6,85,680
Gross Profit (Rs. in lakh)	62,854	54,997.25	70,710.75	1,88,562

* Net Selling price unit = Rs. 8,57,100 – 15% commission on Rs. 8,57,100 = Rs.7,28,535.

ICAI · Old P3 · MTP Nov 2022 · Series II · question p. 4-5 · answer p. 9-10

1108 MTP Old P3 May 2022 S-II Q1(d) 5 marks old scheme

F Ltd. requires you to PREPARE the Master budget for the next year from the following information:

Sales	₹ 1,20,00,000
Direct material cost	60% of sales
Direct wages	20 workers @ ₹ 2,250 per month
Factory overheads:	
Indirect labour –	
Works manager	₹ 7,500 per month
Foreman	₹ 6,000 per month
Stores and spares	2.5% on sales
Depreciation on machinery	₹ 1,89,000
Light and power (fixed)	₹ 45,000
Repairs and maintenance	₹ 1,20,000
Other sundries	10% on direct wages
Administration, selling and distribution expenses	₹ 5,40,000 per year

ICAI'S ANSWER

Master Budget for the year ending _____

Particulars		Amount (₹)	Amount (₹)
Sales			1,20,00,000
Less: Cost of production:			
Direct materials (60% of ₹ 1,20,00,000)		72,00,000	
Direct wages (20 workers × ₹ 2,250 × 12 months)		5,40,000	
Prime Cost		77,40,000	
Fixed Factory Overhead:			
Works manager's salary (7,500 × 12)	90,000		
Foreman's salary (6,000 × 12)	72,000		
Depreciation	1,89,000		
Light and power	45,000	3,96,000	
Variable Factory Overhead:			

Particulars		Amount (₹)	Amount (₹)
Stores and spares (2.5% of ₹ 1,20,00,000)	3,00,000		
Repairs and maintenance	1,20,000		
Sundry expenses (10% of ₹ 5,40,000)	54,000	4,74,000	
Works Cost			86,10,000
Gross Profit (Sales – Works cost)			33,90,000
Less: Adm., selling and distribution expenses			5,40,000
Net Profit			28,50,000

ICAI · Old P3 · MTP May 2022 · Series II · question p. 2 · answer p. 3

1109 MTP **Old P3 Dec 2021 S-II** Q5(b) 10 marks old scheme

T Ltd manufactures and sells a single product and has estimated sales revenue of ₹1,51,20,000 during the year based on 20% profit on selling price. Each unit of product requires 6 kg of material A and 3 kg of material B and processing time of 4 hours in machine shop and 2 hours in assembly shop. Factory overheads are absorbed at a blanket rate of 20% of direct labour. Variable selling & distribution overheads are ₹30 per unit sold and fixed selling & distribution overheads are estimated to be ₹34,56,000.

The other relevant details are as under:

Purchase Price:	Material A	₹80 per kg
	Materials B	₹50 per kg
Labour Rate:	Machine Shop	₹70 per hour
	Assembly Shop	₹35 per hour

	Finished Stock	Material A	Material B
Opening Stock	2,500 units	7,500 kg	4,000 kg
Closing Stock	3,000 units	8,000 kg	5,500 kg

Required

(i) CALCULATE number of units of product proposed to be sold and selling price per unit,

(ii) PREPARE Production Budget in units and

(iii) PREPARE Material Purchase Budget in units.

ICAI'S ANSWER

Workings

Statement Showing "Total Variable Cost for the year"

Particulars	Amount (₹)
Estimated Sales Revenue	1,51,20,000
Less: Desired Profit Margin on Sale @ 20%	30,24,000
Estimated Total Cost	1,20,96,000
Less: Fixed Selling and Distribution Overheads	34,56,000
Total Variable Cost	86,40,000

Statement Showing "Variable Cost per unit"

Particulars	Variable Cost p.u. (₹)
Direct Materials:	
A: 6 Kg. @ ₹80 per kg.	480
B: 3 Kg. @ ₹50 per kg.	150
Labour Cost:	
Machine Shop: 4 hrs. @ ₹70 per hour	280
Assembly Shop: 2 hrs. @ ₹35 per hour	70
Factory Overheads: 20% of (₹280 + ₹70)	70
Variable Selling & Distribution Expenses	30
Total Variable Cost per unit	1,080

(i) Calculation of number of units of product proposed to be sold and selling price per unit:

Number of Units Sold	=	Total Variable Cost / Variable Cost per unit
	=	₹ 86,40,000 / ₹ 1,080
	=	8,000 units
Selling Price per unit	=	Total Sales Value / Number of Units Sold
	=	₹ 1,51,20,000 / 8,000 units
	=	₹ 1,890

(ii) Production Budget (units)

Particulars	Units
Budgeted Sales	8,000
Add: Closing Stock	3,000
Total Requirements	11,000
Less: Opening Stock	(2,500)
Required Production	8,500

(iii) Materials Purchase Budget (Kg.)

Particulars	Material A	Material B
Requirement for Production	51,000 (8,500 units × 6 Kg.)	25,500 (8,500 units × 3 Kg.)
Add: Desired Closing Stock	8,000	5,500
Total Requirements	59,000	31,000
Less: Opening Stock	(7,500)	(4,000)
Quantity to be purchased	51,500	27,000

ICAI · Old P3 · MTP Dec 2021 · Series II · question p. 8-9 · answer p. 17-18

1110 MTP **Old P3 Jul 2021 S-I** Q3(a) 10 marks old scheme

The information of Z Ltd. for the year ended 31st March 2021 is as below:

	Amount (Rs.)
Direct materials	17,50,000
Direct wages	12,50,000
Variable factory overhead	9,50,000
Fixed factory overhead	12,00,000
Other variable costs	6,00,000
Other fixed costs	4,00,000
Profit	8,50,000
Sales	70,00,000

During the year, the company manufactured two products, X and Y, and the output and cost were:

	X	Y
Output (units)	8,000	4,000
Selling price per unit (Rs.)	600	550
Direct material per unit (Rs.)	140	157.50
Direct wages per unit (Rs.)	90	132.50

Variable factory overheads are absorbed as a percentage of direct wages and other variable costs are computed as:

Product X – Rs. 40 per unit and Product Y- Rs. 70 per unit.

For the FY 2021-22, it is expected that demand for product X and Y will fall by 20% & 10% respectively. It is also expected that direct wages cost will raise by 20% and other fixed costs by 10%. Products will be required to be sold at a discount of 20%.

You are required to:

(i) PREPARE profitability statement for the FY 2020-21 and

(ii) PREPARE a budget for the FY 2021-22.

ICAI'S ANSWER

(i) Product-wise Profitability Statement for the FY 2020-21:

Particulars	Product-X (Rs.)	Product-Y (Rs.)	Total (Rs.)
Output (units)	8,000	4,000	
Selling price per unit	600	550	
Sales value	48,00,000	22,00,000	70,00,000
Direct material	11,20,000 (Rs.140 × 8,000 units)	6,30,000 (Rs.157.50 × 4,000 units)	17,50,000
Direct wages	7,20,000 (Rs.90 × 8,000 units)	5,30,000 (Rs.132.5 × 4,000 units)	12,50,000
Variable factory overheads*	5,47,200 (76% of Rs. 7,20,000)	4,02,800 (76% of Rs. 5,30,000)	9,50,000
Other variable costs	3,20,000 (Rs.40 × 8,000 units)	2,80,000 (Rs.70 × 4,000 units)	6,00,000
Contribution	20,92,800	3,57,200	24,50,000
Fixed factory overheads	-	-	12,00,000
Other fixed costs	-	-	4,00,000
Profit			**8,50,000**

* Percentage absorption of variable factory overhead on the basis of direct wages = $\frac{9,50,000}{12,50,000}$ x 100 = 76%

(ii) Preparation of Budget for the FY 2021-22:

Particulars	Product-X (Rs.)	Product-Y (Rs.)	Total (Rs.)
Output (units)	6,400 (8,000 units × 80%)	3,600 (4,000 units × 90%)	
Selling price per unit	480 (Rs.600 × 80%)	440 (Rs.550 × 80%)	
Sales value	30,72,000	15,84,000	46,56,000
Direct material	8,96,000 (Rs.140 × 6,400 units)	5,67,000 (Rs.157.50 × 3,600 units)	14,63,000
Direct wages per unit	6,91,200 (Rs.108 × 6,400 units)	5,72,400 (Rs.159 × 3,600 units)	12,63,600
Variable factory overheads	5,25,312 (76% of Rs.6,91,200)	4,35,024 (76% of Rs.5,72,400)	9,60,336
Other variable costs	2,56,000 (Rs.40 × 6,400 units)	2,52,000 (Rs.70 × 3,600 units)	5,08,000
Contribution	7,03,488	(2,42,424)	4,61,064
Fixed factory overheads	-	-	12,00,000
Other fixed costs (110% of Rs.4,00,000)	-	-	4,40,000
Profit/ (Loss)			**(11,78,936)**

ICAI · Old P3 · MTP Jul 2021 · Series I · question p. 3-4 · answer p. 5-6

1111 MTP **Old P3 Nov 2020 S-I** Q1(c) 5 marks old scheme

'Mirror Look', a high gloss wooden manufacturing company, requires you to PREPARE the Master budget for the next year from the following information:

Sales:	
Acrylic finish wooden sheets	₹ 70,00,000
Lacquer finish wooden sheets	₹ 30,00,000
Direct material cost	65% of sales
Direct wages	25 workers @ ₹ 1,500 per month

Factory overheads:	
Indirect labour –	
Works manager	₹ 5,500 per month
Foreman	₹ 4,500 per month
Stores and spares	2.5% on sales
Depreciation on machinery	₹ 1,26,000
Light and power (fixed)	₹ 30,000
Repairs and maintenance	₹ 80,000
Others sundries	10% on direct wages
Administration, selling and distribution expenses	₹ 3,99,000 p.a.

ICAI'S ANSWER

Master Budget for the year ending _____

Particulars	(₹)	(₹)	(₹)
Sales:			
Acrylic finish wooden sheets			70,00,000
Lacquer finish wooden sheets			30,00,000
Total Sales			1,00,00,000
Less: Cost of production:			
Direct materials (65% of ₹ 1,00,00,000)		65,00,000	
Direct wages (25 workers × ₹ 1,500 × 12 months)		4,50,000	
Prime Cost		69,50,000	
Fixed Factory Overhead:			
Works manager's salary (5,500 × 12 months)	66,000		
Foreman's salary (4,500 × 12 months)	54,000		
Depreciation	1,26,000		
Light and power	30,000	2,76,000	
Variable Factory Overhead:			
Stores and spares (2.5% of ₹ 1,00,00,000)	2,50,000		
Repairs and maintenance	80,000		

Particulars	(₹)	(₹)	(₹)
Sundry expenses	45,000	3,75,000	
Works Cost			76,01,000
Gross Profit (Sales – Works cost)			23,99,000
Less: Adm., selling and distribution expenses			3,99,000
Net Profit			20,00,000

ICAI · Old P3 · MTP Nov 2020 · Series I · question p. 2 · answer p. 2

1112 MTP **Old P3 Nov 2019 S-I** Q3(a) 10 marks old scheme

V Ltd. produces and markets a very popular product called 'X'. The company is interested in presenting its budget for the second quarter of 2019.

The following information are made available for this purpose:

(i) It expects to sell 50,000 bags of 'X' during the second quarter of 2019 at the selling price of Rs. 900 per bag.

(ii) Each bag of 'X' requires 2.5 kgs. of a raw – material called 'Y' and 7.5 kgs. of raw – material called 'Z'.

(iii) Stock levels are planned as follows:

Particulars	Beginning of Quarter	End of Quarter
Finished Bags of 'X' (Nos.)	15,000	11,000
Raw – Material 'Y' (Kgs.)	32,000	26,000
Raw – Material 'Z' (Kgs.)	57,000	47,000
Empty Bag (Nos.)	37,000	28,000

(iv) 'Y' cost Rs.120 per Kg., 'Z' costs Rs.20 per Kg. and 'Empty Bag' costs Rs.80 each.

(v) It requires 9 minutes of direct labour to produce and fill one bag of 'X'. Labour cost is Rs.50 per hour.

(vi) Variable manufacturing costs are Rs.45 per bag. Fixed manufacturing costs Rs.30,00,000 per quarter.

(vii) Variable selling and administration expenses are 5% of sales and fixed administration and selling expenses are Rs.20,50,000 per quarter.

Required

(i) PREPARE a production budget for the said quarter.

(ii) PREPARE a raw – material purchase budget for 'Y', 'Z' and 'Empty Bags' for the said quarter in quantity as well as in rupees.

(iii) COMPUTE the budgeted variable cost to produce one bag of 'X'.

(iv) PREPARE a statement of budgeted net income for the said quarter and show both per unit and total cost data.

ICAI'S ANSWER

(i) Production Budget of 'X' for the Second Quarter

Particulars	Bags (Nos.)
Budgeted Sales	50,000
Add: Desired Closing stock	11,000
Total Requirements	61,000
Less: Opening stock	15,000
Required Production	46,000

(ii) Raw–Materials Purchase Budget in Quantity as well as in Rs. for 46,000 Bags of 'X'

Particulars	'Y' Kgs.	'Z' Kgs.	Empty Bags Nos.
Production Requirements Per bag of 'X'	2.5	7.5	1.0
Requirement for Production	1,15,000 (46,000 × 2.5)	3,45,000 (46,000 × 7.5)	46,000 (46,000 × 1)
Add: Desired Closing Stock	26,000	47,000	28,000
Total Requirements	1,41,000	3,92,000	74,000
Less: Opening Stock	32,000	57,000	37,000
Quantity *to be purchased*	1,09,000	3,35,000	37,000
Cost per Kg./Bag	Rs.120	Rs.20	Rs.80
Cost of Purchase (Rs.)	1,30,80,000	67,00,000	29,60,000

(iii) Computation of Budgeted Variable Cost of Production of 1 Bag of 'X'

Particulars	(Rs.)
Raw – Material	
Y 2.5 Kg @120	300.00
Z 7.5 Kg. @20	150.00
Empty Bag	80.00
Direct Labour(Rs.50× 9 minutes / 60 minutes)	7.50
Variable Manufacturing Overheads	45.00
Variable Cost of Production *per bag*	582.50

(iv) Budgeted Net Income for the Second Quarter

Particulars	Per Bag (Rs.)	Total (Rs.)
Sales Value (50,000 Bags)	900.00	4,50,00,000
Less: Variable Cost:		
Production Cost	582.50	2,91,25,000
Admn. & Selling Expenses (5% of Sales Price)	45.00	22,50,000
Budgeted Contribution	272.50	1,36,25,000
Less: Fixed Expenses:		
Manufacturing		30,00,000
Admn. & Selling		20,50,000
Budgeted Net Income		85,75,000

ICAI · Old P3 · MTP Nov 2019 · Series I · question p. 4-5 · answer p. 5-6

15.3 Flexible budgets

31 questions · asked in 25 ICAI documents · 5 exam papers

1113 **EXAM** **May 2025** Q5(c) **7 marks**

A reputed engineering college in Pune has 20 sections with 60 students per section. The college plans a one day pleasure trip around the city for the students during the weekend to places such as zoo, the amusement park and the technological museum. A private transport operator has agreed to provide the required number of buses at a hire charge of ₹ 6,500 per bus per trip. The bus hire charge is inclusive of special permit fees of ₹ 500 per bus per trip paid to the city municipal corporation.

Each bus has a maximum seating capacity of 54 persons excluding the driver. Four seats are reserved for the teachers who accompany each bus. The college will employ four teachers for each bus and pay ₹ 500 as daily allowance to each teacher for the trip. No other costs in respect of teachers are relevant to the trip. The following are the other cost estimates:

Particulars	Cost per student (₹)
Breakfast	55
Lunch	120
Evening tea with snacks	40
Entrance Fee at amusement park	Fee entry
Entrance Fee at the zoo	25

As regards the technological museum, the authorities charge block entrance fees for group of students depending upon the number of students in the group as enumerated below:

Number of students in the group	Block Entrance Fees (₹)
Upto 300	2,000
301-900	3,000
901 and above	3,500

Cost of prizes that would be distributed to the winners in different games being arranged in the amusement park depends upon the number of students in a trip. The cost of prizes to be distributed is:

Number of students in a trip	Cost of prizes (₹)
Upto - 300	2,200
301- 600	2,400
601 - 900	2,400
901 and above	2,500

Assuming that the college hires the requisite number of buses depending upon the number of students in a trip, you are required to:

(i) Prepare a flexible budget, estimating the total cost for a trip for the levels of 300, 600, 900 and 1,200 students showing each item of cost separately.

(ii) Compute the average cost per student at each of the above levels.

ICAI'S ANSWER

(i) Flexible Budget showing the total cost for a trip:

Particulars	No. of Students			
	300	**600**	**900**	**1200**
	(₹)	**(₹)**	**(₹)**	**(₹)**
Variable Cost				
Breakfast @ ₹ 55 per student	16,500	33,000	49,500	66,000
Lunch @ ₹ 120 per student	36,000	72,000	1,08,000	1,44,000
Evening Tea with snacks @ ₹ 40 per student	12,000	24,000	36,000	48,000
Entrance Fee at amusement park (Free entry)	-	-	-	-
Entrance Fee at the zoo @ ₹ 25 per student	7,500	15,000	22,500	30,000
Total Variable Costs (A)	**72,000**	**1,44,000**	**2,16,000**	**2,88,000**
Semi-Variable Cost				

Particulars	No. of Students			
Hire Charges @ ₹ 6,000 per bus	36,000	72,000	1,08,000	1,44,000
Special Permit Fees @ ₹ 500 per bus	3,000	6,000	9,000	12,000
DA to Teachers @ ₹ 500 × 4 teachers per bus = ₹ 2,000 per bus	12,000	24,000	36,000	48,000
Block Entrance Fees at the Technological Museum	2,000	3,000	3,000	3,500
Prizes to students	2,200	2,400	2,400	2,500
Total Semi-Variable Costs (B)	**55,200**	**1,07,400**	**1,58,400**	**2,10,000**
Total Cost for a Trip (A+B)	**1,27,200**	**2,51,400**	**3,74,400**	**4,98,000**

Working Note:

Particulars				
No. of Students	**300**	**600**	**900**	**1200**
No. of Buses required (50 students per bus)	300/50 = 6 buses	600/50 = 12 buses	900/50 = 18 buses	1200/50 = 24 buses

(ii) Average cost per student at each level:

Particulars				
No. of Students	**300**	**600**	**900**	**1200**
Total Cost (₹)	1,27,200	2,51,400	3,74,400	4,98,000
Average cost per student (₹)	424	419	416	415

ICAI · P4 · May 2025 exam · question p. 26-27 · answer p. 29-30

1114 EXAM **Sep 2024** Q3(b) 6 marks

Savi Limited is currently working at 80% of its capacity level and furnished the following information

for current period:

Production / Sales	96,000 units
Direct Variable Cost	₹ 20 per unit
Factory Overheads	₹ 8,40,000
Administrative Overheads (Fixed)	₹ 20,60,000
Sales Commission	2% of Sales Value
Transportation Expenses	₹ 4,000 per truck
(Loading Capacity 4,000 units)	

The selling price of the product is ₹ 120 per unit and Factory Overheads are 80% variable in nature.

The management of Savi Limited has come.to know that there will be high fluctuations in the demand of the product in upcoming year and it would not be an easy task to predict the demand. Selling price per unit will not be affected by demand fluctuations.

Savi Limited has decided to prepare a flexible budget for. the product at 60%, 80% and 100% capacity level.

You are required to prepare the Flexible Budget showing total cost of the product at each level.

ICAI'S ANSWER

Flexible Budget of Savi Ltd

	60% (72,000 units) (₹)	80% (96,000 units) (₹)	100% (1,20,000 units) (₹)
Sales **(A)**	120.00	120.00	120.00
Variable Costs:			
- Direct Variable Cost	20.00	20.00	20.00
- Variable Factory Overheads (WN1)	7.00	7.00	7.00
- Sales Commission (2%)	2.40	2.40	2.40
- Transportation Expenses	1.00	1.00	1.00
Total Variable Cost **(B)**	30.40	30.40	30.40
Contribution Per Unit **(C) = (A – B)**	89.60	89.60	89.60
Total Contribution **(D)**	64,51,200.00	86,01,600.00	1,07,52,000.00
Fixed Costs:			
- Administrative Overheads (100%)	20,60,000.00	20,60,000.00	20,60,000.00

	60% (72,000 units) (₹)	80% (96,000 units) (₹)	100% (1,20,000 units) (₹)
- Factory Overheads (20%)	1,68,000.00	1,68,000.00	1,68,000.00
Total Fixed Costs **(E)**	22,28,000.00	22,28,000.00	22,28,000.00
Profit **(D-E)**	42,23,200.00	63,73,600.00	85,24,000.00
Total Cost	**44,16,800.00**	**51,46,400.00**	**58,76,000.00**

WN1:

Variable factory Overheads = ₹ 8,40,000 x 80% = ₹ 6,72,000

Variable factory Overheads per unit = ₹ 6,72,000/96,000 units = ₹ 7

ICAI · P4 · Sep 2024 exam · question p. 21 · answer p. 24

1115 EXAM **May 2024** Q3(a) 7 marks

A factory is currently working at 60% capacity and produces 12,000 units of a product. Management is thinking to increase the working capacity either to 70% or 90% level. It is estimated that at both the levels, it will be able to sell all the produced units. The other details are as under:

- At 70% capacity, the cost of raw materials increases by 4% and the selling price falls by 3%.
- At 90% capacity, the cost of raw materials increases by 5% and selling price falls by 4%.
- At 60% capacity, the product cost is ₹ 360 per unit and it is sold at ₹ 400 per unit.
- The unit cost of 360 consists of the following:

Material	₹ 200
Labour	₹ 60
Factory overhead	₹ 60 (50 % fixed)
Administrative & Selling overhead	₹ 40 (60 % fixed)

- Additional advertising cost of ₹ 20,000 is to be incurred for selling the product above 80% capacity.

You are required to:

(i) Calculate the profits of the company when the factory works at 60%, 70% and 90% capacity level.

(ii) Offer your comments regarding increase in the capacity based on profit calculated.

ICAI'S ANSWER

(i) Expense Budget at 60%, 70% & 90% level

	60% (12,000 units)		70% (14,000 units)		90% (18,000 units)	
	Per unit (₹)	**Amount (₹)**	**Per unit (₹)**	**Amount (₹)**	**Per unit (₹)**	**Amount (₹)**
Sales **(A)**	400	48,00,000	388	54,32,000	384	69,12,000
Variable Costs:						

	60% (12,000 units)		70% (14,000 units)		90% (18,000 units)	
Direct Material	200	24,00,000	208	29,12,000	210	37,80,000
Direct Wages	60	7,20,000	60	8,40,000	60	10,80,000
Variable Factory Overheads	30	3,60,000	30	4,20,000	30	5,40,000
Variable Administrative & Selling Overheads	16	1,92,000	16	2,24,000	16	2,88,000
Total Variable Cost **(B)**	306	36,72,000	314	43,96,000	316	56,88,000
Contribution **(C)=(A–B)**	94	11,28,000	74	10,36,000	68	12,24,000
Fixed Costs:						
Fixed Factory Overheads (50%)	--	3,60,000	--	3,60,000	--	3,60,000
Fixed Administrative & Selling Overheads (60%)	--	2,88,000	--	2,88,000	--	2,88,000
Adverting Cost	--	--	--	--	--	20,000
Total Fixed Costs **(D)**	--	6,48,000	--	6,48,000	--	6,68,000
Profit **(C – D)**	--	4,80,000	--	3,88,000	--	5,56,000

(ii) Comment: Increase of production capacity to 90% is likely to increase the profit to maximum of ₹ 5,56,000 due to increase in contribution while fixed cost is slightly increased due to in advertising cost. At 70% capacity, profit is reduced to minimum of ₹ 3,88,00 due to decrease in selling price by 3% along with increase in raw material cost by 4% resulting in decrease of contribution. Therefore, it is recommended that factory should operate at 90% capacity.

ICAI · P4 · May 2024 exam · question p. 11-12 · answer p. 13-14

1116 EXAM **Old P3 Dec 2021** Q1(b) 5 marks old scheme

The Accountant of KPMR Ltd. has prepared the following budget for the coming year 2022 for its two

products 'AYE' and 'ZYE':

Particulars	Product 'AYE'	Product 'ZYE'
Production and Sales (in Units)	4,000	3,000
	Amount (in ₹)	Amount (in ₹)
Selling Price per unit	200	180
Direct Material per unit	80	70
Direct Labour per unit	40	35
Variable Overhead per unit	20	25
Fixed Overhead per unit	10	10

After reviewing the above budget, the management has called the marketing team for suggesting some measures for increasing the sales. The marketing team has suggested that by promoting the products on social media, the sales quantity of both the products can be increased by 5%. Also, the selling price per unit will go up by 10%. But this will result in increase in expenditure on variable overhead and fixed overhead by 20% and 5% respectively for both the products.

You are required to prepare flexible budget for both the products:

(i) Before promotion on social media,

(ii) After promotion on social media.

ICAI'S ANSWER

(i) Flexible Budget (before promotion)

	Particulars	Product 'AYE'	Product 'ZYE'	Total
	Production & Sales (units)	4,000	3,000	
		Amount (₹)	**Amount (₹)**	**Amount (₹)**
A.	Sales Value	8,00,000 (₹ 200×4,000)	5,40,000 (₹ 180×3,000)	13,40,000
B.	Direct Materials	3,20,000 (₹ 80 × 4,000)	2,10,000 (₹70 × 3,000)	5,30,000
C.	Direct labour	1,60,000 (₹ 40 × 4,000)	1,05,000 (₹ 35 × 3,000)	2,65,000
D.	Variable Overheads	80,000 (₹ 20 × 4,000)	75,000 (₹ 25 × 3,000)	1,55,000
E.	Total Variable Cost (B+C+D)	5,60,000	3,90,000	9,50,000
F.	Contribution (A-E)	2,40,000	1,50,000	3,90,000
G.	Fixed Overhead	40,000 (₹10 × 4,000)	30,000 (₹10 × 3,000)	70,000
H.	Profit (F-G)	2,00,000	1,20,000	3,20,000
	Profit per unit	50	40	

(ii) Flexible Budget (after promotion)

	Particulars	Product 'AYE'	Product 'ZYE'	Total
	Production & Sales (units)	4,200 (4,000×105%)	3,150 (3,000×105%)	
		Amount (₹)	**Amount (₹)**	**Amount (₹)**
A.	Sales Value	9,24,000 (₹ 220 × 4,200)	6,23,700 (₹ 198 × 3,150)	15,47,700
B.	Direct Materials	3,36,000 (₹ 80 × 4,200)	2,20,500 (₹ 70 × 3,150)	5,56,500
C.	Direct labour	1,68,000 (₹ 40 × 4,200)	1,10,250 (₹ 35 × 3,150)	2,78,250
D.	Variable Overheads	1,00,800 (₹ 24 × 4,200)	94,500 (₹ 30 × 3,150)	1,95,300
E.	Total Variable Cost (B+C+D)	6,04,800	4,25,250	10,30,050
F.	Contribution (A-E)	3,19,200	1,98,450	5,17,650
G.	Fixed Overhead	42,000 (₹ 40,000 × 105%)	31,500 (₹ 30,000 × 105%)	73,500
H.	Profit (F-G)	2,77,200	1,66,950	4,44,150
	Profit per unit	66	53	

ICAI · Old P3 · Dec 2021 exam · question p. 1-2 · answer p. 3-4

1117 EXAM **Old P3 Nov 2020** Q1(a) 5 marks old scheme

G Ltd. manufactures a single product for which market demand exists for additional quantity. Present sales of ₹ 6,00,000 utilises only 60% capacity of the plant. The following data are available:

(1) Selling price : ₹ 100 per unit

(2) Variable cost : ₹ 30 per unit

(3) Semi-variable expenses : ₹ 60,000 fixed + ₹ 5 per unit

(4) Fixed expenses : ₹ 1,00,000 at present level, estimated to increase by 25% at and above 80% capacity.

You are required to prepare a flexible budget so as to arrive at the operating profit at 60%, 80% and 100% levels.

ICAI'S ANSWER

Flexible Budget

Activity Level	60%	80%	100%
Production (units)	**6,000**	**8,000**	**10,000**
	(₹)	**(₹)**	**(₹)**
Sales @ ₹ 100 per unit	6,00,000	8,00,000	10,00,000

Activity Level	60%	80%	100%
Variable Cost (@ ₹ 35 (₹ 30 + ₹ 5) per unit)	2,10,000	2,80,000	3,50,000
Contribution (A)	3,90,000	5,20,000	6,50,000
Fixed Cost (part of semi-variable cost)	60,000	60,000	60,000
Other Fixed Cost	1,00,000	1,25,000	1,25,000
Total Fixed Cost (B)	1,60,000	1,85,000	1,85,000
Operating Profit (A – B)	**2,30,000**	**3,35,000**	**4,65,000**

ICAI · Old P3 · Nov 2020 exam · question p. 1 · answer p. 3

1118 RTP **Sep 2025** Q18(c)

Fixed Budget does not change with actual volume of activity achieved, however, Flexible Budget can be re-casted on the basis of activity level to be achieved. DISCUSS other differences between these two type of budget classified based on their capacity.

ICAI'S ANSWER

Difference between Fixed and Flexible Budgets:

Sl. No.	Fixed Budget	Flexible Budget
1.	It operates on one level of activity and under one set of conditions. It assumes that there will be no change in the prevailing conditions, which is unrealistic.	It consists of various budgets for different levels of activity.
2.	Here as all costs like - fixed, variable and semi-variable are related to only one level of activity so variance analysis does not give useful information.	Here analysis of variance provides useful information as each cost is analysed according to its behaviour.
3.	If the budgeted and actual activity levels differ significantly, then the aspects like cost ascertainment and price fixation do not give a correct picture.	Flexible budgeting at different levels of activity facilitates the ascertainment of cost, fixation of selling price and tendering of quotations.
4.	Comparison of actual performance with budgeted targets will be meaningless specially when there is a difference between the two activity levels.	It provides a meaningful basis of comparison of the actual performance with the budgeted targets.

ICAI · P4 · RTP Sep 2025 · question p. 23 · answer p. 55-56

1119 RTP May 2025 Q19

The budgets for activity and cost of PQR Ltd. for the first three quarters of operation are shown below:

Period Covered	Budgets Quarters I – III		
	Q – I	**Q – II**	**Q – III**
Months	**1 – 3**	**4 – 6**	**7 – 9**
	('000)	**('000)**	**('000)**
Activity :			
Sales (Units)	9	17	15
Production (Units)	10	20	15
Costs (₹) :			
Direct Material			
A	60	120	90
B	50	100	75
Production Labour	180	285	230
Manufacturing Overheads Excluding Depreciation	90	120	105
Depreciation of Production Machinery	20	20	20
Administration Expenses	25	25	25
Selling & Distribution Expenses	38	54	50

The figures shown above represent the costs structure of PQR Ltd., which have the following major features:

(i) Fixed element of any cost is completely independent of activity levels.

(ii) Any variable element of each cost displays a linear relationship with activity level, except that the variable labour cost become 50% higher for activity in excess of 19,000 units per quarter due to the necessity for overtime working.

(iii) The variable element of selling and distribution expenses is a function of sales. All other costs with a variable element are a function of production volume.

Activity for each quarter is spread evenly throughout that quarter.

In Quarter IV Production level will be set equal to sales level. Production and sales in this quarter is expected to range between 15,000 units and 21,000 units. The most likely volume is 18,000 units. In month 9 it will be possible to accurately estimate the sales for Quarter IV.

Cost structure will remain the same as in Quarters I to III except the following:

(i) Labour cost will rise by 12½%.

(ii) Variable labour input per unit of output will decrease, due to learning curve effect, such that 80% of the previous labour input per unit of output will be required in Quarter IV. The threshold for overtime working remains at 19,000 units per quarter.

(iii) Fixed factory overheads and the fixed element of selling and distribution costs will each rise by 20% (The variable element of selling and distribution costs will be unaltered.)

Required

(i) PREPARE a Statement to show, under each cost classification given in the budgets, the variable cost per unit and fixed costs which will be effective in Quarter IV.

(ii) PREPARE a flexible budget of production costs for the Quarter IV.

ICAI'S ANSWER

(i) Statement of Variable Cost *per unit* and Fixed Costs under Given Cost Classification Effective for Quarter IV

Particulars	Total Fixed Cost (₹)	Variable Cost p.u. (₹)
Direct Materials (W.N.1)		
A	-----	6
B	-----	5
Production Labour (W.N.2)	90,000	9
Manufacturing Overhead Ex. Depreciation (W.N.3)	72,000	3
Depreciation of Production Machinery	20,000	-----
Administration Expenses	25,000	-----
Selling & Distribution Expenses (W.N.4)	24,000	2

(ii) Flexible Budget of Production Costs for the Quarter IV

Particulars	15,000 units (₹)	18,000 units (₹)	21,000 units (₹)
Direct Material			
A	90,000 (15,000 units × ₹ 6)	1,08,000 (18,000 units × ₹ 6)	1,26,000 (21,000 × ₹ 6)
B	75,000 (15,000 units × ₹ 5)	90,000 (18,000 × ₹ 5)	1,05,000 (21,000 units × ₹ 5)
Production Labour	2,25,000 (15,000 units × ₹ 9 + ₹ 90,000)	2,52,000 (18,000 units × ₹ 9 + ₹ 90,000	2,88,000*
Manufacturing Overhead	1,17,000 (15,000 units × ₹ 3 + ₹ 72,000)	1,26,000 (18,000 units × ₹ 3 ₹ 72,000)	1,35,000 (21,000 units × ₹ 3 + ₹ 72,000)
Depreciation	20,000	20,000	20,000

Particulars	15,000 units (₹)	18,000 units (₹)	21,000 units (₹)
Total Production Cost	5,27,000	5,96,000	6,74,000

* Production Labour (21,000 units level)

	₹
Variable Cost (21,000 units × ₹ 9)	1,89,000
Fixed Cost	90,000
Overtime (2,000 units × ₹ 9 × 0.50)	9,000
	2,88,000

Working Notes

1. Direct Material Cost:

A: $\frac{₹\ 60{,}000}{10{,}000\ \text{units}} = ₹\ 6$

B: $\frac{₹\ 50{,}000}{10{,}000\ \text{units}} = ₹\ 5$

Direct material cost (variable cost) for material A and B for all the quarters on computation comes to ₹ 6 /- and ₹ 5 /- for materials A and B respectively.

2. Fixed and Variable Cost Component of *production labour cost:*

Particulars	Quarter I	Quarter III	Change
Production (units)	10,000	15,000	5,000
Production labour (₹)	1,80,000	2,30,000	50,000

$$\text{Variable Cost (per unit)} = \frac{\text{Change in Production Labour Cost}}{\text{Change in Production Units}}$$

$$\text{Change in Production units} = \frac{₹\ 16{,}000}{8{,}000}$$

$$= \frac{₹\ 50{,}000}{5{,}000}$$

$$= ₹\ 10$$

$$\text{Fixed Cost} = ₹\ 1{,}80{,}000 - ₹\ 1{,}00{,}000$$

$$= ₹\ 80{,}000$$

For Quarter II (20,000 units):

	₹
Variable Cost of 20,000 units @ ₹ 10 p.u.	2,00,000
Fixed Cost	80,000
Overtime Premium on 1,000 @ ₹ 5 p.u.	5,000
Total Production Labour Cost	2,85,000

For Quarter IV (18,000 units):

	₹
Variable Cost of 18,000 units @ ₹ 9 p.u. (₹ 10 × 1.125 × 0.80 = ₹ 9)	1,62,000
Fixed Cost (₹ 80,000 × 1.125)	90,000
Total Production Labour Cost	2,52,000

3. Fixed and Variable Cost Component of *manufacturing overhead:*

	Quarter I	Quarter II	Change
Production (units)	10,000	20,000	10,000
Manufacturing Overhead (₹) (Excluding Depreciation)	90,000	1,20,000	30,000

Variable Cost Component of *manufacturing overhead:*

$$= \frac{\text{Change in Manufacturing Overhead Costs}}{\text{Change in Production Units}}$$

$$= \frac{₹\ 30{,}000}{10{,}000\ \text{units}}$$

= ₹ 3 p.u

Fixed Cost Component of *manufacturing overhead:*

= ₹ 1,20,000 – 20,000 units × ₹ 3

= ₹ 60,000

For Quarter IV:

		₹
Fixed Cost	=	60,000
Add: 20% Increase	=	12,000
Total Fixed Cost	=	72,000

4. Fixed and Variable Cost Component of *selling and distribution expenses*

	Quarter I	Quarter II	Change
Sales (units)	9,000	17,000	8,000
Selling & Distribution Expenses	38,000	54,000	16,000

Variable Cost Component of *selling & distribution expenses:*

$$= \frac{\text{Change in selling \& Distribution expenses}}{\text{Change in sales units}}$$

$$= \frac{₹\ 16{,}000}{8{,}000}$$

= ₹ 2 per unit

Fixed Cost Component of *selling & distribution expenses:*

= ₹ 54,000 – 17,000 units × ₹ 2

= ₹ 20,000

Fixed Cost Component for IV Quarter:

= ₹ 20,000 × 1.20

= ₹ 24,000

ICAI · P4 · RTP May 2025 · question p. 23-25 · answer p. 51-55

1120 **RTP** **Jan 2025** **Q18**

BT Ltd. achieves sale of ₹ 73,12,500 with COGS of 40% while operating at 75% of its normal capacity during the current financial year.

The information relating to Administration, Selling and Distribution costs is given below:

Administration costs:

Office salaries	₹ 11,70,000
General expenses	5 per cent of COGS
Depreciation	₹ 97,500
Rates and taxes	₹ 1,13,750

Selling costs:

Salaries	8 per cent of sales
Travelling expenses	5 per cent of COGS
Sales office expenses	2.5 per cent of COGS
General expenses	2.5 per cent of COGS

Distribution costs:

Wages	₹ 1,95,000
Rent	1 per cent of sales
Other expenses	10 per cent of COGS

Considering some of the expenses like office salaries, depreciation, rates and taxes, and wages, to remain the same irrespective of the level of activity, as these expenses are fixed in nature, PREPARE flexible administration, selling and distribution costs budget, operating at 85%, 100% and 115% of normal capacity.

ICAI'S ANSWER

Flexible Budget of BT Ltd.

Particulars	75% (₹)	85% (₹)	100% (₹)	115% (₹)
Sales	73,12,500	82,87,500	97,50,000	1,12,12,500
COGS (40% of Sales)	29,25,000	33,15,000	39,00,000	44,85,000
Administration Costs:				
Office Salaries (fixed)	11,70,000	11,70,000	11,70,000	11,70,000
General expenses (5% of COGS)	1,46,250	1,65,750	1,95,000	2,24,250
Depreciation (fixed)	97,500	97,500	97,500	97,500
Rent and rates (fixed)	1,13,750	1,13,750	1,13,750	1,13,750
(A) Total Adm. Costs	15,27,500	15,47,000	15,76,250	16,05,500
Selling Costs:				
Salaries (8% of sales)	5,85,000	6,63,000	7,80,000	8,97,000
Travelling expenses (5% of COGS)	1,46,250	1,65,750	1,95,000	2,24,250
Sales office (2.5% of COGS)	73,125	82,875	97,500	1,12,125
General expenses (2.5% of COGS)	73,125	82,875	97,500	1,12,125
(B) Total Selling Costs	8,77,500	9,94,500	11,70,000	13,45,500
Distribution Costs:				
Wages (fixed)	195,000	195,000	195,000	195,000
Rent (1% of sales)	73,125	82,875	97,500	1,12,125
Other expenses (10% of COGS)	2,92,500	3,31,500	3,90,000	4,48,500
(C) Total Distribution Costs	5,60,625	6,09,375	6,82,500	7,55,625

Particulars	75% (₹)	85% (₹)	100% (₹)	115% (₹)
Total Costs (A + B + C)	**29,65,625**	**31,50,875**	**34,28,750**	**37,06,625**

ICAI · P4 · RTP Jan 2025 · question p. 19 · answer p. 43-44

1121 **RTP** **May 2024** Q12

M Ltd. is a public sector undertaking (PSU), produces a product A. The company is in process of preparing its revenue budget for the year 2024. The company has the following information which can be useful in preparing the budget:

(i) It has anticipated 12% growth in sales volume from the year 2023 of 4,20,000 tonnes.

(ii) The sales price of ₹ 23,000 per tonne will be increased by 10% provided Wholesale Price Index (WPI) increases by 5%.

(iii) To produce one tonne of product A, 2.3 tonnes of raw material are required. The raw material cost is ₹ 4,500 per tonne. The price of raw material will also increase by 10% if WPI increase by 5%.

(iv) The projected increase in WPI for 2024 is 4%

(v) A total of 6,000 employees works for the company. The company works 26 days in a month.

(vi) 85% of employees of the company are permanent and getting salary as per 5- year wage agreement. The earnings per manshift (means an employee cost for a shift of 8 hours) is ₹ 3,000 (excluding terminal benefits). The new wage agreement will be implemented from 1st July 2024 and it is expected that a 15% increase in pay will be given.

(vii) The casual employees are getting a daily wage of ₹ 850. The wages in linked to Consumer Price Index (CPI). The present CPI is 165.17 points and it is expected to be 173.59 points in year 2024.

(viii) Power cost for the year 2023 is ₹ 42,00,000 for 7,00,000 units (1 unit = 1 Kwh). 60% of power is used for production purpose (directly related to production volume) and remaining are for employee quarters and administrative offices.

(ix) During the year 2023, the company has paid ₹ 60,00,000 for safety and maintenance works. The amount will increase in proportion to the volume of production.

(x) During the year 2023, the company has paid ₹ 1,20,000 for the purchase of diesel to be used in car hired for administrative purposes. The cost of diesel will increase by 15% in year 2024.

(xi) During the year 2023, the company has paid ₹ 6,00,000 for car hire charges (excluding fuel cost). In year 2024, the company has decided to reimburse the diesel cost to the car rental company. Doing this will attract 5% GST on Reverse Charge Mechanism (RCM) basis on which the company will not get GST input credit.

(xii) Depreciation on fixed assets for the year 2023 is ₹ 80,40,00,000 and it will be 15% lower in 2024.

You being an associate to the budget controller of the company, PREPARE Revenue (Flexible) budget for the year 2024 and also show the budgeted profit/ loss for the year.

ICAI'S ANSWER

Revenue Budget (Flexible Budget) of M Ltd. for the Year 2024

	Particulars	PY 2023	CY 2024
A	Sales Volume (Tonnes)	4,20,000	4,70,400 [112%×4,20,000]
B	Selling Price per tonne (₹)	23,000	23,000
		(₹ in lakh)	**(₹ in lakh)**
C	**Sales value [A×B]**	**96,600**	**1,08,192**
D	**Raw material Cost:**		
(i)	Qty. of Material [2.3 tonnes × A] (tonnes)	9,66,000	10,81,920
(ii)	Price per tonne (₹)	4,500	4,500
(iii)	**Total raw material cost [(i)×(ii)]**	**43,470**	**48,686.40**
E	**Wages & Salary Cost:**		
(i)	Wages to casual employees (15%×6,000 = 900 employees)	2,386.80 [900×26×12×₹850]	2,508.47 [900×26×12×₹893.33]
(ii)	Salary to permanent employees (85%×6,000 = 5,100 employees)	47,736 [5100×26×12×₹3,000]	51,316.20 [(5100×26×6×₹3,000) + (5100×26×6×₹3,450)]
(iii)	**Total wages & salary [(i)+(ii)+(iii)]**	**50,122.80**	**53,824.67**
F	**Power cost:**		
(i)	For production (units)	4,20,000 [60%×7,00,000]	4,70,400 [112%×4,20,000]
(ii)	For employees & offices (units) [40%×7,00.000]	2,80,000	2,80,000
(iii)	Total Power consumption (units) [(i)+(ii)]	7,00,000	7,50,400
(iv)	Power rate per unit (₹) [₹42,00,000÷7,00,000]	6.00	6.00
(v)	Total power cost [(iii)×(iv)]	**42**	**45.024**
G	**Safety and maintenance Cost**	**60**	**67.20 [112%×4,20,000]**
H	**Diesel cost**	**1.2**	**-**
I	**Car Hire charge:**		
(i)	Car hire charge	6	6

	Particulars	PY 2023	CY 2024
(ii)	Fuel reimbursement cost	-	1.38 [115%×1.2]
(iii)	GST@5% on RCM basis [5%×(i+ii)]	-	0.369
(iv)	**Total Car hire charge cost [(i)+ (ii)+(iii)]**	**6**	**7.749**
J	**Depreciation**	**8,040**	**6,834 [85%×8040]**
K	Total Cost [Sum of D to J]	1,01,742	1,09,465.043
L	**Profit/ (Loss) [C-L]**	**(5,142)**	**(1273.043)**

ICAI · P4 · RTP May 2024 · question p. 15-16 · answer p. 40-41

1122 RTP **Old P3 May 2022** Q13 old scheme

Maharatna Ltd., a public sector undertaking (PSU), produces product A. The company is in process of preparing its revenue budget for the year 2022. The company has the following information which can be useful in preparing the budget:

(i) It has anticipated 12% growth in sales volume from the year 2021 of 4,20,000 tonnes.

(ii) The sales price of ₹23,000 per tonne will be increased by 10% provided Wholesale Price Index (WPI) increases by 5%.

(iii) To produce one tonne of product A, 2.3 tonnes of raw material are required. The raw material cost is ₹4,500 per tonne. The price of raw material will also increase by 10% if WPI increase by 5%.

(iv) The projected increase in WPI for 2022 is 4%

(v) A total of 6,000 employees works for the company. The company works 26 days in a month.

(vi) 85% of employees of the company are permanent and getting salary as per 5- year wage agreement. The earnings per manshift (means an employee cost for a shift of 8 hours) is ₹ 3,000 (excluding terminal benefits). The new wage agreement will be implemented from 1st July 2022 and it is expected that a 15% increase in pay will be given.

(vii) The casual employees are getting a daily wage of ₹ 850. The wages in linked to Consumer Price Index (CPI). The present CPI is 165.17 points and it is expected to be 173.59 points in year 2022.

(viii) Power cost for the year 2021 is ₹ 42,00,000 for 7,00,000 units (1 unit = 1 Kwh). 60% of power is used for production purpose (directly related to production volume) and remaining are for employee quarters and administrative offices.

(ix) During the year 2021, the company has paid ₹ 60,00,000 for safety and maintenance works. The amount will increase in proportion to the volume of production.

(x) During the year 2021, the company has paid ₹ 1,20,000 for the purchase of diesel to be used in car hired for administrative purposes. The cost of diesel will increase by 15% in year 2022.

(xi) During the year 2021, the company has paid ₹ 6,00,000 for car hire charges (excluding fuel cost). In year 2022, the company has decided to reimburse the diesel cost to the car rental company. Doing this will attract 5% GST on Reverse Charge Mechanism (RCM) basis on which the company will not get GST input credit.

(xii) Depreciation on fixed assets for the year 2021 is ₹ 80,40,00,000 and it will be 15% lower in 2022.

Required:

From the above information PREPARE Revenue (Flexible) budget for the year 2022 and also show the budgeted profit/ loss for the year.

ICAI'S ANSWER

Revenue Budget (Flexible Budget) of Maharatna Ltd. for the Year 2022

	Particulars	PY 2021	CY 2022
A	Sales Volume (Tonnes)	4,20,000	4,70,400 [112%×4,20,000]
B	Selling Price per tonne (₹)	23,000	23,000
		(₹ in lakh)	**(₹ in lakh)**
C	**Sales value [A×B]**	**96,600**	**1,08,192**
D	**Raw material Cost:**		
(i)	Qty. of Material [2.3 tonnes × A] (tonnes)	9,66,000	10,81,920
(ii)	Price per tonne (₹)	4,500	4,500
(iii)	**Total raw material cost (₹ in lakh) [(i)×(ii)]**	**43,470**	**48,686.40**
E	**Wages & Salary Cost:**		
(i)	Wages to casual employees (15% × 6,000 = 900 employees)	2,386.80 [900 × 26 × 12 × ₹ 850]	2,508.47 [900 × 26 × 12 × ₹ 893.33]
(ii)	Salary to permanent employees (85% × 6,000 = 5,100 employees)	47,736 [5100 × 26 × 12 × ₹ 3,000]	51,316.20 [(5100 × 26 × 6 × ₹ 3,000) + (5100 × 26 × 6 × ₹ 3,450)]
(iii)	**Total wages & salary [(i)+(ii)]**	**50,122.80**	**53,824.67**
F	**Power cost:**		
(i)	For production (units)	4,20,000 [60% × 7,00,000]	4,70,400 [112% × 4,20,000]
(ii)	For employees & offices (units) [40% × 7,00.000]	2,80,000	2,80,000
(iii)	Total Power consumption (units) [(i)+(ii)]	7,00,000	7,50,400

	Particulars	PY 2021	CY 2022
(iv)	Power rate per unit (₹) [₹42,00,000 ÷ 7,00,000]	6.00	6.00
(v)	Total power cost [(iii)×(iv)]	**42**	**45.024**
G	**Safety and maintenance Cost**	**60**	**67.20 [112% × 60,00,000]**
H	**Diesel cost**	**1.2**	**-**
I	**Car Hire charge:**		
(i)	Car hire charge	6	6
(ii)	Fuel reimbursement cost	-	1.38 [115% × 1.2]
(iii)	GST@5% on RCM basis [5%×(i+ii)]	-	0.369
(iv)	**Total Car hire charge cost [(i)+ (ii)+(iii)]**	**6**	**7.749**
J	**Depreciation**	**8,040**	**6,834 [85% × 8040]**
K	Total Cost [Sum of D to J]	1,01,742	1,09,465.043
L	**Profit/ (Loss) [C-L]**	**(5,142)**	**(1273.043)**

ICAI · Old P3 · RTP May 2022 · question p. 9-10 · answer p. 30-31

1123 RTP **Old P3 Nov 2020** Q14 old scheme

The information of Z Ltd. for the year ended 31st March 2020 is as below:

	Amount (₹)
Direct materials	17,50,000
Direct wages	12,50,000
Variable factory overhead	9,50,000
Fixed factory overhead	12,00,000
Other variable costs	6,00,000
Other fixed costs	4,00,000
Profit	8,50,000
Sales	70,00,000

During the year, the company manufactured two products, X and Y, and the output and cost were:

	X	Y
Output (units)	8,000	4,000
Selling price per unit (₹)	600	550
Direct material per unit (₹)	140	157.50
Direct wages per unit (₹)	90	132.50

Variable factory overheads are absorbed as a percentage of direct wages and other variable costs are computed as:

Product X – ₹40 per unit and Product Y- ₹70 per unit.

For the FY 2020-21, due to a pandemic, it is expected that demand for product X and Y will fall by 20% & 10% respectively. It is also expected that direct wages cost will raise by 20% and other fixed costs by 10%. Products will be required to be sold at a discount of 20%.

You are required to:

(i) PREPARE product- wise profitability statement on marginal costing method for the FY 2019-20 and

(ii) PREPARE a budget for the FY 2020-21.

ICAI'S ANSWER

(i) Product-wise Profitability Statement for the FY 2019-20:

Particulars	Product-X (₹)	Product-Y (₹)	Total (₹)
Output (units)	8,000	4,000	
Selling price per unit	600	550	
Sales value	48,00,000	22,00,000	70,00,000
Direct material	11,20,000 (₹140×8,000)	6,30,000 (₹157.50×4,000)	17,50,000
Direct wages	7,20,000 (₹90×8,000)	5,30,000 (₹132.5×4,000)	12,50,000
Variable factory overheads	5,47,200 (76%of 7,20,000)	4,02,800 (76%of 5,30,000)	9,50,000
Other variable costs	3,20,000 (₹40×8,000)	2,80,000 (₹70×4,000)	6,00,000
Contribution	20,92,800	3,57,200	24,50,000
Fixed factory overheads	-	-	12,00,000
Other fixed costs	-	-	4,00,000
Profit			**8,50,000**

(ii) Preparation of Budget for the FY 2020-21:

Particulars	Product-X (₹)	Product-Y (₹)	Total (₹)
Output (units)	6,400 (8,000×80%)	3,600 (4,000×90%)	
Selling price per unit	480 (600×80%)	440 (550×80%)	
Sales value	30,72,000	15,84,000	46,56,000
Direct material	8,96,000 (₹140×6,400)	5,67,000 (₹157.50×3,600)	14,63,000
Direct wages per unit	6,91,200 (₹108×6,400)	5,72,400 (₹159×3,600)	12,63,600
Variable factory overheads	5,25,312 (76%of 6,91,200)	4,35,024 (76%of 5,72,400)	9,60,336
Other variable costs	2,56,000 (₹40×6,400)	2,52,000 (₹70×3,600)	5,08,000
Contribution	7,03,488	(2,42,424)	4,61,064
Fixed factory overheads	-	-	12,00,000
Other fixed costs (110%of ₹4,00,000)	-	-	4,40,000
Profit/ (Loss)			**(11,78,936)**

ICAI · Old P3 · RTP Nov 2020 · question p. 11-12 · answer p. 28-29

1124 MTP Sep 2026 S-I MCQ 15 2 marks MCQ

The following extract is taken from the overhead budget of X:

Budgeted activity	50%	75%
Budgeted overhead (₹)	30,00,000	40,00,000

What would be the budgeted overhead for 60% level of activity:

A ₹ 32,00,000

B ₹ 34,00,000

C ₹ 30,00,000

D ₹ 36,00,000

ICAI'S ANSWER

(B) ₹ 34,00,000

Variable overhead for each % of level of activity

$$= \frac{40,00,000 - 30,00,000}{75 - 50} = 40,000$$

Fixed cost = 30,00,000 – (40,000 × 50) = 10,00,000

Total overheads for 60% level of activity

= 10,00,000 + (40,000 × 60) = ₹ 34,00,000

ICAI · P4 · MTP Sep 2026 · Series I · question p. 7 · answer p. 6

1125 MTP **May 2026 S-I** Q2(b) 6 marks

Pentax Limited has prepared its expense budget for 20,000 units in its factory for the year 2026 as detailed below:

	(₹ per unit)
Direct Materials	50
Direct Labour	20
Variable Overhead	15
Direct Expenses	6
Selling Expenses (20% fixed)	15
Factory Expenses (100% fixed)	7
Administration expenses (100% fixed)	4
Distribution expenses (85% variable)	12
Total	129

PREPARE an expense budget for the production of 15,000 units and 18,000 units.

ICAI'S ANSWER

Expense Budget of M/s Pentax Ltd.

Particulars	20,000 Units (₹)	15,000 Units (₹)	18,000 Units (₹)
Direct Material	10,00,000 (20,000 x 50)	7,50,000 (15,000 x 50)	9,00,000 (18,000 x 50)
Direct Labour	4,00,000 (20,000 x 20)	3,00,000 (15,000 x 20)	3,60,000 (18,000 x 20)
Variable Overhead	3,00,000 (20,000 x 15)	2,25,000 (15,000 x 15)	2,70,000 (18,000 x 15)
Direct Expenses	1,20,000 (20,000 x 6)	90,000 (15,000 × 6)	1,08,000 (18,000 x 6)
Selling Expenses (Variable)*	2,40,000 (20,000 x 12)	1,80,000 (15,000 x 12)	2,16,000 (18,000 x 12)
Selling Expenses (Fixed)* (3 x 20,000)	60,000	60,000	60,000
Factory Expenses (Fixed) (7 x 20,000)	1,40,000	1,40,000	1,40,000

Particulars	20,000 Units (₹)	15,000 Units (₹)	18,000 Units (₹)
Administration Expenses (Fixed) (4 × 20,000)	80,000	80,000	80,000
Distribution Expenses (Variable)**	2,04,000 (10.20 x 20,000)	1,53,000 (10.20 x 15,000)	1,83,600 (10.20 x 18,000)
Distribution Expenses (Fixed)** (1.80 × 20,000)	36,000	36,000	36,000
	25,80,000	20,14,000	23,53,600

*Selling Expenses: Fixed cost per unit = ₹ 15 x 20% = ₹ 3

Fixed Cost = ₹ 3 x 20,000 units = ₹ 60,000

Variable Cost Per unit = ₹ 15 – ₹ 3 = ₹ 12

**Distribution Expenses: Fixed cost per unit = ₹ 12 x 15% = ₹ 1.80

Fixed Cost = ₹ 1.80 x 20,000 units = ₹ 36,000

Variable cost per unit = ₹ 12 – ₹ 1.80 = ₹ 10.20

ICAI · P4 · MTP May 2026 · Series I · question p. 9-10 · answer p. 9-10

1126 MTP **May 2026 S-II** Case Scenario 2 · MCQ 6 to MCQ 10 10 marks

CASE SCENARIO 2

A company, ABC Professional Services Ltd., is organizing a one-day corporate training workshop as part of its annual employee development program. The objective is to improve managerial efficiency, team coordination, and leadership skills of its employees.

The workshop will be conducted at a premium venue, and may include interactive sessions, guest lectures, and team-building activities.

Due to uncertainty in the number of participants, management wants to evaluate costs at different activity levels using a flexible budgeting approach.

The company plans to conduct the workshop for different numbers of participants. Each training batch can accommodate a maximum of 50 participants. Two trainers will be engaged for each batch.

The company will pay an honorarium of ₹1,000 per trainer. It will also hire the required number of venue and equipment setup.

The following are the other cost estimates:

Particulars	Cost per participant (₹)
Breakfast	80
Lunch	200
Evening Snacks	70
Training Material Kit	150

Other Costs

Particulars	Amount
Hall rent	₹8,000 per batch
Equipment charges	₹2,000 per batch
Guest speaker fees	₹10,000
Event coordination charges	₹5,000

Other information

- No costs are incurred in respect of trainers except the honorarium of ₹ 1,000 per trainer.
- If the number of participants exceeds 100, an additional lump-sum supervision cost of ₹ 3,000 will be incurred.
- A discount of 10% on hall rent will be available more than 3 batches are hired.

Management asked you to prepare a flexible budget and answer the following questions (MCQs 6 to 10):

MCQ 6 2 marks

What is the total cost and average cost per participant if there is 40 participants?

A Total Cost ₹ 47,000; Avg ₹ 1,175

B Total Cost ₹ 35,000; Avg ₹ 875

C Total Cost ₹ 37,000; Avg ₹ 925

D Total Cost ₹ 40,000; Avg ₹ 1,000

MCQ 7 2 marks

What is the total cost and average cost per participant if there is 80 participants?

A Total Cost ₹ 69,000; Avg ₹ 862.5

B Total Cost ₹ 65,000; Avg ₹ 812.5

C Total Cost ₹ 75,000; Avg ₹ 937.5

D Total Cost ₹ 79,000; Avg ₹ 987.5

MCQ 8 2 marks

What is the total cost and average cost per participant if there is 120 participants?

A Total Cost ₹ 1,14,000; Avg ₹ 950

B Total Cost ₹ 1,05,000; Avg ₹ 875

C Total Cost ₹ 1,08,000; Avg ₹ 900

D Total Cost ₹ 1,00,000; Avg ₹ 833

MCQ 9 2 marks

What is the total cost and average cost per participant if there is 180 participants?

A Total Cost ₹ 1,45,000; Avg ₹ 805.5

B Total Cost ₹ 1,50,000; Avg ₹ 833.3

C Total Cost ₹ 1,52,800; Avg ₹ 848.9

D Total Cost ₹ 1,38,000; Avg ₹ 766.7

MCQ 10 2 marks

What will be your conclusions regarding the break-even level of participants, if the company proposes to charge ₹850 per participant?

A 90 participants

B 180 participants

C 110 participants

D 125 participants

ICAI'S ANSWERS

MCQ 6

(A) Total Cost ₹ 47,000; Avg ₹ 1,175

MCQ 7

(D) Total Cost ₹ 79,000; Avg ₹ 987.5

MCQ 8

(A) Total Cost ₹ 1,14,000; Avg ₹ 950

MCQ 9

(C) Total Cost ₹ 1,52,800; Avg ₹ 848.9

Flexible Budget

Level of participants	40	80	120	180
Variable Costs (₹)				
Breakfast @ ₹ 80	3,200	6,400	9,600	14,400
Lunch @ ₹ 200	8,000	16,000	24,000	36,000
Evening Snacks @ ₹ 70	2,800	5,600	8,400	12,600
Training Material Kit @ ₹ 150	6,000	12,000	18,000	27,000
Variable cost @ ₹ 500	20,000	40,000	60,000	90,000
Semi-variable costs (₹)				
Hall rent	8,000	16,000	24,000	28,800
Equipment charges	2,000	4,000	6,000	8,000
Trainer honorarium	2,000	4,000	6,000	8,000
	12,000	24,000	36,000	44,800
Fixed costs (₹)				
Guest speaker fees	10,000	10,000	10,000	10,000
Event coordination charges	5,000	5,000	5,000	5,000

Level of participants	40	80	120	180
Additional supervision cost	–	–	3,000	3,000
	15,000	15,000	18,000	18,000
Total Cost	47,000	79,000	1,14,000	1,52,800
Average Cost per participant (Total Cost/Number of Participants)	1,175	987.5	950	848.89

MCQ 10

(B) 180 Participants

Calculation of Break-even Level of Participants

Collection per participant = ₹ 850

Variable cost per participant = ₹ 500

Contribution per participant = ₹ 350

Semi-variable & Fixed Costs at Different Levels

Level of participants	50	100	150	200
Semi-variable costs (₹)	12,000	24,000	36,000	44,800
Fixed costs (₹)	15,000	15,000	18,000	18,000
Total Fixed Cost (₹)	27,000	39,000	54,000	62,800
Break- even students [Fixed cost / Contribution per participant]	77	111	154	180

Conclusion: Break-even level of 180 participants is within the relevant capacity of 200 participants. Break-even level of 77, 111, 154 participants is outside the relevant capacity, hence ignored for decision-making.

Note: Semi-variable costs are taken as fixed cost specifically for the purpose of calculating break even participants.

ICAI · P4 · MTP May 2026 · Series II · question p. 3-4 · answer p. 2, 2-3, 3

1127 MTP **Jan 2026 S-II** Q6(c) 4 marks

EXPLAIN the suitability for flexible budget.

ICAI'S ANSWER

Suitability for flexible budget:

1. Seasonal fluctuations in sales and/or production, for example in soft drinks industry;
2. a company which keeps on introducing new products or makes changes in the design of its products frequently;
3. industries engaged in make-to-order business like ship building;
4. an industry which is influenced by changes in fashion; and

5. general changes in sales.

ICAI · P4 · MTP Jan 2026 · Series II · question p. 18 · answer p. 22

1128 **MTP** **Sep 2025 S-II** Q3(b) **6 marks**

S Ltd. has prepared budget for the coming year for its two products A and B.

	Product A (₹)	Product B (₹)
Production & Sales unit	6,000 units	9,000 units
Raw material cost per unit	60.00	42.00
Direct labour cost per unit	30.00	18.00
Variable overhead per unit	12.00	6.00
Fixed overhead per unit	8.00	4.00
Selling price per unit	120.00	78.00

After some marketing efforts, the sales quantity of the Product A & B can be increased by 1,500 units and 500 units respectively but for this purpose the variable overhead and fixed overhead will be increased by 10% and 5% respectively for the both products.

You are required to PREPARE flexible budget for both the products:

(i) Before marketing efforts

(ii) After marketing efforts.

ICAI'S ANSWER

(i) **Flexible Budget before marketing efforts**

	Product A (₹)		Product B (₹)	
	6,000 units		**9,000 units**	
	Per unit	**Total**	**Per unit**	**Total**
Sales	120.00	7,20,000	78.00	7,02,000
Raw material cost	60.00	3,60,000	42.00	3,78,000
Direct labour cost per unit	30.00	1,80,000	18.00	1,62,000
Variable overhead per unit	12.00	72,000	6.00	54,000
Fixed overhead per unit	8.00	48,000	4.00	36,000
Total cost	110.00	6,60,000	70.00	6,30,000
Profit	10.00	60,000	8.00	72,000

(ii) **Flexible Budget after marketing efforts**

	Product A (₹)		Product B (₹)	
	7,500 units		**9,500 units**	
	Per unit	**Total**	**Per unit**	**Total**
Sales	120.00	9,00,000	78.00	7,41,000
Raw material cost	60.00	4,50,000	42.00	3,99,000
Direct labour cost per unit	30.00	2,25,000	18.00	1,71,000
Variable overhead per unit	13.20	99,000	6.60	62,700
Fixed overhead per unit	6.72	50,400	3.98	37,800
Total cost	109.92	8,24,400	70.58	6,70,500
Profit	10.08	75,600	7.42	70,500

ICAI · P4 · MTP Sep 2025 · Series II · question p. 12 · answer p. 16-17

1129 MTP **Sep 2025 S-II** Q6(b) 5 marks

You are the finance manager of a medium-sized manufacturing company. Recently, the management noticed that the actual expenses for the last quarter significantly exceeded the amounts allocated in the fixed budget, primarily due to unexpected fluctuations in production levels and raw material prices. The CEO has asked you to suggest how the budgeting system can be improved to address such deviations in the future.

Required:

IDENTIFY and EXPLAIN which type of budgeting system you would recommend in this situation and justify your answer with at least three key advantages it offers over the current system.

ICAI'S ANSWER

In this situation, It would be recommended to adopt a flexible budgeting system instead of relying solely on a fixed budget.

A flexible budget is a budget which, by recognising the difference in behaviour between fixed and variable costs in relation to fluctuations in output, turnover, or other variable factors, is designed to change appropriately with such fluctuations. According to CIMA, "**a flexible budget is defined as a budget which, by recognizing the difference between fixed, semi-variable and variable costs is designed to change in relation to the level of activity attained.**" Unlike static (fixed) budgets, the flexible budgets show the expected results of a responsibility center for different activity levels.

It will offer following advantages:

1. With the help of flexible budget, the sales, costs and profit may be calculated easily by the business at various levels of production capacity.
2. In flexible budget, adjustment is very simple according to change in business conditions.
3. It also helps in determination of production level as it shows budgeted costs with classification at various levels of activity along with sales. Hence the management can easily select the level of production which shows the profit predetermined by the owners of the business.

4. It also shows the quantity of product to be produced to earn determined profit.

ICAI · P4 · MTP Sep 2025 · Series II · question p. 16-17 · answer p. 25-26

1130 MTP **Sep 2024 S-II** Q5(b) 7 marks

PPP Ltd. is currently operating at 80% of its capacity producing 80,000 units. For the past two years, the production is increasing by 10% of its capacity consistently. The cost details are as follows:

	Year 3 (₹)	Year 2 (₹)	Year 1 (Current year) (₹)
Direct Materials	12,00,000	14,00,000	16,00,000
Direct Labour	6,00,000	7,00,000	8,00,000
Factory Overheads	3,20,000	3,40,000	3,60,000
Selling Overheads	3,40,000	3,80,000	4,20,000
Administrative Overheads	1,60,000	1,60,000	1,60,000
	26,20,000	29,80,000	33,40,000

The company is planning for 90% capacity level for next year.

Additional information:

Due to increase in demand of the raw material, the distributor is expected to increase the price by 10% from the next year.

At the beginning of the current year, the dispute occurred between workers and employees regarding wages which lead them to go on strike. Later on, they settled for 20% increase in wages from next year.

Following increases in overhead cost are expected for next year:

Variable Factory Overheads	5%
Fixed Factory Overheads	10%
Variable Selling Overheads	10%
Fixed Selling Overheads	15%
Administrative Overheads	15%

Profit is estimated @ 25% on total cost.

You are required to PREPARE flexible budget for the next year at 90% level of capacity.

Also ascertain profit and contribution.

ICAI'S ANSWER

PPP Ltd.

Budget for 90% capacity level for the next year

	Per Unit (₹)	Amount (₹)
Budgeted production (units)		90,000
Direct Material (note 2)	22	19,80,000
Direct Labour (note 3)	12	10,80,000
Variable factory overhead (note 4)	2.10	1,89,000
Variable selling overhead (note 5)	4.40	3,96,000
Variable cost	40.50	36,45,000
Fixed factory overhead (note 4)		2,20,000
Fixed selling overhead (note 5)		1,15,000
Administrative overhead (note 6)		1,84,000
Fixed cost		5,19,000
Total cost		41,64,000
Add: Profit 25% on total cost		10,41,000
Sales		52,05,000
Contribution (Sales – Variable cost)		15,60,000

Working Notes:

1. At 80% level of capacity (current year), the production is 80,000 units.
   Thus, total level of capacity is 1,00,000 units.

   Therefore, Year 2 is at 70% capacity and Year 3 is at 60% capacity as the production is increasing by 10% of its capacity consistently.

2. Direct Material

	(₹)		(₹)
80% Capacity	16,00,000	70% Capacity	14,00,000
70% Capacity	14,00,000	60% Capacity	12,00,000
10% change in capacity	2,00,000	10% change in capacity	2,00,000

For 10% increase in capacity, the total direct material cost regularly changes by ₹ 2,00,000

Thus, Direct material cost (variable) = ₹ 2,00,000 ÷ 10,000 = ₹ 20

After 10% increase in price, direct material cost per unit = ₹ 20 × 1.10 = ₹ 22

Direct material cost at 90,000 budgeted units = 90,000 × ₹ 22 = ₹ 19,80,000

3. Direct labour:

	(₹)		(₹)
80% Capacity	8,00,000	70% Capacity	7,00,000
70% Capacity	7,00,000	60% Capacity	6,00,000
10% change in capacity	1,00,000	10% change in capacity	1,00,000

For 10% increase in capacity, direct labour cost regularly changes by ₹ 1,00,000.

Direct labour cost per unit = ₹ 1,00,000 ÷ 10,000 = ₹ 10

After 20% increase in price, direct labour cost per unit = ₹ 10 × 1.20 = ₹ 12

Direct labour for 90,000 units = 90,000 units × ₹ 12 = ₹ 10,80,000.

4. Factory overheads are semi-variable overheads:

	(₹)		(₹)
80% Capacity	3,60,000	70% Capacity	3,40,000
70% Capacity	3,40,000	60% Capacity	3,20,000
10% change in capacity	20,000	10% change in capacity	20,000

Variable factory overhead = ₹ 20,000 ÷ 10,000 units = ₹ 2

Variable factory overhead for 80,000 units = 80,000 × ₹ 2 = ₹ 1,60,000

Fixed factory overhead = ₹ 3,60,000 – ₹ 1,60,000 = ₹ 2,00,000.

Variable factory overhead after 5% increase = ₹ 2 × 1.05 = ₹ 2.10

Fixed factory overhead after 10% increase = ₹ 2,00,000 × 1.10 = ₹ 2,20,000.

5. Selling overhead is semi-variable overhead:

	(₹)		(₹)
80% Capacity	4,20,000	70% Capacity	3,80,000
70% Capacity	3,80,000	60% Capacity	3,40,000
10% change in capacity	40,000	10% change in capacity	40,000

Variable selling overhead = ₹ 40,000 ÷ 10,000 units = ₹ 4

Variable selling overhead for 80,000 units = 80,000 × ₹ 4 = ₹ 3,20,000.

Fixed selling overhead = ₹ 4,20,000 – ₹ 3,20,000 = ₹ 1,00,000

Variable selling overhead after 10% increase = ₹ 4 × 1.10 = ₹ 4.40

Fixed selling overhead after 15% increase = ₹ 1,00,000 × 1.15 = ₹ 1,15,000

6. Administrative overhead is fixed:
   After 15% increase = ₹ 1,60,000 × 1.15 = ₹ 1,84,000

ICAI · P4 · MTP Sep 2024 · Series II · question p. 11-12 · answer p. 17-19

1131 MTP Sep 2024 S-II Q6(c) 4 marks

Fixed budgets are very simple to understand and less time consuming, however, only flexible budgets are more realistic and practicable because it gives due consideration to behaviour of revenue and cost at different levels of activity. But still there are certain demerits of both the budgets. NARRATE the same.

ICAI'S ANSWER

Demerits of Fixed Budget

1. It does not suite a dynamic organization and may give misleading results. A poor or good performance may remain un-noticed.
2. It is not suitable for long period.
3. It is also found unsuitable particularly when the business conditions are changing constantly.
4. Accurate estimates are not possible.

Demerits of Flexible Budget

1. The formulation of flexible budget is possible only when there is proper accounting system maintained, perfect knowledge about the factors of production and various business circumstances is available.
2. Flexible Budget also requires the system of standard costing in business.
3. It is very expensive and labour oriented.

ICAI · P4 · MTP Sep 2024 · Series II · question p. 12 · answer p. 20-21

1132 MTP May 2024 S-I MCQ 4 2 marks MCQ

The following extract is taken from the overhead budget of X:

Budgeted activity	50%	75%
Budgeted overhead (₹)	30,00,000	40,00,000

What would be the budgeted overhead for 60% level of activity:

A ₹ 32,00,0000

B ₹ 34,00,000

C ₹ 30,00,000

D ₹ 36,00,000

ICAI'S ANSWER

Answer key: **(B)**

B Variable overhead for each % of level of activity

$$=\frac{40,00,000-30,00,000}{75-50}=40,000$$

Fixed cost = 30,00,000 – (40,000 × 50) = 10,00,000

Total overheads for 60% level of activity

= 10,00,000 + (40,000 × 60) = 34,00,000

ICAI · P4 · MTP May 2024 · Series I · question p. 5 · answer p. 4

1133 MTP May 2024 S-II Q3(a) 10 marks

Aman International School has a total of 180 students consisting of 6 sections with 30 students per section. The school plans for a picnic around the city during the week-end to places such as Prayag zoo, the Capi Park, Azad planetarium etc. A private transport operator has come forward to lease out the buses for taking the students. Each bus will have a maximum capacity of 50 (excluding 2 seats reserved for the teachers accompanying the students). The school will employ two teachers for each bus, paying them an allowance of ₹ 500 per teacher. It will also lease out the required number of buses. The following are the other cost estimates:

	Cost per student (₹)
Breakfast	50
Lunch	100
Tea	10
Entrance fee at zoo	20

Rent ₹ 6500 per bus.

Special permit fee ₹ 500 per bus.

Block entrance fee at the planetarium ₹ 2500.

Prizes to students for games ₹ 500.

No cost are incurred in respect of the accompanying teachers (except the allowance of ₹ 500 per teacher).

You are required to PREPARE:

(a) A flexible budget estimating the total cost for the levels of 60, 90,120,150 and 180 students. Each item of cost is to be indicated separately.

(b) COMPARE the average cost per student at these levels.

(c) WHAT will be your conclusions regarding the break-been level of student if the school proposes to collect ₹ 400 per student?

ICAI'S ANSWER

(a) **Flexible Budget for different levels**

	₹	₹	₹	₹	₹
No. of Students	60	90	120	150	180
VARIABLE COST					
Breakfast	3000	4500	6000	7500	9000
Lunch	6000	9000	12000	15000	18000
Tea	600	900	1200	1500	1800
Entrance fee	1200	1800	2400	3000	3600

	₹	₹	₹	₹	₹
Sub-total (A)	10800	16200	21600	27000	32400
Variable cost/unit	180	180	180	180	180
SEMI-VARIABLE COST					
Bus rent	13000	13000	19500	19500	26000
Special permit fee	1000	1000	1500	1500	2000
Allowance for teachers	2000	2000	3000	3000	4000
Sub-total (B)	16000	16000	24000	24000	32000
FIXED COST					
Block entrance fee	2500	2500	2500	2500	2500
Prize to students	500	500	500	500	500
Sub total (C)	3000	3000	3000	3000	3000
Total cost (A + B + C)	29,800	35,200	48,600	54,000	67,400

(b)

Cost per student	496.67	391.11	405.00	360.00	374.44

(c)

Break-even level	₹
Collection per students	400
Less Variable Cost	180
Contribution	220

Since semi-fixed costs relate to a block of 50 students, the fixed and semi-variable cost for three level will be:

Level of Student	51–100	101–150	151-200
Fixed + Semi–variable cost (₹)	19,000	27,000	35,000
Contribution per unit (₹)	220	220	220
Break Even level of students	86	123	159

ICAI · P4 · MTP May 2024 · Series II · question p. 7-8 · answer p. 8-9

1134 MTP Old P3 May 2023 S-I Q3(a) 10 marks old scheme

SoyaB Limited is presently operating at 50% capacity and producing 50,000 units. The entire output is sold at a price of Rs. 180 per unit. The cost structure at the 50% level of activity is as under:

	(₹)
Direct Material	60 per unit
Direct Wages	20 per unit
Variable Overheads	20 per unit
Direct Expenses	12 per unit
Factory Expenses (30% fixed)	16 per unit
Selling and Distribution Exp. (85% variable)	10 per unit
Office and Administrative Exp. (100% fixed)	6 per unit

The company anticipates that the variable costs will go up by 20% and fixed costs will go up by 10%.

You are required to prepare an Expense budget, based on marginal cost for the company at 50%,75% and 100% level of activity and find out the profits at respective levels.

ICAI'S ANSWER

Expense Budget of SoyaB Ltd. for the period

	Per unit (₹)	50,000 units	75,000 units	1,00,000 units
		Amount (₹)	**Amount (₹)**	**Amount (₹)**
Sales **(A)**	180	90,00,000	1,35,00,000	1,80,00,000
Less: Variable Costs:				
- Direct Material	72	36,00,000	54,00,000	72,00,000
- Direct Wages	24	12,00,000	18,00,000	24,00,000
- Variable Overheads	24	12,00,000	18,00,000	24,00,000
- Direct Expenses	14.4	7,20,000	10,80,000	14,40,000
- Variable factory expenses (70% of Rs 16 p.u.)x 120%	13.44	6,72,000	10,08,000	13,44,000
- Variable Selling & Dist. exp. (85% of Rs 10 p.u.)x120%	10.2	5,10,000	7,65,000	10,20,000
Total Variable Cost (**B**)	158.04	79,02,000	1,18,53,000	1,58,04,000
Contribution **(C) = (A – B)**	21.96	10,98,000	16,47,000	21,96,000
Less: Fixed Costs:				

	Per unit (₹)	50,000 units	75,000 units	1,00,000 units
- Office and Admin. exp. (100%)	--	3,30,000	3,30,000	3,30,000
- Fixed factory exp. (30%)	--	2,64,000	2,64,000	2,64,000
- Fixed Selling & Dist. exp. (15%)	--	82,500	82,500	82,500
Total Fixed Costs (**D**)	--	6,76,500	6,76,500	6,76,500
Profit (**C – D**)	--	4,21,500	9,70,500	15,19,500

ICAI · Old P3 · MTP May 2023 · Series I · question p. 3 · answer p. 5

1135 MTP **Old P3 May 2023 S-II** Q1(b) 5 marks old scheme

SP Ltd. has prepared budget for the coming year for its two products A and B.

	Product A (₹)	Product B (₹)
Production & Sales unit	6,000 units	9,000 units
Raw material cost per unit	60.00	42.00
Direct labour cost per unit	30.00	18.00
Variable overhead per unit	12.00	6.00
Fixed overhead per unit	8.00	4.00
Selling price per unit	120.00	78.00

After some marketing efforts, the sales quantity of the Product A & B can be increased by 1,500 units and 500 units respectively but for this purpose the variable overhead and fixed overhead will be increased by 10% and 5% respectively for the both products.

You are required to PREPARE flexible budget for both the products:

(i) Before marketing efforts

(ii) After marketing efforts

ICAI'S ANSWER

(i) Flexible Budget before marketing efforts:

	Product A (₹) 6,000 units		Product B (₹) 9,000 units	
	Per unit	**Total**	**Per unit**	**Total**
Sales	120.00	7,20,000	78.00	7,02,000
Raw material cost	60.00	3,60,000	42.00	3,78,000
Direct labour cost per unit	30.00	1,80,000	18.00	1,62,000
Variable overhead per unit	12.00	72,000	6.00	54,000
Fixed overhead per unit	8.00	48,000	4.00	36,000
Total cost	110.00	6,60,000	70.00	6,30,000
Profit	10.00	60,000	8.00	72,000

(ii) Flexible Budget after marketing efforts:

	Product A (₹) 7,500 units		Product B (₹) 9,500 units	
	Per unit	**Total**	**Per unit**	**Total**
Sales	120.00	9,00,000	78.00	7,41,000
Raw material cost	60.00	4,50,000	42.00	3,99,000
Direct labour cost per unit	30.00	2,25,000	18.00	1,71,000
Variable overhead per unit	13.20	99,000	6.60	62,700
Fixed overhead per unit	6.72	50,400	3.98	37,800
Total cost	109.92	8,24,400	70.58	6,70,500
Profit	10.08	75,600	7.42	70,500

ICAI · Old P3 · MTP May 2023 · Series II · question p. 1 · answer p. 1-2

1136 MTP **Old P3 Nov 2022 S-I** Q4(b) 10 marks old scheme

A firm has a total capacity of producing 1,00,000 units of an item. The budgeted expenses at this level of

activity are as under:

	Per unit (₹)
Direct Materials	650
Direct Wages	325
Direct Expenses	125
Variable overheads	50
Fixed Production Overheads	25
Selling and Distribution Overheads (20% fixed)	25
Administrative Expenses (100% fixed)	60
Total	1,260

The selling price is ₹ 1,750 per unit and is anticipated to remain constant.

You are required to PREPARE a flexible budget, on the basis of marginal costing, for 60,000 and 75,000 units of output level showing the profit and P/V Ratio.

ICAI'S ANSWER

Workings -

1. Fixed Production overheads (given) = ₹ 25 per unit

So, at 1,00,000 units capacity, it will be ₹ 25,00,000 (1,00,000 units x ₹ 25)

2. Selling and distribution overheads:

Given (1,00,000 units x ₹ 25) = ₹ 25,00,000

So, Fixed component = ₹ 25,00,000 x 20% = ₹ 5,00,000

Hence, variable component = ₹ 25,00,000 - ₹ 5,00,000 = ₹ 20,00,000

Variable per unit = ₹ 20,00,000/1,00,000 units

= ₹ 20 per unit

Flexible Budget

Particulars	Per unit (₹)	Output Level	
		60,000 units (₹)	**75,000 units (₹)**
Sales (A)	1,750	10,50,00,000	13,12,50,000
Variable costs:			
Direct Material	650	**3,90,00,000**	**4,87,50,000**
Direct Wages	325	**1,95,00,000**	**2,43,75,000**
Direct expenses	125	**75,00,000**	**93,75,000**

Particulars	Per unit (₹)	Output Level	
Variable overheads	50	**30,00,000**	**37,50,000**
Selling and distribution overheads	20	12,00,000	15,00,000
Total Variable cost **(B)**	1,170	7,02,00,000	8,77,50,000
Contribution (C = A - B)		**3,48,00,000**	**4,35,00,000**
Fixed costs:			
Production overheads		25,00,000	25,00,000
Administrative overheads		60,00,000	60,00,000
Selling and distribution overheads		**5,00,000**	**5,00,000**
Total Fixed cost **(D)**		90,00,000	90,00,000
Profit (C-D)		**2,58,00,000**	**3,45,00,000**

P/V Ratio = (₹ 3,48,00,000/₹ 10,50,00,000) x 100 = 33.143%

OR

P/V Ratio = (₹ 4,35,00,000/₹ 13,12,50,000) x 100 = 33.143%

ICAI · Old P3 · MTP Nov 2022 · Series I · question p. 6 · answer p. 8-9

1137 MTP **Old P3 May 2022 S-II** Q6(e) 5 marks old scheme

BRIEF OUT difference between Fixed and Flexible Budget.

ICAI'S ANSWER

S. No.	Fixed Budget	Flexible Budget
1.	It does not change with actual volume of activity achieved. Thus it is known as rigid or inflexible budget.	It can be re-casted on the basis of activity level to be achieved. Thus it is not rigid.
2.	It operates on one level of activity and under one set of conditions. It assumes that there will be no change in the prevailing conditions, which is unrealistic.	It consists of various budgets for different levels of activity.
3.	Here as all costs like - fixed, variable and semi-variable are related to only one level of activity so variance analysis does not give useful information.	Here analysis of variance provides useful information as each cost is analysed according to its behaviour.
4.	If the budgeted and actual activity levels differ significantly, then the aspects like cost ascertainment and price fixation do not give a correct picture.	Flexible budgeting at different levels of activity facilitates the ascertainment of cost, fixation of selling price and tendering of quotations.

S. No.	Fixed Budget	Flexible Budget
5.	Comparison of actual performance with budgeted targets will be meaningless specially when there is a difference between the two activity levels.	It provides a meaningful basis of comparison of the actual performance with the budgeted targets.

ICAI · Old P3 · MTP May 2022 · Series II · question p. 7 · answer p. 13

1138 MTP **Old P3 Jul 2021 S-II** Q5(b) 10 marks old scheme

Tricon Co. furnishes the following information for the month of September, 2020.

Particulars	Budget Details	Static Budget	Actual
Units produced & Sold		4,000	3,200
		(Rs.)	**(Rs.)**
Direct Material	3 kg p.u. @ Rs. 30 per kg.	3,60,000	3,10,000
Direct Labour	1 hr. p.u. @ Rs. 72 per hr.	2,88,000	2,25,600
Variable Overhead	1 hr. p.u. @ Rs. 44 per hr.	1,76,000	1,47,200
Fixed Overhead		1,80,000	1,68,000
Total Cost		10,04,000	8,50,800
Sales		12,00,000	8,96,000
Profit		1,96,000	45,200

During the month 10,000 kg. of materials and 3,100 direct labour hours were utilized.

Required:

(i) PREPARE a flexible budget for the month.

(ii) DETERMINE the material usage variance and the direct labour rate variance for the actual vs the flexible budget.

ICAI'S ANSWER

(i) Statement Showing "Flexible Budget for 3,200 units Activity Level"

Particulars	Amount (Rs.)	Amount (Rs.)
Sales $\left(\frac{\text{Rs. } 12,00,000}{4,000 \text{ units}} \text{ x } 3,200 \text{ units}\right)$		9,60,000
Less: Variable Cost		
Direct Material (3,200 units × 3 kg. p.u. × Rs. 30 per kg.)	2,88,000	
Direct Labour (3,200 units × 1 hr. p.u. × Rs. 72 per hr.)	2,30,400	
Variable Overhead (3,200 units × 1 hr. p.u. × Rs. 44 per hr.)	1,40,800	(6,59,200)
Contribution		3,00,800
Less: Fixed Overhead		1,80,000
Profit		1,20,800

(ii) Computation of Variances

Material Usage Variance = Standard Cost of Standard Quantity for Actual Production – Standard Cost of Actual Quantity

= (SQ × SP) – (AQ × SP)

Or

= (SQ – AQ) × SP

= [(3,200 units × 3 kg.) – 10,000 kg.] × Rs. 30.00

= Rs. 12,000 (A)

Labour Rate Variance = Standard Cost of Actual Time – Actual Cost

= (SR × AH) – (AR × AH)

Or

= (SR – AR) × AH

$= \left[\left(\text{Rs. } 72 - \frac{\text{Rs. } 2,25,600}{3,100 \text{ hrs.}}\right) \text{ x } 3,100 \text{ hrs.}\right]$

= Rs. 2,400 (A)

ICAI · Old P3 · MTP Jul 2021 · Series II · question p. 6-7 · answer p. 10-11

1139 MTP **Old P3 May 2020 S-I** Q4(b) 10 marks old scheme

During the FY 2019-20, GP Limited has produced 30,000 units operating at 50% capacity level. The cost

structure at the 50% level of activity is as under:

Particulars	Rs.
Direct Material	150 per unit
Direct Wages	50 per unit
Variable Overheads	50 per unit
Direct Expenses	30 per unit
Factory Expenses (25% fixed)	40 per unit
Selling and Distribution Exp. (80% variable)	20 per unit
Office and Administrative Exp. (100% fixed)	10 per unit

The company anticipates that in FY 2020-21, the variable costs will go up by 10% and fixed costs will go up by 15%.

The selling price per unit will remain unchanged at Rs.400.

Required:

(i) CALCULATE the budgeted profit/ loss for the FY 2019-20.

(ii) PREPARE an Expense budget on marginal cost basis for the FY 2020-21 for the company at 50% and 60% level of activity and FIND OUT the profits at respective levels.

ICAI'S ANSWER

(i) Calculation of Budgeted profit for the FY 2019-20

		30,000 units	
		Per unit (Rs.)	**Amount (Rs.)**
Sales	**(A)**	400.00	1,20,00,000
Less: Variable Costs:			
- Direct Material		150.00	45,00,000
- Direct Wages		50.00	15,00,000
- Variable Overheads		50.00	15,00,000
- Direct Expenses		30.00	9,00,000
- Variable factory expenses (75% of Rs.40p.u.)		30.00	9,00,000
- Variable Selling & Dist. exp. (80% of Rs.20p.u.)		16.00	4,80,000
Total Variable Cost	**(B)**	326.00	97,80,000
Contribution	**(C) = (A – B)**	74.00	22,20,000
Less: Fixed Costs:			

		30,000 units	
- Office and Admin. exp. (100%)		--	3,00,000
- Fixed factory exp. (25%)		--	3,00,000
- Fixed Selling & Dist. exp. (20%)		--	1,20,000
Total Fixed Costs	**(D)**	--	7,20,000
Profit	**(C – D)**	--	15,00,000

(ii) Expense Budget of GP Ltd. for the FY 2020-21 at 50% & 60% level

		30,000 units		36,000 units	
		Per unit (Rs.)	**Amount (Rs.)**	**Per unit (Rs.)**	**Amount (Rs.)**
Sales	**(A)**	400.00	1,20,00,000	400.00	1,44,00,000
Less: Variable Costs:					
- Direct Material		165.00	49,50,000	165.00	59,40,000
- Direct Wages		55.00	16,50,000	55.00	19,80,000
- Variable Overheads		55.00	16,50,000	55.00	19,80,000
- Direct Expenses		33.00	9,90,000	33.00	11,88,000
- Variable factory expenses		33.00	9,90,000	33.00	11,88,000
- Variable Selling & Dist. exp.		17.60	5,28,000	17.60	6,33,600
Total Variable Cost	**(B)**	358.60	1,07,58,000	358.60	1,29,09,600
Contribution	**(C) = (A – B)**	41.40	12,42,000	41.40	14,90,400
Less: Fixed Costs:					
- Office and Admin. exp. (100%)		--	3,45,000	--	3,45,000
- Fixed factory exp. (25%)		--	3,45,000	--	3,45,000
- Fixed Selling & Dist. exp. (20%)		--	1,38,000	--	1,38,000
Total Fixed Costs	**(D)**	--	8,28,000	--	8,28,000
Profit	**(C – D)**	--	4,14,000	--	6,62,400

15.4 Cash budget

8 questions · asked in 4 ICAI documents

1140 RTP **May 2026** Case scenario · Q1(i) to Q1(v)

CASE SCENARIO

ABC Enterprises is a well-established trading firm specialised in the distribution and sale of a diverse range of products, with a commitment to deliver superior quality to its customers.

Following informations are provided by ABC Enterprises:

(a) The company is engaged in both cash and credit sales, with credit sales being twice of cash sales. Cash sales are eligible for 3% discount.

(b) 60% of credit sales are collected within the same month of sale. The remaining 40% is collected in the following month. All receivables are expected to be fully recovered, with no bad debts anticipated.

(c) The company employs a structured pricing policy, ensuring that the selling price includes a 20% profit margin on the selling price.

(d) Goods are purchased one month before the month in which these are to be sold.

(e) The company procures goods through a combination of cash and credit, with cash purchases comprising 40% of the total purchases. Half of the credit purchases are paid within the same month of purchase, while the remaining 50% is settled in the subsequent month.

(f) The fourth installment of advance tax, amounting to ₹ 50,000, is due for payment by 15th March, 2025.

(g) The opening cash in hand as on 1st February, 2025 stands at ₹ 60,000.

The projected sales figures are as follows:

Month	Sales (₹)
January, 2025	9,00,000
February, 2025	7,50,000
March, 2025	8,25,000
April, 2025	6,75,000
May, 2025	6,00,000

On the basis of above Case Scenario, you are required to answer the following MCQs:

Q1(i)

What is the net cash sales (after discount) for the month of February 2025?

A ₹ 2,42,500

B ₹ 2,50,000

C ₹ 2,66,750

D ₹ 2,75,000

Q1(ii)

What is the amount of total collection from debtors for the month of March 2025?

- A ₹ 5,13,000
- B ₹ 5,40,000
- C ₹ 5,03,500
- D ₹ 5,30,000

Q1(iii)

What is the amount of credit purchases for the month of February 2025?

- A ₹ 3,60,000
- B ₹ 3,96,000
- C ₹ 3,24,000
- D ₹ 4,95,000

Q1(iv)

What is the amount of payment to creditors for the month of March 2025?

- A ₹ 3,42,000
- B ₹ 3,78,000
- C ₹ 3,60,000
- D ₹ 4,50,000

Q1(v)

What is the amount of closing cash in hand as on 31st March, 2025?

- A ₹ 3,61,250
- B ₹ 3,71,250
- C ₹ 3,23,250
- D ₹ 2,00,500

ICAI'S ANSWERS

ICAI's working for this case

Working Note No. 1: Calculation of Sales

Credit Sales = Twice of cash sales

If Cash sales = X, then Credit sales = 2X

Total sales = X + 2X = 3X

Cash Sales = Total Sales / 3

Credit Sales = 2/3 of Total Sales

(In ₹)

Sales	January 2025	February 2025	March 2025	April 2025
Total Sales	9,00,000	7,50,000	8,25,000	6,75,000
Cash Sales	3,00,000	2,50,000	2,75,000	2,25,000
Discount @3%	9,000	7,500	8,250	6,750
Net Cash Sales	2,91,000	2,42,500	2,66,750	2,18,250
Credit Sales	6,00,000	5,00,000	5,50,000	4,50,000

Working Note No. 2: Collection from Debtors

60% of credit sales are collected in the same month

40% of credit sales are collected in the following month

- Current Month Collection = 60% of Credit Sales
- Next Month Collection = 40% of Previous Month's Credit Sales

(In ₹)

Particulars	January 2025	February 2025	March 2025
Credit Sales	6,00,000	5,00,000	5,50,000
60% of credit sale (same month)	3,60,000	3,00,000	3,30,000
40% next month	–	2,40,000	2,00,000
Total Collection	**–**	**5,40,000**	**5,30,000**

Working Note No. 3: Calculation of Purchases

Selling price includes a 20% profit margin on selling price.

Goods are purchased one month before sale.

For sales of February 2025, goods were purchased in January 2025.

Sales for February 2025 = ₹ 7,50,000

Purchases for January 2025 = 80% of Sales

= ₹ 7,50,000 × 80% = ₹ 6,00,000

(In ₹)

Particulars	January 2025	February 2025	March 2025
Purchases	6,00,000	6,60,000	5,40,000
Cash Purchases (40%)	2,40,000	2,64,000	2,16,000
Credit Purchases	3,60,000	3,96,000	3,24,000

Working Note No. 4: Payment to Creditors

(In ₹)

Particulars	January 2025	February 2025	March 2025
Credit Purchases	3,60,000	3,96,000	3,24,000
Payment–Same Month @50%	1,80,000	1,98,000	1,62,000
Balance Next Month @50%		1,80,000	1,98,000
Payment to Creditors		3,78,000	3,60,000

Working Note No. 5: Cash Budget

(In ₹)

Particulars	February 2025	March 2025
Receipts		
Opening Cash in Hand	60,000	2,00,500
Net Cash Sales (W.N.1)	2,42,500	2,66,750
Collection from Debtors (W.N.2)	5,40,000	5,30,000
Total Receipts (A)	8,42,500	9,97,250
Payments		
Cash Purchases (W.N.3)	2,64,000	2,16,000
Payment to Creditors (W.N.4)	3,78,000	3,60,000
Advance Income Tax	–	50,000
Total Payments (B)	6,42,000	6,26,000
Closing Cash in Hand (A – B)	2,00,500	3,71,250

Q1(i)

(A): ₹ 2,42,500

Q1(ii)

(D): ₹ 5,30,000

Q1(iii)

(B): ₹ 3,96,000

Q1(iv)

(C): ₹ 3,60,000

Q1(v)

(B): ₹ 3,17,750

PassBook's note: ICAI's key line reads "(B): ₹ 3,17,750", while option (B) and ICAI's own Working Note No. 5 give the closing cash in hand on 31st March as ₹ 3,71,250. The key is ICAI's (B).

ICAI · P4 · RTP May 2026 · question p. 1-2 · answer p. 21

1141 MTP **Sep 2025 S-II** MCQ 15 2 marks MCQ

PJ Ltd. is currently preparing cash budget for the year 2025. An extract from its sales budget for the same year shows the following sales values:

	(₹ '000)
March	750
April	900
May	1,000
June	600

40% of its sales are expected to be for cash. Of its credit sales, 80% are expected to be realised in the next month and the balance in the second month. The value of sales receipts to be shown in cash budget for May 2025 is (in ₹ '000):

A 840

B 922

C 948

D 1060

ICAI'S ANSWER

(B) 922

Cash sales in May: 40% of ₹ 10,00,000 = ₹ 4,00,000

Credit sales from April realized in May: 80% of (60% of ₹ 9,00,000)

= 0.8 × 0.6 × ₹ 9,00,000 = ₹ 4,32,000

Credit sales from March realized in May: 20% of (60% of ₹750,000)

= 0.2 × 0.6 × ₹ 7,50,000 = ₹ 90,000

Total cash receipts for May = ₹ 4,00,000 + ₹ 4,32,000 + ₹ 90,000

= ₹ 9,22,000

ICAI · P4 · MTP Sep 2025 · Series II · question p. 8 · answer p. 7

1142 MTP **May 2025 S-II** Q5(b) 6 marks

Bricks & Cement Pvt. Ltd. is a building supplies company offering a wide range of materials like bricks, cement, and aggregates to both trade professionals and private customers. The company focuses on providing high-quality products and reliable service for all type of construction projects.

The management provides the following budgeted data for each of the six months in the first half of the year:

Particulars	April (₹)	May (₹)	June (₹)	July (₹)	August (₹)	September (₹)
Credit sales	22,50,000	22,50,000	22,50,000	23,40,000	23,40,000	25,20,000
Cash sales	5,40,000	5,40,000	5,85,000	6,75,000	7,20,000	8,10,000
Credit purchases	15,30,000	16,20,000	16,20,000	18,00,000	18,00,000	18,00,000
Other operating costs (excluding depreciation)	8,10,000	8,10,000	8,10,000	10,98,000	11,07,000	11,07,000

Following table shows the distribution of payments received against credit sales:

Value of credit sales	Payment received
80%	One month after sale
10%	Two months after sale
8%	Three months after sale
Balance	Written off as a bad debt

However, credit purchases are paid in the following manner:

Value of credit purchases	Payment made
75%	One month after purchase
Balance	Two months after purchase

Any remaining operational costs, including overheads, utilities, and day-to-day expenses, are paid in the month they are incurred. This ensures that all routine expenses are settled promptly without delay, maintaining smooth financial operations throughout the period.

During the month of July, the company has also placed an order for four pallet jacks that wiII cost around ₹ 2,25,000 each, the payment of which is divided into four equal instalments following the month of purchase.

The cash balance at the beginning of 2nd quarter is projected to be ₹ 1,35,000.

You are required to PREPARE a cash budget for each of the three months of 2nd Quarter.

ICAI'S ANSWER

Cash Budget

Particulars	July (₹)	August (₹)	September (₹)
Cash sales	6,75,000	7,20,000	8,10,000
Receipts from credit sales (WN1)	22,05,000	22,77,000	22,86,000
Total receipts (A)	28,80,000	29,97,000	30,96,000

Particulars	July (₹)	August (₹)	September (₹)
Payment for purchases (WN2)	(16,20,000)	(17,55,000)	(18,00,000)
Other operating costs paid	(10,98,000)	(11,07,000)	(11,07,000)
Pallet jacks		(2,25,000)	(2,25,000)
Total payments (B)	(27,18,000)	(30,87,000)	(31,32,000)
Net cash (A - B)	1,62,000	(90,000)	(36,000)
Opening balance	1,35,000	2,97,000	2,07,000
Closing balance	**2,97,000**	**2,07,000**	**1,71,000**

Working Notes:

(WN1) Credit sales - receipts

Particulars	Total Sales (₹)	July (₹)	August (₹)	September (₹)
April	22,50,000	1,80,000	-	-
May	22,50,000	2,25,000	1,80,000	-
June	22,50,000	18,00,000	2,25,000	1,80,000
July	23,40,000	-	18,72,000	2,34,000
August	23,40,000	-	-	18,72,000
Total		**22,05,000**	**22,77,000**	**22,86,000**

(WN2) Credit purchases – payments

Particulars	Total purchases (₹)	July (₹)	August (₹)	September (₹)
May	16,20,000	4,05,000	-	-
June	16,20,000	12,15,000	4,05,000	-
July	18,00,000	-	13,50,000	4,50,000
August	18,00,000	-	-	13,50,000
Total		**16,20,000**	**17,55,000**	**18,00,000**

ICAI · P4 · MTP May 2025 · Series II · question p. 13-14 · answer p. 16-17

1143 MTP **Jan 2025 S-I** Q4(b) 8 marks

The following information relates to Anu Limited, a AI enabled toy manufacturing company:

The selling price of a toy is ₹ 3,000, and sales are made on credit and invoiced on the last day of the month.

Variable costs of production per toy are materials (₹ 1,000), labour (₹ 800), and overhead (₹ 400)

The sales manager has forecasted the following volumes:

Month	No. of Toys
November	1,000
December	1,000
January	1,000
February	1,250
March	1,500
April	2,000
May	1,900
June	2,200
July	2,200
August	2,300

Customers are expected to pay 50% One month after the sale and 50% Two months after the sale.

The company produces the toys two months before they are sold and the creditors for materials are paid two months after production.

Variable overheads are paid in the month following production and are expected to increase by 25 % in April; 75% of wages are paid in the month of production and 25% in the following month. A wage increase of 25% will take place on 1st March.

The company needs funds for the running the business and purchase of new machine so it will sell one of its freehold properties in June for ₹ 20,00,000, and buy a new machine in June for ₹ 5,00,000.

Depreciation is currently ₹ 10,000 per month, and will rise to ₹ 15,000 after the purchase of the new machine.

The company's corporation tax of ₹ 1,00,000 is due for payment in March.

The company presently has a cash balance at bank on 31 December 2023, of ₹ 50,000.

You are required to PREPARE a cash budget for the six months from January to June, 2024.

ICAI'S ANSWER

Workings:

1. Sale receipts

Month	Nov	Dec	Jan	Feb	Mar	Apr	May	Jun
Forecast sales (S)	1,000	1,000	1,000	1,250	1,500	2,000	1,900	2,200
	₹	₹	₹	₹	₹	₹	₹	₹
S×3000	30,00,000	30,00,000	30,00,000	37,50,000	45,00,000	60,00,000	57,00,000	66,00,000
Debtors pay:								
1 month 50%		15,00,000	15,00,000	15,00,000	18,75,000	22,50,000	30,00,000	28,50,000
2nd month 50%		-	15,00,000	15,00,000	15,00,000	18,75,000	22,50,000	30,00,000
	-	15,00,000	30,00,000	30,00,000	33,75,000	41,25,000	52,50,000	58,50,000

2. Variable overheads

Month	Nov	Dec	Jan	Feb	Mar	Apr	May	Jun
Qty produced (Q)	1,000	1,250	1,500	2,000	1,900	2,200	2,200	2,300
	₹	₹	₹	₹	₹	₹	₹	₹
Var. overhead (Q×400)	4,00,000	5,00,000	6,00,000	8,00,000	7,60,000			
Var. overhead (Q×500)						11,00,000	11,00,000	11,50,000
Paid one month later		4,00,000	5,00,000	6,00,000	8,00,000	7,60,000	11,00,000	11,00,000

3. Wages payments

Month	Dec	Jan	Feb	Mar	Apr	May	Jun
Qty produced (Q)	1,250	1,500	2,000	1,900	2,200	2,200	2,300
	₹	₹	₹	₹	₹	₹	₹
Wages (Q × 800)	10,00,000	12,00,000	16,00,000				
Wages (Q × 1,000)				19,00,000	22,00,000	22,00,000	23,00,000
75% this month	7,50,000	9,00,000	12,00,000	14,25,000	16,50,000	16,50,000	17,25,000

Month	Dec	Jan	Feb	Mar	Apr	May	Jun
25% next month		2,50,000	3,00,000	4,00,000	4,75,000	5,50,000	5,50,000
	7,50,000	11,50,000	15,00,000	18,25,000	21,25,000	22,00,000	22,75,000

CASH BUDGET – SIX MONTHS ENDED JUNE

	Jan	Feb	Mar	Apr	May	Jun
	₹	₹	₹	₹	₹	₹
Receipts:						
Sales receipts	30,00,000	30,00,000	33,75,000	41,25,000	52,50,000	58,50,000
Freehold property	-	-	-	-	-	20,00,000
	30,00,000	**30,00,000**	**33,75,000**	**41,25,000**	**52,50,000**	**78,50,000**
Payments:						
Materials	10,00,000	12,50,000	15,00,000	20,00,000	19,00,000	22,00,000
Var. overheads	5,00,000	6,00,000	8,00,000	7,60,000	11,00,000	11,00,000
Wages	11,50,000	15,00,000	18,25,000	21,25,000	22,00,000	22,75,000
Machine	-	-	-	-	-	5,00,000
Tax	-	-	1,00,000	-	-	-
	26,50,000	**33,50,000**	**42,25,000**	**48,85,000**	**52,00,000**	**60,75,000**
Net cash flow	3,50,000	(3,50,000)	(8,50,000)	(7,60,000)	50,000	17,75,000
Balance b/f	50,000	4,00,000	50,000	(8,00,000)	(15,60,000)	(15,10,000)
Cumulative cash flow	4,00,000	50,000	(8,00,000)	(15,60,000)	(15,10,000)	2,65,000

ICAI · P4 · MTP Jan 2025 · Series I · question p. 14-15 · answer p. 15-16

15.5 Zero based budgeting and performance budgeting

7 questions · asked in 8 ICAI documents · 2 exam papers

1144 EXAM Old P3 Nov 2020 Q6(b) 5 marks old scheme

What are the important points an organization should consider if it wants to adopt Performance Budgeting?

ICAI'S ANSWER

For an enterprise that wants to adopt Performance Budgeting, it is thus imperative that:

- the objectives of the enterprise are spelt out in concrete terms.
- the objectives are then translated into specific functions, programmes, activities and tasks for different levels of management within the realities of fiscal constraints.
- realistic and acceptable norms, yardsticks or standards and performance indicators should be evolved and expressed in quantifiable physical units.
- a style of management based upon decentralised responsibility structure should be adopted, and
- an accounting and reporting system should be developed to facilities monitoring, analysis and review of actual performance in relation to budgets.

ICAI · Old P3 · Nov 2020 exam · question p. 22 · answer p. 23

1145 EXAM **Old P3 Nov 2019** Q6(e) 5 marks old scheme

Define Zero Base Budgeting and mention its various stages.

ICAI'S ANSWER

Zero-based Budgeting: (ZBB) is an emergent form of budgeting which arises to overcome the limitations of incremental (traditional) budgeting system. Zero- based Budgeting (ZBB) is defined as 'a method of budgeting which requires each cost element to be specifically justified, although the activities to which the budget relates are being undertaken for the first time, without approval, the budget allowance is zero'.

ZBB is an activity based budgeting system where budgets are prepared for each activities rather than functional department. Justification in the form of cost benefits for the activity is required to be given. The activities are then evaluated and prioritized by the management on the basis of factors like synchronisation with organisational objectives, availability of funds, regulatory requirement etc.

ZBB is suitable for both corporate and non-corporate entities. In case of non-corporate entities like Government department, local bodies, not for profit organisations, where these entities need to justify the benefits of expenditures on social programmes like mid -day meal, installation of street lights, provision of drinking water etc.

ZBB involves the following stages:

(i) Identification and description of Decision packages

(ii) Evaluation of Decision packages

(iii) Ranking (Prioritisation) of the Decision packages

(iv) Allocation of resources

ICAI · Old P3 · Nov 2019 exam · question p. 24 · answer p. 26-27

1146 RTP **Old P3 Nov 2020** Q15(c) old scheme

Also asked in Old P3 MTP May 2023 S-I Q6(a)

STATE the advantages of Zero-based budgeting.

ICAI'S ANSWER

The advantages of zero-based budgeting are as follows:

- It provides a systematic approach for the evaluation of different activities and ranks them in order of preference for the allocation of scarce resources.
- It ensures that the various functions undertaken by the organization are critical for the achievement of its objectives and are being performed in the best possible way.
- It provides an opportunity to the management to allocate resources for various activities only after having a thorough cost-benefit-analysis. The chances of arbitrary cuts and enhancement are thus avoided.
- The areas of wasteful expenditure can be easily identified and eliminated.
- Departmental budgets are closely linked with corporation objectives.
- The technique can also be used for the introduction and implementation of the system of 'management by objective.' Thus, it cannot only be used for fulfillment of the objectives of traditional budgeting but it can also be used for a variety of other purposes.

ICAI · Old P3 · RTP Nov 2020 · question p. 12 · answer p. 30

1147 MTP **May 2026 S-I** Q5(b) 5 marks

Briefly EXPLAIN Zero-based Budgeting (ZBB) along with its steps

ICAI'S ANSWER

Zero-based Budgeting (ZBB) is defined as a method of budgeting which requires each cost element to be specifically justified, though the activities to which the budget relates are not being undertaken for the first time. The cost of each activity has to be justified and without justification, the budget allowance is zero.

ZBB involves the following stages:

(i) Identification and description of Decision packages

(ii) Evaluation of Decision packages

(iii) Ranking (Prioritisation) of the Decision packages

(iv) Allocation of resources

ICAI · P4 · MTP May 2026 · Series I · question p. 14 · answer p. 18-19

1148 MTP **May 2024 S-I** MCQ 5 2 marks MCQ

Which of the following statements relating to Zero Based Budgeting (ZBB) is false:

A. It is a method of budgeting whereby all activities are re-evaluated each time a budget is formulated.

B. ZBB attempts to eliminate unnecessary expenditure being retained in budgets.

C. It is probably the least time consuming and least costly approach to budgeting.

D. It requires that budgets are built up from scratch.

ICAI'S ANSWER

Answer key: **(C)**

C

ICAI · P4 · MTP May 2024 · Series I · question p. 5 · answer p. 4

1149 MTP Old P3 Dec 2021 S-I Q6(c) 5 marks old scheme

DEFINE Zero Based Budgeting and mention its various stages.

ICAI'S ANSWER

Zero-based Budgeting: (ZBB) is an emergent form of budgeting which arises to overcome the limitations of incremental (traditional) budgeting system. Zero- based Budgeting (ZBB) is **defined** as 'a method of budgeting which requires each cost element to be specifically justified, although the activities to which the budget relates are being undertaken for the first time, without approval, the budget allowance is zero'.

ZBB is an activity based budgeting system where budgets are prepared for each activities rather than functional department. Justification in the form of cost benefits for the activity is required to be given. The activities are then evaluated and prioritized by the management on the basis of factors like synchronisation with organisational objectives, availability of funds, regulatory requirement etc.

ZBB is suitable for both corporate and non-corporate entities. In case of non-corporate entities like Government department, local bodies, not for profit organisations, where these entities need to justify the benefits of expenditures on social programmes like mid-day meal, installation of street lights, provision of drinking water etc.

ZBB involves the following stages:

(i) Identification and description of Decision packages

(ii) Evaluation of Decision packages

(iii) Ranking (Prioritisation) of the Decision packages

(iv) Allocation of resources

ICAI · Old P3 · MTP Dec 2021 · Series I · question p. 6 · answer p. 12

1150 MTP Old P3 May 2020 S-I Q6(b) 5 marks old scheme

EXPLAIN the stages in Zero-based budgeting.

ICAI'S ANSWER

Zero-based budgeting (ZBB) involves the following stages:

(i) Identification and description of Decision packages

(ii) Evaluation of Decision packages

(iii) Ranking (Prioritisation) of the Decision packages

(iv) Allocation of resources

(i) Identification and description of Decision packages: Decision packages are the programmes or activities for which decision is required to be taken. The programmes or activities are described for technical specifications, financial impact in the form of cost benefit analysis and other issues like environmental, regulatory, social etc.

(ii) Evaluation of Decision packages: Once Decision packages are identified and described, it is evaluated against factors like synchronisation with organisational objectives, availability of funds, regulatory requirement etc.

(iii) Ranking (Prioritisation) of the Decision packages: After evaluation of the decision packages, it is ranked on the basis priority of the activities. Because of this prioritization feature ZBB is also known as ***Priority-based Budgeting***.

(iv) Allocation of resources: After ranking of the decision packages, resources are allocated for decision packages. Budgets are prepared like it is done first time without taking reference to previous budgets.

ICAI · Old P3 · MTP May 2020 · Series I · question p. 6 · answer p. 12-13

15.6 Control ratios, budget variances, motivation, feedback and feed-forward

10 questions · asked in 11 ICAI documents · 2 exam papers

1151 **EXAM** **Jan 2026** Q5(b) **5 marks**

Following data is available for XYZ Ltd. for the month of March 2025:

Standard working hours	7 hours per day of 5 days per week
Number of weeks in the month	4 weeks
Maximum Capacity (No. of employees)	60 employees
Actual working (No. of employees)	42 employees
Actual Capacity Usage Ratio	65%
Activity Ratio	125%

Calculate the following:

(i) Actual Hours worked

(ii) Standard Hours for actual output

(iii) Standard Capacity Usage Ratio

(iv) Actual Usage of Budgeted Capacity Ratio

ICAI'S ANSWER

(i) Actual Hours worked

$$\text{Actual Capacity Usage Ratio} = \frac{\text{Actual Hours worked}}{\text{Max. possible working hours in a period}} \times 100$$

65% = (Actual working hours ÷ Maximum Capacity in a budget period) × 100

Maximum Capacity *in a budget period*

= 60 workers × 7 hours × 5 days × 4 weeks = 8,400 hours

Actual hours= 8,400 × 65% = 5,460 hours

(ii) Standard hours for actual output

$$\text{Activity ratio} = \frac{\text{Standard Hrs}}{\text{Budgeted Hrs}} \times 100$$

125% = (Standard hours ÷ Budgeted Hours) × 100

Budgeted Hours = 42 workers × 7 hours × 5 days × 4 weeks = 5,880 hours

Standard hours for actual output = 5,880 x 125% = 7,350 hours

(iii) Standard capacity usage ratio

$$= \frac{\text{Budgeted Hours}}{\text{Max. possible hours in the budgeted period}} \times 100$$

= (5,880 hours÷ 8,400 hours) × 100

= 70%

(iv) Actual Usage of Budgeted Capacity Ratio

$$= \frac{\text{Actual working Hours}}{\text{Budgeted Hours}} \times 100$$

= (5,460 hours ÷ 5,880hours) × 100

= 92.86%

ICAI · P4 · Jan 2026 exam · question p. 24-25 · answer p. 27

1152 **EXAM** **May 2024** Q1(c) **4 marks**

Following data is available for XYZ Ltd. for the month of February 2024:

Standard working hours	8 hours per day of 6 days per week
No. of weeks in the month	4
Maximum capacity	150 employees
Actual working	125 employees
Actual usage of Budgeted Capacity Ratio	86%
Efficiency Ratio	110%

You are required to calculate the following:

(i) Actual Hours worked.

(ii) Standard Hours for actual output.

(iii) Activity Ratio.

(iv) Standard Capacity Usage Ratio.

ICAI'S ANSWER

(i) Actual Hours worked

$$\text{Actual Usage of Budgeted Capacity Ratio} = \frac{\text{Actual working Hours}}{\text{Budgeted Hours}} \times 100$$

86% = (Actual working hours÷ Budgeted hours) × 100

Budgeted hours=125 workers × 8 hours× 6 days × 4 weeks = 24,000 hours

Actual hours= 24,000 × 86% = 20,640 hours

(ii) Standard hours for actual output

$$\text{Efficiency ratio} = \frac{\text{Standard Hrs}}{\text{Actual Hrs}} \times 100$$

110% = Standard hours÷ Actual hours

Standard hours= 20,640 × 110% = 22,704 hours

(iii) Activity ratio $= \frac{\text{Standard Hrs}}{\text{Budgeted Hrs}} \times 100$

$= (22{,}704 \div 24{,}000) \times 100 = 94.6\%$

(iv) Standard capacity usage ratio

$= \frac{\text{Budgeted Hours}}{\text{Max. possible hours in the budgeted period}} \times 100$

$= \{24{,}000 \text{ hours} \div (150 \text{ workers} \times 8 \text{ hours} \times 6 \text{ days} \times 4 \text{ weeks})\} \times 100$

$= (24{,}000 \div 28{,}800) \times 100 = 83.33\%$

ICAI · P4 · May 2024 exam · question p. 2 · answer p. 6

1153 **EXAM** **May 2024** Q5(c) **4 marks**

Also asked in MTP May 2026 S-II Q6(c)

Discuss Feedback Control and Feedforward Control system of budgetary control.

ICAI'S ANSWER

There are two types of budgetary control system based on timing of action:

Feedback Control: The feedback system of budgetary control, the actual results for the budgeted period are collected and compared with the budgeted figures. The exercise of variance identification is done after the completion of the budget period. The variances are reported and based on the report corrective actions are taken, responsibility is fixed and based on experience, modification in future targets is implemented. As the name suggests, it is an Ex-post Corrective control system of budget.

This system of budgetary control is common in organistions where Management Information System (MIS) is not so robust and where data is obtained only after the finalisation of books of account. Though this type of control system is less expensive to maintain but has limitations.

Feedforward Control: This the opposite of feedback control system of budgetary control. It is Ex-Ante Preventive control mechanism of budgetary control. The budgets are set at the inception of the budgeted period and the actual results are continuously monitored and compared. The targets are kept realistic as far as possible and the targets are reviewed and reset if necessary.

This budgetary control system requires a robust MIS supported by integrated ERP system enabling an entity to get data as and when desired basis. This system is very expensive and beneficial for the organisations where the business environment is dynamic and information has important role in getting edge in competition and todays data warfare.

ICAI · P4 · May 2024 exam · question p. 21 · answer p. 24

1154 RTP Old P3 Nov 2022 Q14 old scheme

Following information is available for DK and Co.:

Standard working hours	9 hours per day of 5 days per week
Maximum capacity	50 employees
Actual working	40 employees
Actual hours expected to be worked per four week	7,200 hours
Std. hours expected to be earned per four weeks	9,000 hours
Actual hours worked in the four- week period	6,750 hours
Standard hours earned in the four- week period	7,875 hours.

The related period is of 4 weeks. In this period there was a one special day holiday due to national event.

You are required to CALCULATE the following ratios:

(i) Efficiency Ratio

(ii) Activity Ratio

(iii) Calendar Ratio

(iv) Standard Capacity Usage Ratio

(v) Actual Capacity Usage Ratio

(vi) Actual Usage of Budgeted Capacity Ratio

ICAI'S ANSWER

Maximum Capacity in a budget period

= 50 Employees × 9 Hrs. × 5 Days × 4 Weeks = 9,000 Hrs.

Budgeted Hours

= 40 Employees × 9 Hrs. × 5 Days × 4 Weeks = 7,200 Hrs.

Actual Hrs.

= 6,750 Hrs.

Standard Hrs. for Actual Output

= 7,875 Hrs.

Budget No. of Days

= 20 Days (4 Weeks x 5 Days)

Actual No. of Days

= 20 – 1 = 19 Days

(i) Efficiency Ratio = $\frac{\text{Standard Hrs}}{\text{Actual Hrs}} \times 100 = \frac{7{,}875 \text{ hours}}{6{,}750 \text{ hours}} \times 100 = 116.67\%$

(ii) Activity Ratio = $\frac{\text{Standard Hrs}}{\text{Budgeted Hrs}} \times 100 = \frac{7{,}875 \text{ hours}}{7{,}200 \text{ hours}} \times 100 = 109.375\%$

(iii) Calendar Ratio = $\frac{\text{Available working days}}{\text{Budgeted working days}} \times 100 = \frac{19 \text{ days}}{20 \text{ days}} \times 100 = 95\%$

(iv) Standard Capacity Usage Ratio = $\frac{\text{Budgeted Hours}}{\text{Max. possible hours in the budgeted period}} \times 100$

$= \frac{7{,}200 \text{ hours}}{9{,}000 \text{ hours}} \times 100 = 80\%$

(v) Actual Capacity Usage Ratio = $\frac{\text{Actual Hours worked}}{\text{Max. possible working hours in a period}} \times 100$

$= \frac{6{,}750 \text{ hours}}{9{,}000 \text{ hours}} \times 100 = 75\%$

(vi) Actual Usage of Budgeted Capacity Ratio = $\frac{\text{Actual working Hours}}{\text{Budgeted Hours}} \times 100$

$= \frac{6{,}750 \text{ hours}}{7{,}200 \text{ hours}} \times 100 = 93.75\%$

ICAI · Old P3 · RTP Nov 2022 · question p. 12-13 · answer p. 33-34

1155 MTP **Sep 2026 S-II** Q1(c) 4 marks

A company manufactures two products X and Y. Product X requires 8 hours to produce while Y requires 12 hours. In April, 2024, of 22 effective working days of 8 hours a day, 1,200 units of X and 800 units of Y were produced. The company employs 100 workers in production department to produce X and Y. The budgeted hours are 1,86,000 for the year.

You are required to Calculate:

(i) Capacity Ratio,

(ii) Activity Ratio and

(iii) Efficiency ratio.

ICAI'S ANSWER

Working Note:

Actual Hours (AH) = 22× 8 ×100 = 17600 Hrs.

Standard Hours (SH) = (1200×8 +800 ×12) =19,200 Hrs.

Budgeted Hours (BH) = 1,86,000/12 = 15,500

i. Capacity Ratio = $\frac{\text{Actual Hours (worked)}}{\text{Budgeted Hours}} \times 100$

= (17600÷15,500) ×100 = **113.55%**

ii. Activity Ratio = $\frac{\text{Standard Hours (for actual production)}}{\text{Budgeted Hours}} \times 100$

= (19,200÷15,500) ×100 = **123.87%**

iii. Efficiency Ratio = $\frac{\text{Standard Hours (for actual production)}}{\text{Actual Hours (worked)}} \times 100$

= (19,200÷17,600) ×100 = **109.09%**

ICAI · P4 · MTP Sep 2026 · Series II · question p. 9 · answer p. 9

1156 MTP **May 2026 S-I** MCQ 12 2 marks MCQ

Also asked in MTP Sep 2024 S-I MCQ 15

Standard hours required for doing a work is 100 hours and budgeted hours is 120 hrs. while the same work is actually completed by workers in 110 hrs. You are required to calculate the activity ratio:

A 109.09%

B 83.33%

C 90.90%

D 110%

ICAI'S ANSWER

(B) 83.33%

$$\text{Activity Ratio} = \frac{\text{Standard Hours}}{\text{Budgeted Hours}} \times 100$$

$= 83.33\%$

ICAI · P4 · MTP May 2026 · Series I · question p. 6 · answer p. 5

1157 MTP Jan 2026 S-II MCQ 11 2 marks MCQ

A manufacturing unit has budgeted standard hours of 15,000 for May 2025. During the month, the capacity utilisation ratio was 85%, while the efficiency ratio stood at 110%. The factory reported smooth operations with no major downtime.

The standard hours allowed for May 2025 was:

A 14,025

B 13,200

C 15,750

D 12,900

ICAI'S ANSWER

(A) 14,025

$$\text{Capacity Utilisation Ratio} = \frac{\text{Actual Hours}}{\text{Budgeted Hours}}$$

$$0.85 = \frac{\text{Actual Hours}}{15{,}000}$$

$\text{Actual Hours} = 15{,}000 \times 0.85 = 12{,}750$

$$\text{Efficiency Ratio} = \frac{\text{Standard Hours Allowed}}{\text{Actual Hours Taken}}$$

$$1.10 = \frac{\text{Standard Hours Allowed}}{12{,}750}$$

$\text{Standard Hours Allowed} = 12{,}750 \times 1.10 = 14{,}025$

ICAI · P4 · MTP Jan 2026 · Series II · question p. 5 · answer p. 5

1158 MTP Sep 2024 S-II MCQ 14 2 marks MCQ

A factory has a capacity utilization ratio of 85% and its activity ratio is 95%. Which one of the following is the efficiency ratio?

A 120%

B 110%

C 112%

D 90%

ICAI'S ANSWER

(C)

Efficiency Ratio = Activity Ratio/Capacity Utilization Ratio

= 0.95/0.85 = 1.117 or 112%

ICAI · P4 · MTP Sep 2024 · Series II · question p. 5 · answer p. 5

1159 MTP **Old P3 May 2023 S-I** Q1(b) 5 marks old scheme

CALCULATE (i) Efficiency ratio (ii) Activity Ratio (iii) Capacity Ratio. The relevant data is as below:

Budgeted Production 1,44,000 units

Standard Hours per unit 12

Actual Production 1,20,000 units

Actual Working Hours 12,00,000

ICAI'S ANSWER

(i) **Efficiency Ratio** = $\frac{\text{Stardard hour (for actual production)}}{\text{Actual hour works}}$ x 100

= $\frac{1,20,000 \text{ units x12 hrs}}{12,00,000 \text{ hrs}}$ x 100 = 120%

(ii) **Activity Ratio** = $\frac{\text{Stardard Hour (for actual production)}}{\text{Budgeted Hours}}$ x 100

= $\frac{14,40,000}{1,44,000 \text{ units x12 hours}}$ x 100 = = 83.34%

(iii) **Capacity Ratio** = $\frac{\text{Actual Hours (worked)}}{\text{Budgeted Hours}}$ x100

= $\frac{12,00,000 \text{ hrs}}{1,44,000 \text{ units x12 hours}}$ x 100 = 69.45%

ICAI · Old P3 · MTP May 2023 · Series I · question p. 1 · answer p. 1

1160 MTP **Old P3 Nov 2019 S-I** Q1(b) 5 marks old scheme

CALCULATE from the following figures:

(i) Efficiency ratio,

(ii) Activity, Ratio and

(iii) Capacity Ratio:

Budgeted Production	88,000 units
Standard Hours per unit	10
Actual Production	75,000 units
Actual Working Hours	6,00,000

ICAI'S ANSWER

$$\text{(i) Efficiency Ratio} = \frac{\text{Standard Hours (for actual production)}}{\text{Actual Hours (worked)}} \times 100$$

$$= \frac{75{,}000 \text{ units} \times 10 \text{ hrs.}}{6{,}00{,}000 \text{ hrs.}} \times 100$$

$$= 125\%$$

$$\text{(ii) Activity Ratio} = \frac{\text{Standard Hours (for actual production)}}{\text{Budgeted Hours}} \times 100$$

$$= \frac{75{,}000 \text{ units} \times 10 \text{ hrs.}}{88{,}000 \text{ units} \times 10 \text{ hrs.}} \times 100$$

$$= 85.23\%$$

$$\text{(iii) Capacity Ratio} = \frac{\text{Actual Hours (worked)}}{\text{Budgeted Hours}} \times 100$$

$$= \frac{6{,}00{,}000 \text{ hrs.}}{88{,}000 \text{ units} \times 10 \text{ hrs.}} \times 100$$

$$= 68.18\%$$

ICAI · Old P3 · MTP Nov 2019 · Series I · question p. 1 · answer p. 2

APPENDIX

Out of syllabus

Old-scheme questions on topics the January 2027 syllabus no longer has, and items ICAI itself marks out of syllabus. Skip them unless you are curious.

A.1 Near Chapter 9: Job Costing

21 questions ICAI marks out of syllabus, or whose topic has left the syllabus. For reference only.

1161 EXAM **Old P3 May 2023** Q2(b) 10 marks old scheme

B Limited has taken a contract for ₹ 70,00,000 and furnishes the following information:

	1st Year	2nd Year
	(Amount in ₹)	**(Amount in ₹)**
Material	12,50,000	13,65,000
Wages	12,50,000	11,44,000
Direct Expenses	4,20,000	3,80,000
Indirect Expenses	2,70,000	2,60,000
Work Certified	32,00,000	70,00,000
Work Uncertified	2,19,000	-

Other Information:

- Plant costing ₹ 3,40,000 was bought at the commencement of the contract.
- Depreciation of ₹ 85,000 per annum is charged on the plant on Straight Line Method (SLM) basis.
- There is a provision for escalation clause in the contract for increase in material rate and wage rate in the second year only.
- Standard material for the first and second year was ₹ 12,000 units each year @ ₹ 90 per unit whereas the actual consumption was 12,500 @ ₹ 100 per unit in the first year and 13,000 units @ ₹ 105 per unit in the 2nd year. Standard labour hours for first year were 10,000 hours and for the second year it was 9,000 hours. Standard wage rate was ₹ 120 per hour. The firm has paid for 10,000 hours @ ₹ 125 per hour in the first year and 8,800 hours @ ₹ 130 per hour in the second year.

Required:

(i) Prepare Contract Account for both years without considering escalation clause.

(ii) Compute the total value of contract by considering the escalation clause.

(iii) Compute the total increase / (decrease) in the cost of material and wages for both the years.

ICAI'S ANSWER

(i) Contract Account (For 1st Year)

Particulars	(₹)	Particulars		(₹)
To Material	12,50,000	By work in Progress		
To Wages	12,50,000	Work certified	32,00,000	
To Direct expense	4,20,000	Work uncertified	2,19,000	34,19,000
To Indirect expense	2,70,000			
To Depreciation (₹3,40,000 – ₹2,55,000)	85,000			
To Costing P&L (Notional Profit b/f)	1,44,000			
	34,19,000			34,19,000

Contract Account (For 2nd Year)

Particulars		(₹)	Particulars	(₹)
To Opening Work in Progress:			By Contractee A/C	70,00,000
Work certified	32,00,000			
Work uncertified	2,19,000	34,19,000		
To Material		13,65,000		
To Wages		11,44,000		
To Direct expenses		3,80,000		
To Indirect expenses		2,60,000		
To Depreciation (₹ 2,55,000 – ₹ 1,70,000)		85,000		
To Costing P&L (b/f)		3,47,000		
		70,00,000		70,00,000

(ii) Computation of total value of contract after escalation clause

Particulars	(₹)
Contract Price	70,00,000
Increase in cost of material 12,000 (105 – 90)	1,80,000
Increase in cost of labour 9,000 (130-120)	90,000
	72,70,000

(iii) Calculation of Increase/Decrease in cost of material and wages

Year 1	Standard Cost	Actual Cost	Increase/Decrease
Material	10,80,000 (12,000 × 90)	12,50,000 (12,500 × 100)	1,70,000
Labour	12,00,000 (10,000 × 120)	12,50,000 (10,000 × 125)	50,000
			2,20,000
Year 2	**Standard Cost**	**Actual Cost**	**Increase/Decrease**
Material	10,80,000 (12,000 × 90)	13,65,000 (13,000 × 105)	2,85,000
Labour	10,80,000 (9,000 × 120)	11,44,000 (88,000 × 130)	64,000
			3,49,000

PassBook's note: Out of syllabus (old scheme): contract costing (the contract account, notional profit on work certified and uncertified, the escalation clause) is not in the new-scheme Paper 4 syllabus or the January 2027 Study Material, which has no Contract Costing chapter.

ICAI · Old P3 · May 2023 exam · question p. 8-9 · answer p. 10-11

1162 EXAM **Old P3 Nov 2022** Q3(a) 10 marks old scheme

XYZ Construction Ltd. has obtained a contract of ₹ 25,00,000 in the Financial Year 2021-22. The work on the contract commenced immediately and it is expected that the contract will be completed by 31st March,2023. Chief accountant of the company has provided following information related to the Contract:

Particulars	2021-22 (Actual) (in ₹)	2022-23 (Estimated) (in₹)
Material issued	4,00,000	3,50,000
Wages: Paid	2,50,000	1,40,000
- Prepaid at the end of the Year	15,000	-
Plant	2,00,000	-
Sundry Expenses: Paid	50,000	35,000
- Outstanding at the end of the year	7,500	5,000
Office Expenses: Paid	65,000	55,000
- Outstanding at the end of the year	12,500	15,000
Contingency Expenses	-	1,25,000

Following additional information is also available:

- A part of plant costing ₹ 12,000 was scrapped and written off in the F.Y.2021-22.
- The value of Plant-at-Site on 31st March,2022 was ₹ 18,000.

- Company would have to spend an additional sum of ₹ 80,000 on the plant in FY. 2022-23 and the residual value of the plant on the completion of contract would be ₹ 10,000.
- A part of material costing ₹ 30,000 was scrapped and sold for ₹ 20,000 in F.Y. 2021-22.
- The value of Material-at-Site on 31st March, 2022 was ₹ 17,000.
- Cash received on account till 31st March,2022 was ₹ 13,50,000 representing 90% of the work certified.
- The cost of work uncertified on 31st March, 2022 was valued at 20% of work certified.

You are required to:

(i) Prepare a Contract Account for the year ended 31st March, 2022

(ii) Calculate Estimated Total Profit on this Contract.

ICAI'S ANSWER

Contract Account (2021-22)

Particulars		(₹)	Particulars		(₹)
To Materials issued		4,00,000	By Costing P & L A/c		12,000
To Wages paid	2,50,000		By Material sold		20,000
Less: Prepaid	15,000	2,35,000	By Plant at site c/d		18,000
To Plant		2,00,000	By Material at site c/d		17,000
To Sundry Expenses	50,000		By Costing P & L A/c (₹ 30,000 – ₹ 20,000)		10,000
Add: Outstanding	7,500	57,500	By Work-in-progress c/d		
To Office Expenses	65,000		Work certified (13,50,000 ÷ 90%)	15,00,000	
Add: Outstanding	12,500	77,500	Work uncertified (15,00,000 × 20%)	3,00,000	18,00,000
To Notional profit (Profit for the year)		9,07,000			
		18,77,000			18,77,000

Calculation of Estimated Profit

			(₹)	(₹)
(1)	Material consumed	(4,00,000-10,000-20,000)	3,70,000	
	Add: Further consumption		3,50,000	7,20,000
(2)	Wages:		2,35,000	
	Add: Further cost	(1,40,000+15,000)	1,55,000	3,90,000

			(₹)	(₹)
(3)	Plant used	(2,00,000-12,000)	1,88,000	
	Add: Further plant introduced		80,000	
	Less: Closing balance of plant		(10,000)	2,58,000
(4)	Sundry expenses		57,500	
	Add: Further expenses	(35,000-7,500)	27,500	
	Add: Outstanding		5,000	90,000
(5)	Office expenses		77,500	
	Add: Further expenses	(55,000 – 12,500)	42,500	
	Add: Outstanding		15,000	1,35,000
(6)	Reserve for contingencies			1,25,000
Estimated profit (balancing figure)				7,82,000
Contract price				25,00,000

Estimated Profit can also be calculated showing cost as per Contract Account for the year 2021-22 and estimated cost for the year 2022-23 in the following manner

Calculation of Estimated Profit

Cost as per contract A/c 2021-22 (A)		8,93,000
Estimated cost for 2022-23		
Material (3,50,000 +17,000)	3,67,000	
Wages (1,40,000 +15,000)	1,55,000	
Sundry Expenses (3,5000-7,500 +5,000)	32,500	
Contingency Expenses	1,25,000	
Office expenses (55,000 +15,000-12,500)	57,500	
Plant (80,000+18,000-10,000)	88,000	
Total estimated cost of 2022-23(B)		8,25,000
C=(A)+(B)		17,18,000
Estimated Profit (D)-(C)		7,82,000

Contract Price(D)	25,00,000

PassBook's note: Out of syllabus (old scheme): contract costing (the contract account, notional profit on work certified and uncertified, estimated total profit) is not in the new-scheme Paper 4 syllabus or the January 2027 Study Material, which has no Contract Costing chapter.

ICAI · Old P3 · Nov 2022 exam · question p. 10 · answer p. 11-13

1163 EXAM **Old P3 May 2022** Q2(b) 5 marks old scheme

Paramount Constructions Limited is engaged in construction and erection of bridges under long term contracts. It has entered into a big contract at an agreed price of ₹ 250 Lakhs subject to an escalation clause for material and labour as spelt out in the contract and corresponding actual are as follows:

	Standard		Actual	
Material	**Quantity Tonnes**	**Rate Per Tonne (₹)**	**Quantity Tonnes**	**Rate Per Tonne (₹)**
P	2,800	1,500	3,000	1,750
Q	3,100	900	2,900	800
R	800	4,500	950	4,350
S	150	32,500	120	34,200
Labour	**Hours**	**Hourly rate (₹)**	**Hours**	**Hourly rate (₹)**
LM	65,000	60	61,500	70
LN	46,000	45	45,000	50

Required:

(i) Prepare a statement showing admissible additional claim of material and labour due to escalation clause.

(ii) Determine the final price payable after admissible escalation claim.

ICAI'S ANSWER

Statement showing Additional claim

	Standard Qty/Hrs.	Standard Rate (₹)	Actual Rate (₹)	Variation in Rate (₹)	Escalation Claim (₹)
	(a)	**(b)**	**(c)**	**(d) = (c)–(b)**	**(e) =(a) × (d)**
Materials					
P	2,800	1,500	1,750	250	7,00,000
Q	3,100	900	800	(100)	(3,10,000)
R	800	4,500	4,350	(150)	(1,20,000)
S	150	32,500	34,200	1,700	2,55,000

	Standard Qty/Hrs.	Standard Rate (₹)	Actual Rate (₹)	Variation in Rate (₹)	Escalation Claim (₹)
				Materials escalation claim: (A)	5,25,000
Wages					
LM	65,000	60	70	10	6,50,000
LN	46,000	45	50	5	2,30,000
				Wages escalation claim: (B)	8,80,000
				Final claim: (A + B)	14,05,000

Statement showing final price payable

	(₹)	(₹)
Agreed price		2,50,00,000
Add: Agreed escalation		
Material cost	5,25,000	
Labour cost	8,80,000	**14,05,000**
Final price payable		**2,64,05,000**

PassBook's note: Out of syllabus (old scheme): the escalation clause of a long-term construction contract (admissible escalation claim and final contract price) is contract costing, which is not in the new-scheme Paper 4 syllabus or the January 2027 Study Material (no Contract Costing chapter; The Study Material does not mention escalation clauses).

ICAI · Old P3 · May 2022 exam · question p. 7 · answer p. 10

1164 EXAM **Old P3 Jul 2021** Q4(c) 5 marks old scheme

Brick Constructions Ltd. commenced a contract on April 1,2020. The contract was for ₹ 10,00,000. The following information relates to the Contract as on 31st March, 2021:

- The value of work completed up to Feb. 28, 2021 was certified by the architect and as a matter of policy, the Contractee has retained ₹ 1,30,000 as retention money which is 20% of the certified work and paid the balance amount.
- The cost of work completed subsequent to the architect's certificate was of ₹ 30,000.
- The expenditure incurred related to material purchase, wages and other chargeable expenses were ₹ 5,10,000
- Materials of the value of ₹ 20,000 were lying on the site.
- A special plant was purchased specifically for this contract at ₹ 40,000 and after use on this contract till 31st March, 2021, it was valued at ₹ 25,000.

You are required to compute the value of Work Certified, Cash received for certified work and Notional profit of the contract for the year ended on 31st March, 2021.

ICAI'S ANSWER

1. Value of Work Certified

$= \frac{₹\ 1,30,000}{20\%}$ = **₹ 6,50,000**

2. Cash Received

= Value of Work certified – Retention Money

= 6,50,000 – 1,30,000 = **₹ 5,20,000**

3. Notional Profit

= Value of Work certified – Cost of work certified

= 6,50,000 - 4,75,000* = **₹ 1,75,000**

***Working Note**

Cost of work certified = Work cost - Cost of work uncertified

= (Expenditure + Plant used – Material at site) - Cost of work uncertified

= [5,10,000 + (40,000 - 25,000) - 20,000] - 30,000 = ₹ 4,75,000

PassBook's note: Out of syllabus (old scheme): contract costing (work certified, retention money, cash received and notional profit on a contract) is not in the new-scheme Paper 4 syllabus or the January 2027 Study Material, which has no Contract Costing chapter.

ICAI · Old P3 · Jul 2021 exam · question p. 69-70 · answer p. 74

1165 EXAM **Old P3 Nov 2020** Q1(d) 5 marks old scheme

W Limited undertook a contract for ₹ 5,00,000 on 1st July, 2019. On 30th June, 2020 when the accounts were closed, the following details about the contract were gathered:

	Amount (₹)
Materials purchased	1,00,000
Wages paid	45,000
General expenses	10,000
Materials on hand (30-6-2020)	25,000
Wages accrued (30-6-2020)	5,000
Work certified	2,00,000
Cash received	1,50,000
Work uncertified	15,000

The above contract contained "Escalation clause" which read as follows :

"In the event of increase in the prices of materials and rates of wages by more than 5%, the contract price would be increased accordingly by 25% of the rise in the cost of materials and wages beyond 5% in each case."

It was found that since the date of signing the agreement, the prices of materials and wage rates increased by 25%. The value of the work certified does not take into account the effect of the above

clause.

Calculate the 'value of work certified' after taking the effect of 'Escalation Clause' as on 30th June, 2020.

ICAI'S ANSWER

Workings:

(i) Percentage of work certified:

$$\frac{\text{Value of work certified}}{\text{Contract price}} \times 100 = \frac{₹\ 2,00,000}{₹\ 5,00,000} \times 100 = 40\%$$

(ii) Value of material and labour used in the contract:

Particulars	Amount (₹)	Amount (₹)
Material purchased	1,00,000	
Less: Material on hand (30-06-2020)	(25,000)	75,000
Wages paid	45,000	
Add: Wages accrued (30-06-2020)	5,000	50,000
		1,25,000

Price of materials and wages has been increased by 25%, the value before price increase is:

$$\frac{₹\ 1,25,000}{125} \times 100 = ₹\ 1,00,000$$

(iii) Calculation of Value of work certified:

The value of the contract would be increased by 25% of the price increased beyond 5%.

Price increased beyond 5% = ₹ 25,000 – 5% of ₹ 1,00,000 = ₹ 20,000

Value of contract would be increased by 25% of ₹ 20,000 = ₹ 5,000

Therefore, the revised contract value = ₹ 5,00,000 + ₹ 5,000 = ₹ 5,05,000

Calculation of the Value of work certified after taking the effect of escalation clause:

Revised contract value × Percentage of work certified

= ₹ 5,05,000 × 40% = ₹ 2,02,000

PassBook's note: Out of syllabus (old scheme): contract costing (the value of work certified under an escalation clause) is not in the new-scheme Paper 4 syllabus or the January 2027 Study Material, which has no Contract Costing chapter.

ICAI · Old P3 · Nov 2020 exam · question p. 2-3 · answer p. 5-6

1166 **RTP** **Old P3 Nov 2023** Q14(a) old scheme

What is cost plus contract? What are its advantages?

ICAI'S ANSWER

Cost plus contract: Under cost plus contract, the contract price is ascertained by adding a percentage of profit to the total cost of the work. Such types of contracts are entered into when it is not possible to estimate the contract cost with reasonable accuracy due to unstable condition of material, labour services etc.

Following are the advantages of cost-plus contract:

(i) The contractor is assured of a fixed percentage of profit. There is no risk of incurring any loss on the contract.

(ii) It is useful specially when the work to be done is not definitely fixed at the time of making the estimate.

(iii) Contractee can ensure himself about the 'cost of contract' as he is empowered to examine the books and documents of the contractor to ascertain the veracity of the cost of contract.

PassBook's note: Out of syllabus (old scheme): the cost plus contract belongs to contract costing, which is not in the new-scheme Paper 4 syllabus or the January 2027 Study Material (no Contract Costing chapter.

ICAI · Old P3 · RTP Nov 2023 · question p. 9 · answer p. 24

1167 RTP **Old P3 May 2023** Q8 old scheme

XYZ LLP, contractors and civil engineers, are building a new wing to a school. The quoted fixed price for the contract is ₹30,00,000. Work commenced on 1st January 20X2 and is expected to be completed on schedule by 30 June 20X3.

Data relating to the contract at the year ended 31st March 20X3 is as follows.

	(₹)
Plant sent to site at commencement of contract	2,40,000
Hire of plant and equipment	77,000
Materials sent to site	6,62,000
Materials returned from site	47,000
Direct wages paid	9,60,000
Wage related costs	1,32,000
Direct expenses incurred	34,000
Supervisory staff salaries - Direct	90,000
- Indirect	20,000
Regional office expenses apportioned to contract	50,000
Head office expenses apportioned to contract	30,000
Surveyor's fees	27,000
Progress payments received from school	18,00,000

Additional information:

1. Plant is to be depreciated at the rate of 25 % per annum following straight line method, with no residual value.
2. Unused materials on site at 31st March are estimated at ₹ 50,000.

3. Wages owed to direct workers total ₹ 40,000
4. Budgeted profit on the contract is ₹ 8,00,000
5. Value of work certified by the surveyor is ₹ 24,00,000.
6. The surveyor has not certified the work costing ₹ 1,80,000

You are required to PREPARE the account for the school contract for the fifteen months ended 31st March 20X3, and CALCULATE the notional profit to date.

ICAI'S ANSWER

School Contract Account

Particulars	(₹)	Particulars	(₹)
To Plant	2,40,000	By Material returned	47,000
To Hire of plant	77,000	By Plant c/d	1,65,000
To Materials	6,62,000	By Materials c/d	50,000
To Direct wages 9,60,000		By WIP c/d:	
Add: Accrued 40,000	10,00,000	Value of work certified	24,00,000
To Wages related costs	1,32,000	Cost of work not certified	1,80,000
To Direct expenses	34,000		
To Supervisory staff:			
Direct 90,000			
Indirect 20,000	1,10,000		
To Regional office expenses	50,000		
To Head office expenses	30,000		
To Surveyors' fees	27,000		
To Notional profit c/d	4,80,000		
	28,42,000		28,42,000

PassBook's note: Out of syllabus (old scheme): contract costing (the contract account, notional profit on work certified and uncertified, depreciation of plant at site) is not in the new-scheme Paper 4 syllabus or the January 2027 Study Material, which has no Contract Costing chapter.

ICAI · Old P3 · RTP May 2023 · question p. 5-6 · answer p. 17-18

1168 RTP **Old P3 Nov 2022** Q8 old scheme

A contractor prepares his accounts for the year ending 31st March each year. He commenced a contract on 1st July, 2021.

The following information relates to the contract as on 31st March, 2022:

	(₹)
Material issued	12,55,000
Wages	28,28,000
Salary to Foreman	4,06,500

A machine costing ₹ 13,00,000 has been on the site for 4.8 months, its working life is estimated at 7 years and its final scrap value at ₹ 75,000.

A supervisor, who is paid ₹ 40,000 p.m. has devoted one-half of his time to this contract.

All other expenses and administration charges amount to ₹ 6,82,500.

Material in hand at site costs ₹ 1,77,000 on 31st March, 2022.

The contract price is ₹ 1,00,00,000. On 31st March, 2022 2/3rd of the contract was completed. The architect issued certificates covering 50% of the contract price, and the contractor had been paid ₹ 37,50,000 on account.

PREPARE Contract A/c and show the notional profit or loss as on 31st March, 2022.

ICAI'S ANSWER

Contract Account

	Particulars	(₹)		Particulars	(₹)
To	Material issued	12,55,000	By	Machine (Working note 1)	12,30,000
"	Wages	28,28,000	"	Material (in hand)	1,77,000
"	Foreman's salary	4,06,500	"	Works cost (balancing figure)	52,45,000
"	Machine	13,00,000			
"	Supervisor's salary (₹ 40,000 × 9)/2	1,80,000			
"	Administrative charges	6,82,500			
		66,52,000			66,52,000
"	Works cost	52,45,000	"	Value of work certified	50,00,000
"	**Costing P&L A/c** (Notional profit)	**10,66,250**	"	Cost of work uncertified (Working Note 2)	13,11,250
		63,11,250			**63,11,250**

Working notes:

1. Written down value of Machine:

$$\text{Depreciation} = \frac{₹\ 13,00,000 - 75,000}{7\ \text{years}} \times \frac{4.8\ \text{months}}{12\ \text{months}} = ₹\ 70,000$$

Hence the value of machine after the period of 4.8 months = ₹ 13,00,000 – ₹ 70,000 = ₹ 12,30,000

2. The cost of 2/3rd of the contract is ₹ 52,45,000

∴ Cost of 100% of the contract is $\frac{₹\ 52,45,000}{2} \times 3 = ₹\ 78,67,500$

∴ Cost of 50% of the contract which has been certified by the architect is ₹ 39,33,750. Also, the cost of 1/3rd of the contract, which has been completed but not certified by the architect is ₹ 13,11,250.

PassBook's note: Out of syllabus (old scheme): contract costing (the contract account, depreciation of plant on site, work certified and uncertified, and the notional profit) is not in the new-scheme Paper 4 syllabus or the January 2027 Study Material, which has no Contract Costing chapter.

ICAI · Old P3 · RTP Nov 2022 · question p. 7-8 · answer p. 26

1169 RTP **Old P3 May 2022** Q14(b) old scheme

Also asked in Old P3 MTP Nov 2023 S-I Q6(b) · Old P3 MTP May 2023 S-II Q6(b)

DEFINE cost plus contract? STATE its advantages.

ICAI'S ANSWER

Cost plus contract: Under cost plus contract, the contract price is ascertained by adding a percentage of profit to the total cost of the work. Such types of contracts are entered into when it is not possible to estimate the contract cost with reasonable accuracy due to unstable condition of material, labour services etc.

Following are the advantages of cost plus contract:

(i) The contractor is assured of a fixed percentage of profit. There is no risk of incurring any loss on the contract.

(ii) It is useful specially when the work to be done is not definitely fixed at the time of making the estimate.

(iii) Contractee can ensure himself about the 'cost of contract' as he is empowered to examine the books and documents of the contractor to ascertain the veracity of the cost of contract.

PassBook's note: Out of syllabus (old scheme): the cost plus contract belongs to contract costing, which is not in the new-scheme Paper 4 syllabus or the January 2027 Study Material (no Contract Costing chapter.

ICAI · Old P3 · RTP May 2022 · question p. 10 · answer p. 31

1170 RTP **Old P3 Dec 2021** Q8 old scheme

RN Builders Ltd. entered into a contract on April 1, 2019. The total contract was for ₹ 2,00,00,000. Actual expenditure for the period April 1, 2019 to March 31, 2020 and estimated expenditure for April 1, 2020 to

December 31, 2020 are given below:

Particulars	2019-20 (actual) (₹)	2020-21 (9 months) (estimated) (₹)
Materials issued	36,00,000	34,30,000
Wages: Paid	30,00,000	34,93,000
Outstanding at the end	2,50,000	3,32,000
Plant purchased	10,00,000	-
Sundry expenses: Paid	2,90,000	2,75,000
Prepaid at the end	25,000	-
Establishment charges	5,85,000	-

A part of the material was unsuitable and thus sold for ₹ 7,25,000 (cost being ₹ 6,00,000) and a part of plant was scrapped and disposed-off for ₹ 1,15,000. The value of plant at site on 31 March, 2020 was ₹ 3,10,000 and the value of material at site was ₹ 1,70,000. Cash received on account to date was ₹ 70,00,000, representing 80% of the work certified. The cost of work uncertified was valued at ₹ 10,95,000.

The contract would be completed by 31st December, 2020 and the contractor estimated further expenditure that would be incurred in completion of the contract:

- A sum of ₹ 12,50,000 would have to be spent on the plant and the residual value of the plant on the completion of the contract would be ₹ 1,50,000.
- Establishment charges would cost the same amount per month as in the previous year.
- ₹ 4,32,000 would be sufficient to provide for contingencies.

Required:

PREPARE a Contract Account for the year ended 31st March, 2020, and CALCULATE estimated total profit on this contract.

ICAI'S ANSWER

RN Builders Ltd.

Contract Account (2019-20)

	Particulars		(₹)		Particulars	(₹)
To	Materials issued		36,00,000	By	Material sold	7,25,000
To	Wages paid	30,00,000		By	Plant sold	1,15,000
	Add: Outstanding	2,50,000	32,50,000	By	Plant at site c/d	3,10,000
To	Plant		10,00,000	By	Material at site c/d	1,70,000
To	Sundry Expenses	2,90,000		By	Work-in-progress c/d	

	Particulars		(₹)	Particulars		(₹)
	Less: Prepaid	(25,000)	2,65,000	Work certified (₹ 70,00,000 ÷ 80%)	87,50,000	
To	Establishment charges		5,85,000	Work uncertified	10,95,000	98,45,000
To	Costing P & L A/c (₹ 7,25,000 – ₹ 6,00,000)		1,25,000			
To	Notional profit (Profit for the year)		23,40,000			
			1,11,65,000			**1,11,65,000**

Calculation of Estimated Profit

	Particulars		(₹)	(₹)
(1)	Material consumed	(36,00,000+ 1,25,000– 7,25,000)	30,00,000	
	Add: Further consumption		34,30,000	64,30,000
(2)	Wages:		32,50,000	
	Add: Further cost	(34,93,000 – 2,50,000)	32,43,000	
	Add: Outstanding		3,32,000	68,25,000
(3)	Plant used	(10,00,000– 1,15,000)	8,85,000	
	Add: Further plant introduced		12,50,000	
	Less: Closing balance of plant		(1,50,000)	19,85,000
(4)	Establishment charges		5,85,000	
	Add: Further charges for nine months	(5,85,000 × 9/12)	4,38,750	10,23,750
(5)	Sundry expenses		2,90,000	
	Add: Further expenses		2,75,000	5,65,000
(6)	Reserve for contingencies			4,32,000

Particulars	(₹)	(₹)
Estimated profit (balancing figure)		**27,39,250**
Contract price		2,00,00,000

PassBook's note: Out of syllabus (old scheme): contract costing (the contract account, notional profit, work certified and uncertified, estimated total profit on a contract) is not in the new-scheme Paper 4 syllabus or the January 2027 Study Material, which has no Contract Costing chapter.

ICAI · Old P3 · RTP Dec 2021 · question p. 7-8 · answer p. 25-26

1171 RTP **Old P3 Jul 2021** Q13(a) old scheme

Same question, different ICAI answer: #1179 Old P3 MTP Dec 2021 S-II

WRITE note on cost-plus-contracts.

ICAI'S ANSWER

These contracts provide for the payment by the contractee of the actual cost of construction plus a stipulated profit, mutually decided between the two parties.

The main features of these contracts are as follows:

(i) The practice of cost-plus contracts is adopted in the case of those contracts where the probable cost of the contracts cannot be ascertained in advance with a reasonable accuracy.

(ii) These contracts are preferred when the cost of material and labour is not steady and the contract completion may take number of years.

(iii) The different costs to be included in the execution of the contract are mutually agreed, so that no dispute may arise in future in this respect. Under such type of contracts, contractee is allowed to check or scrutinize the concerned books, documents and accounts.

(iv) Such a contract offers a fair price to the contractee and also a reasonable profit to the contractor.

The contract price here is ascertained by adding a fixed and mutually pre-decided component of profit to the total cost of the work.

PassBook's note: Out of syllabus (old scheme): cost plus contracts belong to contract costing, which old Paper 3 taught.

ICAI · Old P3 · RTP Jul 2021 · question p. 11 · answer p. 31

1172 RTP **Old P3 Nov 2020** Q15(b) old scheme

DESCRIBE the three advantages of Cost-plus contract.

ICAI'S ANSWER

Advantages of Cost plus contracts are as follows:

(i) The Contractor is assured of a fixed percentage of profit. There is no risk of incurring any loss on the contract.

(ii) It is useful specially when the work to be done is not definitely fixed at the time of making the estimate.

(iii) Contractee can ensure himself about 'the cost of the contract', as he is empowered to examine the books and documents of the contractor to ascertain the veracity of the cost of the contract.

PassBook's note: Out of syllabus (old scheme): cost plus contracts belong to contract costing, which old Paper 3 taught; The new-scheme Paper 4 syllabus and the January 2027 Study Material have no contract costing chapter (the Job Costing chapter does not

treat cost plus contracts).

ICAI · Old P3 · RTP Nov 2020 · question p. 12 · answer p. 30

1173 RTP **Old P3 May 2020** Q13(c) old scheme

DISCUSS the Escalation Clause in a Contract.

ICAI'S ANSWER

Escalation clause in a contract empowers a contractor to revise the price of the contract in case of increase in the prices of inputs due to some macro-economic or other agreed reasons. A contract takes longer period to complete and the factors based on which price negotiation is done at the time of entering into the contract may change till the contract completes. This protect the contractor from adverse financial impacts and empowers the contractor to recover the increased prices. As per this clause, the contractor increases the contract price if the cost of materials, employees and other expenses increase beyond a certain limit. Inclusion of such a clause in a contract deed is called an "Escalation Clause".

PassBook's note: Out of syllabus (old scheme): the escalation clause of a contract is contract costing, which is not in the new-scheme Paper 4 syllabus or the January 2027 Study Material (no Contract Costing chapter; The Study Material does not teach escalation clauses).

ICAI · Old P3 · RTP May 2020 · question p. 10 · answer p. 29

1174 MTP **Old P3 May 2023 S-I** Q4(b) 10 marks old scheme

Ultra Builders Ltd. has started a contract on 1st April 2021. The Trial balance as on 31st March 2022 showed the following balances:

Particulars	Dr. (₹)	Cr. (₹)
Paid up share capital		2,05,75,000
Land and buildings	50,60,000	
Machinery at cost (85% at site)	39,60,000	
Cash and bank	33,000	
Materials at cost	27,78,600	
Creditors for materials		11,33,660
Direct wages	14,60,800	
Site expenses	10,56,000	
Vehicles	40,00,000	
Furniture	7,00,000	
Office equipment	12,00,000	
Postage and Stationery	32,560	
Office expenses	6,88,600	

Particulars	Dr. (₹)	Cr. (₹)
Rates and taxes	28,160	
Fuel and power	9,30,600	
Outstanding wages		2,21,200
Advance rates and taxes	1,540	
	2,19,29,860	2,19,29,860

The contract price is ₹ 2,00,00,000 and work certified is ₹80,00,000. The cost of work uncertified is ₹ 9,60,000. Machinery costing ₹ 1,60,000 was returned to stores at the end of the year. Stock of material at site on 31st March 2022 was of the value of ₹40,000. Depreciation on Machinery, Vehicles and furniture are 10%, 15% and 10% respectively. You are required to calculate the profit from the contract.

ICAI'S ANSWER

Contract Account

Particulars	Amount (₹)	Amount (₹)	Particulars	Amount (₹)	Amount (₹)
To Materials		27,78,600	By material at site		40,000
To Direct wages	14,60,800		By Work in progress:		
Add: outstanding	2,21,200	16,82,000	- Work certified	80,00,000	
To Site expenses		10,56,000	- Work uncertified	9,60,000	89,60,000
To Office expenses		6,88,600			
To Postage and Stationery		32,560			
To Rates and taxes	28,160				
Less: Advance	(1,540)	26,620			
To Fuel and power		9,30,600			
To Depreciation*		10,06,600			
To Notional profit c/d		7,98,420			
		90,00,000			90,00,000

* Depreciation

(i) On Machinery = {10% on (₹39,60,000 × 0.85)} = ₹ 3,36,600

(ii) On Vehicles = 15% on ₹40,00,000 = ₹ 6,00,000

(iii) On Furniture = 10% on ₹7,00,000 = ₹ 70,000

Total = ₹ 9,86,800

PassBook's note: Out of syllabus (old scheme): contract costing (the contract account, notional profit on work certified and uncertified, depreciation of plant at site) is not in the new-scheme Paper 4 syllabus or the January 2027 Study Material, which has no Contract Costing chapter.

ICAI · Old P3 · MTP May 2023 · Series I · question p. 4-5 · answer p. 7-8

1175 MTP **Old P3 Nov 2022 S-II Q1(d)** 5 marks old scheme

Premier Construction Company undertook a contract for ₹50,00,000 on 1st August, 2021. On 31st March, 2022 when the accounts were closed, the following information was available:

Cost of work uncertified	₹12,00,000
Cash received	₹25,00,000 (80 of work certified)
Profit transferred to costing Profit and Loss account at the end of the year on Incomplete contract	₹8,00,000

CALCULATE:

(i) The value of work in progress certified

(ii) Degree of completion of contract

(iii) Notional Profit and

(iv) Cost of contract as on 31-03-2022.

ICAI'S ANSWER

(i) Value of work in progress certified:

Since, Cash Received of Rs. 25,00,000 is 80% of work certified

Therefore, Value of work in progress certified = $\frac{\text{Rs.}25,00,000}{80\%}$ = Rs. 31,25,000

(ii) Degree of completion of contract:

$= \frac{\text{Value of work certified}}{\text{Value of contract}} \times 100 = \frac{\text{Rs.}31,25,000}{\text{Rs.}50,00,000} \times 100 = 62.5\%$

(iii) Notional Profit:

Profit transferred to Costing Profit & Loss A/c = $\frac{2}{3}$ ×Notional Profit × $\frac{\text{Cash Received}}{\text{Value of work certified}}$

(Since contract completion is 62.5% i.e. more than 50%)

Or, Rs. 8,00,000 = $\frac{2}{3}$ × Notional Profit× $\frac{\text{Rs. } 25,00,000}{\text{Rs.}31,25,000}$

Notional Profit = Rs. 15,00,000

(iv) Cost of contract as on 31-03-2022:

= Value of Work certified + Cost of work uncertified – Notional profit

= Rs. 31,25,000 + Rs.12,00,000 – Rs. 15,00,000

= Rs. 28,25,000

PassBook's note: Out of syllabus (old scheme): contract costing (work certified, degree of completion, notional profit and the profit taken on an incomplete contract) is not in the new-scheme Paper 4 syllabus or the January 2027 Study Material, which has no Contract Costing chapter.

ICAI · Old P3 · MTP Nov 2022 · Series II · question p. 2 · answer p. 3

1176 MTP Old P3 May 2022 S-I Q1(b) 5 marks old scheme

The following expenses were incurred on a contract:

	(₹)
Materials purchased	7,20,000
Material drawn from stores	1,20,000
Wages	2,70,000
Plant issued	90,000
Chargeable expenses	90,000
Apportioned indirect expenses	30,000

The contract was for ₹ 24,00,000 and it commenced on April 1, 2021. The value of the work completed and certified upto 28th February, 2022 was ₹ 15,60,000 of which ₹ 12,48,000 was received in cash, the balance being held back as retention money by the contractee. The value of work completed subsequent to the architect's certificate but before 31st March, 2022 was ₹ 72,000. There were also lying on the site materials of the value of ₹ 48,000. It was estimated that the value of plant as at 31st March, 2022 was ₹ 36,000.

You are required to COMPUTE notional profit on the contract till the year ended 31st March, 2022.

ICAI'S ANSWER

Contract Account

	Particulars	(₹)		Particulars	(₹)
To	Material purchased	7,20,000	By	Work-in-progress:	
"	Stores issued	1,20,000		Value of work certified	15,60,000
"	Wages	2,70,000		Cost of work uncertified	72,000
"	Plant	90,000	"	Material unused	48,000
"	Chargeable expenses	90,000	"	Plant less depreciation	36,000
"	Indirect expenses	30,000			
"	Costing P&L A/c (Notional profit) (bal. fig.)	3,96,000			
		17,16,000			**17,16,000**

PassBook's note: Out of syllabus (old scheme): contract costing (the contract account, notional profit on work certified and uncertified, plant at site) is not in the new-scheme Paper 4 syllabus or the January 2027 Study Material, which has no Contract Costing chapter.

ICAI · Old P3 · MTP May 2022 · Series I · question p. 1-2 · answer p. 1-2

1177 MTP Old P3 May 2022 S-II Q1(b) 5 marks old scheme

A contractor prepares his accounts for the year ending 31st March each year. He commenced a contract on 1st July, 2021.

The following information relates to the contract as on 31st March, 2022:

	(₹)
Material issued	7,53,000
Wages	16,96,800
Salary to Foreman	2,43,900

A machine costing ₹ 7,80,000 has been on the site for 146 days, its working life is estimated at 7 years and its final scrap value at ₹ 45,000.

A supervisor, who is paid ₹ 24,000 p.m. has devoted one-half of his time to this contract.

All other expenses and administration charges amount to ₹ 4,09,500.

Material in hand at site costs ₹ 1,06,200 on 31st March, 2022.

The contract price is ₹ 60,00,000. On 31st March, 2022 two-third of the contract was completed. The architect issued certificates covering 50% of the contract price, and the contractor had been paid ₹ 22,50,000 on account.

PREPARE Contract A/c and show the notional profit or loss as on 31st March, 2022.

ICAI'S ANSWER

Contract Account

Particulars	(₹)	Particulars	(₹)
To Material issued	7,53,000	By Machine (Working note 1)	7,38,000
" Wages	16,96,800	" Material (in hand)	1,06,200
" Foreman's salary	2,43,900	" Works cost (balancing figure)	31,47,000
" Machine	7,80,000		
" Supervisor's salary (₹ 24,000 × 9)/2	1,08,000		
" Administrative charges	4,09,500		
	39,91,200		39,91,200
" Works cost	31,47,000	" Value of work certified	30,00,000
" Costing P&L A/c (Notional profit)	6,39,750	" Cost of work uncertified (Working Note 2)	7,86,750
	37,86,750		37,86,750

Working notes:

1. Written down value of Machine:

$$= \frac{₹\ 7{,}80{,}000 - ₹\ 45{,}000}{7 \text{ years}} \times \frac{146 \text{ days}}{365 \text{ days}} = ₹\ 42{,}000$$

Hence, the value of machine after the period of 146 days = ₹ 7,80,000 – ₹ 42,000 = ₹ 7,38,000

2. The cost of 2/3rd of the contract is ₹ 31,47,000

$\therefore$ Cost of 100% " " " " $\frac{₹\ 31{,}47{,}000}{2} \times 3 = ₹\ 47{,}20{,}500$

$\therefore$ Cost of 50% of the contract which has been certified by the architect is ₹ 23,60,250. Also, the cost of the contract, which has been completed but not certified by the architect is ₹ 7,86,750.

PassBook's note: Out of syllabus (old scheme): contract costing (the contract account, work certified and uncertified, notional profit) is not in the new-scheme Paper 4 syllabus or the January 2027 Study Material, which has no Contract Costing chapter.

ICAI · Old P3 · MTP May 2022 · Series II · question p. 1-2 · answer p. 1-2

1178 MTP **Old P3 Dec 2021 S-I** Q1(d) 5 marks old scheme

From the following particulars, COMPUTE Notional profit and estimated profit on a contract (which has been 80% complete):

	(₹)
Total expenditure to date	4,00,000
Estimated further expenditure to complete the contract (including contingencies)	22,000
Contract price	5,44,000
Work certified	4,89,600
Work uncertified	30,200
Cash received	3,91,680

ICAI'S ANSWER

Computation of Notional Profit

	(₹)
Value of work certified	4,89,600
Less: Cost of work certified	
(₹ 4,00,000 – ₹ 30,200)	3,69,800
Notional profit	**1,19,800**

Computation of Estimated Profit

		(₹)
Contract price		5,44,000
Less: Estimated total cost		
Cost of work to date	4,00,000	
Estimated further expenditure to complete the contract	22,000	4,22,000
Estimated profit		**1,22,000**

PassBook's note: Out of syllabus (old scheme): contract costing (notional profit on work certified and estimated profit on a contract) is not in the new-scheme Paper 4 syllabus or the January 2027 Study Material, which has no Contract Costing chapter (it names contract costing only as a method of costing).

ICAI · Old P3 · MTP Dec 2021 · Series I · question p. 2 · answer p. 3

1179 MTP **Old P3 Dec 2021 S-II** Q6(c) 5 marks old scheme

Same question, different ICAI answer: #1171 Old P3 RTP Jul 2021

WRITE NOTE on cost-plus-contracts.

ICAI'S ANSWER

Cost-Plus Contracts: These contracts provide for the payment by the contractee of the actual cost of construction plus a stipulated profit, mutually decided between the two parties.

The main features of these contracts are as follows:

(i) The practice of cost-plus contracts is adopted in the case of those contracts where the probable cost of the contracts cannot be ascertained in advance with a reasonable accuracy.

(ii) These contracts are preferred when the cost of material and labour is not steady and the contract completion may take number of years.

(iii) The different costs to be included in the execution of the contract are mutually agreed, so that no dispute may arise in future in this respect. Under such type of contracts, contractee is allowed to check or scrutinize the concerned books, documents and accounts.

(iv) Such a contract offers a fair price to the contractee and also a reasonable profit to the contractor.

The contract price here is ascertained by adding a fixed and mutually pre-decided component of profit to the total cost of the work.

PassBook's note: Out of syllabus (old scheme): cost plus contracts belong to contract costing, which is not in the new-scheme Paper 4 syllabus or the January 2027 Study Material (no Contract Costing chapter.

ICAI · Old P3 · MTP Dec 2021 · Series II · question p. 9 · answer p. 20

1180 MTP **Old P3 Jul 2021 S-II** Q2(b) 10 marks old scheme

Zed Limited obtained a contract No. 1551 for Rs. 150 lacs. The following details are available in respect of this contract for the year ended March 31, 2021:

	Rs.
Materials purchased	4,80,000
Materials issued from stores	15,00,000

	Rs.
Wages paid	21,00,000
Drawing and maps	1,80,000
Sundry expenses	45,000
Electricity charges	75,000
Plant hire expenses	1,80,000
Sub-contract cost	60,000
Materials returned to stores	90,000
Materials returned to suppliers	60,000

The following balances relating to the contract No. 1551 for the year ended on March 31, 2020 and March 31, 2021 are available:

	as on 31st March, 2020	as on 31st March, 2021
Work certified	36,00,000	1,05,00,000
Work uncertified	60,000	1,20,000
Materials at site	45,000	90,000
Wages outstanding	30,000	60,000

The contractor receives 70% of work certified in cash.

PREPARE Contract Account and Contractee's Account.

ICAI'S ANSWER

Contract No. 1551 Account for the year ended 31st March, 2021

Dr. Cr.

Particulars		Amount (Rs.)	Particulars	Amount (Rs.)
To Work in progress b/d:			By Material returned to stores	90,000
- Work certified		36,00,000	By Material returned to suppliers	60,000
- Work uncertified		60,000	By Stock (Materials) c/d	90,000
To Stock (Materials) b/d		45,000	By Work in progress c/d:	
To Material purchased		4,80,000	- Work certified	1,05,00,000
To Material issued		15,00,000	- Work uncertified	1,20,000
To Wages paid	21,00,000			

Particulars		Amount (Rs.)	Particulars	Amount (Rs.)
Less: Opening O/s	(30,000)			
Add: Closing O/s	60,000	21,30,000		
To Drawing and maps		1,80,000		
To Sundry expenses		45,000		
To Electricity charges		75,000		
To Plant hire expenses		1,80,000		
To Sub-contract cost		60,000		
To Notional profit c/d (balancing figure)		25,05,000		
		1,08,60,000		1,08,60,000

Dr. Contractee's Account Cr.

Particulars	Amount (Rs.)	Particulars	Amount (Rs.)
To Balance c/d (Rs. 1,05,00,000 × 70%)	73,50,000	By Balance b/d (70% of Rs. 36,00,000)	25,20,000
		By Bank A/c	48,30,000
	73,50,000		73,50,000

PassBook's note: Out of syllabus (old scheme): contract costing (the contract account with work certified and uncertified, notional profit and the contractee's account) is not in the new-scheme Paper 4 syllabus or the January 2027 Study Material, which has no Contract Costing chapter.

ICAI · Old P3 · MTP Jul 2021 · Series II · question p. 3 · answer p. 5

1181 MTP **Old P3 Nov 2020 S-I** Q4(b) 10 marks old scheme

A contractor has entered into a long term contract at an agreed price of ₹18,70,000 subject to an escalation clause for materials and wages as spelt out in the contract and corresponding actuals are as

follows:

	Standard		Actual	
Materials	**Qty (tons)**	**Rate (₹)**	**Qty (tons)**	**Rate (₹)**
A	6,000	50.00	6,050	48.00
B	3,000	80.00	2,950	79.00
C	2,500	60.00	2,600	66.00
Wages	**Hours**	**Hourly Rate (₹)**	**Hours**	**Hourly Rate (₹)**
X	3,000	70.00	3,100	72.00
Y	2,500	75.00	2,450	75.00
Z	3,000	65.00	3,100	66.00

Reckoning the full actual consumption of material and wages, the company has claimed a final price of ₹ 18,94,100. Give your ANALYSIS of admissible escalation claim and indicate the final price payable.

ICAI'S ANSWER

	Standard Qty/Hrs.	Standard Rate (₹)	Actual Rate (₹)	Variation in Rate (₹)	Escalation Claim (₹)
	(a)	**(b)**	**(c)**	**(d) = (c)–(b)**	**(e) =(a) × (d)**
Materials					
A	6,000	50.00	48.00	(–) 2.00	(–) 12,000
B	3,000	80.00	79.00	(–) 1.00	(–) 3,000
C	2,500	60.00	66.00	(+) 6.00	15,000
	Materials escalation claim: (A)				**0**
Wages					
X	3,000	70.00	72.00	(+) 2.00	6,000
Y	2,500	75.00	75.00	–	–
Z	3,000	65.00	66.00	(+) 1.00	3,000
	Wages escalation claim: (B)				9,000
	Final claim: (A + B)				9,000

Statement showing final price payable

Agreed price		₹ 18,70,000
Agreed escalation:		
Material cost	--	
Labour cost	₹ 9,000	₹ 9,000
Final price payable		₹ 18,79,000

The claim of ₹ 18,94,100 is based on the total increase in cost. This can be verified as shown below:

Statement showing total increase in cost

	Standard Cost			Actual Cost			Increase/ (Decrease)
	Qty/hrs	**Rate (₹)**	**Amount (₹)**	**Qty/hrs**	**Rate (₹)**	**Amount (₹)**	
	(a)	**(b)**	**(c) = (a)×(b)**	**(d)**	**(e)**	**(f) =(d) × (e)**	**g = (f) – (c)**
I. Materials							
A	6,000	50.00	3,00,000	6,050	48.00	2,90,400	
B	3,000	80.00	2,40,000	2,950	79.00	2,33,050	
C	2,500	60.00	1,50,000	2,600	66.00	1,71,600	
			6,90,000			6,95,050	5,050
II. Wages							
X	3,000	70.00	2,10,000	3,100	72.00	2,23,200	
Y	2,500	75.00	1,87,500	2,450	75.00	1,83,750	
Z	3,000	65.00	1,95,000	3,100	66.00	2,04,600	
			5,92,500			6,11,550	19,050
							24,100

Contract price	₹ 18,70,000
Add: Increase in cost	₹ 24,100
The final price claimed by the company	₹ 18,94,100

This claim is not admissible because escalation clause covers only that part of increase in cost, which has been caused by inflation.

Note: It is fundamental principle that the contractee would compensate the contractor for the increase in costs which are caused by factors beyond the control of contractor and not for increase in

costs which are caused due to inefficiency or wrong estimation.

PassBook's note: Out of syllabus (old scheme): the escalation clause of a long-term contract (admissible escalation claim and final contract price) is contract costing, which is not in the new-scheme Paper 4 syllabus or the January 2027 Study Material (no Contract Costing chapter; The Study Material does not teach escalation clauses).

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A.2 Topics no longer in the syllabus

4 questions ICAI marks out of syllabus, or whose topic has left the syllabus. For reference only.

1182 EXAM Old P3 Nov 2023 Q1(d) 5 marks old scheme

Unique Construction Limited commenced a contract on 01.08.2022. The total contract price was ₹ 96,00,000. The following information was available from their costing records as at 31.03.2023:

Material consumed	₹ 35,91,000
Wages paid	₹ 9,65,000
Wages outstanding as on 31.03.2023	₹ 75,000
Plant issued to site on 01.08.2022	₹ 7,50,000
Direct expenses	₹ 1,96,650
General overheads	₹ 2,08,000

A supervisor who was paid ₹ 18,000 per month, had spent 40% of his time on this contract. Plant costing ₹ 60,000 was transferred to other contracts on 31.12.2022. Plant was to be depreciated at 15% per annum on straight line method (SLM) basis. On 31.03.2023, 60% of the contract was completed. The architect's certificate had been issued covering 50% of the contract price.\

Prepare a Contract account and show the notional profit or loss as on 31.03.2023.

ICAI'S ANSWER

Contract A/c for the year ending 31/03/23

Particulars	(₹)	Particular	(₹)
To Material	35,91,000	By work in Progress:	
To Wages:		Work certified 48,00,000	
Current Wages 9,65,000		Work uncertified 8,61,000	56,61,000
Add: outstanding Wages 75,000	10,40,000	By Plant (Transferred) 60,000	
To Plant	7,50,000	Less: Dep @ 15% for 5 months 3,750	56,250
To Direct Expenses	1,96,650	By Plant at site	6,21,000
To General overheads	2,08,000	(7,50,000 - 60,000-69,000)	
To Supervision Salary (18,000 × 8 × 40%)	57,600		

Particulars	(₹)	Particular	(₹)
To Notional profit c/d	4,95,000		
	63,38,250		**63,38,250**

Working Note:

Calculation of cost of work uncertified:

Particular	(₹)
Cost incurred till date	51,66,000
Estimated total cost (51,66,000/60%)	86,10,000
Cost of work certified (86,10,000 × 50%)	43,05,000
Cost of uncertified work (51,66,000 – 43,05,000)	**8,61,000**

The solution can also be presented in following way and depreciation can be calculated as shown below:

Contract A/c for the year ending 31/03/23

Particulars	(₹)	Particular	(₹)
To Material	35,91,000		
To Wages:			
Current Wages 9,65,000			
Add: outstanding Wages 75,000	10,40,000		
To Depreciation on plant	72,750		
To Direct Expenses	1,96,650		
To General overheads	2,08,000		
To Supervision Salary (18,000 × 8 × 40%)	57,600	By work Cost (Bal Fig.)	51,66,000
	51,66,000		**51,66,000**
To work cost	51,66,000	By work certified	48,00,000
To Notional profit c/d	4,95,000	By Work uncertified	8,61,000
	56,61,000		**56,61,000**

Working Note:

1. Calculation of cost of work uncertified:

Particular	(₹)
Cost incurred till date	51,66,000
Estimated total cost (51,66,000/60%)	86,10,000
Cost of work certified (86,10,000 × 50%)	43,05,000
Cost of uncertified work (51,66,000 – 43,05,000)	**8,61,000**

2. Calculation of Depreciation

Plant value (7,50,000 – 60,000) = ₹ 6,90,000 used for 8 months and plant value ₹ 60,000 used for 5 months.

Depreciation amount for 8 months = (6,90,000 x 15% x 8 months) /12 = ₹ 69,000

Depreciation amount for 5 months = (60,000 x 15% x 5 months) /12 = ₹ 3,750

Total depreciation amount = ₹ 72,750

PassBook's note: Old-scheme contract costing (contract account, work certified and uncertified, notional profit). Contract costing is not in the new-scheme Paper 4 syllabus or the current Study Material (its Job Costing chapter covers job costing only), so the item is out of syllabus and carries no tag. The question ends "covering 50% of the contract price.\" with a stray backslash, as printed.

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1183 EXAM **Old P3 Dec 2021** Q4(a) 10 marks old scheme

A construction company has obtained a contract of ₹ 30 lakhs contract price.

The following details are available in respect of this contract for the year ended March 31, 2021:

Particulars	(₹)
Materials purchased	2,00,000
Materials issued from stores	8,00,000
Wages paid	1,50,000
Plant Supervisor Salary	2,40,000
Drawing and maps	50,000
Sundry expenses	30,000
Electricity charges	40,000
Plant hire expenses paid	75,000
Sub-contract cost	40,000
Materials returned to stores	35,000
Materials returned to suppliers	50,000

The following balances related to the contract for the year ended on March 31, 2020 and March 31, 2021 are available:

	As on 31st March, 2020 (₹)	As on 31st March, 2021 (₹)
Work certified	2,50,000	70% of Contract Price
Work uncertified	10,000	?
Materials at site	35,000	25,000
Wages outstanding	15,000	22,000
Plant hire charges outstanding	20,000	15,000

Further informations are as under:

1. An additional plant was used for 270 days costing ₹ 5,00,000 with a residual value of ₹ 20,000 having life of 4 years.

2. During the year, material costing ₹ 40,000 was sold for ₹ 20,000.

3. Plant supervisor has devoted 1/3rd of his time to this contract.

4. As on 31.03-2021, 80% of the contract was completed.

You are required to prepare Contract Account and show the notional profit or loss as on 31st March, 2021 (Assume 360 days in a year).

ICAI'S ANSWER

Contract A/c

Dr. | Cr.

Particulars		Amount (₹)	Particulars	Amount (₹)
To Opening Work in progress			By Material returned to store	35,000
- Work certified	2,50,000		By Material returned to suppliers	50,000
- Work uncertified	10,000	2,60,000	By Costing P&L (Loss on sale of material)	20,000
To Material at site		35,000	By Material Sold	20,000
To Material purchased		2,00,000	By Material at site	25,000
To Stores		8,00,000	By Works cost (Bal. fig.)	17,02,000
To Wages	1,50,000			
Add: Closing O/s wages	22,000			
Less: Opening O/s wages	(15,000)	1,57,000		

Particulars		Amount (₹)	Particulars	Amount (₹)
To Plant supervisor salary (2,40,000 × 1/3)		80,000		
To Drawing and maps		50,000		
To Sundry expenses		30,000		
To Electricity charges		40,000		
To Plant hire expenses	75,000			
Add: O/s at end	15,000			
Less: O/s at beginning	(20,000)	70,000		
To Sub-contract		40,000		
To Depreciation $\left[\frac{5,00,000-20,000}{4} \times \frac{270}{360}\right]$		90,000		
		18,52,000		18,52,000
To works cost		17,02,000	By work in progress:	
To Costing P& L (Notional profit)		6,10,750	Work certified 21,00,000	
			Work uncertified 2,12,750	23,12,750
		23,12,750		23,12,750

Working Note:

Calculation of Value of work uncertified

Cost incurred till date	17,02,000
Estimate total cost $\left[\frac{17,02,000}{80\%}\right]$	21,27,500
Cost of work certified till date (21,27,500 × 70%)	14,89,250
Cost of uncertified work (17,02,000 – 14,89,250)	2,12,750

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1184 MTP **Old P3 Nov 2023 S-II** Q6(d) 5 marks old scheme

State the advantages and disadvantage of cost- plus contract.

ICAI'S ANSWER

Cost plus contracts have the following advantages and disadvantages:

Advantages:

(i) The Contractor is assured of a fixed percentage of profit. There is no risk of incurring any loss on the contract.

(ii) It is useful specially when the work to be done is not definitely fixed at the time of making the estimate.

(iii) Contractee can ensure himself about 'the cost of the contract', as he is empowered to examine the books and documents of the contractor to ascertain the veracity of the cost of the contract.

Disadvantages - The contractor may not have any inducement to avoid wastages and effect economy in production to reduce cost.

PassBook's note: Out of syllabus (old scheme): cost plus contracts belong to contract costing, which old Paper 3 taught; The new-scheme Paper 4 syllabus and the January 2027 Study Material have no contract costing chapter and never discuss cost plus contracts. No tag. The question's "disadvantage" and "cost- plus contract" are as printed.

ICAI · Old P3 · MTP Nov 2023 · Series II · question p. 7 · answer p. 14

1185 MTP **Old P3 Nov 2022 S-I** Q1(c) 5 marks old scheme

EFF Limited, a construction company, entered into a contract for ₹ 14,50,000 on 1st July, 2021. On 30th June, 2022 when the accounts were closed, the following details were gathered about the contract:

	Amount (₹)
Materials purchased	2,90,000
Wages paid	1,30,500
General expenses	29,000
Materials on hand (30-6-2022)	72,500
Wages accrued (30-6-2022)	14,500
Work certified	5,80,000
Work uncertified	43,500
Cash received	4,35,000

The above contract contained "Escalation clause" which read as follows:

"In the event of increase in the prices of materials and rates of wages by more than 5%, the contract price would be increased accordingly by 25% of the rise in the cost of materials and wages beyond 5% in each case."

It was found that since the date of signing the agreement, the prices of materials and wage rates increased by 25% leading to increase the values from the very beginning. However, the value of the work certified does not take into account the effect of the above clause.

You are required to CALCULATE the 'value of work certified' after taking the effect of 'Escalation Clause' as on 30th June, 2022.

ICAI'S ANSWER

Workings:

(i) Percentage of work certified:

$$\frac{\text{Value of work certified}}{\text{Contract price}} \times 100 = \frac{₹5,80,000}{₹14,50,000} \times 100 = 40\%$$

(ii) Value of material and labour used in the contract:

Particulars	Amount (₹)	Amount (₹)
Material purchased	2,90,000	
Less: Material on hand (30-06-2022)	(72,500)	2,17,500
Wages paid	1,30,500	
Add: Wages accrued (30-06-2022)	14,500	1,45,000
		3,62,500

Price of materials and wages has been increased by 25%, the value before price increase is:

$$\frac{₹3,62,500}{125} \times 100 = ₹\ 2,90,000$$

(iii) Calculation of Value of work certified:

The value of the contract would be increased by 25% of the price increased beyond 5%.

Price increased beyond 5% = ₹ (3,62,500 – 2,90,000) – 5% of ₹ 2,90,000

= ₹ 72,500 – ₹ 14,500 = ₹ 58,000

Value of contract would be increased by 25% of ₹ 58,000 = ₹ 14,500

Therefore, the revised contract value = ₹ 14,50,000 + ₹ 14,500 = ₹ 14,64,500

Calculation of the Value of work certified after taking the effect of escalation clause:

= Revised contract value × Percentage of work certified

= ₹ 14,64,500 × 40% = ₹ 5,85,800

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